Lab Tools & Equipment - SOP Handbook

How to use, safety and care for every tool in the vocational labs

1333 SOPs

Prepared by - Subodh Kumar

Agriculture & Gardening

SOP-0339 Automatic Shade System
SOP-1030 Basket
SOP-0185 Battery Operated Sprayer
SOP-0652 Bonsai Tool Set
SOP-0399 Budding and Grafting Knife
SOP-0202 Combine Harvester
SOP-0788 Cono Weeder
SOP-0075 Cultivator
SOP-0576 Dibbler
SOP-0445 Digging Fork
SOP-0020 Disc Harrow
SOP-0206 Disc Plough
SOP-0419 Drip Irrigation System
SOP-0800 Drum Seeder
SOP-0224 Fertigation Unit
SOP-0979 Fertilizer
SOP-0588 Fertilizer Spreader
SOP-0336 Fogging System
SOP-0300 Foot Sprayer
SOP-1052 Fumigation Cover
SOP-0310 Fumigation Probe
SOP-0522 Garden Knife
SOP-0470 Garden Tool Set
SOP-0231 Garden Tractor
SOP-0593 Grain Sampling Probe
SOP-0234 Grass Cutter
SOP-0677 Greenhouse
SOP-0820 Grow Light
SOP-1135 Growing Medium
SOP-0494 Hand Cultivator
SOP-0679 Hand Rotary Duster
SOP-0296 Hand Sprayer
SOP-0429 Hand Trowel
SOP-0140 Hedge Cutter
SOP-0472 Hedge Shear
SOP-0362 Hoe
SOP-0523 Hori Hori Knife
SOP-1062 Hose Pipe
SOP-0833 Irrigation System
SOP-0015 Knapsack Sprayer
SOP-0526 Lopping Shear
SOP-0840 Manual Transplanter
SOP-0430 Moisture Meter
SOP-0252 Mould Board Plough
SOP-1195 Nursery Bag
SOP-1065 Organic Manure
SOP-0259 Paddy Puddler
SOP-0260 Paddy Transplanter
SOP-0327 Pesticide
SOP-1058 Pheromone Trap
SOP-0931 Plant Growth Regulator
SOP-0702 Planting Board
SOP-0022 Plough
SOP-0029 Power Sprayer
SOP-0010 Power Tiller
SOP-0434 Pruning Knife
SOP-0616 Pruning Saw
SOP-0393 Pruning Secateurs
SOP-0396 Rake
SOP-0618 Rat Trap
SOP-0267 Reaper
SOP-0713 Rocker Sprayer
SOP-0030 Rotavator
SOP-1148 Sample Bag
SOP-0621 Sample Divider
SOP-1304 Seed Box
SOP-0041 Seed Cleaner Grader
SOP-0875 Seed Counter
SOP-1305 Seed Drying Tray
SOP-0269 Seed Germinator
SOP-0876 Seed Testing Kit
SOP-0031 Seed Treating Equipment
SOP-0042 Seed-cum-Fertilizer Drill
SOP-1103 Seedling Tray
SOP-0479 Sickle
SOP-0725 Soil Scoop
SOP-0032 Soil Testing Kit
SOP-0161 Sorting Machine
SOP-0380 Spade
SOP-0727 Sprayer Nozzle
SOP-0453 Sprinkler Irrigation System
SOP-1310 Stake
SOP-0063 Straw Reaper
SOP-1158 Tarpaulin
SOP-0163 Thresher
SOP-0024 Tractor
SOP-0634 Tree Pruner
SOP-0460 Watering Can
SOP-0551 Weeder
SOP-0638 Weeding Fork
SOP-0742 Weeding Hook
SOP-0743 Wheel Hoe
SOP-0744 Wheelbarrow
SOP-0926 Winnowing Fan

Animal Husbandry & Dairy

SOP-0065 AI Cylinder
SOP-0066 AI Gun
SOP-0169 Aeration System
SOP-0552 Animal Awakener
SOP-1075 Animal Feed
SOP-0173 Aquarium Heater
SOP-0758 Automatic Waterer
SOP-0187 Beak Trimmer

SOP-1231 Biosecurity Material
SOP-0771 Boat
SOP-0563 Branding Numbers
SOP-0776 Breeding Tank
SOP-0193 Brooder
SOP-0194 Bulk Milk Cooler
SOP-0071 Bull Exerciser
SOP-0565 Bull Leader
SOP-0566 Bull Nose Punch
SOP-0567 Bull Nose Ring
SOP-0072 Burdizzo Castrator
SOP-0195 Butter Churner
SOP-0568 CMT Paddle
SOP-0780 Chick Box
SOP-0117 Cream Separator
SOP-0204 Curd Incubator
SOP-0578 Drenching Bottle
SOP-0803 Egg Candler
SOP-0213 Egg Packing Machine
SOP-1254 Egg Tray
SOP-0214 Egg Washer
SOP-0077 Electric Dehorner
SOP-1258 Feeding Tray
SOP-0467 Fish Net
SOP-1260 Fish Rearing Tank
SOP-1261 Fish Transport Bag
SOP-0232 Gerber Centrifuge
SOP-0233 Ghee Boiler
SOP-0953 Grooming Brush
SOP-0823 Hatchery Jar
SOP-0599 Hoof Trimmer
SOP-0525 Lactometer
SOP-0086 Liquid Nitrogen
SOP-0951 Milk Can
SOP-0960 Milk Collection Accessories
SOP-0249 Milk Filter Machine
SOP-0841 Milk Storage Tank
SOP-0952 Milk Strainer
SOP-0842 Milk Testing Kit
SOP-0955 Milking Bucket
SOP-0607 Mouth Gag
SOP-0253 Multipurpose Vat
SOP-0956 Muzzle
SOP-0958 Nest Box
SOP-0851 Paneer Press
SOP-0532 Poultry Drinker
SOP-0703 Poultry Feeder
SOP-1206 Semen Straw
SOP-0933 Teat Dip
SOP-0103 Trocar and Cannula
SOP-0912 Vaccinator

Audio-Visual, Camera & Media

SOP-0488 Audio/Video Recorder
SOP-0558 Binoculars
SOP-0770 Bluetooth Device
SOP-0367 CCTV Camera
SOP-0654 CCTV Recorder
SOP-0777 Camera Trap
SOP-0519 Digital Camera
SOP-0381 Display Screen
SOP-0597 Headphones
SOP-0828 Infrared Camera
SOP-0527 Microphone
SOP-0860 Presentation Remote
SOP-0366 Projection Screen
SOP-0353 Projector
SOP-1292 Projector Mounting Kit
SOP-0478 Public Address System
SOP-0542 Smart Board
SOP-0390 Speaker
SOP-0894 Studio Lighting Kit
SOP-0908 Tripod
SOP-0736 Turntable
SOP-0911 VR Headset
SOP-0916 Video Camera
SOP-0737 Video Conferencing System
SOP-0917 Voice Recorder
SOP-0741 Webcam

Automotive

SOP-1048 Air Filter
SOP-0340 Automotive Cooling System
SOP-0044 Automotive Electrical Trainer Kit
SOP-1008 Automotive Spare Part
SOP-0762 Ball Joint Separator
SOP-0562 Body Repair Spoon
SOP-0069 Brake Roller Tester
SOP-0402 Brake System
SOP-0200 Car Lift
SOP-0073 Chassis Dynamometer
SOP-0518 Clutch Assembly
SOP-0782 Clutch Puller
SOP-0569 Compression Gauge
SOP-1129 Cut Section Model
SOP-0573 Demonstration Kit
SOP-0794 Diagnostic Kit
SOP-0121 Diagnostic Scanner
SOP-0577 Drain Pan
SOP-0048 EV Charger
SOP-0039 Electric Vehicle
SOP-0123 Electric Vehicle Kit
SOP-0078 Engine
SOP-0584 Equipment Stand
SOP-0219 Exhaust Gas Analyzer
SOP-0808 Exhaust System

SOP-0080 Four Wheeler
SOP-0330 Fuel
SOP-0815 Fuel Pressure Tester
SOP-0230 Fuel System
SOP-1264 Funnel
SOP-0819 Grease Gun
SOP-0496 Hydrometer
SOP-0683 Inspection Mirror
SOP-0055 Jack
SOP-0085 Lifting & Jacking Equipment
SOP-0604 Lug Wrench
SOP-1140 Magnetic Tool Holder
SOP-1193 Magnetic Tray
SOP-0450 Oil Can
SOP-1057 Oil Filter
SOP-0957 Oil Measure Container
SOP-0529 Oil Pressure Gauge
SOP-1284 Oil Seal
SOP-0709 Radiator
SOP-1099 Radiator Pressure Cap
SOP-0884 Slide Hammer
SOP-1071 Spark Plug
SOP-0098 Spark Plug Cleaner
SOP-0417 Special Tool
SOP-0630 Stud Extractor
SOP-0730 Suspension System
SOP-0547 Tappet Puller
SOP-1111 Thermostat
SOP-0907 Transmission Assembly
SOP-0104 Two-Wheeler
SOP-0486 Tyre Pressure Gauge
SOP-0909 Tyre Repair Kit
SOP-0914 Vehicle Chassis
SOP-0915 Vehicle Dashboard
SOP-0635 Vehicle Ramp
SOP-0291 Wheel Alignment
SOP-0292 Workshop Equipment

Beauty, Wellness & Fitness

SOP-0752 Abdominal Wheel
SOP-0756 Ankle Weights
SOP-0511 Applicator Brush
SOP-1225 Ayurvedic Product
SOP-0761 Balance Board
SOP-0555 Barber Brush
SOP-0184 Bashpa Swedan
SOP-0107 Basti Yantra
SOP-0650 Bendy Roller
SOP-0651 Blackhead Remover
SOP-0773 Body Reformer
SOP-1080 Bolster
SOP-0199 Cable Machine
SOP-0365 Comb
SOP-0570 Cosmetic Pencil Sharpener
SOP-1010 Cosmetic Product
SOP-0046 Curling Iron
SOP-0574 Dermascope
SOP-0669 Dumbbell
SOP-0079 Exfoliation Machine
SOP-1256 Eye Pillow
SOP-0585 Eyelash Brush
SOP-1182 Facial Bed
SOP-0126 Facial Machine
SOP-0127 Facial Steamer
SOP-0673 Foot Scraper
SOP-0130 Foot Spa
SOP-0132 Galvanic Machine
SOP-0948 Hair Brush
SOP-1012 Hair Clip
SOP-0138 Hair Clipper
SOP-1089 Hair Connector
SOP-0051 Hair Curler (Electric)
SOP-0026 Hair Dryer
SOP-1035 Hair Pin
SOP-0982 Hair Roller
SOP-0009 Hair Steamer
SOP-0306 Hair Straightener
SOP-0824 Heart Rate Monitor
SOP-0237 High Frequency Machine
SOP-1055 Highlighting Cap
SOP-0142 Hood Dryer
SOP-0143 Hot Hair Roller
SOP-0835 Kettlebell
SOP-0839 Magnifier Lamp
SOP-0605 Makeup Brush
SOP-0474 Manicure Pedicure Kit
SOP-1141 Massage Bed
SOP-1279 Meditation Cushion
SOP-0254 Nadi Swedana Yantra
SOP-0692 Nail Cutter
SOP-0693 Nail File
SOP-0608 Neti Pot
SOP-0849 Oil Pouring Vessel
SOP-0153 Paraffin Heater
SOP-0477 Perm Curler
SOP-1094 Perming Cap
SOP-0863 Pull-up Bar
SOP-0866 Push-up Bar
SOP-0536 Razor
SOP-0619 Resistance Band
SOP-0436 Shampoo Unit
SOP-0880 Shirobasti Cap
SOP-0722 Shirodhara Equipment
SOP-1104 Shower Cap
SOP-0893 Step Platform
SOP-0319 Stone Therapy Set

SOP-1317 Sutra Neti
SOP-0102 Tattoo Machine
SOP-0483 Thinning Scissors
SOP-0437 Tinting Brush
SOP-1327 Tratak Stand
SOP-0165 Wax Heater
SOP-0922 Weight Plates
SOP-1333 Yantra
SOP-0927 Yoga Block
SOP-1219 Yoga Clothing
SOP-0487 Yoga Mat

Computers, IT & Office

SOP-0321 3D Pen
SOP-0167 3D Printer
SOP-0746 3D Printer Bed Leveling Tool
SOP-0747 3D Printer Build Plate
SOP-0748 3D Scanner
SOP-0559 Biometric Scanner
SOP-0341 Blueprint Machine
SOP-1235 Cable Organizer
SOP-0513 Calculator
SOP-1240 Clipboard
SOP-1082 Computer Cable
SOP-0784 Computer Keyboard
SOP-0492 Computer Peripheral
SOP-0785 Computer Repair Tool Kit
SOP-0352 Desktop Computer
SOP-0420 External Storage Device
SOP-0810 Fax Machine
SOP-0587 Feedback Kiosk
SOP-0594 Graphics Tablet
SOP-0364 Laptop
SOP-1063 Marker
SOP-1281 Mouse Pad
SOP-1196 Optical Disc
SOP-0307 Paper Shredder
SOP-0501 Photocopier
SOP-0356 Printer
SOP-1201 Printer Cartridge
SOP-0347 Printing Equipment
SOP-0387 Scanner
SOP-0877 Server
SOP-0724 Smartphone
SOP-1154 Stapler
SOP-1316 Stylus Pen
SOP-0506 Tablet

Construction & Masonry

SOP-0561 Boaster
SOP-0514 Caulking Tool
SOP-0811 Filling Board
SOP-0468 Float
SOP-0950 Mortar Pan
SOP-1287 Paint Bucket
SOP-0530 Paint Roller
SOP-1288 Paint Stirrer
SOP-0961 Painting Edger
SOP-0615 Plaster Scratcher
SOP-0707 Putty Knife
SOP-0622 Scabbling Hammer
SOP-0624 Scutch Hammer
SOP-0627 Spalling Hammer
SOP-0629 Straight Edge
SOP-0385 Trowel
SOP-0636 Water Level

Consumables & Raw Materials

SOP-1165 3D Printer Consumable
SOP-0987 Adhesive
SOP-1029 Adhesive Tape
SOP-1221 Aggregate
SOP-1116 Aluminium Foil
SOP-1167 Bag
SOP-1119 Bamboo Mat
SOP-1077 Beads
SOP-1122 Brick
SOP-1173 Button
SOP-1049 Cable Tie
SOP-1175 Candle
SOP-1081 Cement
SOP-1124 Chain
SOP-1237 Chalk Powder
SOP-0326 Chemical Reagent
SOP-1238 Clay
SOP-0968 Cleaning Material
SOP-1127 Concrete Block
SOP-1083 Cotton Wool
SOP-1244 Craft Wire
SOP-1245 Culture Media
SOP-0939 Disinfectant
SOP-1251 Distilled Water
SOP-0986 Fabric
SOP-1032 Fastener (Screws/Nails/Bolts)
SOP-1185 Food Ingredient
SOP-0940 Food Preservative
SOP-1187 Hacksaw Blade
SOP-1053 Hand Sanitizer
SOP-1054 Hand Wash
SOP-1017 Hinge
SOP-1189 Hook
SOP-1267 Immersion Oil
SOP-1191 Jute Fibre
SOP-1091 Label
SOP-1272 Lace
SOP-1043 Lubricant
SOP-1275 Marking Flag

SOP-1276 Marking Powder
SOP-1056 Masking Tape
SOP-1282 Naphthalene Ball
SOP-1283 Newspaper
SOP-0973 Packaging Material
SOP-1018 Paint
SOP-1004 Paper
SOP-1289 Personal Care Product
SOP-1097 Poly Bag
SOP-1145 Polythene Sheet
SOP-1294 Raw Material
SOP-0332 Refrigerant Gas
SOP-0941 Resin
SOP-1044 Rope
SOP-1147 Rubber Sheet
SOP-1101 Sand
SOP-1205 Sandpaper
SOP-1301 Sealant
SOP-1109 Sponge
SOP-0963 Stationery
SOP-1155 Stone
SOP-1028 Tracing Paper
SOP-1330 Varnish
SOP-1114 Water
SOP-1217 Workshop Consumable

Drones & Aviation

SOP-0037 Drone
SOP-0579 Drone Frame
SOP-0025 Drone Kit
SOP-0582 Electronic Speed Controller
SOP-0128 Fixed Wing UAV
SOP-0589 Flight Controller
SOP-0672 Flight Simulator
SOP-0816 GPS Module
SOP-0837 LiDAR Sensor
SOP-0084 LiPo Battery
SOP-0088 Motor Thrust Stand
SOP-0932 Propeller
SOP-0705 Propeller Balancer
SOP-0732 Telemetry Module

Electrical & Electronic Components & Kits

SOP-0749 AI Device
SOP-0750 AI Learning Kit
SOP-1222 Appliance Spare Parts
SOP-0653 Breadboard
SOP-1234 Buzzer
SOP-1123 Capacitor
SOP-0312 Circuit Breaker
SOP-1179 Conduit Pipe
SOP-0662 DC Motor
SOP-1247 DIN Rail
SOP-0795 Dimmable Lighting System
SOP-0664 Display Module
SOP-0209 Distribution Board
SOP-0928 Earthing Material
SOP-0804 Electric Bell
SOP-0013 Electric Motor
SOP-1051 Electrical Connector
SOP-1085 Electrical Socket
SOP-0999 Electrical Switch
SOP-0581 Electrical Trainer Kit
SOP-0988 Electrical Wire
SOP-1016 Electronic Component
SOP-0305 Extension Board
SOP-0586 Fan
SOP-1186 Fuse
SOP-0235 HVAC Trainer Kit
SOP-1024 Insulation Tape
SOP-0832 IoT Project Kit
SOP-1190 Jumper Wire
SOP-1268 LED
SOP-1269 LED Light
SOP-1092 Lamp Holder
SOP-0394 Microcontroller Board
SOP-0845 Motor Driver Module
SOP-0150 Motor Starter
SOP-0846 Multivibrator Kit
SOP-0610 Ohm's Law Kit
SOP-0855 Peltier Module
SOP-1096 Plug Top
SOP-1291 Potentiometer
SOP-0704 Power Adapter
SOP-1067 Printed Circuit Board
SOP-0867 RF Module
SOP-0869 Rectifier
SOP-0503 Relay
SOP-1295 Resistor
SOP-0370 Sensor Module
SOP-0878 Servo Motor
SOP-0626 Single Board Computer
SOP-0895 Surge Protector
SOP-0550 Torch
SOP-0325 Transformer
SOP-0740 Water Level Indicator
SOP-0924 Wi-Fi Module

Electrical & Electronic Test Instruments

SOP-0462 Ammeter
SOP-0556 Battery Tester
SOP-0412 Clamp Meter
SOP-0443 Continuity Tester
SOP-0294 Digital Multimeter
SOP-0038 Earth Tester
SOP-0216 Electrical Safety Analyzer
SOP-0314 Energy Meter

SOP-0309 Frequency Meter
SOP-0345 Function Generator
SOP-0682 Infrared Thermometer
SOP-0147 Insulation Resistance Tester
SOP-0687 Leak Detector
SOP-0499 Lux Meter
SOP-0691 Motherboard Diagnostic Card
SOP-0298 Non-contact Voltage Detector
SOP-0089 Oscilloscope
SOP-0302 Phase Sequence Meter
SOP-0400 Phase Tester
SOP-0533 Power Supply
SOP-0708 Radiation Meter
SOP-0881 Signal Detector
SOP-0456 Tachometer
SOP-0631 Temperature Indicator
SOP-0901 Test Equipment
SOP-0033 Voltmeter
SOP-0034 Wattmeter

Food Processing & Kitchen

SOP-0398 Baking Mould
SOP-1168 Baking Paper
SOP-0464 Bar Tool
SOP-0110 Blanching Unit
SOP-0190 Boiler
SOP-1232 Bottle
SOP-0111 Bottle Sealing Machine
SOP-0191 Bottle Washing Machine
SOP-0192 Bread Proofer
SOP-0655 Cake Decorating Tool
SOP-0657 Can Opener
SOP-0112 Can Sealer
SOP-0658 Chikki Cutter
SOP-0114 Chiller
SOP-1178 Chopping Board
SOP-0203 Cookie Depositor
SOP-0295 Cookware
SOP-1180 Cooling Rack
SOP-0118 Crown Corking Machine
SOP-0119 Cup Sealing Machine
SOP-0466 Decanter
SOP-0036 Deep Freezer
SOP-0120 Dehuller
SOP-0207 Dishwasher
SOP-0047 Dough Kneader
SOP-0666 Dough Scraper
SOP-0076 Dough Sheeter
SOP-0210 Drying Chamber
SOP-0215 Electric Stove
SOP-0125 Extruder Machine
SOP-0129 Food Dehydrator
SOP-0227 Fruit Mill
SOP-0228 Fruit Pulper
SOP-0229 Fruit and Vegetable Washer
SOP-0315 Gas Lighter
SOP-0050 Gas Stove
SOP-0134 Grading Machine
SOP-0595 Grater
SOP-0135 Grinding Mill
SOP-0236 Halving Machine
SOP-0082 Hand Blender
SOP-0141 Homogenizer
SOP-0144 Ice Cream Maker
SOP-1266 Ice Cube Tray
SOP-0145 Induction Cooktop
SOP-0244 Juice Extractor
SOP-0247 Khoa Pan
SOP-0524 Kitchen Knife
SOP-0684 Kitchen Tongs
SOP-1139 Ladle
SOP-0944 Measuring Jug
SOP-0945 Measuring Spoon
SOP-0149 Microwave Oven
SOP-0250 Millet Cleaning Machine
SOP-0027 Mixer Grinder
SOP-0946 Mixing Bowl
SOP-0843 Mixing Tank
SOP-0257 Outdoor Cooking Equipment
SOP-0016 Oven
SOP-0258 Packaging Machine
SOP-0057 Pasteurizer
SOP-0962 Pastry Brush
SOP-0612 Peeler
SOP-0262 Peeling Machine
SOP-0700 Piping Bag
SOP-0701 Pizza Cutter
SOP-0154 Planetary Mixer
SOP-0857 Plastic Wrap Dispenser
SOP-0092 Pouch Sealing Machine
SOP-0094 Pressure Cooker
SOP-0156 Pressure Stove
SOP-0265 Puffing Machine
SOP-0422 Refrigerator
SOP-0268 Roaster
SOP-0504 Rolling Pin
SOP-1150 Scoop
SOP-0384 Sieve
SOP-0272 Soy Milk Plant
SOP-0416 Spatula
SOP-0277 Steam Jacketed Kettle
SOP-0279 Steamer
SOP-0990 Storage Container
SOP-0457 Timer
SOP-0903 Tofu Press
SOP-1159 Toothpick Holder

SOP-0282 Transfer Pump
SOP-1212 Tray
SOP-0284 Vacuum Evaporator
SOP-0285 Vacuum Packaging Machine
SOP-0919 Washing Tank
SOP-0290 Weighing and Filling Machine
SOP-0641 Whisk
SOP-0166 Wick Stove

Furniture, Storage & Display

SOP-1120 Bar Counter
SOP-1230 Bed
SOP-1169 Bedside Locker
SOP-1170 Bench
SOP-1233 Business Card Holder
SOP-0974 Chair
SOP-1125 Classroom Desk
SOP-1126 Clock
SOP-0978 Cupboard
SOP-0663 Desk Lamp
SOP-1249 Desk Mat
SOP-1250 Desk Organizer
SOP-1011 Display Board
SOP-1130 Drawing Table
SOP-1253 Drying Rack
SOP-1262 Foot Rest
SOP-1025 Ironing Board
SOP-1274 Locker
SOP-1194 Manicure Table
SOP-1277 Mattress
SOP-0995 Mirror
SOP-0980 Rack
SOP-1146 Reception Desk
SOP-1036 Salon Chair
SOP-1302 Seating Arrangement
SOP-1153 Side Station
SOP-1308 Sofa
SOP-1313 Step Stool
SOP-0997 Stool
SOP-1110 Storage Bin
SOP-1314 Storage Facility
SOP-0970 Table
SOP-1113 Tool Trolley
SOP-1014 Trolley
SOP-1162 Workbench
SOP-1163 Workstation

Hand Tools - Cutting & Shaping

SOP-0754 Adhesive Mixing Tool
SOP-0554 Awl
SOP-0439 Axe
SOP-0765 Bamboo Knife
SOP-0649 Bamboo Splitter
SOP-0768 Bill Hook
SOP-0774 Bow Drill
SOP-0512 Cable Cutter
SOP-0403 Chisel
SOP-0659 Cold Chisel
SOP-0572 Cutter
SOP-0797 Draw Knife
SOP-0799 Drill Guide
SOP-0520 Electrician Knife
SOP-0373 File
SOP-0382 Hacksaw
SOP-0471 Hand Drill
SOP-0596 Hand Plane
SOP-0821 Hand Punch
SOP-0825 Hole Punch Pliers
SOP-0383 Knife
SOP-0686 Laminate Cutter
SOP-0838 Machete
SOP-0611 Paper Cutter
SOP-0859 Post-Processing Tool Kit
SOP-1300 Saw Blade
SOP-0363 Scissors
SOP-0401 Scraper
SOP-0721 Sharpening Stone
SOP-0625 Shears
SOP-0888 Spokeshave
SOP-0410 Tap and Die Set
SOP-0733 Tenon Saw
SOP-0549 Tin Snips
SOP-0510 Utility Knife
SOP-0642 Wire Brush
SOP-0643 Wire Cutter
SOP-0397 Wire Stripper
SOP-0461 Wood Saw

Hand Tools - Fastening & Gripping

SOP-0411 Adjustable Wrench
SOP-0376 Allen Key
SOP-0442 Bench Vice
SOP-0516 Chain Wrench
SOP-0781 Circlip Pliers
SOP-0428 Clamp
SOP-0395 Crimping Tool
SOP-0671 Fish Tape
SOP-0591 Fuse Puller
SOP-0676 Gear Puller
SOP-0498 Locking Pliers
SOP-0528 Nut Driver
SOP-0354 Pliers
SOP-0716 Saw Setting Pliers
SOP-0360 Screwdriver
SOP-0424 Socket Spanner
SOP-0361 Spanner
SOP-0887 Specialty Wrench

SOP-0458 Tool Box
SOP-0388 Tool Kit
SOP-1112 Tool Pouch
SOP-0904 Torque Screwdriver
SOP-0485 Torque Wrench
SOP-0459 Tweezers

Hand Tools - Striking & Digging

SOP-0463 Auger
SOP-0564 Brick Hammer
SOP-0444 Crowbar
SOP-0369 Hammer
SOP-0415 Mallet
SOP-0606 Mattock
SOP-0847 Nail Punch
SOP-0432 Pickaxe
SOP-0706 Punch
SOP-0423 Shovel
SOP-0541 Sledge Hammer

Hospitality & Housekeeping

SOP-0067 Air Conditioner
SOP-1115 Air Freshener
SOP-0375 Air Purifier
SOP-1117 Ash Tray
SOP-0759 Baggage Cart
SOP-1039 Bath Tub
SOP-1040 Bed Linen
SOP-1079 Bidet
SOP-1121 Blanket
SOP-0490 Broom
SOP-1009 Bucket
SOP-0656 Call Bell
SOP-0517 Cleaning Brush
SOP-0391 Cleaning Equipment
SOP-0005 Cleaning Robot
SOP-0074 Climate Control System
SOP-1241 Cloth Drying Stand
SOP-0967 Crockery
SOP-1128 Cruet Set
SOP-0976 Cutlery
SOP-1248 Decoration Item
SOP-0992 Drinking Glass
SOP-1131 Dust Pan
SOP-0985 Dustbin
SOP-1257 Feather Duster
SOP-1133 Finger Bowl
SOP-1033 Garbage Bag
SOP-1088 Guest Kit
SOP-1273 Laundry Basket
SOP-0943 Mop
SOP-1142 Napkin Holder
SOP-1198 Peg
SOP-0328 Pest Control Chemical
SOP-1199 Pillow
SOP-1203 Recycling Station
SOP-1296 Rug
SOP-0872 Safe Deposit Box
SOP-1149 Sanitizer
SOP-1027 Serving Tray
SOP-0162 Spotting Gun
SOP-1072 Spray Bottle
SOP-1156 Straw Holder
SOP-1157 Sugar Sachet Holder
SOP-1059 Table Linen
SOP-0898 Tank Cleaning Tool
SOP-1060 Tissue Paper
SOP-1325 Toilet Roll
SOP-1210 Toothbrush
SOP-1211 Toothpaste
SOP-1007 Towel
SOP-1328 Tub
SOP-0004 Vacuum Cleaner
SOP-1038 Wash Basin
SOP-0286 Washing Machine
SOP-0007 Waste Disposal Equipment
SOP-1214 Water Bottle
SOP-1215 Water Container
SOP-0288 Water Dispenser
SOP-0334 Water Filtration Unit
SOP-0064 Water Heater
SOP-1021 Water Jug
SOP-1161 Wet Wipes

Laboratory Instruments & Glassware

SOP-0043 Autoclave
SOP-0465 Beaker
SOP-0769 Blood Cell Pipette
SOP-0045 Centrifuge
SOP-0786 Concentration Measuring Tool
SOP-0787 Condenser
SOP-0791 DO Meter
SOP-0208 Distillation Unit
SOP-0802 EC Meter
SOP-0212 ESR Kit
SOP-1132 Enamel Tray
SOP-0136 Growth Chamber
SOP-0598 Hectolitre Weight Apparatus
SOP-0239 Hot Air Oven
SOP-0240 Hot Plate
SOP-0054 Incubator
SOP-0994 Lab Consumable
SOP-1270 Lab Infrastructure
SOP-1271 Lab Rack
SOP-0497 Laboratory Glassware
SOP-0083 Laminar Air Flow Cabinet
SOP-0414 Magnifying Glass

SOP-0378 Measuring Cylinder
SOP-0338 Microscope
SOP-0500 Petri Dish
SOP-0613 Physical Balance
SOP-0614 Pipette
SOP-0537 Refractometer
SOP-0873 Sample Collection Tool
SOP-1100 Sample Container
SOP-0274 Spirit Lamp
SOP-0731 TDS Meter
SOP-0348 Tensiometer
SOP-0507 Test Tube
SOP-1321 Test Tube Rack
SOP-0350 Viscometer
SOP-1331 Wash Bottle
SOP-0351 Water Analyser
SOP-0287 Water Bath
SOP-0289 Water Testing Kit
SOP-0639 Weight Box
SOP-0335 pH Meter
SOP-1164 pH Test Strip

Measuring & Marking Tools

SOP-0647 Anemometer
SOP-0766 Barometer
SOP-0557 Bevel
SOP-0515 Centre Punch
SOP-0783 Combination Set
SOP-0660 Cross Staff
SOP-0575 Dial Gauge
SOP-0665 Divider
SOP-0667 Drafting Machine
SOP-1050 Drawing Board
SOP-0798 Drawing Compass
SOP-0668 Drawing Instrument
SOP-0801 Dumpy Level
SOP-0446 Feeler Gauge
SOP-0590 Flow Meter
SOP-0674 French Curve
SOP-0602 Inclinometer
SOP-0448 L-Square
SOP-0836 Levelling Staff
SOP-0421 Line Rope
SOP-0473 Magnetic Compass
SOP-0087 Manifold Gauge
SOP-0475 Measuring Instrument
SOP-0359 Measuring Tape
SOP-0449 Micrometer
SOP-0848 Offset Rod
SOP-0408 Plumb Bob
SOP-0409 Pressure Gauge
SOP-0861 Prismatic Compass
SOP-0534 Protractor
SOP-0710 Radius Gauge
SOP-0868 Rain Gauge
SOP-0711 Ranging Rod
SOP-0379 Ruler
SOP-0714 Ruling Pen
SOP-0717 Scale Ruler
SOP-0718 Scriber
SOP-0720 Set Square
SOP-0372 Spirit Level
SOP-0728 Spring Caliper
SOP-0896 Surveying Chain
SOP-0545 T-Square
SOP-0548 Temperature Gauge
SOP-0481 Thermo-Hygrometer
SOP-0508 Thread Pitch Gauge
SOP-0349 Total Station
SOP-0418 Try Square
SOP-0913 Vacuum Gauge
SOP-0374 Vernier Caliper
SOP-0921 Weather Instrument
SOP-0358 Weighing Scale
SOP-0438 Wire Gauge

Medical, Healthcare & First Aid

SOP-0168 AED
SOP-0342 AED Trainer
SOP-1166 Adult Diaper
SOP-0646 Air Cushion
SOP-0343 Airway Mannequin
SOP-0019 Ambu Bag
SOP-1076 Antiseptic
SOP-0648 Artery Forceps
SOP-0175 Aspiration Pump
SOP-1118 Back Rest
SOP-1227 Bandage
SOP-0767 Bathing Aid
SOP-1078 Bed Pan
SOP-1171 Biomedical Waste Bag
SOP-1172 Biomedical Waste Bin
SOP-0344 Birthing Simulator
SOP-0427 CPR Manikin
SOP-0308 CPR Pocket Mask
SOP-1176 Cardiac Table
SOP-0779 Casualty Evacuation Kit
SOP-0035 Catheterization Kit
SOP-0113 Cervical Collar
SOP-0116 Crash Cart
SOP-0661 Crutch
SOP-0938 Diagnostic Test Strip
SOP-1084 Disposable Gloves
SOP-0211 ECG Machine
SOP-0218 Enema Kit
SOP-0670 Examination Table

SOP-0293 First Aid Kit
SOP-0469 Forceps
SOP-0133 Glucometer
SOP-0137 Haemoglobinometer
SOP-0495 Hospital Bed
SOP-0681 Hot Water Bottle
SOP-0601 IV Stand
SOP-0827 IV Training Arm
SOP-1136 Ice Pack
SOP-0146 Infrared Lamp
SOP-1138 Kidney Tray
SOP-1192 Mackintosh Sheet
SOP-0689 Manikin
SOP-0929 Medical Consumable
SOP-0248 Medical Equipment
SOP-0329 Medicine
SOP-1278 Medicine Container
SOP-0844 Mobility Aid
SOP-0256 Nebulizer
SOP-0056 Needle Destroyer
SOP-0935 Normal Saline
SOP-0694 Oral Care Set
SOP-0090 Oxygen Cylinder
SOP-0695 Oxygen Cylinder Key
SOP-0850 Oxygen Cylinder Trolley
SOP-0853 Patient Call Bell
SOP-0324 Physical Restraint
SOP-1290 Pill Organizer
SOP-0502 Pulse Oximeter
SOP-0266 Rapid Test Kit
SOP-0331 Ryle's Tube
SOP-0333 Scalp Vein Set
SOP-0942 Sharps Container
SOP-0008 Sphygmomanometer
SOP-0099 Spine Board
SOP-0275 Spirometer
SOP-0628 Splint
SOP-0729 Stadiometer
SOP-0017 Sterilizer
SOP-0454 Stethoscope
SOP-0543 Stretcher
SOP-0101 Suction Apparatus
SOP-1209 Surgical Tape
SOP-0320 Syringe
SOP-0371 Thermometer
SOP-0734 Tourniquet
SOP-0906 Transfer Forceps
SOP-1015 Urinal
SOP-1213 Urine Bag
SOP-1329 Vacutainer
SOP-0164 Ventilator
SOP-0738 Walker
SOP-0739 Walking Cane
SOP-0640 Wheelchair

Other

SOP-1252 Drum
SOP-0984 Field Visit Equipment
SOP-1183 Fixture and Fitting
SOP-1263 Frame
SOP-1026 Miscellaneous
SOP-1197 Pad
SOP-1151 Seal
SOP-1311 Steel Ring
SOP-1326 Training Setup
SOP-1073 Uniform

Plumbing & Pipe Fitting

SOP-0796 Drain Auger
SOP-1086 Faucet
SOP-0812 Flaring Tool
SOP-0521 Flushing Tank
SOP-0053 Hydraulic Pressure Testing Machine
SOP-0698 Pipe Cutter
SOP-0856 Pipe Fitting Tool
SOP-0699 Pipe Threading Die
SOP-0058 Pipe Threading Machine
SOP-0452 Pipe Vice
SOP-0433 Pipe Wrench
SOP-1000 Pipe and Fittings
SOP-1105 Shower Head
SOP-1107 Sink
SOP-0060 Smoke Generator
SOP-0280 Submersible Pump
SOP-1322 Thread Seal Tape
SOP-1323 Thread Sealant
SOP-0735 Tube Bender
SOP-0920 Water Meter
SOP-0012 Water Pump
SOP-1332 Water Tank

Power Tools & Workshop Machines

SOP-0106 Air Blow Gun
SOP-0018 Air Compressor
SOP-0755 Air Pump
SOP-0170 Air Ratchet
SOP-0172 Angle Grinder
SOP-0174 Arbor Press
SOP-0176 Bamboo Bending Machine
SOP-0177 Bamboo Planer Machine
SOP-0178 Bamboo Shaving Machine
SOP-0179 Bamboo Sliver Making Machine
SOP-0180 Bamboo Splitting Machine
SOP-0181 Bamboo Strip Sizing Machine
SOP-0182 Bamboo Veneer Slicing Machine
SOP-0108 Bench Drill Machine
SOP-0109 Bench Grinder
SOP-0198 CNC Machine

SOP-0115 Circular Saw
SOP-0205 Cutting Machine
SOP-0404 Drill Bit
SOP-0002 Drill Machine
SOP-0225 Filament Dryer
SOP-0131 Forking Machine
SOP-0052 Hammer Drill
SOP-0139 Heat Gun
SOP-0238 Hose Pump
SOP-0243 Jig Saw
SOP-0251 Milling Machine
SOP-0255 Nail Gun
SOP-0155 Pneumatic Gun
SOP-0263 Pneumatic Press
SOP-0091 Pneumatic Tool
SOP-0093 Power Tool
SOP-0095 Punching Machine
SOP-0157 Sanding Machine
SOP-0270 Shredder
SOP-0276 Spray Gun
SOP-0632 Tensioning Equipment
SOP-0283 UV Curing Station
SOP-0105 Vacuum Pump

Retail, Logistics & Warehouse

SOP-1226 Badge
SOP-0440 Barcode Printer
SOP-0441 Barcode Scanner
SOP-0778 Cash Register
SOP-1061 Crate
SOP-0313 Currency Counting Machine
SOP-1246 Currency Notes
SOP-0792 Delivery System
SOP-0014 Forklift
SOP-1034 Gondola Shelf
SOP-0040 Labelling Machine
SOP-0405 Ladder
SOP-0021 Material Handling Equipment
SOP-0431 POS Terminal
SOP-0476 Packaging Tool
SOP-1013 Pallet
SOP-0028 Pallet Truck
SOP-0989 Product Sample
SOP-1098 Promotional Material
SOP-0059 Reach Truck
SOP-1037 Shopping Cart
SOP-0100 Stacker
SOP-1315 Strap
SOP-1319 Tag
SOP-0546 Tagging Machine

Safety & PPE

SOP-0171 Alarm Panel
SOP-0553 Antistatic Wrist Strap
SOP-0983 Apron
SOP-1228 Barricade Tape
SOP-1229 Barrier Post
SOP-1041 Coverall
SOP-0413 Ear Protection
SOP-0805 Emergency Light
SOP-0220 Explosive Detector
SOP-0809 Eye Shield
SOP-0377 Face Mask
SOP-1259 Field Gear
SOP-0493 Fire Bucket
SOP-0001 Fire Extinguisher
SOP-0447 Full Body Harness
SOP-0322 Gas Leak Detector
SOP-0081 Gas Mask
SOP-0355 Gloves
SOP-1134 Gown
SOP-0822 Handcuffs
SOP-1090 Head Cap
SOP-0311 Insulated Gloves
SOP-0829 Insulated Rescue Pole
SOP-1137 Insulating Mat
SOP-0685 Knee Pads
SOP-0386 Locking System
SOP-0148 Metal Detector
SOP-1285 Outdoor Safety Kit
SOP-0357 PPE Kit
SOP-1286 Padlock
SOP-0299 Phosphine Gas Monitor
SOP-0862 Protective Guard
SOP-1293 Rainwear
SOP-0870 Rescue Hook
SOP-0871 Respirator Mask
SOP-1297 Safety Data Sheet
SOP-0368 Safety Goggles
SOP-0392 Safety Helmet
SOP-1298 Safety Installation
SOP-0389 Safety Shoes
SOP-0715 Safety Station
SOP-1299 Safety Tag
SOP-0538 Safety Vest
SOP-1102 Sand Bag
SOP-1303 Security Guard Belt
SOP-0539 Security System
SOP-1207 Shoe Cover
SOP-0971 Signage
SOP-0885 Smoke Detector
SOP-0897 Survival Kit
SOP-0910 Under Vehicle Inspection Mirror
SOP-0918 Warning Light
SOP-0936 Welding Apron
SOP-0937 Welding Goggles
SOP-0934 Welding Helmet

Sewing, Textile & Handicraft

SOP-0751 Aari Needle
SOP-0753 Adda Frame
SOP-1223 Art Mannequin
SOP-0764 Bamboo Craft Tool
SOP-0183 Bar Tack Machine
SOP-0188 Blind Stitch Machine
SOP-0560 Block Printing Equipment
SOP-0489 Bobbin
SOP-0772 Bobbin Case
SOP-0196 Button Sewing Machine
SOP-0197 Buttonhole Machine
SOP-1174 Calligraphy Pen
SOP-0201 Chain Stitch Machine
SOP-1242 Colour Palette
SOP-1243 Craft Material
SOP-0790 Crochet Needle
SOP-1181 Cutting Mat
SOP-0947 Dress Form
SOP-0930 Dye
SOP-0122 Dyeing Equipment
SOP-0011 Electric Iron
SOP-1042 Embellishment Material
SOP-0583 Embroidery Hoop
SOP-0049 Embroidery Machine
SOP-1023 Embroidery Stabilizer
SOP-1255 Embroidery Template
SOP-0221 Fabric Cutting Machine
SOP-0222 Fabric Inspection Machine
SOP-0223 Fabric Steamer
SOP-0226 Fixation Chamber
SOP-0813 Flattening Tool
SOP-0817 GSM Cutter
SOP-1087 Garment Trims
SOP-1265 Gemstone
SOP-0301 Glue Gun
SOP-0678 Grey Scale
SOP-1188 Hanger
SOP-0680 Hem Marker
SOP-0242 Interlock Sewing Machine
SOP-0245 Jute Article Finishing Machine
SOP-0834 Jute Craft Tool
SOP-0246 Jute Fibre Straightening Machine
SOP-0337 Loom
SOP-0688 Loop Turner
SOP-1280 Modelling Clay
SOP-0323 Needle Detector
SOP-0949 Needle Threader
SOP-0151 Overlock Machine
SOP-0407 Paint Brush
SOP-0696 Palette Knife
SOP-0852 Paper Mache Tool
SOP-0152 Paper Making Machine
SOP-0261 Paper Squeezing Machine
SOP-0531 Pattern Making Ruler
SOP-0854 Pattern Notcher
SOP-1143 Pattern Paper
SOP-1144 Pattern Template
SOP-0697 Pick Glass
SOP-1095 Pin Cushion
SOP-0451 Pinking Shears
SOP-1200 Press Cloth
SOP-0864 Punch Needle
SOP-0617 Raking Needle
SOP-1202 Raw Material Sample
SOP-1204 Ribbon
SOP-0620 Rotary Cutter
SOP-0623 Sculpting Tool
SOP-0435 Seam Ripper
SOP-0719 Seam Roll
SOP-0540 Sewing Gauge
SOP-1152 Sewing Kit
SOP-0023 Sewing Machine
SOP-0879 Sewing Machine Attachment
SOP-1006 Sewing Needle
SOP-1045 Sewing Pin
SOP-1046 Shade Card
SOP-1306 Shading Stump
SOP-1208 Size Chart
SOP-0723 Sleeve Board
SOP-0273 Spinning Machine
SOP-0891 Squeegee
SOP-0278 Steam Press
SOP-1312 Stencil
SOP-0892 Stencil Cutting Tool
SOP-0544 Striking Needle
SOP-1020 Tailor's Chalk
SOP-0482 Thimble
SOP-1001 Thread
SOP-0484 Thread Cutter
SOP-0902 Thread Stand
SOP-0281 Thread Sucking Machine
SOP-0633 Tracing Wheel
SOP-0637 Weaving Beater
SOP-1218 Yarn
SOP-1220 Zipper
SOP-0745 Zipper Guide

Solar, Power & Energy

SOP-0297 Battery
SOP-0304 Battery Charger
SOP-0186 Battery Storage Unit
SOP-0217 Electrolyser
SOP-0241 Hydrogen Purification System
SOP-0603 Load Bank
SOP-0264 Portable Generator

SOP-0535 Pyranometer
SOP-0096 Solar Charge Controller
SOP-0158 Solar Charging Station
SOP-1108 Solar Connector
SOP-0097 Solar Conversion Kit
SOP-0316 Solar Cooker
SOP-0886 Solar Dryer
SOP-0159 Solar E-Rickshaw
SOP-0317 Solar Home Lighting System
SOP-0061 Solar Inverter
SOP-0505 Solar Lantern
SOP-0062 Solar PV System
SOP-0303 Solar Panel
SOP-1047 Solar Panel Mounting Structure
SOP-0726 Solar Power Bank
SOP-0318 Solar Street Light
SOP-0160 Solar Water Heater
SOP-0271 Solar Water Pump
SOP-0006 UPS

Sports & Physical Education

SOP-0645 Agility Ladder
SOP-0757 Arrow
SOP-0760 Balance Beam
SOP-0426 Ball
SOP-0763 Ball Pump
SOP-0775 Boxing Equipment
SOP-1236 Camping Equipment
SOP-1239 Clip
SOP-0571 Cricket Bat
SOP-0789 Cricket Stump Set
SOP-0806 Exercise Ball
SOP-0807 Exercise Bench
SOP-0814 Foam Roller
SOP-0675 Frisbee
SOP-0818 Goal Post
SOP-0959 Hula Hoop
SOP-0600 Hurdle
SOP-0954 Marker Cone
SOP-0690 Medicine Ball
SOP-0858 Plyometric Box
SOP-0865 Punching Pad
SOP-0712 Relay Baton
SOP-0882 Sit and Reach Box
SOP-0883 Skipping Rope
SOP-1309 Sports Bib
SOP-0889 Sports Protective Gear
SOP-0890 Sports Ring
SOP-0455 Stopwatch
SOP-1318 Swimming Ring
SOP-0899 Tent
SOP-0900 Tessel
SOP-0905 Training Arch
SOP-0923 Whistle

Teaching Aids, Charts, Books & Software

SOP-0644 AR Learning Kit
SOP-1074 Anatomical Model
SOP-1224 Audio Guide Tool
SOP-0981 Audio-Visual Content
SOP-0966 Book / Manual
SOP-1022 Brochure
SOP-1177 Case Study
SOP-0991 Computer Based Training
SOP-1003 Demonstration Model
SOP-0998 Display Material
SOP-0972 Document
SOP-1031 Drawing / Plan
SOP-1184 Flash Cards
SOP-0977 Flip Chart
SOP-0975 Form
SOP-1093 Map
SOP-1066 Picture / Photograph
SOP-1005 Presentation Slides
SOP-1069 Reference Sample
SOP-0996 Role-Play Material
SOP-1307 Sheet Metal Sample
SOP-1106 Simulated Environment
SOP-0965 Software
SOP-1320 Teaching Aid
SOP-1324 Time Management Tool
SOP-1160 Training Material
SOP-0964 Wall Chart
SOP-0969 Whiteboard
SOP-1002 Whiteboard Duster

Telecom & Networking

SOP-0491 Cable Tester
SOP-0580 EPABX
SOP-0592 GPS Device
SOP-0826 IP Phone
SOP-0830 Intercom
SOP-0993 Internet Connection
SOP-0831 Internet Dongle
SOP-1064 Network Cable
SOP-0406 Network Switch
SOP-0609 Networking Equipment
SOP-0346 OTDR
SOP-1068 RJ45 Connector
SOP-0874 Satellite Phone
SOP-0480 Telephone
SOP-0509 Two-Way Radio
SOP-0425 Wi-Fi Router
SOP-0925 Wildlife Tracking Device

Welding & Soldering

SOP-0068 Arc Welding Machine
SOP-0189 Blow Lamp

SOP-0070 Brazing Torch
SOP-0793 Desoldering Pump
SOP-0124 Electrode Holder
SOP-1070 Solder Wire
SOP-1019 Soldering Flux
SOP-0003 Soldering Iron
SOP-1216 Welding Cable

Automatic Shade System

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **Medium** 2 labs

Covers these lab items: Automatic shade system

What it is used for

An automatic shade system adjusts shade screens or coverings to control sunlight, temperature, and light exposure in a greenhouse or growing area. Correct setup and sensor control help protect crops and maintain suitable growing conditions.

Main parts

- Shade cloth, screen, or covering material
- Support frame, rails, and mounting brackets
- Motor or actuator and drive mechanism
- Light, temperature, or weather sensors if fitted
- Controller, timer, and limit switches
- Power supply and manual override control

Before you start

1. Read the installation and operating instructions for the specific shade system.
2. Inspect the shade cloth, frame, rails, cables, motor, and mounting points for damage.
3. Confirm that the structure is secure and that the shade can move without obstruction.
4. Check sensor placement and verify that the controller settings suit the crop and growing conditions.
5. Test the emergency stop or manual override if fitted and confirm the movement limits are set.
6. Ensure the power supply and protective devices meet the manufacturer's requirements.

How to use

1. Switch on the controller and select the approved automatic or manual operating mode.
2. Set the required light, temperature, timer, or weather thresholds according to the crop plan.
3. Run a brief movement test to confirm that the shade opens and closes smoothly.
4. Observe the sensors and controller response as environmental conditions change or during a controlled test.
5. Adjust the threshold settings only within the approved growing plan and equipment limits.
6. Check that the shade stops at the end positions without excessive tension or mechanical impact.
7. Inspect the covered area for uniform shading and verify that crops receive the intended light level.
8. Use the manual override only when required and return the system to the approved mode afterward.

9. Stop the system if movement becomes uneven, the frame shifts, or a motor or cable shows a fault.
10. Record settings, test results, and maintenance needs according to the facility procedure.

⚠ Safety

- Keep hands and tools clear of moving screens, rails, cables, and actuator linkages.
- Isolate power before cleaning, adjusting, or repairing the drive mechanism.
- Do not climb on the shade structure or use it as a support platform.
- Keep people away from the movement path while the system operates.
- Stop operation during unsafe wind or weather conditions specified by the manufacturer.
- Do not bypass limit switches, guards, or other fitted safety devices.

Care & storage

- Remove dust, plant debris, and other material that obstructs shade movement.
- Inspect cloth, stitching, rails, fasteners, motors, cables, and sensors regularly.
- Keep sensors clean and verify their readings according to the maintenance schedule.
- Arrange servicing of actuators and controllers according to the manufacturer's instructions.

Common mistakes to avoid

- Incorrect sensor thresholds can expose crops to excessive heat or insufficient light.
- Obstructions in the rails can damage the shade cloth or motor.
- Ignoring wind limits can damage the screen and supporting structure.
- Leaving manual override active can prevent automatic adjustment from working as intended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Basket

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Plastics Baskets

What it is used for

Plastic baskets collect, carry, and organise produce, tools, and gardening supplies.

How to handle

- Inspect the basket for cracks, sharp edges, dirt, and handle damage.
- Choose a basket suitable for the load and intended contents.
- Distribute contents evenly and avoid overfilling or carrying unstable loads.
- Wash baskets used for produce and prevent cross-contamination between clean and dirty items.
- Do not use damaged baskets for heavy loads.

Storage

- Clean and dry baskets after use.
- Stack securely in a designated area away from heat and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Battery Operated Sprayer

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Battery operated sprayer

What it is used for

A battery-operated sprayer uses a rechargeable battery and pump to apply water, fertiliser solutions, or approved crop-treatment liquids through a nozzle. Use only chemicals authorised for the intended application and follow the product label, equipment instructions, and required protective measures to prevent exposure, spills, and environmental contamination.

Main parts

- Chemical tank and filling opening
- Rechargeable battery and battery connector
- Electric pump and pressure switch
- Spray hose, trigger valve, and lance
- Nozzle and filter or strainer
- Harness, control switch, and charger

Before you start

1. Confirm the intended application and verify that the liquid is approved for the crop, site, and sprayer materials.
2. Read the product label and safety data; identify the required dilution, application rate, re-entry restrictions, and PPE.
3. Inspect the tank, cap, seals, hose, trigger valve, lance, nozzle, battery, and electrical connectors for leaks or damage.
4. Check that the battery is securely fitted and adequately charged; use only the specified charger and battery type.
5. Ensure the nozzle and filter are clean and appropriate for the liquid and target application.
6. Wear the chemical-specific PPE, such as gloves, protective clothing, eye protection, and respiratory protection where the label requires it; check wind direction before outdoor spraying.

How to use

1. Place the sprayer on a stable surface in a well-ventilated preparation area away from drains and water sources.
2. Confirm that the pump is switched off before filling the tank.
3. Prepare and measure the liquid exactly according to the product label; never mix incompatible chemicals.
4. Fill the tank without exceeding its marked capacity and close the cap securely.

5. Check the hose and nozzle connections and ensure that the trigger valve is closed before starting.
6. Fit the harness correctly and position the sprayer securely on the back or on its approved support.
7. Switch on the pump and test the spray pattern in a designated safe area using the approved procedure.
8. Adjust pressure or nozzle setting only within the manufacturer's permitted range and product-label requirements.
9. Apply the liquid evenly to the intended target, avoiding drift toward people, animals, neighbouring crops, and water bodies.
10. Monitor the tank, hose, battery, and spray output during operation; stop immediately if leakage, clogging, or abnormal pump behaviour occurs.
11. Switch off the pump before clearing a blockage; release pressure and follow the manual before removing the nozzle or filter.
12. After application, switch off the unit, remove PPE safely, and clean the sprayer according to the chemical label and manufacturer instructions.
13. Handle leftover liquid, rinse water, and empty containers according to the product label and local hazardous-waste requirements; recharge and store the battery in a dry, safe location.

⚠ Safety

- Never spray an unidentified chemical or use a product contrary to its label.
- Do not eat, drink, smoke, or touch the face while mixing or spraying chemicals.
- Avoid spraying in strong winds, during unsuitable weather, or where people and animals may be exposed.
- Do not open the tank or remove the nozzle while the system is pressurised.
- Keep the battery, charger, and electrical connections away from water and chemical spills.
- Do not pour chemical residue or rinse water into drains, wells, ponds, or natural waterways.

Care & storage

- Rinse and clean the tank, hose, filter, and nozzle according to the chemical label and equipment instructions.
- Inspect seals, hose connections, straps, and nozzle components regularly for wear or leakage.
- Clean blocked nozzles only with the approved tool; never blow through a nozzle by mouth.
- Charge the battery with the specified charger and avoid storing it in excessive heat or moisture.
- Store the clean, dry sprayer securely and keep chemicals in their original labelled containers.

Common mistakes to avoid

- Using an incorrect dilution or exceeding the tank capacity.
- Spraying without label-required PPE or ignoring wind direction.
- Clearing a blocked nozzle while the pump is running or the system remains pressurised.
- Disposing of leftover chemicals or rinse water in drains or water bodies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the

manufacturer's original manual / file.

Bonsai Tool Set

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Bonsai Cutter; Bonsai set

What it is used for

To trim, shape, and maintain bonsai plants using suitable cutters and shaping tools.

How to handle

- Inspect cutters and other tools for rust, loose joints, and damaged edges.
- Choose the correct tool for pruning, trimming, or shaping the plant.
- Wear suitable gloves and make controlled cuts without damaging the trunk or healthy branches.
- Avoid cutting more growth than necessary and disinfect tools between plants when disease transmission is possible.
- Remove plant debris and clean tools after use.

Storage

- Clean and dry each tool; protect cutting edges with covers where provided.
- Arrange tools in their designated case in a dry place to prevent rust and accidental injury.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Budding and Grafting Knife

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 13 labs

Covers these lab items: Budding and grafting knife; Budding Knife; Grafting knife; Mixed budding and grafting knife

What it is used for

Use a budding and grafting knife to make controlled cuts when preparing plant buds, scions, or rootstocks for propagation.

How to use

1. Select a clean, sharp knife suitable for the plant and grafting method.
2. Inspect the blade, handle, and locking mechanism if fitted.
3. Wear suitable gloves and secure the plant material before cutting.
4. Make precise cuts using the method appropriate to the selected plant.
5. Keep cut surfaces clean and join the bud or scion promptly as required.
6. Secure the graft with suitable grafting tape or material.
7. Clean and disinfect the blade after use according to plant health guidance.

⚠ Safety

- Cut away from your body and supporting hand.
- Keep the blade covered when not in use.
- Do not use a loose, damaged, or blunt knife.
- Handle disinfectants according to their label instructions.

Care & storage

- Clean, dry, and protect the blade after use.
- Store the knife closed or in a protective sheath away from children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Combine Harvester

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Paddy combiner (small)

What it is used for

A small paddy combine harvester cuts rice plants and combines harvesting operations such as threshing and grain separation in one machine. Correct field preparation, machine adjustment, and safe operation help reduce grain loss, crop damage, and machinery accidents.

Main parts

- Cutting header and crop divider
- Reel or crop-feeding mechanism, if fitted
- Threshing drum and concave
- Straw walkers or grain separation system
- Grain cleaning fan, sieves, and collection tank
- Engine, drive system, controls, guards, and discharge outlet

Before you start

1. Read the operator manual and confirm the machine is suitable for the paddy crop, field conditions, and expected crop height.
2. Inspect the engine, fuel system, belts, chains, cutting blades, threshing parts, tyres or tracks, guards, and controls.
3. Check engine oil, coolant, fuel, lubrication points, and other fluid levels according to the manual.
4. Remove people, animals, stones, wire, and other obstacles from the field and harvesting route.
5. Check soil firmness, water depth, slopes, field access, and turning space; avoid unsafe or excessively soft ground.
6. Adjust cutting height, threshing clearance, fan speed, and sieve settings according to crop moisture and machine instructions.

How to use

1. Park the harvester on firm, level ground and complete the pre-start inspection.
2. Ensure guards are fitted, the discharge area is clear, and all bystanders are at a safe distance.
3. Start the engine using the manufacturer's procedure and check for warning indicators, leaks, or abnormal sounds.
4. Allow the engine and relevant systems to reach the recommended operating condition before harvesting.
5. Engage the cutting and threshing systems using the prescribed sequence and confirm they operate normally.
6. Move into the crop at a controlled speed and keep the header aligned with the standing paddy.

7. Maintain a steady forward speed so the crop feeds evenly without overloading the threshing system.
8. Monitor grain flow, grain cleanliness, straw discharge, engine load, and visible crop losses.
9. Adjust travel speed or approved machine settings if grain loss, unthreshed ears, or excessive broken grain appears.
10. Stop on firm ground before clearing blockages; disengage the drives, switch off the engine, remove the key, and wait for all moving parts to stop.
11. Clear crop material only with approved tools and according to the manual; never reach into moving mechanisms.
12. Unload collected grain at a designated safe location, keeping people clear of the discharge mechanism.
13. At the end of harvesting, disengage the systems, park securely, switch off the engine, and allow hot components to cool.
14. Clean crop residue from accessible areas and inspect the machine for damage before storage or transport.

⚠ Safety

- Only trained operators should operate the harvester; follow the manufacturer's instructions and local agricultural safety requirements.
- Keep all guards and shields fitted and never approach the cutting header, belts, chains, or threshing drum while they are moving.
- Never clear blockages with the engine running or stored mechanical energy present.
- Keep people and animals away from the machine, especially during reversing, turning, and grain unloading.
- Do not operate on steep slopes, unstable embankments, deep water, or ground that cannot safely support the machine.
- Refuel only with the engine stopped and cooled, away from ignition sources; keep suitable fire-fighting equipment accessible.
- Wear suitable footwear, eye protection, and hearing protection as required; keep loose clothing away from moving parts.

Care & storage

- Remove straw, dust, and chaff after shutdown, paying attention to hot engine and exhaust areas only after they have cooled.
- Inspect blades, belts, chains, bearings, sieves, and threshing components for wear or damage.
- Check engine oil, coolant, fuel filters, and lubrication points at the intervals specified in the manual.
- Clean air filters and inspect the grain cleaning system regularly.
- Store the harvester in a dry, secure area and arrange scheduled servicing before the next season.

Common mistakes to avoid

- Operating at excessive speed for the crop density or field condition.
- Using unsuitable cutting height, threshing clearance, or cleaning settings.

- Clearing a blockage before fully stopping and isolating the machine.
- Ignoring grain loss, abnormal vibration, overheating, or damaged guards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cono Weeder

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Cono weeder

What it is used for

A cono weeder removes weeds between rows of transplanted rice or similar crops while helping disturb the soil surface. Use it in suitable field conditions and follow safe operating technique.

How to use

1. Inspect the frame, rollers, handles, and moving parts for damage.
2. Wear suitable footwear and protective clothing before entering the field.
3. Check crop spacing and soil conditions to confirm the tool is suitable.
4. Hold both handles firmly and align the rollers between crop rows.
5. Push and pull the cono weeder steadily along the row without striking plants.
6. Lift the tool carefully when turning at the end of a row.
7. Remove trapped weeds and soil after stopping the tool.

⚠ Safety

- Keep feet and hands away from rollers and moving parts.
- Do not use the tool in unsuitable soil or crowded conditions that risk crop damage.
- Avoid working in deep water or unstable ground without appropriate precautions.
- Carry the tool with its working parts directed away from others.

Care & storage

- Wash off soil, dry the tool, and inspect handles and rollers after use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cultivator

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Cultivator

What it is used for

A cultivator loosens and aerates soil, controls weeds, and prepares the seedbed using tractor-mounted, powered, or manually operated tines. Correct adjustment, suitable soil conditions, and safe operation help improve field preparation while reducing equipment damage and operator injury.

Main parts

- Frame supports the cultivator assembly.
- Tines or shanks loosen soil and disturb weeds.
- Shares or sweeps cut and lift soil where fitted.
- Hitch or mounting points connect the cultivator to a tractor where applicable.
- Depth adjustment controls working depth where provided.
- Drive or power transmission components operate powered models.
- Wheels or skids support the machine and help control depth.
- Guards and shields protect the operator from moving components where fitted.

Before you start

1. Only trained operators should use tractor-mounted or powered cultivators.
2. Read the operator manual and confirm the cultivator is suitable for the tractor or power source.
3. Inspect the frame, tines, shares, fasteners, hitch, wheels, guards, and drive components for damage.
4. For tractor-mounted equipment, park on level ground, apply the parking brake, lower attachments, and shut off the engine before connecting or adjusting.
5. Check the field for stones, wire, buried services, people, and other obstructions.
6. Wear fitted work clothing, safety footwear, eye protection where needed, and hearing protection when required.

How to use

1. Connect and secure the cultivator using the approved hitching method where applicable.
2. Set the working depth and tine arrangement according to the crop, soil, and field conditions.
3. Confirm guards are fitted and no tools or loose items remain near moving parts.
4. Start the tractor or powered unit from the operator position and check for abnormal noise or vibration.
5. Begin at a low, controlled speed and engage the cultivator according to the manual.

6. Maintain a steady path and adjust speed or depth if soil conditions cause excessive bouncing or blockage.
7. Raise or disengage the cultivator before turning or travelling over unsuitable ground as directed by the manual.
8. Stop the machine, isolate power, and wait for all movement to cease before clearing blockages or making adjustments.
9. After work, lower the attachment safely, disconnect power where applicable, and inspect the tines and frame.
10. Clean off soil and plant residue, record faults, and park the machine securely.

⚠ Safety

- Never approach rotating or moving tines while the machine is running.
- Shut off the engine or power source, remove the key where applicable, and wait for all parts to stop before clearing a blockage.
- Never carry passengers on the cultivator or allow bystanders near the working area.
- Keep hands and feet away from pinch points during hitching and adjustment; use approved lifting supports where needed.
- Do not operate on unstable slopes or near ditches beyond the machine's approved limits.
- Use guards and follow tractor PTO and power-transmission safety instructions where relevant.

Care & storage

- Remove soil and plant residue after use, especially around tines and moving parts.
- Inspect tines, shares, fasteners, hitch points, wheels, and guards before and after operation.
- Lubricate designated points and tighten fasteners according to the maintenance schedule.
- Replace worn or damaged parts with compatible components and arrange authorised repairs when required.

Common mistakes to avoid

- Using an unsuitable working depth or speed for the soil conditions.
- Operating with loose tines, damaged guards, or an insecure hitch.
- Clearing blockages while the engine or power source is running.
- Failing to inspect the field for people, stones, wire, or hidden obstructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dibbler

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Dibbler; Manual dibbler

What it is used for

To make planting holes at consistent spacing and depth using a manual dibbler. To place seeds or seedlings accurately while reducing soil disturbance.

Main parts

- Pointed planting tip
- Shaft or body
- Handle or grip
- Depth marker, if fitted

Before you start

1. Select a dibbler suitable for the crop and soil condition.
2. Inspect the tip, shaft, and handle for cracks or looseness.
3. Prepare the bed and remove large stones or debris.
4. Check the required planting spacing and depth.
5. Wear suitable gloves and footwear.

How to use

1. Mark the planting positions according to the crop spacing plan.
2. Hold the dibbler securely by its handle.
3. Place the tip at the first marked position.
4. Press into the soil to the required depth using controlled force.
5. Withdraw the dibbler vertically to leave a neat hole.
6. Repeat at the remaining marked positions.
7. Place the seed or seedling and cover with soil as required.
8. Clean the tool after use.

⚠ Safety

- Keep hands and feet clear of the pointed tip.
- Do not force the tool into soil containing hidden hard objects.
- Maintain stable footing while pressing into the ground.
- Do not use a cracked or loose-handled dibbler.

Care & storage

- Remove soil from the tip and shaft after use.
- Dry metal parts to reduce rust.
- Inspect the handle and tip regularly.
- Store in a dry, secure location.

Common mistakes to avoid

- Making holes at inconsistent depths.
- Ignoring the crop's spacing requirements.
- Tilting the dibbler and widening the hole unnecessarily.
- Using excessive force in unsuitable soil.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Digging Fork

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Digging fork; Fork; Garden/Digging Fork

What it is used for

Use a digging fork to loosen soil, lift roots, turn compost, or break up compacted ground in garden and agricultural work.

How to use

1. Inspect the tines, socket, and handle for cracks, bends, or looseness.
2. Choose a fork suitable for the soil and task.
3. Clear the work area and check for buried utilities or sharp objects.
4. Place the tines into the soil using controlled pressure and stable footing.
5. Push the handle gently to loosen the soil or lift the material.
6. Remove roots or debris carefully without overloading the fork.
7. Clean the tines and store the fork safely after use.

⚠ Safety

- Wear sturdy footwear and suitable gloves.
- Keep feet clear of the tines when pressing the fork into the ground.
- Do not use a fork with bent tines or a loose handle.
- Check for underground cables or pipes before digging.

Care & storage

- Remove soil and dry the tines after use.
- Store upright or securely with the tines protected from accidental contact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Disc Harrow

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: Disc Harrow

What it is used for

A disc harrow breaks soil clods, cuts crop residue, and helps prepare fields after ploughing. Correct tractor attachment, depth adjustment, and safe transport reduce injury and equipment damage.

Main parts

- Disc blades cut and turn the soil.
- Disc gangs hold multiple blades at the required angle.
- Frame supports the disc gangs and connecting components.
- Hitch or linkage connects the harrow to the tractor.
- Scrapers help remove soil and residue from the discs.
- Depth or angle adjustment controls working depth and soil action.
- Transport wheels or supports help move the implement where fitted.

Before you start

1. Read the operator's manual and use the implement only after suitable training.
2. Inspect discs, bearings, frame, hitch, fasteners, scrapers, and transport components for damage.
3. Confirm the tractor is compatible with the harrow's size, weight, and hitch requirements.
4. Attach the harrow on firm, level ground with the tractor secured against movement.
5. Ensure hitch pins and locking devices are correctly installed before moving.
6. Clear people, animals, stones, wires, and other hazards from the field.

How to use

1. Set the tractor parking brake and lower or support the implement safely before adjustments.
2. Connect the harrow using the approved hitch and secure all locking devices.
3. Adjust disc angle and working depth according to the soil and operating instructions.
4. Raise the implement to the transport position before travelling between work areas.
5. Move slowly to the field and lower the harrow only after aligning with the work direction.
6. Engage the tractor drive and operate at the recommended speed for the soil conditions.
7. Maintain straight, controlled passes and monitor soil coverage, vibration, and implement movement.
8. Raise the harrow before turning sharply, reversing, or crossing unsuitable ground.
9. Stop the tractor, secure it, and lower or support the implement before inspection or clearing residue.
10. After work, remove accumulated soil, inspect for damage, and park the harrow securely.

⚠ Safety

- Never stand beneath a raised harrow or between the tractor and implement during hitching.
- Keep hands away from sharp disc edges and moving linkage points.
- Stop the tractor engine and secure the machine before adjusting, cleaning, or repairing the harrow.
- Never carry passengers on the implement or allow bystanders near operating discs.
- Use suitable gloves and footwear when handling stationary discs, while keeping gloves away from moving machinery.
- Follow tractor and implement limits on slopes, speed, and transport.

Care & storage

- Clean soil and crop residue from discs and scrapers after safely stopping the equipment.
- Inspect disc edges, bearings, hitch pins, fasteners, and frame for wear or damage.
- Lubricate bearings and moving joints according to the manufacturer's maintenance schedule.
- Store the harrow on firm, level ground with the implement secured against movement.

Common mistakes to avoid

- Adjusting the harrow while it is raised without proper support.
- Travelling with the implement lowered or at an unsafe speed.
- Ignoring worn discs, loose hitch pins, or damaged bearings.
- Standing between the tractor and harrow without securing the tractor against movement.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Disc Plough

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Disc plough

What it is used for

A disc plough cuts, lifts, and turns soil using rotating concave discs to prepare land for planting. Correct hitching, disc adjustment, and controlled tractor operation help achieve uniform tillage while reducing equipment damage and soil compaction.

Main parts

- Concave plough discs
- Disc hubs, bearings, and scrapers
- Main frame and disc gangs
- Three-point hitch or drawbar connection
- Depth and angle adjustment mechanism
- Safety reflectors, fasteners, and transport locks where fitted

Before you start

1. Read the operator manual and confirm that the plough is compatible with the tractor's power, hitch, and lifting capacity.
2. Inspect discs for cracks, excessive wear, or deformation and check hubs, bearings, scrapers, and fasteners.
3. Check the frame, hitch points, hydraulic connections, and transport locks for damage.
4. Park the tractor and plough on firm, level ground before hitching or adjustment.
5. Confirm that guards, reflectors, and safety devices are fitted where provided.
6. Survey the field for rocks, buried obstacles, steep slopes, overhead hazards, and people or animals.

How to use

1. Position the tractor safely and attach the plough using the approved hitching procedure.
2. Secure all hitch pins and retainers, and connect hydraulic lines if fitted according to the manual.
3. Adjust the top link, disc angle, working width, and depth settings for the soil and tillage requirement.
4. Raise the plough and verify transport clearance before moving to the field.
5. Travel at a controlled speed and lower the plough gradually at the start of the furrow.
6. Engage the plough at the recommended operating speed and maintain a steady tractor path.
7. Monitor disc penetration, soil turning, tractor load, and the uniformity of the furrow.
8. Adjust depth or angle only according to the manual and safe operating procedure if tillage is uneven.

9. Raise the plough before turning at headlands or crossing obstacles where required.
10. Stop on firm, level ground before inspecting or clearing soil and crop residue from the discs.
11. Lower the implement to the ground, stop the tractor, apply the parking brake, and isolate the engine before adjustments or maintenance.
12. At the end of the job, clean soil from the discs and inspect the implement for damage.
13. Park the plough securely with the implement lowered or supported as specified by the manufacturer.

⚠ Safety

- Only trained operators should operate the tractor and plough combination.
- Keep people and animals clear of the implement, especially during hitching, turning, and raising or lowering.
- Never work beneath a raised plough unless it is secured with the manufacturer's approved support.
- Stop the tractor and isolate the engine before clearing blockages or servicing the discs.
- Use the correct hitch pins and retainers; never rely on improvised connections.
- Reduce speed on slopes and uneven ground, and avoid conditions that may cause rollover or loss of traction.

Care & storage

- Clean soil and crop residue from discs, scrapers, and frame after use.
- Inspect discs, hubs, bearings, and fasteners regularly for wear or damage.
- Lubricate designated points and maintain bearings according to the manual.
- Check hydraulic hoses, hitch points, and transport locks before each use.
- Replace worn or damaged discs and arrange scheduled servicing before the next season.

Common mistakes to avoid

- Using a plough that exceeds the tractor's rated capacity.
- Setting incorrect disc angle or working depth for the soil conditions.
- Turning sharply with the plough engaged in the soil.
- Inspecting or clearing the implement while it is raised or the tractor is running.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drip Irrigation System

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 9 labs

Covers these lab items: Drip and sprinkler; Drip and sprinkler system; Drip Irrigation Unit; Drip Irrigation Unit (Lateral pipe 16mm x 20 ft , Elbow , coupler , T-joint , mini valve, gurment teck, end cap , drip punch 5nos each; Drip Irrigation UTender (Lateral pipe 16mm x 20 ft , Elbow , coupler , T- joint , mini valve,

What it is used for

Use a drip irrigation system to deliver water slowly near plant roots and reduce unnecessary water loss.

How to use

1. Inspect the water source, filter, mainline, lateral pipes, fittings, and emitters for damage or blockage.
2. Place lateral pipes near the plant rows and connect fittings securely.
3. Install the filter and pressure regulator where required by the system design.
4. Open the water supply gradually and flush the lines before normal operation.
5. Check each emitter for steady flow, leaks, and blocked outlets.
6. Adjust the schedule and duration according to crop needs, soil, and weather.
7. Close the supply after irrigation and check the system for leaks or damage.

⚠ Safety

- Use pressure ratings suitable for the pipes and fittings.
- Release system pressure before disconnecting pipes or cleaning components.
- Keep electrical pumps and connections protected from water.
- Do not use untreated or unsafe water on edible crops where it creates a health risk.

Care & storage

- Clean filters and flush lines regularly to prevent clogging.
- Repair leaking fittings and store removable parts away from direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drum Seeder

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Paddy drum seeder

What it is used for

A paddy drum seeder places pre-germinated rice seeds in rows to support uniform sowing. Prepare the field and operate the seeder at a steady pace for consistent spacing.

How to use

1. Inspect the drum, seed outlets, frame, handles, and wheels for damage.
2. Prepare a level, puddled field with suitable moisture and surface conditions.
3. Select clean, pre-germinated paddy seeds according to the recommended procedure.
4. Fill each drum compartment evenly without overloading it.
5. Align the seeder with the field rows and push it steadily forward.
6. Monitor seed flow and row spacing, stopping to clear any blocked outlets.
7. Clean the seeder and remove remaining seeds after completing the field.

⚠ Safety

- Wear suitable footwear and avoid unstable or excessively deep mud.
- Keep fingers away from moving wheels and seed outlets during operation.
- Do not force blocked outlets while the seeder is moving.
- Follow local guidance for seed treatment and field hygiene.

Care & storage

- Wash off mud, dry the equipment, and inspect the seed outlets before storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fertigation Unit

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Nutrient pump/ Fertigation Unit

What it is used for

A fertigation unit delivers dissolved fertilisers through an irrigation system to supply nutrients to crops in measured quantities. Correct nutrient preparation, injection-rate control, and irrigation monitoring help improve nutrient distribution while reducing clogging, crop injury, and environmental contamination.

Main parts

- Fertiliser or nutrient mixing tank
- Injection pump or dosing device
- Irrigation water inlet and filter
- Flow meter, pressure gauge, and control valves
- Injection line and non-return valve
- Controller, hoses, and distribution network

Before you start

1. Confirm the crop, growth stage, irrigation schedule, water quality, and approved nutrient recipe.
2. Read the unit manual and fertiliser labels to verify compatibility, mixing order, and required concentration.
3. Inspect the tank, pump, filters, hoses, valves, injection line, and electrical connections for leaks or damage.
4. Check that the irrigation filter is clean and the non-return valve or backflow protection is correctly installed and functional.
5. Prepare fertiliser solution using the specified quantities and safe handling procedures; do not mix incompatible chemicals.
6. Verify that the pump, flow meter, pressure gauge, and controller are functioning and that the field irrigation system is ready.

How to use

1. Place the unit on a stable surface and confirm that the mixing tank is clean and suitable for the nutrient solution.
2. Measure and add water to the tank according to the approved mixing procedure.
3. Add the specified fertilisers in the recommended order while mixing; ensure each component dissolves as instructed.
4. Close the tank securely and inspect the prepared solution for undissolved material or unexpected precipitation.

5. Open the required irrigation valves and start the water flow according to the system's startup procedure.
6. Confirm that irrigation pressure and flow are within the permitted operating range before nutrient injection.
7. Set the dosing rate, target concentration, and injection duration according to the crop plan and equipment instructions.
8. Start the injection pump and monitor solution level, water flow, pressure, and dosing indicators.
9. Check the distribution system for leaks, blocked emitters, unusual pressure changes, or uneven irrigation.
10. Adjust settings only within approved limits if the measured dosing rate differs from the required value.
11. Stop nutrient injection if a leak, backflow risk, pump fault, or unexpected pressure change occurs.
12. At the end of dosing, stop the injection pump and flush the irrigation lines with clean water according to the approved procedure.
13. Close valves in the prescribed sequence and prevent residual concentrated fertiliser from entering water sources or drains.
14. Record nutrient type, concentration, applied volume, field or crop area, and any operating issues.

⚠ Safety

- Only trained personnel should operate and maintain the fertigation unit.
- Wear chemical-compatible gloves, eye protection, and protective clothing when preparing or handling fertiliser solutions.
- Follow product labels and safety data sheets; never mix incompatible fertilisers or add chemicals without an approved procedure.
- Protect drinking-water supplies against backflow using suitable approved prevention devices.
- Never open pressurised lines or remove filters while the system is operating; isolate pressure first.
- Keep electrical components dry and isolate power before servicing pumps or controls.
- Prevent spills and dispose of leftover solution and rinsing water according to product instructions and local environmental requirements.

Care & storage

- Rinse the mixing tank, injection lines, and pump according to the fertiliser and equipment instructions after use.
- Clean irrigation filters and inspect emitters, valves, and non-return devices regularly.
- Check pump seals, hoses, fittings, and electrical connections for wear or leakage.
- Verify dosing accuracy and flow-meter readings at the prescribed intervals.
- Store fertilisers safely and arrange servicing if injection rate, pressure, or pump operation becomes unstable.

Common mistakes to avoid

- Using the wrong nutrient concentration or mixing incompatible fertilisers.

- Starting injection before confirming irrigation flow and pressure.
- Ignoring clogged filters, leaks, or backflow protection failures.
- Failing to flush the system after dosing or disposing of concentrated solution improperly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fertilizer

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 29 labs

Covers these lab items: Ammonium sulphate; Biofertilizers; Complex fertilizers; DAP; Dry and liquid fertiliser; Farmyard Manure; Fertilisers; Fertilizers; Liquid fertilizer; Murate of potash; Potassium sulphate; Urea; Vermicompost

What it is used for

Fertilizers supply plant nutrients and should be selected and applied according to crop needs and product instructions.

How to handle

- Identify the fertilizer type, nutrient composition, crop suitability, and recommended application rate.
- Wear suitable gloves and protective equipment; avoid breathing dust or contacting eyes and skin.
- Measure accurately, prevent runoff, and keep fertilizer away from drains and water bodies.
- Do not mix fertilizers or apply them with other products unless compatibility is confirmed.

Storage

- Keep in original labelled containers or suitable sealed packaging in a dry, ventilated area.
- Separate incompatible materials and protect fertilizer from moisture, heat, and unauthorized access.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fertilizer Spreader

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: A compact spreader; Manual fertilizer broadcaster

What it is used for

To distribute granular fertilizer evenly over a designated area using a manual fertilizer spreader. To apply the correct quantity while minimizing waste, drift, and uneven crop growth.

Main parts

- Fertilizer hopper or container
- Flow control gate or adjustment dial
- Spreading outlet or disc, depending on design
- Handle or frame
- Wheels or carrying straps, if fitted

Before you start

1. Read the fertilizer label and confirm the required application rate.
2. Choose a spreader suitable for the fertilizer granule size and area.
3. Inspect the hopper, flow control, outlet, and moving parts for damage or blockage.
4. Wear gloves, eye protection, and other PPE specified by the fertilizer label.
5. Check wind and weather conditions and keep people and animals away from the application area.

How to use

1. Calibrate the spreader according to the product rate and equipment instructions.
2. Fill the hopper in a well-ventilated area without overfilling.
3. Set the flow control to the calibrated position.
4. Start at the edge of the treatment area and maintain a steady walking pace.
5. Operate the spreader evenly, following the planned path and overlap pattern.
6. Avoid spreading near drains, open water, or areas not approved for treatment.
7. Close the flow gate before stopping or turning where appropriate.
8. Clean the spreader and wash hands after application.

⚠ Safety

- Follow fertilizer label directions and local environmental requirements.
- Avoid breathing fertilizer dust and prevent contact with eyes or skin.
- Do not spread during strong winds or heavy rain.
- Keep fertilizer away from food, drinking water, children, and animals.

Care & storage

- Empty unused fertilizer after use according to the label instructions.
- Clean the hopper and outlet to prevent corrosion or blockages.
- Dry metal parts before storage.
- Inspect wheels, controls, and fasteners regularly.

Common mistakes to avoid

- Applying fertilizer without calibrating the spreader.
- Using an incorrect flow setting or walking speed.
- Overlapping passes excessively or leaving untreated gaps.
- Allowing fertilizer to enter drains or waterways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fogging System

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **Medium** 4 labs

Covers these lab items: Fogging and blackout; Misting

What it is used for

A fogging or misting system disperses fine droplets to provide controlled humidity, cooling, or an approved treatment application. Correct setup and nozzle control help distribute mist evenly while reducing exposure, runoff, and equipment blockage.

Main parts

- Water or approved solution reservoir
- Pump and pressure regulator
- Filter and supply tubing
- Fogging or misting nozzles
- Control valves, timer, and pressure gauge if fitted
- Mounting brackets and electrical controls where applicable

Before you start

1. Read the system instructions and identify the approved liquid and intended application.
2. Inspect the reservoir, pump, tubing, filters, nozzles, and electrical connections for damage or leakage.
3. Confirm that the water or treatment solution is suitable for the system and intended site.
4. Clean the filter and check that nozzles are clear and correctly installed.
5. Verify that the pressure setting and droplet size are suitable for the application and surrounding conditions.
6. Wear the personal protective equipment required for the liquid and ensure people, animals, and sensitive equipment are protected.

How to use

1. Fill the reservoir with clean water or the approved solution according to the instructions.
2. Secure the reservoir lid and check that the supply lines are connected correctly.
3. Open the required valves and set the recommended operating pressure.
4. Start the pump or control system and inspect each nozzle for an even mist pattern.
5. Adjust nozzle direction and flow to cover the intended area without excessive wetting or drift.
6. Set the timer or application duration according to the approved operating plan.
7. Monitor pressure, liquid level, leakage, wind or airflow conditions, and the uniformity of mist distribution.
8. Stop the system if a nozzle blocks, a line leaks, or mist drifts into an unintended area.
9. Switch off and isolate power before cleaning nozzles or servicing the pump.

10. Drain, clean, and store the system according to the manufacturer's instructions and the liquid used.

⚠ Safety

- Use only liquids approved for the system and intended application; never improvise chemical mixtures.
- Follow the safety data sheet for any treatment solution and use the specified protective equipment.
- Avoid inhaling mist and keep people or animals away when the application requires exclusion.
- Depressurize the lines before loosening fittings or removing nozzles.
- Disconnect electrical power before maintenance and keep electrical components protected from moisture.
- Do not operate fogging equipment near ignition sources when using flammable liquids.

Care & storage

- Clean reservoir, filters, and nozzles at the recommended intervals to prevent blockage and contamination.
- Inspect tubing, seals, pump connections, and electrical protection for wear or leaks.
- Flush the system after using a treatment solution as instructed by the manufacturer.
- Store the unit clean and dry, and protect nozzles from dust and physical damage.

Common mistakes to avoid

- Excessive pressure can create unwanted drift or a mist pattern unsuitable for the application.
- Using dirty water or an unfiltered solution can block fine nozzles.
- Applying mist in unsuitable wind or airflow can carry droplets beyond the target area.
- Failing to flush the system after treatment use can cause corrosion, residue, or cross-contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Foot Sprayer

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **High** 5 labs

Covers these lab items: Foot / Hand sprayers; Foot Sprayer

What it is used for

A foot sprayer uses foot-operated pumping to deliver water or an approved spray solution. Operate it on stable ground and use only the liquid specified for the task.

How to use

1. Inspect the tank, foot pump, hose, nozzle, seals, and straps for damage or leaks.
2. Prepare the approved spray solution according to its label and required dilution.
3. Fill the tank without exceeding its marked capacity and close the cap securely.
4. Fit the sprayer correctly and check that the hose and nozzle are connected firmly.
5. Operate the foot pump as instructed to build pressure.
6. Direct the nozzle away from people and apply the spray evenly to the target area.
7. Release pressure safely before opening or cleaning the tank.
8. Empty, rinse, and clean the sprayer according to the product instructions.

⚠ Safety

- Wear the PPE specified for the spray solution, including gloves and eye protection where required.
- Avoid inhaling spray mist and do not spray against the wind.
- Never point the nozzle at people or animals.
- Do not open the tank while it is pressurised.

Care & storage

- Clean the tank, hose, and nozzle after use to prevent residue and blockages.
- Store the sprayer in a dry, secure location away from children and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fumigation Cover

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Multilayered cross laminated fumigation sheets/covers; Multilayered cross laminated fumigation sheets/covers (IS 14611:1998); Thermoplastic fumigation covers; Thermoplastic fumigation covers (IS 13217:1991)

What it is used for

Fumigation covers enclose designated agricultural materials or spaces during approved pest-control procedures.

How to handle

- Use fumigation covers only under an authorised pest-control plan and trained supervision.
- Inspect the cover for tears, punctures, damaged seams, and compatibility with the treatment.
- Wear the PPE specified by the risk assessment and follow all site exclusion rules.
- Never enter a covered fumigation area or handle fumigants without proper authorisation and monitoring.
- Secure the cover as specified and report leaks or damage immediately.

Storage

- Clean and dry reusable covers only when safe under the approved procedure.
- Store folded covers away from sunlight, sharp objects, heat, and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fumigation Probe

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **High** 3 labs

Covers these lab items: Metal Probe; Metal Probe (12 ft long for introducing ALP tablets)

What it is used for

A fumigation probe is used to introduce approved fumigant tablets into designated grain or storage locations when permitted by the applicable procedure. Because aluminium phosphide can release highly toxic phosphine gas, use this equipment only through authorised, trained fumigation personnel.

How to use

1. Verify that the fumigation task is legally authorised and covered by an approved site procedure.
2. Inspect the probe for damage, blockage, or loose parts before use.
3. Confirm the target material, treatment location, and required equipment using the approved fumigation plan.
4. Wear the specified protective equipment and establish the required exclusion zone and warning signs.
5. Use the probe only as directed by the authorised fumigation procedure and product label.
6. Keep untrained people and animals away from the treatment area.
7. After the operation, secure the area and follow the prescribed monitoring and clearance procedure.
8. Clean, inspect, and store the probe only as directed by the authorised procedure.

⚠ Safety

- Never handle aluminium phosphide tablets or perform fumigation without required training and legal authorisation.
- Phosphine gas is highly toxic; never rely on smell to detect its presence.
- Never enter a treated or suspected contaminated area without authorised clearance and required respiratory protection.
- Follow the product label, safety data sheet, emergency plan, and local regulations.

Care & storage

- Inspect the probe after use and report damage or blockage before reuse.
- Store the probe in its designated secure area under the site's fumigation procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Garden Knife

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Garden Knife

What it is used for

To cut, trim, or harvest small plants and garden materials using a garden knife. To make controlled cuts while protecting the user and nearby plants.

Main parts

- Blade
- Handle
- Grip or guard
- Protective sheath

Before you start

1. Select a knife suitable for the intended gardening task.
2. Inspect the blade, handle, and sheath for damage.
3. Wear suitable gloves and closed footwear where appropriate.
4. Clear the work area and identify the intended cut.

How to use

1. Hold the handle firmly with the cutting path directed away from the body.
2. Position the blade carefully at the intended cutting point.
3. Make a controlled cut without twisting or forcing the blade.
4. Keep the free hand away from the cutting path.
5. Check the result and repeat carefully if needed.
6. Clean the blade and fit the sheath after use.

⚠ Safety

- Never cut toward the body or supporting hand.
- Keep the blade away from children and bystanders.
- Do not use a loose, cracked, or blunt damaged blade.
- Carry and store the knife with its blade covered.

Care & storage

- Clean and dry the blade after use.
- Apply suitable rust protection when needed.
- Inspect the handle and fasteners regularly.

- Store in a dry, secure location.

Common mistakes to avoid

- Using the knife for tasks beyond its design.
- Cutting toward the hand or body.
- Leaving the blade wet or dirty.
- Storing the knife without its sheath.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Garden Tool Set

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Garden Hand Tools; Hand tools like shovels, rakes, scythes, picks; Trowels and rake

What it is used for

A garden tool set is used for planting, digging, raking, pruning, and routine garden maintenance. Select each tool according to the task and soil conditions.

How to use

1. Inspect tools for cracks, loose handles, rust, or damaged edges.
2. Choose the suitable tool for digging, planting, raking, or trimming.
3. Wear suitable gloves and secure the work area before starting.
4. Use each tool with controlled movements and keep others at a safe distance.
5. Remove soil and plant debris after finishing the task.
6. Dry tools and return them to their designated storage area.

⚠ Safety

- Wear gloves and closed-toe footwear while using garden tools.
- Keep sharp edges and pointed tools away from your body.
- Do not use tools with loose handles or damaged heads.
- Carry tools with sharp ends pointed downward and away from others.

Care & storage

- Clean and dry tools after use; oil metal parts when appropriate.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Garden Tractor

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Garden tractor

What it is used for

To operate a garden tractor for small-scale agricultural, gardening, and landscaping tasks. To prepare the machine, use suitable attachments, and complete work safely.

Main parts

- Engine and fuel tank
- Steering wheel and controls
- Clutch, brake, and transmission
- Tyres and wheels
- PTO or attachment coupling, if fitted
- Seat and safety switches

Before you start

1. Inspect the tractor for visible damage, loose parts, and fluid leaks.
2. Check engine oil, coolant if applicable, and fuel levels.
3. Check tyre condition and pressure according to the manual.
4. Inspect brakes, steering, lights, and safety switches.
5. Confirm that the attachment is compatible and securely connected.
6. Clear people, animals, stones, and other obstacles from the work area.

How to use

1. Park on level ground and apply the parking brake.
2. Check that the transmission is in neutral and the PTO is disengaged.
3. Adjust the seat and confirm that all guards are installed.
4. Start the engine according to the manufacturer's instructions.
5. Allow the engine to warm up and check for unusual noise or warning lights.
6. Select the appropriate gear and release the parking brake when ready.
7. Move slowly and steer carefully while operating the tractor.
8. Engage an attachment only as instructed and at the specified operating speed.
9. Reduce speed before turning and avoid sudden movements.
10. Disengage the PTO and lower attachments before stopping, as applicable.
11. Park on level ground, disengage the transmission, apply the parking brake, and switch off the engine.
12. Remove the key and inspect the tractor after use.

⚠ Safety

- Only trained operators should use the tractor.
- Keep hands, feet, clothing, and people away from moving parts and PTO shafts.
- Never carry passengers unless the tractor is specifically designed for them.
- Avoid steep slopes, unstable ground, and sudden turns that could cause rollover.
- Use the required guards and rollover protection; wear a seat belt when provided and required.
- Stop the engine and remove the key before clearing blockages or servicing attachments.

Care & storage

- Clean dirt and plant debris after use.
- Check fluid levels and inspect for leaks regularly.
- Lubricate service points according to the maintenance schedule.
- Inspect tyres, belts, blades, and attachments for wear.
- Store in a dry, secure area and follow the service schedule.

Common mistakes to avoid

- Starting with the PTO engaged or transmission in gear.
- Using an incompatible or poorly secured attachment.
- Operating at excessive speed or on unsafe slopes.
- Ignoring warning lights, leaks, or unusual noises.
- Cleaning or adjusting moving parts while the engine is running.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grain Sampling Probe

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Parkhi (Bag trier)

What it is used for

To collect representative grain samples from bags or bulk grain for quality inspection and testing.

How to handle

- Inspect the probe for damage, corrosion, sharp edges, and cleanliness before use.
- Wear suitable gloves and follow the sampling plan to select representative bags or sampling locations.
- Insert the probe carefully into the grain, collect the sample, and withdraw it without spilling or contaminating the material.
- Transfer the sample to a clean, labelled container and record the source, date, and sample identification.
- Clean and dry the probe after sampling, especially before sampling a different grain lot.

Storage

- Store clean and dry in a designated tool rack or protective case.
- Cover sharp edges and keep the probe away from moisture and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grass Cutter

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Grass cutter

What it is used for

To cut grass and light vegetation using a powered grass cutter. To achieve an even cut while keeping the operator and nearby people clear of the cutting zone.

Main parts

- Engine or electric motor
- Cutting blade or trimmer head
- Guard and deflector
- Handle and throttle or speed control
- Shaft and drive mechanism
- Harness, switch, and power connection as applicable

Before you start

1. Read the operating manual and identify the correct cutting attachment for the task.
2. Inspect the blade or trimmer head, guard, handle, shaft, and fasteners for damage.
3. For electric models, inspect the cable, plug, battery, and connectors; for fuel models, check fuel and leaks.
4. Confirm that the cutting attachment is securely fitted and suitable for the machine.
5. Remove stones, wire, glass, and other objects from the cutting area.
6. Wear eye and face protection, hearing protection, sturdy footwear, long trousers, and gloves as recommended.

How to use

1. Place the cutter on stable ground in a clear starting area.
2. For a petrol model, follow the manual's starting procedure; for an electric model, connect the power source safely.
3. Keep the cutting head clear of the ground and people when starting.
4. Hold the machine firmly with both hands or use the prescribed harness and grip.
5. Start the cutter and allow it to reach the appropriate operating speed.
6. Cut grass in controlled, overlapping passes without forcing the cutting head.
7. Maintain a safe distance from people, animals, vehicles, windows, and fragile objects.
8. Stop and isolate the machine before clearing wrapped grass or checking the cutting head.
9. Switch off the engine or disconnect the battery or power source before changing attachments.
10. After work, allow hot components to cool and remove grass debris using a safe method.
11. Inspect the cutting attachment for wear or damage before storage.

2. Store the cutter with its blade guarded in a dry, secure place.

⚠ Safety

- Keep bystanders well away from the cutting area and the direction of thrown debris.
- Never remove or modify the cutting guard.
- Do not operate with a damaged blade, loose head, or faulty switch.
- Avoid cutting in wet, slippery conditions or near hidden cables and pipes.
- Keep hands and feet away from the cutting head and never touch it while moving.
- Refuel petrol models only when switched off and cool, away from flames or sparks.
- Disconnect electrical power or remove the battery before maintenance.

Care & storage

- Clean grass and dirt from the guard, shaft, and air vents after use.
- Inspect and sharpen or replace blades only as directed by the manufacturer.
- Check fasteners, handles, harness, and safety switch regularly.
- Maintain the engine, battery, or power cable according to the model's manual.
- Store fuel safely and keep the machine protected from moisture.

Common mistakes to avoid

- Operating without the required guard or protective equipment.
- Using the wrong blade or an attachment that is not securely fitted.
- Cutting too close to people or hard objects that can throw debris.
- Clearing tangled grass while the cutting head can still move.
- Refuelling while the engine is hot or leaving the machine connected during maintenance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Greenhouse

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Green house; Green house/ Hydroponics Unit

What it is used for

To provide a controlled growing environment for plants, including soil-based or hydroponic cultivation.

How to handle

- Inspect the frame, covering, doors, ventilation, irrigation, and support structures before use.
- Check temperature, humidity, light, and airflow; adjust ventilation or shading according to crop requirements.
- For hydroponics, check the reservoir, pump, tubing, nutrient solution, and water level before starting the system.
- Use clean tools, follow safe handling instructions for fertilizers and plant-protection products, and keep walkways clear.
- Monitor plants and equipment regularly; isolate leaks or electrical faults and report structural damage promptly.

Storage

- Clean reusable tools and safely store nutrient supplies according to their labels.
- Keep the greenhouse orderly, secure chemicals separately, and protect irrigation and electrical components from damage during shutdown.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grow Light

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Grow lights

What it is used for

A grow light supplies artificial light to support plant growth when natural light is insufficient. Set the light intensity, distance, and duration according to the plant and equipment specifications.

How to use

1. Inspect the light, power cable, mounting bracket, and timer for damage.
2. Identify the plant type and recommended light intensity and daily exposure duration.
3. Mount the grow light securely at the recommended distance from the plants.
4. Connect the light to a suitable power source and set the timer if available.
5. Switch on the light and verify even coverage across the growing area.
6. Monitor plant response, temperature, and moisture during use.
7. Switch off and disconnect power before repositioning or cleaning the unit.

⚠ Safety

- Keep water away from electrical connections and use equipment rated for the growing environment.
- Do not stare directly into high-intensity grow lights.
- Prevent overheating by maintaining the recommended clearance and ventilation.
- Do not overload outlets or extension leads.

Care & storage

- Clean the light surface when powered off and inspect cables and mounts regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Growing Medium

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Growing medium (Cocopeat, Vermiculite, perlite etc.); Peat

What it is used for

Growing media such as cocopeat, vermiculite, perlite, and peat support seed germination, root growth, and plant cultivation.

How to handle

- Identify the medium type and confirm it suits the crop, seedling, or planting method.
- Check for contamination, mould, pests, and excessive moisture before use.
- Wear suitable gloves and dust protection when handling dry or dusty media.
- Moisten and mix the medium according to the growing requirements without creating unnecessary dust.
- Keep media away from chemicals, waste, and contaminated tools.

Storage

- Store unused media in clean, labelled, moisture-protected packaging.
- Keep bags sealed and protected from pests, rain, and direct contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Cultivator

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Hand Cultivator

What it is used for

A hand cultivator loosens surface soil, removes small weeds, and helps aerate soil around plants. Use it carefully to avoid damaging roots or buried irrigation lines.

How to use

1. Inspect the cultivator tines and handle for cracks, looseness, or rust.
2. Wear gloves and clear stones or other obstacles from the working area.
3. Hold the handle firmly and place the tines near the soil surface.
4. Pull or push the tool through the soil with controlled shallow movements.
5. Work around plants carefully and avoid deep strokes near roots or irrigation lines.
6. Remove soil from the tines, dry the tool, and store it safely.

⚠ Safety

- Wear gloves and closed-toe footwear while cultivating soil.
- Keep fingers and feet away from the tines.
- Do not use a tool with loose handles or damaged tines.
- Check for buried cables or pipes before digging in unfamiliar ground.

Care & storage

- Clean and dry the metal tines after use and protect them from rust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Rotary Duster

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Hand rotary duster

What it is used for

To distribute suitable dust formulations over plants or target areas using a hand-operated rotary duster, following product-label instructions.

How to handle

- Inspect the hopper, handle, rotating mechanism, outlet, and seals for damage or blockage.
- Read the product label and wear the specified protective equipment before loading any dust formulation.
- Fill the hopper only with a product approved for the intended use and avoid creating airborne dust.
- Operate the handle steadily and direct the outlet away from people, animals, food, and water sources.
- Empty and clean the duster safely after use according to the product label; do not blow dust out by mouth.

Storage

- Clean the device according to the product label and allow it to dry completely.
- Store empty in a labelled, dry, secure area away from food, children, and incompatible chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Sprayer

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **High** 9 labs

Covers these lab items: Hand Sprayer; Sprayer; Spraying , Pump; Water sprayers

What it is used for

A hand sprayer applies water or approved liquid treatments to plants, surfaces, or work areas. Use the correct liquid and spray setting for the intended task.

How to use

1. Inspect the tank, trigger, nozzle, seals, and hose for leaks or damage.
2. Read the product label and prepare only an approved solution at the specified dilution.
3. Fill the tank without exceeding its marked capacity and secure the cap.
4. Adjust the nozzle to the required spray pattern and check it in a safe direction.
5. Pump or pressurise the sprayer according to its design.
6. Spray evenly while avoiding people, animals, electrical equipment, and unintended surfaces.
7. Release pressure safely before opening, emptying, or cleaning the sprayer.
8. Rinse and dry the sprayer as directed after use.

⚠ Safety

- Wear gloves, eye protection, and other PPE required by the liquid label.
- Never spray unknown chemicals or mix products without approved instructions.
- Avoid inhaling mist and spraying against the wind.
- Do not open a pressurised tank or point the nozzle at people.

Care & storage

- Clean the tank and nozzle after use to prevent blockage and residue buildup.
- Store empty or as specified by the manufacturer, away from children and heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Trowel

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Hand Trowel; Transplanting trowel; Trowel (Khurpi)

What it is used for

Use a hand trowel to dig small planting holes, transplant seedlings, loosen soil, or move compost in garden work.

How to use

1. Choose a trowel with a blade size suitable for the task.
2. Inspect the blade and handle for cracks, rust, or looseness.
3. Wear suitable gloves and footwear before working in the soil.
4. Use the blade to loosen soil or make a small planting hole.
5. Lift soil carefully and avoid damaging nearby roots.
6. Place the seedling or soil as required and firm the soil gently.
7. Clean the trowel after use and store it safely.

⚠ Safety

- Keep fingers clear of the blade while digging.
- Check the work area for sharp objects or buried hazards.
- Do not use a damaged trowel or leave it hidden in the soil.
- Wear gloves when handling compost or soil that may contain irritants.

Care & storage

- Remove soil and dry the blade after use.
- Store the trowel in a dry place with its sharp edge protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hedge Cutter

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hedge cutter

What it is used for

A hedge cutter trims hedges, shrubs, and suitable garden growth using reciprocating or rotating cutting blades. Use the correct tool for the plant material and maintain safe distance from people, cables, and hidden objects.

Main parts

- Cutting blade or blade bar
- Front and rear handles
- Trigger, switch, and safety lock
- Electric cord, battery, or engine power unit
- Blade guard or protective cover
- Hand guard and safety controls

Before you start

1. Inspect the blade, guard, handles, switches, and power source for damage.
2. For electric models, check the cord and plug; for battery models, check the battery condition.
3. For engine models, check fuel and engine condition according to instructions.
4. Clear the hedge of visible wire, stones, and other foreign objects.
5. Check the work area for people, pets, traffic, and overhead or hidden cables.
6. Wear eye protection, hearing protection, sturdy footwear, and suitable gloves.

How to use

1. Keep the tool switched off while inspecting or adjusting the blade.
2. Fit the blade guard correctly for transport and remove it only before operation.
3. Hold the tool with both hands using the designated handles.
4. Stand securely with balanced footing and keep the blade away from the body.
5. Start the tool according to the manufacturer's procedure.
6. Begin cutting at a manageable section and use smooth, controlled strokes.
7. Keep the cutting blade clear of soil, walls, fences, and other hard objects.
8. Do not force the blade through branches thicker than the tool's rated capacity.
9. Release the trigger and wait for the blades to stop before repositioning or clearing debris.
10. Switch off and isolate the power source before removing tangled material or making adjustments.
11. After cutting, switch off the tool and wait until all moving parts stop.
12. Clean the blade, inspect for damage, and apply approved blade care product if specified.

3. Fit the blade guard and store the tool securely.

⚠ Safety

- Only trained users should operate powered hedge cutters.
- Keep both hands on the designated handles and never reach toward moving blades.
- Keep people and pets well outside the cutting area.
- Do not use in rain or wet conditions unless the tool is specifically rated for them.
- Never cut near power lines or hidden cables; maintain required clearance and seek qualified help when needed.
- Disconnect the plug, remove the battery, or isolate the engine before maintenance.

Care & storage

- Clean plant residue from the blade after each use.
- Inspect blade teeth, fasteners, guards, and handles regularly.
- Lubricate and sharpen blades only according to the manufacturer's instructions.
- Store with the blade guard fitted in a dry, secure place.
- Report unusual vibration, damaged blades, or faulty switches before reuse.

Common mistakes to avoid

- Cutting without checking for hidden wire or other obstacles.
- Using one hand or standing in an unstable position.
- Clearing tangled branches while the blades are moving.
- Using the cutter on branches beyond its rated cutting capacity.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hedge Shear

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Grass Shear; Hedge Shear

What it is used for

Hedge shears are used to trim hedges, shrubs, and small soft branches. Use them for suitable plant growth rather than thick woody branches.

How to use

1. Inspect blades, pivot, handles, and fasteners for damage or looseness.
2. Wear suitable gloves and eye protection before trimming.
3. Check the work area for people, wires, and hidden objects.
4. Hold both handles firmly and cut small sections with controlled movements.
5. Avoid forcing the blades through thick branches or hard material.
6. Remove plant debris, clean and dry the blades, and store safely.

⚠ Safety

- Keep hands and fingers away from the cutting edges.
- Never cut toward your body or another person.
- Do not use shears with loose handles or damaged blades.
- Carry shears with blades closed and pointed away from others.

Care & storage

- Clean and dry blades after use; lubricate the pivot and protect blades from rust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hoe

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 36 labs

Covers these lab items: Dutch Hand Hoe; Dutch-hoe; Garden hoe; Garden Hoes; Hand hoe (Khurpi); Hoe; Hoe-cum-rack; Khurpi; Long Handle Hoes; Pronged hoes; Straight Blade Hand Hoe; Trenching hoe

What it is used for

A hoe is used to loosen soil, remove weeds, shape planting beds, and perform light garden cultivation. Choose the appropriate hoe type and work with controlled movements in a clear area.

How to use

1. Inspect the blade, head, handle, and attachment for cracks or looseness.
2. Select a hoe suitable for the soil and intended task.
3. Clear the work area of stones, cables, and other obstructions.
4. Hold the handle firmly with both hands and maintain stable footing.
5. Use controlled strokes directed away from the body and nearby people.
6. Avoid striking buried pipes, irrigation lines, or plant roots unnecessarily.
7. Remove soil and debris from the blade after work.
8. Store the hoe safely with the blade protected.

⚠ Safety

- Wear sturdy footwear, gloves, and eye protection when required.
- Keep other people clear of the swinging range.
- Do not use a hoe with a loose head or damaged handle.
- Carry the hoe with the blade pointed down and away from the body.

Care & storage

- Clean and dry the blade after use to reduce rust.
- Store the hoe in a dry location and sharpen or repair it as required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hori Hori Knife

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Hori Hori Knife; Hori Hori Knife (Soil Knife); Hori Knife

What it is used for

To dig, transplant, weed, and cut small roots using a Hori Hori garden knife. To use the blade and measuring markings carefully for controlled gardening tasks.

Main parts

- Steel blade
- Sharp or serrated edge, depending on model
- Handle and grip
- Depth markings, if fitted
- Protective sheath

Before you start

1. Inspect the blade, handle, and sheath for damage or looseness.
2. Check the soil for hidden cables, pipes, stones, or other hazards before digging.
3. Wear suitable gloves and closed footwear.
4. Confirm the tool is appropriate for the soil and task.

How to use

1. Hold the handle securely and keep the blade directed away from the body.
2. Insert the blade into the soil at the required location using controlled pressure.
3. Use the blade to loosen soil or lift small weeds and seedlings carefully.
4. Cut small roots only where safe and appropriate.
5. Use depth markings as a guide when provided, not as a substitute for checking underground hazards.
6. Clean the blade and place it in its sheath after use.

⚠ Safety

- Keep fingers clear of the blade and serrated edge.
- Never use the knife to probe near suspected electrical or utility lines.
- Do not twist excessively or use the tool as a pry bar.
- Keep the sheathed tool away from children and bystanders.

Care & storage

- Remove soil and plant residue after use.

- Dry the blade thoroughly to prevent rust.
- Inspect the edge and handle regularly.
- Store in the supplied sheath in a dry place.

Common mistakes to avoid

- Digging without checking for buried utilities.
- Forcing or twisting the blade in hard soil.
- Holding the soil or plant in the blade path.
- Storing the knife with wet soil on the blade.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hose Pipe

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Hosepipe; Water Hose

What it is used for

Hose pipes deliver water for irrigation, gardening, washing, and other approved tasks.

How to handle

- Inspect the hose, connectors, and fittings for leaks, cracks, or blockages.
- Connect securely to a compatible water supply and avoid excessive water pressure.
- Uncoil the hose before use and keep it clear of walkways to prevent trips.
- Avoid kinks, sharp bends, vehicle traffic, and contact with hot surfaces.
- Turn off the water supply and release pressure before disconnecting.

Storage

- Drain the hose after use and coil it without tight bends.
- Store in a shaded, dry location away from sharp objects and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Irrigation System

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Irrigation system

What it is used for

An irrigation system supplies water to crops through suitable channels, pipes, sprinklers, or drip lines. Operate it according to crop water needs and the system's design specifications.

How to use

1. Inspect the water source, pump, filters, valves, pipes, and emitters for damage or leaks.
2. Confirm the water supply is suitable and available for irrigation.
3. Open the required valves and start the pump according to the operating procedure.
4. Check water pressure and flow at the designated points.
5. Apply water evenly according to crop requirements and the irrigation schedule.
6. Inspect the field for blocked emitters, leaks, pooling, or uneven distribution.
7. Stop the system, close valves as directed, and record any maintenance needs.

⚠ Safety

- Keep electrical pumps and connections protected from water exposure.
- Do not operate pumps without adequate water supply if dry running can cause damage.
- Relieve pressure before opening pressurized lines or filters.
- Use approved precautions when working near ponds, channels, or other water hazards.

Care & storage

- Clean filters and emitters regularly and repair leaks before the next irrigation cycle.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Knapsack Sprayer

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 8 labs

Covers these lab items: Knapsack Sprayer

What it is used for

A knapsack sprayer applies approved liquid treatments to crops, plants, or other designated areas. Correct mixing, pressure control, protective equipment, and cleaning help prevent exposure, leaks, and environmental contamination.

Main parts

- Tank stores the approved spray liquid.
- Filler opening and cap allow safe filling and closure.
- Pump or pressure mechanism builds the required spray pressure.
- Lever or motor control operates the pump on applicable models.
- Hose carries liquid from the tank to the spray lance.
- Lance and nozzle direct the spray onto the target.
- Shoulder straps support the tank during carrying.

Before you start

1. Read the product label and sprayer manual before preparing any spray mixture.
2. Use only approved substances and follow the specified dilution and application instructions.
3. Wear the required PPE, including chemical-resistant gloves, protective clothing, footwear, and eye protection.
4. Inspect the tank, cap, straps, hose, seals, lance, nozzle, and pump for damage or leaks.
5. Prepare the mixture in a ventilated area away from food, drinking water, children, and animals.
6. Check wind direction and avoid spraying in strong wind or toward people and sensitive areas.

How to use

1. Confirm the sprayer is clean and suitable for the intended product.
2. Prepare the spray mixture according to the product label and fill the tank without overfilling.
3. Close the cap securely and check for leaks before carrying the sprayer.
4. Adjust the shoulder straps so the tank sits securely without restricting movement.
5. Build pressure using the approved pump or start the motor according to the manual.
6. Test the spray pattern in a safe designated area before application.
7. Hold the lance toward the target and apply the recommended amount evenly.
8. Keep the nozzle away from your body and avoid breathing spray mist.
9. Stop spraying if leaks develop, the nozzle blocks, or wind changes create exposure risk.
10. Release pressure safely, empty and clean the sprayer as directed, and wash hands after removing PPE.

⚠ Safety

- Never spray without the PPE specified on the product label.
- Do not mix chemicals or use unlabelled substances in the tank.
- Never point the nozzle at people, animals, food, or drinking-water sources.
- Do not eat, drink, or smoke while mixing or applying chemicals.
- Release pressure before opening the tank, removing the nozzle, or clearing a blockage.
- Keep children and animals away from the treated area for the period stated on the product label.

Care & storage

- Rinse and clean the tank, hose, lance, and nozzle according to the product label and manual.
- Use a suitable brush or cleaning method for blocked nozzles; never blow through them by mouth.
- Inspect seals, straps, valves, and hoses regularly and replace damaged parts.
- Store the clean, dry sprayer securely and dispose of leftover mixture and rinse water as directed.

Common mistakes to avoid

- Mixing chemicals without reading the label or measuring the required dilution.
- Opening a pressurised tank or clearing a nozzle with the mouth.
- Spraying against the wind or without the required protective equipment.
- Disposing of leftover chemicals or rinse water into drains or water bodies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lopping Shear

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Looping shear; Loppers or Pruning Saw; Lopping Shear

What it is used for

To cut or prune branches using lopping shears. To make clean cuts safely while reducing damage to the plant and avoiding tool slippage.

Main parts

- Cutting blade
- Counter blade or anvil
- Long handles
- Grip
- Locking catch

Before you start

1. Select loppers suitable for the branch diameter and type.
2. Inspect blades, handles, pivot, and locking catch for damage.
3. Wear gloves and eye protection where appropriate.
4. Check the work area for overhead hazards and stable footing.

How to use

1. Release the locking catch and open the handles fully.
2. Position the branch near the intended cutting point within the blade's rated capacity.
3. Hold both handles securely and keep fingers away from the blades and pivot.
4. Apply steady pressure until the branch is cut.
5. Avoid twisting the tool or forcing it beyond its capacity.
6. Check the cut and reposition only after the blades are clear.
7. Close and lock the tool after use.

⚠ Safety

- Keep hands and face away from the cutting path.
- Do not cut branches above safe reach or near power lines.
- Never use damaged loppers or exceed their rated capacity.
- Keep the tool locked when carrying or storing it.

Care & storage

- Remove sap and plant residue after use.

- Dry the blades and apply suitable rust protection if needed.
- Lubricate the pivot according to the manufacturer's instructions.
- Sharpen or replace blades when required.

Common mistakes to avoid

- Cutting branches thicker than the tool's capacity.
- Twisting the handles during cutting.
- Working from an unstable ladder or unsafe position.
- Leaving the blades unlocked during storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Manual Transplanter

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Manual transplanter

What it is used for

A manual transplanter places seedlings into soil at a controlled depth and spacing. Prepare the field properly and use the tool according to the crop and planting method.

How to use

1. Inspect the transplanter handle, planting mechanism, frame, and moving parts for damage.
2. Prepare moist, workable soil and mark the required planting spacing.
3. Select healthy seedlings suitable for the crop and planting stage.
4. Position the transplanter at the marked planting point.
5. Operate the planting mechanism to open the soil, place the seedling, and release it as designed.
6. Move to the next point while maintaining consistent spacing and planting depth.
7. Check the planted seedlings, clean soil from the tool, and store it after use.

⚠ Safety

- Wear suitable footwear and gloves when working in wet or rough soil.
- Keep fingers away from moving planting jaws or mechanisms.
- Do not force the tool in compacted soil or around hidden obstacles.
- Use safe lifting posture when moving the equipment.

Care & storage

- Wash off soil, dry metal parts, and inspect the planting mechanism before storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Moisture Meter

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Digital moisture meter; Digital moisture meter calibrated for all the commodities to be stored/ handled; Moisture meter; Seed moisture tester

What it is used for

Use a moisture meter to measure moisture content in soil, seeds, grain, or other materials supported by the instrument.

How to use

1. Select the correct meter and mode for the material being tested.
2. Check the probe, sensor, display, and battery condition before use.
3. Confirm calibration or perform the recommended check according to the manual.
4. Collect a representative sample or prepare the material as instructed.
5. Insert the probe or place the sample in the meter correctly.
6. Wait for the reading to stabilize and record the value with the material and units.
7. Clean the sensor and return the meter to storage after testing.

⚠ Safety

- Use the meter only for materials and conditions it is designed for.
- Handle probes carefully and avoid contact with sharp or contaminated samples.
- Keep electronic parts dry unless the meter is rated for wet use.
- Follow the manual for calibration, cleaning, and safe sample handling.

Care & storage

- Clean and dry probes or sample holders after use.
- Store the meter in its case and replace batteries as recommended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mould Board Plough

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: MB Plough

What it is used for

A mould board plough cuts, lifts and turns soil during primary tillage. It helps bury weeds and crop residues while preparing the field for further cultivation.

Main parts

- Identify the mould board that turns the lifted soil slice.
- Locate the share that cuts soil horizontally near the ground.
- Identify the plough standard, frog and supporting frame.
- Locate the hitching points connecting the plough to the tractor.
- Identify the landside that helps resist sideways soil pressure.
- Locate the depth and width adjustment points, where provided.
- Identify the transport lock and safety components, if fitted.

Before you start

1. Inspect the plough frame, share, mould board and fasteners for damage.
2. Check hitching points and tractor compatibility before attachment.
3. Park the tractor on firm, level ground and apply the parking brake.
4. Lower the implement safely and stop the engine before hitching adjustments.
5. Keep bystanders away and use the correct lifting points during attachment.
6. Check the field for stones, buried objects, people and other hazards.
7. Confirm that transport locks and hydraulic connections are in safe condition.

How to use

1. Position the tractor and plough on firm, level ground for attachment.
2. Attach the plough using the designated hitching points and secure all pins.
3. Connect hydraulic components only as specified for the implement.
4. Adjust plough alignment, working depth and width according to field conditions.
5. Raise the plough fully before moving to the field.
6. Remove transport locks only when safely positioned for field operation.
7. Lower the plough gradually at the field entry point.
8. Drive at a suitable low working speed while monitoring soil turning.
9. Raise the plough before turning at headlands or crossing obstacles.
10. Park on stable ground, lower the implement and stop the tractor before inspection.

⚠ Safety

- Keep clear of the space between the tractor and plough during hitching.
- Never work beneath a raised plough without approved mechanical support.
- Stop the engine and lower the implement before clearing obstructions.
- Keep people away from the operating and turning area.
- Use suitable footwear and gloves when handling sharp shares or edges.
- Secure the implement properly before road transport and follow local transport rules.

Care & storage

- Remove soil and crop residue after the implement is safely lowered.
- Inspect shares, mould boards, pins and fasteners for wear or damage.
- Tighten loose fasteners and replace worn parts as instructed.
- Store the plough in a dry, stable location with cutting edges protected.

Common mistakes to avoid

- Attempting to attach the plough on unstable or sloping ground.
- Working with loose hitch pins or damaged structural parts.
- Turning sharply with the plough buried in the soil.
- Leaving the implement raised or unsupported during inspection and maintenance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nursery Bag

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Grafting transplanting bags

What it is used for

Handle nursery and grafting bags to support healthy plant propagation.

How to handle

- Select bags of suitable size and check for tears or blocked drainage holes.
- Fill with the recommended growing medium without excessive compaction.
- Handle seedlings gently and avoid damaging roots or stems.
- Use clean bags and tools to reduce the spread of plant disease.

Storage

- Keep unused bags dry and protected from sunlight and pests.
- Store away from chemicals and contaminated plant waste.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Organic Manure

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Compost; Leaf manure; Neem cake; Oil cake

What it is used for

Organic manure such as compost, leaf manure, and neem or oil cakes adds organic matter and nutrients to soil.

How to handle

- Identify the manure type and confirm it is suitable for the crop and application rate.
- Wear gloves and avoid inhaling dust during handling or mixing.
- Use mature, properly composted material and avoid contaminated or foul-smelling batches.
- Apply according to crop requirements and local agricultural guidance.
- Wash hands and tools after handling and keep manure away from drinking water.

Storage

- Store in a designated area protected from rain runoff and water sources.
- Keep bags or piles covered and prevent contamination or excessive moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paddy Puddler

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Paddy cultivator & puddler

What it is used for

A paddy puddler mixes soil and water to create soft, muddy field conditions for rice cultivation. It helps prepare a level puddled field before transplanting rice seedlings.

Main parts

- Identify the tractor or power-unit hitching frame.
- Locate the puddling blades, tines or rotary working elements.
- Identify the main drive shaft and transmission components, if fitted.
- Locate the depth or working-position adjustment mechanism.
- Identify the supporting frame and protective guards.
- Locate the hydraulic connections or lifting mechanism, where provided.
- Identify the transport lock and manufacturer's operating instructions.

Before you start

1. Inspect the frame, blades, tines, fasteners and guards for damage.
2. Confirm that the puddler is compatible with the tractor or power unit.
3. Park on firm, level ground, apply the parking brake and stop the engine before attachment.
4. Check hitch pins and secure all connections according to instructions.
5. Inspect the field for stones, stumps, buried objects and people.
6. Ensure water depth and soil conditions are suitable for puddling.
7. Keep bystanders away from the attachment and intended working area.

How to use

1. Position the power unit and puddler on stable ground for attachment.
2. Connect the implement using designated hitch points and secure locking pins.
3. Connect hydraulic or drive components according to the manufacturer's instructions.
4. Set the working depth and implement alignment for the field conditions.
5. Raise the implement fully before travelling to the field.
6. Start the power unit and engage the drive only as instructed.
7. Lower the puddler gradually into the flooded or water-softened soil.
8. Move forward at a suitable low speed while monitoring soil mixing.
9. Raise the implement before turning sharply or crossing obstacles.
10. Stop the power unit, lower the implement safely and inspect after operation.

⚠ Safety

- Keep clear of rotating blades, shafts and the tractor-implement connection.
- Never adjust or clear the puddler while the drive is engaged.
- Stop the engine and lower the implement before inspection or maintenance.
- Never stand beneath a raised implement without approved mechanical support.
- Avoid deep mud, unstable field edges and hidden ditches.
- Use suitable footwear and keep people away from flying mud or debris.

Care & storage

- Remove mud and crop residue after stopping and safely lowering the implement.
- Inspect blades, tines, shafts and fasteners for wear or damage.
- Lubricate designated points and check drive components as instructed.
- Store the implement in a stable, dry location protected from corrosion.

Common mistakes to avoid

- Operating with loose hitch pins or damaged protective guards.
- Working too deeply or too quickly for the soil conditions.
- Turning sharply while the puddler remains engaged in the soil.
- Cleaning mud from moving parts without stopping and isolating the drive.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paddy Transplanter

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Paddy transplanter

What it is used for

A paddy transplanter places young rice seedlings into prepared wet or puddled fields at regular spacing. It improves planting consistency and reduces the manual effort required for transplanting.

Main parts

- Identify the seedling tray or mat-seedling loading platform.
- Locate the seedling feeding mechanism and planting fingers.
- Identify the planting unit that places seedlings into the soil.
- Locate the engine or tractor power connection, depending on model.
- Identify the steering controls, gear selector and travel controls.
- Locate the depth, row-spacing and planting adjustments, where provided.
- Identify protective guards, emergency stop and transport locking components.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Inspect the frame, planting fingers, seedling feed mechanism and guards for damage.
3. Check fuel, engine oil and other specified fluids on engine-powered models.
4. Confirm that seedlings are healthy and prepared in the required tray or mat format.
5. Inspect the field for suitable puddling, water depth and obstructions.
6. Check that safety controls, brakes and steering work correctly.
7. Keep bystanders away and wear suitable footwear for wet-field conditions.

How to use

1. Park the transplanter on firm, level ground and stop the engine before loading.
2. Arrange seedling trays or mats correctly on the designated platform.
3. Set planting depth, row spacing and operating speed according to instructions.
4. Start the engine or power unit using the prescribed procedure.
5. Move to the field at low speed with the planting unit raised.
6. Lower the planting unit gradually at the starting point.
7. Engage the planting mechanism and move forward at a steady speed.
8. Monitor seedling feeding, planting depth and spacing during operation.
9. Stop safely to replenish trays or clear a blockage after disengaging the mechanism.
10. Raise the planting unit before turning at field ends.
11. Park on stable ground, lower the implement and stop the engine after work.

⚠ Safety

- Keep hands away from planting fingers, chains and seedling feed mechanisms.
- Never load trays or clear blockages while the planting mechanism is engaged.
- Stop the engine before adjustments, maintenance or cleaning.
- Use designated steps and handholds when mounting or dismounting the machine.
- Avoid steep banks, deep mud and unstable field edges.
- Keep people away from the machine while reversing or turning.

Care & storage

- Remove mud, roots and plant residue after stopping and lowering the unit safely.
- Clean seedling trays, feeding parts and planting fingers after use.
- Inspect chains, belts, planting fingers and fasteners for wear or damage.
- Check fluid levels and lubricate specified points according to instructions.
- Store the transplanter clean and dry with planting units secured.

Common mistakes to avoid

- Loading incorrectly prepared seedlings that feed poorly through the mechanism.
- Operating at excessive speed and producing uneven seedling spacing.
- Adjusting planting fingers while the mechanism is moving.
- Turning sharply with the planting unit lowered in the soil.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pesticide

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **High** 22 labs

Covers these lab items: Bavestin; Biopesticides; Fungicide; Indofil-45; Insect Killer Spray; Insecticide; Non-systemic fungicides; Pesticides; Recommended Pesticides; Recommended Pesticides (for prophylactic and curative treatments) and rat control; Systemic fungicides

What it is used for

Pesticides control specified agricultural pests, insects, or plant diseases when applied according to the approved product label.

How to handle

- Identify the target pest and select only a product approved for the intended crop and use.
- Read the label, wear required PPE, and prepare the application at the specified concentration.
- Avoid inhalation, skin contact, drift, and contamination of food, water, people, and animals.
- Follow label instructions for application, re-entry, harvest intervals, and disposal.

Storage

- Keep in original labelled containers in a locked, dry, ventilated area away from food, feed, and water.
- Store away from children, animals, heat, and incompatible materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pheromone Trap

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Pheromone traps

What it is used for

Pheromone traps monitor or help manage target insect populations in approved agricultural pest-management programs.

How to handle

- Identify the target pest and use the lure and trap type recommended for the crop.
- Read the label and follow placement, timing, and replacement instructions.
- Wear gloves when handling lures and avoid touching the active material directly.
- Place traps securely at the recommended height and spacing, away from unintended interference.
- Record trap counts and dispose of used lures according to label instructions.

Storage

- Keep lures sealed in labelled packaging and store under specified temperature conditions.
- Separate different pheromone types to avoid cross-contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Plant Growth Regulator

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Medium** 3 labs

Covers these lab items: Growth regulator/ harmone; Plant Growth regulator/hormones

What it is used for

Plant growth regulators influence plant growth or development and must be selected for the intended crop and purpose.

How to handle

- Confirm the product, crop suitability, approved use, dose, and application timing from the label.
- Wear specified PPE and use calibrated measuring and application equipment.
- Avoid drift, direct contact, and contamination of water, food, people, and non-target plants.
- Follow label instructions for re-entry, harvest intervals, cleaning, and disposal.

Storage

- Keep in original labelled containers in a secure, dry area away from food, feed, and water.
- Protect from temperature extremes and unauthorized access.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Planting Board

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Planting board

What it is used for

To assist with consistent plant spacing, seed placement, or row marking during planting activities.

How to handle

- Inspect the planting board, marked holes, spacing guides, and handles for cracks or damage.
- Prepare and level the soil or planting surface before positioning the board.
- Align the board with the intended row or planting layout and press gently to mark evenly spaced positions.
- Use the marks or openings to place seeds or seedlings at the recommended depth and spacing for the crop.
- Lift the board carefully, clean off soil, and check that the planting layout is consistent.

Storage

- Remove soil and plant debris, then clean and dry the board.
- Store flat in a dry tool area away from moisture, direct heat, and heavy loads that may warp it.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Plough

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: Improved paddy biasi plough; Plough

What it is used for

A plough cuts, lifts, and turns soil to prepare land for crop planting. Correct adjustment, suitable field conditions, and safe tractor operation improve soil preparation and reduce equipment damage.

Main parts

- Ploughshare cuts into the soil.
- Mouldboard lifts and turns the cut soil on applicable ploughs.
- Frame supports the plough bodies and connecting components.
- Hitch or linkage connects the plough to the tractor.
- Landside helps stabilise the plough against sideways soil pressure.
- Depth or width adjustment controls working conditions on applicable models.
- Wheel or support component helps control depth or transport where fitted.

Before you start

1. Read the operator's manual and use the plough only after suitable training.
2. Inspect the ploughshare, mouldboard, frame, hitch pins, fasteners, and supports for damage.
3. Confirm that the tractor and plough are compatible in size and hitch type.
4. Attach the plough on firm, level ground with the tractor secured against movement.
5. Set the working depth and width according to soil conditions and the manual.
6. Remove stones, wires, people, animals, and other hazards from the field.

How to use

1. Park the tractor securely and lower or support the plough before adjustments.
2. Connect the plough using the approved hitch and secure all locking pins.
3. Adjust the plough alignment and working depth according to the operator's manual.
4. Raise the plough fully to the transport position before moving between fields.
5. Align the tractor with the intended furrow and lower the plough at the correct starting point.
6. Move forward at the recommended speed while maintaining a steady direction.
7. Monitor soil turning, working depth, and tractor performance during each pass.
8. Raise the plough before turning sharply or reversing at the end of a row.
9. Stop the tractor, secure it, and lower or support the plough before clearing blockages.
10. After use, remove soil, inspect wear parts, and park the plough securely.

⚠ Safety

- Never stand beneath a raised plough or between the tractor and plough during hitching.
- Keep hands away from sharp ploughshares and moving linkage points.
- Stop the engine and secure the tractor before adjusting, cleaning, or repairing the plough.
- Keep people and animals away from the working area.
- Do not exceed the tractor's capacity or operate on unsafe slopes.
- Use proper support before working beneath or near any raised implement.

Care & storage

- Clean soil and crop residue from the ploughshare and mouldboard after safe shutdown.
- Inspect cutting edges, hitch pins, fasteners, frame, and supports for wear or damage.
- Lubricate moving joints and protect exposed metal according to the manual.
- Store the plough on stable, level ground with the implement secured.

Common mistakes to avoid

- Adjusting the plough while raised without proper support.
- Operating with loose hitch pins or damaged cutting parts.
- Travelling with the plough lowered or turning sharply while it is engaged.
- Ignoring unsuitable soil conditions or the recommended working depth.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Power Sprayer

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Power sprayer

What it is used for

A power sprayer uses a motor or engine-driven pump to apply liquid over crops, plants, or approved surfaces. Correct chemical handling, pressure control, and maintenance help prevent exposure, leaks, and equipment damage.

Main parts

- Motor or engine drives the spray pump.
- Tank stores the spray liquid on tank-equipped models.
- Pump creates pressure to move liquid through the system.
- Pressure gauge displays operating pressure where fitted.
- Hose carries liquid from the pump to the spray lance.
- Lance and nozzle direct the spray toward the target.
- Pressure regulator and control valve adjust output on applicable models.

Before you start

1. Read the product label and sprayer manual before preparing the spray mixture.
2. Use only approved substances and follow the specified dilution and application rate.
3. Wear the required PPE, including chemical-resistant gloves, protective clothing, footwear, and eye protection.
4. Inspect the pump, hoses, tank, seals, clamps, lance, nozzle, and power source for damage.
5. Check that the suction line and filters are clean and all fittings are secure.
6. Choose a ventilated mixing area away from people, animals, food, and water sources.

How to use

1. Prepare the mixture according to the product label and fill the tank without overfilling.
2. Secure the tank cap and check that all hose connections are tight.
3. Position the sprayer on stable ground and ensure the intake has an adequate liquid supply.
4. Start the motor or engine according to the manual and check for leaks.
5. Adjust pressure to the recommended setting for the nozzle and task.
6. Test the spray pattern in a designated safe area before treating the target.
7. Hold the lance securely and apply the specified amount evenly over the target area.
8. Monitor pressure, flow, and hose condition during operation.
9. Stop the machine if leakage, blockage, abnormal vibration, or chemical exposure occurs.
10. Switch off, release pressure safely, clean the equipment as directed, and wash hands after removing PPE.

⚠ Safety

- Never operate the sprayer without the PPE specified on the product label.
- Do not point the spray at people, animals, food, or drinking-water sources.
- Never open pressurised fittings or clear a nozzle while the pump is running.
- Keep the engine exhaust away from enclosed areas and ignition sources.
- Do not mix incompatible chemicals or use unlabelled substances.
- Dispose of leftover mixture and wash water according to the product label and local requirements.

Care & storage

- Release pressure and isolate the power source before cleaning or maintenance.
- Rinse the tank, pump, hose, and nozzle using the approved cleaning procedure.
- Inspect seals, filters, hoses, clamps, and nozzle openings regularly for wear or blockage.
- Store the clean equipment in a secure, dry area and arrange servicing as instructed.

Common mistakes to avoid

- Running the pump without an adequate liquid supply when dry running is not permitted.
- Setting pressure beyond the rated limit of the hose or equipment.
- Opening pressurised connections or clearing a blockage with the pump running.
- Spraying without PPE or disposing of chemical waste improperly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Power Tiller

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 10 labs

Covers these lab items: Power tiller; Rigid Tillers

What it is used for

A power tiller is a motorised machine used to loosen and prepare soil for agricultural work. Correct setup, controlled operation, and routine maintenance help reduce injury and machine damage.

Main parts

- Engine or motor supplies power to the tiller.
- Handlebars allow the operator to guide and control the machine.
- Throttle or speed control adjusts engine speed on applicable models.
- Clutch or drive control engages and disengages power to the wheels or tines.
- Rotating tines or blades break and loosen the soil.
- Wheels or transport supports help move the machine between work areas.
- Protective guards cover hazardous moving parts and hot components.

Before you start

1. Read the operator's manual and use the tiller only after practical training.
2. Wear sturdy footwear, close-fitting clothing, and required eye and hearing protection.
3. Inspect the engine, fuel system, controls, tyres or wheels, tines, guards, and fasteners.
4. Check engine oil, fuel, and other required fluids according to the manual.
5. Clear stones, wires, branches, and people from the work area.
6. Start the machine outdoors in a well-ventilated area, away from flames and ignition sources.

How to use

1. Position the tiller on stable, level ground before starting the engine.
2. Set the transmission and controls to the starting position specified by the manual.
3. Start the engine using the approved procedure and check for unusual noise or leaks.
4. Hold the handlebars firmly and engage the drive or tines gradually.
5. Walk behind the machine at a controlled pace and keep a stable footing.
6. Guide the tiller in straight, manageable passes without forcing it through hard soil.
7. Disengage the drive before turning, crossing obstacles, or making adjustments.
8. Stop the engine and wait for all moving parts to stop before clearing tangled material.
9. Park the tiller securely, switch off the engine, and allow hot parts to cool.
10. Clean soil from the machine and inspect it before storage.

⚠ Safety

- Never place hands or feet near rotating tines, blades, belts, or drive parts.
- Do not refuel a hot or running engine; stop it and allow it to cool first.
- Keep children, bystanders, and animals away from the operating area.
- Never operate the tiller indoors or where exhaust gases can accumulate.
- Stop the engine before clearing blockages, adjusting parts, or carrying out maintenance.
- Do not operate on steep, unstable, or unsuitable ground beyond the machine's stated limits.

Care & storage

- Clean soil and plant material from tines and guards after the machine is stopped and cooled.
- Check fasteners, tines, wheels, cables, and controls regularly for wear or damage.
- Follow the manual for engine oil, air filter, fuel system, and scheduled servicing.
- Store the tiller in a dry, secure place with fuel handled according to the manual.

Common mistakes to avoid

- Starting the machine before checking the work area and protective guards.
- Trying to clear tangled plants while the tines are rotating.
- Refuelling while the engine is hot or running.
- Forcing the tiller through hard soil or operating beyond safe ground conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pruning Knife

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Pruning knife

What it is used for

Use a pruning knife to make controlled cuts when trimming small branches, stems, or plant material suitable for hand pruning.

How to use

1. Select a sharp pruning knife suitable for the plant and cut size.
2. Inspect the blade, handle, and folding lock or sheath for damage.
3. Secure the plant material and plan the cut before starting.
4. Cut away from your body and keep the supporting hand clear of the blade path.
5. Make a clean cut without twisting or forcing the blade.
6. Close or sheath the knife immediately after use.
7. Clean and disinfect the blade when needed to reduce disease spread between plants.

⚠ Safety

- Wear suitable gloves and keep the blade pointed away from the body.
- Do not use a loose, damaged, or blunt knife.
- Keep the knife closed or covered when not in use.
- Follow safe handling instructions for disinfectants and cutting tools.

Care & storage

- Clean and dry the blade after use to prevent rust.
- Store it closed or in a protective sheath away from children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pruning Saw

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Pruning Saw

What it is used for

To cut branches and woody plant material during pruning and garden maintenance.

How to handle

- Inspect the blade, teeth, handle, and locking mechanism or sheath for damage before use.
- Wear suitable gloves and eye protection and ensure the work area below the branch is clear.
- Secure the branch where practical and use controlled strokes, keeping hands and body clear of the blade path.
- Do not cut branches near power lines or attempt overhead work beyond safe reach; use trained personnel where required.
- Clean the blade after use, dry it thoroughly, and cover the teeth before storage.

Storage

- Store clean and dry in a sheath or tool rack with the blade covered.
- Keep the saw secured away from children and protect the teeth from impact and corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pruning Secateurs

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 16 labs

Covers these lab items: Pruners; Pruning Sheared / C Cutter; Pruning Shears; Secateur; Secateurs; Tree Cutting Shears

What it is used for

Pruning secateurs cut small stems, twigs, and plant growth during gardening and plant maintenance. Select a suitable tool and make controlled cuts to avoid damaging the plant or injuring the user.

How to use

1. Inspect the blades, pivot, spring, handle, and locking catch for damage.
2. Select secateurs suitable for the branch thickness and plant type.
3. Wear suitable gloves and secure the plant or branch where needed.
4. Open the blades and position the branch correctly between the cutting edges.
5. Make a controlled cut without twisting the tool or exceeding its capacity.
6. Lock the secateurs closed immediately after use.
7. Remove plant residue and clean the blades before storage.
8. Store the secateurs with the blades closed in a safe, dry location.

⚠ Safety

- Keep fingers away from the cutting edges and pinch points.
- Do not cut branches thicker than the tool's rated capacity.
- Never leave open secateurs unattended or within children's reach.
- Use eye protection when cutting branches that may spring back.

Care & storage

- Clean and dry the blades after use to reduce corrosion and sap buildup.
- Lubricate the pivot and sharpen or replace blades according to the manufacturer's instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rake

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 14 labs

Covers these lab items: Garden Rake; Leaf Rake; Leaf Rake with Handle; Rake

What it is used for

Use a rake to collect leaves, level loose soil, or gather garden debris according to the rake type.

How to use

1. Choose a rake suited to leaves, soil, or garden debris.
2. Inspect the handle, head, and teeth for cracks or looseness.
3. Clear people and obstacles from the working area.
4. Hold the handle firmly with both hands and keep a balanced stance.
5. Pull the rake toward you using controlled strokes without twisting your back.
6. Collect the gathered material and place it in the designated area.
7. Clean the rake after use and check it for damage.

⚠ Safety

- Wear closed shoes and gloves while working.
- Keep rake teeth pointed away from your body and other people.
- Do not leave the rake on a walkway with its teeth facing upward.

Care & storage

- Remove soil and plant debris before storage.
- Store the rake securely with its teeth protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rat Trap

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Rat cages (wonder traps)

What it is used for

To capture or control rats in an approved pest-management programme while reducing risks to people and non-target animals.

How to handle

- Inspect the trap for damage, corrosion, and proper operation before setting it.
- Place the trap only in approved locations based on signs of rodent activity and away from children, pets, and non-target wildlife.
- Follow the manufacturer's instructions for setting and baiting the trap; use gloves and avoid direct contact with bait or rodent waste.
- Check the trap at the frequency required by local rules and the trap design, and handle captured rodents using appropriate protective equipment.
- Clean, disinfect, reset, or dispose of the trap according to local pest-control and waste-handling procedures.

Storage

- Store unset, clean, and dry in a secure container away from children and non-target animals.
- Keep bait separate and labelled, and follow the manufacturer's storage instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Reaper

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Paddy reaper

What it is used for

A paddy reaper cuts mature rice stalks close to the ground and gathers them for collection. It reduces manual harvesting effort and helps improve harvesting efficiency under suitable field conditions.

Main parts

- Identify the cutting blade or cutter bar used to sever the crop.
- Locate the crop divider that guides standing rice into the cutting area.
- Identify the reel, conveyor or gathering mechanism, depending on design.
- Locate the engine, power transmission and drive components.
- Identify the operator handles, steering controls and speed control.
- Locate the cutting-height adjustment and protective guards.
- Identify the emergency stop, transport lock and manufacturer's instructions.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Inspect blades, guards, fasteners, belts and drive components for damage.
3. Check fuel, engine oil and other specified fluids on engine-powered models.
4. Confirm that the crop is suitable for reaping and the field is accessible.
5. Inspect the field for stones, wires, people and other hidden hazards.
6. Check tyre or wheel condition and verify that operating controls work correctly.
7. Wear suitable footwear, eye protection and close-fitting work clothing.

How to use

1. Park the reaper on level ground and stop the engine before inspection.
2. Adjust the cutting height according to crop and field conditions.
3. Check that all guards are fitted and the cutting area is clear.
4. Start the engine using the prescribed procedure.
5. Move to the crop at a safe, controlled speed.
6. Engage the cutting and gathering mechanisms according to instructions.
7. Guide the reaper through the crop using steady forward movement.
8. Monitor cutting quality and the flow of gathered stalks.
9. Stop safely if the blades clog or crop material becomes entangled.
10. Disengage the cutting mechanism before turning or transporting the machine.
11. Park on stable ground, lower attachments where applicable and stop the engine.

⚠ Safety

- Keep hands and feet away from the cutter bar and moving gathering parts.
- Never clear entangled stalks while the engine or blades are running.
- Stop the engine and isolate the drive before adjustments or cleaning.
- Keep bystanders away from the cutting path and discharge area.
- Refuel only after the engine is switched off and has cooled.
- Do not operate the reaper with missing guards or faulty controls.

Care & storage

- Remove crop residue and soil after stopping and safely isolating the machine.
- Inspect blades, belts, fasteners and guards for wear or damage.
- Sharpen or replace cutting blades according to manufacturer instructions.
- Store the reaper in a dry, secure location with cutting parts protected.

Common mistakes to avoid

- Operating with dull blades that tear or leave stalks uncut.
- Driving too quickly through lodged or uneven crop sections.
- Clearing tangled stalks without stopping and isolating the drive.
- Ignoring damaged guards, loose fasteners or abnormal machine vibration.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rocker Sprayer

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Rocker Sprayer

What it is used for

To apply approved agricultural spray solutions to crops using a manually operated rocker sprayer.

How to handle

- Inspect the tank, pump, hose, lance, nozzle, seals, and connections for leaks or damage.
- Read the chemical label and follow the approved dilution, application, and personal protective equipment requirements.
- Fill the tank only to its marked safe level and secure all caps and connections before pumping.
- Operate the rocker handle steadily and direct the spray away from people, animals, water sources, and unintended areas.
- Release pressure safely before opening the tank; clean the sprayer and manage leftover solution according to the chemical label and local rules.

Storage

- Depressurize, empty, clean, and dry the sprayer according to the manufacturer's instructions.
- Store in a ventilated, locked area away from food, animal feed, children, and incompatible chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rotavator

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Paddy rotrivator; Rotary Tiller; Rotavator

What it is used for

A rotavator uses rotating blades to break soil and mix crop residue into the field during land preparation. Correct tractor attachment, depth setting, and safe operation help reduce injuries and equipment damage.

Main parts

- Rotating blades or tines cut and break the soil.
- Rotor shaft carries the rotating blades.
- Gearbox transfers power from the tractor to the rotor.
- Drive shaft or power take-off connection supplies rotation.
- Frame supports the rotor, gearbox, and protective components.
- Depth skids or adjustment mechanism control working depth.
- Protective guards and rear cover reduce exposure to moving parts and flying debris.

Before you start

1. Read the operator's manual and use the rotavator only after suitable training.
2. Inspect blades, rotor shaft, gearbox, drive shaft, guards, fasteners, and hitch points for damage.
3. Confirm that the tractor and rotavator are compatible in power, hitch, and drive connection.
4. Park the tractor on firm, level ground, switch off the engine, and secure it against movement.
5. Attach the rotavator using the approved hitch and secure the power take-off connection correctly.
6. Clear the field of stones, wires, people, animals, and other hazards.

How to use

1. Set the working depth according to soil conditions and the manufacturer's instructions.
2. Check that all guards are fitted and the power take-off cover is in place.
3. Start the tractor according to its operating instructions and raise the implement to the transport position.
4. Move to the work area slowly and lower the rotavator only when aligned with the intended path.
5. Engage the power take-off at the recommended engine speed and avoid sudden engagement.
6. Travel at the recommended speed while keeping the implement at the set working depth.
7. Monitor soil breakup, vibration, noise, and the movement of the attached implement.
8. Disengage the power take-off before turning sharply, reversing, or approaching obstacles.
9. Stop the engine, remove the key, and wait for all rotating parts to stop before clearing residue.
10. After work, clean soil from the blades, inspect components, and park the implement securely.

⚠ Safety

- Never approach rotating blades or a moving power take-off shaft.
- Never operate with missing guards, loose connections, or damaged blades.
- Switch off the engine, remove the key, and secure the tractor before adjustment or maintenance.
- Never stand beneath a raised rotavator without approved mechanical support.
- Keep people and animals outside the operating area because stones and debris may be thrown.
- Do not exceed the recommended speed, depth, or terrain limits.

Care & storage

- Clean soil and crop residue from blades and guards after safe shutdown.
- Inspect blades, bolts, gearbox, drive shaft, hitch, and protective covers regularly.
- Lubricate the gearbox and moving parts according to the manufacturer's maintenance schedule.
- Replace damaged blades and arrange authorised servicing for unusual vibration, noise, or oil leakage.

Common mistakes to avoid

- Connecting or adjusting the rotavator while the tractor is not secured.
- Operating with missing guards or an incorrectly connected power take-off shaft.
- Clearing crop residue before the rotor has completely stopped.
- Using excessive depth or speed for the soil conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sample Bag

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Sample bags-polythene and cloth

What it is used for

Sample bags collect and transport soil, plant, seed, or other field samples for identification or analysis.

How to handle

- Choose a clean bag made from suitable polythene or cloth for the sample type.
- Label each bag with sample ID, collection location, date, and other required details.
- Use clean tools and avoid mixing samples or contaminating the bag interior.
- Seal bags appropriately and follow any required ventilation, moisture, or temperature conditions.
- Handle potentially contaminated samples with suitable PPE and follow the relevant collection procedure.

Storage

- Keep samples in their specified conditions and separate incompatible sample types.
- Store empty bags clean and dry in a protected area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sample Divider

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Sample Divider

What it is used for

To divide a bulk grain, seed, soil, or other suitable sample into smaller representative portions for testing.

How to handle

- Inspect the divider, channels, hopper, and collection trays for damage and cleanliness.
- Mix the original sample gently as required by the sampling procedure before dividing.
- Feed the sample evenly through the divider without overloading or shaking the apparatus unnecessarily.
- Collect the divided portions in clean, labelled containers and follow the approved procedure for selecting the test portion.
- Brush or clean the divider between samples to prevent cross-contamination and record sample identification.

Storage

- Store clean and dry in a designated sampling or laboratory area.
- Keep channels and collection trays protected from dust, moisture, and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seed Box

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Seed boxes

What it is used for

Handle seed boxes to keep seeds organised and protected before sowing.

How to handle

- Check boxes for cleanliness, dryness, and damage before use.
- Label each box with seed type, variety, batch, and date as applicable.
- Keep different seed lots separate to prevent mixing or cross-contamination.
- Avoid exposing seeds to moisture, direct sunlight, and pests.

Storage

- Store seeds under the temperature and humidity conditions recommended for the species.
- Keep boxes closed, labelled, and raised off damp floors.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seed Cleaner Grader

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Cleaner, grader; Seed Cleaner; Seed cleaning machines; Seed graders / seed separators

What it is used for

A seed cleaner grader removes unwanted materials from seed lots and separates seeds by size or other physical properties. Correct adjustment, clean handling, and regular inspection help improve seed quality and support consistent grading.

Main parts

- Feed hopper receives the uncleaned seed material.
- Feed gate controls the flow of seeds into the machine.
- Screens or sieves separate seeds by size.
- Air blower or aspiration system removes light impurities on equipped models.
- Vibrating or rotating mechanism moves seeds across the screens.
- Collection outlets direct graded seeds and rejected materials into separate containers.
- Drive motor and transmission power the cleaning mechanism.
- Control panel or adjustment levers regulate operating settings.

Before you start

1. Read the operating manual and confirm the machine is suitable for the seed type.
2. Inspect the hopper, screens, guards, belts, motor, and outlets for damage.
3. Ensure screens and air settings match the seed size and cleaning requirements.
4. Clean the machine to prevent mixing different seed varieties or batches.
5. Place labelled collection containers beneath the correct outlets.
6. Wear suitable PPE and keep the work area clear of loose materials.

How to use

1. Ensure the machine is switched off before changing screens or adjusting internal components.
2. Fit the recommended screens and secure all guards and covers.
3. Position labelled containers at the graded-seed and reject outlets.
4. Start the machine using the manufacturer's recommended sequence.
5. Feed a small quantity of seed and check the cleaning result.
6. Adjust feed rate, screen settings, or airflow within the permitted operating range.
7. Feed the remaining seed gradually without overloading the hopper.
8. Monitor seed flow, separation quality, unusual vibration, and dust during operation.
9. Stop the machine before clearing blockages or making adjustments that require access to moving parts.
10. Switch off and isolate the power, collect the output, and clean the machine after processing.

⚠ Safety

- Never put hands into the hopper, screens, or outlets while the machine is running.
- Keep guards fitted and do not bypass safety interlocks.
- Isolate power before removing blockages, changing screens, or cleaning internal parts.
- Use suitable respiratory protection and eye protection where seed dust is present.
- Keep loose clothing, jewellery, and hair away from belts and rotating parts.
- Do not operate the machine if the frame, motor, wiring, or guards are damaged.

Care & storage

- Remove seed residues and dust after each batch using approved cleaning methods.
- Inspect screens for tears, wear, and blockage before storage.
- Check belts, fasteners, bearings, and drive components according to the manual.
- Store the machine in a dry, clean area and record maintenance issues.

Common mistakes to avoid

- Using incorrect screens for the seed type or required grade.
- Feeding too quickly and overloading the machine.
- Allowing seed varieties or batches to mix between collection containers.
- Cleaning or clearing blockages while moving parts are active.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seed Counter

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Seed counters

What it is used for

A seed counter helps count seeds accurately for sowing, sampling, and laboratory work. Select the correct counting method and keep seeds clean and properly identified.

How to use

1. Inspect the seed counter, tray, sensors, and display for damage or contamination.
2. Confirm that the counter supports the seed size and type being tested.
3. Clean the counting surface and prepare a labelled seed sample.
4. Set the required count or operating mode according to the manual.
5. Feed seeds evenly into the counter without overloading the mechanism.
6. Record the displayed count and compare it with the sample requirement.
7. Clear remaining seeds and clean the equipment after counting.

⚠ Safety

- Keep fingers away from moving parts and feed mechanisms.
- Use dry, clean seeds unless the instrument supports other sample conditions.
- Disconnect power before cleaning powered equipment.
- Do not exceed the instrument's specified capacity.

Care & storage

- Remove seed debris after use and store the counter in a clean, dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seed Drying Tray

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Seed drying trays / racks

What it is used for

Handle seed drying trays and racks to dry seeds evenly and hygienically.

How to handle

- Clean and dry trays before spreading seeds.
- Spread seeds in a suitable, even layer and label each batch clearly.
- Protect seeds from rain, dust, pests, and contamination while allowing adequate airflow.
- Turn seeds as required and monitor drying according to the crop-specific procedure.

Storage

- Clean trays and racks after use and dry them completely.
- Store in a dry, ventilated area away from chemicals and pests.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seed Germinator

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Seed germination chamber / incubator

What it is used for

A seed germinator provides controlled temperature, moisture and sometimes light to support seed germination. It helps students and growers observe germination rates and prepare healthy seedlings under suitable conditions.

Main parts

- Identify the insulated chamber and access door.
- Locate the seed trays, racks or germination containers.
- Identify the temperature sensor and display.
- Locate the heating and cooling system, depending on model.
- Identify the humidity control, water tray or misting arrangement, if fitted.
- Locate the lighting controls and lamps, where provided.
- Identify the control panel, power switch and alarm indicators.

Before you start

1. Read the operating instructions and select conditions suitable for the seed species.
2. Inspect the chamber, door seals, trays, sensors and electrical connections for damage.
3. Clean trays and containers using appropriate hygiene procedures.
4. Use viable seeds from a labelled and suitable source.
5. Prepare clean growing medium or germination paper as required.
6. Check that water reservoirs and drainage arrangements are clean and functional.
7. Place the germinator in a stable, ventilated area away from direct sunlight.

How to use

1. Arrange seeds in labelled trays or containers using the recommended method.
2. Add the required moisture without waterlogging the seeds or growing medium.
3. Place the trays evenly inside the chamber without blocking airflow.
4. Close the door and set the specified temperature and humidity conditions.
5. Set the lighting schedule if the selected seeds require light control.
6. Switch on the germinator and allow conditions to stabilise.
7. Monitor temperature, moisture and visible germination at suitable intervals.
8. Add water or adjust settings only as recommended for the seed species.
9. Record germination counts, dates and observations for the learning activity.
10. Remove seedlings carefully when ready and switch off the unit before cleaning.

⚠ Safety

- Keep water away from electrical outlets, controls and damaged cables.
- Do not exceed the manufacturer's temperature or humidity limits.
- Avoid opening the chamber unnecessarily, as this can disturb conditions.
- Use clean hands or suitable gloves when handling seeds and growing materials.
- Disconnect power before cleaning, servicing or accessing internal components.
- Report unusual odours, overheating or faulty sensors to the teacher.

Care & storage

- Clean and dry trays and the chamber after each germination cycle.
- Empty and clean water reservoirs according to the manufacturer's instructions.
- Check sensors, door seals and temperature controls regularly.
- Store the unit dry and protect it from dust and pests.

Common mistakes to avoid

- Setting unsuitable temperature or moisture levels for the seed species.
- Waterlogging seeds and encouraging mould growth or seed rot.
- Overloading trays and blocking airflow inside the chamber.
- Failing to label seed samples and record germination observations.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seed Testing Kit

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Tetrazolium testing kit

What it is used for

A tetrazolium seed testing kit helps assess seed viability through a controlled staining procedure. Follow the validated method for the seed species and use appropriate laboratory precautions when handling reagents.

How to use

1. Read the kit instructions and confirm the method for the seed species being tested.
2. Inspect reagents, containers, cutting tools, and protective equipment before starting.
3. Label the samples and prepare seeds according to the specified preconditioning procedure.
4. Prepare and handle the tetrazolium solution at the concentration stated in the protocol.
5. Expose the required seed tissues using the approved cutting or preparation method.
6. Apply the stain for the specified duration and temperature, protecting samples from unsuitable light conditions.
7. Assess staining patterns against the validated criteria and record the results.

⚠ Safety

- Wear laboratory gloves, eye protection, and a lab coat when handling reagents.
- Avoid skin or eye contact and follow the reagent safety data sheet.
- Do not ingest reagents or use food containers for laboratory chemicals.
- Dispose of used solutions and seed material according to laboratory rules.

Care & storage

- Keep reagents labelled and stored as specified, and clean tools after testing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seed Treating Equipment

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Seed Treating Equipment

What it is used for

Seed treating equipment applies approved protective or conditioning substances to seeds before sowing. Correct dosing, uniform treatment, and safe chemical handling help protect seeds, operators, and the environment.

Main parts

- Treatment hopper or container holds the seeds before processing.
- Metering mechanism controls the amount of treatment applied.
- Mixing chamber or rotating drum distributes treatment over the seeds.
- Feed and discharge openings guide seeds through the equipment.
- Motor, handle, or drive mechanism operates the equipment, depending on the model.
- Control panel or adjustment controls regulate operation on applicable models.
- Protective covers and guards reduce contact with moving parts.

Before you start

1. Read the equipment manual and product label before preparing the treatment.
2. Use only approved seed-treatment products and follow the specified dosage.
3. Wear the required PPE, including gloves, protective clothing, and eye protection.
4. Inspect the hopper, drum, seals, guards, controls, and electrical connections for damage.
5. Set up the equipment in a ventilated area away from food, drinking water, and animals.
6. Check that seeds are suitable for the selected treatment and the equipment is clean.

How to use

1. Measure the seed quantity and treatment product according to the product label.
2. Prepare the treatment mixture only if the product instructions require one.
3. Load the seeds into the designated hopper without exceeding its capacity.
4. Set the metering rate and mixing time according to the equipment instructions.
5. Start the equipment using the approved controls and confirm normal operation.
6. Introduce the treatment at the specified rate while the seeds move through the mixing area.
7. Observe the process to confirm that treatment is distributed evenly without excessive wetting.
8. Stop the equipment before opening covers, clearing blockages, or making adjustments.
9. Collect treated seeds in clearly labelled containers and keep them separate from untreated seeds.
10. Clean the equipment as directed, manage waste safely, and wash hands after removing PPE.

⚠ Safety

- Never handle seed-treatment chemicals without the PPE specified on the product label.
- Do not eat, drink, or smoke while handling treated seeds or chemicals.
- Keep hands away from moving mixers, rollers, and feed mechanisms.
- Do not use treated seeds for food, feed, or direct consumption.
- Prevent treatment chemicals and rinse water from entering drains or water sources.
- Keep treated seeds and chemicals away from children, animals, and unauthorised users.

Care & storage

- Switch off and isolate power before cleaning or servicing the equipment.
- Clean the hopper, mixing chamber, and discharge path according to product instructions.
- Inspect metering parts, seals, guards, and controls regularly for wear or damage.
- Store equipment clean and dry, and arrange authorised maintenance when required.

Common mistakes to avoid

- Applying more treatment product than the label specifies.
- Mixing treated and untreated seeds in the same container.
- Operating equipment with missing guards or blocked discharge openings.
- Handling treated seeds without protective equipment or proper labelling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seed-cum-Fertilizer Drill

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Seed cum fertilizer; Seed-cum-Fertilizer Drill

What it is used for

A seed-cum-fertilizer drill places seeds and fertilizer in rows during a single field operation. Correct calibration, suitable placement, and careful operation help achieve uniform sowing while reducing seed waste and uneven fertilizer distribution.

Main parts

- Seed hopper stores seed for sowing.
- Fertilizer hopper stores fertilizer for application.
- Metering mechanisms regulate seed and fertilizer delivery.
- Seed tubes guide seed from the metering unit to the furrow.
- Fertilizer tubes direct fertilizer to its intended placement point.
- Furrow openers create channels in the soil.
- Drive wheels and transmission operate the metering mechanisms.
- Frame, hitch, and depth adjustment components support the implement.

Before you start

1. Read the manual and identify suitable seed, fertilizer, and field settings.
2. Inspect the frame, hitch, wheels, chains, tubes, openers, and guards for damage.
3. Confirm the tractor or power unit is compatible with the drill.
4. Calibrate seed and fertilizer rates according to the recommended procedure.
5. Check that seed and fertilizer tubes are clear and correctly connected.
6. Wear suitable footwear and PPE, and remove people from the implement's working area.

How to use

1. Park the tractor and implement safely, lower the drill, and stop the power source before adjustments.
2. Set the row spacing and sowing depth according to the crop requirements.
3. Fill the seed and fertilizer hoppers with the correct materials without exceeding capacity.
4. Connect and secure the implement to the compatible tractor hitch.
5. Raise the implement for transport and lower it into working position at the field edge.
6. Engage the drive mechanism according to the manual and begin moving at a suitable speed.
7. Monitor seed placement, fertilizer delivery, furrow coverage, and row spacing during sowing.
8. Stop in a safe location and inspect the sowing pattern if irregular delivery occurs.
9. Make adjustments only after stopping the machine and isolating relevant power sources.

10. At the end, empty remaining materials as appropriate, clean the tubes and hoppers, and inspect the implement.

⚠ Safety

- Keep hands and feet away from metering mechanisms, chains, drive shafts, and furrow openers.
- Never clear blocked tubes while the drill or tractor power take-off is moving.
- Stop the tractor, lower the implement, apply the parking brake, and isolate power before servicing.
- Handle fertilizers according to the product label and avoid skin, eye, and inhalation exposure.
- Keep bystanders away during turning, reversing, hitching, and field operation.
- Use guards and never ride on the drill unless the manufacturer specifically provides an approved operator position.

Care & storage

- Remove seed and fertilizer residues after use to reduce corrosion and cross-contamination.
- Inspect seed tubes, fertilizer tubes, openers, chains, and metering parts for wear or blockage.
- Lubricate approved points and maintain drive components according to the manual.
- Store the implement in a dry area with the unit safely supported.

Common mistakes to avoid

- Skipping calibration and using an unsuitable seed or fertilizer rate.
- Setting the sowing depth incorrectly for the crop and soil conditions.
- Allowing seed and fertilizer tubes to remain blocked or incorrectly connected.
- Adjusting or clearing the drill while drive components are moving.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seedling Tray

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Plug trays; Seeding Trays for Nursery ?(70 Cavity / holes)

What it is used for

Seedling trays hold seeds and young plants in separate cells for germination and nursery management.

How to handle

- Inspect trays for cracks, blocked drainage holes, and leftover soil or plant debris.
- Clean and sanitise reusable trays according to nursery hygiene procedures.
- Fill cells evenly with the appropriate growing medium without compacting it excessively.
- Place seeds at the recommended depth and label trays with crop and sowing details.
- Water gently and handle young seedlings carefully to avoid root damage.

Storage

- Allow trays to dry before stacking where appropriate.
- Store clean trays in a protected area away from pests, direct sunlight, and crushing loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sickle

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Sickle

What it is used for

A sickle is a curved hand tool used to cut grass, weeds, and suitable crop stems. Use it with controlled strokes for harvesting or garden maintenance.

How to use

1. Inspect the blade, handle, and attachment for cracks, looseness, or rust.
2. Wear suitable gloves and closed-toe footwear before cutting.
3. Clear the cutting area and check for stones, wires, or other obstacles.
4. Hold the handle firmly and cut with controlled strokes away from your body.
5. Keep other people at a safe distance while working.
6. Clean and dry the blade after use and store it with the edge protected.

⚠ Safety

- Keep hands and legs away from the cutting path.
- Never swing the sickle near other people or toward your body.
- Do not use a sickle with a loose handle or damaged blade.
- Carry and store the sickle with its blade covered.

Care & storage

- Clean, dry, and lightly oil the blade before storing it safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Soil Scoop

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Soil Scoop

What it is used for

A soil scoop collects small soil samples for observation, testing, and gardening activities. Use it carefully to avoid contamination and injury.

How to use

1. Check the scoop for cracks, rust, loose handles, or sharp damaged edges.
2. Wear suitable gloves and select the required soil sampling location.
3. Push the scoop gently into the soil without excessive force.
4. Collect the required amount and transfer it into a clean labelled container.
5. Remove attached soil and clean the scoop after sampling.

⚠ Safety

- Wear gloves when handling unknown, contaminated, or chemically treated soil.
- Keep fingers away from sharp edges and avoid forceful twisting.
- Do not touch your face or eat while collecting soil samples.

Care & storage

- Dry the scoop thoroughly and store it in a clean, dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Soil Testing Kit

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Equipment for sample collection and environmental monitoring (e.g., soil testers, water quality kits); Soil testing kit

What it is used for

A soil testing kit helps collect samples and measure selected soil properties such as pH, moisture, or nutrient levels. Correct sampling, clean tools, and careful testing improve the reliability of results.

Main parts

- Sample collection tools gather soil from the required depth.
- Sample containers or bags hold collected soil separately.
- Testing strips, reagents, or test chambers support selected chemical tests.
- Measuring device or sensor reads the selected soil property.
- Reference chart or calibration solution helps interpret or check readings.
- Water testing accessories may be included in environmental monitoring kits.
- Protective gloves, labels, and cleaning materials support safe sampling and handling.

Before you start

1. Read the kit instructions and identify which soil properties it can measure.
2. Select representative sampling locations and avoid unusual spots unless specifically testing them.
3. Wear suitable gloves and footwear, especially when soil contamination is possible.
4. Inspect probes, containers, reagents, sensors, and sample tools for damage or expiry.
5. Clean sampling tools between locations to avoid cross-contamination.
6. Label sample containers with location, depth, date, and other required details.

How to use

1. Remove surface litter before collecting soil from the required depth.
2. Collect several representative subsamples according to the sampling instructions.
3. Mix subsamples in a clean container when a composite sample is required.
4. Prepare the sample by removing stones and large debris as instructed.
5. Measure the required soil quantity and add water or reagent only as specified.
6. Insert the probe to the recommended depth or follow the kit's test procedure.
7. Wait for the specified reaction time and read the result under suitable lighting.
8. Compare the reading with the supplied reference chart or instrument scale.
9. Record the result, sample location, date, and testing conditions.
10. Clean reusable tools, dispose of used reagents safely, and store the kit as instructed.

⚠ Safety

- Never taste soil, testing reagents, or water samples.
- Avoid direct contact with unknown or potentially contaminated soil and wear suitable PPE.
- Follow reagent labels and avoid contact with eyes, skin, and clothing.
- Do not mix reagents unless the kit instructions explicitly require it.
- Keep chemical reagents away from children, food, and drinking water.
- Dispose of chemical waste according to the product instructions and local requirements.

Care & storage

- Clean and dry reusable probes, containers, and sampling tools after use.
- Store reagents according to their labels and check expiry dates before testing.
- Protect electronic sensors from moisture, impact, and extreme temperatures.
- Calibrate or verify instruments at the intervals recommended by the manufacturer.

Common mistakes to avoid

- Collecting a sample from only one unrepresentative spot.
- Using dirty tools that contaminate samples from different locations.
- Adding incorrect amounts of water or reagent to the sample.
- Interpreting a reading without checking the kit's reference chart or instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sorting Machine

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Sorting machines

What it is used for

A sorting machine separates agricultural products such as seeds, grains, fruits, or vegetables by size, weight, colour, or other specified characteristics. Use suitable settings for the crop and inspect the output to maintain product quality and reduce damage.

Main parts

- Feed hopper or loading conveyor
- Sorting screens, rollers, belts, or channels
- Drive motor and transmission
- Sensors or grading controls, if fitted
- Collection bins and discharge outlets
- Protective guards and control panel

Before you start

1. Confirm that the machine is suitable for the crop and sorting task.
2. Inspect the hopper, screens, belts, rollers, guards, and drive components.
3. Check that the emergency stop and other safety devices function correctly.
4. Remove stones, metal, and other foreign objects from the incoming material.
5. Ensure collection bins are clean, correctly positioned, and labelled if required.
6. Wear suitable PPE and clear the work area of unnecessary people and obstacles.

How to use

1. Place the machine on a stable surface and secure it as instructed.
2. Fit the required screen, guide, or sorting attachment while the machine is isolated.
3. Set sorting parameters according to the crop and required output grade.
4. Position collection bins under the correct discharge outlets.
5. Close all guards and confirm that feed and discharge paths are clear.
6. Switch on the machine and run it empty briefly to check normal operation.
7. Feed material gradually at the recommended rate.
8. Monitor sorting accuracy, material flow, vibration, and unusual noise.
9. Inspect samples from each outlet and adjust settings using the approved procedure if necessary.
10. Stop feeding and allow remaining material to clear before shutdown.
11. Switch off and isolate power before removing jams or cleaning internal parts.
12. Collect and label sorted products according to the workplace procedure.
13. Clean the machine and record output, settings, and any faults.

⚠ Safety

- Only trained operators should operate or adjust the sorting machine.
- Keep hands and tools away from moving belts, rollers, screens, and drive parts.
- Never bypass guards, interlocks, or emergency stops.
- Isolate power and verify complete stoppage before clearing jams or servicing.
- Control dust and use suitable respiratory protection where required.
- Use safe lifting methods when handling heavy crop containers.

Care & storage

- Remove crop residue and dust after each production run.
- Inspect screens, belts, bearings, fasteners, and guards regularly.
- Keep sensors and sorting surfaces clean and unobstructed.
- Lubricate only specified points according to the maintenance schedule.
- Report damaged components, inaccurate sorting, or abnormal vibration.

Common mistakes to avoid

- Feeding material too quickly and causing blockages or poor sorting.
- Using an incorrect screen or unsuitable grading setting.
- Ignoring worn screens, dirty sensors, or loose fasteners.
- Clearing a blockage while the machine is energised.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spade

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 22 labs

Covers these lab items: Digging Spade; Garden/Digging Spade; Rabbiting Spade; Spade; Spade with wooden handle

What it is used for

A spade is used to dig, turn, and move soil in gardening and landscaping tasks. Choose a suitable spade and use controlled movements to work safely and efficiently.

How to use

1. Inspect the blade, handle, and handle joint for cracks or looseness.
2. Select a spade suitable for the soil and task.
3. Clear the work area of stones, cables, and other obstructions.
4. Hold the handle firmly and position your feet on stable ground.
5. Push the blade into the soil using controlled body movement and foot pressure where designed.
6. Lift and move manageable amounts of soil without twisting the back excessively.
7. Place the soil where required and repeat as needed.
8. Clean the blade and store the spade securely after use.

⚠ Safety

- Wear sturdy footwear and suitable gloves.
- Keep other people clear of the digging area and swing path.
- Do not use a spade with a loose handle or damaged blade.
- Check for buried utilities before digging in unfamiliar areas.

Care & storage

- Remove soil from the blade and dry it after use.
- Store the spade in a dry place and repair or replace damaged parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sprayer Nozzle

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Nozzle set

What it is used for

A sprayer nozzle controls the direction, flow, and spray pattern of liquid. Select the correct nozzle for the sprayer and intended application.

How to use

1. Check the nozzle for cracks, blockage, worn threads, or damaged seals.
2. Switch off and depressurize the sprayer before fitting or removing the nozzle.
3. Select a compatible nozzle suitable for the liquid and required spray pattern.
4. Fit the nozzle securely without overtightening the connection.
5. Test the spray in a safe, designated area and adjust the pattern if permitted.
6. Depressurize the sprayer and clean the nozzle after use.

⚠ Safety

- Wear appropriate gloves, goggles, and other PPE required for the sprayed liquid.
- Never point the nozzle at people, animals, or electrical equipment.
- Never clear a blocked nozzle by blowing through it with your mouth.

Care & storage

- Clean using the manufacturer's recommended method and store nozzles in a dry container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sprinkler Irrigation System

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Sprinkler; Sprinkler Irrigation Unit; Sprinkler Irrigation Unit (Lateral pipe 16mm x 20ft , Micro sprinkler & micro jet 05 nos each); Sprinkler Irrigation UTender (Lateral pipe 16mm x 20ft , Micro sprinkler & micro jet 05 nos each)

What it is used for

Use a sprinkler irrigation system to distribute water over crops or garden areas through controlled spray from sprinkler heads.

How to use

1. Inspect the water source, pump, filter, pipes, valves, and sprinkler heads for damage or blockage.
2. Position sprinklers to provide suitable coverage without unnecessary overlap or dry gaps.
3. Connect pipes and fittings securely and confirm pressure ratings are suitable.
4. Open the water supply gradually and check for leaks or abnormal pressure.
5. Observe the spray pattern and adjust sprinkler direction or spacing as needed.
6. Set watering duration according to crop needs, soil, weather, and local guidance.
7. Shut off the supply after irrigation and inspect the system before storage or reuse.

⚠ Safety

- Release pressure before disconnecting pipes or servicing sprinkler heads.
- Keep electrical pumps and connections protected from water.
- Use fittings and pipes rated for the operating pressure.
- Avoid slippery walking surfaces and keep people clear of pressurized spray components.

Care & storage

- Clean filters and sprinkler nozzles regularly to prevent clogging.
- Drain or protect components as appropriate before long storage or freezing conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stake

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Stakes/pins

What it is used for

Handle stakes and pins safely for gardening and field marking.

How to handle

- Inspect stakes and pins for bending, rust, or sharp damage before use.
- Insert them carefully into suitable ground and avoid buried utilities or irrigation lines.
- Keep pointed ends directed away from the body and wear suitable gloves where needed.
- Remove temporary stakes when no longer required to prevent trip or puncture hazards.

Storage

- Clean and dry reusable stakes before storing.
- Keep pointed ends covered and store securely in a labelled container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Straw Reaper

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Straw Reaper

What it is used for

A straw reaper processes harvested crop residue to recover straw and prepare it for collection or further use. Correct machine adjustment, suitable crop conditions, and safe operation help improve output quality while reducing blockages and injury risks.

Main parts

- Feed chute guides crop residue into the machine.
- Cutting or chopping mechanism processes the straw.
- Rotor or drum carries the cutting components.
- Concave or sieve separates processed material according to the machine design.
- Blower or conveying system moves processed straw toward the outlet.
- Collection chute directs straw into a trailer or collection area.
- Drive system receives power from the tractor or machine engine.
- Guards and adjustment controls protect and regulate operating components.

Before you start

1. Read the manual and confirm the straw reaper is suitable for the crop and operating conditions.
2. Inspect the frame, hitch, guards, belts, blades, fasteners, and drive components for damage.
3. Check blade condition and ensure all guards are correctly fitted.
4. Verify compatibility with the tractor or power source and confirm safe connections.
5. Clear the field of stones, wire, metal, and other foreign objects where practical.
6. Keep bystanders away and wear suitable PPE, including eye and hearing protection.

How to use

1. Park the tractor or machine safely and isolate the power before inspection or adjustment.
2. Connect the machine according to the manufacturer's instructions and secure all required pins and guards.
3. Set the cutting height and other adjustments for the crop and field conditions.
4. Start the power source and engage the drive using the specified sequence.
5. Allow the machine to reach the recommended operating speed before feeding crop residue.
6. Move forward at a controlled speed and monitor the feed, straw quality, and machine load.
7. Stop feeding and disengage the drive if blockage, excessive vibration, or unusual noise occurs.
8. Wait for all moving parts to stop and isolate power before clearing a blockage.
9. Collect the processed straw safely and avoid approaching the discharge outlet while material is being ejected.

10. At the end, disengage the drive, shut down the machine, clean residues, and inspect for damage.

⚠ Safety

- Never place hands or feet into the feed chute or discharge outlet.
- Keep guards fitted and never approach rotating shafts, belts, blades, or drums.
- Switch off the engine, isolate power, and wait for all parts to stop before clearing blockages.
- Do not wear loose clothing or operate the machine while tired or distracted.
- Keep people away from the discharge area and avoid directing material toward bystanders.
- Do not operate with damaged blades, loose fasteners, missing guards, or an unstable tractor connection.

Care & storage

- Remove straw, dust, and crop residue after use using the approved method.
- Inspect blades, belts, bearings, guards, and fasteners for wear or damage.
- Lubricate specified points and maintain the drive system according to the manual.
- Store the machine in a dry, secure location and record faults requiring repair.

Common mistakes to avoid

- Operating with damaged blades or missing guards.
- Feeding crop residue too quickly and overloading the machine.
- Attempting to clear a blockage before all moving parts stop.
- Allowing bystanders near the feed or discharge area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tarpaulin

Category: **Agriculture & Gardening** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Tarpaulin

What it is used for

A tarpaulin provides temporary weather protection or ground covering for equipment, crops, and materials.

How to handle

- Inspect the tarpaulin for tears, holes, weak eyelets, and damaged ropes before use.
- Choose a suitable size and secure all corners to prevent wind displacement.
- Avoid covering hot equipment or materials that require ventilation.
- Keep the sheet clear of machinery, walkways, and sharp edges that may cause damage or tripping.
- Use safe methods when unfolding or folding large sheets, especially in windy conditions.

Storage

- Clean and dry the tarpaulin before folding or rolling.
- Store in a dry, protected location away from pests, heat, and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thresher

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Paddy Thresher; Small-scale seed threshers

What it is used for

A thresher separates grain or seeds from harvested crop stalks, pods, or heads using a rotating threshing mechanism. Use the machine settings and feed rate suitable for the crop to reduce grain damage and maintain safe operation.

Main parts

- Feed chute or crop feeding table
- Threshing drum or rotor with beaters
- Concave, sieve, or separation screen
- Blower and grain collection outlet
- Drive motor or engine and transmission
- Protective guards, belts, and emergency stop

Before you start

1. Confirm that the thresher is suitable for the crop and harvesting condition.
2. Inspect the drum, beaters, screens, belts, guards, and fasteners for damage.
3. Check the drive unit, lubrication, and fuel or electrical supply as applicable.
4. Place the thresher on stable ground and ensure it is properly secured.
5. Clear the work area of stones, metal, ropes, and other foreign objects.
6. Wear close-fitting clothing, eye and hearing protection, suitable footwear, and a dust mask where required.

How to use

1. Keep the power source isolated while inspecting or adjusting the machine.
2. Set the drum speed, concave clearance, and screens according to the crop instructions.
3. Ensure all guards are fitted and the grain collection containers are positioned correctly.
4. Start the machine according to the manufacturer's procedure and allow it to reach normal operating speed.
5. Feed dry or suitably conditioned crop gradually at the recommended rate.
6. Keep hands and tools away from the feed opening and rotating drum.
7. Monitor grain separation, grain damage, chaff discharge, vibration, and unusual noise.
8. Adjust feed rate or approved settings if output becomes uneven or the machine overloads.
9. Stop feeding and allow remaining crop material to clear before shutdown.
10. Switch off the machine and isolate the power before removing blockages or cleaning.
11. Wait until all rotating parts stop completely before opening guards or inspecting the drum.

2. Collect separated grain and remove chaff using safe handling methods.
3. Clean crop residue from the machine and record any faults or maintenance needs.

⚠ Safety

- Only trained operators should operate the thresher.
- Never push crop into the feed opening with hands, feet, or improvised tools.
- Keep guards fitted over belts, pulleys, and rotating components.
- Isolate the power and verify complete stoppage before clearing blockages.
- Keep children and bystanders away from the feeding and discharge areas.
- Control dust and keep sparks, open flames, and smoking away from dry crop residue.

Care & storage

- Remove crop residue from the drum, screens, and discharge paths after use.
- Inspect beaters, screens, belts, bearings, and fasteners regularly.
- Lubricate the specified points and service the drive unit according to instructions.
- Store the machine in a dry, secure location when not in use.
- Arrange repairs for excessive vibration, damaged parts, or poor threshing performance.

Common mistakes to avoid

- Feeding crop too quickly and overloading the drum.
- Using unsuitable drum speed or concave clearance for the crop.
- Clearing a blockage while the machine is running.
- Operating with missing guards or allowing bystanders near the feed opening.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tractor

Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: Tractor

What it is used for

A tractor provides power for agricultural tasks such as tilling, towing, and operating mounted implements. Correct inspection, trained driving, and safe attachment handling help prevent collisions, rollovers, and mechanical injury.

Main parts

- Engine produces the power needed to move the tractor and operate implements.
- Transmission and clutch transfer power to the drive wheels.
- Steering system controls the direction of travel.
- Service brakes and parking brake control stopping and stationary holding.
- Power take-off shaft drives compatible powered implements.
- Three-point linkage or hitch connects mounted or towed implements.
- Operator seat, controls, and protective structure support safe operation.

Before you start

1. Only a trained and authorised operator should drive the tractor.
2. Read the operator's manual and complete the required pre-use inspection.
3. Check tyres, wheels, brakes, steering, lights, horn, mirrors, and safety guards.
4. Check engine oil, coolant, fuel, and other fluids according to the manual.
5. Inspect hitch points, linkage, power take-off cover, and attached implements.
6. Adjust the seat and mirrors, wear the seat belt where fitted, and clear the route.

How to use

1. Mount the tractor using the designated steps and maintain three points of contact.
2. Adjust the seat and confirm the controls are within safe reach.
3. Start the engine according to the manual and check the instrument indicators.
4. Test the brakes, steering, lights, and horn in a safe open area.
5. Select the correct gear and move off slowly after checking the surroundings.
6. Operate at a suitable speed for the terrain, load, and attached implement.
7. Use the approved procedure when connecting or disconnecting implements and isolate power before adjustments.
8. Disengage the power take-off before approaching or servicing an attached implement.
9. Park on firm, level ground, lower implements, apply the parking brake, and switch off.
10. Remove the key when leaving the tractor unattended and report defects promptly.

⚠ Safety

- Never carry passengers unless the tractor is specifically designed and approved for them.
- Use the seat belt with the rollover protective structure where fitted.
- Keep clear of rotating power take-off shafts and other moving components.
- Avoid steep slopes, sharp turns at speed, and sudden braking with a raised load.
- Never start or service the tractor from beneath an unsupported raised implement.
- Refuel only with the engine stopped and away from flames or sparks.

Care & storage

- Clean mud and crop residue from the tractor after safe shutdown.
- Inspect tyres, brakes, hitch points, belts, hoses, and guards regularly.
- Follow the manual for engine servicing, lubrication, filters, and fluid changes.
- Arrange authorised repair for leaks, brake faults, warning lights, or unusual vibration.

Common mistakes to avoid

- Driving without checking brakes, steering, tyres, and attached implements.
- Carrying passengers on a tractor not designed for passengers.
- Approaching a rotating power take-off shaft while it is running.
- Leaving the tractor running or implements raised while unattended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tree Pruner

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Tree Pruner

What it is used for

To prune branches and maintain the healthy growth and shape of trees and shrubs.

How to handle

- Inspect the pruner for damaged blades, loose fasteners, and proper operation.
- Wear suitable gloves and eye protection; keep hands clear of the cutting blades.
- Position the blades around the branch and make a controlled cut without twisting the tool.
- Use suitable equipment and trained assistance for high branches; never overreach.
- Clean, dry, and inspect the blades after use.

Storage

- Clean and dry the pruner before storage.
- Close or lock the blades and store in a designated tool rack away from children and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Watering Can

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Water can; Watering Can; Watering Can , 8 to 10 Ltr.

What it is used for

Use a watering can to deliver water gently to plants, seedlings, or garden soil in a controlled amount.

How to use

1. Inspect the can, handle, spout, and rose attachment for damage or leaks.
2. Fill the can with clean water to a manageable level.
3. Carry it with a secure grip and avoid overfilling.
4. Direct the spout toward the soil near the plant roots.
5. Pour slowly to prevent soil erosion, splashing, or damage to seedlings.
6. Check that the soil receives enough water without prolonged pooling.
7. Empty, rinse, and store the watering can after use.

⚠ Safety

- Do not lift a full can beyond your safe handling capacity.
- Keep water away from electrical equipment and slippery walkways.
- Use clean water suitable for the plants being watered.
- Avoid spilling water where it may cause slips.

Care & storage

- Rinse the can and rose attachment after use.
- Store it empty in a shaded place to reduce damage and algae growth.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weeder

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Hand weeder; Manual weeder

What it is used for

To remove weeds and loosen soil around plants using a manual weeder. To control unwanted growth while minimizing damage to crop roots and nearby plants.

Main parts

- Handle
- Working blade or fork
- Neck or shaft
- Grip, if fitted

Before you start

1. Choose a weeder suitable for the soil and weed type.
2. Inspect the handle and working end for cracks or looseness.
3. Wear suitable gloves and closed footwear.
4. Check the work area for buried cables, pipes, or sharp debris.

How to use

1. Identify the weed and the area around its roots.
2. Hold the handle securely and position the working end near the weed.
3. Insert the blade or fork into the soil with controlled pressure.
4. Loosen the soil and lift the weed, aiming to remove its roots where practical.
5. Avoid damaging nearby plant roots.
6. Collect the removed weeds and dispose of them appropriately.
7. Clean the tool after use.

⚠ Safety

- Keep hands and feet clear of the working end.
- Do not dig near suspected underground utilities.
- Use the tool only for its intended purpose.
- Maintain stable footing on uneven ground.

Care & storage

- Remove soil and plant residue after use.
- Dry metal parts to reduce rust.

- Inspect the handle and working end regularly.
- Store in a dry location.

Common mistakes to avoid

- Using excessive force in compacted soil.
- Cutting nearby crop roots while removing weeds.
- Leaving weed seeds or roots where they can regrow.
- Using a loose or damaged handle.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weeding Fork

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Weeding fork

What it is used for

To loosen soil and remove weeds, including their roots, from garden beds and cultivated areas.

How to handle

- Inspect the tines and handle for cracks, looseness, or bending.
- Wear suitable gloves and sturdy footwear before starting work.
- Insert the fork into the soil beside the weed and loosen the soil carefully around its roots.
- Pull out the weed without damaging nearby plants; avoid excessive force in hard soil.
- Remove soil and plant debris from the fork after use.

Storage

- Clean and dry the fork to prevent corrosion.
- Store with the tines protected in a dry, designated tool area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weeding Hook

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Weeding Hook

What it is used for

A weeding hook removes weeds and loosens soil around plants in gardens and school plots. Use controlled movements to protect plant roots and avoid injuries.

How to use

1. Inspect the hook, handle, and joint for cracks, looseness, or rust.
2. Wear suitable gloves and closed footwear before starting work.
3. Identify weeds and check nearby plants before inserting the hook.
4. Grip the handle firmly and loosen weeds using short, controlled movements.
5. Pull out weeds carefully without damaging nearby roots or stems.
6. Remove soil and plant debris from the hook after use.

⚠ Safety

- Keep the sharp hook away from hands, feet, and other people.
- Do not swing the tool or use excessive force in hard soil.
- Avoid handling unknown thorny or irritating plants with bare hands.

Care & storage

- Clean and dry the hook, then store it with the sharp end protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wheel Hoe

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Wheel hoe

What it is used for

A wheel hoe cuts weeds and loosens surface soil using a rolling wheel and blade. Operate it on suitable ground with controlled movement and clear visibility.

How to use

1. Inspect the wheel, blade, handle, fasteners, and guards for damage.
2. Wear closed footwear and suitable gloves before using the tool.
3. Clear stones, wires, and debris from the working area.
4. Hold the handles firmly and position the blade near the weeds.
5. Push the wheel hoe steadily along the row while avoiding crop stems.
6. Stop before turning, removing debris, or adjusting the blade.
7. Clean the blade and wheel after finishing the task.

⚠ Safety

- Keep hands and feet away from the blade and moving wheel.
- Do not use the tool on steep, slippery, or unstable ground.
- Never clear tangled weeds while the tool is moving.

Care & storage

- Remove soil and plant debris, dry metal parts, and store the tool securely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wheelbarrow

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Wheelbarrow or Garden Cart

What it is used for

A wheelbarrow transports soil, tools, plants, and other materials around the garden or workshop. Load it evenly and move it carefully to prevent tipping and strain.

How to use

1. Inspect the tray, wheel, handles, axle, and supports for damage.
2. Wear closed footwear and plan a clear route before loading.
3. Place the load evenly and keep it within the recommended capacity.
4. Grip both handles firmly and lift using a stable posture.
5. Push the wheelbarrow slowly while watching the path and load balance.
6. Stop on level ground before unloading materials carefully.
7. Remove soil and debris and park the wheelbarrow securely.

⚠ Safety

- Do not overload the wheelbarrow or stack materials above a stable height.
- Keep fingers and feet clear of the wheel, axle, and tipping tray.
- Use assistance for heavy loads and avoid steep or slippery routes.

Care & storage

- Clean and dry the tray, check tyre condition, and store it in a stable place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Winnowing Fan

Category: **Agriculture & Gardening** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Winnowing fans

What it is used for

A winnowing fan uses airflow to separate lighter chaff and debris from heavier grains or seeds. Operate it on a stable surface and adjust airflow to suit the crop and material condition.

How to use

1. Inspect the fan, blades, guards, handle or motor, and frame for damage.
2. Place the equipment on a stable surface in a well-ventilated work area.
3. Confirm the crop or seed material is suitable for the winnowing process.
4. Prepare a clean collection tray or containers for grain and separated debris.
5. Operate the fan according to its design and feed material gradually into the airflow.
6. Adjust airflow or material position until separation is effective without excessive grain loss.
7. Switch off the fan, collect separated materials, and clean the equipment.

⚠ Safety

- Keep hands, clothing, and loose objects away from moving blades.
- Use a guard and never operate equipment with exposed rotating parts.
- Wear suitable respiratory protection when dust levels require it.
- Disconnect power before clearing blockages or cleaning motorized equipment.

Care & storage

- Remove dust and crop residue after use and store the fan in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

AI Cylinder

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Artificial insemination (AI) cylinder

What it is used for

An AI cylinder stores liquid nitrogen for preserving semen straws or other approved biological material at cryogenic temperatures. Correct handling, level monitoring, and secure storage help maintain sample viability while reducing cold-burn and oxygen-displacement risks.

Main parts

- Vacuum-insulated cylinder reduces heat transfer and maintains cryogenic conditions.
- Neck opening provides access to the storage chamber.
- Canisters or racks hold labelled straws or approved sample containers.
- Neck plug or cover limits unnecessary nitrogen loss while allowing safe venting as designed.
- Liquid nitrogen provides the cryogenic cooling medium.
- Level measuring device or monitoring method checks the nitrogen level where provided.
- Identification labels and inventory records track stored samples.
- Protective outer casing supports safe handling and transport.

Before you start

1. Only trained and authorised personnel should handle the cylinder and biological samples.
2. Read the manufacturer's instructions and the facility's cryogenic material handling procedure.
3. Inspect the cylinder, neck, lid, handles, labels, and base for damage or abnormal frost patterns.
4. Check the liquid nitrogen level using the approved measuring method without leaning over the opening.
5. Ensure the room is adequately ventilated and an oxygen-deficiency monitor is available where required.
6. Wear cryogenic gloves, eye protection or a face shield, closed footwear, and suitable protective clothing.

How to use

1. Place the cylinder upright on a stable, level surface in the designated ventilated area.
2. Confirm the sample inventory and identify the required canister or rack before opening the cylinder.
3. Remove the cover carefully and allow the neck area to remain clear of obstructions.
4. Use the approved lifting tool to raise the correct canister without reaching into the neck opening.
5. Locate the labelled sample using the inventory and minimise the time the cylinder remains open.
6. Return the canister to its correct position and ensure stored samples remain securely arranged.

7. Close the cylinder with the specified cover or neck plug without sealing it in a way that prevents designed venting.
8. Measure and record the liquid nitrogen level according to the facility schedule.
9. Refill only according to the approved procedure using compatible transfer equipment and trained personnel.
10. Secure the cylinder upright, update sample records, and report any damage, abnormal frost, or low nitrogen level.

⚠ Safety

- Liquid nitrogen can cause severe cold burns and can displace oxygen in poorly ventilated spaces.
- Never seal the cylinder airtight or block its designed venting path.
- Never look directly into the opening or place the face over the neck.
- Use only cryogenic-rated gloves and face or eye protection; ordinary thin gloves are not sufficient.
- Do not handle the cylinder alone during high-risk transfers or move it by dragging or tipping.
- If an oxygen-deficiency alarm sounds or breathing difficulty occurs, leave the area immediately and follow the emergency procedure.

Care & storage

- Keep the cylinder upright, clean, and protected from impact.
- Monitor nitrogen level at the required frequency and maintain sample inventory records.
- Inspect the neck, cover, handles, and exterior regularly for damage or unusual frost.
- Arrange authorised inspection if vacuum insulation may have failed or nitrogen consumption increases unexpectedly.

Common mistakes to avoid

- Allowing the cylinder to remain open longer than necessary.
- Blocking the vent or sealing the cylinder airtight.
- Handling liquid nitrogen without appropriate cryogenic PPE or ventilation.
- Failing to monitor nitrogen level and sample location records.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

AI Gun

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: AI gun

What it is used for

An artificial insemination (AI) gun is used by trained animal-reproduction personnel to deposit prepared semen at the appropriate site during an approved breeding procedure. Correct hygiene, temperature control, and gentle handling help protect semen quality and animal welfare.

Main parts

- Outer barrel holds the prepared insemination sheath or compatible assembly.
- Plunger or push rod advances the semen dose.
- Handle provides a grip for controlled insertion and delivery.
- Locking or retaining mechanism secures compatible sheaths where fitted.
- Protective cover maintains cleanliness before use.
- Compatible insemination sheath guides the instrument during the procedure.
- Identification label or marking helps confirm the intended semen dose.
- Cleaning and storage accessories support hygienic reuse where permitted.

Before you start

1. Only trained and authorised animal-reproduction personnel should perform insemination.
2. Confirm the animal's identity, breeding record, and suitability for the planned procedure.
3. Check semen identification, expiry or approved use period, and handling requirements.
4. Inspect the AI gun and compatible sheath for damage, cleanliness, and correct fit.
5. Prepare clean gloves, lubricant, and other supplies required by the approved procedure.
6. Keep frozen semen in the approved storage system until it is needed and follow the specified thawing protocol.

How to use

1. Wash hands and put on the required clean protective gloves.
2. Verify the semen identification against the animal's breeding record before preparation.
3. Prepare and load the semen straw and sheath according to the equipment and semen instructions.
4. Secure the sheath and confirm the plunger moves correctly without forcing it.
5. Prepare the animal using the approved hygiene and restraint procedure.
6. A trained operator should position and guide the instrument gently using the species-specific approved technique.
7. Deposit the semen at the appropriate site according to the established breeding protocol.

8. Withdraw the instrument carefully and dispose of single-use components in the designated waste stream.
9. Clean and disinfect reusable components using the approved method, if the device is designed for reuse.
10. Record the semen identification, animal details, date, operator, and relevant procedure observations.

⚠ Safety

- Never perform insemination without appropriate training, authorisation, and species-specific procedure knowledge.
- Use clean, compatible equipment and avoid contaminating the semen or sheath.
- Do not force the instrument or continue if resistance, pain, bleeding, or injury occurs.
- Handle animals with suitable restraint and assistance to prevent injury to the animal and operator.
- Follow approved semen storage and thawing instructions; avoid uncontrolled temperature exposure.
- Dispose of contaminated and single-use materials according to veterinary hygiene and waste procedures.

Care & storage

- Clean and disinfect reusable components immediately after use according to the manufacturer instructions.
- Inspect the barrel, plunger, and retaining parts for wear or damage.
- Store the dry, clean instrument in a protective case away from dust and chemicals.
- Maintain breeding and equipment records according to the facility procedure.

Common mistakes to avoid

- Using semen without verifying its identification and handling requirements.
- Touching or contaminating the prepared sheath or semen dose.
- Forcing the instrument when insertion meets resistance.
- Failing to record the animal, semen dose, and insemination details.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Aeration System

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Oxygenation system

What it is used for

An aeration or oxygenation system supplies air to water in aquaculture or animal-husbandry settings to support dissolved oxygen levels and aquatic animal health. Select the system capacity for the tank or pond, monitor water conditions, and maintain reliable operation, especially during low-oxygen periods.

Main parts

- Air pump, blower, or oxygen supply unit
- Airline tubing and connectors
- Diffusers, air stones, or aeration plates
- Valves and flow regulators
- Electrical supply and protective enclosure
- Dissolved oxygen meter or water-quality monitoring equipment

Before you start

1. Confirm the system type, rated capacity, and suitability for the tank, pond, or holding system.
2. Inspect the pump, tubing, diffusers, valves, electrical cables, and protective covers.
3. Position the pump in a dry, ventilated area protected from water splash and flooding.
4. Check that tubing is securely connected and diffusers are clean and correctly positioned.
5. Verify the electrical protection and earthing according to the installation instructions.
6. Measure dissolved oxygen and check water temperature and animal behaviour before operation.

How to use

1. Ensure the system is installed according to the manufacturer's instructions and local requirements.
2. Open the required valves and confirm that air pathways are clear.
3. Switch on the pump or blower and check that air is flowing through all intended diffusers.
4. Adjust airflow within the rated range to achieve suitable aeration without excessive turbulence or stress to animals.
5. Observe bubble distribution, water circulation, pump noise, and any unusual vibration.
6. Monitor dissolved oxygen and water temperature at suitable intervals, especially during hot weather, at night, or when stocking density is high.
7. If oxygen levels fall or the system fails, follow the site's emergency aeration plan and activate approved backup equipment.

8. Inspect tubing and diffusers for blockages or leaks; isolate power before removing or repairing components.
9. Record operating status, water-quality readings, adjustments, and maintenance actions.
10. At shutdown, follow the site procedure and ensure alternative oxygen support is available if animals depend on continuous aeration.

⚠ Safety

- Only trained personnel should install, adjust, or repair the system.
- Keep electrical equipment dry and use appropriate protective devices and earthing.
- Never handle plugs, switches, or electrical connections with wet hands or while standing in water.
- Do not block air inlets or operate a pump beyond its rated capacity.
- Avoid entering tanks or confined spaces unless the site has an approved entry procedure and appropriate safeguards.
- Maintain backup aeration or oxygen supply where loss of aeration could endanger animals.

Care & storage

- Clean diffusers and air stones regularly to prevent clogging.
- Inspect tubing, valves, seals, and connectors for leaks or deterioration.
- Clean pump air filters and service the blower according to the maintenance schedule.
- Keep the area around the pump dry, clean, and unobstructed.
- Test alarms, backup systems, and dissolved oxygen meters at the required intervals.

Common mistakes to avoid

- Choosing a pump that is too small for the water volume or animal load.
- Ignoring low dissolved oxygen readings or abnormal animal behaviour.
- Allowing diffusers to clog or tubing to leak.
- Performing maintenance on powered equipment without isolation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Animal Awakener

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Animal awakener

What it is used for

To use an approved animal-awakening or stimulation device only for its intended animal-care application. To follow veterinary or manufacturer guidance and avoid unnecessary stress or injury to the animal.

Main parts

- Device body
- Control switch or trigger
- Contact or stimulation component, if fitted
- Power source, if applicable
- Protective housing

Before you start

1. Confirm the exact device type, intended purpose, and approved method of use.
2. Consult a veterinarian or trained animal-care supervisor where required.
3. Inspect the device for damage, exposed wires, or malfunction.
4. Ensure the animal is handled in a calm, controlled environment.
5. Check power settings and accessories according to the instructions.

How to use

1. Prepare the animal and surrounding area according to the approved procedure.
2. Position the device as directed by the manufacturer or veterinary professional.
3. Activate it only for the specified duration and purpose.
4. Observe the animal continuously for signs of distress or adverse reaction.
5. Stop immediately if the animal shows pain, panic, or injury.
6. Clean and store the device according to its instructions.
7. Record the use and any abnormal response when required.

⚠ Safety

- Do not use an unidentified device or an unapproved stimulation method.
- Never apply painful or excessive stimulation to an animal.
- Use only under appropriate veterinary or trained supervision.
- Keep electrical components away from water and wet surfaces.
- Stop use and seek professional help if the animal is injured or distressed.

Care & storage

- Clean contact surfaces after use according to the instructions.
- Inspect controls, casing, and power connections regularly.
- Store securely away from animals and children.
- Report faults and do not use damaged equipment.

Common mistakes to avoid

- Using the device without confirming its intended purpose.
- Applying stimulation for too long or at an excessive setting.
- Ignoring signs of animal distress.
- Using a damaged or unapproved device.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Animal Feed

Category: **Animal Husbandry & Dairy** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Feed; Various feeds and fodders

What it is used for

Animal feed provides nutrients for livestock and other animals according to species, age, and production needs.

How to handle

- Verify feed type, species suitability, expiry date, and any storage instructions.
- Inspect feed for mould, pests, unusual odour, or contamination before feeding.
- Measure portions according to the approved feeding plan and provide clean drinking water.
- Introduce feed changes gradually under appropriate animal-care guidance.
- Wash hands after handling feed and prevent contamination from chemicals or animal waste.

Storage

- Store feed in clean, dry, ventilated containers or designated feed stores.
- Protect from moisture, rodents, insects, chemicals, and direct contact with the floor.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Aquarium Heater

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Aquarium heater

What it is used for

An aquarium heater maintains water temperature within a suitable range for the aquatic species being kept. Use a correctly rated, thermostatically controlled heater and monitor temperature to prevent overheating, thermal shock, and electrical hazards.

Main parts

- Heater element and protective casing
- Thermostat or temperature control
- Temperature indicator or display, if fitted
- Power cord and plug
- Mounting clips or suction cups
- Separate aquarium thermometer

Before you start

1. Confirm the required temperature range for the species and check the heater's rated aquarium volume.
2. Inspect the heater casing, power cord, plug, and mounting accessories for cracks or damage.
3. Check that the aquarium contains enough water to cover the heater's required immersion level.
4. Position the heater according to the manufacturer's instructions, away from gravel pressure and direct contact with delicate animals.
5. Verify that the power outlet, earthing, and residual-current protection are suitable for a wet environment.
6. Allow the heater to acclimatise to water temperature as instructed before switching it on; use a separate thermometer to check current temperature.

How to use

1. Switch off and unplug the heater before installing, repositioning, or removing it.
2. Secure the heater using its mounting clips or suction cups in the recommended position.
3. Confirm that the minimum immersion line, if present, is correctly covered by water.
4. Set the thermostat to the required species-appropriate temperature.
5. Wait for any manufacturer-specified immersion or settling period before connecting power.
6. Switch on the heater and observe its indicator or display, if fitted.
7. Check aquarium temperature with a separate thermometer and compare it with the target range.
8. Monitor temperature regularly and adjust the thermostat in small increments when needed.

9. Observe aquatic animals for signs of heat stress, unusual behaviour, or distress.
10. If water temperature rises unexpectedly or the heater malfunctions, disconnect power safely and use a suitable backup temperature-control method.
11. Before water changes, maintenance, or cleaning, unplug the heater and allow it to cool for the specified period.
12. After maintenance, ensure the heater is correctly submerged and securely mounted before restoring power.

⚠ Safety

- Never operate a heater outside its specified immersion conditions.
- Never touch, remove, or expose a hot heater to air immediately after operation; allow it to cool as directed.
- Keep plugs and electrical connections dry and create a drip loop in the power cord where appropriate.
- Do not switch on a cracked heater or use one with damaged insulation or cable.
- Disconnect power before putting hands into the aquarium or servicing electrical equipment.
- Avoid sudden temperature changes that can harm aquatic animals.

Care & storage

- Remove algae and mineral deposits only after unplugging and cooling the heater, using an approved method.
- Inspect the casing, cable, thermostat, and mounting clips regularly.
- Check actual water temperature with a reliable separate thermometer.
- Replace a heater that is cracked, corroded, unreliable, or unable to maintain the set temperature.
- Store a removed heater safely and dry according to the manufacturer's instructions.

Common mistakes to avoid

- Selecting a heater with insufficient or excessive capacity for the aquarium.
- Operating the heater without the required water immersion.
- Trusting the thermostat display without checking actual water temperature.
- Handling or cleaning the heater while it is hot or connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Automatic Waterer

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Bell Type Automatic Waterer

What it is used for

An automatic waterer supplies drinking water to animals as the water level changes. Check its water supply, valve, and cleanliness before allowing animals to use it.

How to use

1. Inspect the bowl, valve, float mechanism, pipes, and fittings for damage or leaks.
2. Clean and rinse the drinking surface using an animal-safe method.
3. Connect the water supply according to the equipment instructions.
4. Open the supply gradually and check that the valve controls the water level correctly.
5. Check that water flows freely and the bowl does not overflow.
6. Allow animals access only after confirming clean water and stable installation.
7. Close the supply before maintenance and clean the unit regularly.

⚠ Safety

- Never allow animals to drink water contaminated with chemicals or cleaning products.
- Keep electrical equipment away from water unless suitably protected.
- Repair leaks and unstable fittings before use.

Care & storage

- Clean the bowl routinely and inspect the valve, float, and water lines for blockages.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Beak Trimmer

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Beak Trimmer

What it is used for

A beak trimmer is a specialised poultry-care device used by trained personnel to trim or treat the beak according to an approved animal-welfare and veterinary procedure. Correct setup, restraint, equipment calibration, and post-procedure monitoring are essential to minimise pain, bleeding, infection, and stress.

Main parts

- Trimming or treatment head
- Heated blade, cutting blade, or approved treatment element, depending on model
- Handle or machine frame
- Temperature or operating control, if fitted
- Power supply and protective guard
- Animal restraint or positioning equipment

Before you start

1. Confirm that the procedure is legally permitted, necessary, and authorised under the applicable veterinary and animal-welfare protocol.
2. Only trained and competent personnel should perform the procedure; obtain veterinary guidance where required.
3. Inspect the device, blade or treatment head, guard, cables, and controls for damage or contamination.
4. Verify that the correct head, operating temperature, or settings are specified for the equipment and procedure.
5. Prepare a clean, well-lit work area and suitable restraint equipment to minimise bird stress and injury.
6. Prepare appropriate hygiene supplies and a plan for bleeding, injury, abnormal distress, or other complications.

How to use

1. Clean and prepare the equipment according to the manufacturer's instructions and the approved hygiene protocol.
2. Check the blade or treatment head condition and confirm that the device is operating within its specified settings.
3. Handle and restrain the bird gently using the approved method and avoid compressing the chest.

4. Confirm the bird's identity or group and assess its condition before proceeding.
5. Position the beak carefully according to the veterinary or approved procedure; avoid trimming beyond the authorised limit.
6. Activate the device only as instructed and complete the authorised treatment with controlled, consistent technique.
7. Observe the bird immediately for bleeding, burns, breathing difficulty, or unusual distress.
8. Stop the procedure and obtain veterinary assistance if a significant injury, uncontrolled bleeding, or abnormal reaction occurs.
9. Place the bird in a suitable recovery area with access to water and feed according to the care protocol.
10. Monitor feeding, drinking, behaviour, and signs of pain or infection during the required recovery period.
11. Clean and disinfect reusable equipment, dispose of waste safely, and record the procedure and any complications.

⚠ Safety

- Do not perform beak trimming without appropriate training, authorisation, and an approved welfare protocol.
- Follow applicable laws and veterinary guidance; do not perform cosmetic or unnecessary procedures.
- Keep fingers away from blades and hot treatment surfaces, and use the specified PPE.
- Do not use equipment with a damaged blade, faulty temperature control, or missing protective guard.
- Avoid excessive restraint and monitor the bird's breathing and stress throughout handling.
- Seek prompt veterinary care for significant bleeding, burns, infection, or persistent inability to eat or drink.

Care & storage

- Clean and disinfect the device after use according to the manufacturer's instructions.
- Inspect the blade, treatment head, guard, cable, and control system before each session.
- Maintain blade sharpness or replace worn parts only according to the approved service procedure.
- Store the equipment dry, secure, and inaccessible to unauthorised users.
- Review procedure records and animal outcomes to identify equipment or technique problems.

Common mistakes to avoid

- Performing the procedure without training, veterinary guidance, or proper authorisation.
- Removing more beak tissue than permitted by the approved procedure.
- Using incorrect temperature or damaged equipment.
- Failing to monitor birds for bleeding, distress, or difficulty eating after treatment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Biosecurity Material

Category: **Animal Husbandry & Dairy** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Biosecurity materials

What it is used for

Handle biosecurity materials to reduce the spread of animal diseases.

How to handle

- Follow the facility's biosecurity plan and product label instructions.
- Wear the specified PPE when handling disinfectants or contaminated materials.
- Use the correct dilution and contact time for approved disinfectants.
- Keep clean and contaminated equipment separate and report spills promptly.

Storage

- Store chemicals in labelled, compatible containers in a secure area.
- Keep PPE and clean supplies separate from used or contaminated materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Boat

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Droni

What it is used for

A boat or droni is a watercraft used for approved activities on water. Operate it only with suitable supervision, safety equipment, and knowledge of local water conditions.

How to use

1. Inspect the hull, seating, oars or motor, and safety equipment for damage.
2. Check weather, water conditions, permitted access, and the planned route before departure.
3. Ensure every occupant has a correctly fitted life jacket.
4. Load passengers and equipment evenly without exceeding the boat's capacity.
5. Launch and operate the boat under the direction of a qualified supervisor.
6. Keep clear of obstacles and follow instructions throughout the activity.
7. Return safely, secure the boat, and inspect it after use.

⚠ Safety

- Never enter the watercraft without appropriate flotation equipment and supervision.
- Do not overload the boat or operate it in unsafe weather.
- Keep clear of propellers, deep water, and other moving watercraft.
- Follow local boating rules and emergency procedures.

Care & storage

- Rinse and dry the boat after use and store it securely as instructed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Branding Numbers

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Branding numbers

What it is used for

To apply approved identification numbers to livestock using the designated animal-branding procedure. To ensure identification is legible while minimizing pain, burns, and infection risk.

Main parts

- Numbering irons or approved branding device
- Heat source or device control, depending on method
- Restraint equipment
- Cleaning and disinfection supplies
- Protective gloves and other PPE

Before you start

1. Confirm the animal's identity, ownership authorization, and applicable identification rules.
2. Use branding only when legally permitted and approved by a veterinarian or trained livestock professional.
3. Ensure the animal can be safely restrained by trained handlers.
4. Inspect the equipment and prepare the approved cleaning supplies.
5. Confirm that the chosen method, number size, and placement are appropriate for the animal.

How to use

1. Verify the assigned identification number before starting.
2. Prepare and restrain the animal using an approved humane procedure.
3. Prepare the branding device strictly according to its instructions and the approved method.
4. Apply the identification only at the approved site and for the specified contact time.
5. Release the animal safely and inspect the site for injury or abnormal reaction.
6. Provide aftercare according to veterinary guidance.
7. Record the identification and report any complication.

⚠ Safety

- Branding must be performed only by trained, authorized personnel.
- Never improvise with heated metal or apply excessive contact time.
- Use humane handling and avoid unnecessary distress.
- Follow veterinary guidance for pain management and wound care.
- Seek veterinary help for severe burns, bleeding, infection, or abnormal behaviour.

Care & storage

- Clean and disinfect reusable equipment according to its instructions.
- Inspect the branding device for damage before storage.
- Keep records of identification and aftercare.
- Store equipment securely away from animals and children.

Common mistakes to avoid

- Branding without confirming the assigned number.
- Using an unapproved method or location.
- Applying excessive heat or contact time.
- Ignoring wound care or signs of infection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Breeding Tank

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Plastic or glass breeding tanks

What it is used for

A breeding tank provides a controlled environment for approved aquatic breeding or observation activities. Maintain suitable water conditions and follow species-specific care instructions.

How to use

1. Inspect the tank, seams, stand, lid, and fittings for cracks or leaks.
2. Clean and rinse the tank using a method safe for the intended species.
3. Set up suitable water, filtration, aeration, and temperature controls as required.
4. Allow the system to stabilize and check water conditions before introducing organisms.
5. Introduce only approved, compatible organisms according to the care plan.
6. Observe water quality, animal behaviour, and equipment function regularly.
7. Remove waste safely and clean the tank according to the maintenance schedule.

⚠ Safety

- Never use tanks containing cleaning chemical residue or unsuitable water.
- Handle aquatic organisms gently and follow hygiene precautions after contact.
- Keep electrical pumps and heaters protected from splashes and water leaks.
- Do not release captive organisms into natural waterways.

Care & storage

- Remove waste regularly, monitor water conditions, and inspect the tank for leaks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Brooder

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Reflectors/ Hovers

What it is used for

A poultry brooder, using a reflector, hover, or other approved heat source, provides young chicks with a suitable warm area during early growth. Set up the brooder to create a safe temperature gradient and monitor chick behaviour to prevent chilling, overheating, crowding, and fire.

Main parts

- Reflector, hover, or heating unit
- Heating element or approved heat source
- Height adjustment or suspension supports
- Temperature gauge or thermometer
- Protective guard and chick enclosure
- Power supply or fuel connection, depending on model

Before you start

1. Confirm the chick age, number, breed, and recommended brooding temperature from the husbandry plan or veterinary guidance.
2. Inspect the heater, reflector, supports, guards, cables, fuel connections, and controls for damage.
3. Clean and dry the brooding area and provide suitable bedding, water, and feed.
4. Position the heat source securely above the designated area and maintain the specified clearance from bedding and combustible materials.
5. Check ventilation and ensure that chicks cannot touch the hot element or electrical parts.
6. Use a reliable thermometer at chick level and verify that backup heat and emergency arrangements are available.

How to use

1. Install the brooder according to the manufacturer's instructions and secure all supports before energising it.
2. Start the heat source in advance of chick arrival so the floor and brooding area reach the required temperature.
3. Measure temperature at chick level at several positions to confirm an even gradient.
4. Adjust heater height or output according to the equipment design and the recommended temperature range.
5. Introduce chicks gently and observe whether they spread evenly, gather under the heat, or move away from it.

6. Check chick activity, feeding, drinking, and signs of panting, huddling, or distress at regular intervals.
7. Adjust heat and ventilation gradually in response to measured temperature and chick behaviour.
8. Keep bedding dry and remove wet or heavily soiled patches promptly.
9. If the heater fails, temperature becomes unsafe, or smoke or unusual odour appears, activate the emergency plan and provide safe backup heat.
10. During routine checks, confirm that guards, supports, and power or fuel connections remain secure.
11. Reduce or remove supplementary heat gradually according to chick age, feather development, and husbandry guidance.
12. Clean the area and inspect the brooder after the flock has been moved.

⚠ Safety

- Only trained personnel should install or adjust the brooder heat source.
- Keep heat sources away from bedding, dust, curtains, and other combustible materials.
- Never leave an unstable or unguarded heater where chicks can contact it.
- For fuel-burning brooders, provide required ventilation and inspect for gas leaks and carbon monoxide risks.
- Protect electrical connections from moisture and pecking; isolate power before maintenance.
- Do not make sudden temperature changes that could stress or harm chicks.

Care & storage

- Clean the reflector, guard, and surrounding area after shutdown and cooling.
- Inspect suspension chains, supports, guards, heating elements, and cables regularly.
- Check temperature gauges against a reliable reference at specified intervals.
- Replace damaged components and service fuel or electrical systems according to the manufacturer's instructions.
- Keep the brooder dry, secure, and ready for inspection before each new batch.

Common mistakes to avoid

- Setting the brooder too high or too low without checking chick-level temperature.
- Ignoring chick huddling, panting, or uneven distribution.
- Placing the heater too close to bedding or combustible materials.
- Failing to maintain ventilation or provide backup heat during equipment failure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bulk Milk Cooler

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bulk milk Cooler

What it is used for

A bulk milk cooler rapidly cools and stores freshly collected milk at a controlled temperature to slow bacterial growth and preserve quality before collection or processing. Operate it using hygienic milk-handling practices, monitor cooling performance, and follow applicable dairy requirements for temperature and storage time.

Main parts

- Insulated milk tank and lid
- Cooling compressor and refrigeration circuit
- Agitator or milk-stirring system
- Temperature sensor and display
- Milk inlet, outlet valve, and drain
- Cleaning system, control panel, and electrical protection

Before you start

1. Confirm the tank capacity, cooling specification, and required milk storage temperature from the dairy procedure.
2. Inspect the tank, lid, seals, valves, agitator, refrigeration unit, and electrical connections for damage or contamination.
3. Verify that the tank and milk-contact surfaces have been cleaned and sanitised according to the approved procedure.
4. Check that the cooling system, agitator, temperature display, and alarms operate correctly.
5. Ensure the milk inlet and outlet connections are clean, capped when not in use, and protected from contamination.
6. Confirm that the tank is positioned on a stable surface with adequate ventilation around the refrigeration unit.

How to use

1. Switch on the cooling system before or at milk intake according to the approved operating procedure.
2. Verify the displayed temperature and confirm that the tank is ready to receive milk.
3. Transfer freshly collected milk through the clean inlet using hygienic handling procedures.
4. Close the lid promptly and start the agitator according to the tank manufacturer's instructions.
5. Monitor milk temperature, cooling time, agitator operation, and any alarms during the cooling cycle.

6. Confirm that the milk reaches and remains within the temperature required by the dairy procedure and applicable regulations.
7. Avoid unnecessary lid opening and prevent water, cleaning chemicals, or foreign material from entering the tank.
8. Record milk intake time, quantity, temperature readings, and any abnormal cooling delays.
9. Before milk collection, follow the approved procedure for agitation, sampling, and transfer to the collection vehicle or processing line.
10. After emptying the tank, isolate power as required and carry out the prescribed wash, rinse, and sanitising cycle.
11. Inspect the tank and valves after cleaning, allow surfaces to drain as specified, and record cleaning completion or faults.

⚠ Safety

- Only trained personnel should operate, clean, or maintain the cooler.
- Never enter the tank or reach into it while the agitator is energised; isolate and verify power before internal cleaning or maintenance.
- Keep electrical connections dry and use the installed earthing and protective devices.
- Use approved detergents and sanitising agents at the specified concentration; never mix incompatible chemicals.
- Treat refrigerant systems and pressure components as specialist service items; do not open or repair them without qualification.
- If cooling fails or milk temperature exceeds the permitted limit, notify the responsible dairy operator and follow the milk quality disposition procedure.

Care & storage

- Clean and sanitise all milk-contact surfaces after each batch or as required by the dairy schedule.
- Inspect seals, valves, agitator blades, sensors, and the tank outlet regularly.
- Check refrigeration performance, condenser cleanliness, and airflow around the unit.
- Calibrate or verify the temperature sensor at the specified intervals.
- Arrange qualified servicing for refrigerant leaks, abnormal noise, poor cooling, or repeated alarms.

Common mistakes to avoid

- Loading milk into an inadequately cleaned or sanitised tank.
- Ignoring slow cooling or incorrect temperature readings.
- Leaving the lid open or allowing contaminated equipment to contact milk.
- Cleaning or servicing the agitator without isolating power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bull Exerciser

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Bull exerciser

What it is used for

A bull exerciser is used to provide controlled physical activity for breeding bulls under a planned animal-management routine. Correct setup, close supervision, and careful handling help maintain animal fitness while reducing stress and injury risks.

Main parts

- Exercise frame or walking area provides the designated movement space.
- Drive mechanism moves the exercise equipment where fitted.
- Control panel or switch operates the powered system where fitted.
- Speed or movement control adjusts the exercise rate where provided.
- Barriers and gates help manage animal movement.
- Restraint or guiding equipment supports controlled handling where approved.
- Emergency stop halts powered movement where fitted.
- Inspection and maintenance points allow routine equipment checks.

Before you start

1. Only trained animal handlers should operate the bull exerciser.
2. Check the bull's health, temperament, and suitability for exercise with the responsible veterinarian or animal-care professional where needed.
3. Inspect the frame, floor, barriers, gates, drive mechanism, and controls for damage.
4. Confirm that the exercise area is dry, non-slip, free of obstacles, and adequately ventilated.
5. Check that emergency stops and protective guards work correctly on powered models.
6. Prepare suitable assistance and follow the facility's animal-handling and exercise plan.

How to use

1. Guide the bull calmly into the designated exercise area using the approved handling method.
2. Secure gates and barriers according to the equipment instructions without causing distress.
3. Start the exercise system at the prescribed low setting where speed control is available.
4. Observe the bull continuously and allow movement at the rate specified by the animal-care plan.
5. Check for signs of fatigue, lameness, overheating, distress, or abnormal behaviour.
6. Stop the system immediately if the bull becomes distressed, loses balance, or enters an unsafe position.
7. Finish the session according to the planned duration and stop all powered movement before releasing the animal.
8. Guide the bull out through the designated route and check its condition after exercise.

9. Record exercise duration, observations, equipment faults, and any required follow-up.

⚠ Safety

- Never leave a bull unattended during exercise.
- Keep handlers and bystanders outside crush, kick, and entanglement zones.
- Do not use the exerciser on an injured, ill, overheated, or unsuitable animal without veterinary clearance.
- Never force a bull to continue when it shows distress, fatigue, or loss of coordination.
- Keep hands and clothing away from moving parts and use the emergency stop when necessary.
- Do not modify the equipment or use restraints contrary to the approved procedure.

Care & storage

- Remove manure, mud, and debris from the exercise area after use.
- Inspect barriers, gates, flooring, drive components, and controls regularly.
- Clean and disinfect equipment using animal-safe products where required.
- Report unusual noise, damaged parts, or unreliable controls and arrange authorised repairs.

Common mistakes to avoid

- Starting exercise without checking the bull's health and temperament.
- Using a slippery or obstructed exercise area.
- Leaving the bull unsupervised or ignoring signs of fatigue.
- Releasing the bull before all powered movement has stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bull Leader

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Bull leader

What it is used for

To guide and control a bull using an approved livestock leader or handling device. To move the animal with trained handling practices while minimizing stress and injury.

Main parts

- Leader body or frame
- Handle or control section
- Attachment or connection point
- Fastening mechanism, if fitted

Before you start

1. Confirm that the device is designed for the animal and intended handling method.
2. Inspect the leader and attachment points for wear, cracks, or looseness.
3. Ensure handlers are trained in bull behaviour and safe livestock movement.
4. Choose a clear route with secure gates and non-slip footing.
5. Use suitable protective equipment and maintain an escape route.

How to use

1. Approach the bull calmly from a safe, visible position.
2. Fit or connect the leader only according to its design and approved handling procedure.
3. Check that all fastenings are secure without restricting breathing or causing injury.
4. Maintain a safe position and a controlled distance from the animal's head and horns.
5. Guide the bull using calm, consistent movements without sudden pulling.
6. Stop if the animal becomes highly agitated or the equipment slips.
7. After handling, remove the device safely and inspect the animal for injury.

⚠ Safety

- Only trained handlers should manage bulls.
- Never wrap a lead around a hand or body part.
- Do not stand directly in front of the horns or in a crush zone.
- Avoid forceful pulling, hitting, or provoking the animal.
- Follow veterinary and farm safety procedures.

Care & storage

- Clean the device after use.
- Inspect fastenings and load-bearing parts regularly.
- Dry and store it away from moisture and direct sunlight as appropriate.
- Replace worn or damaged components promptly.

Common mistakes to avoid

- Using a device that does not fit the animal.
- Working alone with an agitated bull.
- Wrapping the lead around the hand.
- Ignoring loose fastenings or signs of animal distress.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bull Nose Punch

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Bull nose punch

What it is used for

To use a bull nose punch only as part of an approved veterinary or livestock procedure involving the nose or nose-ring fitting. To avoid tissue damage by following trained professional guidance.

Main parts

- Punch body
- Working tip
- Handle or grip
- Protective storage case

Before you start

1. Confirm the exact tool identity and intended procedure with a veterinarian or trained livestock professional.
2. Do not use the tool for piercing unless the procedure is approved and performed by a qualified person.
3. Inspect the punch for damage, corrosion, or a blunt or irregular tip.
4. Prepare appropriate restraint, hygiene supplies, and protective equipment as directed by the professional.
5. Ensure the animal can be handled humanely and safely.

How to use

1. Verify the animal and the approved procedure before starting.
2. Prepare and restrain the animal using the professional's humane method.
3. Clean or sterilize the instrument as required by the procedure and its material.
4. Position the tool only as directed by the qualified professional.
5. Carry out the procedure only under the professional's supervision.
6. Check the animal for bleeding, swelling, or other adverse signs.
7. Follow the prescribed aftercare and record any complications.

⚠ Safety

- Never attempt an improvised nose-piercing procedure.
- Use only under qualified veterinary or livestock-professional direction.
- Wear appropriate PPE and follow infection-control practices.
- Stop and seek veterinary help if severe bleeding, injury, or distress occurs.

Care & storage

- Clean and sterilize the tool according to approved procedure.
- Inspect the tip and handle before storage.
- Keep the instrument dry and protected.
- Remove damaged tools from service.

Common mistakes to avoid

- Using the punch without confirming its intended purpose.
- Attempting a procedure without qualified supervision.
- Using a contaminated or damaged instrument.
- Ignoring bleeding, swelling, or signs of infection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bull Nose Ring

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Bull nose ring

What it is used for

To provide an approved nose-ring fitting for livestock when required for safe handling. To ensure the ring is suitable, securely fitted by a trained professional, and checked for injury.

Main parts

- Ring body
- Closing or locking joint
- Hinge or connecting section, depending on design
- Fastener or locking mechanism

Before you start

1. Confirm the ring is appropriate for the animal and approved handling procedure.
2. Have a veterinarian or trained livestock professional assess the animal and fit the ring.
3. Inspect the ring for cracks, corrosion, sharp edges, or a faulty lock.
4. Prepare appropriate hygiene and restraint equipment.
5. Check that the ring size and design are suitable for the animal.

How to use

1. Verify the animal and the intended fitting procedure.
2. Restrain the animal humanely under trained supervision.
3. Clean or prepare the fitting equipment according to professional guidance.
4. Allow the qualified professional to fit and secure the ring using the approved procedure.
5. Check that the locking mechanism is secure and the ring is correctly positioned.
6. Observe the animal for pain, bleeding, swelling, or abnormal behaviour.
7. Follow the prescribed aftercare and record the fitting when required.

⚠ Safety

- Do not attempt to pierce or fit a nose ring without qualified training and authorization.
- Never use a ring with sharp edges or a faulty lock.
- Avoid pulling or lifting the animal by the ring unless an approved handling procedure specifically permits it.
- Seek veterinary care for injury or signs of infection.

Care & storage

- Inspect the ring and locking mechanism regularly.
- Clean the area according to veterinary instructions.
- Replace damaged rings only through an approved procedure.
- Keep spare rings clean and securely stored.

Common mistakes to avoid

- Using the wrong ring size or design.
- Fitting the ring without proper training.
- Ignoring a loose lock or sharp edge.
- Using the ring in a way that causes pain or injury.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Burdizzo Castrator

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Burdizzo castrator

What it is used for

A Burdizzo castrator is a specialised clamp used by trained veterinary or livestock professionals to crush the spermatic cords during an approved bloodless castration procedure. Correct animal selection, anatomical knowledge, hygiene, pain management, and aftercare are essential for animal welfare and procedural safety.

Main parts

- Clamp jaws apply controlled compression to the targeted tissue.
- Handle arms provide leverage to operate the clamp.
- Hinge or pivot connects the handle arms.
- Locking mechanism holds the clamp position where fitted.
- Grip surfaces help the operator maintain control.
- Identification or size markings may help select the appropriate instrument.
- Cleaning and storage case protects the instrument between uses.

Before you start

1. Only a veterinarian or specifically trained and legally authorised livestock professional should perform the procedure.
2. Confirm the animal's identity, age, health, suitability, and applicable veterinary requirements before proceeding.
3. Explain the procedure and ensure an approved pain-management and aftercare plan is in place.
4. Inspect the clamp for damaged jaws, loose joints, corrosion, or unreliable movement.
5. Prepare suitable restraint, clean supplies, and a hygienic working area.
6. Confirm the instrument is cleaned, disinfected or sterilised as required by its instructions and the approved veterinary protocol.

How to use

1. Restrain the animal using the approved method to minimise movement and distress.
2. Confirm the correct anatomical landmarks and procedure plan before applying the instrument.
3. Use the prescribed pain-management and hygiene measures before starting.
4. Position the clamp only as directed by the approved veterinary technique and instrument instructions.
5. Apply compression in the specified manner without improvising placement or duration.
6. Complete the procedure according to the veterinary protocol and release the animal only when safe.

7. Observe the animal for distress, bleeding, swelling, abnormal movement, or other complications.
8. Provide the prescribed aftercare and clear instructions for follow-up monitoring.
9. Record the animal identification, procedure date, operator, pain management, and observations.
10. Clean and maintain the instrument according to its instructions after use.

⚠ Safety

- Never use a Burdizzo castrator without formal training and authorisation.
- Do not attempt the procedure on an unsuitable or unassessed animal.
- Do not guess clamp placement, force, or application duration; follow the approved veterinary technique.
- Use appropriate animal restraint and avoid placing hands in crushing zones.
- Stop and seek veterinary assistance if complications, severe distress, or uncertainty occurs.
- Follow local animal-welfare law and veterinary guidance, including pain relief and post-procedure monitoring.

Care & storage

- Clean the instrument promptly after use and disinfect or sterilise it as specified.
- Dry the instrument thoroughly before storage to reduce corrosion.
- Inspect the jaws, hinge, handles, and locking mechanism before each use.
- Store the clamp in a clean, dry, protected case and arrange servicing if it does not operate reliably.

Common mistakes to avoid

- Using the instrument without appropriate training or veterinary authorisation.
- Applying the clamp without confirming anatomical landmarks.
- Ignoring pain management, hygiene, or follow-up care.
- Failing to monitor the animal for complications after the procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Butter Churner

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Butter Churner

What it is used for

A butter churner agitates cream to separate butterfat into butter granules and buttermilk. Use clean equipment, suitable cream conditions, and controlled churning to obtain consistent product quality while preventing contamination and mechanical injury.

Main parts

- Churning vessel or drum
- Lid and locking mechanism
- Agitator, paddles, or rotating churn assembly
- Drive motor, gearbox, and speed control
- Drain or buttermilk outlet
- Control panel, guards, and emergency stop

Before you start

1. Confirm the batch size, cream quality, and required processing conditions from the dairy procedure.
2. Inspect the vessel, lid, seals, agitator, outlet valve, drive system, and guards for damage or residue.
3. Ensure all product-contact surfaces have been cleaned and sanitised using approved food-safe procedures.
4. Check that the lid interlock, emergency stop, and speed controls work correctly.
5. Verify that the cream is within the specified temperature and quality range for churning.
6. Prepare clean collection containers for butter granules and buttermilk, and wear appropriate food-handling PPE.

How to use

1. Switch off and isolate power before opening the vessel, inspecting the agitator, or cleaning the machine.
2. Confirm that the drain valve is closed and that the vessel is clean and ready for the batch.
3. Measure and load the required quantity of cream without exceeding the manufacturer's capacity.
4. Secure the lid and confirm that the safety interlock is engaged where fitted.
5. Select the approved speed or operating setting and start the churner using the normal controls.
6. Observe the cream during churning and monitor the motor, product movement, temperature, and sound.

7. Continue churning until butter granules form and the separation stage meets the dairy procedure's criteria.
8. Stop the machine before opening the lid or draining buttermilk; wait for all moving parts to stop.
9. Drain the buttermilk through the designated outlet into a clean container without contaminating the butter.
10. Collect the butter granules and carry out washing, working, salting, or packaging only as specified by the product procedure.
11. Inspect the finished butter for expected texture and appearance and record yield and batch details.
12. After unloading, isolate power and clean, rinse, and sanitise the vessel, agitator, lid, and outlet according to the sanitation schedule.

⚠ Safety

- Only trained personnel should operate or maintain the churner.
- Never open the lid or reach into the vessel while the agitator is moving.
- Do not bypass lid interlocks, guards, or emergency stop devices.
- Keep hands, loose clothing, and tools away from rotating shafts and drive components.
- Use approved cleaning chemicals at the specified concentration and prevent residues from contacting food.
- Disconnect and isolate power before maintenance, jam clearing, or dismantling the agitator.

Care & storage

- Clean and sanitise the vessel, lid, agitator, and outlet after each batch.
- Inspect seals, bearings, drive belts, fasteners, and interlocks regularly.
- Check speed controls and motor operation according to the maintenance schedule.
- Keep the exterior and surrounding floor clean and dry to reduce slip hazards.
- Arrange servicing if the churner vibrates excessively, leaks, overheats, or produces unusual noise.

Common mistakes to avoid

- Loading too much cream or using cream outside the specified processing conditions.
- Churning at an incorrect speed or stopping before the required separation stage.
- Opening the lid or draining the vessel while the agitator is moving.
- Failing to clean and sanitise the machine between batches.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

CMT Paddle

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: CMT Paddle; CMT Paddle (for detection of mastitis)

What it is used for

To screen dairy animals for possible mastitis using a CMT paddle and the approved California Mastitis Test procedure. To identify abnormal milk reactions for further veterinary assessment, not to make a diagnosis alone.

Main parts

- CMT paddle with separate wells
- Approved CMT reagent
- Milk sample from each quarter
- Sample collection container, if required
- Protective gloves

Before you start

1. Confirm the test kit instructions and reagent expiry date.
2. Wear clean gloves and use clean equipment.
3. Clean the teats and discard the first streams according to the dairy procedure.
4. Collect a separate sample from each udder quarter into the corresponding paddle well.
5. Avoid mixing samples between quarters.

How to use

1. Place an appropriate amount of milk in each labelled paddle well.
2. Add the specified amount of CMT reagent to each well according to the kit instructions.
3. Gently swirl the paddle in a circular motion to mix the contents.
4. Observe the mixture within the time window specified by the kit.
5. Compare the reaction with the supplied interpretation guide.
6. Record each quarter's result and identify samples requiring follow-up.
7. Dispose of the mixture and clean the paddle according to the hygiene procedure.

⚠ Safety

- Wear gloves and avoid contact with the reagent.
- Do not ingest the reagent or allow it to contact eyes or broken skin.
- Do not use expired or unidentified chemicals.
- Seek veterinary assessment for abnormal results or clinical signs of mastitis.

Care & storage

- Clean and dry the paddle after use.
- Store the reagent according to its label.
- Keep the kit away from milk containers and feed.
- Replace damaged paddles and expired reagents.

Common mistakes to avoid

- Mixing milk samples from different udder quarters.
- Using incorrect milk-to-reagent amounts.
- Reading the reaction outside the specified time window.
- Treating a CMT result as a definitive diagnosis without veterinary assessment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chick Box

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Chick box

What it is used for

A chick box provides temporary housing and transport space for young chicks. Maintain suitable ventilation, warmth, cleanliness, and stocking density according to animal-care guidance.

How to use

1. Inspect the box, lid, ventilation openings, and base for damage.
2. Clean and dry the box using a method safe for chicks.
3. Prepare suitable bedding and ensure ventilation openings remain unobstructed.
4. Check the temperature and arrange a safe heat source if required.
5. Place chicks gently inside without overcrowding and secure the lid appropriately.
6. Monitor chick behaviour, warmth, and ventilation throughout use.
7. Remove bedding and waste safely and clean the box after use.

⚠ Safety

- Never block ventilation openings or expose chicks to excessive heat or cold.
- Keep chicks away from drafts, predators, and contaminated surfaces.
- Handle chicks gently with clean hands and follow hygiene precautions.
- Do not place the box near unprotected heaters or open flames.

Care & storage

- Replace wet bedding promptly and clean and dry the box between uses.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cream Separator

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Cream separator; Mini Cream Separator

What it is used for

A cream separator separates cream from milk by using centrifugal force to divide the lower-density fat-rich portion from the skim milk. Correct assembly, hygienic handling, balanced operation, and timely cleaning help maintain separation efficiency and milk quality.

Main parts

- Feed bowl or inlet receives the milk.
- Rotating bowl contains the separation assembly.
- Disc stack increases the effective separation area where fitted.
- Cream outlet discharges the separated cream.
- Skim milk outlet discharges the lower-fat milk.
- Drive motor or hand-operated mechanism rotates the bowl.
- Frame and base support the separator.
- Flow control regulates milk feed where fitted.
- Cover and locking mechanism enclose the rotating assembly.
- Collection containers receive cream and skim milk.

Before you start

1. Only trained personnel should operate the cream separator.
2. Read the manual and confirm that the separator is suitable for the milk volume and intended use.
3. Inspect the bowl, disc stack, shaft, drive, cover, locking mechanism, outlets, and collection containers for damage.
4. Confirm that all food-contact parts are clean, sanitised, and assembled according to the manufacturer's instructions.
5. Check that the rotating bowl is correctly seated and all required locking components are secured.
6. Place the separator on a stable, level surface and verify the power supply or hand drive is ready.
7. Prepare fresh milk according to the process instructions and use clean, labelled collection containers.

How to use

1. Assemble the separator and install the bowl and disc stack according to the manual.
2. Secure the cover and all locking components before operating the drive.
3. Place clean containers beneath the cream and skim milk outlets.
4. Start the drive and allow the bowl to reach its specified operating speed before feeding milk.

5. Feed milk at the recommended temperature and rate for the model and process.
6. Observe the cream and skim milk outlets and adjust feed or settings only as permitted by the manual.
7. Collect the separated products in their designated containers and prevent cross-contamination.
8. Stop feeding milk before stopping the drive, following the manufacturer's shutdown sequence.
9. Wait until the bowl has stopped completely before opening the cover or removing components.
10. Disconnect power where applicable, disassemble the product-contact parts, and remove milk residues promptly.
11. Clean and sanitise the bowl, disc stack, outlets, and containers according to the approved procedure.
12. Record production quantities and any separation or equipment problems according to facility practice.

⚠ Safety

- Never open the cover or reach into the separator while the bowl is rotating.
- Do not operate with a damaged bowl, loose disc stack, faulty locking mechanism, or excessive vibration.
- Keep hands, hair, clothing, and tools away from moving parts and drive components.
- Disconnect electrical power before disassembly, cleaning, or maintenance of powered models.
- Use approved food-contact cleaning chemicals and follow their dilution and rinsing instructions.
- Keep milk within required time and temperature controls to limit microbial growth.

Care & storage

- Clean and sanitise all milk-contact components after each use.
- Inspect the bowl, disc stack, shaft, seals, and outlets for wear or residue buildup.
- Lubricate only designated points using food-compatible lubricant specified by the manufacturer.
- Keep the separator dry and store removable components in a clean, protected location.
- Arrange authorised service if vibration, leakage, unusual noise, or separation quality problems persist.

Common mistakes to avoid

- Operating before the bowl and disc stack are correctly assembled and locked.
- Feeding milk too quickly or outside the recommended temperature range.
- Opening the cover before the bowl has completely stopped.
- Leaving milk residue in the separator or using unclean collection containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Curd Incubator

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Small Incubation Box/Curd Maker

What it is used for

A curd incubator or small curd-making box maintains a controlled temperature that allows a suitable starter culture to ferment milk into curd. Hygienic preparation, accurate temperature control, and adequate incubation time help produce safe, consistent curd.

Main parts

- Insulated incubation chamber
- Heating element or temperature-control system
- Thermostat or digital temperature controller
- Temperature sensor or display
- Shelves, trays, or container supports
- Power cord, switch, and protective housing

Before you start

1. Confirm that the milk has been properly heat-treated and cooled to the starter culture's recommended inoculation temperature.
2. Use clean, sanitised food-grade containers and utensils; wash hands and follow the dairy hygiene procedure.
3. Check that the incubator chamber, shelves, door seal, sensor, and power cable are clean and undamaged.
4. Verify that the temperature controller and sensor are functioning and that the chamber can maintain the required temperature.
5. Prepare the starter culture according to its instructions and check its suitability and expiry date.
6. Place the incubator on a stable, dry, ventilated surface away from direct sunlight and water splashes.

How to use

1. Switch on the incubator and set the temperature recommended for the starter culture and curd process.
2. Allow the chamber to stabilise and verify the displayed temperature before loading the containers.
3. Add the recommended quantity of starter culture to the prepared milk and mix gently using a sanitised utensil.
4. Fill clean containers to a suitable level and cover them to protect against contamination.
5. Arrange containers evenly inside the chamber, leaving space for air circulation where the design requires it.

6. Close the door and start the incubation period according to the starter culture and process instructions.
7. Monitor the chamber temperature at the specified intervals and record the start time and temperature readings.
8. Avoid unnecessary door opening, shaking, or moving the containers during fermentation.
9. Check curd setting after the recommended incubation period using the approved quality check.
10. Stop incubation when the required texture and acidity are reached; do not rely on time alone if the process requires product checks.
11. Remove containers with clean hands or suitable clean handling tools and close them promptly.
12. Cool the curd promptly in a suitable refrigerator to slow further fermentation and maintain quality.
13. Switch off and clean the chamber and removable supports after use, following the manufacturer's instructions.

⚠ Safety

- Use only food-grade containers and utensils and maintain strict dairy hygiene.
- Do not handle electrical switches or plugs with wet hands.
- Keep the incubator away from water, wet floors, and sources of contamination.
- Do not exceed the equipment's specified temperature range or block ventilation openings.
- Use care when handling warm milk and hot surfaces to avoid burns.
- Refrigerate finished curd promptly and discard product when contamination or unsafe processing is suspected.

Care & storage

- Clean and sanitise the chamber, shelves, and container supports after use according to the manufacturer's guidance.
- Check the door seal, power cable, temperature sensor, and controller regularly.
- Verify temperature accuracy periodically using a suitable calibrated thermometer.
- Keep ventilation openings clear and avoid pouring water directly onto electrical components.
- Arrange servicing if temperature fluctuates, heating fails, or the controller gives unreliable readings.

Common mistakes to avoid

- Adding starter culture when milk is too hot or too cold for the culture.
- Using unclean containers or utensils that can contaminate the milk.
- Setting an incorrect incubation temperature or opening the chamber repeatedly.
- Leaving finished curd warm for too long instead of cooling it promptly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drenching Bottle

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Drenching bottles

What it is used for

To administer an oral drench to livestock using a suitable drenching bottle when directed by a veterinarian or approved animal-care procedure. To deliver the prescribed liquid carefully and reduce the risk of choking or aspiration.

Main parts

- Bottle body
- Graduated markings, if fitted
- Neck or nozzle
- Cap
- Dose measuring device, if separate

Before you start

1. Confirm the animal, prescribed product, dose, and route with the veterinary instructions.
2. Check the product label, expiry date, and bottle cleanliness.
3. Measure the prescribed amount using an appropriate measuring device.
4. Ensure the animal is calm and restrained by a trained handler.
5. Inspect the bottle and nozzle for cracks, blockage, or sharp edges.

How to use

1. Prepare the prescribed liquid according to its label and veterinary directions.
2. Hold the bottle securely and position the animal's head naturally.
3. Insert the nozzle carefully at the side of the mouth as directed by the product procedure.
4. Administer the liquid slowly in small amounts, allowing the animal to swallow.
5. Pause immediately if coughing, choking, or breathing difficulty occurs.
6. Check the animal's response and follow the required aftercare.
7. Clean the bottle and record administration according to the farm procedure.

⚠ Safety

- Use only the prescribed product and dose.
- Never force large amounts of liquid into the animal's mouth.
- Do not tilt the animal's head excessively or block its breathing.
- Seek veterinary advice if the animal cannot swallow or shows distress.

Care & storage

- Wash and dry the bottle after use if it is reusable.
- Keep products in their original labelled containers where possible.
- Inspect the nozzle and cap regularly.
- Store according to the product label and keep away from children.

Common mistakes to avoid

- Using an incorrect dose or product.
- Drenching too quickly and risking aspiration.
- Forcing the animal's head into an unsafe position.
- Reusing a contaminated bottle without cleaning.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Egg Candler

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Egg candler

What it is used for

An egg candler uses a light source to examine the internal development and condition of an egg without breaking the shell. Handle eggs gently and follow the recommended candling schedule.

How to use

1. Inspect the candler, light source, power cable, and housing for damage.
2. Wash and dry hands before handling eggs.
3. Switch on the candler and check that the light is working properly.
4. Place the egg gently against the designated viewing opening in a dim area.
5. Rotate the egg slowly and observe the visible internal features.
6. Record findings according to the incubation or inspection procedure.
7. Switch off the device and return eggs carefully to their appropriate conditions.

⚠ Safety

- Do not use a candler with damaged wiring or an exposed electrical part.
- Avoid prolonged exposure to strong heat or light.
- Handle eggs gently to prevent cracks and contamination.
- Follow hygiene precautions when examining eggs from different sources.

Care & storage

- Clean the contact surface according to the manufacturer's instructions and store the candler in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Egg Packing Machine

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Egg Packing Machine

What it is used for

An egg packing machine sorts, orients, and places eggs into trays, cartons, or other packaging according to the machine design. Correct setup, gentle handling, and routine checks help reduce cracked eggs, contamination, and packaging errors.

Main parts

- Egg infeed conveyor or loading table
- Egg orientation and transfer mechanism
- Grading or sorting section where fitted
- Tray or carton feed system
- Packing conveyor and collection area
- Control panel, guards, sensors, and drive system

Before you start

1. Confirm the egg size range, packaging type, batch requirements, and machine capacity.
2. Inspect conveyors, rollers, guides, cups, sensors, guards, and drive components for damage or residue.
3. Verify that food-contact surfaces and egg-contact parts are clean and suitable for use.
4. Check that trays and cartons are clean, dry, undamaged, and appropriate for the product.
5. Confirm that emergency stops and guards are in place and that the operating area is clear.
6. Follow hygiene procedures, wear clean protective clothing, and check that the machine is correctly assembled.

How to use

1. Switch off and isolate power before installing or adjusting removable guides, cups, or packaging components.
2. Fit the required tray or carton format and adjust guides according to the manufacturer's instructions.
3. Check that the infeed path, transfer mechanism, and collection area are clear of obstructions.
4. Switch on the machine and select the approved operating settings for the egg and packaging format.
5. Feed eggs gently and evenly into the designated infeed area without dropping or crowding them.
6. Start at a controlled speed and observe the transfer of eggs through orientation and packing stages.

7. Check that eggs are seated correctly in the trays or cartons and that packaging remains aligned.
8. Inspect early output for cracked eggs, empty positions, misplaced eggs, and damaged packaging.
9. Adjust approved settings or reduce feed rate if breakage, misalignment, or jams occur.
10. Monitor the machine during operation and remove damaged eggs using the approved hygienic procedure.
11. Stop the machine immediately if guards fail, an unsafe jam occurs, or abnormal noise or movement is detected.
12. Isolate power and wait for all moving parts to stop before clearing jams or reaching into the mechanism.
13. At the end of the batch, stop feeding eggs, complete the approved clearing sequence, and switch off the machine.
14. Remove packed products carefully, record batch and quality information, and clean egg-contact surfaces after use.

⚠ Safety

- Only trained personnel should operate or adjust the egg packing machine.
- Keep hands away from moving conveyors, rollers, transfer cups, and pinch points.
- Never bypass guards or reach into the machine while it is running.
- Switch off and isolate power before clearing jams, changing formats, or performing maintenance.
- Handle broken eggs promptly using the site's hygiene and waste procedures to prevent contamination and slipping hazards.
- Keep water and cleaning liquids away from electrical controls unless the equipment is designed for the approved cleaning method.
- Follow site biosecurity and food-safety procedures to prevent contamination between batches.

Care & storage

- Clean egg-contact surfaces, guides, conveyors, and collection areas according to the approved sanitation schedule.
- Inspect cups, rollers, belts, sensors, and guides for wear, damage, or misalignment.
- Check fasteners, guards, and emergency stops at the prescribed intervals.
- Keep electrical components protected during cleaning and follow the manufacturer's instructions.
- Arrange servicing if egg breakage rises, sensors malfunction, or the transfer mechanism becomes unreliable.

Common mistakes to avoid

- Feeding eggs too quickly or dropping them onto the conveyor.
- Using trays or cartons that do not match the selected machine format.
- Ignoring cracked eggs, misalignment, or repeated jams.
- Clearing an obstruction before the machine is isolated and stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Egg Tray

Category: **Animal Husbandry & Dairy** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Hatching egg trays

What it is used for

Handle hatching egg trays carefully to protect eggs and maintain hatchery hygiene.

How to handle

- Clean hands and use appropriate hygiene precautions before handling eggs or trays.
- Check trays for cracks, dirt, or damage before loading.
- Place eggs gently in the correct orientation according to hatchery procedures.
- Keep clean trays separate from used trays and avoid unnecessary movement.

Storage

- Store clean trays in a designated, dry, protected area.
- Clean and disinfect reusable trays according to the hatchery's biosecurity protocol.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Egg Washer

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Egg Washer

What it is used for

An egg washer removes surface dirt from eggs using a controlled washing process designed for the equipment and egg-handling facility. Correct water conditions, approved cleaning agents, gentle handling, and hygienic drying help reduce contamination and egg damage.

Main parts

- Egg infeed conveyor or loading rack
- Wash chamber and spray nozzles or brushes
- Water supply, valves, and temperature controls
- Approved detergent dosing system where fitted
- Rinse section and drainage system
- Outfeed conveyor, guards, and control panel

Before you start

1. Confirm that the washer is approved for the type of eggs and the facility's food-safety process.
2. Inspect the wash chamber, brushes, nozzles, conveyors, guards, hoses, and drainage system for damage or residue.
3. Verify that the water supply meets the facility's approved quality and temperature requirements for egg washing.
4. Use only cleaning agents approved for the equipment and process, following the specified concentration and instructions.
5. Check that drainage is clear and that washed eggs can move to the next approved drying or handling stage without recontamination.
6. Wear clean protective clothing and follow handwashing, biosecurity, and egg-handling procedures.

How to use

1. Switch off and isolate power before adjusting brushes, guides, or other moving components.
2. Set the water temperature, flow rate, and cleaning-agent concentration according to the validated facility procedure and equipment instructions.
3. Confirm that the spray system, brushes, and drainage are operating correctly before loading eggs.
4. Start the washer using the approved control and allow the system to reach the required operating conditions.
5. Place eggs onto the infeed conveyor in the correct orientation and avoid crowding or impact.

6. Feed eggs at a controlled rate so that the washing action reaches the intended surfaces without excessive friction.
7. Monitor water temperature, flow, detergent dosing, brush movement, and drainage during operation.
8. Inspect washed eggs for remaining soil, cracks, damaged shells, and residue; follow the site's rejection criteria.
9. Stop the machine if water conditions leave the approved limits, a nozzle blocks, a leak occurs, or eggs are being damaged.
10. Do not allow dirty wash water to contaminate clean eggs or downstream handling areas.
11. At the end of the batch, stop the infeed and follow the approved shutdown and draining procedure.
12. Remove retained debris and clean the chamber, brushes, nozzles, and conveyors according to the sanitation schedule.
13. Record required operating checks, cleaning completion, and any rejected eggs or process deviations.

⚠ Safety

- Only trained personnel should operate and clean the egg washer.
- Use only the approved water conditions and cleaning agents; unsuitable washing can increase contamination risk or damage egg shells.
- Keep hands away from moving brushes, rollers, and conveyors while the machine is running.
- Switch off and isolate power before clearing blocked nozzles, removing debris, or servicing moving parts.
- Prevent water from entering electrical controls and use the required electrical protection for wet areas.
- Wear appropriate gloves and eye protection when handling cleaning chemicals and follow their safety labels.
- Handle broken eggs and contaminated wash water according to the facility's hygiene and waste procedures.

Care & storage

- Clean and sanitise the chamber, brushes, nozzles, conveyors, and contact surfaces at the required frequency.
- Inspect brushes, seals, hoses, valves, and spray nozzles for wear, leaks, or blockage.
- Check water temperature controls, detergent dosing, and drainage performance regularly.
- Remove scale and deposits using approved products and methods.
- Arrange servicing if water temperature fluctuates, spray coverage becomes uneven, or shell damage increases.

Common mistakes to avoid

- Using water conditions or cleaning agents that are not approved for the process.
- Overloading the infeed and causing eggs to collide or crack.
- Allowing dirty water or contaminated surfaces to contact washed eggs.

- Cleaning or clearing the machine while moving parts remain active.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electric Dehorner

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Electric dehorner

What it is used for

An electric dehorner uses controlled heat to destroy horn-producing tissue in young livestock when the procedure is approved and performed by a trained professional. Correct animal assessment, pain management, temperature control, and aftercare are essential to reduce burns, infection, and animal distress.

Main parts

- Heating tip transfers heat to the treatment area.
- Heating element produces the required heat.
- Handle provides a controlled grip.
- Power cable connects the tool to its supply.
- Power switch controls heating where fitted.
- Temperature control or indicator helps monitor heat where available.
- Protective stand holds the hot tool between uses.
- Power supply or battery powers the model.

Before you start

1. Only trained and legally authorised animal-care personnel should perform dehorning.
2. Confirm the animal's age, health, suitability, and applicable veterinary requirements before treatment.
3. Use an approved pain-management plan and follow veterinary guidance for the species and age.
4. Inspect the heating tip, handle, cable, plug, controls, and protective stand for damage.
5. Prepare appropriate animal restraint, clean equipment, and a safe, ventilated work area.
6. Check the tool's instructions for warm-up time and approved operating temperature.
7. Keep flammable materials away and prepare suitable burn first-aid supplies.

How to use

1. Restrain the animal using an approved method that minimises stress and sudden movement.
2. Confirm the correct treatment site and procedure with the responsible veterinary professional.
3. Apply the prescribed pain-management and hygiene measures before using the tool.
4. Place the dehorner on its protective stand and warm it up according to the manual.
5. Check that the tool has reached the specified operating condition using its indicator or approved method.
6. Apply the heating tip only according to the approved veterinary technique and equipment instructions.

7. Avoid unnecessary contact with surrounding skin and follow the prescribed application duration.
8. Remove the tool carefully and place it back on the protective stand.
9. Observe the animal for excessive bleeding, burns, distress, or other complications and seek veterinary help when needed.
10. Provide aftercare and follow-up monitoring according to the veterinary plan.
11. Switch off, unplug where applicable, and allow the tool to cool completely before cleaning or storage.
12. Record the treatment date, animal identification, operator, and observations.

⚠ Safety

- Never use an electric dehorner without appropriate training and authorisation.
- The heating tip can cause severe burns; never touch it or leave it on an unsuitable surface.
- Do not exceed the approved temperature or application duration.
- Use secure animal restraint and keep hands away from the hot tip and sudden movement zones.
- Do not use damaged cables, plugs, or temperature controls.
- Follow animal-welfare requirements, pain-management guidance, and veterinary aftercare instructions.

Care & storage

- After the tool has cooled completely, remove residue using the approved cleaning method.
- Inspect the heating tip, handle, cable, plug, and controls before and after use.
- Keep the tool dry and store it in a protected location away from combustible materials.
- Arrange authorised servicing if heating becomes uneven or temperature control is unreliable.

Common mistakes to avoid

- Using the tool without checking animal suitability or the veterinary plan.
- Applying excessive heat or holding the tip in place too long.
- Leaving the hot dehorner unattended or without its protective stand.
- Ignoring signs of burns, infection, or distress after treatment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Feeding Tray

Category: **Animal Husbandry & Dairy** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Feeding trays for fish fry

What it is used for

Handle feeding trays for fish fry to support clean and controlled feeding.

How to handle

- Use trays intended for aquaculture and check them for cracks or residue.
- Measure feed as instructed and avoid overfeeding or spilling feed into the water.
- Clean trays between uses to reduce contamination and disease transmission.
- Use separate equipment for different tanks or groups where biosecurity requires it.

Storage

- Rinse and dry trays after cleaning before storage.
- Keep labelled trays in a clean, protected area away from chemicals and pests.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fish Net

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Fish net; Fishing tools; Hand net for transferring fish fry

What it is used for

Use a fish net to catch or transfer fish or fry gently while reducing injury and stress to aquatic animals.

How to use

1. Select a net with suitable mesh size and handle length for the fish.
2. Inspect the mesh, frame, handle, and joints for tears or sharp edges.
3. Prepare a suitable receiving container with water of appropriate quality and temperature.
4. Move the net slowly through the water and avoid chasing fish excessively.
5. Lift the net carefully and keep fish supported in water as much as possible.
6. Transfer fish promptly and gently into the prepared container.
7. Rinse the net and allow it to dry or disinfect it as required.

⚠ Safety

- Use clean equipment to reduce disease transfer between tanks or ponds.
- Avoid handling fish with dry hands or leaving them exposed to air unnecessarily.
- Keep sharp net frames away from people and delicate tank surfaces.
- Follow local rules for protected species and responsible fishing.

Care & storage

- Rinse off organic material after use and disinfect when required.
- Dry and store the net away from sunlight, pests, and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fish Rearing Tank

Category: **Animal Husbandry & Dairy** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Small fish rearing tanks

What it is used for

Handle small fish rearing tanks safely to maintain suitable conditions for fish growth.

How to handle

- Inspect the tank for cracks, leaks, and stable support before filling.
- Use clean, aquatic-safe equipment and maintain suitable water quality, temperature, and aeration.
- Avoid sudden changes in water conditions and prevent overstocking.
- Remove waste and uneaten feed regularly using appropriate procedures.

Storage

- Drain and clean the tank before long-term storage when appropriate.
- Store dry tanks in a secure area protected from impact and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fish Transport Bag

Category: **Animal Husbandry & Dairy** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Fish seed transport bags

What it is used for

Handle fish seed transport bags to transport live fish safely.

How to handle

- Use clean, leak-free bags suitable for aquatic transport.
- Fill with suitable water and oxygen according to the approved transport procedure.
- Avoid overloading bags and minimise handling time and temperature fluctuations.
- Seal securely and keep bags away from sharp objects and direct sunlight.

Storage

- Keep unused bags clean, dry, and sealed in their original packaging.
- Store away from heat, punctures, chemicals, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gerber Centrifuge

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Gerber Centrifuge

What it is used for

To use a Gerber centrifuge to separate the fat layer from prepared milk samples during dairy fat testing. To obtain consistent separation by balancing tubes and following the instrument's operating instructions.

Main parts

- Centrifuge body and lid
- Rotor or sample tube holders
- Gerber centrifuge tubes
- Lid lock and safety interlock
- Speed and timer controls
- Brake and display, if fitted

Before you start

1. Place the centrifuge on a stable, level surface with adequate ventilation.
2. Inspect the rotor, tube holders, lid, and tubes for cracks, corrosion, or damage.
3. Confirm that the rotor and tubes are approved for the instrument and test method.
4. Prepare milk samples and Gerber tubes according to the laboratory procedure.
5. Balance opposing tubes by equal mass and use compatible tubes in matching positions.
6. Check that the lid lock and safety interlock function correctly.

How to use

1. Switch off the centrifuge before loading samples.
2. Place each tube in its holder and arrange balanced loads opposite one another.
3. Ensure all tubes are seated correctly and their closures are secure as required by the method.
4. Close the lid fully and confirm that the lock engages.
5. Set the speed, temperature if available, and running time specified by the test method and instrument manual.
6. Start the centrifuge and observe it from a safe distance for unusual vibration or noise.
7. Stop the run immediately if excessive vibration, unusual noise, or leakage occurs and wait for the rotor to stop completely.
8. When the cycle ends, wait until the rotor has stopped and the lid unlocks.
9. Open the lid carefully and remove tubes without disturbing the separated layers.
10. Read the fat column using the approved Gerber method and record the result.
11. Switch off and clean any contamination according to laboratory procedures.

⚠ Safety

- Use only tubes, stoppers, and rotors rated for the selected speed and temperature.
- Never open the lid or attempt to slow the rotor by hand while it is moving.
- Do not operate with cracked tubes, damaged holders, or an unbalanced load.
- Wear a lab coat, safety glasses, and suitable gloves when handling chemicals and samples.
- Follow the approved method for handling sulfuric acid, amyl alcohol, and hot tubes where applicable.
- Disconnect power before maintenance and allow hot components to cool.

Care & storage

- Clean the chamber and holders after use using a method approved by the manufacturer.
- Dry components thoroughly and check for corrosion or residue.
- Inspect the rotor, lid seal, and locking mechanism regularly.
- Keep ventilation openings clear and arrange periodic servicing.
- Record faults and remove damaged components from service.

Common mistakes to avoid

- Loading tubes without balancing their mass.
- Using tubes or a rotor that are not rated for the operating conditions.
- Opening the lid before the rotor stops completely.
- Using incorrect test settings or an unapproved sample preparation method.
- Leaving corrosive residue or sample spills inside the chamber.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ghee Boiler

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Ghee Boiler

What it is used for

To heat cream or butter in a ghee boiler to remove moisture and develop the desired ghee characteristics. To control heating, monitor the product, and maintain hygienic operation.

Main parts

- Heating vessel or boiler
- Heating element or burner
- Temperature controller and sensor
- Agitator, if fitted
- Lid and outlet valve
- Insulation and control panel

Before you start

1. Ensure the vessel is clean, dry, and suitable for food processing.
2. Inspect the heating source, wiring or fuel connections, temperature sensor, and controls.
3. Check that the outlet valve is closed and the lid and seals are in good condition.
4. Confirm that the boiler is installed on a stable surface with suitable ventilation.
5. Prepare the required quantity of cream or butter and avoid exceeding the marked capacity.
6. Wear suitable heat-resistant gloves, protective clothing, and closed footwear.

How to use

1. Switch on the heating system according to the manufacturer's instructions.
2. Load the prepared cream or butter without exceeding the vessel's capacity.
3. Set the temperature and heating mode specified by the approved process.
4. Start agitation if fitted and required by the process.
5. Heat gradually while monitoring temperature, product appearance, and aroma.
6. Adjust the heat as required to prevent scorching, foaming, or overflow.
7. Continue processing until the product reaches the approved moisture-removal and ghee-finish criteria.
8. Switch off the heat before unloading or opening outlets, following the equipment procedure.
9. Allow the product to settle as required and drain through the designated outlet into a clean, dry container.
10. Filter the ghee using the approved process if filtration is required.
11. Switch off and isolate the power or fuel supply after the operation.
12. Clean the vessel and product-contact parts using the approved food-safe cleaning procedure.

⚠ Safety

- Only trained personnel should operate the boiler.
- Never exceed the manufacturer's fill limit or operating temperature and pressure limits.
- Avoid contact with hot vessel surfaces, hot ghee, steam, and splashes.
- Do not open a pressurized vessel; confirm that pressure is safely released if pressure operation applies.
- Keep water away from hot fat to prevent dangerous splashing.
- For gas-fired units, check for fuel leaks and maintain adequate ventilation.

Care & storage

- Clean and dry all product-contact surfaces after use.
- Inspect seals, valves, agitator, and temperature sensor regularly.
- Remove product residue before it hardens using an approved cleaning method.
- Check electrical connections or burner components as applicable.
- Arrange periodic maintenance and temperature-control checks.

Common mistakes to avoid

- Overfilling the vessel and allowing the product to overflow.
- Heating too rapidly and scorching the product.
- Ignoring temperature readings or unusual smells.
- Allowing water or cleaning chemicals to contaminate finished ghee.
- Cleaning or servicing the unit before it has cooled and been safely isolated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grooming Brush

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Low** 3 labs

Covers these lab items: Grooming brush

What it is used for

A grooming brush removes loose hair, dirt, and debris from an animal's coat while supporting routine coat care. Select a brush suitable for the species, coat type, and animal's condition.

How to use

1. Inspect the brush bristles, handle, and body for damage or sharp edges.
2. Choose a brush suitable for the animal's coat and grooming needs.
3. Secure the animal calmly using approved handling methods.
4. Brush gently in the direction of hair growth, avoiding sensitive areas.
5. Check the coat and skin for wounds, irritation, parasites, or unusual changes.
6. Remove collected hair and debris from the brush after grooming.
7. Clean and store the brush separately from equipment used for other animals when required.

⚠ Safety

- Use trained animal-handling practices and avoid sudden movements that may startle the animal.
- Do not brush over open wounds or severely irritated skin without veterinary guidance.
- Keep fingers away from sharp or broken bristles.
- Use separate or disinfected brushes to reduce disease transmission between animals.

Care & storage

- Remove hair after each use and clean or disinfect the brush according to the animal-care procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hatchery Jar

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Hatchery jars

What it is used for

A hatchery jar provides a controlled container for incubating or observing suitable aquatic eggs and larvae, depending on the system design. Maintain the required water conditions and follow the hatchery protocol.

How to use

1. Inspect the jar, lid, fittings, and support stand for cracks or leaks.
2. Clean and prepare the jar according to the hatchery sanitation procedure.
3. Position the jar securely and connect the specified water or aeration lines if required.
4. Fill with suitable conditioned water and verify temperature and water quality.
5. Introduce eggs at the recommended density and follow the species-specific hatchery protocol.
6. Monitor water flow, aeration, temperature, and egg condition at scheduled intervals.
7. Remove waste and transfer larvae only as directed by the hatchery procedure.

⚠ Safety

- Use clean equipment and follow biosecurity procedures to prevent contamination.
- Avoid overfilling or operating with leaking fittings.
- Keep electrical aeration equipment and connections protected from water.
- Handle eggs and larvae gently and follow species-specific welfare requirements.

Care & storage

- Clean and disinfect the jar between batches using approved methods, then rinse as required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hoof Trimmer

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Hoof trimmer

What it is used for

To trim and maintain livestock hooves as part of appropriate hoof-care practices.

How to handle

- Inspect the trimmer, handles, cutting edges, and locking parts for damage or looseness.
- Restrain the animal using an approved, safe method and have a trained handler assist as needed.
- Wear appropriate protective equipment and clean the hoof before examination.
- Trim only as much hoof material as required by a trained person; avoid sensitive tissue and stop if bleeding or distress occurs.
- Clean and disinfect the tool after use and arrange veterinary assessment for injuries, infection, or abnormal hoof conditions.

Storage

- Store clean, dry, and secured with cutting edges covered.
- Keep in a designated tool cabinet away from children and unauthorised users.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lactometer

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Lactometer

What it is used for

To estimate the specific gravity of milk using a lactometer. To obtain a reliable reading by using a clean instrument and measuring milk under the specified temperature conditions.

Main parts

- Lactometer stem and scale
- Weighted bulb
- Milk sample cylinder
- Thermometer
- Clean sample container

Before you start

1. Clean and dry the lactometer and sample cylinder.
2. Collect a representative milk sample in a clean container.
3. Check the lactometer for cracks and a readable scale.
4. Measure the sample temperature and note the method's reference temperature.

How to use

1. Gently mix the milk sample without creating excessive foam.
2. Pour milk into the cylinder without overflowing it.
3. Lower the lactometer gently into the milk so it floats freely.
4. Ensure the instrument does not touch the cylinder sides or bottom.
5. Allow the lactometer to settle and remove any bubbles near the stem.
6. Read the scale at the specified meniscus position and record the temperature.
7. Apply the approved temperature correction if required by the method.
8. Record the result and clean the instrument after use.

⚠ Safety

- Handle the glass instrument carefully to prevent breakage.
- Use clean sampling practices to avoid milk contamination.
- Do not use a cracked or damaged lactometer.
- Follow laboratory procedures for sample disposal.

Care & storage

- Rinse and dry the lactometer after use.
- Store it in a protective case.
- Check scale visibility and instrument condition regularly.
- Keep the cylinder clean and free from residue.

Common mistakes to avoid

- Reading before the lactometer settles.
- Allowing the instrument to touch the cylinder.
- Ignoring sample temperature when correction is required.
- Using a foamy or unrepresentative sample.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Liquid Nitrogen

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Liquid nitrogen; Liquid nitrogen and semen straw

What it is used for

Liquid nitrogen is used in approved livestock breeding and laboratory procedures, including long-term cryogenic storage of semen straws. Safe handling requires trained personnel, suitable cryogenic PPE, proper ventilation, and controlled storage because liquid nitrogen can cause severe cold burns and displace oxygen.

Main parts

- Insulated cryogenic vessel stores liquid nitrogen at very low temperature.
- Lid or neck plug reduces heat entry while allowing safe venting.
- Storage canister holds semen-storage goblets or racks where fitted.
- Canister rack organises semen straws within the vessel.
- Semen straw contains the preserved semen sample.
- Cryogenic gloves and face shield protect against splashes and cold contact.
- Liquid-level measuring device checks the nitrogen level where provided.
- Tongs or approved retrieval tools help handle frozen straws.

Before you start

1. Only trained and authorised personnel should handle liquid nitrogen or retrieve semen straws.
2. Review the vessel instructions, semen-storage procedure, and emergency response plan.
3. Work in a well-ventilated area and ensure oxygen-deficiency monitoring is available where required by the risk assessment.
4. Inspect the cryogenic vessel, lid, canisters, racks, labels, and handling tools for damage.
5. Wear a face shield over safety glasses, loose-fitting cryogenic gloves, protective clothing, and closed footwear as specified by the procedure.
6. Confirm the semen straw identification, storage location, and required retrieval plan before opening the vessel.
7. Keep the vessel upright, stable, and secured against tipping.

How to use

1. Move the cryogenic vessel using the approved handling method and suitable transport equipment.
2. Check the nitrogen level using the specified method without leaning over the vessel opening.
3. Open the vessel carefully and keep the face and body away from escaping cold vapour.
4. Identify the correct canister and semen straw using the storage record.
5. Raise the canister only as far as necessary and keep other stored material submerged as required by the procedure.

6. Use approved cryogenic tongs to retrieve the required straw without touching it with bare skin.
7. Transfer the straw promptly using the approved thawing procedure for that semen product.
8. Return the canister to its correct storage position and close the vessel according to its instructions.
9. Check the nitrogen level after retrieval and replenish only under the approved procedure when required.
10. Record the straw identification, retrieval time, storage status, and any vessel or sample concerns.

⚠ Safety

- Liquid nitrogen can cause severe cold burns and eye injuries; avoid all direct contact.
- Never seal liquid nitrogen in an airtight container because pressure can build up dangerously.
- Nitrogen gas can displace oxygen; never handle or store it in a poorly ventilated area.
- Never lean over the vessel or look directly into its opening.
- Do not use ordinary gloves as a substitute for suitable cryogenic gloves and face protection.
- Do not use a vessel that is damaged, leaking, or showing abnormal frost or pressure behaviour; follow the site emergency plan.

Care & storage

- Keep the vessel upright, clean externally, and in a designated ventilated storage area.
- Inspect the vessel, neck, lid, canisters, and racks regularly for damage or abnormal frost patterns.
- Monitor and record liquid nitrogen levels at the required interval.
- Use only approved methods for filling, moving, and maintaining the vessel.
- Report suspected vacuum failure, unusual nitrogen loss, damaged labels, or sample-storage discrepancies.

Common mistakes to avoid

- Handling liquid nitrogen without suitable cryogenic PPE and training.
- Allowing stored semen straws to warm during unnecessary canister handling.
- Closing liquid nitrogen inside an airtight container.
- Ignoring low nitrogen levels, abnormal frost, or inadequate room ventilation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Milk Can

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Low** 4 labs

Covers these lab items: Milk can; Milk Cans

What it is used for

A milk can stores and transports milk while helping maintain cleanliness and prevent contamination. Use a food-grade can with a secure lid and follow the dairy's milk-handling and temperature-control procedures.

How to use

1. Inspect the can, lid, handles, and interior for cracks, dents, rust, or residue.
2. Wash and sanitize all milk-contact surfaces before use.
3. Confirm the can is clean, dry, and suitable for food contact.
4. Collect milk using hygienic handling practices and avoid touching the inner surfaces.
5. Fill only to the recommended level and secure the lid properly.
6. Label or identify the can as required and transport it promptly under suitable temperature conditions.
7. Empty, wash, sanitize, and dry the can after use.

⚠ Safety

- Do not use rusty, damaged, or non-food-grade containers for milk.
- Prevent contact with dirty hands, surfaces, and contaminated utensils.
- Follow required milk cooling and storage temperatures.
- Lift filled cans with safe technique and assistance when necessary.

Care & storage

- Clean promptly after emptying and store the can with its lid protected from contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Milk Collection Accessories

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Low** 1 labs

Covers these lab items: Milk collection and handling accessories

What it is used for

Milk collection and handling accessories support hygienic transfer, filtering, measuring, and storage of milk. Use food-grade accessories and follow approved dairy hygiene and temperature-control procedures.

How to use

1. Identify the accessories required for collection, transfer, filtering, or storage.
2. Inspect containers, lids, tubing, strainers, connectors, and seals for damage or residue.
3. Wash and sanitize all milk-contact components before use.
4. Assemble the accessories securely according to their intended function.
5. Collect and transfer milk hygienically while minimizing exposure to dust and contaminants.
6. Maintain required milk temperature and label containers according to the dairy procedure.
7. Disassemble, clean, sanitize, dry, and store reusable accessories after use.

⚠ Safety

- Use only food-grade materials intended for milk contact.
- Prevent contamination from dirty hands, surfaces, animals, or non-food equipment.
- Follow required cooling temperatures and milk-handling time limits.
- Dispose of damaged or contaminated components that cannot be cleaned effectively.

Care & storage

- Inspect seals and tubing regularly and store clean accessories in a covered, hygienic area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Milk Filter Machine

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Milk Filter machine

What it is used for

To remove visible suspended particles from raw milk using a milk filter machine before further dairy processing. To maintain hygienic milk handling and ensure that the filter is correctly installed and replaced when needed.

Main parts

- Machine body and milk inlet
- Filter housing and filter element
- Milk outlet and collection vessel
- Pump or gravity-feed arrangement, if fitted
- Valves and flow controls
- Cover, seals, and support frame

Before you start

1. Confirm that the filter machine is designed for the intended milk flow and operating conditions.
2. Clean and sanitize all milk-contact surfaces using the approved dairy hygiene procedure.
3. Inspect the filter housing, seals, valves, pipes, and collection vessel for damage or residue.
4. Install the correct food-grade filter element and check that it is seated properly.
5. Ensure that milk containers are clean, covered where appropriate, and ready for use.
6. Check that valves are positioned correctly and that the outlet path is clear.

How to use

1. Keep the milk supply isolated while opening the filter housing or installing the filter element.
2. Fit the filter element in the correct direction and close the housing securely.
3. Position a clean collection vessel under the outlet or connect the approved downstream line.
4. Transfer raw milk into the inlet vessel using hygienic handling practices.
5. Open the inlet valve gradually or start the pump according to the machine instructions.
6. Monitor milk flow and inspect the filter housing and connections for leakage.
7. Check the outlet milk and filter condition during operation without opening a pressurized system.
8. Stop the flow if it becomes unusually slow, the filter blocks, or leakage occurs.
9. Isolate the supply and release pressure according to the approved procedure before opening the housing.
10. Remove the used filter element carefully and discard or clean it according to the filter type and food-safety procedure.

1. Clean and sanitize the housing, seals, pipes, and collection vessel after processing.
2. Record filter replacement, cleaning, and any abnormal observations.

⚠ Safety

- Use only food-grade filter elements and materials suitable for milk contact.
- Maintain cold-chain and milk-handling requirements specified by the dairy process.
- Never open a pressurized filter housing or disconnect a line while it is under pressure.
- Keep hands and tools clean and prevent contamination of filtered milk.
- Do not continue operation with damaged seals, cracked housings, or leaking connections.
- Follow the approved procedure for handling and disposing of used filters and contaminated residues.

Care & storage

- Clean and sanitize all milk-contact components after each processing session or batch as required.
- Inspect filter elements, seals, clamps, and valves regularly.
- Replace disposable filter elements at the specified interval or when blocked or damaged.
- Check pumps and flow controls where fitted and maintain them according to the manual.
- Store clean, dry parts in a protected hygienic area.

Common mistakes to avoid

- Installing the filter element in the wrong direction or leaving it poorly seated.
- Using a non-food-grade or damaged filter element.
- Opening the housing while it is pressurized.
- Allowing dirty tools or containers to contaminate filtered milk.
- Continuing to use a clogged filter and ignoring reduced flow or leakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Milk Storage Tank

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Milk storage tanks (50 liter)

What it is used for

A milk storage tank temporarily holds milk under controlled hygienic conditions before collection or processing. Clean and sanitize the tank correctly and maintain the required storage temperature.

How to use

1. Inspect the tank body, lid, outlet valve, seals, and temperature display for damage.
2. Clean and sanitize the tank using the approved dairy sanitation procedure.
3. Rinse if required by the cleaning chemical instructions and allow the tank to drain properly.
4. Verify that the tank is dry or prepared as specified and the cooling system is operational.
5. Transfer milk using clean, food-grade equipment and close the lid securely.
6. Monitor and record milk temperature and storage time according to dairy requirements.
7. Drain the tank after transfer and repeat the cleaning procedure before the next batch.

⚠ Safety

- Use only food-grade materials and approved cleaning or sanitizing chemicals.
- Follow chemical labels and wear the required protective equipment.
- Never enter the tank unless a formal confined-space procedure and authorization apply.
- Do not store milk outside the required temperature limits.

Care & storage

- Clean and sanitize after each batch and inspect valves, seals, and cooling equipment regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Milk Strainer

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Low** 4 labs

Covers these lab items: Milk strainer

What it is used for

A milk strainer removes visible particles from milk during hygienic collection and handling. Use a clean, food-grade strainer and follow the dairy's approved filtration and milk storage procedure.

How to use

1. Inspect the strainer body, mesh, handle, and joints for damage or residue.
2. Wash and sanitize the strainer before use.
3. Place it securely over a clean, suitable milk collection vessel.
4. Pour milk slowly through the mesh without splashing or overflowing.
5. Check the filtered milk and remove accumulated residue safely.
6. Transfer the milk to the required storage container and maintain the specified temperature.
7. Clean, sanitize, and dry the strainer immediately after use.

⚠ Safety

- Do not use torn mesh, rusted parts, or contaminated strainers.
- Prevent hands and non-food objects from contacting the milk or strainer interior.
- Do not treat straining as a substitute for pasteurization or other required milk safety controls.
- Handle hot sanitizing water and cleaning chemicals carefully.

Care & storage

- Clean mesh openings thoroughly and store the strainer in a dry, hygienic location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Milk Testing Kit

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: School-level Milk Testing Kit

What it is used for

A school-level milk testing kit supports basic checks of milk quality using approved test methods. Follow the kit instructions and record results without treating screening tests as a substitute for accredited laboratory analysis.

How to use

1. Check the kit contents, reagent labels, expiry dates, and test instructions.
2. Wear the specified gloves, eye protection, and other required PPE.
3. Collect a representative milk sample in a clean, labelled container.
4. Prepare the sample and reagents according to the kit's instructions.
5. Perform each selected test using the specified quantities, timing, and procedure.
6. Compare the results with the kit's interpretation guide and record the observations.
7. Dispose of used reagents and samples according to the safety instructions.

⚠ Safety

- Do not taste or consume samples used for testing.
- Handle reagents according to their safety labels and avoid skin or eye contact.
- Do not mix reagents from different tests unless the instructions permit it.
- Keep chemicals away from students unless supervised by an authorized instructor.

Care & storage

- Clean reusable equipment and store reagents as labelled, observing temperature and expiry requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Milking Bucket

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Low** 3 labs

Covers these lab items: Milking bucket

What it is used for

A milking bucket collects milk during manual milking or approved dairy training activities. Maintain strict hygiene and use a clean, food-grade bucket to reduce contamination risk.

How to use

1. Inspect the bucket, handle, rim, and interior for damage, rust, or residue.
2. Wash and sanitize the bucket before milking.
3. Prepare a clean milking area and follow the required udder and teat hygiene procedure.
4. Position the bucket securely beneath the animal without touching contaminated surfaces.
5. Collect milk using hygienic milking technique and avoid introducing dirt or hair.
6. Transfer milk promptly to the approved storage container and maintain the required temperature.
7. Wash, sanitize, and dry the bucket immediately after use.

⚠ Safety

- Use only clean, food-grade buckets for milk collection.
- Keep the bucket away from manure, dirty bedding, and contaminated equipment.
- Handle animals calmly and follow safe milking practices.
- Do not use buckets with cracks, rust, or residues that cannot be removed.

Care & storage

- Clean promptly after milking and store inverted or covered in a clean, dry area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mouth Gag

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Mouth gags

What it is used for

To assist trained veterinary personnel in safely opening or maintaining an animal's mouth during an approved examination or procedure.

How to handle

- Use only under the direction of a veterinarian or appropriately trained animal-care professional.
- Inspect the device for cracks, sharp edges, damaged hinges, and cleanliness before use.
- Restrain the animal using an approved method and select a device suitable for the species and animal size.
- Insert and position the gag gently according to veterinary instructions; never force the mouth open or obstruct breathing.
- Monitor the animal continuously, remove the device promptly when directed, and stop if distress or injury occurs.

Storage

- Clean and disinfect according to veterinary infection-control procedures, then dry completely.
- Store in a labelled, secure veterinary instrument cabinet away from unauthorised users.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Multipurpose Vat

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Multipurpose Stainless Steel Vat

What it is used for

A multipurpose stainless steel vat holds, mixes or processes liquids and food materials in dairy operations. Its use depends on the design and whether heating, cooling or agitation is provided.

Main parts

- Identify the stainless steel vessel and its intended working capacity.
- Locate the inlet, outlet valve and drain arrangement, where fitted.
- Identify the lid or cover and its securing arrangement.
- Locate the support frame, legs and any lifting handles.
- Identify the agitator, motor or mixing shaft, if provided.
- Locate temperature indicators, heating or cooling connections, where fitted.
- Identify seals, gaskets and other removable food-contact components.

Before you start

1. Confirm the vat is suitable for the intended food or dairy process.
2. Wash and sanitise all food-contact surfaces using approved procedures.
3. Inspect the vessel, welds, valves, seals and supports for damage or leaks.
4. Ensure the vat is stable and its outlet valve is closed before filling.
5. Check that heating, cooling or agitation systems are correctly connected, if present.
6. Wear clean protective clothing and prepare suitable heat-resistant PPE when needed.

How to use

1. Place the vat on a stable, level surface with adequate working clearance.
2. Confirm that the drain and outlet valve are correctly closed.
3. Add the required material without exceeding the marked or recommended capacity.
4. Secure the lid where required while allowing designed vents to remain unobstructed.
5. Operate the mixing, heating or cooling system only if fitted and required.
6. Monitor the contents and temperature using the provided controls or instruments.
7. Avoid sudden filling, splashing or mixing speeds that cause overflow.
8. Stop agitation and isolate power before opening or emptying where required.
9. Drain or transfer the contents through the designated outlet into a suitable container.
10. Clean and sanitise the vat promptly after the process is complete.

⚠ Safety

- Never exceed the rated capacity or operating temperature of the vat.
- Avoid contact with hot surfaces, steam and heated liquids.
- Keep hands away from agitators and mixing shafts while operating.
- Isolate electrical and heating supplies before maintenance or internal cleaning.
- Do not enter the vat or lean inside while equipment is operating.
- Use suitable lifting assistance when moving heavy or filled vessels.

Care & storage

- Wash away product residue and sanitise food-contact surfaces after use.
- Clean valves, seals and outlet fittings without damaging their surfaces.
- Inspect welds, gaskets, supports and connections regularly for leaks or wear.
- Store the clean vat dry with its lid protected from contamination.

Common mistakes to avoid

- Filling beyond capacity and causing spills during mixing or heating.
- Using a vat with damaged seals, valves or unstable supports.
- Operating an agitator while hands or cleaning tools are inside.
- Assuming every vat is designed for heating, pressure or cooling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Muzzle

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Low** 3 labs

Covers these lab items: Muzzle

What it is used for

An animal muzzle helps reduce biting or unwanted feeding during approved handling and veterinary care. Select a properly fitted muzzle for the species and use it only when necessary under appropriate supervision.

How to use

1. Confirm that the muzzle type is suitable for the animal's species, size, and intended purpose.
2. Inspect straps, fasteners, and material for damage or wear.
3. Introduce the muzzle calmly and allow the animal to become accustomed to it when possible.
4. Fit the muzzle according to manufacturer or veterinary instructions without excessive tightness.
5. Check that breathing remains unobstructed and monitor the animal continuously.
6. Remove the muzzle promptly when the supervised task is complete or distress occurs.
7. Clean and store the muzzle after use.

⚠ Safety

- Use a muzzle only under appropriate supervision and for a limited duration.
- Never obstruct breathing, panting, vomiting, or essential cooling functions.
- Do not leave a muzzled animal unattended or use a muzzle on an unsuitable animal without expert guidance.
- Stop immediately if the animal shows breathing difficulty or severe distress.

Care & storage

- Clean and disinfect the muzzle as appropriate and inspect straps before each use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nest Box

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Low** 2 labs

Covers these lab items: Laying Boxes; Nest Boxes

What it is used for

Nest boxes provide laying or nesting spaces for suitable poultry or other managed animals. Place the boxes in a quiet, accessible location and maintain clean bedding and appropriate hygiene.

How to use

1. Inspect the nest box, supports, edges, and access opening for damage.
2. Confirm the box size and design suit the intended species.
3. Install the box securely in a quiet location with suitable access.
4. Add clean, dry bedding appropriate for the animal and management system.
5. Observe whether animals use the box comfortably and check for eggs or nesting material as required.
6. Remove soiled bedding and clean the box according to the husbandry schedule.
7. Record damage or unusual animal behaviour and repair or replace the box as needed.

⚠ Safety

- Avoid sharp edges, unstable mounting, and openings that may trap animals.
- Use appropriate hygiene practices when handling eggs, bedding, or animal waste.
- Keep the nest area dry and protect it from predators and environmental hazards.
- Follow species-specific welfare and stocking recommendations.

Care & storage

- Replace wet or soiled bedding promptly and disinfect the box when required by the animal-care plan.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paneer Press

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Paneer Press

What it is used for

A paneer press removes excess whey from coagulated milk solids to form paneer with a suitable texture. Maintain food hygiene and use the press according to its design and batch requirements.

How to use

1. Inspect the press frame, plates, handles, fasteners, and drainage outlets for damage.
2. Clean and sanitize all food-contact surfaces before use.
3. Place the prepared curd in a clean, food-grade cloth or approved mould.
4. Position the curd evenly inside the press and fit the top plate.
5. Apply pressure gradually according to the product and equipment instructions.
6. Allow whey to drain into a suitable clean container without contaminating the paneer.
7. Release the pressure safely, remove the paneer, and clean the press after use.

⚠ Safety

- Keep fingers clear of moving plates and pressure points.
- Do not exceed the press's rated load or force.
- Use food-grade materials and prevent contact with contaminated surfaces.
- Release pressure fully before dismantling or cleaning the press.

Care & storage

- Wash, sanitize, and dry the press after each batch and inspect fasteners regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Poultry Drinker

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Linear Waterer/Channel Type Waterer; Manual Drinker; Nipple Drinker; Pan and Jar Type Waterer

What it is used for

To supply clean drinking water to poultry using a suitable drinker system. To maintain reliable water access and reduce spillage and contamination.

Main parts

- Water container or supply line
- Drinker channel, nipple, pan, or jar
- Valve or float mechanism, if fitted
- Mounting supports
- Water filter, if fitted

Before you start

1. Select a drinker suitable for the birds' age and housing system.
2. Wash and disinfect drinker components according to farm procedures.
3. Check the water supply, valves, and fittings for leaks or blockage.
4. Position the drinker at an appropriate height for the birds.
5. Ensure the water source is clean and accessible.

How to use

1. Fill the container or connect the supply line securely.
2. Open the supply valve gradually if applicable.
3. Check that water reaches all drinker points.
4. Adjust height and flow according to the birds' needs and system instructions.
5. Observe birds using the drinker and confirm water is available.
6. Remove spilled feed or litter from the water area.
7. Clean and refill the drinker on the farm's schedule.

⚠ Safety

- Use clean water suitable for poultry.
- Avoid sharp edges and unstable mounting.
- Prevent electrical equipment from contacting spilled water.
- Follow farm biosecurity and sanitation procedures.

Care & storage

- Wash and disinfect the drinker regularly.
- Check nipples, valves, and seals for leaks or blockages.
- Remove algae and mineral deposits as appropriate.
- Replace cracked or damaged parts.

Common mistakes to avoid

- Setting the drinker too high or too low.
- Allowing water to become contaminated.
- Ignoring blocked nipples or leaking valves.
- Failing to clean drinkers routinely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Poultry Feeder

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Circular Feeder; Linear Feeder

What it is used for

To provide poultry birds with feed in a clean, accessible, and controlled manner using a circular or linear feeder.

How to handle

- Inspect the feeder body, edges, supports, and hanging or mounting parts for cracks, sharp edges, or instability.
- Clean and dry the feeder before filling and use feed appropriate for the birds' age and type.
- Position the feeder at a suitable height and spacing so birds can access feed without crowding or excessive spillage.
- Fill to the recommended level and monitor feed condition, moisture, pests, and contamination regularly.
- Remove spoiled feed and clean the feeder on the farm's hygiene schedule; avoid placing it where litter or waste can contaminate feed.

Storage

- Empty, wash, and dry the feeder before long-term storage.
- Store in a clean, dry, pest-protected area away from chemicals and animal waste.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Semen Straw

Category: **Animal Husbandry & Dairy** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Semen straw

What it is used for

Maintain semen straw identification, hygiene, and temperature control in animal breeding work.

How to handle

- Only trained personnel should handle semen straws using the approved procedure.
- Maintain the required storage temperature and minimise exposure during transfer.
- Use clean, appropriate tools and follow biosecurity precautions.
- Verify the label and animal identification before use; never use an unidentified or damaged straw.

Storage

- Store in the designated cryogenic or temperature-controlled system according to supplier instructions.
- Maintain inventory and handling records; follow approved procedures for disposal.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Teat Dip

Category: **Animal Husbandry & Dairy** Type: **Handling note** Priority: **Medium** 3 labs

Covers these lab items: Post milking teat dip

What it is used for

Teat dip is applied to dairy animals' teats after milking to support teat hygiene and reduce infection risk.

How to handle

- Confirm the product is intended for post-milking teat disinfection and check its label instructions.
- Use a clean, dedicated applicator and apply the product at the specified concentration and coverage.
- Avoid contaminating the product container and do not mix products unless explicitly permitted.
- Observe the recommended contact time and any milk or animal-use precautions on the label.

Storage

- Keep in the original labelled container, tightly closed and protected from contamination.
- Follow the manufacturer's temperature and shelf-life requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Trocar and Cannula

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Trocar and cannula

What it is used for

A trocar and cannula assembly is a veterinary instrument used for controlled access to a body cavity in specific clinical situations, such as relieving ruminal bloat in cattle when indicated. It must be used only by trained veterinary professionals because incorrect placement can cause severe injury, infection, leakage, or death.

Main parts

- Trocar is the pointed instrument used to penetrate tissue.
- Cannula is the hollow sleeve that remains in the access opening.
- Handle provides grip and control during insertion.
- Cannula opening allows gas or fluid to pass where indicated.
- Cap or valve controls the opening where fitted.
- Reusable models may include a locking mechanism to retain the trocar.
- Compatible cleaning and sterilisation accessories support reprocessing where permitted.

Before you start

1. Only a qualified veterinarian or specifically trained personnel acting under veterinary direction should perform the procedure.
2. Confirm the clinical indication, animal identity, condition, and appropriate treatment plan before use.
3. Inspect the trocar, cannula, handle, locking mechanism, and any cap for damage, corrosion, or obstruction.
4. Confirm the instrument is the correct size and design for the intended animal and procedure.
5. Verify that reusable equipment has been cleaned and sterilised according to validated instructions, or that the single-use device is sealed and in date.
6. Prepare the approved site-preparation supplies, PPE, restraint equipment, and emergency support required by the veterinary protocol.
7. Use appropriate animal restraint and ensure trained assistance is available to protect the animal and staff.

How to use

1. Explain the planned procedure to the animal owner or responsible handler and obtain required consent.
2. Assess the animal and confirm that the procedure is appropriate under the veterinary treatment plan.

3. Position and restrain the animal using the approved method while monitoring distress and breathing.
4. Prepare the intended site using the veterinary protocol for cleaning, antisepsis, and any indicated local anaesthesia.
5. Open sterile or properly reprocessed equipment without contaminating the working parts.
6. Verify the trocar fits the cannula and that the assembly operates correctly before use.
7. The qualified veterinarian should locate the correct access site and perform insertion using the established clinical technique.
8. Once access is established, manage the trocar and cannula according to the procedure and veterinary protocol.
9. Monitor the animal closely for worsening distress, bleeding, abnormal discharge, or other complications.
10. Remove the instrument or cannula only when clinically appropriate and according to the veterinarian's plan.
11. Provide the prescribed aftercare, observe the animal, and arrange follow-up or further treatment as required.
12. Dispose of single-use instruments and contaminated materials in designated waste streams; reprocess reusable instruments according to validated procedures.
13. Record the indication, procedure, instrument details where required, animal response, and follow-up plan.

⚠ Safety

- Do not use the instrument without veterinary training, appropriate clinical indication, and suitable restraint.
- Incorrect placement can damage internal organs, blood vessels, or other tissues and may cause life-threatening complications.
- Maintain aseptic technique and use suitable PPE to reduce zoonotic and wound infection risks.
- Do not force the trocar if resistance, unexpected pain, bleeding, or an abnormal response occurs; stop and reassess.
- Keep sharp tips protected during handling, transport, cleaning, and storage.
- Follow local veterinary rules for animal welfare, medicines, infection control, and clinical waste disposal.

Care & storage

- Immediately after use, handle contaminated sharps carefully and place them in a secure designated container for transport to reprocessing.
- Clean reusable instruments promptly according to the manufacturer's instructions so organic material does not dry on them.
- Inspect the point, cannula lumen, handle, and locking mechanism for damage, blockage, or corrosion.
- Sterilise reusable instruments using a validated method compatible with the device.

- Store clean, dry, processed instruments in protective packaging and remove damaged items from service.

Common mistakes to avoid

- Using the trocar and cannula without confirming the clinical indication and correct placement site.
- Using inadequately cleaned or non-sterile equipment for an invasive procedure.
- Forcing insertion despite resistance or signs of complications.
- Failing to monitor the animal or provide appropriate aftercare and follow-up.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vaccinator

Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Automatic Vaccinator

What it is used for

An automatic vaccinator delivers a measured vaccine dose for approved animal vaccination procedures. Use only the prescribed vaccine, confirm the device setting, and follow veterinary guidance and vaccine handling requirements.

How to use

1. Verify the animal species, vaccine identity, expiry date, storage condition, and prescribed dose.
2. Inspect the vaccinator, needle, dose setting, seals, and connections for damage or contamination.
3. Perform hand hygiene and wear appropriate protective equipment.
4. Prepare the vaccine according to its label and load the device using the approved method.
5. Set and verify the prescribed dose before administering the vaccine at the approved site and route.
6. Monitor the animal after vaccination and record the vaccine details, dose, date, and animal identification.
7. Dispose of used needles safely and clean or disinfect reusable parts according to instructions.

⚠ Safety

- Vaccines must be administered under veterinary direction and according to label instructions.
- Never use contaminated, expired, or improperly stored vaccine.
- Do not recap used needles by hand; use an approved sharps container.
- Stop use if the device leaks, misdoses, or malfunctions.

Care & storage

- Clean and maintain the vaccinator as instructed and store vaccines within their required temperature range.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Audio/Video Recorder

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Video and audio recorder; Video and audio recorders

What it is used for

An audio/video recorder captures sound, video, or both for documentation, training, interviews, and media production. Check recording settings and available storage before use.

How to use

1. Inspect the recorder, microphone, lens, cables, and battery or power supply.
2. Insert a compatible memory card or confirm available internal storage.
3. Set the date, time, recording format, resolution, and audio level as needed.
4. Position the recorder securely and check the framing and sound.
5. Make a short test recording and play it back to verify quality.
6. Record the session, save the files, and switch off the device after use.

⚠ Safety

- Obtain permission before recording people or private spaces.
- Keep equipment and power connections away from water and excessive heat.
- Use stable mounts and secure cables to prevent falls or trips.
- Do not remove storage media while recording or transferring files.

Care & storage

- Clean the lens and microphone gently, recharge the battery, and store the recorder safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Binoculars

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Binoculars, magnifying glasses, and field microscopes for plant and animal observation; Larger binoculars or spotting scopes for wildlife viewing; Professional binoculars

What it is used for

To observe distant objects, wildlife, or field details using binoculars. To obtain a clear image through correct focusing, stable handling, and suitable eye adjustment.

Main parts

- Objective lenses
- Eyepieces
- Focus wheel
- Diopter adjustment
- Hinged bridge
- Neck strap or protective case

Before you start

1. Inspect the lenses, body, focus wheel, and strap for damage.
2. Clean the lenses with a suitable lens cloth.
3. Adjust the binocular hinge to match the distance between the eyes.
4. Check the viewing area for safe footing and suitable visibility.
5. Use a tripod adapter if required for heavier binoculars or extended viewing.

How to use

1. Hold the binoculars securely using both hands or the neck strap.
2. Look through the eyepieces and adjust the hinge until both fields merge.
3. Close one eye and focus using the central focus wheel as instructed.
4. Adjust the diopter for the other eye if the model provides this feature.
5. Look through both eyepieces and fine-tune focus on the subject.
6. Keep the binoculars steady and avoid sudden movements.
7. Lower the binoculars when moving through uneven terrain.
8. Fit the lens caps and store the binoculars after use.

⚠ Safety

- Never look directly at the sun through binoculars.
- Avoid walking while looking through the eyepieces.
- Keep lenses away from excessive heat, moisture, and dust unless the model is rated for those conditions.
- Use a strap to reduce the risk of dropping the instrument.

Care & storage

- Clean lenses only with suitable optical cleaning materials.
- Keep lens caps fitted when not in use.
- Store dry in a protective case.
- Avoid touching lens surfaces with fingers.

Common mistakes to avoid

- Focusing without adjusting the hinge for eye spacing.
- Using excessive force on the focus wheel.
- Cleaning lenses with rough cloth or paper.
- Looking at the sun through the binoculars.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bluetooth Device

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Bluetooth

What it is used for

A Bluetooth device connects compatible electronics wirelessly over a short distance. Pair devices securely and check that the connection suits the intended activity.

How to use

1. Check that both devices support compatible Bluetooth functions.
2. Switch on Bluetooth on the required devices and keep them nearby.
3. Open the Bluetooth settings and select the correct device name.
4. Confirm the pairing request or code on both devices when prompted.
5. Test the connection by playing audio or transferring an approved file.
6. Disconnect or unpair devices when the activity is finished.

⚠ Safety

- Pair only with trusted devices and avoid unknown pairing requests.
- Do not transfer confidential information over an untrusted connection.
- Keep device volume at a safe level when using headphones or speakers.

Care & storage

- Keep devices charged, dry, and updated, and remove old pairings when no longer needed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

CCTV Camera

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 28 labs

Covers these lab items: Cameras / CCTV; CCTV Cameras; Different types of CCTV cameras; Dome Camera; Pan Tilt and Zoom (PTZ) Camera; Security cameras

What it is used for

A CCTV camera captures video for security, monitoring, or observation through a connected recording system. Use the camera according to site privacy rules and the approved installation plan.

How to use

1. Inspect the camera housing, lens, cable, mount, and connectors for damage.
2. Confirm the camera location is authorised and appropriate for the monitoring purpose.
3. Mount the camera securely and adjust its direction to cover the intended area.
4. Connect power and network or video cables according to the system design.
5. Configure the camera using authorised settings and secure access credentials.
6. Check the live view, image clarity, date, time, and recording status.
7. Verify that the camera covers the intended area without unnecessary obstruction.
8. Report faults and safely disconnect power before authorised maintenance.

⚠ Safety

- Respect privacy laws and do not monitor private areas without proper authorisation.
- Use suitable mounting and fall protection when working at height.
- Do not open powered equipment or touch exposed electrical connections.
- Protect passwords and restrict access to authorised personnel.

Care & storage

- Clean the lens and housing using suitable materials without scratching the lens.
- Inspect mounts, cables, image quality, and recording operation regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

CCTV Recorder

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Digital Video Recorder; DVRs, NVRs, storage devices, and monitors

What it is used for

To record, store, and review CCTV video from compatible cameras using a DVR or NVR system.

How to handle

- Check the recorder, storage drive, cables, monitor, and ventilation for damage or obstruction.
- Connect compatible cameras and network or display cables according to the system manual; switch off equipment before changing connections where required.
- Configure date, time, recording schedule, storage, and authorized user access.
- Verify live view and playback, and confirm that recording and time settings are correct.
- Access footage only for authorized purposes and follow applicable privacy, retention, and security policies.

Storage

- Keep the recorder in a ventilated, secure, dust-controlled location and avoid blocking vents.
- Store configuration records securely; protect accounts with strong credentials and authorized access controls.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Camera Trap

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Camera traps for observing wildlife without human interference

What it is used for

A camera trap records images or video when its motion or heat sensor detects activity. Install it responsibly to observe wildlife while minimizing disturbance and protecting privacy.

How to use

1. Inspect the camera housing, lens, sensor, batteries, and mounting strap.
2. Choose a suitable location based on the observation purpose and permitted access.
3. Mount the camera securely at an appropriate height and angle.
4. Set date, time, sensor sensitivity, and recording mode according to the instructions.
5. Test detection and framing before leaving the camera unattended.
6. Retrieve the camera and review recordings at the planned interval.
7. Switch off the device before changing batteries or removing the memory card.

⚠ Safety

- Do not approach dangerous wildlife or disturb nests and breeding areas.
- Obtain permission before placing cameras on private or restricted land.
- Avoid recording people in private spaces and follow local privacy rules.
- Keep batteries and small parts away from children and animals.

Care & storage

- Clean the lens and sensor gently and store the camera dry with batteries handled as instructed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Digital Camera

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Camera; Camera or Smartphone; Digital Camera; Digital Camera (for documentation)

What it is used for

To capture photographs or videos for documentation, inspection, or media production using a digital camera. To obtain clear images while protecting the equipment and respecting privacy requirements.

Main parts

- Camera body
- Lens
- Battery and memory card
- Shutter and control buttons
- Display or viewfinder
- USB or charging connection

Before you start

1. Check battery charge and available memory.
2. Inspect the lens, screen, body, and memory card for damage.
3. Clean the lens with a suitable lens cloth.
4. Confirm the required image format, resolution, and storage settings.
5. Obtain permission before photographing people or restricted areas.

How to use

1. Insert the charged battery and compatible memory card.
2. Switch on the camera and select photo or video mode.
3. Set focus, exposure, and zoom as needed.
4. Frame the subject and hold the camera steadily.
5. Take the photograph or start recording.
6. Review the image or clip and repeat if necessary.
7. Transfer files using the approved cable, card reader, or wireless method.
8. Back up important files and safely eject the memory card or disconnect the camera.

⚠ Safety

- Do not photograph confidential information without authorization.
- Keep the camera away from water, dust, and excessive heat unless rated for those conditions.
- Use straps or a secure grip to prevent drops.
- Follow battery charging and storage instructions.

Care & storage

- Clean the lens and exterior with approved materials.
- Charge batteries with a compatible charger.
- Keep the lens cap fitted when not in use.
- Store the camera in a dry, protective case.

Common mistakes to avoid

- Shooting with a dirty lens or low battery.
- Using incorrect focus or exposure settings.
- Filling the memory card without checking storage.
- Sharing images containing private or confidential information.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Display Screen

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 21 labs

Covers these lab items: Digital screens displaying travel packages; LCD screens; Liquid Crystal Display screens

What it is used for

A display screen presents digital information, images, videos, or presentations for viewing. Use the correct input source and display settings to ensure clear content and reliable operation.

How to use

1. Inspect the screen, stand or mount, power cable, and connectors for damage.
2. Place or mount the screen securely with adequate ventilation.
3. Connect the approved computer, media player, or signal source.
4. Switch on the screen and select the correct input source.
5. Adjust brightness, contrast, resolution, and orientation as needed.
6. Check that text and images are clear and fully visible.
7. Switch off the screen after use and disconnect cables if required.
8. Keep the display area clear and secure the equipment after use.

⚠ Safety

- Do not operate the screen with damaged cables or a cracked panel.
- Keep liquids away from the screen and electrical connections.
- Secure wall mounts and stands to prevent falling or tipping.
- Do not block ventilation openings or open the screen casing while powered.

Care & storage

- Clean the screen with a soft cloth and manufacturer-approved cleaner.
- Keep vents clear and protect the display from impact and excessive heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Headphones

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Headphones

What it is used for

To listen to audio privately for training, communication, media review, or monitoring.

How to handle

- Inspect the earcups, headband, cable, plug, and controls for damage before use.
- Connect to a compatible audio device and start at a low volume.
- Adjust the fit comfortably and keep the volume at a safe listening level.
- Avoid sharing headphones without cleaning contact surfaces and do not use them where they prevent awareness of nearby hazards.
- Disconnect by holding the plug or connector rather than pulling the cable, then wipe the contact surfaces clean.

Storage

- Store in a clean, dry case or hanger without sharply bending the cable.
- Keep away from moisture, excessive heat, and heavy objects that could damage the earcups.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Infrared Camera

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Infrared Ray based Camera

What it is used for

An infrared camera detects infrared radiation to display thermal patterns or capture images in supported infrared bands. Configure the camera correctly and interpret readings according to its intended application.

How to use

1. Inspect the camera, lens, battery, display, and connectors for damage.
2. Charge or install the approved battery and switch on the camera.
3. Select the correct imaging mode, temperature range, and measurement settings for the task.
4. Focus the lens and position the camera at a suitable distance from the target.
5. Capture the image while keeping the camera steady and following the operating instructions.
6. Review the image and record relevant settings and observations.
7. Switch off the camera, protect the lens, and store the equipment safely.

⚠ Safety

- Do not point the camera directly at the sun unless it is specifically rated for that use.
- Follow laser safety precautions if the device includes a laser pointer.
- Do not touch hot targets or enter hazardous areas just to obtain an image.
- Remember that thermal readings depend on emissivity, distance, and environmental conditions.

Care & storage

- Clean the lens with approved materials and protect the camera from moisture, dust, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Microphone

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Microphone / voice system for lecture and class activities; USB Microphone

What it is used for

To capture and transmit speech using a microphone for lectures, classes, recordings, or presentations. To produce clear audio by positioning the microphone correctly and checking the connected sound system.

Main parts

- Microphone capsule
- Body or handle
- USB or audio connector
- Cable or wireless transmitter
- Power or mute control, if fitted
- Stand or mounting clip

Before you start

1. Check the microphone, cable, connector, and stand for damage.
2. Confirm compatibility with the computer, mixer, speaker, or recording device.
3. Charge the battery or connect USB or audio power as required.
4. Choose a suitable position away from loudspeakers to reduce feedback.
5. Check the recording or sound system input level.

How to use

1. Connect the microphone to the correct input or USB port.
2. Select the microphone as the input source on the connected device if required.
3. Position it at a suitable distance from the speaker's mouth.
4. Set the input gain to a low starting level.
5. Speak normally and check the level through headphones or the sound system.
6. Adjust gain and position until speech is clear without clipping or feedback.
7. Mute the microphone when it is not needed.
8. After use, switch it off and disconnect it safely.

⚠ Safety

- Keep connectors and electrical equipment dry.
- Avoid pointing microphones toward loudspeakers at high volume.
- Secure cables to prevent trips and connector damage.
- Use only compatible power supplies and batteries.

Care & storage

- Wipe the microphone with an approved cleaning method after use.
- Do not immerse the microphone or connector in liquid.
- Coil cables loosely and avoid sharp bends.
- Store in a clean, dry case.

Common mistakes to avoid

- Placing the microphone too far from the speaker.
- Setting gain too high and causing distortion or feedback.
- Using the wrong input or incompatible connector.
- Pulling the cable to disconnect the microphone.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Presentation Remote

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Presentation Remote

What it is used for

A presentation remote allows the presenter to advance slides and control compatible presentation functions from a distance. Check device compatibility and test the controls before the presentation.

How to use

1. Inspect the remote, buttons, battery compartment, and receiver if supplied.
2. Install the correct batteries or charge the device as instructed.
3. Connect the receiver or pair the remote with the presentation device.
4. Open the presentation and test slide advance, back, and pointer functions if available.
5. Stand within the specified operating range and control the slides during use.
6. Switch off the remote after the presentation and disconnect the receiver if required.
7. Store the remote and receiver together for the next use.

⚠ Safety

- Use only the specified battery type and correct polarity.
- Keep small batteries away from children and dispose of them properly.
- Avoid exposing the remote to liquids, excessive heat, or impact.
- Do not point a laser pointer into anyone's eyes.

Care & storage

- Wipe the remote with a dry or approved cloth and remove batteries for long-term storage when recommended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Projection Screen

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 30 labs

Covers these lab items: Projection Screen; Projector Screen

What it is used for

A projection screen provides a suitable surface for displaying images and presentations from a projector. Position and maintain the screen to support clear viewing and safe use.

How to use

1. Inspect the screen surface, frame, stand, and mounting points for damage.
2. Place or mount the screen securely with adequate clearance around it.
3. Position the screen to face the audience and align it with the projector.
4. Extend or lower the screen using its approved mechanism without forcing it.
5. Check image alignment, visibility, and screen stability before the presentation.
6. Keep the screen surface free from obstructions during use.
7. Retract or fold the screen carefully after the presentation.
8. Secure the screen and store it safely if it is portable.

⚠ Safety

- Secure wall mounts and stands to prevent tipping or falling.
- Keep fingers clear of retracting mechanisms and moving supports.
- Do not hang objects from the screen or its frame.
- Keep the screen away from heat, moisture, and walkways.

Care & storage

- Clean the screen using materials approved by the manufacturer.
- Retract or cover the screen when not in use to protect it from dust and damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Projector

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 83 labs

Covers these lab items: A/V Equipment (for presentations); LCD Projector; LCD Projector and Laptop for presentations; LCD projector and projection screen; LCD projector, projection screen, and audio system; Multimedia Projector; Multimedia projector and projector screen; Overhead/ Multimedia projector and Projector Screen;

What it is used for

A projector displays computer or media content on a screen for teaching, meetings, and presentations. Set it up with compatible connections and allow adequate ventilation during operation.

How to use

1. Inspect the projector, power cable, lens, remote control, and connection ports.
2. Place the projector on a stable surface or secure mounting bracket with clear ventilation.
3. Connect the computer or media source using a compatible cable.
4. Switch on the source device and projector, then select the correct input.
5. Adjust focus, image size, and alignment to fit the screen.
6. Check image clarity, brightness, sound, and presentation content before starting.
7. Use the normal shutdown procedure and allow the cooling cycle to finish.
8. Disconnect cables if required and store the projector after it cools.

⚠ Safety

- Never block ventilation openings or touch a hot projector lamp.
- Do not look directly into the projector lens while it is operating.
- Keep liquids away from the projector and its power connections.
- Disconnect power before cleaning or connecting equipment when instructed by the manufacturer.

Care & storage

- Clean the lens and air filter according to the manufacturer's instructions.
- Keep vents clear and store the projector in a dry, protected location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Projector Mounting Kit

Category: **Audio-Visual, Camera & Media** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Projector Mounting Kit

What it is used for

Handle projector mounting kits safely during installation and maintenance.

How to handle

- Verify the kit's compatibility with the projector and mounting surface before installation.
- Inspect brackets, fasteners, and load ratings for suitability and damage.
- Isolate power and use approved access equipment when installing overhead components.
- Secure all fittings according to manufacturer instructions and verify stability before use.

Storage

- Keep hardware grouped in labelled bags or compartments.
- Store the kit in a dry, secure location away from corrosion and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Public Address System

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Audio-visual aids; Effective sound system; Portable PA systems or megaphones for large group management; Public Address System

What it is used for

A public address system amplifies speech or audio for announcements, teaching, and group communication. Use suitable microphones, speakers, and volume settings for the venue.

How to use

1. Inspect the speakers, microphone, cables, connectors, and power supply.
2. Place speakers securely and position them to cover the intended listening area.
3. Connect equipment with power switched off and follow the system layout.
4. Switch on the system in the recommended order and start with low volume.
5. Test the microphone and adjust volume to achieve clear sound without feedback.
6. Switch off equipment, disconnect as instructed, and store cables neatly after use.

⚠ Safety

- Keep electrical connections dry and inspect cables for damaged insulation.
- Do not place speakers where they can fall or block walkways.
- Avoid excessive volume that may harm hearing or cause feedback.
- Turn power off before changing connections unless the equipment instructions state otherwise.

Care & storage

- Wipe equipment with a dry cloth and store microphones and cables safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Smart Board

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Smart Board; White Board/Smart Board

What it is used for

To present lessons, annotations, and digital content using a smart board. To ensure clear display, accurate touch input, and reliable connection to the teaching device.

Main parts

- Interactive display surface
- Power supply
- Computer or presentation device
- USB or touch connection
- HDMI or wireless display connection
- Stylus or touch controls

Before you start

1. Check the display, power cable, and connection ports for damage.
2. Confirm that the computer and smart board are compatible and connected.
3. Ensure the display surface is clean and free from obstructions.
4. Check the selected input source and required software.
5. Prepare presentation files and verify audio if needed.

How to use

1. Switch on the smart board and connected computer.
2. Select the correct input source.
3. Open the presentation or teaching application.
4. Test touch or stylus input before beginning.
5. Adjust display brightness and audio as required.
6. Present content and use annotation tools where appropriate.
7. Save any annotations or changes that need to be retained.
8. Shut down the software and equipment according to the manufacturer's instructions.

⚠ Safety

- Keep liquids away from electrical connections.
- Use only approved styluses and cleaning materials.
- Secure cables to prevent trips.
- Do not press the display with sharp or excessive force.

Care & storage

- Clean the surface using the approved method.
- Keep vents clear and avoid blocking cooling airflow.
- Inspect cables and styluses regularly.
- Switch off the display when not in use for extended periods.

Common mistakes to avoid

- Selecting the wrong input source.
- Using an unapproved marker or sharp tool on the screen.
- Forgetting to save annotations.
- Leaving cables loose across walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Speaker

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 18 labs

Covers these lab items: Bluetooth Speaker; Computer system; Laptop External Speakers; Portable speakers and microphones; Portable speakers for voice clarity during role-play or video demonstrations; Speaker; Speakers

What it is used for

A speaker converts audio signals into sound for presentations, training, communication, or entertainment. Connect it to a compatible source and set a suitable volume for the room.

How to use

1. Inspect the speaker, power cable, battery, ports, and controls for damage.
2. Place the speaker on a stable surface with ventilation and clear sound output.
3. Connect it to the compatible audio source using the approved cable or pairing method.
4. Switch on the speaker and select the correct input or Bluetooth mode.
5. Start audio at a low volume and gradually adjust to a clear listening level.
6. Check sound quality and microphone feedback if a microphone is connected.
7. Lower the volume and switch off the speaker after use.
8. Disconnect cables and store the speaker safely.

⚠ Safety

- Avoid excessive volume that may damage hearing.
- Keep liquids away from electrical speakers and charging ports.
- Do not use damaged batteries or power adapters.
- Keep cables arranged to prevent trips and accidental pulling.

Care & storage

- Clean the speaker exterior with a soft, dry cloth.
- Charge batteries with the approved adapter and store the unit in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Studio Lighting Kit

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Lighting Equipment kit

What it is used for

A studio lighting kit provides controlled illumination for photography, video recording, or presentations. Set up lights securely and adjust brightness and direction to suit the subject and recording environment.

How to use

1. Inspect lights, stands, power cables, connectors, modifiers, and mounting points.
2. Choose a stable location and position stands away from walkways.
3. Extend each stand carefully and secure its locks before mounting the light.
4. Attach the light and modifier using the supplied fittings.
5. Connect power safely and arrange cables to prevent tripping.
6. Switch on the lights and adjust brightness, angle, and diffusion as required.
7. Switch off and unplug equipment before dismantling and storing it.

⚠ Safety

- Use stands rated for the light's weight and secure them against tipping.
- Keep hot lights and ventilation openings clear of fabrics and other combustible materials.
- Do not use damaged cables or connectors.
- Protect electrical equipment from liquids and use appropriate power supplies.

Care & storage

- Allow hot fixtures to cool before packing and store cables loosely without sharp bends.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tripod

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Tripod

What it is used for

A tripod supports a camera, lighting device, or other compatible equipment for stable positioning. Adjust its legs and head securely before mounting equipment and confirm that the setup remains balanced.

How to use

1. Inspect the tripod legs, locks, feet, head, and mounting plate for damage.
2. Choose a level surface with enough space for all tripod legs.
3. Spread the legs fully and lock each leg section securely.
4. Adjust the height and level the tripod head as required.
5. Attach compatible equipment to the mounting plate and tighten the lock.
6. Check stability and frame the shot or position the equipment as needed.
7. Remove the equipment before folding the tripod for storage.

⚠ Safety

- Do not exceed the tripod's rated load.
- Keep legs clear of walkways and secure the setup against accidental tipping.
- Never loosen mounting locks while equipment is unsupported.
- Use extra caution outdoors in wind or on uneven ground.

Care & storage

- Wipe the legs and head clean, dry them, and store the tripod folded in a protective bag.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Turntable

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Turn table

What it is used for

A turntable rotates a record, display, or object at a controlled speed for playback or demonstration. Operate it according to its design and keep rotating parts unobstructed.

How to use

1. Place the turntable on a stable, level surface away from liquids.
2. Inspect the power cable, platter, controls, and accessories for damage.
3. Connect the device using the specified power supply, if required.
4. Position the record or object centrally and select the correct operating mode.
5. Start rotation and use the controls according to the instructions.
6. Stop the turntable before removing the record or adjusting components.
7. Switch off and disconnect the power supply when finished, if appropriate.

⚠ Safety

- Keep fingers, loose clothing, and cables away from moving parts.
- Do not operate the turntable with damaged wiring or exposed electrical components.
- Keep liquids away from electrical connections and the device.

Care & storage

- Clean the surface gently and cover or store the turntable to protect it from dust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

VR Headset

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: VR Standalone Headset

What it is used for

A standalone VR headset provides immersive virtual experiences for learning, simulation, and demonstrations. Check the device fit, play area, and software settings before starting a session.

How to use

1. Inspect the headset, lenses, straps, controllers, and charging cable for damage.
2. Charge the headset and install approved software or updates as required.
3. Adjust the head strap and lens position for a comfortable, clear view.
4. Set up the designated play area and remove obstacles from the user's movement space.
5. Complete the headset's boundary setup and explain controller functions before use.
6. Start the approved application and supervise the user during the session.
7. Exit the application, clean contact surfaces, and store the headset safely.

⚠ Safety

- Do not use a damaged headset or charging cable.
- Keep users away from stairs, sharp objects, and other physical hazards.
- Stop the session if dizziness, nausea, eye strain, or disorientation occurs.
- Follow age guidance, session limits, and hygiene procedures for shared devices.

Care & storage

- Clean lenses with a suitable microfiber cloth and sanitize shared contact surfaces as directed by the manufacturer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Video Camera

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Handy Camera, Microphone / voice system for lecture and class activities

What it is used for

A video camera with a microphone or voice system records lectures, demonstrations, and classroom activities. Set up the camera and audio equipment before recording, and verify image and sound quality with a test.

How to use

1. Inspect the camera, lens, battery, memory card, microphone, and audio cables.
2. Charge the battery and confirm sufficient recording space is available.
3. Mount the camera securely on a stable tripod or designated support.
4. Connect the microphone or voice system and check compatible input settings.
5. Frame the subject, adjust focus and exposure, and test the audio level.
6. Record a short test clip and verify picture clarity and sound quality.
7. Start recording, monitor the recording status, and safely stop and save the file afterward.

⚠ Safety

- Keep cables away from walkways and secure the camera against tipping.
- Do not operate damaged batteries, chargers, or cables.
- Obtain required permission before recording people or classroom activities.
- Avoid recording private or confidential information without authorization.

Care & storage

- Clean the lens with suitable materials, recharge batteries safely, and store equipment in a padded case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Video Conferencing System

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Video conferencing tools; Videoconferencing units

What it is used for

A video conferencing system enables remote audio and video communication for lessons, meetings, and demonstrations. Use the correct settings and protect participants' privacy during sessions.

How to use

1. Check the camera, microphone, speakers, display, cables, and network connection.
2. Place the camera securely and position the microphone for clear sound.
3. Connect the system to power and the approved network or computer.
4. Open the authorized conferencing application and select the correct audio and video devices.
5. Test the microphone, speakers, camera, and screen sharing before the session.
6. Join the scheduled session and mute the microphone when not speaking.
7. End the meeting, sign out if required, and switch off equipment safely.

⚠ Safety

- Obtain permission before recording or sharing participants' images, voices, or screens.
- Do not share meeting links, passwords, or confidential information publicly.
- Keep cables secured to prevent trips and electrical hazards.
- Use only approved applications and avoid opening unknown links during sessions.

Care & storage

- Clean camera and display surfaces with suitable materials and store accessories neatly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Voice Recorder

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Voice recorder

What it is used for

A voice recorder captures spoken audio for lectures, interviews, meetings, and training activities. Check storage, battery level, microphone placement, and recording settings before capturing audio.

How to use

1. Inspect the recorder, built-in microphone, external microphone jack, buttons, and battery compartment.
2. Charge or install the approved batteries and confirm sufficient storage space.
3. Select the required recording format, quality, and input source.
4. Place the recorder near the speaker on a stable surface, away from unwanted noise.
5. Make a short test recording and play it back to verify sound quality.
6. Record the session while checking the recording indicator and remaining storage.
7. Stop recording, save the file, and transfer or back it up as authorized.

⚠ Safety

- Do not record confidential conversations without proper authorization.
- Keep the recorder and cables away from liquids and excessive heat.
- Avoid blocking the microphone or handling the device noisily during recording.
- Use only compatible batteries and approved charging equipment.

Care & storage

- Clean the microphone openings gently and store the recorder in a dry, protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Webcam

Category: **Audio-Visual, Camera & Media** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Video-scopes or web cams and microphones; Webcam/Camera

What it is used for

A webcam captures video and may record audio for online lessons, meetings, and demonstrations. Connect it correctly and check privacy before recording or sharing.

How to use

1. Inspect the webcam, lens, cable, microphone, and mounting clip for damage.
2. Position the webcam securely with the lens facing the intended subject.
3. Connect it to a compatible computer or approved device.
4. Open the authorized application and select the correct camera and microphone.
5. Preview the image and test audio before starting the session.
6. Adjust framing and lighting without blocking the lens or microphone.
7. Close the application and disconnect or cover the webcam when appropriate.

⚠ Safety

- Obtain permission before recording or sharing anyone's image or voice.
- Do not point the webcam at private areas or confidential information.
- Keep cables away from walkways and electrical hazards.

Care & storage

- Clean the lens gently with a suitable microfiber cloth and store the webcam safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Filter

Category: **Automotive** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Air cleaner; Air filters

What it is used for

Automotive air filters remove dust and particles from intake air to help protect engine components.

How to handle

- Confirm the filter type and vehicle compatibility before replacement.
- Inspect the filter housing, seals, and intake area for damage or contamination.
- Switch off the engine and allow hot components to cool before servicing.
- Install the filter in the correct orientation and ensure the housing seals properly.
- Avoid blowing dust into the engine intake or breathing airborne particles.

Storage

- Keep replacement filters sealed in clean, dry packaging.
- Store away from moisture, oil, dust, and crushing pressure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Automotive Cooling System

Category: **Automotive** Type: **Full SOP** Priority: **Medium** 2 labs

Covers these lab items: Automotive Cooling System; Cooling system

What it is used for

An automotive cooling system removes excess engine heat and maintains the engine within its designed operating temperature range. Correct inspection and servicing help prevent overheating, coolant loss, and engine damage.

Main parts

- Radiator and cooling fan
- Water pump and drive belt or electric drive
- Thermostat and coolant passages
- Coolant hoses, clamps, and expansion tank
- Pressure cap and temperature sensor
- Heater core and control components where fitted

Before you start

1. Park the vehicle on level ground, apply the parking brake, and switch off the engine.
2. Allow the engine and cooling system to cool completely before opening the pressure cap.
3. Check the coolant level in the expansion tank and use the specified coolant type.
4. Inspect the radiator, hoses, clamps, water pump, drive belt, and tank for leaks or damage.
5. Check the cooling fan area for obstructions and inspect the belt or electrical connections as applicable.
6. Use the vehicle service manual to identify the correct pressure-test and filling procedures.

How to use

1. Inspect the system visually and identify signs of coolant leakage, corrosion, or damaged components.
2. Check the coolant level only at the designated reservoir when the engine is cold.
3. Pressure-test the system and cap only with approved equipment and the specified test pressure.
4. Repair or replace faulty hoses, clamps, seals, or components according to the service procedure.
5. Drain old coolant into an approved container when the procedure requires replacement.
6. Refill with the specified coolant mixture using the correct filling or bleeding method.
7. Start the engine only after the system is correctly filled and all tools are removed.
8. Monitor engine temperature and check for leaks, fan operation, and proper coolant circulation.
9. Allow the engine to cool and recheck the coolant level according to the service instructions.
10. Dispose of used coolant safely, clean the work area, and record the service performed.

⚠ Safety

- Never remove a pressure cap while the engine or coolant is hot.
- Keep hands, clothing, and tools away from cooling fans because they may start automatically.
- Wear suitable eye and hand protection when handling coolant and pressurized test equipment.
- Use the specified coolant and never mix incompatible coolant types.
- Collect coolant in a closed, labelled container and prevent spills onto soil or drains.
- Keep coolant away from children and animals because some formulations are toxic.

Care & storage

- Check coolant level and condition at the vehicle manufacturer's recommended intervals.
- Inspect hoses, clamps, radiator fins, pump, belt, and cap for wear or corrosion.
- Clean external radiator surfaces carefully without bending the cooling fins.
- Replace coolant and service the system according to the vehicle maintenance schedule.

Common mistakes to avoid

- Opening the pressure cap while hot can cause severe scalding.
- Using plain water or incompatible coolant can reduce corrosion protection and cooling performance.
- Trapped air after refilling can cause overheating or unstable temperature readings.
- Ignoring small leaks or a faulty pressure cap can lead to coolant loss and engine damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Automotive Electrical Trainer Kit

Category: **Automotive** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Electrical, ignition, electronic and air-conditioning system; Electronic Ignition System of An Automobile 4-Wheeler; Instruction Kit for Charging and Starting System; Lighting and Wiring System (two-wheeler)

What it is used for

An automotive electrical trainer kit demonstrates vehicle electrical, ignition, starting, charging, electronic, or air-conditioning circuits in a controlled training setup. Correct connections, supervised testing, and safe fault-finding help learners understand system operation without damaging components.

Main parts

- Trainer panel or board supports the electrical components and circuit layout.
- Battery or approved low-voltage power supply provides electrical power.
- Switches and connectors control and link the training circuits.
- Fuses and circuit protection help limit damage from electrical faults.
- Wiring harnesses and terminals represent vehicle electrical connections.
- Ignition, starting, charging, or electronic modules depend on the trainer model.
- Meters, indicator lamps, and test points support measurement and diagnosis.
- Circuit diagrams and instruction materials explain the intended connections.

Before you start

1. Read the trainer manual and identify the circuits included in the specific kit.
2. Ensure the activity is supervised by a qualified instructor or authorised technician.
3. Inspect the power supply, cables, connectors, fuse ratings, and trainer panel for damage.
4. Keep the power switched off while identifying terminals and preparing connections.
5. Review the circuit diagram and verify the required components and test instruments.
6. Remove metal jewellery and prepare suitable PPE and an uncluttered work area.

How to use

1. Place the trainer on a stable, dry workbench and check the circuit diagram.
2. Identify the supply terminals, ground points, switches, loads, and test points.
3. Connect the circuit with power off, following the diagram and specified polarity.
4. Check wiring continuity and confirm that connections are secure before energising the circuit.
5. Set the meter to the correct function and range before making measurements.
6. Switch on the approved supply and operate the circuit using the designated controls.
7. Observe indicator lamps and measure the specified voltage, current, or continuity values.
8. Introduce only instructor-approved faults using the trainer's designated fault switches or method.

9. Switch off the supply before changing connections, replacing fuses, or correcting faults.
10. After testing, restore the original circuit, switch off power, disconnect leads, and record observations.

⚠ Safety

- Use only the voltage and current ratings specified for the trainer kit.
- Never connect or disconnect wires while the circuit is energised unless the trainer procedure explicitly requires it.
- Do not bypass fuses or replace them with a higher rating.
- Keep liquids away from electrical panels, power supplies, and connectors.
- Never test real vehicle high-voltage EV systems using a low-voltage trainer procedure.
- For air-conditioning modules, do not open refrigerant circuits or handle refrigerants without appropriate training and authorisation.

Care & storage

- Switch off and disconnect the power supply before cleaning the trainer.
- Use suitable dry materials to clean panels, connectors, and test leads.
- Inspect cables, probes, fuses, switches, and terminals for wear or damage.
- Store leads and accessories neatly and report defective components before the next session.

Common mistakes to avoid

- Connecting terminals without checking the circuit diagram or polarity.
- Using the wrong meter function or range for a measurement.
- Bypassing a fuse or energising a circuit with loose wiring.
- Changing circuit connections while power is on.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Automotive Spare Part

Category: **Automotive** Type: **Handling note** Priority: **Low** 11 labs

Covers these lab items: Gaskets; Glow plugs; Seal; Worn out/ defective/ spurious samples

What it is used for

Automotive spare parts such as gaskets, glow plugs, and seals support vehicle maintenance and component replacement.

How to handle

- Verify the part number, vehicle compatibility, specifications, and authenticity before installation.
- Inspect parts for cracks, corrosion, deformation, wear, or signs of counterfeiting.
- Use suitable tools and follow the vehicle manufacturer's service instructions and torque specifications.
- Keep parts clean and avoid damaging sealing surfaces or electrical contacts.

Storage

- Store parts in labelled packaging away from moisture, dust, and extreme temperatures.
- Separate defective, worn, or suspect parts from approved service stock.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ball Joint Separator

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Ball joint separators

What it is used for

A ball joint separator helps disconnect a vehicle's ball joint from its mating component during repair. Use the correct separator type and follow the vehicle service procedure.

How to use

1. Park the vehicle securely, switch off the engine, and prevent movement.
2. Consult the service manual and identify the correct ball joint separator.
3. Use approved lifting points and support the vehicle with rated stands when lifting is required.
4. Inspect the separator, threads, jaws, and contact points for damage.
5. Position the tool correctly around the joint according to its design.
6. Apply force gradually using the specified method until the joint separates.
7. Inspect components and remove the tool carefully after separation.

⚠ Safety

- Never work beneath a vehicle supported only by a jack.
- Wear eye protection and keep hands away from pinch points.
- Do not strike a loaded separator unless its instructions specifically permit it.
- Only trained persons should perform vehicle suspension repairs.

Care & storage

- Clean the tool after use, inspect its threads, and store it dry in its case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Body Repair Spoon

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Spoons

What it is used for

To support or shape automotive body panels during approved dent-repair work using a body repair spoon. To apply controlled pressure while minimizing stretching and surface damage.

Main parts

- Spoon-shaped working blade
- Handle or grip
- Neck or shaft
- Polished contact surface

Before you start

1. Confirm the spoon is suitable for the panel and repair method.
2. Inspect the blade and handle for cracks, burrs, or looseness.
3. Clean the panel and identify the damaged area.
4. Use appropriate eye protection and gloves.
5. Check the repair procedure for nearby wiring, trim, and structural parts.

How to use

1. Position the spoon against the approved support area behind or beside the panel as appropriate.
2. Keep the contact surface aligned with the panel contour.
3. Apply gradual, controlled pressure to support or raise the damaged area.
4. Use small adjustments rather than sudden force.
5. Check the panel contour frequently under suitable lighting.
6. Stop if the panel begins to stretch, crease, or crack.
7. Clean the tool and inspect the repair after finishing.

⚠ Safety

- Avoid working near sharp panel edges without protection.
- Do not apply force near airbags, wiring, or fuel-system parts without following the vehicle procedure.
- Use the spoon only for its intended body-repair method.
- Keep hands clear of pinch points between the tool and panel.

Care & storage

- Wipe the contact surface clean after use.
- Remove corrosion or residue using an approved method.
- Inspect the blade and handle for damage.
- Store separately to protect the smooth working surface.

Common mistakes to avoid

- Using the spoon at an unsuitable angle.
- Applying excessive force and stretching the panel.
- Scratching paint or nearby trim with a dirty tool.
- Ignoring hidden wiring or safety components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Brake Roller Tester

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Brake roller tester

What it is used for

A brake roller tester checks vehicle braking performance by measuring the braking force produced at the wheels on rotating rollers. Correct vehicle positioning, controlled testing, and accurate recording help assess brake condition while reducing equipment and vehicle risks.

Main parts

- Roller assemblies rotate the vehicle wheels during testing.
- Drive motors power the roller assemblies.
- Force sensors or load cells measure braking force.
- Display or control console shows wheel force and test results.
- Vehicle positioning guides help align the wheels with the rollers.
- Safety guards cover moving parts where provided.
- Emergency stop controls halt the test system.
- Test software or recording system stores measurements on equipped models.

Before you start

1. Only trained and authorised personnel should operate the brake roller tester.
2. Read the tester and vehicle instructions and confirm the vehicle is compatible with the equipment.
3. Inspect rollers, guards, sensors, controls, and the test platform for damage or obstruction.
4. Check tyre condition, vehicle weight limits, and any special vehicle test requirements.
5. Ensure the test area is clear and the emergency stop is accessible.
6. Confirm the tester is ready, calibrated or verified as required, and free of fault indicators.

How to use

1. Guide the vehicle slowly onto the rollers following the tester's alignment instructions.
2. Position the test wheels correctly and apply the prescribed vehicle restraint or parking procedure.
3. Confirm that the vehicle transmission and driver controls are set as required by the test method.
4. Start the test using the console and allow the rollers to operate as instructed.
5. Apply the service brake gradually and consistently when prompted by the test procedure.
6. Observe and record braking force, left-right differences, and any displayed imbalance or fault values.
7. Repeat the test for the required axle or braking system according to the approved procedure.
8. Stop the rollers completely before moving the vehicle off the tester.

9. Drive the vehicle off slowly when the operator confirms it is safe to do so.
10. Save the results, report abnormal readings, and clear the test area after the vehicle exits.

⚠ Safety

- Keep people clear of rollers, moving wheels, and vehicle travel paths during testing.
- Never step onto rotating rollers or reach into moving mechanisms.
- Follow the tester's instructions for vehicle restraints, transmission position, and brake application.
- Do not test vehicles with unsafe tyres, leaking fluids, or other conditions prohibited by the procedure.
- Use the emergency stop if the vehicle shifts unexpectedly or a dangerous condition develops.
- Do not move the vehicle until the rollers have stopped and the operator authorises exit.

Care & storage

- Remove dirt and debris from the platform and accessible roller areas after use.
- Inspect roller surfaces, guards, sensors, and emergency stop controls regularly.
- Keep the test software and measurement system maintained according to the manufacturer instructions.
- Arrange authorised calibration and servicing if readings become inconsistent or faults occur.

Common mistakes to avoid

- Misaligning the vehicle wheels on the rollers.
- Applying the brake abruptly instead of following the test procedure.
- Ignoring vehicle compatibility, tyre condition, or displayed fault warnings.
- Driving off before the rollers stop and the operator authorises exit.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Brake System

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 12 labs

Covers these lab items: Brake pads; Brake shoes; Brakes & Controls; Disc & drum brakes system; Old second hand disc brake

What it is used for

Use a brake system assembly or training unit to understand and inspect how braking components slow or stop a vehicle.

How to use

1. Park the vehicle securely or stabilize the training unit before inspection.
2. Identify the brake pedal, master cylinder, lines, calipers or wheel cylinders, and pads or shoes.
3. Check visible components for leaks, cracks, wear, or loose connections.
4. Follow the manufacturer's procedure to inspect or demonstrate brake operation.
5. Use approved tools to measure pad, shoe, disc, or drum wear when required.
6. Confirm all parts are correctly installed before any functional test.
7. Carry out testing only in a controlled area using the approved procedure.

⚠ Safety

- Never operate a vehicle with suspected brake failure.
- Support raised vehicles with approved stands; never rely on a jack alone.
- Avoid inhaling brake dust and use approved cleaning methods.
- Brake hydraulic repairs and road tests must be performed by qualified personnel.

Care & storage

- Keep brake components clean and replace worn parts to specification.
- Record inspection findings and report leaks, unusual noise, or poor braking.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Car Lift

Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Car lift -4Ton

What it is used for

A four-tonne car lift raises and supports a vehicle for inspection, maintenance, or repair. Correct vehicle positioning, rated-capacity checks, lifting-point selection, and mechanical locking are essential to prevent vehicle falls, crushing injuries, and equipment damage.

Main parts

- Lift posts or supporting structure
- Lift arms and adjustable vehicle pads
- Hydraulic cylinders or drive mechanism
- Power unit and operating controls
- Mechanical safety locks and release mechanism
- Vehicle contact pads, anchors, and emergency stop, if fitted

Before you start

1. Confirm that the lift is rated for the vehicle's weight, dimensions, and centre-of-gravity requirements; do not exceed the four-tonne rating.
2. Inspect the posts, arms, pads, cables or hydraulic lines, anchors, safety locks, and controls for damage or wear.
3. Verify that the lift is installed on an approved foundation and that required inspections are current.
4. Check the vehicle manufacturer's lifting points and ensure the floor is level, clean, and free of obstacles.
5. Confirm that doors, roof accessories, and other vehicle features have adequate clearance during lifting.
6. Keep unauthorised people away, wear safety footwear, and ensure the lift operating area is clearly visible.

How to use

1. Drive the vehicle slowly into the lift bay and align it centrally between the posts.
2. Apply the parking brake, select the recommended transmission position, and switch off the engine as required by the procedure.
3. Position the lift arms and pads at the vehicle manufacturer's approved lifting points.
4. Adjust each arm so the pads make secure contact without contacting fragile components or body panels.
5. Check that all arms and pads are locked in position and that the vehicle is correctly balanced.

6. Clear people, tools, and obstructions from under and around the vehicle.
7. Operate the lift to raise the vehicle a short distance and stop to recheck pad contact and vehicle stability.
8. Continue lifting smoothly while observing the vehicle, arms, and lift for movement or abnormal noise.
9. Raise the vehicle to the required height and lower it gently onto the mechanical safety locks according to the lift design.
10. Confirm that the safety locks are engaged before working beneath the vehicle; never rely on hydraulic pressure alone.
11. Before lowering, remove tools and personnel from the danger area and raise the lift slightly if required to release the locks.
12. Disengage the safety locks using the approved release procedure and lower the vehicle smoothly to the ground.
13. Confirm that all pads are clear, retract the arms, and move the vehicle out of the bay when safe.
14. Report any abnormal operation and record inspections or maintenance concerns.

⚠ Safety

- Only trained and authorised personnel should operate the car lift.
- Never exceed the lift's rated capacity or lift a vehicle with unknown weight distribution without an approved assessment.
- Use only the vehicle manufacturer's approved lifting points and the lift manufacturer's operating procedure.
- Never work under a raised vehicle until the mechanical safety locks are engaged and stability is verified.
- Keep people clear of the vehicle and moving arms during lifting and lowering.
- Do not use a lift with damaged anchors, arms, pads, cables, hydraulic lines, or safety locks; isolate it and report the defect.

Care & storage

- Inspect lift arms, pads, anchors, safety locks, cables, and hydraulic lines at the specified intervals.
- Check hydraulic fluid and lubricate designated moving points according to the manufacturer's instructions.
- Keep the lift bay and floor clean, dry, and free of tools or debris.
- Test safety locks, controls, and emergency stop according to the maintenance schedule.
- Arrange qualified inspection and repair for leaks, uneven lifting, abnormal movement, or damaged components.

Common mistakes to avoid

- Lifting a vehicle above the rated capacity or with incorrect weight distribution.
- Positioning lift pads away from approved vehicle lifting points.
- Working under the vehicle before mechanical locks are engaged.
- Lowering the lift while people, tools, or obstructions remain in the danger area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chassis Dynamometer

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Chassis dynamometer

What it is used for

A chassis dynamometer measures a vehicle's power and performance by running its driven wheels on rollers or using a hub-connected test system. Correct vehicle setup, secure restraint, ventilation, and controlled test operation help obtain reliable results while reducing vehicle damage and operator hazards.

Main parts

- Rollers or hub adapters transfer vehicle power to the measurement system.
- Drive or load unit applies the required test load where fitted.
- Speed sensors measure wheel or roller speed.
- Torque sensors measure rotational force.
- Control console sets the test programme and displays results.
- Vehicle restraint straps or fixtures secure the vehicle.
- Cooling fans provide airflow during testing.
- Emergency stop and safety guards help control hazards.

Before you start

1. Only trained automotive testing personnel should operate the dynamometer.
2. Check the vehicle manufacturer's limits and confirm that the vehicle and drivetrain are compatible with the test system.
3. Inspect rollers or hub adapters, restraint fixtures, sensors, guards, cables, and controls for damage.
4. Check tyre condition and pressure where roller testing is used, and inspect the test surface for debris.
5. Secure the vehicle using the specified restraints and verify that they are correctly attached.
6. Provide adequate ventilation and cooling airflow, and ensure the exhaust extraction system is working where required.
7. Confirm emergency stops, interlocks, and test limits are set and functional.

How to use

1. Position the vehicle on the dynamometer using the approved alignment procedure.
2. Secure the vehicle with the specified restraints and verify clearance around wheels and moving parts.
3. Connect required measurement leads or diagnostic equipment according to the test plan.

4. Set the test programme, speed limits, load limits, and other parameters within approved vehicle limits.
5. Start the cooling fans and exhaust extraction before running the engine where required.
6. Start the test at the prescribed low load and follow the dynamometer operating instructions.
7. Increase speed or load only as specified by the test procedure while monitoring live readings.
8. Watch engine temperature, vehicle stability, tyre behaviour, abnormal vibration, and warning indicators throughout the test.
9. Stop the test immediately if restraint movement, tyre damage, overheating, unexpected vibration, or another unsafe condition occurs.
10. Reduce load and bring the system to a complete stop before removing restraints or moving the vehicle.
11. Save the test results and inspect the vehicle and equipment for signs of damage.

⚠ Safety

- Never operate a chassis dynamometer without training, correct restraints, and required guards.
- Keep all people clear of rotating wheels, rollers, belts, and vehicle travel paths.
- Do not stand directly behind or in front of a running vehicle during testing.
- Ensure adequate exhaust extraction and ventilation to prevent harmful exhaust-gas accumulation.
- Do not exceed vehicle, tyre, roller, or dynamometer limits.
- Never attach or remove restraints while rollers or wheels are moving; follow the emergency shutdown procedure if the vehicle shifts.

Care & storage

- Remove dirt and debris from the test area after the system is fully stopped.
- Inspect rollers, hub adapters, restraint straps, sensors, guards, and cooling fans regularly.
- Keep measurement software and sensors maintained according to the manufacturer's instructions.
- Arrange authorised calibration and service when measurements are inconsistent or faults appear.

Common mistakes to avoid

- Running a test without verifying vehicle compatibility and restraint security.
- Ignoring tyre condition, vehicle temperature, or dynamometer limits.
- Testing without adequate ventilation or cooling airflow.
- Releasing restraints before the system and vehicle wheels have completely stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Clutch Assembly

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Clutch assembly; Clutch plate

What it is used for

To inspect and install a clutch assembly or clutch plate during vehicle maintenance. To ensure correct alignment, secure fastening, and reliable power transmission.

Main parts

- Clutch plate or friction disc
- Pressure plate
- Release bearing
- Clutch cover
- Alignment tool
- Mounting bolts

Before you start

1. Confirm the correct clutch parts for the vehicle model.
2. Park the vehicle securely and follow the approved lifting procedure if raised.
3. Disconnect the battery according to the vehicle service manual.
4. Inspect the clutch parts for cracks, wear, oil contamination, or damage.
5. Use the specified tools and alignment equipment.

How to use

1. Remove the transmission and old clutch components according to the service manual.
2. Inspect the flywheel and related components for damage or wear.
3. Clean mating surfaces using an approved method.
4. Align the clutch plate using the correct alignment tool.
5. Fit the pressure plate and tighten bolts in the specified sequence and torque.
6. Reinstall the transmission and release mechanism according to the manual.
7. Reconnect the battery and check clutch operation using the approved test procedure.

⚠ Safety

- Only trained technicians should perform clutch replacement.
- Support the vehicle and transmission with rated lifting equipment.
- Never work beneath an unsupported vehicle or transmission.
- Avoid inhaling clutch dust; do not clean friction parts with compressed air.
- Follow vehicle-specific torque and adjustment instructions.

Care & storage

- Inspect clutch operation during scheduled maintenance.
- Keep friction surfaces free from oil and grease.
- Replace worn or damaged components as a suitable set when specified.
- Record repairs and test results.

Common mistakes to avoid

- Installing the clutch plate backwards.
- Misaligning the clutch plate and input shaft.
- Tightening bolts to incorrect torque.
- Contaminating friction surfaces with oil or grease.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Clutch Puller

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Clutch puller

What it is used for

A clutch puller removes a compatible clutch component from its shaft during mechanical servicing. Use the specified puller and follow the equipment manufacturer's removal procedure.

How to use

1. Switch off the machine or vehicle and isolate the power source as instructed.
2. Consult the service manual to identify the correct puller and attachment points.
3. Inspect the puller threads, bolts, jaws, and contact surfaces for damage.
4. Secure the component and fit the puller squarely according to its design.
5. Tighten the puller gradually using the specified tool and procedure.
6. Stop if the puller slips, bends, or excessive force is required.
7. Remove the separated component and inspect all parts before reassembly.

⚠ Safety

- Wear eye protection and keep hands clear of loaded components.
- Do not use improvised extensions or exceed the puller's rated capacity.
- Never strike a loaded puller unless the manufacturer explicitly permits it.
- Only trained personnel should perform clutch removal and servicing.

Care & storage

- Clean the puller after use, inspect threads, and store it dry in its case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Compression Gauge

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Compressor gauge

What it is used for

To measure engine cylinder compression using a compatible compression gauge during automotive diagnosis. To compare readings with the vehicle manufacturer's specifications and identify possible mechanical problems.

Main parts

- Pressure gauge or digital display
- Pressure hose
- Spark-plug adapter or specified connector
- Release valve
- Fittings and seals

Before you start

1. Confirm the engine type and the correct gauge and adapter.
2. Review the vehicle service manual for the specified test procedure.
3. Park securely, apply the parking brake, and disable ignition and fuel delivery as directed.
4. Allow the engine to reach the specified test temperature if required.
5. Inspect the gauge, hose, fittings, and seals for damage.

How to use

1. Remove the specified spark plugs or access the designated test port according to the service manual.
2. Install the correct adapter securely without cross-threading.
3. Set the throttle and other conditions as specified by the manufacturer.
4. Crank the engine using the approved procedure and observe the gauge reading.
5. Repeat the measurement consistently for each cylinder.
6. Record the readings and compare them with specifications and cylinder-to-cylinder variation limits.
7. Release pressure, remove the gauge, and reinstall components to the specified torque.

⚠ Safety

- Keep hands, clothing, and tools away from moving engine parts.
- Follow the required ignition and fuel-system isolation procedure.
- Do not exceed the gauge or adapter pressure rating.
- Allow hot components to cool where required and wear eye protection.

Care & storage

- Release pressure before disconnecting the hose.
- Clean adapters and fittings after use.
- Inspect the hose, seals, and gauge for leaks or wear.
- Store the gauge in a protective case.

Common mistakes to avoid

- Using the wrong adapter or thread type.
- Failing to disable ignition or fuel delivery as specified.
- Comparing readings taken under different test conditions.
- Ignoring manufacturer limits and cylinder-to-cylinder differences.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cut Section Model

Category: **Automotive** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Cut a section of 2-stroke/ 4-Stroke Engine; Old second hand cut section lead acid battery; Old second-hand cut section leads to acid battery

What it is used for

Cut-section models demonstrate internal parts of engines or lead-acid batteries for technical training and visual inspection.

How to handle

- Identify the model and confirm it is a non-operational training sample before handling.
- Inspect cut edges, exposed components, mounting points, and stability for hazards.
- Wear suitable PPE and avoid contact with sharp edges, residues, or exposed terminals.
- Treat used lead-acid battery sections as hazardous training items; do not open, dismantle, or disturb residues without an approved procedure.
- Keep models secure during demonstrations and report damage or leakage immediately.

Storage

- Store models in a stable, labelled, ventilated area as appropriate to their materials.
- Keep battery-related samples away from children, food, ignition sources, and unauthorised handling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Demonstration Kit

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Demonstration Board of Working Model MPFI System with Motorized Control; Transmission / Gearbox Demonstration Kit; Washer and wiper demonstration kit

What it is used for

To demonstrate the operation of automotive systems such as MPFI, transmission or gearbox, and washer-wiper mechanisms using a training demonstration kit. To explain component functions and system operation safely in a controlled learning environment.

Main parts

- Demonstration board or training frame
- Automotive components or working model
- Motor, switches, and control panel, if fitted
- Power supply and wiring
- Labels, diagrams, or indicators
- Protective covers and emergency stop, if fitted

Before you start

1. Identify the specific system and review the kit's operating manual.
2. Inspect wiring, connectors, moving parts, guards, and the power supply.
3. Place the kit on a stable bench with adequate ventilation and lighting.
4. Confirm the rated supply voltage and required protective devices.
5. Keep liquids, loose clothing, and unnecessary objects away from moving parts.

How to use

1. Explain the system layout and identify the main components.
2. Check that all connections and guards are secure before powering on.
3. Connect the specified power source according to the manual.
4. Switch on the kit and demonstrate the controls in the approved sequence.
5. Observe the indicators and explain the system response to each input.
6. For moving mechanisms, keep hands clear and operate only within the permitted range.
7. Switch off and isolate the power before changing connections or inspecting components.
8. Record observations and return the kit to its safe storage condition.

⚠ Safety

- Only trained instructors or supervised learners should operate the kit.
- Never touch exposed live wiring or moving gears while powered.
- Use the emergency stop if unsafe movement or an electrical fault occurs.
- Do not bypass guards, interlocks, or protective devices.
- Follow the manual for fuel-system, motor, and electrical demonstrations.

Care & storage

- Switch off and disconnect power after use.
- Clean the board and components with suitable materials.
- Inspect cables, connectors, guards, and controls regularly.
- Report faults and store the kit in a dry, secure area.

Common mistakes to avoid

- Powering on before checking wiring and guards.
- Using an incorrect supply voltage or connection.
- Touching moving or live components during operation.
- Changing wiring while the kit is energized.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Diagnostic Kit

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Diagnostic kit

What it is used for

An automotive diagnostic kit helps identify vehicle faults using compatible test tools or electronic scanners. Select the correct equipment and follow the vehicle service procedure.

How to use

1. Park the vehicle safely, apply the parking brake, and switch off the ignition.
2. Identify the vehicle model and confirm the diagnostic tool is compatible.
3. Inspect cables, probes, connectors, and scanner housing for damage.
4. Connect the diagnostic tool to the specified port according to the service manual.
5. Switch on the required vehicle systems and read the available fault codes or measurements.
6. Record the codes and observations before clearing any stored information.
7. Disconnect the tool as instructed and verify system operation after approved checks.

⚠ Safety

- Keep hands, clothing, and cables away from moving engine parts.
- Use ventilation when the engine must run during testing.
- Do not probe airbag or high-voltage hybrid or EV circuits unless trained and authorized.
- Follow battery and electrical isolation instructions in the service manual.

Care & storage

- Clean the kit after use, protect connectors, and store tools in their case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Diagnostic Scanner

Category: **Automotive** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Analyzer Machines; Electronic diagnostic devices

What it is used for

An automotive diagnostic scanner communicates with a vehicle's electronic control units to retrieve fault codes, display sensor data, and perform supported diagnostic or service functions. Correct vehicle identification, safe connection, accurate interpretation, and approved testing procedures help diagnose faults without damaging vehicle electronics.

Main parts

- Display screen presents diagnostic information.
- Control buttons or touchscreen navigate menus.
- Diagnostic connector cable links the scanner to the vehicle port.
- Vehicle communication interface exchanges data with electronic control units.
- Power supply or rechargeable battery powers the scanner.
- Software provides vehicle-specific diagnostic functions.
- USB or wireless interface supports updates or data transfer where fitted.
- Protective housing and cable connectors protect the device.

Before you start

1. Only trained personnel should perform vehicle diagnostics and service functions.
2. Confirm the vehicle make, model, year, powertrain type, and supported communication protocol.
3. Inspect the scanner, cable, connector pins, housing, and power supply for damage.
4. Read the scanner instructions and confirm that the software supports the target vehicle and intended function.
5. Park the vehicle safely, apply the parking brake, select the appropriate gear position, and follow vehicle-specific ignition instructions.
6. Check the vehicle battery condition; low voltage may interrupt communication or affect diagnostic results.
7. For electric and hybrid vehicles, follow high-voltage safety procedures and do not touch orange high-voltage cables or components unless qualified and authorised.

How to use

1. Switch on the scanner and select the correct vehicle identification details or approved automatic identification option.
2. Locate the vehicle's diagnostic port using the service information and inspect it for damage or contamination.
3. Connect the diagnostic cable securely to the vehicle port without forcing the connector.
4. Switch the vehicle ignition or diagnostic mode to the state required by the scanner instructions.

5. Select the required system, such as engine, transmission, ABS, airbag, or battery management, where supported.
6. Read and record stored, pending, and permanent diagnostic trouble codes as applicable.
7. Review freeze-frame data and live sensor values when available, comparing them with vehicle service specifications.
8. Perform supported active tests or service functions only when authorised and when their prerequisites are satisfied.
9. Do not clear fault codes until the original codes and relevant data have been recorded and the repair plan permits clearing.
10. After authorised repairs, clear codes if appropriate and repeat the scan or specified functional test.
11. Turn off the diagnostic session and disconnect the scanner carefully according to the instructions.
12. Save or document results, disconnect the device, and return the vehicle to a safe operating condition.

⚠ Safety

- Never perform coding, programming, actuator tests, or resets unless the function is supported and the procedure is understood.
- Do not disconnect the scanner or interrupt vehicle power during a programming operation.
- Use vehicle-specific procedures for airbag, ABS, immobiliser, and high-voltage system diagnostics.
- Keep cables clear of pedals, steering components, fans, belts, and hot engine parts.
- Do not assume a diagnostic trouble code identifies the failed component; verify the fault using service information and testing.
- Only qualified personnel should access high-voltage EV systems or perform safety-critical vehicle repairs.

Care & storage

- Keep scanner software and vehicle coverage updated using approved update procedures.
- Clean the housing and connectors with a manufacturer-approved method.
- Inspect cable insulation, connector pins, and battery condition regularly.
- Store the scanner in a dry, protected case away from excessive heat and moisture.
- Report recurring communication failures and arrange authorised repair if the device is damaged.

Common mistakes to avoid

- Selecting the wrong vehicle or control system.
- Replacing parts based only on a fault code without confirming the cause.
- Clearing codes before recording diagnostic information.
- Interrupting power or communication during programming or other critical service functions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drain Pan

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Drain pan

What it is used for

To collect used engine oil, coolant, or other automotive fluids during servicing using a suitable drain pan. To prevent spills and ensure fluids are handled and disposed of correctly.

Main parts

- Collection pan or basin
- Pouring lip or spout, if fitted
- Handles or grips
- Drain plug or tap, if fitted

Before you start

1. Select a pan with sufficient capacity for the fluid being drained.
2. Inspect the pan for cracks, leaks, and chemical compatibility.
3. Park the vehicle securely and follow the service manual isolation procedure.
4. Allow hot fluids to cool to a safe handling temperature where appropriate.
5. Wear suitable gloves and eye protection.

How to use

1. Position the pan directly below the drain point.
2. Confirm the pan is stable and can catch the full fluid stream.
3. Open the drain plug or valve according to the service procedure.
4. Allow the fluid to drain without splashing.
5. Reinstall the drain plug or close the valve to the specified requirement.
6. Transfer the collected fluid into a labelled, compatible waste container.
7. Clean spills promptly and dispose of waste through an approved channel.

⚠ Safety

- Avoid contact with hot fluids and moving vehicle parts.
- Never mix incompatible fluids or unknown chemicals.
- Prevent fluids from entering soil, drains, or watercourses.
- Use spill-control materials and follow hazardous-waste procedures.

Care & storage

- Empty and clean the pan after use using an approved method.

- Inspect for cracks and leaks before each use.
- Keep the pan dry and free from debris before storage.
- Store separately from clean-fluid containers.

Common mistakes to avoid

- Using a pan with insufficient capacity.
- Positioning the pan away from the drain stream.
- Mixing different fluids in one container without authorization.
- Pouring used fluids into a drain or onto the ground.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

EV Charger

Category: **Automotive** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: AC EV Charger (Slow); DC fast Charger; DC Fast Charger (Demo Unit); EV Charging Gun (Type-2)

What it is used for

An EV charger supplies electrical energy to a compatible electric vehicle through an approved charging connection. Correct equipment matching, inspection, and charging procedures help maintain reliable charging and reduce electrical hazards.

Main parts

- Charging unit controls and monitors the charging process.
- Charging connector or gun connects the charger to the vehicle inlet.
- Charging cable carries electrical power and communication signals as designed.
- Control and protection circuits monitor faults and charging conditions.
- Display or status indicators show charging state on equipped models.
- AC chargers provide alternating current for the vehicle's onboard charger.
- DC fast chargers supply controlled DC power through compatible vehicle systems.
- Emergency stop and protective devices are provided on applicable charging units.

Before you start

1. Read the charger and vehicle manuals and confirm connector and charging compatibility.
2. Check applicable site rules, charger rating, and required user authorisation.
3. Inspect the charger, cable, connector, plug, and vehicle inlet for damage or contamination.
4. Ensure connectors and hands are dry and the charging area is clear.
5. Check that the charger is free from warning indicators or visible faults.
6. Confirm the vehicle is parked safely and switched off or placed in the required charging state.

How to use

1. Position the vehicle within reach of the charging cable without stretching or crushing it.
2. Activate the charger using the approved access method, if required.
3. Open the vehicle charging port and inspect it for visible debris or damage.
4. Connect the correct charging connector firmly to the vehicle inlet as instructed.
5. Follow the charger's displayed prompts or approved procedure to start the session.
6. Check the display or vehicle indicators to confirm charging has started normally.
7. Monitor charging status and respond to warnings according to the manual.
8. To finish, stop the session using the approved controls before disconnecting the connector.
9. Remove the connector by its handle without pulling the cable and return it to its holder.
10. Close the vehicle charging port, secure the cable, and report any faults.

⚠ Safety

- Never use a charger, cable, or connector with visible damage or exposed conductors.
- Do not touch connectors with wet hands or use them in standing water.
- Never force a connector into the vehicle inlet or modify the plug.
- Do not disconnect the charging connector under load unless the equipment procedure explicitly permits it.
- Do not open charger enclosures or attempt electrical repairs without authorised training.
- If there is smoke, unusual heat, sparking, or an electric shock risk, stop if safe, keep away, and contact emergency or authorised service personnel.

Care & storage

- Keep connectors and cable holders clean and protected from impact.
- Inspect cables, connectors, and visible enclosure parts regularly for wear or damage.
- Keep the charging area clear and follow the manufacturer's cleaning instructions.
- Report repeated charging failures, warning indicators, or damaged equipment for authorised inspection.

Common mistakes to avoid

- Using a connector or charging mode incompatible with the vehicle.
- Starting a session without inspecting the cable and connectors.
- Pulling the cable instead of releasing the connector by its handle.
- Disconnecting before stopping the charging session as instructed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electric Vehicle

Category: **Automotive** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Four wheeler Electric vehicle model; Four wheelers old second hand Electric vehicle; Old second hand Electric two-wheeler

What it is used for

An electric vehicle uses a traction battery, electric motor, and power electronics to provide vehicle movement. Safe inspection, correct charging practices, and approved operating procedures help protect occupants and technicians while maintaining vehicle performance.

Main parts

- Traction battery stores electrical energy for propulsion.
- Electric motor converts electrical energy into mechanical movement.
- Inverter or motor controller manages power supplied to the motor.
- Vehicle control unit coordinates vehicle systems and operating functions.
- Charging port connects the vehicle to compatible charging equipment.
- Onboard charger converts AC charging power when supported by the vehicle.
- Battery management system monitors battery conditions and protection limits.
- Braking system, tyres, lights, and safety restraints support safe driving.

Before you start

1. Confirm the operator has the required driving licence and vehicle authorisation.
2. Read the vehicle manual and identify its charging type and operating instructions.
3. Inspect tyres, brakes, lights, mirrors, windscreen, and visible body damage.
4. Check the dashboard for warning lights and verify the battery state of charge.
5. Inspect the charging cable and connector for damage before charging.
6. Ensure the vehicle has no visible fluid leaks, exposed high-voltage wiring, or unusual damage.

How to use

1. Enter the vehicle and adjust the seat, mirrors, and steering position safely.
2. Fasten the seat belt and ensure all occupants use appropriate restraints.
3. Check the surroundings and start or activate the vehicle using the prescribed procedure.
4. Select the required drive mode and confirm the dashboard shows readiness to drive.
5. Release the parking brake as instructed and move off smoothly.
6. Follow traffic rules, speed limits, and safe driving distances throughout the journey.
7. Monitor battery level, warning indicators, and vehicle response while driving.
8. Park in a suitable location, apply the parking brake, and switch off the vehicle.
9. For charging, use compatible approved equipment and follow the vehicle's charging sequence.
10. After charging, stop the session as instructed, disconnect the connector safely, and secure the charging port.

⚠ Safety

- Never touch exposed high-voltage cables, connectors, or damaged battery components.
- Do not operate the vehicle if critical brake, steering, tyre, or high-voltage warnings appear.
- Use only compatible charging equipment and avoid charging with damaged cables or connectors.
- Keep charging connectors dry and follow the manufacturer's instructions for wet conditions.
- Do not open the traction battery or attempt high-voltage repairs without specialised training and authorisation.
- If smoke, unusual heat, strong chemical odour, or battery damage appears, stop safely, move away, and contact emergency or authorised service personnel.

Care & storage

- Check tyre pressure, tread, brakes, lights, and washer fluid at recommended intervals.
- Keep the charging port and connector clean, dry, and free from debris.
- Follow the manufacturer's battery charging, parking, and maintenance recommendations.
- Arrange inspection by authorised service personnel when warnings, reduced performance, or unusual noises occur.

Common mistakes to avoid

- Driving without checking battery charge, tyres, brakes, or dashboard warnings.
- Using an incompatible or damaged charging cable.
- Ignoring high-voltage warnings or attempting unauthorised battery repairs.
- Leaving the vehicle in an unsuitable charging or parking condition contrary to the manual.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electric Vehicle Kit

Category: **Automotive** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Electric Vehicle Kit (two-wheeler); Electric Vehicle Training Kit

What it is used for

An electric vehicle training kit demonstrates the main electrical, electronic, and mechanical systems used in an electric two-wheeler or EV platform. Correct setup, controlled testing, isolation procedures, and supervision help learners understand EV operation while reducing electric shock, short-circuit, and battery hazards.

Main parts

- Traction battery or low-voltage training power source supplies electrical energy.
- Motor and drive assembly demonstrate propulsion.
- Motor controller or inverter regulates motor operation.
- Throttle or command input provides a driving request.
- DC-DC converter demonstrates voltage conversion where included.
- Wiring harness and connectors link system components.
- Display or instrument panel shows operating information.
- Brake switches and sensors demonstrate safety inputs where included.
- Charging port or charger interface demonstrates charging functions where supplied.
- Emergency stop and isolation device disconnect power where fitted.

Before you start

1. Only trained instructors and authorised learners should use the kit under supervision.
2. Read the kit manual and identify whether the setup uses low-voltage training power or a traction-voltage battery.
3. Inspect the frame, battery, motor, controller, connectors, cables, charger, guards, and emergency stop for damage.
4. Confirm the kit's voltage, current, and power ratings and verify that all components are compatible.
5. Use the specified charger and check battery condition; do not use a swollen, leaking, overheated, or damaged battery.
6. Establish a clear work area and identify the isolation point, emergency stop, and required PPE.
7. For traction-voltage systems, only qualified and authorised personnel may perform high-voltage work using the prescribed isolation and verification procedure.

How to use

1. Place the training kit on a stable surface and secure any moving assembly according to the manual.

2. Confirm the emergency stop and isolation controls are accessible and functional before energising the kit.
3. With power isolated, inspect wiring diagrams and identify the battery, controller, motor, sensors, and communication connections.
4. Connect only the approved low-voltage supply or battery using the specified polarity and sequence.
5. Check connector locking, cable routing, protective covers, and any required fuses before starting.
6. Energise the kit according to the instructor's procedure and observe the display for warnings or fault indicators.
7. Demonstrate throttle input, motor response, sensor readings, and control functions at the approved training level.
8. Keep the motor output and moving components within the kit's specified operating limits during tests.
9. Use diagnostic software or measurement instruments only according to the training instructions and equipment ratings.
10. If abnormal heating, smoke, unusual noise, unexpected motor movement, or a fault occurs, activate the emergency stop and isolate power when safe.
11. After testing, return controls to the safe state, switch off the supply, and follow the specified shutdown sequence.
12. Disconnect and store accessories as instructed, record test results, and report faults before the next session.

⚠ Safety

- Never bypass the emergency stop, interlocks, fuses, or other protective devices.
- Do not touch exposed conductors or connect or disconnect power cables while energised unless the approved procedure explicitly permits it.
- Never short-circuit a battery or use an unapproved charger or power supply.
- Keep hands, clothing, and tools away from rotating motors, wheels, belts, and drive components.
- Do not work on traction-voltage circuits without required qualifications, isolation, voltage verification, and appropriate PPE.
- Stop use immediately if the battery is swollen, leaking, smoking, unusually hot, or physically damaged; follow the site's battery emergency procedure.

Care & storage

- Inspect cables, connectors, battery enclosure, motor mountings, and protective devices before each training session.
- Keep the kit clean, dry, and free from conductive debris or loose metal objects.
- Charge and store batteries according to the manufacturer's temperature and state-of-charge guidance.
- Update training software and maintain measurement equipment according to approved procedures.
- Remove damaged equipment from service and arrange authorised inspection or repair.

Common mistakes to avoid

- Connecting components without checking the wiring diagram, polarity, and voltage ratings.
- Using a battery or charger that is incompatible with the kit.
- Testing motor functions with people or objects inside the moving-component zone.
- Attempting high-voltage repairs without qualification or proper isolation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Engine

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Engines and fuel system; Four stroke working petrol engines or diesel engines model

What it is used for

An engine training model demonstrates how a petrol or diesel engine converts fuel energy into mechanical power and how its fuel system operates. Correct inspection, controlled starting, and safe observation help learners understand engine components and the four-stroke cycle without unnecessary risk.

Main parts

- Cylinder block houses the cylinders and major engine components.
- Cylinder head contains valves and combustion-chamber components.
- Piston and connecting rod transfer combustion force to the crankshaft.
- Crankshaft converts reciprocating motion into rotation.
- Camshaft operates valves in a four-stroke engine.
- Intake and exhaust systems manage airflow and exhaust gases.
- Fuel system supplies petrol or diesel to the engine.
- Cooling and lubrication systems help control temperature and friction.
- Starter, battery, and ignition components support starting where fitted.
- Training controls, gauges, and cutaway panels show operation and measurements.

Before you start

1. Only trained instructors or authorised technicians should operate the engine model.
2. Identify whether the model is petrol or diesel and review its operating manual.
3. Inspect the engine mount, guards, fuel lines, wiring, battery connections, and controls for damage or leakage.
4. Confirm that the area is well ventilated and exhaust extraction is available for a running engine.
5. Check engine oil, coolant, and fuel levels according to the manual.
6. Ensure the transmission or output shaft is in the specified safe state and the work area is clear.
7. Keep a suitable fire extinguisher accessible and remove ignition sources from fuel-handling areas.

How to use

1. Explain the main engine components and identify the intake, compression, power, and exhaust strokes.
2. Confirm that all guards, connections, fluid levels, and safety devices are ready for operation.
3. For a petrol model, follow the specified ignition and starting procedure; for a diesel model, follow its prescribed starting and preheating procedure where fitted.

4. Start the engine from the designated operating position and observe the initial gauges and warning indicators.
5. Allow the engine to run at the approved idle speed while checking for unusual noise, vibration, smoke, or leakage.
6. Use the training controls or cutaway model to observe the four-stroke sequence and fuel-system operation.
7. Record readings such as engine speed, temperature, pressure, or fuel-system measurements where instruments are provided.
8. Change operating settings only within the instructor's approved limits and according to the manual.
9. Stop the engine immediately if a warning indicator, fuel leak, abnormal noise, or unsafe condition appears.
10. Shut down using the prescribed procedure and wait for hot and moving parts to become safe before inspection.
11. Record observations, report faults, and restore the training area after the session.

⚠ Safety

- Never run an engine in an enclosed or poorly ventilated area because exhaust gases can be fatal.
- Keep hands, clothing, hair, and tools away from belts, pulleys, fans, and rotating shafts.
- Do not smoke or create sparks near fuel, fuel vapour, or open fuel-system components.
- Do not touch the exhaust manifold or other hot parts during or immediately after operation.
- Never open a hot cooling system or remove a pressurised cap.
- Isolate the battery and other energy sources as specified before electrical or mechanical servicing.

Care & storage

- Check oil, coolant, fuel lines, filters, and visible connections at the scheduled interval.
- Clean the training model and keep air intakes and cooling passages unobstructed.
- Inspect belts, hoses, wiring, mounts, guards, and battery terminals regularly.
- Use the specified fluids and approved service procedures for the engine type.
- Arrange qualified maintenance for leaks, abnormal smoke, starting faults, or inconsistent readings.

Common mistakes to avoid

- Running the engine without adequate ventilation or exhaust extraction.
- Confusing petrol-engine ignition procedures with diesel-engine starting procedures.
- Touching moving or hot components during inspection.
- Ignoring fluid leaks, warning indicators, or abnormal engine noise.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Equipment Stand

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Equipment stands

What it is used for

To support automotive equipment securely at a suitable working height during inspection or maintenance. To prevent tipping, movement, and equipment damage by using a correctly rated stand.

Main parts

- Base or support feet
- Vertical post or frame
- Equipment mounting platform
- Height adjustment mechanism, if fitted
- Locking pins or fasteners

Before you start

1. Confirm the stand's load rating and compatibility with the equipment.
2. Inspect the base, frame, fasteners, and locking mechanism for damage.
3. Place the stand on a firm, level surface.
4. Check that the equipment's centre of gravity can be supported safely.
5. Use suitable lifting equipment and trained assistance for heavy loads.

How to use

1. Position the stand in the designated work area.
2. Adjust the height before placing the load whenever possible.
3. Secure any adjustable sections with the specified locking mechanism.
4. Place the equipment centrally on the support platform.
5. Fasten the equipment using the approved mounting points.
6. Check stability and confirm that the stand does not rock or shift.
7. Remove the load before adjusting or moving the stand.

⚠ Safety

- Never exceed the rated load capacity.
- Do not work under equipment supported only by an unstable stand.
- Keep hands and feet clear of pinch points during loading.
- Do not move a loaded stand unless it is designed and approved for that purpose.

Care & storage

- Clean the platform and base after use.
- Inspect welds, fasteners, locks, and support feet regularly.
- Lubricate adjustment mechanisms only as specified.
- Store the stand in a dry area and remove damaged units from service.

Common mistakes to avoid

- Using a stand with insufficient load capacity.
- Placing the stand on uneven ground.
- Leaving height-adjustment locks unsecured.
- Moving or adjusting the stand while loaded.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Exhaust Gas Analyzer

Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: CO Machine /Emission check machine

What it is used for

An exhaust gas analyzer measures selected gases in vehicle exhaust, commonly carbon monoxide (CO), carbon dioxide (CO2), oxygen (O2), and hydrocarbons (HC), depending on the instrument. Correct probe placement, engine preparation, and instrument checks help assess emissions and support maintenance or compliance testing under the applicable procedure.

Main parts

- Analyzer unit and display
- Exhaust sampling probe and flexible hose
- Gas sampling filter and water trap
- Calibration or zero-check controls
- Engine speed or temperature input where fitted
- Printer, data interface, or results storage

Before you start

1. Operate the analyzer only if trained and authorised for vehicle emissions testing.
2. Confirm the vehicle fuel type, engine configuration, test procedure, and applicable emission limits.
3. Inspect the analyzer, probe, hose, filters, water trap, power supply, and connectors for damage or blockage.
4. Check calibration status, service date, and the instrument's required warm-up and self-check procedure.
5. Work in a well-ventilated area with a suitable exhaust extraction system; never run an engine in an enclosed space without adequate exhaust removal.
6. Inspect the exhaust system for major leaks and confirm that the vehicle can be tested safely under the prescribed conditions.

How to use

1. Park the vehicle in the designated test area, apply the parking brake, and secure it according to the workshop procedure.
2. Ensure the exhaust extraction system is operating and keep people away from direct exhaust discharge.
3. Switch on the analyzer and complete the manufacturer's warm-up, zero, leak, and readiness checks as applicable.

4. Select the correct vehicle and test mode, including the required measurement units and engine conditions.
5. Prepare the engine according to the approved test procedure and allow it to reach the required operating temperature where specified.
6. Insert the sampling probe into the exhaust outlet to the depth and position specified by the analyzer or test procedure.
7. Ensure the probe and hose are secure and that the sample path is not kinked, blocked, or drawing in excessive ambient air.
8. Allow the reading to stabilise and monitor the displayed gases and any instrument warnings.
9. Follow the prescribed idle or elevated-speed test sequence without exceeding the vehicle manufacturer's operating limits.
10. Record the measured values, engine conditions, vehicle identification, test mode, and any required inspection details.
11. If readings fluctuate abnormally, check for leaks, probe position, filter blockage, water-trap condition, and instrument warnings before repeating the test.
12. Remove the probe carefully after the measurement sequence and allow the analyzer to purge or recover as instructed.
13. Switch off the analyzer according to the manual, then clean or replace filters and empty the water trap using the approved procedure.
14. Report results against the applicable limits and refer failed or suspicious results for further vehicle diagnosis.

⚠ Safety

- Exhaust gases contain harmful substances, including carbon monoxide, which is colourless and potentially fatal; use adequate ventilation and exhaust extraction.
- Never perform engine-running tests in a closed or poorly ventilated area.
- Keep the sampling probe, hose, and hands away from hot exhaust components and moving engine parts.
- Use suitable eye protection and workshop PPE, and secure loose clothing around rotating components.
- Do not smoke or introduce ignition sources near fuel vapours or flammable workshop materials.
- Never exceed the manufacturer's engine-speed or test-duration limits.
- Stop the test if exhaust extraction fails, a gas alarm activates, the operator feels unwell, or unsafe engine behaviour occurs.

Care & storage

- Inspect the probe, hose, filters, water trap, and connectors after use.
- Replace filters and drain collected moisture at the intervals specified by the manufacturer.
- Perform calibration, leak checks, and gas-zero checks according to the instrument's maintenance schedule.
- Store the analyzer and probe in a clean, dry area protected from impact and contamination.

- Arrange servicing if readings drift, response becomes slow, or the instrument repeatedly fails self-checks.

Common mistakes to avoid

- Testing before the analyzer completes its required warm-up or readiness checks.
- Using a leaking or blocked sampling hose or incorrectly positioned probe.
- Ignoring ventilation and carbon monoxide exposure risks.
- Comparing readings with limits for the wrong vehicle type, fuel, or test mode.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Exhaust System

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Emission and exhaust system

What it is used for

An automotive exhaust system carries engine exhaust gases away and may include components that reduce emissions and noise. Inspect it according to the vehicle service manual and repair faults safely.

How to use

1. Park the vehicle in a well-ventilated service area and apply the parking brake.
2. Allow the exhaust system to cool before inspection.
3. Inspect pipes, joints, hangers, muffler, and emission-control components for leaks or damage.
4. Check for unusual noise, visible corrosion, loose fittings, or warning indicators.
5. Use approved diagnostic equipment and service procedures when checking emissions or sensors.
6. Repair or replace faulty components according to the service manual.
7. Run the vehicle only when safe and verify that no exhaust leaks remain.

⚠ Safety

- Never run an engine in an enclosed or poorly ventilated area due to carbon monoxide risk.
- Do not touch hot exhaust pipes, catalytic converters, or related components.
- Use suitable eye protection and gloves during inspection.
- Do not bypass or remove required emission-control equipment.

Care & storage

- Inspect exhaust joints and mounts periodically and follow the vehicle's maintenance schedule.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Four Wheeler

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Four wheelers old second hand vehicle; Four-wheeler old second-hand Vehicle; Serviceable training Vehicle

What it is used for

A four-wheeler training vehicle is used to teach vehicle inspection, basic maintenance, servicing, and automotive system diagnosis on a serviceable older vehicle. Correct workshop preparation, safe lifting and isolation, and manufacturer-guided procedures help learners develop practical skills while reducing injury and vehicle damage.

Main parts

- Engine and fuel system provide propulsion and engine operation.
- Transmission and drivetrain transfer engine power to the wheels.
- Steering system controls vehicle direction.
- Brake system slows and stops the vehicle.
- Suspension and tyres support the vehicle and maintain road contact.
- Battery and electrical system supply and distribute electrical power.
- Lights, instruments, and warning indicators show vehicle status.
- Vehicle lift, jack points, and wheel chocks support safe workshop access where applicable.
- Diagnostic connector provides access to compatible vehicle diagnostic systems.

Before you start

1. Only trained instructors and authorised learners should perform vehicle servicing under supervision.
2. Confirm the vehicle identity, service history, condition, and available manufacturer service information.
3. Park on firm, level ground, apply the parking brake, select the appropriate gear or Park, and chock the wheels as needed.
4. Inspect the vehicle for fluid leaks, damaged tyres, exposed wiring, and other obvious hazards.
5. Remove the ignition key and isolate the battery when the task requires electrical or mechanical isolation.
6. Use a suitable vehicle lift or approved jack points and rated axle stands for underbody work.
7. Wear suitable workshop clothing, safety footwear, and task-specific PPE; ensure the workshop is ventilated.

How to use

1. Conduct a walk-around inspection and record visible body, glass, tyre, and lighting defects.
2. Check fluid levels, battery condition, belts, hoses, and accessible connections according to the service plan.

3. Identify the engine, transmission, brake, steering, suspension, and electrical components before servicing.
4. Connect a compatible diagnostic tool only when required and follow the vehicle manufacturer's instructions.
5. Perform the assigned inspection or maintenance task using the correct tools and service specifications.
6. For underbody work, raise the vehicle only at approved lift points and verify the support is secure before access.
7. Keep removed fasteners and parts organised and prevent dirt from entering open systems.
8. Reinstall all parts, tighten fasteners to specified torque where required, and restore disconnected connections correctly.
9. Check fluid levels, tool clearance, guards, connectors, and the work area before restarting the vehicle.
10. Start the engine only when safe and check for warning lights, leaks, abnormal noise, or other faults.
11. Conduct the required functional checks or supervised road test only after the vehicle is confirmed safe and legally permitted.
12. Record work completed, measurements, faults, replaced parts, and any follow-up maintenance needed.

⚠ Safety

- Never work beneath a vehicle supported only by a jack; use correctly rated stands or an approved lift.
- Keep clear of moving belts, fans, hot exhaust parts, and rotating components when the engine is running.
- Use suitable exhaust extraction and never run the engine in a closed workshop without adequate ventilation.
- Handle fuel, oils, coolant, brake fluid, and batteries according to their safety instructions.
- Isolate the battery and follow manufacturer procedures before work on electrical or restraint systems.
- Do not road-test an unsafe, unregistered, uninsured, or otherwise unauthorised vehicle.
- Use approved lifting points and never exceed the lift, jack, or stand capacity.

Care & storage

- Clean the vehicle and return tools to their designated storage after the task.
- Inspect tyres, brakes, steering, lights, fluid levels, and visible safety-critical parts at the scheduled interval.
- Dispose of used oil, filters, coolant, batteries, and other workshop waste through approved channels.
- Maintain service records and report defects that could affect vehicle safety.
- Arrange qualified repairs when a defect is outside the learner's authorised scope.

Common mistakes to avoid

- Working beneath a vehicle without rated stands or an approved lift.
- Using incorrect service specifications or tightening fasteners without the required torque guidance.
- Restarting the vehicle with tools, loose parts, or disconnected components remaining.
- Ignoring leaks, warning indicators, tyre defects, or brake and steering problems.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fuel

Category: **Automotive** Type: **Handling note** Priority: **High** 3 labs

Covers these lab items: Petrol/diesel

What it is used for

Petrol and diesel are fuels used in compatible engines and equipment and require strict fire and spill precautions.

How to handle

- Confirm the correct fuel type and use approved, compatible containers and transfer equipment.
- Keep away from flames, sparks, smoking, and hot surfaces; shut down equipment before refuelling when required.
- Work in a well-ventilated area and avoid inhalation, skin contact, and fuel spills.
- Clean spills using the approved procedure and report leaks immediately.

Storage

- Store in approved, clearly labelled fuel containers in a designated, ventilated area away from ignition sources.
- Follow applicable fire-safety, quantity, and separation requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fuel Pressure Tester

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: fuel pressure testers

What it is used for

A fuel pressure tester measures fuel-system pressure to help diagnose delivery problems. Use the correct tester and follow the vehicle manufacturer's service procedure.

How to use

1. Identify the vehicle fuel system and confirm tester compatibility and pressure range.
2. Park safely, switch off the ignition, and allow hot components to cool.
3. Wear suitable eye protection and gloves, and keep an appropriate fire extinguisher accessible.
4. Relieve fuel-system pressure using the specified service procedure.
5. Connect the tester to the designated test port with the correct fittings.
6. Restore the system as instructed, check for leaks, and measure pressure under specified conditions.
7. Switch off the engine or ignition, relieve pressure safely, and disconnect the tester as directed.

⚠ Safety

- Work in a well-ventilated area away from flames, sparks, and smoking.
- Never disconnect fuel lines while the system is pressurized.
- Avoid fuel contact with skin and eyes; clean spills promptly using approved procedures.
- Follow special precautions for high-pressure direct-injection and diesel systems.

Care & storage

- Clean the tester, inspect hoses and seals, and store it with fittings protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fuel System

Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fuel supply System & Urea Handling (Diesel)

What it is used for

An automotive fuel supply system stores, filters, and delivers fuel to the engine at the pressure and flow required for operation. Diesel systems may also include a separate urea-based diesel exhaust fluid (DEF/AdBlue) system for emissions control; correct identification, safe handling, and leak checks help prevent fire, contamination, and system damage.

Main parts

- Fuel tank and filler assembly
- Fuel lines, hoses, and connectors
- Fuel filter and water separator where fitted
- Fuel pump and pressure-control components
- Injectors or fuel rail depending on engine design
- Diesel exhaust fluid (DEF/AdBlue) tank, pump, lines, and dosing injector where fitted

Before you start

1. Only trained automotive personnel should inspect or service fuel and diesel exhaust fluid systems.
2. Identify the vehicle and engine type, fuel specification, pressure hazards, and manufacturer's service procedure.
3. Park the vehicle in a ventilated workshop, apply the parking brake, and allow hot components to cool as required.
4. Inspect the fuel tank, lines, connectors, filters, pump, and DEF components for visible leaks, damage, or corrosion without touching suspected high-pressure leaks.
5. Confirm that the correct fuel and DEF specification are available and that containers are clean and clearly labelled.
6. Review the required pressure-relief, electrical-isolation, and spill-response procedures before opening any system.

How to use

1. Identify the system layout and locate the fuel supply components and, where fitted, the separate DEF tank and dosing system.
2. Switch off the engine and follow the manufacturer's procedure to isolate electrical power and relieve fuel pressure before servicing.
3. Place suitable drip trays beneath the work area and keep approved absorbent material available for small spills.

4. Inspect low-pressure fuel hoses, clamps, connectors, filters, and tank fittings for damage or leakage.
5. For high-pressure diesel injection systems, follow the exact manufacturer procedure; never loosen a high-pressure line while the engine is running or pressure remains present.
6. Replace fuel filters using the specified part and cleanliness procedure, preventing dirt or water from entering the fuel system.
7. Prime the fuel system using the approved method and check for leaks before starting the engine.
8. Where fitted, inspect DEF tank, filler cap, pump, lines, heater, and dosing components for leaks, crystallisation, or damage.
9. Fill the DEF tank only with the specified diesel exhaust fluid from a clean, dedicated container; never add DEF to the diesel fuel tank or diesel fuel to the DEF tank.
10. Follow the manufacturer's procedure for DEF service, priming, or dosing-system checks; do not mix DEF with other fluids or add unapproved additives.
11. Start the engine only after all serviced components are correctly reassembled and tools have been removed from the engine bay.
12. Observe system operation and warning indicators, and check for leaks without placing hands near high-pressure fuel components.
13. If a leak, warning, abnormal pressure, or engine fault appears, stop the engine when safe and investigate using the approved diagnostic procedure.
14. Clean small spills using the appropriate material, collect contaminated waste, and dispose of it according to workshop and environmental requirements.
15. Record parts replaced, fluids used, test results, and any faults requiring further repair.

⚠ Safety

- Diesel fuel is combustible; keep flames, sparks, smoking, and other ignition sources away from fuel handling areas.
- High-pressure diesel fuel can penetrate skin and cause serious injury; never search for leaks with bare hands and seek immediate medical attention after suspected injection injury.
- Relieve system pressure and follow the manufacturer's isolation procedure before disconnecting fuel lines.
- Wear suitable gloves and eye protection and avoid skin contact or inhalation of fuel and DEF mist.
- Never mix diesel exhaust fluid with fuel, coolant, oil, or other chemicals; DEF contamination can damage emissions equipment.
- Clean spills promptly, prevent fluids from reaching drains, and keep absorbents and appropriate fire equipment available.
- Do not work beneath a vehicle supported only by a jack; use approved lifting supports and follow workshop safety rules.

Care & storage

- Inspect fuel hoses, connectors, filters, tank mounts, and seals at the recommended service intervals.

- Replace fuel filters and water separators according to the vehicle manufacturer's schedule.
- Keep fuel and DEF containers sealed, labelled, and protected from dirt and contamination.
- Clean DEF crystals using the manufacturer's approved method and prevent contamination of connectors and dosing parts.
- Arrange qualified diagnosis if pressure, flow, emissions warnings, or recurring leaks remain abnormal.

Common mistakes to avoid

- Opening a high-pressure fuel line before following the pressure-relief procedure.
- Allowing dirt or water into the fuel system during filter replacement.
- Filling the DEF tank with diesel or adding DEF to the fuel tank.
- Ignoring leaks, warning lights, or contamination and returning the vehicle to service without checks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Funnel

Category: **Automotive** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Funnel

What it is used for

Use funnels safely for transferring automotive fluids without spills.

How to handle

- Confirm the funnel is compatible with the fluid being transferred.
- Use a clean, dry funnel and avoid mixing different fluids.
- Switch off the engine and allow hot components to cool before pouring.
- Prevent spills, wipe them promptly, and keep fluids away from ignition sources.

Storage

- Clean the funnel after use according to the fluid type and disposal rules.
- Store dry in a labelled area away from dust and incompatible chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grease Gun

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Grease gun

What it is used for

A grease gun delivers lubricant into designated grease fittings on compatible machinery or vehicle components. Use the specified grease and apply the recommended quantity and pressure.

How to use

1. Identify the grease points and confirm the lubricant specification in the service manual.
2. Inspect the grease gun, hose, coupler, and fittings for damage or leaks.
3. Load the gun with the specified grease using the approved filling method.
4. Wipe dirt from the grease fitting before attaching the coupler.
5. Connect the coupler securely and operate the handle to deliver the recommended amount.
6. Stop if abnormal resistance, leakage, or seal damage is observed.
7. Release pressure as instructed, disconnect the coupler, and wipe away excess grease.

⚠ Safety

- Wear suitable gloves and eye protection when handling grease.
- Never point a grease gun at skin; high-pressure grease can cause serious injection injuries.
- Do not exceed the component's recommended pressure or lubricant quantity.
- Keep grease away from hot surfaces and dispose of waste properly.

Care & storage

- Clean the gun and coupler, inspect the hose, and store it in a clean location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hydrometer

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Hydrometer

What it is used for

A hydrometer measures the relative density of a liquid, such as battery electrolyte, when the instrument is designed for that liquid. Use it only with the correct fluid type and manufacturer instructions.

How to use

1. Confirm that the hydrometer is suitable for the liquid being tested.
2. Inspect the body, scale, float, and seals for cracks or damage.
3. Wear the required protective equipment and prepare a clean sampling area.
4. Draw the liquid into the hydrometer as instructed, avoiding bubbles and spills.
5. Allow the float to settle and read the scale at eye level.
6. Return or dispose of the sample according to the procedure and clean the instrument.

⚠ Safety

- For battery electrolyte, wear chemical-resistant gloves and eye protection.
- Avoid skin or eye contact with corrosive liquids and follow the safety data instructions.
- Do not use a damaged hydrometer or mix samples between containers.
- Never return a contaminated sample to the original container.

Care & storage

- Rinse or clean the hydrometer as specified, dry it appropriately, and store it safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Inspection Mirror

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Mirror (inspection)

What it is used for

To inspect hidden, obstructed, or hard-to-see vehicle components without unnecessary disassembly.

How to handle

- Inspect the mirror, handle, extension shaft, joints, and any built-in light for damage.
- Park the vehicle securely, apply the parking brake, and allow hot components to cool before inspection.
- Extend and angle the mirror to view the required area while keeping hands clear of moving or hot parts.
- Use adequate lighting and avoid placing the mirror or hands near belts, fans, or other moving components.
- Clean the mirror after use and retract the handle carefully.

Storage

- Wipe the mirror clean and dry; retract the shaft if adjustable.
- Store in a padded tool holder or designated drawer to protect the mirror surface from scratches.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Jack

Category: **Automotive** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Jack hydraulic; Wheel jack

What it is used for

A jack lifts a vehicle or other approved load for limited access during inspection, wheel changes, or maintenance. Correct load rating, stable positioning, and suitable support stands are essential to prevent the load from falling.

Main parts

- Lifting saddle or pad contacts the approved lifting point.
- Hydraulic cylinder raises the load on hydraulic jack models.
- Pump handle operates the hydraulic lifting mechanism.
- Release valve lowers the load in a controlled manner.
- Base supports the jack on the ground.
- Frame and lifting arm transmit force to the load.
- Wheels or rollers allow positioning on applicable models.
- Load rating label specifies the jack's permitted capacity.

Before you start

1. Read the jack and vehicle manuals and identify the correct lifting points.
2. Check the jack's load rating against the vehicle or load requirements.
3. Inspect the frame, saddle, handle, hydraulic seals, wheels, and release valve for damage or leaks.
4. Park the vehicle on firm, level ground and switch off the engine or drive system.
5. Apply the parking brake and use wheel chocks on wheels remaining on the ground.
6. Keep people clear of the vehicle and prepare correctly rated support stands.

How to use

1. Position the jack on firm, level ground beneath the approved lifting point.
2. Ensure the saddle is centred and makes secure contact with the lifting point.
3. Close the release valve according to the jack instructions before pumping.
4. Operate the handle smoothly and raise the load only to the required height.
5. Check that the load remains stable and the jack stays upright during lifting.
6. Place correctly rated support stands at the manufacturer's approved support points.
7. Lower the load slowly onto the support stands and verify stable contact before working underneath.
8. For lowering, remove people and tools from the hazard area and raise the load slightly to unload the stands.
9. Remove the stands and lower the load slowly using the release valve as instructed.

10. Remove the jack, check for leaks or damage, and store it safely.

⚠ Safety

- Never work under a vehicle supported only by a jack; use correctly rated support stands.
- Never exceed the jack's rated capacity or use it on soft, sloping, or unstable ground.
- Do not place any part of the body under a load while it is being raised or lowered.
- Use only the manufacturer's approved lifting points and keep the saddle correctly positioned.
- Do not use a jack with hydraulic leaks, damaged parts, or an unreadable rating label.
- Keep bystanders clear and never move a vehicle while it is supported by a jack.

Care & storage

- Wipe the jack clean after use and keep the lifting saddle free of debris.
- Inspect hydraulic seals, hoses where fitted, wheels, frame, and release valve regularly.
- Maintain hydraulic fluid only according to the manufacturer's instructions and specified fluid type.
- Store the jack in a dry place and arrange servicing for leaks or faulty lowering.

Common mistakes to avoid

- Lifting from an unapproved vehicle point or on unstable ground.
- Exceeding the jack's load rating.
- Working beneath a vehicle without support stands.
- Opening the release valve too quickly and lowering the load suddenly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lifting & Jacking Equipment

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Lifting and jacking equipment

What it is used for

Lifting and jacking equipment raises and supports vehicles or heavy components for inspection, repair, and maintenance. Correct equipment selection, rated load limits, approved lifting points, and secure support are essential to prevent crushing injuries and equipment damage.

Main parts

- Vehicle lift or jack raises the vehicle or component.
- Hydraulic pump or screw mechanism generates lifting force.
- Lifting arms, pads, or saddle transfer force to approved lifting points.
- Axle stands or support stands hold the raised load where required.
- Release valve or lowering control lowers the load in a controlled manner.
- Wheel chocks prevent unintended vehicle movement.
- Locking mechanism or safety latch helps secure a raised lift.
- Control switch or handle operates the equipment.

Before you start

1. Only trained operators should use lifting and jacking equipment.
2. Identify the vehicle or load weight and select equipment with a suitable rated capacity.
3. Inspect the jack, lift, arms, pads, saddle, hoses, pins, locks, and stands for damage or leakage.
4. Check the floor is firm, level, and suitable for the equipment and load.
5. Locate the manufacturer's approved lifting points and check the vehicle load distribution.
6. Park the vehicle securely, apply the parking brake, select Park or the appropriate gear, and chock the wheels as required.
7. Keep people, tools, and loose items away from the lifting area.

How to use

1. Position the jack, lift, or lifting arms at the approved lifting points.
2. Confirm that pads or saddles contact the intended points securely and are not resting on weak or damaged components.
3. Raise the load slightly and pause to check balance, contact, and stability.
4. Continue lifting slowly while observing the load and equipment for movement, tilting, or unusual sounds.
5. Engage the lift's mechanical locks where provided and follow the manufacturer's instructions.
6. For jacking equipment, lower the load onto correctly rated axle stands at approved support points before working underneath.

7. Verify the load is stable and the supports are correctly positioned before accessing the underside.
8. Keep the lifting area clear and never move the equipment with a raised load unless the equipment is specifically designed for it.
9. When work is complete, remove tools and people from the danger zone and raise the load slightly if needed to release support stands safely.
10. Lower the load slowly using the designated release control and ensure the area is clear.
11. Remove the equipment, store it securely, and report any defects or unusual operation.

⚠ Safety

- Never work under a vehicle supported only by a hydraulic jack or lifting mechanism without the required mechanical supports.
- Never exceed the rated capacity of the jack, lift, stands, or lifting accessories.
- Do not lift a vehicle at unapproved points or on unstable, sloping, or soft ground.
- Keep hands and feet away from pinch points and never place any part of the body under an unsecured load.
- Do not bypass safety locks, overload protection, or interlocks.
- Stop immediately if the load shifts, the equipment leaks, or stability becomes uncertain; clear the area and follow the emergency procedure.

Care & storage

- Clean lifting pads, saddles, and contact points after use.
- Inspect hydraulic hoses, cylinders, seals, wheels, pins, locks, and stands regularly.
- Check hydraulic fluid and lubricate designated points according to the maintenance schedule.
- Keep equipment dry and store it in the designated location with moving parts lowered safely.
- Arrange authorised inspection and repair for leaks, deformation, damaged threads, or unreliable locking mechanisms.

Common mistakes to avoid

- Using a jack or lift with insufficient rated capacity.
- Positioning lifting pads away from the manufacturer's approved points.
- Working underneath a load without correctly rated mechanical support stands.
- Lowering the load while people, tools, or other obstacles remain in the danger zone.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lug Wrench

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Lug wrenches

What it is used for

To loosen and tighten vehicle wheel lug nuts during wheel removal or installation.

How to handle

- Inspect the lug wrench for cracks, bending, or damaged socket ends before use.
- Park the vehicle on firm, level ground, apply the parking brake, and secure the wheels as appropriate.
- Use the correct socket size and loosen lug nuts slightly before lifting the vehicle, following the vehicle manual.
- Support the vehicle with approved jack stands before removing the wheel; never rely on a jack alone.
- Tighten lug nuts in the specified sequence and to the manufacturer's torque using a calibrated torque wrench where required.

Storage

- Wipe clean and store in a dry tool cabinet or designated rack.
- Keep socket ends protected from rust and avoid placing heavy loads on the wrench.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Magnetic Tool Holder

Category: **Automotive** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Magnetic bar

What it is used for

A magnetic tool holder keeps compatible metal tools organised and accessible in workshops.

How to handle

- Inspect the magnetic bar and mounting points for damage or looseness.
- Mount securely on a suitable surface according to the manufacturer's instructions.
- Place compatible tools evenly and ensure they are held firmly.
- Keep tools away from sensitive electronic devices, magnetic media, and equipment affected by magnets.
- Do not overload the holder or place sharp tools where they may fall.

Storage

- Keep the holder clean and dry.
- Remove tools before cleaning or servicing the holder.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Magnetic Tray

Category: **Automotive** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Magnetic Tray

What it is used for

Use a magnetic tray to keep small metal parts organised during automotive work.

How to handle

- Place the tray on a stable, suitable surface away from moving machinery.
- Keep screws, nuts, and other metal parts inside the tray during work.
- Do not use the tray near sensitive magnetic devices unless approved.
- Clean grease and debris from the tray after use.

Storage

- Store clean and dry in the workshop tool area.
- Keep away from electronic equipment that may be affected by magnets.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oil Can

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Oil can; Oil cane

What it is used for

Use an oil can to apply small, controlled amounts of the specified lubricant to suitable machine or vehicle components.

How to use

1. Confirm the lubricant type is correct for the component and task.
2. Inspect the oil can, nozzle, cap, and body for leaks or damage.
3. Fill the can with clean lubricant and close the cap securely.
4. Identify the approved lubrication point and clean away loose dirt.
5. Apply a small amount of oil without overfilling the component.
6. Wipe away excess lubricant and check for leaks or spills.
7. Clean the nozzle and store the oil can after use.

⚠ Safety

- Do not apply oil to hot, moving, or energized parts unless an approved procedure allows it.
- Keep oil away from brakes, belts, tires, and other friction surfaces.
- Wear suitable gloves and clean spills promptly to prevent slips.
- Follow the lubricant label for handling and disposal.

Care & storage

- Wipe the can and nozzle clean after use.
- Store it upright in a labeled place away from heat and contaminants.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oil Filter

Category: **Automotive** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Oil filter; Oil filters

What it is used for

Automotive oil filters remove contaminants from engine oil to help protect lubricated engine components.

How to handle

- Confirm the correct filter specification and vehicle compatibility before replacement.
- Switch off the engine and allow hot parts and oil to cool safely.
- Use suitable tools and follow the vehicle manufacturer's replacement procedure.
- Check the sealing surface and gasket; lubricate the seal if the manufacturer specifies.
- Dispose of used filters and drained oil according to local environmental requirements.

Storage

- Keep new filters sealed in clean, dry packaging.
- Store away from moisture, dust, crushing, and extreme temperatures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oil Measure Container

Category: **Automotive** Type: **Short SOP** Priority: **Low** 3 labs

Covers these lab items: Oil measure container; Oil measure container, funnel

What it is used for

An oil measure container and funnel help measure and transfer automotive oils with reduced spillage. Use clean, clearly marked equipment and follow the vehicle manufacturer's fluid specification.

How to use

1. Inspect the measuring container, markings, funnel, and cap for damage or contamination.
2. Confirm the required oil type, quantity, and specification from the service manual.
3. Place the vehicle on a stable surface and prepare the designated filling area.
4. Measure the oil at eye level using the correct scale and units.
5. Position the funnel securely and pour oil slowly into the specified fill opening.
6. Check the fluid level using the vehicle's prescribed method and add only if required.
7. Clean the container and funnel and dispose of oily waste appropriately.

⚠ Safety

- Keep oil away from hot engine parts and ignition sources.
- Wear suitable gloves and avoid prolonged skin contact with used oil.
- Do not mix incompatible fluids or use contaminated measuring equipment.
- Clean spills promptly to prevent slips and environmental contamination.

Care & storage

- Wipe and store the container and funnel separately from food or drinking-water equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oil Pressure Gauge

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Oil pressure gauges

What it is used for

To measure engine oil pressure using an appropriate oil pressure gauge. To compare the reading with the vehicle manufacturer's specified range.

Main parts

- Pressure gauge
- Connecting hose
- Adapter fittings
- Pressure port connector
- Gauge scale or display

Before you start

1. Park the vehicle securely and consult the service manual.
2. Confirm the gauge range and adapter suit the engine.
3. Inspect the hose, fittings, and gauge for damage.
4. Allow hot engine parts to cool before connecting the gauge.

How to use

1. Locate the specified oil pressure test port.
2. Switch off the engine before fitting the gauge.
3. Connect the correct adapter and tighten securely.
4. Start the engine according to the test procedure.
5. Observe pressure at the specified engine speed and temperature.
6. Compare readings with manufacturer specifications.
7. Switch off the engine before removing the gauge.
8. Reconnect the original fitting and check for leaks.

⚠ Safety

- Keep clear of hot oil and moving engine parts.
- Never remove a pressurized fitting while the engine is running.
- Use the correct pressure-rated hose and adapter.
- Stop the test if oil leaks or the gauge behaves abnormally.

Care & storage

- Clean the fittings after use.
- Inspect the hose and gauge regularly.
- Store the gauge in its protective case.

Common mistakes to avoid

- Using an incorrect adapter.
- Comparing readings without considering engine temperature or speed.
- Removing the gauge while the system is pressurized.
- Ignoring oil leakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oil Seal

Category: **Automotive** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: oil seals

What it is used for

Handle automotive oil seals carefully during inspection and installation.

How to handle

- Verify the seal size, material, and part number against the application.
- Inspect the sealing lip and surface for cuts, deformation, or contamination.
- Use suitable installation tools and avoid striking or twisting the seal excessively.
- Keep sealing surfaces clean and follow the manufacturer's fitting and lubrication instructions.

Storage

- Store seals in labelled packaging away from dust, direct sunlight, and heat.
- Keep away from oils, solvents, or chemicals that may damage the seal material.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Radiator

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Old second hand Radiator

What it is used for

To dissipate heat from engine coolant and help maintain the engine's operating temperature.

How to handle

- Inspect the radiator core, fins, tanks, hoses, mounting points, and cap for leaks, corrosion, or damage.
- Ensure the engine is switched off and completely cool before opening the cooling system; never remove a hot radiator cap.
- Check coolant level and condition according to the vehicle manufacturer's instructions.
- Use the specified coolant and approved filling procedure; keep hands clear of the cooling fan, which may start unexpectedly.
- After servicing, check for leaks and confirm correct operation; handle used coolant as hazardous waste.

Storage

- For a removed radiator, drain coolant into an approved container and cap open ports.
- Store clean and dry in a stable position, protecting fins and connections from impact and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Radiator Pressure Cap

Category: **Automotive** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Old second hand pressure cap

What it is used for

A radiator pressure cap regulates cooling-system pressure and helps maintain the correct coolant-system seal.

How to handle

- Never remove the cap while the engine or coolant is hot or pressurised.
- Allow the engine to cool completely and follow the vehicle manufacturer's service procedure.
- Inspect the cap seal, spring, and locking features for damage or corrosion.
- Use only the specified cap rating and compatible replacement part.
- If the cap is from a used vehicle, treat it as a training sample unless it has been inspected and tested before reuse.

Storage

- Store the cap clean and dry in a labelled container.
- Protect the sealing surface and spring from dirt, corrosion, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Slide Hammer

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: slide hammers

What it is used for

A slide hammer applies controlled impact force to pull suitable automotive components or attachments. Select the correct adapter and follow the vehicle service procedure for the specific repair.

How to use

1. Inspect the hammer shaft, sliding weight, handle, threads, and adapters for damage.
2. Confirm that the tool and adapter are suitable for the component being removed.
3. Secure the vehicle or workpiece and isolate relevant power systems where required.
4. Attach the correct adapter firmly to the slide hammer and component.
5. Keep the shaft aligned with the removal direction and maintain a stable grip.
6. Move the sliding weight in controlled strokes to apply pulling force as specified.
7. Inspect the removed component and tool, then clean and store the equipment.

⚠ Safety

- Wear safety glasses and suitable gloves during impact work.
- Keep hands and body clear of the sliding weight and impact zones.
- Do not exceed the tool's rated load or use damaged adapters.
- Follow vehicle-specific precautions around airbags, fuel systems, and electrical components.

Care & storage

- Clean threads and moving surfaces after use and inspect adapters before storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spark Plug

Category: **Automotive** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: spark plug; Spark plugs

What it is used for

Spark plugs ignite the air-fuel mixture in compatible spark-ignition engines.

How to handle

- Confirm the correct plug type, thread size, heat range, and gap specification for the engine.
- Allow the engine to cool and disconnect the ignition system as specified before servicing.
- Inspect the electrode and ceramic insulator for cracks, deposits, or damage.
- Use the correct socket and tightening torque specified by the vehicle manufacturer.
- Keep removed plugs organised and dispose of damaged plugs safely.

Storage

- Store new spark plugs in their original protective packaging.
- Keep them dry and protect the ceramic insulator and threads from impact or contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spark Plug Cleaner

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Spark plug cleaning machine

What it is used for

A spark plug cleaning machine removes carbon deposits and other suitable contaminants from spark plugs using the cleaning method specified for the equipment. Correct plug inspection, controlled cleaning, and post-cleaning checks help support engine maintenance while reducing dust, eye, and equipment hazards.

Main parts

- Machine housing encloses the cleaning mechanism.
- Spark plug holder secures the plug during cleaning.
- Cleaning chamber contains the cleaning process where fitted.
- Compressed-air inlet supplies air to pneumatic models.
- Abrasive or cleaning medium removes deposits where the machine uses it.
- Control switch or valve starts and stops the cleaning cycle.
- Pressure regulator controls air pressure where fitted.
- Inspection port or work light helps examine the plug where available.

Before you start

1. Only trained personnel should operate the spark plug cleaning machine.
2. Read the machine instructions and confirm the cleaner is suitable for the spark plug type.
3. Inspect the housing, plug holder, lid, hose, fittings, controls, and power cable for damage.
4. Check that the cleaning medium and operating pressure match the manufacturer's requirements.
5. Inspect the spark plug for cracked ceramic, damaged threads, worn electrodes, or other defects that require replacement rather than cleaning.
6. Ensure dust extraction or ventilation is operating where required and wear suitable eye and respiratory protection.
7. Disconnect power or air before opening the chamber or servicing the machine.

How to use

1. Switch off the engine and allow the spark plug to cool before removing it from the engine.
2. Inspect and identify the spark plug and check whether it is suitable for cleaning.
3. Secure the plug in the holder according to the machine instructions.
4. Close the cleaning chamber or lid and confirm the safety interlock is engaged where fitted.
5. Set the approved cleaning duration and pressure or operating mode.

6. Start the cleaning cycle from the designated control and keep the chamber closed during operation.
7. Stop the machine if abnormal noise, leakage, excessive vibration, or a chamber fault occurs.
8. Wait for the cycle to finish and release pressure according to the instructions before opening the chamber.
9. Remove the plug carefully and remove residual cleaning medium using the approved method.
10. Inspect the ceramic insulator, threads, electrodes, and electrode gap against the engine manufacturer's specifications.
11. Replace the spark plug if damaged or outside the service limits; do not return an unserviceable plug to use.
12. Clean the machine, check for residue, and record any equipment or plug defects.

⚠ Safety

- Never operate the machine with the chamber open or required guards disabled.
- Wear suitable eye protection and respiratory protection where dust or cleaning media may become airborne.
- Do not clean a spark plug with cracked ceramic, damaged threads, or other defects that make it unserviceable.
- Disconnect power or air before clearing blockages or opening the chamber for maintenance.
- Keep hands away from moving parts and high-pressure air outlets.
- Follow the manufacturer's restrictions on cleaning media and do not use flammable or unapproved substances.

Care & storage

- Remove residual cleaning medium from the chamber after isolating the machine.
- Inspect the plug holder, lid, seals, hoses, fittings, and controls regularly.
- Replace cleaning media and service filters according to the manufacturer's schedule.
- Keep the machine clean and dry and store accessories in their designated locations.
- Arrange authorised service if pressure control, interlocks, or cleaning performance becomes unreliable.

Common mistakes to avoid

- Attempting to clean a spark plug that should be replaced.
- Opening the cleaning chamber before the cycle and pressure release are complete.
- Using the wrong cleaning medium or excessive operating pressure.
- Returning a plug to service without checking its ceramic, electrodes, and gap.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Special Tool

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 10 labs

Covers these lab items: Security activator; special maintenance tools; Special tools; Suspension activation

What it is used for

Use a designated special automotive tool to perform a specific inspection, adjustment, activation, or maintenance task according to the vehicle manufacturer's instructions.

How to use

1. Identify the exact tool and the vehicle system it is designed for.
2. Read the tool instructions and relevant vehicle service procedure.
3. Inspect the tool, connectors, and accessories for damage or missing parts.
4. Park the vehicle securely and isolate power when the procedure requires it.
5. Connect or position the tool only at the specified service point.
6. Perform the prescribed operation and observe the system response or diagnostic result.
7. Disconnect the tool safely and confirm the vehicle system is returned to the required state.

⚠ Safety

- Use the tool only for its intended vehicle system and function.
- Follow manufacturer instructions for high-voltage, airbag, suspension, or electronic systems.
- Do not activate a system while anyone is near moving parts or stored-energy components.
- Stop and consult a qualified technician if the procedure or tool is unclear.

Care & storage

- Clean the tool and inspect cables or adapters after use.
- Store it with its accessories and instructions in a labeled case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stud Extractor

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Stud extractor; Stud remover; Stud remover with socket handle

What it is used for

To remove damaged, seized, or broken threaded studs from automotive components with minimal damage to surrounding material.

How to handle

- Inspect the extractor, socket handle, gripping mechanism, and compatible socket for wear or damage.
- Confirm the stud size, thread direction, and extractor compatibility before applying force.
- Secure the workpiece and wear eye protection and suitable gloves.
- Fit the extractor squarely onto the stud and apply controlled torque using the specified handle; avoid sudden force that may break the stud further.
- Clean the tool after use and inspect the removed threads and surrounding component for damage.

Storage

- Store clean and dry in a labelled automotive tool case with compatible sockets.
- Protect gripping surfaces from corrosion and keep the tool organised to avoid loss of small components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Suspension System

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Suspension Steering Wheel and Tyre Systems; Suspension system

What it is used for

A suspension system supports the vehicle body and helps absorb road shocks while maintaining tyre contact. Use training models or inspect actual vehicle components only under authorized supervision.

How to use

1. Park the vehicle on level ground, apply the parking brake, and switch off the engine.
2. Identify suspension components using the training model or approved service information.
3. Check visible springs, shock absorbers, linkages, and mounts for damage or leakage.
4. Observe steering and tyre movement only under an instructor's approved procedure.
5. Use specified tools and lifting equipment if any practical inspection requires vehicle access.
6. Report defects and restore all guards and components before returning equipment to service.

⚠ Safety

- Never work beneath a vehicle supported only by a jack.
- Do not dismantle loaded springs or pressurized suspension components without specialist training.
- Keep hands away from moving steering linkages and pinch points.
- Only trained and authorized personnel should lift or repair actual vehicles.

Care & storage

- Keep training models clean, report loose parts, and follow the vehicle maintenance schedule.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tappet Puller

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Tappet puller

What it is used for

To remove or reposition tappets or related valve-train components during approved engine maintenance. To apply controlled force using a compatible tappet puller and avoid damage to engine parts.

Main parts

- Puller body
- Gripping jaws or hooks
- Screw or pulling mechanism
- Handle or drive point
- Adjustment components

Before you start

1. Confirm the engine model and the correct puller type.
2. Switch off the engine and follow the service manual isolation procedure.
3. Allow hot engine components to cool.
4. Inspect the puller for cracks, wear, or damaged threads.
5. Prepare the specified tools and protect nearby components.

How to use

1. Access the tappet or component according to the service manual.
2. Position the puller securely on the intended component.
3. Align the pulling mechanism with the component axis.
4. Apply force gradually using the specified method.
5. Stop if the component binds or the tool slips.
6. Remove the component carefully and inspect it for damage.
7. Clean the tool and reinstall components according to the service procedure.

⚠ Safety

- Only trained technicians should perform engine disassembly.
- Never work on a running engine.
- Keep fingers clear of gripping jaws and pinch points.
- Do not use improvised extensions or excessive force.
- Follow manufacturer instructions for engine timing and reassembly.

Care & storage

- Clean the puller after use.
- Inspect jaws, threads, and moving parts regularly.
- Apply lubricant only where specified.
- Store in a dry tool case.

Common mistakes to avoid

- Using an incompatible puller.
- Applying force at an angle.
- Forcing a seized component and damaging the engine.
- Ignoring damaged puller threads or jaws.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thermostat

Category: **Automotive** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Old second hand thermostat

What it is used for

A thermostat regulates coolant flow or temperature control in a vehicle cooling system, depending on the design.

How to handle

- Treat a used thermostat as a training sample until its condition and function are verified.
- Allow the engine and cooling system to cool completely before inspection or removal.
- Inspect the housing, valve, spring, and sealing surfaces for corrosion, deposits, or damage.
- Use the correct part specification and follow the vehicle manufacturer's service procedure.
- Keep coolant away from skin and dispose of used coolant according to local requirements.

Storage

- Store the thermostat clean and dry in a labelled container.
- Protect the valve and sealing surfaces from dirt, corrosion, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Transmission Assembly

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Transmission /Transaxle

What it is used for

A transmission or transaxle assembly transfers engine power to the vehicle's drive wheels and provides the required gear ratios. Inspect, handle, and demonstrate the assembly using approved workshop procedures and suitable lifting equipment.

How to use

1. Confirm the assembly identity, specifications, and intended training or service task.
2. Inspect the casing, shafts, mounts, connectors, and visible components for damage or leaks.
3. Secure the assembly on a suitable work stand or approved lifting support.
4. Review the relevant service manual before disassembly, inspection, or adjustment.
5. Use appropriate tools to demonstrate component identification and power-transfer functions.
6. Check fluid requirements and inspect seals or connections according to the service procedure.
7. Clean the work area and record any faults or maintenance actions.

⚠ Safety

- Never work beneath an unsupported vehicle or suspended transmission assembly.
- Use rated lifting equipment and approved lifting points.
- Keep hands clear of pinch points and rotating components.
- Drain and handle transmission fluids according to workshop safety and disposal rules.

Care & storage

- Protect machined surfaces and openings from contamination, and store the assembly securely when not in use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Two-Wheeler

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Old second hand motor cycle; Two-wheeler

What it is used for

A two-wheeler is a motorcycle or scooter used for personal or commercial transportation. Pre-ride inspection, correct starting and riding practices, routine maintenance, and protective equipment help improve vehicle reliability and reduce the risk of road injury.

Main parts

- Engine or electric motor provides propulsion.
- Fuel tank or traction battery stores the energy source.
- Throttle controls engine power or motor output.
- Brake levers or pedals operate the braking system.
- Front and rear wheels support movement and road contact.
- Tyres provide grip and absorb road irregularities.
- Headlamp, tail lamp, indicators, and reflectors improve visibility.
- Mirrors help the rider observe surrounding traffic.
- Instrument cluster displays speed, warning lights, and other vehicle information.
- Chain and sprockets or final-drive components transmit power where fitted.

Before you start

1. Only a properly licensed and competent rider should operate the two-wheeler on public roads.
2. Check vehicle registration, insurance, required emission documents, and other applicable legal requirements.
3. Inspect tyres, tread, tyre pressure, wheels, brakes, lights, indicators, mirrors, horn, and visible cables.
4. Check engine oil, coolant where applicable, fuel level, or battery charge according to the vehicle type.
5. Look for fuel, oil, coolant, or battery-fluid leaks and any loose or damaged parts.
6. Confirm that the throttle returns smoothly and both brakes operate correctly before riding.
7. Wear a correctly fitted certified helmet and suitable protective clothing and footwear; secure loose items.

How to use

1. Park the vehicle on a stable surface and follow the owner's manual for starting checks.
2. Confirm that the transmission is in neutral or the drive mode is safe for starting.
3. For an internal-combustion model, check the fuel supply and start the engine using the prescribed procedure.

4. For an electric model, confirm battery status, switch on the vehicle, and check for warning messages.
5. Allow the engine or systems to stabilise as specified by the manufacturer before moving.
6. Check the area around the vehicle, mount safely, adjust mirrors, and maintain a balanced riding position.
7. Release the stand fully and start moving gently after checking traffic and surroundings.
8. Use smooth throttle inputs, appropriate gears where fitted, and a safe speed for road and weather conditions.
9. Maintain a safe following distance and apply both brakes progressively according to road conditions and vehicle design.
10. Signal before turning or changing lanes and check mirrors and blind spots before manoeuvring.
11. Park in a permitted stable location, switch off the engine or power, and secure the vehicle against unintended movement.
12. Record defects and arrange qualified maintenance if a warning light, unusual noise, poor braking, or handling issue appears.

⚠ Safety

- Always wear a properly fastened helmet and follow applicable road traffic laws.
- Never ride under the influence of alcohol, impairing drugs, or medication that makes riding unsafe.
- Do not carry more passengers or load than the vehicle is approved to handle.
- Never use a phone or other distracting device while riding.
- Do not ride with defective brakes, unsafe tyres, steering faults, or serious warning indicators.
- Keep hands, clothing, and tools away from moving or hot parts during inspection and maintenance.

Care & storage

- Follow the owner's manual for engine oil, filters, coolant, brake fluid, and scheduled servicing as applicable.
- Maintain the correct tyre pressure and inspect tyre wear and damage regularly.
- Check brake pads or shoes, brake operation, lights, and battery condition at recommended intervals.
- Lubricate and adjust the drive chain where fitted according to the manufacturer's instructions.
- Keep the vehicle clean, dry, and protected from corrosion, and store it securely.
- Use qualified service personnel for repairs beyond the rider's competence.

Common mistakes to avoid

- Skipping tyre, brake, light, or leak checks before riding.
- Riding with an improperly fitted helmet or loose protective clothing.
- Ignoring warning lights, abnormal noises, or handling changes.
- Using incorrect fluids, incorrect tyre pressure, or unsuitable replacement parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the

manufacturer's original manual / file.

Tyre Pressure Gauge

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Tire pressure gauges; Tyre pressure gauge

What it is used for

A tyre pressure gauge measures the air pressure inside a vehicle tyre. Compare the reading with the vehicle manufacturer's recommended pressure for the tyre condition and load.

How to use

1. Check the gauge, hose, and connector for damage or leakage.
2. Find the recommended tyre pressure from the vehicle label or manual.
3. Park safely, switch off the vehicle, and check the tyre when appropriate.
4. Remove the valve cap and press the gauge connector firmly onto the valve.
5. Read the pressure and compare it with the recommended value.
6. Adjust pressure if required, recheck the reading, and replace the valve cap.

⚠ Safety

- Check pressure on a safe, level surface away from moving traffic.
- Do not measure or inflate a visibly damaged or dangerously hot tyre without professional guidance.
- Keep fingers clear of the valve and avoid sudden air release.

Care & storage

- Keep the gauge clean and dry, protect it from impact, and store it safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tyre Repair Kit

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Tubeless tyre repair kits

What it is used for

A tubeless tyre repair kit provides tools and approved materials for repairing eligible punctures. Confirm that the tyre damage is repairable and follow the kit instructions and tyre manufacturer's repair limits.

How to use

1. Park the vehicle safely on firm, level ground and apply the parking brake.
2. Inspect the tyre and locate the puncture without removing embedded objects unnecessarily.
3. Check whether the puncture is within the permitted repair area and size limits.
4. Use the specified reamer and repair plug or patch according to the kit instructions.
5. Inflate the tyre to the vehicle manufacturer's recommended pressure using a suitable gauge.
6. Check for leakage with an approved method and verify pressure again.
7. Arrange professional inspection or tyre replacement if damage is unsuitable for repair.

⚠ Safety

- Never repair a tyre while the vehicle is supported only by an unstable jack.
- Do not repair sidewall damage, large cuts, or damage outside approved repair limits.
- Use eye protection and gloves when required by the repair method.
- A temporary plug may not be a permanent repair; follow professional tyre-service guidance.

Care & storage

- Keep the kit complete, dry, and within the manufacturer's shelf-life recommendations.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vehicle Chassis

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: TATA ACE – Chassis with Axle and wheels

What it is used for

A vehicle chassis with axle and wheels demonstrates the vehicle's basic structural layout and running gear. Use it for supervised automotive training, inspection, and component identification on a stable support.

How to use

1. Confirm the chassis identity and inspect the frame, axle, wheels, suspension, and mounts for damage.
2. Position the chassis on a stable, level workshop floor or approved support stand.
3. Secure the assembly against rolling or tipping before starting the inspection.
4. Identify major components and explain their functions using the relevant service information.
5. Inspect visible fasteners, joints, and suspension parts without removing safety-critical components unnecessarily.
6. Demonstrate basic inspection points and record any observed defects.
7. Clean the training area and return tools to storage after the session.

⚠ Safety

- Never work beneath a chassis supported only by a jack.
- Use wheel chocks and approved stands to prevent movement.
- Keep hands clear of pinch points and suspended components.
- Do not operate or move the chassis unless all required systems and safeguards are in place.

Care & storage

- Keep the frame and components clean, protect exposed metal from corrosion, and report defects promptly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vehicle Dashboard

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Dashboard with indicators

What it is used for

A vehicle dashboard with indicators demonstrates driver controls, warning lights, and instrument displays. Use the dashboard for supervised learning and verify indicator meanings against the vehicle manual.

How to use

1. Inspect the dashboard, switches, display, indicator lamps, and connectors for visible damage.
2. Confirm the vehicle or training panel is secured and the operating area is safe.
3. Identify the major instruments, switches, and warning indicators using the owner's manual.
4. Switch on the ignition or training power as appropriate and observe the indicator self-check.
5. Demonstrate selected controls and explain the meaning of relevant warning lights.
6. Switch off power after the demonstration and report any abnormal indicator or fault.
7. Return the dashboard or training equipment to its designated storage or parked condition.

⚠ Safety

- Do not ignore warning indicators on a real vehicle; follow the owner's manual.
- Keep hands clear of moving vehicle parts and avoid distraction during live vehicle operation.
- Do not bypass dashboard safety warnings or electrical protections.
- Use a qualified technician for electrical faults or unexpected warning lights.

Care & storage

- Clean dashboard surfaces with a suitable non-abrasive cleaner and protect electrical connectors from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vehicle Ramp

Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Ramp

What it is used for

To raise a vehicle safely for inspection or maintenance when the ramp is designed and rated for that use.

How to handle

- Inspect the ramp for cracks, deformation, oil, and other damage; verify its load rating and compatibility with the vehicle.
- Park on firm, level ground, apply the parking brake, and follow the vehicle and ramp manufacturer instructions.
- Align the wheels carefully and drive onto the ramp slowly with a spotter when appropriate.
- Secure the vehicle as instructed and use wheel chocks where required; never work beneath a vehicle supported only by an unsuitable ramp or jack.
- After work, clear the area and drive down slowly.

Storage

- Clean the ramp and inspect it after use.
- Store on a stable, dry surface in the designated area; do not stack beyond manufacturer instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wheel Alignment

Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Tyres and wheel alignment

What it is used for

A wheel alignment system measures wheel angles and helps adjust them to the vehicle manufacturer's specifications. Correct alignment improves vehicle stability, steering response, tyre life, and driving safety.

Main parts

- Wheel alignment lift or level alignment platform
- Wheel clamps and alignment targets or sensors
- Steering wheel holder
- Brake pedal depressor if required
- Alignment computer and vehicle specification database
- Turn plates and rear slip plates

Before you start

1. Confirm the vehicle is suitable for the alignment equipment and identify its correct model specifications.
2. Inspect the tyres for damage, uneven wear, correct pressure, and suitable tread condition.
3. Check wheel bearings, suspension joints, steering components, and ride height for defects.
4. Position the vehicle centrally on the level alignment platform and apply the parking brake as instructed.
5. Install wheel clamps and sensors or targets according to the equipment instructions.
6. Check that turn plates and slip plates are unlocked and that the alignment system is calibrated.

How to use

1. Select the correct vehicle make, model, year, and suspension configuration in the alignment system.
2. Compensate for wheel runout using the method specified by the equipment manufacturer.
3. Fit the steering wheel holder and brake pedal depressor if required by the procedure.
4. Follow the on-screen instructions to measure camber, caster, toe, and other supported angles.
5. Compare the measured values with the manufacturer's specification ranges.
6. Inspect adjustment points and identify worn or damaged parts before making adjustments.
7. Adjust the specified angles in the recommended sequence while keeping fasteners within approved torque values.
8. Recheck all readings and confirm that the values fall within the permitted ranges.
9. Tighten adjustment fasteners to specification and remove the measuring equipment carefully.

10. Complete the final inspection, perform a road test if authorized, and save or print the alignment report.

⚠ Safety

- Use the alignment lift only within its rated capacity and secure the vehicle at approved lifting points.
- Keep people clear of the vehicle while raising or lowering the lift.
- Do not perform alignment adjustments on damaged suspension or steering components without first addressing the defects.
- Keep cables, sensors, and tools clear of moving wheels and lift mechanisms.
- Use suitable protective equipment and follow workshop vehicle-movement procedures.
- Never rely on alignment readings alone to confirm that unsafe tyres or damaged steering parts are roadworthy.

Care & storage

- Clean wheel clamps, sensors, targets, and turn plates after use.
- Inspect cables, clamps, lift locks, and slip plates for damage or wear.
- Keep alignment software and vehicle specifications updated as authorized.
- Arrange calibration and preventive maintenance according to the equipment manufacturer's schedule.

Common mistakes to avoid

- Using incorrect vehicle specifications can produce unsuitable alignment settings.
- Ignoring tyre pressure or worn suspension parts can make measurements unreliable.
- Adjusting angles before checking for damaged components can hide underlying faults.
- Failing to secure fasteners or perform a final measurement can leave the vehicle incorrectly aligned.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Workshop Equipment

Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Workshop tool/equipment

What it is used for

Automotive workshop equipment supports vehicle inspection, servicing, repair, lifting, measuring, and maintenance tasks. Correct selection, setup, and operation help technicians work efficiently while reducing injury and vehicle damage.

Main parts

- Equipment frame, base, or supporting structure
- Operating controls, handles, or switches
- Power supply, air supply, or hydraulic connections as applicable
- Guards, locks, and safety interlocks where fitted
- Adapters, attachments, and work-holding accessories
- Operating instructions and inspection records

Before you start

1. Identify the equipment type, intended task, rated capacity, and applicable operating instructions.
2. Inspect the frame, handles, guards, cables, hoses, connectors, and moving parts for damage.
3. Check that required inspection, testing, and calibration are current for the equipment type.
4. Confirm the work area is clean, adequately lit, and free from obstructions.
5. Verify that the correct attachment, adapter, and personal protective equipment are available.
6. Ensure the operator is trained and that power, air, or hydraulic supplies meet the equipment requirements.

How to use

1. Position the equipment on a stable surface or at the designated workshop station.
2. Secure the workpiece or vehicle using the approved holding and support points.
3. Connect the required power, air, or hydraulic supply according to the instructions.
4. Set the operating controls and select the correct accessory for the task.
5. Start or activate the equipment using the designated control.
6. Operate at the specified speed, pressure, load, or measurement range for the task.
7. Monitor the work for unusual noise, vibration, overheating, leakage, or unstable movement.
8. Stop the equipment immediately if a fault, unsafe condition, or abnormal result occurs.
9. Switch off and isolate energy sources before adjustments, tool changes, or cleaning.
10. Clean the equipment, inspect it for damage, and return it to its designated storage location.

⚠ Safety

- Use equipment only for its intended purpose and within its rated capacity.
- Wear task-appropriate personal protective equipment and secure loose clothing and hair.
- Never bypass guards, safety locks, or interlocks fitted to the equipment.
- Support raised vehicles and heavy components with approved stands or lifting devices.
- Isolate electrical, pneumatic, and hydraulic energy before servicing or clearing a blockage.
- Keep unauthorized people away from moving equipment and suspended loads.

Care & storage

- Clean the equipment after use and remove oil, grease, and debris safely.
- Inspect cables, hoses, fasteners, guards, and moving parts at the required intervals.
- Lubricate serviceable parts and replace worn accessories according to the manufacturer's instructions.
- Maintain inspection, calibration, repair, and preventive maintenance records where required.

Common mistakes to avoid

- Using equipment beyond its rated capacity can cause failure or serious injury.
- Selecting the wrong attachment can damage the workpiece and equipment.
- Ignoring damaged cables, hoses, guards, or locks creates avoidable hazards.
- Returning equipment to storage without cleaning or reporting defects can affect the next user.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Abdominal Wheel

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Abdominal Wheel

What it is used for

An abdominal wheel is a fitness tool used to train the core and abdominal muscles. Use controlled movements on a stable, non-slip surface and within your fitness ability.

How to use

1. Inspect the wheel, axle, handles, and grips for damage or looseness.
2. Place the wheel on a flat, non-slip surface with enough clear space.
3. Kneel on a suitable exercise mat and grip both handles firmly.
4. Keep the abdomen engaged and roll the wheel forward slowly within a comfortable range.
5. Pull the wheel back under control without abruptly arching the lower back.
6. Start with a short range and repeat only while maintaining correct form.
7. Clean the handles and wheel after exercising.

⚠ Safety

- Stop immediately if you feel sharp pain, dizziness, or unusual discomfort.
- Do not allow the lower back to sag or twist excessively.
- Keep the exercise area clear of furniture, pets, and other people.
- Seek professional guidance if unsure whether the exercise is suitable for you.

Care & storage

- Wipe the wheel and grips dry after use and store away from direct heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ankle Weights

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Ankle/Wrist Weights

What it is used for

Ankle and wrist weights add resistance during selected fitness exercises. Choose a suitable weight and use controlled movements to avoid joint strain.

How to use

1. Inspect the weight straps, fasteners, and outer covering for damage.
2. Select a light, suitable weight based on the exercise and ability level.
3. Secure the weights comfortably around the ankles or wrists without restricting circulation.
4. Begin with gentle movements and maintain correct posture throughout the exercise.
5. Stop and remove the weights if movement becomes uncomfortable or difficult.
6. Remove the weights after exercising and check the skin for irritation.

⚠ Safety

- Do not wear weights that cause numbness, pain, or restricted circulation.
- Avoid running, jumping, or fast movements unless a qualified instructor approves them.
- Keep straps secured and away from moving exercise equipment.

Care & storage

- Wipe weights clean after use and store them dry with straps fastened loosely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Applicator Brush

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Applicator Brush; Plastic applicators

What it is used for

An applicator brush is used to apply creams, gels, powders, or other beauty products in a controlled manner. Select a brush compatible with the product and treatment area.

How to use

1. Inspect the brush or applicator for cleanliness, loose bristles, and damage.
2. Choose an applicator suitable for the product and intended area.
3. Clean hands and prepare the treatment area according to salon procedure.
4. Take a small amount of product using the applicator without contaminating the container.
5. Apply the product evenly using gentle, controlled strokes.
6. Clean or dispose of the applicator as directed and store reusable tools hygienically.

⚠ Safety

- Use a clean applicator for each client and avoid sharing disposable brushes.
- Do not apply products near eyes or broken skin unless the product is intended for that use.
- Stop application if irritation occurs and follow the product instructions.

Care & storage

- Wash reusable brushes with a suitable cleaner, dry fully, and store in a clean container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ayurvedic Product

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Oil, churna, and other material

What it is used for

Handle Ayurvedic oils, churna, and related products hygienically and according to label instructions.

How to handle

- Read the label, intended use, ingredients, and expiry date before use.
- Use clean, dry scoops for powders and avoid contaminating the container.
- Avoid contact with eyes and damaged skin unless the product directions permit it.
- Do not ingest or apply products unless intended for that use; follow qualified professional guidance where needed.

Storage

- Keep products tightly closed in their original labelled containers.
- Store as directed, away from heat, moisture, sunlight, and children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Balance Board

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Balance Board

What it is used for

A balance board challenges balance and coordination by creating an unstable standing surface. Begin with supported exercises and use it only on a clear, non-slip floor.

How to use

1. Inspect the board, base, and surface for cracks, wear, or loose parts.
2. Place the board on a level, non-slip surface away from sharp edges.
3. Stand near a stable support or instructor before stepping onto the board.
4. Place both feet evenly and keep knees slightly bent.
5. Shift weight gently while maintaining control and looking ahead.
6. Step off slowly and rest if balance becomes difficult.
7. Wipe the board clean and return it to storage after use.

⚠ Safety

- Use supervision or nearby support if you are new to balance training.
- Do not use the board on wet floors or near stairs.
- Stop immediately if you feel pain, dizziness, or loss of control.

Care & storage

- Clean the board with a suitable cloth and store it dry away from direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Barber Brush

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Barber Brush

What it is used for

To remove loose hair and distribute grooming products during barbering. To maintain hygiene by cleaning the brush between clients and storing it properly.

Main parts

- Brush handle
- Bristles
- Brush head
- Grip or hanging loop, if fitted

Before you start

1. Inspect the bristles and handle for damage or loose parts.
2. Ensure the brush is clean and dry before use.
3. Use a brush suitable for the intended grooming task.
4. Follow salon hygiene procedures between clients.

How to use

1. Hold the brush by its handle.
2. Use gentle strokes to remove loose hair from the neck or work area as appropriate.
3. For product application, follow the salon's approved procedure.
4. Avoid brushing over irritated or broken skin.
5. Remove loose hair from the brush after use.
6. Clean and disinfect the brush as required and allow it to dry.

⚠ Safety

- Do not use a brush with sharp, broken, or contaminated bristles.
- Avoid sharing a brush without proper cleaning and disinfection.
- Keep the brush away from eyes and irritated skin.
- Follow hygiene and infection-control requirements.

Care & storage

- Remove trapped hair after each use.
- Wash and disinfect according to the brush material and salon procedure.
- Dry completely before storage.

- Replace worn or damaged brushes.

Common mistakes to avoid

- Using a contaminated brush between clients.
- Applying excessive pressure to the skin.
- Storing the brush while wet.
- Ignoring loose or damaged bristles.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bashpa Swedan

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bashpa Swedan

What it is used for

Bashpa Swedan is a steam-based wellness procedure used in traditional practices to expose the body to controlled warmth and steam. Perform it only with appropriate training, suitable equipment, informed consent, and health screening, while carefully controlling temperature, duration, hydration, and emergency access.

Main parts

- Steam generator or heating unit
- Treatment chamber, enclosure, or steam tent
- Steam delivery pipe and outlet
- Temperature control and thermometer
- Timer and ventilation arrangement
- Water reservoir, drain, and emergency shutoff

Before you start

1. Confirm the person's consent, treatment purpose, and suitability for the procedure through the applicable health screening process.
2. Do not proceed if the person has a condition or symptom for which heat or steam exposure is contraindicated; seek qualified clinical advice when uncertain.
3. Inspect the steam generator, pipes, chamber, controls, electrical connections, and emergency shutoff for damage or leaks.
4. Check that the treatment area is clean, ventilated, and equipped with safe access and a quick way to stop steam delivery.
5. Verify that temperature and timer controls work correctly and that water is at the required level.
6. Explain the procedure, expected warmth, possible discomfort, stop signal, and the need to report dizziness, nausea, breathing difficulty, or excessive heat.

How to use

1. Prepare the equipment and treatment area according to the manufacturer's instructions and facility protocol.
2. Fill the reservoir with the specified water and confirm that the drain and steam outlet are clear.
3. Start the steam generator and allow the chamber to reach the approved treatment conditions before the person enters.
4. Verify the actual temperature using the designated measuring device; do not rely only on a control setting.

5. Help the person enter or position themselves safely, keeping the face and airway clear of direct steam jets.
6. Set the treatment duration and temperature within the approved protocol and the person's tolerance.
7. Stay nearby and monitor comfort, skin response, breathing, and general condition throughout the procedure.
8. Stop steam delivery immediately if the person reports excessive heat, dizziness, nausea, weakness, pain, or breathing difficulty.
9. Assist the person out safely, allow gradual cooling, and provide drinking water if appropriate and permitted.
10. Switch off the steam generator and isolate power before draining or cleaning the equipment.
11. Clean and disinfect contact surfaces and reusable accessories according to the facility protocol.
12. Record the treatment duration, settings, person's response, and any adverse event; report concerns to the responsible practitioner.

⚠ Safety

- Only trained and authorised personnel should perform or supervise the procedure.
- Avoid steam exposure that could cause burns; never direct hot steam onto the face or sensitive skin.
- Do not leave the person unattended during treatment, especially if mobility or heat tolerance is limited.
- Stop immediately if faintness, confusion, chest discomfort, breathing difficulty, or signs of heat illness occur; seek emergency assistance when indicated.
- Use extra caution with children, older adults, pregnancy, impaired sensation, cardiovascular conditions, and other heat-sensitive circumstances; obtain qualified medical advice where appropriate.
- Do not open pressurised steam lines or service the heater while hot or connected to power.

Care & storage

- Drain and clean the water reservoir according to the manufacturer's instructions and facility hygiene protocol.
- Remove mineral deposits using an approved descaling method at the specified interval.
- Inspect steam hoses, seals, controls, temperature sensors, and electrical connections regularly.
- Keep the treatment chamber dry and ventilated between uses.
- Remove damaged equipment from service and arrange qualified inspection or repair.

Common mistakes to avoid

- Using excessive temperature or treatment duration without following an approved protocol.
- Leaving the person unattended or ignoring early signs of heat intolerance.
- Allowing direct steam contact to burn the skin.
- Cleaning or opening hot equipment before it has cooled and been isolated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the

manufacturer's original manual / file.

Basti Yantra

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Basti; Basti yantra

What it is used for

A Basti Yantra is a device used in traditional Ayurvedic practice to administer a prescribed enema preparation through the rectal route under appropriate clinical supervision. Correct patient assessment, hygienic preparation, suitable equipment, and careful administration are essential to reduce injury, infection, and other complications.

Main parts

- Reservoir or container holds the prescribed liquid preparation.
- Connecting tube carries the preparation from the reservoir.
- Nozzle or rectal tip is designed for the intended route of administration.
- Clamp or flow control regulates liquid delivery where fitted.
- Connectors join the reservoir, tube, and nozzle.
- Support or stand holds the reservoir where included.
- Disposable or reusable components depend on the device design and instructions.

Before you start

1. Only qualified Ayurvedic or healthcare practitioners trained in the procedure should administer Basti.
2. Confirm the patient's identity, indication, prescribed preparation, volume, temperature, and relevant contraindications.
3. Explain the procedure and obtain informed consent according to clinical policy.
4. Inspect the reservoir, tube, nozzle, connectors, and flow control for damage, blockage, or leakage.
5. Confirm that reusable components have been properly cleaned and processed, or that disposable items are sterile and within expiry.
6. Prepare the prescribed solution using approved ingredients and the correct clinical method; do not use unprescribed substances.
7. Perform hand hygiene, wear appropriate gloves and PPE, and prepare a private, clean treatment area.

How to use

1. Position the patient as directed by the qualified practitioner and maintain privacy and comfort.
2. Prepare and verify the prescribed liquid, ensuring its volume and temperature meet the clinical instructions.

3. Assemble the device according to the manufacturer's instructions and check that connections are secure.
4. Ensure the nozzle or rectal tip is clean, appropriate for the patient, and lubricated with an approved lubricant when indicated.
5. Prime the tubing as required by the device instructions while preventing contamination and excessive air entry.
6. Administer the preparation gently using the prescribed technique and controlled flow under clinical supervision.
7. Monitor the patient for pain, abdominal discomfort, dizziness, bleeding, nausea, or other unexpected reactions.
8. Stop administration immediately if significant pain, bleeding, resistance, or other concerning symptoms occur and seek clinical assessment.
9. Complete the procedure according to the prescribed protocol and assist the patient with safe recovery and hygiene.
10. Dispose of single-use items and clinical waste according to facility policy; clean reusable components using validated procedures.
11. Document the preparation, administered amount, patient response, and any adverse event as required.

⚠ Safety

- Do not perform the procedure without appropriate clinical training, a valid indication, and informed consent.
- Do not administer an unprescribed preparation or exceed the prescribed volume, temperature, or rate.
- Do not force the nozzle or fluid if resistance or significant pain occurs.
- Seek clinical assessment before use when contraindications or warning signs such as rectal bleeding, severe abdominal pain, suspected obstruction, or suspected perforation are present.
- Use appropriate infection-control precautions and never reuse items designated for single use.
- Follow clinical protocols for patients who are medically vulnerable or require additional monitoring.

Care & storage

- Clean and process reusable components immediately after use according to validated instructions.
- Inspect the tube, nozzle, reservoir, and connectors for cracks, residue, blockage, and leakage.
- Dry components completely where required and store them in a clean, protected area.
- Check expiry and packaging integrity of sterile disposable components before use.
- Remove damaged equipment from service and arrange authorised replacement or repair.

Common mistakes to avoid

- Using the device without confirming the patient's suitability and prescribed preparation.

- Using inadequately cleaned or contaminated components.
- Applying excessive force or delivering the preparation too quickly.
- Ignoring pain, bleeding, resistance, or other signs of complications.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bendy Roller

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Bendy rollers

What it is used for

To shape or curl hair using flexible bendy rollers without excessive heat.

How to handle

- Inspect rollers for cracks, sharp edges, residue, or damaged foam.
- Start with clean, slightly damp or dry hair as recommended for the desired style.
- Separate a small hair section, position the roller at the ends, and roll evenly toward the scalp without pulling tightly.
- Bend or secure the roller gently so it stays in place without discomfort.
- Remove rollers carefully and clean them after use; do not share without proper cleaning.

Storage

- Clean and dry rollers completely before storing.
- Keep in a clean, ventilated container away from direct heat and dust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Blackhead Remover

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Blackhead Remover

What it is used for

To assist with the controlled removal of blackheads by a trained beauty professional using an appropriate hygienic method.

How to handle

- Inspect the tool for damage and ensure it has been cleaned and disinfected according to its instructions.
- Clean the skin and hands; use gloves and suitable hygiene precautions where required.
- Use only gentle pressure on appropriate skin areas and follow the tool manufacturer's directions.
- Do not force extraction, puncture skin, or use on inflamed, infected, broken, or irritated areas.
- Stop immediately if pain, bleeding, or significant irritation occurs; clean and disinfect the tool after use.

Storage

- Dry the tool thoroughly after approved cleaning and disinfection.
- Store in a clean, covered container separate from used tools and personal items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Body Reformer

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Can float body reformer

What it is used for

A body reformer is a fitness device designed to support selected body-conditioning or resistance exercises. Identify its specific design and follow the supplied instructions before use.

How to use

1. Inspect the frame, straps, handles, connectors, and support surfaces for damage.
2. Place the equipment on a stable, non-slip surface with enough clear space.
3. Read the operating instructions and select an appropriate exercise setting.
4. Check that all straps, supports, and adjustment locks are secured.
5. Perform slow, controlled movements within a comfortable range of motion.
6. Stop if the equipment shifts or movement causes discomfort.
7. Clean contact surfaces and return adjustable parts to their safe position.

⚠ Safety

- Use the device only for its intended purpose and within its rated limits.
- Keep fingers away from moving joints, tension straps, and pinch points.
- Use instructor supervision when learning unfamiliar exercises.
- Stop immediately if you experience pain, dizziness, or instability.

Care & storage

- Wipe contact surfaces after use and store the equipment clean and dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bolster

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Bolster; Bolsters

What it is used for

Bolsters support body positioning and comfort during approved beauty, wellness, or therapy activities.

How to handle

- Inspect the bolster cover, seams, and filling for damage or contamination.
- Select a bolster suitable for the activity and user's needs.
- Position it securely and check that it will not roll or slip unexpectedly.
- Clean or replace covers between users according to hygiene procedures.
- Do not use damaged bolsters or place excessive loads on them.

Storage

- Keep bolsters clean and fully dry before storage.
- Store on a clean shelf or in a designated cabinet away from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cable Machine

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Cable machine

What it is used for

A cable machine is a fitness station that uses adjustable resistance through cables and pulleys for controlled strength exercises. Set the load and attachments correctly, maintain stable posture, and inspect the cable system before use to reduce the risk of injury or equipment failure.

Main parts

- Machine frame and base
- Weight stack or resistance mechanism
- Cable and pulley assemblies
- Weight selector pin or resistance control
- Handles, bars, and exercise attachments
- Seat, adjustment points, and safety stops, if fitted

Before you start

1. Confirm that the machine and intended exercise are suitable for the user's fitness level and any applicable restrictions.
2. Inspect the frame, base, cables, pulleys, selector pin, attachments, and adjustment locks for damage or wear.
3. Check that cable strands are intact and there are no frays, kinks, or loose connections.
4. Ensure the machine is stable, surrounding floor is dry, and there is sufficient clearance for the exercise.
5. Select a suitable starting resistance and verify that the selector pin is fully inserted.
6. Use suitable exercise clothing and footwear, and remove jewellery that could catch on the equipment.

How to use

1. Adjust the seat, pulley height, and other settings according to the intended exercise.
2. Attach the correct handle or bar and confirm that the connector is fully secured.
3. Select a manageable resistance and test the movement with a controlled, light repetition.
4. Stand or sit in the designated position with stable footing and a neutral posture.
5. Grip the attachment securely and perform the exercise through a comfortable, controlled range of motion.
6. Keep the cable tension controlled and avoid jerking, swinging, or letting the weight stack drop suddenly.
7. Breathe steadily and maintain correct technique throughout each repetition.

8. Return the attachment slowly to its resting position and ensure the weight stack settles fully.
9. Stop immediately if there is unusual cable noise, slipping, pain, or unstable movement.
10. After the set, place the attachment in its designated holder and lower the resistance safely.
11. Report any damage or malfunction and do not use the machine until it has been inspected.
12. Wipe contact surfaces after use and leave the area clear for the next user.

⚠ Safety

- Use the machine only as intended and follow the facility's operating instructions.
- Never use frayed cables, damaged pulleys, loose attachments, or a faulty selector pin.
- Keep fingers clear of pulleys, weight stacks, and moving cable connections.
- Do not exceed the machine's rated load or attach unapproved accessories.
- Avoid sudden release of the handle, which can cause the weight stack or cable to snap back.
- Stop exercising if sharp pain, dizziness, or unusual symptoms occur; seek appropriate assistance.

Care & storage

- Wipe down handles, seats, and contact surfaces after use.
- Inspect cables, pulleys, connectors, and weight-stack guides regularly.
- Check bolts, adjustment locks, and the machine base for looseness or damage.
- Arrange replacement of worn cables and servicing according to the manufacturer's schedule.
- Keep the area around the machine clean, dry, and unobstructed.

Common mistakes to avoid

- Selecting excessive resistance before learning the correct movement.
- Using a damaged cable or an attachment that is not secured.
- Jerking the handle or allowing the weight stack to slam down.
- Ignoring pain, unusual cable sounds, or unstable machine parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Comb

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 31 labs

Covers these lab items: All Propose Comb; All Purpose Comb; Back comb; Bath Comb; Comb; Hair Cutting Comb; Large Tooth Comb; Tail Comb; Tail Comb, cutting comb; Vent Dressing comb; Wide Toothed Comb

What it is used for

A comb separates, arranges, and styles hair for grooming or salon work. Select a comb type suitable for the hair texture and intended styling task.

How to use

1. Inspect the comb for broken teeth, sharp edges, or cracks.
2. Select a clean comb appropriate for the hair type and styling method.
3. Prepare the hair and work area according to the grooming procedure.
4. Hold the comb securely and work gently from the ends toward the roots when detangling.
5. Use the appropriate comb technique for parting, cutting, backcombing, or styling.
6. Avoid pulling the hair or scratching the scalp.
7. Clean and disinfect the comb between clients according to salon hygiene procedures.
8. Dry the comb and return it to clean storage.

⚠ Safety

- Do not use a cracked comb or one with sharp broken teeth.
- Avoid excessive pulling that may damage hair or irritate the scalp.
- Do not share unclean combs between clients.
- Keep combs away from heat sources unless designed for heat exposure.

Care & storage

- Remove trapped hair and wash the comb after use.
- Disinfect reusable combs as appropriate and store them in a clean, dry container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cosmetic Pencil Sharpener

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Eyebrow Pencil Sharpener

What it is used for

To sharpen an eyebrow or cosmetic pencil to a clean, usable point. To maintain hygiene and avoid breaking the pencil core or damaging the skin during use.

Main parts

- Sharpener body
- Cutting blade
- Pencil opening
- Shavings compartment, if fitted
- Protective cover, if fitted

Before you start

1. Check that the sharpener opening matches the pencil size.
2. Inspect the blade for damage, corrosion, or trapped residue.
3. Ensure the sharpener is clean and dry.
4. Use a separate clean sharpener for personal or client hygiene as required.

How to use

1. Remove the protective cover if fitted.
2. Insert the pencil gently into the correct opening.
3. Rotate the pencil slowly without forcing it.
4. Stop when the point reaches the desired shape.
5. Remove the pencil and inspect the tip.
6. Empty the shavings compartment and clean the sharpener.
7. Replace the cover and store safely.

⚠ Safety

- Keep fingers away from the cutting blade.
- Do not sharpen a pencil while it is near the eye or skin.
- Do not use a sharpener with a loose or damaged blade.
- Avoid sharing a contaminated sharpener between clients.

Care & storage

- Remove shavings after each use.

- Clean residue using the manufacturer's recommended method.
- Keep the blade dry and protected.
- Replace the sharpener if the blade becomes dull or damaged.

Common mistakes to avoid

- Using the wrong opening for the pencil size.
- Rotating too forcefully and breaking the core.
- Using a dirty sharpener on another person.
- Touching the cutting blade during cleaning.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cosmetic Product

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 11 labs

Covers these lab items: Hair Oil; Make up Palate; Mild soothing fragrance; Multani Mitti (Fuller's Earth); Shampoo Bottle; Talcum powder

What it is used for

Cosmetic products support grooming and personal-care demonstrations when used according to their labels and hygiene requirements.

How to handle

- Check product identity, ingredients, expiry date, seal, and suitability for the intended user.
- Use clean applicators and avoid sharing products that contact skin, eyes, or lips directly.
- Perform any required patch test and stop use if irritation or an allergic reaction occurs.
- Do not use contaminated, expired, or altered products.

Storage

- Keep products tightly closed and stored within the manufacturer's temperature and light conditions.
- Separate products from cleaning chemicals and keep them away from heat and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Curling Iron

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Curling Iron

What it is used for

A curling iron uses controlled heat to shape hair into curls or waves. Correct temperature selection, careful handling, and proper cleaning help achieve consistent styling while reducing burns and hair damage.

Main parts

- Heated barrel transfers heat to the hair.
- Clamp holds hair against the barrel on clamp-style models.
- Handle provides a grip and helps keep hands away from the heated barrel.
- Temperature control adjusts the heat setting.
- Power cord and plug supply electrical power.
- Indicator light or display shows heating status on equipped models.
- Safety stand or heat-resistant rest supports the hot tool.
- Cool tip provides a designated area for handling on applicable models.

Before you start

1. Read the manufacturer's instructions and identify the available temperature settings.
2. Inspect the power cord, plug, barrel, clamp, and handle for damage.
3. Ensure hair is dry unless the tool is specifically designed for damp hair.
4. Remove tangles and divide hair into manageable sections.
5. Place the tool on a stable, heat-resistant surface near a suitable power outlet.
6. Keep suitable heat-resistant gloves and a heat-safe resting surface available.

How to use

1. Place the curling iron on its heat-resistant stand before switching it on.
2. Connect it to a suitable power outlet and select the appropriate temperature.
3. Wait until the tool reaches the selected temperature, following the indicator or manual.
4. Take a small section of hair and position it carefully near the barrel.
5. Wrap or clamp the hair according to the tool design, keeping the barrel away from the scalp and skin.
6. Hold the hair only for the recommended time and release it gently.
7. Repeat with other sections, avoiding repeated heating of the same section.
8. Switch off and unplug the curling iron immediately after styling.
9. Place the tool on a heat-resistant surface and allow it to cool completely.
10. Clean the cooled barrel as instructed and store the tool safely.

⚠ Safety

- Never touch the heated barrel or leave the hot tool unattended.
- Keep the curling iron away from water, sinks, and wet surfaces.
- Do not use a tool with a damaged cord, plug, or casing.
- Avoid direct contact between the heated barrel and scalp, ears, neck, or fingers.
- Use the lowest effective temperature suitable for the hair type and condition.
- Unplug the tool before cleaning and allow it to cool before storage.

Care & storage

- Clean styling product residue from the cooled barrel using approved materials.
- Check the cord, plug, clamp, and temperature controls regularly.
- Do not wrap the power cord tightly around the hot tool.
- Store the fully cooled curling iron in a dry, safe location.

Common mistakes to avoid

- Using excessive heat for the hair type or condition.
- Holding hair against the barrel longer than recommended.
- Using the curling iron near water or with wet hands.
- Storing the tool before it has cooled completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dermascope

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Derma scope; Derma scope Auto scope

What it is used for

To magnify and examine skin or scalp surfaces during a beauty or skin-care assessment. To obtain a clear view while maintaining hygiene and avoiding irritation.

Main parts

- Magnifying lens
- Light source, if fitted
- Handle or body
- Focus or magnification control, if fitted
- Power source, if applicable

Before you start

1. Inspect the lens, light, and body for damage.
2. Clean the lens and contact surfaces using the approved method.
3. Ensure the device is suitable for the intended examination.
4. Explain the procedure and obtain consent where required.
5. Check that the skin area is clean and suitable for examination.

How to use

1. Switch on the light if the device requires power.
2. Position the dermascope at the recommended distance or contact point.
3. Adjust focus or magnification if available.
4. Examine the area systematically under suitable lighting.
5. Avoid pressing hard against the skin.
6. Record relevant observations according to workplace procedures.
7. Switch off and clean the device after use.

⚠ Safety

- Do not use a damaged lens or electrical device.
- Avoid contact with broken or infected skin unless the device and procedure are approved for it.
- Disinfect contact surfaces between clients according to instructions.
- Do not present observations as a medical diagnosis unless qualified to do so.

Care & storage

- Clean lenses with suitable optical materials.
- Keep the device dry and protected from dust.
- Check batteries and light function where applicable.
- Store in a protective case.

Common mistakes to avoid

- Using a dirty lens or contact surface.
- Holding the device at an unsuitable distance.
- Pressing excessively on the skin.
- Making medical conclusions beyond the operator's qualifications.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dumbbell

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Adjustable Dumbbells; Dumbbells

What it is used for

To perform strength, resistance, and conditioning exercises using fixed or adjustable dumbbells.

How to handle

- Inspect the dumbbell, handle, plates, collars, and locking mechanism for damage or looseness.
- Choose a weight suitable for the user's fitness level and exercise; secure all plates on adjustable models.
- Use a stable stance, controlled movements, and correct posture; avoid swinging or dropping the weights.
- Keep feet and fingers clear when picking up or setting down dumbbells and maintain space from other users.
- Stop if pain, dizziness, or unusual discomfort occurs and return weights carefully after the session.

Storage

- Wipe the dumbbells after use and check adjustable locks before returning them.
- Store on a stable rack with heavier weights placed securely and prevent rolling or falling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Exfoliation Machine

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Exfoliation machine

What it is used for

An exfoliation machine is used in beauty and skincare treatments to remove surface dead skin or cleanse the skin using a method designed for that device. Correct skin assessment, suitable settings, hygienic attachments, and trained operation help reduce irritation, burns, and cross-contamination.

Main parts

- Main unit houses the controls and operating components.
- Handpiece delivers the treatment to the skin.
- Treatment tip or attachment performs the device-specific exfoliation method.
- Power switch starts and stops the machine.
- Intensity or mode control adjusts treatment settings where fitted.
- Display or indicator shows operating status and settings.
- Connecting hose or cable links the handpiece to the main unit where applicable.
- Reusable attachment holder or storage tray keeps accessories organised.

Before you start

1. Only trained beauty or skincare personnel should operate the machine within their authorised scope.
2. Read the device manual and identify its specific exfoliation method and approved attachments.
3. Ask about skin sensitivity, allergies, current skin problems, recent procedures, and relevant contraindications.
4. Do not treat broken, infected, severely irritated, or otherwise unsuitable skin; refer to a qualified healthcare professional when needed.
5. Inspect the handpiece, tips, cable, hose, controls, and housing for damage.
6. Clean or disinfect reusable attachments according to the manufacturer's instructions and prepare single-use items as required.
7. Explain the treatment, obtain informed consent, and follow the approved skin-assessment procedure.

How to use

1. Wash hands and prepare the treatment area using the required hygiene procedure.
2. Select the correct attachment and setting for the device, treatment area, and assessed skin condition.

3. Position the client comfortably and protect clothing, eyes, and nearby sensitive areas as required by the device instructions.
4. Start the machine at the recommended initial setting and check that it operates normally.
5. Apply the handpiece using the approved technique, pressure, movement, and treatment duration for that device.
6. Observe the skin and ask the client about discomfort throughout the treatment.
7. Stop immediately if severe redness, burning, pain, swelling, or another unexpected reaction occurs.
8. Complete the treatment according to the approved protocol and apply suitable aftercare products where permitted.
9. Explain post-treatment care, including sun protection and avoiding irritating products when recommended for that device.
10. Switch off the machine and safely disconnect the handpiece before cleaning or changing attachments.
11. Clean, disinfect, or dispose of attachments according to their instructions and record the treatment where required.

⚠ Safety

- Never use an exfoliation machine without training and knowledge of its specific technology.
- Do not use damaged attachments or unapproved accessories.
- Avoid treating contraindicated skin conditions or areas restricted by the manufacturer.
- Do not increase intensity or treatment duration beyond the approved protocol.
- Use required eye protection and other PPE for the device and treatment method.
- Follow infection-control procedures and never reuse single-use attachments.

Care & storage

- Clean the main unit and handpiece according to the manufacturer's instructions.
- Inspect treatment tips, cables, hoses, and connectors before each session.
- Replace worn or damaged attachments and keep reusable accessories stored hygienically.
- Arrange authorised servicing if performance becomes irregular or controls fail.

Common mistakes to avoid

- Choosing an attachment or setting without assessing the client's skin.
- Applying excessive pressure or repeating passes beyond the approved protocol.
- Reusing single-use items or inadequately disinfecting reusable attachments.
- Ignoring pain, burning, or unexpected skin reactions during treatment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Eye Pillow

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Eye pillows (tissues or washable cloth to cover them)

What it is used for

Handle eye pillows hygienically for relaxation or wellness activities.

How to handle

- Check the pillow cover and filling for tears, leakage, or contamination before use.
- Use a clean cover or tissue barrier for each person where appropriate.
- Avoid placing pressure on injured or irritated eyes and follow activity guidance.
- Clean reusable covers according to the manufacturer's instructions.

Storage

- Keep clean eye pillows in labelled, dry containers or bags.
- Store separately from used covers and protect from moisture and dust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Eyelash Brush

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: All in one Eyelash Brush/Comb

What it is used for

To separate and groom eyelashes or eyebrow hairs using an eyelash brush or comb. To achieve a neat appearance while keeping the tool clean and avoiding eye irritation.

Main parts

- Brush bristles or comb teeth
- Handle
- Brush head
- Protective cap, if fitted

Before you start

1. Inspect the bristles or teeth for damage or loose parts.
2. Ensure the brush is clean and dry.
3. Wash or sanitize hands before use.
4. Use an individual clean brush for each client when required by hygiene procedures.

How to use

1. Hold the handle securely.
2. Ask the person to keep the eyes relaxed and avoid sudden movements.
3. Brush the lashes gently from the base toward the tips without touching the eye surface.
4. Use light strokes to separate clumps or arrange eyebrow hairs as needed.
5. Stop if the person feels pain or irritation.
6. Clean or dispose of the brush according to its type and hygiene procedure.
7. Store the tool with its head protected.

⚠ Safety

- Do not insert the brush into the eye or scrape the eyelid.
- Avoid use on irritated or infected eyes.
- Do not share disposable brushes between people.
- Stop immediately if the tool causes pain or injury.

Care & storage

- Remove cosmetic residue after use.
- Wash and disinfect reusable tools using a compatible method.

- Allow the brush to dry completely before storage.
- Replace bent, damaged, or contaminated brushes.

Common mistakes to avoid

- Brushing too forcefully near the eye.
- Using a contaminated brush on another person.
- Using a damaged comb or brush head.
- Storing the tool while wet or dirty.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Facial Bed

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Facial Bed

What it is used for

Handle a facial bed safely and maintain client hygiene.

How to handle

- Check that the bed frame, hinges, and adjustments are secure before use.
- Apply a clean cover for each client.
- Adjust the bed slowly and keep hands clear of moving parts.
- Clean and disinfect contact surfaces between clients.

Storage

- Return the bed to its recommended position after use.
- Keep the area dry and free of clutter.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Facial Machine

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Brushing Machine 10 in 1

What it is used for

A 10-in-1 facial machine combines multiple beauty functions for skin cleansing and cosmetic treatments. Use only the attachments and settings suitable for the client's skin and the manufacturer's instructions.

Main parts

- Main control unit and power switch
- Treatment handpieces and compatible attachments
- Speed or intensity controls
- Connecting cables and handpiece holders
- Treatment accessories and storage tray
- Power cord and plug

Before you start

1. Check the machine body, cables, plug, and attachments for damage.
2. Confirm the required handpiece is compatible with the machine.
3. Clean and disinfect reusable attachments according to instructions.
4. Check the client's skin condition, allergies, and treatment suitability.
5. Wash hands and prepare clean towels and required PPE.
6. Ensure the treatment area is dry, clean, and well ventilated.

How to use

1. Place the machine on a stable, dry treatment trolley.
2. Connect the required handpiece securely while the machine is switched off.
3. Connect the power plug to a suitable electrical outlet.
4. Switch on the machine and select the required function.
5. Set the intensity to the lowest suitable level before starting.
6. Test the attachment according to the manufacturer's instructions.
7. Perform the treatment using the approved technique and recommended duration.
8. Monitor the client's comfort and skin response throughout treatment.
9. Switch off the function before changing handpieces or attachments.
10. Switch off and unplug the machine after completing the treatment.
11. Clean, disinfect, dry, and store the attachments as instructed.

⚠ Safety

- Only trained beauty personnel should operate the machine.
- Do not use damaged cables, cracked attachments, or faulty controls.
- Avoid treating irritated, injured, infected, or unsuitable skin areas.
- Keep electrical connections and the control unit away from water.
- Do not use electrical, suction, or exfoliation functions beyond their intended purpose.
- Stop immediately if the client experiences pain, burning, or unusual irritation.

Care & storage

- Clean the machine exterior with an approved cleaning product.
- Disinfect compatible reusable attachments after every client.
- Keep connectors dry and inspect cables regularly.
- Store handpieces safely and arrange periodic servicing as recommended.

Common mistakes to avoid

- Using an attachment that is unsuitable for the client's skin.
- Starting with excessive intensity instead of a low setting.
- Sharing attachments without proper cleaning and disinfection.
- Changing handpieces while the machine is operating.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Facial Steamer

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Facial Steamer

What it is used for

A facial steamer produces warm mist to support selected facial treatments. Use it at a suitable distance and temperature to reduce the risk of burns and skin irritation.

Main parts

- Water reservoir or tank
- Heating element and control panel
- Steam outlet or nozzle
- Power switch and power cord
- Reservoir lid and water-level markings
- Stand or adjustable support, if fitted

Before you start

1. Place the steamer on a stable, level, dry surface.
2. Inspect the power cord, plug, reservoir, and nozzle for damage.
3. Fill the reservoir with the water type specified by the manufacturer.
4. Ensure the water level is within the marked limits.
5. Check the client's skin sensitivity and suitability for steam treatment.
6. Keep the steam outlet clear and the surrounding area dry.

How to use

1. Switch off the steamer before filling the reservoir.
2. Fill the tank to the specified level and close the lid securely.
3. Position the unit so the nozzle can direct mist safely toward the client.
4. Connect the plug to a suitable electrical outlet.
5. Switch on the unit and allow it to heat as instructed.
6. Check that the mist output is steady before beginning treatment.
7. Position the client's face at the recommended distance from the nozzle.
8. Use the treatment time specified for the device and client's suitability.
9. Monitor comfort and skin response continuously during treatment.
10. Stop immediately if excessive heat, discomfort, or dizziness occurs.
11. Switch off and unplug the steamer after use.
12. Allow the unit to cool before emptying or cleaning the reservoir.

⚠ Safety

- Only trained personnel should operate the steamer.
- Never direct hot steam at the eyes or too close to the skin.
- Do not overfill the water reservoir or operate it empty.
- Keep the unit and plug away from splashes and wet surfaces.
- Use caution for clients with heat sensitivity or conditions unsuitable for steam.
- Never open a hot reservoir or move the unit while it contains hot water.

Care & storage

- Unplug and cool the unit before cleaning.
- Empty remaining water according to the manufacturer's instructions.
- Remove mineral deposits using only approved descaling methods.
- Inspect the nozzle and power cord regularly.

Common mistakes to avoid

- Positioning the face too close to the steam outlet.
- Using unapproved additives or oils in the reservoir.
- Filling above the maximum level or running the tank empty.
- Cleaning or opening the reservoir while it is hot.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Foot Scraper

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Foot Scraper

What it is used for

To gently remove rough, dead skin from the feet during personal care or a supervised pedicure.

How to handle

- Inspect the scraper surface and handle for cracks, corrosion, or damage; clean and disinfect it before use.
- Soak or soften the feet as appropriate and ensure the skin is intact before starting.
- Use gentle strokes on rough areas only; avoid excessive pressure and do not scrape cuts, inflamed skin, or infected areas.
- Stop if pain, bleeding, or irritation occurs and do not share the tool without proper reprocessing.
- Clean and disinfect the scraper after use according to its material and instructions.

Storage

- Dry the scraper completely after cleaning and disinfection.
- Store in a clean, covered container separate from used personal-care tools.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Foot Spa

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Foot spa

What it is used for

A foot spa provides a warm-water soak for foot comfort and cosmetic care. Clean and disinfect the unit between users and check that water temperature is safe before placing feet in the basin.

Main parts

- Water basin or tub
- Temperature or heat controls, if fitted
- Water jets or massage rollers, if fitted
- Power switch and power cord
- Drain outlet or removable basin
- Splash guard and control panel

Before you start

1. Place the foot spa on a stable, level surface away from electrical hazards.
2. Inspect the basin, cord, plug, controls, and jets for damage.
3. Confirm the unit is clean and disinfected according to instructions.
4. Check the user's skin condition, allergies, and suitability for soaking.
5. Fill the basin only to the marked level with water at a safe temperature.
6. Ensure the plug and electrical connections remain dry.

How to use

1. Keep the unit switched off while filling the basin.
2. Fill with water to the indicated level and check the temperature.
3. If the unit heats water, follow its instructions and verify the temperature before use.
4. Ask the user to sit securely before placing their feet into the basin.
5. Switch on the selected massage or jet function if appropriate.
6. Keep the soak within the recommended duration and monitor comfort.
7. Stop the unit immediately if the user feels pain, burning, numbness, or discomfort.
8. Switch off the unit before removing the feet or draining water.
9. Unplug the unit before cleaning, unless the manufacturer's instructions specify otherwise.
10. Drain the basin carefully and remove visible debris.
11. Clean and disinfect all user-contact surfaces and compatible accessories.
12. Rinse if required, dry completely, and store the unit safely.

⚠ Safety

- Do not use the foot spa with damaged wiring or a cracked basin.
- Keep electrical plugs and controls dry and keep hands dry when operating switches.
- Check water temperature carefully to prevent burns.
- Avoid use on open wounds, active skin infections, or unsuitable skin conditions.
- Users with reduced foot sensation or circulation concerns should seek appropriate professional advice before use.
- Do not immerse the electrical unit or exceed the marked water level.

Care & storage

- Drain and clean the basin after every user.
- Disinfect jets and internal water-contact areas according to the manufacturer's instructions.
- Remove scale and deposits only with approved cleaning products.
- Inspect the cord, plug, basin, and controls regularly.

Common mistakes to avoid

- Using the foot spa without cleaning and disinfecting it between users.
- Filling above the marked level or using excessively hot water.
- Operating the unit with wet electrical connections or damaged wiring.
- Ignoring skin irritation, pain, or other signs of discomfort.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Galvanic Machine

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Galvanic Machine

What it is used for

A galvanic machine uses controlled low-level direct current for selected cosmetic treatments, depending on its design and attachments. Use only the approved electrode, product, intensity, and procedure suitable for the client.

Main parts

- Main control unit and power switch
- Positive and negative electrodes or treatment probes
- Electrode cables and connectors
- Intensity and mode controls
- Compatible treatment accessories
- Power cord and plug

Before you start

1. Inspect the machine, electrode surfaces, cables, and connectors for damage.
2. Confirm the electrodes and accessories are compatible with the machine.
3. Clean and disinfect reusable electrodes according to the manufacturer's instructions.
4. Check the client's skin condition, allergies, and treatment suitability.
5. Ask about relevant contraindications, including implanted electronic devices, as specified by the manufacturer.
6. Prepare only the approved conductive product and keep the work area dry.

How to use

1. Place the machine on a stable, dry treatment surface.
2. Connect the required electrodes while the machine is switched off.
3. Apply the approved conductive product according to its instructions.
4. Position the electrodes as specified for the selected treatment.
5. Connect the machine to a suitable outlet and switch it on.
6. Select the correct mode and begin at the lowest suitable intensity.
7. Increase intensity gradually only within the approved operating range.
8. Move or hold the electrodes using the prescribed technique and treatment time.
9. Monitor the client's comfort and skin response throughout the procedure.
10. Reduce intensity to zero and switch off before removing or changing electrodes.
11. Unplug the machine after treatment if required by the instructions.
12. Clean, disinfect, dry, and store reusable electrodes and accessories.

⚠ Safety

- Only trained beauty personnel should operate the machine.
- Do not use damaged electrodes, cables, or electrical connectors.
- Follow the manufacturer's contraindications and avoid use where the device is not suitable.
- Do not place electrodes across the chest or other prohibited areas.
- Keep electrical connections dry and never adjust electrodes while current is flowing.
- Stop immediately if the client experiences pain, burning, dizziness, or unusual irritation.

Care & storage

- Clean the machine exterior using an approved method.
- Disinfect reusable electrodes after every client.
- Inspect electrode surfaces and cables for corrosion, wear, or damage.
- Store electrodes separately to protect their surfaces and connectors.

Common mistakes to avoid

- Using an unapproved product or incompatible electrode.
- Starting at excessive intensity or exceeding the prescribed treatment time.
- Changing electrode positions while current is flowing.
- Ignoring contraindications or signs of discomfort.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Brush

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Low** 7 labs

Covers these lab items: Hair Brush; Paddle Bristle Brushes; Round Brushes; Round Brushes (various diameters)

What it is used for

Hair brushes detangle, smooth, shape, and style hair using bristles suited to different hair types and techniques. Select the correct brush shape and clean it between users to maintain hygiene.

How to use

1. Inspect the brush body, bristles, handle, and barrel for damage or loose parts.
2. Choose a paddle or round brush appropriate for the hair type and styling task.
3. Remove loose hair from the brush before use.
4. Ensure the hair is in a suitable condition for brushing or heat styling.
5. Brush gently from the ends upward to reduce pulling and breakage.
6. For round brushes, control tension and keep the brush moving during blow-drying.
7. Remove trapped hair and clean the brush after use.

⚠ Safety

- Do not use brushes with broken bristles or sharp damaged edges.
- Avoid pulling hair forcefully or brushing in a way that causes scalp pain.
- Keep brushes away from excessive heat unless designed for heat styling.
- Clean shared brushes between users and avoid use on irritated or infected scalps.

Care & storage

- Remove hair and product buildup regularly and disinfect the brush according to its material and salon hygiene procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Clip

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 11 labs

Covers these lab items: Hair Clips set; Pin curl clips; Pin Curl Clips Boxes; Pins curl clips; Section Clips

What it is used for

Hair clips and section clips hold hair in place during styling, sectioning, and beauty training.

How to handle

- Inspect clips and pins for broken edges, rust, or weak springs before use.
- Select clips suitable for the hair type and styling task.
- Use clean clips and secure hair without pulling the scalp or pinching skin.
- Remove clips gently and disinfect or clean them between clients as appropriate.

Storage

- Store clips in clean, labelled containers separated by type and size.
- Keep small clips secured to prevent loss and accidental swallowing by children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Clipper

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Clippers

What it is used for

A hair clipper trims or cuts hair using powered moving blades and interchangeable guards. Use clean, intact blades and the correct guard setting to achieve an even cut while preventing skin injury.

Main parts

- Clipper body and power switch
- Moving and stationary blade set
- Interchangeable cutting guards or combs
- Battery or power cord and charging connection
- Blade adjustment lever, if fitted
- Cleaning brush and blade lubricant

Before you start

1. Inspect the clipper body, blades, guard, and power connection for damage.
2. Confirm the battery is charged or the power supply is suitable.
3. Clean and disinfect the clipper and compatible guards between clients.
4. Check the blade alignment and ensure the blade set is securely fitted.
5. Ask about scalp sensitivity, skin irritation, and areas requiring special care.
6. Wash hands and prepare a clean workstation and required PPE.

How to use

1. Choose a guard length suitable for the requested haircut.
2. Switch off the clipper before attaching or changing the guard.
3. Fit the guard securely and check that the blade moves freely.
4. Switch on the clipper and verify normal operation and sound.
5. Hold the clipper with a firm, comfortable grip.
6. Trim in controlled strokes, following the hair growth pattern as appropriate.
7. Use shorter, gentle strokes around ears, neckline, and sensitive areas.
8. Remove loose hair from the blade area as needed, with the clipper switched off.
9. Check the cut for evenness and adjust the guard only after switching off.
10. Switch off the clipper when the haircut is complete.
11. Remove hair clippings and clean and disinfect reusable accessories.
12. Apply blade lubricant only as specified by the manufacturer.
13. Store the clipper and guards in a clean, dry place.

⚠ Safety

- Only trained users should operate the clipper professionally.
- Do not use a clipper with damaged blades, guards, or electrical insulation.
- Keep fingers, loose clothing, and cords away from moving blades.
- Do not press blades hard against the skin or use them on irritated skin.
- Switch off and disconnect the clipper before cleaning or blade changes.
- Use a new or properly disinfected compatible attachment for each client.

Care & storage

- Brush away hair after each use.
- Clean and disinfect compatible guards and contact surfaces between clients.
- Lubricate blades according to the manufacturer's instructions.
- Inspect blades, battery, cord, and charging port regularly.
- Replace dull or damaged blades with approved parts.

Common mistakes to avoid

- Using the wrong guard length for the requested cut.
- Applying excessive pressure and irritating or cutting the skin.
- Sharing guards without proper cleaning and disinfection.
- Cleaning or adjusting the blades while the clipper is running.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Connector

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Hair Connector

What it is used for

Hair connectors join or secure hair sections or hair accessories during styling, depending on the product design.

How to handle

- Identify the connector type and confirm its intended use before application.
- Inspect for sharp edges, defects, or damage that could pull hair or irritate the scalp.
- Use the connector gently and follow the styling product instructions.
- Avoid excessive tension and remove the connector carefully to prevent hair breakage.
- Clean or dispose of the item according to its reusable or single-use design.

Storage

- Store connectors in a labelled, closed container.
- Keep clean items separate from used accessories and protect them from dust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Curler (Electric)

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Electric Curler

What it is used for

An electric hair curler uses controlled heat to shape hair into curls or waves. Correct temperature selection, careful handling, and safe cooling help achieve consistent styling while reducing burns and hair damage.

Main parts

- Heated barrel or curling surface transfers heat to the hair.
- Clamp or curling attachment holds hair on applicable models.
- Handle provides a grip away from the heated surface.
- Temperature control adjusts the heating level.
- Power cord and plug supply electrical power.
- Indicator light or display shows heating status on equipped models.
- Heat-resistant stand supports the tool during use.
- Cool tip provides a designated gripping area on applicable models.

Before you start

1. Read the manufacturer's instructions and identify the available temperature settings.
2. Inspect the cord, plug, handle, barrel, and controls for damage.
3. Ensure hair is dry unless the device is specifically designed for damp hair.
4. Remove tangles and divide hair into manageable sections.
5. Place the curler on a stable, heat-resistant surface near a suitable outlet.
6. Keep the tool away from water, flammable materials, and loose fabrics.

How to use

1. Place the curler on its heat-resistant stand before switching it on.
2. Connect it to a suitable power outlet and select an appropriate temperature.
3. Wait until the tool reaches the selected temperature according to its indicator or manual.
4. Take a small section of hair and position it carefully near the barrel.
5. Wrap or clamp the hair according to the device design, keeping the hot surface away from skin.
6. Hold the hair for the recommended time and release it gently.
7. Repeat for other sections without repeatedly heating the same section.
8. Switch off and unplug the curler after styling.
9. Place the tool on a heat-resistant surface and allow it to cool completely.
10. Clean the cooled tool according to the manual and store it safely.

⚠ Safety

- Never touch the heated barrel or leave the hot tool unattended.
- Do not use the curler near water or with wet hands.
- Stop using the device if the cord, plug, or casing is damaged.
- Keep the heated surface away from the scalp, ears, neck, and fingers.
- Use the lowest effective temperature suitable for the hair type and condition.
- Unplug the tool before cleaning and allow it to cool before storage.

Care & storage

- Remove styling product residue from the cooled barrel using approved materials.
- Inspect the cord, plug, handle, and temperature controls regularly.
- Avoid wrapping the power cord tightly around the tool.
- Store the fully cooled curler in a dry, safe place.

Common mistakes to avoid

- Selecting excessive heat for the hair type or condition.
- Holding hair against the heated surface longer than recommended.
- Using the device near water or with wet hands.
- Storing the curler before it has cooled completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Dryer

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Blow dryer; Hair Dryer; Hand held dryer and attachments

What it is used for

A hair dryer uses heated airflow to dry hair and support basic hair styling. Correct heat settings, safe distance, and electrical precautions help prevent burns, hair damage, and electric shock.

Main parts

- Motor drives the fan that produces airflow.
- Heating element warms the passing air.
- Air inlet allows air to enter the dryer.
- Air outlet directs heated airflow toward the hair.
- Handle provides a secure grip during use.
- Control switches adjust heat and airflow settings.
- Concentrator or diffuser attachment shapes airflow on compatible models.

Before you start

1. Read the manufacturer's instructions and select a dryer suitable for the task.
2. Inspect the body, plug, power cord, switches, and attachments for damage.
3. Ensure hands and hair are not excessively wet before handling the electrical appliance.
4. Use the dryer in a dry, well-ventilated area away from water.
5. Remove tangled hair and loose objects from the air inlet and outlet.
6. Select the appropriate attachment and check that it fits securely.

How to use

1. Plug the dryer into a suitable outlet with dry hands.
2. Select a low or moderate heat setting suitable for the hair type.
3. Switch on the dryer and keep the outlet a safe distance from the scalp.
4. Move the dryer continuously rather than directing heat at one spot for too long.
5. Use a concentrator or diffuser only as intended by the manufacturer.
6. Keep hair and fingers away from the air inlet to prevent entanglement.
7. Reduce heat or switch to cool airflow when appropriate.
8. Switch off the dryer before removing or changing attachments.
9. Unplug the dryer after use and allow it to cool on a safe surface.
10. Clean the air inlet filter when cool and store the dryer in a dry place.

⚠ Safety

- Never use the dryer near a bath, sink, or standing water.
- Do not touch the plug or switch with wet hands.
- Keep the air inlet clear and away from hair, fabric, and other objects.
- Stop using the dryer if it smells burnt, sparks, or overheats.
- Do not use an appliance with damaged wiring or a loose attachment.
- Keep the hot outlet away from skin, eyes, and flammable materials.

Care & storage

- Unplug the dryer and allow it to cool before cleaning.
- Remove lint from the air inlet according to the manufacturer's instructions.
- Inspect the cord, plug, housing, and attachments regularly for damage.
- Store the cooled dryer with its cord loosely arranged in a dry place.

Common mistakes to avoid

- Using the dryer near water or with wet hands.
- Holding the hot airflow in one place for too long.
- Blocking the air inlet with hair or fabric.
- Wrapping the cord tightly around the dryer while it is still hot.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Pin

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Hair Pins Boxes; Invisible Pins Boxes

What it is used for

Hair pins and invisible pins secure hairstyles during grooming, styling, and beauty services.

How to handle

- Inspect pins for rust, sharp burrs, bending, or damaged protective tips.
- Choose pins appropriate for the hairstyle and hair type.
- Insert pins carefully and avoid scratching the scalp or skin.
- Collect pins immediately after use and sanitise reusable items according to salon procedures.

Storage

- Keep pins in closed, labelled boxes or designated containers.
- Store separately from clean towels and protect against moisture and rust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Roller

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 23 labs

Covers these lab items: Jumbo Rollers set; Large Size Rollers; Medium Size Rollers; Roller pins; Rollers; Small Size Hair Rollers Wires; Velcro rollers

What it is used for

Hair rollers shape curls or add volume during hairstyling and beauty training.

How to handle

- Inspect rollers, pins, clips, and wires for damage or sharp edges.
- Select a roller size suitable for the hairstyle and hair length.
- Use clean rollers on prepared hair and avoid pulling hair excessively tight.
- If using heated rollers, follow the manufacturer's temperature and cooling instructions.
- Remove rollers gently and clean them after each client.

Storage

- Store rollers and accessories in a clean, dry container separated by size.
- Keep heated equipment away from flammable materials while cooling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Steamer

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 10 labs

Covers these lab items: Climazone; Hair Steamer; Head steamer; Scalp steamer

What it is used for

A hair steamer produces warm vapour to support selected hair and scalp care treatments. Correct temperature, distance, hygiene, and supervision help prevent burns and discomfort.

Main parts

- Water reservoir holds the water used to produce vapour.
- Heating unit warms water to generate steam.
- Hood or nozzle directs vapour toward the hair and scalp.
- Control panel adjusts operating settings on supported models.
- Stand or height adjustment positions the unit for the user.
- Timer controls treatment duration on supported models.
- Power cord and switch supply and control electrical power.

Before you start

1. Read the manufacturer's instructions and confirm the steamer is suitable for the intended treatment.
2. Inspect the cord, plug, housing, reservoir, hood, and controls for damage.
3. Place the unit on a stable surface with clear space around it.
4. Fill the reservoir only with the type and amount of water specified by the manufacturer.
5. Clean reusable contact surfaces and prepare the person's hair and scalp as required.
6. Check the person's comfort and ask about scalp injuries, sensitivity, or discomfort before starting.

How to use

1. Switch off and unplug the unit before filling or adjusting the reservoir.
2. Fill the reservoir to the marked level and secure its cap if provided.
3. Position the user comfortably and adjust the hood or nozzle according to instructions.
4. Keep the recommended distance between the steam outlet and the scalp.
5. Switch on the steamer and select the approved temperature and treatment time.
6. Check comfort frequently and stop if the person reports excessive heat or irritation.
7. Keep the user supervised throughout the treatment and avoid directing steam toward the face.
8. Switch off the unit when the treatment ends and allow steam output to stop.
9. Wait for the unit to cool before opening the reservoir or draining remaining water.
10. Empty, clean, dry, and store the unit according to the manufacturer's instructions.

⚠ Safety

- Never touch the steam outlet or direct hot steam onto the skin or eyes.
- Do not open a hot or pressurised reservoir; allow the unit to cool first.
- Keep water away from the plug, switch, cord connections, and other electrical parts.
- Do not operate the steamer with a damaged cord, unstable stand, or leaking reservoir.
- Stop immediately if the user experiences burning, dizziness, breathing discomfort, or irritation.
- Use the equipment only under trained supervision and follow its specified operating limits.

Care & storage

- Unplug the unit and allow it to cool before cleaning or draining it.
- Remove mineral deposits only using the method approved by the manufacturer.
- Dry the reservoir and external surfaces before storage to reduce residue and contamination.
- Inspect the cord, plug, reservoir, and steam outlet regularly and report defects.

Common mistakes to avoid

- Filling the reservoir beyond its marked maximum level.
- Positioning the steam outlet too close to the scalp or face.
- Opening the reservoir while it is hot or pressurised.
- Leaving water inside the unit for long periods or ignoring mineral buildup.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hair Straightener

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **High** 4 labs

Covers these lab items: Straightening Iron

What it is used for

A hair straightener uses heated plates to straighten hair for styling. Use it on dry hair unless the manufacturer specifically permits another method, and select a suitable temperature for the hair type.

How to use

1. Inspect the power cord, plug, handle, and plates for damage.
2. Place the straightener on a heat-resistant surface and connect it to the approved power supply.
3. Switch on the device and select the recommended temperature setting.
4. Ensure the hair is dry and combed into manageable sections.
5. Separate a small section of hair and place it between the plates near the roots without touching the scalp.
6. Glide the straightener steadily toward the ends without holding it in one place for too long.
7. Switch off and unplug the device after styling.
8. Allow the plates to cool completely before cleaning or storage.

⚠ Safety

- Never touch the heated plates or leave the powered straightener unattended.
- Keep the device away from water, sinks, and wet surfaces.
- Avoid direct contact between hot plates and skin, scalp, or eyes.
- Do not wrap the power cord tightly around the hot device.

Care & storage

- Clean cooled plates with a soft, dry cloth according to the manual.
- Store only after the device has cooled fully and the cord is dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Heart Rate Monitor

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Heart Rate Monitor

What it is used for

A heart rate monitor measures pulse rate during exercise or routine fitness monitoring. Fit the sensor correctly and confirm that readings are stable before using them to guide activity.

How to use

1. Inspect the monitor, sensor, strap, and battery or charging status.
2. Read the manufacturer's instructions and choose the correct wearing position.
3. Fit the chest strap or wrist sensor snugly without restricting movement or circulation.
4. Switch on the monitor and wait for a stable heart rate reading.
5. Begin the activity gradually and observe the displayed heart rate during exercise.
6. Record readings as needed and stop if unusual symptoms occur.
7. Clean the sensor and strap after use and store the monitor safely.

⚠ Safety

- Do not rely on the monitor alone to diagnose a medical condition.
- Stop exercising and seek appropriate help for chest pain, faintness, or severe breathlessness.
- Do not wear damaged equipment or overtighten straps.
- Follow medical or exercise-professional advice when monitoring under special conditions.

Care & storage

- Clean the sensor and strap according to instructions and recharge or replace batteries as required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

High Frequency Machine

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: High frequency

What it is used for

To use a high-frequency beauty device for approved skin-care treatments according to the device manual and professional training. To apply the correct electrode and settings while protecting the client from burns, irritation, and electrical hazards.

Main parts

- Main unit and control dial
- High-frequency handpiece
- Glass electrodes
- Electrode holder or connector
- Power switch and cord
- Timer or intensity control, if fitted

Before you start

1. Confirm the device is intended for the planned treatment and is in good working condition.
2. Inspect the power cord, handpiece, connectors, and glass electrodes for cracks or damage.
3. Check the client's treatment history, skin condition, and contraindications according to professional protocol.
4. Remove metal jewellery from the treatment area and ensure the skin is clean and dry.
5. Explain the procedure and obtain informed consent according to workplace policy.
6. Ensure the intensity control is at minimum and the device is switched off before fitting an electrode.

How to use

1. Wash hands and prepare the treatment area using the approved hygiene procedure.
2. Select the correct intact glass electrode for the intended treatment.
3. Fit the electrode securely into the handpiece while the device is switched off.
4. Position the client comfortably and provide suitable eye protection if required by the device instructions.
5. Connect the device to the approved power source and switch it on at the lowest setting.
6. Increase intensity gradually only within the device instructions and client comfort limits.
7. Keep the electrode in controlled contact or use the specified technique for the selected treatment.
8. Avoid prolonged treatment on one spot and monitor the skin continuously for discomfort or excessive redness.

9. Reduce intensity to zero and switch off before removing or changing the electrode.
10. Disconnect the handpiece and carefully remove the electrode after the device has stopped.
11. Clean reusable components only with manufacturer-approved products and disinfect them as directed.
12. Record the treatment and report any unexpected skin reaction.

⚠ Safety

- Use only by trained beauty or skin-care professionals within their scope of practice.
- Do not use on clients with contraindications listed in the manual, such as certain implanted electronic devices or other specified medical conditions.
- Do not use near flammable products, alcohol vapours, oxygen-enriched environments, or other fire hazards.
- Never fit or remove a glass electrode while the device is energized.
- Do not use cracked electrodes, wet hands, damaged cables, or a device with faulty insulation.
- Stop immediately if the client experiences pain, burning, unusual tingling, or unexpected skin changes.

Care & storage

- Clean and disinfect handpieces and reusable electrodes according to the manual.
- Inspect glass electrodes for cracks before and after every treatment.
- Store electrodes in protective holders to prevent breakage.
- Keep the unit dry and avoid pulling the power cord.
- Arrange inspection and servicing if the unit overheats, sparks abnormally, or malfunctions.

Common mistakes to avoid

- Starting at excessive intensity instead of increasing gradually.
- Using a cracked electrode or fitting it while the device is on.
- Applying the treatment over wet skin or near flammable products.
- Ignoring contraindications or client discomfort.
- Using unapproved cleaning chemicals that can damage electrodes or the handpiece.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Highlighting Cap

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Frosting Cap; Hi/lo-lighting cap

What it is used for

A highlighting cap helps separate hair sections during frosting or highlighting treatments.

How to handle

- Inspect the cap for tears, damaged holes, and a secure fit before use.
- Select a cap suitable for the client's hair and the planned treatment.
- Position the cap comfortably and avoid pulling hair or irritating the scalp.
- Use the specified hook carefully and keep its point away from the skin.
- Clean or dispose of the cap according to its design and salon hygiene procedure.

Storage

- Store reusable caps clean and dry in a labelled container.
- Keep disposable caps sealed until use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hood Dryer

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hood dryer

What it is used for

A hood dryer circulates warm air around the head to dry hair or support selected hairstyling treatments. Use an appropriate heat and airflow setting and monitor client comfort throughout the session.

Main parts

- Hood or dome
- Heating element and airflow fan
- Temperature and timer controls
- Height or position adjustment mechanism
- Base, stand, or mounting assembly
- Power cord and switch

Before you start

1. Inspect the hood, stand, controls, cord, and plug for damage.
2. Place the dryer on a stable surface and check its adjustment mechanism.
3. Clean the hood and any reusable client-contact accessories.
4. Check the client's scalp condition, heat sensitivity, and treatment suitability.
5. Set a suitable temperature and timer for the hair type and treatment.
6. Ensure air inlets and outlets are clear and the area is dry.

How to use

1. Position the hood dryer securely and adjust its height before use.
2. Ask the client to sit comfortably and remove unsuitable accessories near the hood.
3. Arrange the hair and treatment product according to the service procedure.
4. Connect the dryer to a suitable electrical outlet.
5. Switch on the unit and select a low suitable temperature setting.
6. Lower or position the hood so it does not press tightly against the client.
7. Set the timer according to the treatment instructions.
8. Monitor the client's comfort, scalp temperature, and airflow throughout the session.
9. Reduce heat or stop the dryer immediately if discomfort or excessive heat occurs.
10. Switch off the unit before moving the hood away from the client.
11. Allow hot components to cool before cleaning or repositioning the unit.
12. Clean the hood and client-contact surfaces after use.
13. Store or park the dryer securely and report any faults.

⚠ Safety

- Only trained personnel should operate the hood dryer professionally.
- Do not use the dryer with damaged wiring, a loose stand, or faulty controls.
- Keep ventilation openings clear and avoid covering the unit.
- Prevent hair, towels, and clothing from contacting hot components.
- Do not leave the client unattended if the procedure requires monitoring.
- Disconnect power before cleaning or servicing the dryer.

Care & storage

- Clean the hood and contact surfaces between clients.
- Remove dust and hair from air inlets according to instructions.
- Inspect the stand, adjustment mechanism, cord, and plug regularly.
- Arrange servicing if airflow, temperature control, or timer operation becomes abnormal.

Common mistakes to avoid

- Setting excessive heat or using an unsuitable treatment duration.
- Positioning the hood too close or pressing it against the client.
- Blocking airflow openings with towels or other items.
- Cleaning the unit while it is hot or connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hot Hair Roller

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hot rollers with pins

What it is used for

Hot hair rollers use heated rollers to shape curls or waves in hair. Select suitable roller sizes and temperature, handle heated parts carefully, and follow the manufacturer's timing instructions.

Main parts

- Heated roller set and heating base
- Individual rollers in different sizes
- Roller clips or pins
- Power switch and temperature controls, if fitted
- Power cord and plug
- Heat-resistant storage case or holder

Before you start

1. Inspect the heating base, rollers, clips, cord, and plug for damage.
2. Place the heating base on a stable, heat-resistant surface.
3. Check that rollers are clean and suitable for the intended hairstyle.
4. Confirm the rollers are compatible with the heating base.
5. Prepare dry or appropriately prepared hair according to the product instructions.
6. Keep water, sprays, and flammable styling products away from hot surfaces.

How to use

1. Connect the heating base to a suitable electrical outlet.
2. Switch on the base and allow the rollers to heat as instructed.
3. Select roller sizes according to the desired curl or wave.
4. Use suitable heat protection if recommended for the client's hair.
5. Lift one small section of hair and position the roller at the end.
6. Roll the hair evenly without pulling excessively on the scalp.
7. Secure the roller using the supplied clip or pin.
8. Repeat with manageable sections while keeping heated rollers away from the skin.
9. Allow rollers to remain in place for the recommended time.
10. Remove clips and unwind each roller gently without pulling the hair.
11. Switch off and unplug the heating base after use.
12. Allow rollers to cool completely before cleaning or storage.
13. Store rollers and clips in their designated case after cooling.

⚠ Safety

- Use only rollers and accessories compatible with the heating base.
- Avoid direct contact between hot rollers and scalp, ears, neck, or fingers.
- Do not use damaged cords, cracked rollers, or faulty heating bases.
- Keep the heating base away from water and flammable materials.
- Do not leave the heating base switched on unattended.
- Unplug the base before cleaning or handling electrical parts.

Care & storage

- Allow all rollers to cool before cleaning.
- Remove hair and product residue using the approved cleaning method.
- Keep electrical contacts and the heating base dry.
- Inspect clips, rollers, and cord regularly for wear or damage.
- Store the set in a dry location away from children.

Common mistakes to avoid

- Using rollers before they reach the recommended temperature.
- Wrapping hair too tightly and pulling on the scalp.
- Touching hot rollers directly or placing them on an unprotected surface.
- Storing rollers while still hot or leaving the base powered on.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Kettlebell

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Kettlebells

What it is used for

A kettlebell is used for strength, balance, and conditioning exercises. Select an appropriate weight and maintain controlled technique throughout each movement.

How to use

1. Inspect the kettlebell handle and body for cracks, looseness, or surface damage.
2. Choose a weight suitable for your ability and the planned exercise.
3. Place the kettlebell on a stable, nonslip floor with enough clearance around it.
4. Warm up and review the correct movement technique before lifting.
5. Grip the handle securely and perform the exercise with controlled posture and movement.
6. Return the kettlebell to the floor carefully without dropping or throwing it.
7. Check the equipment and clear the exercise area after use.

⚠ Safety

- Keep feet, hands, and other people clear of the kettlebell's movement path.
- Do not use a damaged kettlebell or a weight beyond your ability.
- Avoid sudden twisting and maintain a stable spine and controlled grip.
- Stop if you experience sharp pain, dizziness, or unusual shortness of breath.

Care & storage

- Wipe the handle and body clean and store the kettlebell in a designated rack.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Magnifier Lamp

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Desk Light and Magnifier

What it is used for

A magnifier lamp combines illumination and magnification for close visual work such as inspection or beauty treatments. Position the lamp securely and adjust lighting to suit the task.

How to use

1. Inspect the lens, lamp, arm joints, base, cable, and switch for damage.
2. Place the lamp on a stable surface or secure its mounting clamp.
3. Connect it to the specified power source and switch it on.
4. Adjust the arm and lens to the required working distance and angle.
5. Set brightness and magnification for comfortable, clear viewing.
6. Perform the task while avoiding contact with the hot lamp or lens housing.
7. Switch off and disconnect power before cleaning or repositioning the unit.

⚠ Safety

- Do not look directly into the light source at close range.
- Keep electrical connections away from water and cosmetic liquids.
- Avoid overextending the arm or placing the lamp where it may fall.
- Do not use the lamp if its cable, lens, or mounting is damaged.

Care & storage

- Clean the lens with approved lens material and wipe the housing with power disconnected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Makeup Brush

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Complete Make Up Brushes Set

What it is used for

To apply, blend, and distribute makeup products evenly during beauty services.

How to handle

- Inspect brush bristles, handles, and ferrules for damage or shedding before use.
- Select a brush appropriate for the product and application area.
- Use gentle strokes and avoid contact with irritated skin or the eyes unless the brush is intended for that area.
- Use clean brushes for each client and avoid double-dipping a contaminated brush into shared products.
- Clean brushes with a suitable cleanser, rinse as appropriate, reshape the bristles, and allow them to dry fully.

Storage

- Store completely dry in a clean, covered brush holder or case.
- Keep brushes away from dust, moisture, and used or contaminated tools.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Manicure Pedicure Kit

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Cuticle knife, Buffers, Pedicure rasp (hard skin removal), Nail scissors, Clippers, Hard skin remover, nipper (for pedicure); Manicure Set; Pedicure Set

What it is used for

A manicure pedicure kit is used to clean, shape, and care for fingernails, toenails, and surrounding skin. Select tools according to the treatment and hygiene requirements.

How to use

1. Inspect all tools for cleanliness, damage, rust, and sharp edges.
2. Wash hands and clean the client's hands or feet before treatment.
3. Disinfect or sterilize reusable tools according to the tool type and salon protocol.
4. Use clippers and scissors carefully to trim nails without cutting too deeply.
5. Use files, buffers, and cuticle tools gently; avoid cutting living skin.
6. Clean and process tools after use, then store them in a clean container.

⚠ Safety

- Use clean tools for every client and never share unprocessed instruments.
- Do not cut cuticles deeply or use sharp tools on irritated skin.
- Avoid treatment on open wounds or signs of infection; follow salon procedures.
- Handle sharp nippers and scissors carefully to prevent cuts.

Care & storage

- Clean, dry, disinfect, and store tools according to the manufacturer's instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Massage Bed

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Body Massage Bed

What it is used for

A massage bed provides a stable, comfortable surface for massage and approved wellness treatments.

How to handle

- Inspect the frame, padding, hinges, height adjustments, and locks before use.
- Confirm the bed's weight capacity and position it on a level, stable floor.
- Use clean linens for each client and follow applicable hygiene procedures.
- Adjust the bed safely and assist clients when getting on or off.
- Do not use a bed with loose fittings, unstable legs, or damaged padding.

Storage

- Clean and dry the bed after use.
- Keep it in a designated area and secure folding parts before moving or storing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Meditation Cushion

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Meditation cushions

What it is used for

Use meditation cushions comfortably and hygienically during wellness sessions.

How to handle

- Check the cushion cover and filling for tears, stains, or dampness before use.
- Place the cushion on a stable, clean surface.
- Use a clean cover for shared sessions and avoid sharing cushions when hygiene cannot be maintained.
- Follow the activity leader's guidance and stop if the cushion causes discomfort.

Storage

- Air the cushion after use and ensure it is fully dry before storage.
- Store in a clean, ventilated area away from moisture and pests.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nadi Swedana Yantra

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Nadee Swedan Yantra

What it is used for

A Nadi Swedana Yantra is a traditional Ayurvedic steam apparatus used to direct steam toward a selected body area under trained supervision. It requires careful control of steam flow, temperature and exposure to reduce burn risk.

Main parts

- Identify the steam-generating vessel or boiler.
- Locate the water-filling opening and specified water-level indicator, if fitted.
- Identify the heating element or fuel-fired heating arrangement.
- Locate the steam outlet pipe and directing nozzle.
- Identify the steam-flow control valve, where provided.
- Locate the temperature and pressure indicators, if fitted.
- Identify the support frame, insulated handles and power or fuel controls.

Before you start

1. Use the apparatus only under a trained Ayurvedic or healthcare practitioner's supervision.
2. Read the operating instructions and confirm the equipment is designed for this use.
3. Inspect the vessel, pipe, nozzle, seals and controls for damage or leaks.
4. Fill with the specified water quantity before switching on the heater.
5. Place the apparatus on a stable, heat-resistant surface in a ventilated area.
6. Keep clean towels, suitable protective equipment and first-aid supplies accessible.
7. Explain the procedure and check for heat sensitivity or other relevant precautions.

How to use

1. Position the apparatus securely away from the user and other people.
2. Fill the vessel with the specified amount of clean water.
3. Close the filling opening and confirm all required connections are secure.
4. Switch on the heater or operate the approved heat source as instructed.
5. Allow steam to develop while monitoring the specified operating indicators.
6. Adjust the steam flow using the designated control, if provided.
7. Test steam delivery safely away from the body before beginning exposure.
8. Direct steam toward the intended area from the practitioner-recommended distance.
9. Monitor skin response and comfort continuously throughout the procedure.
10. Stop immediately if excessive heat, discomfort or skin redness develops.
11. Switch off the heat source and allow the apparatus to cool before draining.

⚠ Safety

- Never direct steam toward the face, eyes, wounds or sensitive areas without specific clinical guidance.
- Keep hands and skin away from the nozzle and escaping steam.
- Never open a hot pressurised vessel or block the steam outlet.
- Do not leave the heating apparatus unattended during operation.
- Use only the specified water quantity and never operate the heater dry.
- Stop exposure immediately if burning, dizziness or unusual discomfort occurs.

Care & storage

- Switch off and cool the apparatus completely before cleaning.
- Drain remaining water safely and clean the vessel and nozzle as instructed.
- Inspect pipes, seals, valves and electrical components for damage after use.
- Store the dry apparatus in a clean, ventilated and secure area.

Common mistakes to avoid

- Starting the heater without checking the required water level.
- Directing hot steam too close to skin or for too long.
- Opening the vessel before it has cooled and pressure has safely released.
- Using damaged pipes or blocked nozzles that cause unsafe steam release.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nail Cutter

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Nail Cutter

What it is used for

To trim fingernails or toenails safely and maintain personal grooming hygiene.

How to handle

- Inspect the nail cutter blades, hinge, and lever for rust, looseness, or damage.
- Clean and disinfect the cutter before use, especially when used in a salon or shared setting.
- Trim small sections at a time and avoid cutting too close to the skin or cutting into corners deeply.
- Use extra caution around damaged nails or sensitive skin; do not use a shared tool without proper reprocessing.
- Clean and disinfect the cutter after use and dry it thoroughly.

Storage

- Dry the cutter completely to prevent corrosion.
- Store closed in a clean, dry personal-care case separate from other people's grooming tools.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nail File

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Nail Filer

What it is used for

To smooth, shape, and refine fingernails or toenails during personal grooming or a supervised manicure or pedicure.

How to handle

- Inspect the nail file for cracks, worn abrasive surfaces, or contamination.
- Clean the file before use and ensure the nail is dry unless the file instructions specify otherwise.
- File in controlled strokes from the side toward the centre, shaping the nail without excessive pressure.
- Avoid filing the surrounding skin or using the file on injured, infected, or irritated areas.
- Clean or dispose of the file according to its material and hygiene requirements; do not share single-use files.

Storage

- Remove nail dust and dry the file after cleaning if it is reusable.
- Store in a clean, dry, covered personal-care case; keep single-use files separate for disposal.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Neti Pot

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Jalneti pot; Neti Pot; Rubber Neti

What it is used for

To rinse the nasal passages with saline solution as part of personal nasal hygiene, when appropriate.

How to handle

- Inspect the pot and spout for cracks, residue, or damage before use.
- Use only sterile, distilled, or previously boiled and cooled water with an appropriate saline preparation; never use untreated tap water.
- Wash hands and prepare the solution according to the product instructions.
- Tilt the head over a sink and allow the solution to flow gently without forcing it; stop if pain, significant discomfort, or bleeding occurs.
- Wash the pot thoroughly after use and allow it to air-dry completely.

Storage

- Store completely dry in a clean, ventilated place away from dust.
- Do not share the pot; keep it separate from other personal-care items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oil Pouring Vessel

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Kindi/kumbhika for poring oil

What it is used for

An oil pouring vessel, such as a kindi or kumbhika, holds and dispenses oil in controlled quantities for suitable traditional or wellness applications. Clean the vessel and pour carefully to prevent spills and contamination.

How to use

1. Inspect the vessel, handle, spout, and opening for cracks or sharp edges.
2. Wash and dry the vessel according to the material and intended use.
3. Fill it with the required oil without exceeding a safe handling level.
4. Hold the vessel firmly and position the spout over the intended container or application area.
5. Pour slowly and control the flow to avoid splashing or overflow.
6. Close or cover the vessel after pouring and wipe any spilled oil immediately.
7. Empty, clean, and store the vessel according to its intended use.

⚠ Safety

- Keep oil away from flames and hot surfaces if it is flammable.
- Avoid spills that may make floors slippery.
- Use clean oil and a clean vessel for applications requiring hygiene.
- Do not use a cracked vessel or one with contaminated residue.

Care & storage

- Clean after use, dry thoroughly, and store covered in a clean location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paraffin Heater

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Paraffin Heater

What it is used for

A paraffin heater melts paraffin wax to a controlled temperature for approved beauty or wellness treatments. Verify the wax temperature before use and follow hygiene procedures to prevent burns and contamination.

Main parts

- Heating tank or wax reservoir
- Temperature control and thermostat
- Power switch and indicator light
- Lid and insulated outer housing
- Removable inner container, if fitted
- Power cord and plug

Before you start

1. Confirm the heater is intended for the specific paraffin wax and treatment.
2. Inspect the tank, lid, controls, cord, and plug for damage.
3. Place the heater on a stable, heat-resistant, dry surface.
4. Ensure the tank is clean and free of water or incompatible products.
5. Check the recommended wax level and operating temperature.
6. Assess the client's skin condition, sensitivity, and suitability for the treatment.

How to use

1. Switch off the heater before adding wax.
2. Add the approved paraffin wax without exceeding the maximum fill level.
3. Close the lid and connect the heater to a suitable outlet.
4. Switch on the unit and set the temperature according to the manufacturer's instructions.
5. Allow the wax to melt completely and stabilise at the required temperature.
6. Check the wax temperature using the recommended method before treatment.
7. Prepare the treatment area and ensure the client's skin is clean and dry.
8. Apply the wax using the approved technique and avoid areas unsuitable for treatment.
9. Monitor the client's comfort and skin response throughout the procedure.
10. Stop immediately if burning, pain, or unusual irritation occurs.
11. Switch off and unplug the heater after the session or as instructed.
12. Allow the wax and heater to cool before cleaning or moving the unit.
13. Clean the heater according to the manufacturer's instructions and store it safely.

⚠ Safety

- Only trained personnel should operate the paraffin heater for treatments.
- Never apply wax before confirming that its temperature is safe.
- Do not use on broken skin, active infection, or unsuitable areas.
- Keep water away from molten wax and electrical components.
- Do not overfill the tank or leave the heater unattended while operating.
- Avoid contact with hot wax and heated tank surfaces.

Care & storage

- Switch off, unplug, and cool the heater before cleaning.
- Remove wax residue using only the manufacturer's approved method.
- Keep the control panel, cord, and plug dry.
- Inspect the tank, thermostat, and electrical connections regularly.
- Replace damaged components and arrange authorised servicing when needed.

Common mistakes to avoid

- Applying wax without checking its temperature.
- Using contaminated wax or unsuitable additives.
- Overfilling the reservoir or allowing water into the wax.
- Cleaning or moving the heater while it is hot or connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Perm Curler

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Curling Rods; Curling Rods (Small, Medium, Large); Perm curlers; Perm curlers (various sizes)

What it is used for

Perm curlers are used to wrap hair into sections during a permanent waving treatment. Select curler size according to hair length, desired curl, and the treatment method.

How to use

1. Inspect curlers for cracks, sharp edges, and cleanliness.
2. Wash or prepare the hair according to the treatment instructions.
3. Section the hair evenly and select an appropriate curler size.
4. Comb each section smoothly and wrap it around the curler without excessive tension.
5. Secure the curler without pulling the scalp or creasing the hair.
6. After the prescribed processing time, remove curlers carefully and clean them.

⚠ Safety

- Follow product instructions and avoid contact of perm chemicals with eyes and skin.
- Do not wrap hair so tightly that it causes pain or scalp tension.
- Use clean curlers for each client and follow salon hygiene procedures.

Care & storage

- Clean and dry curlers after use, then store them in a clean container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Perming Cap

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Perming Cap

What it is used for

A perming cap helps cover hair during designated salon perming or treatment procedures.

How to handle

- Confirm the cap is intended for the specific treatment and fits securely without discomfort.
- Inspect the cap for tears, residue, or damage before use.
- Follow the product and salon procedure instructions for application and timing.
- Keep treatment chemicals away from eyes and skin areas not intended for contact.
- Clean reusable caps or dispose of single-use caps according to hygiene procedures.

Storage

- Keep unused caps clean and dry in labelled packaging.
- Store away from direct sunlight, heat, and salon chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pull-up Bar

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Pull-up Bar

What it is used for

A pull-up bar supports upper-body strength exercises using controlled pulling movements. Confirm that the bar and its mounting system can safely support the intended load before exercising.

How to use

1. Inspect the bar, grips, mounting brackets, fasteners, and surrounding structure.
2. Confirm the installation is secure and rated for the intended user and activity.
3. Clear the area beneath and around the bar.
4. Warm up shoulders, arms, and back before beginning.
5. Grip the bar securely and perform controlled repetitions within your ability.
6. Step down carefully and rest between sets.
7. Recheck the bar for movement or damage after use.

⚠ Safety

- Do not use a loose, cracked, or incorrectly installed bar.
- Avoid swinging or jumping onto the bar unless the equipment is designed for it.
- Keep the landing area clear and use assistance when needed.
- Stop exercising if pain, dizziness, or unusual discomfort occurs.

Care & storage

- Wipe grips after use and inspect mounting points and fasteners regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Push-up Bar

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Push-up Bars

What it is used for

Push-up bars provide raised hand grips for push-ups and other suitable bodyweight exercises. Check the bars for stability and use controlled movements on a nonslip surface.

How to use

1. Inspect both bars, grips, bases, and joints for damage or looseness.
2. Place the bars on a flat, nonslip surface with sufficient spacing.
3. Confirm that both bars remain stable under light pressure before use.
4. Warm up wrists, shoulders, and arms before exercising.
5. Grip the bars securely and perform push-ups with controlled body alignment.
6. Avoid excessive range or repetitions beyond your ability.
7. After use, inspect the bars and return them to storage.

⚠ Safety

- Do not use cracked, loose, or unstable push-up bars.
- Keep the floor dry and clear of obstacles.
- Avoid locking joints forcefully or allowing the bars to slide during exercise.
- Stop if pain, dizziness, or unusual discomfort occurs.

Care & storage

- Wipe grips and bases after use and store the bars in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Razor

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Razor with blade

What it is used for

To shave or remove hair using a suitable razor. To reduce skin irritation and cuts through proper preparation, controlled strokes, and hygienic blade handling.

Main parts

- Razor handle
- Blade or cartridge
- Blade guard or cover
- Grip area

Before you start

1. Check that the razor and blade are clean and undamaged.
2. Wash the skin and soften hair with suitable water or shaving product.
3. Use a personal razor and avoid sharing blades.
4. Prepare a clean place for rinsing and disposal.

How to use

1. Apply suitable shaving gel or cream if needed.
2. Hold the razor with a secure grip.
3. Shave using short, gentle strokes in the direction recommended for the skin and hair.
4. Rinse the blade frequently to remove hair and product buildup.
5. Avoid repeated pressure over irritated or damaged skin.
6. Rinse the skin and apply a suitable aftercare product if appropriate.
7. Rinse and dry the razor safely, then cover or store it.

⚠ Safety

- Keep the blade away from children and other users.
- Do not use a rusty, damaged, or blunt blade.
- Avoid shaving over infected wounds or severely irritated skin.
- Dispose of replaceable blades in a suitable sharps container.

Care & storage

- Rinse and dry the razor after use.
- Replace blades when dull or damaged.

- Store in a clean, dry place.
- Do not share razors or cartridges.

Common mistakes to avoid

- Applying excessive pressure.
- Using a dull blade or shaving dry skin without suitable preparation.
- Sharing a razor.
- Leaving the razor wet in an unclean area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Resistance Band

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Exercise Bands; Resistance Bands; Resistance Tubes

What it is used for

To support strength training, stretching, mobility, and rehabilitation exercises using elastic resistance.

How to handle

- Inspect the band or tube for cuts, cracks, thinning, damaged handles, or loose connectors before use.
- Select a resistance level appropriate for the exercise and the user's ability.
- Secure the band only to a suitable anchor and check that it cannot slip or release unexpectedly.
- Use controlled movements, maintain correct posture, and keep the band away from the face and eyes.
- Release tension gradually, stop if the band is damaged or causes pain, and wipe it clean after use.

Storage

- Store clean and dry away from direct sunlight, heat, and sharp objects.
- Keep bands loosely coiled without excessive stretching or twisting.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shampoo Unit

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Shampoo Bowl Set; Shampoo Unit

What it is used for

Use a shampoo unit to wash a client's hair while supporting the head and controlling water flow and temperature.

How to use

1. Clean and disinfect the basin, neck rest, spray hose, and nearby surfaces.
2. Check the water supply, drainage, chair stability, and hose connections.
3. Adjust the chair and neck support to suit the client comfortably.
4. Test water temperature on your wrist or with an approved thermometer before wetting the hair.
5. Wet the hair gently and apply shampoo according to the service procedure.
6. Rinse thoroughly while preventing water from entering the client's eyes or ears.
7. Dry splashes, clean the basin, and reset the unit after the service.

⚠ Safety

- Confirm water temperature with the client and avoid excessively hot water.
- Support the client's neck and avoid forcing the head into an uncomfortable position.
- Keep electrical equipment and plugs away from water.
- Disinfect contact surfaces and follow hygiene procedures between clients.

Care & storage

- Clean the basin and spray attachments after each client.
- Check drainage and hoses regularly for leaks or blockages.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shirobasti Cap

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Shirobasti cap

What it is used for

A Shirobasti cap is used in traditional wellness treatments to contain oil on the head for a specified period under trained supervision. Use hygienic equipment and follow the approved treatment protocol and client precautions.

How to use

1. Inspect the cap, sealing edges, fasteners, and supporting accessories for damage.
2. Confirm that the treatment is appropriate and that relevant client precautions have been reviewed.
3. Clean and prepare the cap according to the approved hygiene procedure.
4. Position the client comfortably and fit the cap securely without excessive pressure.
5. Check the seal and apply oil only according to the trained practitioner's protocol.
6. Monitor comfort and skin condition throughout the treatment and stop if discomfort occurs.
7. Remove the cap carefully, clean the client as required, and sanitize reusable equipment.

⚠ Safety

- Use the cap only under appropriately trained supervision.
- Avoid oil entering the eyes, ears, or respiratory passages.
- Do not use damaged equipment or continue if irritation, dizziness, or discomfort develops.
- Check for known allergies and follow applicable hygiene precautions.

Care & storage

- Clean, dry, and store the cap in a hygienic place after each use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shirodhara Equipment

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Shirodhara Stand equipment; Shirodhara Sweda

What it is used for

To deliver a controlled stream of liquid during a supervised Shirodhara wellness treatment using the designated stand and vessel.

How to handle

- Inspect the stand, vessel, tubing, flow-control parts, and fasteners for cracks, leaks, instability, or contamination.
- Set the equipment on a level surface and secure the stand and vessel according to the manufacturer's instructions.
- Use only approved, suitable liquids and check temperature carefully before treatment to avoid burns or irritation.
- Ensure the client is positioned comfortably and the procedure is performed by a trained practitioner with appropriate consent and hygiene precautions.
- Control the flow and duration according to the treatment protocol; stop immediately if discomfort, dizziness, skin irritation, or equipment instability occurs.

Storage

- Drain the vessel and clean and dry all liquid-contact parts according to the manufacturer's instructions.
- Store disassembled components in a clean, dry area and protect the stand and vessel from impact or contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shower Cap

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Shower Cap

What it is used for

A shower cap covers hair during bathing or designated beauty and personal-care activities.

How to handle

- Check the cap for tears, cleanliness, and suitable fit before use.
- Use the cap for its intended purpose and avoid sharing single-use caps.
- Remove carefully to prevent hair pulling or discomfort.
- Clean reusable caps according to the manufacturer's instructions.
- Discard disposable caps after use or if damaged or contaminated.

Storage

- Store clean caps in a dry, covered container.
- Keep unused caps separate from used personal-care items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Step Platform

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Step Platform

What it is used for

A step platform supports aerobic stepping, balance practice, and fitness exercises. Select an appropriate height and ensure the platform remains stable on a nonslip surface.

How to use

1. Inspect the platform surface, feet, risers, and locking points for damage.
2. Assemble the platform securely and confirm that all risers are locked in place.
3. Place it on a flat, nonslip surface with clear space around it.
4. Choose a height suitable for the user's fitness and coordination level.
5. Warm up and step onto the platform with controlled foot placement.
6. Maintain balance and step down carefully without jumping unnecessarily.
7. After exercise, inspect, clean, and store the platform.

⚠ Safety

- Do not use a cracked, unstable, or wet platform.
- Keep the exercise area clear of obstacles and other participants.
- Wear suitable footwear and avoid movements beyond your ability.
- Stop if pain, dizziness, or loss of balance occurs.

Care & storage

- Wipe the platform after use and check its feet, risers, and locking points regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stone Therapy Set

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Stone Therapy set

What it is used for

A stone therapy set contains stones used for massage or wellness treatments. Use clean, suitable stones at a safe temperature and follow the client's comfort and the treatment protocol.

How to use

1. Inspect stones and accessories for cracks, sharp edges, or damage.
2. Clean and disinfect reusable stones according to the material and product instructions.
3. For heated stones, use an approved heater and check each stone's temperature before application.
4. Explain the procedure and check for sensitivities or contraindications before treatment.
5. Position the client comfortably and test the stone on a suitable area first.
6. Apply stones gently using the approved technique and monitor the client's comfort continuously.
7. Remove stones promptly if discomfort, redness, or excessive heat occurs.
8. Clean, dry, and store stones after the treatment.

⚠ Safety

- Never apply stones that are excessively hot, cracked, or contaminated.
- Avoid placing heated stones on numb skin, injured areas, or areas with reduced sensation.
- Do not perform treatment where contraindications exist; seek qualified medical guidance when uncertain.
- Maintain hygiene and never share unclean stones between clients.

Care & storage

- Clean and disinfect stones after every client according to the product instructions.
- Store stones dry in a clean, protected container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sutra Neti

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Sutra Neti

What it is used for

Handle a Sutra Neti practice cord hygienically and safely for nasal cleansing demonstrations.

How to handle

- Use only a purpose-made, clean cord intended for Sutra Neti and follow qualified instructor guidance.
- Do not share the cord between users; inspect it for fraying or damage before use.
- Never force the cord or perform the practice if there is pain, bleeding, obstruction, or uncertainty.
- Clean and disinfect or dispose of the cord according to the manufacturer's and instructor's guidance.

Storage

- Keep the cord in a clean, sealed, labelled container after it is fully dry.
- Store separately from general personal-care items and contaminated materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tattoo Machine

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Tattooing set

What it is used for

A tattoo machine drives a needle cartridge or needle assembly to deposit tattoo pigment into the skin under controlled conditions. Safe operation requires appropriate training, sterile single-use needles, hygienic work practices, suitable equipment settings, and correct sharps disposal to reduce infection and injury risks.

Main parts

- Machine body houses the drive mechanism.
- Motor or electromagnetic coil drives the needle mechanism depending on the design.
- Grip provides a controlled handhold.
- Needle cartridge or needle assembly carries the needles.
- Power supply or battery controls machine operation.
- Foot pedal operates compatible wired machines where fitted.
- Connecting cable supplies power on wired models.
- Barrier covers help protect machine surfaces from contamination.
- Approved pigment cups hold small amounts of tattoo pigment.

Before you start

1. Tattooing should be performed only by trained personnel in compliance with applicable local laws and hygiene requirements.
2. Inspect the machine, grip, power supply, cable, and controls for damage before use.
3. Confirm that needles or cartridges are sterile, sealed, within expiry, and intended for single use.
4. Clean and disinfect the work area and use suitable barriers on surfaces and equipment.
5. Perform hand hygiene, wear new suitable gloves, and prepare clean supplies without contaminating sterile items.
6. Check that pigments are appropriate for tattooing, traceable, and stored according to their instructions.
7. Explain the procedure, obtain informed consent, check relevant contraindications, and follow appropriate skin-preparation procedures.

How to use

1. Prepare a clean, well-lit work area and organise clean and contaminated zones separately.
2. Cover machine surfaces, cables, and controls with appropriate single-use barriers where practical.

3. Open the sterile needle cartridge or needle assembly immediately before use without touching sterile parts.
4. Fit the needle assembly and grip securely according to the machine instructions.
5. Set the machine voltage, stroke, and needle depth only within the manufacturer's guidance and the operator's training.
6. Dispense pigment into single-use cups without contaminating the original containers.
7. Prepare the skin using an approved skin antiseptic and allow it to dry as directed.
8. Operate the machine with controlled hand movements and avoid excessive pressure or repeated trauma to the skin.
9. Monitor the client's comfort and skin response; stop if there is an unexpected reaction or significant bleeding.
10. After the procedure, switch off the machine and remove the needle assembly without touching exposed sharps.
11. Place used needles and other sharps immediately into an approved puncture-resistant sharps container.
12. Dispose of contaminated consumables according to applicable biomedical waste requirements and clean and disinfect reusable equipment.
13. Provide appropriate aftercare instructions and record the procedure and any adverse event according to local requirements.

⚠ Safety

- Never reuse needles, cartridges, pigment cups, or other items designated for single use.
- Use gloves, hand hygiene, barriers, and appropriate disinfection to prevent cross-contamination and bloodborne infection.
- Do not tattoo over infected, inflamed, or otherwise unsuitable skin; refer health concerns for appropriate assessment.
- Never recap used needles by hand or leave exposed sharps on the work surface.
- Use only pigments permitted for the intended use and follow their safety instructions.
- Stop immediately if the machine overheats, electrical insulation is damaged, or the client develops a serious reaction.

Care & storage

- Switch off and disconnect the machine before cleaning or changing compatible components.
- Clean and disinfect reusable parts using methods approved by the manufacturer; do not immerse electrical components unless expressly permitted.
- Inspect cables, grips, seals, and machine housing regularly for wear or damage.
- Store the machine and sterile supplies in a clean, dry, protected area.
- Arrange authorised servicing if the drive mechanism becomes irregular or the machine overheats.

Common mistakes to avoid

- Using a needle cartridge that is unsealed, expired, or previously used.
- Touching sterile needles or clean supplies with contaminated gloves.

- Using excessive needle depth, pressure, or repeated passes that unnecessarily injure skin.
- Failing to dispose of sharps promptly or disinfect the work area between clients.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thinning Scissors

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Thinning scissor; Thinning Scissors; Thinning Scissors, cutting scissor

What it is used for

Thinning scissors remove some hair bulk while leaving much of the hair length intact. Use them carefully to create a softer blend and avoid excessive thinning.

How to use

1. Inspect the thinning scissors for clean teeth, smooth movement, and secure pivot.
2. Consult the client about the desired result before starting.
3. Work on dry or suitably prepared hair according to the technique being used.
4. Separate a small section and hold it away from the scalp and face.
5. Make controlled cuts with the toothed blade positioned as intended for the technique.
6. Check the result gradually, clean the blades, and store the scissors safely.

⚠ Safety

- Keep fingers and blades away from the client's skin, ears, and eyes.
- Do not use damaged scissors or force the pivot.
- Avoid repeated thinning in the same area, which may leave uneven or sparse hair.

Care & storage

- Remove hair from the teeth, dry the blades, and store scissors in a protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tinting Brush

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Bleach Brush; Bowl and colour Brush; Tinting Brush with Comb

What it is used for

Use a tinting brush to apply hair colour, bleach, or treatment product evenly according to the service instructions.

How to use

1. Select a clean tinting brush suitable for the product and application area.
2. Inspect the bristles and handle for damage or loose parts.
3. Wear gloves and prepare the product according to its instructions.
4. Section the hair and apply the product in controlled, even strokes.
5. Use the comb attachment, if present, only as intended and avoid contact with the eyes.
6. Observe the product's specified processing time and service procedure.
7. Clean or dispose of the brush according to its type and the product instructions.

⚠ Safety

- Wear suitable gloves and protect the client's clothing and skin as required.
- Avoid contact with eyes and irritated or broken skin.
- Follow product instructions, including patch testing where required.
- Do not mix products unless the manufacturer specifically permits it.

Care & storage

- Wash reusable brushes promptly with the recommended cleaner.
- Store clean, dry brushes separately from chemicals and used tools.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tratak Stand

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Tratak Stand

What it is used for

Handle a Tratak stand safely for supervised concentration and wellness practice.

How to handle

- Inspect the stand for stability, loose parts, and sharp edges before use.
- Place it on a level, secure surface at an appropriate viewing height.
- If a flame or candle is used, keep it away from flammable materials and never leave it unattended.
- Follow instructor guidance and stop if eye discomfort, dizziness, or strain occurs.

Storage

- Clean and dry the stand after use.
- Store securely in a designated area away from impact and heat sources.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wax Heater

Category: **Beauty, Wellness & Fitness** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Double Wax heater

What it is used for

A double wax heater melts two types of beauty wax in separate containers so an operator can prepare different waxing products at the same time. Use compatible wax, control each container's temperature, and follow hygiene procedures to prevent burns and contamination.

Main parts

- Heater base and two wax containers
- Independent temperature controls or thermostats
- Power switch and indicator lights
- Container lids and heat-resistant handles
- Power cord and plug
- Wax spatulas and approved application accessories

Before you start

1. Confirm that the heater is designed for the selected wax products and containers.
2. Inspect the base, containers, controls, cord, and plug for damage.
3. Place the heater on a stable, dry, heat-resistant surface.
4. Ensure both containers are clean, dry, and correctly seated in the heater.
5. Check each wax product's recommended melting temperature and fill limits.
6. Assess the client's skin condition, allergies, and suitability for waxing; prepare clean single-use applicators.

How to use

1. Switch off the heater before adding or removing wax containers.
2. Place the approved wax in the correct container without exceeding its fill limit.
3. Close the lids where applicable and connect the heater to a suitable outlet.
4. Switch on the heater and set each container's temperature independently as instructed.
5. Allow the wax to melt fully and reach a stable working temperature.
6. Check each wax temperature using the recommended safe method before application.
7. Clean and dry the treatment area according to the waxing procedure.
8. Test the wax on a suitable small area to confirm comfortable temperature and consistency.
9. Apply wax in the direction and manner recommended for the product and treatment area.
10. Use a clean applicator for each dip and never return used applicators to the wax.
11. Monitor the client's skin response and stop if burning, pain, or unusual irritation occurs.
12. Switch off and unplug the heater after use or as instructed.

3. Allow wax and containers to cool before cleaning or moving the unit.
4. Clean the containers and heater using the manufacturer's approved method.

⚠ Safety

- Only trained beauty personnel should operate the wax heater.
- Never apply wax before checking its temperature and consistency.
- Do not apply wax to broken, infected, or otherwise unsuitable skin.
- Avoid water entering molten wax or electrical components.
- Do not overfill containers or leave the heater operating unattended.
- Use single-use applicators and follow infection-control procedures to prevent cross-contamination.

Care & storage

- Switch off, unplug, and cool the heater before cleaning.
- Remove wax residue using approved products and methods only.
- Keep temperature controls, cord, and plug dry.
- Inspect containers, handles, thermostats, and electrical connections regularly.
- Arrange servicing if temperature control becomes unreliable or the unit shows damage.

Common mistakes to avoid

- Using the same container or settings without checking the requirements of each wax product.
- Applying wax that is too hot or not fully melted.
- Double-dipping a used applicator and contaminating the wax.
- Cleaning or moving the heater while it is hot or connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weight Plates

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Weight plates, bars and other weights

What it is used for

Weight plates, bars, and other weights provide resistance for strength training and fitness exercises. Inspect the equipment, select an appropriate load, and secure all plates before lifting.

How to use

1. Inspect plates, bars, collars, and grips for cracks, bending, rust, or wear.
2. Choose weights appropriate for the user's ability and planned exercise.
3. Place the bar securely on a suitable rack before loading plates.
4. Load plates evenly on both sides and secure them with compatible collars.
5. Check the setup and lifting area for stability and sufficient clearance.
6. Perform controlled movements using correct technique and assistance where needed.
7. Unload plates carefully, wipe equipment, and return weights to storage.

⚠ Safety

- Do not use cracked plates, bent bars, or faulty collars.
- Never exceed the rated capacity of the bar or equipment.
- Keep hands and feet clear when loading or unloading weights.
- Use a spotter for lifts that require assistance and avoid lifting beyond ability.

Care & storage

- Wipe equipment after use and store plates securely on designated racks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Yantra

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Yantra

What it is used for

Handle a yantra used for wellness or traditional practice respectfully and safely.

How to handle

- Identify the specific yantra and follow the supplier's or instructor's directions for its intended use.
- Inspect for sharp edges, breakage, loose components, or material that may cause irritation.
- Place it on a stable, clean surface and avoid unsafe heating, ingestion, or skin application unless explicitly approved for that purpose.
- Clean it using a method compatible with its material and intended use.

Storage

- Keep in a clean, dry, labelled container or designated display area.
- Protect from moisture, impact, direct sunlight, and unauthorised handling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Yoga Block

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Blocks

What it is used for

A yoga block supports balance, alignment, and modified poses during yoga or stretching exercises. Choose a suitable block height and place it securely on a stable, nonslip surface.

How to use

1. Inspect the block for cracks, tears, deformation, or slippery surfaces.
2. Choose a block material and height appropriate for the exercise and user.
3. Place the block on a stable, nonslip surface near the intended position.
4. Demonstrate how to place the hands, feet, or body safely on the block.
5. Move into the pose gradually and maintain controlled breathing and alignment.
6. Adjust the block position if balance or comfort becomes difficult.
7. Wipe the block after use and return it to storage.

⚠ Safety

- Do not use a damaged or unstable block.
- Avoid placing the block on wet or slippery flooring.
- Do not force a stretch or place excessive body weight on an unsuitable block.
- Stop if pain, dizziness, or loss of balance occurs.

Care & storage

- Clean the block according to its material and allow it to dry before storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Yoga Clothing

Category: **Beauty, Wellness & Fitness** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Vastra Dhoti; Yoga dress for male and female

What it is used for

Handle yoga clothing and vastra dhoti hygienically and safely during use and care.

How to handle

- Check garment size, cleanliness, and condition before use.
- Wear clothing that allows safe movement and does not trail near equipment or underfoot.
- Use a clean garment for each person and wash reusable garments regularly.
- Follow fabric care instructions to avoid shrinkage or damage.

Storage

- Store clean, dry clothing folded in labelled shelves or bags.
- Keep used clothing separate from clean garments and damp items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Yoga Mat

Category: **Beauty, Wellness & Fitness** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: EVA Foam Interlocking Mats and Judo Mats; Exercise/Yoga Mats; Gym Mats/Yoga Mats; Yoga Mat; Yoga Mats

What it is used for

A yoga mat provides a stable, cushioned surface for stretching, yoga, and floor exercises. Use a clean, non-slip mat suited to the activity and floor type.

How to use

1. Inspect the mat for tears, worn areas, moisture, or damaged edges.
2. Place it flat on a clean, dry, non-slip floor.
3. Ensure enough clear space around the mat for safe movement.
4. Perform exercises with controlled movements and maintain balance on the mat.
5. After use, wipe away sweat and allow the mat to dry completely.
6. Roll or fold the mat as recommended and store it in a clean area.

⚠ Safety

- Do not use a wet or slippery mat.
- Keep the exercise area free from furniture and other obstacles.
- Stop if the mat shifts or causes loss of balance.

Care & storage

- Clean according to the manufacturer's instructions and store the mat dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

3D Pen

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **High** 1 labs

Covers these lab items: 3d filament pen

What it is used for

A 3D pen heats plastic filament and extrudes it to create raised drawings and small models. Use compatible filament and work on a suitable heat-resistant surface.

How to use

1. Inspect the pen, power adapter, controls, nozzle, and cable for damage.
2. Place the pen on a stable, heat-resistant work surface.
3. Insert the compatible filament according to the manufacturer's instructions.
4. Connect the approved power adapter and select the recommended temperature setting.
5. Wait until the pen indicates that it is ready for extrusion.
6. Feed the filament slowly and draw the design while keeping the nozzle clear of the surface and fingers.
7. Stop extrusion, unload filament if required, and switch off the pen.
8. Allow the nozzle to cool completely before cleaning or storage.

⚠ Safety

- Never touch the hot nozzle or molten plastic.
- Keep the pen away from water and flammable materials.
- Use ventilation suitable for the filament and avoid inhaling fumes.
- Keep children supervised and do not leave the powered pen unattended.

Care & storage

- Remove filament and clean the nozzle only as directed by the manufacturer.
- Store the cooled pen and adapter in a dry, secure place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

3D Printer

Category: **Computers, IT & Office** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: 3D Printer

What it is used for

A 3D printer creates three-dimensional objects by building material layer by layer from a digital model. Use the correct material, printer settings, and operating procedure to produce accurate parts while preventing burns, moving-part injuries, fumes, and electrical hazards.

Main parts

- Printer frame and build platform
- Print head, nozzle, or resin exposure unit, depending on technology
- Filament spool or resin/material feed system
- Motion rails, belts, and motors
- Control panel and display
- Power supply, enclosure, and ventilation system, if fitted

Before you start

1. Confirm the printer technology, material type, and intended print job.
2. Inspect the power cord, frame, platform, motion system, and protective covers.
3. Check that the nozzle, build surface, or resin vat is clean and correctly installed.
4. Use material approved for the printer and review its safety data where applicable.
5. Prepare the digital model and verify orientation, supports, dimensions, and slicer settings.
6. Ensure the printer is on a stable surface with adequate ventilation and clear space around moving parts.

How to use

1. Switch off the printer before installing material or performing manual adjustments that require access to internal parts.
2. Load filament or prepare the resin/material system according to the manufacturer's instructions.
3. Secure the build platform and check its level or calibration as required.
4. Transfer the prepared print file using the approved method and verify the selected material and settings.
5. Close the enclosure or protective cover where required and start the print job.
6. Observe the first layers or initial build stage to confirm proper adhesion and material flow.
7. Monitor progress through the display or approved remote monitoring system.
8. If a print failure, unusual noise, smoke, or material leak occurs, stop the printer when safe and follow the fault procedure.

9. Allow the build platform, nozzle, resin, or printed part to cool or be handled according to the material instructions.
10. Remove the finished part using the approved tool and avoid touching hot or sharp components.
11. For resin printers, use the specified washing and post-curing process with appropriate PPE.
12. Clean the build area, remove waste material safely, and record any maintenance or print issues.

⚠ Safety

- Only trained users should operate the printer and perform maintenance.
- Hot nozzles, heated beds, and cured or uncured materials can cause burns or irritation; follow material-specific precautions.
- Keep fingers, hair, loose clothing, and tools away from moving axes and belts.
- Provide appropriate ventilation, especially for resin, heated plastics, and materials that release fumes.
- Use gloves and eye protection when handling resin, solvents, sharp parts, or uncured prints.
- Disconnect power before electrical servicing; never bypass interlocks or protective covers.

Care & storage

- Clean the build platform and nozzle using the manufacturer's approved procedure after cooling.
- Remove dust and material residue from rails, fans, and accessible surfaces.
- Inspect belts, fasteners, cables, fans, and motion components regularly.
- Replace worn consumables and filters according to the maintenance schedule.
- Store filament or resin according to its manufacturer's moisture, temperature, and light requirements.

Common mistakes to avoid

- Starting a print without checking platform calibration or material settings.
- Touching a hot nozzle or moving mechanism during operation.
- Printing in a poorly ventilated area without material-specific precautions.
- Removing a print with excessive force or handling uncured resin without protection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

3D Printer Bed Leveling Tool

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Bed leveling tools

What it is used for

A 3D printer bed leveling tool helps check the distance or alignment between the nozzle and build plate. Use the tool specified for the printer and follow its leveling procedure.

How to use

1. Check the leveling tool for damage and confirm it suits the printer model.
2. Allow the printer to reach the required temperature if its instructions specify heated leveling.
3. Home the printer axes using the approved control procedure.
4. Move the nozzle to the specified checking positions without touching hot parts.
5. Use the leveling tool or gauge to check the nozzle-to-bed gap.
6. Adjust leveling points gradually according to the manufacturer's procedure.
7. Repeat the checks and confirm the build plate is evenly aligned.

⚠ Safety

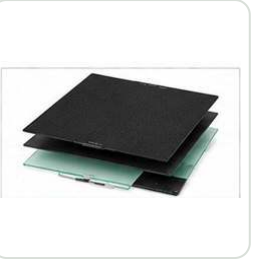
- Keep fingers away from hot nozzles, heated beds, and moving axes.
- Do not force the nozzle against the build plate or leveling tool.
- Follow power isolation instructions before manual adjustments near electrical components.

Care & storage

- Clean the tool after use and store it safely away from heat and dust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

3D Printer Build Plate



Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Build Plates

What it is used for

A 3D printer build plate supports the object while it is printed. Install the compatible plate correctly and keep its surface clean for reliable adhesion.

How to use

1. Switch off the printer and allow the build plate to cool before handling.
2. Check the plate for cracks, warping, residue, or damaged coating.
3. Confirm the plate type and size match the printer model.
4. Install and secure the plate according to the manufacturer's instructions.
5. Clean the printing surface using the recommended method and suitable materials.
6. Recheck plate seating and leveling before starting a print.
7. Remove finished prints only after the plate has cooled sufficiently.

⚠ Safety

- Do not touch a heated build plate with bare hands.
- Avoid scraping the surface with sharp tools unless the manufacturer permits it.
- Do not use damaged or incompatible plates.

Care & storage

- Clean the surface with approved materials and store removable plates flat in a protected place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

3D Scanner

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: 3D Scanners

What it is used for

A 3D scanner captures an object's shape and surface details to create a digital model. Use the scanner and its software according to the equipment instructions.

How to use

1. Inspect the scanner, cables, lens, and accessories for damage.
2. Place the object on a stable surface and prepare it as instructed.
3. Connect the scanner to a compatible computer and open the approved software.
4. Select the appropriate scanning mode and configure the required settings.
5. Move or position the scanner around the object as instructed while maintaining suitable distance.
6. Review the captured model and repeat scanning where details are missing.
7. Save the file in the required format and clean the equipment after use.

⚠ Safety

- Do not stare into scanner lasers or light sources when warnings prohibit exposure.
- Keep cables secured and avoid moving the scanner abruptly near the object.
- Do not scan people or confidential items without appropriate permission.

Care & storage

- Clean the scanner lens with approved materials and store the equipment in its protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Biometric Scanner

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Biometric and document verification tools (simulated setup); Biometric devices; Iris scanner

What it is used for

To verify identity using a biometric scanner in an authorized or simulated verification setup. To capture biometric data accurately while protecting personal information.

Main parts

- Scanner or sensor
- Display or indicator lights
- USB or network connection
- Verification software
- User interface

Before you start

1. Confirm authorization and explain the verification process to the user.
2. Inspect the scanner and connection cables for damage.
3. Ensure the required software is running and the device is recognized.
4. Check that the sensor surface is clean.
5. Use only approved test data in simulated training activities.

How to use

1. Open the authorized verification application.
2. Select the correct verification mode, such as fingerprint or iris scanning.
3. Ask the user to follow the displayed positioning instructions.
4. Capture the biometric sample using the device prompts.
5. Review the quality indicator and repeat only if required.
6. Check the verification result displayed by the authorized system.
7. End the session and secure or clear temporary data according to policy.

⚠ Safety

- Obtain appropriate consent and follow applicable privacy rules.
- Do not collect or share biometric data without authorization.
- Never attempt to bypass identity verification controls.
- Do not scan irritated eyes or use an iris scanner contrary to its instructions.

Care & storage

- Clean the sensor using the manufacturer's approved method.
- Keep the scanner protected from dust and moisture.
- Update software only through authorized procedures.
- Report connection errors or unreliable readings.

Common mistakes to avoid

- Using the wrong verification mode.
- Positioning the finger or eye incorrectly.
- Allowing unauthorized access to biometric records.
- Treating a failed scan as proof of a person's identity or intent.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Blueprint Machine

Category: **Computers, IT & Office** Type: **Full SOP** Priority: **Medium** 2 labs

Covers these lab items: Blueprint machine

What it is used for

A blueprint machine reproduces large-format technical drawings, plans, and engineering documents using the supported copying or printing process. Correct document handling, settings, and ventilation help produce readable copies and maintain safe operation.

Main parts

- Document feed or scanning surface
- Original document guides and alignment marks
- Copying or printing mechanism
- Paper supply roll or sheet tray
- Control panel and size or density settings
- Output tray, ventilation system, and power connection

Before you start

1. Read the machine instructions and confirm the supported document sizes and media types.
2. Inspect the feed path, rollers, guides, output area, power cable, and ventilation openings.
3. Check that the correct paper or approved reproduction media is loaded and aligned.
4. Ensure the original drawing is clean, dry, and free from clips or objects that could damage the feed path.
5. Select the required copy size, scale, orientation, and density settings.
6. Confirm that ventilation is adequate and any required covers or safety interlocks are in place.

How to use

1. Switch on the machine and allow it to complete its startup checks.
2. Select the correct paper size, reproduction mode, scale, and image density.
3. Align the original drawing using the guides or scanning reference marks.
4. Feed or scan the document according to the machine's operating instructions.
5. Preview the image or check the first copy for line clarity, orientation, and scale.
6. Adjust approved settings if the copy is too faint, too dark, cropped, or distorted.
7. Run the required number of copies and collect each sheet from the output area.
8. Check that the reproduced dimensions and drawing details meet the job requirements.
9. Remove any jam only after stopping the machine and following the approved jam-clearing procedure.
10. Switch off the machine when finished and clean the accessible feed and output areas.

⚠ Safety

- Keep hands, hair, loose clothing, and tools away from rollers and moving feed mechanisms.
- Do not reach into the machine while it is operating or bypass fitted safety covers.
- Use only media and cleaning products approved for the machine.
- Avoid inhaling toner, ink mist, or process fumes and maintain required ventilation.
- Disconnect power before cleaning internal areas or performing maintenance.
- Allow hot components to cool before touching them, where applicable.

Care & storage

- Clean the scanning surface, guides, and accessible rollers with approved materials.
- Inspect paper feed components, cables, covers, and ventilation openings regularly.
- Replace consumables according to the manufacturer's instructions and avoid touching sensitive surfaces unnecessarily.
- Arrange servicing and calibration according to the equipment schedule.

Common mistakes to avoid

- Incorrect scale settings can produce drawings with misleading dimensions.
- Poor document alignment can crop borders or distort the reproduced plan.
- Using incompatible paper or media can cause jams and poor image quality.
- Clearing jams while the machine is running can cause injury or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cable Organizer

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Cable Organizers

What it is used for

Organise cables to reduce tangling, damage, and trip hazards.

How to handle

- Group cables by function and label both ends where useful.
- Avoid tight bends, excessive tension, and pinching under furniture.
- Do not bundle power and data cables in ways that conflict with equipment guidance.
- Keep walkways and ventilation openings clear.

Storage

- Use reusable ties or designated cable organisers without overtightening.
- Store spare cables coiled loosely in labelled containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Calculator

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Calculator; Financial Calculators

What it is used for

A calculator performs arithmetic and other mathematical operations for study, office work, and financial calculations. Select the appropriate mode and verify inputs before relying on the result.

How to use

1. Inspect the calculator, display, buttons, and battery or power supply.
2. Switch it on and select the required mode if applicable.
3. Enter the numbers and operation carefully, checking decimal points and signs.
4. Press the calculation key and review the displayed result.
5. Clear the entry and repeat the calculation if an input error is found.
6. Switch off the calculator if needed and store it safely after use.

⚠ Safety

- Keep the calculator dry and away from excessive heat.
- Do not use a device with a damaged battery compartment or exposed wiring.
- Verify important financial or technical calculations independently.

Care & storage

- Clean the keys and display with a soft dry cloth and replace batteries as directed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Clipboard

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Clipboards

What it is used for

Use clipboards to support writing, inspections, and field documentation.

How to handle

- Check that the clip holds paper securely and has no sharp or broken edges.
- Keep documents aligned and avoid overloading the clip mechanism.
- Use clean hands when handling official forms or shared records.
- Protect confidential information from unauthorised viewing.

Storage

- Store flat in a designated office or fieldwork area.
- Keep dry and separate from liquids, chemicals, and heavy objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Computer Cable

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Cables and Connectors; HDMI Cables; VGA Cables

What it is used for

Computer cables connect computers, displays, and peripheral devices for data, video, or other compatible signals.

How to handle

- Identify the cable type and confirm compatibility with both devices.
- Power down equipment when required before connecting or disconnecting cables.
- Inspect connectors, insulation, and pins for damage or bending.
- Insert connectors carefully without forcing them and secure any locking screws or clips.
- Route cables to prevent tripping, pinching, or tension on ports.

Storage

- Coil cables loosely and avoid tight bends.
- Label and store in a dry area away from dust, heat, and heavy objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Computer Keyboard

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Key Boards

What it is used for

A computer keyboard is used to enter text, numbers, and commands into a computer. Use the correct keys and keep the keyboard clean for reliable operation.

How to use

1. Place the keyboard on a stable, comfortable surface.
2. Connect the USB cable or pair the wireless keyboard according to its instructions.
3. Check that the keyboard is recognized by the computer.
4. Type a short test message and verify that the keys respond correctly.
5. Use the appropriate function keys and shortcuts for the required task.
6. Disconnect the keyboard safely before cleaning or moving a wired setup.
7. Wipe the keyboard gently after use and keep it free from debris.

⚠ Safety

- Keep liquids and wet hands away from the keyboard.
- Do not force stuck keys or pull the cable sharply.
- Keep cables arranged to prevent tripping and connector damage.

Care & storage

- Clean with a soft, dry cloth and follow the manufacturer's cleaning instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Computer Peripheral

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Computer speakers; Peripherals

What it is used for

Computer peripherals are external devices such as speakers, keyboards, mice, webcams, and printers that add functions to a computer. Connect and configure each device according to its interface and instructions.

How to use

1. Inspect the peripheral, cable, connector, and power adapter if present.
2. Check that the device is compatible with the computer and operating system.
3. Switch off the equipment when required and connect it to the correct port.
4. Power on the device and install or select the required driver or settings.
5. Test the device to confirm that it works correctly.
6. Disconnect using the recommended method and store accessories neatly.

⚠ Safety

- Keep electrical devices and connectors away from liquids.
- Do not force connectors into ports or use damaged cables.
- Switch off and unplug equipment before cleaning or handling internal connections.

Care & storage

- Clean peripherals with suitable materials and keep cables organized and undamaged.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Computer Repair Tool Kit

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Machine tools for servicing the computer

What it is used for

A computer repair tool kit contains tools for inspecting, assembling, and servicing computer hardware. Use the correct tool for each component and follow safe electrical handling practices.

How to use

1. Switch off the computer and disconnect its power cable and peripherals.
2. Use an antistatic wrist strap or suitable ESD protection before handling internal components.
3. Select the correct screwdriver or tool for the fastener being removed.
4. Keep screws and small components organized in a parts tray.
5. Handle circuit boards by their edges and avoid touching exposed contacts.
6. Reassemble components carefully and check that cables and connectors are secure.
7. Reconnect power and test the computer after closing the case.

⚠ Safety

- Never open a power supply unit unless specifically trained and authorized.
- Do not work on equipment while it is connected to mains power.
- Avoid static discharge and keep tools away from exposed circuit contacts.

Care & storage

- Clean and inspect tools after use and store them in the designated case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Desktop Computer

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 92 labs

Covers these lab items: Computer; Computer / Laptop; Computer and Computer Peripherals; Computer Lab; Computer Lab with 1:1 PC: trainee ratio and having internet connection; Computer Lab with 1:1 PC: trainee ratio and internet connection; Computer Lab with 1:1 PC:trainee ratio and having Internet connection (for IT Lab ses

What it is used for

A desktop computer processes information and runs software for learning, office work, and technical tasks. Use the computer and its peripherals correctly to protect data and maintain reliable operation.

How to use

1. Inspect the computer, monitor, keyboard, mouse, cables, and power connections.
2. Ensure ventilation openings are clear and the workstation is dry and stable.
3. Switch on the power supply and start the computer using its power button.
4. Sign in with your authorised account and open the required application.
5. Save work regularly and use approved storage or backup locations.
6. Shut down applications and the operating system using the normal shutdown procedure.
7. Switch off peripherals as appropriate and leave the workstation tidy.

⚠ Safety

- Keep liquids away from the computer and electrical connections.
- Do not open the computer case or modify internal components while connected to power.
- Use strong passwords and avoid unknown downloads or suspicious links.
- Keep cables arranged to prevent trips and accidental disconnection.

Care & storage

- Clean the screen and peripherals with suitable materials while equipment is powered off as needed.
- Install approved updates and report unusual noise, overheating, or repeated errors.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

External Storage Device

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 9 labs

Covers these lab items: External Hard Drive; External Storage Devices; External Storage Devices (External Hard Drives, USB Flash Drives); USB / Pen drive; USB Drives

What it is used for

Use an external storage device to save, transfer, or back up digital files between compatible computers and devices.

How to use

1. Inspect the storage device and connector for damage before use.
2. Connect it to a compatible USB port or interface.
3. Wait for the computer to recognize the device and confirm the correct drive name.
4. Copy or move the required files and verify that the transfer is complete.
5. Use the operating system's safe-eject option before disconnecting the device.
6. Store the device securely after removal.

⚠ Safety

- Scan unfamiliar storage devices for malware before opening files.
- Do not disconnect the device while files are being transferred.
- Avoid storing confidential data on unapproved or unencrypted devices.
- Keep the device away from moisture, extreme heat, and strong impacts.

Care & storage

- Keep the connector clean and protect the device from dust.
- Maintain a separate backup of important files and store the device safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fax Machine

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Fax Machine

What it is used for

A fax machine sends and receives documents through a telephone line or supported network service. Load documents correctly and verify transmission or receipt before completing the task.

How to use

1. Check the fax machine, power cable, telephone or network connection, and paper supply.
2. Switch on the machine and confirm that it is ready for operation.
3. Load the document in the direction indicated by the feeder markings.
4. Enter the recipient's fax number or select the approved contact.
5. Review the document placement and transmission settings before sending.
6. Start transmission and wait for the confirmation or error report.
7. For received faxes, collect the pages and route them to the intended recipient.

⚠ Safety

- Do not send confidential documents to an unverified number.
- Keep liquids away from electrical parts and paper-feed mechanisms.
- Do not reach into moving rollers while the machine is operating.
- Follow workplace rules for handling and disposing of sensitive documents.

Care & storage

- Remove paper dust, use the specified paper and supplies, and follow the cleaning instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Feedback Kiosk

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Customer feedback kiosks and digital record systems; Digital feedback kiosks or tablets for guest surveys; Digital kiosks for self-check-in/check-out

What it is used for

To collect customer feedback or support digital self-check-in and check-out using a feedback kiosk or tablet. To provide a clear user experience while protecting personal and transaction data.

Main parts

- Touchscreen or display
- Kiosk body or stand
- Computer or tablet unit
- Network connection
- Power supply
- Printer or scanner, if fitted
- Feedback or check-in software

Before you start

1. Inspect the screen, stand, cables, and peripherals for damage.
2. Confirm the kiosk is powered and connected to the approved network.
3. Check that the correct feedback or check-in application is open.
4. Ensure privacy notices and user instructions are visible where required.
5. Verify that the system does not display another user's private information.

How to use

1. Welcome the user or display the starting screen.
2. Ask the user to follow the on-screen instructions.
3. Select the appropriate survey, feedback, or check-in/check-out option.
4. Enter or scan only the information required for the task.
5. Review the displayed details before confirming a transaction or submission.
6. Show a completion message or receipt if the system provides one.
7. Return the kiosk to its starting screen and ensure the previous session is closed.

⚠ Safety

- Do not expose passwords, payment details, or personal information on a public screen.
- Use approved network and software settings only.
- Do not attempt to bypass access controls or tamper with records.
- Report suspicious activity, device damage, or software errors.

Care & storage

- Clean the touchscreen with a compatible method.
- Keep cables secured and the kiosk area unobstructed.
- Check printer paper or scanner function where applicable.
- Install updates and review access settings through authorized administration.

Common mistakes to avoid

- Leaving a previous user's session open.
- Entering information into the wrong user's record.
- Ignoring privacy notices or confirmation screens.
- Disconnecting power during an active transaction.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Graphics Tablet

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Digital Drawing Tablet; Drawing Tablets; Touch-pad and Digital Pen

What it is used for

To create digital drawings, illustrations, annotations, and designs using a stylus or digital pen.

How to handle

- Inspect the tablet, stylus, cable, and connectors for damage before use.
- Connect the tablet to a compatible computer or device and install the approved driver or application if required.
- Configure the pen, pressure sensitivity, and working area according to the task.
- Use the stylus gently and keep liquids, sharp objects, and excessive pressure away from the drawing surface.
- Disconnect safely where applicable and clean the tablet and stylus with a suitable soft cloth after use.

Storage

- Store in a protective sleeve or case in a clean, dry location.
- Keep the stylus in its holder and protect the screen or drawing surface from scratches and pressure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Laptop

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 32 labs

Covers these lab items: Laptop; Laptop / Desktop (Basic); Laptop and Parts; Laptop for presentations; Laptop for Teacher; Laptop with Charger; Laptop with Internet (For Digital Learning); Laptop/Desktop; Laptops; Laptops or desktops with internet access for planning digital itineraries; Laptops/PCs

What it is used for

A laptop is a portable computer used for learning, office work, data processing, and presentations. Operate it correctly, protect stored information, and use compatible power and network accessories.

How to use

1. Inspect the laptop, charger, ports, screen, keyboard, and casing for damage.
2. Place the laptop on a stable surface with clear ventilation openings.
3. Connect the approved charger if needed and switch on the laptop.
4. Sign in using an authorised account and open the required application.
5. Connect to an approved network when internet access is required.
6. Save files regularly and use approved storage or backup locations.
7. Close applications and shut down the operating system properly after use.
8. Disconnect accessories and store the laptop and charger safely.

⚠ Safety

- Keep liquids away from the laptop, charger, and electrical connections.
- Do not block ventilation openings or use a damaged charger.
- Use strong passwords and avoid suspicious links or unapproved downloads.
- Disconnect power before cleaning and follow manufacturer instructions.

Care & storage

- Clean the screen and keyboard with suitable materials while the laptop is powered off.
- Keep the laptop dry, updated, and protected during transport.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Marker

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: White board marker; White/Black Board Marker; Whiteboard Markers

What it is used for

Markers write, label, and annotate boards or surfaces using the appropriate ink type.

How to handle

- Choose a marker compatible with the board or surface.
- Check the cap, tip, and ink condition before use.
- Use only approved markers on whiteboards or blackboards to avoid staining.
- Replace the cap firmly after use and avoid pressing the tip excessively.
- Use in a ventilated area and avoid inhaling vapours.

Storage

- Store capped markers horizontally or as directed by the manufacturer.
- Keep away from heat and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mouse Pad

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Mouse Pads

What it is used for

Use mouse pads to support smooth mouse movement and protect desk surfaces.

How to handle

- Place the pad flat on a clean, dry surface.
- Keep the tracking area free of dust, crumbs, and liquid spills.
- Clean using a method suitable for the pad material and allow it to dry fully.
- Replace pads that are torn, curled, or no longer provide reliable tracking.

Storage

- Store flat in a clean, dry place away from heat and sharp objects.
- Avoid folding pads that may crease or damage the surface.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Optical Disc

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: CD / DVDs; CDROMs

What it is used for

Handle optical discs carefully to preserve readable data.

How to handle

- Hold discs by the outer edge or centre hole.
- Avoid touching the recording surface, scratching, or bending discs.
- Label discs on the approved label area using a suitable marker.
- Check the disc for cracks before inserting it into a drive.

Storage

- Keep discs in protective cases and store vertically where practical.
- Store away from heat, moisture, dust, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paper Shredder

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **High** 4 labs

Covers these lab items: Paper Shredders

What it is used for

A paper shredder cuts documents into smaller pieces to protect confidential information and support secure disposal. Use only the paper types and quantities permitted by the shredder manufacturer.

How to use

1. Inspect the shredder, power cord, waste bin, and feed opening for damage or obstruction.
2. Place the shredder on a stable surface and ensure the waste bin is correctly fitted.
3. Switch on the shredder and select the required operating mode.
4. Remove clips or staples unless the manufacturer confirms they are supported.
5. Feed paper slowly within the specified sheet capacity and keep documents straight.
6. Allow the shredder to finish cutting and use reverse mode only as instructed if a jam occurs.
7. Switch off and unplug the shredder before clearing a persistent jam or emptying the bin.
8. Close the bin and switch off the shredder after use.

⚠ Safety

- Keep fingers, hair, jewellery, and loose clothing away from the feed opening.
- Never insert hands into the cutting area, even when the shredder appears stopped.
- Do not exceed the rated sheet capacity or shred prohibited materials.
- Unplug the machine before maintenance or clearing a jam as directed.

Care & storage

- Empty the waste bin before it becomes overfilled and causes a blockage.
- Clean accessible surfaces and maintain the cutting mechanism according to the manual.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Photocopier

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Photocopier

What it is used for

A photocopier produces paper copies of documents and images. Use the correct paper size and copy settings to obtain clear, accurate copies.

How to use

1. Check the copier for visible damage, paper jams, and sufficient paper and toner.
2. Switch on the machine and wait until it is ready.
3. Place the original document on the glass or in the document feeder as directed.
4. Select paper size, number of copies, colour mode, and other required settings.
5. Press the start button and collect the copies when printing finishes.
6. Remove any leftover originals and report persistent errors or jams.

⚠ Safety

- Keep fingers away from moving parts and hot internal components.
- Do not force open panels or remove jammed paper against machine instructions.
- Keep liquids away from the copier and electrical connections.

Care & storage

- Clean the glass with an approved cloth and refill paper and toner as required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Printer

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 47 labs

Covers these lab items: Color Laser Printer; Color Printer; Colour Printer; Dot Matrix Printer; Dot Matrix Printers; Ink Jet Printers; Inkjet Printer; Inkjet Printers (Colour & Black); Laser Printer; Laser Printer Black; Laser Printers; Multi-Function Laser Printer; Printer; Printer and scanner; Printers; Printers and Scan

What it is used for

A printer produces paper copies of documents, images, labels, or technical drawings. Use compatible supplies and correct print settings to obtain clear output and avoid equipment damage.

How to use

1. Inspect the printer, power cable, paper tray, and access covers for damage.
2. Check that the correct ink or toner is installed and paper is loaded properly.
3. Switch on the printer and wait for its startup checks to finish.
4. Connect to the computer or network and select the correct printer.
5. Set paper size, orientation, print quality, colour mode, and page range.
6. Preview the document and print one test page when appropriate.
7. Check the output for clarity, alignment, and missing content before printing the full job.
8. Remove finished pages and switch off the printer according to its instructions.

⚠ Safety

- Do not touch hot fuser parts inside a laser printer.
- Keep liquids and loose objects away from electrical parts and moving mechanisms.
- Use compatible cartridges and follow instructions when clearing paper jams.
- Disconnect power before maintenance when instructed by the manufacturer.

Care & storage

- Clean accessible surfaces and paper trays regularly.
- Replace cartridges as needed and report recurring jams or print defects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Printer Cartridge

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Printer Cartridges (Compatible Printer Type Required)

What it is used for

Handle printer cartridges without damaging the printer or exposing toner or ink.

How to handle

- Confirm cartridge compatibility with the printer model before installation.
- Follow the manufacturer's instructions and avoid touching electrical contacts or print nozzles.
- Keep toner or ink away from skin, eyes, and clothing; clean spills according to product guidance.
- Do not force a cartridge into its slot.

Storage

- Keep cartridges sealed in original packaging until needed.
- Store upright as directed, away from heat, moisture, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Printing Equipment

Category: **Computers, IT & Office** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Printing Set Up Equipment

What it is used for

Printing equipment is used to produce hard copies of documents, drawings, labels, and training materials. Follow the correct setup, operating, and maintenance procedures to obtain clear prints and prevent equipment damage.

Main parts

- Printer or multifunction printing machine
- Power cable and power switch
- USB or network connection cable
- Paper tray and paper guides
- Ink cartridges or toner cartridges
- Control panel and display
- Output tray and access covers

Before you start

1. Place the printer on a stable, level, well-ventilated surface.
2. Check the power cable, plug, and electrical supply for damage.
3. Confirm that the correct ink or toner is installed.
4. Load the recommended paper size and adjust the paper guides.
5. Check the computer connection, printer driver, and selected printer settings.
6. Ensure access covers are closed and no paper jam is present.

How to use

1. Switch on the printer and wait until it completes initialization.
2. Connect the printer to the computer or approved network if required.
3. Open the document and select the correct printer from the print menu.
4. Set paper size, orientation, colour mode, print quality, and number of copies.
5. Use print preview to check page layout and content.
6. Send one test page and inspect text, images, alignment, and colour.
7. Print the remaining pages after confirming the test result.
8. Collect printed sheets from the output tray and check for missing pages.
9. Switch off the equipment according to the manufacturer's instructions when finished.

⚠ Safety

- Keep hands dry when operating power switches and electrical connections.
- Never touch exposed electrical parts or dismantle the printer while powered.
- Do not touch hot fuser components inside laser printers.
- Keep liquids, loose metal objects, and food away from the equipment.
- Use only compatible paper, ink, toner, and approved accessories.
- Follow the manufacturer's instructions when clearing paper jams.

Care & storage

- Clean accessible surfaces with a soft, dry cloth when the equipment is off.
- Keep paper trays and the output area free from dust and scraps.
- Replace cartridges and maintenance parts according to manufacturer guidance.
- Report repeated jams, unusual noises, streaks, or error messages.

Common mistakes to avoid

- Loading the wrong paper size or placing paper incorrectly in the tray.
- Selecting an incorrect printer, page range, or print orientation.
- Using incompatible cartridges or refilling cartridges against manufacturer guidance.
- Pulling jammed paper forcefully and damaging internal rollers.
- Ignoring low-supply warnings or continuing to print with recurring defects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scanner

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 20 labs

Covers these lab items: Digital document scanners, printers, and secure data management systems; Scanner; Scanner and Printer; Scanners

What it is used for

A scanner converts paper documents, photographs, or other suitable media into digital files. Use the correct scan settings and approved storage location to maintain image quality and data security.

How to use

1. Inspect the scanner glass, feeder, lid, cable, and controls for damage.
2. Switch on the scanner and connect it to the authorised computer or network.
3. Clean the scanning glass and remove dust or loose objects from the feeder.
4. Place the document correctly on the glass or in the approved feeder.
5. Select the required file format, resolution, colour mode, and destination folder.
6. Preview the scan and adjust alignment or cropping if needed.
7. Start scanning and check the resulting file for clarity and completeness.
8. Save the file using an appropriate name and approved storage location.
9. Remove documents and switch off the scanner when finished.

⚠ Safety

- Keep liquids away from the scanner and electrical connections.
- Do not place hands near moving feeder rollers while scanning.
- Scan and store confidential documents only with proper authorisation.
- Disconnect power before cleaning internal areas as instructed.

Care & storage

- Clean the glass and feeder rollers using approved materials.
- Keep the scanner covered when not in use and report recurring feed errors.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Server

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Server

What it is used for

A server provides computing resources, applications, files, or network services to authorized users and devices. Follow the approved deployment, access-control, backup, and maintenance procedures.

How to use

1. Confirm the server's intended role, installation location, and required specifications.
2. Inspect the chassis, power supplies, drives, network ports, and cooling components.
3. Mount the server securely in a suitable rack or approved location.
4. Connect power, network cables, and required peripherals according to the deployment plan.
5. Configure firmware, operating system, network settings, and authorized user access.
6. Apply approved security updates and configure monitoring and backup services.
7. Verify service availability, record the configuration, and monitor system health.

⚠ Safety

- Use approved electrical supplies and proper grounding for rack-mounted equipment.
- Prevent blocked airflow and maintain the required cooling conditions.
- Restrict physical access and protect credentials from unauthorized disclosure.
- Back up important data before major configuration changes or maintenance.

Care & storage

- Monitor temperatures, storage health, updates, backups, and logs according to the maintenance schedule.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Smartphone

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: M-health equipments smart phone, tablet phone; Smart phone

What it is used for

A smartphone is used to access learning apps, record information, communicate, and collect digital data. Use it carefully to protect the device, user privacy, and stored information.

How to use

1. Inspect the phone, screen, charging port, and battery for visible damage.
2. Charge the phone using a compatible charger when needed.
3. Unlock the phone and open the required authorized app or function.
4. Enter information carefully and verify readings, images, or saved records.
5. Save work securely and close apps after completing the activity.
6. Lock the screen and disconnect the charger when charging is complete.

⚠ Safety

- Keep the phone and charging cable away from water and wet hands.
- Do not use a phone with a swollen battery or damaged charger.
- Avoid sharing personal information, passwords, or confidential student records.
- Do not use the phone while walking through the lab or operating machinery.

Care & storage

- Clean the screen with a soft, dry microfiber cloth and store the phone securely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stapler

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Stapler; Stapler (small and big size)

What it is used for

A stapler fastens sheets of paper or light documents using staples.

How to handle

- Check the stapler body, hinge, staple channel, and base for damage.
- Use staples of the correct size and load them according to the manufacturer's instructions.
- Keep fingers away from the staple outlet and moving parts while operating.
- Remove jammed staples carefully with the tool disconnected or unloaded where applicable.
- Do not use the stapler on materials beyond its intended capacity.

Storage

- Store the stapler closed or in a stable position.
- Keep spare staples in a dry container and away from children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stylus Pen

Category: **Computers, IT & Office** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Stylus Pen

What it is used for

Handle a stylus pen correctly for touchscreen, tablet, or digital writing tasks.

How to handle

- Check the stylus tip for damage and confirm compatibility with the device.
- Use gentle pressure and avoid scratching the screen or pressing the tip against hard surfaces.
- Keep the stylus away from liquids and do not dismantle electronic or battery-powered models.
- Clean the tip according to the manufacturer's instructions.

Storage

- Store in a designated holder or protective case.
- Keep dry and secure, away from heat, sharp objects, and heavy pressure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tablet

Category: **Computers, IT & Office** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Electronic tablets with species identification apps; Mobile devices or tablets for simulation of digital guest registration and feedback; Tablets; Tablets for visual aids during storytelling; Tablets or mobile devices

What it is used for

A tablet is a portable touchscreen device used for digital forms, learning apps, communication, and data collection. Use approved applications and protect stored information.

How to use

1. Inspect the tablet, screen, buttons, charging port, and battery condition.
2. Switch on the device and unlock it using the approved method.
3. Connect to an authorized network if required and open the intended application.
4. Enter information carefully and check entries before saving or submitting.
5. Save work and confirm that the information has been stored correctly.
6. Sign out of shared accounts when required, clean the screen, and store the tablet safely.

⚠ Safety

- Keep the tablet away from liquids, extreme heat, and physical impact.
- Use only authorized applications and networks for work-related data.
- Do not share passwords or leave sensitive information visible on a shared device.

Care & storage

- Clean the screen with a suitable cloth, charge as directed, and use a protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Boaster

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Boaster

What it is used for

To dress, shape, or remove excess material from masonry surfaces using a boaster, where this tool type is specified for the task. To work with controlled blows and maintain a stable tool position.

Main parts

- Striking head or working end
- Body or shaft
- Handle or grip, if fitted
- Cutting or dressing edge, depending on design

Before you start

1. Confirm the tool type and its intended masonry application.
2. Inspect the working end, body, and handle for cracks or looseness.
3. Wear safety glasses, suitable gloves, and protective footwear.
4. Secure the masonry workpiece and clear the surrounding area.

How to use

1. Mark the area to be dressed or shaped.
2. Hold the tool securely using its intended grip.
3. Position the working end at the marked location.
4. Apply controlled force appropriate to the material and tool design.
5. Work gradually and check the surface after each series of blows.
6. Remove loose fragments and inspect the finished area.
7. Clean and store the tool after use.

⚠ Safety

- Wear eye protection against flying masonry fragments.
- Keep hands and feet clear of the impact area.
- Do not use a tool with a loose handle or damaged working end.
- Keep other people outside the striking zone.

Care & storage

- Brush away masonry dust and debris after use.
- Inspect the tool for cracks, mushrooming, or other damage.

- Keep metal parts dry to reduce corrosion.
- Store in a secure tool rack.

Common mistakes to avoid

- Using the tool for an unsuitable task.
- Striking too hard and breaking the masonry.
- Working with a loose or damaged tool.
- Ignoring flying fragments and nearby people.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Caulking Tool

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Caulking tools

What it is used for

To apply sealant neatly into joints and gaps using a caulking tool. To achieve a consistent seal while keeping the work surface clean.

Main parts

- Caulking gun or applicator
- Sealant cartridge or tube
- Trigger and plunger
- Nozzle
- Release mechanism

Before you start

1. Select sealant suitable for the surface and application.
2. Check the tool, cartridge, and nozzle for damage.
3. Clean and dry the joint as required.
4. Wear suitable gloves and eye protection.

How to use

1. Load the sealant cartridge according to the tool instructions.
2. Cut the nozzle to the required opening size.
3. Position the nozzle at the joint.
4. Press the trigger steadily while moving along the joint.
5. Maintain an even bead of sealant.
6. Smooth the bead using an approved tool if required.
7. Release pressure before removing the nozzle.
8. Clean the tool according to the sealant instructions.

⚠ Safety

- Follow the sealant label and safety instructions.
- Avoid skin and eye contact with sealant.
- Use ventilation when required.
- Release plunger pressure before removing the cartridge.

Care & storage

- Remove uncured residue using the recommended cleaner.
- Keep the plunger and trigger mechanism clean.
- Store the tool in a dry place.

Common mistakes to avoid

- Using incompatible sealant.
- Applying an uneven bead.
- Leaving the joint dirty or wet when preparation is required.
- Removing a pressurized cartridge carelessly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Filling Board

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Filling Board

What it is used for

A filling board provides a flat support for holding or spreading filling material during suitable construction or finishing work. Check its condition and use it on a stable surface.

How to use

1. Inspect the board for cracks, warping, loose handles, or damaged edges.
2. Clean the board and remove old material that could affect the finish.
3. Place it securely on a stable work surface.
4. Prepare the filling material according to the product instructions.
5. Load or spread the material using the appropriate hand tool.
6. Maintain a consistent working technique and remove excess material as required.
7. Clean the board before the material hardens and store it safely.

⚠ Safety

- Wear suitable gloves and eye protection when handling filling compounds.
- Avoid breathing dust and follow the material's safety data instructions.
- Do not use a broken board or unstable work surface.
- Keep tools and material away from walkways to prevent slips and trips.

Care & storage

- Remove residue promptly, dry the board, and store it flat in a suitable location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Float

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Metal Float; Wooden Float

What it is used for

Use a construction float to smooth, level, or finish fresh mortar, plaster, or concrete surfaces as appropriate for the tool type.

How to use

1. Choose a wooden or metal float suitable for the material and finishing stage.
2. Inspect the working face and handle for damage, loose parts, or hardened residue.
3. Prepare the surface and confirm the material is ready for floating.
4. Hold the float firmly and place its face flat against the surface.
5. Move the float in controlled overlapping strokes without removing excessive material.
6. Check the surface level and finish, adjusting pressure as needed.
7. Clean the float before material hardens and store it safely.

⚠ Safety

- Wear gloves, eye protection, and suitable footwear when handling cement-based materials.
- Avoid direct skin contact with wet cement or mortar.
- Keep the work area clear to prevent slips and trips.
- Do not use a damaged float or work from an unstable platform.

Care & storage

- Wash off mortar or concrete residue immediately after use.
- Dry the float and store it in a clean place to prevent corrosion or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mortar Pan

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Low** 5 labs

Covers these lab items: Ghamelas (MS Steel) (for construction work); Mortar pan; Plastic Ghamale

What it is used for

A mortar pan or ghamela carries mortar, sand, cement, and other construction materials around the worksite. Select a suitable pan, load it within a safe limit, and handle it without spilling materials.

How to use

1. Inspect the pan for cracks, sharp edges, corrosion, or damaged handles.
2. Choose a metal or plastic pan suitable for the material and task.
3. Place the pan on stable ground before filling it.
4. Load the material evenly without exceeding a manageable weight.
5. Grip the handles securely and carry the pan using a safe posture.
6. Empty the material at the designated location without sudden tipping toward others.
7. Remove residue, clean the pan, and store it safely.

⚠ Safety

- Wear gloves, safety footwear, and other required PPE.
- Do not overload the pan or lift beyond your capability.
- Avoid carrying materials through obstructed or slippery paths.
- Keep hands away from sharp edges and damaged handles.

Care & storage

- Clean off cement or mortar before it hardens and store the pan in a dry area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paint Bucket

Category: **Construction & Masonry** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Paint Bucket

What it is used for

Handle paint buckets safely during construction and painting work.

How to handle

- Check the bucket, handle, and base for damage before filling or carrying.
- Read the paint label and use the specified PPE and ventilation precautions.
- Do not overfill the bucket and keep the lid secured during transport when appropriate.
- Clean spills promptly and avoid pouring leftover paint into drains.

Storage

- Close and label containers after use, following the product instructions.
- Store upright in a ventilated area away from heat, children, and incompatible materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paint Roller

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Roller Brush set 5 in 1

What it is used for

To apply paint evenly to walls and other suitable surfaces using a paint roller. To achieve a smooth finish while reducing splashes and paint waste.

Main parts

- Roller frame and handle
- Roller sleeve or cover
- Paint tray
- Extension pole, if required
- Scraper or cleaning tool

Before you start

1. Choose a roller sleeve suitable for the paint and surface texture.
2. Check the roller frame and sleeve for damage or loose parts.
3. Prepare the surface and protect nearby areas from paint splashes.
4. Wear gloves, eye protection, and suitable work clothing.

How to use

1. Pour a suitable amount of paint into the tray.
2. Load the roller evenly and remove excess paint on the tray grid.
3. Apply paint in overlapping strokes over a manageable area.
4. Maintain steady pressure and avoid pressing too hard.
5. Check coverage and smooth out visible roller marks.
6. Allow the coat to dry as directed before applying another coat.
7. Clean the roller or dispose of the sleeve according to the paint instructions.

⚠ Safety

- Ensure adequate ventilation when using paint.
- Keep paint away from eyes and skin.
- Use a stable platform when working at height.
- Keep paint and cleaning solvents away from flames.

Care & storage

- Remove excess paint before cleaning.

- Clean reusable sleeves according to the paint type.
- Dry and store the roller frame properly.
- Replace worn or damaged sleeves.

Common mistakes to avoid

- Overloading the roller and causing drips.
- Using the wrong sleeve for the surface.
- Painting over a dirty or damp surface.
- Leaving paint to harden on reusable equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paint Stirrer

Category: **Construction & Masonry** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Paint Stirrers

What it is used for

Handle paint stirrers safely to mix paint evenly before application.

How to handle

- Choose a stirrer suitable for the paint type and container size.
- Stir slowly to minimise splashing and avoid creating unnecessary air bubbles.
- Wear suitable gloves and eye protection when required by the paint label.
- Clean reusable stirrers according to product instructions and dispose of contaminated waste properly.

Storage

- Clean and dry reusable stirrers before storing.
- Keep unused stirrers in a clean, dry area away from chemicals and heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Painting Edger

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Low** 1 labs

Covers these lab items: Painting Edger

What it is used for

A painting edger helps apply paint neatly along corners, trim, and edges while reducing unwanted marks on adjacent surfaces. Use a clean edger and test the paint application before working on the finished surface.

How to use

1. Inspect the edger pad, rollers, handle, and frame for damage or loose parts.
2. Choose an edger suitable for the paint type and surface.
3. Prepare the surface by cleaning dust and protecting adjacent areas as required.
4. Load the pad or roller with a small amount of paint and remove excess.
5. Position the edger against the edge or trim and move it with steady pressure.
6. Check the finish and reload paint as needed without overloading the tool.
7. Clean the edger according to the paint manufacturer's instructions after use.

⚠ Safety

- Provide adequate ventilation when using paint and cleaning solvents.
- Wear suitable gloves and eye protection when required.
- Keep paint away from flames and follow the product safety label.
- Use a stable ladder or platform when working at height.

Care & storage

- Remove paint residue promptly and store the edger dry with the pad protected from deformation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Plaster Scratcher

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Scratcher

What it is used for

To scratch or roughen a plaster or render surface to improve the mechanical key for a subsequent coat.

How to handle

- Inspect the scratcher teeth, handle, and body for damage or loose parts before use.
- Wear safety glasses, gloves, and a suitable dust mask when working with dry plaster or masonry.
- Use controlled strokes to create an even pattern of grooves while avoiding excessive pressure that may damage the base coat.
- Keep other workers clear of the immediate work area and avoid contact with hidden electrical cables or services.
- Remove plaster residue and inspect the tool after use.

Storage

- Store clean and dry in a designated construction-tool area.
- Protect the teeth from damage and keep the tool away from moisture and corrosive materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Putty Knife

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Filling Knife; Putty Knife

What it is used for

To spread filler, putty, or similar compounds and smooth surfaces during repair or finishing work.

How to handle

- Inspect the blade, handle, and joint for rust, cracks, looseness, or a bent edge.
- Select a blade width suitable for the repair and prepare the surface according to the compound instructions.
- Load a small amount of material onto the blade and spread it using smooth, controlled strokes.
- Keep fingers away from the blade edge and avoid using excessive force that could damage the surface.
- Remove compound residue and clean the blade before the material hardens.

Storage

- Clean and dry the blade thoroughly after use to prevent residue buildup and corrosion.
- Store with the blade protected in a dry tool rack or drawer away from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scabbling Hammer

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Scrabbling Hammer

What it is used for

To roughen or remove surface material from concrete or masonry to prepare it for repair or a subsequent layer.

How to handle

- Inspect the hammer head, striking points, handle, and joints for cracks, wear, or looseness.
- Wear safety goggles or a face shield, hearing protection, gloves, safety footwear, and suitable respiratory protection for dust exposure.
- Secure the workpiece or confirm the surface is stable, and clear other people from the work area.
- Strike the surface with controlled blows, maintaining a secure grip and avoiding hidden services or unstable material.
- Stop if the tool becomes loose or damaged, and remove dust and debris using an appropriate method after work.

Storage

- Clean and dry the tool before placing it in a designated tool rack.
- Keep the striking points protected and store away from moisture and corrosive materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scutch Hammer

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Scutch

What it is used for

To remove mortar, plaster, or surface projections from masonry using an appropriate scutch head or attachment.

How to handle

- Inspect the hammer, handle, head, and scutch attachment for damage, wear, or loose fittings.
- Fit only the correct compatible attachment and ensure it is secured according to the manufacturer's instructions.
- Wear eye and face protection, gloves, safety footwear, hearing protection, and appropriate dust protection.
- Hold the tool firmly and use controlled strokes, keeping other people clear of flying chips and debris.
- Stop work if the head or attachment loosens, then clean the tool and inspect it after use.

Storage

- Store clean and dry in a designated masonry-tool rack with attachments secured.
- Protect striking edges from damage and keep the tool away from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spalling Hammer

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Spall Hammer

What it is used for

To remove loose or damaged concrete and masonry fragments during surface preparation and repair work.

How to handle

- Inspect the hammer head, striking face, handle, and joints for cracks or looseness.
- Wear eye or face protection, hearing protection, gloves, safety footwear, and suitable respiratory protection for dust.
- Check the work area for hidden services and unstable material, and keep bystanders at a safe distance.
- Use controlled blows to remove only the intended loose material and avoid striking near exposed reinforcement or services without authorisation.
- Collect debris safely and inspect the hammer for damage after work.

Storage

- Store clean and dry in a designated construction-tool rack.
- Keep the striking face protected and the handle secured against accidental damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Straight Edge

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Floating Rule

What it is used for

To check surface straightness and levelness or to guide the levelling of plaster, screed, or other construction material.

How to handle

- Inspect the straight edge for bending, cracks, damaged ends, and a clean reference surface.
- Place it against or across the surface being checked without forcing it into position.
- Use a suitable spirit level or measuring method where level or slope verification is required.
- For levelling work, draw the straight edge steadily across the material and remove excess material as appropriate.
- Clean off mortar or plaster before it hardens and recheck the tool for damage after use.

Storage

- Store clean and dry on a flat rack to prevent bending.
- Protect the reference edge from impact, corrosion, and heavy loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Trowel

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 21 labs

Covers these lab items: Big Trowel; Small Trowel; Trowel; Trowel (Thapi- Plane); Trowel (Thapi) Set; Trowels

What it is used for

A trowel spreads, smooths, or applies materials such as mortar, plaster, and soil. Choose a suitable trowel and keep the work area organised during use.

How to use

1. Inspect the blade, handle, and joint for cracks, bending, or looseness.
2. Select a trowel suitable for the material and finishing task.
3. Prepare the work surface and mix the material according to the approved procedure.
4. Hold the handle firmly and collect a manageable amount of material on the blade.
5. Spread or shape the material with smooth, controlled strokes.
6. Avoid excessive pressure that may damage the surface or tool.
7. Remove material residue from the blade after finishing.
8. Clean, dry, and store the trowel safely.

⚠ Safety

- Wear gloves and eye protection when handling cement, plaster, or irritating materials.
- Keep hands away from sharp blade edges.
- Do not use a trowel with a loose handle or damaged blade.
- Keep the work area clear to prevent slips and trips.

Care & storage

- Clean the blade before mortar or plaster hardens.
- Dry the trowel and store it in a dry place to reduce corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Level

Category: **Construction & Masonry** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Water level; Water Level Tube; Water leveller

What it is used for

To check and transfer level between points using a water level tube or water leveller.

How to handle

- Inspect the tube for cracks, leaks, and blockages before use.
- Fill the tube with clean water and remove trapped air bubbles; keep both ends open and upright as appropriate for the design.
- Place the ends at the points being compared and allow the water to settle.
- Read the water surfaces at eye level and mark the reference points accurately.
- Drain and wipe the tube after use; avoid kinks and contamination.

Storage

- Drain, dry, and coil the tube loosely without sharp bends.
- Store in a clean, dry location away from sharp tools and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

3D Printer Consumable

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Extruder Nozzles; Filament for FDM printers

What it is used for

3D printer consumables, including filament and extruder nozzles, support additive manufacturing and routine printer maintenance.

How to handle

- Confirm filament type, diameter, and nozzle compatibility with the printer and print settings.
- Inspect filament for moisture, contamination, tangles, or damage before loading.
- Keep hands away from hot nozzles and heated components during operation and maintenance.
- Switch off and follow the manufacturer's procedure before replacing a nozzle; allow components to cool where required.
- Use suitable ventilation and follow the safety guidance for the filament material.

Storage

- Store filament sealed with desiccant where recommended and protect it from moisture.
- Keep nozzles labelled and organised in a dry container away from dust and damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Adhesive

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 17 labs

Covers these lab items: Adhesive; Conductive Tape or Aluminium Foil; Consumables such as zip ties, adhesives etc.; Fabric Glue; Fabric Glue (to stick the stones); Flour and Rice Paste (adhesive); Glue; Glue Stick/Fevicol/Adhesive, Fabric Glue, Cello-Tape (Transparent and Foam Double sided tapes, Single sided tapes); Glue S

What it is used for

Adhesives and bonding materials join compatible surfaces in craft, textile, electrical, and general workshop tasks.

How to handle

- Check the adhesive type, surface compatibility, curing time, and product instructions before use.
- Wear the specified PPE and provide ventilation when required.
- Apply only the recommended quantity and avoid skin, eye, and unintended surface contact.
- Keep adhesives away from flames when flammable and close containers after dispensing.

Storage

- Store in original labelled containers under the specified temperature conditions.
- Keep away from heat, sunlight, children, and incompatible materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Adhesive Tape

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Adhesive tape; Duct Tape; Floor Marking Tape

What it is used for

Adhesive tapes secure materials, seal surfaces, or mark designated areas according to their specific type.

How to handle

- Select tape suitable for the surface, temperature, load, and intended application.
- Clean and dry the surface before applying tape where required.
- Apply evenly without stretching excessively and press firmly for adhesion.
- Use scissors or a suitable cutter safely when trimming tape.
- Remove tape carefully to avoid surface damage and dispose of used tape properly.

Storage

- Keep rolls clean, dry, and in their original packaging when practical.
- Store away from heat, direct sunlight, and dust.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Aggregate

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Coarse Aggregates (per cu. m)

What it is used for

Handle coarse aggregates safely for construction and practical work.

How to handle

- Check aggregate grading and suitability before use.
- Wear safety glasses, gloves, and a dust mask when handling dusty material.
- Lift manageable loads and use suitable equipment for heavy quantities.
- Keep aggregates clear of walkways to prevent slips and trips.

Storage

- Store in designated, separated piles or bins on a stable surface.
- Protect from contamination and control dust runoff where necessary.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Aluminium Foil

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Aluminum Foil; Aluminum foil Charts

What it is used for

Aluminium foil wraps, protects, separates, or supports chart-making and other approved practical activities.

How to handle

- Use foil appropriate for the intended purpose and check for tears or contamination.
- Handle cut edges carefully because they can be sharp.
- Keep foil away from live electrical parts and avoid using it in unsuitable heating applications.
- For food contact, use only foil intended for food use and follow applicable temperature guidance.
- Collect scraps after work and dispose of them according to local recycling rules.

Storage

- Store foil rolls or sheets in clean, dry packaging.
- Protect from dust, moisture, and crushing or tearing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bag

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Bags; Ball Storage Duffel/Tote Bag

What it is used for

Bags carry, organise, and transport supplies, personal items, or equipment according to their intended design.

How to handle

- Choose a bag with suitable size, capacity, and material for the contents.
- Inspect seams, handles, straps, zips, and base for damage before use.
- Distribute weight evenly and avoid overloading the bag.
- Keep sharp or leaking items safely contained and separate from delicate contents.
- Use safe lifting techniques and keep bags out of walkways.

Storage

- Empty and clean reusable bags after use as appropriate.
- Store dry bags in a clean area and keep them protected from pests and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Mat

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Bamboo mats

What it is used for

Bamboo mats provide a woven surface for floor covering, seating, display, or approved practical activities.

How to handle

- Inspect the mat for broken strands, splinters, mould, and loose edges before use.
- Place it on a flat, dry surface and secure it if it could slip.
- Keep the mat away from flames, heaters, and excessive moisture.
- Clean according to the material's care instructions and allow it to dry fully.
- Remove damaged mats that may cause trips or splinters.

Storage

- Roll or fold the mat as recommended without forcing the weave.
- Store in a dry, ventilated area away from pests and direct moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Beads

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Beads; Pearls

What it is used for

Beads and pearls are decorative materials used in craft, jewellery, embroidery, and textile embellishment.

How to handle

- Inspect beads for sharp edges, cracks, and inconsistent sizes before use.
- Choose materials appropriate for the design and attachment method.
- Use suitable needles, thread, or tools and keep small pieces away from young children.
- Avoid scattering beads on floors or work surfaces where they may cause slips or loss.
- Collect unused pieces and dispose of broken beads safely.

Storage

- Store beads by type, colour, and size in labelled, closed containers.
- Keep containers dry and securely closed to prevent spills.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Brick

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Bricks

What it is used for

Bricks are masonry units used in construction demonstrations, building work, and practical training.

How to handle

- Inspect bricks for cracks, loose fragments, or unstable shapes before use.
- Wear gloves and safety footwear when handling heavy or rough bricks.
- Lift using safe techniques and avoid carrying loads that block visibility.
- Stack bricks on stable, level ground and keep stacks within safe height limits.
- Control dust during cutting or breaking and use suitable eye and respiratory protection.

Storage

- Store bricks on a firm, level surface in stable stacks.
- Keep storage areas organised and away from walkways and emergency exits.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Button

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Button; Buttons

What it is used for

Buttons fasten garments and accessories or serve as components in sewing and craft activities.

How to handle

- Check buttons for cracks, sharp edges, missing holes, or other defects before use.
- Select the correct button size, shape, and material for the garment or craft project.
- Keep small buttons away from young children and prevent accidental swallowing.
- Use needles and cutting tools carefully when attaching or removing buttons.
- Count and collect loose buttons after the task to prevent loss or tripping.

Storage

- Store buttons sorted by size, colour, and type in labelled compartments.
- Keep containers closed and dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cable Tie

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Cable ties; Cable Ties and clamps

What it is used for

Cable ties and clamps bundle, organise, and secure compatible cables, hoses, and lightweight components.

How to handle

- Choose ties or clamps of suitable size, material, and load capacity.
- Inspect ties and clamps for cracks, sharp edges, or damaged locking mechanisms.
- Bundle cables without excessive compression and maintain required bend radius.
- Trim excess tie ends safely and avoid leaving sharp edges exposed.
- Keep ties away from hot surfaces or moving parts unless specifically rated for that use.

Storage

- Store ties and clamps by size and type in labelled containers.
- Keep them dry and away from sunlight and excessive heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Candle

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Candles; Candles/ Lamp

What it is used for

Candles provide a flame-based light source for approved demonstrations or designated uses.

How to handle

- Use candles only where open flames are permitted and under appropriate supervision.
- Place the candle in a stable, heat-resistant holder away from curtains, paper, and other combustibles.
- Keep the flame away from children, pets, drafts, and unattended areas.
- Never leave a burning candle unattended and extinguish it fully after use.
- Do not move a candle while wax is hot; keep matches and lighters secured.

Storage

- Store unlit candles in a cool, dry area away from heat and direct sunlight.
- Keep candles and ignition tools separate from combustible materials where possible.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cement

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Cement; Cement Bag

What it is used for

Cement is a binding material used in approved construction, repair, and practical training activities.

How to handle

- Read the product label and safety information before handling cement.
- Wear gloves, safety goggles, protective clothing, and suitable dust protection as required.
- Avoid inhaling cement dust and prevent wet cement from contacting skin or eyes.
- Mix only the required quantity in a ventilated area and use appropriate tools.
- Wash exposed skin promptly and seek medical attention for eye exposure or persistent irritation.

Storage

- Store cement bags in a dry, covered area on raised pallets.
- Protect from moisture and use stock in accordance with the manufacturer's shelf-life guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chain

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Metal chains

What it is used for

Metal chains connect, secure, suspend, or demonstrate mechanical linkages in approved applications.

How to handle

- Identify the chain type and confirm its rated load and intended use before use.
- Inspect links for cracks, deformation, corrosion, or damaged connecting hardware.
- Never exceed the rated capacity or use a chain for lifting unless it is certified for that purpose.
- Keep fingers clear of pinch points and wear suitable gloves when appropriate.
- Remove damaged chains from service and report them for inspection.

Storage

- Clean and dry chains before storage to reduce corrosion.
- Store coiled or hung securely in a designated area away from moisture and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chalk Powder

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Chalk Powder/Whitening Powder

What it is used for

Handle chalk or whitening powder while reducing dust exposure and contamination.

How to handle

- Read the product label and confirm its intended use before handling.
- Avoid creating airborne dust; wear suitable eye protection and a dust mask when needed.
- Do not inhale or ingest the powder and keep it away from eyes.
- Do not mix with other chemicals unless the instructions explicitly allow it.

Storage

- Keep in a sealed, labelled container in a dry place.
- Store away from food, moisture, children, and incompatible chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chemical Reagent

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **High** 23 labs

Covers these lab items: Buffer; Chemical reagents; Chemicals; CMT reagent; CMT reagent (for detection of mastitis); Crystals; Diluting fluids; Formalin; Hydrogen Peroxide Bottle; Liquid; Sodium benzoate; Sodium hydroxide pellets; Staining solution / reagents / Gram stain / AFB stain / KOH stain; Staining solution / reagent

What it is used for

Chemical reagents are used for laboratory testing, preparation, cleaning, and analysis according to their specific properties.

How to handle

- Read the label and Safety Data Sheet before use; wear appropriate PPE.
- Use correct measuring tools and never mix chemicals unless the procedure specifically requires it.
- Handle corrosive, toxic, volatile, and reactive chemicals in suitable ventilation with appropriate precautions.

Storage

- Keep in compatible, clearly labelled, tightly closed containers, separated by chemical compatibility.
- Follow the specified temperature, light protection, and restricted-access requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Clay

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Clay

What it is used for

Handle clay safely for modelling, craft, and practical activities.

How to handle

- Check clay for foreign objects or contamination before use.
- Use clean tools and a suitable work surface; avoid eating or drinking while working.
- Prevent dry clay dust from becoming airborne and use suitable respiratory protection when necessary.
- Wash hands and clean the work area after the activity.

Storage

- Keep unused clay sealed to maintain workable moisture.
- Store separately from food and protect from contamination and extreme temperatures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cleaning Material

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 75 labs

Covers these lab items: Chlorine bleach (Sodium hypochlorite); Cleaning chemicals; Cleaning Cloths; Cleaning equipment and solutions; Cleaning Solution (Colin); Cleaning solutions; Cleaning Supplies; Computer Cleaning Kit; Cotton waste; Dish towels; Dishwashing Liquid & Sponge; Disinfectants; Disinfectant/Sanitizer; Kitche

What it is used for

Cleaning materials include cloths, equipment, and chemical solutions used to maintain hygiene on approved surfaces.

How to handle

- Identify the surface and select a compatible cleaning product according to its label.
- Wear required PPE and provide ventilation when handling chemical cleaning solutions.
- Never mix cleaning chemicals, especially bleach with acids or ammonia-based products.
- Use separate cloths or equipment for different hygiene zones and clean spills promptly.

Storage

- Keep chemicals labelled and closed in a secure, ventilated area away from food and incompatible products.
- Store cloths and reusable equipment clean, dry, and separated by use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Concrete Block

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Concrete block

What it is used for

Concrete blocks are used for construction practice, masonry demonstrations, and approved building activities.

How to handle

- Inspect blocks for cracks, broken corners, or unstable surfaces before use.
- Wear gloves, safety footwear, and eye protection where required.
- Use suitable lifting methods or mechanical assistance for heavy blocks.
- Place blocks on stable, level ground and build stacks within safe limits.
- Avoid uncontrolled breaking or cutting; control dust and flying fragments during authorised work.

Storage

- Store blocks on a firm, level surface in stable stacks.
- Keep them away from walkways and protect the storage area from vehicle impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cotton Wool

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Cotton Rolls; Cotton wool

What it is used for

Cotton wool is an absorbent material used for approved cleaning, personal care, and first-aid tasks.

How to handle

- Use clean cotton wool appropriate for the intended purpose.
- For wound care, follow the applicable first-aid procedure and avoid using loose fibres where they may remain in the wound.
- Use clean gloves or applicators when required and avoid reusing contaminated cotton.
- Keep the material away from flames and hot surfaces.
- Dispose of used cotton according to the waste category and local procedure.

Storage

- Keep cotton wool sealed in its original clean packaging.
- Store in a dry, dust-free area away from chemicals and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Craft Wire

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Wire (for armatures)

What it is used for

Handle craft wire safely when constructing armatures and models.

How to handle

- Choose a wire gauge and material suitable for the model.
- Wear eye protection when cutting or bending wire and direct cut ends away from people.
- Use suitable pliers and avoid excessive force that may cause wire to snap.
- Bend or cover sharp ends to reduce puncture hazards.

Storage

- Coil wire securely and cover exposed ends where practical.
- Store by type and gauge in labelled containers away from children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Culture Media

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Culture media

What it is used for

Handle culture media according to laboratory hygiene and contamination-control procedures.

How to handle

- Only trained personnel should prepare or handle culture media in a designated work area.
- Check the label, expiry date, preparation instructions, and required storage conditions.
- Use appropriate PPE and aseptic technique; avoid touching sterile surfaces.
- Treat used media as potentially contaminated and dispose of it according to laboratory procedures.

Storage

- Keep prepared and unopened media under the conditions specified by the manufacturer or laboratory protocol.
- Label prepared media with contents, preparation date, and relevant expiry information.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Disinfectant

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Medium** 1 labs

Covers these lab items: Stocks of isopropyl alcohol

What it is used for

Isopropyl alcohol is used as a solvent or disinfectant for suitable applications and is highly flammable.

How to handle

- Check the product concentration, intended use, label, and Safety Data Sheet before handling.
- Use adequate ventilation, avoid inhalation and prolonged skin contact, and wear suitable PPE.
- Keep away from flames, sparks, hot surfaces, and other ignition sources during use.
- Keep containers closed when not in use and clean spills using the approved procedure.

Storage

- Store tightly closed in a cool, ventilated area away from ignition sources and incompatible materials.
- Keep labelled and secured from unauthorized access.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Distilled Water

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Distilled water

What it is used for

Handle distilled water cleanly for laboratory or equipment use.

How to handle

- Verify the required grade and intended application before use.
- Use clean containers and avoid contaminating the water during transfer.
- Label containers clearly and do not drink laboratory-grade water unless specifically approved for that purpose.
- Follow equipment or laboratory instructions for the required water quality.

Storage

- Keep in clean, tightly closed, compatible containers.
- Store away from chemicals, direct sunlight, and sources of contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fabric

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 18 labs

Covers these lab items: Cotton fabric (Meters); Defected fabric, trims, accessories; Different jute fabric swatches; Fabric; Fabric Yardages, surplus fabric, Trim/Accessory buttons, surface embellishments, zippers, assorted needles and Pin Cushion etc.; Fabrics; Jute Fabrics (meter); Kora; Linen; Silk fabric (Meters)

What it is used for

Fabric materials and swatches are used for sewing, textile training, sampling, and product development.

How to handle

- Identify fabric type, fibre content, colour, dimensions, and any special handling requirements.
- Inspect fabric for stains, defects, loose threads, or damage before cutting or use.
- Use clean hands and suitable cutting tools; secure fabric to prevent slipping.
- Keep fabric samples and surplus material labelled to avoid mix-ups.

Storage

- Store clean fabric in a dry, dust-free area away from direct sunlight and moisture.
- Separate defective fabric, trims, and reusable offcuts from approved stock.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fastener (Screws/Nails/Bolts)

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: fasteners; Fasteners (Nuts, Bolts, Washers); Gypsum Screw; Mechanical fixtures required for panel installation; Nail; Nut Bolt

What it is used for

Screws, nails, bolts, nuts, and washers fasten compatible materials and mechanical assemblies.

How to handle

- Select fasteners of the correct type, size, material, and strength rating.
- Inspect threads, heads, shanks, and coatings for damage or corrosion.
- Use the correct driver, wrench, or installation tool and follow specified torque requirements.
- Wear eye protection when driving or removing fasteners and keep fingers clear of pinch points.
- Dispose of bent, stripped, or damaged fasteners rather than reusing them.

Storage

- Sort fasteners by type, size, and rating in labelled containers.
- Keep them dry and protected from corrosion or accidental spillage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Food Ingredient

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Common salt; Sugar

What it is used for

Handle common salt and sugar hygienically as food ingredients.

How to handle

- Use clean, dry scoops and food-grade containers.
- Keep salt and sugar covered to prevent moisture, pests, and contamination.
- Check labels and condition before use; discard contaminated material.
- Avoid touching ingredients directly with bare hands.

Storage

- Store separately in sealed, labelled food-grade containers.
- Keep in a cool, dry area away from chemicals and cleaning products.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Food Preservative

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Medium** 1 labs

Covers these lab items: Potassium meta bisulphite

What it is used for

Potassium metabisulphite is used for approved food-processing or laboratory applications and requires controlled handling.

How to handle

- Verify product identity, purity, intended use, and applicable food-processing or laboratory instructions.
- Wear suitable gloves and eye protection; avoid breathing dust and prevent contact with eyes or skin.
- Measure with dedicated tools in a well-ventilated area and avoid mixing with acids unless the approved procedure allows it.
- Keep the material away from incompatible substances and follow the relevant spill and waste procedures.

Storage

- Store in a tightly closed, labelled container in a cool, dry, ventilated location.
- Protect from moisture and incompatible chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hacksaw Blade

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Hack Saw Blade

What it is used for

Handle hacksaw blades carefully to prevent cuts and damage.

How to handle

- Wear safety glasses and suitable gloves when appropriate.
- Check the blade for cracks, bent sections, or damaged teeth before use.
- Fit the blade securely with teeth facing the correct cutting direction.
- Do not force the blade or touch the teeth unnecessarily.

Storage

- Store blades in their protective packaging or a designated holder.
- Keep dry and separate from loose tools to prevent injury.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Sanitizer

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Hand hygiene measures; Hand Sanitizer; Hand sanitizer stations and hygiene signage

What it is used for

Hand sanitizer supports hand hygiene when soap and running water are not readily available.

How to handle

- Check the product label, expiry date, and dispenser condition before use.
- Apply the recommended amount to hands and rub all surfaces until dry.
- Keep sanitizer away from eyes, mouth, and broken skin unless the label permits use.
- Use only as directed and keep alcohol-based sanitizer away from flames and heat.
- Do not substitute sanitizer for soap and water when hands are visibly dirty or greasy.

Storage

- Keep containers closed and labelled in a cool place away from ignition sources.
- Store out of reach of young children and supervise use where necessary.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Wash

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Bath Soap; Hand Wash; Hand washing; Liquid Soap

What it is used for

Hand wash products remove dirt and support routine hand hygiene in homes, workplaces, and service areas.

How to handle

- Check the product label, expiry date, and dispenser cleanliness before use.
- Wet hands, apply sufficient soap, and rub all hand surfaces thoroughly.
- Rinse well with clean running water and dry hands using a clean method.
- Avoid contact with eyes and stop use if irritation occurs.
- Refill dispensers hygienically and do not top up containers unless the approved procedure allows it.

Storage

- Keep soap containers closed and store within the manufacturer's recommended conditions.
- Store away from contamination and separate from hazardous chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hinge

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 10 labs

Covers these lab items: Butt Hinges); Parliamen t hinge; Piano Hinges; Rising Butt Hinges; T' hinges

What it is used for

Hinges connect movable panels, doors, lids, and frames to allow controlled opening and closing.

How to handle

- Verify hinge type, dimensions, material, load suitability, and mounting compatibility.
- Inspect hinge leaves, pins, and fasteners for corrosion, deformation, or damage.
- Use appropriate tools to mount hinges securely and keep fingers clear of pinch points.
- Test movement and alignment slowly before placing the assembly into regular service.

Storage

- Store hinges and fasteners in labelled containers by type and size.
- Keep metal hinges dry and protected from corrosion and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hook

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Hook; Hooks

What it is used for

Handle hooks safely for their intended fastening or hanging use.

How to handle

- Identify the hook type and load limit before use.
- Check for sharp points, corrosion, or deformation.
- Use suitable fasteners and secure the hook firmly.
- Keep sharp hook ends away from hands and exposed skin.

Storage

- Store in labelled containers separated by size and type.
- Keep dry and secure to prevent punctures or corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Immersion Oil

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Cedarwood oil

What it is used for

Handle cedarwood immersion oil according to laboratory instructions for microscopy.

How to handle

- Check the product label and confirm compatibility with the microscope objective.
- Use only the quantity required and avoid contact with eyes and skin.
- Wipe oil from the lens using approved lens tissue and the recommended cleaning method.
- Do not mix the oil with other chemicals or return unused oil to the original bottle.

Storage

- Keep tightly closed in its labelled original container.
- Store according to the safety data sheet, away from heat and incompatible substances.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Jute Fibre

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Jute; Jute Fibre (per kg)

What it is used for

Handle jute fibre cleanly while preventing fibre loss and dust exposure.

How to handle

- Wear a dust mask if fibres or particles become airborne.
- Keep fibre bundles dry and avoid dragging them across dirty surfaces.
- Use clean hands or suitable gloves when sorting and measuring.
- Remove loose fibres from the work area after handling.

Storage

- Store in dry, ventilated conditions away from moisture and ignition sources.
- Keep bundles off the floor and protected from pests.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Label

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Barcode; Labelling Material; Labels & Stickers; Luggage Tag

What it is used for

Labels and stickers identify products, packages, assets, locations, and luggage for tracking and organisation.

How to handle

- Use accurate, legible information and verify codes or descriptions before applying labels.
- Choose a label material and adhesive suitable for the surface and environment.
- Apply labels to clean, dry surfaces without covering safety information or required markings.
- Protect personal or confidential details on luggage tags and shipping labels from unnecessary exposure.
- Remove obsolete labels where permitted and update records when identifiers change.

Storage

- Keep labels flat or in their original rolls and packaging.
- Store in a dry, clean place away from heat and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lace

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Lace

What it is used for

Handle lace carefully for sewing, trimming, and decorative applications.

How to handle

- Check the lace for defects, loose threads, and colour compatibility before use.
- Measure and cut with suitable tools on a clean surface.
- Avoid pulling or stretching delicate lace excessively.
- Keep lace away from sharp tools and substances that may stain it.

Storage

- Roll or fold gently without creasing delicate patterns.
- Store by colour and width in clean, dry labelled containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lubricant

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Grease; Lubricant; Machine Oil

What it is used for

Lubricants reduce friction, wear, and heat in compatible machinery, tools, and mechanical components.

How to handle

- Identify the lubricant grade and confirm compatibility with the equipment and application.
- Read the product label and Safety Data Sheet before handling.
- Use required PPE and avoid skin contact, splashes, and inhalation of mist.
- Apply the specified quantity and prevent spills from reaching drains or walkways.
- Clean spills promptly using the approved procedure and dispose of waste properly.

Storage

- Keep lubricants in original labelled, sealed containers.
- Store away from ignition sources, heat, and incompatible materials with spill containment where required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Marking Flag

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Marking flag

What it is used for

Use marking flags to identify locations or boundaries during field and site activities.

How to handle

- Confirm the intended location and marking scheme before placing flags.
- Insert flags securely without damaging buried services or nearby plants.
- Wear suitable gloves when working in rough terrain and keep flag tips away from eyes.
- Remove temporary flags when no longer required, following site procedures.

Storage

- Bundle flags by type and colour in labelled containers.
- Keep dry and protect from bending, corrosion, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Marking Powder

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Marking powder

What it is used for

Handle marking powder safely for layout and measurement tasks.

How to handle

- Read the product label and confirm the powder is suitable for the surface and task.
- Avoid creating dust clouds and wear eye protection and a suitable dust mask when needed.
- Keep powder away from eyes, mouth, food, and strong air currents.
- Clean spills using a method that minimises airborne dust.

Storage

- Keep sealed in a labelled container in a dry area.
- Store away from food, moisture, children, and incompatible chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Masking Tape

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Masking tape; Masking tape (Rolls); Masking/Paper Tape

What it is used for

Masking tape temporarily protects surfaces, marks boundaries, and supports painting or craft tasks.

How to handle

- Select tape compatible with the surface and required removal time.
- Ensure the surface is clean and dry before application.
- Apply evenly without excessive stretching and press edges down to reduce paint seepage.
- Remove tape at the recommended stage and angle to minimise surface damage.
- Do not use masking tape for electrical insulation or loads beyond its intended purpose.

Storage

- Keep rolls clean and dry in their original packaging where possible.
- Protect adhesive from dust, heat, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Naphthalene Ball

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Naphthalene balls

What it is used for

Handle naphthalene balls cautiously to prevent exposure and accidental ingestion.

How to handle

- Read the product label and wear the specified protective equipment when handling.
- Avoid inhaling vapours or dust and prevent contact with skin and eyes.
- Never place naphthalene balls near food or use them where children or pets can access them.
- Do not use them contrary to label directions or in unventilated occupied spaces.

Storage

- Keep in the original, tightly closed and labelled container.
- Store securely in a cool, ventilated area away from food, children, pets, and ignition sources.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Newspaper

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Newspapers

What it is used for

Handle newspapers for reading, reference, and craft activities.

How to handle

- Keep newspapers clean and dry, especially when used for educational or craft purposes.
- Use scissors or cutting tools carefully when preparing newspaper materials.
- Wash hands after handling heavily soiled or damp newspapers.
- Recycle or dispose of used newspapers according to local waste procedures.

Storage

- Stack neatly in a dry, ventilated area away from flames and heat sources.
- Keep separate from food and moisture to prevent mould.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Packaging Material

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 40 labs

Covers these lab items: Aluminium/ Poly bags; Carry Bags; Cartons; Different Colour Plastic Bags with Dustbins; Different Colur Plastic Bags; Euro bags; Examples of biodegradable guest kits and eco-packaging materials; Foil; Food Cling Wrap; Neck-Locks for Carry Bags; Packaging bag for storage; Packaging Material; Packagin

What it is used for

Packaging materials protect, contain, carry, and organise products or supplies during storage, handling, and transport.

How to handle

- Select packaging suitable for the product's weight, dimensions, fragility, and hygiene requirements.
- Inspect bags, cartons, and packaging for damage, contamination, or moisture before use.
- Pack items securely without overloading bags or cartons and label contents where required.
- Separate clean packaging from waste and follow applicable reuse, recycling, and disposal instructions.

Storage

- Store packaging in a clean, dry, pest-free area away from chemicals and direct sunlight.
- Keep different sizes and material types sorted and labelled.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paint

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 10 labs

Covers these lab items: Acrylic/Dust/Enamel Paints (variety); Color; Colouring material (Poster colour, Water colour, Acrylics, Pastels, fabric printing dyes and colors and materials etc); Marking paint; Paints; Spray paint/ chalk; Water colour; Water colours, Markers, and Pastels; Watercolor

What it is used for

Paints and colouring materials provide decorative, protective, and educational finishes on compatible surfaces and fabrics.

How to handle

- Check paint type, surface compatibility, label instructions, and required drying or curing time.
- Wear suitable gloves and eye protection, and ensure adequate ventilation when using paints or solvents.
- Stir and apply according to the manufacturer's instructions; avoid skin contact and inhalation of vapours or dust.
- Keep flammable products away from ignition sources and clean spills using the approved procedure.

Storage

- Keep containers tightly closed and labelled in a cool, ventilated area.
- Separate incompatible products and protect paints from freezing, overheating, and direct sunlight as specified.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paper

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 12 labs

Covers these lab items: Backing paper; Brown sheet; Butter Paper; Cardboard sheets (large); Different size of papers; Drawing paper; Drawing sheets; End papers; Various Papers (newspaper, tissue, etc.)

What it is used for

Paper materials support drawing, printing, writing, packaging, craft, and other classroom or workshop activities.

How to handle

- Select paper type, size, thickness, and finish suitable for the intended task.
- Keep sheets clean and dry and check for tears, stains, or moisture damage.
- Use cutters and scissors carefully when trimming paper, following safe cutting practices.
- Separate reusable offcuts from waste and avoid unnecessary paper consumption.

Storage

- Store paper flat or in suitable folders in a dry, clean area.
- Protect from moisture, direct sunlight, dust, and fire hazards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Personal Care Product

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Environmentally friendly insect repellents, sunblock, and sanitizing supplies

What it is used for

Handle insect repellents, sunblock, and sanitising supplies safely for personal protection.

How to handle

- Read labels and follow instructions for application, dosage, contact time, and precautions.
- Avoid contact with eyes and irritated skin; use products only for their intended purpose.
- Keep sanitiser away from flames and heat when alcohol-based, and allow hands to dry before approaching ignition sources.
- Do not mix products and check expiry dates before use.

Storage

- Keep products in original labelled containers with caps secured.
- Store away from heat, direct sunlight, food, children, and incompatible substances.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Poly Bag

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Poly bags; Poly bags (different sizes); Polybags (assortment in sizes and variety)

What it is used for

Poly bags package, protect, separate, and organise lightweight goods and materials.

How to handle

- Choose a bag size and strength suitable for the contents and intended use.
- Inspect bags for holes, contamination, or damage before packing.
- Avoid overfilling and keep sharp objects from puncturing the bag.
- Keep bags away from young children because of suffocation risk.
- Reuse or dispose of bags according to local waste and recycling rules.

Storage

- Store bags sorted by size in a clean, dry area.
- Keep away from heat, sharp objects, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Polythene Sheet

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Polythene film

What it is used for

Polythene sheets provide temporary covering, separation, or protection for materials and work surfaces.

How to handle

- Select a sheet of suitable thickness and size for the intended task.
- Inspect for holes, tears, contamination, and slippery surfaces before use.
- Spread the sheet carefully and secure loose edges to prevent tripping or wind displacement.
- Keep plastic sheets away from flames, hot surfaces, and incompatible chemicals.
- Collect offcuts and dispose of or recycle them according to local requirements.

Storage

- Fold or roll sheets neatly after use if they are reusable.
- Store in a clean, dry area away from heat, sharp objects, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Raw Material

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Raw Material (as per business type)

What it is used for

Handle business-specific raw materials safely and maintain their quality.

How to handle

- Identify each material and review its specification, hazards, and intended use before handling.
- Wear suitable PPE and use appropriate tools for the material's physical properties.
- Keep different materials clearly labelled and prevent cross-contamination.
- Follow approved handling, measuring, and waste-disposal procedures for the business activity.

Storage

- Store materials in compatible, labelled containers or designated racks.
- Follow required temperature and humidity conditions and separate incompatible substances.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Refrigerant Gas

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **High** 1 labs

Covers these lab items: Refrigerant

What it is used for

Refrigerant gases transfer heat in compatible refrigeration and air-conditioning systems and require pressure and exposure controls.

How to handle

- Identify the refrigerant and check equipment compatibility before connecting cylinders or service hoses.
- Use trained personnel, specified PPE, and equipment rated for the refrigerant pressure and type.
- Provide ventilation, avoid direct contact with escaping gas or liquid, and prevent oxygen displacement.
- Recover refrigerant with approved equipment; never intentionally vent it where prohibited.

Storage

- Secure cylinders upright with protective caps where applicable in a cool, ventilated area.
- Keep away from heat and use only approved cylinders for the specific refrigerant.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Resin

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Medium** 1 labs

Covers these lab items: Resin for SLA

What it is used for

SLA resin is a light-sensitive material used in compatible stereolithography 3D printers and requires careful chemical handling.

How to handle

- Confirm resin compatibility with the printer and review the product Safety Data Sheet.
- Wear chemical-resistant gloves and eye protection, and work in suitable ventilation.
- Avoid skin and eye contact; protect resin from unintended light exposure during handling.
- Follow the printer manufacturer's instructions for printing, washing, curing, spill cleanup, and uncured resin disposal.

Storage

- Keep in a sealed, labelled, light-protective container under the specified temperature conditions.
- Store away from heat, sunlight, food, and unauthorized access.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rope

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Rope; Ropes

What it is used for

Ropes secure, tie, lift, or restrain compatible loads during approved tasks and practical training.

How to handle

- Select rope material, diameter, length, and load rating suitable for the task.
- Inspect the rope for fraying, cuts, chemical damage, knots, or weakened sections.
- Use approved knots and attachment points and follow safe load limits.
- Never use an unidentified or damaged rope for lifting or supporting people.
- Keep hands clear of pinch points and avoid standing beneath suspended loads.

Storage

- Coil ropes neatly and keep them dry and free from contamination.
- Store away from sunlight, chemicals, sharp edges, and excessive heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rubber Sheet

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Rubber sheet

What it is used for

Rubber sheets provide a flexible surface for lining, cushioning, protection, or approved practical work.

How to handle

- Check the sheet for tears, cracks, brittleness, contamination, and suitable thickness.
- Confirm material compatibility with the intended application, chemicals, and temperature.
- Place the sheet on a stable surface and secure it if it may slide.
- Use suitable gloves when handling rough, contaminated, or chemically exposed rubber.
- Avoid contact with flames, excessive heat, and sharp cutting edges.

Storage

- Store sheets flat or rolled as recommended by the manufacturer.
- Keep dry and protect from sunlight, oils, chemicals, and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sand

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Sand

What it is used for

Sand is used in construction, practical training, gardening, and other approved material-handling tasks.

How to handle

- Check that the sand type and grade are suitable for the intended application.
- Avoid creating dust when pouring, transferring, or mixing sand.
- Wear suitable gloves, eye protection, and dust protection when required.
- Use safe lifting techniques for bags and keep sand away from drains and walkways.
- Do not use contaminated sand for construction, gardening, or training activities.

Storage

- Store in a designated area protected from contamination and excessive runoff.
- Keep bags sealed and dry where the material requires protection from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sandpaper

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Sand Paper; Sandpaper (assorted grades)

What it is used for

Handle sandpaper safely and select the correct grade for the task.

How to handle

- Choose the appropriate grit for the material and finish required.
- Wear eye protection and a suitable dust mask when sanding creates dust.
- Sand in a controlled direction and avoid excessive pressure.
- Keep sandpaper away from sparks or heat when sanding flammable materials.

Storage

- Store assorted grades in labelled packets or compartments.
- Keep dry and flat to prevent curling or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sealant

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: sealants

What it is used for

Handle sealants safely while maintaining product quality and joint performance.

How to handle

- Read the product label and safety data sheet before use.
- Check compatibility with the surface and application; use suitable gloves and ventilation.
- Apply with the recommended tool and avoid skin or eye contact.
- Close the container after use and clean spills according to the product instructions.

Storage

- Keep in the original, labelled, tightly closed container.
- Store within the manufacturer's specified temperature range, away from heat and incompatible materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sponge

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Sponge; Sponge (soft, for finishing); Sponges

What it is used for

Sponges absorb liquids and support cleaning, finishing, and other approved practical tasks.

How to handle

- Select a sponge suitable for the surface and intended application.
- Inspect for dirt, tearing, residue, or deterioration before use.
- Use separate sponges for different tasks, especially food-contact, cleaning, and personal-care activities.
- Rinse and clean reusable sponges after use and allow them to dry completely.
- Replace sponges that develop odours, mould, or persistent contamination.

Storage

- Store clean, dry sponges in a ventilated area.
- Keep task-specific sponges labelled and separated to prevent cross-contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stationery

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 167 labs

Covers these lab items: 3-Hole Punched Paper; Address Labels; Art Gum Eraser; Basic Stationary Items; Basic Stationery Set; Binder; Binders and Folders; Board marker; Calendar/Planner; Carbon Paper; Cash Receipt Book; Cells/Transparencies; Chalk; Charcol pencil; Chart paper and sketch pens; Chart, Sketch Pens; Clips, Pins

What it is used for

Stationery supports writing, labelling, filing, planning, drawing, and routine office or classroom activities.

How to handle

- Select items suitable for the task and check tools for damage before use.
- Use cutters, staplers, and other sharp stationery carefully and keep them away from children when unsupervised.
- Use markers and adhesives in ventilated areas when required by their labels.
- Return shared items after use and dispose of waste paper and packaging appropriately.

Storage

- Sort items into labelled drawers, folders, or containers for easy access.
- Keep paper and stationery dry, clean, and away from heat or moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stone

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Stone

What it is used for

Stone is used for construction, landscaping, craft work, or practical demonstrations depending on its type.

How to handle

- Identify the stone type and check its suitability for the intended application.
- Inspect for unstable pieces, sharp edges, dust, and loose fragments before handling.
- Wear suitable gloves, safety footwear, and eye protection when required.
- Use safe lifting techniques or assistance for heavy stones.
- Control dust during cutting or grinding and use approved tools and protective equipment.

Storage

- Store stones on stable ground to prevent rolling or collapse.
- Keep materials organised and away from walkways and access routes.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tracing Paper

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 9 labs

Covers these lab items: Design Transferring Paper; Tracing material; Tracing paper; Tracing paper or Butter Paper; Tracing Paper Packs

What it is used for

Tracing paper transfers outlines, drawings, and design details onto suitable surfaces or materials.

How to handle

- Choose the appropriate paper size, transparency, and type for the task.
- Keep paper flat and clean to maintain accurate tracing.
- Secure the original and tracing sheet to prevent shifting during work.
- Use pencils or approved marking tools and avoid excessive pressure that may tear the sheet.
- Check transferred lines for accuracy before proceeding.

Storage

- Store tracing sheets flat in folders or protective sleeves.
- Keep away from moisture, dirt, and excessive heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Varnish

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Varnish/Lacquer (1 litre bottles)

What it is used for

Handle varnish or lacquer safely for surface finishing and coating work.

How to handle

- Read the product label and safety data sheet before use.
- Use suitable gloves, eye protection, and adequate ventilation; avoid inhaling vapours.
- Keep away from flames, sparks, and hot surfaces, as many products are flammable.
- Close the bottle promptly after use and manage spills and waste according to product instructions.

Storage

- Keep in the original, tightly closed and labelled container.
- Store in a cool, ventilated area away from ignition sources and incompatible materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Water; Water as required

What it is used for

Water is used for drinking, cleaning, preparation, irrigation, and other approved activities according to its quality.

How to handle

- Confirm the water source and quality are suitable for the intended use.
- Use potable water for drinking, food preparation, and personal hygiene where required.
- Keep clean water containers covered and use clean dispensing equipment.
- Avoid contaminating stored water with hands, chemicals, waste, or dirty utensils.
- Use water responsibly and report leaks or signs of contamination.

Storage

- Store water in clean, food-grade containers when intended for drinking or food use.
- Keep containers covered, labelled where necessary, and away from chemicals and direct contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Workshop Consumable

Category: **Consumables & Raw Materials** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Consumable items

What it is used for

Handle general workshop consumables safely and minimise waste.

How to handle

- Identify each consumable and read its label or safety information before use.
- Use the correct PPE and follow the specified application method.
- Keep materials in original or clearly labelled containers; never mix unknown substances.
- Report damaged packaging, leaks, or unusual reactions promptly.

Storage

- Separate incompatible materials and keep hazardous items in designated storage.
- Maintain stock rotation and dispose of waste according to local procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drone

Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Commercial drone; Different MODEL of drone; Hexacopter

What it is used for

A drone uses powered rotors, flight controls, and onboard electronics to perform approved aerial tasks. Pre-flight checks, controlled operation, and compliance with applicable aviation rules help reduce injury, equipment damage, and airspace risks.

Main parts

- Frame supports the drone's main components.
- Motors and propellers generate lift and movement.
- Flight controller stabilises the aircraft and manages control inputs.
- Battery supplies electrical power to the motors and electronics.
- Remote controller sends pilot commands to the drone.
- Navigation sensors and positioning systems support flight stability.
- Camera or mission payload performs the intended task when fitted.
- Landing gear supports the drone during take-off and landing.

Before you start

1. Confirm the operator is trained and authorised for the intended flight.
2. Check applicable local aviation rules, airspace restrictions, permissions, and weather conditions.
3. Inspect the frame, propellers, motors, landing gear, and fasteners for damage.
4. Check battery charge, secure mounting, and battery condition.
5. Verify controller operation, communication link, navigation status, and required software settings.
6. Choose a clear take-off and landing area away from people, vehicles, and obstacles.

How to use

1. Install the battery and payload according to the manufacturer's instructions.
2. Place the drone on a level surface and power it on in the specified sequence.
3. Wait for the required system checks and navigation status before flight.
4. Confirm the control response and set the approved flight mode and limits.
5. Take off gradually and hover at a safe, controlled height.
6. Operate the drone within visual and authorised operating limits as required by local rules.
7. Monitor battery level, signal quality, weather, and surrounding airspace throughout the flight.
8. Return to the designated landing area before the battery reaches its required reserve level.
9. Land slowly, disarm the motors, and power off in the recommended sequence.
10. Inspect the drone after flight, record any faults, and store the battery and equipment safely.

⚠ Safety

- Never fly near airports, restricted airspace, or emergency operations without required authorisation.
- Keep people clear of spinning propellers and never handle them while motors are armed.
- Do not fly in weather conditions beyond the drone's rated limits.
- Maintain required visual contact and follow applicable altitude and distance restrictions.
- Do not fly over people or sensitive sites unless specifically permitted under applicable rules.
- Stop the flight and land safely if control, navigation, battery, or communication problems arise.

Care & storage

- Clean the frame and sensors with suitable materials after use.
- Inspect propellers, motors, fasteners, and landing gear before and after flights.
- Store batteries according to the manufacturer's charge and temperature recommendations.
- Update firmware only through approved procedures and maintain flight and maintenance records.

Common mistakes to avoid

- Taking off without completing system and propeller checks.
- Flying with a damaged propeller or an insecure battery.
- Ignoring low-battery warnings, poor signal, or changing weather.
- Operating outside authorised airspace or losing required visual contact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drone Frame

Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Drone base; Foldable Air Frame – Aluminium & Glass Fibre

What it is used for

To assemble and inspect a drone frame before installing electronics and propulsion components. To ensure the frame is correctly aligned, securely fastened, and suitable for the intended drone configuration.

Main parts

- Central frame or body plate
- Arms or booms
- Motor mounting points
- Landing gear, if fitted
- Fasteners and spacers
- Foldable hinges or locking joints, if fitted

Before you start

1. Confirm the frame size and material are compatible with the planned drone components.
2. Inspect the frame, arms, hinges, and mounting points for cracks or deformation.
3. Prepare the correct fasteners and tools.
4. Work on a clean, stable surface with power disconnected from all electronics.
5. Review the frame assembly instructions.

How to use

1. Lay out all frame parts and verify them against the assembly instructions.
2. Attach the arms to the central frame in the specified orientation.
3. Fit the specified fasteners and tighten them evenly to the recommended requirement.
4. Check that motor mounts and electronic mounting points are aligned.
5. For foldable frames, test each hinge and locking mechanism.
6. Inspect the frame for twisting, loose parts, and sharp edges.
7. Install electronics only after the frame passes inspection.

⚠ Safety

- Never power motors while assembling or adjusting the frame.
- Use the specified fasteners and avoid overtightening composite parts.
- Wear eye protection when drilling or cutting is explicitly required.
- Do not fly with cracked arms, loose joints, or damaged locking mechanisms.

Care & storage

- Remove dust and debris after assembly.
- Inspect joints and fasteners before each flight.
- Keep the frame dry and protected from impacts.
- Replace cracked or deformed parts before use.

Common mistakes to avoid

- Installing arms in the wrong orientation.
- Using incorrect fasteners or excessive tightening force.
- Ignoring a loose foldable hinge.
- Installing electronics on an unchecked or damaged frame.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drone Kit

Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Micro Quadcopter kit; Plastic Hexacopter with Propulsion Kit; Quadcopter drone Combo with Pix hawk Kit for beginner

What it is used for

A drone kit provides components to assemble, configure, and operate a small unmanned aircraft for learning or approved tasks. Correct assembly, battery handling, calibration, and responsible flight practices help prevent crashes and injuries.

Main parts

- Frame supports the flight controller, motors, and other components.
- Brushless motors drive the propellers.
- Propellers generate lift and must match the motor specifications.
- Electronic speed controllers regulate motor speed on applicable kits.
- Flight controller processes sensor data and stabilises the aircraft.
- Battery supplies electrical power to the drone.
- Remote controller and receiver provide flight commands and communication.

Before you start

1. Read the kit manual and confirm all components are compatible.
2. Assemble the drone only under suitable supervision and with power disconnected.
3. Inspect the frame, propellers, motors, wiring, connectors, and battery for damage.
4. Check that propellers are fitted in the correct positions and orientations.
5. Charge the battery with a compatible charger on a safe, supervised surface.
6. Check local drone rules, permitted airspace, weather, and the approved flight area.

How to use

1. Secure all frame parts, motors, flight controller, and electronic connections according to the assembly guide.
2. Check motor and propeller compatibility and tighten propeller fittings as instructed.
3. Connect the flight controller and receiver using the specified wiring diagram.
4. Power on the controller and drone in the sequence required by the manufacturer.
5. Complete required calibration and configure failsafe settings according to the manual.
6. Check battery level, control response, sensor status, and communication before flight.
7. Place the drone on a level, open surface and keep people clear of the take-off area.
8. Start the motors and lift off gradually while maintaining visual line of sight.
9. Fly at a controlled height and speed within the permitted area and monitor battery status.
10. Land promptly when battery warnings appear or control becomes unreliable, then disarm the motors.

⚠ Safety

- Never install or remove propellers while the battery is connected.
- Keep hands, hair, and clothing away from rotating propellers.
- Do not fly near people, roads, power lines, airports, or restricted areas without legal authorisation.
- Never fly beyond visual line of sight unless specifically permitted by applicable rules.
- Do not use swollen, leaking, punctured, or overheating batteries.
- Stop flight during unsafe wind, rain, poor visibility, or loss of control.

Care & storage

- Disconnect the battery before cleaning, repairing, or changing components.
- Inspect propellers, motors, frame, wiring, and battery after each flight.
- Store batteries according to the manufacturer's charge and temperature guidance.
- Keep the drone and kit in a dry protective case and replace damaged parts before flying again.

Common mistakes to avoid

- Connecting the battery before checking wiring and propeller installation.
- Flying without checking calibration, battery level, and control response.
- Operating outside permitted airspace or losing visual contact with the drone.
- Continuing flight after battery warnings, damage, or unstable behaviour.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electronic Speed Controller

Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Electronic Speed Controller; ESC (Electronic Speed controllers)

What it is used for

To install and check an electronic speed controller (ESC) in a compatible drone or motor training setup. To ensure correct wiring, configuration, and safe motor response before operation.

Main parts

- ESC electronic unit
- Battery power input leads
- Motor phase output leads
- Signal or control connector
- Cooling surface or heat sink, if fitted

Before you start

1. Confirm the ESC is compatible with the motor, battery voltage, and required current.
2. Inspect the ESC, wiring, connectors, and insulation for damage.
3. Disconnect the battery before making connections.
4. Review the ESC and motor manufacturer's wiring and configuration instructions.
5. Remove propellers before bench testing a drone motor.

How to use

1. Connect the motor leads to the ESC according to the manufacturer's instructions.
2. Connect the signal lead to the correct control output with the specified polarity.
3. Connect the battery input only after verifying polarity and connector compatibility.
4. Secure the ESC so it has appropriate cooling and cannot contact conductive parts.
5. Configure calibration and operating settings as specified by the manufacturer.
6. Perform a low-power test with the propeller removed and the area clear.
7. Disconnect the battery before changing wiring or correcting faults.

⚠ Safety

- Never connect a battery with reversed polarity.
- Keep fingers, clothing, and tools away from motor shafts.
- Do not operate an ESC with exposed damaged conductors or signs of overheating.
- Use a battery and power source within the ESC's rated limits.

Care & storage

- Keep the ESC dry and free of conductive debris.
- Inspect connectors, solder joints, and insulation regularly.
- Ensure cooling airflow is not blocked.
- Stop using the unit if it overheats, swells, or shows electrical damage.

Common mistakes to avoid

- Using an ESC with an incompatible battery or motor.
- Reversing power polarity.
- Testing with propellers attached in an unsecured setup.
- Ignoring overheating or loose electrical connections.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fixed Wing UAV

Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Fixed Wing UAV

What it is used for

A fixed-wing UAV is an unmanned aircraft that uses wings to generate lift for flight. Operate it only with proper training, legal authorisation, approved airspace, and a mission-specific flight plan.

Main parts

- Fixed wings and fuselage
- Propulsion system, motor, and propeller
- Flight controller and navigation sensors
- Battery or approved power system
- Radio transmitter, receiver, and antennas
- Landing gear and payload mount, if fitted

Before you start

1. Confirm required aviation permissions, airspace restrictions, and local operating rules.
2. Inspect wings, fuselage, control surfaces, fasteners, and landing gear.
3. Check the propeller, motor mount, and propulsion system for damage.
4. Inspect the battery, connectors, and wiring for damage or swelling.
5. Verify control links, navigation sensors, failsafe settings, and battery status.
6. Review weather, wind, visibility, flight route, landing area, and emergency plan.

How to use

1. Select a clear approved operating area away from people and obstacles.
2. Install the battery and payload according to the manufacturer's instructions.
3. Secure all covers, fasteners, and connectors before powering on.
4. Place the UAV in the specified preflight position and power it on.
5. Complete controller checks, sensor checks, and navigation readiness checks.
6. Verify control-surface direction and motor operation using the approved procedure.
7. Confirm home position, failsafe behaviour, return procedure, and battery reserve.
8. Start the flight only when the area is clear and all checks pass.
9. Launch using the approved method for that aircraft and conditions.
10. Monitor flight path, battery, telemetry, weather, and surrounding airspace continuously.
11. Abort or return to the landing area if conditions become unsafe.
12. Land using the approved procedure and disarm the propulsion system.
13. Power off the aircraft and controller as instructed, then remove the battery if required.

4. Inspect the UAV after flight and record any faults or unusual behaviour.

⚠ Safety

- Only trained and authorised operators should fly the UAV.
- Follow applicable aviation rules, airspace restrictions, and required permissions.
- Keep people clear of the propeller and never handle it while powered.
- Do not fly in unsuitable weather, poor visibility, or unsafe wind conditions.
- Maintain required separation from people, buildings, aircraft, and restricted areas.
- Do not continue flight with low battery, lost control link, or critical warnings.

Care & storage

- Clean dirt and moisture from the airframe after use.
- Store batteries in a safe, dry location according to battery instructions.
- Inspect propellers, control surfaces, fasteners, and connectors regularly.
- Update firmware only using approved procedures and verify settings afterward.
- Record flights, maintenance, battery condition, and faults.

Common mistakes to avoid

- Skipping preflight checks or failing to verify control-surface direction.
- Launching without checking airspace, weather, or landing conditions.
- Ignoring battery reserve, telemetry warnings, or failsafe alerts.
- Touching or adjusting the propeller while the propulsion system is armed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Flight Controller

Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Arducopter Flight Controller with GPS; FCB (Flight Controller Board)

What it is used for

To process sensor inputs and control drone movement, stability, and flight functions.

How to handle

- Inspect the flight controller board, connectors, wiring, and mounting for damage.
- Confirm compatibility with the drone frame, motors, ESCs, GPS module, and battery system.
- Disconnect the battery before wiring or changing connections.
- Install and configure the controller according to the manufacturer's instructions, including firmware and sensor calibration.
- Verify GPS lock, sensor status, motor direction, and failsafe settings; perform initial checks with propellers removed.

Storage

- Store in an antistatic protective package in a cool, dry location.
- Protect connectors and circuit components from moisture, dust, static discharge, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Flight Simulator

Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: RC Flight Simulator

What it is used for

To practise basic aircraft or RC flight controls and procedures in a simulated environment without operating a real aircraft.

How to handle

- Inspect the computer, controller, cables, software, and display for safe operation.
- Launch the approved simulator and select the correct aircraft model, control layout, and training scenario.
- Calibrate the controller and verify that control inputs respond correctly before beginning the exercise.
- Follow the instructor's scenario and practise take-off, flight control, orientation, and landing procedures as applicable.
- Save training progress if required, exit the software properly, and disconnect equipment safely.

Storage

- Shut down the simulator software and equipment according to the operating procedure.
- Store controllers and accessories in a labelled, dry cabinet and protect cables from bending or crushing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

GPS Module

Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: GPS Module

What it is used for

A GPS module receives satellite signals to estimate position, speed, and time for compatible electronic or drone systems. Install it securely and verify signal reception before operation.

How to use

1. Check the module, antenna, connectors, and mounting hardware for damage.
2. Confirm electrical voltage, interface, and compatibility with the host device.
3. Switch off the host system before connecting the module.
4. Mount the module and antenna according to the manufacturer's orientation requirements.
5. Connect the specified power and data interfaces securely.
6. Power on the system and allow time for satellite acquisition.
7. Verify position and signal status using the approved software or display.

⚠ Safety

- Use the specified voltage and correct connector polarity.
- Keep the antenna clear of obstructions and sources of electrical interference.
- Do not handle exposed circuits while powered unless designed for it.
- For drones, complete GPS checks before flight and follow applicable flight rules.

Care & storage

- Keep connectors clean and dry, and protect the antenna and module during storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

LiDAR Sensor

Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Hovering function using LiDAR sensor

What it is used for

A LiDAR sensor measures distance using laser light and can provide height or obstacle data for drone positioning and hovering systems. Mount and configure it according to the flight controller's specifications.

How to use

1. Inspect the LiDAR sensor, lens or optical window, cables, and mounting points.
2. Confirm compatibility with the flight controller, interface, voltage, and software version.
3. Switch off the drone and disconnect the battery before wiring or mounting the sensor.
4. Mount the sensor securely in the specified orientation with a clear sensing path.
5. Connect power and data lines according to the wiring diagram.
6. Power on the system and verify sensor readings and calibration status using approved software.
7. Test hovering functions in a controlled, obstacle-free area under trained supervision.

⚠ Safety

- Keep the drone powered off while making electrical connections.
- Never look directly into the LiDAR laser aperture or aim it at people's eyes.
- Do not rely on LiDAR alone for safe flight or obstacle avoidance.
- Follow drone operating rules and maintain a safe test area.

Care & storage

- Keep the optical window clean, protect cables from strain, and check mounting security before each flight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

LiPo Battery

Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: LiPo Battery; Lipo Battery and Charger

What it is used for

A lithium-polymer (LiPo) battery supplies electrical energy to drones and other portable equipment and is commonly charged using a compatible balance charger. Correct voltage selection, balanced charging, careful handling, and appropriate storage are essential because damaged or incorrectly charged LiPo batteries can ignite or release hazardous fumes.

Main parts

- Battery cells store electrical energy in a lithium-based chemistry.
- Battery pack casing or wrapping protects the cell assembly.
- Main power connector supplies energy to the equipment.
- Balance connector provides access to individual cell voltages.
- Compatible charger controls the charging process.
- Charger display or indicators show voltage, current, and charge status.
- Battery management or protection circuit is included where fitted.
- Fire-resistant charging surface or approved charging container supports safer charging.

Before you start

1. Only trained users should charge and handle LiPo batteries, following the battery and charger instructions.
2. Check the battery chemistry, cell count, rated voltage, capacity, and permitted charge rate.
3. Inspect the battery for swelling, punctures, crushed areas, damaged insulation, loose wires, or unusual odour.
4. Do not charge a damaged, swollen, wet, or abnormally hot battery.
5. Use a charger with the correct LiPo mode and compatible cell-count and balance functions.
6. Place the battery on a non-combustible surface in a well-ventilated area away from combustible materials.
7. Check the charger, leads, connectors, and balance plug for damage and correct polarity.

How to use

1. Confirm the battery and charger specifications match before connecting anything.
2. Connect the charger to its power source according to the charger instructions.
3. Connect the battery's main lead and balance connector to the correct charger ports without forcing the connectors.
4. Select the correct LiPo chemistry, cell count, and approved charging settings.
5. Verify the displayed cell count and voltage before starting the charge.

6. Start charging and remain nearby, observing the battery and charger throughout the process.
7. Stop charging immediately if the battery swells, becomes unusually hot, leaks, smells abnormal, or the charger reports a fault.
8. When charging is complete, stop the charger and disconnect the battery using the approved sequence.
9. Allow the battery to cool to a suitable temperature before use or storage.
10. For storage, use the manufacturer's recommended storage voltage and a suitable cool, dry location.
11. Record battery condition, charging issues, and any battery removed from service.

⚠ Safety

- Never leave a LiPo battery charging unattended.
- Never use a damaged or swollen battery or attempt to puncture, compress, or repair it yourself.
- Do not charge beyond the battery manufacturer's voltage, current, or temperature limits.
- Prevent short circuits by keeping metal objects away from battery terminals and exposed connectors.
- Do not charge near flammable materials, escape routes, or in direct sunlight or excessive heat.
- If a battery smokes, hisses, swells rapidly, or ignites, move away, warn others, and follow the site's battery-fire emergency procedure; do not handle a failing battery.

Care & storage

- Inspect battery casing, wires, connectors, and balance leads before and after use.
- Keep terminals clean and dry and protect connectors from impact and contamination.
- Store batteries at the recommended storage voltage in a cool, dry, fire-conscious location.
- Separate damaged batteries from serviceable stock and follow approved hazardous-battery disposal procedures.
- Use only compatible chargers and arrange replacement when capacity, condition, or safety becomes unacceptable.

Common mistakes to avoid

- Selecting the wrong cell count or battery chemistry on the charger.
- Charging a swollen, punctured, or otherwise damaged battery.
- Leaving the battery charging without supervision.
- Storing a LiPo battery fully charged or fully discharged for an extended period against manufacturer guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Motor Thrust Stand

Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Motor Thrust Stand; Thrust Measuring Stand

What it is used for

A motor thrust stand measures the thrust produced by an electric motor and propeller or other test assembly. Secure mounting, correct instrumentation, controlled power application, and effective guarding are essential because rotating propellers and high-current electrical systems can cause serious injury.

Main parts

- Rigid frame supports the motor and test assembly.
- Motor mount secures the motor in the specified orientation.
- Load cell measures the force produced by the motor.
- Signal conditioner or amplifier processes the load-cell output where fitted.
- Electronic speed controller regulates motor speed where applicable.
- Power supply or battery provides electrical energy.
- Voltage and current sensors measure electrical input where fitted.
- Data acquisition unit records thrust, speed, voltage, and current.
- Propeller guard or test enclosure limits access to the rotating assembly where fitted.

Before you start

1. Only trained personnel should operate the thrust stand and high-power motor test equipment.
2. Inspect the frame, motor mount, fasteners, load cell, wiring, connectors, and test enclosure for damage.
3. Confirm the motor, propeller, ESC, battery, and stand are compatible and within their rated limits.
4. Secure the stand to a stable test bench and verify that the motor mount cannot shift.
5. Balance and inspect the propeller according to its instructions; do not use cracked, chipped, or damaged propellers.
6. Calibrate or zero the load cell and verify the sensor and data acquisition connections.
7. Clear the test zone of people and loose objects, and use appropriate eye protection and hearing protection.

How to use

1. Mount the motor securely and install the specified propeller only when the power source is disconnected.
2. Connect the ESC, power supply or battery, and sensors according to the test wiring diagram.
3. Check polarity, connector security, sensor zero, and data recording before energising the system.

4. Fit the required propeller guard or enclosure and verify that all personnel are outside the exclusion zone.
5. Start the data acquisition system and confirm that thrust, speed, voltage, and current readings are plausible.
6. Energise the motor and increase throttle gradually according to the approved test plan.
7. Record readings at the specified operating points without exceeding motor, propeller, ESC, battery, or stand limits.
8. Monitor vibration, temperature, current, thrust readings, and unusual sound throughout the test.
9. Reduce throttle to zero and disconnect the power source before approaching the motor or propeller.
10. Wait until the propeller has completely stopped before removing guards, adjusting sensors, or changing components.
11. Save the test data, inspect the assembly for damage, and document any abnormal readings.

⚠ Safety

- Never touch or approach a rotating propeller, even at low throttle.
- Keep people outside the propeller plane and designated exclusion zone during testing.
- Never exceed the rated voltage, current, speed, or thrust of any component.
- Disconnect the battery or power source before changing wiring, mounting, or propeller components.
- Do not operate with a cracked propeller, loose motor mount, damaged wiring, or unprotected rotating parts.
- Stop the test immediately if severe vibration, unexpected movement, smoke, overheating, or sensor failure occurs.

Care & storage

- Remove power before cleaning or inspecting the stand.
- Inspect the load cell, mounts, fasteners, connectors, guards, and wiring before each test session.
- Keep the load cell free from side loading and impact beyond its rated limits.
- Check sensor calibration and data acquisition operation at the scheduled interval.
- Store the stand and test accessories in a dry, secure location and replace damaged components.

Common mistakes to avoid

- Testing with an unbalanced, damaged, or incorrectly installed propeller.
- Approaching the motor before the propeller has fully stopped.
- Ignoring load-cell zero errors or unreliable sensor readings.
- Increasing throttle beyond the ratings of the motor, ESC, propeller, or power source.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Propeller

Category: **Drones & Aviation** Type: **Handling note** Priority: **Medium** 3 labs

Covers these lab items: Propellers

What it is used for

Propellers generate thrust or airflow in compatible drones and aviation equipment and require careful inspection and handling.

How to handle

- Confirm the propeller model, size, rotation direction, and compatibility with the equipment.
- Keep power disconnected and remove batteries before inspection or replacement.
- Check blades and hubs for cracks, bends, loose fittings, or other damage; never use defective propellers.
- Fit and secure propellers according to the manufacturer's instructions and verify clearance before operation.

Storage

- Store in a protective case away from heat, moisture, and heavy objects.
- Keep blades supported to prevent bending or deformation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Propeller Balancer

Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Magnetic Propeller Balancer

What it is used for

To check whether a compatible drone or model-aircraft propeller is balanced before installation and operation.

How to handle

- Inspect the balancer base, shaft, magnetic supports, and propeller for damage or contamination.
- Remove the propeller from the aircraft and ensure the motor or power source cannot start accidentally.
- Mount the propeller on the balancer using the correct adapter and ensure it can rotate freely without rubbing.
- Observe which side settles downward and follow the propeller manufacturer's balancing method to make adjustments gradually.
- Recheck the balance and inspect the propeller for cracks or deformation; never use a damaged propeller.

Storage

- Remove the propeller and clean the balancer and its shaft carefully after use.
- Store in a protective case in a dry place and keep the shaft and magnetic supports free from impact and debris.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Telemetry Module

Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: 433 MHz Telemetry

What it is used for

A telemetry module transmits data between compatible equipment, such as a drone and its ground station. Use the approved frequency, configuration, and operating procedure for the training system.

How to use

1. Inspect the module, antenna, connectors, cables, and mounting for damage.
2. Confirm that the module is compatible with the controller and ground station.
3. Check the approved frequency, configuration, and local operating requirements with the instructor.
4. Power off equipment before connecting the module and antenna.
5. Secure the antenna and mount the module as specified by the manufacturer.
6. Power on the system and verify the link and data readings before operation.
7. Power down the system before disconnecting or storing the module.

⚠ Safety

- Never transmit without a suitable antenna connected when the equipment requires one.
- Keep antennas away from eyes and avoid touching exposed electrical contacts.
- Use radio equipment only on authorized frequencies and within applicable local rules.
- Do not change flight-control or telemetry settings without instructor approval.

Care & storage

- Keep connectors clean and dry and store the module in an anti-static protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

AI Device

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Different AI devices

What it is used for

An AI device uses artificial intelligence to process inputs and provide predictions, classifications, or other outputs. Operate it with approved software, suitable data, and appropriate supervision.

How to use

1. Identify the device's function and read its operating instructions.
2. Inspect the power supply, cables, sensors, and connectors for damage.
3. Connect the device to a compatible computer or approved network if required.
4. Open the authorized software and load a suitable sample dataset or input.
5. Run the activity using the provided settings and observe the output.
6. Check results for errors and compare them with expected or verified information.
7. Save required work and shut down the device using the correct procedure.

⚠ Safety

- Do not connect damaged cables or use an unsuitable power supply.
- Do not upload personal, confidential, or sensitive data without authorization.
- Treat AI outputs as suggestions and verify important results independently.
- Keep devices ventilated and away from liquids and excessive heat.

Care & storage

- Keep the device clean, update software through approved channels, and store accessories safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

AI Learning Kit

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Intel Brand or Equivalent Learning Kit

What it is used for

An AI learning kit helps students explore artificial intelligence through sensors, coding, models, and practical activities. Use compatible components and follow the kit's instructions during experiments.

How to use

1. Check the kit inventory and inspect boards, sensors, cables, and accessories.
2. Read the activity guide and identify the required components.
3. Disconnect power before assembling or changing circuit connections.
4. Connect components according to the approved diagram and check polarity where applicable.
5. Connect the kit to a compatible computer and open the approved software.
6. Run the sample program, observe the output, and troubleshoot using the guide.
7. Switch off the power, disconnect components, and return items to storage.

⚠ Safety

- Use the specified voltage and never short-circuit the power connections.
- Keep liquids away from electronic boards and exposed connectors.
- Handle circuit boards by their edges and follow anti-static precautions where required.
- Do not install unapproved software or share private data with online AI tools.

Care & storage

- Return components to labelled compartments and store the kit in a clean, dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Appliance Spare Parts

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Raw materials and parts of RO Water Purifier, Mixer/ Grinder/ Juicer, and Microwave Oven

What it is used for

Handle spare parts for RO purifiers, mixers, grinders, juicers, and microwave ovens safely.

How to handle

- Identify the appliance model and confirm part compatibility before replacement.
- Disconnect the appliance from power and water supplies before servicing.
- Allow hot components to cool and follow manufacturer instructions.
- Do not handle damaged electrical parts or microwave high-voltage components unless qualified.

Storage

- Keep parts labelled by appliance and model in protective packaging.
- Store dry and secure, away from moisture, dust, and electrostatic damage where relevant.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Breadboard

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Breadboard; Breadboards and Jumper Wires

What it is used for

To assemble and test temporary electronic circuits without soldering, using a breadboard and jumper wires.

How to handle

- Inspect the breadboard, jumper wires, and components for damage before use.
- Keep the circuit power supply switched off while inserting or changing components.
- Follow the circuit diagram and breadboard connection layout; check polarity and pin placement carefully.
- Use only suitable low-voltage supplies within component ratings and verify connections before powering the circuit.
- Switch off power before troubleshooting or dismantling the circuit.

Storage

- Remove power and disconnect loose jumper wires after use.
- Store the breadboard, wires, and components in labelled antistatic-safe containers where appropriate, away from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Buzzer

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Active Buzzer

What it is used for

Handle active buzzers safely in electronic circuits and practical exercises.

How to handle

- Check the buzzer's rated voltage, polarity, and pinout before connection.
- Switch off the circuit power before wiring or changing connections.
- Use current-limited supplies where appropriate and avoid short circuits.
- Do not expose the buzzer to water, excessive heat, or mechanical impact.

Storage

- Keep components in labelled anti-static or protective packaging where appropriate.
- Store in a dry component organiser away from dust and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Capacitor

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Capacitor; Capacitor Kit

What it is used for

Capacitors store electrical energy and are used in compatible electronic circuits and training kits.

How to handle

- Identify the capacitor type, capacitance, voltage rating, and polarity where applicable.
- Disconnect power and safely discharge capacitors using an approved method before handling circuit connections.
- Verify zero voltage with a suitable meter before touching terminals; never short terminals with a tool.
- Inspect the casing and leads for swelling, leakage, cracks, or corrosion.
- Use ESD precautions for sensitive components and follow the circuit diagram.

Storage

- Store components in labelled antistatic packaging where appropriate.
- Keep dry and separate damaged or suspect capacitors from usable components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Circuit Breaker

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Miniature Circuit Breaker

What it is used for

A circuit breaker automatically interrupts electrical current when an overload or fault occurs. Use a correctly rated breaker and follow the approved installation and reset procedure.

How to use

1. Identify the circuit and confirm the breaker's current, voltage, and pole ratings.
2. Inspect the breaker and enclosure for visible damage before installation or operation.
3. Ensure installation or replacement is performed by a qualified electrician with the supply isolated.
4. Verify that the wiring connections and protective enclosure meet the approved design.
5. Close the enclosure and energise the circuit according to the approved procedure.
6. Check operation and confirm that the connected circuit functions normally.
7. If the breaker trips, identify and correct the cause before resetting it.

⚠ Safety

- Never install, remove, or work on an energised breaker.
- Do not repeatedly reset a breaker without identifying the fault.
- Never bypass a breaker or replace it with an incorrect rating.
- Use qualified personnel and follow lockout and electrical safety procedures.

Care & storage

- Keep the enclosure clean, dry, and accessible for inspection.
- Arrange inspection and replacement if the breaker is damaged, overheats, or trips repeatedly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Conduit Pipe

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: PVC Pipes (For conduits)

What it is used for

Conduit pipes protect and route electrical cables through buildings, workshops, and installations.

How to handle

- Confirm the conduit material, size, and rating are suitable for the installation and environment.
- Inspect pipes for cracks, deformation, sharp edges, or damaged ends before use.
- Measure and cut pipes with appropriate tools and remove burrs before inserting cables.
- Keep conduits and cables away from incompatible chemicals, excessive heat, and mechanical damage.
- Electrical installation work must be carried out by authorised personnel with the supply isolated where required.

Storage

- Store pipes on a level, supported surface to prevent bending or rolling.
- Keep fittings organised and protect pipes from sunlight, impact, and contamination as required by the material.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

DC Motor

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

2 labs

Covers these lab items: DC Motor; Small DC Motors and Servo Motors

What it is used for

To convert direct-current electrical energy into mechanical rotation for learning and circuit experiments.

How to handle

- Inspect the motor, shaft, terminals, mounting, and connecting wires for damage.
- Check the rated voltage, current, and polarity from the motor label or datasheet before connecting power.
- Secure the motor and keep fingers, hair, clothing, and loose wires away from the rotating shaft and moving parts.
- Use a suitable current-limited DC supply or driver; switch off power before changing wiring or mechanical attachments.
- Stop immediately if the motor overheats, vibrates excessively, makes unusual noise, or emits a burning smell.

Storage

- Disconnect the power supply and remove attached loads before storing.
- Keep the motor and accessories in a dry, dust-free compartment; protect terminals and the shaft from impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

DIN Rail

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Mounting Rails

What it is used for

Handle DIN rails safely for mounting electrical and control components.

How to handle

- Confirm rail dimensions and compatibility with the enclosure and components.
- Wear suitable gloves when cutting or handling rails with sharp edges.
- Deburr cut edges and secure the rail with appropriate fasteners.
- Isolate electrical power before mounting or servicing components on an installed rail.

Storage

- Store rails horizontally on a stable support to prevent bending.
- Keep dry and protect edges from impact or corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dimmable Lighting System

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Lighting system that allows for dimming

What it is used for

A dimmable lighting system adjusts light output to suit visibility and activity requirements. Use compatible lamps and controls, and check the manufacturer's operating instructions.

How to use

1. Switch off the power before inspecting wiring, lamps, or control connections.
2. Confirm that the lamp, dimmer, and power supply are compatible.
3. Inspect the switch, controller, cables, and lamp fittings for visible damage.
4. Restore power and operate the dimmer using its designated control.
5. Adjust brightness gradually and check for flickering, noise, or unusual heating.
6. Switch the system off after testing or when the activity is complete.
7. Report abnormal operation and isolate the system before authorized inspection.

⚠ Safety

- Electrical installation and wiring changes must be handled by qualified personnel.
- Never touch exposed wiring or open the controller while power is connected.
- Do not exceed the rated load or use incompatible lamps.
- Keep lamps and controls away from water and flammable materials.

Care & storage

- Clean the outer surfaces with power off and replace faulty parts with compatible components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Display Module

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: 7 Segment Display; LCD Display

What it is used for

To display numbers, letters, symbols, or measured values using a seven-segment or LCD display module in electronic circuits.

How to handle

- Inspect the display module, pins, connector, and wiring for damage before use.
- Identify the module pinout, operating voltage, interface, and polarity from its datasheet.
- Switch off power before connecting or removing wires; connect the module to a compatible controller or circuit.
- Use the required current-limiting resistors or driver circuitry and avoid exceeding voltage and current ratings.
- Power the circuit and verify the displayed output; disconnect power before troubleshooting.

Storage

- Disconnect all power and remove the module carefully from the circuit if required.
- Store in an antistatic-safe container in a dry place, protecting pins and the display surface.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Distribution Board

Category: **Electrical & Electronic Components & Kits** Type: **Full SOP** Priority: **High**

1 labs

Covers these lab items: Distribution Box (MCB, RCCB, Isolator)

What it is used for

A distribution board distributes electrical power to outgoing circuits and provides switching and protective devices such as MCBs, RCCBs, and isolators. Correct inspection, circuit identification, and protection testing help maintain safe distribution and reduce electric-shock, overload, and fire risks.

Main parts

- Enclosure and protective cover
- Main isolator or incoming switch
- Miniature circuit breakers (MCBs)
- Residual current circuit breaker (RCCB), where fitted
- Neutral and earth bars
- Busbars, terminals, labels, and cable entries

Before you start

1. Only a qualified and authorised electrician should inspect, install, or test the distribution board.
2. Identify the supply source, nominal voltage, circuit schedule, and board rating from the approved drawings and labels.
3. Check the enclosure, cover, circuit labels, cable entries, and visible wiring for damage, moisture, overheating, or corrosion without touching exposed conductors.
4. Confirm that the work area is dry, adequately lit, and clear of obstructions.
5. For internal inspection or maintenance, isolate all relevant power sources, apply lockout/tagout where applicable, and verify absence of voltage with an appropriate tested instrument.
6. Use the required electrical PPE, insulated tools, and test equipment rated for the installation.

How to use

1. Review the approved circuit schedule and identify the incoming supply and outgoing circuits before any work.
2. Confirm that the distribution board is suitable for the installation and that protective devices match the approved design.
3. With power safely isolated and absence of voltage verified, inspect terminals, busbars, neutral and earth connections for signs of looseness, overheating, or damage.
4. Check that cable sizes, terminations, protective-device ratings, and circuit labels agree with the approved design.
5. Verify that neutral and protective earth conductors are connected to the correct designated bars and are not interchanged.

6. Check enclosure bonding, cable glands, cover fit, and barriers to prevent accidental contact with live parts.
7. Carry out continuity, insulation resistance, polarity, and earthing checks as required by the approved test procedure and applicable electrical rules.
8. Test MCB and RCCB operation using the appropriate test method and instrument; use the RCCB test button only as an additional routine check where specified.
9. Reinstall all covers and barriers and confirm that no tools or loose materials remain inside the enclosure.
10. After inspection is complete and the area is clear, restore supply using the authorised switching procedure.
11. Verify circuit operation and check for abnormal tripping, heating, smell, or noise without opening an energised enclosure.
12. Record test results, identified defects, corrective actions, and any circuits that remain isolated.

⚠ Safety

- Never touch exposed conductors or work inside an energised distribution board unless a formally authorised procedure specifically permits the task and suitable controls are in place.
- Isolate all relevant sources, including backup supplies or generators, and verify absence of voltage before internal work.
- Do not rely solely on an isolator position or indicator lamp to confirm that a circuit is dead.
- Never bypass, oversize, or replace protective devices with incorrect ratings.
- Keep the enclosure closed and secured against access by unauthorised persons.
- Do not allow water, metal objects, or loose debris to enter the board.
- Follow applicable electrical codes, lockout/tagout procedures, and site emergency arrangements.

Care & storage

- Keep the enclosure dry, clean, closed, and clearly labelled.
- Arrange periodic inspection and testing by qualified personnel according to applicable requirements.
- Investigate repeated breaker trips, RCCB operation, burning smells, discoloration, or abnormal heating promptly.
- Maintain an up-to-date circuit schedule and records of test results and modifications.
- Replace damaged breakers, terminals, covers, or wiring only with compatible approved components and after safe isolation.

Common mistakes to avoid

- Working on the board without isolating and verifying absence of voltage.
- Using protective devices with incorrect ratings or bypassing them to stop tripping.
- Mixing neutral and earth conductors or leaving terminals loose.
- Restoring power before covers, barriers, and circuit labels are properly reinstated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the

manufacturer's original manual / file.

Earthing Material

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Medium** 13 labs

Covers these lab items: Earthing Kit (Copper Rod, Plate); Earthing Pipe; Earthing Plate; Earthing rod; Earthing Wire

What it is used for

Earthing materials provide an electrical path to ground to help reduce electric shock and equipment fault risks.

How to handle

- Verify the required material, dimensions, condition, and compatibility with the approved earthing design.
- Inspect rods, plates, pipes, and wires for damage, corrosion, or defective connections.
- Use suitable tools and PPE, and ensure electrical isolation before working on connected systems.
- Install and test the earthing system according to the approved design and qualified personnel's instructions.

Storage

- Keep materials dry, protected from corrosion, and separated from damaging mechanical loads.
- Store connectors and small components in labelled containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electric Bell

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Cell Bell

What it is used for

An electric bell converts electrical energy into an audible signal for alerts or notifications. Install and test it using the specified voltage and approved wiring diagram.

How to use

1. Check the bell, power supply rating, terminals, and mounting hardware for damage.
2. Switch off the electrical supply before making connections.
3. Mount the bell securely at the designated location.
4. Connect the bell, switch, and supply according to the approved wiring diagram.
5. Check terminal tightness and ensure no exposed conductor remains accessible.
6. Restore power and press the switch to test the bell.
7. Switch off the supply if the bell fails or operates abnormally, then report the fault.

⚠ Safety

- Electrical wiring must be completed by a qualified person when required.
- Never touch exposed conductors or terminals while power is on.
- Use only the rated voltage and a suitable protected supply.
- Keep the bell and connections away from water unless rated for that environment.

Care & storage

- Disconnect power before cleaning and inspect mounting and wiring periodically.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electric Motor

Category: **Electrical & Electronic Components & Kits** Type: **Full SOP** Priority: **High**

8 labs

Covers these lab items: AC and DC motors; BLDC Motors; MOTER ELECTRIC 1 PHASE; MOTER ELECTRIC 1 PHASE ISI Mark; Motor

What it is used for

An electric motor converts electrical energy into mechanical rotation or motion. Correct electrical connections, suitable load, and proper cooling help prevent electric shock, overheating, and mechanical damage.

Main parts

- Stator provides the stationary magnetic field or winding assembly.
- Rotor rotates inside the motor to produce mechanical output.
- Shaft transfers rotation to the driven equipment.
- Bearings support the shaft and reduce friction.
- Motor housing protects internal components and supports heat dissipation.
- Terminal box or connectors provide electrical connection points.
- Cooling fan or vents remove heat on applicable models.

Before you start

1. Read the motor nameplate and manual to confirm voltage, phase, speed, and duty rating.
2. Allow only trained and authorised personnel to make electrical connections.
3. Isolate the power supply and verify it is safe before inspection or wiring.
4. Inspect the housing, shaft, mounting bolts, terminal box, cables, and cooling vents.
5. Check that the motor is firmly mounted and the driven equipment is correctly aligned.
6. Confirm guards are fitted over couplings, belts, fans, and other moving parts.

How to use

1. Confirm the motor and connected equipment are suitable for the intended task.
2. Check the nameplate and approved wiring diagram before connecting supply terminals.
3. Ensure earthing, protective devices, and cable ratings are appropriate for the installation.
4. Secure all permitted electrical connections and close the terminal box before energising.
5. Clear tools and loose objects from the motor and surrounding area.
6. Switch on the supply using the approved control and observe the starting behaviour.
7. Check rotation direction, vibration, noise, and temperature within the manufacturer's limits.
8. Stop the motor immediately if it overheats, vibrates excessively, or produces unusual noise or smell.
9. Isolate power and wait for moving parts to stop before adjustment or maintenance.
10. After use, switch off the supply and record any faults for authorised personnel.

⚠ Safety

- Never touch live terminals or internal motor parts while the supply is connected.
- Do not bypass earthing, guards, overload protection, or other safety devices.
- Keep hands, hair, clothing, and tools away from rotating shafts and couplings.
- Never connect a motor to a supply that does not match its nameplate requirements.
- Allow hot motors to cool before touching the housing or nearby components.
- Use appropriate lockout and isolation procedures before maintenance.

Care & storage

- Keep cooling vents and the motor exterior free from dust and obstruction.
- Inspect mounting bolts, couplings, cables, and guards regularly for wear or looseness.
- Follow the manual for lubrication, bearing inspection, and preventive maintenance.
- Arrange authorised inspection for overheating, insulation damage, unusual vibration, or repeated tripping.

Common mistakes to avoid

- Connecting the motor without checking its nameplate and wiring diagram.
- Running the motor with missing guards or loose mounting bolts.
- Ignoring incorrect rotation, excessive vibration, overheating, or unusual smells.
- Opening the terminal box before isolating and verifying the power supply is safe.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrical Connector

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low**

6 labs

Covers these lab items: Battery clips with DC Jack; Battery Holder/Connector; Cable Connector; Connectors

What it is used for

Electrical connectors join wires, batteries, and compatible devices to provide reliable electrical contact.

How to handle

- Identify connector type, polarity, current rating, and compatibility before connection.
- Disconnect the power source before installing or removing connectors.
- Inspect contacts, insulation, locking tabs, and terminals for damage or corrosion.
- Ensure connectors are fully seated and polarity is correct before energising the circuit.
- Do not force mismatched connectors or touch exposed live contacts.

Storage

- Store connectors in labelled protective packaging in a dry location.
- Protect terminals from dust, moisture, short circuits, and physical damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrical Socket

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Sockets (Assorted) 2 Pin; Sockets (Assorted) 3 Pin

What it is used for

Electrical sockets provide connection points for compatible plugs and electrical equipment.

How to handle

- Confirm the socket type, voltage, current rating, and earthing requirements before installation or use.
- Inspect the faceplate, contacts, and casing for cracks, looseness, or signs of overheating.
- Isolate the circuit before installation, repair, or internal inspection; work must be done by a qualified person.
- Keep sockets dry and never insert foreign objects into openings.
- Stop use and report sparking, burning smells, or unusual heat immediately.

Storage

- Store spare sockets in labelled protective packaging in a dry place.
- Keep them away from moisture, dust, and physical impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrical Switch

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low**

13 labs

Covers these lab items: Bell Push Switch; Pull-Push Switch; Rotary Switch; Switches; Switches (Assorted) One way; Switches (Assorted) Two way

What it is used for

Electrical switches control power to compatible circuits and must be selected and installed according to their ratings.

How to handle

- Verify the switch type, pole configuration, current and voltage ratings, and intended circuit.
- Isolate the supply and verify absence of voltage before installation or inspection.
- Inspect terminals and insulation for damage and ensure connections are made by qualified personnel.
- Test switch operation and circuit protection according to the approved electrical procedure.

Storage

- Store switches in original packaging in a dry, clean area.
- Protect terminals from moisture, dust, impact, and electrostatic damage where applicable.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrical Trainer Kit

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Kirchhoff's Law kit

What it is used for

To demonstrate and verify Kirchhoff's current and voltage laws using an electrical trainer kit. To make safe circuit connections and compare measured values with calculated results.

Main parts

- Trainer board or circuit panel
- DC power supply
- Resistors or load components
- Connecting leads
- Ammeter or multimeter
- Voltmeter or multimeter
- Switches and terminals

Before you start

1. Read the trainer manual and identify the circuit diagram.
2. Switch off the supply before making or changing connections.
3. Inspect leads, terminals, meters, and components for damage.
4. Confirm the supply voltage and component ratings.
5. Select the correct meter function, range, and polarity.

How to use

1. Build the specified circuit with the power supply switched off.
2. Check all connections against the circuit diagram.
3. Set the meters to the required current or voltage measurement mode.
4. Switch on the supply and increase it only to the specified level.
5. Measure branch currents and voltage drops at the designated points.
6. Record readings and compare current sums and voltage sums with Kirchhoff's laws.
7. Switch off the supply before changing the circuit or removing leads.

⚠ Safety

- Never connect an ammeter directly across a voltage source.
- Do not exceed component or meter ratings.
- Avoid touching exposed conductive parts when energized.
- Report overheating, smoke, or abnormal readings and switch off the supply.

Care & storage

- Switch off and disconnect the supply after use.
- Return leads and components to their labelled locations.
- Inspect meters and leads for damage after the exercise.
- Keep the trainer board clean and dry.

Common mistakes to avoid

- Changing wiring while the supply is on.
- Using the wrong meter terminal or range.
- Reversing polarity where it matters.
- Recording readings without checking connections and instrument settings.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrical Wire

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 17 labs

Covers these lab items: 3 Core Flexible Cable (For Drilling , Grinding , Welding M/c conectine); Cable; Cables; Connecting wires; Electric Wires; Hook Up Wire; Wire

What it is used for

Electrical wires and cables carry power or signals between compatible equipment and circuits.

How to handle

- Verify wire type, conductor size, insulation rating, voltage rating, and intended application.
- Disconnect power before installation, modification, or inspection of connected wiring.
- Inspect insulation and connectors for cuts, exposed conductors, corrosion, or other damage.
- Use suitable tools and qualified personnel for electrical connections and testing.

Storage

- Keep cables coiled without sharp bends and protect them from moisture and crushing.
- Separate wire types and label reels or bundles for identification.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electronic Component

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 10 labs

Covers these lab items: Components / Dividers; Electrical and electronic components; Electronic Components – Active and Passive; ICs; LED, Resistors, Capacitors (Basic Electronic Components kits); LEDs Multi Color; Light Emission Diode

What it is used for

Electronic components such as ICs, LEDs, resistors, and capacitors are used to assemble, test, and repair circuits.

How to handle

- Identify each component and verify its rating, polarity, package type, and circuit compatibility.
- Disconnect power before inserting, removing, or testing components in a circuit.
- Use ESD precautions where required and avoid touching sensitive leads or contacts unnecessarily.
- Use suitable tools and follow component datasheets to prevent incorrect installation or damage.

Storage

- Store components in labelled anti-static or suitable protective packaging as appropriate.
- Keep leads protected and separate components by type, value, and rating.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Extension Board

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **High**

4 labs

Covers these lab items: Electric Extension Board; Extension Cord; Extension Cords

What it is used for

An extension board distributes power from one outlet to multiple compatible electrical devices. Use it within its rated capacity and inspect it before connecting equipment.

How to use

1. Check the extension board, plug, cable, and sockets for cracks or damage.
2. Confirm the voltage, current rating, and total load requirements of connected devices.
3. Place the board on a dry, stable surface with good ventilation.
4. Switch off the supply before connecting or removing plugs where appropriate.
5. Insert plugs fully and keep cables arranged to avoid tripping or crushing.
6. Switch on the supply and check for normal operation without overloading the board.
7. Switch off and unplug the board before moving or storing it.

⚠ Safety

- Never exceed the board's rated current or power capacity.
- Do not use the board in wet areas or with wet hands.
- Do not connect multiple extension boards in a chain.
- Stop using the board if it becomes hot, sparks, smells burnt, or has loose sockets.

Care & storage

- Keep the board dry and clean, and avoid pulling it by the cable.
- Inspect the cable and sockets regularly and replace damaged boards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fan

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Ceiling Fan / Exhaust Fan; Fan

What it is used for

To operate a ceiling fan or exhaust fan for air circulation or ventilation. To maintain safe operation by checking mounting, electrical connections, and airflow before use.

Main parts

- Motor housing
- Blades or impeller
- Mounting bracket or frame
- Switch or speed controller
- Protective grille, if fitted
- Electrical wiring and terminal cover

Before you start

1. Confirm the fan type, voltage rating, and installation requirements.
2. Inspect the housing, blades, mounting points, and grille for damage.
3. Ensure the fan is securely mounted and the blades have adequate clearance.
4. Check that wiring and earthing are installed by a qualified person.
5. Isolate power before cleaning or inspecting internal electrical parts.

How to use

1. Ensure the area around the fan is clear.
2. Switch on the fan using the designated control.
3. Select an appropriate speed or operating mode.
4. Observe the fan for unusual noise, vibration, or reduced airflow.
5. Switch off immediately if abnormal operation occurs.
6. Isolate power before cleaning or performing maintenance.
7. Restore power only after covers and guards are securely fitted.

⚠ Safety

- Never touch moving blades or reach through a grille.
- Do not operate a fan with loose mounting, damaged blades, or exposed wiring.
- Electrical installation and repairs must be performed by qualified personnel.
- Keep water away from electrical components unless the equipment is specifically rated for that environment.

Care & storage

- Clean blades and grille after isolating power.
- Check mounting hardware and vibration periodically.
- Keep ventilation openings free of dust and obstructions.
- Report electrical faults and replace damaged parts before reuse.

Common mistakes to avoid

- Cleaning the fan while power is connected.
- Operating with loose mounting or damaged blades.
- Ignoring unusual noise or vibration.
- Blocking the exhaust path or air intake.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fuse

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Different Type of Fuse

What it is used for

Handle electrical fuses safely during inspection and replacement.

How to handle

- Switch off and isolate the power supply before handling a fuse.
- Use only the correct fuse type and rated current specified for the equipment.
- Never bypass a fuse or replace it with wire or an incorrect rating.
- Have qualified personnel investigate repeated fuse failures.

Storage

- Keep fuses labelled and separated by type and rating.
- Store in dry packaging away from moisture and metal debris.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

HVAC Trainer Kit

Category: **Electrical & Electronic Components & Kits** Type: **Full SOP** Priority: **High**

1 labs

Covers these lab items: HVAC Demonstration Kit

What it is used for

To demonstrate the basic components and operating principles of heating, ventilation, and air conditioning systems using an HVAC trainer kit. To perform supervised measurements and practice safe troubleshooting on the training setup.

Main parts

- Training panel and frame
- Refrigeration compressor and circuit, if fitted
- Condenser and evaporator sections
- Fans, ducts, and air filters
- Thermostat, sensors, and control switches
- Pressure gauges, temperature meters, and electrical terminals

Before you start

1. Read the trainer manual, circuit diagram, and laboratory instructions.
2. Inspect the power cord, plug, earthing, terminals, guards, and emergency stop if fitted.
3. Check that refrigeration pipes, hoses, and fittings show no visible damage or leakage.
4. Ensure the trainer is placed in a ventilated area with sufficient clearance around air inlets and outlets.
5. Confirm that the refrigerant circuit is sealed and that only authorized personnel service it.
6. Prepare suitable measuring instruments and check their condition before use.

How to use

1. Keep the trainer switched off while checking connections and components.
2. Identify the compressor, condenser, evaporator, fans, sensors, and control devices using the diagram.
3. Verify that electrical connections and protective earthing are correct before energizing.
4. Set the trainer controls to the demonstration settings specified by the instructor or manual.
5. Switch on the main supply and start the system using the approved procedure.
6. Observe the compressor, fans, airflow, indicators, and temperature readings during operation.
7. Allow the system to stabilize before recording inlet and outlet temperatures or other specified readings.
8. Measure voltage, current, pressure, or airflow only with suitable instruments and approved test points.

9. Compare readings with the expected operating behavior and discuss any differences with the instructor.
10. For troubleshooting exercises, switch off and isolate power before inspecting electrical components.
11. Do not adjust refrigerant charge or open the sealed refrigeration circuit during routine training.
12. Switch off the trainer, isolate the supply, and allow moving or hot components to stop and cool.
13. Record results, return controls to their safe positions, and tidy the training area.

⚠ Safety

- Only trained learners under instructor supervision should operate the trainer.
- Use correct earthing and never touch exposed live terminals.
- Isolate power before changing wiring or handling electrical components.
- Fans and compressor components may start automatically; keep hands and tools clear.
- Do not loosen refrigerant fittings or release refrigerant; servicing must be done by authorized personnel.
- Avoid contact with hot pipes and cold refrigerant lines, which may cause burns or frost injury.
- Stop immediately if there is smoke, an electrical smell, unusual vibration, or suspected refrigerant leakage.

Care & storage

- Switch off and isolate the trainer after each session.
- Clean filters, vents, and external surfaces as instructed.
- Inspect cables, terminals, hoses, and guards for wear or damage.
- Keep instruments and accessories in their designated storage places.
- Arrange scheduled maintenance and calibration for gauges and sensors.

Common mistakes to avoid

- Starting the trainer before checking earthing and connections.
- Blocking air inlets or outlets during operation.
- Touching live terminals or moving fan parts during measurements.
- Using unsuitable meters or incorrect measurement points.
- Opening the refrigerant circuit or changing charge without authorization.
- Ignoring unusual readings, noises, or signs of leakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Insulation Tape

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 9 labs

Covers these lab items: Electrical tape; Insulation Tape

What it is used for

Insulation tape covers and insulates compatible electrical joints and helps identify wires where permitted.

How to handle

- Verify the tape's voltage rating, temperature rating, and suitability for the application.
- Isolate electrical power before working on wiring or connections.
- Inspect tape for damage, contamination, or loss of adhesion.
- Apply evenly to clean, dry surfaces according to the manufacturer's instructions.
- Do not use tape as a substitute for proper connectors or damaged cable replacement.

Storage

- Keep tape rolls sealed or covered in a cool, dry place.
- Protect adhesive surfaces from dust, heat, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

IoT Project Kit

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Mini Project Kits

What it is used for

An IoT project kit combines sensors, controllers, and communication modules for learning and prototyping connected devices. Assemble and test the project according to the supplied circuit diagram and specifications.

How to use

1. Check the kit inventory and inspect boards, sensors, wires, and power components.
2. Read the circuit diagram and identify each module and its connection points.
3. Switch off power before assembling or changing circuit connections.
4. Connect sensors, controller, and communication modules according to the project diagram.
5. Install the required software or code using approved tools and libraries.
6. Apply the specified power and test sensor readings and communication functions.
7. Disconnect power after testing and organize all kit components.

⚠ Safety

- Use only the specified voltage and correct polarity.
- Avoid short circuits and never change wiring while power is applied.
- Handle electronic boards using suitable electrostatic discharge precautions.
- Do not connect the kit to mains voltage unless the design specifically requires it and a qualified supervisor authorizes it.

Care & storage

- Store modules in labelled compartments and protect boards from moisture, dust, and static discharge.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Jumper Wire

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Jumper Cable; Receiver cables

What it is used for

Handle jumper and receiver wires without damaging connectors or equipment.

How to handle

- Confirm wire type, connector, and polarity before connection.
- Switch off equipment before connecting or disconnecting wires where required.
- Insert and remove connectors by their housings, not by pulling the cable.
- Inspect insulation and connectors for damage before use.

Storage

- Coil wires loosely without sharp bends or tight knots.
- Store labelled wires in a dry organiser away from heat and metal edges.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

LED

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: RGB LEDs

What it is used for

Handle RGB LEDs safely for electronic circuits and colour-display projects.

How to handle

- Check LED polarity, pin configuration, and rated current before connection.
- Use appropriate current-limiting resistors or a suitable driver circuit.
- Switch off power before changing wiring and avoid short circuits.
- Avoid touching exposed leads unnecessarily and protect components from static discharge.

Storage

- Keep LEDs in labelled anti-static packaging or component organisers.
- Store in a dry place away from heat, moisture, and mechanical damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

LED Light

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: LED Tube Lights (Industrial Area)

What it is used for

Handle LED tube lights safely during inspection, installation, and replacement.

How to handle

- Switch off and isolate the electrical supply before installation or removal.
- Verify tube type, voltage rating, and fixture compatibility before fitting.
- Use a stable ladder or approved access equipment and suitable PPE.
- Do not install cracked tubes or touch exposed electrical contacts.

Storage

- Keep replacement tubes in their protective packaging until needed.
- Store horizontally or as directed by the manufacturer in a dry, secure area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lamp Holder

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Lamp Holders (Assorted) Angle; Lamp Holders (Assorted) Bettle

What it is used for

Lamp holders support lamps and provide electrical contact in compatible lighting fixtures.

How to handle

- Verify holder type, lamp base compatibility, voltage rating, and temperature rating.
- Switch off and isolate electrical power before installation, removal, or inspection.
- Inspect the holder body, contacts, and terminals for cracks, corrosion, or heat damage.
- Install only according to the approved wiring diagram and have electrical work performed by a qualified person.
- Never touch live contacts or use a damaged holder.

Storage

- Store holders in labelled protective packaging in a dry location.
- Keep contacts protected from dust, moisture, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Microcontroller Board

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

15 labs

Covers these lab items: Ardino Uno Micro Controller; Arduino; Arduino Boards (UNO/Nano); Arduino UNO; Arduino Yún; ESP32; LightBlue Bean Punch Through; Microcontroller; Microduino; Microprocessor boards; OpenPicus; Pinoccio; Processors, controllers, sensors and communication hardware; SODAQ

What it is used for

Use the microcontroller board to read inputs, process data, and control electronic devices through programmed instructions.

How to use

1. Inspect the board, USB port, pins, and components for visible damage.
2. Install the correct development software and select the matching board model.
3. Connect the board to the computer using a suitable USB cable.
4. Write or open a compatible program and verify the selected port.
5. Upload the program and observe the board indicators and connected device response.
6. Disconnect power before changing wiring or attaching external components.

⚠ Safety

- Use the correct voltage and polarity for the board and connected components.
- Avoid short circuits between power, ground, and signal pins.
- Disconnect power before changing circuit connections.
- Use suitable drivers and trusted code when connecting external devices.

Care & storage

- Keep the board dry, clean, and protected from static electricity.
- Store it in an antistatic bag or suitable case.
- Avoid bending pins or applying force to connectors.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Motor Driver Module

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: L298P Motor Driver

What it is used for

An L298P motor driver module controls suitable DC motors or compatible loads using low-power control signals. Check the module specifications and connect the motor, supply, and controller according to its datasheet.

How to use

1. Inspect the module, terminals, connectors, and board for visible damage.
2. Read the datasheet to confirm operating voltage, current limits, and supported motor type.
3. Switch off all power before making connections.
4. Connect the motor, motor supply, logic supply, and controller signals according to the wiring diagram.
5. Check polarity, common ground requirements, and secure terminal connections.
6. Apply the specified power and test motor direction and speed using suitable control signals.
7. Switch off power before changing wiring and record any abnormal operation.

⚠ Safety

- Never exceed the module's voltage or current ratings.
- Avoid short circuits and keep exposed conductors separated.
- Use appropriate protection against motor current, heat, and inductive voltage spikes.
- Do not touch hot components while the module is operating.

Care & storage

- Disconnect power after testing and store the module in a dry, static-safe location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Motor Starter

Category: **Electrical & Electronic Components & Kits** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: DOL STARTER

What it is used for

A direct-on-line (DOL) motor starter connects a suitable motor directly to the supply through switching and overload-protection components. Use it only with correctly rated equipment, proper wiring, and approved electrical isolation procedures.

Main parts

- Contactor and main switching contacts
- Thermal or electronic overload relay
- Start and stop push-buttons or control switch
- Enclosure and terminal blocks
- Control circuit and auxiliary contacts
- Supply terminals and motor output terminals

Before you start

1. Confirm the starter rating matches the motor, supply voltage, and application.
2. Inspect the enclosure, contactor, overload relay, terminals, and wiring for damage.
3. Isolate the electrical supply and apply lockout/tagout before opening the enclosure.
4. Verify absence of voltage using an appropriate tester and safe procedure.
5. Check motor earthing, cable sizing, overload settings, and protective-device coordination.
6. Confirm the motor and driven machine can start safely and the work area is clear.

How to use

1. Keep the supply isolated while inspecting or connecting the starter.
2. Check that all power and control terminals are correctly identified.
3. Connect the incoming supply, motor leads, and control circuit according to the approved wiring diagram.
4. Ensure protective earthing and required covers are correctly fitted.
5. Set the overload relay according to the motor nameplate and approved design.
6. Check that the stop control and emergency stop, where fitted, operate correctly.
7. Close the enclosure and remove tools before restoring power.
8. Restore supply only after authorised checks confirm the installation is safe.
9. Press the start control and observe the motor starting and running behaviour.
10. Verify motor rotation, current, noise, and vibration where applicable.
11. Use the stop control to stop the motor and confirm that it de-energises.
12. If an overload trip or abnormal condition occurs, isolate power before investigating.

3. Record test results, settings, and any defects according to site procedure.

⚠ Safety

- Installation, wiring, and maintenance must be performed by qualified electrical personnel.
- Never open or work on an energised starter unless an approved procedure specifically permits it.
- Use lockout/tagout and verify absence of voltage before touching conductors.
- Do not bypass overload relays, interlocks, protective earthing, or short-circuit protection.
- Keep hands and tools clear of exposed terminals and moving machinery.
- Follow electrical ratings and local regulations; a DOL starter can produce high motor starting current.

Care & storage

- Inspect contactor contacts, terminals, enclosure, and wiring at scheduled intervals.
- Check overload relay settings and function according to the maintenance plan.
- Keep the enclosure clean, dry, and free from dust buildup.
- Tighten terminals only to the specified torque with power isolated.
- Replace worn or damaged components with correctly rated approved parts.

Common mistakes to avoid

- Using a starter with an unsuitable motor or supply rating.
- Setting the overload relay incorrectly or bypassing protection.
- Working on terminals without isolation and voltage verification.
- Restoring power with loose wiring, missing covers, or tools inside the enclosure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Multivibrator Kit

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Astable, Monostable, Bistable Multivibrator kit

What it is used for

A multivibrator kit demonstrates astable, monostable, and bistable circuit operation. Assemble the circuit using the supplied diagram and verify its output with suitable test equipment.

How to use

1. Check the kit components, ICs, resistors, capacitors, switches, and power connections.
2. Identify the astable, monostable, or bistable circuit from the supplied diagram.
3. Keep power disconnected while assembling or changing circuit connections.
4. Insert components in the correct positions and check polarity where applicable.
5. Connect the specified supply and output measurement equipment.
6. Apply power and observe the output waveform or state change for the selected mode.
7. Switch off power, disconnect test leads, and return components to their storage positions.

⚠ Safety

- Use only the specified supply voltage and correct polarity.
- Avoid short circuits and do not change wiring while powered.
- Handle ICs and electronic components with suitable electrostatic precautions.
- Use an oscilloscope or other equipment according to its safety instructions.

Care & storage

- Return components to labelled compartments and protect the kit from dust, moisture, and static discharge.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ohm's Law Kit

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Ohm’s Law kit

What it is used for

To demonstrate and verify the relationship between voltage, current, and resistance in an electrical circuit.

How to handle

- Inspect the trainer board, resistors, meters, leads, switches, and power supply for damage.
- Keep the supply switched off while building or modifying the circuit.
- Connect the circuit according to the experiment diagram and check resistor values and meter polarity.
- Apply only the specified low voltage, measure voltage and current using appropriate meter ranges, and record the readings.
- Switch off the supply before disconnecting the circuit and return all components to their designated positions.

Storage

- Store the kit, meters, leads, and resistors in a dry cabinet with components organised.
- Protect terminals and electronic parts from moisture, dust, static discharge, and short circuits.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Peltier Module

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Peltzier

What it is used for

A Peltier module transfers heat from one side to the other when supplied with direct current, enabling small-scale heating or cooling experiments. Use the specified voltage and provide adequate heat dissipation on the hot side.

How to use

1. Inspect the module's ceramic faces, leads, and mounting surfaces for cracks or damage.
2. Check the datasheet for rated voltage, current, polarity, and thermal limits.
3. Switch off the power supply before wiring the module.
4. Mount the module with suitable thermal contact and a heat sink on the hot side.
5. Connect the power supply with correct polarity and current capacity.
6. Apply power gradually and monitor temperature on both sides using suitable sensors.
7. Switch off power and allow the module to return to a safe temperature before handling.

⚠ Safety

- Do not exceed the module's rated voltage, current, or temperature limits.
- Provide adequate heat sinking to prevent overheating and damage.
- Avoid bending, crushing, or applying uneven pressure to the ceramic faces.
- Disconnect power before changing wiring or mounting.

Care & storage

- Keep contact surfaces clean, protect the leads, and store the module in a dry, cushioned container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Plug Top

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Plugs (Assorted) Top 2 Pin; Plugs (Assorted) Top 3 Pin

What it is used for

Plug tops connect compatible electrical cords to matching sockets and power supplies.

How to handle

- Verify plug type, current rating, voltage rating, and earthing requirements before use.
- Inspect the casing, pins, cord entry, and strain relief for damage.
- Disconnect power before assembly, inspection, or replacement; electrical work must be performed by a qualified person.
- Never use a plug with loose, bent, cracked, or overheated parts.
- Ensure the plug is fully inserted into a compatible socket and keep it dry.

Storage

- Store spare plug tops in labelled, dry packaging.
- Keep pins protected from bending, moisture, and physical damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Potentiometer

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Potentiometer

What it is used for

Handle potentiometers safely during electronic circuit assembly and testing.

How to handle

- Check resistance value, taper, power rating, and pin configuration before use.
- Switch off circuit power before wiring or adjusting connections.
- Avoid exceeding the rated voltage, current, or mechanical rotation limits.
- Use appropriate tools and protect component terminals from static discharge where applicable.

Storage

- Store in labelled anti-static packaging or a compartmented component box.
- Keep dry and away from dust, heat, and physical damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Power Adapter

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

2 labs

Covers these lab items: Power Adaptor; Power Supply Adapters (5V/12V DC)

What it is used for

To provide the required regulated DC voltage to a compatible electronic device or training circuit.

How to handle

- Inspect the adapter casing, cable, plug, and output connector for cracks, exposed wires, or other damage.
- Check that the input supply, output voltage, polarity, current rating, and connector match the device requirements.
- Connect the output plug to the device before energizing only if the manufacturer's instructions recommend that sequence.
- Keep the adapter ventilated and dry; do not overload it, cover it during operation, or handle it with wet hands.
- Disconnect power before changing wiring or troubleshooting and stop using the adapter if it overheats, sparks, or smells abnormal.

Storage

- Disconnect the adapter from the supply and device before storage.
- Coil the cable loosely without sharp bends and store in a dry, ventilated location away from heat and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Printed Circuit Board

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: PCB; Printed Circuit Board; Printed Circuit Board (PCB)

What it is used for

Printed circuit boards hold and electrically connect electronic components in a circuit.

How to handle

- Verify the PCB model, orientation, and component specifications before assembly or testing.
- Handle boards by the edges and use ESD protection where required.
- Inspect for cracked tracks, damaged pads, loose components, and solder bridges.
- Disconnect power before inserting, removing, or reworking the board.
- Use suitable tools and avoid bending the PCB or touching exposed conductors unnecessarily.

Storage

- Store boards in antistatic bags or approved ESD-safe packaging.
- Keep dry and protected from dust, impact, and electrostatic discharge.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

RF Module

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: RF Transmitter and receiver

What it is used for

An RF transmitter and receiver send and receive radio-frequency signals for compatible electronic communication experiments. Verify frequency, supply requirements, antenna connections, and applicable radio-use restrictions before testing.

How to use

1. Inspect the transmitter, receiver, antenna connections, power leads, and circuit board.
2. Read the datasheets to confirm frequency band, voltage, pinout, and compatible communication protocol.
3. Switch off power before making wiring connections.
4. Connect the modules to the controller or test circuit according to the wiring diagram.
5. Check antenna placement, grounding requirements, and supply polarity.
6. Power the modules using specified ratings and send a test signal.
7. Verify reception at the receiver and disconnect power after testing.

⚠ Safety

- Do not exceed the modules' voltage or current ratings.
- Use only permitted frequencies and comply with applicable radio regulations.
- Keep antennas and wiring away from hazardous electrical equipment.
- Avoid short circuits and disconnect power before changing connections.

Care & storage

- Store modules in antistatic packaging and protect antenna connectors from damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rectifier

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Rectifier

What it is used for

A rectifier converts alternating current into direct current for suitable electronic circuits or power applications. Identify the rectifier type and verify its electrical ratings before connecting it to a circuit.

How to use

1. Inspect the rectifier, terminals, heat sink, and package markings for damage.
2. Read the datasheet to confirm input limits, output ratings, polarity, and pin configuration.
3. Disconnect power before installing or wiring the rectifier.
4. Connect AC input and DC output terminals according to the circuit diagram.
5. Check polarity, insulation, and load suitability before energizing the circuit.
6. Apply the specified input and measure the DC output using a suitable meter.
7. Switch off and isolate the supply before inspecting or changing connections.

⚠ Safety

- Never exceed the rectifier's voltage, current, or temperature ratings.
- Avoid contact with exposed live terminals and charged capacitors.
- Use suitable fuses, insulation, and heat dissipation where required.
- Only trained personnel should work on hazardous mains-voltage circuits.

Care & storage

- Allow components to cool, inspect terminals, and store spare rectifiers in dry antistatic packaging.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Relay

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

5 labs

Covers these lab items: 4-point relay; 5-point relay; Relay; Relays and Switch Modules

What it is used for

A relay is an electrically operated switch used to control a circuit using a control signal. Identify its coil voltage, contact arrangement, and current rating before connecting it.

How to use

1. Inspect the relay body, pins, markings, and socket for damage.
2. Read the relay diagram to identify coil terminals and contact terminals.
3. Confirm the coil voltage and contact ratings match the circuit requirements.
4. Switch off and isolate the power supply before making connections.
5. Wire the relay according to the circuit diagram and secure all terminals.
6. Restore power under supervision and test the switching operation.

⚠ Safety

- Never exceed the relay's rated coil voltage or contact current.
- Disconnect power before wiring or changing relay connections.
- Use suitable insulation and protective equipment for the circuit voltage.
- Do not touch exposed live terminals during testing.

Care & storage

- Keep relay contacts and terminals clean and replace relays showing overheating or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Resistor

Category: **Electrical & Electronic Components & Kits** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Resistor Kit

What it is used for

Handle resistor kits safely for electronic circuit assembly and learning exercises.

How to handle

- Identify resistance values, tolerance, and power ratings before selecting components.
- Use colour codes or a suitable meter to verify values when necessary.
- Switch off circuit power before replacing or measuring components in-circuit.
- Avoid bending leads excessively and protect components from static discharge where applicable.

Storage

- Sort resistors by value and rating in labelled compartments.
- Keep components dry and protected from dust, heat, and physical damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sensor Module

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

27 labs

Covers these lab items: Basic Sensors; Devices such as humidity sensor, temperature sensor, gyro meter, accelerometer, video surveillance cameras; Distance and Gesture Sensor Module; Gas Sensor; Humidity and Temperature Sensor; Humidity Sensor; Inertia Measurement Unit; Inertial Sensors; IR Sensor; LDR Module; Pulse Rate H

What it is used for

A sensor module detects a physical condition such as temperature, humidity, motion, distance, or orientation and converts it into a usable signal. Connect and test the module according to its specifications and the circuit design.

How to use

1. Identify the sensor type, pin labels, operating voltage, and output specifications.
2. Inspect the module and connecting wires for damage or loose pins.
3. Switch off the power before making circuit connections.
4. Connect power, ground, and signal pins according to the datasheet or circuit diagram.
5. Apply the specified power supply and run the approved test program or measurement procedure.
6. Expose the sensor to a suitable test condition and observe the output.
7. Compare readings with expected behaviour and record the results.
8. Switch off power before changing connections and store the module safely.

⚠ Safety

- Never exceed the module's rated voltage or current.
- Check polarity and pinout before connecting power.
- Avoid touching exposed conductors while the circuit is energised.
- Use appropriate ESD precautions for sensitive electronic modules.

Care & storage

- Keep modules dry, clean, and protected from electrostatic discharge.
- Store sensors in suitable packaging and inspect connectors before reuse.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Servo Motor

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Servo Motor

What it is used for

A servo motor provides controlled angular or linear movement in compatible automation and robotics systems. Confirm the motor's electrical and mechanical ratings and configure its controller before operation.

How to use

1. Inspect the motor body, shaft, connector, mounting points, and cable for damage.
2. Check the datasheet for voltage, current, torque, travel limits, and control signal requirements.
3. Mount the motor securely and connect the load without misalignment or binding.
4. Connect the motor to a compatible controller and specified power supply.
5. Set the required control parameters and verify the movement range without load where appropriate.
6. Apply power and test movement at low speed or within a safe range.
7. Switch off power before adjusting the mounting, load, or wiring.

⚠ Safety

- Do not exceed rated voltage, current, torque, or travel limits.
- Keep fingers and loose clothing away from the rotating shaft and connected mechanism.
- Secure the load to prevent unexpected movement.
- Disconnect power before maintenance or changing connections.

Care & storage

- Keep the motor clean, inspect cables and mounting bolts, and follow the manufacturer's lubrication guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Single Board Computer

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Beagle Board; Raspberry Pi Kit

What it is used for

To run computing, programming, electronics, and embedded-system experiments on a compact single-board computer.

How to handle

- Inspect the board, ports, connectors, and power adapter for damage before use.
- Confirm the required operating system, compatible power supply, storage media, and peripherals.
- Disconnect power before attaching or removing hardware, and observe electrostatic-discharge precautions when handling the board.
- Install approved software or an operating system image, then connect peripherals and power on according to the manufacturer's instructions.
- Shut down the operating system properly before removing power and avoid shorting exposed pins or ports.

Storage

- Store in an antistatic bag or protective case in a dry, dust-free location.
- Keep the board, storage media, power adapter, and accessories organised and protected from static discharge and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Surge Protector

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Surge Protectors

What it is used for

A surge protector helps protect compatible electrical equipment from transient voltage surges. Check its rating and condition, then connect it according to the manufacturer's electrical safety instructions.

How to use

1. Inspect the surge protector, plug, cable, outlets, indicator lights, and casing for damage.
2. Confirm the voltage, current, and load ratings are suitable for the equipment.
3. Plug the protector directly into an appropriate grounded outlet as specified.
4. Connect compatible equipment without exceeding the rated load.
5. Check the status indicator and confirm the protector is operating normally.
6. Arrange cables neatly and keep the unit ventilated during operation.
7. Disconnect equipment safely and replace the protector if its protection status indicates failure.

⚠ Safety

- Do not use a cracked, wet, overheated, or damaged surge protector.
- Never exceed its rated load or daisy-chain multiple power strips.
- Keep the unit away from water and heat sources.
- Do not assume every surge protector provides backup power or protects against all electrical faults.

Care & storage

- Keep outlets free of dust and inspect cables and protection indicators regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Torch

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

4 labs

Covers these lab items: Flashlights; Torch; Torch Light

What it is used for

To provide portable illumination using a torch or flashlight. To improve visibility in dim areas while maintaining safe battery and electrical handling.

Main parts

- Housing and grip
- Lens or light window
- LED or bulb
- Power switch
- Battery or rechargeable cell
- Charging port, if fitted

Before you start

1. Check the torch body, lens, switch, and battery compartment for damage.
2. Confirm that batteries are correctly fitted and charged.
3. Ensure the lens is clean and the beam works properly.
4. Select a torch suitable for the environment and task.

How to use

1. Switch on the torch using its power control.
2. Aim the beam at the area that needs illumination.
3. Adjust brightness or beam focus if the model supports it.
4. Use the torch to inspect the area without touching hazardous components.
5. Switch it off when no longer needed.
6. Recharge or replace batteries according to the instructions.

⚠ Safety

- Do not shine a high-intensity beam directly into anyone's eyes.
- Keep the torch away from water unless it is rated for wet conditions.
- Use only compatible batteries and chargers.
- Do not use a torch with leaking batteries or damaged wiring.

Care & storage

- Wipe the lens and housing clean.

- Remove batteries for long-term storage if recommended.
- Recharge according to the manufacturer's instructions.
- Store in a dry, secure place.

Common mistakes to avoid

- Using the wrong battery type or polarity.
- Leaving the torch switched on until the battery drains.
- Using a damaged torch in a hazardous environment.
- Pointing the beam into someone's eyes.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Transformer

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **High** 1 labs

Covers these lab items: Small size/demonstration units of transformer

What it is used for

A transformer transfers electrical energy between circuits through electromagnetic induction and may change AC voltage levels. Use demonstration units only within their rated voltage, current, and power limits.

How to use

1. Inspect the transformer, terminals, insulation, core, and connecting leads for damage.
2. Read the nameplate to identify input voltage, output voltage, frequency, and rating.
3. Ensure the power supply is switched off before making connections.
4. Connect the primary and secondary circuits according to the demonstration diagram.
5. Check that the load and measuring instruments are suitable for the transformer rating.
6. Switch on the supply and measure the output using correctly rated instruments.
7. Record the readings and switch off the supply before changing connections.
8. Disconnect leads safely and store the transformer after the unit cools if necessary.

⚠ Safety

- Never touch exposed terminals while the transformer is energised.
- Do not exceed rated voltage, current, or power limits.
- Use qualified supervision and appropriate electrical protection during demonstrations.
- Disconnect power immediately if smoke, unusual noise, overheating, or burning smell occurs.

Care & storage

- Keep the transformer dry, clean, and free from obstructed ventilation.
- Inspect leads and insulation regularly and report damaged equipment before use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Level Indicator

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

2 labs

Covers these lab items: Water level indicator

What it is used for

A water level indicator detects or displays the level of water in a tank or container. Install and test it according to its circuit design and operating instructions.

How to use

1. Identify the indicator type and read its wiring or installation instructions.
2. Inspect the sensor, wires, terminals, and housing for damage.
3. Disconnect the power supply before making electrical connections.
4. Connect the sensor and indicator according to the approved circuit diagram.
5. Secure the sensor at the required level and keep connections properly insulated.
6. Restore power and test the indicator using a controlled water-level change.
7. Switch off the supply before adjusting, cleaning, or disconnecting the device.

⚠ Safety

- Never handle exposed electrical connections with wet hands.
- Keep non-waterproof electrical components away from water and splashes.
- Use the correct rated power supply and adult supervision for school projects.
- Disconnect power immediately if wires overheat or insulation becomes damaged.

Care & storage

- Clean the sensor as instructed and store the device with wires protected from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wi-Fi Module

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Wi-Fi Module (ESP8266/ESP32)

What it is used for

A Wi-Fi module such as ESP8266 or ESP32 enables compatible electronics projects to communicate wirelessly. Configure the module using its documentation and verify power, wiring, and network settings before testing.

How to use

1. Identify the module version and check its voltage, pinout, and interface requirements.
2. Inspect the module, development board, USB cable, and connecting wires for damage.
3. Connect the module to a compatible power supply and controller using the documented wiring.
4. Install the required development environment, drivers, and approved firmware tools.
5. Configure the wireless network credentials and application settings securely.
6. Upload or run a simple test program and verify connection status and data exchange.
7. Disconnect power before changing wiring and record the working configuration.

⚠ Safety

- Use the specified supply voltage and avoid short circuits between pins.
- Do not connect GPIO pins directly to incompatible voltage levels.
- Never expose the module to liquids or conductive debris.
- Protect network credentials and avoid sharing sensitive keys in public code.

Care & storage

- Handle the module by its edges and store it in an antistatic, dry environment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ammeter

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

6 labs

Covers these lab items: Ammeter

What it is used for

Use an ammeter to measure electric current in a circuit within the instrument's rated range and measurement method.

How to use

1. Inspect the ammeter, leads, probes, and terminals for damage.
2. Confirm the meter's current rating, fuse condition, and correct measurement sockets.
3. Switch off and isolate the circuit before connecting the ammeter in series.
4. Connect the meter with the correct polarity and range for the circuit.
5. Restore power using the approved procedure and observe the reading.
6. Record the current value and units, then isolate the circuit again.
7. Disconnect the meter and return leads to their safe storage positions.

⚠ Safety

- Never connect an ammeter directly across a voltage source.
- Do not exceed the meter's current rating or bypass its fuse.
- Use appropriate CAT-rated equipment and PPE for the circuit.
- Isolate power before changing connections or measurement ranges.

Care & storage

- Inspect and replace only specified fuses when required.
- Store the meter and leads dry and arrange calibration as specified.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Battery Tester

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Cell tester; LiPo Battery Checker

What it is used for

To check the condition or voltage of a battery or cell using a compatible battery tester. To obtain a safe, meaningful result by selecting the correct test mode and following the device instructions.

Main parts

- Tester body and display
- Test leads or probe contacts
- Battery holder or connector
- Mode or range selector
- Test button or control

Before you start

1. Identify the battery chemistry, nominal voltage, and connector type.
2. Confirm that the tester supports the battery type and voltage range.
3. Inspect test leads, probes, and connectors for damage.
4. Check polarity and ensure the battery is not swollen, leaking, or visibly damaged.
5. Read the tester instructions and select the appropriate test mode.

How to use

1. Switch off the tester before connecting the battery if required by the instructions.
2. Connect the battery with correct polarity and compatible connectors.
3. Select the specified test mode or range.
4. Start the test and observe the displayed reading.
5. Compare the result with the battery manufacturer's limits and the tester guidance.
6. Disconnect the battery after the test is complete.
7. Record the result and switch off the tester.

⚠ Safety

- Do not test swollen, leaking, punctured, or overheated batteries.
- Never short-circuit battery terminals with probes or tools.
- Use the correct tester for lithium polymer and other battery chemistries.
- Stop if the battery becomes hot, emits an odour, or behaves abnormally.

Care & storage

- Keep probes and connectors clean and dry.
- Inspect test leads and insulation regularly.
- Replace damaged leads and connectors.
- Store the tester in a dry protective case.

Common mistakes to avoid

- Connecting the battery with reversed polarity.
- Using an unsupported test mode or voltage range.
- Shorting the battery terminals.
- Assuming a voltage reading alone fully determines battery health.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Clamp Meter

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

10 labs

Covers these lab items: Clamp-meter; Tong tester AC/DC

What it is used for

Use a clamp meter to measure current without disconnecting a conductor, and measure other electrical values when supported by the meter.

How to use

1. Inspect the meter, clamp jaws, leads, probes, and display for damage.
2. Select the correct measurement function, range, and AC or DC mode as required.
3. Clamp around a single conductor only when measuring current.
4. Close the jaws fully and keep the conductor centered where possible.
5. Read the stable value and note the unit shown on the display.
6. Use test leads only in the correct sockets and only for supported measurements.
7. Remove the meter safely and switch it off after testing.

⚠ Safety

- Use a meter with a suitable CAT rating and voltage rating for the circuit.
- Never clamp around both live and return conductors together for current measurement.
- Do not use damaged leads or touch exposed metal probe tips.
- Follow electrical isolation and PPE requirements for the work area.

Care & storage

- Keep the clamp jaws clean and free of debris.
- Store the meter and leads in a dry protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Continuity Tester

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

7 labs

Covers these lab items: Circuit Testers; Continuity Tester

What it is used for

Use a continuity tester to check whether a suitable electrical path is complete when the circuit is safely isolated from power.

How to use

1. Inspect the tester, probes, leads, and insulation for damage.
2. Confirm the tester is suitable for the circuit and test purpose.
3. Disconnect the circuit from all power sources and verify it is de-energized.
4. Test the tester on a known continuity reference or according to its instructions.
5. Connect the probes to the two points being checked.
6. Observe the indicator or sound and interpret it according to the tester manual.
7. Disconnect the probes and restore the circuit only when safe.

⚠ Safety

- Never use a continuity tester on an energized circuit.
- Discharge capacitors and follow isolation procedures before testing.
- Do not use damaged probes, leads, or insulation.
- A continuity result does not prove that a circuit is safe to touch.

Care & storage

- Clean the probes and tester after use.
- Store it dry and check the battery and leads regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Digital Multimeter

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **High**

29 labs

Covers these lab items: Digital multimeter; Multimeter; Multimeter, Clamp-meter; Multimeters; Multi-metre; Multi-tester; Voltmeter / DIGITAL MULTIMETER

What it is used for

A digital multimeter measures electrical quantities such as voltage, current, and resistance, depending on its design. Select the correct function and range before testing a circuit or component.

How to use

1. Inspect the meter, test leads, probes, and insulation for damage.
2. Confirm the meter and leads are rated for the circuit being tested.
3. Insert leads into the correct COM and measurement terminals.
4. Select the required function and range, and choose AC or DC correctly.
5. Connect probes in parallel for voltage or in series for current, following the meter instructions.
6. For resistance or continuity, isolate power and discharge capacitors before testing.
7. Read the display and record the measurement with its units.
8. Remove probes safely, switch off the meter, and store the leads.

⚠ Safety

- Never measure resistance on an energised circuit.
- Never place a meter set to current directly across a voltage source.
- Keep fingers behind probe guards and use suitable PPE.
- Do not use damaged leads or exceed the meter's voltage and current ratings.

Care & storage

- Keep the meter dry and clean, and store it in a protective case.
- Replace weak batteries and damaged leads, and verify calibration when required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Earth Tester

Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Earth tester

What it is used for

An earth tester measures the resistance of an earthing system or electrode to help assess electrical safety. Correct test connections, suitable test conditions, and careful interpretation of readings support reliable results.

Main parts

- Display or scale shows measured earth resistance.
- Test terminals connect the instrument to the electrode and auxiliary test leads.
- Test leads connect the tester to the specified test points.
- Auxiliary spikes or probes provide reference points for applicable test methods.
- Selector switch chooses the test function or range.
- Test button initiates the measurement on applicable models.
- Battery or power source operates the instrument.

Before you start

1. Read the instrument manual and identify the approved earth-testing method.
2. Confirm authorisation and follow site electrical safety procedures before testing.
3. Inspect the tester, leads, clamps, and auxiliary spikes for damage.
4. Check the instrument's battery status and calibration or verification status.
5. Identify the earthing electrode and confirm whether the installation must be isolated for the selected test.
6. Select a suitable test location and check for buried services before inserting auxiliary spikes.

How to use

1. Switch off and isolate the relevant installation when required by the approved procedure.
2. Set up auxiliary spikes and test leads according to the selected measurement method.
3. Connect the tester terminals to the electrode and auxiliary leads as shown in the manual.
4. Check that all connections are secure and the leads are routed safely.
5. Select the correct test function and range.
6. Start the test using the designated control and wait for the reading to stabilise.
7. Record the resistance value, unit, test method, location, and relevant conditions.
8. Repeat the measurement if required by the procedure or if readings are unstable.
9. Disconnect the test leads and remove auxiliary spikes carefully.
10. Restore the installation only after the test area is clear and the approved procedure permits it.

⚠ Safety

- Only authorised personnel should test electrical earthing systems.
- Never disconnect protective earthing on an energised installation unless an approved safe procedure specifically permits it.
- Check for buried electrical cables and services before driving auxiliary spikes into the ground.
- Do not use damaged leads, probes, or instrument insulation.
- Keep test leads away from traffic, sharp edges, and wet areas where possible.
- Follow site isolation, lockout, and personal protective equipment requirements.

Care & storage

- Switch off the tester and disconnect leads before cleaning or storage.
- Clean the casing and test accessories with suitable dry materials.
- Inspect lead insulation, connectors, clamps, and spikes after use.
- Store the tester in its protective case and follow recommended calibration intervals.

Common mistakes to avoid

- Connecting test leads to the wrong terminals or test points.
- Ignoring buried services before placing auxiliary spikes.
- Testing an installation without required authorisation or isolation.
- Recording resistance values without units or the test method used.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrical Safety Analyzer

Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Electrical safety analyzer

What it is used for

An electrical safety analyzer tests electrical safety characteristics of equipment, such as protective earth continuity, insulation resistance, leakage current, and selected functional checks, depending on the model. Correct test selection, safe connections, and accurate recording help identify electrical hazards before equipment is returned to service.

Main parts

- Analyzer main unit and display
- Test leads and probes
- Protective earth continuity test connections
- Insulation resistance and leakage current test connections, where fitted
- Equipment-under-test power outlet or connection
- Test controls, adapters, and calibration label

Before you start

1. Only trained and authorised personnel should operate the analyzer and interpret its test results.
2. Identify the equipment under test (EUT), its rated voltage, protection class, and applicable test procedure.
3. Read the analyzer manual and the EUT manufacturer's instructions to confirm which tests are appropriate.
4. Inspect the analyzer, leads, probes, adapters, and connectors for damage, exposed conductors, or contamination.
5. Verify that the analyzer is within its calibration period and that the correct test limits and settings are selected.
6. Remove the EUT from normal service and disconnect it from other equipment unless the approved test procedure requires a specific connection.

How to use

1. Place the analyzer on a stable, dry surface in a controlled test area.
2. Switch on the analyzer and perform its specified self-test or readiness check.
3. Connect the test leads and EUT according to the manufacturer's connection diagram and the selected test procedure.
4. Confirm that the EUT is correctly identified and that its normal operating controls are set as required for the test.

5. Perform protective earth continuity testing where applicable, using the specified test current and acceptance limit.
6. Perform insulation resistance or leakage current testing only when appropriate for the EUT and permitted by its instructions.
7. Select the correct test configuration for the EUT's protection class, applied parts, and intended use.
8. Start each test using the analyzer's approved sequence and keep personnel clear of exposed conductive parts during energised tests.
9. Observe displayed readings, warnings, and test status; stop if a connection is loose, a fault appears, or the setup becomes unsafe.
10. Compare results with the applicable manufacturer's limits, facility procedures, and relevant standards for the equipment type.
11. If a test fails, label the EUT as failed or not for use, document the result, and refer it for qualified investigation.
12. After testing, follow the analyzer procedure to discharge any stored electrical energy and return outputs to a safe state.
13. Switch off the analyzer before disconnecting leads and remove connections carefully.
14. Record equipment identification, test selection, measured results, acceptance limits, instrument identification, and test date.

⚠ Safety

- Use the analyzer only within its rated voltage, current, and measurement category.
- Follow the analyzer manual because some tests apply hazardous voltage or current to the EUT.
- Never touch exposed conductors or the EUT during an energised test.
- Do not perform insulation resistance or high-voltage tests on sensitive electronics unless the equipment manufacturer and approved procedure permit them.
- Verify leads, adapters, protective earth connections, and calibration status before testing.
- Do not bypass interlocks or ignore warning indicators; stop if the analyzer or EUT behaves abnormally.
- Use appropriate electrical PPE and follow lockout/tagout requirements where applicable.

Care & storage

- Inspect and clean the analyzer, leads, probes, and adapters after use.
- Store leads without sharp bends and protect connectors from impact and moisture.
- Check calibration status and arrange calibration at the required interval or after suspected damage.
- Keep test records and note instrument identification and any failed tests.
- Remove damaged accessories from service and arrange qualified repair when required.

Common mistakes to avoid

- Selecting a test or limit that is unsuitable for the EUT.
- Touching the EUT or exposed conductive parts during an energised test.

- Ignoring calibration status or damaged test leads.
- Returning failed equipment to service without documented assessment and corrective action.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Energy Meter

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **High**

2 labs

Covers these lab items: Energy meter

What it is used for

An energy meter measures electrical energy consumed by a circuit or installation over time. Use the correct meter type and approved testing method to obtain reliable readings.

How to use

1. Inspect the meter, display, terminals, and enclosure for damage.
2. Confirm the meter's voltage, current, phase, and connection ratings suit the installation.
3. Have a qualified electrician isolate the supply before installation or wiring changes.
4. Connect the meter according to its terminal diagram and approved wiring procedure.
5. Restore power safely and check that the display and indicators operate normally.
6. Record the energy reading, units, meter identification, and observation time.
7. Compare readings with expected values and report unusual results.
8. Isolate the supply before any required maintenance or wiring inspection.

⚠ Safety

- Never touch exposed terminals or work on live meter wiring.
- Do not exceed the meter's rated voltage or current.
- Installation and electrical testing must be carried out by qualified personnel.
- Follow isolation, lockout, and applicable electrical safety procedures.

Care & storage

- Keep the meter enclosure clean, dry, and undamaged.
- Arrange periodic inspection and calibration or replacement according to applicable requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Frequency Meter

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **High**

3 labs

Covers these lab items: Frequency meter

What it is used for

A frequency meter measures the frequency of an electrical signal or supply. Select a compatible meter and connection method for the signal being tested.

How to use

1. Inspect the meter, leads, and connectors for damage.
2. Confirm the meter's frequency range and input rating match the signal.
3. Select the frequency measurement function and required range.
4. Connect the meter using the manufacturer's specified terminals and procedure.
5. Apply the signal within the permitted input limits.
6. Wait for the reading to stabilise and record the frequency with units.
7. Remove the input safely and switch off the meter after testing.

⚠ Safety

- Only trained personnel should test hazardous or live electrical circuits.
- Do not exceed the meter's voltage, frequency, or input limits.
- Keep hands dry and use appropriately rated leads and PPE.
- Never connect a meter to an unknown signal without checking its rating.

Care & storage

- Keep the meter clean, dry, and stored in a protective case.
- Inspect leads and check calibration at the recommended intervals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Function Generator

Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **Medium**

1 labs

Covers these lab items: Function generator

What it is used for

A function generator produces electrical waveforms such as sine, square, and triangle signals for testing and demonstrating electronic circuits. Correct amplitude, frequency, and offset settings help verify circuit behaviour safely and accurately.

Main parts

- Function generator unit and display
- Waveform selection and frequency controls
- Amplitude and DC offset controls
- Output terminals or BNC connectors
- Power switch and power supply
- Test leads and compatible connecting cables

Before you start

1. Read the instrument manual and identify the voltage, frequency, and output limits.
2. Inspect the case, display, connectors, power cable, and test leads for damage.
3. Confirm that the circuit under test can tolerate the planned signal amplitude and offset.
4. Check the required output termination and cable type for the test setup.
5. Switch on the generator and allow it to complete its startup checks.
6. Ensure the output is disabled while connecting leads and configuring initial settings.

How to use

1. Select the required waveform, such as sine, square, or triangle.
2. Set the target frequency using the frequency controls or numeric entry.
3. Set the output amplitude and DC offset within the circuit's permitted limits.
4. Select the required output load or termination setting if supported by the instrument.
5. Connect the output cable to the circuit input with the generator output disabled.
6. Check the wiring and confirm the circuit ground and signal connections are correct.
7. Enable the output and observe the circuit response using an oscilloscope or suitable measuring instrument.
8. Adjust frequency, amplitude, or offset gradually while monitoring the response.
9. Disable the output before disconnecting the circuit or changing the wiring.
10. Switch off the generator, remove the leads, and store the accessories safely.

⚠ Safety

- Never exceed the input voltage or signal limits of the circuit or connected instrument.
- Disable the output before connecting, disconnecting, or changing test leads.
- Do not connect the generator output directly to mains voltage or an incompatible power source.
- Use suitable cables and avoid shorting the output terminals.
- Keep the instrument and connectors dry and use appropriate electrical safety practices.
- Disconnect power before opening the case or performing internal maintenance.

Care & storage

- Keep the display, case, and connectors clean using approved materials.
- Inspect output cables and connectors for loose contacts or damaged insulation.
- Store leads neatly and avoid bending cables sharply near their connectors.
- Arrange calibration and functional checks according to the laboratory schedule.

Common mistakes to avoid

- Setting the wrong amplitude or offset can damage the circuit under test.
- Ignoring output termination can produce an incorrect signal amplitude.
- Connecting leads while the output is enabled can create unexpected transients or shorts.
- Using damaged cables can cause unstable signals and unreliable measurements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Infrared Thermometer

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

2 labs

Covers these lab items: Infrared Thermometer; IR Thermometer

What it is used for

To measure surface temperature without direct contact using infrared radiation.

How to handle

- Inspect the thermometer lens, display, trigger, and battery compartment for damage or dirt.
- Select the correct measurement mode and emissivity setting where applicable, following the instrument manual.
- Aim at the intended surface from the recommended distance and keep the lens clean and unobstructed.
- Press the trigger and record the reading with units; avoid measuring through glass or steam unless the instrument is designed for it.
- Do not use a surface-temperature reading as a substitute for a clinical body-temperature measurement unless the device is approved for that purpose.

Storage

- Switch off the thermometer and clean the lens gently with the recommended material.
- Store in its protective case in a dry place away from extreme temperatures, dust, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Insulation Resistance Tester

Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hand driven Megger; Megger

What it is used for

An insulation resistance tester applies a specified test voltage to assess insulation resistance in isolated electrical equipment or circuits. Use the correct test voltage and follow electrical isolation procedures before testing or reconnecting equipment.

Main parts

- Insulation resistance tester or hand-driven megger
- Test leads and insulated probes
- Positive and negative test terminals
- Test voltage selector or crank mechanism
- Display, scale, or resistance indicator
- Protective case and test accessories

Before you start

1. Confirm the circuit or equipment is suitable for insulation resistance testing.
2. Identify the circuit and isolate it from every power source.
3. Apply lockout/tagout procedures and verify absence of voltage with an appropriate tester.
4. Disconnect sensitive electronic devices, control modules, and components that could be damaged by test voltage.
5. Select the test voltage according to the equipment rating and applicable procedure.
6. Inspect the tester, leads, probes, and insulation for damage.

How to use

1. Keep the circuit isolated and ensure it cannot be energised during testing.
2. Connect the test leads to the designated tester terminals.
3. Connect the probes across the specified conductors or between conductor and earth as required.
4. Confirm that the selected test voltage is suitable for the equipment under test.
5. For a hand-driven model, operate the handle at the speed specified by the manufacturer.
6. For an electronic model, activate the test according to its instructions.
7. Maintain the test for the prescribed duration and observe the resistance reading.
8. Record the reading, test voltage, duration, equipment identification, and environmental conditions if required.
9. Release the test control and wait for the instrument to indicate completion.
10. Allow the tested circuit to discharge fully; use the tester's discharge function if provided.

1. Verify that residual voltage has dissipated before touching the conductors.
2. Disconnect the test leads and restore connections only after testing is complete and safe.
3. Remove isolation only under the authorised procedure and record the test result.

⚠ Safety

- Only qualified or authorised personnel should perform electrical insulation tests.
- Never apply test voltage to an energised circuit.
- Do not touch test leads or conductors during a test.
- Use the correct test voltage to avoid damaging insulation or connected components.
- Treat tested cables and capacitive equipment as charged until fully discharged.
- Do not test in wet conditions or use damaged leads and probes.

Care & storage

- Keep the tester and accessories clean, dry, and protected from impact.
- Inspect leads, probes, insulation, and terminals before each use.
- Check calibration or verification status according to the maintenance schedule.
- Replace damaged accessories with compatible approved parts.
- Store the tester in its protective case and record faults.

Common mistakes to avoid

- Testing a circuit without verifying electrical isolation.
- Selecting an unsuitable test voltage for the equipment.
- Touching conductors before the tested circuit has discharged.
- Failing to disconnect sensitive electronics before applying test voltage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Leak Detector

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

2 labs

Covers these lab items: Leak Detector

What it is used for

To identify leaks in compatible systems, such as refrigerant, gas, or pressurized lines, using a detector designed for the specific substance.

How to handle

- Inspect the detector, probe, hoses, display, and battery or power supply for damage.
- Confirm the detector is compatible with the suspected substance and follow its manual and applicable safety procedure.
- Check calibration or perform the specified function test before use, if required by the manufacturer.
- Scan the suspected leak area using the recommended distance and speed; keep the probe away from liquids or surfaces unless the device is designed for contact.
- Treat an alarm or positive reading as an indication requiring further verification; isolate hazards and follow the relevant emergency or maintenance procedure.

Storage

- Switch off the detector, clean the probe according to the manual, and replace protective caps.
- Store in its case in a dry, secure place; maintain calibration records and batteries as recommended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lux Meter

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

5 labs

Covers these lab items: Light meter; Lux meter; Lux meter/Light meter

What it is used for

A lux meter measures illuminance, or the amount of light falling on a surface. Use it to check lighting levels in workplaces, classrooms, laboratories, and other areas.

How to use

1. Inspect the meter, sensor, display, and battery before use.
2. Switch on the meter and select the appropriate measurement range if required.
3. Place the light sensor at the point where illumination needs to be measured.
4. Keep your body and hands from casting shadows on the sensor.
5. Wait for the reading to stabilize and record the lux value.
6. Switch off the meter and store it safely after use.

⚠ Safety

- Keep the meter dry and protect the sensor from impact.
- Do not stare directly into intense light sources while taking measurements.
- Use the instrument within its specified measurement range.

Care & storage

- Clean the sensor gently and store the meter in its protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Motherboard Diagnostic Card

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

2 labs

Covers these lab items: Motherboard Diagnoser

What it is used for

To assist in diagnosing computer motherboard startup or hardware faults by displaying diagnostic codes on compatible systems.

How to handle

- Inspect the diagnostic card, connector, display, and pins for damage; confirm compatibility with the motherboard and slot type.
- Shut down the computer, unplug the power supply, and follow electrostatic discharge precautions before installing the card.
- Insert the card into the specified compatible slot in the correct orientation without forcing it.
- Reconnect power and start the system only after checking that the card and all connections are secure.
- Record displayed codes and consult the motherboard or card manual; power down and unplug before removing the card.

Storage

- Remove the card using ESD-safe handling and protect its connector pins.
- Store in an antistatic bag or case in a dry location away from dust and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Non-contact Voltage Detector

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **High**

6 labs

Covers these lab items: Non-contact Voltage Detector

What it is used for

A non-contact voltage detector indicates the possible presence of AC voltage without touching a conductor directly. Use it as a preliminary screening tool, not as the sole proof that a circuit is safe.

How to use

1. Inspect the detector body, tip, and battery condition for damage.
2. Read the manufacturer's instructions and confirm the detector's voltage range.
3. Test the detector on a known live source to verify operation.
4. Bring the sensing tip near the insulated wire or test point as instructed.
5. Observe the indicator light, sound, or display for a voltage indication.
6. Recheck the detector on a known live source after testing.
7. Use an appropriately rated tester and approved isolation procedure to confirm absence of voltage before electrical work.

⚠ Safety

- Do not touch bare conductors or rely on the detector alone to prove a circuit is de-energised.
- Keep hands dry and use suitable electrical PPE.
- Do not use a damaged detector or one outside its rated range.
- Follow lockout and isolation procedures before working on electrical equipment.

Care & storage

- Keep the detector clean and dry, and replace batteries as required.
- Store it safely and check operation before each use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oscilloscope

Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Cathode Ray Oscilloscope (CRO); CRO; Oscilloscope

What it is used for

An oscilloscope displays electrical signals as waveforms over time, allowing users to measure signal level, timing, frequency, and noise. Correct probe selection, grounding, voltage limits, and measurement setup are essential for reliable results and protection against electric shock or instrument damage.

Main parts

- Display shows the measured waveform and measurement information.
- Input channels connect probes to the instrument.
- Probe attenuates or transfers the signal to the input channel.
- Vertical controls adjust voltage scale and channel position.
- Time-base controls adjust the horizontal time scale.
- Trigger controls stabilise repeating waveforms on the display.
- Probe ground lead provides the reference connection where applicable.
- Measurement controls and menu functions calculate signal parameters.
- Power input and protective earth connection supply power and provide grounding as designed.

Before you start

1. Only trained users should measure electrical circuits with an oscilloscope.
2. Read the instrument and probe manuals and identify the circuit voltage, signal type, and measurement category.
3. Inspect the oscilloscope, probes, leads, connectors, and power cable for damage.
4. Confirm the probe attenuation setting matches the instrument setting and the probe is rated for the expected voltage and frequency.
5. Use a suitable differential or isolated probe when required; never assume a standard ground clip is isolated.
6. Confirm the circuit reference and protective-earth requirements before connecting any probe ground.
7. Keep the bench dry and remove unnecessary conductive objects from the measurement area.

How to use

1. Switch on the oscilloscope and allow it to complete its startup checks.
2. Connect the correct probe to the selected input channel and set the matching attenuation factor.
3. Configure the vertical scale, time base, coupling, and trigger according to the signal being measured.

4. With the circuit de-energised where required, connect the probe tip to the test point and the reference lead only to an approved reference point.
5. Energise the circuit only after checking all connections and confirming the setup is safe.
6. Adjust the trigger and scales until the waveform is stable and clearly visible.
7. Use cursors or automatic measurement functions to read voltage, period, frequency, or other required parameters.
8. Compare results with the expected signal range and note any clipping, noise, distortion, or unexpected transients.
9. De-energise the circuit before moving probe connections when required by the risk assessment.
10. Remove the probe tip and reference lead safely, then switch off the instrument if no longer needed.
11. Save waveform data or record readings and report abnormal results or instrument faults.

⚠ Safety

- Never exceed the voltage, frequency, or measurement-category ratings of the oscilloscope or probe.
- Do not connect a standard earth-referenced probe ground clip to a live point that is not at earth potential.
- Never remove the oscilloscope protective earth to make an unsafe measurement.
- Use correctly rated differential probes or approved isolation methods for floating or high-side measurements.
- Keep hands away from exposed live conductors and use suitable PPE when required.
- Disconnect power and discharge stored energy before changing circuit connections whenever the procedure permits.

Care & storage

- Switch off and disconnect the instrument before cleaning its exterior.
- Clean the display and housing with manufacturer-approved materials; keep liquids away from openings.
- Inspect probes, tips, ground leads, connectors, and cables regularly.
- Store probes without sharp bends and protect the instrument from dust, moisture, and impact.
- Arrange calibration or service if readings are unreliable, controls fail, or the instrument is damaged.

Common mistakes to avoid

- Using the wrong probe attenuation or an unsuitable voltage rating.
- Connecting the probe ground clip to an unsafe reference point.
- Changing probe connections on an energised circuit without an approved procedure.
- Misreading an unstable waveform because trigger and time-base settings are incorrect.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Phase Sequence Meter

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **High**

5 labs

Covers these lab items: Phase sequence meter; Sequence Meter

What it is used for

A phase sequence meter identifies the phase order of a three-phase electrical supply. Use a correctly rated meter to verify phase sequence before connecting equipment that depends on rotation direction.

How to use

1. Inspect the meter, test leads, and probes for damage and confirm their ratings.
2. Check the manufacturer's instructions and identify the required phase connections.
3. Verify that the meter is suitable for the system voltage and frequency.
4. Connect the test leads to the designated three-phase terminals following the instrument markings.
5. Energise or test the circuit only as permitted by the instrument instructions and site procedure.
6. Observe the phase sequence indication and record the result.
7. Disconnect the leads safely, switch off the meter if applicable, and store it.

⚠ Safety

- Only trained personnel should test live three-phase circuits.
- Use appropriate PPE and keep hands behind probe guards.
- Never touch exposed conductors or change connections while energised.
- Do not use damaged leads or a meter with an insufficient voltage rating.

Care & storage

- Keep the meter clean and dry and store the leads without sharp bends.
- Inspect leads regularly and arrange calibration or servicing as specified.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Phase Tester

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

13 labs

Covers these lab items: Electricity tester; Line tester; Phase tester; Tester

What it is used for

Use a phase tester to check for the presence of voltage at suitable points in an electrical circuit, following the tester's design and rating.

How to use

1. Inspect the tester body, insulation, and tip for cracks or damage.
2. Confirm the tester type and voltage rating are suitable for the circuit.
3. Test the tester on a known live source or approved proving unit where applicable.
4. Use the tester only at an appropriate accessible test point.
5. Observe the indicator according to the manufacturer's instructions.
6. Recheck the tester on the proving source after testing when applicable.
7. Keep the circuit isolated before touching conductors or making repairs.

⚠ Safety

- A basic neon line tester is not a substitute for a rated voltage detector or meter.
- Never touch exposed metal parts of the tester tip during use.
- Do not use a damaged tester or exceed its rated voltage.
- Follow safe isolation procedures before working on electrical wiring.

Care & storage

- Wipe the tester clean and keep it dry.
- Store it safely and inspect its insulation before each use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Power Supply

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

4 labs

Covers these lab items: Power supplies, PoE switches, adapters, and UPS systems; Power supply; Regulated power supply kit

What it is used for

To provide the required electrical power to compatible equipment using a suitable power supply. To verify output settings and connections before powering the connected device.

Main parts

- Power supply unit
- Input power connector
- Output terminals or connector
- Voltage and current controls, if fitted
- Status indicators
- Fuse or overload protection

Before you start

1. Confirm input and output ratings match the equipment requirements.
2. Inspect the unit, cable, connectors, and casing for damage.
3. Check polarity and grounding requirements where applicable.
4. Ensure output is switched off before connecting the load.

How to use

1. Place the power supply on a stable, dry surface.
2. Connect the input power using the approved cable.
3. Set the required output voltage and current limit where adjustable.
4. Check polarity and connect the load securely.
5. Switch on the power supply and enable the output.
6. Monitor indicators and confirm the equipment operates normally.
7. Switch off the output before disconnecting the load.
8. Disconnect the input supply before cleaning or storing the unit.

⚠ Safety

- Do not exceed the rated voltage or current.
- Keep ventilation openings clear.
- Do not touch exposed live terminals.
- Use the correct fuse and approved power cable.
- Disconnect power if overheating, smoke, or abnormal noise occurs.

Care & storage

- Keep the unit clean and dry.
- Inspect cables and connectors regularly.
- Do not block cooling vents.
- Store accessories safely and arrange service for faults.

Common mistakes to avoid

- Using incorrect voltage or polarity.
- Connecting the load while output is enabled when the procedure prohibits it.
- Exceeding the current rating.
- Blocking ventilation or using damaged cables.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Radiation Meter

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

2 labs

Covers these lab items: Radiation measuring instrument

What it is used for

To measure or survey radiation levels using an instrument appropriate for the radiation type and intended application.

How to handle

- Inspect the meter, probe, display, battery, and cables for damage before use.
- Confirm that the instrument is suitable for the radiation type, measurement range, and intended survey; check calibration status.
- Follow the radiation-safety plan and wear required personal protective equipment; do not handle unknown radioactive sources without authorization.
- Switch on the meter, allow it to stabilize as specified, and measure from the recommended position and distance.
- Record readings with units, location, time, and instrument details; follow site procedures if readings exceed established limits or an alarm occurs.

Storage

- Switch off the meter, clean it according to the manual, and protect the probe and connectors.
- Store in its designated case in a secure, dry location; maintain calibration and service records.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Signal Detector

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Signal detector

What it is used for

A signal detector identifies or indicates the presence of specified electrical, radio-frequency, or other signals depending on its design. Verify the detector's operating range and intended use before testing.

How to use

1. Inspect the detector, probe, antenna, display, battery, and leads for damage.
2. Read the manual to confirm the signal type, detection range, and operating limits.
3. Select the correct mode and sensitivity for the signal being checked.
4. Connect probes or position the antenna as specified by the manufacturer.
5. Power on the detector and perform any required self-test or calibration.
6. Scan the intended area or circuit and observe the displayed indication.
7. Record relevant results, switch off the device, and disconnect accessories after use.

⚠ Safety

- Do not use the detector as proof of electrical isolation unless specifically rated and approved for that purpose.
- Keep probes and hands away from exposed live conductors.
- Use only the specified accessories and operating range.
- Follow site authorization and applicable radio-frequency restrictions.

Care & storage

- Clean the detector with an approved method and store it with probes protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tachometer

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

7 labs

Covers these lab items: Laser Tachometer; Tachometer

What it is used for

Use a tachometer to measure the rotational speed of a compatible rotating shaft, wheel, or machine component.

How to use

1. Inspect the tachometer, sensor, display, and accessories for damage.
2. Select the correct measurement mode and units according to the instrument manual.
3. For a laser tachometer, apply a suitable reflective mark to the stopped target if required.
4. Position the sensor at the specified distance and angle for the instrument type.
5. Keep clear of rotating parts and take the reading according to the manufacturer's procedure.
6. Record the stable speed value and units shown on the display.
7. Switch off the instrument and remove any temporary reflective mark if appropriate.

⚠ Safety

- Never touch or approach rotating machinery while taking a measurement.
- Do not aim a laser at anyone's eyes.
- Install reflective tape only when the machine is stopped and safely isolated.
- Use suitable guards and follow site machine-safety procedures.

Care & storage

- Clean the sensor and display with a suitable soft cloth.
- Store the tachometer in its case and protect it from dust and impacts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Temperature Indicator

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Digital electronic temperature indicator

What it is used for

To display or monitor temperature measured by a compatible sensor in equipment, processes, or laboratory setups.

How to handle

- Inspect the indicator, display, terminals, sensor, and cables for damage before use.
- Confirm that the sensor type, temperature range, input configuration, and power supply match the indicator specifications.
- Isolate power before wiring the sensor or changing electrical connections.
- Install the sensor at the intended measurement point and configure units, limits, and alarm settings as required.
- Compare the displayed value with a suitable reference when accuracy matters, and disconnect power before maintenance.

Storage

- Store in a clean, dry location protected from dust, moisture, and impact.
- Keep sensors and leads labelled and avoid bending cables sharply or damaging probe tips.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Test Equipment

Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Test equipment

What it is used for

Electrical test equipment measures or verifies electrical quantities and circuit performance. Identify the required measurement, select a suitable rated instrument, and follow its operating instructions before testing.

How to use

1. Inspect the instrument, probes, leads, connectors, and casing for damage.
2. Confirm the measurement category, voltage rating, current rating, and correct test function.
3. Check the instrument's calibration or verification status where required.
4. De-energize the circuit before connecting leads whenever the test procedure permits.
5. Connect probes to the correct terminals and select the appropriate range or mode.
6. Perform the measurement using the approved procedure and record the result.
7. Remove probes safely, switch off the instrument, and store the accessories.

⚠ Safety

- Use instruments and probes rated for the circuit and measurement category.
- Never touch exposed live conductors or exceed instrument ratings.
- Do not measure resistance on an energized circuit.
- Only qualified personnel should perform hazardous voltage or current measurements.

Care & storage

- Clean the instrument, inspect leads after use, and store it in a protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Voltmeter

Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Voltmeter

What it is used for

A voltmeter measures the electrical potential difference between two points in a circuit. Correct meter settings and parallel connection help obtain reliable readings while reducing the risk of electric shock and instrument damage.

Main parts

- Display or scale shows the measured voltage.
- Selector switch chooses voltage type and measurement range.
- Positive input terminal accepts the red test lead.
- COM terminal accepts the black common test lead.
- Test leads and probes connect the meter to the circuit.
- Internal protection helps protect the meter within its rated limits.
- Battery supplies power to digital models.

Before you start

1. Read the meter instructions and confirm its voltage rating suits the circuit.
2. Inspect the meter, probes, insulation, leads, and terminals for damage.
3. Select AC or DC voltage according to the circuit being tested.
4. Start with a range above the expected voltage when using a manual-range meter.
5. Insert the black lead into COM and the red lead into the voltage terminal.
6. Keep hands behind the probe finger guards and wear appropriate PPE for the task.

How to use

1. Switch off circuit power before connecting probes whenever the task permits.
2. Set the selector to the correct voltage type and suitable range.
3. Connect the probes across the two points being measured, in parallel with the circuit component.
4. For DC measurements, observe polarity and connect red to the more positive point when known.
5. Restore power only when the connections are secure and the procedure permits live measurement.
6. Read the display or scale without touching exposed conductive parts.
7. If the reading exceeds the selected range, disconnect safely and select a higher range.
8. Record the reading with the unit and whether it is AC or DC.
9. Switch off the circuit before removing probes when possible.
10. Remove the leads, switch off the meter, and store it safely.

⚠ Safety

- Never connect a voltmeter in series as a current-measuring device.
- Do not exceed the meter's voltage or measurement-category rating.
- Never use damaged probes, exposed wires, or cracked meter insulation.
- Avoid touching metal probe tips or live conductive parts during measurement.
- Use appropriate isolation and authorised supervision when testing mains or high-voltage circuits.
- Never change exposed wiring or terminals while the circuit is energised.

Care & storage

- Disconnect the leads and switch off the meter before cleaning it.
- Wipe the case with a dry or manufacturer-approved cloth.
- Store probes without sharp bends and protect their insulation from damage.
- Check the battery, leads, and calibration status according to the manual.

Common mistakes to avoid

- Selecting AC when measuring DC, or selecting DC when measuring AC.
- Connecting the meter in series instead of parallel.
- Using a range below the expected voltage on a manual-range meter.
- Touching exposed probe tips or using damaged test leads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wattmeter

Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Power Meter; Watt meter, pressure meters; Wattmeter; Wattmeter and Clamp meter

What it is used for

A wattmeter measures electrical power consumed or delivered by a circuit or device. Correct voltage and current connections, suitable ratings, and careful interpretation help prevent shock and inaccurate measurements.

Main parts

- Display or scale shows measured power and related values.
- Voltage input terminals connect across the measured supply or circuit.
- Current input terminals connect in series or through the specified current sensor.
- Current coil or sensor measures current in applicable meter designs.
- Selector controls choose measurement mode or range.
- Test leads, probes, or clamp sensor connect the instrument to the circuit.
- Protective casing and rated insulation help reduce electrical hazards.

Before you start

1. Read the instrument manual and confirm its voltage, current, and power ratings suit the circuit.
2. Inspect the casing, test leads, probes, connectors, and clamp sensor for damage.
3. Identify the supply type and the instrument's required voltage and current connection method.
4. Switch off and isolate the circuit before making connections whenever the procedure permits.
5. Use the correct terminals and range according to the wiring diagram.
6. Wear suitable PPE and obtain authorised supervision for mains or high-energy circuits.

How to use

1. Set the instrument to the required power measurement mode and range.
2. Connect the voltage inputs across the specified circuit points according to the manual.
3. Connect current inputs in series or position the clamp sensor as specified for the model.
4. Check polarity, phase, and connection direction where required by the instrument.
5. Verify all connections are secure before energising the circuit for measurement.
6. Read the wattage and other displayed values after the reading stabilises.
7. Check that the reading is within the instrument's rated range and expected operating conditions.
8. Record the value with units and relevant AC or DC details.
9. Switch off and isolate the circuit before disconnecting leads whenever possible.
10. Remove the connections carefully and store the instrument and accessories safely.

⚠ Safety

- Never exceed the instrument's voltage, current, or measurement-category ratings.
- Do not connect current terminals directly across a voltage source unless the manual explicitly permits it.
- Never handle exposed conductors or damaged leads on an energised circuit.
- Keep fingers behind probe guards and use suitable protective equipment.
- Use clamp sensors only according to their specified conductor and current limits.
- Do not change wiring or terminals while the circuit is live.

Care & storage

- Switch off and disconnect the instrument before cleaning or inspecting accessories.
- Clean the casing with a dry, suitable cloth and keep terminals free from dirt.
- Inspect leads, connectors, clamp jaws, and insulation regularly for wear or damage.
- Follow the manufacturer's instructions for battery replacement, calibration, and servicing.

Common mistakes to avoid

- Connecting voltage and current inputs to the wrong terminals.
- Exceeding the instrument's rated current or voltage range.
- Using a clamp sensor incorrectly or around multiple conductors when a single conductor is required.
- Recording a reading without checking units, mode, or connection direction.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Baking Mould

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 13 labs

Covers these lab items: Aluminium Cake pan; Baking Trays & Molds; Baking Trays, Moulds, Pans; Bread tin/ mould; Cake rings; Moulds/cutter; Muffin tray; One-piece and two-piece molds

What it is used for

Use a baking mould to shape batter or dough during baking while maintaining the required portion and product form.

How to use

1. Select a mould suitable for the recipe, oven, and required product size.
2. Inspect the mould for cracks, sharp edges, rust, or damaged coating.
3. Clean and dry the mould before use.
4. Apply suitable grease or lining if required by the recipe.
5. Fill the mould evenly without exceeding its recommended capacity.
6. Place the mould securely on an oven tray or rack using oven-safe protection.
7. Bake at the recipe's specified temperature and time.
8. Allow the mould to cool safely before removing the baked product.

⚠ Safety

- Use oven mitts when handling hot moulds and trays.
- Do not place moulds in an oven unless their material is oven-safe.
- Avoid sudden temperature changes that may warp or crack the mould.
- Follow food hygiene practices and avoid damaged non-stick coatings.

Care & storage

- Wash using a suitable cleaner and non-abrasive tools.
- Dry thoroughly and store without stacking in a way that damages the coating.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Baking Paper

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Parchment paper

What it is used for

Baking paper provides a suitable non-stick lining for baking trays and approved food-preparation tasks.

How to handle

- Confirm the paper is food-grade and suitable for the intended oven temperature and appliance.
- Cut to fit the tray and keep loose edges away from heating elements or open flames.
- Do not use baking paper in applications prohibited by the manufacturer, including unsuitable direct-flame exposure.
- Keep paper away from hot surfaces when placing or removing trays and use heat-resistant protection.
- Discard paper that is burnt, heavily soiled, or otherwise unsuitable for reuse.

Storage

- Store unused paper in its original packaging in a clean, dry area.
- Keep away from heat, moisture, and open flames.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bar Tool

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Bar tool kit; Cocktail Shaker

What it is used for

Use a bar tool kit or cocktail shaker to prepare and mix beverages according to the approved recipe and service procedure.

How to use

1. Inspect the shaker, jigger, strainer, spoon, and other tools for damage or residue.
2. Wash and sanitize food-contact tools before preparing a drink.
3. Measure ingredients using the recipe and appropriate measuring tool.
4. Add ingredients to the shaker only when the recipe calls for shaking.
5. Secure the lid and shake with controlled movements away from other people.
6. Strain or pour the drink into the correct serving vessel.
7. Clean, dry, and return the bar tools to storage after use.

⚠ Safety

- Keep glassware and sharp tools away from the edge of the work surface.
- Do not shake a damaged or poorly sealed container.
- Use clean ice and ingredients handled according to food safety rules.
- Follow workplace policies and legal requirements for beverage service.

Care & storage

- Wash and sanitize reusable tools promptly after use.
- Store tools dry and keep measuring equipment free from residue.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Blanching Unit

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Blanching Unit; Vegetable bleacher

What it is used for

A blanching unit briefly heats vegetables or other suitable food products in hot water or steam before cooling or further processing. Correct temperature, time, loading, and hygiene control help maintain product quality, support enzyme inactivation, and reduce food safety risks.

Main parts

- Heating chamber or tank holds the blanching medium or food product.
- Heating element, burner, or steam inlet supplies heat.
- Product basket, conveyor, or loading tray holds the food.
- Temperature sensor or thermometer monitors process temperature.
- Timer or controller sets the blanching cycle where fitted.
- Water inlet and drain support filling and emptying.
- Lid or cover helps retain heat and reduce contamination.
- Cooling or transfer arrangement moves the product to the next process where included.

Before you start

1. Only trained food-processing personnel should operate the blanching unit.
2. Read the operating manual and confirm the product-specific temperature, time, and loading requirements.
3. Inspect the tank, basket, lid, temperature sensor, controls, drains, and heating system for damage or leaks.
4. Clean and sanitise food-contact surfaces according to the facility's approved hygiene procedure.
5. Check water quality, steam supply, and any food-contact additives against applicable food safety requirements.
6. Verify that temperature and timing instruments are working and that guards and emergency controls are available.
7. Wear heat-resistant gloves, apron, non-slip footwear, and eye protection appropriate to the process.

How to use

1. Fill the unit with the specified water level or prepare the steam system according to the manual.
2. Start the heating system and allow the medium to reach the validated target temperature.
3. Prepare and inspect the food, removing unsuitable pieces and visible soil as required by the process.
4. Load the product basket or conveyor evenly without exceeding the rated capacity.

5. Lower or feed the product carefully to avoid splashing, burns, or mechanical trapping.
6. Start the timer when the process reaches the defined condition specified by the validated procedure.
7. Maintain the required temperature and exposure time; monitor the display and product flow throughout the cycle.
8. Remove the product using the designated basket, conveyor, or handling tool while avoiding contact with hot surfaces.
9. Transfer the blanched product promptly to the specified cooling or next processing stage.
10. Check representative product samples and record temperature, time, batch identity, and any deviations.
11. Stop heating, drain the unit where required, and allow equipment to cool safely before cleaning.

⚠ Safety

- Beware of hot water, steam, hot metal surfaces, and splashing; never reach into a running unit.
- Do not open pressurised steam equipment or remove guards while the system is operating.
- Use approved lifting tools and safe handling methods for heavy baskets or hot product loads.
- Keep electrical connections and controls protected from water and follow lockout procedures before maintenance.
- Prevent cross-contamination by separating raw, blanched, and cooled product handling areas.
- Follow the validated food safety process; blanching alone does not guarantee sterilisation or eliminate all pathogens.

Care & storage

- Clean and sanitise food-contact surfaces after production according to the approved sanitation schedule.
- Remove food residues from baskets, drains, and chamber areas without damaging sensors or seals.
- Inspect heating elements, steam lines, temperature sensors, gaskets, and controls regularly.
- Calibrate temperature and timing instruments at the prescribed intervals.
- Report leaks, inconsistent temperature, damaged guards, or control faults and arrange authorised repair.

Common mistakes to avoid

- Using an unvalidated temperature or blanching time for the product.
- Overloading the basket and causing uneven heat treatment.
- Delaying the required cooling or next processing step after blanching.
- Ignoring temperature deviations or failing to record batch process data.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Boiler

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Boiler

What it is used for

A boiler heats water or another approved fluid to provide hot water, steam, or process heat for kitchen, food-processing, or other industrial applications. Operate it only within its rated pressure, temperature, and fluid limits, and follow the approved startup, monitoring, and shutdown procedures to prevent scalding, pressure-related failures, and equipment damage.

Main parts

- Boiler vessel or heating chamber
- Burner, heating element, or heat source
- Feedwater inlet and level control
- Pressure gauge and temperature indicator
- Safety relief valve and pressure controls
- Blowdown or drain valve, piping, and control panel

Before you start

1. Confirm the boiler type, rated pressure, operating temperature, and intended process use.
2. Inspect the vessel, piping, valves, gauges, insulation, burner or heating element, and visible connections for damage or leaks.
3. Verify that the water level, feedwater supply, and water quality meet the manufacturer's requirements.
4. Check that the safety relief valve, pressure controls, alarms, and low-water protection are available and not obstructed.
5. Ensure the exhaust or ventilation system and required protective guards are in place.
6. Confirm that only trained, authorised operators are present and that the operating log and emergency procedure are available.

How to use

1. Review the previous operating log and confirm that no unresolved fault or maintenance restriction remains.
2. Check the feedwater supply, required fluid level, fuel or electrical supply, and control settings.
3. Confirm that all required valves are in their designated startup positions and that vents and drains are configured according to the approved procedure.
4. Start the boiler using the manufacturer's prescribed sequence and monitor the initial heating stage.

5. Observe pressure, temperature, water level, burner status, and alarms as the boiler approaches operating conditions.
6. Increase load gradually and remain within the rated operating limits and approved process requirements.
7. Check accessible piping and connections for leakage, abnormal vibration, unusual noise, or steam release.
8. Maintain the required water level and follow the specified water-treatment and blowdown schedule.
9. If a low-water condition, pressure excursion, abnormal noise, leak, or alarm occurs, follow the emergency procedure and shut down when required.
10. For normal shutdown, reduce the load and stop the heat source using the approved sequence.
11. Allow pressure and temperature to fall to the specified safe levels before opening drains, access covers, or connected equipment.
12. Record operating readings, water treatment, blowdown, alarms, inspections, and maintenance actions in the log.

⚠ Safety

- Only qualified and authorised boiler operators should start, operate, or service the equipment.
- Never exceed the boiler's rated pressure or temperature and never bypass safety interlocks or the relief valve.
- Never operate a boiler with inadequate water level, defective pressure controls, or a blocked relief path.
- Treat steam, hot water, surfaces, and piping as burn hazards; use suitable PPE and keep unauthorised people away.
- Do not open pressurised fittings, drains, or access covers; confirm safe pressure and temperature before opening as specified by the procedure.
- If a serious leak, pressure-control failure, or suspected vessel damage occurs, keep clear, isolate the unit only if safe, and contact qualified personnel or emergency responders.

Care & storage

- Inspect the vessel, burner or heating element, gauges, valves, piping, and insulation according to the maintenance schedule.
- Test safety devices, alarms, and low-water protection at the specified intervals using approved procedures.
- Maintain feedwater quality and water treatment to limit scale, corrosion, and deposits.
- Perform blowdown only at the prescribed intervals and according to the approved procedure.
- Arrange qualified inspection, cleaning, and servicing; keep records of tests, repairs, and operating hours.

Common mistakes to avoid

- Starting the boiler without checking water level or safety controls.
- Operating beyond rated pressure or temperature.

- Ignoring leaks, unusual noises, alarms, or unstable water level.
- Opening a drain or access cover before pressure and temperature have reached safe levels.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bottle

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Bottle

What it is used for

Handle bottles hygienically for their intended food or kitchen use.

How to handle

- Check that the bottle is clean, intact, and suitable for the intended contents.
- Use food-grade bottles for food or drinking water.
- Avoid filling hot liquids into containers not rated for high temperatures.
- Close caps securely and do not use bottles with cracks or contamination.

Storage

- Store clean, dry bottles in a designated area.
- Keep food-use bottles separate from chemical containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bottle Sealing Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Automatic bottle sealing unit; Soy milk bottle sealing machine manual

What it is used for

A bottle sealing machine closes bottles using a compatible cap, liner, or sealing film to help prevent leakage, contamination, and product spoilage. Correct setup, seal selection, hygienic handling, and seal-quality checks help maintain packaging integrity during storage and transport.

Main parts

- Machine frame supports the sealing assembly.
- Bottle holder or conveyor positions bottles for sealing.
- Capping head or sealing head applies the closure.
- Heating element or induction unit activates compatible sealing liners where fitted.
- Temperature and time controls regulate heat sealing where applicable.
- Pressure or height adjustment sets the sealing force or position.
- Control panel and switches operate the machine.
- Safety guards protect operators from moving or hot parts.
- Power supply and cables provide electrical connection.

Before you start

1. Only trained personnel should operate the bottle sealing machine.
2. Read the machine manual and confirm compatibility with the bottle, cap, liner, film, and product.
3. Inspect the frame, sealing head, guards, cables, controls, and bottle holder for damage.
4. Clean and sanitise food-contact parts according to the approved hygiene procedure.
5. Check that bottles and closures are clean, undamaged, and suitable for the intended product.
6. Verify the specified temperature, pressure, sealing time, and alignment settings for the packaging materials.
7. Wear appropriate PPE and keep the work area dry, clean, and free of obstructions.

How to use

1. Place the machine on a stable surface and connect power according to the manufacturer's instructions.
2. Install the correct sealing head, cap guide, or film holder for the packaging format.
3. Set the required temperature, sealing time, pressure, and height according to the validated packaging procedure.
4. Allow the heating system to reach its specified operating condition where applicable.
5. Fill and prepare bottles according to the product process, leaving the required headspace where specified.

6. Position the cap, liner, or sealing film correctly on the bottle opening.
7. Place the bottle in the holder or conveyor and align it with the sealing head.
8. Activate the sealing cycle using the designated control and keep hands clear of moving parts.
9. Remove the sealed bottle after the cycle finishes and the head has returned to its safe position.
10. Inspect the closure for alignment, wrinkles, incomplete bonding, leakage, or other defects.
11. Perform the prescribed seal-integrity test on representative samples and record process settings and results.
12. Stop the machine, isolate power, and allow hot components to cool before cleaning or adjustment.

⚠ Safety

- Keep hands and fingers away from the sealing head, moving parts, and hot surfaces.
- Never bypass guards, interlocks, or emergency stops.
- Do not seal containers that are cracked, unsuitable for the process, or contaminated around the sealing area.
- Use only approved packaging materials and settings compatible with the bottle and product.
- Disconnect power before clearing jams, changing components, or performing maintenance.
- Follow product-specific hygiene requirements and avoid contaminating bottle openings or closures.

Care & storage

- Clean and sanitise product-contact components after use according to the approved procedure.
- Remove adhesive, film, or product residue using manufacturer-approved cleaning methods.
- Inspect sealing heads, holders, guides, cables, and guards regularly.
- Check temperature and sealing controls at the prescribed maintenance intervals.
- Arrange authorised servicing if seals become inconsistent or the machine overheats.

Common mistakes to avoid

- Using an incompatible cap, liner, or sealing film.
- Applying incorrect temperature, pressure, or sealing time.
- Misaligning the bottle or closure during sealing.
- Skipping seal-integrity checks or cleaning between production runs.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bottle Washing Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bottle washing machine

What it is used for

A bottle washing machine cleans reusable bottles by using water, approved detergents, rinsing, and mechanical or spray action. Operate it according to the bottle type and sanitation procedure to remove residues and reduce contamination before filling.

Main parts

- Machine frame and bottle holders or conveyor
- Wash and rinse tanks or spray nozzles
- Water supply and drain connections
- Detergent dosing system, if fitted
- Pump, motor, and drive mechanism
- Control panel, guards, and emergency stop

Before you start

1. Confirm the bottle type, washing capacity, and required cleaning or sanitation procedure.
2. Inspect the conveyor, bottle holders, spray nozzles, hoses, tanks, guards, and electrical connections.
3. Remove broken glass, loose debris, and unsuitable bottles from the work area.
4. Check that water quality, detergent, and any sanitising agent are approved for the intended food-contact application.
5. Verify that wash and rinse tanks contain the required water and that drains are clear.
6. Wear suitable gloves, safety footwear, and eye protection when handling chemicals or glass bottles.

How to use

1. Switch off and isolate the machine before opening guards, clearing jams, or cleaning internal components.
2. Remove remaining liquid and loose debris from bottles using the approved pre-rinse procedure, where required.
3. Load bottles into the designated holders or conveyor with stable spacing and correct orientation.
4. Prepare detergent or cleaning solution at the concentration specified by the manufacturer and site procedure.
5. Select the correct wash cycle, temperature, pressure, and conveyor speed for the bottle type.
6. Close guards and start the machine using the approved controls.
7. Observe bottle movement, spray coverage, water flow, and machine operation during the cycle.

8. Check that bottles receive the required wash and rinse stages without damage or tipping.
9. Stop the machine if bottles jam, glass breaks, leakage occurs, or abnormal noise develops.
10. Allow the cycle to finish and unload bottles carefully into the designated clean area.
11. Inspect a representative sample for visible residue and verify required sanitation checks according to the facility procedure.
12. Drain and clean the machine as specified, then record cleaning results and any faults.

⚠ Safety

- Only trained operators should operate or maintain the machine.
- Never reach into moving conveyors, bottle holders, or spray mechanisms.
- Isolate power and any relevant water or chemical supply before clearing jams or servicing equipment.
- Use appropriate PPE when handling cleaning chemicals, hot water, or broken glass.
- Do not mix cleaning chemicals unless the approved procedure explicitly permits it.
- Keep electrical components protected from water and report damaged cables or guards immediately.

Care & storage

- Clean wash tanks, filters, spray nozzles, and bottle holders according to the sanitation schedule.
- Inspect hoses, seals, pumps, conveyor components, and chemical dosing equipment regularly.
- Remove mineral deposits and detergent residue using approved cleaning methods.
- Check temperature controls and dosing accuracy at specified intervals.
- Arrange servicing for leaks, poor cleaning performance, or repeated conveyor faults.

Common mistakes to avoid

- Loading bottles incorrectly or overloading the conveyor.
- Using the wrong detergent concentration or wash settings.
- Ignoring blocked spray nozzles or dirty rinse water.
- Handling broken glass without proper protection or reaching into a moving machine.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bread Proofer

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bread Proofer (Optional)

What it is used for

A bread proofer provides controlled warmth and humidity to help yeast dough rise before baking. Use the correct settings and proofing time for the dough and recipe to achieve consistent volume while preventing overproofing, contamination, and burns.

Main parts

- Insulated cabinet or proofing chamber
- Trays, racks, or trolley supports
- Heating element and temperature controller
- Humidity or water reservoir system, if fitted
- Circulation fan and ventilation openings
- Control panel, door seal, and power supply

Before you start

1. Confirm the dough type, batch size, and proofing requirements from the recipe or production procedure.
2. Inspect the chamber, racks, trays, door seal, controls, fan, and electrical cable for damage or contamination.
3. Clean and sanitise food-contact surfaces using approved food-safe methods.
4. Check the water reservoir or humidity system, if fitted, and ensure the drain is clear.
5. Verify that temperature and humidity controls operate correctly and that the chamber is free of cleaning chemicals.
6. Wash hands and use clean food-handling clothing and suitable PPE.

How to use

1. Switch on the proofer and set the temperature and humidity according to the recipe or approved process.
2. Allow the chamber to reach the required conditions before loading the dough, unless the process specifies otherwise.
3. Place covered or appropriately protected dough pieces on clean trays with enough space for expansion.
4. Load trays or racks evenly without blocking air circulation or overloading the chamber.
5. Close the door securely and start the proofing timer or follow the monitored production schedule.
6. Observe temperature, humidity, and dough development at the specified intervals.

7. Check dough size and readiness using the approved visual or process criteria rather than relying only on time.
8. Adjust the process only within the permitted recipe and equipment settings.
9. If abnormal temperature, excessive condensation, contamination, or equipment failure occurs, stop the process and notify the responsible operator.
10. Remove trays carefully using suitable heat protection if surfaces are warm.
11. Transfer proofed dough to the next production stage without unnecessary delay or contamination.
12. Switch off the unit when required, clean and dry the chamber, and record process deviations or maintenance needs.

⚠ Safety

- Only trained personnel should operate or clean the proofer.
- Do not touch heating elements, hot racks, or hot internal surfaces with bare hands.
- Keep water away from exposed electrical parts and disconnect power before maintenance.
- Use clean hands, utensils, and trays to prevent food contamination.
- Do not block ventilation openings or operate the unit with damaged electrical parts.
- Follow food-safety procedures for dough holding times and temperatures.

Care & storage

- Clean trays, racks, chamber walls, door seals, and water reservoirs regularly.
- Remove crumbs and moisture after shutdown to prevent mould and odours.
- Inspect door seals, fans, temperature sensors, and humidity controls for wear or faults.
- Descale or clean the water reservoir according to the manufacturer's instructions.
- Arrange servicing if temperature or humidity becomes unstable.

Common mistakes to avoid

- Loading dough before the chamber reaches the required conditions when preheating is specified.
- Setting temperature or humidity incorrectly for the recipe.
- Overloading trays or blocking air circulation.
- Leaving food residue or standing water inside the chamber.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cake Decorating Tool

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Icing comb

What it is used for

To create even patterns, textures, and decorative finishes in icing or frosting using an icing comb.

How to handle

- Inspect the icing comb for cracks, sharp damage, or residue before use.
- Wash and sanitize the tool using food-safe procedures, then dry it before decorating.
- Apply frosting evenly and hold the comb at a suitable angle against the cake surface.
- Rotate the cake or move the tool steadily to create a consistent pattern without excessive pressure.
- Clean the tool immediately after use and prevent contact with non-food chemicals.

Storage

- Wash, sanitize, and dry completely before storage.
- Store in a clean, covered food-grade utensil container away from dust and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Can Opener

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Can opener

What it is used for

To open food cans safely while minimizing sharp edges and food contamination.

How to handle

- Inspect the opener, cutting wheel, handle, and turning mechanism for damage or residue.
- Wash hands and ensure the can and opener are clean before use.
- Position the cutting wheel securely on the can rim and operate the handle steadily according to the design.
- Handle the lid carefully because its edges may be sharp; do not touch the cut edge with bare fingers.
- Transfer food using a clean utensil and wash the opener after use.

Storage

- Clean and dry the opener thoroughly to prevent residue buildup and corrosion.
- Store in a clean, dry utensil drawer with the cutting edge protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Can Sealer

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Can sealer

What it is used for

A can sealer closes compatible metal or other approved cans by forming a secure seam or closure. Correct can and lid matching, machine adjustment, hygienic handling, and seam inspection help prevent leakage, contamination, and product spoilage.

Main parts

- Machine frame supports the seaming mechanism.
- Can platform or chuck holds and positions the can.
- Seaming rollers form the closure seam.
- Chuck or seaming head supports the lid during sealing.
- Drive motor powers the sealing mechanism.
- Can height adjustment accommodates approved container sizes.
- Control switch or pedal starts and stops the machine.
- Safety guard protects the operator from moving components.
- Seam inspection tools check closure quality where provided.

Before you start

1. Only trained personnel should operate the can sealer.
2. Read the manual and confirm that the can body, lid, and seam tooling are compatible.
3. Inspect the frame, rollers, chuck, guards, controls, cables, and adjustment mechanisms for damage.
4. Clean and sanitise food-contact surfaces according to the facility hygiene procedure.
5. Check cans and lids for dents, corrosion, deformation, contamination, or other defects.
6. Verify roller settings, chuck size, can height, and operating speed against approved packaging specifications.
7. Ensure the machine is stable and that required guards and emergency stops function correctly.

How to use

1. Prepare the can and lid according to the approved filling and packaging procedure.
2. Check that the filled can has the specified headspace and that its rim is clean and undamaged.
3. Install the correct chuck and seaming tooling for the can and lid format with power isolated.
4. Set the can height, roller positions, and other adjustments according to the machine manual and packaging specification.
5. Place the lid squarely on the can without damaging the rim or contaminating the sealing surfaces.

6. Position the can securely on the platform or chuck and confirm correct alignment.
7. Close the guard and activate the seaming cycle using the designated control.
8. Keep hands clear while the rollers form the seam and allow the cycle to finish completely.
9. Remove the can only after the mechanism has returned to its safe position.
10. Inspect the seam for folds, sharp edges, gaps, looseness, or other visible defects.
11. Perform specified seam measurements or integrity tests at the required sampling frequency and record results.
12. Stop the machine and isolate power before cleaning, adjustment, jam clearing, or maintenance.

⚠ Safety

- Never place fingers near rotating rollers, the chuck, or the moving can platform.
- Do not operate with missing guards, faulty controls, or damaged seaming tooling.
- Never adjust rollers or clear a jam while the machine is running.
- Use only can bodies, lids, and tooling approved for the machine and product.
- Handle sharp or damaged can edges carefully and dispose of defective containers safely.
- Follow validated food-processing and closure-integrity requirements; a visually acceptable seam alone may not prove hermetic integrity.

Care & storage

- Clean and sanitise product-contact areas after production according to the approved procedure.
- Remove can fragments and residue only after isolating the machine.
- Inspect rollers, chuck, bearings, guards, and adjustment mechanisms for wear.
- Lubricate designated components only with the specified lubricant and method.
- Arrange authorised maintenance if seam quality changes or the machine develops abnormal noise or vibration.

Common mistakes to avoid

- Using mismatched can bodies, lids, chuck, or seaming rollers.
- Incorrect roller adjustment causing a loose or defective seam.
- Removing a can before the seaming cycle finishes.
- Relying only on visual inspection without required seam testing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chikki Cutter

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Hand Operated Chikki Slicer (Roller), cutter

What it is used for

To slice prepared chikki or similar brittle confectionery into uniform portions using a hand-operated roller cutter.

How to handle

- Inspect the cutter, roller, handle, guards, and cutting surface for damage and cleanliness.
- Place the equipment on a stable, clean, food-safe work surface and secure it as designed.
- Use the cutter only with prepared material suitable for the machine; keep fingers clear of the roller and cutting path.
- Operate the handle steadily and never push material by hand near moving cutting parts.
- Stop before clearing jams or cleaning, then wash and sanitize food-contact parts according to the manufacturer's instructions.

Storage

- Clean, sanitize, and dry all food-contact parts before storage.
- Store the cutter in a dry, covered area with moving and cutting parts protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chiller

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Chiller

What it is used for

A chiller removes heat from water, a process fluid, or an enclosed space to maintain a specified temperature. Correct operating settings, clean heat-transfer surfaces, proper flow, and routine monitoring help maintain cooling performance and protect equipment and product quality.

Main parts

- Compressor raises refrigerant pressure in vapour-compression chillers.
- Condenser rejects heat from the refrigerant.
- Evaporator transfers heat from the process fluid or air.
- Expansion device regulates refrigerant flow into the evaporator.
- Pump circulates chilled water or process fluid where fitted.
- Temperature sensors monitor supply, return, and system conditions.
- Control panel sets operating parameters and displays alarms.
- Fans or cooling-water components remove heat from the condenser where fitted.
- Filters and strainers help prevent debris from entering fluid circuits.

Before you start

1. Only trained personnel should operate or service the chiller.
2. Read the equipment manual and confirm the intended application, setpoint, fluid type, flow requirements, and operating limits.
3. Inspect the housing, piping, hoses, valves, electrical connections, insulation, and visible refrigerant lines for damage or leakage.
4. Check fluid level, fluid concentration where applicable, and the condition of filters or strainers.
5. Ensure the condenser has adequate airflow or cooling-water supply and that ventilation paths are clear.
6. Verify that pumps, fans, guards, emergency stops, and alarms are available and in working condition.
7. Confirm that electrical supply and protective earth meet equipment requirements; refrigerant-system service must be performed by qualified personnel.

How to use

1. Place the chiller in its approved operating location with the required clearance and ventilation.
2. Check that the process circuit is filled, vented, and ready for operation according to the manual.
3. Open the required valves and confirm that the fluid circuit has no visible leaks.

4. Switch on the main supply and inspect the control panel for alarms before starting the cooling cycle.
5. Set the required temperature, flow conditions, and other permitted parameters according to the process specification.
6. Start the pump or circulation system in the sequence specified by the manufacturer.
7. Start the cooling cycle and verify that the compressor, fans, or cooling-water system operate as intended.
8. Monitor supply and return temperatures, fluid flow, pressure indicators, and alarm messages during operation.
9. Check the cooled process or product for the required temperature and record operating readings.
10. If abnormal noise, leakage, low flow, overheating, or a persistent alarm occurs, stop the unit safely and investigate using approved procedures.
11. At shutdown, follow the manufacturer's sequence for stopping cooling, pumps, and the main supply.
12. Isolate power before cleaning, opening panels, clearing filters, or performing maintenance.

⚠ Safety

- Do not bypass pressure switches, flow interlocks, guards, or other protective devices.
- Do not touch electrical components with wet hands or operate damaged electrical equipment.
- Refrigerant leaks may create toxicity, flammability, frostbite, or oxygen-displacement hazards depending on refrigerant type; follow the specified emergency procedure.
- Do not open a pressurised refrigerant circuit or attempt repairs without appropriate qualifications and equipment.
- Avoid contact with hot surfaces, rotating fans, and moving pump components.
- Use compatible process fluids and follow site procedures for chemical handling, spills, and disposal.

Care & storage

- Clean air filters, condenser surfaces, and strainers at the prescribed intervals.
- Inspect pumps, fans, valves, hoses, insulation, and visible pipework for wear or leaks.
- Check fluid levels and quality according to the manufacturer's schedule.
- Verify temperature sensors, alarms, and control settings at the recommended service intervals.
- Arrange qualified maintenance for refrigerant-system checks and investigate recurring faults promptly.

Common mistakes to avoid

- Operating with insufficient fluid flow or an incorrectly filled circuit.
- Blocking condenser airflow or neglecting filter cleaning.
- Setting a temperature outside the equipment's rated operating range.
- Ignoring alarms, refrigerant leakage signs, or repeated compressor trips.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chopping Board

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Chopping Boards (Colour Coded); Cutting board

What it is used for

A chopping board provides a stable surface for cutting and preparing food ingredients.

How to handle

- Inspect the board for deep grooves, cracks, warping, and signs of contamination.
- Use colour-coded boards according to the kitchen's food-safety system to separate raw and ready-to-eat foods.
- Place the board on a stable surface and use a non-slip base where needed.
- Clean and sanitise the board between tasks and after handling raw meat, poultry, seafood, or allergens.
- Use knives carefully and replace boards that cannot be cleaned effectively.

Storage

- Wash, sanitise, and dry boards before storage.
- Store upright or in a ventilated rack so surfaces dry fully and remain protected from contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cookie Depositor

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Cookie Depositor / Moulds

What it is used for

A cookie depositor portions and places cookie dough onto trays or moulds in consistent shapes and quantities. Correct hopper loading, nozzle or mould setup, and portion adjustment help maintain product quality and hygienic food handling.

Main parts

- Food-grade dough hopper
- Dosing piston, pump, or portioning mechanism
- Depositing nozzle or mould assembly
- Tray conveyor or tray support
- Control panel and speed or portion controls
- Drive mechanism, guards, and removable food-contact parts

Before you start

1. Confirm the recipe, dough consistency, required portion weight, shape, and production quantity.
2. Verify that the machine is suitable for the dough type and that food-contact surfaces are clean and sanitised according to the food safety procedure.
3. Inspect the hopper, seals, nozzles, moulds, conveyor, guards, and electrical connections for damage or residue.
4. Ensure removable parts are assembled correctly and that seals and fittings are secure.
5. Check that the trays or moulds are clean, dry, correctly positioned, and suitable for the product.
6. Wear clean protective clothing and follow handwashing and allergen-control procedures.

How to use

1. Confirm that the machine is switched off before fitting nozzles, moulds, or other removable parts.
2. Install the required nozzle or mould and verify that it is securely locked in place.
3. Set the portion size, depositing speed, and conveyor settings according to the product specification.
4. Prepare the dough to the specified consistency and load it into the hopper without introducing foreign material.
5. Close the hopper cover and ensure the discharge path is clear before starting.
6. Place trays or moulds in the correct position and check that they align with the depositing outlet.
7. Start the machine using the approved control and produce a small test batch.
8. Inspect the first portions for weight, shape, spacing, and clean release from the nozzle or mould.
9. Adjust approved settings if portions are inconsistent, then repeat the test before full production.

10. Monitor hopper level, portion consistency, conveyor movement, and product accumulation during operation.
11. Stop the machine if a blockage, leak, unusual noise, or unsafe movement occurs.
12. Switch off and isolate power before removing a blockage or reaching into the depositing mechanism.
13. At the end of the run, stop the machine and transfer the product to the next process using hygienic handling practices.
14. Disassemble designated food-contact parts and clean and sanitise them according to the approved procedure.

⚠ Safety

- Only trained personnel should operate or clean the depositor.
- Keep hands away from moving pistons, nozzles, conveyors, and pinch points during operation.
- Never bypass guards or reach into the hopper while the machine is running.
- Switch off and isolate power before clearing blockages, changing moulds, or performing maintenance.
- Use food-safe cleaning chemicals as directed and prevent cleaning residues from contacting ready-to-use product.
- Follow allergen controls and prevent cross-contamination between recipes.

Care & storage

- Clean and sanitise the hopper, nozzles, moulds, seals, and other food-contact parts after use.
- Inspect seals, hoses, fasteners, and dosing components for wear or leakage.
- Check conveyor alignment and portioning accuracy at scheduled intervals.
- Store clean, dry removable parts in a protected hygienic area.
- Arrange servicing if portioning becomes inconsistent, the drive overheats, or product leaks from seals.

Common mistakes to avoid

- Using dough with unsuitable consistency for the selected nozzle or mould.
- Failing to clean or sanitise food-contact surfaces correctly.
- Starting production before verifying portion weight and spacing.
- Removing blockages while the machine is energised.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cookware

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **High** 10 labs

Covers these lab items: 2 litre vessel; Cooking Utensils; Frying Pan (medium)- With Handle; Frying Pans (Non-stick & Cast Iron); Ladles, Spoons, Tongs; Rasgulla/Rabri Cooking Vessel; S.S. Patila; Stainless Steel Cooking Pots & Pans; Stainless Steel Utensils

What it is used for

Cookware is used to prepare, heat, fry, and cook food in kitchens or training laboratories. Choose suitable cookware for the food, heat source, and cooking method.

How to use

1. Inspect cookware, handles, and surfaces for cracks, loose parts, or damage.
2. Wash cookware and utensils with suitable detergent and clean water.
3. Place the cookware securely on a compatible cooking surface.
4. Add ingredients and use appropriate heat settings for the recipe.
5. Use suitable utensils and handle hot cookware with heat-resistant protection.
6. Check that food is cooked properly and avoid overheating empty cookware.
7. Turn off the heat source and allow cookware to cool before cleaning.
8. Wash, dry, and return cookware to its designated storage place.

⚠ Safety

- Use oven mitts or heat-resistant gloves when handling hot cookware.
- Keep handles away from flame edges and walkways.
- Do not use damaged non-stick coatings or cookware unsuitable for the heat source.
- Keep water away from hot oil to prevent dangerous splashing.

Care & storage

- Clean cookware using methods suitable for its material and coating.
- Dry thoroughly and store neatly to prevent rust and surface damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cooling Rack

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Cooling racks

What it is used for

Handle cooling racks safely and maintain food hygiene.

How to handle

- Place the rack on a stable, heat-resistant surface.
- Use heat-resistant gloves when handling hot trays or racks.
- Clean and dry the rack before and after use.
- Avoid overloading or placing unstable items on the rack.

Storage

- Store clean and dry in a designated kitchen area.
- Keep away from chemicals and floor-level contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Crown Corking Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Crown corking machine

What it is used for

A crown corking machine seals compatible glass bottles using metal crown caps to create a secure closure. Correct cap and bottle matching, controlled capping pressure, hygienic handling, and closure inspection help reduce leakage, contamination, and breakage.

Main parts

- Machine frame supports the capping assembly.
- Capping head presses the crown cap onto the bottle finish.
- Cap holder or magnetic head positions the crown cap where fitted.
- Bottle support or platform holds the bottle upright.
- Height adjustment accommodates approved bottle sizes.
- Lever, pneumatic actuator, or motor provides capping force.
- Control switch or foot pedal operates powered models.
- Safety guard protects against moving parts where fitted.
- Crown caps provide the bottle closure.

Before you start

1. Only trained personnel should operate the crown corking machine.
2. Read the manual and confirm that the crown caps and bottle finish are compatible with the machine.
3. Inspect the frame, capping head, bottle platform, controls, guards, and drive components for damage.
4. Clean and sanitise bottle-contact surfaces according to the approved hygiene procedure.
5. Check bottles for cracks, chips, contamination, and damaged neck finishes; reject defective bottles.
6. Confirm that caps are clean, undamaged, and suitable for the intended beverage and closure requirements.
7. Adjust the capping height and pressure according to the machine instructions and bottle specifications.

How to use

1. Place the machine on a stable surface and prepare the required bottle and crown cap.
2. Position the bottle upright on the designated support and align its neck under the capping head.
3. Place the crown cap squarely on the bottle mouth using the approved cap-loading method.
4. Keep hands clear of the capping head and activate the lever, pedal, or control as designed.

5. Apply the capping cycle until the cap is fully seated and the machine returns to its safe position.
6. Remove the bottle carefully and inspect the cap for uniform seating, crimping, and visible damage.
7. Check for leakage or closure defects using the approved quality-control method.
8. Repeat the process while monitoring cap placement, machine performance, and bottle condition.
9. Stop the machine if a bottle cracks, a cap jams, or the capping mechanism behaves abnormally.
10. At the end of the run, switch off and isolate power before clearing jams or cleaning the machine.
11. Clean and sanitise appropriate surfaces and record inspection results according to production procedures.

⚠ Safety

- Keep fingers away from the capping head, press mechanism, and bottle support during operation.
- Do not use cracked or chipped bottles because they may break under capping force.
- Never bypass guards or reach into a powered machine to clear a jam.
- Use the specified cap type and machine settings; excessive force may damage bottles or closures.
- Handle broken glass with suitable tools and protective equipment, never with bare hands.
- Follow beverage hygiene requirements and prevent contamination of bottle mouths and caps.

Care & storage

- Clean and sanitise bottle-contact surfaces after production using approved methods.
- Inspect the capping head, cap holder, adjustment mechanism, guards, and fasteners regularly.
- Remove cap fragments and broken glass only after isolating the machine and following site procedures.
- Keep the machine dry and protect it from corrosion and impact.
- Arrange authorised servicing if cap seating becomes inconsistent or the mechanism wears.

Common mistakes to avoid

- Using caps that do not match the bottle finish or machine head.
- Misaligning the bottle or cap before applying pressure.
- Applying excessive force and cracking the bottle neck.
- Failing to inspect closures or safely remove broken glass.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cup Sealing Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Cup Sealer; Leg cup sealing machine

What it is used for

A cup sealing machine closes compatible cups with a heat-sealed film or lid to help prevent spills, contamination, and product leakage. Correct film selection, temperature and time settings, hygienic filling, and seal inspection help maintain packaging quality for food and beverages.

Main parts

- Machine frame supports the sealing assembly.
- Cup holder or mould positions the cup.
- Film roll or lid holder supplies sealing material.
- Heating plate or sealing head bonds the film to the cup rim.
- Temperature controller regulates sealing heat.
- Timer or cycle control sets sealing duration where fitted.
- Film feed and cutting mechanism advances and trims the film on applicable models.
- Control panel and switches operate the machine.
- Safety guard protects against moving parts and hot surfaces where fitted.

Before you start

1. Only trained personnel should operate the cup sealing machine.
2. Read the manual and confirm compatibility between cup material, rim shape, film, and product.
3. Inspect the frame, sealing head, heater, film feed, cutter, guards, and controls for damage.
4. Clean and sanitise food-contact surfaces according to the approved hygiene procedure.
5. Check that cups are undamaged, correctly filled, and free of product residue around the rim.
6. Verify the recommended sealing temperature, time, pressure, and film alignment for the packaging materials.
7. Wear suitable heat-resistant PPE where required and keep hands clear of the hot sealing head.

How to use

1. Place the machine on a stable, dry surface and connect power according to the manual.
2. Install the compatible film roll or lid and thread it through the feed mechanism as instructed.
3. Set the sealing temperature, time, and other permitted parameters according to the validated packaging procedure.
4. Allow the heating plate to reach its specified operating temperature where applicable.
5. Fill the cup according to the product process and keep the sealing rim clean and dry.
6. Position the cup correctly in the holder or mould and align the film over the rim.
7. Activate the sealing cycle and keep hands away from the heater, cutter, and moving components.

8. Allow the cycle to finish before removing the sealed cup from the holder.
9. Inspect the seal for wrinkles, incomplete bonding, misalignment, film damage, and visible leakage.
10. Perform the specified seal-integrity checks on representative cups and record the results.
11. Stop the machine and isolate power before clearing film jams, cleaning the sealing head, or servicing components.
12. Clean and sanitise the required surfaces after production and store unused packaging material in a clean, protected area.

⚠ Safety

- Keep fingers away from the hot sealing plate, cutter, and moving film-feed parts.
- Do not operate the machine with damaged wiring, faulty controls, or missing guards.
- Never reach into the machine to clear a jam while it is powered.
- Use only cups and films rated for the product and sealing process.
- Avoid overheating packaging materials and follow manufacturer limits for temperature and cycle time.
- Maintain food hygiene and prevent contamination of cup rims, film, and finished products.

Care & storage

- Switch off, isolate power, and allow hot parts to cool before cleaning.
- Remove film or product residue using manufacturer-approved methods that do not damage the sealing surface.
- Inspect the heater, cutter, film rollers, cup holder, guards, and cables regularly.
- Check temperature controls and sealing performance at the prescribed intervals.
- Arrange authorised maintenance if sealing becomes inconsistent or the heater fails to regulate temperature.

Common mistakes to avoid

- Using film or cups that are incompatible with the sealing process.
- Allowing liquid or food residue to remain on the cup rim.
- Setting incorrect temperature, time, or pressure and producing weak seals.
- Removing a cup before the cycle finishes or cleaning while the heater is hot.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Decanter

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Decanter Large; Decanter Small

What it is used for

Use a decanter to transfer, serve, or temporarily hold suitable liquids in a clean container designed for that purpose.

How to use

1. Inspect the decanter for cracks, chips, residue, or an unstable base.
2. Wash and dry the decanter before filling it.
3. Confirm the decanter material is suitable for the liquid being handled.
4. Pour the liquid slowly using a stable grip and avoid overfilling.
5. Place the decanter on a level, stable surface away from edges.
6. Pour or transfer the contents carefully when required.
7. Empty, wash, and dry the decanter after use.

⚠ Safety

- Do not use cracked or chipped glass decanters.
- Keep the decanter away from children, walkways, and unstable surfaces.
- Do not use a decanter for chemicals or food unless it is approved for that purpose.
- Handle hot liquids only in a container rated for the temperature.

Care & storage

- Wash with a suitable cleaner and rinse thoroughly.
- Store dry in a secure place where it cannot tip or break.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Deep Freezer

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Deep Freezer; Freezers; Vertical four door Freezer

What it is used for

A deep freezer keeps food and other approved materials at low temperatures to maintain quality and slow spoilage. Correct temperature control, hygienic loading, and regular monitoring help protect stored items and ensure reliable operation.

Main parts

- Insulated cabinet and door maintain the cold storage environment.
- Compressor circulates refrigerant through the cooling system.
- Evaporator removes heat from the freezer compartment.
- Condenser releases heat from the refrigerant.
- Thermostat or digital controller regulates the set temperature.
- Shelves, baskets, or racks organise stored items.
- Door gasket and latch help maintain an airtight seal.
- Temperature display and alarm indicate operating conditions on equipped models.

Before you start

1. Read the user manual and identify the required storage temperature for the stored materials.
2. Check that the freezer is level, ventilated, and positioned away from heat sources.
3. Inspect the power cord, plug, door gasket, shelves, and cabinet for damage.
4. Confirm the interior is clean, dry, and free from unwanted odours.
5. Verify the temperature display and alarm function where available.
6. Prepare labelled, sealed containers and ensure items are suitable for freezer storage.

How to use

1. Connect the freezer to a suitable power supply according to the manual.
2. Switch on the freezer and set the recommended temperature.
3. Allow the compartment to reach the required temperature before loading items.
4. Arrange labelled containers with space for air circulation and avoid blocking vents.
5. Close the door firmly and check that the gasket seals properly.
6. Record the temperature at the required monitoring intervals.
7. Minimise door opening and return items promptly after each retrieval.
8. Remove frost using the approved defrosting procedure when necessary.
9. If cleaning or defrosting requires shutdown, transfer stored items to suitable temperature-controlled storage first.

10. Clean and dry the compartment, restore power if required, and verify the temperature before reloading.

⚠ Safety

- Never touch electrical plugs or controls with wet hands.
- Do not use the freezer if the cord, plug, or cabinet wiring is damaged.
- Keep ventilation openings clear and do not obstruct the condenser area.
- Never chip ice with sharp tools or use open flames to defrost the freezer.
- Store food in sealed, labelled containers and separate incompatible materials.
- Follow safe lifting practices when loading heavy containers or shelves.

Care & storage

- Clean interior surfaces and removable shelves using approved cleaning materials.
- Inspect the door gasket regularly and clean it to maintain a proper seal.
- Keep the condenser and ventilation areas clear of dust according to the manual.
- Arrange authorised servicing if cooling performance, alarms, or electrical operation becomes abnormal.

Common mistakes to avoid

- Loading warm items before the freezer reaches its required temperature.
- Overfilling shelves and blocking internal air circulation.
- Leaving the door open or ignoring temperature alarms.
- Removing ice with sharp tools or unsuitable heating methods.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dehuller

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: De-huller; Dehulling Machine

What it is used for

A dehuller removes the outer husk or hull from suitable grains, pulses, or seeds using a controlled mechanical process. Correct crop preparation, machine adjustment, feed control, and safe separation of product and husk help improve processing quality and reduce waste and injury risks.

Main parts

- Feed hopper receives the crop material.
- Feed gate or metering device regulates material flow.
- Dehulling chamber contains the working mechanism.
- Abrasive surface, rollers, blades, or impact mechanism loosens the hull depending on the model.
- Drive motor and transmission power the working mechanism.
- Outlet discharges the dehulled product.
- Husk outlet or separator removes separated hulls where fitted.
- Screen or sieve separates product fractions where fitted.
- Guards and emergency stop protect the operator.

Before you start

1. Only trained personnel should operate the dehuller.
2. Read the manual and confirm that the machine is suitable for the crop type, moisture condition, and required output.
3. Inspect the hopper, chamber, drive, belts, rollers, screens, outlets, guards, and electrical connections for damage.
4. Ensure the machine is clean and free from residue or foreign material from previous batches.
5. Confirm that all guards are fitted, the emergency stop functions, and the machine is securely mounted.
6. Set the working clearance, speed, and feed rate according to the crop and manufacturer guidance.
7. Wear suitable eye protection, hearing protection, safety footwear, and respiratory protection where dust exposure is possible.

How to use

1. Prepare and inspect the crop, removing stones, metal, and other foreign objects using the approved pre-cleaning method.
2. Place suitable collection containers at the product and husk outlets.
3. Ensure the discharge paths are open and all covers and guards are secured.

4. Start the dust extraction system where fitted and switch on the machine according to the manual.
5. Allow the drive to reach its specified operating condition before feeding material.
6. Feed the crop steadily through the hopper without exceeding the rated capacity.
7. Observe the dehulling process, product flow, husk separation, noise, and vibration during operation.
8. Adjust feed rate or approved machine settings only when permitted and using the safe procedure.
9. Stop feeding before shutdown and allow material inside the chamber to clear as instructed.
10. Switch off and isolate power, then wait for all moving parts to stop completely.
11. Collect and inspect the dehulled product; check hull removal, broken kernels, and product quality against process requirements.
12. Clean the hopper, chamber, screens, and outlets after isolation and record production results or faults.

⚠ Safety

- Never reach into the hopper, chamber, or outlet while the machine is operating.
- Do not operate with missing guards, damaged belts, loose parts, or a faulty emergency stop.
- Isolate and lock out power before clearing blockages, changing screens, or performing maintenance.
- Control grain dust using suitable extraction and housekeeping; airborne dust may cause respiratory irritation and, in some conditions, a combustible dust hazard.
- Keep hands, hair, clothing, and tools away from rotating and moving components.
- Do not process wet, contaminated, or unsuitable material beyond the machine's approved limits.

Care & storage

- Remove crop residue and dust after each batch using an approved cleaning method.
- Inspect rollers, abrasive surfaces, screens, belts, bearings, and fasteners for wear or damage.
- Lubricate designated bearings and drive components according to the manufacturer's schedule.
- Keep the machine dry and protect electrical parts from dust and moisture.
- Arrange authorised maintenance if dehulling quality drops or abnormal noise, vibration, or overheating occurs.

Common mistakes to avoid

- Feeding material faster than the machine can process it.
- Using incorrect clearance or speed for the crop type.
- Allowing stones or metal to enter the dehulling chamber.
- Cleaning or clearing blockages before isolating power and confirming all parts have stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dishwasher

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Dishwasher

What it is used for

A dishwasher cleans dishes, utensils, and food-service equipment using controlled washing, rinsing, and, where fitted, sanitising cycles. Correct loading, detergent selection, and maintenance help achieve hygienic results while protecting dishes and the machine.

Main parts

- Wash chamber and racks
- Spray arms or wash jets
- Water inlet, filters, and drain system
- Detergent and rinse-aid dispensers
- Heating element and temperature controls
- Control panel, door latch, and safety interlocks

Before you start

1. Confirm that the dishwasher is intended for the type and quantity of items being washed.
2. Check that the chamber, racks, spray arms, filters, and drain area are clean and correctly fitted.
3. Verify that water supply, drainage, power connection, and door seal are in good condition.
4. Use the detergent and rinse aid recommended for the machine and follow the product instructions.
5. Scrape large food scraps from dishes and separate items according to material and washing requirements.
6. Check that items are dishwasher-safe and that sharp knives or fragile glassware are positioned securely.

How to use

1. Switch on the dishwasher and select the appropriate wash programme for the load and soil level.
2. Load dishes into the correct racks without overcrowding or blocking spray arms.
3. Place cups, bowls, and containers so water can drain and reach all required surfaces.
4. Add the specified detergent and rinse aid to their designated compartments.
5. Close the door securely and confirm that no item prevents the door from latching.
6. Start the selected cycle and allow it to complete without opening the door unnecessarily.
7. Observe any warning indicators and stop the machine if a leak, electrical fault, or unusual noise occurs.
8. When the cycle finishes, wait for hot items and steam to settle before opening the door carefully.

9. Unload clean items with clean hands, handling them by appropriate surfaces to avoid recontamination.
10. Inspect items for remaining food, detergent residue, or water deposits; rewash items that are not clean.
11. Remove and clean the filter as instructed, then check the drain area for trapped debris.
12. Switch off the machine when required and leave it in the recommended condition for ventilation and drying.

⚠ Safety

- Follow the manufacturer's instructions and use only detergents suitable for the machine.
- Do not touch heating elements or handle hot dishes carelessly after a cycle.
- Keep electrical connections and controls dry; never operate the machine with damaged wiring or plugs.
- Do not bypass the door latch or safety interlocks.
- Switch off and isolate power before servicing or accessing electrical components.
- Use care when handling detergent and rinse aid; avoid contact with eyes and skin and follow the product label.
- Do not mix cleaning chemicals unless the product instructions explicitly permit it.

Care & storage

- Clean the filters, spray arms, door seal, and chamber at the recommended frequency.
- Check inlet hoses, drain hoses, and connections for leaks or blockage.
- Remove scale using a suitable descaling product when required by water conditions and manufacturer guidance.
- Keep detergent dispensers and rinse-aid compartments clean and correctly filled.
- Arrange servicing if washing performance declines, heating fails, or drainage becomes unreliable.

Common mistakes to avoid

- Overloading racks or blocking spray arms with dishes.
- Using unsuitable detergent or excessive detergent quantity.
- Ignoring clogged filters or blocked drainage.
- Opening the machine carelessly while hot steam or water may escape.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dough Kneader

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Dough Kneading Machine; Dough Mixer / Blender; Spiral mixer

What it is used for

A dough kneader mixes and develops dough ingredients to achieve a consistent texture for food preparation. Correct loading, speed selection, and hygienic cleaning help produce uniform dough while reducing mechanical and food-safety risks.

Main parts

- Mixing bowl holds the flour, water, and other ingredients.
- Kneading hook or mixing tool combines and develops the dough.
- Drive motor supplies power to the mixing mechanism.
- Speed selector or control panel sets the operating speed.
- Bowl guard or safety cover helps prevent access to moving parts.
- Bowl locking mechanism secures the bowl during operation.
- Timer controls the mixing duration on equipped models.
- Frame and base support the machine during kneading.

Before you start

1. Read the operating manual and confirm the machine capacity for the planned batch.
2. Inspect the bowl, kneading tool, guard, power cord, and controls for damage.
3. Ensure the bowl and food-contact parts are clean and dry.
4. Measure ingredients according to the approved recipe and batch size.
5. Check that the kneading tool and bowl are correctly fitted and locked.
6. Clear the work area and keep loose clothing, jewellery, and hands away from moving parts.

How to use

1. Switch off the machine before installing the bowl or kneading tool.
2. Add ingredients in the order specified by the recipe or manufacturer.
3. Secure the bowl and fit the safety guard before operation.
4. Select the recommended speed and mixing time for the dough type.
5. Start the machine and monitor the dough through the guard without reaching inside.
6. Stop the machine before scraping the bowl or adjusting ingredients.
7. Check dough consistency and continue kneading only for the required time.
8. Switch off and isolate the power before removing the bowl or kneading tool.
9. Transfer the dough using clean food-safe utensils and avoid contact with non-food surfaces.
10. Clean and dry the bowl, kneading tool, guard, and food-contact areas after use.

⚠ Safety

- Never put hands or utensils into the bowl while the machine is running.
- Do not bypass the bowl guard or safety interlock.
- Switch off and disconnect power before clearing jams or changing attachments.
- Keep the machine away from water near electrical components.
- Do not exceed the manufacturer's batch capacity or recommended operating time.
- Use safe lifting practices when handling heavy bowls or dough batches.

Care & storage

- Wash food-contact parts using approved food-safe cleaning methods.
- Dry components thoroughly before reassembly or storage.
- Inspect the kneading tool, bowl lock, guard, and power cord regularly.
- Arrange authorised servicing if the motor overheats, makes unusual noise, or fails to stop correctly.

Common mistakes to avoid

- Overfilling the bowl beyond the rated batch capacity.
- Starting the machine before securing the bowl and guard.
- Changing ingredients or clearing dough while the machine is running.
- Leaving moisture or dough residue on parts after cleaning.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dough Scraper

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Dough scrapers and plastic; Dough scrappers and plastic

What it is used for

To divide, lift, scrape, and transfer dough or other suitable food ingredients during preparation.

How to handle

- Inspect the scraper for cracks, sharp damage, residue, or a loose handle.
- Wash hands and clean the scraper using food-safe procedures before use.
- Use the flat edge to portion dough or scrape a work surface with controlled movements.
- Avoid forceful scraping that could damage the worktop; keep fingers clear of the edge.
- Wash, rinse, sanitize when required, and dry the scraper after use.

Storage

- Ensure the scraper is completely clean and dry before storage.
- Store in a clean, food-grade utensil holder or drawer away from chemicals and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dough Sheeter

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Dough sheeter; Sheeter / Cutter

What it is used for

A dough sheeter rolls dough into a uniform thickness for baking, pastry, and other food preparation tasks. Correct adjustment, hygienic handling, and safe operation help produce consistent sheets while reducing contamination and injury risks.

Main parts

- Rollers flatten dough to the selected thickness.
- Thickness adjustment sets the gap between rollers.
- Feed tray supports dough as it enters the machine.
- Conveyor belt moves dough through the rollers where fitted.
- Cutting attachment cuts dough into required shapes where fitted.
- Drive motor powers the rollers on powered models.
- Control switch starts and stops the machine.
- Safety guard and emergency stop protect the operator where fitted.

Before you start

1. Only trained personnel should operate the dough sheeter.
2. Read the operating manual and confirm the machine is suitable for the dough being processed.
3. Inspect rollers, feed tray, guards, conveyor, controls, and power cable for damage.
4. Ensure the machine is stable and all guards and safety devices are fitted.
5. Clean food-contact surfaces using approved food-safe cleaning procedures before use.
6. Keep hands, jewellery, loose clothing, and tools away from the roller feed area.

How to use

1. Wash hands and prepare the dough according to the recipe or work instructions.
2. Check that the rollers and feed tray are clean, dry, and free of foreign material.
3. Set the initial roller gap according to the dough type and required thickness.
4. Switch on the machine using the designated control.
5. Place a suitable portion of dough on the feed tray without reaching into the rollers.
6. Guide the dough using approved handling tools or the designated feed method.
7. Adjust the roller gap gradually between passes until the required thickness is reached.
8. Use the cutting attachment only when fitted, guarded, and approved for the task.
9. Stop the machine if dough jams, the rollers behave abnormally, or the guard shifts.
10. Switch off and isolate power before cleaning, clearing jams, or changing attachments.
11. Collect the finished dough sheets hygienically and inspect them for consistent thickness.

2. Clean the machine and record any faults according to workplace procedures.

⚠ Safety

- Never push dough into rollers with fingers or hands.
- Do not bypass guards, interlocks, or the emergency stop.
- Switch off and isolate power before clearing jams or cleaning near rollers and cutters.
- Keep loose clothing, jewellery, and long hair away from moving parts.
- Do not operate the machine with damaged rollers, exposed wiring, or faulty controls.
- Follow food hygiene rules and prevent cleaning chemicals from contaminating food-contact surfaces.

Care & storage

- Remove dough residue promptly using the approved cleaning method.
- Clean and sanitise food-contact surfaces according to the food safety procedure.
- Inspect rollers, belts, cutters, guards, and cables regularly.
- Arrange authorised maintenance if rollers become uneven, noisy, or difficult to adjust.

Common mistakes to avoid

- Setting the roller gap too narrow for the initial dough pass.
- Pushing dough into moving rollers by hand.
- Cleaning the machine while it remains connected to power.
- Using the machine with dirty food-contact surfaces or missing guards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drying Chamber

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Drying chambers or controlled rooms

What it is used for

A drying chamber or controlled drying room removes moisture from materials by maintaining specified temperature, airflow, and sometimes humidity conditions. Correct loading, monitoring, and process control help achieve uniform drying while protecting product quality and preventing overheating or contamination.

Main parts

- Insulated chamber or drying-room enclosure
- Heating system and temperature controller
- Air circulation fans and ducts
- Temperature and humidity sensors, where fitted
- Shelves, racks, trays, or trolleys
- Door seals, ventilation outlets, and control panel

Before you start

1. Confirm the material type, batch size, target moisture level, and approved drying temperature and duration.
2. Check that the chamber is clean, dry, and suitable for the product; verify that previous residues or contaminants have been removed.
3. Inspect heaters, fans, ducts, door seals, sensors, electrical connections, and safety controls for damage.
4. Verify that temperature and humidity indicators are functioning and calibrated where required.
5. Arrange material evenly on clean trays or racks, leaving enough space for airflow and avoiding overloading.
6. Confirm that the ventilation route is clear and that the material is safe to heat under the selected process conditions.

How to use

1. Switch on the chamber and set the temperature, airflow, humidity, and timer according to the approved process specification.
2. Allow the chamber to reach the required operating conditions and verify the displayed readings before loading, where the procedure requires preheating.
3. Load the prepared trays or racks carefully and ensure that air can circulate around the material.
4. Close the door securely and confirm that door seals and safety interlocks are working.
5. Start the drying cycle and record the batch identification, start time, setpoints, and initial readings.

6. Monitor temperature, humidity where applicable, airflow, and product condition at the specified intervals.
7. Rotate or reposition trays only if the approved procedure requires it and it can be done safely.
8. Adjust settings only within the authorised process limits if drying is uneven or readings deviate from the specification.
9. Stop the process if there is smoke, unusual odour, overheating, fan failure, electrical fault, or unexpected temperature rise.
10. At the end of the cycle, switch off heating and follow the procedure for cooling and ventilation before opening the chamber.
11. Remove the material using suitable gloves or handling tools if trays or products remain hot.
12. Check final moisture content or other specified quality parameters and record the results before releasing the batch.
13. Switch off the chamber when required, remove residues after cooling, and clean the chamber and trays using the approved method.

⚠ Safety

- Only trained personnel should operate the drying chamber and set process parameters.
- Do not exceed the chamber's rated temperature or use it for materials not approved for the equipment.
- Keep air inlets, outlets, and circulation fans unobstructed.
- Use suitable heat-resistant gloves and other required PPE when handling hot trays or materials.
- Do not bypass door interlocks, over-temperature cut-outs, or other safety controls.
- Isolate power before maintenance, cleaning near electrical components, or servicing heaters and fans.
- For flammable, reactive, or solvent-containing materials, use only equipment and procedures specifically approved for those hazards.

Care & storage

- Clean trays, racks, chamber surfaces, and air passages at the approved frequency.
- Inspect door seals, fans, heaters, sensors, and control wiring for wear or damage.
- Verify temperature and humidity sensors periodically using suitable calibrated instruments.
- Keep ventilation paths clear and remove dust or residue that could obstruct airflow.
- Arrange servicing if temperature control becomes unstable, airflow weakens, or safety cut-outs fail.

Common mistakes to avoid

- Overloading trays and restricting airflow through the chamber.
- Using incorrect temperature or drying time for the material.
- Ignoring sensor readings, uneven drying, or signs of overheating.
- Opening the chamber before following the required cooling and safe-handling procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the

manufacturer's original manual / file.

Electric Stove

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Electric Stove

What it is used for

An electric stove converts electrical energy into heat for cooking or heating food in suitable cookware. Correct cookware selection, controlled heat settings, and safe electrical handling help ensure efficient cooking while reducing burn, shock, and fire hazards.

Main parts

- Heating element or hot plate
- Cooking surface or burner assembly
- Temperature and power controls
- Indicator light or display
- Power cord and plug or fixed electrical connection
- Housing, feet, and protective thermal components

Before you start

1. Confirm that the stove is designed for the available electrical supply and is placed on a stable, level, heat-resistant surface.
2. Inspect the heating surface, controls, power cord, plug, and housing for cracks, exposed wires, or other damage.
3. Ensure the surrounding area is dry, well ventilated, and clear of combustible materials.
4. Select cookware suitable for the stove type, with a flat base and an appropriate size for the heating zone.
5. Check that cookware handles are secure and positioned away from walkways and hot surfaces.
6. Keep a suitable lid, heat-resistant gloves, and appropriate fire safety equipment accessible.

How to use

1. Place the stove on a stable, level surface and keep its ventilation openings clear where applicable.
2. Connect the stove to the approved electrical supply without overloading sockets or using damaged extension leads.
3. Place the filled cookware centrally on the designated heating zone before switching on where the model requires it.
4. Switch on the stove and select a suitable heat setting for the cooking task.
5. Allow the cookware to heat gradually and monitor the food to prevent overheating or boiling over.
6. Adjust the heat as required and use suitable utensils to stir or handle the food.

7. Keep pot handles turned inward where safe and avoid leaving hot cookware near the edge of the surface.
8. If liquid spills onto the stove, switch it off and disconnect power if safe to do so; allow the unit to cool before cleaning.
9. If the stove emits smoke, sparks, an unusual smell, or trips the electrical protection, switch it off and stop using it.
10. When cooking is complete, switch the controls off and disconnect the power if required by the manufacturer's instructions.
11. Treat the cooking surface and cookware as hot until fully cooled; use suitable heat-resistant gloves when needed.
12. After cooling, clean food spills using the method approved for the stove type and avoid wetting electrical parts.
13. Store the stove in a dry location and inspect it before the next use.

⚠ Safety

- Use the stove according to the manufacturer's instructions and its rated electrical supply.
- Never operate the stove with damaged wiring, plugs, controls, or a cracked cooking surface.
- Keep water and wet hands away from plugs, sockets, and electrical controls.
- Do not leave cooking unattended when heat, oil, or combustible materials are present.
- Keep flammable items, paper, and loose cloth away from hot surfaces.
- Do not touch heating elements or hot cookware with bare hands; allow sufficient cooling time.
- Do not immerse the appliance or its electrical components in water unless the manufacturer explicitly permits it.
- If an electrical fire occurs, disconnect power only if safe and never throw water onto energised electrical equipment or burning oil.

Care & storage

- Switch off, unplug where applicable, and allow the stove to cool before cleaning.
- Remove food residue using a soft cloth and a cleaning method suitable for the surface type.
- Inspect the power cord, plug, heating element, and controls regularly for wear or damage.
- Keep ventilation openings clear and avoid abrasive tools that may damage the cooking surface.
- Arrange qualified servicing if heating becomes uneven, controls fail, or electrical protection trips repeatedly.

Common mistakes to avoid

- Using cookware unsuitable for the stove type or too small for the heating zone.
- Operating the stove near water or with damaged electrical connections.
- Leaving the stove unattended or setting heat higher than necessary.
- Cleaning the appliance while hot or allowing water to enter electrical components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Extruder Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Cold Extrusion Machine; Hot Extruder Machine

What it is used for

An extruder machine forces a prepared material through a shaped die to produce a continuous form, such as food products, pellets, or other process-specific outputs. Correct material preparation, temperature and feed control, die selection, and safe shutdown help maintain product consistency and reduce pressure, heat, and entanglement hazards.

Main parts

- Feed hopper receives raw material.
- Feed screw or conveying mechanism moves and compresses material.
- Barrel contains the material during processing.
- Heating or cooling zones control material temperature where fitted.
- Drive motor and gearbox power the screw or extrusion mechanism.
- Die shapes the material as it exits the machine.
- Temperature sensors monitor barrel and die conditions.
- Control panel regulates speed, temperature, and operating settings.
- Cutting or discharge mechanism handles the extruded product where fitted.
- Guards and emergency stop help protect the operator.

Before you start

1. Only trained personnel should operate the extruder machine.
2. Read the manual and confirm whether the machine is configured for cold extrusion, hot extrusion, or food processing.
3. Verify that the raw material, moisture level, feed rate, die, and operating parameters are suitable for the product.
4. Inspect the hopper, barrel, die, drive, guards, controls, sensors, and electrical connections for damage.
5. Confirm that the die and downstream equipment are correctly installed and securely fastened.
6. For food applications, clean and sanitise product-contact parts and verify that ingredients and processing conditions meet the approved food safety procedure.
7. Wear suitable PPE, including heat-resistant gloves for hot processes, eye protection, and other protection required by the risk assessment.

How to use

1. Position the machine on a stable base and ensure the discharge area is clear and ready.

2. Install the correct die and any approved cutting or downstream attachments with the machine isolated.
3. Prepare and measure the raw material according to the approved process recipe.
4. Check that guards are fitted, emergency stop functions, and the hopper and discharge path are unobstructed.
5. Set the feed rate, screw speed, die configuration, and temperature zones according to the validated procedure.
6. For hot extrusion, allow the barrel and die to reach the specified operating temperature before feeding material.
7. Start the drive according to the manufacturer's sequence and verify that the screw operates normally.
8. Feed material gradually and monitor motor load, pressure indicators, temperature, and product flow.
9. Adjust only permitted parameters using the approved procedure and remain alert for blockages, unusual noise, or overheating.
10. Collect or cut the extruded product using the designed discharge system and keep hands clear of moving parts.
11. If pressure rises unexpectedly, material stops flowing, or a fault occurs, stop feeding and follow the manufacturer's safe shutdown and pressure-release procedure.
12. At completion, stop feeding and clear the machine only as specified by the manufacturer.
13. Switch off, isolate power, and allow hot components and material to cool or depressurise before disassembly and cleaning.
14. Clean, inspect, and record production results, material changes, and any faults.

⚠ Safety

- Never reach into the hopper, barrel, die, or cutter while the machine is operating.
- Do not loosen a die, flange, or barrel connection while material is hot or the system remains pressurised.
- Keep guards, interlocks, and emergency stops in working condition; never bypass them.
- Avoid contact with hot barrels, dies, and freshly extruded material; use suitable handling tools.
- Isolate electrical and mechanical energy before clearing blockages or servicing the screw and drive system.
- For food extrusion, prevent cross-contamination and follow the approved allergen, sanitation, and temperature-control procedures.

Care & storage

- Clean the hopper, die, barrel-accessible surfaces, and product-contact components using the approved procedure after safe shutdown.
- Inspect the screw, die openings, seals, bearings, drive coupling, and guards for wear or damage.
- Check temperature sensors, controllers, and pressure indicators at the recommended intervals.
- Lubricate designated drive components according to the manufacturer's instructions and keep lubricant away from food-contact areas.

- Store dies and removable parts clean and dry, and arrange authorised service for persistent pressure, temperature, or product-quality problems.

Common mistakes to avoid

- Using an incorrect die or unsuitable material preparation.
- Feeding too quickly or exceeding the machine's rated capacity.
- Opening a hot or pressurised assembly to clear a blockage.
- Cleaning or disassembling the machine before isolating power and allowing pressure and temperature to reach safe levels.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Food Dehydrator

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Dehydrator

What it is used for

A food dehydrator removes moisture from food using controlled airflow and heat to support drying and storage. Follow the equipment instructions and validated food-safety process for the specific product.

Main parts

- Drying chamber and trays
- Heating element
- Fan and airflow vents
- Temperature and timer controls
- Door or lid and seals
- Power cord and removable drip tray, if fitted

Before you start

1. Place the dehydrator on a stable, level surface with sufficient ventilation.
2. Inspect the cord, plug, door, trays, fan vents, and chamber.
3. Clean and sanitise food-contact surfaces using approved methods.
4. Select fresh, suitable food and wash it using safe food-handling practices.
5. Cut food into uniform pieces and pretreat only as required by the recipe.
6. Check the recommended temperature, loading arrangement, and drying time for the food.

How to use

1. Ensure the unit is switched off before loading trays.
2. Arrange food in a single layer with gaps for airflow.
3. Do not overload trays or block the fan and air vents.
4. Insert trays evenly and close the door or lid securely.
5. Connect the plug to a suitable electrical outlet.
6. Set the temperature and timer according to the approved process.
7. Start the dehydrator and confirm that airflow and heating operate normally.
8. Check progress at suitable intervals using clean utensils and safe handling practices.
9. Rotate trays only if recommended by the manufacturer or process instructions.
10. Verify dryness using the appropriate product-specific check; do not rely only on appearance.
11. Switch off and unplug the unit when drying is complete.
12. Allow hot trays to cool safely before handling.
13. Cool the dried food as required and pack it in clean, suitable containers.

4. Clean the chamber, trays, and removable parts after use.

⚠ Safety

- Keep the unit away from water and wet work surfaces.
- Use heat-resistant protection when handling hot trays and food.
- Do not block ventilation openings or operate the unit with damaged wiring.
- Use safe food-handling practices to prevent cross-contamination.
- Follow validated temperature and drying guidance; drying alone may not destroy all pathogens.
- Unplug the unit and allow it to cool before cleaning or maintenance.

Care & storage

- Wash and dry removable trays and accessories after use.
- Clean the chamber with an approved method; do not immerse electrical parts.
- Keep fan vents clear of dust and food particles.
- Inspect seals, wiring, controls, and fan operation regularly.

Common mistakes to avoid

- Overloading trays and restricting airflow.
- Using unsuitable temperature or drying time for the food.
- Packing food before it has cooled as required by the process.
- Assuming food is safe for long-term storage based on appearance alone.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fruit Mill

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fruit mill

What it is used for

A fruit mill crushes or grinds prepared fruit into smaller particles or a coarse mash for further processing. Correct preparation, machine adjustment, and hygienic operation help produce a consistent feed while reducing contamination and equipment damage.

Main parts

- Fruit feed hopper
- Grinding or crushing rotor, blades, or rollers
- Screen or sizing plate where fitted
- Motor and drive assembly
- Discharge outlet and collection container
- Protective guards, control switch, and emergency stop

Before you start

1. Confirm the fruit type, required particle size, batch quantity, and intended downstream process.
2. Inspect the hopper, grinding parts, screen, discharge outlet, guards, and drive components for damage.
3. Ensure all food-contact surfaces are clean, sanitised, and suitable for food processing.
4. Check that the collection container is clean, food-grade, and correctly positioned under the outlet.
5. Remove stones, metal fragments, damaged fruit, and other foreign material; wash and trim the fruit according to the process.
6. Verify that guards, emergency stop, electrical connections, and other safety controls are functioning.

How to use

1. Switch off and isolate power before opening the mill, changing screens, or adjusting grinding parts.
2. Fit the approved screen or grinding configuration for the required product consistency.
3. Close all covers and confirm that guards are securely fitted before starting.
4. Position the collection container and check that the discharge path is clear.
5. Switch on the mill and allow it to reach normal operating speed before feeding fruit, as directed by the manufacturer.
6. Feed prepared fruit gradually through the hopper without exceeding the machine's rated capacity.

7. Use only the approved feed mechanism or pusher; never insert hands or tools into the hopper while the mill is running.
8. Monitor the discharge consistency, machine load, vibration, and sound during processing.
9. Reduce the feed rate or stop the mill if the outlet blocks, the motor overloads, or abnormal vibration occurs.
10. Stop the machine and isolate power before removing a blockage or checking internal parts.
11. Collect the processed fruit in clean containers and transfer it promptly to the next processing stage.
12. Inspect the product for consistency and any signs of foreign material or contamination.
13. At the end of the batch, switch off and isolate the machine before removing remaining fruit residue.
14. Disassemble only designated removable parts and clean and sanitise them according to the approved procedure.

⚠ Safety

- Only trained personnel should operate or clean the fruit mill.
- Never bypass guards or reach into the hopper, rotor, blades, or rollers while the machine is running.
- Isolate power before clearing jams, changing screens, or servicing the mill.
- Do not exceed the machine's rated capacity or use it for materials not approved by the manufacturer.
- Keep hands, clothing, and utensils away from moving parts and pinch points.
- Use suitable PPE and follow safe handling procedures when working with sharp blades or heavy removable parts.
- Prevent water from entering electrical components unless the equipment is specifically designed for the approved washdown method.

Care & storage

- Clean and sanitise the hopper, grinding parts, screen, and discharge outlet after use.
- Inspect blades, rollers, screens, seals, and fasteners for wear or damage.
- Check motor ventilation, drive components, and emergency-stop function regularly.
- Keep the machine dry and protect electrical components during cleaning.
- Arrange servicing if particle size becomes inconsistent, the motor overheats, or vibration increases.

Common mistakes to avoid

- Feeding fruit with stones or other foreign material still present.
- Overloading the hopper and causing blockages or motor strain.
- Opening the mill or clearing a blockage while it is energised.
- Using unclean containers or failing to sanitise food-contact parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fruit Pulper

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fruit pulper

What it is used for

A fruit pulper separates edible pulp from seeds, skins, and other coarse material using a rotating beater and screen. Correct fruit preparation, screen selection, and hygienic handling help maximise pulp recovery and maintain product quality.

Main parts

- Fruit feed hopper
- Rotating beater or pulping rotor
- Perforated screen or sieve
- Motor and drive assembly
- Pulp outlet and residue discharge outlet
- Protective guards, control panel, and emergency stop

Before you start

1. Confirm the fruit type, required pulp texture, production capacity, and intended downstream process.
2. Inspect the hopper, rotor, screen, seals, outlets, guards, and electrical connections for damage.
3. Ensure all food-contact parts are clean, sanitised, and assembled according to the manufacturer's instructions.
4. Check that the pulp and residue collection containers are clean, food-grade, and correctly positioned.
5. Wash fruit and remove unsuitable foreign material, hard stones, metal, or damaged portions according to the approved preparation method.
6. Verify that the screen is suitable for the fruit and that the emergency stop and guards function correctly.

How to use

1. Switch off and isolate power before installing a screen, opening the pulper, or adjusting the rotor assembly.
2. Fit the approved screen and verify that it is correctly seated and secured.
3. Close the covers, secure the guards, and ensure that both discharge outlets are clear.
4. Position separate clean containers for pulp and rejected residue.
5. Start the machine using the approved control and allow it to reach normal operating speed.
6. Feed prepared fruit gradually and evenly through the hopper without exceeding the rated capacity.

7. Use only the manufacturer's approved feed pusher where supplied; never reach into the hopper during operation.
8. Monitor pulp flow, residue discharge, motor load, vibration, and any unusual noise.
9. Check that pulp consistency and recovery meet the process requirements and that the screen is not blocked.
10. Stop the machine if the rotor jams, the screen shifts, the motor overloads, or product leaks from an unintended location.
11. Isolate power and wait for all moving parts to stop before clearing a blockage or removing internal material.
12. Collect the pulp in clean, covered containers and transfer it promptly to the next processing stage.
13. Inspect the pulp for unwanted seeds, skin fragments, or foreign material according to the quality procedure.
14. At the end of processing, stop and isolate the machine before disassembly and cleaning.
15. Clean and sanitise the screen, rotor, hopper, and outlets using the approved procedure.

⚠ Safety

- Only trained personnel should operate the fruit pulper.
- Keep hands and tools away from the rotor, screen, and feed mechanism while the machine is running.
- Never bypass guards or operate the machine with loose covers.
- Isolate power before clearing jams, changing screens, or performing maintenance.
- Do not force oversized fruit or hard materials into the pulper.
- Handle sharp screens and heavy removable components using appropriate PPE and safe lifting methods.
- Prevent cleaning water from entering electrical parts unless the machine is designed for the approved washdown procedure.

Care & storage

- Clean and sanitise the hopper, screen, rotor, seals, and discharge outlets after each production run.
- Inspect the screen for holes, wear, deformation, or blockage.
- Check rotor blades or beaters, bearings, seals, and fasteners for damage or looseness.
- Keep the motor and electrical controls dry and clean ventilation openings regularly.
- Arrange servicing if pulp recovery declines, the screen becomes damaged, or the machine vibrates excessively.

Common mistakes to avoid

- Using an unsuitable screen that produces poor pulp quality or low recovery.
- Feeding fruit too quickly and overloading the rotor.
- Allowing residue or pulp to remain in the machine and cause contamination.
- Clearing a blocked screen before stopping and isolating the machine.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fruit and Vegetable Washer

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fruit and vegetable washer

What it is used for

A fruit and vegetable washer removes soil and surface debris from produce using controlled water flow, sprays, brushes, or agitation. Correct water quality, loading, and sanitation help clean produce effectively while reducing bruising and cross-contamination.

Main parts

- Produce feed conveyor or loading basket
- Wash tank or chamber
- Spray nozzles, brushes, or agitation mechanism
- Water inlet, valves, and level controls
- Drainage and debris collection system
- Outfeed conveyor, control panel, and protective guards

Before you start

1. Confirm the produce type, required cleaning method, batch size, and downstream processing requirements.
2. Inspect the wash tank, brushes, spray nozzles, conveyors, guards, hoses, and drainage system for damage or residue.
3. Verify that the water supply meets the facility's approved potable-water and food-safety requirements.
4. Clean and sanitise food-contact surfaces according to the approved sanitation schedule.
5. Remove rotten produce, stones, packaging fragments, and other foreign material before loading.
6. Check that water flow, drainage, guards, and emergency-stop controls are functioning correctly.

How to use

1. Prepare the washer according to the manufacturer's instructions and fill the tank to the specified level if applicable.
2. Set the water flow, spray pressure, conveyor speed, and other controls according to the approved process.
3. Where the process requires a sanitiser, use only an approved product at the specified concentration and contact time.
4. Load produce evenly without overfilling the tank, basket, or conveyor.
5. Start the washing system and observe water coverage, movement, and removal of visible soil.

6. Adjust feed rate or approved settings if produce accumulates, collides excessively, or fails to receive adequate washing.
7. Monitor water cleanliness, drainage, spray performance, and produce condition during operation.
8. Replace or refresh wash water at the required intervals and whenever it no longer meets the approved quality criteria.
9. Inspect washed produce for remaining soil, damaged items, chemical residue, or other signs of inadequate cleaning.
10. Stop the machine if a blockage, leak, unsafe movement, or electrical fault occurs.
11. Isolate power before removing trapped produce, cleaning blocked nozzles, or reaching into moving mechanisms.
12. At the end of the batch, stop the feed and drain the system according to the approved shutdown procedure.
13. Remove accumulated debris and clean and sanitise the tank, brushes, nozzles, conveyors, and contact surfaces.
14. Record batch details, water or sanitiser checks where required, and any quality or equipment deviations.

⚠ Safety

- Only trained personnel should operate and clean the washer.
- Use water and sanitising agents approved for food processing and follow the specified concentration and contact time.
- Keep hands away from moving brushes, rollers, conveyors, and agitation mechanisms.
- Isolate power before clearing blockages or servicing moving components.
- Prevent cleaning chemicals and dirty wash water from contaminating clean produce or potable-water supplies.
- Use chemical-compatible gloves and eye protection when handling sanitising agents.
- Keep electrical controls protected from water and follow the equipment's approved washdown procedure.

Care & storage

- Clean and sanitise the tank, brushes, spray nozzles, conveyors, and drainage areas after use.
- Inspect brushes, belts, seals, hoses, and valves for wear, blockage, or leakage.
- Remove food debris and mineral deposits using approved cleaning methods.
- Check water-flow controls and emergency stops at the specified intervals.
- Arrange servicing if washing becomes uneven, drainage slows, or produce damage increases.

Common mistakes to avoid

- Using water that does not meet the required food-safety standard.
- Overloading the washer and preventing adequate contact between water and produce.
- Reusing dirty wash water beyond the approved limits.
- Failing to separate clean produce from dirty surfaces or contaminated equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gas Lighter

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Gas lighter

What it is used for

A gas lighter produces a spark or flame to ignite a compatible gas appliance. Use it only according to its design and the appliance manufacturer's instructions.

How to use

1. Inspect the lighter for cracks, gas leakage, damage, or loose parts.
2. Check that the appliance is correctly connected and its controls are in the required position.
3. Keep your face, hands, and clothing away from the ignition point.
4. Operate the lighter according to its instructions at the designated ignition point.
5. Confirm that the appliance flame lights correctly and remains stable.
6. If ignition fails, turn off the gas supply and follow the appliance's relighting procedure after the required waiting period.
7. Switch off the gas supply after use and store the lighter safely.

⚠ Safety

- Never use a leaking or damaged lighter.
- Keep the lighter away from children, flammable materials, and hot surfaces.
- Do not search for gas leaks using a flame.
- If you smell gas, do not ignite anything or operate electrical switches; leave the area and follow emergency guidance.

Care & storage

- Keep the lighter clean and dry and store it away from heat.
- Replace or service it if ignition becomes unreliable or damage is found.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gas Stove

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Cooking Range; Cooking Range (Gas/Electric); Single Burner Stove

What it is used for

A gas stove burns a compatible fuel to provide heat for cooking or food preparation. Correct ignition, ventilation, and leak checks help ensure efficient operation and reduce fire, burn, and gas exposure risks.

Main parts

- Burner mixes fuel and air to produce a controlled flame.
- Burner cap and head distribute the flame around the burner.
- Control knob regulates the gas flow to the burner.
- Ignition system creates a spark on equipped models.
- Gas inlet and hose or pipe deliver fuel to the stove.
- Regulator controls gas supply pressure where applicable.
- Pan supports hold cookware above the flame.
- Drip tray or stove surface collects spills and food residue.

Before you start

1. Confirm the stove is designed for the available gas type and supply system.
2. Inspect the gas hose, regulator, connections, burner parts, and control knobs for damage.
3. Check the area for gas odour and ensure ventilation is adequate.
4. Keep flammable materials, paper, and loose cloth away from the stove.
5. Ensure burners and pan supports are correctly fitted and dry.
6. Use a suitable, stable cooking vessel and keep the cooking area clear.

How to use

1. Ensure all burner control knobs are in the off position before opening the gas supply.
2. Open the gas supply according to the appliance instructions.
3. For manual ignition, use the approved ignition method while turning the selected burner control as directed.
4. For automatic ignition, operate the control according to the manufacturer's instructions.
5. Confirm that the burner lights promptly and the flame is stable and appropriately coloured.
6. Adjust the flame to suit the cookware and cooking task.
7. Stay nearby and monitor cooking to prevent boil-over, overheating, or unattended flames.
8. Turn the burner control off after cooking and close the gas supply when required by the appliance instructions.
9. Allow the stove to cool before removing pan supports or cleaning spills.

10. Clean and dry the stove, check that all controls are off, and report any faults.

⚠ Safety

- If gas is smelled, do not ignite flames or operate electrical switches; turn off the gas if safe, ventilate, leave the area, and contact the gas supplier or emergency services.
- Never leave a lit burner unattended or use the stove to heat a room.
- Keep children and bystanders away from hot surfaces and open flames.
- Do not use a stove with a damaged hose, loose connection, or suspected gas leak.
- Never check for leaks with an open flame; use an approved method or qualified technician.
- Keep ventilation openings clear and avoid using the stove in an inadequately ventilated space.

Care & storage

- Clean burner caps and heads only after the stove has cooled and according to the manual.
- Keep burner ports clear of food residue and moisture.
- Inspect the hose, regulator, and connections at recommended intervals and replace parts as specified.
- Arrange qualified servicing for ignition problems, uneven flames, gas odour, or damaged components.

Common mistakes to avoid

- Opening the gas supply while burner controls are not in the required position.
- Using the stove despite gas odour or damaged connections.
- Leaving a burner lit without supervision.
- Cleaning or handling burner parts before they cool completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grading Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Fruit grader; Grading Machine

What it is used for

A grading machine sorts fruits or other suitable food products by size, weight, colour, or another specified quality parameter. Use the correct grading settings and handling method to reduce product damage and maintain consistent output.

Main parts

- Feed hopper or loading conveyor
- Sorting belt, rollers, screens, or grading channels
- Size, weight, colour, or optical sensors, if fitted
- Drive motor and transmission
- Control panel and adjustment controls
- Collection bins or discharge conveyor

Before you start

1. Confirm the machine is suitable for the product being graded.
2. Inspect belts, rollers, screens, guards, sensors, and collection bins.
3. Clean and sanitise food-contact surfaces according to the hygiene procedure.
4. Check the emergency stop, guards, and other safety devices.
5. Set grading criteria and verify the relevant sensors or measuring system.
6. Remove foreign objects and ensure collection bins are clean and correctly positioned.

How to use

1. Keep the machine switched off while cleaning or adjusting internal parts.
2. Install the required screens, guides, or approved product-handling attachments.
3. Set the grading parameters according to the product specification.
4. Position clean collection bins at the correct discharge points.
5. Close guards and confirm that the feed and discharge paths are clear.
6. Power on the machine and run it empty briefly to check normal operation.
7. Feed the product gradually at the recommended rate.
8. Observe the sorting process and check that products move freely without excessive impact.
9. Inspect samples from each grade for consistency and visible damage.
10. Adjust settings only according to the approved procedure if grading is inaccurate.
11. Stop feeding and allow the machine to clear the product flow before shutdown.
12. Switch off and isolate the power before removing jams or cleaning internal parts.
13. Collect and label graded products according to the workplace procedure.

4. Clean the machine and record faults or unusual grading results.

⚠ Safety

- Only trained operators should operate or adjust the machine.
- Keep hands and tools away from moving belts, rollers, screens, and drive components.
- Never bypass guards, interlocks, or emergency stops.
- Isolate and verify zero energy before clearing jams or performing maintenance.
- Use safe lifting methods when handling heavy product bins.
- Follow food hygiene requirements and keep cleaning chemicals away from exposed products.

Care & storage

- Clean and sanitise food-contact parts after the production run.
- Remove product debris from belts, rollers, screens, and sensors.
- Inspect belts, bearings, fasteners, guards, and drive components regularly.
- Calibrate or verify grading sensors according to the maintenance schedule.
- Lubricate only specified points and keep lubricant away from food-contact surfaces.

Common mistakes to avoid

- Feeding products too quickly and causing jams or inaccurate grading.
- Using incorrect grading settings for the product batch.
- Ignoring dirty sensors, worn belts, or damaged screens.
- Clearing jams while the machine remains energised.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grater

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Grater; Grater/ Shredder

What it is used for

To grate, shred, or finely reduce food ingredients for preparation and cooking.

How to handle

- Inspect the grater for damage, rust, loose parts, and cleanliness before use.
- Place the grater on a stable surface or hold it securely by its handle.
- Use the appropriate grating surface and move food carefully, keeping fingers away from sharp holes or blades.
- Use a food holder or protective guard when the remaining piece becomes small.
- Wash and dry the grater thoroughly after use, handling sharp edges carefully.

Storage

- Store clean and completely dry in a designated kitchen drawer or utensil rack.
- Use a protective cover where available and keep sharp surfaces away from hands.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grinding Mill

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hammer Mill / Pulverizer; Stone Mill / Roller Mill

What it is used for

A grinding mill reduces suitable materials into smaller particles using a hammer, stone, roller, or other grinding mechanism. Select the correct mill setup for the material and follow the operating instructions to maintain product quality and safe operation.

Main parts

- Feed hopper and feed control
- Grinding chamber with hammers, stones, rollers, or grinding plates
- Drive motor and transmission
- Screens or particle-size adjustment mechanism
- Discharge outlet and collection container
- Protective guards and emergency stop, if fitted

Before you start

1. Identify the mill type and confirm that the material is suitable for processing.
2. Inspect the chamber, grinding elements, screens, belts, fasteners, and guards.
3. Check the motor, electrical connections, and emergency stop.
4. Ensure the mill and collection container are clean and dry as required.
5. Confirm the correct screen, roller gap, or grinding setting for the target particle size.
6. Remove metal, stones, and other foreign objects from the feed material.
7. Wear required PPE, including eye and hearing protection and suitable dust protection.

How to use

1. Keep the mill switched off and isolated while inspecting or changing components.
2. Fit the approved screen or set the grinding gap according to the product requirement.
3. Secure all covers, guards, and collection containers before starting.
4. Power on the mill according to the manufacturer's procedure.
5. Allow the machine to reach normal operating speed before feeding material, if specified.
6. Feed material gradually at the recommended rate without overloading the chamber.
7. Monitor motor sound, vibration, temperature, dust, and discharge flow.
8. Reduce or stop feeding if the mill becomes overloaded or output becomes irregular.
9. Collect the processed material in a suitable clean container.
10. Stop feeding and allow the remaining material to discharge according to the instructions.
11. Switch off and isolate the machine before opening covers or clearing a blockage.
12. Wait for all moving parts to stop completely before inspection or cleaning.

3. Remove remaining material and clean the machine using the approved method.
4. Record output, settings, and any abnormal noise, vibration, or faults.

⚠ Safety

- Only trained operators should operate the mill or change grinding components.
- Never insert hands or tools into the hopper or grinding chamber while the machine is connected to power.
- Keep all guards and covers fitted during operation.
- Isolate the power and verify complete stoppage before clearing blockages or maintenance.
- Control dust and use appropriate respiratory protection where the risk assessment requires it.
- Do not grind flammable, explosive, or hazardous materials unless the mill is specifically designed and approved for them.

Care & storage

- Clean the hopper, chamber, screens, and discharge path after use.
- Inspect hammers, stones, rollers, screens, bearings, and belts for wear.
- Lubricate bearings and service points according to the maintenance schedule.
- Check fasteners, guards, and electrical connections regularly.
- Keep the mill dry and store removable components safely.

Common mistakes to avoid

- Feeding material faster than the mill can process it.
- Using a damaged screen or incorrect grinding setting.
- Opening the chamber before the moving parts have stopped completely.
- Ignoring excessive dust, vibration, unusual noise, or overheating.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Halving Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Halving machine

What it is used for

To cut suitable food products into two equal or specified portions using a halving machine. To maintain consistent cuts while following hygiene and machine safety procedures.

Main parts

- Machine frame and base
- Cutting blade or halving mechanism
- Product feed tray or guide
- Guard and safety interlock
- Start and stop controls
- Collection tray and drive system

Before you start

1. Read the operating manual and confirm the machine is suitable for the food product.
2. Inspect the blade, feed guide, guards, fasteners, and power connections for damage.
3. Ensure the machine and food-contact surfaces are clean and sanitized as required.
4. Check that the blade and guards are securely installed and safety interlocks work.
5. Prepare food products of suitable size and remove unsuitable hard objects where required.
6. Keep the work area dry, clear, and well lit.

How to use

1. Place the machine on a stable surface and ensure the stop control is accessible.
2. Confirm that the machine is switched off before loading or adjusting parts.
3. Install the correct cutting attachment and adjust the guide according to the product size.
4. Position the collection tray securely below the cutting area.
5. Connect the power supply according to the manufacturer's instructions.
6. Switch on the machine and verify that it runs without unusual noise or vibration.
7. Feed one product at a time using the designated tray or pusher.
8. Keep hands away from the blade and never push products directly by hand into a hazardous cutting zone.
9. Collect the cut portions and check their size and quality.
10. Stop the machine before clearing jams or changing settings.
11. Switch off and isolate power before cleaning or removing the blade.
12. Clean and sanitize food-contact parts according to the approved procedure, then dry them before storage.

⚠ Safety

- Only trained personnel should operate the machine.
- Never bypass guards, interlocks, or other safety devices.
- Keep hands, loose clothing, jewellery, and hair away from moving cutting parts.
- Use only the supplied pusher or feeding tool where required.
- Switch off and disconnect power before clearing a jam or handling the blade.
- Follow food hygiene requirements and use cut-resistant protection only where the risk assessment and manufacturer permit it.

Care & storage

- Clean food residue from the blade, guides, and collection tray after use.
- Inspect the blade for dullness, cracks, or damage before each session.
- Check guards, fasteners, and safety interlocks regularly.
- Lubricate only designated machine parts using suitable food-grade lubricant where required.
- Arrange blade replacement and servicing according to the manual.

Common mistakes to avoid

- Feeding products that exceed the machine's rated size or capacity.
- Pushing food toward the blade with bare hands.
- Operating with loose guards or a damaged blade.
- Clearing a jam while the machine is running or connected to power.
- Skipping cleaning and sanitation between food-processing tasks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Blender

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Hand Blender; Hand blender cum mixer

What it is used for

A hand blender is used to blend, puree, mix, or emulsify food directly in a suitable container. Correct attachment selection, hygienic handling, and safe control of the rotating blade help achieve consistent results while reducing splashes, cuts, and electrical hazards.

Main parts

- Motor housing contains the electric motor and controls.
- Blending shaft connects the motor to the blade assembly.
- Blade guard covers the rotating blade area.
- Blade assembly cuts and blends food.
- Speed control adjusts blending speed where fitted.
- Power switch starts and stops the blender.
- Detachable whisk or mixer attachment supports other mixing tasks where provided.
- Power cable or battery supplies energy to the unit.

Before you start

1. Read the operating instructions and use the blender only for approved tasks.
2. Inspect the motor housing, shaft, blade guard, attachment lock, and cable or battery for damage.
3. Confirm the attachment is clean, correctly fitted, and suitable for the food and task.
4. Use a stable, deep container that can accommodate the ingredients without overflowing.
5. Cut large or hard ingredients into suitable pieces and add sufficient liquid when required by the recipe.
6. Keep the motor housing and electrical connections dry, and ensure hands are dry before plugging in.

How to use

1. Wash hands and prepare the ingredients according to the recipe.
2. Place ingredients in a suitable container and keep the container stable on the work surface.
3. Fit and lock the required blending or mixing attachment while the blender is switched off and unplugged where applicable.
4. Position the blade guard fully within the ingredients before starting the motor.
5. Start at a low speed and increase gradually if required.
6. Move the blender gently through the mixture without lifting the running blade above the food surface.

7. Pause the motor before repositioning the blender, scraping the container, or changing speed if the manual requires it.
8. Switch off and unplug the unit before removing the attachment or clearing food from the blade area.
9. Transfer the blended food hygienically and inspect the result for the required consistency.
10. Clean the shaft and attachments promptly according to the manufacturer's instructions.
11. Dry all parts thoroughly and store the blender safely after use.

⚠ Safety

- Never touch the blade or place fingers in the blade guard while the unit is connected to power.
- Switch off and unplug the blender before changing attachments or clearing a jam.
- Do not immerse the motor housing or power connection in water.
- Do not operate the blender with a damaged cable, loose attachment, or broken blade guard.
- Avoid overfilling the container and use care with hot liquids to prevent splashes and burns.
- Keep children and untrained persons away from the operating blender.

Care & storage

- Wash detachable food-contact parts using the approved cleaning method after each use.
- Wipe the motor housing with a suitable damp cloth without immersing it.
- Inspect blades, attachment locks, seals, and cables regularly.
- Allow cleaned parts to dry completely before storage and arrange authorised repair for faults.

Common mistakes to avoid

- Starting the blender before the blade guard is immersed in the ingredients.
- Changing attachments or clearing food while the motor is running.
- Submerging the motor housing in water.
- Using an unsuitable container or overfilling it, causing spills and splashes.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Homogenizer

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Homogenizer; Small Homogenizer

What it is used for

A homogenizer reduces particle or droplet size and helps create a more uniform mixture in suitable food products or laboratory samples. Use the correct homogenizing attachment, speed, and operating procedure for the material being processed.

Main parts

- Main motor and drive unit
- Homogenizing head, rotor, or probe
- Processing vessel or sample container
- Speed and power controls
- Shaft, coupling, and protective guard, if fitted
- Power cord and compatible accessories

Before you start

1. Confirm that the homogenizer is suitable for the material and batch size.
2. Inspect the motor, probe, shaft, cable, and attachments for damage.
3. Clean and sanitise food-contact parts according to the hygiene procedure.
4. Check that the processing vessel is compatible and securely positioned.
5. Select the required speed and processing time according to the product procedure.
6. Wear suitable PPE and ensure the work area is stable and uncluttered.

How to use

1. Switch off and disconnect the unit before fitting the homogenizing attachment.
2. Secure the correct probe or head according to the manufacturer's instructions.
3. Place the material in a compatible vessel without exceeding the recommended fill level.
4. Position the homogenizing head correctly within the material before starting.
5. Connect the unit to a suitable power supply and switch it on.
6. Start at the recommended low speed and increase gradually if required.
7. Maintain the required probe position and avoid contact with the vessel walls.
8. Monitor the mixture for uniformity, splashing, excessive heating, or unusual vibration.
9. Stop the unit before lifting the probe out of the material.
10. Switch off and disconnect the power before removing the attachment.
11. Transfer the processed material using clean, suitable equipment.
12. Clean and sanitise the probe and compatible parts after use.
13. Record processing settings and report any equipment faults.

⚠ Safety

- Only trained personnel should operate the homogenizer.
- Never touch a moving probe or rotor during operation.
- Keep hands, loose clothing, and utensils away from moving parts.
- Do not operate with a damaged shaft, loose attachment, or faulty electrical connection.
- Use suitable containment and PPE when processing hazardous or potentially infectious samples.
- Disconnect power before cleaning, changing attachments, or clearing a blockage.

Care & storage

- Clean and sanitise product-contact parts after each batch.
- Prevent liquid from entering the motor housing or electrical connections.
- Inspect the shaft, coupling, seals, and probe for wear or damage.
- Store attachments clean and dry in a protected location.
- Arrange servicing according to the manufacturer's maintenance schedule.

Common mistakes to avoid

- Starting the homogenizer before the probe is correctly positioned.
- Exceeding the recommended vessel fill level or batch capacity.
- Removing the probe while it is still rotating.
- Using incompatible cleaning chemicals or immersing non-waterproof parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ice Cream Maker

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Ice Cream Maker

What it is used for

An ice cream maker freezes and mixes a prepared mixture to produce ice cream or other frozen desserts. Follow the machine-specific instructions and maintain strict food hygiene and temperature control throughout preparation.

Main parts

- Mixing and freezing bowl or cylinder
- Mixing paddle, dasher, or beater
- Motor or drive unit
- Lid and ingredient feed opening, if fitted
- Control panel, timer, or speed selector
- Power cord and removable accessories

Before you start

1. Confirm the machine type and follow its preparation instructions.
2. Inspect the bowl, paddle, lid, motor, and power cord for damage.
3. Clean and sanitise all food-contact parts before use.
4. Prepare ingredients using safe food-handling practices and keep perishables chilled.
5. Pre-freeze the bowl if required by the machine design.
6. Ensure the machine is stable, correctly assembled, and has adequate ventilation.

How to use

1. Switch off and unplug the machine before assembling its parts.
2. Install the bowl and paddle according to the manufacturer's instructions.
3. Add the prepared mixture without exceeding the marked capacity.
4. Secure the lid and confirm that the paddle is correctly positioned.
5. Connect the machine to a suitable power supply.
6. Select the recommended mixing or freezing setting and start the machine.
7. Add extra ingredients through the designated opening only if the model permits it.
8. Monitor the mixing process and stop if the motor overheats or abnormal noise occurs.
9. Allow the mixture to reach the required consistency according to the recipe and instructions.
10. Switch off the machine before removing the lid or finished product.
11. Transfer the ice cream using a clean utensil into a suitable food-safe container.
12. Cover and store the product at the required frozen-storage temperature.
13. Unplug the machine and allow removable parts to reach a safe temperature before cleaning.

4. Clean, sanitise, dry, and store all compatible components.

⚠ Safety

- Only trained users should operate commercial ice cream-making equipment.
- Keep hands and utensils away from the paddle while the machine is running.
- Never bypass lid interlocks or other safety features.
- Do not exceed the specified capacity or run the machine with incorrect assembly.
- Keep electrical connections dry and avoid water entering the motor unit.
- Follow food-safety guidance for ingredient handling, storage, and allergen separation.

Care & storage

- Clean and sanitise the bowl, paddle, lid, and food-contact parts after use.
- Do not immerse the motor unit unless it is specifically designed for immersion.
- Inspect seals, blades or paddles, fasteners, and electrical parts regularly.
- Store the machine clean and dry with removable parts properly arranged.
- Arrange servicing if freezing performance or motor operation becomes abnormal.

Common mistakes to avoid

- Adding more mixture than the machine's rated capacity.
- Starting the machine with the paddle or bowl incorrectly fitted.
- Opening or reaching into the machine while the paddle is moving.
- Allowing prepared mixture or finished ice cream to remain at unsafe temperatures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ice Cube Tray

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Ice Cube Trays

What it is used for

Handle ice cube trays hygienically for food and beverage preparation.

How to handle

- Wash the tray before first use and after each use.
- Use food-safe trays and fill them with potable water or approved ingredients.
- Avoid touching ice-making surfaces with dirty hands.
- Do not use trays that are cracked, damaged, or contaminated.

Storage

- Allow the tray to dry or store it in the freezer as appropriate.
- Keep clean trays separate from chemicals and non-food items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Induction Cooktop

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Induction cooker; Induction Cooktop

What it is used for

An induction cooktop heats compatible cookware using an electromagnetic field for cooking or warming food. Use suitable cookware, keep the cooking surface clear and dry, and follow the manufacturer's operating instructions.

Main parts

- Glass-ceramic cooking surface
- Induction coil and internal electronics
- Power control and display panel
- Temperature or power-level controls
- Timer and safety lock, if fitted
- Power cord and plug

Before you start

1. Place the cooktop on a flat, stable, heat-resistant surface.
2. Inspect the cooking surface, power cord, plug, and controls for damage.
3. Confirm the cookware is induction-compatible and has a flat base.
4. Ensure the air inlet and outlet are clear and the surrounding area is dry.
5. Keep paper, cloth, plastic, and other combustible items away from the cooking zone.
6. Check the required power supply and review the manufacturer's instructions.

How to use

1. Place compatible cookware centrally on the designated cooking zone.
2. Add the food or liquid to the cookware before heating as appropriate.
3. Connect the cooktop to the recommended power supply.
4. Switch on the unit and select the required cooking zone, if applicable.
5. Set a suitable power level or temperature using the controls.
6. Monitor the cooking process and adjust settings as needed.
7. Use suitable utensils and avoid touching the hot cookware or cooking surface.
8. Use the timer or safety lock when appropriate and available.
9. Switch off the cooking zone when cooking is complete.
10. Remove cookware carefully using suitable heat protection.
11. Switch off and unplug the unit if required before cleaning or storage.
12. Allow the surface to cool, then clean it using an approved method.
13. Check that the cooking area is dry and store the unit safely.

⚠ Safety

- Use only cookware compatible with induction cooking.
- Do not operate the cooktop with cracked glass or damaged wiring.
- Keep water away from electrical connections and do not immerse the unit.
- The cooking surface and cookware may remain hot after switching off.
- Do not place metal objects, foil, or unsuitable containers on the cooking zone.
- Keep ventilation openings clear and follow the specified electrical load requirements.

Care & storage

- Clean spills after the surface has cooled sufficiently.
- Use cleaning products approved for the glass-ceramic surface.
- Keep ventilation openings free from dust and grease.
- Inspect the cord, plug, and surface regularly for damage.
- Disconnect power before cleaning areas near electrical components.

Common mistakes to avoid

- Using cookware that is not induction-compatible.
- Blocking the ventilation openings or placing the unit on an unstable surface.
- Touching the cooking surface immediately after use.
- Cleaning with abrasive tools or pouring water over the electrical unit.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Juice Extractor

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fruit juice extractor

What it is used for

To extract juice from suitable fruits or vegetables using a juice extractor. To produce juice efficiently while maintaining food hygiene and preventing contact with moving cutting or pressing parts.

Main parts

- Motor and machine housing
- Feed chute and pusher
- Cutting or grating assembly
- Strainer or filter basket
- Juice outlet and collection container
- Pulp container, lid, and safety interlock

Before you start

1. Read the manual and confirm that the extractor is suitable for the intended produce.
2. Inspect the power cord, motor housing, lid, pusher, filter, and cutting parts for damage.
3. Wash the produce and remove stones, hard pits, or unsuitable parts according to the product requirements.
4. Clean and sanitize food-contact components using an approved procedure.
5. Assemble the filter, lid, containers, and safety locks correctly.
6. Place the machine on a stable, dry surface and ensure the juice and pulp containers are correctly positioned.

How to use

1. Ensure the switch is off before assembling or loading the extractor.
2. Fit the filter or cutting assembly and lock the lid securely.
3. Place a clean collection container under the juice outlet and the pulp container in its position.
4. Connect the power supply according to the manual.
5. Switch on the extractor and allow it to reach normal operating speed.
6. Feed produce gradually through the chute using the supplied pusher only.
7. Do not force oversized pieces into the feed opening or insert fingers or utensils into the chute.
8. Monitor juice flow, pulp collection, and motor sound during operation.
9. Switch off the machine if the filter becomes blocked, vibration increases, or the motor overheats.
10. Wait for all moving parts to stop before opening the lid or removing the filter.
11. Disconnect power before clearing blockages or dismantling parts for cleaning.

2. Transfer the juice to a clean container and refrigerate or serve it according to food-safety requirements.
3. Clean and sanitize all food-contact components promptly after use.

⚠ Safety

- Never bypass the lid lock or safety interlock.
- Keep hands, hair, jewellery, and loose clothing away from the feed chute and moving parts.
- Use only the supplied pusher to guide produce into the machine.
- Do not operate with a cracked filter, damaged cord, loose blade assembly, or missing lid.
- Switch off and unplug the extractor before clearing a blockage or touching internal parts.
- Keep the motor base dry and never immerse it in water.

Care & storage

- Remove pulp and rinse food-contact parts promptly after use.
- Clean the filter carefully with a suitable brush and inspect its mesh for damage.
- Wipe the motor housing with a damp cloth without immersing it.
- Inspect seals, locks, blades, and the power cord regularly.
- Store clean, dry components in a protected location.

Common mistakes to avoid

- Operating the extractor without securing the lid and interlock.
- Pushing produce into the chute with fingers or unsuitable tools.
- Overloading the feed chute and causing blockage or motor strain.
- Opening the machine before the cutting assembly stops.
- Leaving pulp and juice residue on components after use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Khoa Pan

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Khoa/Mawa Making Pan

What it is used for

To heat milk or a milk-based mixture in a khoa pan until the required moisture is removed and khoa or mawa is formed. To maintain uniform heating and stirring while preventing scorching, overflow, and contamination.

Main parts

- Heating pan or vessel
- Gas burner or electric heating system
- Temperature or heat control
- Agitator or manual stirring tool
- Support frame and heat insulation
- Discharge or product collection arrangement

Before you start

1. Ensure the pan and all milk-contact surfaces are clean and suitable for food processing.
2. Inspect the pan, support frame, heating system, controls, and fuel or electrical connections.
3. Confirm that the pan is stable and the discharge area is prepared.
4. Use fresh milk or an approved milk mixture and check its quality before processing.
5. Prepare clean, dry food-grade tools and containers for collecting the khoa.
6. Wear suitable heat-resistant gloves, protective clothing, and closed footwear.

How to use

1. Place the pan securely and confirm that the heating source is switched off before loading.
2. Pour the required quantity of milk into the pan without exceeding its safe working capacity.
3. Start the heating system according to the manufacturer's instructions.
4. Heat the milk gradually and stir continuously or at the frequency required by the process.
5. Monitor the heating rate and scrape the bottom and sides using an approved food-safe tool.
6. Adjust heat to maintain controlled evaporation and prevent scorching or excessive foaming.
7. Continue heating and stirring as the milk thickens and moisture reduces.
8. Check the product texture and finish against the approved process requirements.
9. Reduce or switch off heat when the required khoa consistency is reached.
10. Transfer the hot product carefully into a clean, dry, food-grade container using suitable tools.
11. Switch off the heating source and isolate fuel or power as applicable.
12. Clean the pan and product-contact tools promptly after they have cooled sufficiently.

⚠ Safety

- Only trained personnel should operate the heating equipment.
- Avoid contact with hot milk, steam, the pan surface, and hot metal tools.
- Never overfill the pan; boiling milk can foam and overflow suddenly.
- For gas heating, check for leaks and maintain suitable ventilation.
- Keep water away from hot fat or concentrated milk residue to reduce splashing risks.
- Do not leave the heating process unattended and keep combustible materials away from the burner.

Care & storage

- Clean the pan and food-contact tools after every batch using approved food-safe procedures.
- Inspect the pan surface, support frame, burner, and controls regularly.
- Remove burnt residue using a method that does not damage the food-contact surface.
- Keep gas lines or electrical connections in safe condition.
- Arrange periodic maintenance and check temperature controls where fitted.

Common mistakes to avoid

- Heating too quickly and scorching milk at the bottom.
- Failing to stir and allowing the product to stick or burn.
- Overfilling the pan and causing overflow.
- Using unclean containers or tools that contaminate the khoa.
- Touching hot surfaces or transferring the product without suitable protection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Kitchen Knife

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Peeling knife; Serrated Bread Knife

What it is used for

To cut, slice, or peel food using a suitable kitchen knife. To prepare ingredients efficiently while maintaining food hygiene and preventing cuts.

Main parts

- Blade
- Handle
- Cutting edge
- Knife tip
- Protective storage or knife rack

Before you start

1. Choose a knife suitable for the food and cutting task.
2. Check that the blade is clean, sharp, and securely attached to the handle.
3. Use a stable cutting board and keep it from slipping.
4. Wash hands and clean the food as required.

How to use

1. Hold the knife by the handle with a secure grip.
2. Place the food firmly on the cutting board.
3. Use controlled slicing or peeling motions suited to the knife type.
4. Keep fingertips curled away from the cutting edge while guiding the food.
5. Use a serrated bread knife with a gentle sawing motion for bread.
6. Place the knife down safely when not in use.
7. Wash, dry, and store the knife after the task.

⚠ Safety

- Always cut away from the body where practical.
- Never try to catch a falling knife.
- Do not leave knives hidden in sinks or under food.
- Carry a knife with the point down and blade facing away from others.

Care & storage

- Wash and dry the knife after use.

- Sharpen and hone the blade using an appropriate method.
- Avoid using the knife on hard surfaces that can damage the edge.
- Store in a knife block, rack, or protective cover.

Common mistakes to avoid

- Using a dull knife that requires excessive force.
- Cutting on an unstable board.
- Using a peeling knife for tasks beyond its design.
- Leaving a knife loose in a sink or drawer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Kitchen Tongs

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Tong /

What it is used for

To grip, turn, lift, and serve food hygienically while keeping hands away from hot surfaces.

How to handle

- Inspect the tongs, gripping ends, hinge, and locking mechanism for damage or residue.
- Wash hands and clean the tongs before food handling; use heat-resistant tongs for hot foods.
- Grip food securely and use controlled movements to avoid splashing, dropping, or cross-contamination.
- Use separate clean tongs for raw and ready-to-eat foods and avoid placing the gripping ends on unclean surfaces.
- Wash, sanitize when required, and dry the tongs after use.

Storage

- Clean, sanitize if required, and dry completely before storage.
- Store in a clean utensil holder or drawer with food-contact ends protected from contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ladle

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Set of cooking spoon; S.S. Karchi

What it is used for

A ladle transfers and serves liquids or cooked food during food preparation and kitchen service.

How to handle

- Inspect the ladle bowl and handle for cracks, sharp edges, looseness, or corrosion.
- Use a ladle made from material suitable for the food and cooking temperature.
- Keep the handle secure and avoid leaving it where it may fall into hot food or cause burns.
- Clean and sanitise the ladle according to food-safety procedures between uses as required.
- Use heat-resistant protection when handling a hot ladle and avoid cross-contamination between foods.

Storage

- Wash, rinse, and dry the ladle before storing.
- Keep it in a clean utensil rack or designated kitchen storage area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Measuring Jug

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Low** 12 labs

Covers these lab items: Measuring cup; Measuring cup and Measuring jug; Measuring Cup sets; Measuring cups; Measuring Cups & Spoons; Measuring jug; Measuring Jugs; Measuring Mug / Cup

What it is used for

Measuring cups and jugs measure liquid or dry ingredients for cooking and food preparation. Use the correct scale and measuring technique to achieve consistent quantities.

How to use

1. Inspect the jug or cup for cracks, stains, and readable measurement markings.
2. Select a capacity and unit suitable for the ingredient and recipe.
3. Place the measuring vessel on a level surface before measuring.
4. Pour liquid to the required mark and read the scale at eye level.
5. Fill dry ingredients using the specified method and level off when required.
6. Check the measurement before adding the ingredient to the recipe.
7. Wash, rinse, and dry the measuring vessel after use.

⚠ Safety

- Do not use cracked vessels or unreadable measuring scales.
- Use heat-resistant vessels for hot liquids when specified.
- Avoid overfilling to prevent spills and burns.
- Keep measuring tools clean to prevent cross-contamination.

Care & storage

- Wash according to the material instructions and store measuring cups and jugs dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Measuring Spoon

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Low** 8 labs

Covers these lab items: Measuring spoon; Measuring Spoon sets; Measuring Spoons

What it is used for

Measuring spoons measure small quantities of ingredients for cooking, baking, and food preparation. Use the correct spoon size and level ingredients as required by the recipe.

How to use

1. Inspect the spoons for damage, residue, and readable size markings.
2. Select the required spoon size and unit specified in the recipe.
3. Ensure the spoon is clean and dry before measuring.
4. Fill the spoon with the ingredient using the appropriate method.
5. Level off dry ingredients when the recipe requires a level measure.
6. Add the measured ingredient carefully to the preparation vessel.
7. Wash, rinse, and dry the spoons after use.

⚠ Safety

- Do not use cracked, corroded, or contaminated spoons.
- Keep measuring spoons separate from cleaning chemicals and non-food materials.
- Avoid tasting ingredients with a spoon that will be returned to the container.
- Use clean spoons to prevent cross-contamination.

Care & storage

- Wash and dry the spoons thoroughly and store the set together in a clean location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Microwave Oven

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Microwave; Microwave Oven

What it is used for

A microwave oven heats or cooks food using microwave energy inside a closed cooking cavity. Use microwave-safe containers, follow the food instructions, and check that food reaches a safe temperature before serving.

Main parts

- Cooking cavity and door
- Door seal, hinges, and safety interlocks
- Control panel, timer, and power settings
- Turntable and roller ring, if fitted
- Magnetron and internal electrical components
- Power cord and plug

Before you start

1. Place the oven on a stable surface with the required ventilation clearance.
2. Inspect the door, hinges, seal, turntable, cord, and plug for damage.
3. Confirm that the oven cavity is clean and free of metal objects.
4. Use only containers labelled safe for microwave cooking.
5. Check food packaging and follow the recommended cooking or reheating instructions.
6. Ensure the door closes fully and the safety interlocks are not damaged.

How to use

1. Place the food in a suitable microwave-safe container.
2. Cover the food with a microwave-safe lid or cover that allows steam to escape.
3. Position the container in the centre of the turntable or cooking area.
4. Close the door securely.
5. Select the required power level and cooking time.
6. Start the oven and monitor operation without opening the door unnecessarily.
7. Pause and stir or rotate food when the instructions recommend it.
8. Use suitable heat protection when removing hot containers.
9. Check that the food is heated evenly and reaches the recommended safe temperature.
10. Allow standing time when required by the recipe or package instructions.
11. Switch off the oven and remove the food carefully.
12. Wipe up spills after the cavity has cooled sufficiently.
13. Switch off and unplug the oven before cleaning or when required by the instructions.

⚠ Safety

- Never operate the oven with a damaged door, seal, or safety interlock.
- Do not operate the oven when empty unless the manufacturer specifically permits it.
- Do not place metal utensils, foil, or unsuitable containers inside the oven.
- Open sealed containers and pierce suitable food skins as instructed to prevent pressure build-up.
- Beware of steam when uncovering food and liquids, which may become superheated.
- Never remove the outer casing or attempt magnetron repairs; servicing must be done by qualified personnel.

Care & storage

- Clean the cavity, door seal, and turntable regularly using a suitable cleaner.
- Keep the door seal and interlock areas free from food debris.
- Avoid abrasive tools that may damage the interior or door surface.
- Inspect the power cord and door operation regularly.
- Arrange qualified service if the oven sparks unexpectedly, overheats, or the door fails to close properly.

Common mistakes to avoid

- Using metal or non-microwave-safe containers.
- Operating the oven with a damaged door seal or interlock.
- Heating food without checking for hot spots or required standing time.
- Attempting to repair internal high-voltage components without qualification.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Millet Cleaning Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Millet Cleaning Machine

What it is used for

To remove dust, stones, chaff, and other unwanted material from millet before further processing. To obtain clean grain while maintaining consistent feeding and preventing blockages or contamination.

Main parts

- Machine frame and feed hopper
- Screens or sieves
- Cleaning chamber and airflow system, if fitted
- Vibrating or rotating cleaning mechanism
- Grain outlet and waste outlet
- Motor, drive, and control panel

Before you start

1. Read the manual and confirm that the machine is suitable for the millet variety and batch size.
2. Inspect the hopper, screens, cleaning chamber, outlets, guards, and drive components for damage.
3. Ensure all product-contact surfaces are clean and free from residue from previous batches.
4. Check that the correct screen size is installed and securely fitted.
5. Confirm that grain and waste collection containers are clean and correctly positioned.
6. Remove foreign objects from the feed area and check that the waste outlet and airflow path are clear.

How to use

1. Switch off and isolate power before opening the chamber or adjusting screens.
2. Close access panels and ensure all guards are fitted before starting.
3. Set the feed rate and airflow, if available, according to the grain condition and machine instructions.
4. Place the raw millet in the feed hopper without exceeding its capacity.
5. Start the machine and allow it to reach normal operating conditions.
6. Feed millet gradually and maintain a steady flow to avoid overloading.
7. Observe the cleaned grain output and the removal of dust, chaff, and other impurities.
8. Check for unusual noise, vibration, blockage, or grain loss during operation.
9. Adjust feed or airflow only within the manufacturer's specified limits.
10. Stop the machine if screens block, grain flow becomes irregular, or a mechanical fault occurs.

1. Isolate power and wait for moving parts to stop before clearing blockages or changing screens.
2. Collect and inspect the cleaned millet, then clean the machine and record any issues.

⚠ Safety

- Only trained personnel should operate the cleaning machine.
- Keep hands and tools away from moving screens, rotating parts, belts, and feed mechanisms.
- Do not operate with missing guards, damaged screens, or exposed drive components.
- Use suitable dust protection and ensure ventilation or dust extraction where necessary.
- Keep the floor around the machine dry and free from spilled grain to prevent slips.
- Disconnect and isolate power before clearing blockages, removing screens, or performing maintenance.

Care & storage

- Remove grain, dust, and chaff from the hopper, screens, chamber, and outlets after use.
- Inspect screens for tears, wear, and loose fittings regularly.
- Check belts, bearings, fasteners, and drive components according to the maintenance schedule.
- Keep motors and electrical controls free from dust buildup using approved methods.
- Store the machine clean and dry and record maintenance or abnormal operation.

Common mistakes to avoid

- Overloading the hopper and feeding millet too quickly.
- Using the wrong screen size for the grain or required cleaning level.
- Operating with blocked screens or an obstructed waste outlet.
- Cleaning the machine while moving parts are still running.
- Ignoring excessive grain loss, unusual vibration, or poor cleaning performance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mixer Grinder

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Mixer and grinder; Mixer Grinder; Mixer (Mixture- Grinder)

What it is used for

A mixer grinder blends, mixes, and grinds suitable food ingredients using rotating blades. Correct jar selection, secure locking, and safe cleaning help prevent cuts, spills, and motor damage.

Main parts

- Motor unit provides the power to rotate the blades.
- Jar or container holds ingredients during mixing or grinding.
- Blade assembly cuts, blends, or grinds the ingredients.
- Lid and lid lock help prevent spills during operation.
- Jar handle supports safe holding and pouring.
- Speed control or pulse switch regulates operation.
- Power cord and plug supply electricity to the appliance.

Before you start

1. Read the manufacturer's instructions and choose the correct jar for the task.
2. Inspect the motor unit, jar, lid, blade assembly, plug, and cord for damage.
3. Place the mixer grinder on a stable, dry surface near a suitable outlet.
4. Ensure the jar, lid, and blade assembly are clean and correctly fitted.
5. Cut large ingredients into suitable pieces and check that they are permitted for the jar.
6. Keep hands, utensils, and loose objects away from the blade assembly.

How to use

1. Switch off the appliance before fitting or removing a jar.
2. Place the ingredients inside the jar without exceeding its marked capacity.
3. Add the required liquid if the recipe and appliance instructions permit it.
4. Fit the lid securely and lock the jar onto the motor unit.
5. Connect the plug with dry hands and select an appropriate speed.
6. Use short pulse intervals when suitable and avoid running the motor longer than recommended.
7. Switch off and wait until the blades stop completely before opening the lid.
8. Disconnect power before scraping ingredients or removing the jar.
9. Pour the contents carefully and handle the blade assembly without touching its sharp edges.
10. Unplug the appliance and clean the jar, lid, and blade parts according to instructions.

⚠ Safety

- Never operate the appliance without a properly secured jar and lid.
- Never insert fingers or utensils into a jar while the motor is running.
- Do not exceed the jar's capacity or operate it with unsuitable ingredients.
- Keep the motor unit, plug, and cord away from water.
- Do not open a hot jar or touch blades immediately after operation.
- Stop using the appliance if it vibrates excessively, leaks, or produces a burning smell.

Care & storage

- Unplug the appliance before cleaning or removing blade assemblies.
- Wash compatible jars and lids according to the manufacturer's instructions and dry them fully.
- Wipe the motor unit with a slightly damp cloth without immersing it in water.
- Inspect the cord, jar, seals, and blades regularly and replace damaged parts.

Common mistakes to avoid

- Running the mixer grinder with a loose lid or incorrectly fitted jar.
- Filling the jar beyond its maximum mark.
- Opening the jar before the blades stop completely.
- Immersing the motor unit or touching sharp blades carelessly during cleaning.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mixing Bowl

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Low** 8 labs

Covers these lab items: Mixing Bowles set; Mixing bowls; Mixing Bowls (Different Sizes); Mixing Bowls set; Mixing bowl/vessels

What it is used for

Mixing bowls hold ingredients for combining, whisking, kneading, and other food preparation tasks. Choose a suitable bowl size and material for the recipe and maintain food hygiene throughout use.

How to use

1. Inspect the bowl for cracks, chips, corrosion, or residue.
2. Select a bowl suitable for the ingredient quantity and preparation method.
3. Wash and sanitize the bowl before food contact as required.
4. Place the bowl on a stable, clean work surface.
5. Add ingredients carefully and mix using suitable utensils.
6. Avoid overfilling and transfer the mixture to the next preparation stage as needed.
7. Wash, rinse, sanitize, and dry the bowl after use.

⚠ Safety

- Do not use chipped, cracked, or food-unsafe bowls.
- Use heat-resistant bowls only within their specified temperature limits.
- Keep raw and ready-to-eat foods separate to prevent cross-contamination.
- Handle heavy or hot bowls carefully to avoid spills and burns.

Care & storage

- Clean bowls promptly after use and store them dry without stacking in a way that causes damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mixing Tank

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Syrup mixing tank

What it is used for

A syrup mixing tank combines syrup ingredients into a uniform mixture for suitable food-processing applications. Follow the approved recipe, equipment instructions, and food hygiene requirements.

How to use

1. Inspect the tank, lid, agitator, valves, seals, and control panel for damage.
2. Clean and sanitize all food-contact surfaces according to the approved procedure.
3. Confirm the recipe, ingredient quantities, and batch identification before starting.
4. Add ingredients in the specified order and quantity, following the recipe.
5. Close the lid as required and start the agitator at the recommended speed.
6. Monitor mixing time, temperature, and consistency according to the process instructions.
7. Stop the agitator, transfer the syrup safely, and clean the tank after the batch.

⚠ Safety

- Keep hands, utensils, and loose clothing away from the agitator while it is operating.
- Do not open guards or reach into the tank until the machine is isolated and stopped.
- Use suitable PPE when handling hot syrup or cleaning chemicals.
- Follow food safety and allergen-control procedures for each batch.

Care & storage

- Clean and sanitize the tank after use and inspect seals, valves, and agitator components regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Outdoor Cooking Equipment

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Outdoor cooking equipment, such as stoves, pots, and reusable utensils

What it is used for

Outdoor cooking equipment includes portable stoves, pots and reusable utensils used to prepare food outside a kitchen. Safe setup, clean handling and proper fuel management help prevent burns, fires and food contamination.

Main parts

- Identify the stove body, burner and designated fuel connection.
- Locate the fuel cylinder, cartridge or supply line, depending on the stove.
- Identify the ignition control, flame adjustment and shut-off valve.
- Locate the cooking pot, pan, lid and heat-resistant handles.
- Identify reusable spoons, ladles, tongs and serving utensils.
- Locate the stable supporting stand and wind protection, if provided.
- Identify cleaning supplies, heat-resistant gloves and suitable fire-extinguishing equipment.

Before you start

1. Choose a stable, level outdoor location away from dry vegetation and combustible materials.
2. Check local fire restrictions and ensure adequate ventilation.
3. Inspect the stove, fuel hose, regulator and connections for damage or leaks.
4. Keep fuel containers upright and away from direct heat.
5. Wash hands and clean pots and reusable utensils before cooking.
6. Prepare fresh ingredients and keep raw foods separate from cooked foods.
7. Keep a suitable fire extinguisher or other recommended fire-control equipment accessible.

How to use

1. Set up the stove on a stable, non-combustible surface outdoors.
2. Connect the approved fuel supply according to the stove instructions.
3. Check fuel connections for leaks using the manufacturer's recommended method.
4. Place the pot securely on the stove support before ignition where required.
5. Ignite the burner according to the instructions and adjust the flame safely.
6. Add ingredients using clean utensils and avoid overfilling the pot.
7. Cook food thoroughly while monitoring the flame and pot stability.
8. Use heat-resistant gloves or suitable tools when handling hot cookware.
9. Turn off the burner and close the fuel supply after cooking.
10. Allow cookware to cool before washing and packing the equipment.

⚠ Safety

- Never use outdoor fuel-burning stoves inside enclosed rooms or poorly ventilated spaces.
- Keep children, loose clothing and combustible materials away from the flame.
- Never leave a lit stove unattended or move it while hot.
- Do not use equipment with leaking fuel connections or damaged hoses.
- Never pour water onto burning cooking oil or grease.
- If a fuel leak or uncontrolled fire occurs, shut off fuel only if safe, move away and call emergency services.

Care & storage

- Allow the stove and cookware to cool completely before cleaning.
- Wash reusable utensils with suitable detergent and safe water, then dry them.
- Inspect burners, hoses and connectors before storage.
- Store clean, dry equipment and fuel containers separately in appropriate locations.

Common mistakes to avoid

- Setting up the stove on an unstable surface or near dry vegetation.
- Ignoring fuel leaks or using an unsuitable fuel connection.
- Leaving cooked food exposed to insects, dust or raw-food contamination.
- Packing hot equipment or storing fuel near heat sources.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oven

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 8 labs

Covers these lab items: Baking oven; Commercial Oven (Deck/Convection); Convection Oven; Deck Oven; Oven; Rotary Oven; Tandoor/OTG

What it is used for

An oven uses controlled heat to bake, roast, or cook suitable food and materials. Correct temperature settings, safe loading, and regular cleaning help prevent burns, fire, and uneven cooking.

Main parts

- Cooking chamber contains the food during heating.
- Heating elements or burners produce heat.
- Temperature control sets the required cooking temperature.
- Thermostat or sensor monitors the chamber temperature.
- Door and insulated handle help contain heat and provide access.
- Racks, trays, or shelves support the items being heated.
- Timer and control panel manage cooking duration and operating settings.

Before you start

1. Read the manufacturer's instructions and confirm the oven suits the intended cooking task.
2. Inspect the door, handle, racks, controls, power cord, or gas connections for damage.
3. Place the oven on a stable surface with adequate ventilation and clearance.
4. Keep the chamber clean and remove grease, loose food, and unnecessary objects.
5. Use suitable heat-resistant gloves and check that trays and utensils are oven-safe.
6. Preheat the oven to the temperature specified in the recipe or operating instructions.

How to use

1. Switch on the oven using the approved controls and select the required heating mode.
2. Set the required temperature and preheating time according to the recipe or instructions.
3. Prepare food in suitable trays and arrange items with enough space for heat circulation.
4. Open the door carefully, keeping your face and hands away from escaping hot air.
5. Place trays on the appropriate rack and close the door securely.
6. Set the cooking timer and monitor the process through the window when available.
7. Use heat-resistant gloves when checking, turning, or removing hot trays.
8. Check that the food is cooked adequately using the appropriate food-safety method.
9. Switch off the oven after use and remove food carefully.
10. Allow the oven to cool before cleaning the chamber, racks, and accessible surfaces.

⚠ Safety

- Never touch heating elements, hot trays, or internal surfaces with bare hands.
- Keep children and unauthorised users away from the hot oven.
- Never block ventilation openings or leave flammable materials near the oven.
- Do not use an oven with damaged wiring, gas connections, door seals, or controls.
- For gas ovens, shut off the supply if you smell gas and follow emergency procedures.
- Do not pour cold water onto hot glass or heating components.

Care & storage

- Switch off and disconnect electrical power before cleaning, where safe and applicable.
- Remove food residue and grease using manufacturer-approved cleaning methods.
- Inspect door seals, racks, controls, and connections regularly for damage.
- Arrange authorised servicing if heating becomes uneven or unusual smells, sparks, or faults occur.

Common mistakes to avoid

- Loading food before the oven reaches the required preheating temperature.
- Overloading trays or blocking airflow in a convection oven.
- Touching hot racks or removing trays without heat-resistant gloves.
- Cleaning the oven while it remains hot or connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Packaging Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Ice Bar Packaging machine

What it is used for

An ice bar packaging machine wraps or seals ice bars in suitable packaging material for handling and distribution. Correct operation helps protect product hygiene, package integrity and consistent sealing.

Main parts

- Identify the packaging material roll, film holder or feed arrangement.
- Locate the product feeding area and guiding mechanism.
- Identify the sealing jaws, heat-sealing unit or closure mechanism.
- Locate the cutting blade or package separation mechanism, if fitted.
- Identify the conveyor or discharge tray for packaged products.
- Locate the control panel, temperature setting and speed controls.
- Identify emergency stop buttons, guards and electrical isolation points.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Wash hands and follow the required hygiene procedures for food packaging.
3. Inspect electrical cables, guards, sealing surfaces and cutting components for damage.
4. Confirm that the packaging material is food-grade and suitable for the product.
5. Check that the film roll is correctly loaded and aligned.
6. Verify sealing temperature and speed settings against the manufacturer's instructions.
7. Ensure the work area is clean, dry and free from obstructions.

How to use

1. Switch off and isolate power before loading film or adjusting the machine.
2. Install the packaging film in the specified direction and align its feed path.
3. Thread the film through the guides and sealing mechanism as instructed.
4. Close all guards and ensure the emergency stop is accessible.
5. Switch on the machine and allow the sealing unit to reach its set temperature.
6. Run a test package and inspect seal quality and cutting alignment.
7. Feed ice bars into the designated area using the approved feeding method.
8. Monitor film movement, package alignment and seal consistency during operation.
9. Stop feeding if packages jam, seals fail or film tracks incorrectly.
10. Switch off and isolate power before removing jams or cleaning the machine.

⚠ Safety

- Keep fingers away from hot sealing jaws and cutting blades.
- Never bypass guards or reach into moving feed mechanisms.
- Disconnect electrical power before changing film, clearing jams or servicing parts.
- Use heat-resistant protection when handling hot components after shutdown.
- Keep the control panel and electrical connections dry.
- Follow food hygiene procedures and reject damaged or improperly sealed packages.

Care & storage

- Isolate power and allow hot sealing components to cool before cleaning.
- Remove packaging scraps and product residue using approved cleaning methods.
- Inspect sealing surfaces, blades, film guides and guards for wear.
- Store unused packaging material in clean, dry and protected conditions.

Common mistakes to avoid

- Loading packaging film in the wrong direction or leaving it misaligned.
- Using incorrect sealing settings that cause weak or burnt seals.
- Reaching into the machine before isolating power.
- Packaging ice bars with contaminated material or damaged seals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pasteurizer

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Bottle flesh pasteurizer; Fruit juice pasteurizer; Pasteurizer; Small Pasteurizer

What it is used for

A pasteurizer heats suitable liquids or food products to a controlled time and temperature combination to reduce harmful microorganisms. Correct process settings, hygienic handling, and prompt cooling where required help maintain food safety and product quality.

Main parts

- Heating vessel or product chamber holds the material being processed.
- Heating element or heat exchanger transfers heat to the product.
- Temperature sensor measures product or process temperature.
- Controller regulates the heating cycle and target temperature.
- Timer controls the required holding period on equipped models.
- Agitator circulates the product in models designed for mixing.
- Inlet and outlet valves control product flow.
- Cooling system or heat exchanger cools the product where provided.

Before you start

1. Read the manual and use a validated process suitable for the specific product.
2. Confirm the pasteurizer is clean, sanitised, and suitable for food contact.
3. Inspect seals, valves, pipes, sensors, controls, and the heating system for damage.
4. Check calibration or verification status for temperature and time controls.
5. Prepare the product and confirm batch size is within the equipment's rated capacity.
6. Ensure clean receiving containers and any required cooling equipment are ready.

How to use

1. Switch on the pasteurizer and select the approved product-specific process settings.
2. Load the measured product without exceeding the permitted capacity.
3. Start heating and monitor the temperature according to the validated procedure.
4. Ensure the product reaches the required temperature throughout the specified holding period.
5. Use the agitator only where the model and process instructions require it.
6. Record the batch details, temperature, holding time, and relevant monitoring results.
7. If the required temperature or holding time is not achieved, follow the approved corrective procedure before releasing the batch.
8. Cool the product promptly using the specified method when the process requires cooling.
9. Transfer the processed product into clean, suitable containers and apply the required labels.
10. Switch off the equipment and clean and sanitise product-contact parts according to the manual.

⚠ Safety

- Use only validated time and temperature settings appropriate to the product and intended process.
- Avoid contact with hot product, steam, heated surfaces, and hot pipes.
- Do not open pressurised vessels or loosen fittings while the system is operating.
- Never bypass temperature controls, alarms, or safety interlocks.
- Use suitable PPE and follow chemical instructions when handling cleaning or sanitising agents.
- Do not release a batch if required process records or safety checks are incomplete.

Care & storage

- Clean and sanitise product-contact surfaces after each batch using approved procedures.
- Inspect seals, valves, filters, and pipes for wear, leaks, or residue.
- Maintain temperature sensors and timers according to the calibration schedule.
- Arrange authorised servicing for heating failures, temperature instability, or persistent leakage.

Common mistakes to avoid

- Using an unvalidated temperature and holding time for the product.
- Starting the timer before the product reaches the required temperature.
- Overfilling the vessel or failing to mix when the process requires it.
- Allowing processed product to cool or remain exposed under unsuitable conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pastry Brush

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Low** 1 labs

Covers these lab items: Pastry Brushes

What it is used for

A pastry brush spreads egg wash, butter, glaze, oil, or other suitable coatings evenly over food. Select a food-safe brush and clean it thoroughly to prevent contamination between ingredients.

How to use

1. Inspect the bristles, handle, and brush head for damage or loose parts.
2. Choose a food-safe brush suitable for the ingredient and temperature.
3. Wash the brush before use and ensure it is free of residue.
4. Dip the bristles into the coating and remove excess liquid to prevent dripping.
5. Brush the food gently and evenly without damaging the surface.
6. Keep separate brushes for ingredients that may cause cross-contamination when required.
7. Wash, sanitize, and dry the brush after use.

⚠ Safety

- Do not use brushes with loose bristles or damaged handles.
- Keep food-contact brushes separate from paint or cleaning brushes.
- Use heat-resistant brushes only within their rated temperature range.
- Follow food hygiene procedures when handling allergens and raw egg mixtures.

Care & storage

- Clean promptly after use, dry thoroughly, and store in a clean, covered location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Peeler

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Peeler; Vegetable Peeler

What it is used for

To remove thin outer skins from vegetables or fruits during food preparation.

How to handle

- Inspect the peeler blade, handle, and joint for damage, rust, or looseness.
- Wash the produce and place it on a stable cutting board before peeling.
- Hold the produce securely and peel away from fingers and the body using controlled strokes.
- Keep fingers clear of the blade and use a stable grip, especially when peeling small pieces.
- Wash, rinse, and dry the peeler after use, taking care around the sharp edge.

Storage

- Store clean and dry in a designated kitchen drawer or utensil rack.
- Use a blade cover where available and keep it separate from loose items that could damage the edge.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Peeling Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Lye peeling machine

What it is used for

A lye peeling machine removes the outer skin of suitable fruits or vegetables using a controlled alkaline solution. Correct operation helps loosen the peel while protecting food quality and operator safety.

Main parts

- Identify the product feeding hopper or loading basket.
- Locate the lye solution tank and designated filling point.
- Identify the heating system and temperature control, if fitted.
- Locate the product conveyor or treatment chamber.
- Identify the rinsing section used to remove residual lye and loosened peel.
- Locate the discharge outlet and waste collection arrangement.
- Identify chemical-resistant seals, protective guards and emergency controls.

Before you start

1. Read the operating instructions and chemical safety information before use.
2. Operate only under trained supervision with appropriate chemical-handling procedures.
3. Wear chemical-resistant gloves, goggles or face protection, apron and suitable footwear.
4. Inspect tanks, pipes, valves, seals and guards for leaks or damage.
5. Confirm that the lye concentration and temperature follow the approved process instructions.
6. Ensure adequate ventilation and access to an appropriate eyewash and emergency rinse facility.
7. Check that the rinsing system and waste collection arrangements are ready.

How to use

1. Confirm the machine is clean and the chemical tank is correctly prepared.
2. Load suitable fruits or vegetables into the designated feeding arrangement.
3. Set the approved treatment time, temperature and solution concentration.
4. Start the machine according to the operating instructions.
5. Introduce the product into the treatment chamber using the designated feed system.
6. Monitor treatment conditions and prevent product overexposure to the lye solution.
7. Transfer treated products through the designated rinsing section.
8. Rinse thoroughly using the validated process to remove residual lye and loosened peel.
9. Inspect a sample to confirm acceptable peeling and rinsing results.
10. Stop the machine and isolate power before draining, cleaning or servicing.

⚠ Safety

- Lye is corrosive and can cause severe skin and eye burns.
- Never touch the solution or open a chemical tank during operation.
- Never mix lye with acids or other chemicals unless an approved procedure specifically requires it.
- Follow the chemical supplier's instructions for safe preparation and handling; add lye to water slowly, never water to concentrated lye.
- Keep emergency eyewash and rinsing facilities accessible and seek urgent medical assistance after exposure.
- Never release alkaline waste without following approved handling and disposal requirements.

Care & storage

- Isolate power and follow the approved procedure before draining the solution.
- Clean product-contact surfaces and rinse sections according to the sanitation instructions.
- Inspect tanks, pipes, valves and seals regularly for corrosion or leakage.
- Store chemicals in labelled, closed containers away from food and unauthorised users.

Common mistakes to avoid

- Using an unverified lye concentration or treatment temperature.
- Failing to rinse treated produce adequately before further processing.
- Handling alkaline solution without appropriate PPE.
- Mixing incompatible chemicals or disposing of lye waste carelessly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Piping Bag

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Piping bag

What it is used for

To dispense icing, cream, or other suitable food mixtures in controlled shapes and portions.

How to handle

- Inspect the piping bag, nozzle, and coupler for tears, cracks, or residue before use.
- Wash hands and use a clean, food-safe bag and nozzle; ensure reusable parts are cleaned and sanitized as required.
- Fit the nozzle securely, fold the top of the bag outward, and fill it only to a manageable level.
- Twist or close the top and apply steady pressure while keeping the nozzle directed away from people.
- After use, empty the bag safely and wash, rinse, sanitize when required, and dry reusable components.

Storage

- Ensure reusable bags and nozzles are fully clean and dry before storage.
- Store in a clean, covered food-equipment container; discard damaged or single-use bags as appropriate.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pizza Cutter

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Pizza cutter

What it is used for

To cut pizza or similar prepared food into portions using a rolling or handled cutting wheel.

How to handle

- Inspect the cutting wheel, handle, guard, and axle for damage, looseness, or food residue.
- Wash hands and clean the cutter before food preparation or service.
- Place the pizza on a stable, suitable cutting surface and hold the handle firmly with fingers away from the wheel.
- Roll the cutter through the food using controlled pressure and keep the cutting path clear of hands and other people.
- Clean and sanitize the cutter after use, taking care when handling the sharp wheel.

Storage

- Clean, sanitize, and dry the cutter thoroughly after use.
- Store with the wheel guarded or covered in a clean, dry utensil drawer or holder.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Planetary Mixer

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Planetary mixer

What it is used for

A planetary mixer mixes, whips, or kneads food ingredients using an attachment that rotates around its own axis while moving around the bowl. Use the correct attachment, speed, and batch capacity for the recipe and machine design.

Main parts

- Mixing bowl and bowl-locking mechanism
- Planetary mixing head and drive motor
- Beater, whisk, or dough hook attachments
- Speed selector and control panel
- Safety guard or bowl shield, if fitted
- Power switch and emergency stop, if fitted

Before you start

1. Confirm the mixer and attachment are suitable for the recipe and batch size.
2. Inspect the bowl, attachment, guard, locking mechanism, and power cable.
3. Ensure the bowl and attachments are clean and sanitised for food use.
4. Check that the machine is stable and all guards and safety devices work.
5. Verify the maximum capacity and recommended speed for the ingredients.
6. Keep utensils, loose clothing, and unnecessary items away from the mixing area.

How to use

1. Switch off the mixer before fitting or changing attachments.
2. Secure the bowl in its correct position and lock it as instructed.
3. Fit the appropriate beater, whisk, or dough hook securely.
4. Add the measured ingredients without exceeding the bowl capacity.
5. Lower the mixing head or close the guard according to the machine design.
6. Select the lowest suitable speed to begin mixing.
7. Increase speed gradually if the recipe and attachment permit it.
8. Observe the mixture and stop the machine if excessive vibration or unusual noise occurs.
9. Switch off and wait for the attachment to stop before scraping the bowl.
10. Use a suitable scraper only when the mixer is fully stopped and safe to access.
11. Switch off and isolate power before removing the bowl or attachment if required.
12. Transfer the mixture using clean utensils and follow food-safety requirements.
13. Clean and sanitise the bowl, attachments, and food-contact surfaces after use.

⚠ Safety

- Only trained personnel should operate commercial planetary mixers.
- Never place hands or utensils into the bowl while the attachment is moving.
- Do not bypass bowl guards, interlocks, or emergency stops.
- Switch off and isolate power before changing attachments or clearing jams.
- Do not exceed the specified capacity or use an attachment at an unsuitable speed.
- Keep the floor and controls dry and report damaged cables immediately.

Care & storage

- Clean and sanitise the bowl and attachments after each batch.
- Wipe the motor housing with an approved method; do not immerse it.
- Inspect the drive, attachment fittings, guards, and locking mechanism regularly.
- Lubricate service points only as specified by the manufacturer.
- Arrange servicing for abnormal noise, vibration, or speed-control problems.

Common mistakes to avoid

- Overloading the bowl or using the wrong attachment.
- Starting at excessive speed and causing splashing or ingredient overflow.
- Scraping or reaching into the bowl while the attachment is moving.
- Removing guards or changing attachments without isolating power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Plastic Wrap Dispenser

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Plastic Wrap Dispenser

What it is used for

A plastic wrap dispenser holds and cuts food-grade plastic film for covering suitable food containers or work items. Load the correct film roll and use the cutting edge carefully to prevent injury and contamination.

How to use

1. Inspect the dispenser body, roll holder, cutting edge, and cover for damage.
2. Confirm that the plastic wrap is suitable for the intended food-contact use.
3. Fit the roll according to the dispenser design and keep it aligned.
4. Pull out the required length of film without touching food-contact surfaces unnecessarily.
5. Wrap the item securely and avoid excessive film contact with unsuitable hot surfaces.
6. Cut the film using the designated cutter while keeping fingers away from the edge.
7. Return the film end to the holder and clean the dispenser as needed.

⚠ Safety

- Keep fingers clear of the cutting edge and moving parts.
- Do not use damaged film or wrap that is not approved for food contact.
- Keep the dispenser away from open flames and hot equipment.
- Follow the film manufacturer's temperature and microwave-use restrictions.

Care & storage

- Remove film residue, clean the dispenser according to its material, and replace damaged cutting parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pouch Sealing Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Hand Sealer/Pouch Sealer; Sealing Machine

What it is used for

A pouch sealing machine seals compatible packaging film or pouches using heat, pressure, or another specified sealing method. Correct temperature, dwell time, pressure, and hygienic handling help produce consistent seals and protect the packaged product.

Main parts

- Heating element produces sealing heat where fitted.
- Sealing jaws or bars press the pouch closure area.
- Temperature controller sets the sealing temperature on heated models.
- Timer controls the sealing duration where fitted.
- Pressure handle or pneumatic mechanism applies sealing pressure.
- Work surface supports the pouch during sealing.
- Power switch starts and stops the machine.
- Indicator lights show heating or operating status.

Before you start

1. Only trained personnel should operate the pouch sealer.
2. Read the manual and confirm the machine is suitable for the pouch material and product.
3. Inspect the power cable, plug, sealing jaws, heating surface, handle, and controls for damage.
4. Check that the temperature, pressure, and sealing time settings match the packaging material instructions.
5. Clean the work surface and sealing area using the approved hygiene procedure.
6. Keep flammable materials away and ensure the machine is placed on a stable, heat-resistant surface.
7. Confirm that the sealing area is free of product residue, moisture, and foreign material.

How to use

1. Switch on the machine and allow the sealing element to reach the specified operating temperature where applicable.
2. Set the sealing temperature, dwell time, and pressure according to the pouch material and product requirements.
3. Prepare the pouch and ensure the sealing edge is clean, dry, and free of product.
4. Place the pouch edge flat between the sealing jaws or at the designated sealing position.
5. Activate the sealing cycle using the approved handle, switch, or control.

6. Maintain the pouch in position for the specified dwell time without touching the hot sealing surface.
7. Release the handle or control and remove the pouch once the sealing jaws have opened safely.
8. Inspect the seal for continuity, wrinkles, gaps, burn-through, or contamination.
9. Adjust settings only within the manufacturer's limits if the seal quality is unsuitable.
10. Switch off the machine after use and allow hot parts to cool before cleaning.
11. Clean the sealing area, record any faults, and store the machine safely.

⚠ Safety

- Do not touch the heated sealing bar or jaws during operation or while they remain hot.
- Keep fingers away from closing jaws and pinch points.
- Do not use the machine with damaged wiring, exposed heating elements, or faulty controls.
- Do not seal containers or materials that may release hazardous fumes when heated.
- Keep water and wet hands away from electrical components.
- Isolate power before cleaning, maintenance, or removing jammed packaging.

Care & storage

- Remove packaging residue from the sealing area after the machine has cooled.
- Clean food-contact or product-contact surfaces using approved methods where applicable.
- Inspect sealing jaws, heating surfaces, handles, cables, and controls regularly.
- Arrange authorised maintenance if temperature control, timing, or sealing pressure becomes unreliable.

Common mistakes to avoid

- Using settings that are unsuitable for the pouch material.
- Sealing over product residue, moisture, or wrinkles.
- Touching hot sealing jaws or placing fingers in pinch points.
- Cleaning or clearing packaging while the machine remains powered.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pressure Cooker

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Pressure Cooker; Pressure Cooker ISI Certified; Pressure Cookers (Different Sizes)

What it is used for

A pressure cooker cooks food in a sealed vessel using controlled steam pressure to increase cooking temperature and reduce cooking time. Correct filling, vent cleaning, gasket condition, and pressure release are essential to prevent burns and pressure-related accidents.

Main parts

- Cooking body holds the food and cooking liquid.
- Lid closes and seals the cooking vessel.
- Gasket or sealing ring creates the pressure seal.
- Pressure regulator controls operating pressure on compatible models.
- Vent tube or steam outlet allows steam to escape through the regulator.
- Safety valve or pressure-release device provides additional pressure protection.
- Lid-locking mechanism helps prevent opening under pressure.
- Handles allow safe carrying when the cooker is cool or depressurised.

Before you start

1. Only users familiar with the pressure cooker's instructions should operate it.
2. Check the cooker body, lid, handles, gasket, regulator, vent tube, and safety valve for damage or blockage.
3. Confirm that the gasket is correctly fitted and is not cracked, stretched, or hardened.
4. Ensure the vent tube and pressure-release components are clean and unobstructed.
5. Add the required food and cooking liquid without exceeding the manufacturer's maximum fill line.
6. For foods that expand or foam, follow the reduced fill limit stated in the manual.
7. Place the cooker on a stable, suitable heat source and keep the handle clear of flames where applicable.

How to use

1. Place the prepared ingredients and required liquid inside the cooker without exceeding the permitted fill level.
2. Check the sealing ring and lid alignment before closing the cooker.
3. Lock the lid using the manufacturer's specified method.
4. Set the pressure regulator or valve according to the recipe and cooker instructions.
5. Heat the cooker and monitor it until the required pressure is reached.
6. Reduce the heat as instructed to maintain the appropriate operating pressure.

7. Observe the cooker during operation and check for unusual leakage, blocked vents, or abnormal sounds.
8. Turn off the heat when cooking is complete and follow the specified pressure-release method.
9. Wait until pressure has fully released and the pressure indicator confirms it is safe to open.
10. Open the lid away from the face and body to avoid escaping steam.
11. Remove food carefully, clean the cooker after it cools, and inspect components before storage.

⚠ Safety

- Never force the lid open while pressure remains inside the cooker.
- Never block, cover, or tamper with the vent tube, regulator, or safety valve.
- Do not fill above the manufacturer's maximum level or ignore special limits for foaming foods.
- Keep face, hands, and body away from escaping steam and the lid opening.
- Do not use a cooker with a damaged gasket, blocked vent, faulty safety valve, or deformed lid.
- Do not leave the cooker operating unattended when the manufacturer's instructions or cooking conditions require supervision.

Care & storage

- Clean the cooker body, lid, regulator, and vent tube after use according to the manual.
- Inspect the gasket and safety devices regularly and replace worn parts with compatible components.
- Keep the vent and pressure-release passages clear of food residue.
- Dry the cooker completely and store the lid according to the manufacturer's instructions.
- Arrange authorised repair or replacement if pressure regulation or locking becomes unreliable.

Common mistakes to avoid

- Overfilling the cooker or using too little cooking liquid.
- Opening the lid before pressure has fully released.
- Using a cooker with a blocked vent tube or damaged gasket.
- Attempting to modify or disable pressure safety devices.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pressure Stove

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Pressure Stove

What it is used for

A pressure stove is a cooking appliance that uses a fuel burner or heating system to operate a compatible pressure-cooking vessel. Follow the specific stove and cooker instructions, ensure safe ventilation, and prevent fuel leaks or pressure-related accidents.

Main parts

- Burner or heating unit
- Fuel supply connection or control valve, where applicable
- Ignition system or lighting mechanism
- Pot support and stable base
- Flame or heat control
- Safety shutoff and ventilation openings, where fitted

Before you start

1. Identify the stove type and confirm the approved fuel or power source.
2. Place the stove on a stable, level, heat-resistant surface.
3. Inspect the burner, fuel hose, regulator, valves, and connections for damage.
4. For gas models, check for leaks using an approved method; never use a flame to test.
5. Ensure adequate ventilation and keep combustible materials away.
6. Check that the pressure cooker is compatible and its safety valves are clean and unobstructed.

How to use

1. Keep the stove switched off while connecting or checking the fuel supply.
2. Position the compatible pressure cooker securely on the pot support.
3. Add the required food and liquid to the cooker according to its instructions.
4. Close and lock the cooker lid correctly before heating.
5. Open the fuel valve only according to the stove's operating procedure.
6. Ignite the burner using the built-in igniter or approved lighting method.
7. Adjust the flame or heat to the recommended level.
8. Monitor the cooking process and check for abnormal flame, smell, or leakage.
9. Reduce the heat when the cooker reaches the required pressure, following its instructions.
10. Switch off the burner after cooking and close the fuel valve, where applicable.
11. Allow the cooker to depressurise fully according to its instructions before opening.
12. After the stove cools, clean spills and check that the burner is clear.
13. Close the fuel supply and store the appliance safely.

⚠ Safety

- Operate the stove only in a well-ventilated area and follow fuel-specific instructions.
- Never use a stove with a suspected gas leak or damaged fuel connection.
- Keep children, loose clothing, and combustible materials away from the flame.
- Never open a pressure cooker while it is pressurised.
- Do not block the burner's air openings or modify safety devices.
- If gas is smelled, shut off the supply if safe, avoid flames and electrical switches, leave the area, and follow emergency procedures.

Care & storage

- Clean the burner and pot support after the appliance has cooled.
- Inspect fuel hoses, regulators, valves, and connections at recommended intervals.
- Keep ignition components and burner openings free from food residue.
- Arrange qualified servicing for gas leaks, unstable flames, or faulty controls.

Common mistakes to avoid

- Using the wrong fuel or an incompatible pressure cooker.
- Lighting the burner before checking the fuel connections.
- Opening the pressure cooker before pressure has fully released.
- Ignoring a gas smell, blocked burner, or unstable flame.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Puffing Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Flaking / Puffing Machine

What it is used for

A flaking or puffing machine processes suitable grains or food materials using controlled heat, pressure or mechanical action. It produces flakes or puffed products for further preparation or consumption according to the machine design.

Main parts

- Identify the feed hopper or material loading opening.
- Locate the heating chamber, roasting section or pressure vessel, depending on design.
- Identify the rollers, dies or expansion chamber used to form the product.
- Locate the motor, drive mechanism and transmission components, where fitted.
- Identify the temperature, pressure and speed controls, where provided.
- Locate the discharge outlet and product collection tray.
- Identify protective guards, emergency stop and cleaning access points.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Inspect the frame, guards, electrical connections and moving parts for damage.
3. Confirm the machine is designed for the intended grain or food material.
4. Clean and dry food-contact surfaces and prepare suitable collection containers.
5. Check the heating system, pressure indicators and safety devices, where fitted.
6. Ensure ventilation and dust control are adequate for the material being processed.
7. Wear suitable protective clothing, hair covering and heat-resistant protection where required.

How to use

1. Check that the machine is clean, correctly assembled and ready for operation.
2. Select the appropriate operating settings for the material and product type.
3. Load the specified quantity of prepared grain or food material into the feed hopper.
4. Close all covers and ensure safety guards are correctly fitted.
5. Start the machine according to the manufacturer's procedure.
6. Allow the heating or processing system to reach the specified operating condition.
7. Feed material gradually and monitor the product formation and discharge.
8. Adjust feed rate or settings only as permitted by the operating instructions.
9. Stop feeding if abnormal noise, overheating, blockage or irregular product formation occurs.
10. Switch off and isolate power before opening the machine or removing trapped material.
11. Collect the finished product in a clean container and allow it to cool safely.

⚠ Safety

- Keep hands away from rollers, dies, hot surfaces and moving feed mechanisms.
- Never open a pressurised chamber or bypass pressure-relief devices.
- Use heat-resistant gloves and eye protection where hot material may splash or escape.
- Isolate power before clearing blockages, cleaning or adjusting internal components.
- Do not operate the machine with damaged guards or faulty safety controls.
- Follow the specified moisture, temperature and pressure limits for the machine and material.

Care & storage

- Disconnect power and allow hot components to cool before cleaning.
- Remove grain residue and food deposits using approved cleaning tools.
- Inspect rollers, dies, guards and seals for wear or damage.
- Maintain lubrication and service schedules according to manufacturer instructions.

Common mistakes to avoid

- Overloading the hopper and causing uneven processing or blockage.
- Using unsuitable moisture levels or settings for the selected grain.
- Opening hot or pressurised components before safe shutdown and cooling.
- Cleaning near moving parts or leaving food residue inside the machine.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Refrigerator

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 9 labs

Covers these lab items: Display Cabinet/Showcase; Refrigerator; Refrigerator and freezer; Refrigerator (Double Door); Refrigerator two door

What it is used for

Use a refrigerator to keep food and other approved items at a controlled low temperature for safe storage and display.

How to use

1. Place the refrigerator on a stable, level surface with adequate ventilation.
2. Check the power cord, plug, door seals, and interior for visible damage.
3. Set the temperature according to the manufacturer's guidance and stored items.
4. Allow the unit to reach the required temperature before loading perishable food.
5. Store food in clean, covered containers and avoid blocking air vents.
6. Check and record the temperature regularly, especially for perishable items.
7. Close the door promptly and report unusual noise, leaks, or temperature changes.

⚠ Safety

- Connect the refrigerator to a properly grounded outlet with the correct electrical rating.
- Keep hands dry when handling the plug and avoid damaged cords.
- Do not block ventilation openings or use sharp tools to remove ice.
- Follow food safety temperature guidance and discard food held at unsafe temperatures.

Care & storage

- Clean shelves, door seals, and the interior regularly using suitable cleaners.
- Defrost and service the unit according to the manufacturer's instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Roaster

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Roaster / Dryer

What it is used for

A roaster or dryer uses controlled heat and airflow to roast or remove moisture from suitable food materials. Correct settings help achieve consistent processing while reducing scorching, contamination and fire risks.

Main parts

- Identify the loading hopper, tray or rotating drum, depending on design.
- Locate the heating element, burner or heat source.
- Identify the airflow fan, vents and exhaust arrangement, where fitted.
- Locate the temperature control and temperature indicator.
- Identify the timer, speed control and operating panel, if provided.
- Locate the product discharge opening and collection container.
- Identify guards, insulation, emergency stop and cleaning access points.

Before you start

1. Read the operating instructions and obtain supervised training before use.
2. Confirm that the machine is suitable for the intended food material.
3. Inspect heating elements, burners, wiring, guards and ventilation passages for damage.
4. Clean food-contact surfaces and prepare clean, heat-resistant collection containers.
5. Check that exhaust outlets and airflow passages are unobstructed.
6. Select the appropriate temperature, time and loading quantity for the product.
7. Wear suitable protective clothing and heat-resistant gloves where required.

How to use

1. Place the machine on a stable, heat-resistant surface with sufficient clearance.
2. Load the prepared food material without exceeding the recommended capacity.
3. Close the chamber or drum and secure all required guards.
4. Select the prescribed temperature, time and airflow settings.
5. Start the heating system according to the operating instructions.
6. Allow the machine to reach the specified operating temperature before processing if required.
7. Monitor the product colour, moisture removal and processing progress through safe observation points.
8. Adjust settings only within the approved operating procedure.
9. Stop heating if smoke, burning smell or abnormal temperature occurs.
10. Switch off the machine and allow hot components to cool before unloading.

1. Transfer the finished product into a clean, suitable container.

⚠ Safety

- Keep hands away from hot chambers, trays, drums and exhaust outlets.
- Never block ventilation or operate the machine with damaged guards.
- Keep combustible materials away from burners and hot surfaces.
- Disconnect electrical power before cleaning or servicing internal components.
- Use fuel-fired equipment only with appropriate ventilation and safe fuel connections.
- Do not spray water onto hot electrical components or burning oil.

Care & storage

- Switch off and cool the machine before cleaning food-contact surfaces.
- Remove crumbs, residue and accumulated grease using approved methods.
- Inspect heaters, burners, fans, seals and electrical connections regularly.
- Store the clean machine in a dry area protected from dust and pests.

Common mistakes to avoid

- Loading too much material and preventing uniform heating or drying.
- Using excessive heat that burns or discolours the product.
- Ignoring blocked airflow or ventilation passages.
- Cleaning the machine while surfaces remain hot or power is connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rolling Pin

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Polpat and Latan (Roller); Rolling pin; Rolling Pin & Board (Chakla-Belan)

What it is used for

A rolling pin is used to flatten and shape dough for cooking or baking. Use a clean, dry pin on a suitable work surface.

How to use

1. Inspect the rolling pin for cracks, splinters, or loose handles.
2. Wash and dry the rolling pin before preparing food.
3. Lightly flour the work surface or dough if the recipe requires it.
4. Place the dough on the surface and roll with even, controlled pressure.
5. Turn or reposition the dough to achieve the required thickness and shape.
6. Clean the rolling pin after use and allow it to dry fully.

⚠ Safety

- Keep fingers clear of pinch points near handles and board edges.
- Use the rolling pin only on a stable, clean surface.
- Do not use a cracked or splintered wooden pin for food preparation.

Care & storage

- Wash according to the material instructions and store completely dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scoop

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Scoops (different sizes)

What it is used for

Scoops measure, transfer, or portion ingredients and materials in food preparation and other designated tasks.

How to handle

- Select a scoop suitable for the material and required quantity.
- Inspect the scoop for cracks, sharp edges, residue, or contamination before use.
- Use food-grade scoops for edible ingredients and keep them separate from non-food materials.
- Avoid touching the scoop's working surface and prevent cross-contamination between materials.
- Clean and dry reusable scoops after use according to applicable hygiene procedures.

Storage

- Store clean, dry scoops in a designated holder or container.
- Keep food-use scoops protected from dust and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sieve

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 21 labs

Covers these lab items: Colanders & Strainers; Flour sifter/ sieves; Hand rotary sieve; Hand Screens/Sieves; Hand sieves; Sand Sieve; Sieve Set; Sieves); Stainer (strainer for adhesives); Strainer; Tea Strainer

What it is used for

A sieve separates materials by particle size or removes solid pieces from liquids. Select a suitable mesh size and clean the sieve before and after processing.

How to use

1. Inspect the mesh, frame, handle, and joints for damage or blockage.
2. Choose a sieve with suitable mesh size for the material.
3. Place the sieve securely over a clean collection container.
4. Add a manageable quantity of material and spread it evenly across the mesh.
5. Shake or rotate the sieve gently as appropriate for its design.
6. Collect the material passing through and the retained fraction separately if required.
7. Dispose of unwanted residue and clean the sieve after use.
8. Dry the sieve fully and store it in a clean location.

⚠ Safety

- Wear suitable gloves and eye protection when handling dusty or hazardous materials.
- Avoid creating dust clouds while sifting powders.
- Do not use a sieve with torn mesh or loose parts.
- Use food-grade, clean sieves for food preparation.

Care & storage

- Brush or rinse the mesh using a method suitable for the material.
- Dry thoroughly and store without bending or damaging the mesh.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Soy Milk Plant

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Soy milk plant

What it is used for

A soy milk plant processes soybeans into soy milk through cleaning, soaking, grinding, heating and filtration. Proper operation supports consistent product quality, food hygiene and safe handling of hot liquids and machinery.

Main parts

- Identify the soybean receiving hopper and cleaning section.
- Locate the soaking tank and water supply arrangement.
- Identify the grinder or wet milling unit.
- Locate the heating vessel or cooking system.
- Identify the filtration unit, filter cloth or separation mechanism.
- Locate the transfer pumps, pipes and product collection tank.
- Identify temperature controls, emergency stops and cleaning access points.

Before you start

1. Read the operating instructions and follow food hygiene procedures.
2. Inspect tanks, grinders, pipes, valves, guards and electrical connections for damage.
3. Clean and sanitise all food-contact surfaces and utensils.
4. Select suitable soybeans and remove stones, damaged beans and foreign materials.
5. Soak beans for the approved time using clean water and hygienic containers.
6. Check that the heating, filtration and transfer systems are correctly assembled.
7. Wear clean protective clothing and heat-resistant gloves where required.

How to use

1. Clean and sort the soybeans using the designated cleaning section.
2. Soak the beans according to the approved processing instructions.
3. Drain soaking water and rinse beans where the process requires it.
4. Feed prepared beans and the specified quantity of water into the grinder.
5. Operate the grinder with all guards secured and collect the resulting slurry.
6. Transfer the slurry to the designated heating vessel.
7. Heat the mixture according to the validated process and stir as required.
8. Filter the processed mixture using the specified filtration arrangement.
9. Transfer the soy milk into a clean, sanitised collection container.
10. Cool or further process the product promptly according to food safety instructions.
11. Stop the equipment and isolate power before dismantling or cleaning components.

⚠ Safety

- Keep hands away from grinders, agitators and other moving parts.
- Avoid contact with hot soy milk, steam and heated vessel surfaces.
- Never open hot pressurised equipment unless the approved procedure permits it.
- Disconnect power before clearing blockages or cleaning internal machinery.
- Prevent cross-contamination by separating raw beans, waste and finished soy milk.
- Follow validated heat treatment and hygienic storage requirements; do not assume filtration alone makes the product safe.

Care & storage

- Remove soybean residue promptly from grinders, filters, pipes and tanks.
- Wash, rinse and sanitise all food-contact components after processing.
- Inspect blades, seals, hoses and electrical connections for wear or damage.
- Dry removable parts and store them in a clean, protected area.

Common mistakes to avoid

- Using spoiled beans or contaminated soaking water.
- Overloading the grinder or using an unsuitable water ratio.
- Failing to apply the required heat treatment or hygienic cooling.
- Leaving soy residue in pipes, filters or tanks after production.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spatula

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 10 labs

Covers these lab items: Spatula; Spatula and pastry brush; Spatulas

What it is used for

Use a spatula to mix, spread, scrape, lift, or turn food ingredients according to the utensil type and recipe.

How to use

1. Choose a spatula suitable for the food, cookware, and task.
2. Inspect the handle and working end for cracks, looseness, or damage.
3. Wash and dry the spatula before food preparation.
4. Use the spatula to mix, scrape, spread, or turn ingredients as required.
5. Avoid scraping non-stick cookware with a spatula that may scratch its surface.
6. Place the spatula on a clean rest between uses.
7. Wash it thoroughly after use and allow it to dry.

⚠ Safety

- Use heat-resistant spatulas for hot pans and cooking tasks.
- Keep hands away from hot surfaces, steam, and splattering oil.
- Do not use damaged utensils that may shed fragments into food.
- Follow food hygiene practices and prevent cross-contamination.

Care & storage

- Wash promptly using a suitable detergent and clean water.
- Store dry in a clean, protected utensil holder.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Steam Jacketed Kettle

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Steam jacket kettles

What it is used for

A steam jacketed kettle heats food or process materials using steam circulating in a jacket around the vessel. It supports controlled cooking, heating, mixing, and batch processing when operated within its rated limits.

Main parts

- Inner kettle and outer steam jacket
- Steam inlet and condensate outlet
- Pressure gauge and temperature indicator
- Pressure relief valve and control valves
- Lid, handles, and tilting or discharge mechanism if fitted
- Agitator and motor if fitted

Before you start

1. Confirm the kettle is installed securely and the operator is trained for the model.
2. Inspect the vessel, jacket, lid, valves, gauges, and safety devices for damage or leaks.
3. Verify the pressure relief valve and gauges are within inspection and calibration requirements.
4. Check that the steam supply, condensate drain, and discharge route are unobstructed.
5. Confirm the kettle and agitator are clean and suitable for the intended product.
6. Load the approved quantity and secure the lid only as specified by the manufacturer.

How to use

1. Open the required condensate drain or steam trap arrangement according to the operating instructions.
2. Open the steam supply gradually and monitor the pressure and temperature indicators.
3. Heat the product at the specified temperature and pressure without exceeding rated limits.
4. Stir or operate the agitator as required, keeping hands and tools away from moving parts.
5. Monitor the product for sticking, foaming, boiling over, or uneven heating.
6. Reduce or close the steam supply when the required process conditions are reached.
7. Allow pressure to return to a safe level before opening or tilting the kettle.
8. Discharge the product using the designated outlet or tilting mechanism.
9. Isolate steam and allow the vessel to cool before cleaning.
10. Clean and rinse food-contact surfaces using the approved sanitation procedure.

⚠ Safety

- Never exceed the vessel's rated pressure or temperature.
- Do not bypass, block, or adjust the pressure relief valve without authorization.
- Open lids and discharge outlets cautiously because hot steam and product can cause severe burns.
- Keep clear of the tilting path and secure the kettle before loading or unloading.
- Do not operate an agitator with guards removed or reach into a moving vessel.
- Isolate steam and follow lockout procedures before maintenance.

Care & storage

- Clean the kettle and food-contact surfaces after each batch using approved chemicals.
- Inspect steam traps, condensate drains, valves, and seals according to the maintenance schedule.
- Arrange periodic inspection of the pressure jacket and safety devices by qualified personnel.
- Record operating pressure, temperature, cleaning, and maintenance as required.

Common mistakes to avoid

- Opening the lid while pressure remains can release dangerous steam.
- Opening the steam valve suddenly can cause water hammer or unstable heating.
- Overfilling can cause foaming, spills, and uneven processing.
- Ignoring blocked condensate drains can reduce heating performance and create hazards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Steamer

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Steamer

What it is used for

A steamer uses hot steam to cook food, soften materials, or carry out an approved heating process. Correct loading and temperature control help achieve consistent results and reduce burn risks.

Main parts

- Steaming chamber or cooking vessel
- Lid or chamber door
- Steam generator or steam inlet
- Water reservoir and level indicator if fitted
- Temperature and timer controls
- Trays, racks, and condensate drain

Before you start

1. Read the equipment instructions and confirm the intended use and capacity.
2. Inspect the chamber, lid, seals, controls, power cable, and steam connections.
3. Check the water level and use the specified water quality.
4. Ensure drains and steam outlets are clear and the equipment stands securely.
5. Arrange food or materials on trays without blocking steam circulation.
6. Wear suitable protection and keep the work area dry.

How to use

1. Add water or connect the approved steam supply as specified by the manufacturer.
2. Switch on the steamer and allow it to reach the required operating condition.
3. Load trays carefully and leave sufficient space for steam circulation.
4. Close the lid or door securely using the designed locking method.
5. Set the required temperature and time for the product or process.
6. Monitor the cycle and avoid opening the chamber unnecessarily.
7. When the cycle ends, switch off steam generation and wait for pressure to release where applicable.
8. Open the lid or door slowly, directing escaping steam away from the body.
9. Remove trays using suitable heat-resistant protection and place them on a stable surface.
10. Switch off, cool, drain if required, and clean the chamber after use.

⚠ Safety

- Hot steam can cause severe burns; keep face and hands away from escaping steam.
- Never force open a pressurized chamber or bypass its interlocks.
- Do not operate the steamer without the required water level or steam supply.
- Use heat-resistant gloves and suitable tools when removing hot trays.
- Keep electrical connections dry and disconnect power before cleaning electrical parts.
- Report damaged seals, blocked drains, and abnormal pressure or temperature readings.

Care & storage

- Clean the chamber, trays, lid, and seals after the equipment cools.
- Remove scale according to the manufacturer's descaling instructions and approved chemicals.
- Inspect seals, hoses, drains, and safety controls regularly.
- Drain and store the unit as instructed when it will remain unused.

Common mistakes to avoid

- Opening the lid too quickly can expose the operator to hot steam.
- Overloading trays can prevent uniform steaming.
- Operating with insufficient water can damage heating components.
- Using unsuitable descaling chemicals can damage internal surfaces and seals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Storage Container

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 17 labs

Covers these lab items: Glass container -; hindalium container with lid - 3 lit; hindalium container with lid - 3 lit - 3 3; Moisture-proof storage containers; Plastic Container -; Plastic recipients; Reusable containers for guest use; Stainless steel Containers with Cover (Pot); Storage containers; Storage Containers (Air

What it is used for

Storage containers hold food, ingredients, materials, or supplies while helping maintain cleanliness and prevent mix-ups.

How to handle

- Check container material, intended use, cleanliness, lid fit, and condition before filling.
- Use food-grade containers for food and avoid using damaged or unsuitable materials.
- Label contents and dates where needed, and keep raw and ready-to-eat foods separate.
- Avoid overfilling and secure lids before moving containers.

Storage

- Store containers clean and dry with lids protected from contamination.
- Follow material-specific temperature limits and keep food containers away from chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Timer

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Digital timer; Simple timer; Timer

What it is used for

Use a timer to measure or control a set duration during food preparation, cooking, or other timed tasks.

How to use

1. Check the timer display, buttons, and battery or power supply.
2. Set the required duration according to the recipe or task instructions.
3. Confirm the displayed time before starting the activity.
4. Start the timer when the timed process begins.
5. Respond to the alarm when the set time expires and check the task result.
6. Reset the timer before the next use.
7. Clean and store the timer after the activity.

⚠ Safety

- Do not rely on a timer alone to verify food safety or doneness.
- Keep non-waterproof timers away from liquids and hot surfaces.
- Follow safe cooking and food-handling procedures while the timer runs.

Care & storage

- Wipe the timer with a soft cloth and keep it dry.
- Store it safely and replace batteries when required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tofu Press

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Tofu hand press

What it is used for

A tofu hand press removes excess liquid from tofu to achieve the required firmness and texture. Use clean food-contact surfaces and apply pressure gradually according to the tofu and press instructions.

How to use

1. Inspect the press frame, plates, screws, drainage tray, and handles for damage.
2. Clean and sanitize all food-contact parts before use.
3. Place the tofu block inside the press using a clean container or liner if required.
4. Position the top plate evenly and ensure the press is stable.
5. Tighten the screw or apply pressure gradually according to the product instructions.
6. Allow liquid to drain into the designated tray and monitor the pressing process.
7. Release pressure safely, remove the tofu, and clean the press after use.

⚠ Safety

- Keep fingers clear of moving plates and screw mechanisms.
- Do not exceed the press's rated capacity or apply sudden excessive force.
- Use clean food-grade materials and prevent cross-contamination.
- Release pressure fully before disassembling or cleaning the press.

Care & storage

- Wash, sanitize, and dry all components after use and inspect the screw mechanism regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Toothpick Holder

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Tooth Pick Holder

What it is used for

A toothpick holder stores toothpicks hygienically for approved food-service use.

How to handle

- Inspect the holder for cleanliness, cracks, and a secure lid where fitted.
- Use clean, dry toothpicks and avoid touching the ends that users may place in their mouths.
- Dispense toothpicks without contaminating the remaining supply.
- Clean and dry the holder regularly according to food-service hygiene procedures.
- Discard any toothpicks that become wet, damaged, or contaminated.

Storage

- Keep the holder covered and in a clean, dry service area.
- Store spare toothpicks in sealed packaging away from moisture and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Transfer Pump

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Juice and pulp transfer pump stainless steel

What it is used for

A stainless-steel transfer pump moves juice, pulp, and other compatible food liquids between tanks or processing stages. Correct setup and hygienic operation help protect product quality and prevent contamination.

Main parts

- Stainless-steel pump housing and impeller
- Food-grade inlet and discharge connections
- Motor and drive assembly
- Suction hose and discharge hose or pipes
- Priming port or seal arrangement if fitted
- Control switch and flow-control valves

Before you start

1. Confirm the pump is approved for the product, temperature, viscosity, and required flow rate.
2. Inspect the pump housing, hoses, seals, clamps, and electrical connections for damage.
3. Verify that product-contact surfaces are clean and sanitized according to the food safety procedure.
4. Check that the suction line is connected securely and the discharge line leads to the correct destination.
5. Confirm that valves are in the required positions and the inlet is not blocked.
6. Prime the pump when required by its design and verify the motor rotation if specified.

How to use

1. Ensure the receiving tank has sufficient capacity for the expected transfer volume.
2. Open the required inlet and discharge valves according to the operating instructions.
3. Switch on the pump and observe the initial flow through the discharge line.
4. Adjust the flow using approved controls without exceeding the pump's operating limits.
5. Monitor the product for leakage, excessive foaming, unusual noise, vibration, or overheating.
6. Prevent the pump from running dry and maintain the required inlet supply.
7. Stop the pump before changing hoses, opening connections, or switching the product route.
8. When transfer is complete, stop the pump and close the valves in the recommended sequence.
9. Drain remaining product safely and perform the approved cleaning and sanitizing procedure.
10. Disconnect and store hoses and accessories hygienically after cleaning and inspection.

⚠ Safety

- Use only food-compatible hoses, seals, and fittings for product contact.
- Never open the pump housing or loosen a pressurized connection while the pump is operating.
- Keep electrical connections dry and use suitable earthing and electrical protection.
- Isolate power before clearing blockages, cleaning the pump, or performing maintenance.
- Avoid touching hot motor surfaces and keep hands away from rotating components.
- Follow food hygiene procedures and wear suitable protection when handling hot or acidic products.

Care & storage

- Clean and sanitize all product-contact parts after use and between incompatible products.
- Inspect impellers, seals, clamps, and hoses for wear, cracks, or contamination.
- Follow the manufacturer's instructions for lubrication and servicing of non-product-contact components.
- Store the pump and hoses clean, dry, and protected from dust and pests.

Common mistakes to avoid

- Running the pump without adequate priming can prevent flow or damage components.
- Using non-food-grade hoses or damaged seals can contaminate the product.
- Operating against a closed discharge valve can cause pressure buildup and overheating.
- Skipping cleaning between batches can cause microbial growth and flavour carryover.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tray

Category: **Food Processing & Kitchen** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Steel Tray

What it is used for

Handle steel trays hygienically for food preparation and service.

How to handle

- Inspect trays for sharp edges, dents, rust, or residue before use.
- Wash and sanitise trays according to food-safety procedures.
- Use suitable gloves or tray-handling methods for hot trays.
- Avoid stacking unstable or overloaded trays.

Storage

- Dry thoroughly before stacking to prevent water spots and corrosion.
- Store in a clean, designated area away from chemicals and floor contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vacuum Evaporator

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Vacuum evaporator

What it is used for

A vacuum evaporator removes solvent or water from a liquid by evaporation under reduced pressure. It is used to concentrate compatible products at lower temperatures while controlling heat and vapour removal.

Main parts

- Evaporation vessel or boiling flask
- Vacuum pump and vacuum line
- Condenser and vapour collection receiver
- Heating bath or jacket
- Temperature and vacuum pressure gauges
- Seals, valves, and control panel

Before you start

1. Read the operating instructions and verify the liquid is compatible with the equipment and seals.
2. Inspect the vessel, glassware if fitted, hoses, clamps, seals, and condenser for cracks or leaks.
3. Confirm the vacuum pump, cooling supply, and condensate receiver are ready for operation.
4. Check that the vessel is rated for vacuum and that all connections are secure.
5. Confirm the liquid volume is within the vessel's working capacity and is unlikely to foam excessively.
6. Wear suitable eye protection, gloves, and any additional protection required by the liquid's safety data sheet.

How to use

1. Load the approved liquid into the evaporation vessel without exceeding the recommended fill level.
2. Secure the vessel and confirm that the condenser and collection receiver are connected correctly.
3. Start cooling water or the approved condenser cooling system before heating.
4. Switch on the vacuum pump and reduce pressure gradually according to the procedure.
5. Set the heating temperature and vacuum level appropriate for the product and solvent.
6. Monitor temperature, pressure, evaporation rate, foaming, and condensate collection throughout the run.
7. Adjust heating or vacuum gradually if bumping, foaming, or unstable boiling occurs.
8. When the target concentration is reached, stop heating and allow the system to cool as required.

9. Release vacuum slowly through the designated valve before opening the vessel or disconnecting components.
10. Switch off the pump and cooling system in the prescribed sequence, then clean and inspect the equipment.

⚠ Safety

- Never use cracked glassware or a vessel that is not rated for vacuum.
- Use a protective shield when required for glass vessels operating under vacuum.
- Do not open the system while it is under vacuum or while hot solvent remains inside.
- Use suitable ventilation and keep flammable solvent vapours away from ignition sources.
- Verify solvent compatibility with the pump, seals, tubing, and condenser.
- Isolate electrical power before maintenance and follow procedures for hazardous solvents.

Care & storage

- Clean the vessel, condenser, and receiver using methods compatible with the processed liquid.
- Inspect seals, vacuum tubing, clamps, and glassware regularly for wear or damage.
- Maintain the vacuum pump and change its oil or filters according to the manufacturer's instructions.
- Record operating conditions, cleaning, and maintenance where required.

Common mistakes to avoid

- Applying vacuum too quickly can cause bumping, foaming, or liquid carryover.
- Heating without adequate condenser cooling can release solvent vapours.
- Overfilling the vessel can contaminate the vacuum line and collection system.
- Releasing vacuum suddenly or opening hot equipment can cause splashing and injury.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vacuum Packaging Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Vacuum packaging machine

What it is used for

A vacuum packaging machine removes air from a package and seals it to help protect compatible products during storage and transport. Correct packaging settings and hygienic handling can improve shelf life, but vacuum packaging does not replace required food safety controls.

Main parts

- Vacuum chamber or external suction nozzle
- Vacuum pump and air extraction line
- Heat-sealing bar and sealing strip
- Sealing temperature and time controls
- Lid, gasket, and chamber locking arrangement
- Vacuum pressure indicator and emergency stop if fitted

Before you start

1. Read the operating instructions and confirm the machine is suitable for the product and packaging material.
2. Inspect the chamber, lid gasket, sealing bar, cables, and controls for damage or residue.
3. Check that the sealing strip is clean and the vacuum pump is ready for use.
4. Use clean, food-grade vacuum bags or approved packaging material of suitable thickness.
5. Ensure the product is at the required safe temperature and the package rim is clean and dry.
6. Keep liquids and powders within the machine's permitted packaging method and capacity.

How to use

1. Place the product inside the approved bag and leave sufficient space at the open end for sealing.
2. Position the bag opening flat across the sealing bar without folds or trapped product.
3. For chamber models, place the bag inside the chamber and close the lid securely.
4. Select the required vacuum level, sealing temperature, and sealing time for the packaging material.
5. Start the cycle and allow the machine to evacuate air and complete the seal automatically.
6. For external models, position the suction nozzle according to the machine instructions and keep the bag opening correctly aligned.
7. Wait until the cycle finishes and the chamber returns to safe opening conditions.
8. Remove the package and inspect the seal for continuity, wrinkles, contamination, or leaks.
9. Check the vacuum and seal quality using the approved inspection method.
10. Switch off the machine and clean product-contact surfaces and the sealing area after use.

⚠ Safety

- Keep fingers away from the hot sealing bar during and immediately after operation.
- Never bypass lid interlocks or force open a chamber under vacuum.
- Use only packaging materials approved for the machine's temperature and sealing method.
- Keep electrical components dry and disconnect power before cleaning or maintenance.
- Do not package hazardous chemicals or incompatible materials unless the machine and procedure are specifically approved.
- Follow food safety requirements for temperature control, sanitation, storage, and shelf life.

Care & storage

- Clean the chamber, lid, gasket, and sealing bar after use according to the manufacturer's instructions.
- Inspect the gasket, sealing strip, vacuum hose, and pump for wear or damage.
- Replace worn sealing elements with compatible approved parts.
- Maintain the vacuum pump and record cleaning, maintenance, and seal checks as required.

Common mistakes to avoid

- A wet or contaminated bag opening can cause an incomplete seal.
- Incorrect sealing temperature or time can weaken or burn the packaging material.
- Overfilling the bag can trap product in the seal and cause leaks.
- Assuming vacuum packaging makes food sterile or safe at room temperature can cause foodborne illness.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Washing Tank

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Washing and Soaking Tanks

What it is used for

Washing and soaking tanks are used to clean, rinse, or soak food ingredients before processing. Follow the approved food-handling procedure and maintain hygienic water and tank conditions.

How to use

1. Inspect the tank, drain, taps, and surfaces for damage or residue.
2. Clean and sanitize food-contact surfaces before use.
3. Confirm the tank is suitable for the ingredient and intended process.
4. Fill with the required quantity and quality of water.
5. Add ingredients carefully and follow the specified soaking or washing time.
6. Drain the water safely and remove ingredients using clean utensils.
7. Clean, sanitize, and dry the tank after use.

⚠ Safety

- Use potable water for food washing and soaking.
- Avoid mixing raw and ready-to-eat ingredients in the same tank without proper cleaning.
- Do not use tanks with cracks, corrosion, or chemical contamination.
- Keep floors dry where possible to prevent slips.

Care & storage

- Clean drains regularly and inspect the tank for residue, leaks, or corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weighing and Filling Machine

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Weighing and Filling Machine

What it is used for

A weighing and filling machine measures a target quantity of product and dispenses it into containers or packages. Correct calibration, product setup, and filling control help maintain consistent pack weight and reduce material waste.

Main parts

- Product hopper or supply tank
- Weighing platform, load cell, or weighing bucket
- Filling nozzle or discharge gate
- Feeding mechanism such as auger, pump, or vibratory feeder
- Control panel and target-weight settings
- Container holder, conveyor, and safety guards

Before you start

1. Read the operating instructions and confirm the machine is suitable for the product and container type.
2. Inspect the hopper, weighing system, filling nozzle, conveyor, guards, and electrical connections for damage.
3. Clean and sanitize product-contact surfaces according to the required hygiene procedure.
4. Check the weighing system calibration using approved test weights or the specified calibration method.
5. Set the target weight, allowable tolerance, feed speed, and filling mode for the product.
6. Ensure containers are clean, correctly positioned, and compatible with the filling method.

How to use

1. Load the approved product into the hopper or supply tank without exceeding its rated capacity.
2. Switch on the machine and allow the weighing system to stabilize as instructed.
3. Run the required zero or tare function with the empty weighing platform or container in position.
4. Place or feed an empty container into the designated filling position.
5. Start the filling cycle using the approved control or automatic sequence.
6. Monitor the product feed and confirm the machine stops or slows at the target quantity.
7. Check the filled container's weight using the machine display or an approved verification scale.
8. Adjust settings only within authorized limits if repeated checks show a consistent deviation.
9. Remove the filled container and transfer it to the next approved packaging or inspection stage.
10. Stop the machine, isolate power, and clean product-contact parts after the production run.

⚠ Safety

- Keep hands away from moving conveyors, augers, pumps, and filling mechanisms.
- Never bypass safety guards or reach into the hopper while the machine is operating.
- Disconnect and isolate power before clearing blockages, changing parts, or cleaning moving components.
- Use suitable lifting methods when handling heavy product containers or supply bags.
- Follow food hygiene and allergen-control procedures when changing products.
- Do not operate the machine if the load cell, guards, controls, or emergency stop are faulty.

Care & storage

- Clean the hopper, nozzle, weighing bucket, and other product-contact parts after use.
- Inspect the load cell, feed mechanism, seals, and nozzle for wear or buildup.
- Verify weighing accuracy at the scheduled frequency and after maintenance or relocation.
- Record calibration checks, weight deviations, cleaning, and maintenance as required.

Common mistakes to avoid

- Skipping tare or zeroing can cause every package to have an incorrect weight.
- Changing settings without verification can create overfilling or underfilling.
- Allowing product buildup around the nozzle can affect flow and fill accuracy.
- Ignoring calibration drift can cause out-of-tolerance packages and material losses.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Whisk

Category: **Food Processing & Kitchen** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Whisks; Wire whiskers

What it is used for

To mix, beat, aerate, or blend food ingredients evenly during preparation.

How to handle

- Inspect the whisk wires and handle for looseness, corrosion, or damage.
- Use a clean whisk and a suitable bowl or container for the ingredients.
- Hold the handle securely and whisk with controlled movements to avoid splashing.
- Keep hands away from powered mixing equipment if using an attachment; switch off before adjusting or removing it.
- Wash, rinse, and dry the whisk after use.

Storage

- Ensure the whisk is completely clean and dry before storage.
- Store in a clean utensil holder or drawer away from contamination and crushing pressure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wick Stove

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Wick Stove

What it is used for

A wick stove uses a fuel-soaked wick to produce a flame for cooking or heating. Operate it on a stable, ventilated surface with the correct fuel and keep it away from combustible materials to prevent fire, burns, and harmful fumes.

Main parts

- Stove body and fuel reservoir
- Wick and wick holder
- Wick-height adjustment knob
- Burner cover or flame control, if fitted
- Fuel filling opening and cap
- Stable base and supporting frame

Before you start

1. Confirm the stove type and use only the fuel specified by the manufacturer.
2. Inspect the fuel reservoir, wick, cap, seals, and stove body for damage or leakage.
3. Place the stove on a level, stable, non-combustible surface in a well-ventilated area.
4. Ensure curtains, paper, cooking oil containers, and other combustible items are kept away.
5. Check that the stove is cool and fully extinguished before refuelling.
6. Keep a suitable fire extinguisher or fire blanket accessible.

How to use

1. Ensure the stove is switched off and completely cool before filling.
2. Fill the reservoir with the specified fuel without overfilling or spilling.
3. Close the fuel cap securely and wipe away any spilled fuel before lighting.
4. Allow the wick to absorb fuel as directed by the manufacturer.
5. Adjust the wick to the recommended starting height.
6. Light the wick using a suitable long-reach lighter or approved lighting method.
7. Observe the flame and adjust the wick gradually to obtain a steady, controlled flame.
8. Place the cooking vessel centrally and ensure it is stable.
9. Monitor the flame continuously while cooking and keep children away.
10. Reduce the flame or extinguish it using the stove's approved control method when cooking is complete.
11. Confirm that the flame is completely out and allow the stove to cool before moving or cleaning it.

2. Clean the stove, check for leakage, and store fuel in an approved labelled container away from heat.

⚠ Safety

- Never refuel a hot or burning stove.
- Never use petrol or any fuel not approved for the stove.
- Do not use the stove in an enclosed or poorly ventilated room because of fire and carbon monoxide risks.
- Never leave a lit stove unattended or move it while burning.
- If fuel leaks, do not light the stove; keep ignition sources away and clean up safely.
- If a fire cannot be controlled immediately, evacuate and contact emergency services.

Care & storage

- Keep the wick trimmed and replace it when damaged or excessively worn, following the manual.
- Remove soot and food residue only after the stove has cooled completely.
- Inspect the reservoir, cap, seals, and fuel lines if fitted for leakage.
- Store the stove dry and keep fuel away from sunlight, flames, and heat sources.
- Arrange repair or replacement if the stove body, controls, or fuel system is damaged.

Common mistakes to avoid

- Refuelling before the stove has cooled completely.
- Using an incorrect fuel or overfilling the reservoir.
- Operating in a poorly ventilated space.
- Leaving the stove unattended or moving it while lit.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bar Counter

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Bar Counter

What it is used for

A bar counter provides a designated work surface for service, display, or training activities.

How to handle

- Inspect the counter surface, edges, supports, and fittings for damage or instability.
- Keep the worktop clean and use it only for its intended purpose.
- Distribute equipment and materials evenly and avoid placing excessive loads on edges.
- Clean spills immediately to prevent slipping or surface damage.
- Keep access routes and emergency exits clear of stools, supplies, and equipment.

Storage

- Remove loose items and clean the counter after use.
- Keep stored supplies organised and protect the surface from moisture and heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bed

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Bed 6X4 feet

What it is used for

Use and maintain a 6-by-4-foot bed safely and hygienically.

How to handle

- Check the bed frame, joints, and supports for stability before use.
- Do not exceed the manufacturer's stated weight capacity.
- Use clean bedding and change it according to hygiene procedures.
- Move the bed carefully and keep access paths clear.

Storage

- Keep the bed in a dry, ventilated room with adequate clearance.
- Store bedding separately in a clean, covered area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bedside Locker

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Bedside Locker

What it is used for

A bedside locker stores personal belongings and care essentials near a bed in healthcare or accommodation settings.

How to handle

- Inspect the locker, drawers, handles, wheels, and edges for damage or instability.
- Position it so it is accessible without blocking movement, emergency access, or medical equipment.
- Keep personal belongings organised and protect confidential documents or valuables.
- Clean high-touch surfaces regularly and follow infection-control procedures where applicable.
- Do not overload drawers or use the locker as a step or seat.

Storage

- Keep the locker clean, dry, and arranged in its designated position.
- Secure stored items and keep essential access routes clear.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bench

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Mobile benches

What it is used for

A bench provides seating for students, visitors, staff, or users in designated indoor or outdoor areas.

How to handle

- Inspect the bench, joints, fasteners, and legs or wheels for damage and stability.
- Place the bench on a level surface and secure mobile benches before use.
- Observe the intended seating capacity and avoid standing or jumping on the bench.
- Keep the area around the bench clear to prevent trips and blocked access.
- Clean surfaces regularly and report sharp edges or loose parts.

Storage

- Park mobile benches in their designated locations and engage brakes if fitted.
- Protect the bench from weather or moisture when its material requires it.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Business Card Holder

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Business Card Holder

What it is used for

Keep business cards organised and accessible for office or training use.

How to handle

- Place cards neatly with names and contact details facing the same direction.
- Avoid bending, marking, or handling cards with wet or dirty hands.
- Remove outdated or damaged cards from the holder.
- Use personal information only for its intended and authorised purpose.

Storage

- Keep the holder clean and dry on a stable desk or counter.
- Store spare cards in a labelled drawer or protective box.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chair

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 39 labs

Covers these lab items: Arm Chair; Chair; Chair (POS); Chairs; Classroom chairs; Comfortable seats with adequate lighting, controlled temperature and acoustics for training and learning; Computer Chairs; Dining Chairs; Ergonomic Chairs; Furniture (Ergonomic Chair and Adjustable Desk); Gaming Chair for Virtual Reality Labor

What it is used for

Chairs provide seating for learners, staff, visitors, and other users in classrooms, offices, and work areas.

How to handle

- Inspect the seat, backrest, legs, fasteners, and wheels for damage or instability.
- Adjust the chair where possible to support a comfortable and appropriate sitting posture.
- Use chairs only for seating and avoid standing on them or exceeding their load capacity.
- Keep chairs positioned safely and move them without dragging or blocking walkways.

Storage

- Arrange chairs in designated rows or storage areas without obstructing exits.
- Keep upholstery clean and protect chairs from moisture and damaging loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Classroom Desk

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Class room chairs and desks

What it is used for

Classroom desks and chairs provide workspaces and seating for students and training activities.

How to handle

- Inspect desks and chairs for loose fasteners, sharp edges, cracks, and instability.
- Adjust or arrange furniture to support comfortable posture and safe movement.
- Do not stand on desks or chairs or overload them beyond their intended capacity.
- Keep aisles, exits, and access to safety equipment clear.
- Clean surfaces regularly and report damaged furniture for repair.

Storage

- Arrange furniture in stable rows or designated layouts when not in use.
- Keep desks and chairs dry and avoid stacking them unless designed for stacking.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Clock

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Clock; Wall clock

What it is used for

Clocks display time to support scheduling, coordination, and time management.

How to handle

- Check the clock casing, mounting, display, and power source for damage.
- Install wall clocks using secure fittings suitable for the wall type and clock weight.
- Use the correct battery or power supply and follow the manufacturer's instructions.
- Keep clocks away from excessive moisture, heat, and impact.
- For wall-mounted clocks, use safe access equipment when installing or replacing batteries.

Storage

- Store spare clocks and accessories in protective packaging.
- Keep batteries in their original packaging and separate from metal objects where appropriate.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cupboard

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 32 labs

Covers these lab items: Almirah; Cabinet with drawer; Cupboard; Cupboard for Storage of tools, Raw material and other Resources; Cupboards; File Cabinet; Filing Cabinets; Filling cabinet; Furniture's Storage Cabinets (open and shuttered) for Lab setup; Steel almirah big size; Steel almirah small size; Steel cupboards drawe

What it is used for

Cupboards and cabinets organise and secure tools, documents, supplies, and other workplace materials.

How to handle

- Inspect the body, shelves, hinges, locks, and handles for damage or instability.
- Place heavy items on lower shelves and distribute loads within the cabinet's capacity.
- Close doors and drawers after use and secure hazardous or confidential contents where required.
- Keep the cabinet stable and avoid climbing on shelves or drawers.

Storage

- Position the cabinet on a level surface and secure it against tipping when necessary.
- Keep contents labelled, dry, organised, and within shelf load limits.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Desk Lamp

Category: **Furniture, Storage & Display** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Desk Lamp; Desk Lamps

What it is used for

To provide localized lighting for reading, writing, inspection, or desk-based tasks.

How to handle

- Inspect the lamp, plug, cord, switch, bulb, and base for damage before use.
- Place the lamp on a stable, dry surface away from water and flammable materials.
- Use the correct bulb type and wattage specified by the manufacturer; avoid touching a hot bulb.
- Route the cord safely to prevent tripping, and do not overload sockets or use damaged electrical parts.
- Switch off and unplug the lamp before cleaning, changing the bulb, or adjusting electrical components.

Storage

- Switch off, unplug, and allow the bulb to cool before storing or moving the lamp.
- Coil the cord loosely and store in a dry location, protecting the shade and bulb from impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Desk Mat

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Desk Mats

What it is used for

Use desk mats to protect work surfaces and keep work areas tidy.

How to handle

- Place the mat flat on a clean, stable desk surface.
- Check for curled edges that may catch objects or cause snagging.
- Clean spills promptly using a method suitable for the mat material.
- Avoid placing very hot items on mats not designed for heat resistance.

Storage

- Roll or lay flat according to the manufacturer's guidance.
- Store clean and dry away from direct sunlight and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Desk Organizer

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Desk Organizers

What it is used for

Use desk organisers to keep stationery and small office supplies accessible.

How to handle

- Place items in suitable compartments and avoid overloading the organiser.
- Check for sharp edges, cracks, or unstable bases.
- Keep frequently used supplies accessible without obstructing the workspace.
- Clean the organiser regularly and remove unnecessary items.

Storage

- Keep on a stable desk or shelf away from the edge.
- Store spare organisers in a clean, dry area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Display Board

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 11 labs

Covers these lab items: Display Board; Display board for tools display and easy access to students; Display Corner; Pin-up Board; Resource Corner; Room readiness boards and status tracking systems; Safety Notice board; Safety Wall

What it is used for

Display boards organise notices, tool displays, learning resources, and workplace status information for easy reference.

How to handle

- Inspect the board, frame, supports, and mounting hardware for damage or instability.
- Arrange tools and notices clearly and secure items to prevent falling.
- Use suitable pins or fasteners and keep sharp points controlled during installation.
- Ensure the display does not obstruct exits, walkways, switches, or emergency equipment.

Storage

- Store removable boards securely and keep spare pins and accessories in a labelled container.
- Protect the board surface from moisture, impact, and excessive loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drawing Table

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Drafting table; Drawing table

What it is used for

Drawing tables support technical drawing, drafting, design, and other precision-based learning activities.

How to handle

- Inspect the tabletop, frame, adjustment mechanism, and locking parts for damage or instability.
- Adjust the table to a comfortable working angle and secure all locks before use.
- Keep drawing sheets and tools organised and avoid overloading the surface.
- Use blades, compasses, and other sharp drawing tools carefully.
- Report loose fittings or damaged adjustment mechanisms before further use.

Storage

- Return the table to its designated position and secure movable parts.
- Keep the surface clean and dry and store drawing tools separately.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drying Rack

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Drying rack

What it is used for

Use drying racks safely to dry materials without damage or contamination.

How to handle

- Check rack stability and load capacity before placing items on it.
- Distribute items evenly and avoid blocking airflow between layers.
- Keep the rack away from walkways, heat sources, and electrical hazards.
- Clean the rack between uses when it is shared or used for hygiene-sensitive items.

Storage

- Allow the rack to dry completely before folding or storing.
- Store securely in a dry area to prevent corrosion, mould, or deformation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Foot Rest

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Foot Rest

What it is used for

Use a foot rest to support comfortable posture during seated work.

How to handle

- Place the foot rest on a stable, non-slip surface.
- Adjust its position so both feet are comfortably supported.
- Check for cracks, loose parts, or sharp edges before use.
- Keep it clear of walkways to prevent tripping.

Storage

- Clean the surface regularly and allow it to dry.
- Store in a designated area away from excessive weight and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ironing Board

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 9 labs

Covers these lab items: Iron Table; Ironing board; Ironing table; Ironing Workstation with stools

What it is used for

An ironing board provides a stable heat-resistant surface for pressing garments and textile samples.

How to handle

- Inspect the board surface, legs, locking mechanism, and height adjustment before use.
- Set the board on a level surface and engage all locks securely.
- Keep the iron stable and follow its temperature settings for the fabric.
- Avoid touching hot surfaces and keep the power cord away from the ironing area.
- Allow the board and iron to cool before folding or moving them.

Storage

- Fold and secure the board only after cooling completely.
- Store upright in a stable location away from moisture and walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Locker

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Locker

What it is used for

Use lockers to store personal belongings and work materials securely.

How to handle

- Check the locker, hinges, handle, and locking mechanism for damage.
- Keep keys or access codes secure and do not share them with unauthorised people.
- Avoid storing food, hazardous chemicals, or wet items unless the locker is designed for them.
- Keep the area around lockers clear and report faulty locks.

Storage

- Keep locker contents organised and within the permitted load.
- Clean the locker periodically and leave it dry when not in use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Manicure Table

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Manicure Table

What it is used for

Use and maintain a manicure table safely and hygienically.

How to handle

- Check that the table is stable and its drawers or attachments are secure.
- Disinfect the work surface between clients.
- Keep manicure tools organised and liquids away from electrical equipment.
- Wipe spills promptly to prevent slipping or surface damage.

Storage

- Remove products and tools before cleaning or moving the table.
- Keep the table dry and store supplies in closed, labelled containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mattress

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Mattress

What it is used for

Handle mattresses hygienically and safely for sleeping or training-room use.

How to handle

- Inspect the mattress for tears, stains, moisture, or structural damage before use.
- Use a clean mattress cover and change bedding according to hygiene procedures.
- Lift or move the mattress with assistance if it is bulky or heavy.
- Allow moisture to dry fully and do not use a mattress with suspected mould.

Storage

- Store in a clean, dry, ventilated area away from pests.
- Keep supported and protected from moisture, sharp objects, and heavy loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mirror

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 14 labs

Covers these lab items: Face Mirror; Full-Length Mirror; Hand Mirrors; Mirror; Mirror for visualizing the complete body; Mirrors; Mirrors (plastic, Glass); Modular Mirrors

What it is used for

Mirrors support grooming, fitting, visual checks, and educational demonstrations when securely installed and clean.

How to handle

- Inspect the mirror surface, edges, backing, and mounting for cracks or looseness.
- Carry mirrors with both hands and use assistance for large or heavy mirrors.
- Place or mount mirrors on stable supports and keep them away from impact-prone areas.
- Clean with a compatible non-abrasive product and avoid excessive pressure on the glass.

Storage

- Store portable mirrors in protective covers or padded compartments.
- Secure large mirrors upright with suitable support and protect them from falling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rack

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 26 labs

Covers these lab items: bins/ racks; Display racks for promotional material, maps, and tourist information; Rack; Racks; Shelf designs; Shelves for Stacking Products; Storage Rack; Storage Rack / Shelves; Storage racks; Storage Solutions; Utensil Rack

What it is used for

Racks and shelves organise, store, and display materials while keeping items accessible and orderly.

How to handle

- Inspect the rack, shelves, joints, and supports for damage or instability.
- Place the rack on level flooring and secure it against tipping where required.
- Distribute loads evenly, observe shelf weight limits, and keep heavy items on lower shelves.
- Keep aisles and emergency exits clear; retrieve items without climbing on racks.

Storage

- Keep stored items labelled and arranged by category.
- Protect racks from moisture, corrosion, impact, and overloading.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Reception Desk

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Hostess Desk

What it is used for

A reception desk supports visitor greeting, enquiries, registration, and front-desk administrative activities.

How to handle

- Inspect the desk, drawers, edges, and fittings for damage or instability.
- Keep the work surface clean and organise stationery and equipment neatly.
- Protect visitor records, identification details, and other confidential information from unauthorised access.
- Keep cables arranged safely and ensure exits and access routes remain clear.
- Report loose fittings or damaged electrical accessories for repair.

Storage

- Secure documents and equipment in designated drawers or cabinets.
- Clean the desk and leave it organised at the end of the shift.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Salon Chair

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Hair Dresser Chair/ Stool; Hair Dresses Chair/Stool; Hydraulic Chair

What it is used for

Salon chairs provide stable, adjustable seating for clients during grooming, cutting, and beauty treatments.

How to handle

- Inspect the chair base, upholstery, footrest, and adjustment mechanism before use.
- Position the chair on a level floor and engage locks where provided.
- Adjust height slowly and ensure the client is seated securely before operating controls.
- Keep fingers, clothing, and tools clear of moving or hydraulic parts.
- Clean and disinfect contact surfaces between clients using compatible products.

Storage

- Lower the chair to a stable position when not in use.
- Keep the area around the chair clean and free from trip hazards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seating Arrangement

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Seating arrangement in full / half circle format for one or more teams as per planned team composition (for team discussions)

What it is used for

Arrange seating to support organised team discussions and participation.

How to handle

- Arrange chairs in the planned full-circle or half-circle format according to team size.
- Keep aisles, doors, and emergency exits unobstructed.
- Ensure chairs are stable, evenly spaced, and accessible to all participants.
- Check visibility, audibility, and safe movement before the session begins.

Storage

- Return chairs and tables to their designated locations after the session.
- Keep furniture clean, dry, and arranged without blocking walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Side Station

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Side Station

What it is used for

A side station provides organised storage and a work surface for service supplies and operational equipment.

How to handle

- Inspect the station, shelves, drawers, and wheels or supports for stability and damage.
- Arrange supplies so frequently used items remain accessible without overloading the surface.
- Keep clean items separate from used or contaminated items.
- Clean spills promptly and maintain clear walkways around the station.
- Move mobile stations carefully and engage brakes where fitted.

Storage

- Return supplies to their assigned locations after use.
- Keep the station clean, dry, and organised at the end of service.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sofa

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Sofa Set

What it is used for

Handle sofa furniture safely while maintaining cleanliness and condition.

How to handle

- Inspect the sofa for loose legs, unstable frames, damaged upholstery, or exposed fasteners.
- Keep the sofa on a level surface and avoid standing or jumping on it.
- Use sufficient assistance when moving large or heavy sofa sections.
- Clean spills promptly using a method suitable for the upholstery material.

Storage

- Keep the sofa in a dry, ventilated area away from direct sunlight and moisture.
- Cover during long periods of non-use without trapping dampness.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Step Stool

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Foot Step

What it is used for

Use a step stool safely to reach low-height work areas.

How to handle

- Inspect the stool for cracks, loose joints, and secure non-slip steps before use.
- Place it on a level, dry surface and fully open or lock it as designed.
- Keep both feet centred on the steps and do not stand on the top unless it is approved for standing.
- Do not overreach; reposition the stool and maintain three points of contact where applicable.

Storage

- Clean and dry the stool after use.
- Fold or store it as designed in a secure area away from walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stool

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 14 labs

Covers these lab items: Drafting stool; Large Stools; Small Stools; Stools for sitting

What it is used for

Stools provide seating or task support in classrooms, workshops, studios, and service areas.

How to handle

- Inspect the seat, legs, joints, and height-adjustment mechanism for damage or instability.
- Place the stool on a level surface and adjust height only as designed.
- Sit centrally and avoid standing on the stool or exceeding its rated load.
- Move stools carefully and keep walkways and work areas clear.

Storage

- Arrange stools in designated locations without blocking exits or access routes.
- Keep stools dry and report loose joints or damaged feet promptly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Storage Bin

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Boxes/baskets For Storage; Storage bins; Storage Bins / Silos; Storage Boxes

What it is used for

Storage bins, boxes, baskets, and silos organise materials and protect them during storage or handling.

How to handle

- Select a container suitable for the contents, load, and environmental conditions.
- Inspect the bin for cracks, sharp edges, contamination, and damaged handles or lids.
- Label contents clearly and avoid mixing incompatible or unidentified materials.
- Do not exceed the rated load or stack containers unsafely.
- Use safe lifting techniques and keep aisles and emergency exits clear.

Storage

- Keep containers clean, dry, and arranged in stable stacks where permitted.
- Separate chemicals, food-related materials, and other incompatible items as required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Storage Facility

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Packaging and storage facilities

What it is used for

Use packaging and storage facilities to protect materials and maintain organised inventory.

How to handle

- Inspect storage areas for cleanliness, ventilation, pest activity, and structural safety.
- Label stored items clearly and separate incompatible, fragile, or hazardous materials.
- Use appropriate lifting aids and keep heavy items on lower shelves within rated limits.
- Maintain clear aisles, emergency access, and safe stacking practices.

Storage

- Keep materials in designated areas with inventory records updated.
- Monitor temperature, humidity, and security where required and remove waste promptly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Table

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 66 labs

Covers these lab items: Activity Table; Big Table; Computer Table; Computer Tables; Cutting Table; Desks; Doctors Table; Faculty Table and Chair; Feeding table; Finishing/Cleaning Table; Fruit sorting table; Furniture (Working Table, cabinets, Stools, etc.), For 15 students; Furniture (Working Table, Stools, etc.); Kitchen

What it is used for

Tables provide stable work surfaces for classroom activities, computer work, clinical tasks, and material preparation.

How to handle

- Inspect the tabletop, legs, fasteners, and supports for damage or instability.
- Place the table on a level surface and lock wheels where provided.
- Distribute loads evenly and keep equipment within the table's rated capacity.
- Clean spills promptly and avoid dragging heavy tables across floors.

Storage

- Keep tables in designated areas without blocking exits, walkways, or emergency equipment.
- Protect surfaces from moisture, excessive heat, and heavy impacts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tool Trolley

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Tool trolley; Tools trolley

What it is used for

A tool trolley moves and organises tools and equipment within workshops or training areas.

How to handle

- Inspect wheels, brakes, handles, shelves, and fasteners before moving the trolley.
- Distribute tools evenly and keep heavy items on lower shelves.
- Secure tools and close drawers before moving to prevent falling objects.
- Push at a controlled speed and keep the route clear of people, cables, and obstacles.
- Engage brakes when parked and never ride on or overload the trolley.

Storage

- Return tools to their assigned compartments after use.
- Park the trolley on stable ground in a designated area with brakes engaged.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Trolley

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 11 labs

Covers these lab items: Trolley; Trolleys; Working and facial Trolleys

What it is used for

Trolleys transport tools, supplies, equipment, and service items while reducing manual carrying effort.

How to handle

- Inspect the frame, wheels, handles, brakes, and shelves for damage before use.
- Check the rated load and distribute items evenly without obstructing visibility.
- Push the trolley at a controlled speed and use ramps only when safe and permitted.
- Keep hands and feet clear of wheels and secure items that may slide or fall.

Storage

- Park the trolley on a level surface and apply brakes where fitted.
- Clean after use and store without blocking exits or access routes.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Workbench

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Two-wheeler workbench; Workbench with vice; Workbenches and workstations

What it is used for

A workbench provides a stable work surface for repairs, assembly, inspection, and practical training.

How to handle

- Inspect the bench, frame, worktop, vice, and fasteners for damage or instability.
- Secure the bench and clamp workpieces properly before cutting, drilling, or assembly.
- Keep tools organised and avoid placing heavy loads near unsupported edges.
- Use suitable PPE and keep hands clear of moving tools and pinch points.
- Clean spills promptly and report loose fixtures or damaged work surfaces.

Storage

- Remove tools and debris after use and leave the surface clean.
- Store the bench in a designated area and secure mobile units before leaving.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Workstation

Category: **Furniture, Storage & Display** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Front-desk workstations, headsets, and office organization systems; Garment Checking Workstation; Painting and finishing stations

What it is used for

A workstation provides an organised area for office, garment inspection, painting, finishing, or other designated tasks.

How to handle

- Inspect the workstation, furniture, tools, electrical connections, and accessories before use.
- Arrange equipment ergonomically and keep frequently used items within safe reach.
- For painting or finishing stations, provide required ventilation and follow chemical safety instructions.
- Keep cables, materials, and waste clear of walkways and emergency exits.
- Clean the work area after use and report defective equipment.

Storage

- Return tools and accessories to designated storage locations.
- Store chemicals, garments, records, and equipment according to their specific requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Adhesive Mixing Tool

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Adhesive preparation and mixing tools

What it is used for

An adhesive mixing tool combines compatible adhesive components into a uniform mixture. Use the correct tool and follow the adhesive manufacturer's mixing instructions.

How to use

1. Read the adhesive label and identify required components, proportions, and mixing time.
2. Wear the gloves, goggles, and other PPE specified for the adhesive.
3. Select a clean, compatible mixing container and suitable mixing tool.
4. Measure each component accurately and add them in the specified order.
5. Mix slowly and thoroughly, scraping the container sides when appropriate.
6. Use the mixture within its stated working time and clean tools as instructed.

⚠ Safety

- Work in a well-ventilated area and avoid inhaling fumes or dust.
- Avoid skin and eye contact with adhesive and hardener components.
- Never mix unknown chemicals or use food containers for adhesive preparation.
- Follow product instructions for spills, disposal, and emergency exposure.

Care & storage

- Clean or dispose of mixing tools according to the adhesive instructions and local waste rules.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Awl

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Pokar; Tailor's Awl

What it is used for

To pierce, mark, or guide small holes in fabric, leather, or other suitable materials using an awl. To make controlled openings while keeping hands clear of the pointed tip.

Main parts

- Pointed shaft
- Handle
- Grip surface
- Protective cap or sheath, if supplied

Before you start

1. Select an awl suitable for the material and task.
2. Check the point and handle for damage or looseness.
3. Secure the workpiece on a stable surface.
4. Keep the supporting hand away from the intended piercing point.

How to use

1. Mark the intended hole location.
2. Hold the awl firmly by its handle.
3. Place the tip at the marked point.
4. Apply controlled pressure while guiding the tip through the material.
5. Rotate gently only if the material and tool require it.
6. Withdraw the awl carefully and inspect the opening.
7. Cover the point or store the tool safely after use.

⚠ Safety

- Keep the pointed tip directed away from the body.
- Do not hold the workpiece in the path of the awl.
- Do not use a bent or damaged awl.
- Keep the tool covered and away from children.

Care & storage

- Wipe the shaft clean after use.
- Dry the tool to prevent corrosion.

- Inspect the handle and point regularly.
- Store with a protective cap or in a tool holder.

Common mistakes to avoid

- Piercing toward the supporting hand.
- Using excessive force and damaging the material.
- Using an unstable workpiece.
- Leaving the pointed tip exposed during storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Axe

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Axe; Axe with wooden handle; Garden hatchet

What it is used for

Use an axe or garden hatchet to cut suitable wood or trim branches with controlled, deliberate strokes.

How to use

1. Inspect the blade, head, and handle for cracks, looseness, or damage.
2. Choose an axe suitable for the material and task.
3. Clear the work area and ensure others are outside the swing zone.
4. Place the wood securely on a stable chopping surface.
5. Grip the handle firmly and direct each controlled stroke away from your body.
6. Stop if the head loosens or the blade becomes damaged.
7. Clean the blade and fit its protective cover after use.

⚠ Safety

- Wear safety glasses, sturdy footwear, and suitable gloves.
- Keep hands and feet clear of the blade path.
- Never use an axe with a loose head or cracked handle.
- Do not chop toward your body or work near other people.

Care & storage

- Clean and dry the blade after use to prevent rust.
- Store with the blade covered in a secure, dry location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Knife

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Bamboo Knife

What it is used for

A bamboo knife cuts or trims bamboo during craft and workshop activities. Use it only with proper training, a stable workpiece, and controlled cutting movements.

How to use

1. Inspect the blade, handle, and fastening points for damage or looseness.
2. Wear suitable eye protection and cut-resistant protection where appropriate.
3. Secure the bamboo on a stable work surface using a suitable clamp.
4. Hold the knife firmly and cut away from your body using controlled strokes.
5. Keep the supporting hand outside the blade path at all times.
6. Place the knife safely down before repositioning the bamboo or clearing offcuts.
7. Clean the blade and store the knife with its edge protected.

⚠ Safety

- Never cut toward your body or hold bamboo in your palm while cutting.
- Keep the blade away from fingers and other people.
- Do not use a loose, cracked, or damaged knife.
- Only trained and supervised students should use sharp bamboo knives.

Care & storage

- Wipe the blade dry after use and store it in a secure protective sheath.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Splitter

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Bamboo Hand Splitter; Bamboo Splitter

What it is used for

To split bamboo into strips or smaller sections for craft, construction, and training activities.

How to handle

- Inspect the splitter for cracks, loose parts, and damaged cutting edges before use.
- Wear cut-resistant gloves and eye protection; secure the bamboo on a stable work surface.
- Position the bamboo correctly and keep hands away from the cutting edge and splitting path.
- Apply controlled force using the tool as designed; never direct the blade toward the body or other people.
- Remove bamboo splinters carefully and clean the tool after use.

Storage

- Clean and dry the splitter after use.
- Cover the cutting edge and store in a secure, dry tool cabinet away from children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bill Hook

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Bill hook

What it is used for

A bill hook cuts and trims branches, brush, and suitable plant material. Use it only for appropriate cutting tasks with a stable work area and controlled strokes.

How to use

1. Inspect the blade, handle, and attachment for cracks, looseness, or rust.
2. Wear eye protection, sturdy footwear, and suitable gloves.
3. Clear the working area and ensure other people remain well away.
4. Secure the material where possible and identify a clear cutting direction.
5. Grip the handle firmly and make controlled cuts away from your body.
6. Place the tool down safely before moving material or clearing debris.
7. Clean the blade and store the bill hook with its edge protected.

⚠ Safety

- Keep the curved blade away from hands, legs, and other people.
- Never swing the tool near another person or work in a crowded area.
- Do not use a damaged handle or loose blade.
- Use additional supervision and avoid overhead cutting.

Care & storage

- Clean and dry the blade after use and store it in a secure sheath.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bow Drill

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Bow hand drill

What it is used for

A bow drill rotates a drill bit using a bow and cord to create holes in suitable materials. Secure the workpiece and use controlled motion to prevent slipping.

How to use

1. Inspect the bow, cord, spindle, drill bit, and bearing surfaces for damage.
2. Select a suitable bit and confirm it is securely fitted.
3. Mark the hole location and clamp the workpiece firmly.
4. Position the drill bit at the mark and hold the tool upright.
5. Move the bow steadily back and forth while maintaining controlled pressure.
6. Stop before adjusting the bit or clearing chips from the hole.
7. Remove the drill carefully and clean the tool after use.

⚠ Safety

- Wear eye protection and keep fingers clear of the rotating bit.
- Secure the workpiece rather than holding small pieces in your hand.
- Keep loose clothing and hair away from the moving cord and spindle.
- Do not use a cracked bow, frayed cord, or damaged bit.

Care & storage

- Remove dust and chips, inspect the cord, and store the drill in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cable Cutter

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Cable cutter

What it is used for

A cable cutter is designed to cut electrical or communication cables cleanly with suitable cutting jaws. Select a cutter rated for the cable material and size.

How to use

1. Inspect the cutter jaws, handles, pivot, and insulation for damage.
2. Confirm the cable type and choose a cutter rated for its size and material.
3. Ensure the cable is de-energized and isolated before cutting electrical conductors.
4. Position the cable fully within the cutting jaws and keep it steady.
5. Squeeze the handles smoothly to complete the cut without twisting the tool.
6. Inspect the cut end, clean the cutter, and store it safely.

⚠ Safety

- Never cut an energized cable unless using a specifically rated method under authorized procedures.
- Do not use ordinary cable cutters on steel wire rope unless designed for it.
- Keep fingers away from the jaws and cutting area.
- Do not use cutters with damaged insulation or cracked handles for electrical work.

Care & storage

- Wipe the jaws clean, lubricate the pivot if needed, and store the cutter dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chisel

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 12 labs

Covers these lab items: Chisel; Chisel and Hammer; Chisel set; Firmer Chisel; Mortise / Chisel /; Pairing Chisel)

What it is used for

Use a chisel with a suitable hammer or hand pressure to cut, shape, or remove material during workshop tasks.

How to use

1. Select a chisel type and size suitable for the material and task.
2. Inspect the cutting edge, shaft, and striking head for damage or mushrooming.
3. Secure the workpiece firmly on a stable support.
4. Wear eye protection and position the chisel at the required angle.
5. Strike the chisel with a suitable hammer using controlled force, where applicable.
6. Remove material in small cuts and check the result regularly.
7. Clean the chisel and protect its cutting edge after use.

⚠ Safety

- Wear safety glasses and keep bystanders clear of flying chips.
- Do not use a chisel with a cracked shaft or mushroomed head.
- Keep the supporting hand away from the cutting path.
- Use the correct chisel for the material and task.

Care & storage

- Keep the cutting edge sharp and free from rust.
- Store chisels with cutting edges covered or separated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cold Chisel

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Flat Chisel

What it is used for

To cut, chip, or shape metal, masonry, or other suitable hard materials using a hammer and cold chisel designed for the task.

How to handle

- Inspect the chisel edge and striking head for cracks, mushrooming, or other damage; remove damaged tools from service.
- Wear safety goggles or a face shield, suitable gloves, and protective footwear; secure the workpiece firmly.
- Select the correct chisel for the material and position it securely on the intended cutting point.
- Strike the chisel with a suitable hammer using controlled blows, keeping bystanders clear of flying chips.
- Do not use a chisel with a mushroomed head; clean the tool and inspect it after use.

Storage

- Wipe the chisel clean and dry after use.
- Store with the cutting edge protected in a designated tool rack away from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cutter

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Cutters

What it is used for

To cut small wires, plastic parts, cable ties, or other materials using a suitable cutter. To make controlled cuts while protecting hands and nearby components.

Main parts

- Cutting jaws or blades
- Pivot joint
- Handles
- Insulated grips, if fitted
- Locking catch or spring, if fitted

Before you start

1. Identify the material and choose a cutter rated for that task.
2. Inspect the cutting edges, pivot, and handles for damage.
3. Disconnect electrical power before cutting wires.
4. Secure the workpiece and keep bystanders clear.
5. Wear eye protection when cut pieces may fly.

How to use

1. Open the cutter handles or release the lock if fitted.
2. Position the material fully within the appropriate cutting area.
3. Keep fingers away from the blades and the likely path of the cut piece.
4. Squeeze the handles steadily to complete the cut.
5. Avoid twisting the cutter or exceeding its rated capacity.
6. Inspect the cut and remove offcuts safely.
7. Close and store the cutter after use.

⚠ Safety

- Never cut live electrical wires.
- Do not use ordinary cutters on hardened metal or materials beyond their rating.
- Keep fingers clear of the cutting jaws and pinch points.
- Do not use cutters with cracked handles or damaged blades.

Care & storage

- Wipe the blades clean after use.
- Keep the pivot clean and lubricate as specified.
- Inspect cutting edges and handle insulation regularly.
- Store with the jaws closed and protected.

Common mistakes to avoid

- Cutting a live wire or an unknown circuit.
- Using the wrong cutter for the material.
- Placing fingers close to the cutting point.
- Twisting or overloading the tool.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Draw Knife

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Draw Knife

What it is used for

A draw knife removes and shapes wood by pulling its blade toward the user along the workpiece. Secure the wood and use controlled strokes with the blade directed away from the body.

How to use

1. Inspect the blade and handles for cracks, looseness, rust, or damage.
2. Clamp the workpiece securely on a stable support.
3. Wear eye protection and position yourself with a clear working path.
4. Grip both handles firmly and keep the blade angle suitable for the wood.
5. Pull the tool in short, controlled strokes while keeping the blade away from your body.
6. Stop before repositioning the workpiece or clearing wood shavings.
7. Clean the blade carefully and store the tool with its edge protected.

⚠ Safety

- Keep hands and other body parts out of the blade path.
- Never cut toward your body or work on an unsecured workpiece.
- Do not use a loose-handled or damaged draw knife.
- Keep the sharp edge covered when the tool is not in use.

Care & storage

- Remove wood residue, dry the blade, and apply suitable rust protection if required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drill Guide

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Hand drill guide

What it is used for

A hand drill guide helps keep a drill aligned at the required angle and position. Secure the workpiece and use the guide to improve drilling accuracy.

How to use

1. Inspect the guide, bushings, clamps, and adjustment parts for damage.
2. Select a guide bushing suitable for the drill bit diameter.
3. Mark the drilling point and secure the workpiece firmly.
4. Position the guide over the mark and align it to the required angle.
5. Clamp the guide securely so it cannot shift during drilling.
6. Drill slowly while maintaining alignment and following the drill manufacturer's instructions.
7. Stop the drill, remove the guide, and inspect the hole.

⚠ Safety

- Wear safety glasses and keep hands away from the rotating bit.
- Secure loose workpieces before drilling.
- Do not adjust or reposition the guide while the drill is running.
- Keep clothing, hair, and cables away from rotating parts.

Care & storage

- Remove chips, clean the guide, and store it in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrician Knife

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Electrician knife

What it is used for

To cut or strip suitable electrical cable insulation using an electrician knife. To prepare cable ends carefully while avoiding damage to conductors and preventing cuts or electrical injury.

Main parts

- Insulated or shaped handle
- Sharp blade
- Blade guard or sheath
- Grip surface

Before you start

1. Confirm the cable is isolated, de-energized, and verified safe before cutting.
2. Inspect the blade and handle for damage or looseness.
3. Select a knife suitable for the cable and task.
4. Wear appropriate eye protection and follow electrical work procedures.

How to use

1. Secure the cable on a stable work surface.
2. Hold the knife firmly with the blade directed away from the body.
3. Make a shallow controlled cut through the insulation without cutting deeply into the conductor.
4. Rotate or reposition the cable as needed instead of forcing the blade.
5. Remove the insulation carefully and inspect the conductor for nicks or damage.
6. Fit the blade guard or sheath immediately after use.

⚠ Safety

- Never cut or strip an energized cable.
- Keep fingers and the free hand out of the blade path.
- Do not use a blunt, cracked, or loose-blade knife.
- Use an insulated tool only as specified; it does not replace isolation and verification.

Care & storage

- Wipe the blade clean and dry after use.
- Inspect the blade and handle regularly.
- Replace or secure damaged parts before further use.

- Store the knife with its blade covered.

Common mistakes to avoid

- Working on a live cable.
- Cutting too deeply and nicking the conductor.
- Cutting toward the body or supporting hand.
- Leaving the blade exposed during storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

File

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 23 labs

Covers these lab items: File Handle (Wooden; File set; Flat Basted File; Flat file; Flat file Rasp); Flat file Rough); Flat Smooth File; Half Round (Wood rasp File); RasW; Round Basted File; Round file; Triangle file; Triangular files

What it is used for

A file removes small amounts of material to smooth, shape, or finish a workpiece. Select the correct file profile and cut for the material and required finish.

How to use

1. Inspect the file teeth, tang, and handle for wear or damage.
2. Select a suitable flat, round, half-round, or other file for the task.
3. Secure the workpiece firmly in a vice or approved holder.
4. Hold the file by its handle and support it safely with the other hand where appropriate.
5. Apply controlled forward strokes with suitable pressure and lift or ease pressure on the return stroke.
6. Check the workpiece frequently to avoid removing too much material.
7. Clean the file using a suitable file brush after use.
8. Store the file with its teeth protected.

⚠ Safety

- Never use a file without a secure handle fitted to its tang.
- Wear eye protection and keep hands away from sharp workpiece edges.
- Do not use a file as a lever or striking tool.
- Secure the workpiece before filing to prevent slipping.

Care & storage

- Remove filings with a file brush rather than bare fingers.
- Keep files dry and store them separately to protect their teeth.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hacksaw

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 21 labs

Covers these lab items: ADJUSTAB LE HACKSAW FRAME; Hacksaw; Hacksaw blade with frame and one spare blade; Hacksaw Frame; Hacksaw frame with blade; Hacksaw frame with the blade; Hacksaw, Hammer, Scissor

What it is used for

A hacksaw cuts metal, plastic, or other suitable materials using a tensioned toothed blade. Select the correct blade and secure the workpiece before cutting.

How to use

1. Inspect the frame, handle, blade, and tension mechanism for damage.
2. Select a blade suited to the material and required cut.
3. Fit the blade securely with its teeth facing the correct cutting direction.
4. Adjust blade tension according to the frame manufacturer's instructions.
5. Secure the workpiece firmly in a vice or approved holder.
6. Align the blade with the cutting line and use smooth, controlled strokes.
7. Avoid twisting the blade and check progress periodically.
8. Release tension if required, clean the saw, and store it safely.

⚠ Safety

- Wear safety goggles to protect against flying particles.
- Keep hands away from the blade path and sharp cut edges.
- Do not use a cracked frame or loose, damaged blade.
- Do not force the saw or touch the blade immediately after prolonged cutting.

Care & storage

- Clean the frame and blade after use and remove metal debris safely.
- Release blade tension for storage if recommended by the manufacturer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Drill

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Hand Drill; Hand Drill Machine; Hand Driller Kit

What it is used for

A hand drill is used to make holes in wood, plastic, metal, or other suitable materials with an appropriate bit. Select the bit and drilling method according to the material and task.

How to use

1. Inspect the drill, chuck, handle, and drill bit for damage.
2. Choose a suitable sharp bit and secure it firmly in the chuck.
3. Mark the hole location and clamp the workpiece securely.
4. Align the drill perpendicular to the surface unless the task requires another angle.
5. Apply steady pressure and rotate the handle smoothly while drilling.
6. Remove the bit carefully, clear debris, and clean the drill after use.

⚠ Safety

- Wear safety glasses and secure loose clothing and hair.
- Clamp the workpiece; never hold small pieces directly in your hand while drilling.
- Keep fingers away from the rotating bit and cutting area.
- Do not use damaged bits or force the drill excessively.

Care & storage

- Clean the drill and store it with bits in a dry, secure place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Plane

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Hand Plane; Iron Plane

What it is used for

To smooth, flatten, or shape wooden surfaces by removing thin shavings.

How to handle

- Inspect the plane body, handle, blade, and adjustment mechanism for damage or looseness.
- Secure the workpiece firmly on a stable workbench before planing.
- Adjust the blade for a suitable shaving depth and check that it is properly secured.
- Push the plane along the wood grain with controlled, even strokes while keeping hands clear of the blade.
- Retract or cover the blade when finished and remove wood shavings from the tool.

Storage

- Store clean and dry with the blade retracted or covered.
- Keep the cutting edge protected and apply suitable rust prevention if specified for the tool.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hand Punch

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Hand punch

What it is used for

A hand punch creates holes or marks in compatible materials using manual pressure. Select the correct punch size and secure the workpiece before use.

How to use

1. Inspect the punch tip, handle, and striking surface for damage.
2. Choose a punch type and size suitable for the material and required hole.
3. Mark the intended position and secure the workpiece on a stable support.
4. Align the punch tip precisely with the mark.
5. Apply controlled pressure or use the specified striking method for the punch type.
6. Check the hole or mark and remove any burrs using an appropriate tool.
7. Clean the punch and return it to storage after use.

⚠ Safety

- Wear eye protection against flying fragments.
- Keep fingers clear of the punch tip and striking area.
- Do not use a mushroomed, cracked, or blunt punch.
- Use a suitable backing surface and never hold small workpieces by hand near the impact point.

Care & storage

- Wipe the punch clean, protect the tip, and store it in a dry tool holder.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hole Punch Pliers

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Plier punch

What it is used for

Hole punch pliers create small holes in compatible materials such as leather, fabric, or thin sheet material. Choose the correct punch size and keep the material securely positioned.

How to use

1. Inspect the pliers, punch wheel or tip, hinge, and handles for damage.
2. Select the punch size suitable for the material and required hole diameter.
3. Mark the hole location and position the material between the punch tip and backing surface.
4. Align the punch with the mark and hold the material securely.
5. Squeeze the handles steadily until the punch cuts through the material.
6. Release the handles, remove the material, and inspect the hole edges.
7. Clear scraps from the tool and store it safely.

⚠ Safety

- Keep fingers away from the punch tip and closing jaws.
- Wear eye protection when working with materials that may produce flying fragments.
- Do not punch material that exceeds the tool's rated capacity.
- Do not use pliers with damaged tips or loose hinges.

Care & storage

- Remove scraps, wipe the tool clean, and protect the hinge from corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Knife

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 21 labs

Covers these lab items: Chisel edged cleaning knife; Chisel Knife; Coring knife; Cutting knife; Double edged cleaning knife; Indian Foldable knife; Indian Splitting knife; Knife; Knife Set; Knife Set (Chef, Paring, Serrated); Knives; Knives (straight and curved)

What it is used for

A knife cuts, trims, cleans, or shapes materials according to its design. Choose the correct knife for the task and use controlled cuts on a stable work surface.

How to use

1. Inspect the blade, handle, locking mechanism, and sheath for damage.
2. Select a knife designed for the material and intended task.
3. Secure the workpiece on a stable surface before cutting.
4. Hold the handle firmly and cut in a controlled direction away from the body.
5. Keep the free hand clear of the cutting path and support the material safely.
6. Close or sheath a foldable knife immediately after use.
7. Clean and dry the knife and return it to secure storage.

⚠ Safety

- Never use a loose, cracked, or blunt blade that may slip unexpectedly.
- Do not cut toward your body or another person.
- Keep knives away from children and never leave them unattended.
- Use appropriate cut-resistant protection when the task requires it.

Care & storage

- Clean and dry the blade after use to prevent corrosion.
- Store the knife with its blade covered and sharpen or replace it safely when needed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Laminate Cutter

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Sun mica Cutter

What it is used for

To cut or trim laminate sheets, such as sunmica, to the required size using a suitable laminate cutter.

How to handle

- Inspect the cutter blade, guide, handle, and adjustment mechanism for damage or looseness.
- Measure and mark the laminate accurately and secure the sheet on a stable work surface.
- Wear eye protection and suitable cut-resistant gloves; keep fingers clear of the blade path.
- Align the cutter with the marked line and make controlled cuts according to the tool design without forcing the blade.
- Deburr or smooth edges using a suitable method and clean the tool after use.

Storage

- Clean the cutter and protect the blade with its guard or cover.
- Store in a secure, dry tool cabinet with the cutting edge inaccessible to accidental contact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Machete

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Machete/Chopper

What it is used for

A machete or chopper is a heavy cutting tool used for vegetation and other suitable materials. Use it only for its intended purpose with a clear work area and controlled cutting technique.

How to use

1. Inspect the blade, handle, and fastening points for cracks, looseness, or damage.
2. Wear suitable protective clothing, sturdy footwear, and eye protection.
3. Clear the work area and ensure that other people remain well outside the cutting range.
4. Hold the handle firmly and maintain a stable stance before each cut.
5. Cut only suitable material using controlled strokes directed away from the body.
6. Stop before moving, clearing debris, or changing the cutting position.
7. Clean the blade, fit its sheath, and store the tool securely.

⚠ Safety

- Never swing the machete near another person or use it as a throwing tool.
- Keep hands and legs outside the blade's path.
- Do not use a damaged, loose-handled, or blunt tool that may slip unpredictably.
- Carry the machete with its blade sheathed and directed safely.

Care & storage

- Clean and dry the blade after use and apply suitable rust protection when recommended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paper Cutter

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Paper cutter

What it is used for

To cut paper, card, or other approved sheet material to the required size and shape.

How to handle

- Inspect the cutter blade, guard, handle, base, and locking mechanism for damage before use.
- Place the material flat on a stable cutting surface and align it with the required measurement marks.
- Keep fingers away from the cutting path and use the guard or hold-down mechanism where provided.
- Cut with controlled movement and never attempt to catch a falling blade or reach across the cutting edge.
- Lock or retract the blade after use and remove scraps carefully.

Storage

- Store in a designated tool drawer with the blade retracted or locked.
- Keep the cutting edge covered and the tool away from children and unauthorised users.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Post-Processing Tool Kit

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Post processing tools

What it is used for

A post-processing tool kit is used to trim, smooth, clean, and finish fabricated parts or craft items. Select tools suitable for the material and follow the required finishing procedure.

How to use

1. Inspect files, cutters, scrapers, sanding tools, and handles for damage.
2. Secure the workpiece on a stable work surface.
3. Select the appropriate tool for trimming, deburring, smoothing, or cleaning.
4. Wear suitable protective equipment before starting the finishing task.
5. Work gradually with controlled strokes and check the surface frequently.
6. Remove dust and burrs using an appropriate cleaning method.
7. Clean, inspect, and return all tools to their storage locations.

⚠ Safety

- Wear safety glasses and suitable gloves when the task requires them.
- Keep hands away from sharp edges and cutting paths.
- Secure small workpieces before trimming or sanding.
- Follow material-specific precautions for dust, fumes, and sharp debris.

Care & storage

- Remove residue after use and store sharp tools in protective holders.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Saw Blade

Category: **Hand Tools - Cutting & Shaping** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Saw Blade (For basic cutting)

What it is used for

Handle saw blades safely for basic cutting tasks.

How to handle

- Check the blade for cracks, missing teeth, rust, or other damage before use.
- Select a blade compatible with the saw and material being cut.
- Wear safety glasses and suitable gloves when handling loose blades; keep fingers away from cutting edges.
- Secure the workpiece and follow the tool manufacturer's fitting and cutting instructions.

Storage

- Keep blades clean and dry with cutting edges protected.
- Store in labelled sleeves or a dedicated tool holder away from moisture and unauthorised access.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scissors

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 33 labs

Covers these lab items: Button hole scissor; Cutting Scissors; Embroidery scissors; Embroidery Snips; Fabric Scissors; Paper Scissors; Scissor; Scissors; Scissors (plastic handled scissor for cutting paper), Pinking Shears; Scissors (sharp, medium size); Small Scissors

What it is used for

Scissors cut paper, fabric, thread, and other materials suited to their design. Select the correct scissors for the material and use them with controlled movements.

How to use

1. Inspect the blades, pivot, handles, and tips for damage or looseness.
2. Choose scissors designed for the material, such as fabric, paper, or embroidery work.
3. Hold the handles correctly and keep the material stable on the work surface.
4. Align the blades with the intended cutting line and cut using smooth strokes.
5. Keep fingers clear of the blades and avoid forcing thick material through unsuitable scissors.
6. Close the blades when carrying or setting the scissors down.
7. Clean the blades and store the scissors safely after use.

⚠ Safety

- Never run or move quickly while carrying open scissors.
- Cut away from the body and keep other people clear of the cutting area.
- Do not use damaged scissors or use them as a prying tool.
- Pass scissors to another person closed and handle-first.

Care & storage

- Wipe blades clean and dry after use, especially after cutting damp materials.
- Store scissors closed and arrange sharpening or replacement when cutting performance declines.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scraper

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 13 labs

Covers these lab items: Carbon scrapers; Gasket scrapers; Round Scrapping Knife; Scrapers; Scrappers; Scrapping Knife; Trim or moulding tools

What it is used for

Use a scraper to remove residue, old gasket material, paint, or deposits from a surface without damaging the underlying material.

How to use

1. Select a scraper with a blade and shape suitable for the surface.
2. Inspect the handle and blade for looseness, cracks, or damage.
3. Secure the workpiece and remove nearby obstructions.
4. Hold the scraper firmly and use controlled strokes at a suitable angle.
5. Work away from your body and avoid gouging the surface.
6. Collect removed residue and check the surface after scraping.
7. Clean the scraper and replace or protect the blade as needed.

⚠ Safety

- Wear safety glasses and suitable gloves when required.
- Keep hands away from the blade edge and cutting path.
- Do not use excessive force or a damaged scraper.
- Confirm the surface can tolerate scraping before starting.

Care & storage

- Remove residue and dry the tool after use.
- Store sharp blades with a protective cover.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sharpening Stone

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Carbound um Stone (Emery)

What it is used for

To sharpen compatible hand-tool edges using a suitable abrasive stone or emery stone.

How to handle

- Inspect the stone for cracks, chips, uneven wear, or looseness in its holder.
- Confirm the stone type and grit are suitable for the tool and follow the manufacturer's instructions for dry or lubricated use.
- Secure the stone on a stable surface and wear eye protection; use gloves only when they do not create an entanglement or loss-of-control hazard.
- Move the tool edge across the stone at the recommended angle using controlled strokes, keeping fingers clear of the edge.
- Clean away abrasive residue and inspect the tool edge carefully after sharpening.

Storage

- Clean and dry the stone as recommended and allow any used lubricant to drain safely.
- Store in a protective holder or case in a dry place away from impact and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shears

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Shears

What it is used for

To cut suitable materials such as fabric, paper, thin sheet material, or plant stems using the appropriate type of shears.

How to handle

- Inspect the blades, pivot, handles, and locking mechanism for damage or looseness before use.
- Choose shears designed for the material being cut and keep the material stable.
- Keep fingers away from the blades and cut with controlled movements, directing the cutting edge away from the body.
- Do not use damaged shears or exceed their intended cutting capacity.
- Wipe the blades clean after use, dry them thoroughly, and engage the lock or cover where provided.

Storage

- Store clean and dry in a tool drawer or protective sheath with blades closed.
- Keep sharp edges protected and store away from children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spokeshave

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Spokeshave

What it is used for

A spokeshave trims and shapes wood by removing thin shavings from curved or flat surfaces. Adjust the blade carefully and secure the workpiece before making controlled passes.

How to use

1. Inspect the spokeshave body, handles, blade, and adjustment screws for damage.
2. Secure the wood firmly on a stable workbench or suitable holding fixture.
3. Check blade sharpness and adjust the blade projection for a light cut.
4. Hold both handles securely and align the tool with the intended cutting direction.
5. Make controlled passes along the grain where appropriate and inspect the surface frequently.
6. Adjust blade projection gradually if the cut is too heavy or uneven.
7. Remove shavings, clean the tool, and store it with the blade protected.

⚠ Safety

- Keep fingers away from the blade edge and cutting path.
- Wear safety glasses and avoid loose clothing near the workpiece.
- Do not cut toward the body or use a poorly secured workpiece.
- Do not adjust or clean the blade while the tool is moving.

Care & storage

- Remove wood shavings after use, protect the blade from corrosion, and store the tool dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tap and Die Set

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 11 labs

Covers these lab items: Die set complete; Die wrench; Tap & Die Set; Tap or die; Tap Wrench

What it is used for

Use a tap and die set to cut or repair internal threads in holes and external threads on rods or bolts.

How to use

1. Select the correct tap or die for the thread size, pitch, and material.
2. Inspect the cutting edges and handles for damage or wear.
3. Secure the workpiece firmly and prepare the hole or rod to the required size.
4. Apply suitable cutting fluid when appropriate for the material.
5. Align the tap or die squarely with the workpiece before turning.
6. Turn the tool gradually, reversing occasionally as required to clear chips.
7. Remove the tool carefully and inspect the finished thread with a suitable fastener or gauge.
8. Clean the tools and remove metal chips after use.

⚠ Safety

- Wear safety glasses and keep hands clear of sharp cutting edges.
- Clamp the workpiece securely; do not hold it by hand while cutting threads.
- Use the correct tool size and avoid excessive turning force.
- Remove chips with a brush, not bare fingers.

Care & storage

- Clean and lightly protect cutting tools against rust after use.
- Store taps, dies, and handles in their labeled case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tenon Saw

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Tenon Saw

What it is used for

A tenon saw makes controlled straight cuts in wood, especially for joints and detailed carpentry. Use it with a stable workpiece and suitable cutting technique.

How to use

1. Inspect the blade, handle, teeth, and fastening points for damage.
2. Wear eye protection and secure the wood firmly on a workbench.
3. Mark the cutting line clearly and position the saw at the start point.
4. Start with gentle strokes to establish a straight kerf.
5. Continue cutting with controlled forward and backward strokes without forcing the blade.
6. Support the offcut and keep hands clear as the cut finishes.
7. Place the saw safely and remove wood dust after cutting.

⚠ Safety

- Keep fingers and hands away from the blade path and cutting edge.
- Wear safety goggles and do not hold the workpiece in your hand while cutting.
- Never use a saw with a loose handle or damaged blade.
- Carry and store the saw with its teeth protected.

Care & storage

- Brush off sawdust, keep the blade dry, and store it with a suitable blade guard.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tin Snips

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Tin Cutter -Scissor; Tin Cutter (Tin cutter with spring)

What it is used for

To cut thin sheet metal using tin snips. To produce controlled cuts while keeping the sheet secure and reducing sharp-edge injuries.

Main parts

- Cutting blades
- Handles
- Pivot joint
- Spring or opening mechanism, if fitted
- Locking catch, if fitted

Before you start

1. Select snips suitable for the sheet thickness and cutting direction.
2. Inspect blades, pivot, handles, and spring for damage.
3. Mark the cutting line clearly.
4. Secure the sheet and wear safety glasses and cut-resistant gloves suitable for the task.

How to use

1. Open the snips and position the sheet between the blades.
2. Align the blades with the marked cutting line.
3. Hold the handles firmly and cut using steady squeezes.
4. Advance along the line without twisting the blades excessively.
5. Support the sheet to prevent it bending or dropping unexpectedly.
6. Deburr or safely handle the cut edge as required.
7. Close and lock the snips after use.

⚠ Safety

- Keep fingers clear of the cutting blades and sharp sheet edges.
- Do not cut material thicker than the tool's rating.
- Wear eye protection against flying metal fragments.
- Do not use snips with loose pivots or damaged blades.

Care & storage

- Wipe metal debris from the blades after use.
- Keep the pivot clean and lubricate as specified.

- Inspect cutting edges regularly.
- Store with the blades closed and locked.

Common mistakes to avoid

- Using snips unsuitable for the material thickness.
- Twisting the tool and damaging the blades.
- Holding sharp sheet edges with bare hands.
- Leaving snips open during storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Utility Knife

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Cutter (utility knife); Razor Blade Knife (Utility Knife); Safety Knife; Utility Knife

What it is used for

A utility knife is used to cut cardboard, packaging, and other suitable materials. Select the correct blade and use the safety mechanism to reduce the risk of injury.

How to use

1. Inspect the knife body, blade, locking mechanism, and handle for damage.
2. Choose a blade suitable for the material and extend only the required length.
3. Place the material on a stable cutting surface and keep the cutting path clear.
4. Hold the knife firmly and cut away from your body with controlled pressure.
5. Retract or lock the blade immediately after cutting.
6. Dispose of worn blades in a designated sharps container and store the knife safely.

⚠ Safety

- Keep fingers and other body parts away from the blade path.
- Do not cut toward yourself or hold the material in the cutting path.
- Never use a loose, broken, or blunt blade.
- Retract the blade before carrying or storing the knife.

Care & storage

- Replace worn blades safely and keep the knife clean and dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wire Brush

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Wire Brush; Wire Brushes

What it is used for

To remove rust, scale, paint residue, and loose debris from suitable surfaces.

How to handle

- Inspect the brush for loose, bent, or missing wires and a damaged handle.
- Wear safety goggles, suitable gloves, and other required protective equipment.
- Secure the workpiece and brush with controlled strokes in a safe direction, keeping the bristles away from the body.
- Do not use a hand brush on a powered tool unless it is specifically designed and rated for that use.
- Clean the brush after use and check for loose bristles.

Storage

- Remove debris and dry the brush before storage.
- Store in a designated tool holder with bristles protected and away from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wire Cutter

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Cutting Pliers; Wire Cutter; Wire Cutters

What it is used for

To cut suitable wires and small conductors using cutting pliers or a wire cutter.

How to handle

- Inspect the cutting edges, pivot, and insulated handles for damage or looseness.
- Choose a cutter rated for the wire material and size; never use it on live electrical conductors.
- Wear eye protection and keep the wire end directed away from people.
- Position the wire near the cutting edge and squeeze the handles steadily; do not twist or overload the tool.
- Wipe the cutter clean after use and check the cutting edges.

Storage

- Clean and dry the cutter; apply a light protective lubricant to the pivot if recommended by the manufacturer.
- Close the jaws and store in a dry tool drawer away from children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wire Stripper

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 14 labs

Covers these lab items: Wire stripper; Wire Stripper (Automatic); Wire strippers

What it is used for

Use a wire stripper to remove insulation from a wire while preserving the conductor for safe electrical connections.

How to use

1. Choose a stripper suitable for the wire type and gauge.
2. Inspect the cutting edges, adjustment setting, and handles for damage.
3. Disconnect the wire from every power source before stripping.
4. Measure the required stripping length and position the wire in the correct notch.
5. Close the handles carefully to cut the insulation without damaging the conductor.
6. Pull away the insulation and inspect the exposed wire strands or core.
7. Adjust the setting or tool size if the conductor has been nicked.

⚠ Safety

- Never strip a wire while it is energized.
- Keep fingers away from cutting edges and moving parts.
- Wear eye protection when cut insulation or wire strands may fly.

Care & storage

- Remove insulation scraps and wipe the tool after use.
- Store the tool dry with the cutting edges protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wood Saw

Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Crosscut Saw; Hand saw; Saw; Saw small size

What it is used for

Use a wood saw to cut timber or other suitable wood materials along a marked line during workshop or carpentry tasks.

How to use

1. Select a saw type and blade length suitable for the wood and cut.
2. Inspect the blade, teeth, handle, and fixing points for damage.
3. Mark the cutting line and secure the workpiece on a stable support.
4. Hold the saw firmly and begin with short, controlled strokes along the line.
5. Continue with steady strokes while keeping the blade aligned with the mark.
6. Keep the offcut supported and avoid pinching the blade as the cut finishes.
7. Clean the saw and store it with the blade protected.

⚠ Safety

- Wear safety glasses and keep hands clear of the cutting path.
- Clamp the workpiece securely instead of holding it close to the blade.
- Do not use a cracked handle or damaged blade.
- Keep bystanders clear and avoid forcing the saw through the wood.

Care & storage

- Remove sawdust and wipe the blade dry after use.
- Store the saw in a dry place with teeth covered or protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Adjustable Wrench

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 10 labs

Covers these lab items: Adjustable wrench; Adjustable Wrench (Adjustabl e Spanner); Slide wrench

What it is used for

Use an adjustable wrench to tighten or loosen compatible nuts and bolts by setting its jaws to the correct size.

How to use

1. Inspect the jaws, adjustment screw, and handle for damage or excessive wear.
2. Turn the adjustment screw until the jaws fit the fastener closely.
3. Place the wrench fully onto the nut or bolt head.
4. Position the movable jaw on the side that receives the main load where possible.
5. Pull the handle steadily in the correct direction without jerking.
6. Reposition the wrench if the fastener cannot be turned safely.
7. Clean and return the wrench to storage after use.

⚠ Safety

- Use the correct wrench size and ensure the jaws fit tightly.
- Do not use extension pipes or strike the wrench with a hammer.
- Keep hands clear of pinch points and slipping surfaces.
- Do not use a damaged wrench or use it on live electrical equipment.

Care & storage

- Wipe the wrench clean and keep the adjustment screw moving freely.
- Store it dry and inspect the jaws before use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Allen Key

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 22 labs

Covers these lab items: Allen key set; Allen key set (Insulated); Allen wrench; Allen Wrench Set; Allen Wrench Set (Hex Set)

What it is used for

An Allen key tightens or loosens fasteners with hexagonal socket heads. Select the correct key size and use it carefully to prevent rounding the socket or damaging the tool.

How to use

1. Inspect the Allen key for bending, worn ends, rust, or damage.
2. Select the correct size that fits fully into the hexagonal socket.
3. Clean the fastener socket if dirt prevents proper engagement.
4. Insert the key fully and keep it aligned with the fastener.
5. Turn the key smoothly with controlled force and avoid excessive leverage.
6. Stop if the key slips or the socket begins to round off.
7. Remove the key and return it to the correct set after use.

⚠ Safety

- Do not use a bent or worn Allen key.
- Avoid adding an unapproved extension to increase leverage.
- Keep fingers clear of pinch points and sharp workpiece edges.
- Use insulated keys for electrical tasks only when correctly rated and inspected.

Care & storage

- Wipe keys clean and dry after use.
- Store the set in an organised holder and replace worn or damaged keys.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bench Vice

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Bench vice; Bench Vice (Code-U - 301); Machine vice

What it is used for

Use a bench vice to hold a workpiece firmly during filing, cutting, drilling, or other workshop operations.

How to use

1. Inspect the vice body, jaws, screw, handle, and mounting bolts for damage.
2. Confirm the vice is securely fixed to a stable workbench.
3. Open the jaws and position the workpiece between them.
4. Place soft jaw covers when needed to protect finished surfaces.
5. Tighten the handle until the workpiece is secure without excessive force.
6. Check that the workpiece cannot move before starting the operation.
7. Loosen the vice carefully and remove the workpiece after the task.

⚠ Safety

- Keep fingers clear of the jaws and screw mechanism.
- Do not use the vice handle as a hammer or add an extension pipe.
- Avoid overtightening, which may damage the workpiece or vice.
- Do not use a loose, cracked, or poorly mounted vice.

Care & storage

- Remove metal chips and dirt from the jaws and screw.
- Keep the screw lightly lubricated and inspect mounting bolts regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chain Wrench

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Chain wrench; Chain wrenches

What it is used for

To grip and turn large pipes or cylindrical fittings using a chain wrench. To apply controlled torque while minimizing damage to the workpiece.

Main parts

- Handle
- Chain
- Chain pins or links
- Hook or gripping teeth
- Adjustment or locking mechanism

Before you start

1. Confirm that the wrench size suits the pipe or fitting.
2. Inspect the chain, links, pins, and handle for damage.
3. Clean excess oil or grease from the gripping area where safe to do so.
4. Secure the pipe or fitting against movement.

How to use

1. Wrap the chain around the pipe or fitting.
2. Engage the chain securely with the wrench head.
3. Position the handle for a controlled pull.
4. Apply force gradually in the required direction.
5. Reposition the chain if a new grip is needed.
6. Release the chain carefully after the fitting is secure.

⚠ Safety

- Keep hands clear of pinch points in the chain.
- Do not use extension bars unless explicitly approved by the manufacturer.
- Do not use a damaged or stretched chain.
- Stand in a stable position and avoid sudden force.

Care & storage

- Clean the chain and gripping teeth after use.
- Inspect links, pins, and handle regularly.

- Apply suitable lubricant to designated moving points.
- Store in a dry location.

Common mistakes to avoid

- Using the wrong wrench size.
- Applying sudden excessive force.
- Gripping a pipe that is not secured.
- Using a damaged chain or handle.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Circlip Pliers

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Circlip pliers

What it is used for

Circlip pliers install or remove retaining rings from shafts or bores. Use the correct plier type and size to prevent ring damage or sudden release.

How to use

1. Identify whether the circlip is internal or external and select matching pliers.
2. Inspect the tips, handles, and spring mechanism for damage.
3. Secure the workpiece and wear eye protection before handling the circlip.
4. Fit the plier tips fully into the circlip holes.
5. Squeeze or spread the handles gradually to compress or expand the ring.
6. Position or remove the ring carefully without overstretching it.
7. Release the pliers slowly and verify that the ring is seated correctly.

⚠ Safety

- Wear eye protection because circlips can spring free unexpectedly.
- Keep the ring pointed away from your face and other people.
- Do not use pliers with damaged tips or an incompatible size.
- Never exceed the tool's intended opening range.

Care & storage

- Wipe the pliers clean, check the spring and tips, and store them dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Clamp

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: C-Clamps (for carpentry use); Clamp; Clamp (C-Clamp); Clamps and Hangers

What it is used for

Use a clamp to hold workpieces securely in position during cutting, drilling, gluing, assembly, or other workshop tasks.

How to use

1. Choose a clamp type and size suitable for the workpiece and task.
2. Inspect the frame, screw, pads, and handle for damage or wear.
3. Position the clamp so its pads contact the workpiece securely.
4. Tighten the screw gradually until the workpiece is held firmly without damage.
5. Check that the workpiece cannot shift before starting the task.
6. Release the clamp carefully when the work is complete.
7. Clean and store the clamp after use.

⚠ Safety

- Keep fingers clear of the clamp jaws and pinch points.
- Do not overtighten and crush or deform the workpiece.
- Do not use a cracked or damaged clamp.
- Secure heavy or awkward workpieces with enough suitable clamps.

Care & storage

- Remove dust, glue, and residue after use.
- Store clamps dry with screws loosened slightly when appropriate.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Crimping Tool

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 14 labs

Covers these lab items: Crimpers; Crimping Tool; Crimping tools; Hand crimping tools; Stripping & Crimping Tools (1 set); Wire Crimpers; Wire Simpers

What it is used for

Use a crimping tool to secure a suitable terminal or connector onto a wire for a reliable electrical or mechanical connection.

How to use

1. Select a crimping tool and terminal that match the wire size and connector type.
2. Inspect the tool jaws and handles for damage or excessive wear.
3. Strip the required length of insulation without cutting the wire strands.
4. Insert the wire fully into the correct terminal barrel.
5. Place the terminal in the matching crimping die and position it correctly.
6. Squeeze the handles fully as instructed by the tool manufacturer.
7. Inspect the crimp and gently pull the wire to check that it is secure.

⚠ Safety

- Keep fingers clear of the crimping jaws and moving handles.
- Use the correct die and terminal for the wire size.
- Do not crimp live electrical wires; isolate power first.
- Wear eye protection when wire strands or cut ends may spring free.

Care & storage

- Wipe the tool clean and store it in a dry place.
- Check jaws and pivots regularly for wear or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fish Tape

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Fish Tape

What it is used for

To guide and pull suitable wires through conduit or enclosed wiring routes during electrical installation work.

How to handle

- Inspect the fish tape, reel, leader, and handle for kinks, fraying, corrosion, or damage.
- Confirm the circuit is de-energized and follow electrical safety procedures before working in conduit.
- Feed the leader carefully through the conduit without forcing it around obstructions.
- Attach the wire securely to the leader and pull steadily while a helper manages the cable where appropriate.
- Avoid contact with energized conductors and retract the tape carefully after the pull.

Storage

- Wipe the tape clean and retract it fully into its reel after use.
- Store in a dry tool cabinet away from moisture, corrosive chemicals, and sharp bends.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fuse Puller

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Fuse puller

What it is used for

To safely remove and install compatible fuses without damaging the fuse or holder.

How to handle

- Inspect the fuse puller for cracks, deformation, or damaged gripping tips.
- Switch off and isolate the equipment's power supply before removing a fuse; verify de-energisation where required.
- Use a fuse puller that matches the fuse type and size.
- Grip the fuse securely and pull it straight out without twisting or touching exposed electrical contacts unnecessarily.
- Install the correct replacement fuse rating and type, then return the puller clean and dry.

Storage

- Store in a designated tool organiser in a dry location.
- Keep gripping tips clean and separate from sharp or heavy tools that may damage them.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gear Puller

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: bearing pullers; gear puller tools

What it is used for

To remove gears, pulleys, bearings, or similar press-fitted parts using a suitable mechanical puller.

How to handle

- Inspect the puller arms, centre screw, tips, and threads for cracks, wear, or damage.
- Select a puller with the correct capacity and configuration for the component; isolate power and secure the workpiece.
- Position the arms evenly behind the component and align the centre screw with the shaft centre.
- Tighten the centre screw gradually with the appropriate tool, keeping hands and face out of the potential release path.
- Stop if the puller slips, bends, or excessive force is required; never use impact methods unless the tool is designed for them.

Storage

- Clean and lightly protect metal surfaces against corrosion if recommended by the manufacturer.
- Retract the screw, arrange the arms safely, and store in a dry tool case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Locking Pliers

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Locking wrenches; Vise Grip

What it is used for

Locking pliers grip and hold objects securely using an adjustable locking mechanism. Choose the correct jaw size and pressure for the workpiece to avoid damage.

How to use

1. Inspect the jaws, adjustment screw, release lever, and pivot for damage.
2. Adjust the jaw opening to fit the workpiece.
3. Position the jaws fully around the object and keep fingers clear of the closing area.
4. Squeeze the handles until the pliers lock securely without crushing the workpiece.
5. Perform the task while checking that the object remains stable.
6. Operate the release lever carefully, remove the pliers, and clean them after use.

⚠ Safety

- Keep fingers away from the jaws and locking mechanism.
- Do not use locking pliers as a substitute for a proper wrench when torque accuracy is required.
- Avoid excessive clamping force that may damage the workpiece.
- Do not use pliers with cracked jaws or a faulty release mechanism.

Care & storage

- Wipe the pliers clean, lightly lubricate the pivot when needed, and store them dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nut Driver

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Nut Driver; Nut driver set

What it is used for

To tighten or loosen nuts using a nut driver of the correct size. To apply controlled torque while reducing the risk of damaging the fastener or surrounding parts.

Main parts

- Handle or driver body
- Socket or nut-driving tip
- Shaft
- Interchangeable bits, if supplied
- Storage case

Before you start

1. Select a nut driver that matches the nut size and type.
2. Inspect the handle, shaft, and tip for cracks, wear, or looseness.
3. Confirm that the fastener and surrounding components are secure.
4. Use an insulated nut driver only where its rating and work procedure permit.

How to use

1. Fit the correct bit or socket securely if the driver is interchangeable.
2. Seat the driver fully over the nut.
3. Keep the shaft aligned with the fastener axis.
4. Turn the handle steadily to loosen or tighten the nut.
5. Use a specified torque tool if a precise torque value is required.
6. Remove the driver after the nut is secure and inspect the fastener.

⚠ Safety

- Keep fingers clear of pinch points.
- Do not use a damaged or incorrectly sized tip.
- Do not use the nut driver as a pry bar or hammer.
- For electrical work, isolate and verify the circuit is de-energized unless an approved procedure explicitly states otherwise.

Care & storage

- Wipe the driver clean after use.

- Inspect bits and tips for wear regularly.
- Replace damaged bits and handles.
- Store the set in its case to keep sizes organized.

Common mistakes to avoid

- Using the wrong size and rounding the nut.
- Holding the driver at an angle.
- Overtightening without a specified torque requirement.
- Using the tool on an energized electrical circuit without an approved safe-work procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pliers

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 58 labs

Covers these lab items: Combination Plier; Combination pliers; Combination pliers (1 set); Grip-on- pliers; Insulated monkey plier; Insulated outer inner plier; Insulated Plier; Linesman Pliers (Combination Pliers); Linesman Pliers ISI Mark,RoHS; Long Nose Pliers; Long Nose Pliers ISI Mark,RoHS; Monkey plier; Nose pliers;

What it is used for

Pliers grip, bend, twist, or cut suitable materials depending on their design. Select the correct type and size for the task and inspect the tool before use.

How to use

1. Inspect the jaws, pivot, handles, and insulation for wear or damage.
2. Select pliers suited to the material and task, such as gripping or cutting.
3. Hold the handles firmly and position the jaws securely on the workpiece.
4. Apply controlled force without twisting the tool beyond its intended use.
5. Keep the workpiece secured and direct cutting ends away from the body.
6. Release the handles carefully and check the workpiece after adjustment.
7. Clean the pliers and return them to the designated storage location.

⚠ Safety

- Never use ordinary pliers on live electrical conductors.
- Use insulated pliers for electrical work only when correctly rated and inspected.
- Keep fingers away from cutting edges and pinch points.
- Do not use pliers as a hammer or apply excessive force.

Care & storage

- Wipe the jaws clean and lightly protect pivot points against corrosion when appropriate.
- Store pliers dry and replace tools with damaged jaws or handles.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Saw Setting Pliers

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Diwad Plier / Saw teeth setting plier

What it is used for

To adjust the set of compatible saw teeth so the saw cuts cleanly with appropriate clearance.

How to handle

- Inspect the pliers, adjustment stops, jaws, and pivot for wear, cracks, or looseness.
- Confirm that the pliers are suitable for the saw type and tooth size; consult the saw manufacturer's guidance.
- Secure the saw blade in a stable holder with the teeth accessible and wear eye protection.
- Adjust teeth gradually and consistently, following the required alternating set pattern; avoid excessive bending.
- Check the tooth alignment and test the saw cautiously on scrap material before normal use.

Storage

- Wipe the pliers clean and dry and check the adjustment mechanism after use.
- Close the jaws if appropriate and store in a dry tool cabinet away from moisture and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Screwdriver

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 41 labs

Covers these lab items: Screw driver (1 set); Screw Driver Set; Screw Driver Set / Electrician-Wiremen Set; Screw driver (Star & minus); Screw Drivers; Screw Drivers (Insulated); Screw drivers set; Screw type juice extractor; Screwdriver; Screwdriver and pliers; Screwdrivers and nut drivers

What it is used for

A screwdriver tightens or loosens screws with a matching tip and suitable handle. Select the correct type and size to prevent damage to the screw and workpiece.

How to use

1. Inspect the tip, shaft, handle, and insulation for damage.
2. Choose a screwdriver tip that matches the screw head and size.
3. Secure the workpiece and align the tip with the screw head.
4. Turn the screwdriver with controlled force while keeping the tool aligned.
5. Use an insulated screwdriver for electrical tasks only when correctly rated and inspected.
6. Stop if the tip slips or the screw head becomes damaged.
7. Clean the screwdriver and return it to the tool storage area.

⚠ Safety

- Never use an ordinary screwdriver on live electrical circuits.
- Do not use the screwdriver as a chisel, pry bar, or punch.
- Keep the free hand away from the screw path to avoid injury.
- Do not use tools with cracked handles or damaged insulation.

Care & storage

- Wipe the shaft and tip clean after use.
- Store screwdrivers dry and replace damaged or worn tips.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Socket Spanner

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 9 labs

Covers these lab items: Socket (Goti) spanner; Socket spanner (Insulates); Socket spanner set; T spanner set insulated; T spanner (tommy) set

What it is used for

Use a socket spanner to tighten or loosen compatible nuts and bolts with a secure fit around the fastener.

How to use

1. Select the correct socket size and handle for the fastener.
2. Inspect the socket, handle, and ratchet or extension for damage.
3. Fit the socket fully onto the nut or bolt head.
4. Keep the socket aligned with the fastener and turn the handle steadily.
5. Use a suitable extension or torque wrench only when required.
6. Remove the socket carefully and check the fastener is secure.
7. Clean and store the set in its case after use.

⚠ Safety

- Use the correct socket size to prevent slipping or damaging the fastener.
- Do not use cracked sockets or damaged handles.
- Do not extend the handle with an unapproved pipe.
- Use insulated tools only when rated for the electrical task; isolate power whenever possible.

Care & storage

- Wipe sockets and handles clean after use.
- Store sockets by size in a dry, organized case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spanner

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 40 labs

Covers these lab items: Alignment wrenches; D.E spanner (insulated); Double End Ring spanner; Double ended flat spanner; Double ended ring spanner; Double Ended spanner set; Double-ended flat and ring spanner; Double-ended flat spanner (1 set); Double-ended ring spanner (1 set); Nozzel/Nippl e spanner; Open and Close (Fix)

What it is used for

A spanner tightens or loosens nuts and bolts using a matching size and suitable jaw or ring profile. Choose the correct spanner to avoid slipping and damaging fasteners.

How to use

1. Inspect the spanner for cracks, worn jaws, bending, or other damage.
2. Select the correct type and size for the nut or bolt.
3. Fit the spanner fully onto the fastener before applying force.
4. Pull smoothly in a controlled direction where possible and keep stable footing.
5. Use the correct ring or open-end profile for the fastener and available access.
6. Stop if the spanner slips or the fastener begins to round off.
7. Clean the tool and return it to the designated storage location.

⚠ Safety

- Do not use a spanner with worn jaws or a cracked handle.
- Do not extend the handle with an unapproved pipe or use excessive force.
- Keep fingers clear of pinch points and avoid sudden loss of balance.
- Use insulated spanners for electrical work only when correctly rated and inspected.

Care & storage

- Wipe the spanner clean and dry after use.
- Store spanners in an organised, dry location and replace damaged tools.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Specialty Wrench

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Specialty wrenches

What it is used for

Specialty wrenches are designed for specific fasteners or restricted-access applications. Identify the fastener type and choose the correct wrench to prevent damage and ensure controlled tightening or loosening.

How to use

1. Inspect the wrench, jaws, drive end, handle, and markings for damage or wear.
2. Identify the fastener and confirm the wrench size and type are compatible.
3. Secure the workpiece and ensure the fastener is accessible and correctly seated.
4. Fit the wrench fully onto the fastener before applying force.
5. Turn steadily in the required direction while maintaining alignment.
6. Use a calibrated torque tool when a specified torque value is required.
7. Remove the wrench, inspect the fastener, and clean the tool after use.

⚠ Safety

- Do not use a cracked, bent, or badly worn wrench.
- Avoid makeshift extensions unless the tool manufacturer permits them.
- Keep hands clear of pinch points and likely slip paths.
- Use eye protection where fasteners or surrounding parts may release debris.

Care & storage

- Wipe the wrench clean, protect precision surfaces, and store it in a dry tool organizer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tool Box

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Basic Tool Box; Tool box; Tool Box (for minor repairs of pesticide application equipment)

What it is used for

Use a toolbox to carry, organize, and access suitable hand tools for routine inspection, maintenance, and minor repairs.

How to use

1. Inspect the toolbox, handle, hinges, latch, and compartments for damage.
2. Select tools appropriate for the planned repair or maintenance task.
3. Check each tool for wear, damage, or missing parts before use.
4. Arrange tools securely in their designated compartments.
5. Carry the closed toolbox by its handle and keep the load manageable.
6. Return cleaned tools to their assigned places after completing the task.
7. Close the toolbox and store it in the designated area.

⚠ Safety

- Do not carry a toolbox with a broken handle or unsecured contents.
- Use the correct tool for each task and avoid makeshift substitutions.
- Keep sharp tools covered and chemicals separate from general tools.
- Follow equipment isolation procedures before repairs, especially on electrical or pesticide equipment.

Care & storage

- Remove dirt and moisture from tools before returning them.
- Keep the toolbox dry, organized, and locked when required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tool Kit

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 20 labs

Covers these lab items: Hand tools; Hand tools kits; Machine tool kit (screw driver, screw etc.); Machine toolkit; Other tools; Tool kit; Tool kit like hand tools; Tool Kit (Spanners, Screwdrivers, Pliers); Tool kits; Tools and equipment

What it is used for

A tool kit keeps hand tools and accessories organised for maintenance, assembly, and repair tasks. Select the correct tool for each job and return equipment to the kit after use.

How to use

1. Open the kit and check the inventory for missing or damaged tools.
2. Select the correct tool size and type for the task.
3. Inspect the selected tool for cracks, wear, or loose parts.
4. Secure the workpiece and use the tool according to its intended function.
5. Apply controlled force and avoid using tools for unintended purposes.
6. Clean tools after use and check for damage before returning them.
7. Return each tool to its designated compartment and close the kit.

⚠ Safety

- Use suitable PPE for the task and material being handled.
- Do not use damaged tools or tools with loose handles.
- Keep sharp tools covered and avoid leaving tools on walkways.
- Use insulated tools for electrical tasks only when correctly rated and inspected.

Care & storage

- Keep the kit clean, dry, and organised after every use.
- Replace missing or damaged tools and store the kit in a secure location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tool Pouch

Category: **Hand Tools - Fastening & Gripping** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: A Tool Pouch

What it is used for

A tool pouch carries and organises hand tools and small accessories for convenient access during work.

How to handle

- Inspect the pouch, seams, belt loops, and fasteners for wear or damage.
- Carry only tools suitable for the pouch's capacity and secure sharp tools safely.
- Distribute weight evenly and ensure the pouch is firmly attached before moving.
- Avoid overloading the pouch or carrying tools in a way that restricts movement.
- Remove damaged tools and keep the pouch free of loose debris.

Storage

- Empty and clean the pouch periodically.
- Store it dry in a designated tool area away from heat and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Torque Screwdriver

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Torque Screwdriver

What it is used for

A torque screwdriver tightens screws to a specified torque to help prevent loose joints and over-tightening. Select the correct torque range and follow the manufacturer's instructions for adjustment and use.

How to use

1. Inspect the screwdriver, handle, bit, and adjustment mechanism for damage.
2. Select a compatible bit that fits the screw head correctly.
3. Set the required torque according to the equipment specification.
4. Secure the workpiece and align the bit with the screw axis.
5. Turn the screwdriver smoothly until the torque indicator or release mechanism signals the set value.
6. Stop applying force when the specified torque is reached.
7. Reset and store the tool according to the manufacturer's instructions.

⚠ Safety

- Use the correct torque range and never exceed the tool's rating.
- Keep the bit fully seated to prevent slipping and screw damage.
- Do not use a damaged tool or improvised extensions.
- Wear eye protection when required by the task.

Care & storage

- Keep the tool clean, store it securely, and arrange calibration at the recommended interval.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Torque Wrench

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Torque wrench

What it is used for

A torque wrench tightens bolts and nuts to a specified torque value. Use the correct wrench range and setting to prevent under-tightening or over-tightening.

How to use

1. Inspect the wrench, socket, and adjustment mechanism for damage.
2. Confirm the required torque value from the equipment specification.
3. Select a suitable socket and set the torque value according to the manufacturer's instructions.
4. Fit the socket fully onto the fastener and keep it aligned.
5. Apply steady force at the handle until the wrench signals the set torque.
6. Reset and store the wrench according to the manufacturer's instructions.

⚠ Safety

- Do not use the wrench beyond its rated torque range.
- Avoid extension pipes or extra leverage unless specifically approved.
- Use the correct socket and keep hands clear of pinch points.
- Do not use a torque wrench as a breaker bar.

Care & storage

- Store at the recommended setting and arrange periodic calibration as specified by the manufacturer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tweezers

Category: **Hand Tools - Fastening & Gripping** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Bent neck, metallic Tweezer; Sharp pointed tweezers; Tweezer; Tweezers

What it is used for

Use tweezers to grip, hold, or position small objects that are difficult to handle safely with fingers.

How to use

1. Select tweezers with a tip shape suitable for the object and task.
2. Inspect the tips and arms for damage, misalignment, or contamination.
3. Clean the tweezers before use, especially for laboratory or healthcare tasks.
4. Grip the object gently without applying excessive pressure.
5. Keep the tips directed away from your eyes and other people.
6. Place the object in the required position and release the tweezers carefully.
7. Clean and store the tweezers after use.

⚠ Safety

- Keep sharp tips away from eyes and skin.
- Use suitable protective equipment when handling hot, contaminated, or hazardous items.
- Do not use damaged or contaminated tweezers for delicate work.
- Use separate, clean tweezers when required to prevent cross-contamination.

Care & storage

- Clean and dry the tips after use.
- Store in a protective case to prevent tip damage and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Auger

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Auger

What it is used for

Use an auger to bore holes in suitable soil or material for planting, posts, or other approved tasks.

How to use

1. Select an auger type and diameter suitable for the material and hole depth.
2. Inspect the bit, shaft, handle, and drive connection for damage.
3. Check the work area for buried cables, pipes, rocks, or other hazards.
4. Mark the hole location and secure the material or tool as appropriate.
5. Position the auger vertically and turn or operate it according to its design.
6. Remove soil or chips periodically and avoid forcing a jammed auger.
7. Withdraw the auger carefully, check the hole, and clean the tool after use.

⚠ Safety

- Wear eye protection, gloves, and sturdy footwear.
- Keep hands clear of rotating parts and stop powered augers before clearing blockages.
- Confirm underground utilities are located before drilling into the ground.
- Do not use an auger with a loose handle or damaged drive parts.

Care & storage

- Remove soil and moisture from the bit and shaft after use.
- Store the auger dry with the cutting edge protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Brick Hammer

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Brick hammer

What it is used for

To split, trim, or shape bricks and masonry using a brick hammer. To produce controlled breaks while reducing the risk of flying fragments and tool damage.

Main parts

- Striking face
- Chisel or cutting edge
- Hammer head
- Handle
- Grip

Before you start

1. Inspect the head and handle for cracks, looseness, or damage.
2. Choose a hammer suitable for the masonry task.
3. Mark the intended cut or break line.
4. Place the brick on a stable, supportive surface.
5. Wear safety glasses, gloves, and suitable protective footwear.

How to use

1. Position the brick securely on the work surface.
2. Hold the hammer firmly at the end of its handle for control.
3. Use the chisel edge to score the intended line if appropriate.
4. Strike with controlled blows along the marked line.
5. Increase force gradually rather than using one excessive blow.
6. Check the break and remove loose fragments safely.
7. Clean the hammer and work area after use.

⚠ Safety

- Wear eye protection against flying chips.
- Keep hands and feet clear of the striking area.
- Do not use a hammer with a loose head or split handle.
- Keep bystanders outside the work zone.

Care & storage

- Remove dust and masonry fragments after use.
- Inspect the head and handle regularly.
- Keep the tool dry and store it securely.
- Replace damaged handles or tools before further use.

Common mistakes to avoid

- Striking an unsupported brick.
- Using excessive force and causing uncontrolled breakage.
- Holding the brick too close to the impact point.
- Ignoring cracks or looseness in the hammer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Crowbar

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Crow-Bar

What it is used for

Use a crowbar to pry apart, lift, or loosen suitable materials during controlled dismantling or maintenance work.

How to use

1. Inspect the bar for cracks, bends, sharp damage, or heavy corrosion.
2. Choose a bar length and tip shape suitable for the task.
3. Secure the workpiece and identify a stable fulcrum point.
4. Position the tip carefully in the gap or under the material.
5. Apply gradual leverage while keeping your body out of the likely slip path.
6. Reposition the bar as needed instead of forcing it excessively.
7. Clean the bar and store it safely after use.

⚠ Safety

- Wear safety glasses, gloves, and sturdy footwear.
- Keep hands and feet away from pinch points and the moving load.
- Do not use a damaged bar or an unstable fulcrum.
- Ensure no one is in the path of material that may shift or fall.

Care & storage

- Remove dirt and moisture after use to reduce corrosion.
- Store the crowbar securely with its tip away from walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hammer

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 27 labs

Covers these lab items: Ball pin; Claw Hammer; Hammer; Hammer & Pliers; Hammer (1 set); Hammer C.P (hammer cross pain) with handle forged steel; Hammer (set of 4); Hammers; Hammers B.P (Hammers ball pain)

What it is used for

A hammer delivers controlled impact for tasks such as driving nails, fitting parts, or shaping materials. Select the correct hammer type and use it only for its intended purpose.

How to use

1. Inspect the hammer head, handle, and joint for cracks, looseness, or damage.
2. Select a hammer suitable for the material and task.
3. Secure the workpiece on a stable surface before striking.
4. Hold the handle firmly and position the striking face accurately.
5. Use controlled strokes and keep the work area clear of other people.
6. Stop if the head loosens, the handle cracks, or the tool slips.
7. Clean the hammer and store it safely after use.

⚠ Safety

- Wear safety goggles when striking materials that may produce flying fragments.
- Never use a hammer with a loose head or damaged handle.
- Keep fingers away from the striking area and support points.
- Do not use a hammer as a pry bar unless designed for that purpose.

Care & storage

- Wipe the head and handle clean and dry after use.
- Store the hammer in a dry location and replace damaged handles or heads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mallet

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 10 labs

Covers these lab items: Mallet Plastic; Mallet Wooden; PVC mallet; Wooden hammer; Wooden Mallet

What it is used for

Use a mallet to apply controlled impact when fitting, shaping, or adjusting parts that may be damaged by a metal hammer.

How to use

1. Choose a wooden, plastic, or PVC mallet suitable for the task and material.
2. Inspect the head and handle for cracks, looseness, or heavy wear.
3. Secure the workpiece on a stable surface before striking.
4. Wear eye protection and hold the mallet firmly by the handle end.
5. Strike the target with controlled force and keep the mallet face aligned.
6. Check the workpiece and mallet between repeated strikes.
7. Clean the mallet and store it safely after use.

⚠ Safety

- Keep fingers clear of the striking area and pinch points.
- Do not use a cracked head, loose handle, or unsuitable mallet.
- Avoid striking hardened metal or other surfaces unless the mallet is designed for it.
- Keep bystanders clear of the work area.

Care & storage

- Remove dirt and residue from the head and handle.
- Store in a dry place away from heat and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mattock

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Kodali; Kudali

What it is used for

To break hard soil, loosen compacted ground, and dig or shape soil during agricultural and construction tasks.

How to handle

- Inspect the head, cutting edges, handle, and head-to-handle joint for cracks or looseness.
- Wear safety footwear, gloves, and eye protection, and ensure the work area is clear of other people.
- Grip the handle firmly and use controlled swings, keeping feet and hands clear of the striking path.
- Do not use a mattock with a loose head, cracked handle, or damaged cutting edge.
- Remove soil from the tool and check it for damage after use.

Storage

- Store clean and dry in a designated tool rack with the head secured.
- Keep cutting edges protected and store the tool away from walkways and children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nail Punch

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Nail Punch

What it is used for

A nail punch drives a nail head slightly below a surface or helps position small nails during woodworking. Select the correct punch size and strike it with a suitable hammer under control.

How to use

1. Inspect the punch tip and body for mushrooming, cracks, or deformation.
2. Select a punch that fits the nail head appropriately.
3. Secure the workpiece on a stable work surface.
4. Position the punch tip squarely on the nail head.
5. Hold the punch securely and strike it carefully with a suitable hammer.
6. Check the nail depth and repeat gently if required.
7. Store the punch safely after use.

⚠ Safety

- Wear safety glasses to protect against flying metal fragments.
- Keep fingers clear of the hammer striking area.
- Do not use a cracked or mushroomed punch.
- Strike with controlled force and keep the workpiece stable.

Care & storage

- Wipe the punch clean and store it dry to prevent corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pickaxe

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Pick Axe; Pickaxe and spade; Pickaxe (Tikav) with wooden handle

What it is used for

Use a pickaxe to break compacted soil, loosen hard ground, or remove small rocks during approved digging tasks.

How to use

1. Inspect the head, eye, and handle for cracks, looseness, or damage.
2. Choose the pickaxe type suitable for the ground and task.
3. Check the work area for buried cables, pipes, and other hazards.
4. Stand with stable footing and keep people well clear of the swing area.
5. Grip the handle firmly and swing under control toward the target area.
6. Break material gradually and stop to remove loose pieces safely.
7. Clean the tool and inspect it after use.

⚠ Safety

- Wear safety glasses, gloves, and sturdy protective footwear.
- Never use a pickaxe with a loose head or damaged handle.
- Keep others outside the full swing and rebound zone.
- Do not strike near buried utilities until their location is confirmed.

Care & storage

- Remove soil and moisture from the head after use.
- Store the pickaxe securely with the head protected and handle supported.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Punch

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Punches; Straight or ejecto punch

What it is used for

To mark, align, drive, or eject pins and small components using a punch suitable for the task.

How to handle

- Inspect the punch tip and striking head for mushrooming, cracks, bending, or other damage.
- Select the correct punch type and size for the material and task; secure the workpiece on a stable support.
- Wear eye protection and position the punch securely on the intended mark or pin.
- Strike the punch with a suitable hammer using controlled blows, keeping hands clear of the impact area.
- Stop if the punch slips or deforms; do not use a damaged punch or strike hardened punches with an unsuitable hammer.

Storage

- Wipe the punch clean and inspect the tip and striking head after use.
- Store by size in a dry tool rack or case with tips protected from damage and accidental contact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shovel

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 9 labs

Covers these lab items: Shovel; Shovel (Small); Shovels and Specialty Spades; Shovel/Trowel/Bucket

What it is used for

Use a shovel to dig, lift, move, or spread soil, sand, compost, or other suitable loose materials.

How to use

1. Choose a shovel shape and size suitable for the material and task.
2. Inspect the blade, handle, and joint for cracks or looseness.
3. Clear the work area and ensure stable footing.
4. Place the blade into the material and grip the handle firmly with both hands.
5. Use your legs and keep your back as neutral as possible when lifting.
6. Move the material to the designated area without twisting suddenly.
7. Remove dirt from the shovel and store it safely after use.

⚠ Safety

- Wear sturdy footwear and gloves appropriate for the task.
- Do not overload the shovel or lift more than you can manage safely.
- Keep other people clear of the digging and swinging area.
- Check for buried cables, pipes, or other hazards before digging.

Care & storage

- Clean and dry the blade after use to reduce rust.
- Store the shovel securely with the blade protected and the handle supported.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sledge Hammer

Category: **Hand Tools - Striking & Digging** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Mash Hammer; Sledge Hammer

What it is used for

To strike, break, or drive suitable materials using a sledge hammer. To apply controlled impact while maintaining a secure grip and stable working position.

Main parts

- Hammer head
- Handle
- Head-to-handle joint
- Grip surface

Before you start

1. Select a hammer with suitable head weight and handle length.
2. Inspect the head, handle, and joint for cracks or looseness.
3. Check that the striking surface and workpiece are suitable.
4. Clear the swing area and ensure bystanders are at a safe distance.
5. Wear safety glasses, safety footwear, and other required PPE.

How to use

1. Stand with stable footing and maintain clear space around the body.
2. Hold the handle firmly with both hands where appropriate.
3. Position the hammer head over the intended striking point.
4. Swing in a controlled motion and strike squarely.
5. Allow the head to contact the workpiece without glancing blows where possible.
6. Reposition the workpiece or tool only after the hammer is lowered safely.
7. Inspect the workpiece and hammer after the task.

⚠ Safety

- Never use a hammer with a loose head or cracked handle.
- Keep hands and feet away from the impact zone.
- Do not strike hardened steel surfaces unless the tool is designed for that use.
- Wear eye protection against flying fragments.
- Do not swing when anyone is within the striking area.

Care & storage

- Clean the head and handle after use.
- Inspect the joint and handle regularly.
- Replace damaged handles or remove unsafe tools from service.
- Store securely in a dry place.

Common mistakes to avoid

- Using an unsuitable hammer weight.
- Striking at an angle and causing the head to glance off.
- Working with a loose head or damaged handle.
- Failing to clear bystanders from the swing area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Conditioner

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Air Conditioner; Air Conditioner 1.5 tonne

What it is used for

An air conditioner removes heat from an indoor space and transfers it outdoors to maintain comfortable conditions. Correct settings, unobstructed airflow, and routine cleaning help improve performance and reduce energy use.

Main parts

- Indoor unit distributes conditioned air into the room.
- Outdoor unit releases heat to the outside air.
- Compressor circulates refrigerant through the cooling system.
- Evaporator coil absorbs heat from indoor air.
- Condenser coil releases heat outdoors.
- Fan and blower move air through the indoor and outdoor units.
- Air filter captures dust and airborne particles.
- Thermostat or remote controller sets the temperature and operating mode.

Before you start

1. Read the user manual and identify the available operating modes and controls.
2. Inspect the visible units, power cord or isolator, and remote control for damage.
3. Ensure indoor and outdoor air inlets and outlets are not blocked.
4. Check that the air filter is correctly installed and reasonably clean.
5. Close doors and windows where practical to reduce unwanted heat entry.
6. Confirm there is no visible water leakage, damaged wiring, or unusual smell.

How to use

1. Switch on the air conditioner using the approved control or remote.
2. Select the required cooling or other supported operating mode.
3. Set a suitable target temperature and fan speed for the room conditions.
4. Allow the unit to start and check that air flows from the indoor outlet.
5. Adjust the air direction to distribute airflow without blowing directly on people unnecessarily.
6. Keep doors and windows closed where appropriate and avoid blocking the air filter or vents.
7. Monitor cooling performance, unusual noise, water leakage, and any fault indicators.
8. Switch the unit off using the control before isolating power for cleaning or maintenance.
9. Clean the filter according to the manual and allow it to dry before reinstalling.
10. Record or report persistent cooling problems, unusual sounds, or recurring error codes.

⚠ Safety

- Do not touch electrical parts or switches with wet hands.
- Never operate the unit if wiring, plugs, or electrical covers are damaged.
- Do not insert fingers or tools into fans or air outlets.
- Do not open refrigerant lines or attempt refrigerant repairs without qualified authorisation.
- Keep outdoor units clear of obstructions and do not stand on unstable supports to reach them.
- Isolate power according to the manual before cleaning accessible components or arranging maintenance.

Care & storage

- Clean or replace air filters at the recommended interval.
- Keep indoor and outdoor air inlets and outlets clear of dust and obstructions.
- Wipe accessible external surfaces with suitable materials and avoid spraying water into electrical areas.
- Arrange qualified servicing for refrigerant issues, persistent leakage, reduced cooling, or electrical faults.

Common mistakes to avoid

- Setting the temperature unnecessarily low and expecting faster cooling.
- Operating with a clogged filter or blocked airflow.
- Ignoring water leakage, unusual noises, or fault indicators.
- Cleaning internal electrical parts without isolating power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Freshener

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Air Freshener; Air fresheners and basic cleaning supplies for the front desk area; Room Freshener

What it is used for

Air fresheners reduce or mask unwanted odours in designated indoor areas.

How to handle

- Read the product label and use the air freshener only as directed.
- Provide suitable ventilation and avoid spraying toward people, eyes, food, or electrical equipment.
- Keep aerosols and liquids away from flames, heat, and incompatible chemicals.
- Check for sensitivity or irritation and discontinue use if adverse effects occur.
- Do not use fragrance products to mask hazardous chemical, gas, or smoke odours; investigate the source.

Storage

- Store in the original labelled container in a cool, ventilated area.
- Keep away from ignition sources and out of reach of children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Purifier

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Medium** 22 labs

Covers these lab items: Air Purifier; Air purifiers

What it is used for

An air purifier filters airborne particles and, depending on its design, certain odours or gases from indoor air. Operate it in a suitable location and maintain its filters according to the manufacturer's instructions.

How to use

1. Inspect the purifier, power cord, air inlet, outlet, and housing for damage.
2. Place the unit on a stable surface with clear space around the air inlet and outlet.
3. Check that the filter is correctly installed and protective packaging has been removed.
4. Connect the unit to an approved power supply and switch it on.
5. Select the required fan speed or operating mode for the room.
6. Keep doors, windows, and nearby obstructions in line with the intended use instructions.
7. Check indicator lights and replace filters when the unit signals or the schedule requires.
8. Switch off and unplug the purifier before cleaning or filter replacement.

⚠ Safety

- Do not operate the purifier with damaged cables or a wet housing.
- Never block air inlets or outlets.
- Keep the unit away from water and excessive heat.
- Use only filters approved for the model and follow replacement instructions.

Care & storage

- Clean the exterior and reusable pre-filter according to the manual.
- Replace filters as recommended and keep the unit dry and dust-free.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ash Tray

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Ash Tray

What it is used for

An ash tray collects ash and cigarette waste where smoking is permitted and designated facilities are provided.

How to handle

- Use ash trays only in designated areas where smoking is permitted.
- Ensure cigarette ends are fully extinguished before disposal.
- Do not place hot ash or burning materials into general waste bins.
- Empty only when contents are completely cool and use appropriate PPE if required.
- Keep ash trays away from flammable materials and maintain a fire-safe area.

Storage

- Keep ash trays clean and stable in designated locations.
- Store empty trays away from combustible materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Baggage Cart

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Guest service desks, baggage carts, and luggage storage systems

What it is used for

A baggage cart transports luggage safely between reception areas, rooms, and storage locations. Load it evenly and follow safe handling practices to protect guests and staff.

How to use

1. Inspect the cart, wheels, handles, brakes, and platform for damage.
2. Confirm the route is clear and the cart can pass through doorways.
3. Place heavier luggage at the bottom and distribute the load evenly.
4. Secure bags to prevent slipping and keep the load within capacity.
5. Push the cart slowly while watching corners, doorways, and nearby people.
6. Apply brakes before loading or unloading if the cart has brakes.
7. Return the cart to its designated storage area after use.

⚠ Safety

- Do not overload the cart or block the operator's view.
- Keep hands and feet away from wheels and moving parts.
- Use team assistance for heavy or awkward luggage.
- Never leave a loaded cart unattended on a slope.

Care & storage

- Clean the platform and handles regularly and report damaged wheels or brakes.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bath Tub

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Bath Tub

What it is used for

A bath tub holds water for bathing and must be kept clean and safe for users.

How to handle

- Inspect the tub surface, drain, taps, and surrounding floor for damage or leaks.
- Clean and disinfect the tub between users according to the facility procedure.
- Check water temperature before use to reduce scalding risk.
- Use non-slip precautions and assist users when required by the care plan.
- Report loose fittings, drainage issues, or slippery surfaces promptly.

Storage

- Keep the tub empty and clean when not in use.
- Store cleaning products separately and maintain the surrounding area dry and ventilated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bed Linen

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Bed Sheet; Bed Sheet, Blanket, Pillow with Pillow Cover; Bed Sheets; Draw Sheet; Pillow Covers

What it is used for

Bed linen provides clean bedding for comfort, hygiene, and proper room or clinical care.

How to handle

- Inspect sheets, blankets, and pillow covers for stains, tears, and contamination before use.
- Use clean linen for each new user and follow infection-control procedures where applicable.
- Handle used or contaminated linen carefully and avoid shaking it.
- Replace wet or visibly soiled linen promptly and wear PPE when required.

Storage

- Store clean, fully dry linen in a clean, covered cupboard.
- Keep used linen in designated laundry bags separate from fresh bedding.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bidet

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Bidet

What it is used for

A bidet provides a water-based personal hygiene facility and should be maintained clean and functional.

How to handle

- Inspect the bidet, taps, spray fittings, drain, and water connections for leaks or damage.
- Clean and disinfect contact surfaces according to the facility hygiene procedure.
- Operate controls gently and check water temperature before use where applicable.
- Avoid splashing contaminated water onto nearby clean surfaces.
- Report leaks, blockages, loose fittings, or damaged controls to maintenance staff.

Storage

- Keep the fixture clean and the surrounding area dry when not in use.
- Store cleaning products separately and follow their label instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Blanket

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Blanket; Blanket / Quilt; Blankets

What it is used for

Blankets and quilts provide warmth and comfort in accommodation, care, and hospitality settings.

How to handle

- Inspect blankets for stains, tears, odours, and signs of pests before use.
- Use clean bedding for each person and follow applicable laundry and infection-control procedures.
- Keep blankets dry and avoid placing them on contaminated floors or surfaces.
- Follow the care label when washing or drying and report damaged items.
- For care settings, ensure blankets do not obstruct breathing, mobility, or necessary equipment.

Storage

- Store clean blankets fully dry in a clean, covered linen area.
- Keep used or soiled bedding separate from fresh linen.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Broom

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Broom; Soft Broom (Indian)

What it is used for

A broom is used to sweep dust, dirt, and loose debris from floors and other suitable surfaces. Choose a broom suited to the surface and type of debris.

How to use

1. Inspect the broom handle and bristles for looseness or damage.
2. Remove large obstacles from the area before sweeping.
3. Hold the handle comfortably and sweep debris in controlled strokes toward one collection point.
4. Use short strokes in corners and along edges.
5. Collect the debris with a dustpan and dispose of it in the correct bin.
6. Shake or clean the bristles as appropriate and return the broom to storage.

⚠ Safety

- Use suitable footwear and avoid creating excessive dust in poorly ventilated areas.
- Keep the handle and broom head away from walkways when not in use.
- Do not sweep hazardous materials without the required protective equipment and procedure.

Care & storage

- Remove trapped debris and store the broom with bristles protected from bending.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bucket

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 11 labs

Covers these lab items: Bucket; Bucket (MS steel bucket); Bucket (20 Ltr.); Plastic Bucket; Plastic Buckets , 10 Ltr; Plastic buckets and containers; Small Bucket (For mixing)

What it is used for

Buckets hold, carry, or mix suitable liquids and materials for cleaning, hospitality, and workshop tasks.

How to handle

- Select a bucket made from material suitable for the contents and intended use.
- Inspect the body, base, handle, and rim for cracks, sharp edges, or leaks.
- Do not overfill; carry heavy loads carefully and use assistance when needed.
- Keep food-use buckets separate from cleaning chemicals and waste-handling equipment.

Storage

- Empty, clean, and dry reusable buckets after use.
- Store upright in a designated area away from heat, contamination, and walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Call Bell

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Call bell

What it is used for

To alert staff or request assistance using a manual or electronic call bell.

How to handle

- Inspect the bell, button, cable, or battery compartment for damage and ensure it is accessible to the intended user.
- For a manual bell, press or ring the mechanism as designed; for an electronic bell, press the call button and confirm the alert is received.
- Place the bell within easy reach and explain its operation to the user.
- Do not use the bell as a toy or repeatedly trigger alerts unnecessarily.
- Clean contact surfaces according to the facility's hygiene procedure and test functionality periodically.

Storage

- Wipe and dry the bell before storing; remove batteries for long-term storage if recommended.
- Keep in the designated clean area and protect electronic models from moisture and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cleaning Brush

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Cleaning Brush; Cleaning Brushes; Dusting Brush; Funner (cleaning brush)

What it is used for

To remove dust, dirt, and loose debris from surfaces using a suitable cleaning brush. To clean effectively without scratching surfaces or spreading contamination.

Main parts

- Brush handle
- Bristles
- Brush head
- Hanging hole or grip, if fitted

Before you start

1. Choose bristle type suitable for the surface and dirt.
2. Check that the bristles and handle are clean and intact.
3. Wear gloves or a mask if the cleaning task requires them.
4. Separate brushes used for different areas or hygiene zones.

How to use

1. Remove loose objects from the surface where practical.
2. Hold the brush securely by its handle.
3. Brush from the cleaner area toward the dirtier area when appropriate.
4. Use controlled strokes suited to the surface.
5. Collect loosened dust and debris for proper disposal.
6. Wash or disinfect the brush when required by the cleaning procedure.
7. Allow the brush to dry before storage.

⚠ Safety

- Avoid brushing near eyes or exposed skin with stiff bristles.
- Use appropriate protection for dusty or contaminated areas.
- Do not use damaged brushes that may shed bristles.
- Follow hygiene rules for food, healthcare, and sanitation areas.

Care & storage

- Remove trapped debris after use.
- Wash and disinfect as appropriate for the task.

- Dry thoroughly and store away from contamination.
- Replace worn or shedding brushes.

Common mistakes to avoid

- Using the same brush across incompatible hygiene zones.
- Using stiff bristles on delicate surfaces.
- Spreading dust instead of collecting it.
- Storing a wet or contaminated brush.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cleaning Equipment

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Medium** 17 labs

Covers these lab items: Cleaning & Maintenance Tools; Cleaning equipment; Cleaning Equipment (Mop, Broom, Bucket); Cleaning tools; Housekeeping equipments; Tools for other tasks such as cleaning of vehicles, tools, equipment and workshop

What it is used for

Cleaning equipment such as brooms, mops, buckets, and brushes removes dirt and helps maintain hygienic work areas. Select suitable tools and cleaning products for the surface and type of contamination.

How to use

1. Inspect cleaning tools for damage, cleanliness, and secure handles.
2. Select the appropriate tool and approved cleaning solution for the task.
3. Place warning signs where wet floors or cleaning work may create hazards.
4. Remove loose debris before wet cleaning when appropriate.
5. Clean the area using the correct method and avoid spreading contamination.
6. Rinse or replace cleaning materials as needed and collect waste safely.
7. Wash, disinfect, and dry reusable equipment according to site procedures.
8. Store tools and chemicals in their designated locations after use.

⚠ Safety

- Wear suitable gloves, footwear, and other PPE for the cleaning chemicals used.
- Never mix cleaning chemicals unless the product instructions explicitly permit it.
- Keep wet-floor areas marked and restrict access until safe.
- Store chemicals in labelled containers away from food and unauthorised users.

Care & storage

- Clean and dry tools after each use to prevent odour and microbial growth.
- Replace worn mop heads, damaged handles, and ineffective cleaning materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cleaning Robot

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 20 labs

Covers these lab items: Cleaning Robots

What it is used for

A cleaning robot moves across suitable floors to collect dust or perform programmed cleaning tasks. Correct setup, obstacle removal, and supervision help prevent collisions, falls, and equipment damage.

Main parts

- Main body contains the drive system and control electronics.
- Cleaning brushes sweep dirt toward the suction inlet or collection area.
- Wheels and sensors help the robot move and detect obstacles.
- Dust bin or collection container stores collected debris.
- Battery supplies power to the robot.
- Charging contacts or dock recharge the battery on supported models.
- Control buttons or an app start cleaning and set operating modes.

Before you start

1. Read the user instructions and confirm the robot suits the floor and cleaning area.
2. Inspect the body, wheels, brushes, sensors, dust bin, battery, and charging dock.
3. Remove loose cables, small objects, liquids, and other hazards from the floor.
4. Keep stairs, raised edges, and drop-offs protected according to the manufacturer's instructions.
5. Empty the dust bin and check that brushes and filters are fitted correctly.
6. Ensure the charging dock is stable, dry, and connected to a suitable power outlet.

How to use

1. Place the robot on the floor or charging dock as instructed by the manufacturer.
2. Charge the battery sufficiently before starting the cleaning cycle.
3. Select the cleaning mode using the control buttons or supported app.
4. Move people, pets, loose cables, and small objects away from the cleaning path.
5. Start the robot and observe its initial movement to confirm normal operation.
6. Allow the robot to clean within the prepared area and monitor it where necessary.
7. Pause or stop the robot if it becomes trapped, collides repeatedly, or behaves unusually.
8. Switch off the robot before removing tangled hair, threads, or other obstructions.
9. Empty the dust bin and clean the brushes and sensors according to instructions.
10. Return the robot to its dock or storage location after cleaning.

⚠ Safety

- Do not operate the robot on wet floors unless the model is designed for that task.
- Keep fingers, hair, loose clothing, and hands away from moving brushes and wheels.
- Keep the robot and charging dock away from water and wet electrical connections.
- Protect stairs and open edges; do not rely only on sensors to prevent falls.
- Do not use a damaged battery, charging cable, dock, or robot housing.
- Supervise operation around young children, pets, and crowded areas.

Care & storage

- Empty the dust bin and clean brushes and filters at the recommended intervals.
- Wipe sensors and charging contacts gently with a dry, suitable cloth.
- Check wheels, brushes, battery, and dock for wear or damage regularly.
- Store the robot in a dry place and follow the manufacturer's battery care instructions.

Common mistakes to avoid

- Starting cleaning before removing loose cables and small objects from the floor.
- Allowing the robot to approach unprotected stairs or open edges.
- Cleaning moving brushes while the robot is powered on.
- Using water or cleaning chemicals on sensors or electrical contacts without approval.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Climate Control System

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Temperature & humidity control system; Temperature control, fans, portable heaters if needed

What it is used for

A climate control system maintains suitable indoor temperature and humidity using controls, fans, and heating or cooling equipment. Correct settings, airflow management, and routine monitoring help support comfort, indoor air quality, and energy efficiency.

Main parts

- Temperature sensor measures indoor air temperature.
- Humidity sensor measures moisture level where fitted.
- Controller or thermostat sets target conditions and operating modes.
- Fan or blower circulates indoor air.
- Heating unit provides heat where installed.
- Cooling unit removes heat where installed.
- Air filter captures dust and particles where fitted.
- Display, switches, and control panel show settings and system status.

Before you start

1. Read the operating instructions and identify the system's temperature and humidity controls.
2. Inspect the controller, visible wiring, plugs, fans, filters, and heating or cooling units for damage.
3. Check that air inlets and outlets are clear and the room is not obstructed.
4. Confirm portable heaters, if used, are approved for the location and placed on a stable surface.
5. Check for unusual smells, visible leakage, damaged cables, or fault indicators.
6. Identify the recommended temperature and humidity range for the room's intended use.

How to use

1. Switch on the system using the approved controller or switch.
2. Select the required heating, cooling, ventilation, or automatic mode as available.
3. Set the target temperature and humidity within the recommended range for the space.
4. Adjust fan speed and airflow direction to distribute air evenly.
5. If a portable heater is needed, switch it on according to its instructions and monitor the room temperature.
6. Observe the displayed temperature, humidity, airflow, and system status during operation.
7. Adjust settings gradually if conditions move outside the required range.
8. Report unusual noise, repeated cycling, condensation, excessive dryness, or persistent humidity problems.
9. Switch off equipment that is not needed and follow the approved shutdown procedure.

⚠ Safety

- Never cover fans, air outlets, heaters, or ventilation grilles.
- Keep portable heaters away from curtains, bedding, paper, water, and other combustible materials.
- Do not use portable heaters with damaged cables, loose plugs, or signs of overheating.
- Avoid extension leads unless explicitly permitted by the equipment instructions and facility rules.
- Keep hands dry when operating electrical controls and do not open electrical panels without authorisation.
- Do not set extreme temperatures or humidity levels that could endanger occupants or damage stored materials.

Care & storage

- Clean or replace filters at the recommended interval.
- Keep sensors unobstructed and avoid placing them next to direct sunlight, heaters, or cooling outlets.
- Wipe accessible external surfaces with a suitable dry or lightly damp cloth as permitted by the manual.
- Inspect fans, vents, and portable heater cables regularly and arrange qualified service for faults.
- Record temperature and humidity readings when required by the facility procedure.

Common mistakes to avoid

- Setting unsuitable temperature or humidity targets for the room's use.
- Blocking air inlets or outlets with furniture or stored items.
- Leaving a portable heater close to combustible materials or unattended when prohibited.
- Ignoring persistent condensation, unusual smells, or system fault indicators.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cloth Drying Stand

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Cloth Drying Stand

What it is used for

Use cloth drying stands safely and allow laundry to dry hygienically.

How to handle

- Check the frame, joints, and feet for stability before loading clothes.
- Distribute wet garments evenly and observe the stand's weight limit.
- Keep the stand away from electrical outlets and clear of walkways.
- Fold or move the stand carefully to avoid trapped fingers.

Storage

- Allow the stand to dry before folding and storing.
- Store folded in a dry area away from corrosive chemicals and excessive weight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Crockery

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 80 labs

Covers these lab items: B&B Plate; Bowles; Bowls; Bowls and Containers; Bowls (Soup, Dessert); Cereal Bowl; Chutney Bowl Small; Cups (Tea/Coffee); Dessert Plate 9"; Dinner Plate 11"; Dinner Set; Entree Dish Round With Lid (1 Portion); Entree Dish Round With Lid (2 Portion); Mug; Oval Platter; Pasta Plate 11"; Plastic Jar -

What it is used for

Crockery such as plates, bowls, and cups is used to serve food and beverages hygienically.

How to handle

- Inspect each item for chips, cracks, sharp edges, and cleanliness before use.
- Handle stacked crockery carefully and avoid overloading trays or shelves.
- Use food-safe washing and sanitizing procedures suitable for the material.
- Remove broken or chipped items from service immediately.

Storage

- Store clean, dry crockery on stable shelves in a protected area.
- Separate fragile items and keep storage surfaces free from dust and pests.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cruet Set

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Salt And Pepper Set

What it is used for

Cruet sets hold seasonings such as salt and pepper for controlled serving at dining tables.

How to handle

- Check containers, lids, and dispensers for cracks, cleanliness, and correct operation.
- Fill only with the intended seasoning and label contents where identification is unclear.
- Keep contents dry and avoid using wet or contaminated utensils to refill them.
- Clean and dry the set according to food-service hygiene procedures.
- Remove chipped, cracked, or damaged containers from service.

Storage

- Store clean, dry cruet sets in a designated food-service area.
- Keep refills sealed and separate from cleaning chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cutlery

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 34 labs

Covers these lab items: Cutlery Set; Cutlery (Spoons, Forks, Knives); Dessert (A.P) Fork; Dessert (A.P) Spoon; Dessert Knife; Soup Spoon; Soup Spoon Chinese; Spoon; Spoon (big and small); Table Service Fork; Table Service Spoon; Table Spoons; Tea Spoon

What it is used for

Cutlery such as spoons, forks, and knives is used for food preparation or eating and requires hygienic handling.

How to handle

- Inspect cutlery for cleanliness, rust, cracks, loose handles, and sharp-edge damage.
- Wash and sanitize according to the food-service procedure before use.
- Handle knives by their handles, carry them safely, and keep sharp edges away from hands.
- Remove damaged items from service and separate clean cutlery from used items.

Storage

- Store clean, dry cutlery in designated covered drawers or containers.
- Keep knives secured separately with blades protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Decoration Item

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Office Decorations

What it is used for

Handle office decoration items safely while maintaining a tidy workspace.

How to handle

- Inspect decorations for sharp edges, loose parts, or breakage before use.
- Use suitable supports and approved methods when mounting items.
- Keep decorations away from heat sources, electrical panels, and emergency exits.
- Use safe access equipment rather than unstable furniture when placing decorations at height.

Storage

- Pack reusable decorations carefully in labelled containers.
- Store fragile items in protective packaging in a dry area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drinking Glass

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 16 labs

Covers these lab items: Glassware (Water, Juice); Hi Ball; Pilsner; Steel Glass; Tom Collins; Wine Glass

What it is used for

Drinking glasses and glassware are used to serve water, juice, and other beverages hygienically.

How to handle

- Inspect each glass for chips, cracks, cleanliness, and stability before use.
- Handle glassware by the body or stem as appropriate and avoid carrying excessive quantities at once.
- Use suitable washing and sanitizing procedures and keep clean glasses separate from used ones.
- Remove broken glass from service and follow the designated glass-breakage cleanup procedure.

Storage

- Store clean, dry glassware in stable racks or cabinets with rims protected from damage.
- Keep fragile glasses separated and away from shelf edges.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dust Pan

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Dust Pan

What it is used for

A dust pan collects swept dust and debris for safe transfer to a waste container.

How to handle

- Inspect the pan for cracks, sharp edges, and a secure handle before use.
- Use with a suitable broom and collect debris without spreading dust unnecessarily.
- Wear gloves and dust protection when handling potentially hazardous or irritating debris.
- Do not pick up hot ash, broken glass, or hazardous substances without the correct procedure and equipment.
- Empty the pan into the appropriate waste container and clean it after use.

Storage

- Store the dust pan clean and dry in a designated cleaning area.
- Keep it separate from food-contact equipment and away from walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dustbin

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 18 labs

Covers these lab items: Compost bins and signage for green behavior; Dust bean (dustbin); Dustbin; Dustbin, Boxes and pouches for storing Items; Dustbins; Dustbins (Dry & Wet Waste); Garbage Bin; Plastic Dustbin; Waste Disposal Bins; Waste disposal bins, recycling stations

What it is used for

Dustbins collect and separate waste to support cleanliness, recycling, and responsible disposal.

How to handle

- Select the bin according to the waste category and local segregation instructions.
- Use designated bins for wet, dry, recyclable, compostable, and hazardous waste as applicable.
- Do not overfill bins; close lids where provided and avoid direct contact with waste.
- Empty and clean bins according to the site schedule, using appropriate PPE.

Storage

- Keep bins stable, labelled, and accessible without blocking walkways or exits.
- Store waste in designated areas protected from pests, leakage, and weather.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Feather Duster

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Feather Duster - Big & Small

What it is used for

Handle large and small feather dusters safely for routine cleaning.

How to handle

- Check that the duster is intact and does not shed excessive feathers or fibres.
- Dust gently to avoid spreading particles into the air.
- Wear suitable protection if dust may contain allergens or irritants.
- Do not use the same duster on food-contact surfaces without appropriate cleaning.

Storage

- Shake out or clean the duster in a designated area after use.
- Store dry in a ventilated holder away from food and clean linen.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Finger Bowl

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Finger Bowl Large With Under Liner

What it is used for

A finger bowl provides water for rinsing fingertips during designated dining or hospitality service.

How to handle

- Inspect the bowl and underliner for chips, cracks, and cleanliness before service.
- Use only clean, food-service-appropriate bowls and fresh water as required.
- Present and handle the bowl carefully to avoid spills or contact contamination.
- Remove used bowls promptly and clean them according to food-service hygiene procedures.
- Do not use chipped or cracked bowls for guest service.

Storage

- Store clean, dry bowls and underliners in a protected service area.
- Keep them separate from used dishes and cleaning chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Garbage Bag

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Biodegradable trash bags; Garbage Bags; Garbage Bags Black; Garbage Bin/Bags

What it is used for

Garbage bags contain waste for hygienic collection, segregation, transport, and disposal.

How to handle

- Choose a bag suitable for the waste type, weight, and bin size.
- Check the bag for tears or leaks before lining the bin.
- Separate waste according to local rules and never place incompatible hazardous waste in ordinary bags.
- Avoid compressing waste by hand and tie bags securely before removal.
- Use appropriate PPE and clean spills promptly.

Storage

- Store unused bags in a dry, clean location away from heat.
- Keep bags organised by size and designated waste category.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Guest Kit

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Branded guest support kits used by local tourism departments; Eco-friendly guest kits with reusable water bottles, biodegradable utensils, and trash bags; Guest hospitality items (e.g., welcome kits); Guest kits including lanyards, multilingual brochures, and city maps

What it is used for

Guest kits provide selected hygiene, comfort, and travel essentials for visitors or guests.

How to handle

- Check kit contents against the approved list before distribution.
- Verify seals, cleanliness, and expiry dates for toiletries or other dated products.
- Ensure reusable bottles and utensils are clean and ready for use.
- Do not distribute damaged, contaminated, or expired items.
- Provide kits according to guest needs and applicable hygiene or sustainability guidelines.

Storage

- Store kits in a clean, dry, labelled area.
- Keep toiletries sealed and separate from cleaning chemicals and waste.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Laundry Basket

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Laundry Basket

What it is used for

Handle laundry baskets hygienically when collecting and moving used linen.

How to handle

- Check the basket for cracks, sharp edges, and stability before use.
- Separate clean laundry from used or contaminated linen.
- Do not overfill the basket or carry loads that obstruct visibility.
- Clean and disinfect the basket as required by the facility's hygiene procedure.

Storage

- Store empty baskets in a clean, dry, designated area.
- Keep clean baskets separate from contaminated laundry and waste.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mop

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Low** 28 labs

Covers these lab items: Mop; Mop (Indian/Western); Mops

What it is used for

A mop cleans floors by removing dust, spills, and surface dirt using water or an approved cleaning solution. Select the correct mop type for the floor and follow the site's cleaning and hygiene procedure.

How to use

1. Inspect the mop head, handle, and wringer or bucket for damage.
2. Choose a mop suitable for the floor type and cleaning task.
3. Place warning signs around the work area and prepare the approved cleaning solution.
4. Dip the mop into the solution and wring out excess liquid as appropriate.
5. Clean the floor in controlled sections, moving dirt toward the collection area.
6. Rinse or replace the mop head when dirty and change solution when required.
7. Allow the floor to dry, clean the equipment, and store it properly.

⚠ Safety

- Use wet-floor signs to prevent slips and falls.
- Wear gloves when handling cleaning chemicals and follow label instructions.
- Never mix cleaning chemicals unless the product instructions explicitly allow it.
- Keep dirty mop water away from food preparation and clean surfaces.

Care & storage

- Wash and dry reusable mop heads thoroughly and store them in a ventilated area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Napkin Holder

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Napkin Holder

What it is used for

A napkin holder keeps napkins organised, clean, and accessible for dining or service.

How to handle

- Inspect the holder for sharp edges, cracks, and stability.
- Use clean, dry napkins of a suitable size.
- Avoid touching the napkin contact area with dirty hands.
- Clean and dry the holder regularly according to service hygiene procedures.
- Remove damaged holders that may tear napkins or cause injury.

Storage

- Store clean holders in a dry, designated service area.
- Protect unused napkins from dust and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Peg

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Pegs

What it is used for

Handle pegs safely for hanging and housekeeping tasks.

How to handle

- Check pegs for cracks, sharp edges, or loose parts before use.
- Use pegs only for their intended purpose and suitable load.
- Keep fingers clear of spring-loaded or clamping sections.
- Collect dropped pegs promptly to prevent slipping hazards.

Storage

- Store in a labelled container or designated holder.
- Keep dry and away from heat that may deform plastic pegs.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pest Control Chemical

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **High** 5 labs

Covers these lab items: Termite treatment

What it is used for

Pest control chemicals are used to control termites and other specified pests in approved treatment areas.

How to handle

- Use only a product approved for the target pest and treatment site.
- Read the label and Safety Data Sheet; wear specified PPE and ensure adequate ventilation.
- Keep people, children, pets, and unprotected workers away during application and required re-entry periods.
- Prevent spills and contamination of food, drinking water, soil, and drains; follow label disposal instructions.

Storage

- Keep securely locked in original labelled containers, separate from food and household supplies.
- Follow product-specific temperature and ventilation requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pillow

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Pillow with Pillow Cover; Pillows

What it is used for

Maintain pillow cleanliness, comfort, and condition.

How to handle

- Use a clean pillow cover for each user or change it according to hygiene procedures.
- Inspect pillows for stains, moisture, odour, or damage.
- Follow approved cleaning instructions for the pillow filling and cover.
- Keep pillows away from wet floors and contaminated linen.

Storage

- Store clean, dry pillows in a ventilated linen cupboard.
- Keep clean and used pillows separate.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Recycling Station

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Public recycling stations and eco-toilets on tourist routes; Recycling stations and composting bins for waste management in natural areas

What it is used for

Handle recycling stations and composting bins safely to support waste segregation.

How to handle

- Separate waste according to clearly marked categories and local procedures.
- Wear suitable gloves and closed footwear when handling waste.
- Do not place hazardous, medical, or sharp waste in general recycling bins.
- Keep lids, access paths, and signs clear; clean spills promptly.

Storage

- Keep bins in designated, ventilated locations with secure lids where required.
- Clean and maintain bins regularly and keep recyclables dry when possible.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rug

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Rugs

What it is used for

Handle rugs safely and maintain cleanliness in hospitality or housekeeping areas.

How to handle

- Inspect rugs for loose fibres, curled edges, stains, and damage before placing them.
- Use non-slip backing or suitable fixing to reduce slipping and tripping risks.
- Vacuum or clean according to the rug material and avoid over-wetting.
- Keep rugs clear of doorways and emergency routes.

Storage

- Clean and dry rugs fully before rolling or folding.
- Store in a dry, ventilated area protected from pests, moisture, and heavy objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safe Deposit Box

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Safe deposit boxes

What it is used for

A safe deposit box provides controlled storage for guests' or authorized users' valuables. Follow the property's access, identification, key control, and recordkeeping procedures for every transaction.

How to use

1. Inspect the safe, lock, hinges, keypad, keys, and access area for visible damage.
2. Verify the user's identity and authorization according to the property procedure.
3. Explain the approved deposit and retrieval process without requesting disclosure of the safe combination.
4. Assign or open the designated box using the authorized access procedure.
5. Confirm the box is secured and provide the required receipt or access instructions.
6. For retrieval, re-verify authorization and follow the required opening and recordkeeping steps.
7. Close and lock the box, secure keys, and record any faults or irregularities.

⚠ Safety

- Never disclose combinations, master keys, or access codes to unauthorized people.
- Do not open a user's box without proper authorization and required verification.
- Keep keys and access records secured and restrict access to authorized staff.
- Report suspected tampering, lost keys, or access discrepancies immediately.

Care & storage

- Clean external surfaces with approved products and check locks and access controls regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sanitizer

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Sanitization supplies; Sanitizers

What it is used for

Sanitizers reduce microorganisms on hands or suitable surfaces when used according to product instructions.

How to handle

- Check the label, expiry date, and intended use before application.
- For hand sanitizer, apply the recommended amount and rub hands until dry; follow product instructions.
- For surface sanitizer, confirm compatibility and required contact time before use.
- Keep away from eyes, ingestion, flames, and ignition sources, especially alcohol-based products.
- Do not mix sanitizers with other cleaning chemicals; discontinue use if irritation occurs.

Storage

- Keep in the original labelled container with the lid closed.
- Store away from heat, direct sunlight, ignition sources, and children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Serving Tray

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 9 labs

Covers these lab items: Rectangular Service Tray; Refreshment trays, concierge tools; Round Service Tray; Serving Tray Set; Serving Trays

What it is used for

Serving trays transport food, beverages, and service items safely and neatly.

How to handle

- Inspect the tray for cleanliness, cracks, sharp edges, and stability.
- Choose a tray size suitable for the load and balance items evenly.
- Keep hot liquids and heavy items stable and avoid overloading.
- Carry the tray with a secure grip and clear the walking route first.
- Clean and sanitise trays according to the food-service procedure.

Storage

- Store clean, dry trays in a designated rack or cabinet.
- Keep trays protected from contamination and avoid unstable stacking.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spotting Gun

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Spotting Gun; Stain Removing Machines (Spot Gun)

What it is used for

A spotting gun applies a controlled jet of cleaning solution or compressed air to treat localised stains on fabric and upholstery. Use the correct solution, pressure, and technique for the material to avoid damage, chemical exposure, or spreading the stain.

Main parts

- Gun body and trigger
- Solution reservoir or chemical feed connection
- Nozzle and spray outlet
- Air or fluid inlet and hose
- Pressure regulator, if fitted
- Power or compressed-air connection, depending on model

Before you start

1. Identify the spotting gun type and read the operating instructions.
2. Check the fabric care label and identify the stain and material compatibility.
3. Inspect the gun, nozzle, hose, reservoir, and connections for damage or leakage.
4. Select a compatible stain-removal solution and follow its safety instructions.
5. Test the solution on a hidden area of the fabric before treatment.
6. Wear suitable gloves and eye protection and ensure adequate ventilation.

How to use

1. Switch off or disconnect the gun before filling the reservoir or changing attachments.
2. Prepare the cleaning solution at the concentration specified by the product instructions.
3. Fill the reservoir or connect the approved solution supply without overfilling.
4. Connect the air or power supply as required by the model.
5. Set the pressure or spray setting within the recommended operating range.
6. Place an absorbent cloth beneath or behind the stained area where appropriate.
7. Hold the nozzle at the recommended distance from the fabric.
8. Apply short, controlled bursts while working from the outside of the stain inward when suitable.
9. Avoid soaking delicate fabric or forcing liquid deep into the material.
10. Blot or rinse the treated area according to the cleaning procedure.
11. Check the result and repeat only if the material and product instructions permit.
12. Switch off and release pressure before disconnecting hoses or cleaning the nozzle.
13. Empty and clean the reservoir and nozzle according to the manufacturer's instructions.

⚠ Safety

- Only trained users should operate the spotting gun.
- Never direct compressed air or cleaning spray toward people, skin, or eyes.
- Use only approved chemicals and never mix cleaning products unless expressly permitted.
- Do not exceed the rated pressure of the gun, hose, or fittings.
- Keep the work area ventilated and avoid inhaling spray mist.
- Depressurise and disconnect the equipment before maintenance or clearing a blockage.

Care & storage

- Empty and rinse the reservoir after use as instructed.
- Clean the nozzle and fluid passages to prevent clogging.
- Inspect hoses, seals, trigger, and connectors regularly.
- Store chemicals in labelled containers and the gun in a clean, dry place.
- Arrange servicing for leakage, inconsistent spray, or faulty controls.

Common mistakes to avoid

- Using a stain remover that is unsuitable for the fabric.
- Applying excessive pressure or soaking the material.
- Spraying without testing the solution on a hidden area.
- Disconnecting a pressurised hose or cleaning a blocked nozzle while connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spray Bottle

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Spray Bottle

What it is used for

Spray bottles dispense compatible liquids for cleaning, sanitising, or other labelled applications.

How to handle

- Label each bottle clearly with its contents and intended use.
- Check the bottle, trigger, and nozzle for leaks or damage before use.
- Follow the liquid's label instructions and wear required PPE.
- Never mix chemicals in the bottle unless the product instructions explicitly permit it.
- Avoid spraying toward people, eyes, food, or electrical equipment.

Storage

- Store upright in a labelled, secure location.
- Keep chemicals in their original containers where required and away from children and incompatible substances.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Straw Holder

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Straw Holder

What it is used for

A straw holder keeps drinking straws organised and protected in food-service environments.

How to handle

- Inspect the holder for cracks, dirt, and sharp edges before use.
- Use clean, dry straws and avoid touching the drinking ends.
- Keep the holder covered or protected from dust and splash contamination.
- Clean and dry the holder regularly according to food-service hygiene procedures.
- Discard damaged or contaminated straws and remove holders that cannot be cleaned safely.

Storage

- Store the holder in a clean, dry service area.
- Keep spare straws sealed in their original packaging.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sugar Sachet Holder

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Sugar Sachet Holder

What it is used for

A sugar sachet holder organises individually packed sugar sachets for beverage service.

How to handle

- Inspect the holder for cleanliness, stability, and damage before use.
- Use sealed, undamaged sachets and discard any that are wet or contaminated.
- Arrange sachets neatly without overfilling the holder.
- Keep hands clean when replenishing supplies and avoid direct contact with sachet openings.
- Clean and dry the holder regularly.

Storage

- Store in a clean, dry food-service area.
- Keep spare sachets sealed and away from moisture, pests, and cleaning chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Table Linen

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Table Cloths; Table Napkins

What it is used for

Table linen protects table surfaces and supports hygienic, attractive food and hospitality service.

How to handle

- Check tablecloths and napkins for stains, tears, and cleanliness before setting the table.
- Use clean linen for each service and keep it away from floor contact.
- Arrange linen evenly and ensure it does not create trip or spill hazards.
- Remove used linen carefully and separate heavily soiled items according to laundry procedures.
- Replace stained or damaged linen before serving guests.

Storage

- Store clean, dry linen in a covered, labelled cupboard or shelf.
- Keep used linen separate from fresh stock and protect all linen from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tank Cleaning Tool

Category: **Hospitality & Housekeeping** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Basic tools for cleaning of tanks

What it is used for

Tank cleaning tools remove deposits and residues from accessible tank surfaces during approved cleaning activities. Identify the tank contents and hazards before selecting tools and following the cleaning procedure.

How to use

1. Confirm the tank's previous contents, isolation status, and approved cleaning method.
2. Inspect brushes, scrapers, hoses, pumps, and other tools for damage and suitability.
3. Empty, isolate, and drain the tank according to the site procedure.
4. Use required PPE and prepare the work area to contain wastewater and residues.
5. Clean accessible surfaces with suitable tools and approved cleaning agents.
6. Rinse or remove residues as specified and check that the required cleanliness level is achieved.
7. Collect waste, decontaminate tools, and document completion of the cleaning task.

⚠ Safety

- Never enter a tank or confined space without the required permit, atmospheric testing, and rescue arrangements.
- Do not mix cleaning chemicals unless the procedure explicitly permits it.
- Isolate electrical, mechanical, and fluid sources before cleaning.
- Follow chemical safety data sheets and applicable waste disposal rules.

Care & storage

- Clean and dry reusable tools after use and store them separately from contaminated equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tissue Paper

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Paper Napkins; Paper Towel; Paper towels (Packs); Tissue Paper; Tissue papers

What it is used for

Tissue paper and paper towels support personal hygiene, cleaning, and food-service needs.

How to handle

- Use clean paper products appropriate for the intended task.
- Dispense without touching unused sheets unnecessarily.
- Use each sheet once where hygiene requires and avoid cross-contamination.
- Dispose of used tissue according to site waste procedures; do not flush unless approved.

Storage

- Keep packs sealed or covered in a clean, dry place.
- Store away from chemicals, moisture, and contaminated items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Toilet Roll

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Toilet Roll

What it is used for

Handle toilet rolls hygienically for washroom use.

How to handle

- Keep rolls protected from splashes, dust, and contact with contaminated surfaces.
- Use clean, dry hands when replenishing dispensers.
- Do not place rolls directly on wet floors or near cleaning chemicals.
- Discard rolls that are wet, visibly soiled, or contaminated.

Storage

- Store in a clean, dry cupboard or dispenser area.
- Keep sealed packs off the floor and away from moisture and strong odours.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Toothbrush

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Tooth Brush

What it is used for

Handle toothbrushes hygienically for personal or guest use.

How to handle

- Use a separate toothbrush for each person.
- Rinse thoroughly after use and allow it to air-dry upright without touching other brushes.
- Do not share toothbrushes or use a brush with damaged bristles.
- Replace regularly and after illness according to hygiene guidance.

Storage

- Store upright in a clean, ventilated holder.
- Keep separate from toilet splash, cleaning chemicals, and used brushes.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Toothpaste

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Toothpaste

What it is used for

Handle toothpaste hygienically for personal or guest use.

How to handle

- Use a separate tube or clean dispensing method for each user where required.
- Avoid touching the tube opening directly to the toothbrush or mouth.
- Close the cap after use and check the product label before use.
- Do not use toothpaste from damaged, contaminated, or expired packaging.

Storage

- Store capped in a cool, dry place away from direct sunlight.
- Keep separate from cleaning chemicals and clearly distinguish guest supplies from used items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Towel

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 12 labs

Covers these lab items: Bath Towels; Hand towels; Kitchen Towels; Small Towel; Towel; Towels; Towels (White and brown)

What it is used for

Towels absorb moisture and support personal care, kitchen hygiene, and hospitality services.

How to handle

- Use towels appropriate for the task and keep clean towels separate from used ones.
- Inspect towels for stains, odours, damage, and visible contamination before issuing.
- Avoid sharing personal towels between users and use dedicated towels for different hygiene areas.
- Handle contaminated towels with appropriate PPE and follow the approved laundry process.

Storage

- Store clean, fully dry towels in a clean, covered area.
- Keep used towels in designated laundry containers away from clean linen.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tub

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Small Tub

What it is used for

Handle small tubs hygienically for cleaning, soaking, or general housekeeping tasks.

How to handle

- Inspect the tub for cracks, leaks, and cleanliness before use.
- Use tubs only for their intended purpose and avoid mixing cleaning chemicals unless instructed.
- Do not overfill; carry with a secure grip and watch for slippery floors or spills.
- Clean and disinfect when required, especially if the tub is shared.

Storage

- Empty, rinse, and dry the tub after use.
- Store upside down or in a designated clean area away from chemicals and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vacuum Cleaner

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 21 labs

Covers these lab items: Vacuum cleaner; Vacuum cleaners

What it is used for

A vacuum cleaner removes dust and loose dirt from suitable floors, furniture, and other surfaces. Correct attachments, filters, and operating methods help maintain cleaning performance and reduce electrical hazards.

Main parts

- Motor creates suction to draw in dust and debris.
- Hose or inlet carries dirt toward the collection unit.
- Floor head or nozzle contacts the surface being cleaned.
- Dust bag or collection bin stores collected dirt.
- Filter traps dust before air leaves the machine.
- Handle and wheels help move the cleaner around the area.
- Power switch and cord or battery control the power supply.

Before you start

1. Read the operating instructions and confirm the cleaner suits the surface and cleaning task.
2. Inspect the housing, hose, nozzle, filter, dust bag or bin, and power supply.
3. Empty the collection bin or replace the dust bag if it is full.
4. Check that the filter is correctly fitted and suitable for the material being collected.
5. Remove large objects, sharp debris, and tangled cords from the cleaning path.
6. Keep hands dry and disconnect the power before fitting attachments or opening the unit.

How to use

1. Choose the correct floor head or attachment for the surface being cleaned.
2. Fit the hose, nozzle, filter, and dust bag or bin securely.
3. Connect the plug with dry hands or check that the battery is properly fitted.
4. Switch on the cleaner and move the nozzle slowly across the surface.
5. Overlap cleaning passes slightly to collect dust from the whole area.
6. Use the appropriate attachment for corners, upholstery, or other suitable surfaces.
7. Switch off immediately if suction drops, the motor overheats, or unusual noise occurs.
8. Switch off and unplug the machine before removing blockages or emptying the bin.
9. Empty the bin or replace the bag carefully without scattering dust.
10. Clean and store the machine after use, following the manufacturer's instructions.

⚠ Safety

- Never vacuum liquids unless the cleaner is specifically designed for wet cleaning.
- Do not vacuum hot ash, burning material, flammable liquids, or hazardous substances with an ordinary cleaner.
- Keep the cord away from the nozzle, wheels, sharp edges, and hot surfaces.
- Never operate a cleaner with a damaged cord, plug, housing, or wet electrical parts.
- Disconnect power before clearing blockages, changing filters, or performing maintenance.
- Keep fingers and loose clothing away from moving brush rollers.

Care & storage

- Empty the dust bin or replace the bag before it becomes overfilled.
- Clean or replace filters according to the manufacturer's instructions and dry washable filters fully.
- Check the hose, brush, wheels, and cord regularly for blockages or damage.
- Store the cleaner and accessories in a dry place with the cord loosely arranged.

Common mistakes to avoid

- Using an ordinary dry vacuum cleaner to collect water or wet waste.
- Operating the machine with a full bin or blocked filter.
- Pulling the cleaner by its power cord instead of its handle.
- Cleaning or opening the unit before switching off and disconnecting power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wash Basin

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Enamel basin; Steel Basin; Wash basin; Water basins

What it is used for

Wash basins support handwashing, cleaning, and personal hygiene in hospitality and service areas.

How to handle

- Inspect the basin, taps, drain, and supports for cracks, leaks, or blockages.
- Clean and sanitise the basin according to the required hygiene schedule.
- Use suitable cleaning products and never mix incompatible cleaning chemicals.
- Keep the basin area dry where possible and remove spills promptly.
- Report leaks, damaged fittings, or drainage problems to maintenance staff.

Storage

- Keep portable basins clean and dry in a designated area.
- Store cleaning chemicals separately from hygiene supplies and toiletries.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Washing Machine

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Washing Machine

What it is used for

A washing machine cleans textiles by using water, detergent, and mechanical movement in a controlled wash cycle. Correct loading, program selection, and maintenance help improve cleaning results and protect fabrics and equipment.

Main parts

- Wash drum and outer tub
- Detergent and fabric conditioner dispenser
- Water inlet hose and inlet valve
- Drain hose and drain pump
- Program selector and control panel
- Door or lid lock and electrical power connection

Before you start

1. Read the operating instructions and check the machine's rated load capacity.
2. Inspect the power cable, water hoses, door seal, and drain connection for damage or leakage.
3. Separate laundry by colour, fabric type, and care-label instructions.
4. Empty pockets and remove loose objects, pins, and items that could damage the drum.
5. Use the recommended detergent type and quantity for the load and water conditions.
6. Ensure the machine is level, the drain is clear, and the water supply is available.

How to use

1. Load laundry evenly without exceeding the rated capacity.
2. Add detergent and fabric conditioner to the designated compartments as required.
3. Close the door or lid securely and confirm the locking mechanism engages.
4. Select the appropriate wash program, water temperature, and spin speed for the fabric.
5. Start the cycle and check that water enters and the drum operates normally.
6. Monitor the machine for unusual noise, excessive vibration, leaks, or error indicators.
7. Allow the complete wash and spin cycle to finish before opening the door.
8. Remove the laundry promptly and check that no items remain inside the drum.
9. Clean the detergent dispenser and accessible lint or debris filters according to the instructions.
10. Switch off the machine and leave the door slightly open when appropriate to allow drying.

⚠ Safety

- Keep hands away from the drum while it is moving or spinning.
- Do not operate the machine with damaged cables, leaking hoses, or a faulty door lock.
- Keep the floor around the machine dry to reduce slip and electric-shock risks.
- Do not exceed the rated load or climb on the machine.
- Disconnect power and close the water supply before maintenance or hose replacement.
- Use only approved detergents and never mix cleaning chemicals unless the product instructions explicitly permit it.

Care & storage

- Clean the detergent dispenser, door seal, and drum regularly.
- Inspect inlet and drain hoses for cracks, bulges, blockages, or leaks.
- Clean filters and perform the recommended drum-cleaning cycle at the specified interval.
- Keep the exterior dry and record faults or maintenance as required.

Common mistakes to avoid

- Overloading the drum can cause poor washing and excessive vibration.
- Using too much detergent can leave residue and create excessive foam.
- Selecting an unsuitable temperature or spin speed can damage delicate fabrics.
- Ignoring blocked filters or leaking hoses can cause drainage problems and water damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Waste Disposal Equipment

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 13 labs

Covers these lab items: Waste disposal equipment such as scrap bins, recyclers, waste processing machines, etc.; Waste disposal equipment's; Waste disposal equipment's such as scrap bins, recyclers, waste processing machines etc

What it is used for

Waste disposal equipment helps collect, separate, process, or transfer waste safely. Correct waste segregation, equipment selection, and hygiene reduce injuries, contamination, and environmental harm.

Main parts

- Collection bins or containers hold waste before disposal or processing.
- Lids and labels identify waste categories and help contain odours or spills.
- Feed opening or loading area receives suitable waste for processing.
- Controls and switches start, stop, or adjust powered equipment.
- Guards and interlocks help prevent access to moving parts.
- Processing unit may shred, compact, sort, or recycle suitable materials.
- Discharge outlet or collection tray receives processed material.

Before you start

1. Identify the waste type and follow the school's waste segregation instructions.
2. Read the equipment instructions and confirm the machine is suitable for that waste.
3. Wear task-appropriate PPE such as gloves, closed footwear, and eye protection when required.
4. Inspect bins, lids, guards, cables, switches, and emergency stops for damage.
5. Check that the work area is clean, dry, ventilated, and free from obstacles.
6. Ensure only trained and authorised users operate powered waste-processing equipment.

How to use

1. Separate general, recyclable, organic, and hazardous waste using the designated containers.
2. Check labels and never mix incompatible waste types or unknown chemicals.
3. Place waste in the correct bin without compressing it by hand.
4. For powered equipment, confirm guards are fitted and the discharge area is clear.
5. Start the machine using its approved controls and follow the operating instructions.
6. Feed only permitted material at the specified rate; never reach into the feed opening.
7. Monitor the equipment and stop it if unusual noise, vibration, blockage, or leakage occurs.
8. Switch off and isolate power before clearing blockages or cleaning powered equipment.
9. Transfer processed waste to the designated collection point using suitable handling methods.
10. Clean the work area, wash hands, and record or report faults as required.

⚠ Safety

- Never place hands or tools inside moving, feeding, shredding, or compacting mechanisms.
- Do not process batteries, chemicals, medical waste, or pressurised containers unless the equipment is specifically approved.
- Keep guards and interlocks in place; never bypass safety devices.
- Use correct lifting techniques or assistance when moving heavy bins and containers.
- Treat sharp scrap, broken glass, leaking waste, and unknown substances as hazards.
- Disconnect and isolate power before maintenance, cleaning, or jam removal.

Care & storage

- Empty and clean bins regularly using the approved method and suitable PPE.
- Inspect wheels, lids, labels, guards, cables, and controls for damage or wear.
- Keep equipment dry and arrange authorised servicing according to the manufacturer's instructions.
- Report spills, pest activity, unusual odours, leaks, and equipment faults promptly.

Common mistakes to avoid

- Mixing recyclable, organic, hazardous, or unknown waste in one container.
- Reaching into a machine to push material or clear a blockage.
- Operating equipment with damaged guards or malfunctioning emergency stops.
- Leaving waste spills or contaminated bins uncleaned after use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Bottle

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Water Bottle; Water Bottles

What it is used for

Handle water bottles hygienically for safe drinking-water storage and use.

How to handle

- Wash bottles before first use and clean them regularly.
- Use food-grade bottles suitable for the intended liquid and temperature.
- Keep caps closed and avoid sharing personal drinking bottles.
- Do not use bottles with cracks, persistent odour, or visible contamination.

Storage

- Store clean, dry bottles with caps off when fully drying.
- Keep away from direct sunlight, chemicals, and unclean surfaces.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Container

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Water storage can

What it is used for

Handle water storage cans to maintain water quality and prevent contamination.

How to handle

- Use only clean, food-grade containers intended for drinking water where applicable.
- Keep the lid closed and avoid dipping hands or unclean utensils into the water.
- Inspect the can for cracks, leaks, odour, or visible contamination.
- Clean and sanitise at a suitable interval according to local water-safety procedures.

Storage

- Keep elevated from the floor in a clean, shaded area.
- Store away from fuels, chemicals, waste, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Dispenser

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Water Dispenser / Cooler

What it is used for

A water dispenser or cooler provides drinking water at ambient, chilled, or heated temperature depending on its design. Correct installation, sanitation, and temperature control help maintain water quality and safe operation.

Main parts

- Water bottle holder or inlet connection
- Cold-water reservoir and cooling system if fitted
- Hot-water reservoir and heating element if fitted
- Dispensing taps or push buttons
- Drip tray and drain arrangement
- Power cable, thermostat, and indicator lights

Before you start

1. Read the manufacturer's instructions and identify the dispenser's hot, cold, and ambient-water functions.
2. Inspect the body, taps, bottle support, power cable, and water connections for damage or leaks.
3. Confirm that the unit is positioned on a stable, ventilated surface away from heat sources.
4. Clean and sanitize the reservoir and water-contact parts according to the approved procedure.
5. Use a sealed, approved drinking-water bottle or a verified potable-water supply.
6. After installation or cleaning, allow the unit to settle and cool or heat as specified before dispensing.

How to use

1. Install the bottle or connect the approved water supply according to the model instructions.
2. Ensure all water connections are secure and the bottle is seated correctly.
3. Switch on the cooling or heating functions only after the required water level is present.
4. Allow the water to reach the specified operating temperature.
5. Place a suitable cup beneath the selected dispensing tap.
6. Operate the tap or button and collect the required amount of water.
7. Close the tap fully and check for drips or leakage.
8. Empty and clean the drip tray when needed.
9. Switch off the unit before cleaning external surfaces or carrying out approved routine care.
10. Record sanitation or maintenance checks according to the facility procedure.

⚠ Safety

- Keep the power plug, cable, and electrical components dry and away from standing water.
- Do not touch the hot-water outlet or dispense hot water where children can reach it unsupervised.
- Never operate a heating function without the required water level.
- Use only potable water and approved food-contact cleaning products.
- Disconnect power before cleaning internal parts or servicing the unit.
- Do not use the dispenser if the cable, plug, taps, or reservoir are damaged or leaking.

Care & storage

- Clean and sanitize water-contact parts at the frequency specified by the facility procedure.
- Empty and clean the drip tray regularly to prevent residue and microbial growth.
- Inspect the water bottle, inlet connection, taps, and seals for leaks or contamination.
- Maintain the cooling and heating systems according to the manufacturer's schedule.

Common mistakes to avoid

- Using an unsealed or contaminated bottle can compromise drinking-water quality.
- Switching on the heater before confirming adequate water can damage the unit.
- Leaving the drip tray dirty can cause odours and microbial growth.
- Using harsh or unapproved chemicals can contaminate water-contact surfaces.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Filtration Unit

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **Medium** 22 labs

Covers these lab items: Filtering machines; Filtration System; Mini water filtration unit

What it is used for

A water filtration unit removes selected suspended particles, impurities, or other contaminants from water using suitable filter media and treatment stages. Correct operation and timely filter replacement help maintain consistent water quality, but filtration alone does not guarantee drinking-water safety.

Main parts

- Raw-water inlet and isolation valve
- Filter housing and filter cartridges or media
- Pre-filter and sediment filter where fitted
- Activated carbon, membrane, or other treatment stage as designed
- Treated-water outlet and storage tank if fitted
- Pressure gauge, flow indicator, and drain connection where fitted

Before you start

1. Read the manufacturer's instructions and confirm the unit's intended water source and treatment capability.
2. Inspect the housing, pipes, valves, seals, electrical connections, and fittings for damage or leakage.
3. Verify that filter cartridges and media are approved for the model and are within their service life.
4. Confirm the raw-water supply pressure and quality are within the unit's operating limits.
5. Flush new filters according to the manufacturer's instructions before collecting treated water.
6. Check that the treated-water tank and outlet are clean and that required water-quality testing is current.

How to use

1. Open the raw-water inlet valve according to the operating instructions.
2. Start the pump or powered treatment stage if the unit requires it.
3. Check the pressure and flow indicators and inspect the system for leaks.
4. Allow the specified startup flush to pass through the filters and send it to the approved drain if required.
5. Collect treated water only after the flushing procedure is complete.
6. Monitor flow rate, pressure drop, unusual odour, colour, or other signs of reduced performance.
7. Record operating readings and water-quality checks according to the facility procedure.

8. Close the inlet valve and isolate power before replacing filters or opening the housing.
9. Replace exhausted cartridges or media using the approved procedure and reseal the housing correctly.
10. Flush the unit after filter replacement and verify that it operates normally before returning it to service.

⚠ Safety

- Depressurize the filter housing before loosening its cap or removing cartridges.
- Use only filter media and replacement parts approved for the unit and water application.
- Disconnect electrical power before servicing pumps, controls, or electrical components.
- Do not drink treated water unless its quality meets the applicable drinking-water requirements.
- Handle chemical cleaning or disinfection agents according to their safety instructions.
- Repair leaks promptly and keep electrical equipment away from standing water.

Care & storage

- Replace cartridges and filter media according to the manufacturer's schedule or verified performance limits.
- Clean and sanitize the housing and storage tank at the specified intervals.
- Inspect valves, seals, pipes, pressure indicators, and connections regularly.
- Maintain records of filter changes, flushing, servicing, and water-quality tests.

Common mistakes to avoid

- Delaying filter replacement can reduce flow and allow treatment performance to deteriorate.
- Failing to flush new filters can leave manufacturing residues or loose media in the water.
- Using incompatible cartridges can damage the unit or reduce contaminant removal.
- Assuming clear-looking water is automatically safe can expose users to invisible contaminants.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Heater

Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Geyser; Geyser/ Water heater

What it is used for

A water heater heats water for domestic, hospitality, or other approved uses and stores it in tank-type models. Correct temperature settings, safe installation checks, and regular maintenance help provide reliable hot water while reducing scalding and electrical or gas hazards.

Main parts

- Tank or heating chamber stores or heats water.
- Heating element or burner supplies heat depending on the model.
- Thermostat or temperature controller regulates water temperature.
- Temperature and pressure relief valve protects compatible storage heaters from excessive conditions.
- Cold-water inlet and hot-water outlet direct water through the unit.
- Insulation reduces heat loss from storage models.
- Power switch or gas control operates the heater.
- Indicator light or display shows heating status on equipped models.

Before you start

1. Read the manual and confirm the heater type, rated supply, and installation requirements.
2. Inspect the casing, water connections, power cord or gas line, and controls for damage or leakage.
3. Confirm that the heater is installed securely and that required earthing or gas connections are approved.
4. For storage models, ensure the tank is filled and air is purged before energising the heating element.
5. Check that required relief valves and discharge pipes are correctly installed and unobstructed.
6. Keep the surrounding area clear and ensure ventilation is adequate for fuel-burning models.

How to use

1. Confirm all water and energy connections are secure before operation.
2. Open the water supply and allow the storage tank to fill completely where applicable.
3. Open a hot-water tap as instructed to release trapped air, then close it when water flows steadily.
4. Switch on the heater or ignite the burner using the manufacturer's specified procedure.
5. Set the temperature within the recommended range for the intended use.
6. Wait for the heater to complete its heating cycle and check the indicator or display.
7. Test the hot-water flow cautiously and adjust the outlet to avoid scalding.

8. Switch off the heater when required by the operating procedure or before extended maintenance.
9. Allow the unit to cool and isolate energy before cleaning or servicing.
10. Inspect for leaks or abnormal operation and report faults to authorised service personnel.

⚠ Safety

- Never energise an electric storage heater before confirming the tank is full of water.
- Do not touch electrical controls with wet hands or operate damaged wiring.
- Do not block or tamper with the temperature and pressure relief valve.
- Avoid direct contact with hot water and verify outlet temperature before use.
- Keep gas-fired heaters adequately ventilated and stop use if gas odour or combustion problems occur.
- Electrical, gas, and pressure-system repairs must be performed by qualified personnel.

Care & storage

- Clean the external casing with suitable materials and keep ventilation openings clear.
- Inspect visible water connections and the relief valve discharge path at recommended intervals.
- Descale or flush the tank only according to the manufacturer's maintenance procedure.
- Arrange authorised servicing for leaks, unusual noises, repeated tripping, or unstable temperature.

Common mistakes to avoid

- Switching on an electric heater before filling the tank.
- Setting the water temperature too high and creating a scalding risk.
- Blocking the relief valve discharge path or ignoring leaks.
- Attempting electrical, gas, or pressure-system repairs without proper qualification.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Jug

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 10 labs

Covers these lab items: Steel Jug; Water Container (Jug); Water Jug; Water pot/ Mug/ Glass

What it is used for

Water jugs hold and serve drinking water or other approved beverages hygienically.

How to handle

- Inspect the jug for cracks, leaks, cleanliness, and a secure handle.
- Wash and rinse the jug before filling it with potable water.
- Avoid touching the inside or pouring spout with hands.
- Do not overfill and pour carefully to prevent spills.
- Clean the jug between uses and when changing beverages.

Storage

- Store clean, dry jugs in a hygienic designated area.
- Keep covered where possible to protect contents from contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wet Wipes

Category: **Hospitality & Housekeeping** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Wet tissues; Wet Wipes; Wet Wipes -Packet

What it is used for

Wet wipes provide convenient cleaning of hands or suitable surfaces when compatible with the product instructions.

How to handle

- Check the packaging, expiry date, and intended use before opening.
- Use wipes only for the purpose specified by the manufacturer; do not assume they are disinfectants.
- Use a fresh wipe where hygiene requires it and avoid sharing personal wipes.
- Avoid contact with eyes, and discontinue use if irritation occurs.
- Dispose of used wipes in the appropriate waste bin; do not flush unless explicitly labelled flushable and permitted locally.

Storage

- Reseal the packet after use to prevent drying.
- Store in a cool, dry area away from direct sunlight and children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Autoclave

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Autoclave; Autoclave - separate for media sterilization and waste disposal; Sterilizers, autoclave

What it is used for

An autoclave uses pressurised saturated steam to sterilise suitable laboratory materials or treat designated biological waste. Correct loading, validated cycles, and careful pressure release are essential for effective treatment and operator safety.

Main parts

- Sterilisation chamber holds the materials being treated.
- Door and locking mechanism seal the chamber during operation.
- Steam generator or steam supply provides the required steam.
- Temperature and pressure sensors monitor cycle conditions.
- Control panel selects the cycle and displays operating status.
- Drain and exhaust system remove condensate and release steam through the designed pathway.
- Gaskets and safety interlocks help maintain a secure seal.
- Trays, racks, and containers support the load.

Before you start

1. Only trained, authorised personnel should operate the autoclave under the laboratory's approved procedures.
2. Read the manual and select the validated cycle for the specific load.
3. Inspect the chamber, door gasket, locking mechanism, drain, and indicators for damage or blockage.
4. Check that water supply and other required utilities are available and correct.
5. Use autoclave-compatible containers and arrange loads to allow steam circulation.
6. Confirm biological waste is packaged and segregated according to laboratory procedures and applicable rules.

How to use

1. Wear the specified PPE, including heat-resistant gloves and eye or face protection as required.
2. Prepare and load materials in compatible containers without overfilling or blocking steam pathways.
3. Place liquid containers in suitable secondary containment and loosen closures as required by the validated procedure.
4. Close and lock the door using the manufacturer's specified method.
5. Select the approved cycle for the load and verify the displayed settings.

6. Start the cycle and monitor temperature, pressure, and cycle status without opening the chamber.
7. Wait until the cycle completes and pressure returns to the permitted safe level.
8. Open the door cautiously from the side, allowing residual steam to escape away from the face.
9. Remove the load with suitable heat protection and allow items to cool in a designated area.
10. Document the cycle, indicators, and any required chemical or biological monitoring results; report failures before using the load.

⚠ Safety

- Never open the door while the chamber is pressurised or the cycle is active.
- Do not bypass door locks, pressure controls, or safety interlocks.
- Use suitable heat-resistant PPE and keep exposed skin away from steam and hot surfaces.
- Do not autoclave sealed containers, volatile chemicals, flammable materials, or incompatible substances.
- Treat liquid loads carefully because hot liquids may boil over after the cycle.
- If the cycle fails or an alarm occurs, isolate the unit as instructed and notify authorised personnel.

Care & storage

- Clean the chamber and trays after cooling, following the manufacturer's procedure.
- Inspect the gasket, drain filter, and door mechanism at recommended intervals.
- Keep records of maintenance, cycle verification, and required performance testing.
- Arrange authorised servicing for leaks, unusual noises, failed indicators, or repeated cycle errors.

Common mistakes to avoid

- Selecting a cycle that is unsuitable for the load or container type.
- Overloading the chamber or blocking steam circulation.
- Opening the door before pressure has returned to a safe level.
- Using treated materials without checking required cycle records or monitoring results.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Beaker

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

6 labs

Covers these lab items: Beaker; Beaker / glass flask; Beaker 250ml; Glass beakers; Glassware/ Beakers

What it is used for

Use a beaker to hold, mix, heat, or measure approximate volumes of liquids during suitable laboratory procedures.

How to use

1. Select a beaker size and material suitable for the liquid and procedure.
2. Inspect the glass or plastic body for cracks, chips, or contamination.
3. Place the beaker on a stable, level surface before filling.
4. Pour the liquid carefully and use markings only for approximate volume measurement.
5. Use compatible heating equipment only if the beaker is rated for heating.
6. Label contents when required and avoid filling beyond a safe working level.
7. Dispose of contents according to laboratory procedures and clean the beaker after use.

⚠ Safety

- Wear suitable eye protection, gloves, and a lab coat when required.
- Do not use cracked or chipped glassware.
- Use heat-resistant protection when handling hot beakers and never assume glass is cool by touch.
- Do not heat a sealed beaker or use incompatible chemicals.

Care & storage

- Wash with a suitable laboratory detergent and rinse as required.
- Allow the beaker to dry and store it safely to prevent breakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Blood Cell Pipette

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: WBC Pipette

What it is used for

A blood cell pipette measures and transfers small quantities of blood or diluting fluid for laboratory procedures. Handle it carefully and follow the laboratory's approved method.

How to use

1. Inspect the pipette for cracks, chips, blocked openings, or damaged markings.
2. Wear laboratory coat, gloves, and eye protection as required.
3. Use the specified pipette aid and never draw blood by mouth.
4. Collect or transfer the required liquid according to the approved laboratory procedure.
5. Avoid air bubbles and read the marked volume at eye level.
6. Dispose of contaminated tips or pipettes in the designated biohazard container.
7. Clean or disinfect reusable equipment according to laboratory instructions.

⚠ Safety

- Treat all blood samples as potentially infectious and avoid direct contact.
- Never pipette by mouth or use a cracked pipette.
- Handle broken glass with suitable tools, never bare hands.

Care & storage

- Store clean reusable pipettes in a protective holder and dispose of single-use types appropriately.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Centrifuge

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Centrifuge

What it is used for

A centrifuge separates components of a sample by spinning it at a controlled speed. Correct balancing, compatible tubes, and proper rotor handling help produce reliable results and reduce the risk of spills or mechanical failure.

Main parts

- Rotor holds the sample tubes or containers during rotation.
- Lid and locking mechanism secure the chamber during operation.
- Drive motor rotates the rotor at the selected speed.
- Speed and time controls set the operating cycle.
- Display shows speed, time, and operating status on equipped models.
- Tube holders or adapters support compatible sample containers.
- Safety interlocks prevent operation when the lid is not properly secured.
- Chamber contains the rotor and helps limit exposure to moving parts.

Before you start

1. Only trained personnel should operate the centrifuge under the relevant laboratory procedure.
2. Read the manual and confirm the rotor, tubes, and adapters are compatible with the sample and speed.
3. Inspect the rotor, chamber, lid, seals, and tubes for cracks, corrosion, or damage.
4. Balance sample tubes by matching their mass and positions according to the rotor instructions.
5. Confirm the sample containers are closed and suitable for centrifugation.
6. Set the centrifuge on a stable, level surface and check that ventilation openings are clear.

How to use

1. Switch off the centrifuge before installing or removing the rotor or adapters.
2. Load balanced tubes into the correct rotor positions and secure them as instructed.
3. Close the lid fully and confirm the locking mechanism engages.
4. Set the required speed, time, and temperature if the model supports temperature control.
5. Verify the settings against the approved method before starting the run.
6. Start the centrifuge and observe it from a safe distance for unusual vibration or noise.
7. Stop the run immediately if severe vibration, unusual noise, or a warning occurs, following the manual.
8. Wait until the rotor has completely stopped before opening the lid.

9. Remove tubes carefully and inspect them for leakage, breakage, or abnormal sample appearance.
10. Clean the chamber and accessories as required and record the run or any observed fault.

⚠ Safety

- Never open the lid or attempt to slow the rotor by hand while it is spinning.
- Do not operate the centrifuge with unbalanced loads, damaged rotors, or cracked tubes.
- Use only the rotor speed and tube ratings approved by the manufacturer.
- Handle potentially infectious or hazardous samples according to the laboratory's containment procedures.
- If a tube breaks or leaks, keep the lid closed and follow the laboratory's spill response procedure.
- Disconnect power before maintenance and never bypass the lid interlock.

Care & storage

- Clean the chamber, rotor, and adapters using materials compatible with the equipment and sample type.
- Inspect the rotor regularly for corrosion, cracks, and wear, and remove damaged parts from service.
- Dry components as instructed and store the rotor in the recommended position.
- Maintain service, inspection, and rotor-use records according to laboratory procedures.

Common mistakes to avoid

- Loading tubes without balancing their mass and positions.
- Using tubes or adapters that are incompatible with the rotor or sample.
- Opening the lid before the rotor has completely stopped.
- Ignoring unusual vibration, noise, leakage, or rotor damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Concentration Measuring Tool

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: KOH concentration measuring tools

What it is used for

A KOH concentration measuring tool helps determine potassium hydroxide concentration using the method supported by the instrument. Select a compatible method and follow laboratory chemical-handling procedures.

How to use

1. Identify the instrument type and confirm its range is suitable for the KOH solution.
2. Wear chemical-resistant gloves, goggles, and other required PPE.
3. Inspect the instrument and check calibration or verification status.
4. Prepare the sample according to the approved laboratory procedure.
5. Measure the sample using the instrument's specified method and record the result with units.
6. Rinse or clean reusable parts as instructed to prevent sample contamination.
7. Dispose of KOH waste according to the laboratory's chemical waste procedure.

⚠ Safety

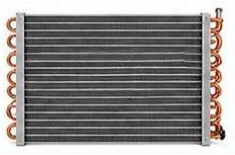
- KOH is corrosive; avoid contact with skin and eyes.
- Never mix chemicals or prepare samples without the approved procedure and supervision.
- Do not use damaged instruments or unlabelled solutions.
- If a spill occurs, alert the supervisor and follow the laboratory spill procedure.

Care & storage

- Clean the instrument as specified and store it dry and protected from chemical exposure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Condenser



Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Condenser

What it is used for

A laboratory condenser cools vapour and converts it into liquid during suitable experimental procedures. Use the correct condenser type and maintain controlled coolant flow.

How to use

1. Identify the condenser type and check the glass body and joints for cracks.
2. Secure the condenser vertically or in the required position using suitable clamps.
3. Connect coolant hoses to the designated inlet and outlet ports.
4. Check hose connections and start a gentle, steady coolant flow.
5. Begin the approved experiment and monitor the coolant flow and apparatus.
6. Stop heating before shutting down the coolant, following the procedure.
7. Allow the apparatus to cool, then disconnect hoses and clean as instructed.

⚠ Safety

- Wear goggles and suitable PPE when working with glassware and hot liquids.
- Do not use cracked glassware or force tight glass joints.
- Secure hoses to prevent leaks and keep water away from electrical equipment.
- Never seal an apparatus unless the experiment specifically requires a rated closed system.

Care & storage

- Rinse and dry the condenser as directed and store it in a padded glassware rack.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

DO Meter

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Dissolved Oxygen (DO) meter

What it is used for

A dissolved oxygen (DO) meter measures the oxygen dissolved in a water sample. Calibrate the meter as instructed and handle the probe carefully for reliable readings.

How to use

1. Inspect the meter, probe, cable, and sensor membrane or cap for damage.
2. Switch on the meter and check battery status and measurement units.
3. Calibrate the meter according to the manufacturer's procedure and required conditions.
4. Rinse the probe with clean water and gently remove excess droplets as instructed.
5. Place the probe in the sample at the required depth without trapping air bubbles.
6. Wait for the reading to stabilize and record DO, temperature, units, and sample details.
7. Rinse the probe after measurement and store it according to its probe type.

⚠ Safety

- Follow laboratory precautions for unknown or contaminated water samples.
- Keep electrical connectors dry and do not immerse parts beyond their rated limit.
- Do not touch or damage the sensor membrane or sensitive probe tip.

Care & storage

- Clean the probe and maintain its storage cap or solution according to the manufacturer's instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Distillation Unit

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Distillation Plant

What it is used for

A distillation unit separates components of a liquid mixture based on differences in volatility by heating, vaporising, and condensing the vapour. Proper setup, controlled heating, cooling-water flow, and safe collection help improve separation while reducing exposure and equipment hazards.

Main parts

- Distillation flask or feed vessel
- Heating mantle, hot plate, or approved heater
- Distillation head and thermometer
- Condenser and cooling-water connections
- Receiving flask or collection vessel
- Clamps, stands, tubing, and protective equipment

Before you start

1. Review the experiment or process procedure, safety data sheets, and the hazards of the liquid being distilled.
2. Confirm that the glassware is compatible with the chemicals, clean, dry where required, and free from cracks or chips.
3. Assemble the flask, distillation head, thermometer, condenser, and receiver using suitable clamps and supports.
4. Check that the apparatus is not sealed into a closed system and that the vapour path and any required pressure-relief route remain unobstructed.
5. Verify that cooling-water hoses are secure, the drain route is safe, and the condenser has adequate water flow.
6. Check the heating equipment, electrical connections, ventilation, and required PPE; prepare suitable spill-response materials.

How to use

1. Place the unit in a suitable laboratory or process area with appropriate ventilation and access control.
2. Add the specified quantity of feed liquid to the distillation flask, leaving adequate headspace for boiling.
3. Add boiling chips or an approved boiling aid before heating when required by the procedure; never add them to a hot, superheated liquid.

4. Secure all joints, clamps, thermometer positions, and the receiving vessel before starting the process.
5. Start condenser cooling water and confirm steady flow and leak-free connections.
6. Switch on the approved heater and increase heat gradually to avoid bumping, foaming, or sudden vapour release.
7. Monitor the temperature, boiling behaviour, condenser performance, and collected distillate throughout operation.
8. Adjust heating only within the procedure's limits and collect fractions in labelled, compatible receiving vessels when required.
9. Do not allow the distillation flask to boil dry; stop heating before the liquid level becomes unsafe.
10. If vapour escapes, cooling water fails, glassware cracks, or abnormal pressure is suspected, stop heating and follow the emergency procedure without touching hot apparatus.
11. At the end of the run, turn off the heater and allow the apparatus to cool while maintaining condenser cooling for as long as the procedure requires.
12. Stop cooling water when vapour generation has ceased and the apparatus has cooled sufficiently, according to the procedure.
13. Transfer and label the collected material as specified, then dispose of residues according to laboratory rules.
14. When fully cool, disassemble the apparatus carefully and clean the equipment using the approved method.

⚠ Safety

- Operate the unit only if trained and authorised for the chemicals and process involved.
- Wear a lab coat, chemical-compatible gloves, and splash goggles; use a fume hood for hazardous or volatile vapours when required.
- Never heat a sealed distillation system; ensure the vapour path is not blocked.
- Keep flammable liquids away from flames and use only an approved heating method for the material.
- Do not touch hot glassware or tighten joints while the apparatus is hot.
- Never leave active distillation unattended unless the approved procedure specifically provides for supervised automatic operation.
- Do not distil unknown mixtures or materials that may decompose dangerously without an approved risk assessment.
- Follow emergency procedures for spills, fire, cooling failure, and chemical exposure.

Care & storage

- Inspect glassware, joints, clamps, tubing, and condenser connections before each use.
- Clean the flask, condenser, and receiving vessels after use using compatible cleaning methods.
- Check cooling-water hoses and electrical heater cables for wear, leaks, or damage.
- Store clean glassware securely and protect it from impact and sudden temperature changes.
- Arrange repair or replacement for cracked glassware, damaged seals, or faulty temperature controls.

Common mistakes to avoid

- Heating the mixture too rapidly or overfilling the distillation flask.
- Operating with blocked vapour paths or a sealed apparatus.
- Starting heating before confirming condenser water flow.
- Allowing the flask to boil dry or handling hot glassware without protection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

EC Meter

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: EC meter

What it is used for

An electrical conductivity (EC) meter measures how well a liquid conducts electricity, helping estimate dissolved ionic content. Calibrate the meter and use a clean probe for reliable results.

How to use

1. Inspect the meter, probe, cable, and sensor for damage.
2. Switch on the meter and confirm the selected measurement units.
3. Calibrate the meter using the recommended standard solution and procedure.
4. Rinse the probe with clean distilled or deionized water as appropriate.
5. Immerse the probe in the sample to the specified depth without touching the container.
6. Wait for the reading to stabilize and record EC, temperature, units, and sample details.
7. Rinse the probe and store it according to the manufacturer's instructions.

⚠ Safety

- Follow laboratory precautions for unknown or contaminated samples.
- Keep electrical connectors dry and do not immerse the meter beyond its rated limit.
- Handle calibration solutions according to their labels and safety instructions.

Care & storage

- Clean the probe after use and replace or maintain it according to the manufacturer's guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

ESR Kit

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Westergren tubes & sodium citrate reagent

What it is used for

An ESR kit measures the erythrocyte sedimentation rate by observing how far red blood cells settle in anticoagulated blood over a specified time. Correct blood collection, anticoagulant ratio, tube positioning, and timing help produce a reliable laboratory result.

Main parts

- Westergren tube or approved ESR tube
- Westergren stand and vertical holder
- Sodium citrate reagent or approved anticoagulant system
- Graduated measurement scale
- Blood collection tube and transfer equipment
- Timer, specimen labels, and required PPE

Before you start

1. Perform the test only if trained and authorised for clinical laboratory work.
2. Confirm the test request, patient identification, specimen requirements, and the laboratory's approved ESR method.
3. Check that the Westergren tube, stand, scale, and accessories are clean, intact, and suitable for the method.
4. Verify the sodium citrate reagent identity, concentration, expiry date, and storage conditions against the approved procedure.
5. Prepare the correct blood-to-anticoagulant ratio specified by the validated method; do not estimate volumes.
6. Wear a laboratory coat, suitable gloves, and eye protection, and prepare a clean, stable work surface.

How to use

1. Identify and label the specimen using the laboratory's approved patient-identification procedure.
2. Collect the required blood specimen using the approved collection method and appropriate precautions.
3. Mix the specimen gently as required to ensure the anticoagulant is distributed evenly; avoid vigorous shaking.
4. If using sodium citrate, combine blood and reagent in the exact ratio specified by the validated method.
5. Check the specimen for clots, leakage, insufficient volume, or other rejection criteria before testing.

6. Transfer the prepared specimen into the approved Westergren tube without introducing bubbles or spilling material.
7. Place the tube vertically in its stand on a stable, level surface, away from vibration and direct sunlight.
8. Start the timer immediately after the tube is positioned, following the laboratory method's timing instructions.
9. Keep the tube undisturbed and maintain the required ambient conditions throughout the test.
10. At the specified endpoint, read the sedimentation level using the correct scale and units for the method.
11. Record the result, method, relevant specimen details, and any required quality-control observations.
12. Review results that appear inconsistent with specimen quality or laboratory acceptance criteria before reporting.
13. Dispose of used tubes, tips, and contaminated materials in the designated biohazard waste stream.
14. Clean the work area and complete the required documentation according to laboratory policy.

⚠ Safety

- Treat all blood specimens as potentially infectious and follow standard precautions.
- Wear gloves, a laboratory coat, and eye protection when handling blood or reagents.
- Never mouth-pipette; use an approved pipette or transfer device.
- Avoid splashes and aerosol generation, and clean spills using the laboratory's approved procedure.
- Dispose of sharps and biohazard waste in designated containers.
- Do not report results from clotted, incorrectly anticoagulated, leaking, or otherwise unacceptable specimens; follow the laboratory's rejection policy.
- Use only validated procedures and quality-control requirements established by the laboratory.

Care & storage

- Clean and dry the Westergren stand and reusable accessories after use according to laboratory policy.
- Check tubes, holders, and measuring scales regularly for cracks, corrosion, or readability problems.
- Store sodium citrate reagent according to its label and laboratory requirements.
- Maintain reagent lot details, expiry records, and required quality-control documentation.
- Arrange inspection or replacement of damaged equipment and report recurring measurement problems.

Common mistakes to avoid

- Using an incorrect blood-to-citrate ratio or unsuitable anticoagulant.
- Allowing the Westergren tube to tilt or vibrate during sedimentation.
- Starting or reading the test at the wrong time.

- Testing a clotted specimen or recording the result with incorrect units.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Enamel Tray

Category: **Laboratory Instruments & Glassware** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Enamel plates with smooth white surface; Enamel plates with smooth white surface (30 cm dia with raised rims)

What it is used for

Enamel trays and plates hold materials or samples for approved laboratory, demonstration, or practical activities.

How to handle

- Inspect the enamel surface for chips, cracks, rust, and sharp exposed edges.
- Use trays only for their designated purpose and confirm chemical compatibility where applicable.
- Carry trays level and avoid overloading them or placing heavy items on fragile surfaces.
- Clean and dry trays using approved methods and avoid abrasive tools that damage the enamel.
- Remove damaged trays from service if the surface cannot be safely cleaned.

Storage

- Store trays dry and stack carefully with protective separation where needed.
- Keep them in a designated area away from impact and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Growth Chamber

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Growing chamber

What it is used for

A growth chamber provides controlled environmental conditions such as temperature, humidity, and lighting for growing plants or conducting laboratory studies. Use the chamber according to the approved experimental protocol and the manufacturer's instructions.

Main parts

- Insulated chamber and access door
- Temperature control system and sensors
- Humidity control system, if fitted
- Lighting system and timer
- Shelves, trays, or sample racks
- Control panel, display, and power connection

Before you start

1. Confirm the chamber is suitable for the intended plant, sample, or experiment.
2. Inspect the chamber, door seals, shelves, sensors, and power cord.
3. Clean the interior and shelves using an approved method.
4. Check calibration or verification status of temperature and humidity sensors.
5. Confirm the required temperature, humidity, lighting, and timing settings.
6. Ensure ventilation openings are clear and samples are correctly labelled.

How to use

1. Place the chamber in a suitable location with the required clearance for ventilation.
2. Connect the unit to the recommended power supply.
3. Switch on the chamber and allow it to stabilise before loading samples.
4. Set temperature, humidity, lighting, and timer according to the approved protocol.
5. Verify displayed conditions using the available monitoring system.
6. Arrange labelled samples on shelves without blocking airflow or lighting.
7. Close the door securely and confirm the chamber maintains the required conditions.
8. Monitor readings and record environmental conditions at the required intervals.
9. Respond to alarms or deviations according to the laboratory procedure.
10. Avoid unnecessary door opening during the growth or test period.
11. At completion, remove samples using appropriate handling and PPE.
12. Switch off the chamber if required by the protocol and clean the interior.
13. Record the run details, deviations, and maintenance needs.

⚠ Safety

- Only trained personnel should configure and operate the chamber.
- Follow laboratory procedures for biological, chemical, or plant materials stored inside.
- Do not block ventilation paths or overload shelves.
- Avoid touching hot, cold, or electrical components without appropriate protection.
- Do not bypass alarms, interlocks, or temperature protections.
- Disconnect power before internal cleaning or maintenance when required by the manufacturer.

Care & storage

- Clean spills promptly using compatible cleaning products.
- Inspect door seals, sensors, lighting, and ventilation openings regularly.
- Keep a record of calibration, temperature checks, and maintenance.
- Arrange servicing if readings drift, alarms persist, or controls malfunction.

Common mistakes to avoid

- Setting incorrect environmental conditions for the experiment.
- Overloading shelves and restricting airflow.
- Ignoring alarms or failing to record environmental readings.
- Cleaning with chemicals that damage chamber surfaces or sensors.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hectolitre Weight Apparatus

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Hectolitre weight apparatus

What it is used for

To determine the bulk density or hectolitre weight of grain using a standardised filling and weighing procedure.

How to handle

- Inspect the measuring container, filling funnel, strike-off tool, balance, and accessories for cleanliness and damage.
- Place the apparatus on a stable, level surface and verify the balance is zeroed or calibrated as required.
- Fill the measuring container using the prescribed method without compressing the grain or causing unintended settling.
- Level the grain surface using the specified strike-off procedure, then weigh the filled container as instructed.
- Record the result and test conditions, empty the apparatus, and clean it before the next sample.

Storage

- Store clean and dry in a protected laboratory cabinet or designated location.
- Keep weighing and measuring components together and protect them from impact, corrosion, and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hot Air Oven

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Hot Air Oven

What it is used for

To heat and dry laboratory materials or perform approved thermal procedures using a hot air oven. To maintain the required temperature and exposure time while protecting samples and users from heat hazards.

Main parts

- Insulated oven chamber
- Heating elements
- Temperature controller and sensor
- Shelves or trays
- Door and heat-resistant seal
- Ventilation openings and control panel

Before you start

1. Read the oven manual and confirm that the intended material is suitable for hot-air heating.
2. Inspect the chamber, door seal, shelves, power cable, plug, and controls for damage.
3. Ensure the oven is clean and free of spilled chemicals or combustible residue.
4. Confirm that the temperature range and timer settings suit the approved procedure.
5. Use suitable heat-resistant containers and arrange samples so that air can circulate.
6. Ensure ventilation is adequate and the oven is placed away from flammable materials.

How to use

1. Place the oven on a stable, heat-resistant surface with sufficient clearance around it.
2. Switch off the oven before loading samples or adjusting shelf positions.
3. Load samples evenly without blocking ventilation openings or touching heating elements.
4. Close the door securely and set the required temperature and time.
5. Switch on the oven and allow it to reach the set temperature.
6. Monitor the displayed temperature and use an approved independent thermometer if the procedure requires verification.
7. Keep the door closed during heating unless the procedure specifically requires access.
8. When the cycle is complete, switch off the heater and allow the chamber to cool as required.
9. Use heat-resistant gloves and suitable tools to remove hot trays and containers.
10. Place hot items on a heat-resistant surface and label or handle them according to the procedure.
11. Switch off and disconnect the oven before cleaning or maintenance when safe to do so.
12. Clean the chamber only after it has cooled and follow the manufacturer's cleaning instructions.

⚠ Safety

- Do not heat flammable, explosive, volatile, or chemically incompatible materials unless the oven is specifically designed and approved for that use.
- Never use a hot air oven for materials that require a specialized sterilization or controlled-atmosphere system unless approved for that purpose.
- Avoid overloading the chamber and keep samples away from heating elements.
- Wear suitable heat-resistant gloves, eye protection, and laboratory clothing when handling hot items.
- Do not touch the chamber or trays with bare hands while hot.
- If smoke, unusual odour, or overheating occurs, switch off the unit if safe, keep clear, and follow laboratory emergency procedures.

Care & storage

- Clean the chamber and shelves after cooling using approved materials.
- Inspect the door seal, shelves, and electrical cord regularly.
- Keep air vents clear and avoid blocking the temperature sensor.
- Check temperature accuracy according to the laboratory schedule.
- Record faults and arrange qualified maintenance when needed.

Common mistakes to avoid

- Setting the wrong temperature or exposure time.
- Overloading the oven and restricting air circulation.
- Heating unsuitable flammable or volatile substances.
- Removing hot trays without protective equipment.
- Cleaning the oven while it is hot or connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hot Plate

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Hot plate/stirrer

What it is used for

To heat laboratory samples or solutions using a hot plate, with stirring when a magnetic stirrer is fitted. To control heating and mixing while preventing burns, spills, and electrical hazards.

Main parts

- Heating surface or ceramic plate
- Temperature control
- Power switch and indicator
- Magnetic stirrer motor, if fitted
- Stir bar and sample vessel
- Support stand or probe connection, if fitted

Before you start

1. Read the instrument manual and confirm the vessel is suitable for heating.
2. Inspect the power cord, plug, heating surface, controls, and casing for damage.
3. Place the unit on a stable, dry, heat-resistant surface away from flammable materials.
4. Check that the sample vessel is intact and suitable for the expected temperature.
5. Add a compatible magnetic stir bar if stirring is required and position the vessel centrally.
6. Confirm that the temperature probe, if used, is suitable and correctly connected.

How to use

1. Ensure the controls are set to off before connecting the unit to power.
2. Add the sample to the vessel without overfilling it.
3. Place the vessel centrally on the heating surface and keep the area around it clear.
4. Connect power and switch on the unit according to the manual.
5. Set a low initial heating level and increase it gradually to the required setting.
6. If stirring is needed, start at a low speed and increase until mixing is steady without splashing.
7. Monitor the sample temperature using the approved sensor or thermometer when required.
8. Watch for boiling, foaming, bumping, or unexpected changes and reduce heat if needed.
9. Avoid touching the hot plate, vessel, or probe with bare hands during operation.
10. After heating, switch off the heater and stirrer and allow the vessel and surface to cool.
11. Disconnect power before cleaning or moving the instrument, once it is safe to do so.
12. Clean spills only after the unit has cooled and follow the laboratory's approved cleaning procedure.

⚠ Safety

- Do not heat sealed containers or vessels that are not rated for the operating temperature.
- Keep flammable solvents and vapours away unless the equipment is specifically approved for that application.
- Use suitable heat-resistant gloves, eye protection, and laboratory clothing.
- Assume the plate remains hot after switching off; do not touch until cooled.
- Keep liquids away from electrical controls and connectors.
- Never leave a hazardous heating process unattended and follow the laboratory's emergency procedures.

Care & storage

- Clean the surface after it cools using manufacturer-approved materials.
- Inspect the power cord, housing, controls, and probe for damage regularly.
- Keep ventilation openings clear and avoid abrasive tools that can damage the surface.
- Check temperature control and stirring operation periodically.
- Store stir bars and accessories clean and dry.

Common mistakes to avoid

- Setting the heater too high at the start and causing sudden boiling or splashing.
- Using a vessel that is unsuitable for the temperature or chemical.
- Touching the hot plate immediately after switching it off.
- Allowing spills to enter electrical controls or ventilation openings.
- Cleaning or moving the unit while it is hot or still connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Incubator

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Incubator; Incubators

What it is used for

A laboratory incubator maintains a controlled temperature for growing or holding approved biological samples or cultures. Correct temperature settings, suitable loading, and contamination control help maintain sample integrity and reliable experimental conditions.

Main parts

- Insulated chamber provides the controlled environment.
- Heating element raises the chamber temperature.
- Temperature sensor measures the internal temperature.
- Thermostat or controller regulates the set temperature.
- Display shows the setpoint and measured temperature on equipped models.
- Shelves or racks hold sample containers.
- Door and gasket help maintain chamber conditions.
- Fan circulates air in forced-air models.

Before you start

1. Only trained personnel should use the incubator according to the laboratory procedure.
2. Read the manual and confirm the required temperature and operating range for the sample.
3. Inspect the chamber, shelves, door gasket, power cord, and controls for damage.
4. Ensure the chamber is clean and suitable for the intended sample type.
5. Check calibration or temperature verification status and confirm the display is functioning.
6. Prepare clearly labelled, compatible containers and follow the required containment procedure.

How to use

1. Place the incubator on a stable surface with adequate ventilation around it.
2. Connect the unit to the specified power supply and switch it on.
3. Set the approved temperature and allow the chamber to stabilise before loading samples.
4. Arrange labelled containers with adequate space for air circulation and temperature uniformity.
5. Close the door securely and avoid unnecessary opening during incubation.
6. Monitor and record temperature at the required intervals.
7. Check samples according to the approved schedule without disturbing other containers unnecessarily.
8. If a temperature alarm or deviation occurs, follow the laboratory procedure and assess sample validity.

9. After use, remove samples using the required containment practices and switch off the unit if instructed.
10. Clean and dry the chamber as required and record maintenance or temperature deviations.

⚠ Safety

- Never incubate unknown or hazardous samples without approved containment and handling procedures.
- Do not use the incubator if electrical parts, the door, or the temperature controls are damaged.
- Avoid overcrowding the chamber or blocking ventilation openings.
- Use appropriate PPE when handling biological samples and contaminated containers.
- Do not use the incubator for flammable or volatile materials unless specifically designed and approved for them.
- Report temperature deviations, spills, or equipment faults promptly.

Care & storage

- Clean the chamber and shelves with approved materials compatible with the incubator.
- Inspect the door gasket, shelves, sensor, and ventilation openings regularly.
- Keep records of temperature checks, calibration, cleaning, and maintenance.
- Arrange authorised servicing for persistent temperature instability, alarms, or electrical faults.

Common mistakes to avoid

- Loading samples before the chamber reaches the required temperature.
- Setting an incorrect temperature or ignoring temperature deviations.
- Overloading shelves and restricting air circulation.
- Using unlabelled samples or failing to follow contamination control procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lab Consumable

Category: **Laboratory Instruments & Glassware** Type: **Handling note** Priority: **Low** 15 labs

Covers these lab items: Antibiotic discs; Blood Concentrate; Blotting paper for BT; Capillary tube; Capillary tube for CT; Cell culture vessels; Filter papers; Glass Slides, coverslips & mounting media; Pasteur Pipette; Small specimen jars; Sterile Loops; Tissue paper / Filter paper / Cotton; Urine Testing strips

What it is used for

Laboratory consumables support testing, sample preparation, microscopy, culture work, and other approved laboratory procedures.

How to handle

- Verify the consumable type, intended procedure, sterility, expiry date, and package integrity before use.
- Follow the laboratory protocol and wear the PPE required for the sample or reagent.
- Use sterile consumables aseptically when required and avoid cross-contamination between samples.
- Dispose of biological, chemical, and sharp waste in the appropriate designated containers.

Storage

- Store items according to their material, sterility, temperature, and light requirements.
- Keep sealed consumables labelled and separate expired or damaged stock.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lab Infrastructure

Category: **Laboratory Instruments & Glassware** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Room for plant tissue culture lab

What it is used for

Maintain plant tissue culture laboratory infrastructure for clean and controlled work.

How to handle

- Restrict access to authorised personnel and follow laboratory entry procedures.
- Keep work surfaces clean and use designated areas for media preparation, culture handling, and waste.
- Operate ventilation, lighting, water, and sterilisation equipment according to approved procedures.
- Report equipment faults, contamination incidents, or environmental deviations promptly.

Storage

- Keep laboratory supplies in labelled, designated storage areas.
- Maintain suitable temperature, humidity, cleanliness, and security for the room and stored materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Lab Rack

Category: **Laboratory Instruments & Glassware** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Tube racks / slide racks

What it is used for

Handle tube racks and slide racks safely to organise laboratory items.

How to handle

- Check racks for cracks, deformation, and chemical residue before use.
- Use racks compatible with the tube or slide size and intended procedure.
- Place racks on stable, level surfaces and avoid overloading them.
- Treat racks used with biological or hazardous materials as potentially contaminated.

Storage

- Clean and dry reusable racks according to laboratory procedures.
- Store by type in a designated area away from contamination and physical damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Laboratory Glassware

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

5 labs

Covers these lab items: Glass wares; Glass wares (Beakers 500 ml, Funnels, Flasks conical 200 ml & 500 ml); Glassware

What it is used for

Laboratory glassware is used to hold, mix, transfer, or measure liquids during laboratory work. Choose glassware suitable for the substance, temperature, and required measurement accuracy.

How to use

1. Inspect each item for cracks, chips, contamination, or damaged markings.
2. Select the correct glassware for the task and confirm its volume or capacity.
3. Wash and rinse glassware according to the laboratory procedure before use.
4. Place it on a stable surface and handle it carefully while filling or pouring.
5. Use suitable tools or heat-resistant glassware when heating is required.
6. After use, dispose of contents correctly, clean the glassware, and return it to storage.

⚠ Safety

- Wear suitable eye protection and other required laboratory PPE.
- Do not use cracked or chipped glassware, especially for heating or pressure.
- Never heat closed glassware or use ordinary glassware over a flame unless designed for it.
- Dispose of broken glass in a designated sharps or broken-glass container.

Care & storage

- Clean, dry, and store glassware in its designated area without stacking it unsafely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Laminar Air Flow Cabinet

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Laminar Air Flow (LAF) Cabinet; Laminar Flow; Laminar flow transfer hood

What it is used for

A laminar air flow cabinet provides filtered, unidirectional airflow over a work surface to reduce particle contamination during suitable laboratory procedures. It is not a substitute for a biological safety cabinet and does not protect the operator from hazardous biological agents, toxic chemicals, or infectious aerosols.

Main parts

- Cabinet enclosure defines the protected work area.
- HEPA filter removes particles from the supplied airflow.
- Blower moves air through the filtration system.
- Work surface supports materials during procedures.
- Front opening provides access to the work zone.
- Airflow grille or diffuser distributes filtered air.
- Control panel operates the blower and fitted lighting.
- Pressure gauge or filter indicator monitors filter condition where provided.

Before you start

1. Only trained laboratory personnel should operate the cabinet for approved procedures.
2. Confirm that the cabinet type is appropriate for the intended work and that hazardous biological or chemical work is not being performed in an unsuitable unit.
3. Check the certification or maintenance status and confirm that the airflow system is operating within its approved specifications.
4. Inspect the work surface, front opening, grille, glass panels, and power cable for damage or obstruction.
5. Remove unnecessary items and disinfect the work surface using a compatible approved disinfectant.
6. Wear the required laboratory PPE and arrange materials before starting work.
7. If UV lighting is fitted, ensure it is switched off before work and follow the manufacturer's UV safety instructions.

How to use

1. Switch on the blower and allow the cabinet to stabilise for the time specified by the manufacturer.
2. Check airflow or filter indicators and report any alarm or abnormal reading before proceeding.
3. Disinfect the work surface and allow the disinfectant to dry as required.

4. Place only the materials needed for the procedure inside the work zone without blocking the airflow grille.
5. Arrange clean and used materials to reduce cross-contamination and avoid unnecessary movement across the airflow path.
6. Perform the approved procedure using slow, controlled movements and keep hands and equipment within the designated work area.
7. Keep the front opening unobstructed and avoid rapid movements that disturb airflow.
8. Monitor the cabinet for unusual noise, airflow changes, or warning indicators throughout the task.
9. Remove materials safely and disinfect the work surface after the procedure.
10. Keep the blower running for the specified purge period if required by the cabinet procedure.
11. Switch off the cabinet according to the approved shutdown procedure and record any faults.

⚠ Safety

- Do not use a laminar air flow cabinet as a substitute for a biological safety cabinet.
- Do not handle infectious agents, hazardous chemicals, or toxic vapours unless the cabinet is specifically designed and approved for that purpose.
- Never block the airflow grille or place large items where they disrupt the filtered airflow.
- Never look directly at or expose skin to an operating UV lamp.
- Do not work in the cabinet if airflow alarms, filter faults, or certification issues are unresolved.
- Follow laboratory biosafety, chemical safety, and waste-disposal procedures for all materials.

Care & storage

- Clean and disinfect the work surface after each session using compatible products.
- Keep air grilles and intake areas free of paper, packaging, and debris.
- Check the blower, lights, controls, airflow indicators, and visible seals regularly.
- Arrange scheduled certification, HEPA filter testing, and maintenance through authorised personnel.
- Report abnormal airflow, filter alarms, or cabinet damage and do not use until cleared.

Common mistakes to avoid

- Using a laminar flow cabinet for hazardous work that requires operator protection.
- Blocking the airflow path with equipment or materials.
- Starting work before the airflow stabilises or after an unresolved alarm.
- Using UV lighting while personnel are exposed to direct UV radiation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Magnifying Glass

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

10 labs

Covers these lab items: Magnifying Glass; Magnifying glass (magnification 10X); Magnifying lens(bi- concave lens)

What it is used for

Use a magnifying glass to enlarge small details for visual inspection, reading, or simple laboratory and craft tasks.

How to use

1. Check the lens and handle for cracks, scratches, or looseness.
2. Hold the object steady in a well-lit area.
3. Position the magnifying glass between your eyes and the object.
4. Adjust the distance until the details appear clear and focused.
5. Move the lens or object slowly to inspect different areas.
6. Put the magnifying glass away after use and avoid touching the lens surface.

⚠ Safety

- Never focus sunlight through the lens onto skin, eyes, or flammable materials.
- Do not use a damaged lens with sharp edges.
- Avoid looking directly at intense light sources through the lens.
- Keep the glass away from children when unsupervised.

Care & storage

- Clean the lens with a soft lens cloth and suitable cleaner.
- Store it in a protective case away from heat and scratches.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Measuring Cylinder

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

22 labs

Covers these lab items: Graduated cylinders; Measuring 250ml Cylinder; Measuring cylinder; Measuring Cylinders (5ml to 50ml) graduated; Measuring cylinders and beakers; Measuring Cylinders graduated; Measuring Glass; Measuring Glass sets

What it is used for

A measuring cylinder measures liquid volume using graduated markings. Select a suitable cylinder size and read the liquid level correctly to obtain an accurate measurement.

How to use

1. Inspect the cylinder for cracks, chips, cleanliness, and readable graduations.
2. Select a cylinder with suitable capacity and scale divisions for the required volume.
3. Place the cylinder upright on a level surface.
4. Pour the liquid slowly to avoid splashing and excessive bubbles.
5. Allow the liquid to settle before reading the level.
6. Read the bottom of the meniscus at eye level for most clear liquids.
7. Record the measured volume with units and sample identification.
8. Empty and clean the cylinder according to the liquid and laboratory procedure.

⚠ Safety

- Handle glass cylinders carefully and do not use cracked or chipped glassware.
- Wear appropriate eye protection and gloves when handling hazardous liquids.
- Never pipette or draw liquid by mouth.
- Do not heat a measuring cylinder unless it is specifically designed for that use.

Care & storage

- Rinse and clean the cylinder using suitable laboratory methods after use.
- Allow it to dry and store it upright in a protected location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Microscope

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 3 labs

Covers these lab items: Microscope

What it is used for

A microscope magnifies small objects and structures so they can be examined in greater detail. Correct slide preparation, focusing, and lens care help produce clear images and reliable observations.

Main parts

- Eyepiece or ocular lens
- Objective lenses and revolving nosepiece
- Stage and slide holder
- Coarse and fine focus controls
- Illuminator or mirror and brightness control
- Base, arm, and power switch where applicable

Before you start

1. Place the microscope on a stable, clean surface away from vibration and direct sunlight.
2. Inspect the eyepiece, objective lenses, stage, power cable, and focus controls for damage.
3. Clean lenses only with approved lens paper and suitable lens-cleaning solution.
4. Prepare a correctly labelled slide with a coverslip when required by the specimen method.
5. Start with the lowest-power objective and ensure the stage and lens are clear.
6. Switch on the illuminator and adjust the light level to a suitable starting setting.

How to use

1. Place the slide on the stage and secure it with the slide holder.
2. Centre the specimen over the stage opening using the stage controls.
3. Look through the eyepiece and adjust the light or diaphragm for suitable contrast.
4. Use the coarse focus control carefully to bring the specimen into initial focus at low magnification.
5. Use the fine focus control to sharpen the image.
6. Adjust the illumination and diaphragm to improve image contrast without excessive brightness.
7. Rotate to a higher-power objective only after centring and focusing the specimen at low power.
8. Use fine focus for higher magnification and avoid allowing the objective lens to touch the slide.
9. Record observations and magnification as required by the laboratory procedure.
10. After examination, return to the lowest-power objective, remove the slide, switch off the light, and clean the work area.

⚠ Safety

- Carry the microscope with one hand supporting the arm and the other supporting the base.
- Do not touch glass slides with bare hands when specimens or reagents may be hazardous.
- Keep liquids away from electrical parts and disconnect power before approved maintenance.
- Avoid forcing focus controls or rotating the nosepiece against resistance.
- Dispose of broken slides and contaminated specimens in the designated containers.
- Do not clean lenses with tissues, clothing, or unapproved solvents.

Care & storage

- Clean eyepiece and objective lenses with approved lens paper after use when needed.
- Keep the microscope covered and store it in a dry, dust-free location.
- Inspect the stage, focus controls, illuminator, and power cable regularly.
- Arrange servicing and optical alignment checks according to the manufacturer's recommendations.

Common mistakes to avoid

- Using coarse focus at high magnification can cause the objective to strike the slide.
- A dirty lens or slide can make the image blurry or misleading.
- Excessive illumination can reduce contrast and hide specimen details.
- Carrying the microscope by the eyepiece or stage can damage the instrument.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Petri Dish

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

5 labs

Covers these lab items: Culture dishes; Petri dish; Petri dish (80X70 mm)

What it is used for

A Petri dish is a shallow dish used to hold samples or culture microorganisms in laboratory work. Use sterile dishes when required by the test procedure.

How to use

1. Inspect the dish for cracks, chips, contamination, or damaged lids.
2. Select a clean or sterile dish according to the laboratory procedure.
3. Label the dish before adding the sample or culture medium.
4. Open the lid only as much and for as long as necessary to reduce contamination.
5. Add the sample using the approved laboratory method and close the lid carefully.
6. Dispose of used dishes or clean reusable ones according to laboratory protocol.

⚠ Safety

- Wear suitable laboratory PPE and follow the required aseptic technique.
- Do not touch sterile inner surfaces with unclean hands or tools.
- Treat used culture dishes as potentially infectious and dispose of them correctly.

Care & storage

- Keep clean dishes covered and store them in a dry, protected area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Physical Balance

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Physical balance(01mg) / Counter Balance (01gm sensitivity)

What it is used for

To determine the mass of a sample by balancing it against known standard weights.

How to handle

- Place the balance on a stable, level surface away from vibration and air currents.
- Check that the pans, beam or pointer, arrest mechanism, and weights are clean and undamaged.
- Ensure the balance is at zero before measurement and use suitable standard weights with clean forceps where appropriate.
- Place the sample carefully on the designated pan and add weights gradually until equilibrium is reached.
- Record the mass and balance identification, then remove the sample and weights and return the balance to its resting position.

Storage

- Keep the balance clean, covered, and level in a dry laboratory location.
- Store standard weights in their labelled case and protect the balance from shock, dust, and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pipette

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Glass pipettes; Pipettes Glass - (1 mL, 2 mL, 5 mL, 10 mL); RBC pipette

What it is used for

To measure and transfer a specified volume of liquid accurately during laboratory procedures.

How to handle

- Inspect the pipette for cracks, chips, contamination, and clear graduation marks before use.
- Select the correct pipette type and volume range for the task and use a compatible pipette aid where required.
- Wear appropriate laboratory PPE and draw liquid without using mouth suction.
- Read the meniscus at eye level where applicable and dispense the liquid according to the pipette's calibration and instructions.
- Clean or dispose of the pipette according to laboratory procedures, and never return unused liquid to the original container.

Storage

- Store clean and dry in a pipette rack or protective case, away from breakage risks.
- Keep pipettes separated by type and volume and protect graduation marks from abrasion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Refractometer

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

4 labs

Covers these lab items: Hand Refractometer; Refractometer

What it is used for

To measure the refractive index or concentration of a suitable liquid using a refractometer. To obtain accurate results by calibrating the instrument and preparing the sample correctly.

Main parts

- Prism or sample surface
- Cover plate
- Eyepiece or digital display
- Calibration control
- Sample dropper

Before you start

1. Check the refractometer type and measurement range for the sample.
2. Clean and dry the prism surface.
3. Calibrate using the specified reference solution or distilled water as instructed.
4. Mix the sample gently and bring it to the required temperature if specified.
5. Check that the scale or display is readable.

How to use

1. Place a small amount of sample on the prism surface.
2. Close the cover plate gently if the model has one.
3. Aim the instrument toward a suitable light source if required for an optical model.
4. Focus the eyepiece and read the boundary line, or read the digital display.
5. Apply temperature correction if required by the method.
6. Record the result and sample details.
7. Clean the prism immediately after measurement using an approved method.
8. Store the instrument safely.

⚠ Safety

- Follow sample-specific precautions, especially for corrosive or hazardous liquids.
- Do not look directly at the sun through an optical instrument.
- Avoid scratching the prism surface.
- Use clean droppers to prevent cross-contamination.

Care & storage

- Clean and dry the prism after each sample.
- Store with the cover closed and protective case fitted.
- Check calibration regularly.
- Avoid dropping or exposing the instrument to excessive heat.

Common mistakes to avoid

- Skipping calibration.
- Using a dirty prism or contaminated dropper.
- Reading before focusing correctly.
- Ignoring required temperature correction.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sample Collection Tool

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Sample collection tools (where permitted)

What it is used for

Sample collection tools are used to collect permitted specimens or environmental materials for testing and assessment. Follow the approved collection protocol, contamination controls, and applicable permissions for the sample type.

How to use

1. Confirm the sample type, purpose, collection location, and required authorization.
2. Review the sampling protocol, hazard information, and required personal protective equipment.
3. Inspect collection tools and containers for cleanliness, integrity, and suitability.
4. Label the sample container with the required identification and collection details before or immediately after sampling.
5. Collect the sample using the approved technique while avoiding cross-contamination.
6. Seal and transport the sample under the specified temperature and handling conditions.
7. Decontaminate or dispose of used tools appropriately and complete the sample record.

⚠ Safety

- Collect only samples that are legally permitted and within your training and authorization.
- Use suitable PPE and follow biosafety or chemical safety procedures for the material.
- Never collect unknown hazardous material without an approved risk assessment and trained supervision.
- Handle sharps and potentially infectious samples using the designated precautions.

Care & storage

- Clean reusable tools according to the validated procedure and store them separately from clean supplies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sample Container

Category: **Laboratory Instruments & Glassware** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Sample Collection Bottle; Sample Container / Swab sticks / culture bottles / Syringe & Needles; Uro Bag Sample Collection Bottle

What it is used for

Sample containers collect, hold, and transport specimens or materials for testing and analysis.

How to handle

- Select a container suitable for the sample type and required test.
- Check container integrity, cleanliness, sterility, expiry date, and closure before use.
- Label each container with the required identification and collection details.
- Use appropriate PPE and avoid touching the inside of sterile containers or lids.
- Handle biological specimens according to the applicable biosafety and disposal procedures; needles and syringes require approved sharps handling.

Storage

- Keep sterile containers sealed until use.
- Store samples under the specified temperature and transport conditions in a secure, labelled area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spirit Lamp

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Spirit Lamp

What it is used for

A spirit lamp burns an approved liquid fuel to provide a small flame for laboratory heating or demonstrations. It must be used only under teacher supervision with suitable fire precautions.

Main parts

- Identify the fuel reservoir or lamp body.
- Locate the wick and its adjustable holder, if provided.
- Identify the wick opening and flame area.
- Locate the cap used to extinguish and cover the lamp.
- Identify the stable, heat-resistant base.
- Locate suitable ignition equipment and approved fire-control materials.

Before you start

1. Use the spirit lamp only when the teacher authorises the activity.
2. Check the lamp body, wick holder and cap for damage or leakage.
3. Use only the fuel specified for the lamp and laboratory procedure.
4. Fill the reservoir only to the recommended level while the lamp is cool and extinguished.
5. Wipe away any spilled fuel and allow vapours to disperse before lighting.
6. Place the lamp on a stable, heat-resistant surface away from paper and chemicals.
7. Tie back long hair and wear suitable eye protection and laboratory clothing.

How to use

1. Confirm that the wick is fitted correctly and is not excessively long.
2. Ensure the outside of the lamp is clean and dry.
3. Use an approved dropper or filling method to add the specified fuel if needed.
4. Close the fuel container and move it away from the lighting area.
5. Ignite the wick using an approved long-reach ignition tool.
6. Adjust the flame only if the lamp design permits safe adjustment.
7. Keep laboratory materials at a safe distance while heating.
8. Extinguish the flame by placing the designated cap over the wick.
9. Confirm the flame is completely extinguished before moving the lamp.
10. Allow the lamp to cool fully before refilling or storing it.

⚠ Safety

- Never refill a lit or hot spirit lamp.
- Never blow on the flame or attempt to extinguish it with water.
- Keep fuel containers tightly closed and away from ignition sources.
- Never leave a burning lamp unattended or carry it while lit.
- Do not use a leaking lamp or a damaged wick holder.
- If fuel ignites outside the lamp, alert the teacher and use the approved fire response procedure.

Care & storage

- Allow the lamp to cool before cleaning or handling the wick.
- Wipe away fuel residue and inspect the reservoir for cracks or leaks.
- Replace a damaged or heavily contaminated wick as instructed.
- Store the empty or approved fuel-filled lamp according to laboratory rules, away from heat and unauthorised access.

Common mistakes to avoid

- Overfilling the reservoir or spilling fuel near the flame area.
- Refilling the lamp while it is hot or still burning.
- Leaving the lamp uncovered or unattended while lit.
- Using an unsuitable fuel or a damaged wick assembly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

TDS Meter

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

2 labs

Covers these lab items: Water testing instrument (TDS meter)

What it is used for

A TDS meter estimates the concentration of dissolved substances in water through electrical conductivity. Use it for comparative water testing, not as proof that water is safe to drink.

How to use

1. Check the meter, display, probe, and battery for damage or low power.
2. Calibrate the meter if required by its instructions.
3. Collect a water sample in a clean container and avoid touching the probe tip.
4. Switch on the meter and rinse the probe as instructed.
5. Immerse the probe to the specified level without touching the container bottom.
6. Wait for the reading to stabilize and record the value with units.
7. Rinse the probe after testing and switch off the meter.

⚠ Safety

- Never immerse the meter body or display unless the device is rated for it.
- Do not taste test samples or assume a low TDS reading means safe drinking water.
- Follow lab hygiene rules when handling unknown or contaminated water.

Care & storage

- Rinse and dry the probe as instructed and store the meter in its protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tensiometer

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Tensiometer

What it is used for

A tensiometer measures surface tension or interfacial tension of liquids, depending on the instrument type. Use the specified measurement method and a clean measuring probe to obtain reliable results.

Main parts

- Tensiometer main unit and display
- Force sensor or balance mechanism
- Ring, plate, or probe appropriate to the method
- Sample container and sample holder
- Calibration weights or reference standard, if specified
- Control buttons and measurement software, if available

Before you start

1. Place the instrument on a stable, level surface away from vibration and airflow.
2. Check the instrument, probe, cable, and sample holder for damage.
3. Confirm the instrument is suitable for the required surface or interfacial tension measurement.
4. Clean and dry the probe using the approved method.
5. Prepare a sufficient quantity of sample in a clean container.
6. Switch on the instrument and allow it to stabilize as recommended.
7. Verify calibration or zero setting according to the manufacturer's instructions.

How to use

1. Select the correct measurement mode and units on the instrument.
2. Install the clean ring, plate, or probe securely.
3. Place the sample container on the holder and position it correctly.
4. Set the probe height or sample stage as required by the method.
5. Zero the instrument or confirm the baseline reading.
6. Lower the probe into contact with the liquid using the prescribed procedure.
7. Start the measurement and allow the instrument to complete its reading.
8. Record the result, units, sample identification, and test conditions.
9. Repeat the measurement if required and compare the readings for consistency.
10. Clean the probe and sample holder after testing, then switch off the instrument.

⚠ Safety

- Handle glass sample containers carefully to prevent cuts and spills.
- Wear suitable gloves and eye protection when handling chemical samples.
- Keep liquids away from electrical controls and instrument connectors.
- Do not exceed the instrument's specified load or probe travel limits.
- Use only approved cleaning solvents and follow their safety instructions.

Care & storage

- Clean and dry the probe after every test using the recommended procedure.
- Store probes safely to prevent bending, contamination, or surface damage.
- Keep the instrument level and protect it from vibration and dust.
- Arrange calibration and servicing at the intervals specified by the manufacturer.

Common mistakes to avoid

- Using a contaminated or wet probe that changes the measured surface tension.
- Selecting the wrong measurement mode or units.
- Touching or moving the instrument during measurement and causing unstable readings.
- Using an unsuitable sample container or incorrect immersion depth.
- Recording readings without sample identification or test conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Test Tube

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

5 labs

Covers these lab items: Culture tubes; Glass tubes

What it is used for

A test tube holds, mixes, or heats small quantities of laboratory materials when suitable for the procedure. Choose a tube made from material compatible with the sample and temperature.

How to use

1. Inspect the test tube for cracks, chips, contamination, or damaged rims.
2. Select a clean tube suitable for the substance and intended procedure.
3. Label the tube and place it securely in a rack before filling.
4. Add the sample carefully, leaving sufficient space to prevent splashing.
5. If heating is required, use suitable glassware and heat it according to laboratory instructions.
6. After use, dispose of contents correctly and clean or discard the tube according to protocol.

⚠ Safety

- Wear suitable eye protection and laboratory PPE.
- Never point an open test tube toward yourself or another person.
- Do not heat a sealed test tube or use cracked glassware.
- Use a suitable holder when handling hot test tubes.

Care & storage

- Wash reusable tubes, allow them to dry, and store them upright in a rack.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Test Tube Rack

Category: **Laboratory Instruments & Glassware** Type: **Handling note** Priority: **Low**

1 labs

Covers these lab items: Culture tube racks

What it is used for

Handle culture tube racks safely for organised laboratory work.

How to handle

- Check racks for cracks, deformation, residue, or instability before use.
- Use racks compatible with the culture tube size and intended procedure.
- Place racks on a stable, level surface and avoid overloading them.
- Treat racks exposed to biological cultures as potentially contaminated and follow laboratory decontamination procedures.

Storage

- Clean and dry reusable racks according to laboratory protocols.
- Store in a designated area separated from clean consumables and contaminated equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Viscometer

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Viscometer

What it is used for

A viscometer measures the viscosity of liquids, indicating their resistance to flow. Select the appropriate instrument and method for the sample to obtain consistent and meaningful results.

Main parts

- Viscometer main unit and display
- Spindle, rotor, capillary, or measuring probe as applicable
- Sample container or measurement tube
- Temperature sensor or temperature-controlled bath, if fitted
- Speed control and measurement buttons
- Stand, clamp, or sample holder
- Calibration fluid and cleaning accessories

Before you start

1. Place the instrument on a stable, level surface away from vibration.
2. Inspect the instrument, spindle, probe, and cables for damage.
3. Select the correct spindle or measuring system for the sample and expected viscosity.
4. Prepare a clean sample container with sufficient sample volume.
5. Bring the sample to the specified test temperature and allow it to stabilize.
6. Check calibration status and verify zero setting where applicable.
7. Confirm that the sample is mixed uniformly without introducing air bubbles.

How to use

1. Switch on the viscometer and select the required measurement mode.
2. Enter the test temperature, spindle type, speed, or method settings as applicable.
3. Attach the clean spindle or probe securely before starting the measurement.
4. Place the sample container in the correct position and set the required immersion depth.
5. Start the instrument at the recommended speed and allow the reading to stabilize.
6. Check that the displayed torque or measurement indicator is within the instrument's valid range.
7. Record viscosity, units, temperature, spindle or probe, and operating speed.
8. Repeat the measurement if required using the same controlled test conditions.
9. Stop the instrument and remove the spindle or probe carefully.
10. Clean and dry the measuring parts according to the manufacturer's instructions.

⚠ Safety

- Wear suitable gloves and eye protection when handling chemical or hot samples.
- Keep liquids away from electrical controls and connectors.
- Do not touch or adjust a moving spindle during operation.
- Avoid contact with hot samples, baths, or heated components.
- Use only approved cleaning agents compatible with the instrument and sample.

Care & storage

- Clean the spindle, probe, and sample holder immediately after use.
- Handle measuring spindles carefully to prevent bending or surface damage.
- Store the instrument in a dry, dust-free location with accessories protected.
- Verify calibration using suitable standards at the recommended intervals.

Common mistakes to avoid

- Measuring at an incorrect or unstable sample temperature.
- Selecting the wrong spindle, speed, or measurement range.
- Introducing air bubbles or failing to mix the sample uniformly.
- Recording readings before they stabilize or outside the valid torque range.
- Leaving residue on the spindle and affecting subsequent measurements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wash Bottle

Category: **Laboratory Instruments & Glassware** Type: **Handling note** Priority: **Low**

1 labs

Covers these lab items: Wash bottles

What it is used for

Handle laboratory wash bottles safely for dispensing water or approved laboratory solutions.

How to handle

- Label each bottle clearly with its contents and any applicable hazard information.
- Use only compatible liquids and never fill a bottle with an unidentified substance.
- Direct the nozzle away from the face and others; avoid creating aerosols when dispensing hazardous solutions.
- Inspect the bottle, cap, and nozzle for leaks or damage before use.

Storage

- Empty, clean, and dry bottles according to laboratory procedures when appropriate.
- Store labelled bottles upright in a designated area, separated by chemical compatibility.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Analyser

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Water and juice analysis machine

What it is used for

A water analyser measures selected water-quality parameters, and some models also analyse juice or other liquid samples. Use the correct test method, sensor, and calibration standards for each parameter to obtain reliable results.

Main parts

- Water analyser main unit and display
- Parameter-specific sensors, electrodes, or probes
- Sample cells, cuvettes, or test vessels
- Reagents and test kits where required
- Calibration standards and reference solutions
- Sample holder and temperature sensor, if fitted
- Power supply, batteries, and data recording system

Before you start

1. Identify the sample type and the parameters to be tested.
2. Read the instrument manual to confirm the approved method and sample compatibility.
3. Inspect the instrument, sensors, cables, and sample cells for damage or contamination.
4. Prepare fresh reagents and calibration standards according to the instructions.
5. Calibrate each required parameter using the specified standards.
6. Collect a representative sample in a clean, suitable container.
7. Bring samples and standards to the required temperature if the method specifies it.

How to use

1. Switch on the analyser and select the required test parameter or method.
2. Rinse reusable probes or sample cells as specified by the method.
3. Prepare the sample and add reagents when required, following the test-kit instructions.
4. Insert the sample cell or immerse the probe to the specified depth.
5. Wait for the required reaction time or sensor stabilisation period.
6. Read the result and confirm that it falls within the instrument's measurement range.
7. Record the sample identification, parameter, result, units, temperature, and test method.
8. Repeat the test or use a suitable dilution when the result is outside the valid range.
9. Dispose of used samples and reagents according to laboratory procedures.
10. Clean the probes and sample cells, then switch off and store the instrument correctly.

⚠ Safety

- Wear laboratory coat, suitable gloves, and eye protection when handling samples or reagents.
- Never taste test samples, including juice or water of unknown quality.
- Follow reagent labels and safety data sheets for handling and disposal.
- Keep liquids away from electrical connections and the instrument body.
- Do not mix reagents unless the approved test method instructs you to do so.
- Treat potentially contaminated water samples as hazardous and wash hands after testing.

Care & storage

- Rinse and dry sensors or sample cells as recommended for each parameter.
- Store electrodes and probes in the specified storage solution when required.
- Keep reagents within their expiry dates and follow storage instructions.
- Check calibration and perform routine quality-control checks using reference standards.

Common mistakes to avoid

- Using expired reagents or incorrect calibration standards.
- Testing a sample with the wrong method or incompatible sensor.
- Contaminating samples through dirty cells, probes, or containers.
- Ignoring reaction time, temperature requirements, or instrument measurement limits.
- Recording results without units, sample identification, or calibration checks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Bath

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Water Bath

What it is used for

A laboratory water bath maintains samples at a controlled temperature using heated water. It is used for incubation, warming, and temperature-dependent procedures that require uniform heating.

Main parts

- Insulated bath tank
- Heating element
- Temperature sensor and controller
- Sample racks or holders
- Lid or cover
- Drain outlet and power switch

Before you start

1. Read the operating instructions and confirm the required temperature range for the procedure.
2. Inspect the tank, power cable, controller, sensor, lid, and drain for damage.
3. Place the water bath on a stable, level, dry surface away from splashes.
4. Fill the tank with the recommended water to the specified operating level.
5. Check that the heating element is covered to the required level before switching on.
6. Verify the temperature sensor and any required calibration status.

How to use

1. Switch on the water bath and set the target temperature using the controller.
2. Allow the water to heat and stabilize at the set temperature.
3. Check the displayed temperature against an independent thermometer when required by the procedure.
4. Place samples in suitable sealed or approved containers and arrange them in the rack.
5. Lower the rack carefully so water does not enter sample containers.
6. Ensure the water level remains within the operating range during use.
7. Monitor the temperature and sample exposure time throughout the procedure.
8. Remove samples using suitable tools or heat-resistant gloves when the procedure is complete.
9. Switch off the heater and allow the water to cool before draining or cleaning.
10. Drain, clean, and dry the tank according to the laboratory procedure.

⚠ Safety

- Do not operate the bath without sufficient water or with an exposed heating element unless designed for it.
- Avoid touching hot water, tank surfaces, or sample containers with bare hands.
- Keep electrical connections and the control panel dry.
- Use suitable racks and containers that tolerate the operating temperature.
- Disconnect power before draining, cleaning, or servicing the equipment.
- Do not use the bath for flammable or hazardous liquids unless specifically approved by the manufacturer and laboratory procedure.

Care & storage

- Replace bath water at the frequency required by the laboratory procedure.
- Clean the tank and remove scale using approved materials.
- Inspect the sensor, heater, cable, and controller for damage or abnormal operation.
- Record temperature checks, calibration, cleaning, and maintenance as required.

Common mistakes to avoid

- Operating with too little water can overheat and damage the heater.
- Overfilling the tank can cause spills when racks or samples are inserted.
- Using an unsealed sample container can allow water to contaminate the sample.
- Ignoring temperature drift can invalidate temperature-sensitive test results.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Testing Kit

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Water testing kit

What it is used for

A water testing kit measures selected physical or chemical parameters in a water sample using test strips, reagents, meters, or comparison charts. Correct sampling and test procedures help produce reliable results for water-quality screening.

Main parts

- Sample collection bottle or container
- Test strips, reagent bottles, or test tablets
- Colour comparison chart or digital meter
- Sample tubes, cuvettes, or measuring accessories
- Dropper, scoop, or measuring syringe
- Instruction sheet and result record form

Before you start

1. Read the kit instructions and identify the parameters and measurement ranges it supports.
2. Check expiry dates, reagent condition, meter battery, and calibration requirements.
3. Use clean sample containers suitable for the parameter being tested.
4. Collect a representative sample using the prescribed sampling method and avoid touching the inside of the container.
5. Label the sample with its source, location, date, and collection time.
6. Wear suitable gloves and eye protection when handling reagents, and review their safety information.

How to use

1. Allow the sample to reach the temperature required by the kit instructions if necessary.
2. Measure the specified sample volume using the supplied or approved measuring accessory.
3. Add the correct test strip, tablet, or reagent quantity in the required order.
4. Mix or wait for the reaction for the exact time specified by the manufacturer.
5. For strip tests, compare the colour within the stated reading window and under suitable lighting.
6. For digital meters, calibrate as required, rinse the probe, and immerse it to the specified depth.
7. Record the displayed value or colour result with the correct units and parameter name.
8. Repeat the test when quality-control instructions require confirmation or the result appears abnormal.
9. Dispose of used strips, reagents, and samples according to the kit instructions and laboratory rules.

10. Clean the equipment, close reagent containers, and store the kit under the recommended conditions.

⚠ Safety

- Never taste or directly smell a water sample or chemical reagent.
- Wear gloves and eye protection when the reagent safety information requires them.
- Do not mix reagents from different kits or use expired chemicals.
- Keep reagents away from children, food, and drinking-water containers.
- Follow laboratory procedures for handling samples suspected of containing biological or chemical hazards.
- Do not assume a screening kit proves water is safe for drinking; use an approved laboratory when required.

Care & storage

- Keep test strips and reagents sealed and protected from heat, moisture, and sunlight.
- Rinse probes and sample tubes with the recommended water after use.
- Calibrate digital meters and replace batteries according to the manufacturer's schedule.
- Record reagent lot numbers, expiry dates, calibration, and quality-control results where required.

Common mistakes to avoid

- Using contaminated containers can change the measured result.
- Reading a colour strip too early or too late can produce an incorrect value.
- Using expired reagents or an uncalibrated meter can reduce accuracy.
- Recording results without units or sample identification can make them unreliable or unusable.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weight Box

Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Weight Box (1mg to 200gm)

What it is used for

To provide standard masses for weighing, balance checking, and laboratory measurement activities.

How to handle

- Inspect the weight box, labels, and individual weights for damage or contamination.
- Use clean forceps or the supplied handling tool for small precision weights; avoid touching precision surfaces with bare fingers.
- Select the required weight and place it gently at the centre of the balance pan.
- Follow the balance procedure and record the reading; do not drop or drag weights across the pan.
- Return each weight to its labelled slot immediately after use.

Storage

- Keep weights clean, dry, and in their original labelled compartments.
- Close the box after use and store in a secure, low-vibration area away from chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

pH Meter

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 6 labs

Covers these lab items: pH meter; Ph Paper Strips/Small pH Meter

What it is used for

A pH meter measures how acidic or alkaline a liquid sample is by detecting the response of a pH electrode. Proper calibration, rinsing, and storage of the electrode help produce reliable measurements.

Main parts

- pH meter display and control buttons
- pH electrode or combined probe
- Temperature sensor or automatic temperature compensation feature if fitted
- Electrode storage solution and protective cap
- Calibration buffer solutions
- Sample beakers, rinse bottle, and lint-free tissue

Before you start

1. Read the instrument instructions and confirm the expected pH range of the sample.
2. Inspect the meter, electrode cable, probe body, and connector for damage.
3. Check the electrode condition and ensure its sensing bulb or junction is not dry or cracked.
4. Prepare fresh, suitable calibration buffers that bracket the expected sample pH.
5. Switch on the meter and allow it to stabilize as specified by the manufacturer.
6. Wear suitable gloves and eye protection when handling samples or buffer chemicals.

How to use

1. Rinse the electrode with distilled or deionized water and gently blot it without rubbing the sensing bulb.
2. Calibrate the meter using the required buffer solutions and follow the displayed prompts.
3. Rinse the electrode between buffer solutions to prevent cross-contamination.
4. Pour a representative sample into a clean beaker without contaminating the main sample container.
5. Immerse the electrode and temperature probe to the recommended depth without touching the beaker bottom.
6. Gently stir if required and wait for the reading to stabilize.
7. Record the pH value, sample identification, temperature, and any required measurement conditions.
8. Rinse the electrode after measurement and repeat the test if the result is unstable or outside expected limits.

9. Store the electrode in the recommended storage solution and replace its protective cap.
10. Switch off the meter and clean the work area after testing.

⚠ Safety

- Never handle electrodes by pulling their cables or strike the glass sensing bulb against hard surfaces.
- Wear suitable protection when testing corrosive, toxic, or biologically hazardous samples.
- Do not store the electrode in distilled water unless the manufacturer specifically permits it.
- Keep electrical connectors dry and disconnect the meter before cleaning non-waterproof components.
- Do not return tested samples to their original containers.
- Dispose of buffer solutions and samples according to laboratory procedures.

Care & storage

- Rinse the electrode after every sample and remove deposits using approved cleaning solution.
- Store the electrode in the specified solution and keep its sensing element hydrated as instructed.
- Calibrate at the required frequency and whenever results or conditions indicate drift.
- Inspect electrode response, cable condition, and meter battery or power supply regularly.

Common mistakes to avoid

- Skipping calibration can produce inaccurate pH readings.
- Wiping the sensing bulb harshly can create static effects or damage the electrode.
- Storing the electrode dry or in an unsuitable liquid can shorten its service life.
- Using contaminated or expired buffers can make calibration unreliable.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

pH Test Strip

Category: **Laboratory Instruments & Glassware** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: pH strip; Ph strips

What it is used for

pH test strips estimate the acidity or alkalinity of a liquid by comparing the strip colour with a reference chart.

How to handle

- Check the strip type, expiry date, packaging integrity, and measurement range before use.
- Use a clean sample container and avoid touching the reactive test area.
- Follow the specified immersion time and reading instructions.
- Compare the colour under suitable lighting and record the result with the sample identification.
- Do not reuse strips or return used strips to the original container.

Storage

- Keep unused strips sealed in their original container.
- Store dry and within the temperature and humidity limits specified by the manufacturer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Anemometer

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Anemometer

What it is used for

To measure air or wind speed for educational, environmental, or workplace monitoring activities.

How to handle

- Inspect the anemometer, sensor, display, and battery compartment for damage.
- Select the appropriate measurement mode and units according to the instrument manual.
- Place the sensor in the airflow as specified, away from obstructions and body interference where possible.
- Allow the reading to stabilize, then record the value, units, location, and relevant conditions.
- Switch off the instrument after use and clean the sensor gently without damaging moving parts.

Storage

- Switch off the instrument and remove batteries for long-term storage if recommended by the manufacturer.
- Store in its protective case in a clean, dry place away from dust, moisture, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Barometer

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Barometer

What it is used for

A barometer measures atmospheric pressure, which helps observe weather changes. Place it correctly and allow the reading to stabilize before recording measurements.

How to use

1. Identify whether the barometer is digital, aneroid, or another type.
2. Inspect the instrument, display, dial, and housing for damage.
3. Place the barometer in a stable location away from direct heat and strong airflow.
4. Allow the instrument to settle according to its instructions.
5. Read the pressure value and record the units, date, and time.
6. Compare readings over time without moving the instrument unnecessarily.
7. Switch off a digital barometer after use if required by its instructions.

⚠ Safety

- Do not open a sealed barometer or handle any hazardous internal liquid.
- Keep the instrument away from water, impact, and extreme temperatures.
- Use only the specified power supply for digital models.

Care & storage

- Dust the housing gently and store the barometer in a dry, protected place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bevel

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Bevel

What it is used for

To set out or transfer an angle using an adjustable bevel. To reproduce an angle accurately by aligning the blade with a reference edge and locking it in position.

Main parts

- Stock or handle
- Adjustable blade
- Pivot joint
- Locking screw or clamp

Before you start

1. Inspect the blade, stock, pivot, and locking screw for damage.
2. Check that the blade moves smoothly and locks securely.
3. Prepare the workpiece and a suitable marking tool.
4. Identify the reference edge and required angle.

How to use

1. Loosen the locking screw enough to adjust the blade.
2. Align the stock with the reference edge or existing angle.
3. Rotate the blade to match the required angle.
4. Tighten the locking screw to hold the setting.
5. Transfer the angle to the workpiece and mark the line.
6. Check the alignment before cutting or machining.

⚠ Safety

- Keep fingers clear of sharp blade edges.
- Do not force a stiff or damaged pivot.
- Lock the blade before transferring the angle.
- Store the bevel with the blade safely folded or secured.

Care & storage

- Wipe the blade and stock clean after use.
- Keep the pivot free from debris.
- Check the locking screw for wear.

- Store in a dry, protected place.

Common mistakes to avoid

- Failing to tighten the locking screw.
- Aligning the stock against the wrong reference edge.
- Allowing the blade to shift during marking.
- Using a damaged or loose pivot.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Centre Punch

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Centre punch

What it is used for

To create a small indentation that guides a drill bit on a marked point. To improve drilling accuracy and reduce bit slipping.

Main parts

- Punch tip
- Steel body
- Striking head
- Marking point

Before you start

1. Mark the drilling location clearly.
2. Check the punch tip and striking head for damage.
3. Secure the workpiece on a stable surface.
4. Wear safety glasses.

How to use

1. Place the punch tip on the marked point.
2. Hold the punch upright and steady.
3. Strike the head gently with a suitable hammer.
4. Check the indentation and repeat with controlled force if needed.
5. Move the punch away before drilling.

⚠ Safety

- Keep fingers clear of the hammer striking area.
- Do not use a cracked or mushroomed punch head.
- Secure small workpieces before striking.
- Wear eye protection against flying metal fragments.

Care & storage

- Wipe the punch clean after use.
- Inspect the tip and head regularly.
- Store with other marking tools in a dry place.

Common mistakes to avoid

- Holding the punch at an angle.
- Striking too hard and damaging the workpiece.
- Using a damaged punch.
- Failing to secure the workpiece.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Combination Set

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Combination set

What it is used for

A combination set provides tools such as a rule, square head, protractor, and centre head for measuring and marking workpieces. Assemble the required components correctly and verify alignment before use.

How to use

1. Inspect the rule, heads, locking screws, and graduations for damage or wear.
2. Select the required head and slide it onto the rule correctly.
3. Tighten the locking screw enough to hold the component securely.
4. Place the tool against the workpiece reference edge or surface.
5. Read measurements at eye level and mark lines using a suitable marking tool.
6. Check squareness or angles against the required reference before cutting or machining.
7. Loosen and remove components carefully and return the set to its case.

⚠ Safety

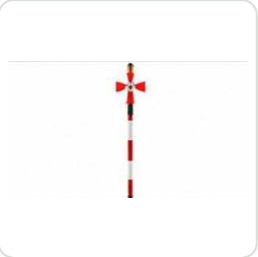
- Keep fingers away from sharp edges and marking points.
- Do not use a damaged rule or unreadable graduations for precision work.
- Secure the workpiece before marking or checking near machinery.

Care & storage

- Wipe components clean, protect the graduations, and store the set in its case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cross Staff



Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Cross staff

What it is used for

To establish right-angle offsets and assist with basic surveying and alignment work.

How to handle

- Inspect the cross staff, sighting slots or vanes, and mounting arrangement for damage.
- Set the staff vertically on a suitable pole or support at the required survey point.
- Sight through the appropriate openings to align the reference line and identify the right-angle offset.
- Keep the instrument steady and record the measured or marked points clearly.
- Avoid forcing the fittings and clean the instrument after use.

Storage

- Clean and dry the instrument before storage.
- Store in its protective case or designated rack to prevent bending and damage to sighting parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dial Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Dial gauge

What it is used for

To measure small linear movement, runout, or dimensional variation using a dial gauge. To obtain repeatable readings with correct alignment and a stable setup.

Main parts

- Dial face and pointer
- Graduated scale
- Contact plunger or probe
- Mounting stem or clamp
- Magnetic base or fixture, if supplied

Before you start

1. Inspect the dial face, probe, and mounting parts for damage.
2. Confirm the gauge range and resolution suit the measurement.
3. Clean the gauge contact point and workpiece surface.
4. Secure the gauge stand and workpiece firmly.
5. Check zero setting and calibration status where required.

How to use

1. Mount the dial gauge securely on a stable stand.
2. Position the probe against the measurement surface at the recommended angle.
3. Apply the specified preload without exceeding the gauge travel.
4. Set the dial to zero or record the initial reading.
5. Move or rotate the workpiece slowly as required.
6. Observe and record the pointer movement or maximum variation.
7. Remove the gauge carefully and compare results with the required tolerance.

⚠ Safety

- Keep fingers clear of rotating or moving machinery.
- Never exceed the probe travel limit.
- Do not force the probe sideways.
- Stop machinery and isolate it before mounting or adjusting the gauge when required.

Care & storage

- Wipe the probe and gauge body clean after use.
- Retract and protect the probe during storage.
- Avoid shocks, drops, and moisture.
- Arrange calibration checks according to workplace requirements.

Common mistakes to avoid

- Mounting the gauge on an unstable base.
- Misaligning the probe with the measurement direction.
- Exceeding the gauge travel range.
- Forgetting to zero the dial or record the initial reading.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Divider

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Divider

What it is used for

To transfer measurements, compare distances, and mark arcs or circles on suitable workpieces.

How to handle

- Inspect the divider points, legs, hinge, and adjustment mechanism for damage or looseness.
- Set the required distance using a ruler or drawing scale before transferring measurements.
- Hold the divider steadily and place its points carefully on the reference marks or workpiece.
- Keep sharp points away from fingers and avoid using excessive force that could damage the surface.
- Wipe the tool clean and close or protect the points after use.

Storage

- Clean and dry the divider and protect its pointed ends.
- Store in the designated instrument case or drawer away from moisture and accidental contact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drafting Machine

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Drafting machine

What it is used for

To draw accurate parallel lines, perpendicular lines, and angles on technical drawing sheets.

How to handle

- Inspect the drafting machine, ruler heads, scales, locking knobs, and mounting clamps for damage or looseness.
- Secure the machine firmly to the drawing board and check that the board and scales are aligned.
- Set the required angle or direction and lock the mechanism before drawing.
- Use a suitable pencil or drawing instrument and move the ruler smoothly without applying excessive pressure.
- Release locks carefully, clean the scales, and check the mechanism after use.

Storage

- Clean the scales and moving parts with a soft, dry cloth.
- Lock or secure the machine as designed and store the board in a dry area to prevent impact or misalignment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drawing Board

Category: **Measuring & Marking Tools** Type: **Handling note** Priority: **Low** 6 labs

Covers these lab items: Drawing board; Drawing Board (any size wooden); Drawing boards; Portable drafting board

What it is used for

Drawing boards provide a stable surface for technical drawing, drafting, and accurate layout work.

How to handle

- Inspect the board surface, clamps, hinges, and support stand for damage or looseness.
- Place the board on a stable level surface and secure adjustable parts.
- Fix paper firmly without tearing it or obstructing drawing movement.
- Use drafting tools carefully and keep sharp instruments away from the board edges.
- Adjust the board angle gradually and avoid placing excessive loads on it.

Storage

- Clean the board surface after use and store it in a dry location.
- Secure portable boards to prevent slipping, bending, or accidental falls.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drawing Compass

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Drawing Compass

What it is used for

A drawing compass draws circles and arcs and transfers distances on paper or suitable drawing surfaces. Set the required radius carefully and keep the point and pencil secure.

How to use

1. Inspect the compass hinge, legs, needle point, and pencil holder for damage.
2. Loosen the adjustment screw and set the required distance between the point and pencil.
3. Tighten the screw so the selected radius remains fixed.
4. Place the needle point on the marked centre of the circle.
5. Hold the compass vertically and rotate it smoothly to draw the circle or arc.
6. Check the shape and radius, then adjust the compass if needed.
7. Close or cover the point and store the compass safely after use.

⚠ Safety

- Keep the sharp needle point away from fingers and eyes.
- Do not leave the compass open where someone may touch the point accidentally.
- Avoid excessive force that may break the pencil lead or damage the paper.

Care & storage

- Wipe the compass clean, protect the needle point, and store it in its case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drawing Instrument

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Erasing shield

What it is used for

To assist with precise technical drawing and protect selected areas of a drawing when using an erasing shield.

How to handle

- Inspect the erasing shield for bent edges, burrs, or damage before use.
- Place the shield flat over the drawing and align the required opening with the line or area to be erased.
- Use a suitable eraser gently through the opening to avoid disturbing adjacent lines.
- Keep sharp or damaged edges away from fingers and avoid pressing hard enough to tear the paper.
- Remove eraser debris and clean the shield after use.

Storage

- Wipe the shield clean and dry before storage.
- Store flat in a drawing instrument case to prevent bending or damage to the openings.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dumpy Level

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Dumpy Level

What it is used for

A dumpy level is used with a staff to determine differences in elevation and establish level points on a site. Set it up securely and take readings carefully to reduce measurement errors.

How to use

1. Inspect the instrument, tripod, levelling screws, and staff for damage.
2. Set the tripod on firm ground and mount the dumpy level securely.
3. Use the levelling screws and bubble to level the instrument.
4. Focus the eyepiece and objective until the crosshairs and staff image are clear.
5. Position the staff vertically at the required survey points.
6. Take and record staff readings, following the survey procedure and calculation method.
7. Pack the instrument carefully after completing the measurements.

⚠ Safety

- Keep the tripod stable and away from vehicle paths and excavation edges.
- Never look directly at the sun through the telescope.
- Carry the instrument securely and avoid touching exposed optical surfaces.

Care & storage

- Clean lenses with suitable lens tissue and store the instrument in its protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Feeler Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Feeler gauges; Tappet gauge

What it is used for

Use a feeler gauge to measure small gaps or clearances between two surfaces or machine components.

How to use

1. Select the gauge leaf or blade range suitable for the required clearance.
2. Inspect the blades for bends, rust, dirt, or damage.
3. Ensure the machine is stopped and safely isolated where required.
4. Clean the surfaces being measured before inserting the blade.
5. Slide the selected blade gently into the gap without forcing it.
6. Choose the blade that fits with slight, even resistance and record the size.
7. Clean and fold the gauge leaves back into the holder after use.

⚠ Safety

- Never measure moving or energized machinery unless a specifically approved safe method permits it.
- Do not force the blade into a tight gap.
- Handle thin blades carefully to avoid cuts or bending.
- Follow isolation procedures before working near moving components.

Care & storage

- Wipe blades clean and dry after use.
- Apply suitable rust protection when needed and store the blades folded.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Flow Meter

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Flow meter

What it is used for

To measure the rate or quantity of liquid or gas flowing through a pipe or system.

How to handle

- Inspect the flow meter body, display, connectors, and seals for damage or leakage.
- Confirm that the meter is suitable for the fluid, flow range, pressure, and temperature.
- Install the meter in the specified direction and position according to the manufacturer's instructions.
- Start the flow gradually and record readings after they stabilise, using the correct units.
- Stop the flow safely, relieve pressure where required, and clean or maintain the meter according to its manual.

Storage

- Store clean and dry in a protected location within the manufacturer's specified conditions.
- Protect measuring ports, seals, and display from dust, impact, and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

French Curve

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: French curve

What it is used for

To draw smooth, accurate curved lines in technical drawings, patterns, and design work.

How to handle

- Inspect the French curve for cracks, warping, scratches, or damaged edges.
- Identify the curve segment that best matches the required line between the marked points.
- Place the curve flat on the sheet and hold it steady without covering the reference marks.
- Draw along the edge using a suitable pencil or drafting pen with consistent pressure.
- Clean the edge after use and avoid bending or scratching the template.

Storage

- Wipe the curve clean and dry before storage.
- Store flat in a drawing instrument case away from heat and heavy objects to prevent warping.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Inclinometer

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Inclinometer

What it is used for

To measure the angle of inclination, slope, or tilt of a surface or component.

How to handle

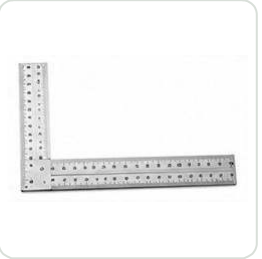
- Inspect the inclinometer, display or scale, base, and calibration status before use.
- Place the instrument on the intended surface or align it with the component being measured.
- Zero or calibrate the device as required by its instructions and select the correct measurement mode or units.
- Allow the reading to stabilise, read the angle without disturbing the instrument, and record the result with units.
- Clean the instrument after use and avoid dropping it or exposing it to conditions outside its rated limits.

Storage

- Store in a protective case in a dry, dust-free place.
- Protect the measuring base, scale, and sensor from impact, moisture, and extreme temperatures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

L-Square



Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: L-Square

What it is used for

Use an L-square to check or mark right angles and straight reference lines on suitable workpieces during layout and fabrication.

How to use

1. Select an L-square with a size and accuracy suitable for the task.
2. Inspect both arms and their joint for bends, damage, or looseness.
3. Clean the reference edge and the square's contact surfaces.
4. Place one arm firmly against the workpiece reference edge.
5. Check the second arm for alignment or mark a line using a suitable pencil or scribe.
6. Repeat the check from another position if necessary.
7. Clean and store the square safely after use.

⚠ Safety

- Keep fingers clear of sharp edges and marking tools.
- Do not use a bent or damaged square for precision work.
- Secure the workpiece before marking or checking.
- Use a suitable marking tool and avoid excessive force.

Care & storage

- Wipe the square clean and keep it dry to prevent rust.
- Store it flat or in a protective holder to prevent bending.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Levelling Staff

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Levelling staff

What it is used for

A levelling staff provides a graduated scale for readings with a level instrument during surveying. Keep the staff vertical and use the correct scale for accurate measurements.

How to use

1. Inspect the staff sections, graduations, locking joints, and base for damage.
2. Extend the staff fully and secure all locking sections.
3. Position the staff at the survey point on firm ground.
4. Hold the staff vertical using its bubble or guidance from the surveyor.
5. Allow the instrument operator to focus and take the reading.
6. Record the reading and move the staff carefully to the next point.
7. Retract the staff, clean it, and store it after surveying.

⚠ Safety

- Keep the staff away from overhead power lines and electrical hazards.
- Do not stand on unstable ground or near unprotected excavation edges.
- Avoid obstructing vehicles, pedestrians, or active work areas.
- Secure extended sections before carrying or moving the staff.

Care & storage

- Wipe off dirt and moisture, protect the graduations, and store the staff in its case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Line Rope

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 9 labs

Covers these lab items: Line and pins; Line dori; Line Rope

What it is used for

Use a line rope to establish straight reference lines for marking, alignment, layout, or measurement tasks.

How to use

1. Inspect the rope and pins for fraying, knots, bends, or damage.
2. Identify the start and end points for the required line.
3. Secure the pins firmly at the marked points where suitable.
4. Tie the rope tightly between the points and keep it clear of obstacles.
5. Check the line alignment and mark along the rope as required.
6. Remove the rope and pins carefully after marking.
7. Clean and coil the rope for storage.

⚠ Safety

- Keep the rope clear of walkways to prevent trips.
- Do not pull the rope across people or fragile objects.
- Use stable pins and avoid placing them where they can injure someone.
- Wear gloves if the rope or pins have sharp or rough parts.

Care & storage

- Remove dirt and allow the rope to dry before storage.
- Coil it neatly and store it away from sharp edges and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Magnetic Compass

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Compass; Compasses

What it is used for

A magnetic compass indicates direction by aligning with Earth's magnetic field. Use it for basic navigation and direction marking away from magnetic interference.

How to use

1. Place the compass on a level surface or hold it horizontally.
2. Move away from magnets, steel objects, electrical equipment, and vehicles.
3. Allow the needle to settle before reading the direction.
4. Rotate the compass housing or align the direction mark as required.
5. Read the indicated direction and compare it with the map if needed.
6. Store the compass safely after use.

⚠ Safety

- Keep the compass away from strong magnets and magnetic objects for reliable readings.
- Do not use a damaged compass or one with a stuck needle.
- Use a map and other navigation aids when direction is critical.

Care & storage

- Keep the compass dry, clean, and protected from impact and magnetic interference.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Manifold Gauge

Category: **Measuring & Marking Tools** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Manifold Gauge; manifold gauge sets

What it is used for

A manifold gauge set measures pressure in compatible fluid systems and can support diagnosis, evacuation, and charging when used with the correct service procedure. Correct gauge selection, hose connections, pressure limits, and refrigerant-specific handling are essential to prevent leaks, equipment damage, and injury.

Main parts

- Low-pressure gauge displays pressure on the low side of the system.
- High-pressure gauge displays pressure on the high side.
- Manifold body connects gauges, valves, and service ports.
- Control valves regulate flow through the manifold.
- Service hoses connect the manifold to approved system ports or service equipment.
- Couplers or adapters provide compatible connections.
- Sight glass shows fluid movement where fitted.
- Protective caps help keep service ports clean.

Before you start

1. Only trained personnel should use the manifold gauge set for the relevant system.
2. Identify the system type, working fluid or refrigerant, and manufacturer service specifications.
3. Confirm the gauges, hoses, seals, and couplers are rated for the expected pressure and compatible with the fluid.
4. Inspect gauge faces, hoses, valve stems, fittings, and seals for damage, wear, or leakage.
5. Use required PPE and ensure adequate ventilation, especially when working with refrigerants or other pressurised fluids.
6. Confirm that the manifold valves are in the required initial position and service ports are accessible.
7. Use only the approved refrigerant recovery and charging equipment where applicable; do not vent refrigerant intentionally.

How to use

1. Switch off or prepare the system according to the manufacturer's service procedure.
2. Identify the correct low- and high-pressure service ports before connecting hoses.
3. Connect the hoses using compatible couplers and the prescribed connection sequence.
4. Check that connections are secure and avoid cross-connecting the high and low sides.

5. Read the static or operating pressures only under the conditions specified by the service procedure.
6. Operate manifold valves only when the procedure requires it and understand which flow path each valve controls.
7. Monitor pressure changes and compare readings with the system specifications and operating conditions.
8. Stop the procedure if a hose leaks, a gauge behaves abnormally, or pressure approaches the equipment limit.
9. Recover or transfer fluid using approved equipment and the specified procedure where required.
10. Close valves, isolate the system as instructed, and disconnect hoses carefully to minimise fluid release.
11. Refit service-port caps, inspect for leaks using an approved method, and record the readings and work performed.

⚠ Safety

- Never exceed the rated pressure of any gauge, hose, coupler, or adapter.
- Do not connect a gauge set to an unknown or incompatible fluid system.
- Do not loosen pressurised connections or disconnect hoses before following the required isolation and recovery procedure.
- Never release refrigerant deliberately into the atmosphere; follow applicable recovery and environmental rules.
- Keep the set away from ignition sources when working with flammable refrigerants.
- Wear suitable eye and hand protection and avoid contact with escaping cold fluid or vapour.

Care & storage

- Depressurise and store the set according to the manufacturer's instructions after use.
- Keep hoses capped and protect gauge faces, fittings, and couplers from impact and contamination.
- Inspect seals, hoses, and valve operation regularly and replace worn components with compatible parts.
- Arrange calibration when required or when readings are inconsistent with a reliable reference.
- Store the set in a clean, dry case away from excessive heat and corrosive chemicals.

Common mistakes to avoid

- Using gauges or hoses with an inadequate pressure rating or incompatible fluid rating.
- Connecting the high-pressure hose to the wrong service port.
- Opening manifold valves without understanding the system flow path.
- Disconnecting hoses without the prescribed isolation or recovery steps.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Measuring Instrument

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Measuring appliances; Measuring equipment; Measuring Tools (ruler, scale, calipers)

What it is used for

Measuring instruments are used to determine dimensions, distances, thickness, or other physical quantities. Choose an instrument with suitable range and accuracy for the measurement.

How to use

1. Select the correct instrument for the quantity and required accuracy.
2. Inspect the instrument for damage, dirt, or worn measuring surfaces.
3. Check zero setting or calibration status before taking measurements.
4. Position the instrument correctly against or around the object.
5. Read the scale or display at the correct viewing angle.
6. Record the result with the correct unit and clean the instrument after use.

⚠ Safety

- Handle precision instruments carefully and avoid dropping or striking them.
- Do not force calipers or measuring jaws against the workpiece.
- Use instruments only within their specified measuring range.

Care & storage

- Clean and store instruments in a dry protective case; calibrate as scheduled.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Measuring Tape

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 42 labs

Covers these lab items: Inch Tape; Measuring 30 meter Tape; Measuring Tape; Measuring Tape (5m); Measuring tape and scale; Measuring tape and/or a ruler; Measuring Tape ISI Mark; Measuring tape, plumb; Measuring tape/ruler; Retractable Tape Measure; Tape; Tape Measure

What it is used for

A measuring tape measures length, distance, and dimensions for layout, inspection, or construction tasks. Use the correct tape length and read the scale carefully for accurate measurements.

How to use

1. Inspect the tape, markings, hook, lock, and casing for damage.
2. Select a tape with suitable length and units for the task.
3. Secure the tape's end at the starting point and align it with the measurement line.
4. Extend the tape straight without twists or unnecessary slack.
5. Read the measurement at eye level and use the correct reference edge.
6. Record the value with units and identify the measured dimension.
7. Retract the tape slowly using the lock and return it to storage.

⚠ Safety

- Keep fingers away from the tape edge when retracting it quickly.
- Do not use a metal tape near live electrical equipment.
- Avoid overstretching the tape or using a damaged hook or blade.
- Keep the work area clear to prevent trips and cuts.

Care & storage

- Wipe the tape clean and dry it before storage if it becomes wet.
- Retract gently and replace tapes with damaged markings or faulty locks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Micrometer

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Micrometer; Out - Side Micromet er; Out - Side Micromet er Digital

What it is used for

Use a micrometer to measure small external dimensions, such as wire, shaft, or component thickness, with high precision.

How to use

1. Select a micrometer with a range and resolution suitable for the part.
2. Inspect the anvil, spindle, frame, ratchet, and display or scale for damage.
3. Clean the measuring faces and the workpiece before measurement.
4. Check zero setting or calibration according to the instrument instructions.
5. Place the workpiece between the anvil and spindle without tilting it.
6. Turn the ratchet or friction device gently until the measuring force is consistent.
7. Read the scale or digital display and record the measurement with units.
8. Open the spindle, remove the part, and store the micrometer carefully.

⚠ Safety

- Do not force the spindle or overtighten the ratchet.
- Keep fingers clear of pinch points between the measuring faces.
- Do not measure moving or energized components.
- Use the micrometer only within its specified range and avoid dropping it.

Care & storage

- Wipe the measuring faces clean and dry after use.
- Store in its case and arrange calibration as specified.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Offset Rod

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Offset rod

What it is used for

An offset rod helps transfer or mark offset positions during surveying, layout, or alignment tasks. Use it with the relevant measuring equipment and follow the project's measurement procedure.

How to use

1. Inspect the rod for bending, damaged markings, loose fittings, or worn tips.
2. Confirm the required offset and measurement reference before starting.
3. Position the rod at the designated point according to the layout plan.
4. Align the rod with the required reference line or measuring instrument.
5. Take and record the reading or mark the offset as specified.
6. Repeat the measurement to confirm accuracy when required.
7. Clean the rod and store it securely after use.

⚠ Safety

- Keep the rod clear of overhead electrical lines and nearby machinery.
- Use stable footing and avoid unprotected edges or uneven ground.
- Do not use a bent or damaged rod for precision measurements.
- Keep the work area clear of pedestrians and moving vehicles.

Care & storage

- Wipe the rod clean, protect its markings, and store it in a dry location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Plumb Bob

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 11 labs

Covers these lab items: Plumb bob; Plumb rule and Bob

What it is used for

Use a plumb bob to establish a true vertical reference when checking alignment or transferring a point vertically.

How to use

1. Inspect the bob, cord, and attachment point for damage or wear.
2. Secure the cord to a stable point directly above the required reference.
3. Allow the bob to hang freely without touching the wall, floor, or nearby objects.
4. Wait until the bob stops swinging before taking a reading or marking a point.
5. Compare the cord with the surface or transfer the point vertically as required.
6. Record the result and carefully retrieve the tool after use.

⚠ Safety

- Keep the bob clear of people and fragile objects while it hangs.
- Do not stand directly beneath a suspended bob unnecessarily.
- Use a stable support and avoid working near moving machinery.

Care & storage

- Wipe the bob clean and dry the cord before storage.
- Wind the cord neatly and store the tool in a protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pressure Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 11 labs

Covers these lab items: Electronic pressure gauge; Pressure gauge; Pressure indicators; Pressure meter

What it is used for

Use a pressure gauge to measure and monitor pressure in a suitable liquid or gas system within the gauge's rated range.

How to use

1. Check the gauge body, dial or display, connector, and scale for damage.
2. Confirm the gauge range and pressure rating suit the system.
3. Ensure the system is isolated and depressurized before fitting or removing the gauge.
4. Connect the gauge securely using the correct fitting and seal.
5. Open the pressure supply gradually according to the equipment procedure.
6. Read and record the pressure once the indication stabilizes.
7. Close the supply and release pressure safely before removing the gauge.

⚠ Safety

- Never exceed the gauge or system pressure rating.
- Use a gauge compatible with the fluid, gas, and operating temperature.
- Do not loosen a pressurized connection or stand in front of it.
- Use suitable PPE and follow the equipment isolation procedure.

Care & storage

- Clean the gauge exterior and keep the display readable.
- Store it safely and arrange calibration or replacement when required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prismatic Compass

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Prismatic and Surveyor compass

What it is used for

A prismatic compass measures magnetic bearings for basic surveying and field orientation. Set it up away from magnetic interference and record bearings using the required survey procedure.

How to use

1. Inspect the compass, prism, sighting system, needle, and protective cover.
2. Choose a stable observation point away from vehicles, steel structures, and electrical equipment.
3. Hold or mount the compass level according to its design.
4. Release the needle lock and allow the magnetic needle to settle.
5. Sight the target through the designated sighting system and read the bearing.
6. Record the bearing, station, and relevant field conditions in the survey notes.
7. Lock the needle if provided and protect the compass after use.

⚠ Safety

- Keep clear of traffic, unstable ground, and exposed edges during surveying.
- Avoid magnetic objects and electrical equipment that may distort readings.
- Do not use a damaged compass for critical measurements.
- Follow site rules when working near roads or construction areas.

Care & storage

- Clean optical surfaces gently and store the compass in its protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Protractor

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Protractor

What it is used for

To measure and draw angles using a protractor. To mark accurate angles by aligning the baseline and reading the correct scale.

Main parts

- Protractor body
- Angle scale
- Centre point
- Baseline
- Reference markings

Before you start

1. Choose a protractor with a suitable range and readable markings.
2. Check for cracks or worn scale markings.
3. Place the drawing or workpiece on a flat surface.
4. Keep a pencil or marking tool ready.

How to use

1. Mark the vertex of the required angle.
2. Align the protractor centre with the vertex.
3. Place the baseline along one arm of the angle.
4. Read the correct scale from the zero aligned with the baseline.
5. Mark the required angle value.
6. Remove the protractor and draw the second arm through the mark.
7. Check the angle and measurement units.

⚠ Safety

- Handle sharp marking tools carefully.
- Keep the protractor away from heat that may distort it.
- Avoid using a damaged or unclear scale.

Care & storage

- Wipe the protractor clean after use.
- Store it flat in a protective place.

- Keep the scale markings readable.

Common mistakes to avoid

- Aligning the centre away from the vertex.
- Reading the wrong scale.
- Allowing the protractor to move while marking.
- Confusing degrees with other measurement units.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Radius Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Radius Gauge

What it is used for

To check the radius of curved edges, corners, and profiles during inspection or fabrication.

How to handle

- Inspect the gauge leaves for rust, bends, burrs, or damaged edges.
- Select the leaf with the required radius and ensure the workpiece is clean.
- Place the leaf gently against the curve and check for gaps or rocking without forcing it.
- Compare the fit under suitable lighting and record the result if required.
- Wipe the gauge clean after use and avoid using it as a scraper or cutting tool.

Storage

- Wipe dry and apply a light protective coating if recommended for the gauge material.
- Fold the leaves into their holder and store in a dry case to prevent rust and bending.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rain Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Rain gauges and soil sampling kits for environmental assessments

What it is used for

A rain gauge measures rainfall, while soil sampling kits collect soil for environmental assessment. Install the gauge in a suitable open location and collect soil samples using the required sampling protocol.

How to use

1. Inspect the rain gauge, measuring scale, collection container, and soil sampling tools.
2. Choose an open, representative site away from buildings, trees, and obstructions for the rain gauge.
3. Install the gauge upright and record its location and installation details.
4. After rainfall, measure the collected water using the prescribed reading method and record the result.
5. For soil sampling, identify the sampling locations and required depth according to the assessment plan.
6. Use clean tools to collect, label, and seal samples in suitable containers.
7. Clean equipment and record dates, locations, weather, and sample handling details.

⚠ Safety

- Wear suitable gloves and footwear when collecting environmental samples.
- Avoid sampling near hazardous waste, contaminated water, or unsafe ground without authorization.
- Use clean tools between sampling points to prevent cross-contamination.
- Follow site permits and environmental sampling procedures.

Care & storage

- Clean and dry equipment after use and store labelled samples according to the testing protocol.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ranging Rod

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Ranging rod

What it is used for

To mark survey stations and help establish straight lines or alignment during land surveying.

How to handle

- Inspect the rod, painted bands, tip, joints, and fittings for damage or looseness.
- Place the rod vertically at the designated survey point using a suitable support when needed.
- Keep the rod visible to the survey team and align it according to the surveyor's instructions.
- Avoid overhead power lines, vehicle routes, and unstable ground when carrying or positioning the rod.
- Clean off soil after use and inspect the rod before returning it to storage.

Storage

- Wipe the rod clean and dry, especially at joints and metal fittings.
- Store horizontally on suitable supports or in its designated rack, away from bending and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ruler

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 22 labs

Covers these lab items: Adhesive- Backed Rulers; Clear Ruler; Meter Stick/Yardstick; Ruler; Ruler or Scale; Rulers; Scale; Steel rule; Steel Ruler; Steel scale; Yardstick/Meter-stick

What it is used for

A ruler measures length and helps draw straight lines or mark dimensions on materials. Select the correct ruler type and use its scale carefully for accurate measurements.

How to use

1. Inspect the ruler, scale markings, and edges for damage or wear.
2. Choose a ruler with suitable length and units for the task.
3. Place the ruler flat against the workpiece and align its zero mark with the starting point.
4. Hold the ruler steady and mark or read the required measurement.
5. Read the scale at eye level to reduce reading errors.
6. Record the measurement with the correct units.
7. Clean the ruler and return it to its storage location.

⚠ Safety

- Keep sharp steel edges away from fingers and skin.
- Do not use a damaged ruler with sharp burrs or broken edges.
- Keep the ruler clear of moving machinery and live electrical equipment.

Care & storage

- Wipe the ruler clean and dry after use.
- Store it flat and protect its markings from abrasion or corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ruling Pen

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Ruling pen

What it is used for

To draw fine, controlled lines with ink or suitable marking fluid in drafting and technical drawing.

How to handle

- Inspect the pen body, adjustable nibs, screw, and tip for damage or dried residue.
- Clean the nib and adjust the gap according to the required line width and the manufacturer's guidance.
- Load a small amount of suitable ink between the nibs without overfilling.
- Hold the pen at a consistent angle and draw a test line on scrap paper before working on the drawing.
- Clean the nib promptly after use and avoid pressing hard enough to scratch the drawing surface.

Storage

- Clean and dry the nib and body thoroughly after use.
- Close the pen securely and store it in a protective case away from moisture and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scale Ruler

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Triangular scales

What it is used for

To measure or draw scaled dimensions on technical drawings using the correct scale face.

How to handle

- Inspect the ruler edges, scale markings, and body for cracks, chips, fading, or distortion.
- Identify the drawing scale and select the matching scale face before measuring.
- Align the zero or reference mark with the drawing point and read the graduation straight on to avoid parallax error.
- Use a sharp pencil or suitable drawing instrument without applying excessive pressure to the scale markings.
- Keep the ruler clean and avoid using it as a cutting guide unless specifically designed for that purpose.

Storage

- Wipe clean with a soft cloth and keep the scale markings dry and legible.
- Store flat in a drawing case away from heat, moisture, and heavy objects that could warp it.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scriber

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Scriber

What it is used for

To mark fine layout lines on suitable metal or other compatible work surfaces before machining or fabrication.

How to handle

- Inspect the scriber tip, handle, and body for bending, cracks, or damage.
- Secure the workpiece and clean the marking surface before use.
- Hold the scriber firmly and draw a light, controlled line along a suitable straightedge or template.
- Keep the sharp tip directed away from hands and body; do not use the scriber as a punch or pry tool.
- Use a suitable protective cap when finished and avoid leaving the sharp point exposed.

Storage

- Wipe the scriber clean and dry after use and fit its protective cap.
- Store in a labelled tool case or rack where the point cannot cause accidental injury.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Set Square

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Triangles (30°-60° and 45°)

What it is used for

To draw and check angles, perpendicular lines, and parallel lines in technical drawing or layout work.

How to handle

- Inspect the set square edges and angle markings for chips, cracks, warping, or worn graduations.
- Choose the appropriate 30°–60° or 45° set square for the required angle.
- Place the set square flat against the drawing surface and align its reference edge accurately.
- Draw along the edge using a suitable pencil or drawing instrument without excessive pressure.
- Keep the tool away from cutting operations unless it is specifically designed for that use.

Storage

- Wipe clean and dry after use to keep edges and markings clear.
- Store flat in a drawing case or rack to prevent bending, scratches, and breakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spirit Level

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 24 labs

Covers these lab items: Hand Leveller; Levelers; Levellers; Spirit level; Spirit Level Bottle; Spirit level/water level; Sprit level

What it is used for

A spirit level checks whether a surface or component is horizontal or vertical using a liquid-filled vial and bubble. Use the correct reference surface and read the bubble carefully.

How to use

1. Inspect the level body, vial, and measuring edges for damage.
2. Clean the reference surface and the level's contact edges.
3. Place the level flat on the surface or component being checked.
4. Wait for the bubble to settle inside the vial.
5. Compare the bubble position with the marked centre lines.
6. Adjust the workpiece or installation until the bubble indicates the required level.
7. Check again from another suitable position when necessary.
8. Wipe the level clean and store it safely.

⚠ Safety

- Do not use a level with a cracked vial or bent body.
- Keep the level secure when working at height or near edges.
- Do not use the tool as a hammer or pry bar.

Care & storage

- Wipe the level clean and dry after use.
- Store it in a protective location and avoid dropping or striking it.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spring Caliper

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Spring Caliper Inside /Outside /Divider

What it is used for

A spring caliper transfers or compares internal and external dimensions without directly displaying a measurement. Use it with a suitable scale or measuring instrument to determine dimensions.

How to use

1. Identify whether the caliper is an inside, outside, or divider type.
2. Inspect the spring, adjustment nut, joints, and contact points for damage.
3. Adjust the legs gently to fit the dimension being checked.
4. Place the contact points correctly against the inside or outside surfaces.
5. Lock or hold the adjustment and remove the caliper without changing its setting.
6. Compare the setting with a suitable ruler or measuring instrument.

⚠ Safety

- Keep sharp points away from fingers, eyes, and other people.
- Do not force the legs against a workpiece or over-tighten the adjustment nut.
- Secure the workpiece before measuring near sharp or moving parts.

Care & storage

- Wipe clean after use, lightly protect metal surfaces if required, and store closed safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Surveying Chain

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Chain and tape

What it is used for

A surveying chain or measuring tape measures distances for land surveying, layout, and field measurements. Use the correct measurement method and account for alignment and site conditions when recording results.

How to use

1. Inspect the chain or tape for stretched links, damaged hooks, worn markings, or kinks.
2. Identify the required distance and confirm the correct measuring unit.
3. Mark the starting and ending points clearly.
4. Align the chain or tape along the measurement line and keep it straight.
5. Apply consistent tension and avoid twists or sag where practical.
6. Read and record the measurement, repeating it if verification is required.
7. Clean the equipment, rewind or coil it carefully, and store it safely.

⚠ Safety

- Wear suitable footwear and remain aware of traffic and nearby hazards.
- Keep hands clear of sharp hooks and moving chain links.
- Do not stretch measuring equipment across active vehicle routes without approved controls.
- Avoid measuring near overhead electrical lines with conductive equipment.

Care & storage

- Wipe the chain or tape dry and store it without knots, kinks, or corrosion exposure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

T-Square

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: T-square

What it is used for

To draw accurate horizontal or vertical lines and right angles on drawing sheets using a T-square. To maintain alignment with the drawing board edge for consistent technical drawings.

Main parts

- Long blade
- Crosshead
- Straight working edge
- Graduation markings, if fitted

Before you start

1. Check the blade and crosshead for damage or looseness.
2. Place the drawing sheet securely on the board.
3. Ensure the board edge is straight and clean.
4. Keep suitable pencils and measuring tools ready.

How to use

1. Place the crosshead firmly against the board edge.
2. Slide the T-square to the required drawing position.
3. Hold it steady against the board.
4. Draw the line along the working edge.
5. Use a suitable set square against the blade when drawing perpendicular lines.
6. Check alignment and measurements before continuing.

⚠ Safety

- Keep fingers clear of sharp drawing tools.
- Do not use a warped or damaged T-square.
- Handle long blades carefully to avoid striking nearby people.

Care & storage

- Wipe the working edge clean after use.
- Avoid bending or dropping the blade.
- Store flat in a protected location.

Common mistakes to avoid

- Allowing the crosshead to move away from the board edge.
- Using a warped blade.
- Drawing while the T-square is moving.
- Using the wrong edge as the reference.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Temperature Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Temp gauge; Temperature gauge

What it is used for

To measure or monitor temperature using a suitable temperature gauge. To obtain a readable and reliable measurement within the instrument's operating range.

Main parts

- Temperature display or dial
- Sensing element or probe
- Connection stem or cable
- Mounting fitting, if fitted
- Calibration markings

Before you start

1. Select a gauge with a suitable temperature range for the application.
2. Inspect the display, probe, cable, and fittings for damage.
3. Check calibration status where accuracy is important.
4. Confirm the sensor is suitable for the material or environment being measured.

How to use

1. Install or position the sensor at the designated measurement point.
2. Secure the probe without damaging the sensing element.
3. Switch on the device if it requires power.
4. Allow the reading to stabilize according to the instrument instructions.
5. Read and record the temperature with units.
6. Compare the result with the required operating range.
7. Remove or switch off the gauge as appropriate after measurement.

⚠ Safety

- Avoid direct contact with hot or extremely cold surfaces.
- Do not exceed the gauge's rated temperature range.
- Isolate equipment before installing a gauge in a hazardous system.
- Use suitable PPE for the measured material or environment.

Care & storage

- Clean the probe using a method compatible with the sensor.

- Store the gauge safely and protect the sensing element.
- Check calibration and battery condition as applicable.
- Replace damaged probes or cables.

Common mistakes to avoid

- Using a gauge outside its rated range.
- Recording a reading before it stabilizes.
- Ignoring units or calibration requirements.
- Damaging the probe during installation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thermo-Hygrometer

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Hygrometer; Thermo-hygrometer

What it is used for

A thermo-hygrometer measures air temperature and relative humidity. Use it to monitor room conditions in classrooms, laboratories, storage areas, or workplaces.

How to use

1. Place the instrument in a shaded, ventilated location away from direct sunlight.
2. Keep it away from heaters, air conditioners, moisture sources, and direct airflow.
3. Allow the reading to stabilize according to the manufacturer's instructions.
4. Read the temperature and relative humidity from the display or scale.
5. Record the values with the date and time if monitoring conditions.
6. Return or leave the instrument in its designated monitoring position.

⚠ Safety

- Keep the instrument away from water unless it is rated for wet conditions.
- Do not expose it to temperatures or humidity beyond its specified range.
- Handle the sensor and display gently; do not open the casing unnecessarily.

Care & storage

- Keep the sensor area clean and check calibration or battery status as recommended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thread Pitch Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Thread wear indicator; Threading Gauge (Engineerin g); Threading Gauge (Plumbing)

What it is used for

A thread pitch gauge identifies the pitch of a screw thread by matching its teeth to the thread profile. Use the correct gauge leaf for metric or imperial threads as required.

How to use

1. Inspect the gauge leaves for wear, damage, rust, or bent teeth.
2. Clean the external or internal thread surface enough to see its profile.
3. Select a gauge leaf that matches the expected thread type and pitch range.
4. Place the leaf against the thread and check that the teeth align closely.
5. Try adjacent leaves until the best fit is found without forcing the gauge.
6. Read the marked pitch and clean and fold the gauge after use.

⚠ Safety

- Do not force the gauge into a thread or use it as a cutting tool.
- Keep fingers away from sharp or damaged gauge edges.
- Do not rely on a worn gauge for critical measurements.

Care & storage

- Wipe the leaves clean, keep them dry, and store the gauge folded in its case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Total Station

Category: **Measuring & Marking Tools** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Total station with prism and prism pole

What it is used for

A total station measures horizontal angles, vertical angles, and distances to determine positions and elevations in surveying. Use the instrument with a correctly positioned prism and prism pole to collect accurate field data.

Main parts

- Total station with telescope and display
- Tripod and mounting screw
- Tribrach and levelling screws
- Optical or laser plummet
- Prism reflector and prism holder
- Prism pole with bubble level
- Battery, charger, and data storage device

Before you start

1. Inspect the total station, tripod, prism, pole, and accessories for damage.
2. Choose firm ground with a clear line of sight to the survey points.
3. Set up the tripod securely and adjust its height for comfortable operation.
4. Mount the total station and centre it over the survey station.
5. Level the instrument using the levelling screws and electronic or circular level.
6. Check battery charge, date, time, units, and required survey settings.
7. Measure or confirm the instrument height and enter the correct station coordinates.
8. Check the prism constant and confirm the prism pole bubble is working.

How to use

1. Switch on the total station and select the appropriate survey job.
2. Enter the station coordinates, instrument height, and required reference information.
3. Sight the known backsight point and set the horizontal orientation as instructed.
4. Place the prism pole vertically over the target point using its bubble level.
5. Aim the telescope at the prism and focus the target clearly.
6. Use the correct prism mode and measure the distance and angles.
7. Record the point number, prism height, coordinates, and relevant field notes.
8. Repeat observations where required and verify results against known control points.
9. Save the survey data and back it up using the approved transfer method.
10. Power off the instrument, remove it carefully, and pack all components in their cases.

⚠ Safety

- Set up the tripod away from traffic, unstable edges, and moving equipment.
- Use high-visibility clothing and warning markers when surveying near roads.
- Never look directly into the sun through the telescope.
- Secure the tripod legs and hold the prism pole firmly in windy conditions.
- Do not leave the instrument unattended in public or exposed locations.
- Protect the instrument from rain, dust, impact, and sudden temperature changes.

Care & storage

- Clean lenses with approved lens tissue and suitable cleaning materials only.
- Keep the instrument and prism dry before storing them in their cases.
- Charge batteries with the recommended charger and avoid deep discharge.
- Check calibration, collimation, and levelling accuracy at recommended intervals.

Common mistakes to avoid

- Setting up the instrument without accurate centring or levelling.
- Entering incorrect station coordinates, instrument height, or prism height.
- Using the wrong prism constant, target mode, or measurement units.
- Holding the prism pole tilted or ignoring its bubble level.
- Saving incorrect point names or failing to back up survey data.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Try Square

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 10 labs

Covers these lab items: Square; Tri-Square (Casting base) (construction and Carpentry); Tri-square (Welding); Try Square

What it is used for

Use a try square to check or mark a right angle between two edges on wood, metal, or other suitable workpieces.

How to use

1. Select a try square suitable for the workpiece and required accuracy.
2. Inspect the blade and stock for bends, damage, or loose joints.
3. Clean the reference edge and the square's contact surfaces.
4. Press the stock firmly against the reference edge of the workpiece.
5. Check the blade for a gap along the edge or mark a line using a sharp pencil or scribe.
6. Reposition the square and repeat the check if needed.
7. Wipe the tool clean and store it safely after use.

⚠ Safety

- Keep fingers clear of sharp blade edges and scribes.
- Do not use a bent or damaged square for accurate marking.
- Secure the workpiece before marking or checking.
- Use a suitable marking tool and avoid excessive force.

Care & storage

- Keep the blade clean and free from rust.
- Store the square flat or in a protective holder to prevent bending.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vacuum Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Micron Gauge

What it is used for

A micron vacuum gauge measures very low absolute pressure in vacuum systems, including refrigeration and related service applications. Connect it correctly and follow the gauge manufacturer's instructions for accurate readings.

How to use

1. Inspect the gauge, sensor, connectors, battery, and display for damage.
2. Confirm the gauge's measurement range and compatibility with the system and refrigerant service setup.
3. Ensure the system is isolated and follow the approved service procedure before connecting the gauge.
4. Attach the sensor to the designated measurement port using clean, suitable connections.
5. Switch on the gauge and allow the reading to stabilize as specified.
6. Monitor and record the pressure reading, checking for unexpected fluctuations or leaks.
7. Disconnect the gauge safely, fit protective caps, and store the sensor clean and dry.

⚠ Safety

- Follow approved refrigeration or vacuum-system service procedures and applicable refrigerant rules.
- Do not expose the gauge to pressure or chemicals beyond its rating.
- Avoid contaminating the sensor with oil, moisture, or debris.
- Isolate equipment before servicing connections where required.

Care & storage

- Clean the sensor according to the manufacturer instructions and replace batteries when needed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vernier Caliper

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 23 labs

Covers these lab items: Calipers; Calipers and measuring tools; Digital Calliper; Vanier calliper; Vernier; Vernier caliper; Vernier Caliper- Digital; Vernier caliper (electronics); Vernier Caliper (manual/Digital); Vernier calliper

What it is used for

A vernier caliper measures external dimensions, internal dimensions, and depth with precision. Choose the correct measuring surfaces and read the scale or digital display carefully.

How to use

1. Inspect the jaws, depth rod, sliding mechanism, and scale or display for damage.
2. Clean the measuring faces and the workpiece before measurement.
3. Close the jaws gently and check the zero reading.
4. Select the appropriate external, internal, or depth measuring feature.
5. Position the caliper squarely on the workpiece without excessive force.
6. Lock the slider if provided and read the scale or digital display.
7. Record the measurement with units and required precision.
8. Clean the caliper and store it in its protective case.

⚠ Safety

- Do not force the jaws or use the caliper as a clamp.
- Keep fingers clear of sharp workpiece edges.
- Do not measure moving or dangerously hot components.
- Handle the depth rod and measuring edges carefully to avoid damage.

Care & storage

- Wipe the caliper clean and dry after use.
- Store it in its case and check zero and calibration at recommended intervals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weather Instrument

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Thermometers, barometers, and anemometers for recording weather conditions

What it is used for

Weather instruments such as thermometers, barometers, and anemometers measure temperature, atmospheric pressure, and wind speed. Position instruments correctly and record readings with the appropriate units and observation time.

How to use

1. Inspect each instrument, sensor, display, and mounting point for damage.
2. Identify the measurement range, units, and operating instructions for each instrument.
3. Place the thermometer in a suitable shaded, ventilated location as specified.
4. Position the barometer away from disturbances and follow its installation guidance.
5. Install the anemometer in an exposed location with unobstructed airflow.
6. Allow readings to stabilize and record values with date and time.
7. Compare readings with expected ranges and report faulty or inconsistent measurements.

⚠ Safety

- Secure instruments and mounts against wind or accidental falling.
- Do not climb unsafe structures to reach measurement locations.
- Keep instruments away from electrical hazards and severe weather exposure when unsafe.
- Follow manufacturer guidance when handling glass thermometers or sensitive sensors.

Care & storage

- Clean sensors gently and store portable instruments in a dry, protected place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weighing Scale

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 43 labs

Covers these lab items: Digital Kitchen Balance; Digital Weighing Balance; Digital Weighing Scale; Electronic Hanging Balance; Electronic top pan balance; Kitchen Weighing Scale; Platform Scales; Simple Balance; Weighing 10 kg , Balance Capacity with minimum Division of 0.1 kg.; Weighing balance; Weighing Balance; Weighing

What it is used for

A weighing scale measures the mass of materials, ingredients, or objects within its rated capacity. Place the scale on a stable surface and use the correct units and weighing procedure.

How to use

1. Inspect the scale, platform, display, and power supply for damage.
2. Place the scale on a firm, level surface away from vibration and airflow.
3. Switch on the scale and allow it to stabilise.
4. Check the selected units and zero the display with the platform empty.
5. Place the item centrally on the platform or suspend it correctly for a hanging scale.
6. Use the tare function for containers when required.
7. Wait for the reading to stabilise and record the value with units.
8. Remove the load and switch off the scale after use.

⚠ Safety

- Do not exceed the scale's rated capacity.
- Keep hands and feet clear of suspended loads and moving hooks.
- Do not weigh hot, wet, or hazardous materials unless the scale is designed for them.
- Avoid placing loads suddenly or unevenly on the platform.

Care & storage

- Clean the platform with suitable materials and keep liquids away from electronics.
- Check zero and calibration according to the manufacturer's schedule.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wire Gauge

Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Standard wire gauge; Wire gauge

What it is used for

Use a wire gauge to compare a wire's diameter or thickness with standardized sizes for identification and selection.

How to use

1. Select the correct gauge type and confirm its measurement standard.
2. Inspect the gauge markings and measuring slots or openings for damage or wear.
3. Clean the wire and gauge contact surfaces before measurement.
4. Place the wire in the matching slot or opening without forcing it.
5. Compare the fit or reading with the gauge markings and note the size standard.
6. Repeat the measurement at another point if the wire is irregular.
7. Clean and store the gauge after use.

⚠ Safety

- Do not measure energized or exposed live conductors by hand.
- Handle wire ends carefully to avoid cuts or punctures.
- Do not force wire into a slot that is too small.
- Use a suitable gauge and standard for the wire type.

Care & storage

- Wipe the gauge clean and keep its markings legible.
- Store it dry and protect the measuring edges from damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

AED

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: AED (Automated External defibrillator)

What it is used for

An Automated External Defibrillator (AED) analyses a person's heart rhythm and, when indicated, advises or delivers an electric shock to help treat certain cardiac arrests. Use it promptly with CPR according to the device prompts and local emergency procedures.

Main parts

- AED unit and display or voice prompts
- Electrode pads and connecting cable
- Battery or power module
- Shock button on semi-automatic models
- Protective case and quick-reference instructions
- Optional child pads or paediatric mode, depending on model

Before you start

1. Confirm that the scene is safe for the rescuer and patient.
2. Check responsiveness and normal breathing; if the person is unresponsive and not breathing normally, call emergency services and request an AED.
3. Ask another person to bring the AED and begin CPR immediately if indicated.
4. Turn on the AED and follow its voice or visual prompts.
5. Expose the patient's chest and dry it if wet; remove medication patches from pad sites using gloves and wipe the skin clean.
6. Check for implanted medical devices or jewellery under pad positions and place pads as directed by the AED instructions.

How to use

1. Open the AED case, turn on the unit, and follow every voice or screen instruction.
2. Attach the pads to the bare chest in the positions shown on the pad illustrations.
3. Connect the pad cable if it is not already attached.
4. Ensure nobody touches the patient while the AED analyses the heart rhythm.
5. If a shock is advised, loudly tell everyone to stand clear and visually confirm that nobody is touching the patient.
6. Press the flashing shock button when instructed on a semi-automatic AED; a fully automatic AED may deliver the shock itself.
7. Immediately resume CPR when prompted, starting with chest compressions, and follow the AED's instructions for the next analysis.

8. Continue CPR and AED-guided care until the person shows signs of life, trained responders take over, the scene becomes unsafe, or you are physically unable to continue.
9. If no shock is advised, immediately resume CPR as directed; do not delay care.
10. When emergency responders arrive, report the time of collapse if known, CPR performed, and any shocks delivered.
11. After use, arrange replacement pads and battery as needed, clean the device according to the manufacturer's instructions, and report the use.

⚠ Safety

- Use the AED as soon as available for a suspected cardiac arrest and follow its prompts.
- Do not touch the patient during rhythm analysis or shock delivery.
- Keep the patient and chest dry; do not use the AED in standing water or an unsafe environment.
- Do not place pads directly over medication patches or implanted device bulges; follow pad-placement instructions.
- Use paediatric pads or mode for children when available and indicated; follow the AED instructions if these are unavailable.
- Do not delay CPR while waiting for an AED or emergency responders.

Care & storage

- Check the readiness indicator and battery status at the schedule specified by the manufacturer.
- Inspect pad packaging and expiry dates regularly and replace used, expired, or damaged pads.
- Keep the AED and accessories in their designated, accessible storage location.
- After use, restore the device to ready status and document checks or faults.
- Arrange service promptly if the AED shows an error or fails its self-test.

Common mistakes to avoid

- Waiting for the AED instead of starting CPR when indicated.
- Failing to ensure everyone is clear during analysis or shock delivery.
- Placing pads over clothing or without adequate skin contact.
- Stopping CPR for too long or ignoring the device prompts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

AED Trainer

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: AED Trainer with Adult Pad

What it is used for

An AED trainer simulates the operation of an automated external defibrillator for first-aid and CPR training. It teaches learners to follow voice prompts, place training pads correctly, and maintain scene safety without delivering a therapeutic shock.

Main parts

- AED trainer unit with control buttons
- Adult training electrode pads
- Training pad cable and connector
- Speaker and voice-prompt system
- Training remote control if supplied
- Carrying case and batteries or power adapter

Before you start

1. Confirm that the unit is an AED trainer and not a clinical defibrillator.
2. Inspect the trainer, pad cables, connectors, controls, and casing for damage.
3. Check the battery level or approved power supply before starting the session.
4. Select the appropriate training scenario and language for the learners.
5. Explain scene safety, emergency response, CPR, and the purpose of AED training.
6. Check that adult training pads are clearly marked and that no therapeutic shock can be delivered.

How to use

1. Switch on the AED trainer and select the required training scenario.
2. Explain that learners must first check scene safety and responsiveness in a real emergency.
3. Demonstrate calling emergency services and asking someone to obtain an actual AED.
4. Position the training mannequin or practice partner as directed by the scenario.
5. Expose the chest area and demonstrate correct placement of the adult training pads.
6. Connect the training pad cable to the trainer if it is not already connected.
7. Follow the voice prompts and demonstrate checking that nobody touches the mannequin during simulated analysis.
8. When prompted, demonstrate the simulated shock sequence or the no-shock response without using a real defibrillator.
9. Resume CPR when the scenario instructs and follow the trainer's compression and ventilation prompts where available.
10. Reset the scenario, remove the training pads carefully, and switch off the trainer after practice.

⚠ Safety

- Use only designated training pads and accessories with the AED trainer.
- Never connect training pads to a person for therapeutic use or to a clinical defibrillator unless explicitly approved for that purpose.
- Make clear that the trainer does not provide a real shock and is not emergency treatment equipment.
- Keep cables arranged to prevent tripping and protect connectors from liquids.
- Use an actual AED and contact emergency services in a real cardiac emergency.
- Clean reusable accessories according to the manufacturer's instructions between learners.

Care & storage

- Wipe the trainer and reusable accessories with approved cleaning materials after each session.
- Inspect pad cables, connectors, buttons, and casing for wear or damage.
- Replace batteries and worn training pads according to the manufacturer's instructions.
- Store the trainer and pads in a clean, dry case and check completeness after use.

Common mistakes to avoid

- Using a clinical defibrillator instead of a trainer can create serious danger during practice.
- Placing pads incorrectly can teach unsafe or ineffective technique.
- Skipping the clear-area instruction can lead learners to overlook shock safety in real emergencies.
- Using damaged cables or depleted batteries can interrupt the training scenario.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Adult Diaper

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Adult Diaper; Adult diapers

What it is used for

Adult diapers support continence care and help maintain hygiene, comfort, and dignity.

How to handle

- Select the correct size and absorbency for the person's needs and follow the care plan.
- Check packaging integrity and product condition before use.
- Perform hand hygiene and wear appropriate gloves when assisting with personal care.
- Change the diaper promptly when soiled or as directed by the care plan; clean and dry the skin gently.
- Dispose of used diapers according to facility waste procedures and report skin irritation or unusual symptoms to the responsible caregiver.

Storage

- Keep unused diapers sealed in a clean, dry area.
- Separate clean supplies from used products and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Cushion

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Air Cushion

What it is used for

To provide cushioning and support for comfort or pressure redistribution when appropriate for the person's needs.

How to handle

- Inspect the cushion for leaks, tears, contamination, and valve damage before use.
- Confirm the cushion type and recommended use; follow clinical guidance when used for pressure management or patient care.
- Position the cushion correctly on a stable, compatible seat or surface and adjust inflation only as directed by the manufacturer.
- Help the user sit or lie in a safe, comfortable position and check stability and skin comfort as appropriate.
- Clean and disinfect according to the product instructions and facility policy.

Storage

- Clean and dry the cushion completely before storage; keep valves closed as directed.
- Store in a clean, dry area away from sharp objects, excessive heat, and heavy loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Airway Mannequin

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Airway Mannequin

What it is used for

An airway mannequin provides hands-on practice for airway assessment and approved airway management techniques. It helps learners develop correct positioning, ventilation, and device-placement skills without practising invasive procedures on a person.

Main parts

- Mannequin head, neck, and torso assembly
- Mouth and nasal airway passages
- Replaceable airway or lung insert if fitted
- Ventilation bag or compatible practice device
- Training airway adjuncts such as masks or tubes
- Cleaning supplies and replacement consumables

Before you start

1. Read the mannequin instructions and identify the airway procedures supported by the model.
2. Inspect the head, neck, mouth, airway passages, connectors, and replaceable parts for damage.
3. Confirm that the correct training accessories and consumables are available.
4. Clean the mannequin and accessories according to the infection-control procedure before use.
5. Explain the learning objectives, approved techniques, and limits of the training model.
6. Wear suitable gloves and ensure the practice area is clean and well lit.

How to use

1. Place the mannequin on a stable training surface in the position required for the exercise.
2. Demonstrate the approved airway assessment and positioning technique for the scenario.
3. Practise opening the airway using the technique appropriate to the training objective.
4. Insert the designated practice airway adjunct only as instructed and with the correct technique.
5. Use the compatible ventilation device to practise gentle ventilation and observe the model's response.
6. Check that the simulated chest movement or other model indicator matches the expected result.
7. Remove training devices carefully and reset replaceable airway components when required.
8. Repeat the exercise with supervised feedback on positioning, technique, and device handling.
9. After practice, remove disposable items and clean reusable components using approved products.
10. Inspect the mannequin, dry it as instructed, and return it to storage.

⚠ Safety

- Use the mannequin only for the procedures approved by its manufacturer.
- Never perform invasive airway procedures on a real person as a practice exercise.
- Do not force airway devices or apply excessive pressure that could damage the model.
- Use single-use training consumables only as permitted and dispose of them appropriately.
- Follow infection-control procedures when sharing the mannequin between learners.
- Keep cleaning chemicals away from learners' eyes and follow product safety instructions.

Care & storage

- Clean and disinfect mouth-contact surfaces after each learner or session as specified.
- Replace damaged airway inserts, masks, or other consumables with compatible parts.
- Inspect joints, airway passages, connectors, and skin-like surfaces for wear or tears.
- Store the mannequin clean and dry in its protective case or designated area.

Common mistakes to avoid

- Using an unsupported device can damage the airway passages or give misleading feedback.
- Excessive force during insertion can tear replaceable components.
- Poor cleaning between learners can spread contamination.
- Incorrect head positioning can prevent effective simulated ventilation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ambu Bag

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: Ambu Bag; Ambu Bag with Mask (Adult); Ambu Mask (Adult); Resuscitator

What it is used for

An Ambu bag, also called a bag-valve-mask resuscitator, provides manual ventilation to a person who is not breathing adequately. It should be used by trained responders with appropriate airway support and emergency assistance.

Main parts

- Self-inflating bag provides manual airflow when squeezed.
- Patient valve directs airflow toward the patient and supports exhalation.
- Face mask forms a seal over the nose and mouth.
- Oxygen inlet and reservoir may supply additional oxygen on equipped models.
- Air intake valve allows the bag to refill after compression.
- Tubing connects compatible oxygen sources when required.
- Pressure relief or safety features may be included in some models.

Before you start

1. Call emergency services and activate the facility's emergency response procedure immediately.
2. Use the device only if trained in resuscitation and bag-mask ventilation.
3. Check responsiveness and breathing, and begin CPR when indicated by current training.
4. Inspect the bag, mask, valves, connectors, reservoir, and tubing for damage or obstruction.
5. Select an appropriate mask size and ensure oxygen supplies are available when indicated.
6. Use infection-control PPE and a suitable barrier where available without delaying urgent care.

How to use

1. Position the person and open the airway using the method appropriate to the situation and training.
2. Place the mask over the nose and mouth and obtain a secure seal.
3. Connect oxygen and the reservoir if indicated and compatible with the device.
4. Squeeze the bag smoothly to deliver only enough air for visible chest rise.
5. Allow the bag to refill fully between breaths and observe chest movement.
6. Ventilate at the rate and volume taught in current resuscitation training.
7. Watch for chest rise, air leaks, vomiting, or signs of airway obstruction.
8. If the chest does not rise, reposition the airway and mask and seek trained assistance.
9. Continue ventilation and CPR as indicated until normal breathing returns or emergency responders take over.

10. After use, dispose of single-use components and clean reusable parts according to infection-control instructions.

⚠ Safety

- Never use the device without appropriate training except as directed by an emergency dispatcher.
- Avoid excessive ventilation or forceful bag squeezing, which can cause harm.
- Ensure the mask seal and airway position are correct without pressing excessively on the face.
- Use oxygen safely and keep it away from flames, sparks, and oil or grease.
- Do not reuse single-use components or equipment that cannot be safely cleaned and disinfected.
- Do not delay emergency services, CPR, or defibrillator use when indicated.

Care & storage

- Check the bag, valves, mask, and connectors before each use according to the checklist.
- Clean and disinfect reusable components according to the manufacturer's instructions and facility policy.
- Store the complete device in a clean, dry, accessible emergency kit.
- Replace damaged, expired, or missing components and document inspection as required.

Common mistakes to avoid

- Squeezing the bag too forcefully or delivering breaths too quickly.
- Failing to create a mask seal or maintain an open airway.
- Ignoring absent chest rise or possible airway obstruction.
- Using contaminated, damaged, or expired equipment during an emergency.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Antiseptic

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Anti-septic; Betadine Solution Bottle; Liquid Antiseptic

What it is used for

Antiseptics are used on skin or minor wounds only as directed by the product label or healthcare professional.

How to handle

- Check the product name, concentration, expiry date, and intended application before use.
- Follow label directions and use appropriate clean applicators where required.
- Avoid eyes, swallowing, and use on unsuitable wounds or body areas unless specifically directed.
- Do not mix antiseptics with other products unless the instructions permit it.
- Stop use and seek appropriate medical advice if severe irritation or an allergic reaction occurs.

Storage

- Keep the original labelled container tightly closed.
- Store according to label instructions and out of reach of children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Artery Forceps

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Artery Forceps

What it is used for

To clamp tissue or blood vessels during clinical or training procedures when used by qualified personnel.

How to handle

- Inspect the forceps for cleanliness, alignment, smooth movement, secure locking, and damage before use.
- Use only for the intended procedure and under appropriate clinical supervision; do not use on a person during training without authorization.
- Maintain aseptic technique and use the required personal protective equipment in clinical settings.
- Grip the intended material gently, engage the locking mechanism only as required, and avoid excessive force.
- After use, handle contaminated instruments as clinical sharps or reusable instruments according to facility protocol and send for approved reprocessing.

Storage

- After approved cleaning, disinfection, and sterilization as required, dry and store the instrument in its designated clean instrument tray or sterile pack.
- Keep sterile and non-sterile instruments separate and follow facility infection-control procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Aspiration Pump

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Aspiration pump (peristaltic or vaccum)

What it is used for

An aspiration pump removes fluids, mucus, or other secretions through a suction system for clinical or laboratory use, depending on the device design. Operate it only for its intended purpose and follow the manufacturer's instructions, infection-control procedures, and the relevant clinical protocol.

Main parts

- Pump unit and motor
- Vacuum regulator and pressure gauge or display
- Collection canister or fluid trap
- Suction tubing and patient-side connector, where applicable
- Bacterial or overflow filter, where fitted
- Power supply, control switch, and optional foot switch

Before you start

1. Confirm the device type, intended application, and authorised procedure; patient use must be handled by trained healthcare personnel.
2. Inspect the pump housing, power cable, controls, vacuum gauge, canister, tubing, and connectors for damage.
3. Check that the collection canister is clean, correctly fitted, and below its maximum fill line.
4. Confirm that the correct filter and approved single-use or reprocessed accessories are installed as specified.
5. Verify the electrical supply and any required alarm or battery status before use.
6. For patient care, confirm the clinical order, required suction setting, patient identity, and infection-control precautions.

How to use

1. Place the pump on a stable, dry surface with ventilation openings unobstructed.
2. Connect the collection canister, filter, and tubing according to the device diagram.
3. Confirm that tubing connections are secure and that the fluid trap or overflow protection is correctly fitted.
4. Switch on the pump and check that the vacuum indicator responds normally.
5. Set the suction pressure only to the level prescribed by the authorised clinical protocol or specified procedure.

6. For patient use, trained personnel should explain the procedure, position the patient appropriately, and connect the approved patient interface using aseptic technique as required.
7. Apply suction only according to the relevant clinical protocol and the device's intended use; continuously observe the patient where applicable.
8. Monitor the collection canister and stop suction before the liquid reaches the maximum fill line.
9. If suction is weak, the gauge is abnormal, fluid enters the pump, or an alarm occurs, stop use and follow the troubleshooting procedure.
10. After use, switch off the unit and disconnect power before dismantling components for cleaning.
11. Dispose of collected material and single-use accessories according to infection-control and biomedical-waste procedures.
12. Clean and disinfect reusable components as instructed, replace required filters, and document use or faults.

⚠ Safety

- Patient aspiration procedures must be performed only by trained and authorised healthcare personnel under the applicable clinical protocol.
- Never exceed the device's specified vacuum range or an authorised clinical suction setting.
- Do not use damaged tubing, an overfilled canister, or a missing or incorrectly fitted filter.
- Prevent collected fluid from entering the pump; stop immediately if overflow or contamination is suspected.
- Use appropriate PPE and standard precautions when handling secretions or contaminated components.
- Disconnect power before cleaning or servicing and never immerse the electrical pump unit in liquid.

Care & storage

- Empty and clean the collection canister according to the manufacturer's instructions and local infection-control policy.
- Inspect tubing, seals, connectors, filters, and the overflow protection regularly.
- Replace single-use accessories and filters at the specified intervals or when contaminated.
- Check vacuum performance, alarms, and battery condition where applicable.
- Arrange qualified servicing if suction is unreliable, the gauge fails, or fluid enters the pump.

Common mistakes to avoid

- Using the pump without checking the canister, filter, or tubing connections.
- Selecting a suction pressure without following the relevant clinical protocol.
- Allowing the collection canister to overfill.
- Reusing single-use accessories or cleaning the electrical unit by immersing it in liquid.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Back Rest

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Back Rest; Back Rest, Foot Rest

What it is used for

Back rests and foot rests support posture and comfort during seating, care, or approved clinical activities.

How to handle

- Inspect the rest, frame, padding, and adjustment mechanism for damage or instability.
- Select and position the support according to the user's needs and equipment instructions.
- Ensure the support is secure before use and does not obstruct movement or access.
- Clean contact surfaces between users according to hygiene procedures.
- Do not use broken or unstable supports; report defects promptly.

Storage

- Store clean, dry supports in a designated area.
- Keep them away from moisture and avoid placing heavy objects on padded surfaces.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bandage

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Bandage

What it is used for

Handle bandages hygienically for basic wound care.

How to handle

- Wash hands and wear suitable gloves before assisting with wound care.
- Use sterile bandages for wounds when required and check packaging integrity.
- Avoid touching the wound-contacting surface of a sterile dressing.
- Dispose of used or contaminated bandages according to healthcare waste procedures.

Storage

- Keep unopened bandages sealed in clean, dry packaging.
- Check expiry dates and protect supplies from moisture, dust, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bathing Aid

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Bathing aids (e.g., shower chair, sponge, no-rinse wipes)

What it is used for

Bathing aids support personal hygiene for people who need assistance with washing. Choose the appropriate aid and maintain privacy, comfort, and fall prevention throughout use.

How to use

1. Select a bathing aid suitable for the person's needs and mobility.
2. Inspect shower chairs, handles, non-slip feet, and other parts for damage.
3. Prepare a warm, safe area and place supplies within easy reach.
4. Explain each step, obtain consent, and protect the person's privacy.
5. Use the aid according to its instructions and provide support as needed.
6. Clean the skin gently, avoiding excessive rubbing and sensitive areas.
7. Dry the person, clean reusable aids, and store supplies safely.

⚠ Safety

- Never leave a person at risk of falling unattended during bathing.
- Check water temperature before contact to prevent burns.
- Use no-rinse wipes and cleaning products according to their instructions.
- Keep floors dry where possible and use non-slip support.

Care & storage

- Clean reusable aids after every use and discard single-use wipes appropriately.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bed Pan

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Bed pan; Bedpans

What it is used for

A bed pan collects bodily waste for individuals who cannot easily use a toilet.

How to handle

- Use gloves and other required PPE when handling the bed pan or its contents.
- Explain the procedure respectfully, maintain privacy, and assist the user safely.
- Check the pan for cracks and position it carefully to avoid discomfort or spills.
- Empty contents into the designated toilet or disposal system according to facility procedure.
- Clean and disinfect reusable bed pans after each use and perform hand hygiene.

Storage

- Store cleaned, dry bed pans in a designated clean area.
- Keep clean items separate from used equipment and waste containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Biomedical Waste Bag

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Bio hazard bags for Waste Disposal; Bio hazard bags for Waste Disposal / Blue sharps container for waste disposal

What it is used for

Biomedical waste bags collect designated healthcare waste for segregation and disposal under applicable biomedical waste rules.

How to handle

- Use only the bag colour and category specified by the applicable biomedical waste segregation rules and facility procedure.
- Check the bag for leaks, punctures, and damage before use; do not overfill.
- Wear appropriate PPE and avoid compressing waste or reaching inside the bag.
- Seal and label the bag as required and place it in the designated collection area.
- Place sharps in an approved puncture-proof sharps container, never loose in a waste bag; report spills immediately.

Storage

- Keep waste bags in designated, secure areas away from public access and incompatible waste.
- Follow facility time limits and authorised collection procedures; do not store beyond permitted limits.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Biomedical Waste Bin

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Biomedical dustbin; Bio-Medical waste bins

What it is used for

Biomedical waste bins support segregation and temporary collection of healthcare waste according to applicable rules.

How to handle

- Use bins with the correct colour, label, and category specified by current regulations and facility procedures.
- Inspect the bin, lid, and liner for damage, leakage, or contamination before use.
- Keep lids closed when not actively depositing waste and never compress contents by hand.
- Place sharps in approved puncture-proof sharps containers rather than ordinary waste bins.
- Clean and disinfect reusable bins according to facility procedures and report spills or damage immediately.

Storage

- Keep bins in designated, secure, well-maintained collection areas.
- Empty and service bins according to authorised waste-handling schedules and applicable regulations.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Birthing Simulator

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Birthing Simulator

What it is used for

A birthing simulator provides practical training for normal childbirth, delivery assistance, and selected obstetric emergency scenarios. It allows learners to practise approved techniques on a model under supervision without involving a real patient.

Main parts

- Maternal pelvis and torso model
- Birth canal and replaceable vaginal insert
- Fetal mannequin with cord and placenta model if supplied
- Positioning supports and assembly connectors
- Lubricant and compatible training accessories
- Cleaning supplies and replacement components

Before you start

1. Read the simulator instructions and confirm which birth and emergency scenarios the model supports.
2. Inspect the torso, pelvis, birth canal, fetal model, cord, and connectors for damage.
3. Assemble the simulator securely according to the manufacturer's instructions.
4. Prepare only the approved lubricants and training accessories for the planned exercise.
5. Clean the model and accessories using the required infection-control procedure.
6. Explain the training objectives, supervision requirements, and limits of the simulator to participants.

How to use

1. Position the simulator securely on the designated training table or support.
2. Assemble the maternal and fetal components according to the selected scenario.
3. Demonstrate the approved assessment and preparation steps for the simulated birth.
4. Apply compatible training lubricant as specified to reduce friction on replaceable components.
5. Guide the fetal model through the birth canal using the supervised technique for the scenario.
6. Practise supporting the simulated delivery and handling the cord or placenta model only as instructed.
7. For emergency scenarios, follow the approved training protocol and instructor guidance.
8. Stop if components become stuck, misaligned, or subject to excessive force.
9. After the exercise, disassemble the reusable components according to the instructions.
10. Clean, dry, inspect, and reset the simulator for the next session.

⚠ Safety

- Use the simulator only for training procedures approved by the manufacturer.
- Do not use excessive pulling, twisting, or pressure on the fetal model or birth canal.
- Do not use unapproved oils or lubricants that may damage the model material.
- Follow infection-control requirements when components are shared between learners.
- Use sharp instruments only if the model and supervised training procedure explicitly allow them.
- Explain that real childbirth and obstetric emergencies require qualified clinical care and local emergency protocols.

Care & storage

- Clean and disinfect reusable surfaces and inserts after each session using compatible products.
- Inspect the birth canal, fetal model, cord, connectors, and supports for wear or tears.
- Replace damaged or worn components with manufacturer-approved parts.
- Store the simulator clean, dry, and assembled or disassembled as recommended.

Common mistakes to avoid

- Incorrect assembly can interfere with the simulated delivery and damage the components.
- Using incompatible lubricant can degrade the birth canal material.
- Pulling the fetal model with excessive force can tear inserts or connectors.
- Treating simulation practice as a substitute for supervised clinical training can create unsafe expectations.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

CPR Manikin

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: CPR Manikin; CPR Manikins and Choking Response Trainers; CPR Mannequin; Manikin; Mannequin (Adult); Simulation Equipment - Mannequins; Simulation tools for CPR and minor injury handling; Training mannequins or simulation tools for CPR demonstrations

What it is used for

Use a CPR manikin to practise chest compressions, rescue-breathing techniques, and basic life-support skills in a controlled training setting.

How to use

1. Place the manikin on a stable, flat training surface.
2. Inspect the manikin, airway parts, and feedback system for damage or missing components.
3. Explain the practice procedure and demonstrate the correct hand position.
4. Practise chest compressions at the recommended rate and depth for the manikin type.
5. Practise rescue breaths only with the approved airway or face-shield arrangement.
6. Follow the trainer's instructions when using feedback indicators or choking-response features.
7. Clean and reset the manikin after the session.

⚠ Safety

- Use a manikin for training, not on a real person in an emergency.
- Use individual face shields or approved airway barriers for each learner.
- Clean contact surfaces between users according to the manufacturer's instructions.
- Stop practice if a component becomes loose, damaged, or unsafe.

Care & storage

- Clean and disinfect reusable parts with approved products after training.
- Store the manikin and accessories in a clean, dry case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

CPR Pocket Mask

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **High** 3 labs

Covers these lab items: Pocket Mask

What it is used for

A CPR pocket mask provides a barrier for rescue breaths during CPR and helps reduce direct contact with the casualty. Use it according to current first aid training and the manufacturer's instructions.

How to use

1. Check the mask, one-way valve, and case for damage or contamination.
2. Call emergency services and assess the scene before approaching the casualty.
3. Wear gloves and other available protective equipment.
4. Place the casualty on their back and open the airway using the trained technique.
5. Position the mask securely over the nose and mouth, following its instructions.
6. Provide rescue breaths as trained and watch for visible chest rise.
7. Continue CPR according to current training and emergency dispatcher guidance.
8. Dispose of single-use parts correctly or clean reusable components as instructed.

⚠ Safety

- Do not delay CPR or emergency assistance while searching for a mask.
- Do not use a damaged or contaminated mask.
- Follow infection-control precautions and avoid direct contact with body fluids.
- Use the mask only within your training and follow emergency dispatcher instructions.

Care & storage

- Store the mask in its clean, closed case and check expiry or condition regularly.
- Replace single-use masks and maintain reusable components according to the manufacturer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cardiac Table

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Cardiac Table

What it is used for

A cardiac table provides a supportive surface for patients or users during approved care and clinical activities.

How to handle

- Inspect the table, frame, height adjustment, wheels, and locking mechanism for damage or instability.
- Position and adjust the table according to the care plan and manufacturer's instructions.
- Engage brakes before placing items or assisting a user and keep the table within safe reach.
- Clean and disinfect contact surfaces according to clinical hygiene procedures.
- Do not use the table for lifting or transferring a person unless it is specifically designed and approved for that purpose.

Storage

- Return the table to its designated position and engage brakes where fitted.
- Keep it clean, dry, and free from unnecessary items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Casualty Evacuation Kit

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Drill Kits for Casualty Evacuation and Triage Practice

What it is used for

A casualty evacuation kit supports training in casualty movement, triage, and emergency response procedures. Use training equipment under instructor supervision and follow the approved drill plan.

How to use

1. Check the kit inventory, stretchers, straps, labels, and training supplies for damage.
2. Review the drill objectives, roles, evacuation route, and communication procedure.
3. Wear the required PPE and prepare the designated practice area.
4. Use training mannequins or approved simulated casualties for lifting and movement practice.
5. Apply triage labels according to the instructor's scenario and training protocol.
6. Move simulated casualties using the approved technique and coordinated team commands.
7. Count all equipment after the drill and report missing or damaged items.

⚠ Safety

- Do not practise casualty lifting on real injured people without qualified supervision.
- Use safe lifting posture and request enough helpers for heavy loads.
- Keep evacuation routes clear and stop if anyone feels pain or unsafe.
- Clearly identify simulated supplies so they are not mistaken for clinical equipment.

Care & storage

- Clean reusable training equipment after drills and replenish consumables according to the inventory.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Catheterization Kit

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Advanced Male and Female Catheterization Kit; Folley catheter; Red rubber catheter for Basti Administration; Rubber Catheter

What it is used for

A catheterization kit contains sterile instruments and supplies used by trained healthcare professionals for urinary catheterization or other specified clinical procedures. Correct aseptic technique, patient identification, and appropriate catheter selection help reduce infection and injury.

Main parts

- Catheter drains or delivers fluid through the intended anatomical passage.
- Sterile gloves help maintain aseptic technique during the procedure.
- Sterile drape provides a clean working field.
- Antiseptic or cleansing supplies prepare the specified area.
- Lubricating gel reduces friction during catheter insertion when indicated.
- Drainage bag and tubing collect urine in urinary catheter kits.
- Syringe, sterile water, or other accessories are included only when specified for the kit and procedure.

Before you start

1. Only trained and authorised healthcare personnel should perform catheterization.
2. Confirm the patient's identity, procedure, indication, consent, and relevant allergies according to clinical policy.
3. Explain the procedure and provide privacy and appropriate positioning.
4. Check the package seal, expiry date, sterility indicators where provided, and kit contents.
5. Gather the correct catheter type and size and all required sterile supplies.
6. Perform hand hygiene and prepare a clean area using the facility's infection-control procedure.

How to use

1. Open the kit using aseptic technique and avoid touching sterile items with unclean hands.
2. Put on the required PPE and sterile gloves according to clinical protocol.
3. Prepare the patient and clean the specified anatomical area using the approved procedure.
4. Arrange the sterile drape and supplies without contaminating the sterile field.
5. Apply the approved lubricant or antiseptic only as directed for the procedure.
6. Insert the catheter only using the approved clinical technique and without forcing it.
7. For urinary catheterization, follow the device instructions for confirming placement before any specified balloon inflation.
8. Connect the appropriate drainage or delivery system using aseptic technique when required.

9. Secure the catheter, maintain appropriate drainage, and monitor the patient for discomfort or complications.
10. Dispose of single-use items safely, document the procedure, and clean reusable equipment according to policy.

⚠ Safety

- Never perform catheterization without appropriate clinical training, authorisation, and a valid indication.
- Do not use a package that is open, damaged, expired, or no longer sterile.
- Never force the catheter if resistance, severe pain, or unexpected bleeding occurs; stop and seek clinical assistance.
- Use aseptic technique to reduce the risk of introducing infection.
- Inflate a catheter balloon only when the specific device instructions and confirmed placement permit it.
- Follow clinical policy for sharps, contaminated materials, and medical waste disposal.

Care & storage

- Store unopened kits in a clean, dry area within the specified storage conditions.
- Check package integrity and expiry dates before each use.
- Dispose of single-use catheters and supplies after use; do not reuse them.
- Report missing, damaged, contaminated, or expired kit components and replace the kit as required.

Common mistakes to avoid

- Using a non-sterile or expired catheterization kit.
- Touching sterile components with unclean hands or contaminating the sterile field.
- Forcing insertion despite resistance, pain, or bleeding.
- Inflating a urinary catheter balloon before confirming correct placement according to device instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cervical Collar

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Cervical Collar; Cervical Colour Set

What it is used for

A cervical collar supports the neck and may help restrict cervical spine movement when clinically indicated. It should be selected, fitted, and monitored by trained healthcare or emergency personnel as part of an appropriate patient assessment and spinal motion restriction plan.

Main parts

- Rigid or semi-rigid shell provides structural support.
- Adjustable height or sizing mechanism helps fit the patient's neck where provided.
- Chin support helps position the lower jaw.
- Occipital support contacts the back of the head where included.
- Hook-and-loop straps or fastening tabs secure the collar.
- Padding improves comfort and helps distribute pressure.
- Sizing guide or measuring reference assists collar selection where supplied.

Before you start

1. Only trained healthcare or emergency personnel should fit a cervical collar.
2. Assess the scene and the patient according to the applicable clinical protocol; address airway, breathing, circulation, and immediate life threats first.
3. Select a collar size suitable for the patient's anatomy and the device instructions.
4. Inspect the shell, padding, straps, fasteners, and adjustment mechanism for damage or contamination.
5. Explain the procedure to the patient when possible and obtain assistance to maintain appropriate head and neck alignment.
6. Check for conditions requiring special assessment, including breathing difficulty, neck deformity, or pressure-sensitive skin.
7. Perform hand hygiene and use appropriate PPE when exposure to bodily fluids is possible.

How to use

1. Position the patient and maintain manual stabilisation when indicated by the clinical protocol.
2. Measure or estimate the required collar size using the manufacturer's sizing method without forcing the neck into a new position.
3. Adjust the collar height or configuration before fitting when the device design requires it.
4. Place the collar around the neck using the trained technique while maintaining the patient's alignment.
5. Position the chin support under the chin without forcing the head or jaw into an uncomfortable position.

6. Secure the fastening straps evenly so the collar is stable but does not obstruct breathing or circulation.
7. Check that the collar does not press excessively on the throat, jaw, ears, or skin.
8. Reassess airway access, breathing, comfort, skin condition, and neurological status as clinically appropriate.
9. Continue monitoring the patient and maintain the wider spinal motion restriction plan where indicated.
10. Remove or adjust the collar only when directed by the applicable clinical protocol or qualified clinician.
11. Clean and disinfect reusable components after use according to the manufacturer's instructions and facility policy.
12. Document fitting, patient response, skin checks, and any changes according to clinical requirements.

⚠ Safety

- A cervical collar does not replace airway management, clinical assessment, or other indicated spinal precautions.
- Do not delay lifesaving care to fit a collar.
- Do not force the neck into alignment or tighten the collar enough to impair breathing or circulation.
- Monitor for breathing difficulty, increasing pain, pressure injury, vomiting, or neurological changes and seek immediate clinical reassessment.
- Do not use a cracked, deformed, contaminated, or incorrectly sized collar.
- Follow local protocols for spinal motion restriction, patient movement, infection control, and equipment cleaning.

Care & storage

- Clean and disinfect reusable parts using approved products and manufacturer instructions.
- Inspect the shell, padding, straps, and fasteners for cracks, wear, and loss of secure fastening.
- Check adjustable parts for reliable operation before returning the collar to service.
- Store the collar clean, dry, and protected from deformation.
- Discard single-use components after use and remove damaged equipment from service.

Common mistakes to avoid

- Selecting an incorrect collar size or fitting it without trained assistance.
- Forcing the head or neck into a position during application.
- Tightening the collar enough to restrict breathing or cause excessive pressure.
- Failing to reassess the patient or check for skin pressure after fitting.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Crash Cart

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Crash Card; Crash cart trolley

What it is used for

A crash cart is a mobile trolley that stores emergency medicines, airway equipment, monitoring devices, and resuscitation supplies for rapid response to medical emergencies. Proper organisation, routine checks, equipment readiness, and secure medication management help clinical teams respond quickly and safely.

Main parts

- Trolley frame and wheels allow rapid movement between care areas.
- Drawers and compartments organise emergency supplies.
- Defibrillator or monitor is stored on the cart where assigned.
- Bag-mask ventilation equipment supports emergency breathing assistance.
- Airway supplies include approved airways and related accessories.
- Emergency medicines are stored in labelled compartments according to facility policy.
- Suction equipment and related accessories are available where assigned.
- Oxygen equipment and delivery accessories are included where approved.
- Sharps container and clinical waste supplies support safe disposal.
- Checklist, seal, or lock helps track cart readiness and access.

Before you start

1. Only trained healthcare personnel should access and use crash cart equipment within their clinical roles.
2. Check the cart location, accessibility, wheel locks, drawer security, and tamper-evident seal according to facility policy.
3. Verify the documented daily or shift check and inspect for missing, expired, or damaged supplies.
4. Confirm that the defibrillator or monitor, cables, electrodes, battery, and power supply are ready for use.
5. Check bag-mask equipment, suction, oxygen supplies, and airway accessories according to the assigned checklist.
6. Verify medicine labels, expiry dates, storage conditions, and required emergency stock without delaying urgent care.
7. Ensure the cart is connected to power for charging where required and is not obstructed by other equipment.

How to use

1. Move the cart to the emergency area and lock the wheels when positioned, if appropriate.

2. Activate the emergency response system and call for the required clinical team.
3. Open the cart and organise access so the team can reach critical equipment without delay.
4. Provide emergency medicines and supplies only under the direction of authorised clinical personnel and applicable protocols.
5. Use the defibrillator or monitor according to the device instructions and current resuscitation protocol.
6. Use airway, ventilation, suction, and oxygen equipment only within the responder's training and clinical scope.
7. Monitor equipment readiness during the response and replace used supplies when safe to do so.
8. After the emergency, account for used medicines, equipment, and disposable supplies.
9. Replace used or missing items, check expiry dates, and replenish stock according to facility policy.
10. Clean and disinfect reusable cart surfaces and equipment using approved methods.
11. Complete the post-event checklist, record equipment use and medicine reconciliation, and apply a new security seal where required.
12. Return the cart to its designated accessible location and confirm readiness for the next emergency.

⚠ Safety

- Do not delay immediate life-saving care while completing routine cart checks.
- Only authorised clinical personnel should administer emergency medicines or operate specialised equipment.
- Follow current resuscitation protocols, medicine policies, and device-specific instructions.
- Check electrical cables, defibrillator pads, oxygen equipment, and suction systems for damage before use when time permits.
- Secure sharps immediately in an approved puncture-resistant container and dispose of clinical waste correctly.
- Maintain patient confidentiality and document emergency interventions according to facility requirements.

Care & storage

- Inspect the cart and critical equipment at the frequency specified by facility policy.
- Check medicines, sterile packs, batteries, electrodes, and consumables for expiry or damage.
- Clean and disinfect the trolley and reusable equipment after use and according to the infection-control schedule.
- Charge electronic equipment as required and verify that it passes its self-test.
- Report missing stock, broken seals, equipment faults, and expired items promptly.

Common mistakes to avoid

- Leaving the crash cart inaccessible, unlocked contrary to policy, or incompletely stocked.
- Failing to check defibrillator readiness, oxygen supplies, or suction function.
- Using expired supplies or administering medicines without appropriate authorisation.
- Failing to replenish, document, and reseal the cart after an emergency.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Crutch

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Crutch

What it is used for

To support mobility and reduce weight-bearing on an injured or weak leg when recommended by a healthcare professional.

How to handle

- Inspect the crutch body, handgrip, cuff or underarm pad, adjustment pins, and rubber tip for damage.
- Adjust the crutch to the user's size according to clinical guidance; do not place body weight directly on an underarm pad.
- Teach the user the recommended walking pattern and safe use on level surfaces before attempting stairs or uneven ground.
- Move slowly, check that the rubber tips grip the floor, and avoid wet or slippery surfaces where possible.
- Stop and seek professional adjustment if the crutch slips, causes pain, or feels unstable.

Storage

- Clean and dry the handgrip and contact surfaces according to facility guidance.
- Store upright in a designated dry area where the crutch cannot fall or obstruct walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Diagnostic Test Strip

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Medium** 1 labs

Covers these lab items: Occult blood strips

What it is used for

Diagnostic test strips detect specified substances in a sample when used with the correct test procedure and interpretation criteria.

How to handle

- Verify the strip type, intended sample, expiry date, and package integrity before testing.
- Follow the manufacturer's instructions for sample collection, timing, and result interpretation.
- Use appropriate PPE and avoid touching the reactive test area or contaminating the sample.
- Dispose of used strips and sample materials according to the relevant clinical or laboratory waste procedure.

Storage

- Keep strips in their original container with the lid tightly closed and desiccant retained where provided.
- Follow the specified temperature, humidity, and expiry requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Disposable Gloves

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Gloves; Gloves (disposable), Gloves (surgical); Gloves (Surgical)-Packet; Gloves(disposable)-Packet

What it is used for

Disposable gloves provide a single-use barrier for hygiene, examination, cleaning, and other approved tasks.

How to handle

- Choose gloves suitable for the task, chemical exposure, and required protection.
- Check the size, expiry date, and packaging integrity before use.
- Perform hand hygiene before wearing and after removing gloves.
- Change gloves between users or tasks and immediately if torn or contaminated.
- Do not wash or reuse single-use gloves; avoid touching clean surfaces with contaminated gloves.

Storage

- Keep gloves sealed in their original packaging until needed.
- Store in a cool, dry place away from sunlight and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

ECG Machine

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: ECG machine

What it is used for

An ECG machine records the electrical activity of the heart through electrodes attached to the patient's skin. Correct electrode placement, patient preparation, and signal quality checks help produce a reliable tracing for interpretation by qualified healthcare personnel.

Main parts

- ECG recorder and display
- Patient lead wires and electrode connectors
- Disposable or reusable electrodes
- Patient cable and lead set
- Control buttons and recording settings
- Power supply, battery, and printer or digital output

Before you start

1. Use the ECG machine only if trained and authorised to perform the procedure.
2. Confirm the patient's identity, explain the procedure, and obtain consent according to facility policy.
3. Check the machine, patient cable, lead wires, electrodes, power supply, and accessories for damage.
4. Verify that the machine is clean, functional, and within its required maintenance and calibration schedule.
5. Prepare a private, comfortable area and ensure the patient can lie still and relaxed.
6. Clean and dry electrode sites as required, and prepare the correct electrodes and lead configuration.

How to use

1. Position the patient comfortably, usually lying on their back, and ask them to relax and remain still.
2. Switch on the ECG machine and select the appropriate patient and recording settings.
3. Enter the required patient details accurately, following the facility's privacy procedures.
4. Attach electrodes to the correct anatomical locations using the approved lead-placement diagram.
5. Connect the lead wires to the corresponding electrodes and check that each connection is secure.
6. Ensure the patient cable is arranged to minimise pulling, movement, and electrical interference.

7. Check the displayed signal for a stable baseline and acceptable waveform quality before recording.
8. Ask the patient to remain still and breathe normally during acquisition unless otherwise instructed.
9. Start the recording and verify that all required leads and waveform segments have been captured.
10. Repeat the recording if movement artefact, loose electrodes, or signal interference makes the tracing unsuitable.
11. Label, save, or print the ECG with the correct patient identification, date, and time.
12. Provide the tracing to the authorised healthcare professional for interpretation; do not independently diagnose from the recording.
13. Disconnect the lead wires gently, remove the electrodes, and help the patient return to a comfortable position.
14. Clean reusable accessories and the machine's patient-contact surfaces according to infection-control procedures.

⚠ Safety

- ECG recording should be performed by trained healthcare personnel and interpreted by an appropriately qualified professional.
- Maintain patient privacy, dignity, and consent throughout the procedure.
- Inspect cables and electrical accessories before use; do not use equipment with exposed wires or damaged connectors.
- Keep the machine and electrical connections away from liquids and wet surfaces.
- Use single-use electrodes for one patient only and follow infection-control procedures for reusable accessories.
- Do not use the ECG procedure to delay emergency care; chest pain, severe breathlessness, fainting, or other urgent symptoms require immediate clinical assessment.
- Follow manufacturer instructions and facility procedures when using the machine with other connected medical equipment.

Care & storage

- Clean the machine and reusable patient-contact accessories after each patient according to approved instructions.
- Inspect lead wires, clips, connectors, and cables regularly for wear or damage.
- Check battery condition, paper supply, and printer function where applicable.
- Arrange preventive maintenance and electrical safety testing according to facility requirements.
- Store the machine in a clean, dry, secure location and report recurring signal or equipment faults.

Common mistakes to avoid

- Placing electrodes in incorrect anatomical positions.
- Recording with loose electrodes, dry contact surfaces, or poorly connected lead wires.
- Allowing patient movement or cable interference to degrade the tracing.

- Saving or printing a tracing with incorrect patient identification.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Enema Kit

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Enema can be done with tubing

What it is used for

An enema kit is used to introduce a prescribed liquid into the rectum through tubing for a specific clinical purpose. It must be used only under appropriate clinical direction, with correct patient assessment, hygienic equipment handling, and careful monitoring to reduce the risk of injury, infection, and fluid-related complications.

Main parts

- Enema container or prescribed solution bag
- Tubing and flow-control clamp
- Patient-end nozzle or rectal tip
- Connector and seal components
- Support hook or stand, where supplied
- Prescribed solution, gloves, lubricant, and waste supplies

Before you start

1. Use an enema kit only when prescribed or specifically advised by an authorised healthcare professional and when trained to perform the procedure.
2. Verify the patient's identity, prescribed solution, intended volume, route, and relevant allergies or contraindications.
3. Assess for warning symptoms or conditions requiring clinical review, such as severe abdominal pain, rectal bleeding, suspected bowel obstruction, or recent relevant surgery.
4. Check the kit packaging, expiry date where applicable, tubing, clamp, connectors, and nozzle for damage or contamination.
5. Prepare the prescribed solution at the specified temperature and concentration; do not substitute solutions or add medicines without authorisation.
6. Provide privacy, explain the procedure, obtain consent, perform hand hygiene, and wear appropriate gloves and protective clothing.

How to use

1. Prepare a clean working area and place suitable absorbent protection beneath the patient as needed.
2. Assemble the kit according to its instructions and confirm that the tubing and nozzle are correctly connected.
3. Fill the container only with the prescribed solution and specified volume, avoiding contamination.
4. Close the clamp and prime the tubing as directed to remove trapped air before use.

5. Position the patient according to the clinician's instructions, commonly on the left side with knees comfortably bent if appropriate.
6. Apply a suitable water-soluble lubricant to the nozzle if instructed by the approved procedure.
7. Explain each step and insert the nozzle gently only as directed by the clinical protocol; never force it against resistance.
8. Open the clamp gradually and administer the solution at the prescribed rate while monitoring the patient's comfort and response.
9. Stop the flow immediately if the patient develops significant pain, bleeding, dizziness, distress, or unexpected resistance.
10. Do not increase pressure, force the solution, or continue beyond the prescribed volume or clinical instructions.
11. After administration, follow the prescribed retention and toileting instructions and provide assistance as needed.
12. Dispose of single-use components safely and clean reusable components only if they are specifically designed for reuse and approved for that purpose.
13. Perform hand hygiene, document the solution, volume, time, patient response, and any complications, and report concerns to the responsible clinician.

⚠ Safety

- This procedure should be performed only under appropriate clinical direction by trained personnel or by a patient who has been specifically instructed by a healthcare professional.
- Do not use an enema when contraindicated or when severe abdominal pain, unexplained rectal bleeding, suspected obstruction, or another urgent warning sign requires medical assessment.
- Never force the nozzle or solution against resistance; stop immediately if significant pain or bleeding occurs.
- Use only the prescribed solution, concentration, volume, and administration rate; some solutions can cause serious electrolyte or fluid complications if misused.
- Use aseptic or clean technique according to the clinical protocol and never share single-use components between patients.
- Do not improvise tubing, connectors, or nozzle attachments, and never connect the kit to pressurised equipment.
- Seek urgent clinical help if severe pain, fainting, significant bleeding, breathing difficulty, or other serious symptoms occur.

Care & storage

- Check packaging integrity, expiry dates, and component condition before use.
- Store the kit and prescribed supplies according to the manufacturer's instructions and away from contamination.
- Discard single-use items after one use and dispose of clinical waste according to local procedures.
- Clean and disinfect reusable components only according to the manufacturer's validated instructions.

- Report damaged kits, missing components, or suspected contamination and replace them before use.

Common mistakes to avoid

- Using an unprescribed solution, incorrect volume, or unsuitable temperature.
- Forcing the nozzle or increasing flow when resistance or pain occurs.
- Reusing single-use components or failing to maintain hygiene.
- Ignoring contraindications, warning symptoms, or unexpected patient reactions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Examination Table

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Examination table; Patient Examination Table

What it is used for

To provide a stable, clean surface for patient examination and routine clinical assessment.

How to handle

- Inspect the table frame, legs, height-adjustment mechanism, upholstery, and accessories for damage or instability.
- Clean and disinfect the patient-contact surfaces according to facility policy before use.
- Explain the procedure and assist the patient onto the table safely; use step supports or additional staff when required.
- Apply brakes or secure the table where applicable and keep the patient supported during position changes.
- After examination, assist the patient off safely and disinfect the contact surfaces again.

Storage

- Keep the table clean and dry and return adjustable parts to the recommended position.
- Park in the designated clinical area with brakes applied where available, leaving access routes clear.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

First Aid Kit

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **High** 49 labs

Covers these lab items: Basic first aid kit; Basic first aid kits; First Aid; First Aid box; First Aid box with basic medicines; First Aid Equipment; First aid kit; First aid stations and rescue kits equipped for wilderness settings; Portable first aid stations, evacuation stretchers, and fire extinguishers

What it is used for

A first aid kit provides basic supplies for immediate care of minor injuries until further medical help is available. Use suitable items according to the injury and the instructions provided.

How to use

1. Check the kit contents, expiry dates, and packaging before use.
2. Wash hands and wear disposable gloves when available.
3. Select the appropriate dressing or first aid item for the injury.
4. Clean minor wounds using suitable clean water or approved wound-cleaning supplies.
5. Apply a sterile dressing or bandage without restricting circulation.
6. Seek professional medical help for serious injuries, severe bleeding, burns, or breathing problems.
7. Replace used items and close the kit after use.

⚠ Safety

- Do not give medicines unless authorised and appropriate for the person.
- Do not use expired, damaged, or contaminated supplies.
- Call emergency services for life-threatening injuries and follow trained first aid procedures.
- Avoid direct contact with blood or other body fluids.

Care & storage

- Store the kit in a clean, dry, clearly marked, accessible location.
- Inspect supplies regularly and replace used or expired items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Forceps

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Dissecting Forceps; Forceps; Forceps and brushes

What it is used for

Forceps are used to grip, hold, or manipulate small objects and materials during medical or laboratory tasks. Use the correct type for the intended procedure.

How to use

1. Inspect the forceps for cleanliness, damage, rust, or loose tips.
2. Wash and disinfect or sterilize according to the intended use and facility protocol.
3. Hold the forceps securely and grip the target material gently.
4. Use sterile forceps only when required for sterile procedures.
5. After use, clean and process reusable forceps according to protocol.
6. Check the tips and place the forceps in their designated storage area.

⚠ Safety

- Use sterile forceps for sterile procedures and avoid touching their working tips.
- Handle sharp or pointed tips carefully to prevent punctures and injuries.
- Do not use damaged, rusty, or contaminated forceps.
- Follow infection-control procedures when handling medical instruments.

Care & storage

- Clean, dry, and store forceps properly after use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Glucometer

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Glucometer

What it is used for

A glucometer measures blood glucose using a compatible test strip and a small blood sample. Follow the device instructions and clinical procedures for accurate results, infection prevention, and appropriate result reporting.

Main parts

- Glucometer meter and display
- Compatible glucose test strips
- Lancing device and single-use lancets
- Control solution, if specified
- Battery or power source
- Approved sharps container and cleaning supplies

Before you start

1. Verify the user's identity and explain the test procedure.
2. Wash hands and ensure the user's finger is clean and completely dry.
3. Check the meter, test-strip expiry date, and strip storage condition.
4. Confirm the strips match the meter model and coding requirements, if applicable.
5. Inspect the lancing device and use a new sterile single-use lancet.
6. Prepare gloves, if required by the clinical procedure, and a sharps container.

How to use

1. Switch on the meter according to the manufacturer's instructions.
2. Insert a compatible test strip in the indicated direction.
3. Confirm that the meter is ready for the blood sample.
4. Prepare the lancing device with a new single-use lancet.
5. Use the approved technique to obtain a small blood drop from the fingertip.
6. Apply the blood to the test strip's designated sample area.
7. Wait for the meter to display the result.
8. Read and record the result with the date and time, following clinical procedure.
9. Compare the result with the user's care plan or applicable clinical guidance.
10. Report critical, unexpected, or concerning results to the appropriate healthcare professional promptly.
11. Dispose of the lancet immediately in an approved sharps container.
12. Remove and discard the used strip according to local clinical waste procedures.

3. Clean the meter as directed by the manufacturer and store it safely.

⚠ Safety

- Use a new sterile lancet for every person and every test.
- Never share a lancing device, even if the lancet is changed.
- Follow standard precautions and applicable infection-control procedures.
- Do not use expired, damaged, or improperly stored test strips.
- Do not make medication changes based only on a reading without appropriate clinical guidance.
- Follow the device's instructions for cleaning, quality control, and result interpretation.

Care & storage

- Store the meter and strips within the specified temperature and humidity range.
- Keep the strip container closed and follow its storage instructions.
- Perform control-solution checks when recommended by the manufacturer or clinical procedure.
- Clean and disinfect the meter using only compatible approved products.
- Replace batteries and report faulty readings or equipment problems.

Common mistakes to avoid

- Using expired strips or strips incompatible with the meter.
- Testing with wet or contaminated fingers.
- Reusing lancets or sharing a lancing device.
- Recording the result incorrectly or ignoring an unexpected reading.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Haemoglobinometer

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Haemoglobin meter; Sahli's Hemoglobinometer

What it is used for

A haemoglobinometer measures haemoglobin concentration in a blood sample using the device-specific method. Operate it according to the manufacturer's instructions and approved clinical procedures, including quality control and infection prevention.

Main parts

- Haemoglobinometer or Sahli comparator
- Graduated tube or compatible cuvette
- Pipette, capillary, or specified blood-sampling accessory
- Reagent or diluent specified for the method
- Stirring rod or mixing accessory, if applicable
- Gloves, lancet, and approved sharps container

Before you start

1. Verify the patient's identity and explain the test procedure.
2. Check the instrument, measuring tube, pipette, and accessories for cleanliness or damage.
3. Check reagent identity, expiry date, and storage requirements.
4. Confirm the instrument is calibrated or quality-checked as required.
5. Perform hand hygiene and prepare appropriate PPE and waste containers.
6. Collect the blood sample using the approved clinical procedure.

How to use

1. Place the instrument on a stable, clean work surface.
2. Prepare the sample and reagent according to the specified method.
3. For a Sahli method, add the specified reagent to the graduated tube as instructed.
4. Measure the required blood volume accurately using the specified sampling accessory.
5. Add the blood to the reagent and mix according to the method instructions.
6. Allow the reaction to develop for the specified time.
7. For a visual comparator method, add diluent gradually and mix until the prescribed colour match is reached.
8. For an electronic model, insert the prepared sample or cuvette as directed by the manufacturer.
9. Read the haemoglobin result using the instrument's scale or display.
10. Record the result with the correct patient identifier, units, date, and time.
11. Report results outside the expected range or any concerning result according to clinical procedure.

2. Dispose of blood-contaminated materials and sharps in approved waste containers.
3. Clean the instrument and reusable accessories according to the manufacturer's instructions.

⚠ Safety

- Only trained healthcare personnel should perform or interpret the test.
- Treat all blood samples as potentially infectious and follow standard precautions.
- Use a sterile single-use lancet for each patient when capillary sampling is performed.
- Handle reagents according to their safety instructions and avoid contact with skin or eyes.
- Do not reuse disposable cuvettes, capillaries, or contaminated accessories.
- Follow quality-control procedures and report unreliable or unexpected results.

Care & storage

- Clean and disinfect compatible surfaces after use.
- Store reagents and accessories within the specified conditions.
- Check instrument condition and perform quality control at the required frequency.
- Keep optical or comparison surfaces clean and free of scratches.
- Document maintenance, quality-control results, and equipment faults.

Common mistakes to avoid

- Using incorrect blood volume or reagent quantity.
- Reading the result before the required reaction time or colour match.
- Using expired reagents or contaminated accessories.
- Recording results against the wrong patient or without units.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hospital Bed

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Bed(ICU); Hospital Bed; Hospital bed or adjustable bed for home care simulation; ICU Bed with Mattress

What it is used for

A hospital bed supports patients and may provide adjustable height, backrest, and side rails for care and positioning. Operate the bed according to the patient's needs and the manufacturer's instructions.

How to use

1. Inspect the bed frame, mattress, wheels, brakes, controls, and side rails.
2. Explain the movement to the patient and check that lines or tubes will not be pulled.
3. Lock the wheels before transferring or positioning the patient.
4. Use the controls to adjust height or backrest slowly while observing the patient.
5. Raise side rails when indicated by the care plan and facility procedure.
6. After use, return the bed to the required safe position and clean contact surfaces.

⚠ Safety

- Use brakes during transfers and whenever the bed must remain stationary.
- Keep hands, feet, and bedding clear of moving joints and mechanisms.
- Do not leave a patient at risk of falling unattended; follow the care plan for rails.
- Only trained staff should use powered adjustments and patient-lifting features.

Care & storage

- Clean and disinfect contact surfaces according to facility protocol and report mechanical or electrical faults.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hot Water Bottle

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Hot Water bag; Hot Water Bottle

What it is used for

To provide gentle external warmth for comfort when appropriate and safe for the user.

How to handle

- Inspect the bottle, stopper, seams, and cover for cracks, leaks, or deterioration before every use.
- Use hot but not boiling water, following the manufacturer's instructions; fill only to the recommended level and expel excess air as directed.
- Close the stopper securely and check for leakage before placing the bottle inside its cover.
- Wrap the bottle in its protective cover and apply it to the intended area without direct prolonged skin contact.
- Check the skin regularly and remove the bottle if discomfort, excessive heat, redness, or irritation occurs; use extra caution with people who have reduced sensation.

Storage

- Empty, rinse if appropriate, and dry the bottle completely after use; follow the manufacturer's storage instructions.
- Store with the stopper loosened or removed if specified by the manufacturer, away from direct sunlight and heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

IV Stand

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: IV Stand

What it is used for

To support an intravenous fluid bag or infusion container at a suitable height during clinical care.

How to handle

- Inspect the stand, base, wheels, pole, hooks, and adjustment locks for damage or instability.
- Place the stand on a level surface and lock the wheels when it is in position.
- Adjust the height and secure the infusion container using the designated hook without exceeding the stand's rated load.
- Keep the stand clear of walkways and prevent tubing from becoming tangled or creating a trip hazard.
- Clean the stand according to healthcare facility procedures and report defects before further use.

Storage

- Store clean and dry in a designated clinical equipment area with the pole secured.
- Keep hooks and wheels protected from damage and avoid placing heavy items on the stand.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

IV Training Arm

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Male Multi-Veno Intravenous Arm

What it is used for

An IV training arm simulates venous access procedures for healthcare skills practice without using a real patient. Use the model with approved training fluids and instructor supervision.

How to use

1. Inspect the arm model, simulated veins, tubing, skin, and base for damage or leakage.
2. Place the model securely on a stable training surface.
3. Prepare the required training supplies and approved simulation fluid.
4. Fill or prime the simulated venous system according to the manufacturer's instructions.
5. Demonstrate hand hygiene, site selection, and aseptic technique under instructor guidance.
6. Practice venipuncture or IV insertion using approved training devices and procedures.
7. Dispose of used training sharps in an approved sharps container and clean the model.

⚠ Safety

- Use the arm only for simulation, never for actual patient infusion.
- Handle needles and other sharps carefully and dispose of them immediately after use.
- Do not use damaged tubing or unapproved fluids that may harm the model.
- Follow infection-control procedures and instructor directions throughout practice.

Care & storage

- Drain and clean the simulated system as instructed and replace worn skin or veins when required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ice Pack

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Ice Cap; Ice caps; Ice pack

What it is used for

Ice packs or ice caps provide controlled cold application for approved first-aid or care procedures.

How to handle

- Inspect the pack or cap for leaks, cracks, and damaged closures before use.
- Follow the product instructions and any applicable first-aid or clinical procedure.
- Wrap the cold pack in a suitable cloth where required; do not place extreme cold directly on skin unless instructed.
- Apply for the recommended duration and check the skin regularly for discomfort or injury.
- Do not use a leaking pack or one with unknown contents; clean reusable packs after use.

Storage

- Store according to the manufacturer's temperature and handling instructions.
- Keep clean packs separate from used or contaminated items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Infrared Lamp

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Infrared Lamp

What it is used for

An infrared lamp emits infrared radiation to provide controlled surface warming for approved therapeutic or wellness applications. Use only for the intended purpose and follow the device instructions, prescribed exposure time, and required safety precautions.

Main parts

- Infrared bulb or heating element
- Reflector and protective lamp housing
- Adjustable stand or positioning arm
- Power switch and intensity control, if fitted
- Timer, if fitted
- Power cord, plug, and stable base

Before you start

1. Confirm that the device is approved and suitable for the intended application.
2. Inspect the bulb, housing, stand, cord, plug, and controls for damage.
3. Position the lamp on a stable surface with adequate clearance.
4. Check the required distance, exposure time, and settings in the device instructions or care plan.
5. Assess the user's skin sensitivity and ability to sense heat, following clinical guidance.
6. Remove flammable materials and ensure the treatment area is dry.

How to use

1. Switch off the lamp before adjusting its position or replacing accessories.
2. Position the lamp at the recommended distance from the treatment area.
3. Secure the stand or positioning arm to prevent accidental movement.
4. Connect the device to a suitable electrical outlet.
5. Switch on the lamp and select the approved setting.
6. Set the exposure time according to the device instructions or authorised care plan.
7. Keep the treated area within the recommended exposure zone.
8. Monitor the user's comfort and skin response throughout the session.
9. Stop immediately if excessive heat, pain, burning, or unusual redness occurs.
10. Switch off the lamp at the end of the session.
11. Allow the bulb and housing to cool before moving or cleaning the device.
12. Disconnect the power when required and store the lamp safely.
13. Record the treatment and report any equipment faults or adverse response.

⚠ Safety

- Use the lamp only for its approved purpose and under appropriate supervision.
- Do not look directly at the infrared source or exceed the recommended exposure.
- Keep the lamp away from water, flammable materials, and combustible surfaces.
- Avoid use on areas with reduced heat sensation unless specifically authorised by a qualified clinician.
- Do not leave the lamp operating unattended during treatment.
- Do not touch the bulb or housing while hot; disconnect power before maintenance.

Care & storage

- Clean the housing only after the lamp has cooled and been disconnected.
- Inspect the bulb, reflector, stand, and electrical cable regularly.
- Replace bulbs only with the specified compatible type.
- Keep vents and reflector surfaces free from dust.
- Arrange authorised servicing if the lamp flickers, overheats, or becomes unstable.

Common mistakes to avoid

- Placing the lamp too close to the skin.
- Exceeding the authorised exposure time or intensity.
- Leaving the user unattended during heating.
- Moving or cleaning the lamp before it has cooled.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Kidney Tray

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Kidney Tray

What it is used for

A kidney tray holds dressings, instruments, or waste during approved medical, first-aid, or care procedures.

How to handle

- Inspect the tray for cracks, sharp edges, corrosion, and cleanliness before use.
- Use the tray for its intended purpose and maintain separation between clean and contaminated items.
- Wear appropriate PPE when handling clinical materials or waste.
- Clean and disinfect reusable trays according to facility procedures after each use.
- Dispose of contaminated contents according to the relevant clinical waste policy.

Storage

- Store cleaned, dry trays in a designated clean area.
- Keep clean trays separate from used instruments and waste.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mackintosh Sheet

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Mackintosh

What it is used for

Handle a mackintosh sheet hygienically for protective bedding or surface coverage.

How to handle

- Inspect the sheet for tears, holes, or damaged edges before use.
- Use a clean sheet for each person or task as required.
- Clean and disinfect according to the manufacturer's instructions.
- Replace the sheet if it becomes damaged or cannot be adequately cleaned.

Storage

- Dry completely before folding and storing.
- Keep in a clean, dry cupboard away from direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Manikin

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Human Body Manikin; Little Anne

What it is used for

To practise clinical skills, patient assessment, or CPR techniques on a training manikin without using a real patient.

How to handle

- Inspect the manikin body, joints, airway components, removable parts, and accessories for damage and cleanliness.
- Set up the manikin according to the training scenario and the manufacturer's instructions.
- Use only approved training procedures and compatible accessories; follow instructor guidance during CPR or clinical simulations.
- Avoid sharp instruments or fluids unless the manikin is designed for them, and do not apply excessive force to joints or connectors.
- After training, clean and disinfect contact surfaces and reusable components according to the manufacturer and institutional protocol.

Storage

- Clean, disinfect, and dry the manikin and accessories thoroughly before storage.
- Store in the designated clean area, protect joints and airway parts, and keep removable components together.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Medical Consumable

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Medium** 7 labs

Covers these lab items: Gauze Piece (4X4); Suction Catheter; Syringe 50 cc/ml; Syringe 50cc/ml (Asepto); Syringe and droppers

What it is used for

Medical consumables such as gauze, suction catheters, syringes, and droppers support clinical care when handled hygienically.

How to handle

- Check product identity, size, sterility, expiry date, and packaging integrity before use.
- Use aseptic technique and follow the relevant clinical procedure and manufacturer's instructions.
- Do not reuse single-use items or share items between patients unless specifically designed and approved for reuse.
- Dispose of contaminated consumables and used sharps in the appropriate clinical waste stream.

Storage

- Keep sterile items sealed in clean, dry storage away from moisture, dust, and direct sunlight.
- Separate expired, damaged, or recalled stock from usable supplies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Medical Equipment

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Medical Equipment

What it is used for

To inspect, prepare, and operate medical equipment according to its intended use and the manufacturer's instructions. To support safe patient care by verifying equipment condition, cleanliness, function, and required checks before use.

Main parts

- Main equipment unit and housing
- Power supply, battery, or charging system
- Controls, display, and indicators
- Sensors, probes, cables, or accessories as applicable
- Alarms and safety features, if fitted
- Cleaning, storage, and maintenance accessories

Before you start

1. Identify the exact device, intended use, and authorized operator requirements.
2. Read the manufacturer's instructions and the facility's relevant clinical procedure.
3. Inspect the casing, power cord, battery, cables, connectors, and accessories for damage.
4. Confirm that the equipment has the required inspection, maintenance, and calibration status where applicable.
5. Clean and disinfect the device and reusable accessories using the approved procedure for that device.
6. Check that the device is configured for the intended patient or task and that necessary supplies are available.
7. Verify alarm settings, self-test results, and functional checks where the manufacturer requires them.

How to use

1. Place the equipment on a stable surface or secure it to its approved mounting system.
2. Connect the approved power supply or confirm sufficient battery charge.
3. Switch on the device and allow its self-test or initialization sequence to finish.
4. Review any displayed errors, alerts, calibration prompts, or maintenance warnings.
5. Connect only compatible accessories and verify that they are fitted correctly.
6. Perform the required functional check using the manufacturer's approved method before patient use.
7. Set operating parameters only within the authorized clinical procedure and operator's training.

8. Explain the procedure to the patient when appropriate and confirm identity according to facility policy.
9. Use the equipment while observing the patient and monitoring relevant indicators or alarms.
10. Stop use and seek clinical or technical assistance if readings are implausible, alarms persist, or the device malfunctions.
11. After use, switch off the equipment and disconnect accessories according to the instructions.
12. Clean, disinfect, and store the device as required, and document faults or maintenance needs.

⚠ Safety

- Only trained and authorized personnel should operate medical equipment or use it for patient care.
- Never use equipment with damaged cables, cracked housing, failed safety checks, or unresolved alarms.
- Follow infection-prevention procedures and the manufacturer's approved cleaning and disinfection instructions.
- Do not change clinical settings or treatment parameters beyond the operator's authorization.
- Use only compatible accessories and approved power supplies.
- Keep liquids away from electrical connections unless the device is specifically designed for that exposure.
- Remove faulty equipment from service, label it clearly, and report it through the facility's maintenance process.

Care & storage

- Clean and disinfect the device and reusable accessories after use according to policy.
- Inspect cables, sensors, probes, connectors, and housing regularly.
- Charge and store batteries according to the manufacturer's instructions.
- Maintain calibration, electrical safety testing, and preventive maintenance records where applicable.
- Store equipment in a clean, dry, secure location and report damage promptly.

Common mistakes to avoid

- Using a device without checking its identity, condition, or intended purpose.
- Ignoring failed self-tests, alarms, or overdue maintenance.
- Using incompatible or unclean accessories.
- Changing clinical parameters without authorization or adequate training.
- Returning faulty equipment to service without inspection and clearance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Medicine

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **High** 4 labs

Covers these lab items: Drugs/Medicines; Medicines

What it is used for

Medicines are used for treatment, prevention, or symptom management only according to authorized clinical directions.

How to handle

- Verify the medicine name, strength, expiry date, packaging integrity, and patient details before administration.
- Follow the prescription, approved dosage, route, and timing; never administer unidentified medicines.
- Check allergies and relevant warnings, and maintain required hygiene during handling.
- Report damaged packaging, unexpected reactions, errors, or suspected contamination to the responsible healthcare professional.

Storage

- Store according to the label, temperature requirements, and light or moisture precautions.
- Keep secured from unauthorized access and separate expired or recalled medicines.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Medicine Container

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Medicine Containers

What it is used for

Handle medicine containers safely to maintain medicine identity and quality.

How to handle

- Check the label, contents, expiry date, and container integrity before use.
- Keep medicines in their original labelled containers unless an authorised process requires otherwise.
- Avoid touching tablets or capsules directly when dispensing if a clean tool is available.
- Keep medicines out of reach of children and follow the required dispensing procedure.

Storage

- Store according to the product label and applicable temperature requirements.
- Keep tightly closed, clearly labelled, and separate from food and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Mobility Aid

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Mobility aids for the handicapped person

What it is used for

Mobility aids such as walkers, crutches, and wheelchairs help people move more safely and independently. Select and adjust the aid to the user's needs and follow professional guidance when required.

How to use

1. Inspect the mobility aid for damage, loose joints, worn grips, and faulty brakes.
2. Confirm the aid is appropriate for the user's height, ability, and environment.
3. Adjust handles, supports, or footrests according to manufacturer instructions.
4. Explain safe use and help the user adopt a stable starting position.
5. Support the user during initial movement and check balance and comfort.
6. Use the aid on a suitable, clear surface and avoid obstacles or slippery areas.
7. After use, park or fold the aid as instructed and report any faults.

⚠ Safety

- Do not use an unstable, damaged, or incorrectly adjusted mobility aid.
- Use assistance when transferring or walking if the user needs support.
- Lock wheelchair brakes before transfers and keep feet clear of wheels.
- Follow clinician guidance for weight-bearing restrictions and safe movement.

Care & storage

- Clean contact surfaces regularly and inspect grips, wheels, tips, and brakes before use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nebulizer

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Nebulizer

What it is used for

A nebulizer converts prescribed liquid medicine into a mist that can be inhaled through a mask or mouthpiece. It helps deliver medication to the airways when used according to medical instructions.

Main parts

- Identify the compressor or ultrasonic nebulizer unit, depending on the model.
- Locate the medicine cup or medication chamber.
- Identify the air tubing connecting the compressor to the medicine chamber, if fitted.
- Locate the patient mask or mouthpiece used for inhalation.
- Identify the power switch, power adapter or battery compartment.
- Locate the air filter and replacement components, where provided.
- Identify the manufacturer's instructions and prescribed medication guidance.

Before you start

1. Use the nebulizer only with medication prescribed or approved by a qualified healthcare professional.
2. Wash and dry hands before handling the device and medication.
3. Inspect the medicine chamber, tubing, mask and power connections for damage.
4. Confirm the device is clean and all components are compatible.
5. Measure the prescribed medication dose using the recommended method.
6. Place the device on a clean, dry and stable surface.
7. Ensure the user is seated comfortably in a suitable position.

How to use

1. Assemble the nebulizer according to the manufacturer's instructions.
2. Pour the prescribed medication into the medicine chamber without exceeding its marked limit.
3. Secure the chamber lid and connect the tubing and patient interface correctly.
4. Connect the power supply and position the mask or mouthpiece appropriately.
5. Switch on the device and confirm that mist is being produced.
6. Breathe normally through the mouthpiece or mask as instructed by the clinician.
7. Continue until the prescribed treatment is complete or mist production stops.
8. Switch off the device and disconnect the power supply.
9. Discard leftover medication according to medical instructions and clean reusable components.
10. Allow components to dry completely before storing them in a clean container.

⚠ Safety

- Never use unprescribed medication, oils or homemade mixtures in the nebulizer.
- Do not share masks or mouthpieces without appropriate cleaning and disinfection.
- Keep electrical connections dry and use the device according to its instructions.
- Never block the air vents or operate a damaged device.
- Stop treatment and seek medical help for severe breathing difficulty or serious adverse reactions.
- Follow the clinician's instructions for medication dose, frequency and treatment duration.

Care & storage

- Clean and disinfect reusable components according to the manufacturer's instructions.
- Replace air filters and worn accessories at the recommended intervals.
- Inspect tubing and the medicine chamber regularly for cracks or blockages.
- Store the dry device in a clean, dust-free location.

Common mistakes to avoid

- Using an incorrect medication or dose in the medicine chamber.
- Failing to clean reusable components between treatments.
- Using cracked tubing or a damaged medicine chamber.
- Stopping prescribed treatment early without appropriate medical guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Needle Destroyer

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Needle burner; Syringe destroyer; Syringe Destroyer and Needle Burner

What it is used for

A needle destroyer or syringe destroyer is used to disable or safely process used needles and syringes according to the device design and approved healthcare waste procedures. Correct handling and immediate disposal help reduce needlestick injuries and infection risks.

Main parts

- Device housing encloses the operating mechanism.
- Needle insertion port accepts compatible needles on applicable models.
- Cutting, melting, or destroying mechanism disables the needle according to the device design.
- Syringe cutting or disabling slot is provided on compatible models.
- Power switch activates the mechanism on powered models.
- Indicator light shows operating status on equipped models.
- Waste collection container receives processed sharps or separated components where provided.
- Power cord or battery supplies energy on powered models.

Before you start

1. Only trained and authorised healthcare personnel should operate the device.
2. Read the manual and confirm which needle and syringe types the device accepts.
3. Inspect the housing, insertion port, switch, power supply, and waste container for damage.
4. Wear the required PPE, including gloves and eye protection under local procedures.
5. Place the device on a stable, cleanable surface in the designated waste-handling area.
6. Prepare an approved sharps container and follow the facility's biomedical waste segregation rules.

How to use

1. Switch on the device according to the manufacturer's instructions, if it requires power.
2. Confirm the device is ready and all guards or covers are correctly positioned.
3. Hold the used syringe by its barrel and keep fingers away from the needle.
4. Insert only the compatible needle into the designated port without forcing it.
5. Activate the destruction mechanism using the approved control where required.
6. Verify that the needle has been processed as intended without touching the sharp end.
7. Place the remaining syringe and all processed components into the correct approved waste stream.
8. Repeat only for compatible items and keep the work area organised.
9. Switch off and disconnect the device when the task is complete, following the manual.

10. Close the sharps container when its fill limit is reached and clean the device exterior using the approved procedure.

⚠ Safety

- Never touch, bend, or manually remove a used needle unless the approved device procedure specifically requires a safe method.
- Never force an incompatible needle or syringe into the device.
- Keep hands away from cutting or heating elements and never bypass guards.
- Do not use the device if the housing, insertion port, or power supply is damaged.
- Follow local biomedical waste rules; processed sharps may still require approved sharps disposal.
- If a needlestick injury or blood exposure occurs, follow the facility's immediate exposure-response procedure and seek prompt medical assessment.

Care & storage

- Switch off and disconnect power before cleaning or maintenance.
- Clean and disinfect external surfaces using manufacturer-approved materials.
- Inspect the insertion port, guards, and waste container regularly for damage or blockage.
- Arrange authorised servicing for abnormal operation, overheating, or failed destruction.

Common mistakes to avoid

- Using a needle or syringe type not supported by the device.
- Touching the needle tip or processed sharp components by hand.
- Putting processed sharps into general waste instead of the approved waste stream.
- Cleaning or clearing a blockage while the device is powered.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Normal Saline

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Medium** 2 labs

Covers these lab items: Normal Saline; Normal Saline Bottle

What it is used for

Normal saline is a sterile saline solution used for approved medical purposes according to clinical instructions.

How to handle

- Verify the solution concentration, product label, expiry date, seal integrity, and intended clinical use.
- Use aseptic technique and open the container only when required by the procedure.
- Do not use cloudy, leaking, damaged, or expired containers; do not reuse single-use containers.
- Follow the approved clinical protocol for administration, irrigation, or other intended use.

Storage

- Store within the manufacturer's specified temperature and light conditions.
- Keep sealed containers clean and separate expired or compromised stock.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oral Care Set

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Oral care Set

What it is used for

To support routine oral hygiene and demonstrate basic mouth-care procedures safely.

How to handle

- Inspect the toothbrush, oral-care accessories, and storage case for damage or contamination.
- Wash hands and prepare clean, individual-use items for each person.
- Use the toothbrush and other accessories gently according to the user's needs and the product instructions.
- Do not share toothbrushes or items that contact saliva; use gloves when providing care to another person.
- Clean reusable items after use and dispose of single-use items safely.

Storage

- Clean, dry, and protect reusable items after use; replace worn toothbrushes or damaged accessories.
- Store personal items separately in a clean, ventilated container away from contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oxygen Cylinder

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Oxygen Cylinder; Oxygen Cylinder with Connector, Key, Face Mask and Tubing

What it is used for

An oxygen cylinder supplies medical oxygen through a compatible regulator, flow control, tubing, and delivery interface when prescribed or authorised. Correct cylinder handling, leak checks, flow settings, and fire precautions are essential because oxygen supports rapid combustion and a high-pressure cylinder can cause serious injury if mishandled.

Main parts

- Oxygen cylinder stores oxygen under high pressure.
- Cylinder valve controls the gas outlet.
- Regulator reduces cylinder pressure to a usable delivery pressure.
- Flowmeter or flow control sets the prescribed oxygen flow.
- Outlet connector links the regulator to the delivery tubing.
- Oxygen tubing carries oxygen to the delivery interface.
- Face mask delivers oxygen to the patient where prescribed.
- Cylinder key or wrench operates the valve where the design requires one.
- Cylinder trolley or restraint secures the cylinder during movement and use.

Before you start

1. Only trained and authorised personnel should set up or administer medical oxygen.
2. Confirm the patient's identity, prescribed oxygen delivery method, and flow rate with the authorised clinician.
3. Check the cylinder label, oxygen identification, pressure status, service condition, and available contents.
4. Inspect the cylinder, valve, regulator, flowmeter, connectors, tubing, and face mask for damage or contamination.
5. Confirm the regulator is designed for the cylinder and that all connectors are compatible.
6. Secure the cylinder upright using an approved stand or trolley before opening the valve.
7. Keep the area free from flames, smoking, oil, grease, and other ignition or combustible hazards.

How to use

1. Wash hands and prepare the prescribed oxygen delivery equipment.
2. Secure the cylinder and confirm the regulator and flow control are fitted correctly.
3. Ensure the flow control is in the specified initial position before opening the cylinder valve.
4. Open the cylinder valve slowly according to the regulator instructions and check for leaks using an approved method.

5. Set the prescribed flow rate and confirm oxygen is flowing through the delivery tubing.
6. Connect the tubing to the prescribed face mask or other authorised delivery interface.
7. Fit the interface to the patient as directed by the clinician and check comfort and correct positioning.
8. Monitor the patient's breathing, oxygen saturation where indicated, and response to therapy according to the care plan.
9. If the supply fails, a leak occurs, or the patient's condition worsens, follow the clinical emergency procedure immediately.
10. When therapy ends, close the cylinder valve according to the instructions and release downstream pressure only as directed by the manufacturer.
11. Disconnect and clean or dispose of patient-contact items according to infection-control procedures.
12. Record the flow rate, start and stop times, patient response, cylinder status, and any faults.

⚠ Safety

- Never use oxygen near smoking, open flames, sparks, or heat sources.
- Never apply oil, grease, petroleum products, or unapproved lubricants to oxygen equipment.
- Keep the cylinder secured upright and protect the valve from impact or damage.
- Do not use a leaking, damaged, unidentified, or out-of-service cylinder or regulator.
- Do not change the prescribed oxygen flow rate without clinical authorisation.
- Do not attempt to repair the cylinder valve or regulator; follow the emergency procedure for leaks or equipment failure.

Care & storage

- Store cylinders in a designated, ventilated area away from heat and combustible materials.
- Keep full and empty cylinders identified and stored according to facility procedures.
- Inspect the cylinder, regulator, flowmeter, tubing, and connectors before each use.
- Clean reusable equipment using approved methods and follow infection-control requirements for patient-contact components.
- Arrange authorised servicing and return cylinders through the approved supplier or facility process.

Common mistakes to avoid

- Using an incompatible regulator or connector on the oxygen cylinder.
- Leaving the cylinder unsecured or opening its valve too quickly.
- Changing oxygen flow without clinical authorisation.
- Allowing oil, grease, flames, or smoking near oxygen equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oxygen Cylinder Key

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Oxygen Key

What it is used for

To operate a compatible oxygen-cylinder valve only when authorized and trained, following the facility's oxygen safety procedure.

How to handle

- Inspect the key for cracks, deformation, corrosion, or incorrect fit before use.
- Confirm the cylinder is correctly identified, secured upright, and fitted with the appropriate regulator and approved accessories.
- Keep oil, grease, flames, sparks, and smoking materials away from oxygen equipment; never lubricate the key or valve.
- Use only the manufacturer-approved key and operate the valve as instructed by trained personnel; do not force a stuck valve.
- Close or secure the valve according to the prescribed procedure and report leaks, damage, or abnormal resistance immediately.

Storage

- Wipe the key clean and dry after use and inspect it for wear.
- Store in its designated labelled location away from oils, grease, heat, and unauthorized access.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Oxygen Cylinder Trolley

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Oxygen Cylinder Trolley

What it is used for

An oxygen cylinder trolley supports the safe movement of oxygen cylinders between approved locations. Secure the cylinder properly and follow medical gas handling procedures at all times.

How to use

1. Inspect the trolley frame, wheels, handle, straps, and restraints for damage.
2. Confirm that the trolley is suitable for the cylinder size and load.
3. Keep the cylinder valve closed and the protective cap fitted when required for transport.
4. Position the cylinder securely in the trolley and fasten the approved restraint.
5. Move the trolley slowly using the handle, avoiding obstacles and uneven surfaces.
6. Park the trolley on stable ground and secure the cylinder against tipping.
7. Report damaged restraints, wheels, or other defects before further use.

⚠ Safety

- Never drag, roll, or lift a cylinder by its valve or regulator.
- Keep oxygen cylinders away from oil, grease, flames, and ignition sources.
- Do not use the trolley if the cylinder cannot be secured properly.
- Follow facility procedures for oxygen cylinder transport and storage.

Care & storage

- Keep the trolley clean and dry and inspect wheels, restraints, and frame regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Patient Call Bell

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Patient Remote Bell

What it is used for

A patient call bell allows a patient to request assistance from nursing or support staff. Install and test the call system so that alerts are clearly received at the designated station.

How to use

1. Inspect the call bell, remote, cable, connector, and indicator for damage.
2. Confirm that the unit is connected to the correct patient station or call system.
3. Position the call button within easy reach of the patient.
4. Power the system according to the manufacturer's instructions and check its status indicator.
5. Press the call button and verify that the alert appears at the designated receiving station.
6. Reset or cancel the alert using the approved procedure and confirm normal status.
7. Report faults and ensure the patient can reach the working call button.

⚠ Safety

- Never disconnect or disable a patient call bell without authorization and an alternative support plan.
- Keep cables arranged to prevent tripping or entanglement.
- Do not use damaged electrical components near patients.
- Follow facility procedures for urgent patient assistance.

Care & storage

- Clean the button and remote with approved disinfectant and inspect cables regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Physical Restraint

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **High** 1 labs

Covers these lab items: Physical restraints

What it is used for

Physical restraints are specialised devices used in healthcare only when legally permitted and clinically justified to prevent immediate harm. Their use requires authorised, trained staff, the least restrictive approach, and continuous monitoring under approved policy.

How to use

1. Assess the immediate risk and consider de-escalation and less restrictive alternatives first.
2. Obtain required clinical authorisation and follow facility policy and applicable law.
3. Explain the process to the person when safe and appropriate, preserving dignity and privacy.
4. Use only approved restraint equipment and follow the facility's trained application procedure.
5. Ensure the person's breathing, circulation, positioning, and access to urgent help are not compromised.
6. Monitor the person continuously or at the policy-required interval and document observations.
7. Release the restraint as soon as the authorised criteria for continued use no longer apply.
8. Document the reason, authorisation, monitoring, duration, and any adverse effects.

⚠ Safety

- Never use restraints for punishment, convenience, or staff discipline.
- Never apply restraints without required training, authorisation, and clinical oversight.
- Do not obstruct breathing, compress the neck or chest, or leave the person unmonitored.
- Follow emergency procedures immediately if distress, injury, or breathing difficulty occurs.

Care & storage

- Inspect restraint equipment for damage and clean it according to facility policy.
- Store restraints securely and review each use according to clinical governance procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pill Organizer

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Medication organizer boxes

What it is used for

Use medication organiser boxes to arrange medicines according to an authorised care plan.

How to handle

- Use only under an authorised medication-management procedure and verify medicine identity before filling.
- Check labels, expiry dates, and prescribed instructions; do not guess when information is unclear.
- Keep medicines in original packaging when required by product instructions or care policy.
- Keep filled organisers away from children and return any errors or uncertainties to the responsible professional.

Storage

- Store organisers in a secure place under the medicines' required conditions.
- Keep dry, labelled, and separate from food and cleaning chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pulse Oximeter

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Pulse Oximeter; Pulse Oximeter (Finger)

What it is used for

A finger pulse oximeter estimates blood oxygen saturation (SpO2) and pulse rate. Use it as a screening aid, not as a replacement for clinical assessment.

How to use

1. Check the device, display, battery, and finger sensor for cleanliness and damage.
2. Ask the person to rest briefly and keep the hand warm and still.
3. Remove nail polish or artificial nails if they interfere with the sensor reading.
4. Place the device on the finger as instructed and keep the hand relaxed.
5. Wait for a stable reading and note SpO2 and pulse rate.
6. Remove the device, clean it according to instructions, and record results if required.

⚠ Safety

- Do not rely on one reading alone if symptoms or clinical concerns are present.
- Seek urgent medical help for severe breathlessness, blue lips, confusion, or other serious symptoms.
- Cold fingers, movement, poor circulation, and nail products can affect accuracy.

Care & storage

- Clean the sensor between users and store the device in a dry, safe place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rapid Test Kit

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Rapid test kits for Malaria, Dengue, HIV, HCV, Typhoid, Pregnancy

What it is used for

A rapid test kit detects specific markers in a sample to support screening for selected infections or pregnancy. Results must be interpreted according to the kit instructions and confirmed by appropriate healthcare professionals when required.

Main parts

- Identify the test device, cassette, strip or reaction card supplied with the kit.
- Locate the sample collection device, such as a dropper, swab or lancet, if included.
- Identify the supplied buffer, reagent or extraction solution, where applicable.
- Locate the test buffer wells and result window, if present.
- Identify the control and test result markings specified by the manufacturer.
- Locate the sterile gloves, sample container and biohazard waste materials.
- Identify the package insert, expiry date and batch number.

Before you start

1. Use only an authorised kit appropriate for the intended screening purpose.
2. Read the complete instructions and check storage conditions and expiry date.
3. Wear appropriate PPE and prepare a clean, well-lit testing area.
4. Confirm the required sample type and collection method for the specific kit.
5. Collect and label samples using approved procedures and correct identification practices.
6. Check that all kit components are present and undamaged.
7. For blood-borne infection testing, ensure privacy, informed consent and trained supervision.

How to use

1. Allow the sealed kit and reagents to reach the specified temperature if instructed.
2. Prepare the sample using the collection method specified for that particular test.
3. Open the test device only when ready to use it.
4. Add the specified sample quantity and buffer using the supplied equipment.
5. Start timing according to the manufacturer's instructions.
6. Keep the device level and undisturbed during the reaction period.
7. Read the result only within the specified time window.
8. Check the control indicator to determine whether the test is valid.
9. Record the result confidentially and explain that screening may require confirmatory testing.
10. Dispose of used devices, lancets and contaminated materials in appropriate biomedical waste containers.

⚠ Safety

- Treat all human samples as potentially infectious and wear appropriate PPE.
- Never reuse lancets, needles, swabs or other single-use collection items.
- Do not interpret an invalid test as positive or negative.
- A screening result alone does not establish a definitive diagnosis for infections such as HIV or HCV.
- Follow approved procedures for consent, confidentiality, sample handling and biomedical waste disposal.
- Seek qualified medical guidance for positive, negative or inconclusive results according to the test and clinical situation.

Care & storage

- Store unopened kits within the manufacturer's specified temperature and humidity limits.
- Keep reagents sealed and protect test components from contamination and direct sunlight.
- Record batch numbers, expiry dates and quality-control checks where required.
- Clean the work area and replenish supplies after each testing session.

Common mistakes to avoid

- Using expired kits or kits stored outside specified conditions.
- Adding an incorrect sample volume or the wrong buffer.
- Reading results too early or after the permitted time window.
- Treating a screening result as a final diagnosis without required follow-up.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ryle's Tube

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **High** 2 labs

Covers these lab items: Ryle's tube

What it is used for

A Ryle's tube is a medical tube used for specified gastric access procedures by trained healthcare personnel.

How to handle

- Verify the tube type, size, packaging integrity, expiry date, and required clinical indication.
- Use aseptic technique and follow the facility's approved insertion and verification protocol.
- Never force insertion; stop and seek clinical assessment if resistance, distress, or breathing difficulty occurs.
- Confirm placement using approved clinical methods before administering feeds, fluids, or medicines.

Storage

- Keep unused tubes sealed in their original packaging under the manufacturer's specified conditions.
- Dispose of single-use tubes after use according to clinical waste procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Scalp Vein Set

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **High** 1 labs

Covers these lab items: Scalp vein set No 20-23 for Siravyadha

What it is used for

A scalp vein set provides vascular access for specified clinical procedures when used by trained healthcare professionals.

How to handle

- Verify the set size, sterile packaging, expiry date, and clinical suitability before use.
- Maintain aseptic technique and follow the approved clinical procedure for insertion and use.
- Never reuse single-use sets or touch sterile needle and tubing connections with unclean hands.
- Dispose of the used needle and set immediately in an approved sharps or clinical waste container.

Storage

- Keep sealed in original packaging in a clean, dry area within the specified storage conditions.
- Protect from damage, moisture, and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sharps Container

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Medium** 1 labs

Covers these lab items: Sharp container

What it is used for

A sharps container safely collects used needles, blades, and other designated sharp medical items to reduce injury and infection risks.

How to handle

- Use an approved, puncture-resistant container positioned upright and within easy reach at the point of use.
- Place sharps directly into the container without bending, breaking, recapping, or manually removing needles.
- Never reach inside, force items through the opening, or fill beyond the marked limit.
- Close and replace the container according to facility policy when the fill line is reached.

Storage

- Keep upright, secure, and inaccessible to children or unauthorized persons.
- Store closed containers in the designated regulated clinical waste area awaiting collection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sphygmomanometer

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 12 labs

Covers these lab items: Blood Pressure Measuring (Sphygmomanometer)- Mercury Free; Blood Pressure Monitor; Blood pressure monitor (manual and/or digital); BP Machine; B.P. Monitoring Machine; Digital BP machine; Electronic Blood Pressure Monitoring Machine; Sphygmomanometer

What it is used for

A sphygmomanometer measures blood pressure using a cuff and a manual or digital measurement system. Correct cuff placement, patient preparation, and careful reading help produce more reliable results.

Main parts

- Cuff wraps around the upper arm or another approved measurement site.
- Inflation bulb or pump inflates the cuff on manual models.
- Valve releases cuff pressure gradually on manual models.
- Pressure gauge displays pressure on aneroid or mercury-free manual models.
- Digital display shows blood pressure and pulse on electronic models.
- Tubing connects the cuff to the pump or measuring unit.
- Power button and batteries operate many digital models.

Before you start

1. Use the device under teacher or healthcare professional supervision and follow its instructions.
2. Explain the procedure and allow the person to sit quietly for about five minutes.
3. Ask the person to avoid talking and keep feet flat with the back supported.
4. Use a correctly sized cuff on a bare upper arm unless the device specifies otherwise.
5. Check the cuff, tubing, connectors, display, and battery condition for damage.
6. Avoid measuring over clothing or immediately after exercise, caffeine, or smoking when possible.

How to use

1. Position the person's arm supported at heart level with the palm facing upward.
2. Wrap the cuff snugly around the bare upper arm according to the cuff markings.
3. For a manual device, place the stethoscope over the brachial artery if trained to do so.
4. Inflate the cuff according to the device instructions and the trained measurement method.
5. Release pressure slowly on manual models while listening for Korotkoff sounds if trained.
6. For digital models, press the start button and keep the person still during measurement.
7. Wait for the reading to appear and note systolic pressure, diastolic pressure, and pulse if displayed.
8. Record the result with the time and relevant observations according to school procedure.

9. If a reading seems unusual, allow a short rest and repeat the measurement correctly.
10. Remove the cuff, switch off the device, and clean reusable surfaces as instructed.

⚠ Safety

- Do not use a cuff on an injured arm or an arm with a medical restriction.
- Keep the cuff and tubing clean and avoid excessive tightening or prolonged inflation.
- Never diagnose a medical condition from one reading; refer concerning results to a healthcare professional.
- Stop if the person experiences pain, dizziness, or distress during measurement.
- Do not share cuffs without cleaning them according to infection-control instructions.
- Keep mercury-free equipment intact and dispose of damaged electronic parts and batteries appropriately.

Care & storage

- Clean the cuff and device using only manufacturer-approved cleaning methods.
- Store the device in a dry protective case away from heat and moisture.
- Check batteries, tubing, cuff fabric, and connectors regularly for damage.
- Arrange calibration or accuracy checks at the intervals recommended by the manufacturer or healthcare facility.

Common mistakes to avoid

- Measuring over clothing or using an incorrectly sized cuff.
- Allowing the person to talk, move, or keep the arm unsupported.
- Placing the arm below or above heart level during measurement.
- Recording the reading incorrectly or interpreting one result as a diagnosis.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spine Board

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Spine Board

What it is used for

A spine board is a rigid patient-transfer device used by trained emergency responders when a patient requires specific movement or extrication precautions. Correct patient assessment, coordinated handling, secure positioning, and prompt transfer to appropriate medical care help reduce avoidable movement and further injury.

Main parts

- Rigid board supports the patient during transfer.
- Handholds allow responders to grip and manoeuvre the board.
- Patient straps secure the patient to the board.
- Head immobilisation blocks or supports help limit head movement where indicated.
- Head straps or forehead and chin restraints help secure the head where appropriate.
- Padding supports comfort and helps fill gaps where indicated.
- Compatible lifting or transfer accessories assist trained responders where approved.

Before you start

1. Only trained emergency or healthcare personnel should use the spine board.
2. Assess the scene for hazards and follow the applicable emergency response protocol.
3. Assess the patient's airway, breathing, circulation, and immediate life threats before moving when feasible.
4. Use a spine board only when indicated by the applicable protocol; it is not a routine long-duration support surface.
5. Inspect the board, straps, buckles, handholds, and head supports for damage or contamination.
6. Prepare enough trained responders to coordinate the move and explain the process to the patient when possible.
7. Wear appropriate PPE and prepare the destination stretcher or transfer surface before lifting.

How to use

1. Approach the patient using the approved scene-safety and patient-assessment procedure.
2. Determine whether a spine board is indicated and select the appropriate transfer method under the response protocol.
3. Assign clear roles and a coordinated command for each movement.
4. Maintain airway access and monitor breathing, circulation, and neurological status throughout the move.

5. Position the board using the approved technique while minimising unnecessary twisting or movement.
6. Transfer the patient onto the board with coordinated assistance and maintain alignment as clinically indicated.
7. Apply appropriate padding and secure straps without restricting breathing or circulation.
8. Use head supports only as indicated and ensure they do not obstruct airway assessment or access.
9. Recheck the patient's condition, strap security, comfort, and circulation after securing.
10. Lift and transfer the board with sufficient trained responders using coordinated commands.
11. Move the patient promptly to the prepared stretcher or appropriate care destination and follow handover procedures.
12. Clean and inspect the board and accessories after use, and document the transfer according to protocol.

⚠ Safety

- Do not move a patient on a spine board unless required for immediate safety or directed by the applicable clinical protocol.
- Airway, breathing, circulation, and other life-threatening conditions take priority over immobilisation.
- Do not leave the patient unattended or secured in a position that compromises breathing or circulation.
- Never lift the board by straps or accessories not designed as lifting points.
- Use enough trained responders to manage the patient and board safely.
- Follow local protocols for spinal motion restriction, patient monitoring, infection control, and equipment decontamination.

Care & storage

- Clean and disinfect the board, straps, and accessories using approved products after each use.
- Inspect the board for cracks, sharp edges, damaged handholds, and structural deformation.
- Check straps, buckles, and head supports for wear and reliable fastening.
- Store the board dry and in its designated location, ready for emergency use.
- Remove damaged equipment from service and arrange authorised replacement or repair.

Common mistakes to avoid

- Using a spine board for prolonged routine transport when an appropriate alternative is indicated.
- Moving the patient without coordinated commands or sufficient trained responders.
- Securing straps too tightly or obstructing airway access.
- Failing to reassess the patient after movement and securing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spirometer

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Spirometer

What it is used for

A spirometer measures lung volumes or airflow during breathing tests. It is used for supervised respiratory assessment or training, and results must be interpreted by a qualified healthcare professional.

Main parts

- Identify the spirometer body and display or measurement scale.
- Locate the mouthpiece and breathing tube or flow sensor.
- Identify the filter or disposable breathing accessory, where provided.
- Locate the start, stop and reset controls.
- Identify the calibration syringe connection or calibration function, if applicable.
- Locate the power supply or battery compartment.
- Identify the manufacturer's instructions and patient-use accessories.

Before you start

1. Use the spirometer under appropriate healthcare or trained instructor supervision.
2. Read the instructions and confirm the device is suitable for the intended test.
3. Inspect the mouthpiece, tubing, sensor and housing for damage or contamination.
4. Clean hands and use a new disposable mouthpiece or filter when specified.
5. Check battery level, device settings and calibration status according to instructions.
6. Explain the procedure and obtain the person's consent before testing.
7. Ensure the person is seated comfortably and understands the breathing instructions.

How to use

1. Switch on the spirometer and complete the required startup checks.
2. Attach a clean mouthpiece and filter according to the device instructions.
3. Select the prescribed test mode and enter required patient details accurately.
4. Position the person upright and fit a nose clip if the test protocol requires one.
5. Ask the person to seal their lips around the mouthpiece without obstructing airflow.
6. Demonstrate and instruct the breathing manoeuvre specified for the test.
7. Start recording and encourage the person to follow the clinician's breathing instructions.
8. Repeat the manoeuvre only as directed, allowing rest between attempts when needed.
9. Review the recorded values and check for errors or an invalid test indication.
10. Save the results securely and remove the mouthpiece after testing.
11. Clean reusable components and switch off the device according to instructions.

⚠ Safety

- Do not perform forced breathing tests without appropriate supervision and screening.
- Stop the test if the person develops chest pain, severe breathlessness, dizziness or distress.
- Never share mouthpieces without appropriate cleaning or a new disposable accessory.
- Use filters and infection-control procedures according to device instructions.
- Do not interpret spirometry values as a diagnosis without qualified clinical assessment.
- Follow healthcare guidance for people with relevant medical risks or recent procedures.

Care & storage

- Clean and disinfect reusable components according to manufacturer instructions.
- Replace disposable mouthpieces and filters after each user as required.
- Check calibration and perform quality-control procedures at the recommended intervals.
- Store the device and accessories in a clean, dry and protected case.

Common mistakes to avoid

- Using a dirty or incorrectly fitted mouthpiece or filter.
- Failing to explain the required breathing manoeuvre before recording.
- Accepting poor-quality or invalid test results without review.
- Changing calibration settings or interpreting results without appropriate training.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Splint

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Malleable Splint Set; Splint; Splints

What it is used for

To support and immobilise a suspected injured limb or body part until appropriate medical assessment and treatment are available.

How to handle

- Use the splint only within the user's first-aid training and applicable clinical procedures.
- Assess the scene for safety and check circulation, sensation, and movement before and after application when appropriate.
- Choose a suitable splint size and shape; pad rigid edges and avoid unnecessary movement of the injured area.
- Secure the splint without excessive tightness and do not attempt to straighten a visibly deformed limb.
- Seek urgent medical care for suspected fractures or serious injuries and monitor the person while awaiting assistance.

Storage

- Store clean and dry in a labelled first-aid or clinical equipment kit.
- Keep reusable splints disinfected according to facility procedures and separate from contaminated supplies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stadiometer

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Height measuring rod; Height Measuring Scale

What it is used for

A stadiometer measures a person's standing height using a vertical scale and headpiece. Use correct posture and positioning for a reliable reading.

How to use

1. Place the stadiometer on a stable, level surface and check its alignment.
2. Ask the person to remove shoes and bulky headwear, if appropriate.
3. Position the person upright with heels together and feet flat.
4. Align the head in a level position and ask the person to stand naturally.
5. Lower the headpiece gently until it touches the top of the head.
6. Read the scale at eye level and record the height with units.
7. Help the person step away and clean contact surfaces as needed.

⚠ Safety

- Ensure the base is stable and the measuring rod cannot tip over.
- Lower the headpiece gently and never press down on the person's head.
- Provide assistance to anyone who may have difficulty standing safely.

Care & storage

- Clean contact surfaces regularly and store the stadiometer upright in a secure place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sterilizer

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 8 labs

Covers these lab items: Sterilizer; Sterilizers; Syringe Sterilizer

What it is used for

A sterilizer uses a validated process to destroy or remove microorganisms from suitable instruments and materials. Correct loading, cycle selection, and verification are essential for safe reuse in healthcare and training settings.

Main parts

- Chamber holds the instruments or materials being processed.
- Door and locking mechanism keep the chamber closed during operation.
- Control panel selects the cycle and displays operating status.
- Heating element or steam generator provides the required treatment conditions.
- Temperature and pressure sensors monitor conditions in applicable models.
- Tray or rack holds items for effective exposure to the sterilizing process.
- Drain, vent, or filter system manages moisture or exhaust according to the design.

Before you start

1. Read the equipment instructions and confirm it is designed for the items being processed.
2. Only trained and authorised personnel should operate the sterilizer.
3. Clean instruments first according to the approved infection-control procedure.
4. Inspect the chamber, door seal, trays, controls, and power or utility connections.
5. Use compatible packaging and arrange items to allow the sterilizing agent to reach surfaces.
6. Check water, indicator supplies, and other requirements specified for the selected cycle.

How to use

1. Load cleaned items according to the manufacturer's instructions without overpacking the chamber.
2. Separate incompatible materials and ensure packages do not obstruct vents or sensors.
3. Close and lock the door using the specified procedure.
4. Select the validated cycle suitable for the load and instrument material.
5. Start the cycle and monitor displayed temperature, pressure, time, and alarms as applicable.
6. Wait until the full cycle finishes and the equipment indicates safe completion.
7. Allow pressure to return to a safe level and the load to cool as instructed.
8. Use appropriate protective equipment when unloading hot items or handling residual moisture.
9. Check cycle records and required chemical, physical, or biological indicators according to local procedures.
10. Store processed items in the designated clean area and report any failed cycle.

⚠ Safety

- Never open the door while the chamber is pressurised or the cycle is active.
- Beware of hot surfaces, steam, hot liquids, and sharp instruments during unloading.
- Do not process sealed containers or incompatible materials unless the equipment specifically permits them.
- Never reuse instruments after an incomplete or failed cycle until they are reprocessed correctly.
- Do not bypass door locks, alarms, sensors, or other safety interlocks.
- Follow the facility's infection-control procedures for contaminated items and failed loads.

Care & storage

- Switch off and isolate the equipment before routine cleaning when required by the manual.
- Clean the chamber, trays, and seals using approved methods and compatible products.
- Inspect seals, filters, drains, and indicators at the required intervals.
- Arrange authorised servicing and calibration according to the manufacturer's schedule.

Common mistakes to avoid

- Overloading the chamber or blocking the sterilizing agent's circulation.
- Selecting a cycle unsuitable for the instrument or material.
- Opening the door before pressure has returned to a safe level.
- Assuming an item is sterile without checking cycle completion and required indicators.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stethoscope

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Stethoscope

What it is used for

Use a stethoscope to listen to body sounds during authorized health assessments or first-aid training.

How to use

1. Inspect the earpieces, tubing, chest piece, and diaphragm for damage.
2. Clean the parts that contact the user or patient according to hygiene procedures.
3. Place the earpieces correctly and ensure a comfortable fit.
4. Position the chest piece on the approved listening area as instructed by a trained professional.
5. Listen carefully and note observations without making an unsupported diagnosis.
6. Clean the contact surfaces after use and store the stethoscope safely.

⚠ Safety

- Use the stethoscope only within your training and authorized role.
- Clean it between users to reduce cross-contamination.
- Do not use damaged tubing or contaminated contact surfaces.
- Seek qualified medical help when symptoms or findings cause concern.

Care & storage

- Wipe contact surfaces with a compatible cleaner and dry them.
- Store in a clean, dry case away from heat and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stretcher

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Auto-loading Stretcher; Scoop Stretcher; Stretcher

What it is used for

To move or transfer a person safely using a suitable stretcher. To maintain patient alignment and reduce movement during transport according to the patient's condition and emergency protocol.

Main parts

- Stretcher frame and patient surface
- Wheels or carrying handles
- Safety straps
- Side rails or locking mechanism, if fitted
- Height adjustment or loading mechanism, if fitted

Before you start

1. Assess the patient and follow the applicable clinical or emergency protocol.
2. Inspect the stretcher, wheels, straps, locks, and frame for damage.
3. Confirm the stretcher's rated load capacity and suitability for the patient.
4. Prepare enough trained helpers and clear the transport route.
5. Explain the procedure to the patient when possible and prepare required support equipment.

How to use

1. Position the stretcher close to the patient and lock the wheels where applicable.
2. Use the approved transfer method and sufficient trained helpers to move the patient onto it.
3. Maintain appropriate head, neck, and spine support according to the patient's condition.
4. Secure the patient with the provided straps without restricting breathing.
5. Check that side rails and locking mechanisms are engaged where fitted.
6. Move the stretcher smoothly and communicate clearly among helpers.
7. Monitor the patient during transport and stop if the patient deteriorates or equipment becomes unsafe.
8. Transfer the patient to the receiving surface using the approved method.
9. Clean and disinfect the stretcher after use according to facility policy.

⚠ Safety

- Only trained personnel should perform patient transfers.
- Do not exceed the stretcher's rated load capacity.
- Use enough helpers and approved lifting techniques to prevent injury.
- Secure the patient before moving the stretcher.
- Do not use a stretcher with faulty wheels, straps, locks, or frame.
- Follow spinal-motion precautions and emergency protocols when indicated.

Care & storage

- Clean and disinfect patient-contact surfaces after each use.
- Inspect straps, wheels, locks, and frame regularly.
- Check loading and height mechanisms before use where fitted.
- Report damage and remove unsafe equipment from service.

Common mistakes to avoid

- Moving the patient before securing the straps.
- Using too few helpers for the patient's condition.
- Failing to lock wheels during transfer when required.
- Ignoring damaged straps or faulty locking mechanisms.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Suction Apparatus

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Suction Apparatus

What it is used for

A suction apparatus removes blood, mucus, secretions, or other fluids from a patient's airway or a designated clinical site when clinically indicated. Correct setup, appropriate suction settings, infection control, and patient monitoring help provide effective suction while reducing tissue injury and contamination risks.

Main parts

- Vacuum pump generates negative pressure.
- Collection canister collects aspirated fluids.
- Patient tubing connects the suction source to the collection system.
- Suction catheter or approved clinical attachment removes fluids from the intended site.
- Pressure regulator adjusts suction strength.
- Vacuum gauge indicates the available negative pressure.
- Bacterial filter helps protect the pump where fitted.
- Power switch, battery, or foot control operates the unit depending on the model.

Before you start

1. Only trained healthcare personnel should operate the apparatus within their clinical scope and local protocols.
2. Confirm the intended clinical use and select the correct suction device and approved accessories.
3. Inspect the housing, power cable, tubing, canister, seals, filter, and connectors for damage or contamination.
4. Check that the collection canister is correctly fitted and below its maximum fill level.
5. Verify the power supply or battery condition and test the vacuum gauge and controls.
6. Set the prescribed suction level according to the patient, procedure, and device instructions.
7. Perform hand hygiene and wear appropriate PPE; use sterile accessories when required by the procedure.

How to use

1. Place the apparatus on a stable surface near the patient without obstructing emergency access.
2. Connect the collection canister, tubing, filter, and approved patient attachment according to the device diagram.
3. Confirm that all connections are secure and the canister lid seals correctly.
4. Switch on the apparatus and occlude the patient end briefly as instructed to check vacuum function.

5. Adjust the pressure regulator to the prescribed setting and verify the gauge reading.
6. Explain the procedure to the patient when feasible and position the patient appropriately for the clinical indication.
7. Insert and use the suction catheter only according to the relevant clinical protocol and training.
8. Apply suction for the prescribed duration while observing the patient and maintaining appropriate technique.
9. Monitor oxygenation, breathing, heart rate, comfort, and other clinically relevant signs throughout the procedure.
10. Stop suction immediately if significant distress, bleeding, abnormal response, or equipment failure occurs.
11. After use, switch off the unit, disconnect accessories safely, and dispose of single-use items in designated waste streams.
12. Clean and disinfect reusable parts according to the manufacturer's instructions and document the procedure as required.

⚠ Safety

- Never use suction settings or durations beyond the prescribed clinical limits or device instructions.
- Use the correct catheter size and technique to reduce mucosal injury and other complications.
- Do not allow the collection canister to overfill or fluid to enter the pump.
- Use appropriate infection-control precautions and never reuse single-use accessories.
- Do not use damaged electrical components or operate the device in an unsuitable wet environment.
- If suction is needed for an airway emergency, follow the applicable emergency protocol and obtain appropriate clinical assistance.

Care & storage

- Empty and dispose of collected fluids according to clinical infection-control and waste procedures.
- Replace single-use canisters, filters, tubing, and catheters as specified by the manufacturer and facility policy.
- Clean and disinfect reusable components after each use using approved products.
- Inspect seals, tubing, connectors, and the vacuum gauge regularly.
- Arrange preventive maintenance and remove the device from service if suction performance is unreliable.

Common mistakes to avoid

- Setting suction pressure without considering the patient and clinical indication.
- Allowing the collection canister to overfill or the filter to become wet.
- Reusing disposable accessories or neglecting infection-control procedures.
- Failing to monitor the patient or verify suction performance before use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Surgical Tape

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Micropore

What it is used for

Handle micropore surgical tape hygienically for appropriate dressing support.

How to handle

- Use only for its intended external application and follow clinical procedures.
- Clean and dry the skin as directed before applying tape.
- Avoid touching the adhesive surface unnecessarily and do not apply excessive tension.
- Stop use and seek appropriate advice if irritation occurs; use a fresh piece for each application.

Storage

- Keep the roll in clean, dry packaging with the protective cover in place.
- Store according to the manufacturer's instructions and check expiry where applicable.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Syringe

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Disposable Glycerine syringe; Plastic syringes and hormone applicators

What it is used for

A syringe measures and delivers a specified volume of liquid for an approved medical, laboratory, or application task. Use the correct syringe type and follow hygiene and disposal procedures appropriate to its intended use.

How to use

1. Check the syringe size, markings, packaging, and intended use before use.
2. Confirm that the syringe is clean, undamaged, and within its expiry date if sterile.
3. Prepare the liquid and work area according to the approved procedure.
4. Draw the required volume carefully and remove air bubbles when the method requires it.
5. Administer or apply the liquid only by the authorised route and intended use.
6. Read the scale at eye level and verify the volume before delivery.
7. Dispose of single-use syringes and needles in the designated sharps or medical waste container.
8. Clean reusable non-medical applicators according to the manufacturer's instructions.

⚠ Safety

- Never reuse a single-use syringe or needle.
- Do not use a syringe with damaged markings, packaging, or barrel.
- Use sterile technique for medical procedures and follow trained clinical guidance.
- Never recap or discard exposed needles in ordinary waste.

Care & storage

- Keep unused syringes sealed in clean, dry storage.
- Dispose of used medical syringes promptly and maintain hygiene for reusable applicators.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thermometer

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 25 labs

Covers these lab items: Animal Digital Thermometer; Digital Candy Thermometer (for taking food temperature; Digital thermometer; Food Thermometer; Temperature meter; Thermometer; Thermometers

What it is used for

A thermometer measures the temperature of a person, animal, food, liquid, or surrounding environment, depending on its design. Select the correct thermometer type and follow the specified measurement method.

How to use

1. Inspect the thermometer, probe, display, and battery or power supply.
2. Confirm that the thermometer is suitable for the material or subject being measured.
3. Clean or sanitise the probe using the approved method before use.
4. Switch on the thermometer and select the required mode or unit.
5. Position the probe at the specified measurement location without damaging the subject or sample.
6. Wait for the reading to stabilise or for the completion signal.
7. Record the temperature with units and identify the sample or subject.
8. Clean the probe after use and switch off the thermometer.

⚠ Safety

- Do not use a damaged thermometer or a probe unsuitable for the sample.
- Use a separate or properly sanitised probe for food and clinical measurements as required.
- Avoid cross-contamination between samples or people.
- Follow the manufacturer's temperature limits and safe handling instructions.

Care & storage

- Clean and store the thermometer according to its design and instructions.
- Replace batteries when needed and check calibration where required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tourniquet

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Tourniquet

What it is used for

A tourniquet is a specialized device used by trained responders to control life-threatening bleeding from an arm or leg. Use it only for severe bleeding and follow approved first-aid training.

How to use

1. Ensure the scene is safe and call emergency services for severe bleeding.
2. Expose the injured limb and locate the bleeding site.
3. Place the tourniquet around the limb above the bleeding site, not over a joint.
4. Tighten the device according to its instructions until severe bleeding stops.
5. Secure the tourniquet and note the time of application.
6. Keep the injured person still, monitor their condition, and await emergency responders.
7. Do not loosen or remove the tourniquet; leave this to medical professionals.

⚠ Safety

- Use a tourniquet for life-threatening limb bleeding when indicated by training.
- Never place it over a joint, wound dressing, or object embedded in the limb.
- Do not cover the tourniquet; keep it visible for responders.
- Do not delay emergency care or attempt to remove an embedded object.

Care & storage

- Use a single-use tourniquet only once and replace damaged training or clinical equipment as required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Transfer Forceps

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Transfer forceps

What it is used for

Transfer forceps are used to grasp and transfer suitable items or materials in healthcare and laboratory settings. Select the correct forceps type and follow the required hygiene or sterilization procedure for the intended use.

How to use

1. Inspect the forceps tips, hinge, alignment, and grip for damage or contamination.
2. Confirm the forceps are appropriate for the task and intended contact type.
3. Perform hand hygiene and wear the required protective equipment.
4. Sterilize or disinfect the forceps as required by the procedure and manufacturer.
5. Grip the item gently and transfer it without touching unintended surfaces.
6. Place the transferred item in the designated location and avoid cross-contamination.
7. Dispose of single-use forceps or reprocess reusable instruments according to policy.

⚠ Safety

- Use sterile forceps for procedures requiring sterility.
- Do not use damaged, corroded, or contaminated forceps.
- Never reuse single-use instruments.
- Follow infection-control procedures and approved clinical protocols.

Care & storage

- Clean, disinfect, or sterilize reusable forceps according to the instrument instructions and facility policy.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Urinal

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 11 labs

Covers these lab items: Urinal; Urinal flush; Urinal (Male & Female); Urinal Set

What it is used for

Urinals and related fixtures collect urine for personal care or clinical use and require appropriate hygiene practices.

How to handle

- Check the urinal, lid, seals, and fittings for cleanliness, cracks, or leakage before use.
- Use the correct fixture or collection device and follow the applicable clinical or sanitation procedure.
- Wear gloves when handling used urinals and avoid splashing or direct contact with contents.
- Empty, clean, and disinfect reusable items promptly using the approved procedure.

Storage

- Store clean urinals in a designated hygienic area, separate from food and clean supplies.
- Keep unused items protected from dust and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Urine Bag

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Uro bag; Urobags (Urine collecting bag)

What it is used for

Handle urine collection bags hygienically to reduce leakage and infection risk.

How to handle

- Use only under appropriate healthcare guidance and follow the device instructions.
- Check packaging, tubing, connectors, and bag for damage before use.
- Maintain the prescribed positioning and avoid kinks in the tubing.
- Use standard precautions; empty and dispose of the bag according to clinical procedures and local rules.

Storage

- Keep unused bags sealed in clean, dry packaging.
- Store away from direct sunlight, excessive heat, and contaminated supplies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vacutainer

Category: **Medical, Healthcare & First Aid** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Vaccutainer (red/black/violet)

What it is used for

Handle vacutainer blood collection tubes safely in clinical or training settings.

How to handle

- Use only by trained personnel following approved blood collection procedures.
- Verify tube type, additive, expiry date, and patient identification before collection.
- Wear appropriate PPE, avoid direct contact with blood, and never reuse single-use tubes.
- Do not use cracked, leaking, or incorrectly labelled tubes; dispose of used tubes in approved sharps or biohazard systems as applicable.

Storage

- Store unopened tubes according to the manufacturer's temperature and light requirements.
- Keep in original packaging, organised by type and expiry date, away from contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ventilator

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Ventilator

What it is used for

A medical ventilator supports or controls a patient's breathing by delivering a prescribed breathing pattern and oxygen concentration. It must be operated by trained healthcare professionals using a patient-specific clinical plan and continuous monitoring.

Main parts

- Ventilator unit and display
- Patient breathing circuit and tubing
- Breathing interface or endotracheal/tracheostomy connection
- Air and oxygen supply connections, as applicable
- Humidifier and compatible chamber, if prescribed
- Alarms, sensors, filters, and power backup

Before you start

1. Verify the patient's identity and the prescribed ventilation plan.
2. Only trained and authorised clinicians should prepare the ventilator and circuit.
3. Inspect the unit, breathing circuit, filters, humidifier, connectors, and power supply.
4. Check oxygen and compressed-air supplies or internal gas sources, as applicable.
5. Complete the manufacturer's self-test and circuit leak test before connecting to the patient.
6. Confirm alarm limits, battery backup, emergency ventilation equipment, and monitoring are available.

How to use

1. Assemble the circuit and patient interface using the approved infection-control procedure.
2. Select the prescribed ventilation mode and enter patient-specific settings only as authorised by the clinical team.
3. Verify settings, alarm limits, gas supply, and circuit connections before use.
4. Connect the ventilator to the patient only under the responsible clinician's direction.
5. Confirm effective ventilation using appropriate clinical observations and monitoring, including oxygenation and carbon dioxide where indicated.
6. Observe chest movement, patient comfort, airway pressure, delivered volumes, and displayed waveforms as clinically appropriate.
7. Respond promptly to alarms by assessing the patient first and checking the circuit and gas supply.

8. If ventilation fails, follow the emergency protocol and use approved backup ventilation when directed.
9. Document settings, observations, alarms, interventions, and changes according to clinical policy.
10. When ventilation is discontinued, follow the authorised weaning or disconnection plan.
11. Disconnect and dispose of or reprocess circuit components according to infection-control instructions.
12. Clean the device exterior and prepare the ventilator for the next approved use.

⚠ Safety

- Never use a ventilator without appropriate clinical training and supervision.
- Ventilator settings must be prescribed and adjusted by authorised healthcare professionals.
- Do not silence or disable alarms without an approved clinical reason and monitoring plan.
- Maintain a backup ventilation method and emergency response plan.
- Use infection-control precautions and patient-specific or appropriately reprocessed circuits and accessories.
- If a critical alarm occurs, assess the patient immediately and follow the emergency protocol.

Care & storage

- Clean and disinfect external surfaces using manufacturer-approved products.
- Replace filters and circuit components according to the instructions and infection-control policy.
- Check battery backup, alarms, sensors, and self-test results at the required intervals.
- Arrange preventive maintenance and calibration by authorised biomedical personnel.
- Remove faulty equipment from clinical use and report the defect promptly.

Common mistakes to avoid

- Using settings that do not match the authorised patient-specific plan.
- Ignoring alarms or failing to assess the patient when they sound.
- Using an incorrectly assembled, leaking, or contaminated breathing circuit.
- Failing to check backup ventilation, gas supplies, or power backup.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Walker

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Walker

What it is used for

A walker provides support and improves stability for a person who has difficulty walking. Select the correct type and height, and use it according to professional guidance.

How to use

1. Inspect the frame, handles, rubber tips, wheels, and adjustment locks for damage.
2. Adjust the walker to the recommended height with all locks secured.
3. Place the walker on a stable surface and stand upright with assistance if needed.
4. Move the walker a short distance forward while keeping all required contact points stable.
5. Step forward carefully into the walker without leaning excessively on the frame.
6. Turn slowly using small steps and ask for assistance when necessary.
7. After use, park the walker securely and report any defects.

⚠ Safety

- Use a walker suited to the person's mobility and follow healthcare professional guidance.
- Do not use a walker with loose locks, damaged tips, or unstable wheels.
- Avoid stairs, slippery floors, and uneven surfaces unless specifically trained and supported.
- Provide assistance if the user feels dizzy, weak, or unsteady.

Care & storage

- Clean handles and frame regularly, inspect rubber tips, and store the walker in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Walking Cane

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Cane

What it is used for

A walking cane provides additional support and balance while walking. Choose the correct height and use it according to healthcare professional guidance.

How to use

1. Inspect the cane, handle, shaft, and rubber tip for damage.
2. Adjust the cane to the recommended height and secure any adjustment lock.
3. Hold the handle firmly and stand upright before beginning to walk.
4. Move the cane forward a short distance and keep its tip firmly on the ground.
5. Step forward carefully, following the walking pattern recommended by your healthcare professional.
6. Turn slowly, maintain balance, and ask for assistance if needed.
7. After use, place the cane securely within easy reach.

⚠ Safety

- Do not use a cane with a loose handle or worn rubber tip.
- Avoid slippery floors and uneven surfaces whenever possible.
- Seek assistance if you feel dizzy, weak, or unstable.

Care & storage

- Clean the handle and shaft regularly and replace worn rubber tips promptly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wheelchair

Category: **Medical, Healthcare & First Aid** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Wheel Chair

What it is used for

To support safe mobility and transport of a person who has difficulty walking.

How to handle

- Inspect wheels, tyres, brakes, footrests, armrests, and seat before use.
- Explain the procedure to the user, engage the brakes during transfers, and move footrests out of the way as appropriate.
- Help the user sit securely with feet supported and clothing clear of the wheels.
- Release the brakes only when ready to move; push at a controlled speed and watch for obstacles, slopes, and doorways.
- Apply the brakes before stopping or transferring the user; follow the wheelchair manufacturer's instructions.

Storage

- Clean and disinfect contact surfaces according to facility policy; dry thoroughly.
- Park in the designated accessible area with brakes applied and footrests secured.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drum

Category: **Other** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Drum

What it is used for

Handle drums safely according to their material, contents, and intended use.

How to handle

- Identify the drum contents and check labels before moving or opening it.
- Inspect the drum for leaks, corrosion, damage, or unstable lids.
- Use suitable lifting or moving equipment for heavy drums and keep hands clear of pinch points.
- Do not reuse a chemical drum for another purpose unless it has been properly assessed and approved.

Storage

- Keep drums upright and secured in a designated area.
- Store contents according to their safety requirements and separate incompatible materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Field Visit Equipment

Category: **Other** Type: **Handling note** Priority: **Low** 20 labs

Covers these lab items: Any other automatic and semi-automatic equipment used on site; Any other equipment mentioned in the student's manual can be observed during the field visit; Any other equipment mentioned in the student's manual can be sighted during the field visit; Regular field visits

What it is used for

Field visit equipment supports supervised observation, measurement, documentation, and practical learning at approved sites.

How to handle

- Confirm site permission, equipment requirements, and the activity plan before departure.
- Inspect equipment, batteries, cables, and protective accessories before use.
- Follow site induction, PPE requirements, restricted-area rules, and supervisor instructions.
- Use equipment only for its intended purpose and record observations accurately.
- Report damage, hazards, incidents, or missing items to the supervisor promptly.

Storage

- Clean and inventory equipment after the visit before returning it to storage.
- Protect instruments and accessories in suitable cases and report items requiring repair.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fixture and Fitting

Category: **Other** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Special fixtures and fittings

What it is used for

Handle special fixtures and fittings without damage or injury.

How to handle

- Identify the fixture and check its intended use before installation.
- Use the correct tools and fasteners for fitting.
- Wear suitable PPE when drilling, cutting, or fastening.
- Do not use damaged or incompatible fittings.

Storage

- Keep small parts labelled in suitable containers.
- Store fixtures in a dry area to prevent corrosion or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Frame

Category: **Other** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Frame

What it is used for

Handle frames safely for display, mounting, or practical activities.

How to handle

- Check the frame for cracks, loose joints, broken glass, or sharp edges.
- Use both hands when carrying large frames and keep them upright where appropriate.
- Secure wall-mounted frames with suitable fittings and supports.
- Handle glazed frames carefully to prevent breakage.

Storage

- Store frames upright with protective padding between them.
- Keep dry and secure away from impact and high-traffic areas.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Miscellaneous

Category: **Other** Type: **Handling note** Priority: **Low** 9 labs

Covers these lab items: Miscellaneous; Reserved; Spark; Super cut; Take-off tool

What it is used for

Miscellaneous tools and supplies support specific tasks when their identity and intended use are confirmed.

How to handle

- Identify each item and confirm its intended function before use.
- Check labels, compatibility, condition, and any supplied instructions.
- Use appropriate PPE and follow task-specific safety procedures.
- Do not use unidentified, damaged, or unsuitable items.
- Report missing instructions, defects, or uncertain applications to the supervisor.

Storage

- Label items clearly and store them according to their material and use.
- Keep unidentified items separate until verified by an authorised person.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pad

Category: **Other** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Pad

What it is used for

Handle pads safely according to their intended use and material.

How to handle

- Identify the pad type and intended application before use.
- Inspect for tears, contamination, or other damage.
- Use clean hands or suitable gloves where hygiene is important.
- Do not reuse disposable pads intended for single use.

Storage

- Store clean pads in labelled packaging or containers.
- Keep dry and separate from chemicals or contaminated materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seal

Category: **Other** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Sample seal

What it is used for

A sample seal closes or secures a sample package to help protect its contents and indicate possible tampering.

How to handle

- Confirm the seal type and its compatibility with the sample container or package.
- Inspect the seal for damage, contamination, or missing identification details.
- Apply the seal according to the approved sampling or chain-of-custody procedure.
- Record seal numbers or identifiers where required and avoid reusing single-use seals.
- Do not rely on a seal alone to guarantee sample safety or integrity.

Storage

- Keep unused seals clean, dry, and protected in their original packaging.
- Store numbered seals securely to prevent loss or unauthorised use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Steel Ring

Category: **Other** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Steel rings

What it is used for

Handle steel rings safely for demonstrations, assembly, or practical activities.

How to handle

- Inspect rings for cracks, sharp edges, corrosion, or deformation before use.
- Select rings of suitable size and load rating for the intended activity.
- Avoid forcing rings onto parts or using them to lift loads unless specifically rated for that purpose.
- Wear appropriate hand protection when edges may cause cuts.

Storage

- Clean and dry rings after use to reduce corrosion.
- Store by size in a labelled container away from moisture and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Training Setup

Category: **Other** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Training observation on live customer interactions and consultation setups

What it is used for

Set up supervised training observations of live customer interactions and consultation activities.

How to handle

- Obtain appropriate permission before observing or recording customer interactions.
- Protect customer confidentiality and do not collect personal data unless authorised and necessary.
- Position trainees so they do not disrupt service, compromise safety, or obstruct staff and customers.
- Use a clear observation checklist and discuss feedback respectfully after the interaction.

Storage

- Store observation notes securely and restrict access to authorised trainers.
- Retain or dispose of records according to the organisation's privacy and retention procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Uniform

Category: **Other** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Comfortable clothing; Duty Uniform; Proper guide uniforms, ID badges, hydration packs; Uniform/Name Badge

What it is used for

Uniforms and work clothing provide role identification, professional appearance, and task-appropriate coverage.

How to handle

- Wear the assigned uniform in the required condition and follow workplace dress rules.
- Check garments, fasteners, ID badges, and required accessories before duty.
- Keep clothing clean, correctly fitted, and suitable for the task and weather.
- Secure loose clothing and accessories around machinery or other hazards.
- Report damaged, unsuitable, or missing required uniform items.

Storage

- Store clean uniforms separately from dirty workwear.
- Keep uniforms dry and store ID badges securely when not in use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drain Auger

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Sink auger

What it is used for

A drain auger clears blockages in sink and drain pipes using a flexible cable or coil. Use it carefully to remove obstructions without damaging the pipe or fixture.

How to use

1. Inspect the auger cable, handle, drum, and tip for damage or sharp defects.
2. Wear gloves and eye protection and clear the area around the drain.
3. Remove the drain cover or strainer if required by the tool instructions.
4. Feed the cable slowly into the drain while turning the handle as directed.
5. Advance gently until resistance indicates a blockage; do not force the cable.
6. Retract the cable slowly and remove trapped debris safely.
7. Flush the drain with suitable water flow and clean the tool after use.

⚠ Safety

- Do not use an auger in a drain containing hazardous chemical cleaner.
- Keep hands clear of the rotating cable and wear suitable protective gloves.
- Do not force the cable against fragile pipes or fittings.
- Stop if the cable jams or the pipe appears damaged.

Care & storage

- Remove debris, clean and dry the cable, and inspect it before storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Faucet

Category: **Plumbing & Pipe Fitting** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Taps/faucets

What it is used for

Faucets control water flow in sinks, wash basins, and plumbing fixtures.

How to handle

- Inspect the faucet body, handle, aerator, and connections for leaks or damage.
- Operate handles gently and avoid forcing stiff controls.
- Keep the outlet clean and prevent contact with contaminated materials.
- Report persistent dripping, low flow, or loose fittings for repair.
- Use suitable tools and isolate the water supply before maintenance.

Storage

- Store replacement faucets and fittings in clean, dry packaging.
- Keep seals and small components together in labelled containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Flaring Tool

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Flaring tool

What it is used for

A flaring tool forms a flared end on compatible tubing to create a secure connection with the correct fitting. Use the tool and flare type specified for the tubing material and application.

How to use

1. Identify the tubing material, diameter, wall thickness, and required flare type.
2. Inspect the flaring tool, clamp, cone, and tubing for damage.
3. Cut the tubing squarely and deburr the end according to the service procedure.
4. Fit the correct flare nut onto the tubing before forming the flare.
5. Position the tubing at the specified projection in the tool clamp.
6. Operate the flaring tool evenly to form the required flare without cracks or distortion.
7. Inspect the flare and assemble it with the compatible fitting to the specified torque.

⚠ Safety

- Wear eye protection and gloves suitable for handling cut tubing.
- Secure the tool and keep fingers clear of the forming mechanism.
- Do not use cracked tubing, damaged tools, or incompatible fittings.
- For refrigerant, gas, or high-pressure systems, follow the applicable service standards and leak-test procedure.

Care & storage

- Remove metal debris, clean the tool, and lightly protect moving parts as recommended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Flushing Tank

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Flushing tanks

What it is used for

To inspect and maintain a flushing tank that stores and releases water for toilet flushing. To ensure reliable refilling, proper flushing, and leak-free operation.

Main parts

- Tank body and lid
- Inlet valve and float mechanism
- Flush valve or outlet
- Flush handle or button
- Overflow pipe
- Seals and connecting pipe

Before you start

1. Confirm that the water supply is isolated if internal servicing is needed.
2. Inspect the tank, lid, inlet connection, and outlet pipe for cracks or leaks.
3. Check that the flush handle or button moves freely.
4. Ensure the tank is securely mounted and the overflow path is clear.
5. Prepare suitable tools and replacement seals if required.

How to use

1. Remove the lid carefully where applicable.
2. Turn on the water supply and observe the tank filling.
3. Check whether the float shuts off the inlet at the correct level.
4. Operate the flush control and observe the water discharge.
5. Check that the outlet valve closes and the tank refills correctly.
6. Adjust the float or linkage according to the manufacturer's instructions if needed.
7. Inspect connections and seals for leakage.
8. Close the lid and test the flush again.

⚠ Safety

- Avoid forcing brittle plastic parts or ceramic lids.
- Isolate the water supply before replacing valves or seals.
- Clean spills promptly to prevent slips.
- Use plumbing components compatible with the tank.

Care & storage

- Clean the tank and lid with suitable non-damaging products.
- Inspect valves, seals, and connections periodically.
- Replace worn washers or faulty mechanisms promptly.
- Check for continuous running or hidden leaks.

Common mistakes to avoid

- Setting the water level too high and causing overflow.
- Forcing the flush handle or button.
- Ignoring a continuously running inlet valve.
- Using incompatible replacement seals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hydraulic Pressure Testing Machine

Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Hydraulic Testing Machine; Hydraulic water pressure Machine

What it is used for

A hydraulic pressure testing machine applies controlled water pressure to pipes, fittings, or other approved components to check for leakage and pressure integrity. Correct setup, gradual pressurisation, and controlled depressurisation help produce reliable results and reduce the risk of sudden failure.

Main parts

- Water reservoir or supply connection provides the test liquid.
- Test pump raises the water pressure.
- Pressure gauge or digital sensor displays the applied pressure.
- Pressure regulator or control valve adjusts the pressure.
- Test hoses and fittings connect the pump to the test item.
- Isolation valves control water flow through the test setup.
- Relief valve or protective device limits excessive pressure where fitted.
- Frame and mounting fixtures support the test equipment and component.

Before you start

1. Only trained personnel should conduct tests using the approved procedure and rated equipment.
2. Read the machine manual and identify the test item's specified pressure and hold time.
3. Inspect hoses, couplings, gauges, valves, seals, and the test item for damage.
4. Confirm the machine, hoses, fittings, and test item are rated for the planned pressure.
5. Secure the test item and verify all connections are correctly fitted.
6. Use water or the specified compatible test liquid and remove trapped air where the procedure requires it.

How to use

1. Switch off the pump and release any residual pressure before connecting the test item.
2. Connect the test item securely using the correct fittings and approved test arrangement.
3. Fill the test circuit with the specified liquid and vent trapped air according to the procedure.
4. Close the required valves and check that all connections are secure.
5. Start the pump and raise pressure gradually to the specified test value.
6. Observe the gauge and inspect the test setup from a safe position for leakage or abnormal movement.
7. Maintain the specified pressure for the approved hold time and record readings.

8. If pressure drops or leakage appears, stop the test and follow the approved depressurisation procedure.
9. Release pressure slowly through the designated valve and confirm the gauge reads zero before disconnecting.
10. Drain the test item as required, inspect it, and record the result and any faults.

⚠ Safety

- Never exceed the rated pressure of the machine, hoses, fittings, or test item.
- Never loosen a connection while the system is pressurised.
- Keep people clear of hose ends, fittings, and the line of possible component failure.
- Do not use damaged hoses, gauges, valves, or test fixtures.
- Use water for hydrostatic testing unless the approved procedure specifies another compatible liquid; never substitute compressed gas without a specifically engineered procedure.
- If a leak, bulging hose, unusual noise, or sudden pressure change occurs, stop the pump and depressurise from a safe position.

Care & storage

- Drain and dry the equipment as recommended to reduce corrosion and contamination.
- Inspect hoses, seals, couplings, and valves after each test.
- Keep pressure gauges and sensors within their calibration or verification schedule.
- Store the machine and accessories in a clean, dry location and record maintenance.

Common mistakes to avoid

- Increasing pressure too quickly or exceeding the specified test limit.
- Leaving trapped air in the circuit when the procedure requires venting.
- Attempting to tighten or disconnect fittings while pressure remains.
- Failing to record the test pressure, hold time, and observed result.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pipe Cutter

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Pipe cutter; Tube cutter

What it is used for

To cut compatible pipes or tubes cleanly to the required length for plumbing or installation work.

How to handle

- Inspect the cutter wheel or blade, rollers, frame, handle, and adjustment screw for damage or wear.
- Confirm the cutter is suitable for the pipe material and diameter; secure the pipe on a stable support.
- Measure and mark the cutting location and wear suitable eye and hand protection.
- Align the cutter with the mark, tighten gradually, and rotate or operate it according to the tool design without excessive force.
- Remove burrs with a suitable deburring tool and check the cut edge before fitting the pipe.

Storage

- Wipe the cutter clean, remove metal or plastic shavings, and protect the blade or wheel.
- Retract or lock the cutting mechanism as designed and store in a dry tool cabinet.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pipe Fitting Tool

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Tools required for laying water pipelines

What it is used for

Pipe fitting tools are used to measure, cut, thread, join, and secure water pipeline components. Select tools compatible with the pipe material and follow the approved plumbing layout and pressure requirements.

How to use

1. Review the pipeline layout and identify pipe sizes, materials, fittings, and valve positions.
2. Inspect pipe cutters, wrenches, measuring tools, threading tools, and jointing equipment.
3. Shut off and verify isolation of the water supply before modifying an existing line.
4. Measure and mark the pipe, then cut it squarely using the suitable cutter.
5. Deburr the cut ends and prepare threads or joints according to the pipe system.
6. Assemble fittings using the specified sealant, joining method, and tightening procedure.
7. Test the completed line for leaks according to the approved procedure and clean the work area.

⚠ Safety

- Wear suitable eye protection and gloves when cutting or threading pipes.
- Secure pipes before cutting and keep hands clear of cutting edges.
- Use tools within their rated capacity and avoid excessive force on fittings.
- Check for concealed electrical cables and other services before drilling or cutting.

Care & storage

- Clean tools after use, dry metal parts, and store cutters and threading tools safely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pipe Threading Die

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Pipe Die Set - with D/E Handle(pi pe wrench)

What it is used for

To cut external threads on compatible pipe ends so that threaded fittings can be installed securely.

How to handle

- Inspect the die, die head, handle, and cutting edges for damage, wear, or missing components.
- Select the correct die size and type for the pipe material and thread specification.
- Secure the pipe in a suitable pipe vice and wear eye protection and appropriate gloves where safe for the task.
- Align the die squarely with the pipe end, apply the recommended cutting lubricant, and turn the handle steadily according to the tool design.
- Remove the die carefully, clean the threads, and check thread quality and fit without forcing incompatible fittings.

Storage

- Clean the die and remove metal chips and lubricant residue after use.
- Dry and lightly protect cutting surfaces as recommended; store the die and handle together in a labelled case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pipe Threading Machine

Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Powered pipe threading machine

What it is used for

A powered pipe threading machine cuts external threads on compatible pipes so that fittings can be securely connected. Correct die selection, stable pipe clamping, and controlled operation help produce accurate threads while reducing injury and equipment damage.

Main parts

- Motor drives the threading head and rotating pipe or die mechanism.
- Chuck or clamping jaws hold the pipe securely.
- Threading head holds the dies that cut the threads.
- Die set determines the thread size and type.
- Pipe support or stand supports longer pipe sections.
- Cutting oil system lubricates and cools the cutting area where fitted.
- Foot switch or control switch operates the machine on applicable models.
- Chip tray collects metal chips and excess cutting oil.

Before you start

1. Read the manual and confirm the machine supports the pipe material, diameter, and thread type.
2. Inspect the power cable, chuck, dies, guards, foot switch, and oil system for damage.
3. Select a compatible, sharp die set of the correct size.
4. Check that the pipe is straight enough, free from serious damage, and suitable for threading.
5. Secure the machine on a stable surface and support long pipes with suitable stands.
6. Wear safety goggles and close-fitting work clothing; remove jewellery and secure long hair.

How to use

1. Switch off and isolate power before fitting dies or adjusting the chuck.
2. Install the correct die set and set the threading head according to the manual.
3. Insert the pipe into the chuck and clamp it securely with sufficient length supported.
4. Position the pipe support so the pipe remains level and stable.
5. Check that guards are in place and the cutting oil supply is ready where fitted.
6. Start the machine using the designated control and apply cutting oil as instructed.
7. Guide the threading head into the pipe end without forcing the dies.
8. Allow the machine to cut the thread at the recommended feed and speed.
9. Stop the machine and wait for all rotating parts to stop before inspecting the thread.

10. Release the pipe only after rotation has stopped, remove chips safely, and check thread fit with the specified gauge or fitting.

⚠ Safety

- Never wear gloves, loose clothing, or jewellery near rotating pipe or chuck components.
- Keep hands away from dies, chuck jaws, and rotating pipe during operation.
- Do not hold long pipe sections by hand alone; use suitable pipe supports.
- Switch off and isolate power before clearing chips, changing dies, or servicing the machine.
- Never operate the machine with missing guards, damaged switches, or loose clamping jaws.
- Use a brush or approved tool to remove metal chips; do not use bare hands.

Care & storage

- Clean metal chips and cutting oil from the machine after use.
- Inspect dies for wear, chipped cutting edges, and correct fit.
- Check chuck jaws, pipe supports, guards, and oil delivery regularly.
- Store dies clean and dry and maintain the machine according to the manual.

Common mistakes to avoid

- Using the wrong die size or thread type.
- Clamping the pipe loosely or failing to support its free end.
- Touching or removing chips while the pipe or chuck is rotating.
- Forcing the threading head instead of using the recommended feed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pipe Vice

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Pipe vice; Yoke vise

What it is used for

Use a pipe vice to hold a pipe securely during cutting, threading, or other plumbing and pipe-fitting operations.

How to use

1. Choose a pipe vice suitable for the pipe diameter and material.
2. Inspect the jaws, frame, screw, and mounting arrangement for damage.
3. Securely mount the vice to a stable workbench or approved support.
4. Position the pipe so the working section is accessible and adequately supported.
5. Tighten the vice until the pipe is held firmly without crushing or deforming it.
6. Check the pipe for movement before cutting or threading.
7. Release the pipe carefully and clean the vice after use.

⚠ Safety

- Keep fingers clear of the jaws and screw mechanism.
- Support long pipes to prevent tipping or sudden movement.
- Do not exceed the vice's rated capacity or use damaged equipment.
- Depressurize and drain pipes before cutting or opening them.

Care & storage

- Remove metal chips, dirt, and moisture after use.
- Keep the screw mechanism maintained and store the vice securely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pipe Wrench

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Pipe wrench; Pipe Wrench BIS certified

What it is used for

Use a pipe wrench to grip and turn compatible pipes or threaded fittings during plumbing and pipe-fitting work.

How to use

1. Choose a pipe wrench size suitable for the pipe diameter and task.
2. Inspect the jaws, adjustment nut, frame, and handle for damage.
3. Adjust the jaws so they grip the pipe securely without excessive looseness.
4. Position the wrench on the pipe with enough clearance for a controlled turn.
5. Apply steady force in the intended direction and avoid sudden jerks.
6. Reposition the wrench as needed and inspect the fitting after turning.
7. Clean the wrench and store it after use.

⚠ Safety

- Do not use a pipe wrench on nuts or finished surfaces where jaw marks may cause damage.
- Keep hands clear of pinch points and slipping pipes.
- Do not use extension pipes or strike the wrench unless the manufacturer permits it.
- Secure the pipe and release system pressure before loosening pressurized fittings.

Care & storage

- Remove dirt and moisture from the jaws and frame after use.
- Store the wrench dry and check jaw teeth and adjustment parts regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pipe and Fittings

Category: **Plumbing & Pipe Fitting** Type: **Handling note** Priority: **Low** 13 labs

Covers these lab items: Advance fixtures and fittings; Advance pipe fittings and fixtures; Associated plumbing accessories; End Cap; fittings; Fittings and accessories; Inter connecting piping; Pipes

What it is used for

Pipes and fittings connect plumbing components and support controlled transport of water or other compatible fluids.

How to handle

- Identify pipe material, diameter, pressure rating, and fitting compatibility before assembly.
- Inspect pipes, threads, seals, and fittings for cracks, deformation, or contamination.
- Use the specified joining method and tools, and avoid overtightening threaded connections.
- Test completed connections for leakage according to the approved procedure before use.

Storage

- Store pipes on stable supports and prevent bending, crushing, or rolling.
- Keep fittings sorted by size and type in clean, labelled containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shower Head

Category: **Plumbing & Pipe Fitting** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Shower head complete

What it is used for

A shower head distributes water for bathing and personal hygiene.

How to handle

- Inspect the shower head, hose, seals, and fittings for leaks, blockages, or damage.
- Confirm fittings are compatible before installation or replacement.
- Clean the spray openings according to the manufacturer's instructions.
- Check water temperature before use and avoid directing hot water toward others.
- Report persistent leaks, poor flow, or loose fittings for maintenance.

Storage

- Store replacement shower heads and fittings in clean, dry packaging.
- Protect threads and seals from dust, impact, and damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sink

Category: **Plumbing & Pipe Fitting** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Sink

What it is used for

A sink provides a facility for washing hands, utensils, tools, or other items appropriate to its installation.

How to handle

- Inspect the basin, tap, drain, and plumbing connections for cracks, leaks, or blockages.
- Use the sink only for its designated purpose and avoid disposing of unsuitable waste into the drain.
- Clean and disinfect the basin and taps according to the applicable hygiene procedure.
- Keep the surrounding floor dry and report persistent drainage problems.
- Use suitable PPE when handling contaminated items or cleaning chemicals.

Storage

- Keep the sink and surrounding area clean and dry when not in use.
- Store cleaning supplies separately in labelled containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Smoke Generator

Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Smoke Generator for testing of pipes and joints

What it is used for

A smoke generator introduces controlled smoke into a suitable pipe system to help identify leaks at joints and connections. Correct sealing, controlled smoke delivery, and suitable ventilation help make the test easier to interpret and safer to perform.

Main parts

- Smoke generation unit produces visible test smoke.
- Air pump or pressure source moves smoke through the test system.
- Smoke outlet hose connects the generator to the pipe system.
- Adapter or sealing plug closes the test connection.
- Flow or pressure control regulates smoke delivery on equipped models.
- Indicator or gauge shows operating status or pressure where provided.
- Power switch controls the unit on powered models.
- Protective casing encloses the generator components.

Before you start

1. Read the equipment manual and confirm the generator is suitable for the pipe material and test method.
2. Inspect the generator, hoses, adapters, seals, controls, and power supply for damage.
3. Identify the pipe section to be tested and isolate it from incompatible connected equipment.
4. Confirm the test system can safely withstand the specified smoke pressure.
5. Ensure adequate ventilation and clear the area of people not involved in testing.
6. Wear suitable PPE and check that the smoke fluid, if used, is approved for the device.

How to use

1. Switch off the generator before connecting hoses or adapters.
2. Seal the pipe openings using suitable adapters or plugs according to the test procedure.
3. Connect the smoke outlet hose securely to the designated test port.
4. Set the pressure or flow limit according to the pipe system and equipment instructions.
5. Switch on the generator and introduce smoke gradually into the isolated pipe section.
6. Observe accessible joints, seals, and connections for smoke escaping from unintended locations.
7. Mark suspected leak points and record the test conditions and observations.
8. Stop smoke delivery if the pressure limit is reached or abnormal conditions appear.
9. Switch off the generator, release residual pressure safely, and disconnect the hose only after pressure has returned to a safe level.

10. Remove test plugs, restore the pipe system as required, and inspect and store the equipment.

⚠ Safety

- Never exceed the pressure rating of the pipe system, seals, or generator.
- Do not use smoke testing on a live gas line or hazardous system unless an approved procedure specifically permits it.
- Keep smoke away from eyes and avoid inhalation; provide ventilation and follow the fluid safety instructions.
- Do not use the generator near flammable atmospheres or ignition sources unless the equipment is specifically approved for that environment.
- Never disconnect a pressurised hose or remove a sealing plug while pressure remains.
- Stop immediately if a hose bulges, a connection moves, or unexpected pressure develops.

Care & storage

- Switch off and disconnect power before cleaning or maintenance.
- Clean adapters and hoses according to the manufacturer's instructions and allow them to dry.
- Inspect hoses, seals, connectors, and casing for wear or damage after use.
- Store the generator and approved smoke fluid in a clean, dry area according to the manual.

Common mistakes to avoid

- Applying more pressure than the pipe system is rated to withstand.
- Leaving untested openings unsealed so smoke escapes before reaching the test section.
- Disconnecting hoses or plugs before releasing residual pressure.
- Using unapproved smoke fluid or testing in a poorly ventilated area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Submersible Pump

Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Submersible pump

What it is used for

A submersible pump operates while immersed in water or another manufacturer-approved liquid to move it through a discharge pipe. Correct installation, electrical protection, and dry-running prevention help maintain safe and reliable operation.

Main parts

- Pump body and sealed motor
- Water inlet and strainer
- Discharge outlet and pipe connection
- Power cable and plug or terminal connection
- Float switch or level sensor if fitted
- Check valve and lifting rope or chain where specified

Before you start

1. Confirm the liquid, temperature, depth, and pump capacity are within the manufacturer's limits.
2. Inspect the pump body, cable, plug, seals, and discharge connection for damage.
3. Check the supply voltage, earthing, and residual-current protection as required.
4. Verify the intake strainer and discharge pipe are clear and correctly connected.
5. Ensure the pump is fully submerged to the required level before starting, unless designed otherwise.
6. Secure a suitable lifting rope or chain to the designated lifting point; never use the cable.

How to use

1. Place the pump on a stable base clear of mud, debris, and loose sediment.
2. Connect the discharge pipe and secure all required clamps or fittings.
3. Lower the pump using the designated lifting rope or chain without dropping it.
4. Confirm the pump's minimum immersion and cooling requirements are satisfied.
5. Set the float switch or level sensor so it can move freely, if fitted.
6. Switch on the supply using the approved control and observe the initial discharge.
7. Check for abnormal noise, vibration, leaks, overheating, or poor flow.
8. Stop the pump if the water level falls below its permitted operating level.
9. Switch off and isolate the electrical supply before removing or repositioning the pump.
10. After use, clean the strainer and exterior, inspect the cable, and store as instructed.

⚠ Safety

- Never lift, lower, or carry the pump by its electrical cable.
- Do not touch the pump, plug, or water with wet hands while connected to power.
- Use appropriate earthing and residual-current protection installed by a qualified person.
- Never enter a tank, pit, or confined space to retrieve a pump without an approved safe-work procedure.
- Do not run the pump dry unless the model is specifically designed for dry operation.
- Isolate and verify the electrical supply before maintenance or clearing a blockage.

Care & storage

- Clean the intake strainer and pump exterior after use as required.
- Inspect the power cable, seals, and discharge fittings for wear or damage.
- Follow the manufacturer's schedule for servicing bearings, seals, and other serviceable parts.
- Store the pump in a clean, dry, frost-protected place when required by its design.

Common mistakes to avoid

- Running the pump dry can overheat and damage the motor or seals.
- Lifting the pump by its cable can damage insulation and create electric-shock hazards.
- Placing the intake in mud or debris can block flow and overload the pump.
- Using an unsuitable extension lead or missing electrical protection increases shock risk.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thread Seal Tape

Category: **Plumbing & Pipe Fitting** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Teflon tape

What it is used for

Handle thread seal tape for sealing compatible threaded pipe joints.

How to handle

- Verify that the tape is suitable for the pipe material, thread type, and service fluid.
- Clean and dry the threads before application and wrap the tape in the recommended direction.
- Keep tape out of the pipe opening and avoid excessive layers that may damage fittings.
- Follow the manufacturer's instructions and inspect the joint for leaks after assembly.

Storage

- Keep the roll in its original packaging or a clean container.
- Store dry and away from dust, heat, and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thread Sealant

Category: **Plumbing & Pipe Fitting** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: GI threaded joint's synthetic compound

What it is used for

Handle synthetic thread sealant for GI threaded joints safely and correctly.

How to handle

- Read the product label and safety data sheet before use.
- Confirm compatibility with the pipe material, thread type, and service conditions.
- Wear the specified PPE, ensure ventilation, and avoid skin or eye contact.
- Apply the recommended quantity and allow the required curing time before pressurising the system.

Storage

- Keep tightly closed in the original labelled container.
- Store under the manufacturer's recommended conditions away from heat and incompatible chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tube Bender

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Tube Bender; Tube Bender, Tube cutter

What it is used for

A tube bender forms compatible tubing into curves without excessive flattening or kinking. Use the correct bender and tube cutter for the tube material, diameter, and wall thickness.

How to use

1. Check the bender, former, guide, handle, and tube cutter for damage.
2. Measure and mark the required tube length and bend location.
3. Cut the tube squarely with a compatible tube cutter when required.
4. Remove burrs from the cut end and verify the tube fits the bender.
5. Position the tube at the correct mark and align it with the former.
6. Bend gradually to the required angle without forcing the handle.
7. Inspect the bend for cracks, flattening, or kinks before use.

⚠ Safety

- Wear safety glasses and suitable gloves when handling metal tubing.
- Keep fingers away from moving handles, rollers, and pinch points.
- Secure the tool and tube before bending or cutting.
- Do not use damaged cutters or apply excessive force to the tube.

Care & storage

- Remove metal chips, wipe the tool dry, and store it in a clean place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Meter

Category: **Plumbing & Pipe Fitting** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Water meter

What it is used for

A water meter measures the volume of water flowing through a supply line. Install, read, and inspect the meter according to the manufacturer's instructions and applicable plumbing requirements.

How to use

1. Inspect the meter body, dial or display, connectors, and seals for damage.
2. Confirm the meter size, flow direction, and rating match the installation requirements.
3. Shut off the water supply before installing or removing the meter.
4. Install the meter in the specified orientation and secure compatible connections.
5. Restore the water supply gradually and check for leaks.
6. Read and record the meter reading using the displayed units.
7. Report unusual readings, leaks, or suspected meter faults for inspection.

⚠ Safety

- Depressurize the line before opening plumbing connections.
- Use suitable tools and protective equipment for the installation environment.
- Do not force damaged fittings or exceed the meter's rated pressure.
- Follow local plumbing codes and authorized utility procedures.

Care & storage

- Keep the meter accessible for reading and inspect connections and seals periodically.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Pump

Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 9 labs

Covers these lab items: Pump sets; Water Pump; WATER PUMP 1 PHASE; WATER PUMP 1 PHASE ISI Mark

What it is used for

A water pump moves water from a source to a required location through pipes or hoses. Correct installation, priming where required, and safe electrical operation help prevent dry running, leaks, and electric shock.

Main parts

- Electric motor supplies power to drive the pump.
- Pump casing contains the impeller or pumping mechanism.
- Inlet connection draws water into the pump.
- Outlet connection delivers water into the discharge pipe.
- Shaft and seals transfer motion while helping prevent leakage.
- Control switch or starter starts and stops the pump.
- Base, mounting feet, or supports hold the pump securely.

Before you start

1. Read the pump manual and confirm its voltage, capacity, and intended water use.
2. Ensure installation and electrical connections are handled by trained personnel.
3. Inspect the casing, pipes, hoses, seals, cable, plug, and switch for damage.
4. Place the pump on a stable, dry, secure base near suitable pipe connections.
5. Check that the inlet is clear and the required valves are correctly positioned.
6. Fill or prime the pump when required by the manufacturer's instructions.

How to use

1. Switch off and isolate the power supply before checking connections.
2. Connect the inlet and outlet pipes securely using compatible fittings and seals.
3. Open the required valves and confirm that the water source is available.
4. Complete priming or filling according to the pump design and manual.
5. Ensure the pump and surrounding electrical connections are dry before energising.
6. Switch on the pump using its approved control and observe the initial operation.
7. Check the water flow, discharge pressure if indicated, vibration, and visible leaks.
8. Stop the pump immediately if it runs dry, overheats, leaks heavily, or makes unusual noise.
9. Switch off and isolate power before clearing blockages or adjusting pipework.
10. After use, shut down the pump according to the manual and leave the area tidy.

⚠ Safety

- Never touch plugs, switches, or electrical connections with wet hands.
- Do not operate a pump with damaged wiring, exposed terminals, or waterlogged electrical parts.
- Never run a pump dry unless its design specifically permits it.
- Do not open the casing or touch internal parts while the pump is powered.
- Keep children away from the pump, moving parts, and electrical supply.
- Use only pumps approved for the intended water type; do not assume drinking-water suitability.

Care & storage

- Disconnect power before cleaning the exterior or checking accessible components.
- Inspect pipes, seals, fittings, and cables regularly for leaks and damage.
- Follow the manual for lubrication, priming, frost protection, and scheduled servicing.
- Report repeated tripping, overheating, reduced flow, or unusual vibration for authorised inspection.

Common mistakes to avoid

- Starting a pump without priming when priming is required.
- Running the pump with a blocked inlet or closed discharge valve when not permitted.
- Ignoring leaks, overheating, unusual vibration, or electrical faults.
- Handling electrical connections with wet hands or while power is connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Water Tank

Category: **Plumbing & Pipe Fitting** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Reservoir/ Water tank

What it is used for

Handle water tanks safely to support clean water storage and reliable plumbing operation.

How to handle

- Inspect the tank body, lid, inlet, outlet, and supports for cracks, leaks, or damage.
- Use only tanks approved for the intended water quality and application.
- Keep the lid closed to prevent contamination and avoid entering the tank unless an authorised confined-space procedure is in place.
- Ensure the base and supports are rated for the full tank weight and arrange repairs by qualified personnel.

Storage

- Keep the tank covered and protected from sunlight, pests, and external contamination.
- If stored unused, drain and clean it according to the manufacturer's guidance and local water-hygiene requirements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Blow Gun

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Air pressure gun

What it is used for

An air blow gun directs compressed air to remove loose dust, chips, and debris from suitable work surfaces and equipment. Correct pressure control, safe nozzle use, and appropriate protective equipment help prevent eye injuries, skin penetration, airborne contamination, and equipment damage.

Main parts

- Gun body provides the main hand grip and air passage.
- Trigger controls the airflow.
- Air inlet connects to the compressed-air hose.
- Nozzle directs the air stream.
- Pressure regulator controls supply pressure where fitted.
- Safety nozzle or pressure-limiting feature reduces specified hazards where provided.
- Hose coupling connects the gun to the air supply.
- Hanging hook or holder supports storage where fitted.

Before you start

1. Only trained personnel should use the air blow gun.
2. Inspect the gun, nozzle, trigger, hose, couplings, and pressure regulator for damage or leakage.
3. Confirm the air supply pressure is within the gun and accessory ratings.
4. Check that the nozzle is suitable for the task and has no blockage.
5. Secure the workpiece and remove nearby loose objects that could become projectiles.
6. Wear safety goggles or a face shield and suitable hearing and respiratory protection based on the risk assessment.
7. Use local extraction or another approved cleaning method where airborne dust or hazardous particles may be generated.

How to use

1. Confirm that the compressor and air line are operating within their specified pressure limits.
2. Connect the gun securely to the approved compressed-air hose.
3. Set the regulator to the minimum effective pressure permitted by the task and equipment instructions.
4. Position the nozzle toward the work surface and away from people, yourself, and exposed skin.
5. Hold the gun firmly and operate the trigger in short, controlled bursts.

6. Direct debris toward a suitable collection area or extraction system rather than into the surrounding workspace.
7. Keep the nozzle at a safe distance from the surface and avoid damaging sensitive components.
8. Release the trigger whenever the gun is not actively being used.
9. Close the air supply and depressurise the hose before disconnecting or changing the nozzle.
10. Inspect the cleaned area and equipment, then store the gun and hose safely.

⚠ Safety

- Never direct compressed air at a person, skin, eyes, or clothing.
- Never use compressed air to clean the body or remove dust from clothing.
- Do not use compressed air to clean hazardous dust unless an approved contained procedure specifically permits it.
- Never exceed the rated pressure or bypass a pressure-limiting safety feature.
- Do not use the gun near flammable vapours or in an atmosphere where air movement could create an additional hazard.
- Isolate and depressurise the line before clearing blockages, repairing leaks, or servicing the gun.

Care & storage

- Wipe the exterior with an approved cleaning method after use.
- Inspect the nozzle, trigger, hose, couplings, and seals regularly.
- Keep the nozzle clear without inserting fingers or unapproved tools.
- Store the gun in a clean, dry location away from heat and impact.
- Remove damaged equipment from service and arrange authorised repair.

Common mistakes to avoid

- Using excessive air pressure for the task.
- Directing the air stream toward skin or other people.
- Blowing hazardous dust into the air without suitable containment or extraction.
- Disconnecting a pressurised hose or using a damaged nozzle.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Compressor

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: air compressor; Air Compressor (Small); Compressor

What it is used for

An air compressor takes in air and delivers it at increased pressure for compatible tools and equipment. Correct pressure settings, drainage, and safe hose handling help prevent injury and equipment failure.

Main parts

- Electric motor or engine drives the compressor mechanism.
- Pump compresses incoming air.
- Air receiver tank stores compressed air on tank-equipped models.
- Pressure gauge displays tank or outlet pressure.
- Pressure regulator adjusts delivery pressure for connected equipment.
- Safety relief valve releases excess pressure when required.
- Air outlet, hose, and couplings deliver compressed air to the tool.

Before you start

1. Read the compressor manual and confirm its pressure and air-flow ratings suit the task.
2. Allow only trained users to operate or service the compressor.
3. Inspect the tank, hoses, couplings, power cord, plug, gauges, and guards for damage.
4. Place the compressor on a stable, ventilated surface with sufficient cooling clearance.
5. Check the oil level if the model requires lubrication and verify the air filter is clear.
6. Ensure the drain valve is closed before starting and the safety relief valve is unobstructed.

How to use

1. Switch off the compressor and release stored pressure before connecting or changing accessories.
2. Connect the hose securely using compatible couplings and check the tool's pressure requirement.
3. Set the regulator to the pressure specified for the connected tool.
4. Connect the compressor to the correct power supply or prepare its approved engine start procedure.
5. Start the compressor and allow the receiver tank to build pressure within its rated limit.
6. Check the gauges and listen for unusual noise, vibration, or air leaks.
7. Operate the connected tool using controlled movements and the recommended air pressure.
8. Stop the compressor if pressure behaves abnormally, the motor overheats, or a hose is damaged.

9. Switch off and isolate power before maintenance, and release pressure according to the manual.
10. Drain moisture from the receiver tank as instructed and store hoses neatly after use.

⚠ Safety

- Never exceed the rated pressure of the tank, hose, regulator, or connected tool.
- Never point compressed air at people, skin, or clothing.
- Do not use compressed air to clean skin or blow dust from the body.
- Never repair, weld, or modify a pressurised air receiver tank.
- Keep hands away from moving belts, pulleys, and hot compressor surfaces.
- Depressurise the system before disconnecting hoses or clearing blockages.

Care & storage

- Drain condensate at the interval specified by the manufacturer to reduce corrosion.
- Inspect hoses, couplings, gauges, guards, and power cables regularly.
- Clean or replace the air filter and service lubrication systems as directed.
- Arrange authorised inspection if the tank corrodes, the relief valve fails, or pressure leaks persist.

Common mistakes to avoid

- Setting the regulator above the connected tool's required pressure.
- Disconnecting a hose while the system remains pressurised.
- Ignoring water accumulation, air leaks, or damaged hoses.
- Using compressed air to clean clothing or skin.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Pump

Category: **Power Tools & Workshop Machines** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Air pumps

What it is used for

An air pump moves air to inflate items or supply air to compatible equipment. Check its pressure rating and intended use before operation.

How to use

1. Inspect the pump, hose, power cable, connectors, and gauge for damage.
2. Confirm that the pump suits the item and required pressure.
3. Connect the correct nozzle or hose securely to the valve or inlet.
4. Switch on the pump or operate the manual handle steadily.
5. Monitor the pressure gauge or item firmness throughout inflation.
6. Stop at the specified pressure and disconnect the hose carefully.
7. Switch off and store the pump after use.

⚠ Safety

- Never exceed the pressure limit of the item or pump.
- Keep fingers, hair, and loose clothing away from moving parts.
- Do not use damaged electrical cables or operate electrical pumps in wet conditions.
- Allow the pump to cool if it becomes unusually hot.

Care & storage

- Keep air inlets clear and store the pump and hose clean and dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Air Ratchet

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Air rachet

What it is used for

An air ratchet uses compressed air to tighten or loosen fasteners in automotive and workshop tasks. Use the correct socket, air pressure, and tool setting to work efficiently while preventing hand injuries, fastener damage, and flying debris.

Main parts

- Air ratchet body and drive square
- Forward/reverse selector
- Trigger or throttle control
- Air inlet and quick-connect fitting
- Air hose and coupling
- Compatible sockets and optional exhaust deflector

Before you start

1. Confirm that the air ratchet is suitable for the fastener and task.
2. Inspect the tool body, drive square, selector, trigger, air inlet, hose, and couplings for damage.
3. Check that the socket is the correct size, securely fitted, and rated for powered use.
4. Verify that the compressor and regulator supply clean, dry air within the tool manufacturer's pressure range.
5. Disconnect the air supply before fitting a socket, adjusting the tool, or performing inspection.
6. Wear safety glasses and hearing protection; secure loose clothing and tie back long hair.

How to use

1. Select the correct socket and push it fully onto the drive square.
2. Connect the tool to the air hose using the approved coupling.
3. Set the forward or reverse direction before operating the trigger.
4. Position the socket squarely on the fastener and keep fingers clear of pinch points.
5. Squeeze the trigger gradually and allow the ratchet to turn without forcing the tool.
6. Use the tool's rated capability and avoid applying excessive force to a stuck fastener.
7. For final tightening, use a calibrated torque wrench when the specified torque is critical.
8. Release the trigger fully before changing direction or moving to another fastener.
9. Disconnect the air supply before removing a jammed socket or inspecting the drive mechanism.
10. After use, disconnect the tool, clean external surfaces, and apply manufacturer-recommended air-tool lubricant if required.
11. Inspect the hose and fittings, then store the tool in a dry, secure location.

⚠ Safety

- Only trained users should operate pneumatic tools.
- Never exceed the tool's specified air pressure or use damaged hoses and couplings.
- Keep hands away from the ratchet head, rotating socket, and pinch points.
- Do not use cracked, worn, or unsuitable sockets.
- Disconnect the air supply before maintenance, socket changes, or clearing jams.
- Never direct exhaust air or flying debris toward people; use eye and hearing protection.

Care & storage

- Clean the tool after use and keep the air inlet free of debris.
- Lubricate with the specified air-tool oil only when required by the manufacturer.
- Inspect the drive square, selector, trigger, hose, and couplings regularly.
- Replace damaged hoses, worn sockets, and faulty fittings promptly.
- Arrange servicing if the tool loses power, leaks air, or operates irregularly.

Common mistakes to avoid

- Using excessive air pressure to increase tool power.
- Using the wrong socket size or a socket not rated for powered tools.
- Over-tightening fasteners without checking the specified torque.
- Changing sockets or clearing a jam while the air supply is connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Angle Grinder

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Angle Grinder

What it is used for

An angle grinder uses a rotating abrasive or cutting disc to grind, cut, or finish suitable materials. Select the correct disc and speed rating, secure the workpiece, and use the guard and appropriate PPE to reduce the risk of disc failure, sparks, and serious injury.

Main parts

- Grinder body and electric motor
- Spindle and disc mounting flange
- Compatible grinding or cutting disc
- Adjustable protective guard
- Auxiliary side handle
- Power switch, cord, and spindle lock

Before you start

1. Confirm that the grinder and disc are suitable for the material and intended operation.
2. Inspect the grinder, power cord, plug, guard, handle, and disc for damage or wear.
3. Check the disc for cracks, chips, moisture damage, and its maximum speed rating.
4. Ensure the disc's rated speed is at least the grinder's no-load speed and that the disc fits correctly.
5. Secure the workpiece in a vice or suitable fixture; remove flammable materials from the spark path.
6. Wear safety glasses and a face shield, hearing protection, suitable gloves, and appropriate protective clothing; use respiratory protection when required.

How to use

1. Disconnect the grinder from power before fitting or changing a disc.
2. Install the correct disc and flange in the orientation specified by the manufacturer.
3. Tighten the mounting nut using the supplied tool without excessive force.
4. Fit and position the guard between the disc and the operator, and attach the auxiliary handle.
5. Check that the disc rotates freely by hand while the tool is disconnected.
6. Connect power and hold the grinder firmly with both hands using the intended grip.
7. Position yourself so that sparks and debris are directed away from people and flammable materials.
8. Start the grinder and allow it to reach operating speed before contacting the workpiece.
9. Apply light, controlled pressure and keep the disc at the angle recommended for the task.

10. Use the correct disc face for grinding; do not side-load a cutting disc.
11. Avoid twisting, jamming, or forcing the disc through the material.
12. Release the switch and wait for the disc to stop completely before setting the tool down.
13. Disconnect power before inspecting the disc, clearing debris, or changing accessories.
14. After the tool cools, clean it and inspect the disc and guard before storage.

⚠ Safety

- Never operate the grinder without the correct guard and auxiliary handle where designed for use.
- Do not use damaged, incorrectly rated, or incompatible discs.
- Keep hands, cables, clothing, and body parts away from the rotating disc.
- Beware of sparks, hot metal, kickback, and flying fragments; keep bystanders clear.
- Never force a cutting disc sideways or use a cutting disc for side grinding.
- Disconnect power before adjustments, accessory changes, or maintenance; follow lockout procedures where applicable.

Care & storage

- Clean ventilation openings and the tool exterior after use, with power disconnected.
- Inspect the guard, handle, spindle, cord, and switch regularly.
- Replace worn or damaged discs and accessories; store discs dry and protected from impact.
- Arrange servicing if the grinder vibrates excessively, sparks internally, or makes unusual noises.
- Store the grinder securely with its cord protected from cuts and crushing.

Common mistakes to avoid

- Removing the guard or holding the grinder with only one hand.
- Using a disc with an unsuitable speed rating or visible damage.
- Applying excessive pressure or twisting the disc during a cut.
- Setting the grinder down before the disc stops completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Arbor Press

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Arbour press

What it is used for

An arbor press applies controlled manual force to press-fit, assemble, straighten, or remove suitable small components using a lever and ram. Secure the press and workpiece, select the correct tooling, and keep hands clear of pinch points to prevent crushing injuries and component ejection.

Main parts

- Rigid frame and base
- Lever or operating handle
- Ram or sliding arbor
- Ram return spring, if fitted
- Pressing attachment, punch, or adapter
- Work table, support blocks, and workpiece fixture

Before you start

1. Confirm that the press capacity and tooling are suitable for the operation and component.
2. Inspect the frame, base, lever, ram, return spring, and mounting bolts for damage or looseness.
3. Secure the press to a stable workbench or approved mounting surface.
4. Inspect the workpiece and tooling for cracks, wear, misalignment, or damage.
5. Select correctly sized supports and attachments so the force is applied along the intended axis.
6. Wear safety glasses and keep the work area clear of loose objects.

How to use

1. Position the workpiece on the press base using suitable supports or a fixture.
2. Align the component and pressing attachment so the ram force acts straight through the intended contact point.
3. Adjust the work height or support blocks with the ram fully raised and hands clear of the pinch area.
4. Keep fingers away from the ram, attachment, workpiece, and support surfaces.
5. Move the lever slowly to bring the ram into contact with the workpiece.
6. Apply force gradually and observe the component for movement, deformation, or misalignment.
7. Stop immediately if the workpiece shifts, the tooling tilts, or unusual resistance occurs.
8. Release the lever carefully and allow the ram to return as designed.
9. Confirm the component is seated or removed as intended before releasing the fixture.
10. Remove the workpiece only after the ram is fully raised and the applied force is released.

1. Inspect the finished component and clean the press and work area after use.

⚠ Safety

- Never exceed the rated capacity of the press or its attachments.
- Keep hands and body parts out of the ram's travel path and all pinch points.
- Do not use cracked tooling, unstable supports, or damaged press components.
- Do not extend the lever with a pipe or other improvised device to increase force.
- Keep the workpiece aligned and restrained to prevent it from being ejected under pressure.
- Only trained users should operate the press; do not leave it loaded or unattended.

Care & storage

- Wipe the ram, base, and frame clean after use.
- Lubricate moving parts only at the points and intervals specified by the manufacturer.
- Inspect mounting bolts, lever pivots, return spring, and ram alignment regularly.
- Replace worn or damaged attachments and keep tooling labelled and organised.
- Remove the press from service if the frame bends, the ram sticks, or the return mechanism fails.

Common mistakes to avoid

- Applying force to a misaligned workpiece or unsupported component.
- Placing fingers under the ram or near pressing surfaces.
- Exceeding the press rating by using an extended lever.
- Continuing to press when the component tilts, deforms unexpectedly, or jams.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Bending Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bamboo Bending Machine

What it is used for

A bamboo bending machine bends bamboo strips, slats, or poles into a required shape using controlled pressure, heat, steam, or moisture, depending on the machine design. Operate it with the correct material preparation, tooling, and clamping method to prevent splintering, burns, crushing injuries, and unexpected movement.

Main parts

- Machine frame and base
- Bending die, mould, or former
- Clamping and holding mechanism
- Hydraulic, pneumatic, mechanical, or motorised drive
- Heating or steaming unit, if fitted
- Control panel, guards, and emergency stop

Before you start

1. Confirm the machine type, permitted bamboo dimensions, and required bend radius.
2. Inspect the frame, dies, clamps, drive system, guards, cables, and controls for damage.
3. Check that the correct mould and supports are installed and securely fastened.
4. Prepare bamboo according to the specified moisture, heating, or steaming procedure.
5. Remove loose dirt, metal pieces, and damaged sections from the bamboo workpiece.
6. Wear safety glasses, suitable gloves for handling bamboo, protective footwear, and other required PPE; tie back long hair.

How to use

1. Ensure the machine is stopped before adjusting the mould, clamps, or supports.
2. Install and secure the correct bending die or former for the required shape.
3. Set the permitted operating parameters according to the machine manual and bamboo specifications.
4. If heating or steaming is used, prepare and operate that system according to its instructions.
5. Position the bamboo correctly against the die and support points.
6. Secure the workpiece with the designated clamps, keeping hands away from pinch points.
7. Check that the workpiece and surrounding area are clear before starting the bending cycle.
8. Activate the machine using the normal controls and apply bending force gradually.
9. Observe the bamboo for cracking, splitting, slipping, or abnormal deformation during bending.
10. Stop the cycle if the workpiece shifts, the machine vibrates abnormally, or a fault occurs.

1. Release pressure and clamps only after the machine has stopped and the load is controlled.
2. Remove the bent bamboo carefully and allow hot or steamed material to cool safely.
3. Inspect the shape and surface, then record any defects and clean the work area.

⚠ Safety

- Only trained operators should use the machine and adjust its drive or heating system.
- Keep hands and body parts away from the die, clamps, and moving mechanisms.
- Never bypass guards, interlocks, or the emergency stop.
- Use heat-resistant gloves and face protection where hot, steamed, or heated bamboo is handled.
- Do not exceed the machine's rated capacity or force a workpiece into an unsuitable mould.
- Isolate the power source before clearing jams, changing tooling, or performing maintenance.

Care & storage

- Remove bamboo dust and splinters from the machine after safe shutdown.
- Inspect dies, clamps, fasteners, and moving parts regularly.
- Lubricate the drive and moving components according to the manufacturer's instructions.
- Check hydraulic or pneumatic lines and heating or steaming equipment, if fitted, for leaks or damage.
- Report worn tooling, cracked guards, or unreliable controls before further use.

Common mistakes to avoid

- Using unsuitable bamboo moisture conditions or an incorrect bending radius.
- Placing hands near clamps or the die during operation.
- Applying bending force too quickly and causing cracks or splits.
- Releasing the workpiece before pressure is safely removed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Planer Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bamboo Planer Machine

What it is used for

A bamboo planer machine removes uneven surface material from bamboo strips, boards, or prepared pieces to achieve a smoother and more uniform thickness. Use the correct feed direction and cutting settings while keeping hands away from the cutter and feed rollers.

Main parts

- Machine frame and work table
- Planing cutter head and blades
- Feed rollers or feed mechanism
- Thickness or depth adjustment
- Motor and drive system
- Safety guard, dust outlet, and emergency stop

Before you start

1. Confirm the machine is suitable for the bamboo size, shape, and required finish.
2. Inspect the cutter blades, guards, feed rollers, table, power cord, and emergency stop.
3. Check that blades are securely fitted, undamaged, and suitable for the machine.
4. Remove nails, metal fragments, stones, and other foreign objects from the bamboo.
5. Adjust the required cutting depth and feed setting according to the manual and material condition.
6. Connect suitable dust extraction and wear safety glasses, hearing protection, and respiratory protection where required.

How to use

1. Switch off and isolate the machine before adjusting the cutter, changing blades, or clearing a blockage.
2. Set the table, thickness guide, and feed mechanism to the required dimensions.
3. Ensure guards and dust extraction connections are fitted correctly.
4. Start the machine and allow the cutter head to reach normal operating speed.
5. Stand clear of the direct feed path and keep hands away from the cutter and feed rollers.
6. Feed the bamboo in the direction specified by the machine, keeping it properly supported and aligned.
7. Use steady feed pressure and do not force material that binds or stalls.
8. Observe the output for uneven thickness, tear-out, excessive vibration, or unusual noise.
9. Stop the machine if the bamboo shifts, jams, or causes abnormal operation.

10. Wait for all moving parts to stop before removing jammed material or inspecting the cutter area.
11. Measure the finished thickness and repeat with a smaller cut if required and permitted.
12. Switch off, isolate power, clean the table and dust collection area, and record any faults.

⚠ Safety

- Never operate the planer with missing guards or exposed cutting blades.
- Keep hands away from the cutter head, feed rollers, and material exit points.
- Use push blocks or approved feeding aids for short or narrow pieces where specified.
- Do not plane bamboo that is unstable, excessively curved, cracked, or outside the machine's permitted dimensions.
- Wear appropriate eye, hearing, and respiratory protection for dust and noise exposure.
- Disconnect and isolate power before blade changes, cleaning near cutters, or maintenance.

Care & storage

- Clean bamboo dust from the table, guards, and dust extraction system after shutdown.
- Inspect blades for wear, chips, and secure mounting before use.
- Check feed rollers, belts, fasteners, and adjustment mechanisms regularly.
- Sharpen or replace blades according to the manufacturer's procedure.
- Arrange servicing if the machine vibrates excessively, feeds unevenly, or overheats.

Common mistakes to avoid

- Setting the cutting depth too large for the material or machine capacity.
- Feeding bamboo with hands too close to the cutter or rollers.
- Operating with dull or loose blades.
- Clearing a jam before the cutter and feed mechanism have completely stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Shaving Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bamboo Shaving Machine

What it is used for

A bamboo shaving machine removes thin outer layers or excess material from bamboo to produce smoother, thinner, or dimensionally consistent pieces. Operate it with the correct blade setting and feeding method to reduce splintering, cuts, entanglement, and material wastage.

Main parts

- Machine frame and work support
- Shaving blade or cutter head
- Bamboo holding and feed mechanism
- Depth or thickness adjustment
- Motor and transmission system
- Protective guards, waste outlet, and emergency stop

Before you start

1. Confirm the machine type and permitted bamboo dimensions and thickness range.
2. Inspect the blade, cutter head, work supports, feed mechanism, guards, and controls for damage.
3. Check that the blade is sharp, securely mounted, and correctly aligned.
4. Prepare bamboo by removing dirt, loose fibres, and unsuitable damaged sections.
5. Set the shaving depth and feed rate according to the material and machine manual.
6. Ensure waste collection and ventilation are adequate; wear safety glasses, hearing protection, and other required PPE.

How to use

1. Switch off and isolate power before adjusting the blade, feed mechanism, or work supports.
2. Set the required shaving thickness and secure the bamboo using the designated holding method.
3. Confirm that the guards are fitted and the waste outlet is clear.
4. Start the machine and allow the cutter to reach its normal operating speed.
5. Feed the bamboo steadily through the approved feed path, keeping hands clear of the blade and rollers.
6. Maintain the required alignment and avoid forcing, twisting, or pushing material by hand near the cutter.
7. Observe the shaved surface for unevenness, deep cuts, splinters, or signs of blade wear.
8. Stop the machine if material slips, jams, or produces unusual vibration or noise.

9. Wait until all moving parts stop before removing waste or clearing a blockage.
10. Measure the finished piece and repeat the process only if needed and within the permitted limits.
11. Switch off and isolate the machine, clean accumulated shavings, and inspect the equipment after use.

⚠ Safety

- Do not operate the machine with a missing guard or exposed blade.
- Keep fingers, loose clothing, hair, and jewellery away from the cutter and feed mechanism.
- Never reach into the feed path to guide material while the machine is running.
- Use approved jigs or feeding aids for small or awkward pieces where required.
- Disconnect and isolate power before cleaning near the blade, changing settings, or servicing.
- Wear suitable eye and respiratory protection when bamboo fibres or dust may become airborne.

Care & storage

- Remove bamboo shavings and dust after shutdown using the approved cleaning method.
- Inspect blades, fasteners, guides, and feed rollers for wear or damage.
- Maintain the blade sharpness and alignment according to the manufacturer's instructions.
- Check motor ventilation, belts, and electrical connections regularly.
- Replace damaged components and report recurring vibration, overheating, or uneven shaving.

Common mistakes to avoid

- Setting the shaving depth too aggressively and splitting the bamboo.
- Feeding irregular material without proper support or alignment.
- Using a dull, loose, or incorrectly aligned blade.
- Removing waste or adjusting the machine while moving parts are still running.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Sliver Making Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bamboo Sliver Making Machine

What it is used for

A bamboo sliver making machine converts bamboo strips or prepared sections into narrow slivers used for weaving, crafts, mats, and other products. Use suitable splitting and sizing settings and guide the material with approved supports to obtain consistent slivers while preventing cuts and splinter injuries.

Main parts

- Machine frame and feed table
- Slitting or sliver-forming blade assembly
- Guide plates and sizing channels
- Feed rollers, pusher, or holding fixture
- Drive motor and transmission
- Protective guards, waste tray, and emergency stop

Before you start

1. Confirm the machine's capacity, permitted bamboo dimensions, and required sliver width and thickness.
2. Inspect the blades, guides, feed mechanism, frame, guards, and controls for damage.
3. Ensure the blades are sharp, securely fitted, and aligned with the selected sliver size.
4. Prepare bamboo by removing dirt, loose fibres, knots, and damaged areas that could disrupt feeding.
5. Set the required sliver dimensions and install the correct guides or tooling.
6. Wear safety glasses, suitable gloves for handling raw bamboo, protective footwear, and other required PPE.

How to use

1. Switch off and isolate the machine before changing blades, guides, or settings.
2. Install the correct sliver-forming tooling and verify that it is secured according to the manual.
3. Adjust the guides and feed mechanism to the required dimensions.
4. Place the bamboo in the designated feed position using the approved holder or pusher.
5. Check that the bamboo is aligned with the cutting path and that the discharge area is clear.
6. Start the machine and allow the drive system to reach normal operating speed.
7. Feed the bamboo at a steady rate using the designed mechanism; keep hands away from blades and rollers.
8. Collect emerging slivers from the designated output area without reaching into the cutting zone.

9. Inspect the slivers for consistent width, thickness, straightness, and excessive splintering.
10. Stop the machine if the bamboo shifts, slivers jam, or the drive system operates abnormally.
11. Wait for all motion to stop before clearing jams, removing waste, or making adjustments.
12. Switch off and isolate power, clean the machine, and record production quality or maintenance issues.

⚠ Safety

- Only trained operators should use or adjust the sliver making machine.
- Never reach toward the blade or feed rollers while the machine is running.
- Use the designated pusher or holding fixture rather than fingers to feed short pieces.
- Do not bypass guards or operate with loose, damaged, or incorrectly fitted blades.
- Keep the output area clear and handle sharp bamboo slivers carefully to prevent cuts and splinters.
- Isolate power before clearing a jam, changing tooling, or performing maintenance.

Care & storage

- Remove bamboo fibres and slivers from the machine after shutdown.
- Inspect blades, guides, fasteners, and feed rollers regularly.
- Keep the cutting assembly clean, sharp, and correctly aligned.
- Lubricate moving parts only as specified by the manufacturer.
- Arrange servicing if sliver dimensions become inconsistent or the machine develops unusual vibration or noise.

Common mistakes to avoid

- Using incorrect guide settings for the required sliver dimensions.
- Feeding bamboo by hand too close to the cutting assembly.
- Allowing slivers or fibres to accumulate in the feed and discharge areas.
- Clearing a jam or changing a blade without isolating the machine.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Splitting Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bamboo Splitting Machine

What it is used for

A bamboo splitting machine divides bamboo poles or sections lengthwise into smaller strips for further processing. Use the correct splitting blade and secure material alignment to produce uniform sections while reducing the risk of kickback, sharp-edge injuries, and crushing.

Main parts

- Machine frame and support table
- Splitting blade, wedge, or splitting head
- Bamboo holding guide or clamp
- Manual, mechanical, hydraulic, or motorised feed system
- Drive mechanism and controls
- Protective guards and emergency stop

Before you start

1. Confirm the machine type, permitted bamboo diameter and length, and required strip size.
2. Inspect the splitting blade, wedge, guides, clamps, frame, drive system, and guards for damage.
3. Check that the blade is securely mounted, sharp, and suitable for the intended split.
4. Inspect bamboo for severe cracks, loose branches, dirt, stones, or metal that could affect operation.
5. Adjust the guides and supports for the bamboo dimensions and verify that the output path is clear.
6. Wear safety glasses or a face shield, protective footwear, and suitable gloves for handling raw bamboo; keep loose clothing secured.

How to use

1. Switch off and isolate the machine before adjusting the blade, guides, or holding mechanism.
2. Install the appropriate splitting head or blade and confirm that all fasteners are secure.
3. Set the bamboo guides and clamps to support the material along the intended splitting line.
4. Place the bamboo in the designated loading position and align it with the blade or splitting wedge.
5. Secure the bamboo using the designed clamp or holding mechanism; keep hands away from the splitting line.
6. Check that no person is standing in the path of the bamboo or emerging strips.
7. Start the machine or engage the manual feed according to the operating instructions.
8. Apply the splitting force steadily and allow the blade or wedge to follow the intended path.

9. Observe the material for twisting, jamming, unexpected cracking, or movement out of the guide.
10. Stop operation if the bamboo slips, the blade binds, or the machine behaves abnormally.
11. Wait until the blade and feed mechanism have stopped and the force is released before removing split pieces or clearing a jam.
12. Inspect the resulting strips for uniformity and defects, then clean the machine and record any faults.

⚠ Safety

- Only trained operators should use the splitting machine.
- Keep hands clear of the blade, wedge, clamp, and moving feed mechanism.
- Never force bamboo beyond the machine's rated dimensions or capacity.
- Do not bypass guards or use a damaged blade, loose clamp, or unstable support.
- Stand clear of the expected path of split sections and flying splinters.
- Isolate all energy sources before clearing jams, changing blades, or servicing the machine.

Care & storage

- Remove bamboo fibres, chips, and dust after safe shutdown.
- Inspect the splitting blade, wedge, clamps, guides, and fasteners regularly.
- Keep the blade sharp and aligned according to the manufacturer's instructions.
- Check the feed mechanism and hydraulic or pneumatic components, if fitted, for wear or leakage.
- Report unusual vibration, uneven splitting, or damaged safety components before further use.

Common mistakes to avoid

- Loading bamboo that exceeds the machine's permitted size or is poorly supported.
- Holding the bamboo by hand near the blade during splitting.
- Using a blunt or misaligned splitting blade.
- Removing jammed material before the machine is stopped and stored energy is released.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Strip Sizing Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bamboo Strip Sizing Machine

What it is used for

A bamboo strip sizing machine trims or calibrates bamboo strips to a specified width, thickness, or length for consistent downstream processing. Use the correct guides and cutting settings, secure the material, and keep hands clear of cutting and feeding mechanisms.

Main parts

- Machine frame and feed table
- Sizing cutter, trimming blade, or thicknessing head
- Material guides and adjustable stops
- Feed rollers or clamping mechanism
- Motor and drive system
- Protective guards, dust outlet, and emergency stop

Before you start

1. Confirm the required strip dimensions and the machine's permitted material size.
2. Inspect the blades, guides, feed rollers, guards, power cable, and controls for damage.
3. Check that the correct cutting or sizing tooling is securely installed and aligned.
4. Remove dirt, loose fibres, metal fragments, and damaged sections from the bamboo strips.
5. Set the width, thickness, or length guides according to the job specification.
6. Wear safety glasses, hearing protection, protective footwear, and respiratory protection where dust exposure requires it.

How to use

1. Switch off and isolate power before adjusting guides, changing blades, or clearing jams.
2. Set the sizing guides and stops to the required dimensions using the approved adjustment method.
3. Secure the workpiece support and confirm that the guards and dust extraction are correctly fitted.
4. Start the machine and allow the cutter and feed mechanism to reach normal operating speed.
5. Place the bamboo strip in the designated feed position and align it with the guides.
6. Feed the strip steadily using the machine's designed feed mechanism or approved push aid.
7. Keep hands away from the cutting head, rollers, and material exit points throughout operation.
8. Inspect the output for correct dimensions, straight edges, splintering, and surface defects.
9. Stop the machine if the strip shifts, jams, or causes unusual vibration or noise.
10. Wait until all moving parts stop before removing jammed material or making adjustments.

1. Measure the finished strips and repeat the sizing operation only if required and permitted.
2. Switch off and isolate the machine, remove dust and offcuts, and record quality or maintenance issues.

⚠ Safety

- Only trained operators should operate or adjust the machine.
- Never operate with missing guards or exposed cutting blades.
- Do not guide short or narrow strips by placing fingers close to the cutting area; use approved feeding aids.
- Keep loose clothing, hair, jewellery, and cables away from moving components.
- Do not exceed the machine's rated capacity or force material through a jam.
- Isolate all energy sources before cleaning near cutters, changing tooling, or performing maintenance.

Care & storage

- Remove bamboo dust and offcuts after safe shutdown.
- Inspect blades, guides, rollers, fasteners, and guards regularly.
- Keep cutting tools sharp and correctly aligned according to the manufacturer's instructions.
- Check belts, bearings, and electrical connections at the specified maintenance intervals.
- Arrange servicing if sizing accuracy deteriorates, vibration increases, or the feed mechanism becomes unreliable.

Common mistakes to avoid

- Setting incorrect sizing dimensions or failing to verify the first finished strip.
- Feeding twisted or poorly supported bamboo into the cutter.
- Using dull blades that cause splintering or uneven edges.
- Clearing a jam before the machine has fully stopped and been isolated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Veneer Slicing Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bamboo Veneer Slicing Machine

What it is used for

A bamboo veneer slicing machine cuts prepared bamboo material into thin sheets or veneers for decorative surfaces, panels, and other manufactured products. Correctly prepare and secure the material, use the specified blade settings, and follow guarding and handling procedures to prevent cuts, entanglement, and material ejection.

Main parts

- Machine frame and material bed
- Slicing knife or blade assembly
- Material holder, clamp, or feed carriage
- Thickness adjustment and positioning guides
- Drive system and operating controls
- Protective guards, lubrication system, and emergency stop

Before you start

1. Confirm the machine type, required veneer thickness, and permitted bamboo dimensions.
2. Inspect the blade, clamps, material bed, guides, guards, and drive system for damage.
3. Ensure the slicing blade is sharp, securely mounted, and aligned according to the manual.
4. Prepare the bamboo block or section to the required condition and remove loose debris or foreign objects.
5. Set the thickness adjustment and material positioning guides for the intended veneer size.
6. Check lubrication and any hydraulic or pneumatic system according to the manufacturer's instructions.
7. Wear safety glasses, protective footwear, and suitable cut-resistant gloves for handling material only when hands are away from moving blades.

How to use

1. Switch off and isolate all relevant energy sources before adjusting the blade or loading the material where required by the machine design.
2. Position the prepared bamboo block securely on the material bed or carriage.
3. Engage the designated clamps or holding system and verify that the material cannot shift during slicing.
4. Set the required veneer thickness and confirm the blade path is clear.
5. Close all guards and ensure the emergency stop is accessible.

6. Start the machine using the approved operating controls and allow the drive system to reach normal operating conditions.
7. Initiate the slicing cycle and monitor material movement from the designated safe position.
8. Collect finished veneer using the designed collection method without reaching into the blade zone.
9. Check veneer thickness, surface finish, and continuity for cracks, tears, or uneven sections.
10. Stop the machine immediately if the material shifts, the blade binds, or abnormal vibration occurs.
11. Wait until the blade and carriage have stopped and stored energy is controlled before removing offcuts or clearing jams.
12. Switch off and isolate the machine, clean the material bed, and document production or maintenance issues.

⚠ Safety

- Only trained operators should set up and operate the slicing machine.
- Never reach across or near an exposed slicing blade while the machine is running.
- Keep guards, clamps, and material restraints in place during operation.
- Do not use a chipped, cracked, loose, or incorrectly aligned blade.
- Keep the veneer output and material handling area clear to prevent cuts and trips.
- Isolate electrical, hydraulic, pneumatic, and other relevant energy sources before maintenance or jam clearing.

Care & storage

- Remove bamboo residue and veneer offcuts after safe shutdown.
- Inspect the slicing blade, clamps, guides, and carriage for wear or damage.
- Maintain blade sharpness and alignment using the approved procedure.
- Check lubrication points, drive components, and hydraulic or pneumatic lines if fitted.
- Arrange qualified servicing for uneven slicing, blade vibration, fluid leaks, or faulty safety controls.

Common mistakes to avoid

- Using incorrect thickness settings or an inadequately secured bamboo block.
- Handling veneer close to the blade while the machine is running.
- Operating with a dull or misaligned slicing blade.
- Clearing jams without isolating energy and confirming all motion has stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bench Drill Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Bench Drill Machine (With belt safety guard & on/off switch facility)

What it is used for

A bench drill machine uses a rotating drill bit to make controlled holes in suitable workpieces secured on a worktable. Correct bit selection, secure clamping, guarded belt operation, and controlled feed help improve drilling accuracy and reduce entanglement, impact, and cutting injuries.

Main parts

- Base supports the machine and worktable assembly.
- Column supports the drill head and table.
- Drill head houses the spindle and drive mechanism.
- Electric motor powers the spindle through the drive system.
- Belt and pulleys transmit drive and provide speed selection.
- Belt safety guard covers moving belts and pulleys.
- Chuck holds the drill bit.
- Feed handle lowers and raises the spindle.
- Worktable supports the workpiece and may be adjustable.
- On/off switch controls electrical operation.

Before you start

1. Only trained personnel should operate the bench drill machine.
2. Inspect the machine, belt guard, chuck, drill bit, switch, power cable, and table for damage.
3. Confirm the drill bit is sharp, correctly fitted, and suitable for the material and hole size.
4. Ensure the belt guard is closed and secured before switching on.
5. Select the correct spindle speed according to the material, bit diameter, and manufacturer's guidance; change belt positions only with power isolated.
6. Clamp the workpiece securely to the table or suitable machine vice; never hold it by hand while drilling.
7. Remove the chuck key before starting and wear eye protection and other task-appropriate PPE.

How to use

1. Secure the machine on a stable bench and ensure the work area is clear and well lit.
2. Isolate electrical power before fitting the drill bit or adjusting the belt and pulleys.
3. Insert the correct drill bit into the chuck and tighten it securely using the approved method.
4. Remove the chuck key and check that the belt guard is properly closed and secured.
5. Adjust the worktable height and position, then clamp the workpiece securely.

6. Set the appropriate spindle speed and verify that all adjustments are locked.
7. Connect power and switch on the machine while standing clear of the rotating spindle.
8. Check for abnormal vibration, noise, or runout before contacting the workpiece.
9. Lower the drill bit gradually using the feed handle and apply steady pressure appropriate to the material.
10. Withdraw the bit periodically when needed to clear chips, using a safe procedure and keeping hands away from the rotating bit.
11. Switch off the machine and wait for the spindle to stop completely before measuring or handling the workpiece.
12. Remove chips with a brush or suitable tool after the machine has stopped; never clear them by hand.
13. Isolate power before adjustments, bit changes, cleaning, or maintenance.

⚠ Safety

- Never operate the machine with the belt safety guard open, removed, or damaged.
- Tie back long hair and secure loose clothing, jewellery, and other items that could become entangled.
- Do not wear gloves near rotating drill bits or spindles.
- Never hold a workpiece by hand or attempt to stop a rotating chuck by hand.
- Keep hands away from the rotating bit and remove the chuck key before starting.
- Disconnect and isolate power before changing belts, adjusting pulleys, or servicing the machine.

Care & storage

- Clean chips and dust from the table and machine after isolating power.
- Inspect the drill bit, chuck, belt, pulleys, guard, and switch regularly.
- Lubricate and maintain moving parts only as specified by the manufacturer.
- Keep the table and machine surfaces clean, dry, and free from loose tools.
- Report unusual noise, vibration, belt damage, or switch faults and arrange authorised maintenance.

Common mistakes to avoid

- Starting the machine with the chuck key still inserted.
- Drilling an unsecured workpiece or using an incorrect spindle speed.
- Opening the belt guard or changing belt positions while power is connected.
- Wearing loose clothing or gloves near rotating parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bench Grinder

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Bench Grinder

What it is used for

A bench grinder uses rotating abrasive wheels to grind, sharpen, deburr, or shape suitable metal workpieces. Correct wheel selection, secure guards and tool rests, controlled contact, and proper eye protection help reduce wheel breakage, flying-particle, and entanglement hazards.

Main parts

- Electric motor drives the grinding spindle.
- Abrasive grinding wheels remove material from the workpiece.
- Wheel guards contain fragments and limit contact with the wheel.
- Adjustable tool rests support the workpiece during grinding.
- Spark deflectors direct sparks and debris away from the operator.
- Eye shields provide an additional barrier where fitted.
- On/off switch controls the motor.
- Mounting base secures the grinder to the bench.
- Wheel flanges and retaining nuts secure the wheels to the spindle.

Before you start

1. Only trained personnel should operate the bench grinder.
2. Inspect the grinder, power cable, switch, wheel guards, tool rests, and eye shields for damage.
3. Check each wheel for cracks, chips, moisture damage, or other defects; use the manufacturer's approved inspection method.
4. Confirm that the wheel type, diameter, bore, and maximum speed are compatible with the grinder.
5. Ensure wheel guards are secure and tool rests are adjusted close to the wheel according to the manual.
6. Check that spark deflectors and eye shields are correctly positioned and the bench grinder is securely mounted.
7. Wear safety glasses and a face shield where required; secure long hair and loose clothing, and remove jewellery.

How to use

1. Confirm the workpiece material is suitable for the installed grinding wheel.
2. Stand to one side of the wheel during startup and switch on the grinder.
3. Allow the wheel to reach operating speed and observe for abnormal vibration, noise, or wobble.
4. Hold the workpiece firmly with both hands where appropriate and rest it securely on the tool rest.

5. Bring the workpiece into gentle contact with the wheel using controlled pressure.
6. Move the workpiece across the appropriate wheel surface to avoid local overheating and uneven wear.
7. Keep fingers clear of the wheel and do not force the workpiece into the abrasive surface.
8. Cool the workpiece as needed using an appropriate method and avoid touching hot metal with bare hands.
9. Switch off the grinder and wait until the wheels stop completely before leaving the machine or making adjustments.
10. Remove debris with an approved brush or tool after the wheel has stopped.
11. Isolate electrical power before changing wheels, adjusting guards, or performing maintenance.

⚠ Safety

- Never use a cracked, damaged, incorrectly rated, or overspeed wheel.
- Do not grind on the side of a wheel unless it is specifically designed for side grinding.
- Keep guards and tool rests correctly installed and adjusted; never operate with missing safety components.
- Do not wear gloves, loose clothing, or jewellery that could become caught in rotating parts.
- Keep flammable materials away from sparks and direct sparks away from people and combustible substances.
- Never stop a wheel by pressing an object against it or by hand.

Care & storage

- Isolate power before cleaning dust and debris from the grinder and surrounding bench.
- Inspect wheels, guards, flanges, tool rests, and mounting bolts regularly.
- Replace worn or damaged wheels only with compatible wheels installed according to the manual.
- Keep ventilation openings clear and maintain the grinder according to the manufacturer's schedule.
- Remove the grinder from service if unusual vibration, wheel damage, or electrical faults occur.

Common mistakes to avoid

- Using a wheel with the wrong speed rating or intended application.
- Grinding with a loose tool rest or missing wheel guard.
- Applying excessive pressure and overheating the workpiece.
- Leaving the machine before the wheel has completely stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

CNC Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: CNC Machine

What it is used for

A CNC machine uses computer-controlled motion and tooling to machine parts to specified dimensions through operations such as milling, turning, drilling, or cutting. Follow the approved program, tooling setup, workholding method, and machine limits to achieve accurate parts while preventing collision, tool breakage, and serious injury.

Main parts

- Machine frame and work table or chuck
- Spindle and tool holder or turret
- Cutting tools and tool magazine, if fitted
- Axis drives, guides, and feed system
- CNC controller and operator panel
- Coolant or lubrication system, enclosure, and safety interlocks

Before you start

1. Confirm the machine type, material, drawing, tolerances, and approved machining program.
2. Inspect the enclosure, doors, interlocks, emergency stop, workholding, tools, and coolant system for damage.
3. Verify that the correct workpiece, fixtures, tool numbers, offsets, and program revision are selected.
4. Check that the workpiece and tools are securely clamped and that tool clearance has been verified.
5. Confirm coolant, lubrication, air supply, and chip-removal systems are ready where required.
6. Wear safety glasses and suitable footwear; secure long hair and remove loose clothing and jewellery.

How to use

1. Switch on the machine using the manufacturer's startup sequence and check the controller for active alarms.
2. Reference or home the axes using the approved procedure if required by the machine.
3. Load or select the authorised program and verify the program number, work coordinate system, and tool offsets.
4. Secure the workpiece and verify that clamps, chuck jaws, and fixtures cannot interfere with machine travel.

5. Check tool condition, tool length or radius offsets, spindle limits, and the intended cutting parameters.
6. Use simulation, graphics, or approved dry-run mode where available to verify the tool path and clearance.
7. Close the enclosure and confirm that all guards and interlocks are engaged before starting automatic motion.
8. Start the cycle and monitor the initial movements from the designated operator position.
9. Observe spindle load, coolant flow, cutting sound, chip formation, and any alarm indications.
10. Stop the cycle using the approved control if vibration, collision risk, tool breakage, or abnormal operation occurs.
11. Wait for all motion to stop before opening the enclosure or measuring the workpiece; isolate energy before reaching into hazardous areas.
12. Inspect the finished part against the drawing and record dimensions, offsets, tool changes, and quality results.
13. At completion, stop the machine, remove chips using approved tools, and leave the machine in the specified safe state.

⚠ Safety

- Only trained and authorised operators should set up, program, or operate the CNC machine.
- Never reach into the work area while the spindle or axes are moving; do not bypass door interlocks.
- Keep hands, clothing, hair, and measuring tools away from rotating tools and chucks.
- Use the correct rated tooling and workholding; never exceed machine or tool limits.
- Do not remove chips by hand or use compressed air in a way that can propel chips toward people.
- Apply lockout and tagout procedures before maintenance, tool changes requiring hazardous access, or clearing trapped material.

Care & storage

- Clean chips and residue from the work table, enclosure, and chip tray after safe shutdown.
- Inspect tools, holders, chuck or fixtures, guards, and coolant filters regularly.
- Check lubrication levels, coolant concentration, and fluid condition according to the maintenance schedule.
- Back up approved programs and record changes to tools, offsets, and machine settings.
- Arrange qualified service for repeated alarms, axis errors, abnormal vibration, or faulty safety interlocks.

Common mistakes to avoid

- Running an unverified program or incorrect work coordinate or tool offset.
- Starting automatic motion with unsecured workholding or incorrect tooling.
- Opening the enclosure or measuring a part before all motion has stopped.
- Ignoring alarms, tool wear, coolant failure, or dimensional deviations.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Circular Saw

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: ELECTRIC CIRCULAR SAW

What it is used for

An electric circular saw uses a rotating toothed blade to cut suitable materials along a straight or guided line. Correct blade selection, workpiece support, guard function, and controlled cutting technique help produce accurate cuts while reducing kickback, laceration, and flying-debris hazards.

Main parts

- Electric motor drives the circular blade.
- Circular blade cuts the workpiece.
- Fixed and retractable blade guards cover the blade where designed.
- Base plate or shoe supports the saw on the material.
- Depth adjustment sets blade exposure below the base plate.
- Bevel adjustment sets the cutting angle where fitted.
- Handle and trigger provide grip and motor control.
- Power cable or battery supplies electrical energy.
- Spindle lock supports blade changes where provided.
- Guide or fence helps maintain a straight cutting line where fitted.

Before you start

1. Only trained personnel should operate the electric circular saw.
2. Inspect the saw, blade, guards, base plate, trigger, cable or battery, and adjustment locks for damage.
3. Select a sharp, undamaged blade suitable for the material, cutting task, and saw speed rating.
4. Confirm that the blade is installed in the correct direction and that the arbor and flange are compatible.
5. Check that the lower guard moves freely and returns to its protective position without sticking.
6. Set cutting depth and bevel angle with the saw disconnected from power or the battery removed.
7. Secure the workpiece on stable supports, keeping the intended cut path clear of clamps and the supporting surface.
8. Wear safety glasses or a face shield, hearing protection, and suitable respiratory protection where dust is generated; secure loose clothing and long hair.

How to use

1. Mark the cutting line and confirm that the workpiece is supported on both sides as appropriate without pinching the blade.
2. Check the planned cut for hidden nails, screws, wiring, or other obstructions.

3. Adjust the cutting depth so the blade extends only as far below the workpiece as recommended by the manufacturer.
4. Set the bevel angle and guide or fence if the cut requires them, then lock all adjustments.
5. Hold the saw firmly with both hands using the designated handles and position your body outside the likely kickback path.
6. Keep the blade clear of the workpiece and start the saw before bringing it into the cut.
7. Allow the blade to reach operating speed, then advance smoothly along the marked line without forcing the saw.
8. Keep the base plate flat against the workpiece and maintain control throughout the cut.
9. Release the trigger at the end of the cut and hold the saw still until the blade stops completely.
10. Wait for the blade to stop before lifting the saw, setting it down, or checking the cut.
11. Disconnect power or remove the battery before changing the blade, adjusting the saw, or clearing debris.
12. Inspect the cut and the saw, then clean and store the tool safely.

⚠ Safety

- Never disable or tie back the blade guard; do not operate if it fails to return properly.
- Keep hands away from the blade path and never reach beneath the workpiece while cutting.
- Do not hold small workpieces by hand or cut material that is unstable or inadequately supported.
- Avoid twisting the saw or forcing the blade, as this can cause binding and kickback.
- Do not use a damaged, cracked, incorrect, or blunt blade.
- Disconnect power or remove the battery before blade changes, adjustments, or maintenance, and wait for the blade to stop.

Care & storage

- Remove dust and debris from the saw after disconnecting power or removing the battery.
- Inspect the blade, guard mechanism, base plate, cable or battery contacts, and adjustment locks regularly.
- Replace damaged or blunt blades with compatible blades according to the manual.
- Keep ventilation openings clear and store the saw in a dry, secure location.
- Arrange authorised service if the guard sticks, the trigger malfunctions, or abnormal vibration develops.

Common mistakes to avoid

- Starting a cut with the blade touching the workpiece.
- Using a blade with the wrong type, direction, or speed rating.
- Cutting unsupported material or allowing the blade to bind in the kerf.
- Lifting or setting down the saw before the blade has stopped completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cutting Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Cutting machine

What it is used for

A cutting machine cuts material such as metal, wood, plastic, fabric, or other workpieces using a blade, disc, or other cutting tool. Select the correct machine and cutting accessory for the material, secure the workpiece, and follow the approved operating procedure to obtain a clean cut safely.

Main parts

- Machine frame and base
- Cutting blade, disc, or tool
- Motor and drive assembly
- Work table, vice, clamp, or material support
- Operating switch and speed or feed controls
- Protective guard, handle, and emergency stop where fitted

Before you start

1. Identify the material, required cut dimensions, and correct machine type for the job.
2. Inspect the blade or disc for cracks, wear, damage, and correct speed rating; confirm compatibility with the machine.
3. Check the guard, handle, switch, power cable, work support, and emergency stop where fitted.
4. Secure the workpiece using the correct vice, clamp, or support and confirm that the cut-off section can move without binding.
5. Remove flammable materials from the cutting area and ensure adequate ventilation and suitable spark control where required.
6. Wear task-appropriate eye and face protection, hearing protection, suitable footwear, and other required PPE; secure hair and loose clothing.

How to use

1. Switch off and isolate power before installing or changing the cutting tool or making adjustments.
2. Fit the correct blade or disc according to the manufacturer's instructions and confirm that all retaining parts are secure.
3. Adjust the guard and cutting depth or angle as required for the job without removing protective devices.
4. Position and clamp the workpiece so it cannot shift, rotate, or pinch the cutting tool during the cut.
5. Check that the cutting path is clear of clamps, supports, cables, and other obstructions.

6. Stand in a stable position with a firm grip and keep hands outside the cutting path.
7. Start the machine using the approved control and allow the tool to reach its specified operating speed before engaging the material, where applicable.
8. Feed the cutting tool into the material steadily without forcing it or twisting the blade or disc.
9. Maintain control of the machine and observe for abnormal vibration, overheating, excessive sparks, or unusual noise.
10. Stop the machine immediately if the tool binds, the workpiece shifts, the guard fails, or abnormal operation occurs.
11. Wait until the blade or disc stops completely before setting the machine down, removing offcuts, or making adjustments.
12. Switch off and isolate power before clearing jammed material or replacing a damaged cutting tool.
13. After the cut, inspect the workpiece and remove sharp burrs using an appropriate safe method.
14. Clean debris from the work area after the machine has stopped and store the tool safely.

⚠ Safety

- Only trained operators should use the cutting machine and its accessories.
- Never operate the machine with a missing, damaged, or incorrectly positioned guard.
- Keep hands, hair, clothing, and body parts away from the cutting path and rotating components.
- Never hold small workpieces by hand when they can be secured with a clamp or vice.
- Use only cutting tools rated for the machine speed and intended material; never use a cracked or damaged blade or disc.
- Do not force the tool, stand in the direct line of a potentially failing disc, or leave the machine running unattended.
- Disconnect power before changing blades, clearing jams, or performing maintenance; follow lockout procedures where applicable.
- Control sparks and dust, use appropriate respiratory protection when required, and keep suitable fire precautions in place.

Care & storage

- Clean dust, chips, and offcuts from the machine and work area after shutdown.
- Inspect the cutting tool, guard, power cable, clamps, and work supports before each use.
- Replace worn or damaged cutting tools with the correct approved type.
- Check fasteners, alignment, and lubrication points according to the manufacturer's maintenance schedule.
- Arrange servicing if the motor overheats, vibration increases, guards malfunction, or cutting accuracy deteriorates.

Common mistakes to avoid

- Using the wrong blade or disc for the material or machine speed.
- Failing to clamp the workpiece securely before cutting.
- Forcing the tool or twisting the blade during the cut.

- Removing offcuts or adjusting the machine before the cutting tool stops completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drill Bit

Category: **Power Tools & Workshop Machines** Type: **Short SOP** Priority: **Medium**

12 labs

Covers these lab items: Auger Bit; Center Drill; Drill bit set; HSS Drill set; Masonry Drill (Concrete Drill)

What it is used for

Use a drill bit with a compatible drill to make holes of the required diameter and depth in suitable materials.

How to use

1. Choose the correct bit type and diameter for the material and hole size.
2. Inspect the bit for cracks, bending, blunt edges, or damage.
3. Secure the workpiece and mark the hole location.
4. Fit the bit firmly into the compatible drill chuck and remove the chuck key.
5. Select an appropriate speed and operating mode for the material.
6. Wear eye protection and drill steadily without forcing the tool.
7. Stop the drill before removing the bit or clearing debris.

⚠ Safety

- Disconnect power or remove the battery before changing bits.
- Clamp the workpiece; never hold small parts by hand while drilling.
- Use suitable eye protection and keep loose clothing away from rotating parts.
- Use masonry bits and appropriate dust controls for concrete or brick.

Care & storage

- Clean the bit after use and store it dry in an organized case.
- Replace worn or damaged bits before the next task.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drill Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 22 labs

Covers these lab items: Drill machine; Drill machine and torque wrench; Drill machine with drill bits of different sizes; Drill machines; Drilling Machine; Electric Drill Machine; Electrical hand drill machine; Electrical hand drill machine MII Class 1 Compliant; Power drills and drivers; Powered Hand drill

What it is used for

A drill machine makes round holes in suitable materials using a rotating drill bit. Correct bit selection, secure clamping, and controlled operation help prevent injury and damage.

Main parts

- Motor or power unit rotates the drill bit.
- Chuck holds the drill bit securely.
- Drill bit cuts the hole in the material.
- Trigger or switch controls the machine.
- Handle provides grip and control during drilling.
- Direction selector, if fitted, selects forward or reverse rotation.
- Battery or power cord supplies electrical power, depending on the model.

Before you start

1. Read the machine instructions and confirm you are trained to use the drill.
2. Wear safety goggles and suitable footwear; tie back long hair and secure loose clothing.
3. Inspect the body, chuck, bit, switch, battery, or power cord for damage.
4. Choose a drill bit suitable for the material, hole size, and machine rating.
5. Disconnect power or remove the battery before fitting or changing a bit.
6. Clamp the workpiece firmly and clear the drilling area of unnecessary objects.

How to use

1. Mark the required hole position and check the workpiece for hidden wires or pipes.
2. Select the correct bit and fit it fully into the chuck.
3. Tighten the chuck securely using its key or the keyless mechanism.
4. Remove the chuck key completely before switching on the machine.
5. Set the appropriate speed and direction according to the material and instructions.
6. Hold the drill firmly with both hands where the design allows.
7. Position the bit on the mark and keep it aligned with the surface.
8. Start the drill and apply steady, controlled pressure without forcing the bit.
9. Release the trigger before withdrawing the bit from the hole.
10. Switch off, wait for rotation to stop, and disconnect power before adjustments.

⚠ Safety

- Never wear gloves near a rotating drill bit or chuck because they can become entangled.
- Wear safety goggles and use hearing protection when required by the task.
- Keep hands away from the rotating bit and secure small workpieces with clamps.
- Do not use damaged bits, frayed cords, or a cracked machine body.
- Never leave a running drill unattended or touch the bit immediately after drilling.
- Keep liquids away from electrical parts and disconnect power before clearing jams.

Care & storage

- Clean dust and chips from the machine after switching off and disconnecting power.
- Inspect bits for wear, cracks, or bending and replace damaged bits.
- Store bits in a dry organiser and keep the drill in a dry, secure place.
- Arrange servicing for unusual noise, overheating, sparking, or damaged electrical parts.

Common mistakes to avoid

- Changing a bit while the machine remains connected to power.
- Holding the workpiece by hand instead of clamping it securely.
- Applying excessive pressure or using the wrong speed for the material.
- Leaving the chuck key in place before starting the drill.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Filament Dryer

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Filament dryer

What it is used for

A filament dryer removes moisture from moisture-sensitive 3D-printing filament, such as nylon, PETG, or other compatible polymers, before use. Correct temperature, drying time, and spool handling help improve print consistency while preventing filament damage and overheating.

Main parts

- Insulated drying chamber
- Heating element and temperature controller
- Air circulation fan or airflow system
- Temperature sensor and display
- Spool holder or filament support
- Power cable, door or lid, and safety controls

Before you start

1. Confirm the filament material, manufacturer drying recommendation, spool type, and required drying duration.
2. Check that the dryer is suitable for the filament and that its temperature range will not damage the spool or material.
3. Inspect the chamber, heater, fan, sensor, controls, door seal, and power cable for damage.
4. Ensure the chamber is clean and free of incompatible filament fragments or foreign material.
5. Verify that the temperature display and safety cut-outs function correctly.
6. Place the dryer on a stable, heat-resistant surface with adequate clearance around ventilation openings.

How to use

1. Switch off the dryer before loading or unloading a spool or adjusting the spool holder.
2. Prepare the filament spool according to the manufacturer's recommendations and confirm that it can tolerate the selected drying temperature.
3. Place the spool securely in the chamber and arrange the filament path to prevent tangling or contact with hot components.
4. Close the door or lid and ensure that ventilation openings remain unobstructed.
5. Switch on the dryer and set the temperature and duration recommended for the filament material.
6. Allow the chamber to reach the set temperature and verify the displayed reading before relying on the drying cycle.

7. Monitor the temperature and dryer status throughout the process, especially during the initial heating period.
8. Stop the dryer if the temperature rises unexpectedly, the fan fails, smoke appears, or an unusual smell develops.
9. Do not exceed the filament or spool manufacturer's temperature limits; reduce the setting if the spool begins to deform.
10. At the end of the cycle, switch off the heater and allow the spool to cool according to the material instructions before handling.
11. Remove the spool carefully and keep the dried filament in a sealed, moisture-resistant container or suitable dry storage.
12. Feed the filament into the 3D printer according to the printer and material instructions.
13. Switch off and unplug the dryer before cleaning or performing maintenance.

⚠ Safety

- Use the dryer only for compatible filament materials and within the manufacturer's temperature limits.
- Do not leave the unit operating if the manufacturer prohibits unattended operation; follow the specified supervision requirements.
- Keep the dryer away from water, flammable materials, and blocked ventilation areas.
- Do not touch the heating element or internal surfaces while hot; allow adequate cooling time.
- Never cover the ventilation openings or modify the thermostat and safety cut-outs.
- Disconnect power before cleaning, opening service panels, or inspecting electrical components.
- Stop using the dryer if there is smoke, burning smell, damaged wiring, or abnormal heating.

Care & storage

- Allow the dryer to cool and disconnect power before cleaning the chamber.
- Remove filament fragments and dust using a method approved by the manufacturer.
- Inspect the fan, temperature sensor, door seal, and power cable regularly.
- Keep air passages clear and check that the spool holder rotates freely where applicable.
- Arrange servicing if temperature control becomes inaccurate, airflow weakens, or the dryer overheats.

Common mistakes to avoid

- Using the same drying temperature for every filament type.
- Exceeding the safe temperature of the filament or its spool.
- Blocking airflow or allowing the filament to tangle during drying.
- Leaving dried filament exposed to humid air instead of storing it in a sealed container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Forking Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Forking machine

What it is used for

A forking machine is used to lift, move, position, or handle materials using forks or fork-shaped attachments. Operate it only within its rated capacity and according to the specific machine design and manufacturer instructions.

Main parts

- Forks or fork-shaped material-handling attachment
- Mast, lifting mechanism, or support frame
- Hydraulic or mechanical operating controls
- Drive wheels, base, or supporting structure
- Power unit or drive mechanism, if fitted
- Safety guards, handles, and emergency stop, if fitted

Before you start

1. Identify the machine type and read its operating instructions.
2. Inspect forks, frame, fasteners, wheels, and lifting components for damage.
3. Check hydraulic lines, electrical cables, or mechanical parts as applicable.
4. Confirm the rated capacity and suitability for the intended load.
5. Ensure the route and work area are level, clear, and well lit.
6. Wear suitable PPE and keep unauthorised people away from the work area.

How to use

1. Position the machine on a stable surface and apply its parking or stabilising system, if fitted.
2. Complete the pre-use inspection and confirm all controls operate correctly.
3. Check that the load is stable, within capacity, and suitable for the forks.
4. Approach the load slowly and align the forks with the designated lifting points.
5. Insert the forks fully beneath or into the approved load support points.
6. Secure the load using the machine's approved method, where applicable.
7. Lift the load gradually and check for instability before moving.
8. Move at a controlled speed while keeping the load as low as the design allows.
9. Avoid sudden turns, impacts, and movements that could shift the load.
10. Position the load over the designated drop-off area.
11. Lower the load gently until it is fully supported and stable.
12. Withdraw the forks carefully without dragging or tipping the load.
13. Return the machine to its designated parking area and lower the forks.

4. Switch off or isolate the machine as instructed and report any faults.

⚠ Safety

- Only trained and authorised operators should use the machine.
- Never exceed the rated load or use damaged or modified forks.
- Keep hands, feet, and other body parts clear of moving forks and mechanisms.
- Never stand or walk beneath a raised load.
- Do not carry people unless the machine is specifically designed and approved for that purpose.
- Use the required stabilisers and follow site rules for slopes and uneven surfaces.

Care & storage

- Clean dirt and debris from forks, wheels, and moving parts after use.
- Inspect lifting components, fasteners, and controls regularly.
- Check hydraulic or electrical systems according to the maintenance schedule.
- Lubricate service points only as specified by the manufacturer.
- Record defects and remove unsafe equipment from service.

Common mistakes to avoid

- Lifting a load beyond the machine's rated capacity.
- Moving with a raised or unstable load.
- Using the machine without inspecting forks and lifting components.
- Standing beneath a raised load or allowing others into the hazard area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hammer Drill

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Hammer/drills

What it is used for

A hammer drill rotates a drill bit and provides a hammering action for drilling suitable masonry and other approved materials. Correct bit selection, secure workholding, and controlled operation help produce accurate holes while reducing injury and tool damage.

Main parts

- Motor drives the rotating drill mechanism.
- Chuck holds the drill bit or approved accessory.
- Hammer mechanism produces repeated impact for suitable materials.
- Speed trigger controls the rotation speed.
- Forward and reverse selector changes rotation direction on equipped models.
- Mode selector chooses drilling, hammer drilling, or other supported modes.
- Side handle provides additional grip and control.
- Depth stop limits drilling depth on equipped models.

Before you start

1. Read the manual and confirm the drill is suitable for the material and task.
2. Inspect the body, chuck, side handle, cable or battery, and switches for damage.
3. Select a compatible, undamaged bit of the correct type and size.
4. Check for hidden electrical wires, pipes, and other services before drilling.
5. Secure the workpiece and clear people from the drilling area.
6. Wear safety goggles, hearing protection, and suitable dust protection where required.

How to use

1. Disconnect the power or remove the battery before fitting or changing the bit.
2. Insert the bit fully into the chuck and tighten it according to the manual.
3. Fit the side handle and set the depth stop if required.
4. Select the correct drilling mode and speed for the material.
5. Position the bit at the marked point and hold the drill firmly with both hands.
6. Start at a controlled speed and keep the bit aligned with the intended hole.
7. Apply steady, appropriate pressure without forcing the motor or bit.
8. Withdraw the bit periodically when needed to clear dust and reduce overheating.
9. Release the trigger and wait until the bit stops completely before setting the tool down.
10. Disconnect power or remove the battery before cleaning and inspecting the drill.

⚠ Safety

- Wear eye protection and keep hands away from the rotating bit.
- Never change bits or make adjustments while the drill is powered.
- Do not use damaged bits, loose accessories, or a cracked tool body.
- Keep the side handle fitted where provided and maintain a stable stance.
- Stop immediately if the bit jams, the tool overheats, or unusual vibration occurs.
- Use suitable dust extraction or respiratory protection when drilling materials that generate harmful dust.

Care & storage

- Clean dust from the chuck, vents, and exterior after use.
- Inspect bits for wear, cracks, or deformation and replace damaged bits.
- Check the cord, battery, chuck, side handle, and switches regularly.
- Store the drill and accessories in a dry case according to the manual.

Common mistakes to avoid

- Using a bit unsuitable for the material or drill mode.
- Holding the drill with one hand when firm two-handed control is needed.
- Forcing the drill when the bit binds or progress slows.
- Changing bits while the tool remains connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Heat Gun

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Heat Gun; Heat Gun/Torch

What it is used for

A heat gun produces a controlled stream of hot air for tasks such as paint softening, heat-shrink work, and material forming. Use the correct temperature and nozzle for the material and keep the hot airflow away from people and combustible items.

Main parts

- Heat-resistant gun body and handle
- Heating element and air outlet
- Temperature and airflow controls
- Power switch and power cord
- Interchangeable nozzle attachments, if supplied
- Stand or heat-resistant resting support

Before you start

1. Read the instructions and confirm that the heat gun suits the intended task.
2. Inspect the cord, plug, casing, air inlet, and nozzle for damage.
3. Clear the work area of flammable materials, liquids, and unnecessary clutter.
4. Select a nozzle and temperature suitable for the material.
5. Wear safety glasses and heat-resistant gloves where appropriate.
6. Ensure adequate ventilation, especially when heating coatings, plastics, or adhesives.

How to use

1. Place the material on a stable, heat-resistant work surface.
2. Fit the correct nozzle while the gun is switched off and cool.
3. Connect the plug to a suitable electrical outlet.
4. Switch on the heat gun and select a low suitable temperature setting.
5. Allow the tool to warm up according to the manufacturer's instructions.
6. Keep the nozzle at the recommended distance from the workpiece.
7. Move the hot airflow continuously unless the task instructions require otherwise.
8. Monitor the material for softening, shrinking, or other expected changes.
9. Reduce heat or stop if the material discolours, smokes, bubbles, or burns.
10. Switch off the tool after completing the task.
11. Place the tool on its designated heat-resistant stand while it cools.
12. Unplug the tool after use and allow it to cool completely before storage.
13. Inspect the work area for hot surfaces or smouldering material.

⚠ Safety

- Never direct hot air at people, animals, skin, or clothing.
- Do not use near flammable vapours, fuel, or explosive atmospheres.
- Keep the air inlet and outlet clear during operation.
- Avoid touching the nozzle or recently heated material; severe burns can occur.
- Do not leave the tool operating unattended.
- Disconnect power before changing accessories or performing maintenance.

Care & storage

- Allow the tool to cool completely before cleaning.
- Keep the air inlet and outlet free from dust and debris.
- Inspect the power cord, casing, and nozzle attachments regularly.
- Store the heat gun in a dry place away from combustible materials.
- Remove damaged tools from service and arrange authorised repair.

Common mistakes to avoid

- Holding the hot airflow too close to the material.
- Using excessive temperature and damaging the workpiece.
- Placing the hot tool on an unsuitable surface.
- Leaving the tool plugged in or unattended while hot.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hose Pump

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: S-hose pump

What it is used for

To operate an S-hose pump for transferring compatible liquids using the pump's moving hose and compression mechanism. To maintain steady flow while preventing hose damage, leakage, and unsafe exposure to the pumped fluid.

Main parts

- Pump casing and drive motor
- Rotor or hose-compression assembly
- Pump hose or tube
- Inlet and outlet connections
- Speed controller and power switch
- Mounting frame and protective cover

Before you start

1. Read the pump manual and confirm that the hose material is compatible with the liquid.
2. Inspect the hose, fittings, casing, drive, and power cable for wear or damage.
3. Check that the pump is mounted securely and the protective cover is fitted.
4. Verify that the inlet and outlet lines are correctly connected and not kinked or blocked.
5. Confirm that the liquid temperature, viscosity, and chemical properties are within the pump's rated limits.
6. Ensure the discharge line is open and the receiving container has enough capacity.

How to use

1. Switch off and isolate the pump before connecting or adjusting hoses.
2. Secure the inlet and outlet hoses and check that all connections are leak-tight.
3. Place the inlet line in the source liquid and the outlet line in the receiving vessel as required.
4. Confirm that the pump is primed if the model or process requires priming.
5. Set the speed controller to a low starting setting.
6. Switch on the pump and check the direction of flow and initial delivery.
7. Increase the speed gradually to the required flow rate within the rated operating range.
8. Monitor the hose, connections, motor, and discharge for leakage, abnormal vibration, or overheating.
9. Adjust the flow only using the approved controller; do not squeeze or obstruct the hose by hand.
10. Stop the pump if the hose collapses, leaks, slips, or the flow becomes irregular.
11. Switch off and isolate the power before removing the hose or clearing a blockage.

2. Drain and flush the pump with a compatible cleaning fluid if required by the process.
3. Clean the external surfaces and inspect the hose before returning the pump to service.

⚠ Safety

- Operate only within the manufacturer’s pressure, speed, temperature, and chemical-compatibility limits.
- Never operate with a missing cover or reach into the moving rotor or hose-compression area.
- Do not run the pump dry unless the manufacturer explicitly permits it.
- Depressurize and isolate the system before disconnecting any hose or fitting.
- Use suitable gloves, eye protection, and other PPE for the pumped liquid.
- Keep electrical connections dry and use appropriate protection when pumping hazardous or conductive liquids.

Care & storage

- Inspect the hose regularly for flattening, cracks, swelling, hardening, or wear.
- Replace the hose at the recommended interval or when damage appears.
- Clean and flush fluid-contact parts using compatible materials and approved procedures.
- Check mounting bolts, fittings, and drive components for looseness or wear.
- Keep the pump dry, clean, and protected during storage.

Common mistakes to avoid

- Using a hose material incompatible with the pumped liquid.
- Running the pump dry or against a blocked discharge line.
- Operating above the rated speed or pressure.
- Opening the pump while the drive is energized or the line is pressurized.
- Ignoring hose wear, leaks, or abnormal vibration.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Jig Saw

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Jig Saw

What it is used for

To cut wood, plastic, or other approved sheet materials using a jigsaw and a suitable blade. To follow the marked cutting line while controlling the tool and limiting the risk of kickback, blade breakage, and injury.

Main parts

- Electric motor and housing
- Reciprocating blade and blade clamp
- Base plate or shoe
- Trigger switch and lock, if fitted
- Cutting-angle adjustment
- Power cord or battery and blade guard

Before you start

1. Read the tool manual and confirm that the blade is suitable for the material and thickness.
2. Inspect the blade, clamp, base plate, cord or battery, and switch for damage.
3. Disconnect power or remove the battery before changing the blade or adjusting the tool.
4. Secure the workpiece firmly using clamps and support the material near the cutting line.
5. Mark the cut line and check for hidden nails, screws, cables, or pipes.
6. Wear safety glasses and hearing protection; use suitable respiratory protection when dust requires it.

How to use

1. Fit the correct blade securely and verify that the blade clamp is locked.
2. Adjust the base plate and cutting angle as required, then tighten all adjustment locks.
3. Set the cutting speed and orbital action, if available, according to the material and blade.
4. Place the base plate flat on the workpiece with the blade clear of the material.
5. Hold the tool firmly with both hands where the design permits and keep the cord clear of the cutting path.
6. Start the tool and allow the blade to reach operating speed before beginning the cut.
7. Guide the jigsaw along the marked line without forcing or twisting the blade.
8. Support offcuts so they do not pinch the blade or fall unexpectedly.
9. Release the trigger and wait for the blade to stop before setting the tool down.
10. Disconnect power before clearing a jam, changing settings that expose the blade, or replacing the blade.

1. Inspect the cut edge and remove dust and debris using a safe method.
2. Store the jigsaw with the blade protected and the power source disconnected.

⚠ Safety

- Keep hands away from the blade path and never reach beneath the workpiece while cutting.
- Do not use damaged blades or operate with a loose blade clamp.
- Never cut a workpiece that is not securely supported.
- Wait for the blade to stop completely before placing the tool down.
- Keep the cord away from the blade and check for concealed electrical wiring before cutting.
- Do not use the tool in wet conditions unless specifically rated for them.
- Secure loose clothing and hair, and keep bystanders away from flying chips and dust.

Care & storage

- Clean dust from the vents and base plate after use.
- Inspect blades for cracks, dullness, or damaged teeth and replace as needed.
- Check the base plate, blade clamp, switch, and power cable regularly.
- Use only manufacturer-approved replacement parts and accessories.
- Store in a dry place and remove the battery or disconnect the cord before storage.

Common mistakes to avoid

- Using the wrong blade for the material or cut type.
- Forcing the tool and bending the blade away from the cut line.
- Failing to clamp the workpiece securely.
- Changing the blade while the tool is connected to power.
- Setting the tool down before the blade has stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Milling Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Milling machine

What it is used for

A milling machine removes material from a workpiece using a rotating cutting tool. It produces slots, flat surfaces, grooves and other shapes in metal or suitable materials.

Main parts

- Identify the machine base, column and worktable supporting the workpiece.
- Locate the spindle that holds and rotates the milling cutter.
- Identify the milling cutter, tool holder and spindle locking arrangement.
- Locate the table movement controls for longitudinal, cross and vertical travel.
- Identify the workholding vice, clamps and T-slots on the table.
- Locate the speed selector, feed controls and emergency stop.
- Identify protective guards, coolant equipment and lubrication points, where fitted.

Before you start

1. Obtain teacher supervision and review the machine operating instructions.
2. Wear safety goggles and suitable workshop clothing; secure long hair.
3. Remove jewellery and avoid loose clothing near rotating components.
4. Inspect the cutter, tool holder, guards and electrical connections for damage.
5. Secure the workpiece firmly in a vice or with suitable clamps.
6. Check cutter clearance, spindle speed, feed setting and workpiece alignment.
7. Remove spanners and loose objects from the table before starting.

How to use

1. Select a cutter suitable for the material and required operation.
2. Switch off the machine before installing or changing the cutter.
3. Secure the cutter and workpiece using the correct holding arrangements.
4. Set the appropriate spindle speed, feed rate and cutting depth.
5. Position the cutter and workpiece with the machine stopped where required.
6. Close the guards and ensure tools are clear of moving parts.
7. Start the spindle and allow it to reach the selected speed.
8. Engage the feed gradually while monitoring cutting conditions.
9. Use approved coolant where required and keep hands away from the cutter.
10. Stop the machine before measuring, adjusting or removing the workpiece.

⚠ Safety

- Never touch the rotating cutter or remove chips by hand.
- Never wear gloves near rotating cutters or exposed spindle components.
- Keep guards in place and never bypass safety interlocks.
- Use a brush or chip hook only after the machine stops.
- Do not measure or adjust the workpiece while the cutter rotates.
- Switch off and isolate power before changing cutters or clearing jams.

Care & storage

- Remove chips with a brush after switching off and isolating power.
- Clean the table, vice and exposed machine surfaces after use.
- Lubricate designated points and maintain coolant according to instructions.
- Inspect cutters, belts and guards regularly and report any damage.

Common mistakes to avoid

- Clamping the workpiece loosely, allowing it to shift during cutting.
- Selecting an unsuitable speed or feed for the cutter and material.
- Measuring the workpiece while the spindle or cutter is moving.
- Removing metal chips with bare hands or while the machine runs.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Nail Gun

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Nail Gun

What it is used for

A nail gun drives nails into suitable materials using a specified power source. It speeds up fastening work but can cause serious puncture injuries if handled incorrectly.

Main parts

- Identify the nosepiece or contact tip where nails are discharged.
- Locate the nail magazine that holds the fasteners.
- Identify the trigger and contact safety mechanism.
- Locate the power source, such as battery, electrical cable or compressed-air connection.
- Identify the depth adjustment and fastener loading controls, where provided.
- Locate the air-pressure control or battery indicator, depending on model.
- Identify the safety lock and manufacturer's operating instructions.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Wear safety goggles and suitable hearing protection where required.
3. Check the tool body, trigger, nosepiece and safety mechanism for damage.
4. Confirm the nail type and length are suitable for the material.
5. Disconnect the battery, power supply or air hose before loading nails or adjustments.
6. Inspect the workpiece and ensure the fastening area contains no hidden cables or pipes.
7. Secure the workpiece and keep other people outside the working area.

How to use

1. Select the correct nails and load them into the magazine as instructed.
2. Reconnect the specified power source only after loading and inspection are complete.
3. Adjust the depth setting and air pressure if the model permits.
4. Hold the nail gun firmly against the designated fastening point.
5. Keep fingers away from the nosepiece and intended nail exit path.
6. Align the tool perpendicular to the surface unless the instructions specify otherwise.
7. Engage the contact safety mechanism and trigger according to the model design.
8. Check the fastener placement and adjust settings only after disconnecting power.
9. Release the trigger and disconnect the power source before clearing jams.
10. Store the nail gun with the safety engaged and power source disconnected.

⚠ Safety

- Never point the nail gun at yourself or another person.
- Keep hands and body parts away from the nail exit area.
- Do not carry the tool with the trigger pressed or safety defeated.
- Disconnect power before loading, unloading, clearing jams or servicing the tool.
- Never use the nail gun near hidden electrical wires or pipes without checking.
- Use only compatible nails and the specified power or air pressure.

Care & storage

- Disconnect the power source and clean dust from the tool exterior.
- Inspect the magazine, nosepiece and safety mechanism after use.
- Lubricate pneumatic models only as specified by the manufacturer.
- Store the tool dry and secure, away from children and unauthorised users.

Common mistakes to avoid

- Pressing the trigger before the nosepiece is correctly positioned.
- Using incompatible nails or excessive air pressure.
- Clearing a jam while the tool remains connected to power.
- Disabling the contact safety mechanism to speed up work.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pneumatic Gun

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Pneumatic gun

What it is used for

A pneumatic gun uses compressed air to perform tasks such as blowing debris, applying sealant, fastening, or other operations depending on the gun type. Use the correct accessory and pressure rating, and follow the tool-specific operating instructions.

Main parts

- Gun body and grip
- Trigger or operating lever
- Air inlet and quick-connect fitting
- Nozzle, barrel, or task-specific attachment
- Pressure regulator or adjustment control, if fitted
- Safety lock and protective accessories, if fitted

Before you start

1. Identify the pneumatic gun type and confirm its intended application.
2. Inspect the body, trigger, nozzle, air hose, and connectors for damage.
3. Confirm the compressor and hose pressure ratings match the tool requirements.
4. Check that couplings are securely fitted and the air supply is regulated correctly.
5. Fit the correct nozzle or attachment and inspect it for blockage or wear.
6. Wear appropriate PPE, including eye protection and hearing protection where required.

How to use

1. Disconnect the air supply before fitting accessories or inspecting the gun.
2. Secure the approved nozzle or attachment according to the manufacturer's instructions.
3. Connect the gun to the correct regulated compressed-air line.
4. Set the air pressure within the tool's specified operating range.
5. Check the connection for leakage using an approved method.
6. Hold the gun securely and keep the nozzle directed toward the work area.
7. Operate the trigger using the technique specified for the tool's purpose.
8. Maintain a safe distance from people and sensitive equipment.
9. Monitor the tool, hose, and workpiece for abnormal vibration, leakage, or malfunction.
10. Release the trigger and disconnect the air supply before clearing a blockage or changing settings that require isolation.
11. After use, shut off the air supply and release residual pressure according to the procedure.
12. Disconnect the hose safely and store the tool and accessories in a dry place.

⚠ Safety

- Only trained users should operate the pneumatic gun for its intended task.
- Never direct compressed air at people, skin, or clothing.
- Do not exceed the tool, hose, or accessory pressure rating.
- Never use damaged hoses, loose couplings, or a faulty trigger.
- Depressurise and disconnect the air supply before maintenance or clearing blockages.
- Use task-specific PPE and follow additional safety requirements for fastening or impact-type guns.

Care & storage

- Clean the nozzle and external surfaces after use.
- Inspect the hose, couplings, trigger, and attachments regularly.
- Lubricate only if required by the manufacturer and use the specified lubricant.
- Store the gun disconnected from compressed air in a dry location.
- Remove faulty tools from service and arrange repair by authorised personnel.

Common mistakes to avoid

- Setting air pressure above the tool's rated limit.
- Directing compressed air at skin or another person.
- Changing attachments while the air supply remains connected and pressurised.
- Ignoring air leaks, damaged hoses, or a sticking trigger.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pneumatic Press

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Pneumatic press

What it is used for

A pneumatic press uses compressed air to apply force for pressing, fitting or forming suitable components. Safe operation requires correct pressure settings, secure workholding and protection from moving press parts.

Main parts

- Identify the press frame, bed and pressing head.
- Locate the pneumatic cylinder that converts compressed air into movement.
- Identify the ram or pressing tool that contacts the workpiece.
- Locate the compressed-air inlet, hose and regulator.
- Identify the pressure gauge and air control valve.
- Locate the operating pedal or two-hand control, depending on design.
- Identify guards, emergency stop and air-isolation valve.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Inspect the frame, ram, air hose, fittings and guards for damage.
3. Check that the compressed-air supply matches the manufacturer's specifications.
4. Confirm the pressure regulator and gauge operate correctly.
5. Select suitable pressing tooling and secure the workpiece on the bed.
6. Keep hands clear of the pressing zone and adjust the guard as required.
7. Remove loose objects and ensure the emergency stop is accessible.

How to use

1. Isolate the air supply before installing tooling or adjusting the press.
2. Fit the correct pressing tool and secure it according to instructions.
3. Position the workpiece centrally and support it firmly on the press bed.
4. Set the working pressure within the approved operating range.
5. Reconnect the air supply and check for leaks or abnormal sounds.
6. Close the guards and select the designated operating control.
7. Activate the press using the approved pedal or control arrangement.
8. Observe the pressing cycle from a safe position without reaching into the work area.
9. Release the control and wait until the ram returns to its safe position.
10. Isolate the air supply before removing the finished workpiece or changing settings.

⚠ Safety

- Never place fingers or hands between the ram and press bed.
- Never exceed the rated pressure or load capacity of the press.
- Keep guards and safety controls functional; never bypass them.
- Isolate compressed air and release trapped pressure before servicing.
- Never use damaged hoses, loose fittings or an unstable press frame.
- Stop operation immediately if the press moves unexpectedly or leaks air.

Care & storage

- Isolate air and release stored pressure before cleaning or maintenance.
- Clean the press bed and remove debris after each session.
- Inspect hoses, fittings, ram and guards regularly for wear or leaks.
- Follow manufacturer instructions for lubrication and scheduled maintenance.

Common mistakes to avoid

- Positioning the workpiece off-centre and causing unstable pressing.
- Increasing air pressure beyond the specified operating range.
- Reaching into the pressing zone before the ram has stopped.
- Adjusting tooling while compressed air remains connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pneumatic Tool

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Pneumatic tool

What it is used for

A pneumatic tool uses compressed air to perform tasks such as drilling, fastening, grinding, cutting, or impact work, depending on the tool type. Correct air-pressure settings, secure hose connections, suitable attachments, and safe handling help prevent injury and equipment damage.

Main parts

- Tool body houses the operating mechanism.
- Trigger or control lever starts and stops the tool.
- Air inlet connects the tool to the compressed-air supply.
- Air hose carries compressed air to the tool.
- Coupler and fittings connect the hose securely.
- Working attachment performs the specific task.
- Air regulator adjusts supply pressure where fitted.
- Exhaust port releases used air away from the operating mechanism.

Before you start

1. Only trained personnel should operate the pneumatic tool for the assigned task.
2. Read the tool manual and confirm its permitted applications and operating pressure.
3. Inspect the tool body, trigger, hose, couplers, fittings, and attachment for damage or leakage.
4. Check that the compressor, regulator, and hose are suitable for the tool's air demand and pressure rating.
5. Confirm that the attachment is compatible, correctly secured, and in good condition.
6. Wear task-appropriate eye protection, hearing protection, safety footwear, and other required PPE.
7. Secure the workpiece and clear the working area of people and loose objects.

How to use

1. Disconnect the air supply before fitting or changing the attachment.
2. Connect the hose using compatible couplers and verify that the connection is secure.
3. Set the regulator to the pressure specified by the tool manufacturer.
4. Check that the trigger is released before opening the air supply.
5. Open the air supply and check for leaks or unintended tool movement.
6. Hold the tool firmly using the designated grip and position the attachment correctly on the workpiece.
7. Operate the tool with controlled pressure and follow the recommended working technique.

8. Monitor tool performance, hose movement, vibration, and unusual noise during operation.
9. Release the trigger and shut off the air supply when the task is complete or an abnormal condition occurs.
10. Disconnect the air hose and release trapped pressure according to the tool instructions before changing attachments or cleaning.
11. Inspect the tool and workpiece, then clean and store the equipment safely.

⚠ Safety

- Never point a pneumatic tool or its exhaust toward yourself or another person.
- Disconnect the air supply and release trapped pressure before servicing or changing attachments.
- Never exceed the tool, hose, coupler, or accessory pressure rating.
- Do not use damaged hoses, loose fittings, or attachments with cracks or excessive wear.
- Keep hands and clothing away from rotating, cutting, or impact parts.
- Use guards and required PPE; do not disable safety controls or use compressed air to clean skin or clothing.

Care & storage

- Wipe the tool clean after use and remove debris from accessible areas.
- Inspect hoses, couplers, seals, triggers, and attachments regularly.
- Lubricate the tool only if required and only with the specified lubricant.
- Store the tool disconnected from the air supply in a clean, dry location.
- Arrange authorised repair if the tool leaks, loses power, or operates irregularly.

Common mistakes to avoid

- Operating at a pressure above the manufacturer's specified limit.
- Changing attachments while the air supply remains connected.
- Using damaged hoses or insecure couplers.
- Ignoring unusual vibration, air leakage, or a malfunctioning trigger.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Power Tool

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Power tools

What it is used for

A power tool uses electrical power or a battery to perform tasks such as drilling, cutting, grinding, sanding, or fastening. Correct tool selection, secure workholding, suitable accessories, and power isolation help improve work quality and reduce injury risks.

Main parts

- Tool housing protects the motor and internal components.
- Handle and grip support controlled operation.
- Trigger or switch starts and stops the tool.
- Power cable, battery, or power inlet supplies energy.
- Chuck, collet, or spindle holds the working accessory where applicable.
- Guard or shield helps contain debris and protect the operator.
- Speed or torque control adjusts operation where fitted.
- Auxiliary handle or depth guide supports control on applicable models.

Before you start

1. Only trained users should operate power tools appropriate to their task.
2. Read the manual and confirm the tool is suitable for the material and operation.
3. Inspect the housing, handle, cable, plug, battery, switch, guard, and accessory for damage.
4. Confirm that the accessory is compatible, correctly fitted, and rated for the tool's speed.
5. Secure the workpiece with a suitable clamp or vice and clear the work area.
6. Wear task-appropriate eye protection, hearing protection, and other required PPE.
7. Check the power supply, battery condition, and any required dust extraction before starting.

How to use

1. Disconnect the power source or remove the battery before fitting or changing accessories.
2. Install the correct accessory and tighten the chuck, collet, or fastening system according to the manual.
3. Adjust the guard, speed, torque, and depth settings for the task where applicable.
4. Secure the workpiece and position the tool so the accessory will not contact unintended objects.
5. Reconnect the power source and check that the trigger is released before starting.
6. Hold the tool firmly using the designated grip and auxiliary handle where provided.
7. Start the tool and allow it to reach the appropriate operating speed before applying it to the workpiece.
8. Apply controlled pressure and let the accessory perform the work without forcing the tool.

9. Release the trigger and wait for all moving parts to stop before setting the tool down.
10. Disconnect power or remove the battery before adjustments, cleaning, or clearing a jam.
11. Inspect the workpiece and tool, then clean and store the equipment safely.

⚠ Safety

- Never operate a power tool with damaged wiring, a defective switch, or a missing required guard.
- Keep hands, clothing, hair, and loose objects away from moving accessories.
- Do not exceed the rated speed of the tool or accessory.
- Disconnect power before changing accessories or making adjustments.
- Do not use an electric tool in wet conditions unless specifically designed and approved for them.
- Keep bystanders clear and use suitable dust extraction or respiratory protection where required.

Care & storage

- Remove dust and debris from the tool after use using the approved cleaning method.
- Inspect accessories, guards, cables, battery connections, and handles regularly.
- Keep ventilation openings clear and avoid exposing the tool to excessive moisture.
- Store batteries and tools according to manufacturer instructions.
- Arrange authorised repair if the tool overheats, sparks abnormally, vibrates excessively, or fails to stop correctly.

Common mistakes to avoid

- Using an accessory that is incompatible or rated for a lower speed than required.
- Holding the workpiece by hand instead of securing it.
- Removing guards or using the tool with damaged cables.
- Changing accessories while the power source remains connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Punching Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Punching Machine

What it is used for

A punching machine creates holes or cut-outs in sheet material by driving a punch through a matching die. Correct tooling alignment, secure material positioning, and functioning guards help produce accurate results while reducing crushing, cutting, and machine damage risks.

Main parts

- Machine frame supports the punching mechanism.
- Punch tool penetrates the work material.
- Die supports the material and defines the hole or cut-out shape.
- Ram or slide drives the punch toward the die.
- Work table supports and positions the material.
- Clamping mechanism holds the material during punching where fitted.
- Control pedal, lever, or switch activates the machine.
- Guard, interlock, or emergency stop reduces operating hazards where fitted.

Before you start

1. Only trained and authorised operators should use the punching machine.
2. Read the operating manual and confirm the machine is suitable for the material thickness and type.
3. Inspect the punch, die, ram, work table, clamps, guards, and controls for damage or wear.
4. Confirm the punch and die are compatible, correctly aligned, and securely installed.
5. Check that the emergency stop and safety interlocks function correctly where fitted.
6. Secure the workpiece and remove loose tools or scrap from the work area.
7. Wear suitable eye protection and task-specific PPE; avoid gloves near exposed moving parts if they could become caught.

How to use

1. Switch off and isolate power before changing the punch, die, or tooling setup.
2. Install the correct punch and matching die according to the machine instructions.
3. Adjust the tooling clearance and material stop or guide for the specified material.
4. Confirm the guard is in place and the workpiece is positioned securely on the table.
5. Restore power and perform any required no-load check from the designated operating position.
6. Place the material in the specified punching position without putting fingers beneath the punch or ram.

7. Activate the machine using the approved switch, lever, or pedal while keeping hands outside the danger zone.
8. Check the punched hole or cut-out for position, shape, burrs, and edge quality.
9. Stop the machine if the material shifts, tooling binds, or abnormal noise occurs.
10. Isolate power and wait for all movement to stop before clearing scrap or making adjustments.
11. Remove finished material and scrap safely, then clean the work area and record any faults.

⚠ Safety

- Never place fingers or hands between the punch and die during operation.
- Never bypass guards, interlocks, or emergency stops.
- Isolate power before changing tooling, clearing jams, or reaching into the machine's danger zone.
- Do not punch material beyond the machine's rated thickness or capacity.
- Keep hands away from sharp punched edges and use approved tools to remove scrap.
- Stop operation immediately if the punch, die, clamp, or safety device becomes loose or damaged.

Care & storage

- Remove scrap and debris after the machine is isolated and fully stopped.
- Inspect the punch, die, ram, clamps, guides, and fasteners regularly.
- Lubricate designated points and maintain tooling according to the manufacturer's schedule.
- Keep guards and safety devices clean and functional.
- Arrange authorised maintenance if alignment, punching force, or control operation becomes unreliable.

Common mistakes to avoid

- Using mismatched or misaligned punch and die tooling.
- Positioning material by hand inside the punch danger zone.
- Clearing scrap while the machine remains powered or moving.
- Exceeding the machine's rated material thickness or capacity.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sanding Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Sanders/Grinders; Sanding Machine

What it is used for

A sanding machine smooths surfaces, removes coatings, or prepares materials using an abrasive belt, disc, pad, or wheel. Select the correct abrasive and operating speed for the workpiece and control dust and contact hazards throughout the task.

Main parts

- Motor and machine housing
- Sanding belt, disc, pad, or abrasive wheel
- Work support or backing plate
- Power switch and speed control, if fitted
- Dust extraction port or collection bag
- Protective guard and handle

Before you start

1. Identify the sanding machine type and read its operating instructions.
2. Inspect the abrasive, backing pad, guards, handle, cord, and plug for damage.
3. Confirm the abrasive grade and speed rating suit the workpiece and tool.
4. Secure the workpiece using a clamp or suitable fixture.
5. Connect approved dust extraction and ensure the work area is ventilated.
6. Wear eye protection, hearing protection, and suitable respiratory protection for the dust.

How to use

1. Disconnect power before installing or changing the abrasive attachment.
2. Fit the correct abrasive securely and align it according to the instructions.
3. Check that guards and dust extraction components are fitted correctly.
4. Position the machine firmly and keep the abrasive clear of the workpiece before starting.
5. Connect the tool to a suitable power supply and switch it on.
6. Allow the machine to reach its normal operating speed before contacting the workpiece.
7. Apply steady, controlled pressure and keep the machine moving as required.
8. Avoid excessive pressure that may overheat the motor or damage the surface.
9. Monitor the finish, dust extraction, vibration, and tool temperature during operation.
10. Switch off and wait for all moving parts to stop before setting the tool down.
11. Disconnect power before clearing a blockage, changing abrasive, or performing maintenance.
12. Remove dust safely and inspect the finished surface and machine.

⚠ Safety

- Only trained users should operate the sanding machine.
- Never touch the moving abrasive or remove guards during operation.
- Do not use damaged, cracked, incorrectly fitted, or over-speed-rated abrasives.
- Secure loose workpieces and keep hands away from the sanding area.
- Control dust, especially from wood, coatings, metals, or other hazardous materials.
- Disconnect power before adjustment, cleaning, or changing accessories.

Care & storage

- Remove dust from vents and the machine exterior after use.
- Inspect the abrasive, backing pad, belt tracking, and guards regularly.
- Replace worn abrasives and damaged components with compatible approved parts.
- Keep electrical connections dry and store the machine in a clean location.
- Arrange servicing if the tool overheats, vibrates excessively, or produces unusual noise.

Common mistakes to avoid

- Using an abrasive with the wrong grade or speed rating.
- Applying excessive pressure and overheating the workpiece.
- Operating without suitable dust extraction or respiratory protection.
- Changing abrasives or clearing jams while the machine is connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shredder

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Shredder

What it is used for

A shredder cuts or breaks suitable materials into smaller pieces for disposal, recycling or further processing. Safe operation depends on using the correct machine type and keeping hands away from cutting mechanisms.

Main parts

- Identify the feed hopper or material entry opening.
- Locate the cutting blades, shredding rollers or cutting shaft inside the housing.
- Identify the motor and drive transmission components.
- Locate the control panel, start switch and stop switch.
- Identify the emergency stop and overload protection, where fitted.
- Locate the discharge chute or collection container.
- Identify the protective covers, interlocks and maintenance access points.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Confirm that the shredder is designed for the material being processed.
3. Inspect blades, guards, cables, fasteners and housing for damage.
4. Remove unsuitable items such as metal pieces or stones where the machine requires it.
5. Wear suitable eye protection and close-fitting clothing; secure long hair.
6. Check that the discharge area and collection container are clear.
7. Ensure the machine is stable and emergency controls are accessible.

How to use

1. Switch off and isolate power before inspecting or preparing the feed opening.
2. Check that guards and covers are fitted correctly.
3. Select the approved operating setting for the material, where adjustable.
4. Start the shredder using the designated control and allow it to run normally.
5. Feed small, manageable quantities through the designated opening.
6. Use only the supplied feeding tool when the design requires one.
7. Monitor the discharge and stop feeding if material jams or the motor strains.
8. Stop the machine before changing material types or collection containers.
9. Switch off, isolate power and wait for all cutting parts to stop before clearing blockages.
10. Collect the shredded material and clean the machine according to instructions.

⚠ Safety

- Never put hands or fingers into the feed opening or discharge chute.
- Never push material into the blades using hands or improvised tools.
- Keep guards in place and never bypass safety interlocks.
- Disconnect power before clearing jams, cleaning or servicing the shredder.
- Do not process materials beyond the machine's specified capacity or intended use.
- Stop immediately if unusual noise, vibration, smoke or overheating occurs.

Care & storage

- Isolate power and confirm all cutting parts have stopped before cleaning.
- Remove material residue using approved tools without touching sharp blades.
- Inspect blades, fasteners, guards and electrical connections regularly.
- Follow manufacturer instructions for blade maintenance and safe storage.

Common mistakes to avoid

- Overloading the feed opening and causing a jam or motor overload.
- Using the shredder for materials it was not designed to process.
- Pushing material into the cutting mechanism with bare hands.
- Clearing blockages before disconnecting power and confirming complete stoppage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spray Gun

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Lacquering/Polishing Tools

What it is used for

A spray gun applies paint, lacquer, polish, or protective coatings evenly using compressed air or a powered spraying system. Correct setup produces a smooth finish while reducing overspray and material waste.

Main parts

- Gun body with handle and trigger
- Paint or coating cup or container
- Nozzle and fluid needle
- Air cap and adjustment controls
- Air hose or power connection, depending on model
- Cleaning brush and compatible cleaning solution

Before you start

1. Read the manufacturer instructions and identify the coating approved for the gun.
2. Inspect the gun body, trigger, cup, nozzle, and hoses for damage or leaks.
3. Confirm the coating viscosity and strain it if the manufacturer recommends filtering.
4. Check ventilation, remove ignition sources, and wear suitable eye, hand, and respiratory protection.
5. Connect the correct air supply or power source and verify secure fittings.
6. Test the spray pattern on scrap material before coating the workpiece.

How to use

1. Prepare the surface by cleaning, drying, and masking areas that must remain uncoated.
2. Mix and thin the coating according to the product instructions.
3. Fill the container without exceeding its marked capacity and close it securely.
4. Adjust air pressure, fluid flow, and spray pattern according to the gun instructions.
5. Hold the gun at a consistent distance and keep it perpendicular to the surface.
6. Move the gun smoothly across the surface with overlapping passes.
7. Release the trigger at the end of each pass when the gun design permits.
8. Apply additional coats only after the specified drying or flash-off time.
9. Switch off the supply, relieve pressure, and empty unused coating safely.
10. Clean the cup, nozzle, air cap, and fluid passages using approved materials.

⚠ Safety

- Use the coating's safety data sheet to select suitable respiratory and skin protection.
- Operate only in adequate ventilation or an approved spray booth.
- Keep flames, sparks, smoking, and hot surfaces away from flammable coatings and solvents.
- Never point the spray gun at people or test it against skin.
- Depressurize the system before loosening fittings or removing the nozzle.
- Disconnect power or air supply before servicing or clearing blockages.

Care & storage

- Clean the gun immediately after use with a compatible cleaner.
- Use the correct tools to clean the nozzle without enlarging its opening.
- Inspect seals, needles, hoses, and fittings regularly for wear.
- Store the clean, dry gun in a protected location with accessories secured.

Common mistakes to avoid

- Using excessive pressure can cause overspray and uneven coating.
- Holding the gun too close can create runs and heavy deposits.
- Using incompatible thinner or cleaner can damage seals or the coating.
- Leaving coating inside the gun can block passages and damage components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tensioning Equipment

Category: **Power Tools & Workshop Machines** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Tensioning equipment

What it is used for

To apply and maintain the specified tension in belts, cables, straps, wires, or other compatible components during assembly or maintenance.

How to handle

- Inspect the tensioning equipment, handles, gauges, jaws, straps, and locking components for wear or damage.
- Confirm that the equipment is suitable for the component, required tension, and rated load.
- Secure the workpiece and set up the equipment according to the manufacturer's instructions before applying tension.
- Increase tension gradually while monitoring the gauge or specified indicator; keep hands and body clear of components that may recoil or release suddenly.
- Lock or release tension using the designated mechanism, then inspect the assembly and equipment before removing it.

Storage

- Release stored tension completely and store clean and dry in a designated equipment area.
- Protect gauges, jaws, straps, and locking parts from corrosion, deformation, and accidental loading.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

UV Curing Station

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: UV Curing station for SLA

What it is used for

A UV curing station exposes compatible UV-sensitive resin or coating to controlled ultraviolet light to harden it. Correct exposure settings and handling help achieve complete curing while reducing UV exposure risks.

Main parts

- UV light source or LED array
- Curing chamber or enclosed exposure area
- Turntable or build-platform holder if fitted
- Timer and exposure controls
- Door, lid, or safety interlock
- Protective eyewear and resin-handling accessories

Before you start

1. Read the resin manufacturer's curing instructions and confirm the required UV wavelength.
2. Inspect the chamber, light source, door, timer, and safety interlocks for damage.
3. Ensure the station is clean and the reflective surfaces are free of residue.
4. Wear suitable gloves and UV-rated eye protection when handling uncured resin or servicing as instructed.
5. Remove uncured resin from the part using the approved washing procedure before curing.
6. Confirm the part fits inside the chamber and can receive light on the required surfaces.

How to use

1. Place the cleaned part on the designated holder or turntable.
2. Select the exposure time and other settings specified for the resin and part geometry.
3. Close the chamber door or lid fully before activating the UV source.
4. Start the curing cycle and allow the part to receive the specified UV exposure.
5. Use the turntable or reposition the part only according to the station instructions.
6. Observe the cycle from outside the chamber without looking directly at the UV source.
7. Wait for the UV source to switch off and the interlock to release before opening the chamber.
8. Remove the part using suitable gloves and check the surface and curing result.
9. Repeat exposure only when the resin instructions permit and the part requires it.
10. Switch off the station, clean any residue using approved methods, and record settings if required.

⚠ Safety

- Never look directly at an operating UV source or expose skin to its radiation.
- Do not bypass door interlocks or operate an open curing chamber unless specifically designed and shielded for it.
- Wear suitable gloves when handling uncured resin and follow its safety data sheet.
- Provide appropriate ventilation when resin products may release vapours during curing.
- Disconnect power before cleaning internal components or replacing the UV source.
- Allow hot lamps or components to cool before handling them.

Care & storage

- Keep the chamber window, reflective surfaces, and turntable clean according to the instructions.
- Inspect the UV source, wiring, seals, and interlocks regularly.
- Replace UV lamps or LED modules according to the manufacturer's service guidance.
- Store uncured resin in its recommended container away from sunlight and unauthorized access.

Common mistakes to avoid

- Using the wrong wavelength or exposure time can leave resin incompletely cured.
- Opening the chamber during exposure can expose eyes and skin to UV radiation.
- Curing a part with excess uncured resin can produce a sticky or uneven finish.
- Touching optical surfaces with unsuitable cleaners can reduce light output or damage components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Vacuum Pump

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Vaccum pump; Vacuum Pump

What it is used for

A vacuum pump removes gas or air from a sealed system to create a reduced-pressure environment for industrial, laboratory, maintenance, or process applications. Correct setup, suitable pump selection, leak checks, and controlled operation help achieve the required vacuum while reducing contamination, overheating, and equipment damage.

Main parts

- Pump housing contains the pumping mechanism.
- Inlet port connects the pump to the vacuum system.
- Exhaust port releases discharged gas or vapour.
- Electric motor or drive system powers the pump.
- Oil reservoir and sight glass are fitted on oil-lubricated models.
- Gas ballast valve helps manage condensable vapours on compatible models.
- Inlet filter or trap protects the pump from particles or liquid carryover where fitted.
- Pressure gauge or vacuum gauge measures system pressure where supplied.
- Control switch and overload protection operate and protect the drive system.

Before you start

1. Only trained personnel should operate the vacuum pump for the intended application.
2. Read the pump manual and confirm compatibility with the gas, vapour, process, required vacuum level, and connected system.
3. Inspect the housing, power cable, inlet and exhaust ports, hoses, fittings, seals, and controls for damage.
4. Check oil level and condition on oil-lubricated pumps; use only the specified oil type.
5. Confirm that filters, traps, exhaust routing, and ventilation are suitable for the process being handled.
6. Check that hoses and vessels are vacuum-rated and that glassware or chambers are protected against implosion where relevant.
7. Verify electrical supply, protective earth, and any required pressure or temperature monitoring before starting.

How to use

1. Place the pump on a stable, level surface in a ventilated location suitable for its exhaust and process hazards.
2. Connect the inlet to the vacuum system using compatible hoses, seals, and fittings.

3. Connect the exhaust to a safe, adequately ventilated discharge point or approved treatment system where required.
4. Install the specified inlet filter, liquid trap, or other protection to prevent contamination of the pump.
5. Check all connections for tightness and ensure valves are set according to the operating procedure.
6. Open required cooling or ventilation paths and verify oil level where applicable.
7. Switch on the pump according to the manufacturer's starting procedure.
8. Monitor the vacuum gauge, pump sound, vibration, temperature, and exhaust during operation.
9. Allow the system to reach the required pressure without exceeding the pump's operating limits.
10. If performance is poor, isolate the system and check for leaks, blocked filters, incorrect valve positions, or unsuitable process conditions.
11. At the end of the process, isolate the system and vent it in the sequence specified by the pump and system manufacturers.
12. Switch off the pump, allow it to cool where needed, and disconnect power before maintenance.
13. Record operating readings, oil condition, maintenance performed, and any faults.

⚠ Safety

- Do not pump flammable, explosive, corrosive, toxic, or reactive vapours unless the pump and system are specifically rated and approved for them.
- Never block the exhaust or discharge hazardous gases into an occupied workspace.
- Use vacuum-rated vessels, hoses, and fittings; implosion can cause serious injury from flying fragments.
- Do not touch hot pump surfaces or exposed moving parts during operation.
- Disconnect and isolate power before opening the pump, changing oil, clearing blockages, or servicing components.
- Use appropriate PPE and follow the relevant chemical, biological, and process-specific safety procedures.

Care & storage

- Inspect hoses, seals, filters, traps, electrical cables, and fittings regularly for wear or leakage.
- Check and change pump oil according to the manufacturer's schedule and contamination limits.
- Clean the inlet filter and liquid trap as specified and dispose of collected materials safely.
- Keep cooling vents clear and maintain the required ventilation around the pump.
- Arrange authorised service if vacuum performance falls, unusual noise develops, or the pump overheats.

Common mistakes to avoid

- Using a pump that is incompatible with the gas, vapour, or required vacuum level.
- Operating with low or contaminated oil on an oil-lubricated model.
- Allowing liquid or particles to enter the pump without suitable protection.
- Blocking the exhaust or using vessels and hoses that are not rated for vacuum.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Badge

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Badges and identifiers for role-playing (e.g., Tour Guide, Guest, Vendor)

What it is used for

Use role-playing badges to identify participants during training activities.

How to handle

- Assign the correct badge to each role before the activity begins.
- Ensure names and role labels are readable and appropriate.
- Attach badges securely without using sharp pins in unsafe ways.
- Collect badges after the activity and check that all are returned.

Storage

- Store badges in labelled envelopes or compartments by role.
- Keep clean and dry to prevent fading or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Barcode Printer

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Bar Code Machine

What it is used for

Use a barcode printer to print readable barcode labels for products, stock, packages, or other approved items.

How to use

1. Check the printer, power cable, label roll, and ribbon if required.
2. Load the correct label stock and ribbon according to the printer instructions.
3. Connect the printer to the approved computer or network.
4. Switch on the printer and confirm that it is ready.
5. Select the correct label size and barcode template in the software.
6. Print a test label and scan it to verify readability.
7. Print the required batch and check label alignment and print quality.

⚠ Safety

- Use only the specified power supply and compatible consumables.
- Keep fingers clear of moving rollers and the print head while operating.
- Allow the print head to cool before cleaning if required by the manual.
- Disconnect power before servicing the printer.

Care & storage

- Clean the print head and rollers using approved materials.
- Store label rolls and ribbons in a dry, dust-free place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Barcode Scanner

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Barcode Scanner; Barcode scanners; Barcode scanners and card readers for guest identification and room keys; Handheld Inventory Scanner (RFID/Barcode); IT Systems

What it is used for

Use a barcode scanner to read product, inventory, or identification codes and transfer the data to an approved system.

How to use

1. Inspect the scanner, cable, battery, and scanning window for damage.
2. Connect or pair the scanner with the approved computer or POS system.
3. Open the correct inventory or identification application.
4. Aim the scanner at the barcode from a suitable distance and angle.
5. Trigger the scan and confirm that the correct code appears on screen.
6. Repeat the scan if the code is not read correctly.
7. Return the scanner to its dock or charging station after use.

⚠ Safety

- Do not stare into a laser scanner beam or point it at anyone's eyes.
- Use only approved chargers and cables.
- Do not scan or access personal information without authorization.
- Keep the scanning window clean and avoid using a damaged device.

Care & storage

- Wipe the scanning window with a soft, suitable cloth.
- Charge and store the scanner according to the manufacturer's instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cash Register

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Cash Register or Cash Drawer

What it is used for

A cash register records sales, calculates totals, and stores cash securely during transactions. Use authorized access and follow the workplace procedure for every sale and cash movement.

How to use

1. Inspect the register, display, keypad, receipt printer, and cash drawer.
2. Log in using your assigned credentials or authorized access method.
3. Enter each item or transaction accurately and verify the displayed total.
4. Confirm the payment method and enter the amount received correctly.
5. Issue the receipt and return the correct change when applicable.
6. Close the transaction and secure the cash drawer.
7. Reconcile the recorded totals and hand over cash according to workplace rules.

⚠ Safety

- Never share passwords or leave an unlocked register unattended.
- Verify cash amounts carefully and follow procedures for refunds or voids.
- Keep cash drawers closed when not in use and report discrepancies promptly.

Care & storage

- Clean the keypad and display gently and report printer, drawer, or software faults.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Crate

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Crate; Wooden crates/polycrates

What it is used for

Crates store, organise, and transport goods, materials, and equipment.

How to handle

- Inspect the crate for cracks, splinters, protruding nails, and structural damage.
- Select a crate suitable for the item's weight, size, and fragility.
- Distribute the load evenly and avoid overloading or unstable stacking.
- Use safe lifting techniques and suitable handling equipment for heavy crates.
- Keep damaged crates out of service.

Storage

- Stack crates securely within their design limits.
- Store in a dry, designated area away from walkways and emergency exits.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Currency Counting Machine

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Money Currency Counting Machine with Built-In-Fake Note Detector

What it is used for

A currency counting machine counts banknotes and may identify suspicious notes using built-in detection features. Prepare the notes correctly and verify unusual results before accepting or recording the count.

How to use

1. Inspect the machine, power cable, hopper, and stacker for damage or obstructions.
2. Switch on the machine and allow its self-check to complete.
3. Select the correct currency and counting or detection mode, if available.
4. Straighten the banknotes and remove clips, rubber bands, and foreign objects.
5. Place the notes neatly in the hopper within the machine's capacity.
6. Start counting and watch for jams, errors, or suspicious-note alerts.
7. Separate flagged notes and verify them using approved banknote-checking procedures.
8. Compare the displayed count with the expected total and record the result.
9. Switch off the machine and clear the hopper and stacker after use.

⚠ Safety

- Keep fingers, jewellery, and loose clothing away from moving rollers.
- Do not force notes into the machine or reach inside while it is running.
- Treat detection alerts as warnings, not definitive proof that a note is counterfeit.
- Disconnect power before cleaning or clearing a persistent jam.

Care & storage

- Clean sensors and rollers according to the manufacturer's instructions.
- Keep the machine dry and arrange servicing if counting errors or alerts persist.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Currency Notes

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Currency Notes of different Denominations

What it is used for

Handle training currency notes responsibly for denomination and transaction exercises.

How to handle

- Use clearly identified training notes where possible and never present them as genuine currency.
- Sort notes by denomination and keep sets complete during exercises.
- Avoid writing on, tearing, or unnecessarily folding the notes.
- Explain that training notes have no monetary value.

Storage

- Keep notes in labelled envelopes or secure training cash boxes.
- Store separately from genuine currency and account for notes after use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Delivery System

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Delivery system

What it is used for

A delivery system coordinates the preparation, dispatch, tracking, and handover of goods to their destination. Follow the approved workflow and maintain accurate delivery records.

How to use

1. Review the delivery order, destination, contact details, and delivery deadline.
2. Verify the item description, quantity, condition, and package labels against the order.
3. Pack goods securely and attach the correct shipping or tracking documents.
4. Assign the delivery to an authorized person or approved carrier.
5. Update the system with dispatch details and tracking information.
6. Confirm delivery with the authorized recipient and obtain proof of receipt.
7. Record exceptions, damaged goods, or failed delivery attempts according to procedure.

⚠ Safety

- Protect customer information and use only authorized system access.
- Handle packages according to their weight, fragility, and hazard labels.
- Do not dispatch unverified goods or release packages to an unauthorized recipient.

Care & storage

- Keep delivery records updated and report system errors or missing tracking information promptly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Forklift

Category: **Retail, Logistics & Warehouse** Type: **Full SOP** Priority: **High** 8 labs

Covers these lab items: Fork Lift; Fork lifts; Forklift etc

What it is used for

A forklift lifts, carries, and places palletised or approved loads in warehouses and other work areas. Correct load handling, operator training, and controlled driving help prevent tip-over, collisions, and crushing injuries.

Main parts

- Forks support and lift the load.
- Mast guides the forks vertically during lifting and lowering.
- Hydraulic system powers lifting and tilting functions.
- Operator seat or platform provides the designated operating position.
- Steering wheel or control handle directs the forklift.
- Service brake and parking brake control stopping and stationary holding.
- Overhead guard and seat belt, where fitted, help protect the operator.

Before you start

1. Only a trained and authorised operator may use the forklift.
2. Complete the required pre-use inspection according to the operator's manual.
3. Check tyres, forks, mast, chains, hydraulics, brakes, steering, lights, horn, and alarms.
4. Confirm the load weight, load centre, and forklift capacity from its rating plate.
5. Ensure the route is clear, the floor is stable, and overhead clearance is sufficient.
6. Wear the seat belt where fitted and adjust the seat and controls before moving.

How to use

1. Mount the forklift using the designated steps and maintain three points of contact.
2. Start the forklift according to the manufacturer's instructions and check controls.
3. Keep forks low and tilt the mast as recommended while travelling without a load.
4. Approach the load slowly and align the forks with the pallet openings.
5. Insert the forks fully and evenly under the load before lifting.
6. Raise the load only enough to clear the floor and tilt it back as instructed.
7. Travel at a controlled speed with the load low and visibility maintained.
8. Use a spotter or approved travel direction when the load blocks forward visibility.
9. Stop on level ground, lower the load fully, and position it securely before withdrawing the forks.
10. Park in the designated area, lower the forks, apply the parking brake, and switch off.

⚠ Safety

- Never exceed the rated capacity or carry unstable, unsecured loads.
- Never allow anyone to stand or walk beneath raised forks or a lifted load.
- Do not carry passengers unless the forklift is specifically designed for them.
- Avoid sharp turns, sudden braking, and travel on slopes beyond the manufacturer's limits.
- Keep hands and feet inside the operator area and away from the mast.
- Never use the forklift to lift people unless an approved personnel platform and procedure permit it.

Care & storage

- Park safely and report defects before the forklift is returned to service.
- Check fluid leaks, tyres, forks, chains, and safety devices during routine inspections.
- Keep the operator area, steps, and controls clean and free from loose objects.
- Arrange scheduled servicing and battery or fuel-system maintenance through authorised personnel.

Common mistakes to avoid

- Travelling with forks raised too high or a load too high.
- Lifting a load without checking its weight and the forklift rating plate.
- Turning quickly or driving too fast in narrow or crowded areas.
- Parking without lowering the forks, applying the parking brake, or switching off.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gondola Shelf

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Gondola; Gondolas

What it is used for

Gondola shelves display and organise retail products for accessible shopping and efficient stock management.

How to handle

- Inspect shelves, uprights, brackets, and base supports for damage or instability.
- Assemble and secure the shelving according to manufacturer instructions.
- Check load limits and distribute products evenly across shelves.
- Place heavy items on lower shelves and keep aisles and exits clear.
- Remove damaged components and report instability before restocking.

Storage

- Keep spare shelves and fittings in a labelled, dry area.
- Avoid stacking components where they may fall or obstruct walkways.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Labelling Machine

Category: **Retail, Logistics & Warehouse** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Labeling Machine or Tag Gun; Labelling machines

What it is used for

A labelling machine prints, applies, or attaches labels to products, packages, or containers for identification and tracking. Correct setup, label verification, and safe operation help maintain readable information and consistent placement.

Main parts

- Label roll or label magazine supplies labels to the machine.
- Feed mechanism moves labels or products through the machine.
- Print head or printing unit prints variable information on applicable models.
- Applicator or dispensing mechanism attaches labels to products or packages.
- Control panel sets label format, speed, and operating options.
- Sensors detect labels or product position on equipped models.
- Guarding and emergency stop controls help protect the operator.
- Power supply or drive system operates the machine.

Before you start

1. Read the operating manual and confirm the machine is suitable for the label and product.
2. Inspect the power cable, guards, rollers, applicator, and emergency stop for damage.
3. Check label size, adhesive type, print quality requirements, and roll direction.
4. Verify the correct product information, batch details, date, and barcode where applicable.
5. Load compatible labels and approved ink or ribbon if required by the machine.
6. Clear the work area and keep hands, loose clothing, and tools away from moving parts.

How to use

1. Switch off the machine before loading labels, ink, ribbon, or making adjustments.
2. Install the label roll according to the feed diagram and secure the roll holder.
3. Thread the label material through the guides and sensors as instructed.
4. Enter or select the approved label layout and verify all printed information.
5. Set the label position, feed rate, and application settings for the product.
6. Run a test label or a small trial batch before full production.
7. Check that labels are readable, correctly positioned, and firmly attached.
8. Start production and monitor label feed, print quality, and product alignment.
9. Stop the machine before clearing a jam or correcting a misfeed.
10. At the end of the task, switch off the machine, remove waste labels, and record any faults.

⚠ Safety

- Keep fingers away from rollers, cutters, applicators, and other moving parts.
- Never bypass guards or disable the emergency stop.
- Isolate power before clearing jams, cleaning internal parts, or performing maintenance.
- Use only approved ink, ribbon, labels, and cleaning materials for the machine.
- Clean adhesive spills promptly using the recommended method and avoid skin or eye contact with chemicals.
- Follow safe lifting practices when handling heavy label rolls or product cartons.

Care & storage

- Clean print heads, sensors, rollers, and label paths according to the manual.
- Remove adhesive residue using approved cleaning materials only.
- Inspect belts, rollers, guards, and label guides regularly for wear or damage.
- Store unused labels, ink, and ribbon in clean, dry conditions.

Common mistakes to avoid

- Loading the label roll in the wrong direction.
- Printing incorrect product, batch, date, or barcode information.
- Starting full production without checking a test label.
- Clearing a label jam while the machine is running or powered.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ladder

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **Medium** 12 labs

Covers these lab items: Ladder; Ladder to climb Dhroni; Ladders and Step Stools

What it is used for

Use a ladder to reach elevated work areas safely when a suitable ladder is appropriate for the task.

How to use

1. Choose a ladder with the correct height, load rating, and type for the task.
2. Inspect the rails, rungs, feet, locks, and hinges for damage.
3. Place the ladder on a firm, level, non-slip surface.
4. Fully open and lock a step ladder, or secure an extension ladder at the correct angle.
5. Maintain three points of contact while climbing and keep your body centered between the rails.
6. Carry tools in a suitable belt or raise them separately rather than climbing with full hands.
7. Descend carefully and store the ladder after use.

⚠ Safety

- Do not use a damaged ladder or place it on boxes or unstable surfaces.
- Keep ladders away from overhead electrical wires and live equipment.
- Do not overreach; climb down and reposition the ladder instead.
- Follow the ladder's load rating and avoid climbing in unsafe weather or conditions.

Care & storage

- Clean the ladder and inspect it after use.
- Store it securely in a dry area where it cannot fall.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Material Handling Equipment

Category: **Retail, Logistics & Warehouse** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: Goods moving equipments; Material Handling Equipment; MHE

What it is used for

Material handling equipment helps lift, move, store, or position goods safely in workplaces. Correct equipment selection, load control, and safe movement reduce injuries and damage to materials.

Main parts

- Frame or platform supports the load during movement.
- Wheels or rollers allow equipment to move across suitable surfaces.
- Handle or steering control guides the equipment.
- Lifting mechanism raises loads on applicable equipment.
- Brakes or locking devices prevent unintended movement.
- Controls and switches operate powered equipment where fitted.
- Load rating label identifies the permitted working capacity.

Before you start

1. Identify the load weight, size, shape, and handling requirements.
2. Choose equipment rated for the load and suitable for the floor and route.
3. Inspect wheels, handles, forks, brakes, lifting parts, controls, and frame for damage.
4. Check that load rating labels are readable and the rated capacity is sufficient.
5. Clear the travel route of obstacles, spills, and people.
6. Wear suitable footwear and other task-specific PPE before handling goods.

How to use

1. Position the equipment close to the load on stable ground.
2. Secure the equipment against movement before loading or adjusting it.
3. Place the load evenly and keep it within the rated capacity.
4. Use straps, restraints, or guards when required to prevent load shifting.
5. Raise or lower loads using the designated controls and approved method.
6. Move at a controlled speed and maintain a clear view of the route.
7. Use assistance when the load blocks visibility or requires team handling.
8. Park on a level surface and apply brakes or locking devices where fitted.
9. Lower lifting parts fully and remove the load using a safe method.
10. Return the equipment to its designated storage area and report any defects.

⚠ Safety

- Never exceed the rated capacity or carry unstable loads without securing them.
- Keep hands and feet away from wheels, rollers, forks, and moving mechanisms.
- Never stand beneath a raised load or allow people into the danger zone.
- Do not ride on equipment unless it is specifically designed for passengers.
- Use correct lifting technique and team assistance for heavy or awkward items.
- Stop using equipment with defective brakes, damaged parts, or unstable movement.

Care & storage

- Clean equipment after use and remove debris from wheels and moving parts.
- Inspect handles, wheels, brakes, lifting mechanisms, and restraints regularly.
- Store equipment in a dry, designated area without blocking emergency exits.
- Arrange authorised maintenance and report defects before returning equipment to service.

Common mistakes to avoid

- Using equipment whose capacity is lower than the load weight.
- Moving goods without securing them against slipping or falling.
- Pushing equipment through cluttered routes or at excessive speed.
- Ignoring damaged wheels, brakes, handles, or lifting mechanisms.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

POS Terminal

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Card Swiping Machine; Credit/Debit Card Machine (EDC Machine); Point of Sale (POS) Systems; Point of Sale (POS) Terminal/System; POS/ Computer; POS terminals for billing and payment processing

What it is used for

Use a POS terminal to process authorized sales transactions and record payments through the approved payment or retail system.

How to use

1. Inspect the terminal, display, card reader, cables, and power supply for damage.
2. Power on the terminal and connect it to the approved network or POS system.
3. Sign in using authorized credentials and confirm the correct merchant or store profile.
4. Enter the transaction amount and verify it with the customer before payment.
5. Follow the terminal prompts for card, contactless, or other supported payment methods.
6. Wait for the approval or decline result and provide the receipt as required.
7. Close or log out of the session according to store procedure.

⚠ Safety

- Never ask customers to disclose their PIN or password.
- Do not record or store full card details outside approved systems.
- Use only authorized software and protect terminal access credentials.
- Report suspected tampering, unusual prompts, or payment discrepancies immediately.

Care & storage

- Clean the terminal with approved materials and keep liquids away from openings.
- Charge or power the device as instructed and keep it in a secure location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Packaging Tool

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Instruments used for opening packages and resealing packages; Instruments used for opening packages and resealing packages like cutters and strapping machines, clips etc.

What it is used for

Packaging tools are used to open packages, cut packaging materials, seal cartons, or secure loads with straps. Choose the correct tool for the package and packaging material.

How to use

1. Inspect cutters, tape dispensers, sealers, or strapping tools before use.
2. Choose the correct tool and confirm that its safety features work.
3. Place the package on a stable surface and keep the cutting path clear.
4. Cut tape or packaging away from your body using controlled movements.
5. For strapping tools, follow the equipment instructions and secure the strap correctly.
6. Collect scraps, retract or cover blades, and return tools to storage.

⚠ Safety

- Use safety cutters where available and keep fingers away from blades.
- Wear suitable eye protection when tensioning or cutting strapping.
- Never point a blade or tensioned strap toward another person.
- Do not use damaged tools or bypass safety guards.

Care & storage

- Clean tools after use, cover blades, and store them in a dry designated place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pallet

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 11 labs

Covers these lab items: Pallets, Totes, Storage Bins; Unitization devices; Unitization devices such as pallets and packing net

What it is used for

Pallets and unitization devices support the safe grouping, storage, and movement of goods in warehouses and logistics areas.

How to handle

- Inspect the pallet, deck boards, blocks, and load restraints for cracks or damage.
- Confirm load capacity, dimensions, and compatibility with the handling equipment.
- Distribute goods evenly, secure the load, and keep it within the pallet edges where practicable.
- Move pallets using trained operators and approved forklifts or pallet trucks; keep people clear of suspended loads.

Storage

- Stack pallets only in approved stable arrangements and within height limits.
- Keep pallets dry and segregate damaged units for repair or disposal.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pallet Truck

Category: **Retail, Logistics & Warehouse** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Battery Operated Pallet Truck; BOPT; Hand Operated Pallet Truck; HOPT

What it is used for

A pallet truck moves palletised goods over suitable warehouse floors using manual or battery-powered lifting. Correct load placement, route checks, and controlled movement help prevent crushing injuries and load instability.

Main parts

- Forks support the pallet and its load.
- Hydraulic lifting unit raises and lowers the forks.
- Handle controls steering and lifting functions on manual models.
- Drive controls operate travel and lifting on powered models.
- Wheels and rollers allow movement across the floor.
- Battery and charger supply power on battery-operated models.
- Load rating plate identifies the permitted capacity.

Before you start

1. Read the operating instructions and confirm the pallet truck suits the load and route.
2. Inspect forks, wheels, rollers, handle, hydraulic unit, controls, battery, and brakes where fitted.
3. Check the load weight and ensure it is within the rated capacity.
4. Confirm the pallet is intact and the goods are stable and secured.
5. Clear the route of obstacles, spills, people, and uneven surfaces.
6. For powered models, check battery charge and confirm the emergency stop works.

How to use

1. Position the pallet truck squarely in front of the pallet on level ground.
2. Lower the forks and insert them fully into the pallet openings.
3. Use the manual handle or approved powered controls to raise the pallet just enough for travel.
4. Check the load remains stable and does not block safe visibility.
5. Move at a controlled walking pace and keep the route clear.
6. Use the designated handle and controls to steer without sudden turns or stops.
7. Keep the load low while travelling and avoid slopes beyond the equipment's limits.
8. Stop on level ground and lower the pallet fully before withdrawing the forks.
9. Park with forks lowered and isolate the power on battery-operated models as instructed.
10. Return the truck to its storage or charging area and report faults.

⚠ Safety

- Never exceed the rated capacity or carry people on the pallet truck.
- Keep feet away from the wheels and forks, especially during turning and lowering.
- Never stand beneath raised forks or between the truck and a fixed object.
- Do not ride on a pallet truck unless it is specifically designed for riding.
- Avoid steep slopes, sharp turns, and sudden braking with a raised load.
- Stop using equipment with hydraulic leaks, damaged forks, faulty controls, or unstable wheels.

Care & storage

- Clean the forks, wheels, and handle after use and remove debris safely.
- Inspect hydraulic leaks, rollers, wheels, controls, and battery connectors regularly.
- Charge batteries only with the compatible charger in the designated ventilated area.
- Arrange authorised maintenance and report defects before returning the truck to service.

Common mistakes to avoid

- Using a damaged pallet or carrying a load beyond the rated capacity.
- Travelling with the load raised higher than necessary.
- Turning sharply or moving too quickly in crowded aisles.
- Leaving powered equipment on or parking with forks raised.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Product Sample

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 17 labs

Covers these lab items: Products; Products for display; Products for display (Dummy Cameras and Mobiles); Samples of bamboo products; Samples of different types of labels; Samples of finished paper mache products (e.g., decorative items, idols, utility objects); Samples of hand block printed textiles; Samples of inventory;

What it is used for

Product samples support demonstration, comparison, inspection, and display in retail, logistics, and training activities.

How to handle

- Confirm sample identity, condition, and purpose before placing it on display.
- Handle fragile, electronic, or sharp-edged samples according to their specific requirements.
- Label samples clearly and distinguish display-only or dummy products from working equipment.
- Keep samples clean and report missing, damaged, or misleading items.

Storage

- Store samples in labelled bins or protective cases organised by category.
- Secure valuable or fragile samples and protect them from moisture, heat, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Promotional Material

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Poster (POS); Promotional Material

What it is used for

Promotional materials and point-of-sale posters communicate approved offers, products, services, and brand information.

How to handle

- Verify campaign details, dates, prices, claims, and approval status before display or distribution.
- Use materials only in authorised locations and follow brand or venue guidelines.
- Display posters securely without blocking signs, exits, walkways, or safety equipment.
- Remove outdated, damaged, or misleading promotional materials promptly.
- Dispose of or recycle materials according to local requirements.

Storage

- Keep unused materials clean, dry, and organised by campaign or version.
- Store posters flat or rolled safely to prevent creases and damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Reach Truck

Category: **Retail, Logistics & Warehouse** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Reach trucks

What it is used for

A reach truck is a powered warehouse vehicle designed to lift, move, and place palletised loads in storage aisles. Pre-use inspection, stable load handling, and controlled driving help prevent collisions, dropped loads, and tip-over incidents.

Main parts

- Operator compartment and controls allow the driver to operate the truck.
- Mast raises and lowers the load.
- Forks support pallets and approved loads.
- Reach mechanism extends or retracts the forks and load.
- Drive wheel and motor move the truck.
- Steering system controls travel direction.
- Battery supplies power on electric models.
- Load capacity plate states permitted load limits and conditions.

Before you start

1. Only trained and authorised operators should use the reach truck.
2. Check the load capacity plate and confirm the load weight and centre are within limits.
3. Inspect forks, mast, reach mechanism, tyres, brakes, steering, and overhead guard for damage.
4. Check the battery, controls, warning devices, and hydraulic system for leaks or faults.
5. Ensure the travel route and storage aisle are clear and the floor is suitable.
6. Wear required PPE and adjust the seat or operator position according to the manual.

How to use

1. Mount the truck using the designated access points and secure the operator restraint where fitted.
2. Switch on the truck and check warning indicators and control functions.
3. Approach the pallet slowly with the forks lowered and aligned with the openings.
4. Insert the forks fully and evenly beneath the pallet without striking the load.
5. Lift the load only enough to clear the floor and retract the reach mechanism as instructed.
6. Travel at a controlled speed with the load low and the mast in the recommended travel position.
7. Slow down at corners, intersections, doorways, and areas with limited visibility.
8. Stop in front of the destination, extend the reach mechanism as required, and raise the load to the correct shelf height.

9. Place the load securely, lower the forks clear of the pallet, and retract the reach mechanism before reversing.
10. Park in the designated area, lower the forks, apply the parking brake, switch off, and report any faults.

⚠ Safety

- Never exceed the rated capacity or lift an unstable or damaged pallet.
- Keep people away from the truck's operating area and never allow anyone under raised forks.
- Do not travel with the load raised high or turn sharply at speed.
- Use the horn and follow site traffic routes, speed limits, and pedestrian separation rules.
- Never carry passengers unless the truck is specifically designed and approved for them.
- If the truck becomes unstable or a fault occurs, follow operator training and emergency procedures; do not jump from a tipping truck.

Care & storage

- Inspect forks, tyres, mast, chains, and reach mechanism before and after use.
- Keep the operator compartment, sensors, and visibility areas clean.
- Charge and maintain the battery according to the manufacturer's instructions.
- Report hydraulic leaks, unusual noise, braking problems, or warning indicators promptly.

Common mistakes to avoid

- Operating without checking the capacity plate and load weight.
- Travelling with forks or the load raised too high.
- Turning sharply or travelling too fast in narrow aisles.
- Leaving the forks raised or the truck powered on when parked.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shopping Cart

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Shopping Cart

What it is used for

Shopping carts carry merchandise through retail areas and support organised customer shopping and stock movement.

How to handle

- Inspect the cart frame, wheels, handle, basket, and child seat if fitted.
- Check that wheels move freely and the cart does not have sharp or broken parts.
- Load items evenly and keep the load within the cart's capacity.
- Push at a controlled speed and avoid blocking aisles, ramps, or emergency exits.
- Remove damaged carts from service and report them for repair.

Storage

- Return carts to the designated parking or storage area.
- Nest or stack carts only as designed and secure them against rolling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stacker

Category: **Retail, Logistics & Warehouse** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Stacker

What it is used for

A stacker lifts and moves palletised goods over short distances in warehouses, stores, and loading areas. Correct load assessment, stable stacking, controlled travel, and safe parking help prevent tipping, collisions, falling loads, and operator injuries.

Main parts

- Mast guides the lifting carriage vertically.
- Forks support the pallet or load.
- Hydraulic lifting system raises and lowers the forks.
- Load backrest supports the load where fitted.
- Steering handle or tiller controls direction on powered models.
- Drive wheel and load wheels allow movement.
- Control levers or buttons operate lifting and travel functions.
- Battery and charger power electric stackers where fitted.
- Emergency stop and braking system help stop movement.

Before you start

1. Only trained and authorised operators should use the stacker.
2. Inspect the stacker, forks, mast, wheels, chains, hydraulic lines, controls, and safety devices for damage or leakage.
3. Check the nameplate for rated capacity, load centre, and maximum lift height.
4. Confirm that the pallet is sound and the load is stable, secured, and within the stacker's rated capacity.
5. Check the travel route for people, uneven flooring, ramps, overhead obstructions, and narrow passages.
6. Verify battery charge or fuel status and confirm brakes, horn, steering, and emergency stop function correctly.
7. Wear suitable safety footwear and any site-required PPE.

How to use

1. Approach the pallet slowly and align the stacker with the load.
2. Lower the forks and insert them fully under the pallet without striking the load.
3. Confirm the load is centred and distributed evenly across both forks.
4. Raise the load only enough to clear the floor and check stability before travelling.
5. Travel at a controlled speed with the load low and the route in view.

6. Use the horn at blind corners and intersections according to site rules.
7. Avoid sudden turns, sharp braking, and travel across slopes beyond the equipment's limits.
8. Position the stacker squarely at the destination and stop before raising the load.
9. Raise the load to the required height and place it gently on the designated stable surface.
10. Lower the forks until the load is fully supported, then withdraw the forks carefully.
11. Lower the forks to a safe travel position, park in the designated area, apply the brake, and switch off the unit.
12. Connect the charger only in the designated charging area and according to the equipment instructions.

⚠ Safety

- Never exceed the rated capacity or lift a load that blocks safe visibility without an approved travel method.
- Keep people clear of the forks, mast, load, and turning area.
- Never allow anyone to stand or pass under raised forks or a suspended load.
- Do not carry passengers or use the stacker to lift people unless it is specifically designed and approved for that purpose.
- Do not operate with damaged forks, faulty brakes, leaking hydraulics, or malfunctioning controls.
- Lower the forks and isolate the equipment before inspection or maintenance; follow battery charging and handling precautions.

Care & storage

- Inspect forks, mast, wheels, chains, brakes, and hydraulic components before each shift.
- Keep the stacker clean and remove debris from wheels and moving parts.
- Check battery condition and charge it using the approved charger where applicable.
- Report leaks, unusual noises, steering problems, and safety-device faults immediately.
- Arrange scheduled maintenance and remove unsafe equipment from service until repaired.

Common mistakes to avoid

- Lifting a load beyond the rated capacity or with an unstable pallet.
- Travelling with the load raised too high.
- Turning sharply or travelling too fast on an uneven surface.
- Leaving the stacker parked with raised forks or without applying the brake.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Strap

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Straps

What it is used for

Handle straps safely for bundling, securing, or moving goods.

How to handle

- Inspect straps for fraying, cuts, damaged buckles, or worn stitching before use.
- Choose a strap rated for the load and intended securing method.
- Keep hands clear of pinch points and avoid sudden release of tension.
- Do not use damaged straps or stand in the path of a loaded strap under tension.

Storage

- Clean and dry reusable straps before storing.
- Coil neatly in a labelled area away from sunlight, sharp edges, chemicals, and excessive heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tag

Category: **Retail, Logistics & Warehouse** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Tags

What it is used for

Handle tags for identifying, pricing, tracking, or labelling goods.

How to handle

- Use tags suitable for the item and intended information.
- Write or print clear, accurate details and verify codes or prices before attaching.
- Attach tags securely without damaging the product or obstructing its use.
- Avoid mixing blank, completed, and obsolete tags.

Storage

- Sort unused tags by type and size in labelled containers.
- Keep clean and dry away from moisture, dust, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tagging Machine

Category: **Retail, Logistics & Warehouse** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Tagging Machine; Tagging set

What it is used for

To attach identification tags to products, garments, or items using a tagging machine. To secure tags consistently without damaging the item or injuring the operator.

Main parts

- Tagging tool body
- Needle or fastener applicator
- Trigger or lever
- Tag fasteners
- Product tags
- Needle guard, if fitted

Before you start

1. Confirm that the tagging tool is suitable for the product and tag type.
2. Inspect the tool and needle for damage or looseness.
3. Load compatible fasteners according to the instructions.
4. Position the item so the tag can be attached at an approved location.

How to use

1. Hold the item securely without stretching or damaging it.
2. Place the tag at the intended attachment point.
3. Align the tool needle with the approved tag location.
4. Press the trigger or lever according to the tool instructions.
5. Release the trigger and remove the tool carefully.
6. Check that the tag is secure and the item is undamaged.
7. Engage the needle guard or safety lock after use.

⚠ Safety

- Keep fingers away from the needle path.
- Never point the needle toward the body or another person.
- Do not use a bent or damaged needle.
- Keep tagging tools away from children and unauthorized users.

Care & storage

- Remove tag fragments and debris after use.
- Inspect the needle and trigger mechanism regularly.
- Store with the needle guard engaged.
- Replace damaged parts before further use.

Common mistakes to avoid

- Attaching tags in an unsuitable or visible damage-prone area.
- Placing fingers behind the material in the needle path.
- Using incompatible fasteners.
- Leaving the needle exposed during storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Alarm Panel

Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Alarm Panels

What it is used for

An alarm panel monitors connected sensors and activates audible or visual alerts when a fire, intrusion, gas leak, or other monitored condition is detected. Operate and test the panel according to the installed system's instructions so that alarms remain reliable and emergency signals are not accidentally disabled.

Main parts

- Control panel and display
- Status indicators and control buttons
- Power supply and backup battery
- Connected sensors and detector circuits
- Alarm sounders, beacons, or notification outputs
- Communication module and event log, if fitted

Before you start

1. Confirm the alarm system type, monitored zones, and authorised operating procedure.
2. Check the panel enclosure, display, indicators, and visible wiring for damage.
3. Verify that mains power and backup battery indicators show normal status.
4. Review any active faults, alarms, or disabled zones before testing.
5. Notify the responsible person and affected occupants before a planned alarm test.
6. Confirm that the test will not interfere with emergency response or other safety systems.

How to use

1. Approach the panel and check its displayed status and indicator lights.
2. Authenticate using the authorised access method if the system requires it.
3. Review active alarms, faults, and zone information on the display.
4. For a test, select the approved test mode or procedure specified by the manufacturer.
5. Test only the designated sensors, alarm sounders, indicators, or communication functions as authorised.
6. Confirm that expected audible, visual, and panel indications occur during the test.
7. Check that the event log records the test or event correctly, where supported.
8. Investigate any fault indication using the approved troubleshooting procedure.
9. Do not bypass a sensor or disable a zone unless authorised and required by the approved procedure.
10. Restore the system to its normal monitoring state after testing.

1. Confirm that all required zones and alarm outputs are enabled and that no unresolved fault remains.
2. Record the test results, faults, corrective actions, and any temporary impairment; notify the responsible person.

⚠ Safety

- Only authorised and trained personnel should change settings, isolate zones, or reset protected systems.
- Never ignore a genuine alarm or assume it is a test without confirmation.
- Follow the site's emergency plan and evacuate when required.
- Do not disconnect power, remove batteries, or open electrical compartments unless authorised and safe to do so.
- If a safety-critical alarm system is impaired, notify the responsible person immediately and implement approved temporary safeguards.
- Do not reset or silence an alarm until the cause has been assessed and the site's procedure permits it.

Care & storage

- Inspect the panel enclosure, indicators, labels, and accessible cables regularly.
- Test backup batteries and alarm functions at the intervals specified by the manufacturer and applicable requirements.
- Keep the panel clean, dry, unobstructed, and accessible to authorised responders.
- Review event logs and resolve recurring faults promptly.
- Arrange qualified servicing for wiring faults, battery failures, or persistent system errors.

Common mistakes to avoid

- Silencing or resetting an alarm before checking its cause.
- Leaving a zone disabled after maintenance or testing.
- Ignoring fault indicators or a low backup-battery warning.
- Conducting an alarm test without notifying affected people or coordinating emergency monitoring.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Antistatic Wrist Strap

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Antistatic wrist wrap

What it is used for

To safely discharge static electricity from a user when handling electrostatic-sensitive electronic components. To reduce electrostatic discharge risk by using a compatible wrist strap and verified grounding point.

Main parts

- Conductive wrist band
- Adjustable strap
- Grounding cord
- Resistor or safety element
- Grounding connector

Before you start

1. Check the wrist strap, cord, connector, and band for damage.
2. Confirm that the workstation has an approved ESD grounding point.
3. Test the strap using an approved wrist-strap tester when required.
4. Ensure the strap is suitable for the electronics work area and procedure.

How to use

1. Place the band snugly against bare skin as instructed.
2. Connect the grounding cord to the designated ESD point.
3. Verify the connection or test result according to the workstation procedure.
4. Handle sensitive components by their edges and avoid touching contacts.
5. Keep the strap connected while working in the designated ESD area.
6. Disconnect and store the strap after completing the task.

⚠ Safety

- Never connect an ESD wrist strap to mains live, neutral, or protective earth terminals except through an approved ESD grounding system.
- Do not use a wrist strap when working on energized electrical equipment where it creates a shock hazard.
- Replace damaged straps or cords.
- Follow the site's ESD and electrical safety procedures.

Care & storage

- Clean the band according to the manufacturer's instructions.
- Inspect the cord and resistor regularly.
- Store without sharp bends or heavy objects on the cord.
- Retest according to the workplace schedule.

Common mistakes to avoid

- Connecting to an unapproved grounding point.
- Wearing the band over clothing instead of against skin.
- Using a damaged cord or failed strap.
- Using the strap around energized equipment without assessing shock risk.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Apron

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 21 labs

Covers these lab items: Apron; Apron & head gear; Apron and cape; Apron (Cloth); Apron self and client; Aprons; Aprons & Chef Hats; Aprons, Gloves, Head Caps; Flame proof aprons; Overcoat - (Apron- Blue color); Working Apron/Coat

What it is used for

Aprons and associated protective garments help protect clothing and the wearer during cooking, cleaning, beauty services, and other work.

How to handle

- Select an apron suitable for the task, material hazards, and required hygiene level.
- Inspect ties, fasteners, fabric, and protective layers before wearing.
- Wear the apron securely and combine it with gloves, headgear, or other PPE when required.
- Replace contaminated, torn, or heat-damaged aprons before continuing work.

Storage

- Launder reusable aprons according to the fabric and workplace hygiene procedure.
- Store clean aprons separately from used or contaminated garments.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Barricade Tape

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Barricading/caution tape

What it is used for

Use barricade tape to clearly mark restricted or hazardous areas.

How to handle

- Identify the hazard and place tape so the restricted area is clearly visible.
- Secure the tape to stable supports without creating trip hazards.
- Do not use tape as a physical load-bearing barrier or fall protection.
- Remove or update tape when the hazard is cleared, following site procedures.

Storage

- Keep unused tape rolls dry and in their original packaging where practical.
- Store with safety supplies away from sharp edges and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Barrier Post

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Post and chain

What it is used for

Handle barrier posts and chains safely to control access to designated areas.

How to handle

- Inspect posts, bases, and chains for damage before setup.
- Place posts on stable ground and ensure chains are securely attached.
- Keep barriers visible and arrange them to avoid obstructing emergency exits.
- Lift and move heavy posts using safe manual-handling techniques.

Storage

- Detach and stack posts securely after use where appropriate.
- Store chains neatly to prevent tangling and trip hazards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Coverall

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Boiler suit; Dangris; Dangris (dungarees); Flame proof overalls buttoned to neck

What it is used for

Coveralls and overalls protect clothing and skin from workplace dirt, splashes, and task-specific hazards.

How to handle

- Select coveralls with protection suitable for the task, exposure, and workplace risk assessment.
- Inspect fabric, seams, fasteners, cuffs, and protective features before wearing.
- Wear the coverall correctly and secure fasteners; combine with other PPE as required.
- Keep loose fabric away from moving machinery and hot surfaces.
- Remove contaminated garments safely and replace damaged or unsuitable coveralls.

Storage

- Launder reusable coveralls according to the manufacturer's instructions.
- Store clean garments separately from used or contaminated workwear.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ear Protection

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 10 labs

Covers these lab items: Ear plug; Ear plugs; Ear Protection

What it is used for

Use suitable ear protection to reduce exposure to harmful noise in workshops, factories, and other noisy work areas.

How to use

1. Select earplugs or earmuffs suitable for the noise level and task.
2. Inspect the ear protection for damage, dirt, or worn parts.
3. Insert disposable or reusable earplugs according to the manufacturer's instructions.
4. Fit earmuffs so the cushions seal fully around both ears.
5. Check the fit before entering the noisy area and keep protection in place.
6. Replace or clean the equipment according to its type and instructions.

⚠ Safety

- Use the correct protection for the noise level; ask the safety supervisor if unsure.
- Do not insert dirty earplugs or share personal ear protection without proper cleaning.
- Do not remove protection in noisy areas unless it is safe to do so.
- Ear protection does not replace machine guards or other noise controls.

Care & storage

- Clean reusable protection as directed and allow it to dry.
- Store ear protection in a clean case away from dust and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Emergency Light

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Emergency Flood Lights

What it is used for

An emergency light provides illumination during a power outage or emergency. Check its charging, battery condition, and automatic or manual operation regularly.

How to use

1. Inspect the light, casing, cable, plug, and mounting bracket for damage.
2. Charge the unit according to the manufacturer's instructions.
3. Check the charge indicator and confirm the battery is ready for use.
4. Test the light using its approved test switch or simulated power failure procedure.
5. Verify that the light illuminates and remains stable during the test.
6. Return the unit to its designated location and keep access unobstructed.
7. Record the test result and report any battery or lighting fault.

⚠ Safety

- Do not use damaged cables, plugs, or swollen batteries.
- Keep the unit away from water unless it is rated for wet conditions.
- Do not open the battery compartment or electrical housing unless authorized.
- Keep emergency lights and exit routes clear of stored items.

Care & storage

- Clean the housing and perform periodic battery and functional checks as specified.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Explosive Detector

Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Explosive Vapour Detector (EVD)

What it is used for

An explosive vapour detector checks the surrounding atmosphere for vapours or gases associated with specified explosive or hazardous substances, depending on its sensor technology. Correct calibration checks, approved sampling methods, and careful interpretation help identify potential hazards without treating a negative reading as proof that an area is safe.

Main parts

- Detector unit and display
- Sensor or sampling inlet
- Sampling probe or tubing where fitted
- Battery or approved power supply
- Alarm indicators and audible alert
- Calibration accessories and manufacturer-approved test materials

Before you start

1. Use the detector only if trained and authorised for the specific instrument and site safety procedure.
2. Confirm the detector's intended target substances, sensor technology, detection limits, and operating environment from the manual.
3. Inspect the unit, sensor inlet, sampling tubing, battery, alarm indicators, and housing for damage or contamination.
4. Verify calibration status, sensor service life, and the required self-test or bump-test procedure.
5. Perform the prescribed functional check using only manufacturer-approved test materials and methods.
6. Review site emergency procedures, required PPE, access restrictions, and communication arrangements before entering the test area.

How to use

1. Switch on the detector in the designated safe area and allow it to complete its startup and self-test sequence.
2. Confirm that the display, alarms, battery status, and sensor readiness indicators show normal operation.
3. Perform the required bump test or calibration check according to the manual and record the result.

4. Select the correct detection mode or target setting when the instrument supports multiple modes.
5. Approach the inspection area according to the site's approved hazardous-material response procedure and maintain a safe position.
6. Sample the atmosphere at the locations and heights specified by the instrument manual and site procedure, allowing the required response time.
7. Move the sampling inlet slowly and avoid blocking it against surfaces or exposing it to liquids, dust, or incompatible materials.
8. Monitor readings, trend changes, and visual or audible alarms continuously during the survey.
9. If an alarm activates or a suspicious reading appears, stop the survey, avoid further exposure, and follow the site's evacuation and incident-response procedure.
10. Do not touch, collect, open, or move any suspected explosive material; leave investigation and handling to the designated specialist team.
11. Record the location, time, detector identification, readings, alarm events, and environmental conditions as required.
12. After the survey, move to the approved safe area and follow the manufacturer's procedure to shut down and inspect the detector.
13. Clean the exterior and sampling accessories only using approved methods and store the unit securely.
14. Report abnormal readings, failed checks, suspected contamination, or sensor faults to the responsible safety officer.

⚠ Safety

- A detector reading is an aid to risk assessment, not a guarantee that an area is safe; follow the site hazard assessment and specialist instructions.
- Never approach or handle a suspected explosive device or suspicious package to obtain a reading.
- If an alarm or suspected explosive hazard is encountered, do not investigate further; withdraw to a safe location and notify the designated emergency or specialist team.
- Use only detectors approved for the target hazard and environment; a general-purpose vapour detector may not detect every explosive substance.
- Do not use unapproved solvents, test chemicals, or improvised calibration methods on the sensor.
- Observe hazardous-area restrictions and use only equipment approved for the area's ignition-risk classification.
- Do not enter a potentially hazardous area alone or without the required PPE, communication, and emergency arrangements.

Care & storage

- Clean the detector exterior and sampling inlet according to the manufacturer's instructions after use.
- Inspect tubing, filters, sensors, seals, and connectors for wear, blockage, or contamination.
- Charge or replace batteries as specified and confirm adequate power before deployment.

- Maintain calibration, bump-test, maintenance, and sensor-replacement records.
- Store the detector in its approved case under the specified temperature, humidity, and transport conditions.

Common mistakes to avoid

- Assuming a negative reading proves that no explosive hazard is present.
- Using a detector that is out of calibration, has a failed sensor, or has insufficient battery power.
- Sampling at inappropriate locations or failing to allow the sensor response time.
- Approaching, touching, or attempting to move a suspected explosive object instead of withdrawing and reporting it.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Eye Shield

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Eye-caps/shield

What it is used for

An eye shield protects the eyes from specified splashes, particles, or other workplace hazards when suitable for the task. Select the correct protection type and inspect it before use.

How to use

1. Identify the hazard and select an eye shield rated for that task.
2. Inspect the shield for cracks, scratches, loose fittings, or damaged straps.
3. Clean the shield using the manufacturer's approved method before wearing it.
4. Position the shield securely so it covers the required eye area.
5. Check that it fits properly and does not obstruct safe vision.
6. Wear it throughout the task and replace it if visibility or protection is compromised.
7. Clean and store the shield after use.

⚠ Safety

- Use a shield appropriate to the hazard; it may not replace safety goggles or glasses.
- Do not use cracked, badly scratched, or poorly fitting protection.
- Keep the shield in place while exposure hazards remain.
- Follow workplace PPE rules and the manufacturer's limitations.

Care & storage

- Clean with compatible materials and store in a protective case away from heat and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Face Mask

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 22 labs

Covers these lab items: Dust Mask; Dust Mask/Respirator; Dust Masks; Face mask; Face Mask (Reusable); Mask; Mask (Canvas Mask); Mask-Packet; Masks; Nose mask

What it is used for

A face mask provides a barrier against specified droplets, dust, or other airborne particles depending on its design and certification. Choose the correct mask for the hazard and wear it according to the manufacturer's instructions.

How to use

1. Check the mask type, packaging, condition, and expiry date where applicable.
2. Wash or sanitise hands before handling the mask.
3. Position the mask over the nose, mouth, and chin with the correct side facing outward.
4. Secure the straps or ties and adjust the nose piece if provided.
5. Check the fit and ensure there are no obvious gaps for the intended use.
6. Avoid touching the front surface while wearing the mask.
7. Remove the mask using its straps and dispose of or clean it as instructed.
8. Clean hands after removal and store reusable masks in a clean, dry place.

⚠ Safety

- Do not use damaged, wet, or contaminated masks.
- A loose-fitting face mask does not replace a required certified respirator for hazardous airborne exposure.
- Do not share used masks between people.
- Follow the mask's limitations and the site's infection-control or dust-control procedures.

Care & storage

- Replace single-use masks after use or when wet or soiled.
- Wash reusable masks as directed and store them clean and dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Field Gear

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Sturdy hiking boots, hats, and gloves suitable for fieldwork

What it is used for

Use sturdy hiking boots, hats, and gloves safely during fieldwork.

How to handle

- Inspect boots, hats, and gloves for damage and correct fit before leaving for the field.
- Choose footwear with suitable grip and protection for the terrain and weather.
- Wear gloves appropriate to the task and avoid using damaged protective gear.
- Clean equipment after fieldwork and check for ticks, thorns, or contamination.

Storage

- Dry boots and clothing thoroughly before storing.
- Keep clean gear in a ventilated, designated area separate from contaminated field equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fire Bucket

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Fire buckets; Sand Bucket

What it is used for

A fire bucket holds sand or another designated extinguishing material for use during suitable small fires. Keep it accessible and use it only for fire types the material can safely control.

How to use

1. Check that the bucket is clearly marked and contains the designated material.
2. Keep the bucket in its assigned, accessible fire-safety location.
3. When a suitable small fire occurs, raise the alarm and follow the site's emergency procedure.
4. Use the appropriate protective equipment and approach only if the escape route remains clear.
5. Apply sand carefully to smother a compatible small fire without spreading burning material.
6. Report use, replace contaminated material, and restore the bucket to its location.

⚠ Safety

- Raise the alarm and contact emergency services according to the site procedure.
- Do not use sand on fires involving energized electrical equipment unless trained and the procedure permits it.
- Never use water or sand on a fire if the material or fire type makes it unsafe.
- Do not attempt to fight a spreading fire; evacuate immediately.

Care & storage

- Keep the bucket clean, filled as designated, dry where appropriate, and unobstructed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fire Extinguisher

Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 60 labs

Covers these lab items: Fire extinguisher; Fire Extinguisher (ABC Type); Fire Extinguisher CO2; Fire extinguishers; Fire extinguishers and warning signs; Fire extinguishers, emergency exit maps, and signage; Fire extinguishers, first aid boxes, and workplace emergency signage; Fire Safety Equipment; Workshop Safety: Fire e

What it is used for

A fire extinguisher helps control small, early-stage fires when it is safe to act. Students and teachers must identify the correct extinguisher type and know when to evacuate instead.

Main parts

- Cylinder containing the fire-extinguishing agent under pressure.
- Safety pin prevents accidental discharge.
- Tamper seal indicates whether the extinguisher may have been used.
- Handle and operating lever release the extinguishing agent.
- Hose or nozzle directs the agent toward the fire.
- Pressure gauge indicates operating pressure on applicable models.
- Instruction label identifies fire classes and operating directions.

Before you start

1. Identify the nearest fire extinguisher and emergency exit before starting practical work.
2. Check the instruction label and confirm the extinguisher suits the fire type.
3. Inspect the cylinder, hose, nozzle, pin, seal, and body for damage.
4. Check the pressure gauge is in the operable zone, if fitted.
5. Ensure the access route is clear and the extinguisher is securely mounted.
6. Raise the alarm and call emergency services when a fire is discovered.

How to use

1. Raise the alarm immediately and inform the teacher or responsible staff member.
2. Assess the fire from a safe distance and identify the burning material.
3. Use an extinguisher only for a small, contained fire with a clear escape route.
4. Select the correct extinguisher; never use water on electrical or flammable-liquid fires.
5. Remove the safety pin while keeping the nozzle pointed away from people.
6. Stand at a safe distance and aim the nozzle at the fire's base.
7. Squeeze the operating lever to discharge the extinguishing agent.
8. Sweep the nozzle across the base of the fire until it is controlled.
9. Back away immediately if flames spread, smoke increases, or the extinguisher becomes ineffective.
10. Evacuate through the safe exit and do not re-enter until authorities permit.

⚠ Safety

- Never fight a large, spreading, or smoke-filled fire; evacuate immediately.
- Keep your back toward a clear escape route while using an extinguisher.
- Never touch electrical equipment or approach live electrical hazards during a fire.
- Use CO2 extinguishers carefully in confined spaces because they can reduce breathable oxygen.
- Never direct the discharge toward a person's face or bare skin.
- Only trained and authorised people should operate extinguishers during practical demonstrations.

Care & storage

- Keep extinguishers visible, accessible, securely mounted, and free from obstruction.
- Arrange scheduled inspection and servicing by authorised personnel according to the label and applicable requirements.
- Report damaged seals, low pressure, corrosion, leakage, or missing safety pins immediately.
- After any discharge, arrange prompt professional inspection, recharging, or replacement before reuse.

Common mistakes to avoid

- Choosing an extinguisher without checking the fire type and instruction label.
- Aiming at the flames instead of the base of the fire.
- Attempting to fight a fire after the escape route becomes unsafe.
- Returning to a fire area before authorised personnel declare it safe.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Full Body Harness

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Full body harness; Safety belt; Safety harness

What it is used for

Use a full body harness as part of an approved fall-protection system when working at height where fall protection is required.

How to use

1. Inspect the harness webbing, stitching, buckles, D-rings, and labels before each use.
2. Confirm the harness fits the user's size and is within its service life.
3. Put on the harness and fasten all buckles according to the manufacturer's instructions.
4. Adjust straps so the harness fits snugly without restricting movement or breathing.
5. Connect the designated attachment point to an approved compatible lanyard or fall-arrest system.
6. Verify the anchor point and system are approved and suitable for the task.
7. After use, inspect the harness and report any fall event or damage.

⚠ Safety

- Only trained users should use a harness as part of a fall-protection system.
- Never use a harness with cuts, damaged stitching, altered parts, or an unreadable label.
- Use an approved anchor and compatible fall-arrest equipment; a harness alone does not prevent a fall.
- Maintain a rescue plan and do not work at height without required protection.

Care & storage

- Store clean and dry away from sunlight, chemicals, heat, and sharp edges.
- Remove the harness from service after a fall or when inspection identifies damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gas Leak Detector

Category: **Safety & PPE** Type: **Short SOP** Priority: **High** 1 labs

Covers these lab items: Gas leakage detector

What it is used for

A gas leak detector identifies or signals the presence of specified combustible or hazardous gases. Use a detector suitable for the target gas and follow its operating and alarm procedures.

How to use

1. Inspect the detector, sensor inlet, display, battery, and accessories for damage.
2. Confirm that the detector is suitable for the gas being checked and within its calibration period.
3. Perform the prescribed bump test or function check before use.
4. Switch on the detector in a safe area and allow its startup checks to finish.
5. Check the display and alarm status before approaching the suspected leak area.
6. Survey the area using the manufacturer's specified method while avoiding direct exposure.
7. If an alarm activates, leave the area and follow the site emergency procedure.
8. Arrange authorised testing, calibration, or servicing if the detector fails its checks.

⚠ Safety

- Never use a gas detector as a reason to enter a suspected hazardous atmosphere without authorisation and protection.
- Do not ignore alarms or rely on smell to identify gas safety.
- Keep away from ignition sources when a flammable gas leak is suspected.
- Only trained personnel should perform calibration and sensor maintenance.

Care & storage

- Keep the detector clean and store it within the specified temperature and humidity limits.
- Charge or replace batteries and complete scheduled function checks and calibration.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gas Mask

Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Gas Mask (full vision facemask) & Canister for use with ALP

What it is used for

A full-face gas mask with a compatible canister helps protect the wearer from specified airborne contaminants when used under an approved respiratory-protection programme. Correct selection, fit testing, inspection, and use are essential because a canister cannot protect against every gas, vapour, or oxygen-deficient atmosphere.

Main parts

- Full-face mask seals around the face and protects the eyes and respiratory system.
- Visor provides a wide field of vision.
- Inhalation valve directs incoming air through the approved airflow path.
- Exhalation valve allows exhaled air to leave the mask.
- Compatible canister or cartridge filters or treats specified contaminants.
- Head harness holds the mask in the correct position.
- Face seal forms the protective seal against the skin.
- Communication or voice diaphragm supports speech where fitted.

Before you start

1. Use the mask only under an approved respiratory-protection programme and after suitable training.
2. Identify the contaminant, concentration, oxygen level, and required protection before selecting a mask and canister.
3. Never use an air-purifying canister in an oxygen-deficient or immediately dangerous atmosphere.
4. Confirm the canister is approved for the specific contaminant and is within its service life.
5. Inspect the face seal, visor, valves, harness, connectors, and canister for damage or contamination.
6. Complete the required fit test and user seal check; facial hair must not interfere with the sealing surface.
7. Confirm the wearer can safely use the mask and that an emergency plan is available.

How to use

1. Wash and dry hands before handling the mask and canister.
2. Inspect the mask and fit the specified canister according to the manufacturer's instructions.
3. Check that valves, connectors, and seals are correctly installed and unobstructed.
4. Put on the mask and adjust the head harness evenly without overtightening.
5. Perform the prescribed user seal check before entering the designated work area.

6. Enter the area only after the responsible safety person confirms the atmosphere and protection are suitable.
7. Keep the mask correctly positioned and monitor for leakage, breathing resistance, visor fogging, or discomfort.
8. Leave the area immediately if leakage, unusual odour, dizziness, breathing difficulty, or another warning sign occurs.
9. Move to a safe area before removing the mask and follow the site decontamination procedure where required.
10. Remove the canister and clean, inspect, and store the mask according to the approved procedure.
11. Record inspections, canister changes, defects, and any use-related concerns.

⚠ Safety

- Never rely on a gas mask for an unknown atmosphere, oxygen deficiency, or conditions beyond its approved limits.
- Use only a canister approved for the identified contaminant and exposure conditions.
- Do not use damaged, expired, saturated, or incorrectly fitted canisters.
- A user seal check does not replace formal fit testing.
- Do not remove the mask in a contaminated area unless an authorised emergency procedure specifically directs it.
- If breathing becomes difficult or contamination is suspected, leave immediately and follow the emergency plan.

Care & storage

- Clean and disinfect the mask using manufacturer-approved products and methods.
- Inspect valves, face seal, visor, harness, and connectors after each use.
- Store the mask in a clean, dry, sealed container away from chemicals, sunlight, heat, and deformation.
- Replace canisters according to the approved change schedule and service-life limits.
- Remove damaged equipment from service and arrange authorised inspection or replacement.

Common mistakes to avoid

- Using a canister that is unsuitable for the contaminant or exposure level.
- Entering a hazardous atmosphere without confirming oxygen levels and protection requirements.
- Skipping fit testing or the required user seal check.
- Continuing to use a damaged mask or an exhausted canister.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gloves

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 51 labs

Covers these lab items: Bamboo Gloves; Cotton hand glove; Cotton Hand Gloves (sheet metal work); Disposable gloves; ESD Gloves; Gloves; Gloves and Goggles; Hand gloves; Hand gloves and aprons; Hand Gloves (Disposable); Hand Gloves (Heat & Chemical Resistant); Hand Gloves (Long); Hand gloves pairs (constricti on); Hand Glov

What it is used for

Gloves protect hands from hazards such as dirt, abrasion, chemicals, heat, or electrostatic discharge, depending on their type. Select gloves suited to the specific task and follow their usage limits.

How to use

1. Identify the task hazards and select gloves made for those hazards.
2. Check glove size, material, condition, and any expiry or inspection markings.
3. Wash and dry hands before putting on clean gloves where appropriate.
4. Wear gloves correctly and check that they fit securely without restricting movement.
5. Perform the task while avoiding contact with hazards beyond the glove's rating.
6. Remove gloves carefully to avoid transferring contamination to skin or other surfaces.
7. Dispose of single-use gloves correctly or clean reusable gloves according to instructions.

⚠ Safety

- Do not use gloves that are torn, contaminated, or unsuitable for the hazard.
- Do not assume ordinary gloves provide electrical insulation or chemical protection.
- Replace disposable gloves between contaminated tasks and follow hygiene procedures.
- Keep gloves away from machinery where they could become caught in moving parts.

Care & storage

- Store gloves clean and dry away from sunlight, chemicals, and excessive heat.
- Inspect reusable gloves regularly and replace them when protection is reduced.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gown

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Gown

What it is used for

A gown provides body coverage for hygiene, protection, or task-specific use in care, laboratory, or workplace settings.

How to handle

- Choose a gown appropriate for the task and required protection level.
- Inspect the gown for tears, contamination, and secure ties or fasteners before use.
- Put on and remove the gown according to the applicable PPE or infection-control procedure.
- Change the gown when contaminated or when moving between tasks that require clean protective clothing.
- Dispose of single-use gowns or launder reusable gowns according to site procedures.

Storage

- Keep unused gowns clean and protected in their original packaging or designated clean storage.
- Separate clean gowns from used or contaminated garments.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Handcuffs

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Handcuffs

What it is used for

Handcuffs are restraint devices intended for use by trained and legally authorized personnel. Follow applicable law, approved training, and the manufacturer's instructions for safe handling.

How to use

1. Confirm that use is legally authorized and consistent with the applicable procedure.
2. Inspect the cuffs, locking mechanism, hinges, and key for damage.
3. Explain the procedure where appropriate and follow approved training requirements.
4. Apply restraints only as trained, without excessive force or unnecessary delay.
5. Check fit and circulation according to the approved procedure and monitor the restrained person's condition.
6. Use the authorized key or release mechanism when removal is permitted.
7. Inspect and document the equipment's condition after use according to policy.

⚠ Safety

- Never use handcuffs for punishment, intimidation, or unauthorized restraint.
- Do not overtighten; monitor for pain, numbness, swelling, or changes in circulation.
- Seek appropriate medical assistance if injury or distress occurs.
- Keep keys controlled and follow the organization's reporting and storage procedures.

Care & storage

- Clean and inspect the cuffs according to the manufacturer's instructions and store them securely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Head Cap

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Hair Cap; Hair Cap-Packet; Head Caps (Cloth)

What it is used for

Head caps cover hair for hygiene, cleanliness, or task-specific protection.

How to handle

- Choose a cap suitable for the task and required hygiene standard.
- Inspect reusable caps for cleanliness, tears, and secure fit before use.
- Cover hair fully and replace disposable caps if torn or contaminated.
- Avoid sharing single-use caps and follow site infection-control rules where applicable.
- Remove caps carefully and perform hand hygiene when required.

Storage

- Keep disposable caps sealed until use.
- Wash reusable cloth caps according to instructions and store them clean and dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Insulated Gloves

Category: **Safety & PPE** Type: **Short SOP** Priority: **High** 3 labs

Covers these lab items: Dielectric safety gloves; Electrical Hand Gloves

What it is used for

Insulated gloves protect workers from electrical hazards when the gloves are correctly rated and used within their approved limits. Select the correct voltage class and follow the required inspection and testing procedures.

How to use

1. Check the glove class, voltage rating, certification, and test validity before use.
2. Inspect both gloves for cuts, punctures, cracks, moisture, and deterioration.
3. Perform the prescribed air-leak or inspection test according to the manufacturer's instructions.
4. Wear compatible leather protectors when required by the glove system.
5. Use gloves only for tasks covered by your electrical safety training and procedure.
6. Remove gloves carefully after work and inspect them again for damage.
7. Report defects and send gloves for required testing or replacement.

⚠ Safety

- Only qualified and authorised personnel should perform electrical work requiring insulating gloves.
- Never use wet, damaged, expired, or untested gloves.
- Do not exceed the gloves' rated voltage or use them outside their intended application.
- Follow isolation, lockout, and other electrical safety procedures; gloves alone do not make a circuit safe.

Care & storage

- Keep gloves clean and dry and store them away from sunlight, heat, oils, and sharp objects.
- Arrange electrical retesting and replacement according to applicable standards and manufacturer instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Insulated Rescue Pole

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Insulated rescue pole

What it is used for

An insulated rescue pole is designed to help trained personnel move a person or object away from an electrical hazard while maintaining an appropriate separation distance. Use only a correctly rated pole under the approved emergency procedure.

How to use

1. Check the pole's voltage rating, inspection label, and service status before use.
2. Inspect the insulating surface for cracks, contamination, moisture, or other damage.
3. Confirm that the pole is suitable for the electrical hazard and rescue situation.
4. Wear the required electrical PPE and follow the emergency response procedure.
5. Maintain the specified safe approach distance and use the pole only as trained.
6. After the rescue, isolate the hazard if authorized and arrange medical assessment for the affected person.
7. Remove the pole from service if damaged and report the incident.

⚠ Safety

- Only trained and authorized personnel should use the rescue pole.
- Never use a wet, dirty, damaged, or unrated pole near electrical hazards.
- Do not assume the pole makes every electrical situation safe.
- Follow site emergency procedures and contact emergency services as required.

Care & storage

- Clean, inspect, and store the pole according to the manufacturer's instructions and applicable electrical safety standards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Insulating Mat

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Insulated Rubber mating; Rubber Matting

What it is used for

Insulating rubber mats provide a protective standing surface for designated electrical or industrial work when correctly rated.

How to handle

- Verify that the mat is rated and approved for the specific electrical application.
- Inspect for cuts, cracks, punctures, moisture, oil, or other contamination before use.
- Keep the mat clean, dry, flat, and free from folds or obstructions.
- Do not treat a mat as a substitute for isolation, lockout procedures, or other required electrical PPE.
- Remove damaged or uncertified mats from service and arrange inspection or replacement.

Storage

- Store flat or rolled as directed by the manufacturer in a clean, dry location.
- Protect from sunlight, oils, chemicals, sharp objects, and heavy loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Knee Pads

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Additional protective equipment, such as knee pads and gaiters for rugged terrain; Knee pad

What it is used for

To protect the knees from impact, abrasion, and discomfort during kneeling or work on rough surfaces.

How to handle

- Inspect the pads, straps, fasteners, and protective shells for tears, cracks, or worn areas.
- Select pads suitable for the task and ensure they fit securely without restricting circulation or movement.
- Position the protective area over each kneecap and fasten the straps comfortably.
- Check the fit during work and reposition the pads if they slip; do not use damaged pads for protection.
- Clean and dry the pads after use, especially if exposed to dirt, sweat, or chemicals.

Storage

- Clean and dry the pads thoroughly before storing.
- Store flat or hang in a ventilated PPE area away from direct heat and chemical contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Locking System

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 20 labs

Covers these lab items: Locking systems

What it is used for

A locking system secures equipment, doors, storage areas, or access points against unauthorised use. Operate the specific lock according to its design and the site's access-control procedure.

How to use

1. Identify the lock type and confirm you are authorised to operate it.
2. Inspect the lock, key, latch, and mounting for damage.
3. Check that the door, cover, or equipment is in the correct position before locking.
4. Engage the lock using the specified key, code, or locking control.
5. Verify that the lock is fully engaged and the secured item cannot open unintentionally.
6. For authorised access, release the lock using the approved method.
7. Record access or lock status when required by site procedure.
8. Return keys or access devices to their designated secure location.

⚠ Safety

- Never force a jammed lock or bypass access controls without authorisation.
- Keep keys and access codes restricted to authorised users.
- Follow lockout and tagout procedures when locking equipment for maintenance.
- Report damaged locks or signs of tampering promptly.

Care & storage

- Keep locks clean and maintain them using the manufacturer's recommended method.
- Inspect keys, latches, and mounting points regularly and replace defective parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Metal Detector

Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Door Frame Metal Detector (DFMD); Hand Held Metal Detector (HHMD)

What it is used for

A metal detector identifies metal objects using electromagnetic sensing, either through a walk-through frame or a handheld scanner. Operate it according to the security procedure, check its function, and conduct screening respectfully and consistently.

Main parts

- Walk-through detector frame or handheld detector body
- Detection coils or sensing panels
- Control unit and status display
- Sensitivity and alarm settings
- Power supply, battery, or charging connection
- Audible alarm and indicator lights

Before you start

1. Confirm the detector type and follow the site's screening procedure.
2. Inspect the frame or handheld unit, cable, battery, and controls for damage.
3. Check the battery or power supply and confirm the unit starts correctly.
4. Perform the required function check using the approved test method.
5. Set sensitivity and alarm parameters according to authorised site settings.
6. Ensure the screening area is organised, accessible, and clearly marked.

How to use

1. Switch on the detector and allow any required self-test to finish.
2. Verify that the ready indicator or status display shows normal operation.
3. For a walk-through frame, direct the person to remove items as required by site policy.
4. Ask the person to walk through the frame at the specified pace and direction.
5. Observe alarm indicators and follow the site's secondary-screening procedure if an alarm occurs.
6. For a handheld detector, explain the process and maintain respectful screening distance.
7. Move the handheld unit slowly and evenly around the approved screening areas without touching the person unnecessarily.
8. Keep the detector at the recommended distance from the body or item being checked.
9. If an alarm occurs, follow authorised procedures to identify the source and resolve the alert.
10. Repeat the check when required by the screening protocol.
11. Record equipment faults or screening incidents according to site policy.
12. Switch off the device when no longer required and store it securely.

⚠ Safety

- Only trained and authorised personnel should conduct security screening.
- Follow privacy, consent, accessibility, and respectful-treatment requirements in site policy.
- Do not use a faulty detector or ignore failed function checks.
- Do not touch or search a person beyond the authority and procedure permitted at the site.
- Keep handheld detectors away from water and protect them from impact.
- Follow site emergency procedures if a prohibited or dangerous item is suspected.

Care & storage

- Clean the detector surfaces using a compatible approved method.
- Check batteries, charging connections, and cables regularly.
- Perform function checks at the intervals required by site policy.
- Protect sensing panels and handheld coils from impact or moisture.
- Report inconsistent alarms, damaged parts, or repeated faults.

Common mistakes to avoid

- Skipping the required function check before screening.
- Using incorrect sensitivity settings or ignoring site procedures.
- Moving the handheld detector too quickly or too far from the target.
- Treating an alarm as proof of a threat without following the secondary-check procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Outdoor Safety Kit

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: UV protection and hydration tools for outdoor safety during different weather conditions

What it is used for

Use outdoor safety supplies to reduce weather-related risks during field activities.

How to handle

- Check sunscreen, insect repellent, hats, and hydration supplies before departure.
- Follow product labels for application, reapplication, and precautions; avoid contact with eyes.
- Drink water regularly and take rest breaks according to weather and activity intensity.
- Watch for signs of heat illness and seek assistance promptly when symptoms occur.

Storage

- Store personal care products sealed and within their labelled temperature range.
- Keep reusable hats and equipment clean and dry, separate from used items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

PPE Kit

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 45 labs

Covers these lab items: Personal Protection Equipment; Personal Protection Equipment Kit; Personal Protective Equipment; Personal Protective Equipment Kit (PPE); Personal Protective Equipment (PPE); Personal protective equipment (PPE) like gloves, face masks, and hand sanitizers; Personal protective equipment (PPE) like gl

What it is used for

A personal protective equipment kit contains items selected to reduce exposure to workplace hazards. Choose the correct PPE for the task and inspect each item before use.

How to use

1. Identify the task hazards and check the required PPE list or risk assessment.
2. Inspect the kit and each item for damage, contamination, or expiry.
3. Select PPE that fits correctly and matches the identified hazards.
4. Put on the equipment in the recommended order and adjust it securely.
5. Check that protective items do not interfere with each other or safe movement.
6. Perform the task while following the workplace safety procedure.
7. Remove PPE carefully to avoid transferring contamination and clean or dispose of items as required.
8. Return reusable equipment to its designated storage location.

⚠ Safety

- Do not use damaged, expired, or unsuitable protective equipment.
- PPE does not replace hazard controls, safe work procedures, or required training.
- Use respirators and other specialised PPE only with appropriate training and fit checks.
- Report missing equipment or any defect before starting hazardous work.

Care & storage

- Clean reusable PPE according to the manufacturer's instructions and store it dry.
- Replace disposable items after use and inspect reusable equipment regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Padlock

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Padlocks together with chains

What it is used for

Handle padlocks and chains safely for securing equipment or restricted areas.

How to handle

- Check the lock body, shackle, chain links, and key or combination mechanism for damage.
- Use locks only for their intended security purpose and verify that the anchor point is suitable.
- Keep keys and combinations controlled and accessible to authorised personnel only.
- Do not use damaged chains or padlocks that may fail under load.

Storage

- Keep padlocks and keys in a labelled, secure location when not in use.
- Store chains clean and dry to reduce corrosion and tangling.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Phosphine Gas Monitor

Category: **Safety & PPE** Type: **Short SOP** Priority: **High** 6 labs

Covers these lab items: Phosphine Alert personal monitor; Phosphine gas Monitor

What it is used for

A phosphine gas monitor detects or warns of phosphine gas in the surrounding atmosphere. Use a suitable, functioning monitor to assess potential exposure and follow the site's emergency procedure.

How to use

1. Inspect the monitor, sensor openings, battery, display, and alarm indicators.
2. Confirm that the monitor is suitable for phosphine and within its calibration period.
3. Perform the prescribed bump test or function check before use.
4. Switch on the monitor in a safe area and allow its startup checks to finish.
5. Wear or position the monitor as specified by the manufacturer and site procedure.
6. Observe readings and alarms while working in the monitored area.
7. If an alarm activates, leave the area immediately and follow the site emergency plan.
8. Report alarms and arrange testing, calibration, or servicing before returning the monitor to service.

⚠ Safety

- Never enter a suspected contaminated area to test the gas without authorised training and protection.
- Do not ignore alarms or assume an area is safe from a single reading.
- Use the site's emergency response and evacuation procedures when an alarm occurs.
- Only trained personnel should perform calibration and sensor maintenance.

Care & storage

- Keep the monitor clean, dry, and within the specified operating conditions.
- Charge or replace batteries and complete scheduled bump tests and calibration.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Protective Guard

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Guard

What it is used for

A protective guard helps shield users from moving parts, flying debris, or other identified workplace hazards. Install and inspect the correct guard before operating the associated equipment.

How to use

1. Identify the hazard and confirm the guard type required for the equipment.
2. Inspect the guard, mounting points, fasteners, and locking features for damage.
3. Switch off and isolate the equipment before fitting or adjusting the guard.
4. Install the guard according to the manufacturer's instructions and secure all fasteners.
5. Check that the guard covers the hazard without obstructing intended operation.
6. Restore power only after confirming the guard is correctly fitted and secure.
7. Report damaged or missing guards and prevent operation until corrected.

⚠ Safety

- Never operate equipment with a required guard removed or bypassed.
- Keep hands and clothing away from hazardous moving parts.
- Do not modify the guard without approved authorization.
- Follow lockout procedures before maintenance or guard adjustment.

Care & storage

- Clean the guard regularly and inspect its fixings and condition before use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rainwear

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Rain ponchos, sun hats, and comfortable walking shoes

What it is used for

Use rainwear, sun hats, and walking shoes to improve comfort and safety outdoors.

How to handle

- Check rainwear for tears and footwear for suitable fit, grip, and damage before use.
- Choose clothing and shoes appropriate to the weather, terrain, and activity.
- Keep visibility high in low-light conditions and avoid footwear that slips easily.
- Clean and dry gear after use; replace damaged items that no longer provide protection.

Storage

- Dry rainwear and shoes fully before storage.
- Keep clean gear in a ventilated area away from direct sunlight and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rescue Hook

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Safety rescue hook

What it is used for

A safety rescue hook helps trained responders move or separate a person or object from a hazard while maintaining distance. Use it only for approved rescue situations and follow the site's emergency procedure.

How to use

1. Inspect the hook, shaft, handle, insulation, and attachments for damage.
2. Confirm that the hook is rated for the intended rescue hazard and application.
3. Assess the scene from a safe position and alert the emergency response team.
4. Wear required protective equipment and maintain a clear escape route.
5. Use the hook only from a safe position and follow trained rescue procedures.
6. Move the affected person or object only when the action is safe and authorized.
7. After use, isolate the equipment if needed, inspect it, and report any damage.

⚠ Safety

- Do not use a rescue hook near live electrical hazards unless it is specifically rated and trained procedures permit it.
- Never enter an unsafe scene without proper authorization and protection.
- Do not use the hook as a lifting device unless designed and rated for lifting.
- Call emergency services and follow the site's rescue plan.

Care & storage

- Clean and inspect the hook after use and store it in its designated emergency location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Respirator Mask

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Chemical Mask

What it is used for

A chemical respirator helps reduce inhalation exposure when equipped with the correct approved filter or cartridge for the identified hazard. Select it through the workplace respiratory protection program and follow fit-testing and manufacturer instructions.

How to use

1. Identify the chemical hazard and consult the safety data sheet and risk assessment.
2. Select a respirator and cartridge approved for the specific contaminant and exposure conditions.
3. Inspect the facepiece, straps, valves, seals, and cartridge expiry date before use.
4. Fit the respirator according to manufacturer instructions and perform the required user seal check.
5. Enter the work area only after confirming suitable protection and required authorization.
6. Leave the area if breathing becomes difficult, leakage occurs, or contamination is suspected.
7. Remove the respirator safely, clean reusable parts as directed, and store it properly.

⚠ Safety

- A respirator does not protect against every chemical or oxygen-deficient atmosphere.
- Do not use a cartridge beyond its approved service life or for an unknown hazard.
- Use fit testing and medical clearance where required by the respiratory protection program.
- For unknown, immediately dangerous, or oxygen-deficient atmospheres, do not enter without a trained response team and appropriate supplied-air equipment.

Care & storage

- Replace filters and cartridges according to the approved schedule and store the respirator sealed in a clean, dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safety Data Sheet

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Material Safety Data Sheet

What it is used for

Use Safety Data Sheets to identify hazards and follow safe handling procedures.

How to handle

- Locate the current SDS for the exact product or chemical before use.
- Review hazard identification, PPE, first aid, spill response, storage, and disposal sections.
- Make the SDS accessible to staff and trainees who may handle the substance.
- Follow the SDS and local procedures during incidents; seek emergency help when required.

Storage

- Keep printed or digital SDS documents organised and readily accessible.
- Maintain current versions and remove obsolete copies from active reference points.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safety Goggles

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 28 labs

Covers these lab items: Goggles; Safety glasses; Safety Goggle; Safety goggles

What it is used for

Safety goggles protect the eyes from specified hazards such as flying particles, splashes, or dust. Select goggles suitable for the task and ensure they fit securely before work begins.

How to use

1. Check the lens, frame, strap, and seals for cracks, scratches, or damage.
2. Confirm that the goggles are rated for the task hazard.
3. Clean the lenses with an approved method before wearing them.
4. Position the goggles securely over the eyes and adjust the strap for a proper fit.
5. Check that the goggles provide suitable coverage and do not obstruct vision.
6. Wear them throughout the task and replace them if damaged or visibility becomes poor.
7. Remove goggles carefully and clean or store them after use.

⚠ Safety

- Do not use cracked, badly scratched, or unsuitable goggles.
- Do not assume ordinary glasses provide the same protection as safety goggles.
- Use additional face or respiratory protection when the task requires it.
- Keep goggles on while the hazard remains present.

Care & storage

- Clean lenses using suitable materials to avoid scratches.
- Store goggles in a clean protective case away from heat and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safety Helmet

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 17 labs

Covers these lab items: Head protection; Helmets/hard hats; Safety helmet; Safety helmets

What it is used for

A safety helmet protects the head from specified workplace hazards such as falling objects or impact. Select a helmet suited to the hazard and ensure it is correctly fitted before entering the work area.

How to use

1. Inspect the shell, suspension, chin strap, and adjustment mechanism for cracks or damage.
2. Confirm the helmet type and rating are suitable for the worksite hazards.
3. Adjust the suspension and headband to obtain a secure, comfortable fit.
4. Fasten the chin strap when required by the task or site rules.
5. Check that the helmet sits correctly and does not obstruct vision.
6. Wear the helmet in designated areas and replace it after a significant impact as required.
7. Clean and store the helmet according to the manufacturer's instructions.

⚠ Safety

- Do not use a cracked, modified, expired, or impact-damaged helmet.
- Do not drill holes, paint, or attach unapproved accessories to the helmet.
- Keep the helmet away from chemicals, excessive heat, and direct sunlight during storage.
- A helmet does not replace other required PPE or safe work practices.

Care & storage

- Clean the shell and suspension with approved materials and inspect them regularly.
- Store the helmet in a clean, dry location away from impact and heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safety Installation

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Safety installations

What it is used for

Handle and maintain installed safety systems to protect people and property.

How to handle

- Inspect safety installations for visible damage, obstruction, and signs of malfunction.
- Do not disable, alter, or bypass safety systems without authorised approval.
- Keep emergency exits, alarms, extinguishers, and protective installations accessible.
- Report defects promptly and arrange inspection or repair by qualified personnel.

Storage

- Keep installation records, inspection logs, and maintenance instructions in a designated location.
- Store spare safety components in labelled, secure conditions according to manufacturer guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safety Shoes

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 18 labs

Covers these lab items: Boots; Gumboots; Rubber Chappals; Safety footwear; Safety shoes

What it is used for

Safety shoes protect the feet from hazards such as impact, compression, punctures, or slipping, depending on their design and rating. Select footwear suitable for the workplace and inspect it before use.

How to use

1. Check the footwear type, size, sole, toe protection, and condition.
2. Confirm that the footwear rating matches the workplace hazards.
3. Wear the shoes securely and fasten laces or closures properly.
4. Check that the soles provide suitable grip and the fit allows safe movement.
5. Use the footwear throughout the task and avoid removing it in hazardous areas.
6. Clean the footwear after use and inspect it for damage.
7. Dry and store the shoes in a suitable place.

⚠ Safety

- Do not use footwear with damaged soles, exposed toe caps, or broken closures.
- Use electrical hazard-rated footwear only where appropriate and as part of a broader safety system.
- Keep soles free from grease, mud, and materials that reduce grip.
- Replace footwear when protective performance is compromised.

Care & storage

- Clean footwear with materials suitable for its construction.
- Dry naturally away from direct heat and inspect regularly for wear.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safety Station

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Lab Safety/Emergency Booth

What it is used for

To provide quick access to designated laboratory emergency equipment and safety supplies during an incident.

How to handle

- Identify the station's contents, such as first-aid supplies, eyewash, safety shower, fire equipment, or emergency instructions, according to its actual setup.
- Keep the station clearly marked, unobstructed, and accessible at all times.
- Check inspection tags, expiry dates, seals, pressure indicators, and supply levels according to the site's schedule.
- Use each item only for its intended purpose and follow posted instructions and the laboratory emergency plan.
- Raise the alarm and contact trained emergency personnel when required; report missing, used, expired, or damaged supplies promptly.

Storage

- Return supplies to their labelled positions and keep emergency access routes clear.
- Maintain a documented inspection schedule and replenish or service items according to site policy and manufacturer instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safety Tag

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Safety tags

What it is used for

Use safety tags to communicate equipment status, hazards, or restricted operation.

How to handle

- Select a tag appropriate to the hazard and follow the site's tagging procedure.
- Write clear, accurate information and include required dates or authorised identification.
- Attach the tag securely where it is clearly visible without obstructing controls or access.
- Never remove or alter another person's safety tag unless authorised under the formal procedure.

Storage

- Keep unused tags clean, dry, and organised by type.
- Store tags and attachment ties in a labelled location near the designated safety supplies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Safety Vest

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: High-visibility vests, mosquito nets, and sun protection gear; Reflective jacket; Reflective jackets

What it is used for

To improve wearer visibility in work areas by wearing a high-visibility safety vest. To remain visible to vehicle operators and coworkers in low-light or busy environments.

Main parts

- High-visibility fabric
- Reflective strips
- Front closure
- Pockets, if fitted
- Size and fit labels

Before you start

1. Select a vest suitable for the task and required visibility level.
2. Check the vest for tears, damaged closures, and worn reflective strips.
3. Ensure the vest fits over the intended clothing without restricting movement.
4. Confirm that the vest is clean enough for reflective features to work.

How to use

1. Put on the vest and close the front fastener.
2. Ensure reflective strips are visible and not covered by bags or equipment.
3. Adjust the vest so it fits securely without obstructing movement.
4. Wear it throughout the designated work activity.
5. Replace or report a vest that becomes damaged or heavily soiled.
6. Remove and store the vest after the task.

⚠ Safety

- A vest does not replace other required PPE.
- Use task-appropriate visibility protection near moving vehicles and machinery.
- Do not wear a torn or poorly visible vest in hazardous areas.
- Follow site-specific PPE requirements.

Care & storage

- Clean according to the garment label.
- Inspect reflective strips and closures regularly.

- Store in a clean, dry place away from chemicals.
- Replace when visibility performance is reduced.

Common mistakes to avoid

- Wearing the vest open or hidden beneath outer clothing.
- Using a dirty or damaged vest.
- Assuming the vest alone protects against traffic or machinery hazards.
- Choosing a size that restricts movement.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sand Bag

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Sand Bag; Sand snakes

What it is used for

Sand bags provide ballast, support, or temporary barriers for approved safety and training applications.

How to handle

- Inspect bags for tears, leakage, weak seams, or damaged handles before use.
- Fill only to a manageable weight and close securely to prevent sand leakage.
- Use safe lifting techniques and team assistance for heavy bags.
- Stack bags on stable ground and avoid unstable piles or blocked access routes.
- Keep sand bags away from sharp objects that may puncture them.

Storage

- Store in a dry, designated location protected from weather where required.
- Arrange bags securely and inspect periodically for deterioration.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Security Guard Belt

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Security Guard Belts

What it is used for

Handle security guard belts safely for uniform and duty use.

How to handle

- Inspect the belt, buckle, stitching, and attachments for wear or damage.
- Adjust the belt securely without restricting breathing or movement.
- Attach only approved accessories and ensure they do not create snagging hazards.
- Keep buckles and fittings away from sharp edges and excessive force.

Storage

- Clean according to the material care instructions and dry fully before storage.
- Store flat or hung in a clean, dry area away from direct sunlight and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Security System

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Security alarms and locking systems; Security Systems; Security systems: cameras, guest ID scanners

What it is used for

To operate a security system such as cameras, alarms, access control, or guest identification equipment. To monitor activity and manage access according to authorized security procedures.

Main parts

- Camera or sensor
- Control panel or recorder
- Alarm and status indicators
- Access reader or identification scanner
- Power supply and network connections
- Locks and door contacts, if fitted

Before you start

1. Confirm authorization and identify the system components to be used.
2. Check power, network connections, camera views, and alarm status.
3. Verify that date, time, and recording settings are correct where applicable.
4. Ensure access permissions and user accounts are approved.
5. Check that cameras and sensors are not obstructed or tampered with.

How to use

1. Log in using an authorized account.
2. Review system status and confirm connected devices are online.
3. Arm or disarm alarms only according to the approved procedure.
4. Monitor camera feeds or access events as required.
5. Verify guest identity using the authorized process when applicable.
6. Record and report suspicious activity or system faults.
7. Log out after use and secure the control area.

⚠ Safety

- Protect passwords and access credentials.
- Do not disable alarms, cameras, or access controls without authorization.
- Handle personal data according to privacy policy and applicable law.
- Report suspected tampering or unauthorized access promptly.

Care & storage

- Keep cameras and sensors clean and unobstructed.
- Check system alerts and device connections regularly.
- Update software and credentials according to IT policy.
- Arrange authorized maintenance for faults.

Common mistakes to avoid

- Sharing login credentials.
- Ignoring offline devices or alarm faults.
- Recording or sharing personal data without authorization.
- Leaving the system logged in and unattended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shoe Cover

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Shoe Cover; Shoe Cover-Packet

What it is used for

Use shoe covers to reduce contamination in designated areas.

How to handle

- Wear shoe covers before entering the area where they are required.
- Check for tears and ensure the covers fit securely over footwear.
- Avoid walking on wet or slippery surfaces if the covers reduce traction.
- Remove covers without touching the contaminated outer surface and dispose of single-use covers appropriately.

Storage

- Keep unused covers sealed in their original packet.
- Store in a clean, dry place away from used PPE and waste.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Signage

Category: **Safety & PPE** Type: **Handling note** Priority: **Low** 58 labs

Covers these lab items: Display of emergency contact numbers (including hospitals having dedicated treatment facility for hydrogen-related accidents); Energy & Environment (EE); Flags, markers, and trail signs for navigation practice; Food Processing Techniques (FPT); Gardening, Nursery & Agriculture Techniques (GNAT); Off

What it is used for

Signage communicates directions, hazards, emergency contacts, environmental messages, and other important site information.

How to handle

- Verify wording, contact details, symbols, and visibility before installing the sign.
- Use the correct sign type and place it at the designated, clearly visible location.
- Secure signs firmly and ensure they do not obstruct exits, walkways, or equipment controls.
- Replace damaged, faded, outdated, or incorrect signage promptly.

Storage

- Store spare signs clean and flat or in protective packaging to prevent damage.
- Keep installation accessories labelled and organised.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Smoke Detector

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Smoke Detectors

What it is used for

A smoke detector identifies smoke and provides an audible or connected alarm to warn occupants of a possible fire. Test and maintain the detector according to the manufacturer's instructions and local fire safety requirements.

How to use

1. Inspect the detector housing, mounting, indicator, and surrounding area for damage or obstruction.
2. Confirm the detector type and check the installation location against the approved fire safety plan.
3. Check the power source and replace batteries if required by the instructions.
4. Use the designated test function to verify the alarm and connected notification system.
5. Confirm that the alarm can be heard or received in the intended area.
6. Record the test result and report any fault or failed alarm.
7. Reset the detector if required and ensure it returns to normal status.

⚠ Safety

- Never disable a smoke detector without authorized controls and an approved alternative fire safety arrangement.
- Do not test with open flames or unapproved smoke sources.
- Follow electrical isolation procedures before servicing hardwired detectors.
- Treat any genuine alarm as an emergency and follow evacuation procedures.

Care & storage

- Keep vents free of dust, follow scheduled cleaning, and replace the detector at its specified service life.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Survival Kit

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Emergency shelters, survival kits, and portable stretchers

What it is used for

A survival kit, emergency shelter, and portable stretcher support basic emergency preparedness, temporary protection, and casualty transport. Check the kit contents and use each item according to its intended purpose and training.

How to use

1. Inspect the kit, shelter fabric, poles, fasteners, stretcher handles, and straps for damage.
2. Confirm that essential supplies are present, in date, and suitable for the intended emergency.
3. Choose a safe location for deploying the shelter and clear the setup area.
4. Assemble the shelter according to its instructions and secure it against expected conditions.
5. For stretcher use, assess the casualty and request trained assistance before moving them.
6. Secure the casualty using the approved straps and coordinate lifting with the trained team.
7. After use, clean reusable items, replenish supplies, and record missing or damaged equipment.

⚠ Safety

- Do not enter unsafe areas without appropriate training, authorization, and protective equipment.
- Do not move a casualty with suspected spinal injury unless immediate danger requires it or trained responders direct the action.
- Never exceed the stretcher's rated load.
- Follow emergency service procedures and protect yourself from environmental hazards.

Care & storage

- Replace expired supplies, dry shelter components fully, and store the kit ready for inspection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Under Vehicle Inspection Mirror

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Under Chassis Inspection Mirror (UCIM)

What it is used for

An under-vehicle inspection mirror helps inspect areas beneath a vehicle that are difficult to view directly. Use it from a safe position to check for visible leaks, damage, or loose components.

How to use

1. Inspect the mirror, handle, adjustment joint, and any built-in light for damage.
2. Park the vehicle on a level surface, apply the parking brake, and secure it against movement.
3. Switch off the engine and allow hot components to cool where practical.
4. Adjust the mirror angle and lighting before positioning it beneath the vehicle.
5. Inspect visible areas from outside the vehicle's underside clearance.
6. Record any suspected leaks, damage, or loose components for qualified follow-up.
7. Clean the mirror and return it to its storage location.

⚠ Safety

- Never crawl beneath a vehicle supported only by a jack.
- Keep hands and the mirror clear of moving parts and hot exhaust components.
- Do not inspect beneath a running vehicle unless an approved procedure specifically requires it.
- Use suitable lighting and personal protective equipment where necessary.

Care & storage

- Clean the mirror with a suitable cloth and store it where the reflective surface will not be scratched.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Warning Light

Category: **Safety & PPE** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Emergency Warning Lights

What it is used for

Emergency warning lights alert people to hazards, restricted areas, or emergency situations. Place and operate the light according to the site emergency plan and ensure it remains visible to approaching people.

How to use

1. Inspect the warning light, lens, casing, switch, battery, and power cable for damage.
2. Confirm the light type and power requirements are suitable for the intended location.
3. Choose a visible, stable position away from traffic obstruction and other hazards.
4. Secure the light and aim it toward the intended approach or warning area.
5. Switch on the light and verify that the signal is clearly visible.
6. Monitor the warning light during the emergency or restricted activity.
7. Switch off, collect, inspect, and store the light after the area is safe.

⚠ Safety

- Do not use damaged lights, batteries, or power cables.
- Do not place warning lights where they create a trip hazard or block emergency access.
- Use approved traffic-control procedures when warning people near vehicle routes.
- Do not rely on a warning light as the only emergency safeguard.

Care & storage

- Recharge or replace batteries as required and test the light periodically according to site procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Welding Apron

Category: **Safety & PPE** Type: **Handling note** Priority: **Medium** 2 labs

Covers these lab items: Welding Apron

What it is used for

A welding apron protects the torso and clothing from sparks, spatter, and heat during suitable welding tasks.

How to handle

- Inspect the apron for holes, tears, damaged fasteners, and worn protective material.
- Select an apron suitable for the welding process and associated heat and spatter hazards.
- Wear it securely over suitable work clothing with other required welding PPE.
- Replace the apron if damage compromises protection or contamination makes it unsafe.

Storage

- Store clean and dry away from flames, chemicals, sharp tools, and excessive heat.
- Keep it hanging or folded without damaging protective layers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Welding Goggles

Category: **Safety & PPE** Type: **Handling note** Priority: **Medium** 2 labs

Covers these lab items: Welding Goggle

What it is used for

Welding goggles protect the eyes from specified welding radiation, sparks, and flying particles when correctly selected.

How to handle

- Inspect lenses, frames, seals, and straps for cracks, damage, or poor fit.
- Use the correct approved lens shade for the welding or cutting process.
- Wear goggles correctly and use additional face protection when required by the task.
- Stop work and replace damaged or unsuitable goggles immediately.

Storage

- Store in a clean protective case away from heat, chemicals, and scratching.
- Keep lenses clean and dry between uses.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Welding Helmet

Category: **Safety & PPE** Type: **Handling note** Priority: **Medium** 3 labs

Covers these lab items: Hand shields; Welding Screen with Head Band

What it is used for

A welding helmet or hand shield protects the face and eyes from welding radiation, sparks, and hot particles.

How to handle

- Inspect the helmet, headband, shield, lens, and filter rating before each use.
- Select the correct filter shade for the welding process and current according to safety guidance.
- Wear the helmet or shield with suitable protective clothing and eye protection as required.
- Position the shield correctly before striking an arc and keep bystanders protected with suitable screens.

Storage

- Store clean and dry in a protective location away from heat, chemicals, and impact.
- Replace damaged lenses, shields, or headbands before further use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Aari Needle

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Aari Needles

What it is used for

An Aari needle is used for decorative embroidery with a hooked needle and suitable fabric setup. Handle the sharp hook carefully and use the correct needle for the embroidery work.

How to use

1. Inspect the needle and handle for damage, bending, or looseness.
2. Select a needle size and thread suitable for the fabric and design.
3. Secure the fabric in the embroidery frame or hoop.
4. Fit the needle into its handle or holder according to the design.
5. Guide the hook through the fabric using small, controlled movements.
6. Keep thread tension even and check the embroidery regularly.
7. Place the needle safely aside and collect loose threads after finishing.

⚠ Safety

- Keep the sharp hook pointed away from fingers and other people.
- Do not leave the needle loose on a chair, table edge, or floor.
- Use adequate lighting and avoid rushing during embroidery.

Care & storage

- Wipe the needle dry after use and store it in a protective needle case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Adda Frame

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Adda Frame

What it is used for

An adda frame holds fabric stretched for embroidery, stitching, or decorative textile work. Set it up securely so the fabric remains stable without being damaged.

How to use

1. Inspect the frame, joints, supports, and fasteners for damage or looseness.
2. Place the frame on a stable surface or secure it in its intended stand.
3. Position the fabric evenly across the frame without twisting it.
4. Tighten the frame gradually until the fabric is firm but not overstretched.
5. Check that the working area is well lit and comfortably positioned.
6. Work with suitable needles and tools while keeping the fabric secure.
7. Release the fabric gently and dismantle the frame after finishing if needed.

⚠ Safety

- Keep sharp needles, scissors, and tools away from the face and fingers.
- Do not overtighten joints or lean heavily on the frame.
- Secure the frame to prevent slipping or tipping during use.

Care & storage

- Remove dust and loose threads, wipe the frame dry, and store it in a clean place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Art Mannequin

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Art Mannequins

What it is used for

Handle art mannequins carefully for drawing, design, and display activities.

How to handle

- Place the mannequin on a stable surface before adjusting its pose.
- Move joints gently and do not force stiff or restricted parts.
- Keep the mannequin away from edges where it could fall.
- Clean off paint or drawing materials using manufacturer-approved methods.

Storage

- Store upright or in a supported position in a designated area.
- Keep dry and protected from impact, excessive heat, and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bamboo Craft Tool

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Equipment used for developing bamboo products

What it is used for

Bamboo craft tools cut, split, shape, smooth, or join bamboo for craft products. Select the correct tool for the task and secure bamboo before working.

How to use

1. Inspect each tool's handle, cutting edge, fasteners, and moving parts for damage.
2. Select the appropriate tool for cutting, splitting, shaping, or smoothing bamboo.
3. Wear suitable eye protection and gloves appropriate to the tool and task.
4. Secure the bamboo on a stable work surface before cutting or shaping.
5. Work with controlled strokes and keep hands clear of cutting paths.
6. Remove splinters and offcuts using a safe method after the tool stops.
7. Clean the tools and return them to their designated storage.

⚠ Safety

- Keep sharp edges and pointed tools away from fingers and other people.
- Do not hold small bamboo pieces in the cutting path by hand.
- Wear a suitable dust mask when sanding or creating fine bamboo dust.

Care & storage

- Remove bamboo dust, dry metal parts, and store sharp tools with protective covers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bar Tack Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bar-tack machine

What it is used for

A bar tack machine sews short, dense reinforcement stitches at stress points on garments and textile products. Use the correct stitch program, thread, needle, and fabric positioning to produce secure tacks while preventing needle injuries and machine damage.

Main parts

- Machine head and sewing needle
- Presser foot and fabric guide
- Needle and bobbin thread system
- Stitch pattern and length controls
- Foot pedal or electronic start control
- Motor, needle guard, and emergency stop, if fitted

Before you start

1. Confirm the required bar-tack pattern, stitch density, fabric type, and reinforcement location.
2. Inspect the needle, presser foot, thread path, bobbin, guards, and power cable for damage.
3. Ensure the needle is correctly fitted, sharp, and suitable for the fabric and thread.
4. Thread the machine and install the bobbin according to the manufacturer's instructions.
5. Select the approved stitch program and verify the settings on a sample piece.
6. Keep the work area clear and wear suitable eye protection where needle breakage or flying fragments are a risk.

How to use

1. Switch off the machine before changing the needle, threading near the needle, or clearing a jam.
2. Select the correct bar-tack program and set the approved stitch length, width, and density.
3. Position the fabric so the reinforcement area lies under the needle and within the designated guide.
4. Lower the presser foot and ensure the fabric is held flat without folds or excessive tension.
5. Keep fingers away from the needle path and activate the machine using the approved control.
6. Observe the first tack for correct placement, stitch formation, and secure thread locking.
7. Stop the machine if the needle breaks, thread tangles, fabric shifts, or abnormal noise occurs.
8. Raise the presser foot and remove the fabric only after the needle has stopped in a safe position.
9. Inspect the tack for skipped stitches, loose threads, incorrect length, and fabric damage.
10. Trim loose threads with suitable scissors while keeping hands clear of the needle.
11. Switch off the machine and clean lint from the needle plate, bobbin area, and accessible guides.

2. Record quality issues and arrange adjustments or maintenance if the stitch pattern is inconsistent.

⚠ Safety

- Only trained users should operate or adjust the bar-tack machine.
- Keep fingers away from the needle and presser foot while the machine is running.
- Never bypass needle guards or other safety devices.
- Switch off and isolate power before replacing a needle, servicing electrical parts, or clearing a jam.
- Do not pull fabric forcibly while the needle is engaged.
- Keep hair, jewellery, and loose clothing away from moving mechanisms.

Care & storage

- Remove lint and thread waste from the needle plate and bobbin area after shutdown.
- Inspect and replace bent, blunt, or damaged needles promptly.
- Check thread tension, bobbin condition, presser foot, and needle alignment regularly.
- Lubricate designated points only as directed by the manufacturer.
- Arrange servicing if stitch density, timing, or machine movement becomes irregular.

Common mistakes to avoid

- Using the wrong stitch program or needle for the fabric.
- Placing fingers too close to the needle during sewing.
- Pulling the fabric while the needle is engaged, causing needle breakage.
- Ignoring skipped stitches, loose thread locking, or repeated thread breakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Blind Stitch Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Blind stitch machine

What it is used for

A blind stitch machine creates nearly invisible stitches that secure folded fabric edges, hems, or linings without showing prominent stitches on the outer surface. Correct needle selection, stitch depth, and fabric positioning are important to achieve a neat finish while preventing needle injuries and fabric damage.

Main parts

- Machine head and needle
- Blind-stitch presser foot and fabric guide
- Thread spool and tension assembly
- Stitch-depth or bite adjustment
- Feed mechanism and work table
- Motor, pedal or start control, and protective guard

Before you start

1. Confirm the fabric type, hem construction, and required blind-stitch finish.
2. Inspect the needle, presser foot, thread path, feed mechanism, guard, and power cable for damage.
3. Select a needle and thread suitable for the fabric weight and fibre type.
4. Thread the machine and set the tension and stitch depth according to the manual.
5. Fold and position the hem so the needle catches the intended amount of inner fabric without making visible outer stitches.
6. Test the setup on a matching fabric scrap and check stitch visibility and strength.

How to use

1. Switch off the machine before changing the needle, threading near the needle, or adjusting internal components.
2. Install the correct needle and presser foot, ensuring they are secure and aligned.
3. Set the blind-stitch depth and thread tension according to the sample result and job specification.
4. Position the folded hem against the guide and place the fabric under the presser foot.
5. Lower the presser foot and ensure the fabric lies flat without pulling or twisting.
6. Start the machine using the approved pedal or control and guide the fabric steadily along the hem.

7. Observe the first stitches to confirm that they catch the fold correctly and remain inconspicuous on the outer side.
8. Stop the machine if the needle breaks, thread tangles, fabric slips, or unusual noise occurs.
9. Raise the presser foot and remove the fabric only after the needle has stopped safely.
10. Inspect the hem for skipped stitches, visible outer stitches, puckering, and uneven tension.
11. Trim loose thread ends with suitable scissors and avoid placing fingers near the needle.
12. Switch off the machine, clean lint from the needle plate and feed area, and record any quality issues.

⚠ Safety

- Only trained users should operate or adjust the sewing machine.
- Keep fingers away from the needle and presser foot while the machine is running.
- Never bypass needle guards or other safety devices.
- Switch off and isolate power before replacing the needle, clearing a jam, or performing maintenance.
- Do not pull or twist fabric forcibly while the needle is engaged.
- Keep loose clothing, hair, and jewellery away from the moving parts.

Care & storage

- Remove lint and thread waste from the needle plate, feed mechanism, and accessible areas after shutdown.
- Replace bent, blunt, or damaged needles promptly.
- Check thread tension, guides, presser foot, and feed movement regularly.
- Lubricate designated points only as instructed by the manufacturer.
- Arrange servicing if stitch depth becomes inconsistent, the machine skips stitches repeatedly, or unusual vibration occurs.

Common mistakes to avoid

- Setting the stitch depth too deep and creating visible outer stitches.
- Using unsuitable needle or thread for the fabric.
- Pulling the fabric while the needle is engaged.
- Ignoring skipped stitches, puckering, or incorrect thread tension.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Block Printing Equipment

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Equipment used for developing hand block printed products; Wooden blocks used for hand block printing; Wooden printing blocks

What it is used for

To create printed patterns on fabric or paper using hand block printing equipment. To achieve clear, repeatable designs through even colour application and careful block alignment.

Main parts

- Carved wooden printing blocks
- Printing table or padded surface
- Fabric or paper
- Printing ink or fabric dye
- Ink tray or pad
- Roller or application tool

Before you start

1. Wash or prepare the fabric and allow it to dry if required.
2. Cover the work surface and secure the material flat.
3. Inspect the wooden blocks for cracks, loose pieces, and blocked details.
4. Prepare the approved printing colour and check its consistency.
5. Wear suitable gloves or apron when required by the colour product.

How to use

1. Pour a small amount of printing colour onto the tray or pad.
2. Spread the colour evenly across the block's raised design.
3. Position the block over the intended print location.
4. Press the block straight down with even pressure.
5. Lift the block vertically to avoid smearing.
6. Align the next impression with the previous pattern and repeat.
7. Allow the print to dry or cure according to the colour instructions.
8. Clean the blocks and work area after printing.

⚠ Safety

- Use printing colours according to their safety instructions.
- Avoid contact with eyes and prolonged skin contact.
- Keep knives and carving tools away from the printing area when not in use.
- Maintain ventilation when required by the product instructions.

Care & storage

- Remove colour residue before it dries.
- Clean wooden blocks gently without soaking them unnecessarily.
- Dry blocks fully before storage.
- Store blocks flat in a dry place to prevent warping.

Common mistakes to avoid

- Applying too much colour to the block.
- Moving the block while pressing or lifting.
- Misaligning repeated impressions.
- Storing damp blocks or leaving colour residue on them.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bobbin

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Bobbin; Bobbins and other Sewing Machine Attachments; Bobbins, Bobbin case

What it is used for

A bobbin holds the lower thread used by a sewing machine to form stitches. Use the bobbin type and thread winding recommended for the machine.

How to use

1. Switch off the sewing machine before opening the bobbin area.
2. Remove the bobbin and bobbin case according to the machine instructions.
3. Check the bobbin for cracks, burrs, and tangled thread.
4. Wind the correct thread evenly onto the bobbin without overfilling it.
5. Insert the bobbin in the correct direction and route the thread through its guide.
6. Reassemble the bobbin area, test stitches on scrap fabric, and adjust if needed.

⚠ Safety

- Keep fingers away from moving machine parts while threading or testing.
- Do not use cracked, bent, or incompatible bobbins.
- Switch off the machine before removing or installing the bobbin.

Care & storage

- Remove lint regularly and store spare bobbins in a clean, dry container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Bobbin Case

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Bobbin case

What it is used for

A bobbin case holds the bobbin and helps feed lower thread in a sewing machine. Install the correct case and check thread movement before sewing.

How to use

1. Switch off the sewing machine before opening the bobbin area.
2. Remove the bobbin case and inspect it for lint, damage, or distortion.
3. Insert the correctly wound bobbin in the direction specified by the machine.
4. Guide the thread through the case slot and tension spring as instructed.
5. Fit the bobbin case securely into its holder.
6. Turn the handwheel slowly to check thread movement before sewing.
7. Close the bobbin cover and test stitching on scrap fabric.

⚠ Safety

- Keep fingers away from the needle and moving machine parts.
- Never remove or adjust the bobbin case while the machine is running.
- Do not force an incompatible or damaged bobbin case into place.

Care & storage

- Remove lint with a suitable brush and store spare cases in a clean container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Button Sewing Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Button sewing machine

What it is used for

A button sewing machine attaches buttons to garments using a programmed or mechanically controlled stitch pattern. Select the correct needle, thread, button clamp, and stitch settings to ensure secure attachment while preventing needle injuries and fabric damage.

Main parts

- Machine head and sewing needle
- Button clamp or button holder
- Needle and bobbin thread system
- Stitch pattern and tension controls
- Presser mechanism and fabric support
- Motor, start pedal, and protective guard

Before you start

1. Confirm the button type, size, hole pattern, fabric, and required attachment strength.
2. Inspect the needle, button clamp, thread path, bobbin, guards, and power cable for damage.
3. Fit a needle suitable for the fabric and thread and ensure it is securely mounted.
4. Thread the machine and install the correct button holder or clamp.
5. Select the appropriate stitch pattern and settings according to the machine manual.
6. Test the setup on a sample fabric and check button security before production.

How to use

1. Switch off the machine before changing the needle, threading near the needle, or adjusting the clamp.
2. Install the correct needle and button holder, then confirm that the holder is securely fixed.
3. Set the stitch pattern, stitch count, and thread tension according to the button and fabric specifications.
4. Position the button in the designated holder and ensure its holes align with the needle path.
5. Place the garment under the clamp with the button attachment point correctly aligned.
6. Lower the clamp and confirm that the fabric and button are held securely without excessive tension.
7. Keep fingers clear of the needle and start the machine using the approved control.
8. Observe the first sewing cycle to confirm that stitches pass through the intended button holes.
9. Stop the machine if the button shifts, the needle breaks, thread tangles, or unusual noise occurs.
10. After the needle stops safely, release the clamp and remove the garment carefully.

1. Check that the button is secure, correctly positioned, and free from loose or excessive thread.
2. Switch off the machine, remove lint and thread waste, and record any quality issues.

⚠ Safety

- Only trained operators should use or adjust the machine.
- Keep fingers away from the needle, button clamp, and moving mechanisms during operation.
- Never bypass needle guards or safety interlocks.
- Switch off and isolate power before changing needles, clearing jams, or performing maintenance.
- Do not hold or reposition a button by hand while the needle is moving.
- Keep loose clothing, hair, and jewellery away from moving parts.

Care & storage

- Clean lint and thread waste from the needle plate, clamp, and bobbin area after shutdown.
- Inspect the needle, button holder, thread guides, and clamp alignment regularly.
- Replace bent or blunt needles promptly.
- Lubricate only the designated points according to the manufacturer's instructions.
- Arrange servicing if stitches become inconsistent, the clamp slips, or the machine develops unusual noise.

Common mistakes to avoid

- Using a button holder or stitch pattern unsuitable for the button.
- Misaligning the button holes with the needle path.
- Placing fingers close to the needle during sewing.
- Ignoring loose stitches or failing to check button security after sewing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Buttonhole Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Button hole sewing machine

What it is used for

A buttonhole machine creates reinforced openings in fabric for buttons using a programmed cutting and stitching cycle. Correctly set the buttonhole length, stitch pattern, and cutting mechanism to achieve consistent openings while preventing needle injuries and fabric damage.

Main parts

- Machine head and sewing needle
- Buttonhole presser foot or fabric clamp
- Stitch-forming mechanism and thread system
- Buttonhole length and width controls
- Cutting knife or slit mechanism
- Motor, start control, and safety guard

Before you start

1. Confirm the fabric type, button size, required buttonhole dimensions, and finish.
2. Inspect the needle, presser foot, clamps, thread path, cutter, guards, and power cable for damage.
3. Fit the correct needle and thread for the fabric and ensure proper threading.
4. Set the buttonhole length and width according to the button and garment specification.
5. Check that the cutting knife is securely installed and correctly positioned, if the machine uses an integrated cutter.
6. Test the stitching and cutting sequence on a matching fabric scrap before production.

How to use

1. Switch off the machine before changing the needle, adjusting the cutter, or clearing a jam.
2. Select the correct buttonhole program or stitch pattern using the approved controls.
3. Set the buttonhole dimensions and thread tension according to the job specification.
4. Position the fabric so the intended buttonhole location is correctly aligned with the clamp and cutting path.
5. Secure the fabric using the presser foot or clamp without stretching or folding it unintentionally.
6. Close the guard where fitted and start the machine using the approved control.
7. Observe the first cycle to confirm that stitching forms evenly around the opening.
8. If the machine cuts automatically, verify that the knife operates only at the intended stage of the cycle.

9. Stop the machine if the fabric shifts, the needle breaks, thread tangles, or the cutter behaves abnormally.
10. Wait for the needle and cutter to stop before releasing the clamp and removing the fabric.
11. Inspect the buttonhole for correct length, clean edges, even stitches, and secure ends.
12. Switch off the machine, remove lint and thread waste, and record any quality or maintenance issues.

⚠ Safety

- Only trained operators should operate or adjust the buttonhole machine.
- Keep fingers away from the needle, clamp, and cutting knife during operation.
- Never bypass guards, interlocks, or cutter safety mechanisms.
- Switch off and isolate power before changing needles, adjusting the cutter, or performing maintenance.
- Do not reach under the presser foot while the machine is running.
- Keep loose clothing, hair, and jewellery away from moving components.

Care & storage

- Clean lint and thread waste from the needle plate, clamp, and cutter area after shutdown.
- Inspect the needle, cutting knife, clamp, and thread guides regularly.
- Replace blunt or damaged needles and maintain the cutter according to the manufacturer's instructions.
- Check stitch formation, cutting alignment, and tension settings periodically.
- Arrange servicing if the cutter jams, buttonholes become uneven, or the machine develops unusual noise.

Common mistakes to avoid

- Setting the buttonhole length incorrectly for the button.
- Misaligning the fabric with the cutting path.
- Reaching near the needle or knife while the machine is running.
- Ignoring uneven stitches, frayed edges, or a dull cutting knife.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Calligraphy Pen

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Speedball pens

What it is used for

A calligraphy pen creates controlled lettering and decorative strokes for writing, design, and craft activities.

How to handle

- Inspect the pen body, nib, reservoir, and any replaceable parts before use.
- Use ink compatible with the pen and follow the manufacturer's filling instructions.
- Hold the pen securely and avoid excessive pressure that may damage the nib.
- Keep ink away from eyes, clothing, and surfaces that may stain; clean spills promptly.
- Clean the nib and pen after use according to the product instructions.

Storage

- Store the pen clean and dry with the nib protected.
- Keep inks capped, labelled, and away from heat or direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Chain Stitch Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Chain stitch machine

What it is used for

A chain stitch machine forms interlocking chain-shaped stitches to join or decorate fabric in garment and textile production. Correct threading, stitch settings, and fabric feeding help produce consistent seams while reducing needle injuries and thread breakage.

Main parts

- Machine head and sewing needle
- Looper or chain-forming mechanism
- Thread stand and thread guides
- Tension controls and stitch-length regulator
- Presser foot and feed dogs
- Motor, foot pedal, and protective guards

Before you start

1. Confirm the fabric type, seam requirement, thread type, and required stitch length.
2. Inspect the needle, presser foot, looper, thread guides, guards, and power cable for damage.
3. Fit the correct needle and confirm that it is securely mounted.
4. Thread the machine according to its threading diagram, including the looper where applicable.
5. Adjust thread tension, stitch length, and differential feed if fitted, according to the fabric specification.
6. Test the stitch on a fabric scrap and check seam formation, tension, and thread security.

How to use

1. Switch off the machine before changing the needle, threading near moving parts, or clearing a jam.
2. Raise the presser foot and position the fabric beneath it at the starting point.
3. Lower the presser foot and ensure the fabric is aligned with the intended seam path.
4. Start the machine using the approved control and guide the fabric gently without pulling or pushing it forcibly.
5. Maintain a steady sewing speed and keep the fabric flat as it passes through the feed mechanism.
6. Observe the stitches for consistent loops, correct tension, and even spacing.
7. Stop the machine if the thread breaks, stitches skip, fabric jams, or unusual noise occurs.
8. Wait for the needle and moving parts to stop before adjusting the fabric or removing tangled thread.

9. At the end of the seam, follow the machine's approved method for securing or finishing the chain stitches.
10. Remove the fabric carefully and inspect the seam for skipped stitches, loose loops, and uneven tension.
11. Switch off the machine and clean lint and thread waste from the needle plate and feed area.
12. Report persistent stitch defects or mechanical problems before further production.

⚠ Safety

- Only trained operators should use or adjust the machine.
- Keep fingers away from the needle, looper, and feed mechanism while the machine is running.
- Do not bypass needle guards or other safety devices.
- Switch off and isolate power before replacing the needle, clearing jams, or performing maintenance.
- Keep loose clothing, hair, and jewellery away from moving parts.
- Use the correct needle and never operate the machine with a bent or damaged needle.

Care & storage

- Remove lint and thread waste after shutdown, especially around the feed dogs and looper area.
- Inspect the needle, looper, presser foot, and thread guides regularly.
- Lubricate only designated points according to the manufacturer's instructions.
- Check thread tension and stitch formation at the start of each production run.
- Arrange servicing if the machine repeatedly skips stitches, produces abnormal noise, or breaks thread.

Common mistakes to avoid

- Threading the looper or needle incorrectly.
- Using incorrect thread tension or stitch length for the fabric.
- Pulling the fabric forcefully while sewing.
- Adjusting the needle or clearing tangled thread while moving parts are active.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Colour Palette

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Palette / Colour mixing plates

What it is used for

Use colour palettes for controlled mixing of paints, dyes, or craft colours.

How to handle

- Check that the palette is clean before mixing colours.
- Use separate tools or clean them between colours to prevent unwanted mixing.
- Avoid contact with skin or eyes when handling products with safety warnings.
- Clean the palette promptly using the recommended method.

Storage

- Dry thoroughly before storing.
- Keep in a clean, dry container away from food and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Craft Material

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Raw materials like recycled paper pulp, natural adhesives (gond, wheat paste), colors, binding agents

What it is used for

Handle recycled paper pulp, natural adhesives, colours, and binding agents safely for craft work.

How to handle

- Read product labels and check suitability for the intended craft activity.
- Use clean tools and separate containers for different materials.
- Avoid inhaling dry powders and prevent adhesives or colours from contacting eyes.
- Prepare only the quantity needed and clean spills promptly.

Storage

- Seal containers after use and label any prepared mixtures.
- Store dry materials in a cool, dry area away from food and incompatible substances.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Crochet Needle

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Crochet Needle Set

What it is used for

A crochet needle or hook draws yarn through loops to create fabric and decorative patterns. Select a suitable hook size and maintain controlled hand movements.

How to use

1. Choose a crochet hook size appropriate for the yarn and pattern.
2. Check the hook tip and handle for damage or rough edges.
3. Secure the yarn and create the starting loop according to the pattern.
4. Hold the hook comfortably and draw yarn through loops with controlled movements.
5. Follow the pattern to make the required stitches and maintain even tension.
6. Check the fabric regularly for missed loops or uneven stitches.
7. Finish the work and store the hook and yarn safely.

⚠ Safety

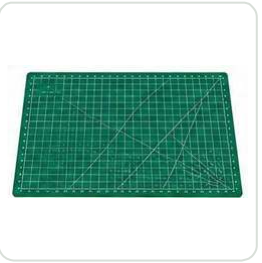
- Keep the hook pointed away from your eyes and other people.
- Do not use a bent, cracked, or sharp-edged hook.
- Take breaks if your hands or wrists become uncomfortable.

Care & storage

- Wipe hooks clean and store them in a case to prevent loss or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cutting Mat



Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Cutting Mat

What it is used for

Use cutting mats safely for sewing and craft work.

How to handle

- Place the mat on a flat, stable work surface.
- Cut away from the body using an appropriate cutting tool.
- Keep fingers clear of the blade path.
- Clean loose fibres and scraps after use.

Storage

- Store flat or as recommended by the manufacturer.
- Keep away from direct sunlight, heat, and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dress Form

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Low** 7 labs

Covers these lab items: Dress Model; Full Body Mannequin – Basic; Mannequin; Mannequin (Male or female, Size M); Mannequins; Sewing Mannequin

What it is used for

A dress form or mannequin provides a body-shaped support for garment fitting, draping, pattern development, and sewing demonstrations. Adjust the form to the required size and secure it before pinning or fitting fabric.

How to use

1. Inspect the dress form, stand, adjustment controls, and base for damage.
2. Place the mannequin on a stable, level surface with adequate working space.
3. Adjust height and body measurements according to the required design or size.
4. Lock all adjustment points and check that the form is stable.
5. Drape fabric or position the garment on the form and align key seams and markings.
6. Use pins carefully to secure fabric and check fit from multiple angles.
7. Remove pins and garments safely, then return the form to its storage position.

⚠ Safety

- Do not use an unstable stand or damaged adjustment mechanism.
- Keep fingers clear of sharp pins and remove pins after the task.
- Do not lean heavily on the mannequin or exceed its designed load.
- Keep the working area clear to prevent trips and falls.

Care & storage

- Wipe the form with a suitable cloth and store it away from moisture and direct heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dye

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Medium** 3 labs

Covers these lab items: Dyes (kg); Natural dyes (kg); Synthetic dyes (kg)

What it is used for

Dyes colour textile, craft, and other suitable materials when prepared and applied according to their type.

How to handle

- Identify the dye type, material compatibility, label instructions, and required protective equipment.
- Measure and prepare dye using dedicated tools in a ventilated work area.
- Avoid skin or eye contact, inhalation of powder, and contamination of food or drinking water.
- Follow specified processing, rinsing, and wastewater disposal instructions.

Storage

- Keep dyes sealed in labelled containers away from moisture, heat, and direct sunlight.
- Separate incompatible dyes and keep them away from food and drink.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Dyeing Equipment

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Dyeing Set Up Equipment; Dyeing units

What it is used for

Dyeing equipment applies dyes to fibres, yarn, fabric, or garments under controlled process conditions to achieve the required colour and shade. Correct dye preparation, temperature and time control, chemical handling, and colour-quality checks help produce consistent results while reducing exposure and environmental risks.

Main parts

- Dye bath, vessel, or tank holds the dye liquor and textile material.
- Heating system controls bath temperature where fitted.
- Agitator or circulation pump helps distribute dye evenly.
- Fabric carrier, basket, reel, or loading frame holds the textile.
- Dosing system measures and introduces dye or auxiliaries where fitted.
- Temperature sensor and controller monitor process conditions.
- Drain and rinse connections remove spent liquor and support cleaning.
- Control panel manages operating parameters.
- Exhaust or ventilation system removes vapours or aerosols where required.

Before you start

1. Only trained personnel should operate dyeing equipment and handle dye chemicals.
2. Read the machine instructions and confirm the approved dye recipe, fabric type, liquor ratio, temperature, and processing time.
3. Inspect the tank, seals, valves, agitator, pumps, heating system, guards, and controls for damage or leaks.
4. Verify that ventilation and any required chemical extraction are operating properly.
5. Check dye and auxiliary chemical labels, safety data sheets, compatibility, and required protective equipment.
6. Prepare the dye solution using the approved recipe and chemical-handling procedure; never mix unknown substances.
7. Wear suitable chemical-resistant gloves, eye protection, protective clothing, and respiratory protection when required by the risk assessment.

How to use

1. Confirm that the vessel, textile loading arrangement, and drain route are clean and ready for use.
2. Weigh and prepare the textile material according to the approved batch instructions.

3. Measure water and chemicals accurately according to the authorised dye recipe.
4. Fill the vessel to the specified level and add dye and auxiliaries in the prescribed sequence.
5. Load the textile evenly and avoid overloading or obstructing circulation.
6. Set the temperature, time, agitation, and other process parameters according to the recipe.
7. Start the dyeing cycle and monitor temperature, circulation, chemical dosing, and machine operation.
8. Take process samples only using the approved sampling method and assess shade or process progress where required.
9. Stop the process if leakage, overheating, abnormal vibration, or a chemical handling incident occurs.
10. Complete dyeing and carry out rinsing, neutralisation, or after-treatment according to the approved process.
11. Drain the bath through the authorised waste-treatment route and never discharge untreated chemicals into an unauthorised drain.
12. Unload the textile safely, inspect colour uniformity, and record batch details and deviations.
13. Clean the vessel and relevant components according to the chemical and machine instructions.

⚠ Safety

- Follow chemical safety data sheets and never mix incompatible dyes, acids, alkalis, or auxiliaries.
- Avoid skin and eye contact with dye liquor and chemical concentrates.
- Never open a hot or pressurised vessel until it is safe and pressure has been released as designed.
- Keep hands away from agitators, rollers, pumps, and other moving parts.
- Isolate energy sources before cleaning, clearing blockages, or maintenance.
- Treat dye effluent according to local environmental rules and the facility's approved wastewater procedure.

Care & storage

- Clean the vessel, filters, dosing lines, and textile-contact components after each batch as required.
- Inspect valves, seals, hoses, heating elements, sensors, and pumps for leaks or wear.
- Calibrate dosing and temperature controls at the prescribed intervals.
- Remove dye deposits using approved cleaning methods compatible with equipment materials.
- Report persistent shade variation, flow problems, or control faults and arrange authorised maintenance.

Common mistakes to avoid

- Using the wrong dye recipe, liquor ratio, temperature, or processing time.
- Adding chemicals in the wrong sequence or measuring them inaccurately.
- Overloading the vessel and causing uneven dye circulation.
- Discharging untreated dye liquor or cleaning chemicals into unauthorised drains.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electric Iron

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 9 labs

Covers these lab items: Iron; Iron and Iron Board; Iron and Ironing Board; Iron box; Press/iron (any type) with table; Steam iron

What it is used for

An electric iron uses heat, and sometimes steam, to remove wrinkles from suitable fabrics. Correct temperature selection, safe handling, and proper storage help prevent burns, electric shock, and fabric damage.

Main parts

- Soleplate transfers heat to the fabric.
- Heating element produces the required heat.
- Handle provides a safe grip during ironing.
- Temperature control selects the setting for different fabrics.
- Power cord and plug supply electrical power.
- Water reservoir stores water in steam iron models.
- Steam control and spray outlets release moisture on supported models.

Before you start

1. Read the manufacturer's instructions and check that the iron suits the fabric.
2. Inspect the soleplate, handle, cord, plug, and controls for damage.
3. Place the ironing board on a stable, level surface away from walkways.
4. Check the fabric care label and select the recommended temperature.
5. Fill a steam iron only as instructed, with the iron unplugged and cool.
6. Keep water away from the plug, socket, and other electrical connections.

How to use

1. Set up the ironing board securely and arrange the fabric flat on it.
2. Select the temperature according to the fabric care label.
3. Plug the iron into a suitable electrical outlet with dry hands.
4. Switch on the iron and wait until it reaches the selected temperature.
5. Place the fabric flat and move the iron steadily across it.
6. Use steam or spray only if the fabric and iron instructions allow it.
7. Keep the hot soleplate moving and avoid leaving it on one spot.
8. Stand the iron upright or use its designated rest when pausing.
9. Switch off and unplug the iron when finished.
10. Allow the iron to cool completely before emptying the reservoir or storing it, as instructed.

⚠ Safety

- Never touch the hot soleplate or direct steam toward skin.
- Never leave a switched-on iron unattended, even briefly.
- Keep the cord away from the hot soleplate and edges of the ironing board.
- Do not use an iron with a damaged cord, plug, or loose electrical connection.
- Keep the iron away from water except for the intended reservoir.
- Keep children away from the iron and hot ironing board.

Care & storage

- Unplug the iron and allow it to cool before cleaning.
- Clean the soleplate using only the method recommended by the manufacturer.
- Empty the water reservoir after use if the instructions require it.
- Store the cooled iron securely with its cord loosely arranged.

Common mistakes to avoid

- Using a high temperature on delicate or synthetic fabric.
- Leaving the hot soleplate resting on fabric or the ironing board cover.
- Filling the reservoir while the iron is hot or plugged in.
- Wrapping the power cord tightly around the iron while it is hot.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Embellishment Material

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Embellishment Material Sequins, mirrors, beads, etc.; Gotta; Sequins; Stones; Surface ornamentation material (Beads, Sequins)

What it is used for

Embellishment materials such as beads, sequins, mirrors, and gota decorate garments and textile products.

How to handle

- Check embellishments for sharp edges, loose pieces, and colour or finish defects.
- Select materials compatible with the fabric and intended stitching or bonding method.
- Use suitable needles, adhesives, or tools and keep small pieces away from children.
- Secure decorations firmly and inspect the finished work for loose parts.
- Collect spills and offcuts to prevent slips or accidental ingestion.

Storage

- Store embellishments in labelled, closed containers sorted by type and colour.
- Keep small pieces dry and separate from food and general waste.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Embroidery Hoop

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Embroidery frame; Embroidery Frames or Hoop; Embroidery hoop

What it is used for

To hold fabric taut for hand embroidery or stitching using an embroidery hoop. To maintain even fabric tension without stretching or marking the material.

Main parts

- Inner ring
- Outer ring
- Adjusting screw or clamp
- Fabric surface

Before you start

1. Select a hoop size suitable for the design and fabric.
2. Inspect both rings and the adjustment mechanism for cracks or rough edges.
3. Ensure the fabric is clean and correctly positioned.
4. Loosen the adjustment screw enough to separate the rings.

How to use

1. Place the inner ring on a flat surface.
2. Lay the fabric evenly over the inner ring.
3. Position the outer ring over the fabric and inner ring.
4. Press the rings together without trapping folds.
5. Tighten the adjustment screw gradually.
6. Pull the fabric edges gently until the surface is taut but not stretched.
7. Check tension before stitching and readjust if necessary.
8. Release the fabric and loosen the hoop after finishing.

⚠ Safety

- Keep fingers clear of the closing gap between the rings.
- Do not overtighten the screw or stretch delicate fabric excessively.
- Avoid using hoops with splinters or sharp edges.

Care & storage

- Remove loose threads and dust after use.
- Wipe the hoop clean and dry it if needed.

- Loosen the hoop when storing fabric for a long period.
- Store flat in a dry location.

Common mistakes to avoid

- Trapping folds between the rings.
- Tightening the screw too quickly or excessively.
- Stretching the fabric until its shape changes.
- Leaving delicate fabric tightly hooped during long storage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Embroidery Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Brother PE800 Embroidery Machine; Embroidery machine; Embroidery machine with stool and table

What it is used for

An embroidery machine stitches decorative designs, lettering, or patterns onto fabric using programmed instructions and thread. Correct hooping, design setup, and thread handling help produce accurate embroidery while protecting the fabric and machine.

Main parts

- Needle penetrates the fabric and forms stitches.
- Embroidery hoop holds the fabric taut during stitching.
- Thread spool and thread path supply thread to the needle.
- Bobbin supplies lower thread for stitch formation.
- Needle bar and presser foot guide needle movement and fabric control.
- Embroidery arm or carriage moves the hoop according to the design.
- Display or control panel selects designs and operating settings.
- Table and stool support the machine and operator where provided.

Before you start

1. Read the machine manual and identify the supported embroidery format and hoop size.
2. Inspect the needle, presser foot, hoop, thread path, bobbin, and power cord for damage.
3. Choose fabric, stabiliser, needle, and thread suitable for the design.
4. Transfer or select the approved design and check its size and placement.
5. Wind and install the bobbin and thread the machine according to the manual.
6. Place the machine on a stable table and keep the work area clear.

How to use

1. Switch off the machine before changing the needle, hoop, or other attachments.
2. Secure the fabric and stabiliser evenly in the hoop without stretching the fabric excessively.
3. Attach the hoop to the embroidery carriage and verify that it locks correctly.
4. Thread the upper thread and bobbin following the machine's threading path.
5. Select the design and position it using the display or approved controls.
6. Use the machine's design trace or positioning check if available to verify the stitching area.
7. Lower the presser foot as instructed and start stitching at the recommended speed.
8. Monitor thread tension, stitch quality, fabric movement, and machine sounds during operation.
9. Pause and switch off the machine before correcting tangles, thread breaks, or fabric misalignment.
10. After completion, remove the hoop, trim loose threads carefully, and inspect the finished design.

⚠ Safety

- Keep fingers away from the needle and moving embroidery carriage during operation.
- Switch off and unplug the machine before replacing a needle or performing maintenance.
- Do not operate the machine with damaged wiring, a loose needle, or a cracked hoop.
- Use the correct needle type and size for the fabric and thread.
- Keep loose clothing, jewellery, and hair away from moving parts.
- Do not pull or push the fabric while the machine is stitching.

Care & storage

- Remove lint and thread scraps from the bobbin area using the recommended method.
- Clean the needle plate and other accessible areas according to the manual.
- Replace bent or blunt needles and inspect hoops and thread guides regularly.
- Cover the machine when not in use and store threads in a clean, dry place.

Common mistakes to avoid

- Hooping fabric loosely or using an unsuitable stabiliser.
- Selecting a design larger than the installed hoop allows.
- Threading the upper thread or bobbin incorrectly.
- Pulling fabric or adjusting thread while the machine is stitching.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Embroidery Stabilizer

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 9 labs

Covers these lab items: Cut Away Machine Embroidery Stabilizer; Fusible No-show Polymesh Plus Embroidery Stabilizer; Iron-On Tear-Away Stabilizer; Nonwoven Non-fusible Backing Paper; Over the Back Fusible Interface; Self-Adhesive Sticky Peel N Stick Tear Away Embroidery Stabilizer; Stabilizer; Water Soluble Embroidery Stab

What it is used for

Embroidery stabilizers support fabric during stitching to reduce puckering, stretching, and design distortion.

How to handle

- Choose stabilizer type according to fabric weight, embroidery method, and design complexity.
- Check product instructions for fusible, tear-away, or cut-away application.
- Position the stabilizer evenly and secure fabric without excessive tension.
- Use appropriate scissors when trimming and keep blades away from the stitched design.
- Remove excess stabilizer according to manufacturer guidance and fabric requirements.

Storage

- Keep stabilizers flat, dry, and protected from dust.
- Store fusible types away from excessive heat and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Embroidery Template

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Embroidery Design template

What it is used for

Use embroidery design templates accurately while protecting the original pattern.

How to handle

- Check the design scale, orientation, and fabric compatibility before tracing.
- Secure the template to prevent shifting during marking.
- Use suitable marking tools and avoid damaging reusable templates.
- Keep pins, needles, and cutting tools handled safely during the process.

Storage

- Store templates flat in labelled sleeves or folders.
- Keep dry and protected from creasing, fading, and sharp tools.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fabric Cutting Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Jute Fabric Cutting Machine

What it is used for

A fabric cutting machine cuts jute or other approved textile materials into specified lengths and shapes for sewing, packaging, or production. Correct blade selection, fabric alignment, and secure handling help achieve accurate cuts while reducing injuries and material waste.

Main parts

- Machine frame and cutting table
- Cutting blade, knife, or rotary cutter
- Fabric guides and measuring scale
- Clamps or fabric holding mechanism
- Motor, drive system, and operating controls
- Blade guard, emergency stop, and dust extraction where fitted

Before you start

1. Confirm the fabric type, thickness, required dimensions, and cutting pattern.
2. Inspect the blade, guard, cutting table, power cable, controls, and emergency stop for damage.
3. Verify that the blade is suitable for jute fabric and is securely installed according to the manual.
4. Ensure the cutting table is clean, stable, and free from pins, tools, and other obstructions.
5. Spread and align the fabric without folds, excessive tension, or overlapping layers beyond machine capacity.
6. Wear appropriate eye protection, suitable footwear, and other required PPE; secure loose clothing and hair.

How to use

1. Switch off and isolate power before changing the blade, adjusting guides, or performing setup work.
2. Set the cutting length, width, guide position, and other required settings according to the cutting plan.
3. Secure the fabric using the approved holding method and confirm that the cutting path is clear.
4. Check that the blade guard and other safety devices are correctly positioned.
5. Switch on the machine and perform a test cut on a suitable fabric sample if required.
6. Feed or guide the fabric at the recommended speed without forcing it or placing hands near the blade.
7. Maintain alignment with the measuring scale or cutting guide throughout the cut.

8. Observe the cutting quality and stop the machine if the fabric shifts, the blade binds, or unusual vibration occurs.
9. Switch off the machine and wait for all moving parts to stop before repositioning fabric or removing offcuts.
10. Inspect the cut edges and dimensions against the production specification.
11. Repeat the cut only after correcting any alignment or setting issues.
12. At the end of the job, switch off and isolate power before removing scraps and cleaning the cutting area.
13. Record material usage, rejected pieces, and any machine faults according to workplace procedures.

⚠ Safety

- Only trained operators should use the fabric cutting machine.
- Keep fingers and hands away from the blade and cutting path; never guide fabric directly in front of an exposed blade.
- Do not operate the machine with a missing or damaged blade guard or disabled emergency stop.
- Isolate power before blade replacement, jam removal, or maintenance.
- Do not cut fabric that exceeds the machine's rated thickness or capacity.
- Keep the floor and work area free from loose fabric, threads, and scraps that may cause slips or entanglement.
- Use suitable dust control and respiratory protection if cutting generates significant fibres or dust.

Care & storage

- Remove fabric scraps, lint, and dust after shutdown using the approved cleaning method.
- Inspect blade sharpness, blade mounting, guards, guides, and clamps regularly.
- Replace worn or damaged blades with the correct approved type.
- Check drive components, power cables, and emergency-stop function according to the maintenance schedule.
- Arrange servicing if cut quality deteriorates, the motor overheats, or abnormal vibration occurs.

Common mistakes to avoid

- Using a dull or unsuitable blade for the fabric thickness.
- Failing to align or secure the fabric before cutting.
- Reaching near the blade while the machine is running or coasting.
- Ignoring damaged guards or continuing after the fabric becomes jammed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fabric Inspection Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fabric checking machine

What it is used for

A fabric inspection machine examines textile rolls for defects, shade variation, stains, holes, weaving faults, and dimensional irregularities. Correct lighting, controlled fabric movement, and systematic defect recording help maintain product quality and reduce downstream rejection.

Main parts

- Fabric unwinding roller and roll holder
- Inspection table or illuminated viewing surface
- Fabric take-up roller or winding unit
- Drive motor, speed controls, and tension adjustment
- Lighting system and inspection scale
- Emergency stop, guards, and defect-marking supplies

Before you start

1. Confirm the fabric specification, roll identification, inspection criteria, and acceptable defect limits.
2. Inspect the rollers, guides, inspection surface, lighting, guards, and drive components for damage.
3. Ensure the inspection area is clean and the lighting is suitable for identifying the fabric's specified defects.
4. Check that the fabric roll is securely mounted and compatible with the machine's width and load capacity.
5. Verify that the emergency stop and speed controls function correctly.
6. Prepare the defect log, marking tools, and required PPE; remove jewellery or loose items that could catch in moving fabric.

How to use

1. Switch off the machine before mounting the roll or adjusting roller alignment and fabric guides.
2. Load the fabric roll onto the unwinding holder and secure it using the approved method.
3. Thread the fabric through the guides and inspection area according to the machine diagram.
4. Set fabric tension and inspection speed to suit the material and inspection procedure.
5. Switch on the inspection lighting and confirm that the viewing surface is evenly illuminated.
6. Start the machine at a low speed and check that the fabric tracks centrally without wrinkles or excessive tension.

7. Increase speed gradually to the approved inspection rate while maintaining clear visibility of the fabric.
8. Inspect the full fabric width for holes, stains, slubs, broken yarns, shade differences, and other specified defects.
9. Mark defects using the approved method without damaging the fabric and record their positions or lengths.
10. Stop the machine safely before correcting tracking problems, removing trapped material, or adjusting tension.
11. Continue inspection until the full roll has passed the viewing area and the required length has been checked.
12. Review the defect record and classify the roll according to the applicable quality criteria.
13. Stop the machine, isolate power where required, and remove the inspected roll safely.
14. Clean the inspection surface and record roll status, defect details, and any equipment problems.

⚠ Safety

- Only trained operators should use the fabric inspection machine.
- Keep hands, hair, and loose clothing away from rollers, winding shafts, and pinch points.
- Never reach between moving rollers or attempt to remove trapped fabric while the machine is running.
- Use the emergency stop if the fabric becomes dangerously tangled or an unsafe condition develops.
- Do not exceed the machine's rated roll weight, width, or operating speed.
- Isolate power before maintenance, cleaning near moving components, or correcting mechanical faults.
- Use safe lifting methods or suitable lifting equipment for heavy fabric rolls.

Care & storage

- Clean the inspection table, rollers, and viewing surface after use.
- Inspect rollers, belts, tension controls, and guards for wear or damage.
- Keep inspection lights clean and replace faulty lamps according to the manufacturer's instructions.
- Check speed controls and emergency stops at the required intervals.
- Arrange servicing if fabric tracking becomes unstable, winding tension varies, or inspection lighting fails.

Common mistakes to avoid

- Inspecting fabric under poor or uneven lighting.
- Running the fabric too quickly to identify defects accurately.
- Applying excessive tension and stretching or distorting the fabric.
- Failing to record or mark defects consistently.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fabric Steamer

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Laboratory Fabric Streamer

What it is used for

A fabric steamer applies controlled steam to textile samples or materials to assess or improve properties such as shrinkage, colour fastness, crease removal, or dimensional stability, depending on the equipment and test method. Correct temperature, steam exposure, and sample handling help achieve repeatable results while reducing burn and fabric-damage risks.

Main parts

- Steam chamber or steaming enclosure
- Steam generator or approved steam supply
- Fabric sample holder, rack, or tray
- Temperature and pressure controls or indicators
- Condensate drain and steam outlet
- Timer, lid, seals, and protective housing

Before you start

1. Confirm the intended process or test method, fabric type, sample dimensions, and required steam exposure.
2. Read the equipment manual and verify that the steamer is suitable for the fabric and the planned procedure.
3. Inspect the chamber, seals, sample holders, drain, steam hoses, and controls for damage or leaks.
4. Check the water supply or approved steam connection and confirm that the drain is clear.
5. Verify that temperature and pressure indicators, interlocks, and over-temperature protection are functioning as specified.
6. Wear suitable heat-resistant gloves, eye protection, and other required PPE; prepare a heat-resistant surface for handling samples.

How to use

1. Place the steamer on a stable surface with adequate ventilation and clear access to its controls.
2. Fill the reservoir with the specified water or connect the approved steam supply according to the manual.
3. Set the required temperature, steam level, and exposure time according to the approved procedure.
4. Switch on the unit and allow it to reach the specified operating conditions before loading the sample, where required.

5. Prepare and label the fabric sample, then position it on the designated holder without blocking steam outlets.
6. Close the chamber or lid securely and confirm that seals and safety interlocks are engaged.
7. Start the steaming cycle and monitor the indicated temperature, pressure where applicable, and remaining time.
8. Avoid opening the chamber during the cycle unless the procedure specifically requires it and the equipment can be opened safely.
9. Stop the process if a steam leak, abnormal pressure, control fault, or unexpected temperature rise occurs.
10. At the end of the cycle, switch off steam generation and follow the manufacturer's depressurisation and cooling instructions.
11. Open the chamber only when the indicated conditions and safety interlocks permit; stand clear of escaping residual steam.
12. Remove the sample using heat-resistant gloves or approved tools and place it on a safe surface to cool or condition as required.
13. Inspect the sample and record the test conditions, exposure time, and observed changes according to the procedure.
14. Drain and clean the unit after cooling as instructed, then switch off and isolate power if required.

⚠ Safety

- Only trained personnel should operate the fabric steamer and interpret formal test results.
- Steam and hot condensate can cause severe burns; keep hands and face away from steam outlets and the chamber opening.
- Never open a pressurised chamber or bypass lid interlocks and pressure-relief devices.
- Do not exceed the equipment's rated temperature, pressure, or exposure limits.
- Use only the specified water and approved steam supply; keep electrical connections dry.
- Allow the unit to cool and depressurise before cleaning, draining, or maintenance.
- Stop use and report any damaged seals, leaking hoses, faulty controls, or abnormal pressure.

Care & storage

- Drain condensate and clean the chamber, sample holders, and accessible surfaces after cooling.
- Inspect seals, hoses, fittings, and the drain system regularly for wear, scale, or leaks.
- Remove scale using only products and methods approved by the manufacturer.
- Check temperature and pressure indicators and arrange calibration where required.
- Store the steamer clean and dry, and arrange servicing if steam delivery or temperature control becomes unreliable.

Common mistakes to avoid

- Using an incorrect steam temperature or exposure time for the fabric or test method.
- Opening the chamber before steam pressure and temperature reach a safe condition.
- Placing samples where they block steam circulation or contact hot surfaces unexpectedly.
- Ignoring scale build-up, blocked drainage, or leaking seals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Fixation Chamber

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fixation chamber

What it is used for

A fixation chamber applies controlled heat, steam, or other specified conditions to fix dyes, pigments, prints, or textile finishes onto fabric. Correct process settings, fabric preparation, and exposure control help improve colour durability and finishing quality.

Main parts

- Insulated fixation chamber or enclosure
- Heating or steam-generation system
- Fabric conveyor, rack, or sample holder
- Temperature and humidity controls where fitted
- Timer, sensors, and control panel
- Ventilation system, door seals, and safety interlocks

Before you start

1. Confirm the fabric type, dye or printing process, finishing chemistry, and approved fixation recipe.
2. Read the equipment manual and chemical safety information for the materials being processed.
3. Inspect the chamber, heating or steam system, conveyor, seals, sensors, and guards for damage or leaks.
4. Verify that temperature controls, timers, ventilation, and safety interlocks function correctly.
5. Ensure that the fabric is prepared and loaded according to the process specification and that incompatible materials are not mixed.
6. Wear suitable heat-resistant gloves, eye protection, and other required PPE; ensure the work area is adequately ventilated.

How to use

1. Switch off the chamber before loading samples or adjusting internal racks and fabric guides.
2. Set the temperature, humidity or steam conditions, conveyor speed, and fixation duration according to the approved recipe.
3. Allow the chamber to reach the specified operating conditions before loading, where required by the process.
4. Arrange the fabric evenly on the designated rack or conveyor without folds, overlapping that restricts treatment, or contact with hot surfaces.
5. Start the fixation cycle and verify that the fabric moves or remains positioned as intended.

6. Monitor temperature, humidity or steam indicators, exposure time, and fabric condition during operation.
7. Check for uneven heating, condensation problems, fabric scorching, colour bleeding, or unusual odours.
8. Stop the process if the temperature exceeds limits, steam escapes unexpectedly, ventilation fails, or the fabric shows signs of damage.
9. Do not open the chamber during operation unless the equipment and approved procedure specifically allow safe access.
10. At the end of the cycle, stop heating or steam supply and follow the prescribed cooling and ventilation procedure.
11. Open the chamber only when conditions are safe, then remove the fabric using appropriate heat-resistant protection.
12. Allow the fabric to cool or condition as required and inspect colour fixation, appearance, and other specified quality parameters.
13. Record the recipe, operating readings, batch identification, and inspection results.
14. Switch off and isolate power before cleaning or maintenance, and clean the chamber according to the manufacturer's instructions.

⚠ Safety

- Only trained personnel should operate the fixation chamber and select process settings.
- Treat chamber surfaces, steam, and processed fabric as hot until confirmed safe to handle.
- Do not exceed the equipment's rated temperature or pressure limits.
- Keep guards, door interlocks, and pressure-relief devices in place and functional.
- Maintain ventilation when processing dyes, coatings, or chemicals that may release hazardous vapours.
- Use chemical-compatible PPE when handling treated fabric or process chemicals.
- Isolate electrical and steam supplies before maintenance, cleaning near hazardous components, or repairing leaks.

Care & storage

- Remove textile residue and deposits after the chamber has cooled, using approved cleaning methods.
- Inspect seals, sensors, heating elements, steam lines, and conveyor components regularly.
- Check temperature control accuracy and arrange calibration at the required intervals.
- Keep ventilation paths clear and inspect the exhaust system for obstructions.
- Arrange servicing if temperature uniformity deteriorates, interlocks fail, or fabric quality becomes inconsistent.

Common mistakes to avoid

- Using incorrect temperature, humidity, steam conditions, or fixation time for the textile process.
- Overloading the chamber and preventing uniform treatment.
- Opening the chamber before it has cooled or reached a safe condition.
- Ignoring uneven colour fixation, scorching, chemical odours, or faulty ventilation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Flattening Tool

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Pedi (flattening tool)

What it is used for

A pedi flattening tool is used in suitable textile or craft work to press, smooth, or flatten material as required. Confirm the tool's intended use and apply controlled pressure to avoid damage.

How to use

1. Inspect the tool surface, handle, and moving parts for damage or rough edges.
2. Prepare the material on a clean, stable work surface.
3. Position the material according to the craft or textile procedure.
4. Hold the tool securely and apply even, controlled pressure to flatten the material.
5. Check the result and repeat gently if further flattening is required.
6. Remove the material carefully and clear any residue from the tool.
7. Clean and store the tool after completing the task.

⚠ Safety

- Keep fingers away from pinch points and moving parts.
- Do not apply excessive force that could damage the material or tool.
- Wear suitable protection if the process involves heat, sharp edges, or chemical treatments.
- Do not use a damaged or unstable tool.

Care & storage

- Clean the tool after use and store it dry to prevent damage or corrosion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

GSM Cutter

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: GSM Cutter

What it is used for

A GSM cutter prepares fabric specimens of a specified area for measuring grams per square metre (GSM). Use the correct specimen area and handle the cutting edge carefully.

How to use

1. Inspect the cutter, circular blade, handle, and safety lock for damage.
2. Confirm the required specimen area and compatible cutting mat.
3. Place the fabric flat on the cutting mat without wrinkles or folds.
4. Position the cutter on the fabric according to the manufacturer's instructions.
5. Unlock the cutter and rotate or press it as designed to cut the specimen.
6. Lock the blade and remove the specimen carefully using a suitable tool if needed.
7. Measure the specimen mass with a calibrated balance and calculate GSM using the specified method.

⚠ Safety

- Keep fingers away from the blade and never cut toward your body.
- Use the cutter only on a suitable cutting mat.
- Lock or cover the blade immediately after use.
- Do not use a cutter with a loose or damaged blade.

Care & storage

- Remove fabric fibres, clean the tool, and store it with the blade secured.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Garment Trims

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Pant hooks; Trims; Trims and Laces

What it is used for

Garment trims such as hooks, laces, and decorative accessories finish, fasten, and enhance textile products.

How to handle

- Check trims for sharp edges, broken hooks, colour defects, and consistent size.
- Choose trims compatible with the garment design, fabric, and washing requirements.
- Use appropriate sewing or fastening tools and secure attachments firmly.
- Keep small parts organised to prevent loss and avoid leaving sharp hooks exposed.
- Inspect the finished garment for loose or unsafe trim components.

Storage

- Sort trims by type, colour, and size in labelled containers.
- Keep fabric laces clean and dry and protect metal components from moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Gemstone

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Precious stones, Semi precious stones

What it is used for

Handle precious and semi-precious stones carefully for craft and design work.

How to handle

- Inspect stones for chips, cracks, and loose settings before use.
- Handle small stones over a tray to prevent loss.
- Use appropriate tools and avoid excessive pressure that may damage stones.
- Keep stones away from children and avoid placing them near the mouth.

Storage

- Store stones in separate labelled compartments or padded pouches.
- Keep secure, dry, and away from abrasive materials and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Glue Gun

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **High** 5 labs

Covers these lab items: Glue Gun; Glue Gun/Adhesive Tools; Hot glue gun

What it is used for

A glue gun heats a compatible adhesive stick and dispenses molten glue for crafts, repairs, and assembly work. Use the correct glue type and allow the adhesive to cool before handling the bonded item.

How to use

1. Inspect the glue gun, power cord, trigger, stand, and nozzle for damage.
2. Insert a compatible glue stick into the feed opening.
3. Place the glue gun upright on its heat-resistant stand.
4. Connect the gun to the approved power supply and allow it to heat as instructed.
5. Test a small amount of glue on scrap material before applying it to the workpiece.
6. Apply glue carefully using the trigger and keep fingers away from molten adhesive.
7. Release the trigger, return the gun to its stand, and allow the bond to cool.
8. Disconnect the gun after use and let it cool completely before storage.

⚠ Safety

- Never touch the hot nozzle or molten glue, which can cause burns.
- Keep the gun away from water and flammable materials.
- Do not leave a powered glue gun unattended.
- Use suitable ventilation and keep the nozzle away from skin and eyes.

Care & storage

- Wipe cooled external residue only as directed and never pull hot glue by hand.
- Store the fully cooled gun in a dry place with its cord undamaged.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Grey Scale

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Greyscale

What it is used for

To assess or compare grey-scale shades for colour-change or staining evaluation in textile testing, according to the applicable test method.

How to handle

- Inspect the grey-scale cards or strips for fading, stains, tears, and surface damage before use.
- Confirm whether the scale is intended for colour change or staining and follow the relevant test standard.
- View the reference scale and test specimen under the specified lighting and viewing conditions.
- Compare the specimen with the scale carefully and record the rating according to the approved method.
- Avoid touching the comparison surfaces and keep the scale away from direct sunlight and contamination.

Storage

- Keep the grey scale clean, dry, and protected from light exposure when not in use.
- Store flat in its original protective case away from chemicals, moisture, and abrasion.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hanger

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Hanger; Hangers

What it is used for

Use hangers to support garments without deforming them.

How to handle

- Choose a hanger suitable for the garment size and weight.
- Hang garments evenly and avoid overloading rails.
- Check for sharp edges or broken clips that may snag fabric.
- Keep hangers clean before placing garments on them.

Storage

- Arrange hangers in a designated rack or storage area.
- Keep dry and away from heat that may deform plastic hangers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hem Marker

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Skirt- Hem Marker

What it is used for

To mark a consistent hemline on garments, such as skirts, before stitching or alteration.

How to handle

- Inspect the hem marker, measuring scale, adjustment mechanism, and marking attachment for damage.
- Place the garment on a suitable surface or have the wearer stand safely in the required posture.
- Set the desired hem height using a measuring tape and adjust the marker accordingly.
- Mark the fabric evenly around the garment without pulling or distorting it; use suitable marking material.
- Check the marked hemline for consistency before cutting or stitching.

Storage

- Clean off marking residue and dry the marker before storage.
- Store upright or in its protective case in a dry sewing area, protecting the measuring and marking parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Interlock Sewing Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: High speed interlock machine

What it is used for

To sew and finish fabric seams using a high-speed interlock sewing machine. To produce consistent stitches while maintaining correct thread tension, fabric feeding, and operator safety.

Main parts

- Machine head and needle bar
- Needles and looper assembly
- Thread guides and tension controls
- Presser foot and feed mechanism
- Motor, speed control, and foot pedal
- Lubrication system and safety guards

Before you start

1. Confirm that the machine is installed securely and is suitable for the fabric and seam type.
2. Inspect needles, loopers, presser foot, thread guides, and guards for damage.
3. Check thread type, needle size, and threading path against the machine instructions.
4. Verify oil level where applicable and check for leaks or unusual residue.
5. Ensure the needle area and table are clear of tools and loose materials.
6. Test the foot pedal, speed control, and safety devices before sewing.

How to use

1. Switch off the machine before threading, changing needles, or making adjustments.
2. Thread the machine through the specified guides and tension controls according to the threading diagram.
3. Fit the correct needle and confirm that it is seated and tightened properly.
4. Place the fabric under the presser foot with the seam aligned to the guide.
5. Lower the presser foot and start sewing at a low speed to verify stitch formation.
6. Increase speed gradually while guiding the fabric without pulling or pushing it forcefully.
7. Monitor stitch formation, thread tension, fabric feeding, and machine noise during operation.
8. Stop the machine if stitches skip, thread breaks repeatedly, or the fabric jams.
9. Raise the needle and presser foot as specified before removing the sewn material.
10. Switch off and isolate power before clearing a jam or accessing the needle and looper area.
11. Trim loose threads with suitable scissors and inspect the seam for uniformity.
12. Clean lint from accessible areas and follow the manufacturer's lubrication schedule.

⚠ Safety

- Only trained operators should use a high-speed interlock machine.
- Keep fingers away from the needle, loopers, and moving feed mechanism.
- Do not operate with missing guards, a bent needle, or damaged electrical controls.
- Switch off and isolate power before threading, needle replacement, jam clearing, or maintenance.
- Tie back long hair and avoid loose clothing, jewellery, or dangling accessories.
- Keep hands clear of the presser foot and do not leave the machine running unattended.

Care & storage

- Remove lint and thread fragments from the needle and looper area after use.
- Check needle condition and replace bent, blunt, or damaged needles.
- Maintain the lubrication system according to the manual.
- Inspect the belt, pedal, guards, and electrical cable regularly.
- Arrange periodic servicing to maintain stitch quality and safe operation.

Common mistakes to avoid

- Threading the machine incorrectly or using unsuitable thread and needle sizes.
- Pulling fabric forcefully and causing needle breakage or uneven stitches.
- Operating at high speed before confirming correct stitch formation.
- Clearing a thread jam while the machine is running.
- Allowing lint to accumulate around the loopers and feed mechanism.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Jute Article Finishing Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Jute Article Finishing Machine

What it is used for

To finish jute articles by trimming loose fibres, smoothing surfaces, and performing approved edge or appearance finishing operations. To improve product consistency while protecting the operator from blades, moving parts, and airborne fibres.

Main parts

- Machine frame and work surface
- Finishing or trimming head
- Feed guide or work holder
- Drive motor and speed control
- Guard and emergency stop, if fitted
- Dust collection or fibre removal arrangement

Before you start

1. Read the manual and identify the finishing operation supported by the machine.
2. Inspect the cutting or finishing head, guards, work holder, drive belt, and power connections for damage.
3. Confirm that the correct tool or attachment is installed and securely fastened.
4. Check that dust or fibre collection equipment is connected and working where provided.
5. Prepare the jute article and remove pins, staples, or other objects that could damage the tool.
6. Wear suitable eye protection, a dust mask or respirator where required, and close-fitting work clothing.

How to use

1. Place the machine on a stable surface and ensure the emergency stop is accessible.
2. Switch off and isolate power before fitting attachments or adjusting guides.
3. Set the feed guide and finishing settings according to the jute article and desired finish.
4. Secure the workpiece using the designated holder or guide.
5. Switch on the machine and check for unusual vibration, noise, or movement.
6. Feed the jute article steadily through the finishing area without forcing it.
7. Trim loose fibres or finish edges in controlled passes while keeping hands clear of the tool head.
8. Inspect the finished surface and repeat only where necessary to meet the specified quality.
9. Stop the machine before clearing fibre buildup or repositioning a jammed workpiece.
10. Switch off and disconnect power before cleaning, replacing a tool, or performing maintenance.
11. Remove fibre waste using the approved cleaning method and avoid blowing dust into the air.

2. Record defects or machine issues and store the equipment in a clean, dry location.

⚠ Safety

- Only trained operators should use the finishing machine.
- Never remove guards or reach into the moving finishing head.
- Keep fingers, loose clothing, jewellery, and hair away from the feed and cutting area.
- Isolate power before clearing jams, removing waste near moving parts, or changing attachments.
- Use appropriate dust protection and ventilation to reduce inhalation of airborne fibres.
- Do not operate with damaged blades, loose tools, faulty switches, or exposed drive components.

Care & storage

- Remove jute fibres and dust from the work area and accessible machine surfaces after use.
- Inspect the finishing head and replace worn or damaged tools as specified by the manufacturer.
- Check guards, work holders, belts, and fasteners regularly.
- Maintain the motor and lubrication points according to the manual.
- Keep dust extraction paths clear and arrange periodic servicing.

Common mistakes to avoid

- Using an unsuitable finishing tool or incorrect setting for the jute article.
- Forcing the material through the machine and causing uneven finishing.
- Holding the article too close to an unguarded moving tool.
- Clearing accumulated fibres while the machine is running.
- Ignoring excessive dust, vibration, or a damaged finishing head.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Jute Craft Tool

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Equipments used for developing jute products

What it is used for

Jute craft tools support cutting, measuring, joining, decorating, and finishing jute-based products. Select tools suited to the material and follow safe craft-work practices.

How to use

1. Inspect scissors, cutters, needles, measuring tools, and other equipment for damage.
2. Prepare a clean, well-lit work surface and gather the required jute material.
3. Measure and mark the material according to the product design.
4. Cut the jute carefully using the appropriate tool and a stable cutting surface.
5. Join, stitch, weave, or decorate the pieces according to the craft plan.
6. Check dimensions, seams, and finish, then trim loose fibres safely.
7. Clean the tools and organize the materials after completing the product.

⚠ Safety

- Keep fingers clear of blades, needles, and cutting edges.
- Use a cutting mat and never cut toward your body.
- Avoid inhaling jute dust and work in a well-ventilated area.
- Store sharp tools safely when not in use.

Care & storage

- Remove fibres and dust, dry tools as needed, and store them in designated containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Jute Fibre Straightening Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Jute Fibre Straightening Machine

What it is used for

To straighten and align jute fibres for subsequent processing, spinning, or product manufacture. To achieve uniform fibre alignment while minimizing fibre damage and keeping the machine free from blockages.

Main parts

- Machine frame and feed table
- Feed rollers or fibre guides
- Straightening combs or rollers
- Drive motor and transmission
- Speed controls and emergency stop
- Protective guards and fibre collection area

Before you start

1. Read the operating manual and confirm the machine is suitable for the jute fibre being processed.
2. Inspect feed rollers, combs, guides, guards, belts, and electrical connections for damage.
3. Ensure the machine is securely installed and all protective covers are fitted.
4. Check that the fibre feed area and discharge path are free from previous material and obstructions.
5. Prepare jute fibres by removing stones, metal pieces, strings, and other foreign objects.
6. Wear suitable eye protection, close-fitting clothing, and respiratory protection where airborne dust is present.

How to use

1. Switch off and isolate power before adjusting guides or inspecting moving parts.
2. Set the feed guides and operating speed according to the fibre condition and machine instructions.
3. Confirm that the discharge and fibre collection areas are ready.
4. Start the machine using the approved procedure and observe it for unusual noise or vibration.
5. Feed a small quantity of jute fibres evenly into the designated feed area.
6. Allow the rollers or combing mechanism to straighten the fibres without forcing the material.
7. Increase the feed gradually while checking fibre alignment and machine loading.
8. Monitor the output for knots, uneven alignment, excessive breakage, or blockage.
9. Stop the machine if fibres wrap around rollers or the feed becomes irregular.

10. Isolate power and wait for all moving parts to stop before clearing any blockage.
11. Collect the processed fibres and inspect them for consistency and damage.
12. Switch off the machine and clean fibre residues using the approved method.

⚠ Safety

- Only trained operators should use the machine.
- Never reach into feed rollers, combs, belts, or other moving parts.
- Do not operate the machine with missing guards or damaged safety controls.
- Tie back long hair and avoid loose clothing, jewellery, or dangling accessories.
- Isolate power before removing tangled fibres or performing adjustments and maintenance.
- Use appropriate dust protection and ventilation where fine fibres or dust are generated.

Care & storage

- Remove fibre waste from the feed table, rollers, and collection area after use.
- Inspect combs, rollers, belts, and guards regularly for wear or damage.
- Lubricate only specified points according to the manual.
- Keep electrical components dry and clean ventilation openings.
- Record faults and arrange servicing according to the maintenance schedule.

Common mistakes to avoid

- Feeding excessive fibre at once and overloading the machine.
- Using fibres containing hard foreign objects that can damage rollers or combs.
- Pulling tangled fibres out while the machine is running.
- Operating with damaged guards or worn feed components.
- Ignoring uneven output, excessive vibration, or repeated blockages.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Loom

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **Medium** 3 labs

Covers these lab items: Bamboo Mat Weaving Loom; Handloom Set Up; Handloom/Power loom

What it is used for

A loom holds warp threads under controlled tension while weft threads are interlaced to produce woven fabric, mats, or other textile products. Correct setup and operation improve weave quality, productivity, and operator safety.

Main parts

- Loom frame and supporting structure
- Warp beam and cloth beam
- Heddles, harnesses, or thread guides
- Reed and beater mechanism
- Shuttle or weft insertion system
- Tension controls, pedals, or drive mechanism depending on loom type

Before you start

1. Identify whether the loom is hand-operated, power-driven, or designed for bamboo mat weaving.
2. Read the operating instructions and confirm the permitted material, width, and production settings.
3. Inspect the frame, beams, heddles, reed, shuttle, drive system, and guards for damage.
4. Check that the loom is stable, correctly installed, and clear of obstructions.
5. Arrange warp threads in the required order and secure them to the warp beam.
6. Set thread tension evenly and check that the weft material feeds without snagging.

How to use

1. Set the required fabric or mat width using the loom's approved adjustment method.
2. Thread the warp through the designated guides, heddles, and reed according to the weave pattern.
3. Secure the warp ends to the cloth beam and verify even tension across the width.
4. Load the shuttle or prepare the approved weft insertion mechanism.
5. For handlooms, operate the pedals or levers in the correct sequence to form the shed.
6. Insert the weft across the warp and use the beater to pack it evenly.
7. For power looms, start the drive using the designated control and observe the initial operation.
8. Inspect the developing weave for gaps, uneven tension, broken threads, or pattern errors.
9. Stop the loom before correcting thread breaks, jams, or alignment problems.

10. At completion, stop and isolate the drive if fitted, remove the woven material, and clean the work area.

⚠ Safety

- Keep fingers, hair, and loose clothing away from moving shafts, gears, belts, and the beater.
- Never reach into a moving loom to repair broken threads or remove a jam.
- Keep guards fitted on powered drives and do not bypass safety switches.
- Use safe lifting methods when handling heavy beams or material rolls.
- Switch off and isolate power before cleaning, adjustment, or maintenance on powered looms.
- Keep the floor clear of loose threads, bamboo strips, and other tripping hazards.

Care & storage

- Remove lint, loose threads, and material debris after use.
- Inspect warp beams, heddles, reed, bearings, belts, and drive components regularly.
- Lubricate approved moving parts according to the manufacturer's instructions.
- Store weaving materials in a clean, dry place and report damaged components promptly.

Common mistakes to avoid

- Uneven warp tension can produce distorted fabric or mats.
- Incorrect threading through the heddles or reed can create weave defects.
- Operating with damaged guards or loose drive components increases injury risk.
- Trying to clear a jam while the loom is moving can cause serious hand injuries.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Loop Turner

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Loop turner

What it is used for

To turn narrow fabric tubes, straps, and sewn loops right-side out during sewing and garment construction.

How to handle

- Inspect the loop turner hook, latch, shaft, and handle for damage or sharp burrs.
- Select a loop turner suitable for the width and fabric type of the sewn tube.
- Insert the tool gently into the tube and attach the fabric edge securely to the hook or latch.
- Pull the tool through slowly while guiding the fabric to prevent snagging, tearing, or overstretching.
- Release the fabric carefully and clean the tool after use.

Storage

- Wipe the tool clean and dry after use.
- Store in a sewing-tool case with the hook protected to prevent bending and accidental injury.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Modelling Clay

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Polymer Clay

What it is used for

Handle polymer modelling clay safely for textile and craft projects.

How to handle

- Read the manufacturer's instructions and use clay only for its intended purpose.
- Work on a clean, dedicated surface and wash hands after handling.
- Do not eat or drink while working, and keep clay away from young children and pets.
- Bake or cure only as instructed; never heat clay in food-use equipment unless the manufacturer explicitly approves it.

Storage

- Seal unused clay to prevent drying or contamination.
- Store in a labelled container away from heat, sunlight, food, and incompatible materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Needle Detector

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **High** 1 labs

Covers these lab items: Needle Detector

What it is used for

A needle detector checks garments, fabrics, or other textile products for broken needles and metallic fragments. Use the detector according to the product size and the manufacturer's inspection procedure.

How to use

1. Inspect the detector, sensor area, power supply, and conveyor if fitted.
2. Switch on the detector and complete its self-check or calibration procedure.
3. Test the detector using the approved test piece supplied for the machine.
4. Place the garment or textile item in the specified inspection area.
5. Move the item through the detector at the recommended speed and orientation.
6. Observe audible, visual, or automatic reject alerts for detected metal.
7. Recheck flagged items according to the site's needle-control procedure.
8. Record inspection results and switch off the detector after use.

⚠ Safety

- Keep hands and loose clothing away from moving conveyor parts.
- Do not bypass alarms or ignore failed test-piece checks.
- Follow the site's procedure for isolating and handling suspected needle-contaminated items.
- Disconnect power before cleaning or servicing the equipment.

Care & storage

- Keep the sensor area clean and free from metal scraps.
- Perform routine test-piece checks and servicing as recommended by the manufacturer.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Needle Threader

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Low** 6 labs

Covers these lab items: Needle Threader

What it is used for

A needle threader helps guide sewing thread through the eye of a needle. Use the correct threader type and handle the needle carefully to avoid bending or breaking delicate parts.

How to use

1. Inspect the threader wire, handle, and needle for damage.
2. Select a threader compatible with the needle eye and thread thickness.
3. Hold the needle securely with its eye accessible.
4. Insert the threader wire gently through the needle eye.
5. Pass the thread through the threader loop or hook as designed.
6. Pull the threader carefully back through the needle eye to draw the thread through.
7. Remove the threader and store it safely after use.

⚠ Safety

- Keep fingers away from sharp needle points.
- Do not force a thick threader through a small needle eye.
- Replace bent or damaged threaders to avoid injury or thread damage.

Care & storage

- Store the threader in a dry container and protect its wire from bending.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Overlock Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Over lock Machine (5 thread over lock machine); Over-lock machine

What it is used for

An overlock sewing machine stitches fabric edges, trims excess fabric, and forms an overedge seam using multiple threads. Use the correct needle, thread arrangement, and settings for the fabric and required seam.

Main parts

- Needle assembly and presser foot
- Upper and lower loopers
- Thread guides and tension controls
- Fabric feed dogs and differential feed control
- Cutting knife and knife guard
- Motor, foot pedal, and power switch

Before you start

1. Confirm the machine is suitable for the fabric and seam type.
2. Inspect the needle, presser foot, loopers, knife, guards, and power cord.
3. Ensure the machine is switched off before threading or changing the needle.
4. Thread the machine according to the threading diagram and required thread count.
5. Check needle size, thread type, tension, and stitch settings for the fabric.
6. Clear the worktable and keep fingers away from the cutting knife.

How to use

1. Place the fabric under the presser foot with the edge aligned to the cutting guide.
2. Lower the presser foot to hold the fabric securely.
3. Switch on the machine and test the stitch on a fabric scrap.
4. Adjust stitch length, differential feed, and thread tension if required.
5. Guide the fabric steadily without pulling or pushing it forcefully.
6. Keep hands clear of the needle, loopers, and cutting knife while sewing.
7. Observe stitch formation, edge trimming, and fabric feeding during operation.
8. Stop the machine before correcting thread tangles or adjusting the fabric near the needle.
9. At the end of the seam, allow a short thread chain to form if required.
10. Raise the presser foot and remove the fabric carefully after the machine stops.
11. Switch off and disconnect power before cleaning or changing machine parts.
12. Remove lint and thread waste and store the machine safely.

⚠ Safety

- Only trained users should operate the overlock machine.
- Never place fingers near the moving needle, loopers, or cutting knife.
- Switch off and disconnect power before threading, needle changes, or maintenance.
- Keep guards fitted and do not operate with damaged safety covers.
- Tie back long hair and secure loose clothing away from moving parts.
- Use the correct needle and replace bent or damaged needles immediately.

Care & storage

- Remove lint and thread waste from the feed area after use.
- Clean and lubricate specified points according to the manufacturer's instructions.
- Inspect the knife, needle, loopers, and thread guides regularly.
- Arrange servicing if the machine skips stitches, vibrates excessively, or makes unusual noise.

Common mistakes to avoid

- Threading the machine in the wrong sequence.
- Using incorrect thread tension or needle size for the fabric.
- Pulling the fabric forcefully and causing uneven seams.
- Cleaning or adjusting the machine while it is running.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paint Brush

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 11 labs

Covers these lab items: Brush; Brushes; Brushes (Various sizes); Colour brushes; Flat Brushes; Knotting Brush; Paint Brushes (different sizes), colour mixing plates; Paint Brushes (various sizes)

What it is used for

Use a paint brush to apply paint, colour, or coating evenly to a suitable surface or craft material.

How to use

1. Choose a brush size and shape suitable for the paint and task.
2. Inspect the bristles and handle for damage or loose parts.
3. Prepare the paint and work surface according to the product instructions.
4. Dip only the required portion of the bristles into the paint.
5. Apply smooth, even strokes and avoid excessive paint buildup.
6. Rinse or clean the brush with the appropriate cleaner after use.
7. Reshape the bristles and allow the brush to dry before storage.

⚠ Safety

- Use paints and cleaners in a well-ventilated area.
- Avoid skin and eye contact with paints, solvents, and cleaning agents.
- Keep brushes and chemicals away from food and drinking water.
- Follow the product label for handling and disposal.

Care & storage

- Clean the brush thoroughly and remove dried paint from the bristles.
- Store dry brushes with bristles protected and not crushed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Palette Knife

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Pallet knife straight and bent

What it is used for

To mix, spread, scrape, or shape suitable art and craft materials using a straight or bent palette knife.

How to handle

- Inspect the blade and handle for cracks, corrosion, loose joints, or sharp burrs.
- Select a straight or bent knife suitable for the material and task.
- Hold the handle securely and use controlled movements to mix or spread material without excessive pressure.
- Keep the blade directed away from the body and fingers; do not use it for cutting unless specifically designed for that purpose.
- Remove material residue promptly and clean the knife using a method suitable for the material used.

Storage

- Clean and dry the blade and handle thoroughly after use.
- Store in a tool case with the blade protected from accidental contact and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paper Mache Tool

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Tools and equipment used for paper mache crafting such as sculpting tools, molds, brushes

What it is used for

Paper mache tools such as sculpting tools, moulds, and brushes help shape, texture, and finish paper-based craft items. Select suitable tools and work with adhesives and materials according to their instructions.

How to use

1. Inspect sculpting tools, moulds, brushes, and work surfaces for damage or residue.
2. Cover the work surface and prepare paper pulp, strips, adhesive, and water as required.
3. Select a mould or build a base shape suitable for the design.
4. Apply paper layers or pulp evenly and shape details using suitable tools.
5. Allow each layer to dry as required before adding more layers or finishing details.
6. Apply paint or finish only after the piece is sufficiently dry.
7. Clean reusable tools and organize materials after completing the craft.

⚠ Safety

- Use adhesives, paints, and finishes in a well-ventilated area.
- Avoid contact with eyes and wash hands after handling craft materials.
- Use blunt-edged sculpting tools where appropriate and handle sharp tools carefully.
- Do not use moulds or materials contaminated with mould or unknown chemicals.

Care & storage

- Remove dried residue promptly and store brushes, moulds, and tools clean and dry.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paper Making Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hollander Beater, Hydra Pulper, Paper Squeezing Machine, Klin Drying Machine etc; Hydra Pulper (Breaking/recycling paper)

What it is used for

Paper-making equipment processes paper pulp or recycled paper through operations such as pulping, beating, pressing, and drying. Follow the procedure for the specific machine and maintain safe handling of wet pulp, moving parts, and heated surfaces.

Main parts

- Pulping tank or hydra pulper with rotor
- Beater or refining unit, where fitted
- Screen, strainer, or pulp transfer system
- Pressing rollers or squeezing press
- Drying unit, rack, or heated chamber
- Control panel, guards, and emergency stops

Before you start

1. Identify the equipment type and read its operating instructions.
2. Inspect tanks, blades, rollers, belts, guards, hoses, and electrical connections.
3. Check that emergency stops and protective guards are functional.
4. Confirm the pulp mixture, water level, and batch size meet the process requirements.
5. Ensure drains and transfer paths are clear and collection containers are ready.
6. Wear suitable PPE, including gloves and eye protection, and keep the floor dry where possible.

How to use

1. Switch off and isolate power before opening covers or inspecting internal components.
2. Prepare the pulp or recycled paper mixture according to the process instructions.
3. Load the specified amount of material without exceeding the machine capacity.
4. Close covers and secure guards before starting the machine.
5. Select the correct pulping, beating, pressing, or drying mode for the operation.
6. Start the machine and observe its operation for abnormal noise, vibration, leakage, or blockage.
7. Transfer pulp between process stages using the designated equipment and safe handling methods.
8. For sheet forming, distribute pulp evenly according to the machine's forming procedure.
9. Operate the press only with hands and tools clear of rollers and pressure zones.
10. Dry the formed paper using the specified temperature and time, where applicable.
11. Stop the machine before removing material from hazardous areas or clearing a jam.
12. Switch off and isolate power before cleaning or maintenance.

3. Remove pulp residue, clean the equipment, and dispose of waste according to site procedures.
4. Record batch details, operating settings, and any faults.

⚠ Safety

- Only trained personnel should operate pulping, pressing, and drying equipment.
- Never place hands or tools inside a running pulper or between moving rollers.
- Keep guards and interlocks fitted and never bypass emergency stops.
- Isolate and verify zero energy before clearing blockages or servicing machinery.
- Use caution around hot drying surfaces, steam, and heated rollers.
- Handle wet floors and pulp spills promptly to prevent slips; follow chemical safety instructions for additives.

Care & storage

- Clean pulp tanks, screens, transfer lines, and rollers after use.
- Inspect blades, seals, hoses, belts, and guards regularly.
- Remove fibre buildup before it hardens or obstructs the machine.
- Check electrical systems and emergency stops according to the maintenance schedule.
- Arrange servicing for leaks, abnormal vibration, or damaged parts.

Common mistakes to avoid

- Overloading the pulper or using the wrong pulp consistency.
- Clearing blockages while machinery is energised.
- Placing hands near rollers during pressing or sheet removal.
- Using incorrect drying settings or ignoring wet-floor hazards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Paper Squeezing Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Paper Squeezing Machine (Water removal from pulp)

What it is used for

A paper squeezing machine removes excess water from wet paper pulp or formed paper sheets by applying pressure. It helps prepare the material for drying and further paper-making processes.

Main parts

- Identify the pressing rollers or compression plates that squeeze the wet pulp.
- Locate the feed area where wet pulp or sheets are placed.
- Identify the discharge area for pressed pulp or sheets.
- Locate the motor, drive mechanism and transmission components, where fitted.
- Identify the pressure adjustment and operating controls.
- Locate protective guards and the emergency stop mechanism.
- Identify the water collection tray, drainage channel and supporting frame.

Before you start

1. Read the operating instructions and obtain teacher supervision before use.
2. Inspect rollers, plates, guards, electrical cables and drainage channels for damage.
3. Ensure the machine is stable and correctly assembled.
4. Check that the water collection tray and drainage outlets are clear.
5. Prepare wet pulp or sheets according to the machine's capacity.
6. Tie back long hair and remove jewellery and loose clothing.
7. Keep the working area dry and clear of unnecessary materials.

How to use

1. Switch off and isolate power before making adjustments or preparing the feed area.
2. Check that the pressure setting suits the material and machine instructions.
3. Position the wet pulp or sheet evenly in the designated feed area.
4. Ensure the material does not extend into unsafe areas around the rollers.
5. Close guards and confirm that the emergency stop is accessible.
6. Start the machine using the designated control.
7. Feed the material gradually using the approved feeding arrangement.
8. Monitor water drainage and the uniformity of the pressing operation.
9. Stop feeding if material bunches, slips or causes abnormal noise.
10. Switch off and wait for all moving parts to stop before removing material.
11. Collect the pressed material and transfer it for further drying.

⚠ Safety

- Never place fingers or hands between rollers or pressing plates.
- Do not wear loose gloves near rotating rollers or exposed moving parts.
- Keep protective guards in place throughout operation.
- Isolate power before clearing jams, cleaning or adjusting the machine.
- Keep water away from electrical controls and damaged connections.
- Report unusual vibration, noise or pressure changes immediately.

Care & storage

- Disconnect power before removing pulp residue and accumulated debris.
- Clean rollers, plates and drainage channels using approved methods.
- Inspect guards, fasteners and drive components regularly for damage.
- Store the machine clean and dry after completing maintenance.

Common mistakes to avoid

- Feeding uneven or oversized pulp batches that cause jamming.
- Applying excessive pressure that damages sheets or machine components.
- Reaching between rollers to reposition material while the machine runs.
- Allowing drained water and pulp residue to accumulate around the machine.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pattern Making Ruler

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Hip Curve; Scale Variety

What it is used for

To draw and measure garment curves and pattern lines using a pattern-making ruler. To improve pattern accuracy and maintain consistent measurements.

Main parts

- Curved ruler or hip curve
- Straight scale
- Measurement markings
- Edges and reference points

Before you start

1. Select the ruler shape suitable for the pattern line.
2. Check the markings for readability and the ruler for cracks or warping.
3. Place the pattern paper on a flat surface.
4. Keep a suitable pencil or fabric marker ready.

How to use

1. Identify the pattern points to be connected.
2. Align the ruler with the required measurements and curve.
3. Hold the ruler steady without covering the markings.
4. Draw the line smoothly along the edge.
5. Check the curve and dimensions against the pattern measurements.
6. Label the pattern line or measurement if required.

⚠ Safety

- Handle sharp ruler edges carefully.
- Keep cutting tools separate while marking.
- Store rulers flat to prevent bending.

Care & storage

- Wipe the ruler clean after use.
- Avoid bending or dropping it.
- Keep markings visible and store it in a safe place.

Common mistakes to avoid

- Using the wrong curve for the pattern.
- Misaligning reference points.
- Drawing while the ruler is moving.
- Ignoring measurement units or scale markings.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pattern Notcher

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Hand notcher (pattern notcher)

What it is used for

A hand pattern notcher makes small notches in pattern paper or suitable marking material to identify construction points. Use the correct notch type and keep the tool aligned with the pattern marks.

How to use

1. Inspect the notcher, cutting edges, handle, and pivot for damage or looseness.
2. Place the pattern on a stable, suitable cutting surface.
3. Identify the required notch positions and confirm the pattern orientation.
4. Align the notcher with each marked position and hold the pattern securely.
5. Squeeze or operate the tool to make the required notch without damaging nearby lines.
6. Check the notches for consistency and confirm that pattern markings remain readable.
7. Clean the tool and store it safely after use.

⚠ Safety

- Keep fingers clear of the cutting edges and moving joints.
- Do not use a damaged or loose notcher.
- Use the tool only on suitable pattern materials, not on skin or unsupported hard objects.
- Store the tool closed or protected when possible.

Care & storage

- Remove paper fibres and dust and keep the cutting edges dry and protected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pattern Paper

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Pattern Drafting Paper, Paper; Pattern making paper

What it is used for

Pattern paper is used to draft, trace, and transfer garment or craft patterns before cutting materials.

How to handle

- Check paper for moisture, tears, folds, and suitable dimensions before use.
- Work on a clean, flat surface and secure the paper to prevent shifting.
- Mark patterns clearly using appropriate pencils or drafting tools.
- Use scissors or cutting tools carefully and keep fingers clear of cutting paths.
- Label pattern pieces and avoid wasting usable paper.

Storage

- Store patterns flat or rolled without creasing, as appropriate.
- Keep paper dry, labelled, and protected from dust and direct sunlight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pattern Template

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Design templates; Templates

What it is used for

Pattern templates provide reusable shapes and guides for drawing, drafting, sewing, and design work.

How to handle

- Inspect templates for cracks, warping, sharp edges, or missing markings.
- Choose a template suitable for the required shape, scale, and material.
- Hold the template securely while tracing to prevent slipping.
- Use cutting tools carefully and avoid cutting directly against a template unless it is designed for that purpose.
- Clean templates after use without damaging printed markings.

Storage

- Store templates flat in a labelled folder or protective case.
- Keep away from excessive heat, moisture, and heavy loads that may bend them.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pick Glass

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Pick glass

What it is used for

To inspect or magnify small details in textile, craft, or other close-up work, according to the instrument design.

How to handle

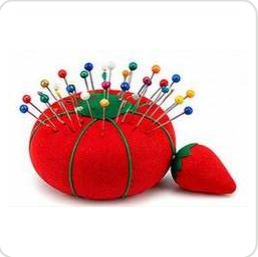
- Inspect the glass, lens, frame, and handle for cracks, chips, loose parts, or distortion.
- Hold the instrument by its handle or frame and position it over the area to be examined.
- Adjust the distance and lighting until the detail is clearly visible; avoid touching delicate surfaces with the glass.
- Keep the glass away from eyes at unsafe distances and do not focus sunlight through a magnifying lens.
- Wipe the lens gently with an appropriate lens-cleaning cloth after use.

Storage

- Clean the lens gently and place a protective cap or cover over it if provided.
- Store in a padded case in a dry place to prevent scratches, breakage, and dust buildup.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pin Cushion



Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Pin Cushions; Pincushion

What it is used for

A pin cushion holds sewing pins safely and keeps them accessible during fabric work.

How to handle

- Inspect the cushion for wear, loose filling, or exposed pins before use.
- Insert pins fully into the cushion and keep the sharp ends safely contained.
- Place the cushion in a stable, visible location on the worktable.
- Count and collect pins after work and avoid leaving them in fabric or on the floor.
- Replace a damaged cushion that cannot hold pins securely.

Storage

- Store the cushion in a designated sewing area away from children.
- Keep it dry and ensure pins are secured before moving it.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pinking Shears

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Pinking shears; Zigzag/ Pinking shear

What it is used for

Use pinking shears to cut fabric with a zigzag edge that helps reduce fraying on suitable textile materials.

How to use

1. Inspect the blades, pivot, and handles for damage or looseness.
2. Choose fabric suitable for pinking shears and confirm the cutting line.
3. Place the fabric flat on a stable cutting surface.
4. Align the blades with the marked line and keep fingers clear of the cutting edge.
5. Cut with smooth, full strokes without twisting the shears.
6. Check the zigzag edge and trim any missed sections carefully.
7. Clean and close the shears after use.

⚠ Safety

- Keep fingers away from the serrated cutting edges.
- Do not use the shears to cut paper, wire, or unsuitable materials.
- Do not force stiff or thick materials through the blades.
- Close the shears when not in use and store them safely.

Care & storage

- Remove fabric fibres from the blades and pivot after use.
- Store dry with the blades closed to protect the cutting edges.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Press Cloth

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Press cloth

What it is used for

Use press cloths to protect fabric during ironing and pressing.

How to handle

- Check the cloth for cleanliness and loose debris before use.
- Select a cloth suitable for the fabric and pressing temperature.
- Test on a small inconspicuous area when fabric compatibility is uncertain.
- Keep the cloth clear of the iron's steam vents and follow iron safety instructions.

Storage

- Wash or clean after use and dry completely.
- Store flat or folded in a clean, dry sewing area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Punch Needle

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Punch Needle

What it is used for

A punch needle creates loops or stitches in fabric for embroidery and textured textile designs. Thread the needle correctly and use a suitable fabric held securely in an embroidery hoop or frame.

How to use

1. Inspect the punch needle, handle, threading channel, and tip for damage.
2. Select compatible yarn or thread and suitable fabric for the needle size.
3. Secure the fabric tightly in a hoop or frame.
4. Thread the needle according to the tool instructions and adjust the yarn flow.
5. Hold the tool at the recommended angle and punch evenly into the fabric.
6. Follow the design pattern and maintain consistent stitch spacing and depth.
7. Finish the yarn securely, release the fabric, and clean the tool after use.

⚠ Safety

- Keep the needle tip pointed away from fingers and body.
- Do not use a bent or damaged needle.
- Keep yarn and loose materials clear of walkways and moving equipment.
- Store the needle with its protective cover when available.

Care & storage

- Remove fibres from the needle and store it dry in a protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Raking Needle

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Racking Needle; Raking Needle

What it is used for

To assist with raking, gathering, or arranging textile fibres or fabric during the relevant textile or handicraft process.

How to handle

- Inspect the needle for bending, rust, sharp damage, or loose attachment to its handle if fitted.
- Select the correct needle for the material and task, and secure the fabric or fibre workpiece before use.
- Guide the needle carefully through or along the material using controlled movements.
- Keep fingers clear of the needle point and do not force it through material that resists penetration.
- Clean the needle after use and account for it before leaving the workstation.

Storage

- Store in a labelled needle case or secure organiser when clean and dry.
- Keep the point covered and separate from loose textiles to prevent injury or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Raw Material Sample

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Samples of bamboo and sliver; Samples of jute fibre and yarn

What it is used for

Handle bamboo, sliver, jute fibre, and yarn samples without contamination or damage.

How to handle

- Label each sample with its material type and source where known.
- Handle fibres gently to preserve their structure and appearance.
- Use a dust mask if loose fibres or particles become airborne.
- Keep different samples separated to prevent mixing.

Storage

- Place samples in labelled bags, sleeves, or sealed containers as appropriate.
- Store dry and away from direct sunlight, pests, and chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ribbon

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Ribbon

What it is used for

Handle ribbons carefully for sewing, decoration, and craft activities.

How to handle

- Check ribbon width, material, and colour before cutting or attaching.
- Use clean scissors and avoid pulling ribbon excessively.
- Keep ribbon away from open flames and hot tools unless the material is heat-rated.
- Prevent tangling by winding ribbon neatly after use.

Storage

- Store on spools or in labelled containers.
- Keep clean and dry, away from direct sunlight and sharp edges.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Rotary Cutter

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Rotary Cutter

What it is used for

To cut fabric, paper, or other suitable sheet materials accurately using a rotating circular blade.

How to handle

- Inspect the blade, handle, blade guard, and locking mechanism for damage before use.
- Place the material on a suitable cutting mat and align it with the required cutting marks.
- Keep fingers and other body parts away from the blade path and use a ruler or guide designed for rotary cutting where appropriate.
- Roll the cutter away from the body using controlled pressure; never leave the blade exposed when not cutting.
- Engage the blade guard or lock immediately after use and replace or secure the blade only according to the manufacturer's instructions.

Storage

- Store with the blade guard engaged in a designated tool case or drawer.
- Keep spare blades covered and out of reach of children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sculpting Tool

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Clay Modelling Tools; Clay Tool Kit; Sculpting hand tools

What it is used for

To shape, carve, smooth, and add details to clay or other suitable modelling materials.

How to handle

- Inspect each tool for cracks, sharp damage, loose handles, or contamination before use.
- Select a tool appropriate for the material and intended shaping or detailing task.
- Secure the workpiece on a stable surface and use controlled movements, directing sharp edges away from the body.
- Avoid excessive force and keep pointed tools away from eyes and fingers.
- Remove material residue and clean the tools after use according to their material and manufacturer's instructions.

Storage

- Store clean and dry in a labelled tool kit with sharp tips protected.
- Keep tools organised by type and away from moisture or heavy objects that may cause damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seam Ripper

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Seam ripper

What it is used for

Use a seam ripper to remove unwanted stitches or open seams without unnecessarily damaging the fabric.

How to use

1. Inspect the seam ripper tip, blade, handle, and protective cap for damage.
2. Place the fabric on a flat, well-lit surface.
3. Identify the stitches to remove and keep the fabric supported.
4. Slide the pointed end beneath a stitch and cut carefully with the blade.
5. Remove loose threads gradually without pulling or tearing the fabric.
6. Check the fabric for accidental cuts and finish removing thread fragments.
7. Cover the tip and store the tool after use.

⚠ Safety

- Keep fingers away from the sharp blade and cutting path.
- Cut stitches away from your body where practical.
- Do not use a bent or damaged seam ripper.
- Keep the protective cap on when the tool is not in use.

Care & storage

- Remove loose threads from the blade and handle after use.
- Store with the tip covered in a dry sewing kit.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Seam Roll

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Seam roll

What it is used for

To roll and press seams or fabric layers smoothly during sewing and garment construction.

How to handle

- Inspect the seam roll for tears, loose covers, stains, or an unstable shape.
- Position the fabric seam over the roll so the surrounding fabric is supported without unwanted creases.
- Press or roll the seam using the appropriate hand pressure or an iron only if the roll's material is heat-rated.
- Keep fingers clear of a hot iron and follow the garment fabric's temperature requirements.
- Remove lint and loose threads after use and check the roll remains clean and dry.

Storage

- Remove lint and allow the roll to dry fully before storing.
- Store in a clean, dry sewing area away from moisture, dust, and direct heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sewing Gauge

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Hem Gauge; Sewing Gauge

What it is used for

To measure small distances, hems, seam allowances, and fold widths during sewing. To maintain consistent garment measurements and accurate marking.

Main parts

- Gauge body
- Measurement scale
- Sliding marker or adjustable guide, if fitted
- Reference edge

Before you start

1. Check that the scale is readable and the gauge is not damaged.
2. Confirm the required unit and measurement.
3. Place the fabric on a flat surface.
4. Keep suitable marking tools ready.

How to use

1. Set the gauge to the required measurement if adjustable.
2. Align its reference edge with the fabric edge or seam line.
3. Mark the measurement at the required points.
4. Move the gauge along the fabric and repeat the marking consistently.
5. Check the spacing before sewing or cutting.

⚠ Safety

- Keep the gauge away from cutting blades when not in use.
- Handle pointed marking tools carefully.
- Do not use a cracked or unreadable gauge.

Care & storage

- Wipe the gauge clean after use.
- Store it flat in a sewing kit.
- Keep scale markings visible.

Common mistakes to avoid

- Measuring from the wrong reference edge.

- Changing the gauge setting accidentally.
- Marking on folded or uneven fabric.
- Ignoring the seam allowance measurement.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sewing Kit

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Sewing Box; Types of Scales: normal straight big ruler, hip curve, leg shaper, pattern master, French curve etc.; Sewing kit (Includes thread clipper/hand trimmer, seam ripper, fabric cutting scissors, tracing wheel, measuring tape etc.)

What it is used for

A sewing kit contains tools and supplies for stitching, repairs, measuring, and pattern-making activities.

How to handle

- Check threads, needles, scissors, clippers, rulers, and curves for damage or missing parts.
- Use the correct tool for each task and handle needles and cutting tools carefully.
- Keep pins and needles in a secure holder when not in use.
- Pass sharp tools handle-first and keep fingers clear when cutting or trimming.
- Return tools after use and count sharp items to prevent accidental injury.

Storage

- Keep the kit organised in a closed box or case.
- Store needles, scissors, and small accessories in designated compartments away from children.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sewing Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: Feed of arm machine; High speed sewing machine; Lock stitch machine; Sewing Machine; Sewing Machines with tables; Zigzag sewing machine

What it is used for

A sewing machine joins fabric pieces using stitches formed by a needle and thread. Correct threading, fabric control, and safe machine operation help produce consistent seams and prevent needle injuries.

Main parts

- Needle carries the upper thread through the fabric.
- Presser foot holds the fabric against the feed mechanism.
- Feed dogs move the fabric beneath the presser foot.
- Needle plate supports the fabric around the needle opening.
- Thread spool and guides direct thread toward the needle.
- Bobbin and bobbin case supply the lower thread on applicable machines.
- Handwheel and speed control allow controlled needle movement and sewing speed.

Before you start

1. Read the machine instructions and identify the controls before starting.
2. Place the machine on a stable table with adequate lighting and workspace.
3. Inspect the needle, presser foot, thread path, bobbin, plug, and power cord for damage.
4. Switch off the machine before threading, changing the needle, or adjusting the presser foot.
5. Select a needle, thread, and stitch suitable for the fabric and task.
6. Keep fingers, loose clothing, and long hair away from the needle area.

How to use

1. Raise the needle and presser foot using the designated controls or handwheel.
2. Thread the upper thread and bobbin according to the machine's threading diagram.
3. Place the fabric beneath the presser foot and lower the foot onto the fabric.
4. Turn the handwheel slowly to confirm the needle moves freely before powered sewing.
5. Connect power and select a suitable stitch and speed for the material.
6. Hold the thread ends briefly and guide the fabric without pulling it.
7. Use the foot pedal or speed control to sew at a steady pace.
8. Stop with the needle in a safe position and raise the presser foot before removing fabric.
9. Cut the threads using the designated cutter or suitable scissors.
10. Switch off and disconnect power before cleaning lint or carrying out adjustments.

⚠ Safety

- Keep fingers away from the moving needle and presser foot during sewing.
- Never sew over pins or use a bent or damaged needle.
- Switch off power before threading, changing needles, or clearing tangled thread.
- Do not operate the machine with damaged wiring or missing safety guards.
- Use the handwheel only in the direction recommended by the manufacturer.
- Keep the machine and floor around the pedal dry and free from clutter.

Care & storage

- Remove lint from the bobbin area and feed dogs after switching off and disconnecting power.
- Replace blunt, bent, or damaged needles with the correct type.
- Follow the manual for lubrication, cleaning, and servicing requirements.
- Cover the machine and store accessories neatly in a dry place.

Common mistakes to avoid

- Threading the machine while it remains powered on.
- Pulling fabric forcefully while the needle is moving.
- Using the wrong needle or incorrect thread path for the fabric.
- Leaving lint or tangled thread in the bobbin area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sewing Machine Attachment

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Attachments for sewing machine

What it is used for

Sewing machine attachments provide specialized functions such as hemming, gathering, zipper insertion, or edge finishing. Confirm attachment compatibility and install it according to the sewing machine instructions.

How to use

1. Switch off the sewing machine and disconnect power before changing attachments.
2. Identify the attachment type and confirm compatibility with the machine model.
3. Remove the existing presser foot or accessory using the approved procedure.
4. Fit the attachment securely and check needle clearance by turning the handwheel manually.
5. Adjust stitch settings, needle position, and thread tension as recommended.
6. Test the attachment on scrap fabric and inspect stitch quality.
7. Switch off the machine, remove the attachment if needed, and clean the work area.

⚠ Safety

- Keep fingers away from the needle and moving parts during operation.
- Never change attachments while the machine is powered.
- Use the correct needle and settings for the attachment and fabric.
- Do not operate if the needle strikes the attachment.

Care & storage

- Remove lint and thread debris after use and store attachments in labelled containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sewing Needle

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 12 labs

Covers these lab items: Hand Needle; Hand Needles; Machine oil, Machine Needle and hand needles (Various Gauges), embroidery needles; Needle; Needle and Thread (different colour); Needles

What it is used for

Sewing needles are used for hand stitching, embroidery, and machine sewing with compatible fabrics and equipment.

How to handle

- Select needle type, size, point, and gauge suitable for the fabric and sewing method.
- Inspect needles for bends, rust, blunt points, or damage before use.
- Handle sharp needles carefully and use the correct machine needle installation procedure.
- Do not force a needle through fabric or leave loose needles on work surfaces.

Storage

- Keep needles in a labelled case, pincushion, or designated sharps holder.
- Separate machine needles and hand needles by size and type and count them after use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sewing Pin

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Beaded Pins; Dress Maker's Pin, Fabric Pins; Pins; Pins and pin cushion

What it is used for

Sewing pins hold fabric layers and pattern pieces in place during cutting and stitching.

How to handle

- Inspect pins for bends, rust, damaged tips, or loose beads before use.
- Use a pincushion or pin holder and keep pins away from the edge of the worktable.
- Insert pins carefully and remove them before machine stitching when required.
- Count and collect pins after use to prevent injury or fabric damage.
- Never leave pins hidden in finished garments or fabric scraps.

Storage

- Store pins in a closed labelled box or secure pincushion.
- Keep pins dry and separated by type where necessary.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shade Card

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Color Swatches and Pantone Guides; Pantone Shade Card, Fashion Forecasting Books/Journals/Magazines; Swatch File; Swatch File, Buyer Requirement; Trims and Accessories sample swatch file; Trims/Accessory samples; Types of Stains Swatches file or samples

What it is used for

Shade cards and swatch files help compare colours, fabrics, trims, and design choices accurately.

How to handle

- Check the shade card name, reference code, version, and material information.
- Compare colours under consistent lighting to reduce visual mismatch.
- Handle swatches with clean hands and avoid staining, bending, or fading them.
- Confirm the required shade against the approved buyer or project reference before ordering materials.
- Report missing, damaged, or outdated samples.

Storage

- Store shade cards and swatches in labelled folders away from sunlight and moisture.
- Keep reference codes visible and retain approved versions for comparison.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Shading Stump

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Shading stump

What it is used for

Handle a shading stump carefully for drawing, shading, and craft activities.

How to handle

- Inspect the stump for damage, loose parts, or sharp edges before use.
- Use it only for its intended shading or craft purpose.
- Keep it away from flames and avoid inhaling dust produced during use.
- Wash hands after handling if residue remains on the skin.

Storage

- Keep clean and dry in a labelled stationery or art-supply container.
- Store away from heat, moisture, and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Size Chart

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Measurement Sheet, spec sheet other, size chart; Measurement Sheet/Size Chart, Trims/Accessory/fabric and embroideries samples/swatch file

What it is used for

Use measurement sheets, size charts, and swatch files accurately for garment specification and selection.

How to handle

- Verify the garment style, measurement units, and size range before use.
- Measure at the specified points and record values clearly.
- Keep fabric, trim, accessory, and embroidery swatches labelled and linked to the correct specification.
- Avoid marking or damaging original reference charts and samples.

Storage

- File charts and sheets by style or project in labelled folders.
- Keep swatches clean, dry, and protected from fading or contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sleeve Board

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Sleeve board

What it is used for

To support narrow garment areas such as sleeves and cuffs during pressing and ironing.

How to handle

- Inspect the sleeve board surface, padding, cover, and support legs for damage or instability.
- Place the board on a stable, heat-resistant work surface and ensure it cannot slip during use.
- Position the sleeve or cuff over the narrow end and smooth the fabric before pressing.
- Set the iron temperature to suit the fabric and keep fingers clear of the hot soleplate and steam.
- Allow the board to cool before folding or moving it, and remove lint or loose threads after use.

Storage

- Clean the cover and allow the board to cool and dry completely before storage.
- Fold or secure the board according to its design and store in a dry place away from moisture and heavy objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Spinning Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Spinning Machine

What it is used for

A spinning machine twists fibres together to form yarn of the required thickness and strength. It may be operated manually or by motor, depending on the design and textile process.

Main parts

- Identify the fibre feeding area and drafting or fibre preparation rollers.
- Locate the spindle, rotor or twisting mechanism used to form yarn.
- Identify the bobbin, spool or yarn collection arrangement.
- Locate the drive belt, motor and transmission components, where fitted.
- Identify the speed control and start-stop switches.
- Locate the thread guides, tension controls and protective guards.
- Identify the emergency stop and cleaning access points.

Before you start

1. Read the operating instructions and obtain teacher supervision before use.
2. Inspect the spindle, rollers, belts, thread guides and guards for damage.
3. Ensure fibres are clean and suitable for the machine type.
4. Tie back long hair and remove jewellery and loose clothing.
5. Keep hands clear of rotating components and avoid loose gloves.
6. Check that bobbins are correctly fitted and the work area is uncluttered.
7. Confirm that electrical connections and emergency controls are functioning properly.

How to use

1. Switch off the machine before threading fibres or fitting a bobbin.
2. Position the prepared fibres in the designated feeding arrangement.
3. Guide the fibre through the specified rollers and thread guides.
4. Attach the fibre end securely to the empty bobbin or collection point.
5. Set the recommended speed and tension for the fibre and yarn type.
6. Close guards and confirm that loose tools are removed.
7. Start the machine and observe the initial yarn formation.
8. Feed fibres steadily and monitor yarn thickness and thread tension.
9. Stop the machine before correcting broken yarn or adjusting the thread path.
10. Switch off and wait for rotating parts to stop before removing full bobbins.

⚠ Safety

- Never touch rotating spindles, rollers or drive belts.
- Do not wear loose gloves, jewellery or loose sleeves near moving parts.
- Switch off and isolate power before clearing tangled fibres or servicing.
- Keep guards fitted and never bypass safety switches.
- Avoid inhaling accumulated fibre dust and maintain suitable ventilation.
- Report unusual vibration, noise or overheating to the teacher.

Care & storage

- Remove fibre lint and dust after switching off and isolating power.
- Clean thread guides and rollers using approved tools.
- Inspect belts, bearings, spindles and guards for wear or damage.
- Store bobbins, tools and the machine in a clean, dry area.

Common mistakes to avoid

- Feeding fibres unevenly and producing inconsistent yarn thickness.
- Operating with tangled fibres or incorrectly fitted bobbins.
- Reaching into rotating parts to repair broken yarn.
- Ignoring worn belts, damaged guards or excessive fibre accumulation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Squeegee

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Squeegees

What it is used for

A squeegee removes excess liquid, smooths suitable surfaces, or spreads material during cleaning and craft work. Select the correct blade type and use even pressure to avoid damaging the surface.

How to use

1. Inspect the handle, blade, and attachment points for cracks or wear.
2. Choose a blade suitable for the surface and material being handled.
3. Prepare the work area and remove loose debris where necessary.
4. Hold the handle firmly and place the blade against the surface at a suitable angle.
5. Move the squeegee with even, controlled strokes toward the intended collection area.
6. Wipe the blade between passes if residue builds up.
7. Clean the tool and allow it to dry after use.

⚠ Safety

- Keep fingers away from sharp blade edges and damaged parts.
- Use suitable gloves when handling cleaning chemicals or contaminated liquids.
- Keep wet floors marked or restricted to prevent slips.
- Do not use a damaged blade on delicate surfaces.

Care & storage

- Rinse off residue, dry the blade, and store the squeegee without bending the rubber edge.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Steam Press

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Steam press

What it is used for

A steam press uses controlled heat, pressure, and steam to remove wrinkles and shape garments or textile materials. Correct operation improves finish quality while protecting fabric and the operator.

Main parts

- Pressing head or heated plate
- Lower pressing buck or table
- Steam supply and steam control
- Temperature and pressure controls
- Foot pedal or operating handle
- Safety guard and condensate collection system if fitted

Before you start

1. Read the operating instructions and identify the fabric's heat and steam limits.
2. Inspect the pressing surfaces, controls, hoses, guards, and power connections for damage.
3. Check the steam supply and condensate system for leaks or blockage.
4. Set the temperature and pressure suitable for the fabric and garment.
5. Ensure the pressing area is dry, clean, and free from loose objects.
6. Wear suitable protection and keep the foot pedal or controls unobstructed.

How to use

1. Switch on the machine and allow the pressing surfaces to reach the set temperature.
2. Arrange the garment smoothly on the lower buck without stretching or trapping seams.
3. Use a pressing cloth when required for delicate, dark, or heat-sensitive fabrics.
4. Keep hands clear of the pressing area and activate the press using its designated control.
5. Apply steam and pressure for the time recommended for the fabric.
6. Release the press fully before repositioning the garment.
7. Check the finish and repeat only if necessary, adjusting settings when needed.
8. Place the pressed garment on a safe surface or hanger to cool and dry.
9. Switch off the machine and isolate the power and steam supply after use.
10. Allow hot surfaces to cool before cleaning or carrying out routine maintenance.

⚠ Safety

- Keep fingers, hands, and clothing away from the closing pressing surfaces.
- Never defeat guards, two-hand controls, or other fitted safety devices.
- Do not press garments while wearing them or direct steam toward people.
- Use heat-resistant protection when handling hot fabrics or components as required.
- Report damaged hoses, electrical cables, controls, or steam leaks immediately.
- Isolate power and steam before servicing or clearing faults.

Care & storage

- Wipe pressing surfaces with approved materials after they cool.
- Inspect steam hoses, electrical cables, controls, and safety devices regularly.
- Drain or maintain the steam system according to the manufacturer's instructions.
- Keep the press clean, dry, and covered when not in use.

Common mistakes to avoid

- Using excessive heat can scorch or shrink fabric.
- Placing hands near the closing head can cause crushing or burns.
- Applying too much pressure can leave marks or damage seams.
- Ignoring steam leaks or damaged controls can create serious hazards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stencil

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Stencil

What it is used for

Handle stencils carefully for accurate tracing, marking, and textile decoration.

How to handle

- Check the stencil for cracks, bent sections, or blocked cut-outs before use.
- Secure it flat on the work surface and use compatible marking materials.
- Apply colour or marking medium evenly without excessive pressure or seepage.
- Clean carefully after use to avoid damaging fine details.

Storage

- Dry completely before storing flat or in a protective sleeve.
- Keep labelled stencils away from heat, bending, and heavy objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stencil Cutting Tool

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Stencil cutting tools

What it is used for

Stencil cutting tools create precise shapes and openings in suitable stencil sheets or craft materials. Select the correct blade and cut on a stable protective mat using controlled movements.

How to use

1. Inspect the cutter, blade, handle, and blade-locking mechanism for damage.
2. Choose a blade suitable for the stencil material and required detail.
3. Place the stencil sheet on a stable cutting mat.
4. Mark the design clearly and secure the material to prevent movement.
5. Hold the tool firmly and cut along the marked lines using controlled strokes.
6. Rotate the material when practical rather than forcing awkward cutting angles.
7. Collect offcuts safely, cover or retract the blade, and store the tool.

⚠ Safety

- Cut away from the body and keep the free hand outside the cutting path.
- Use a cutting mat and never cut directly on an unprotected surface.
- Do not use loose, blunt, or damaged blades.
- Wear suitable eye protection when small pieces may fly.

Care & storage

- Remove adhesive and material residue carefully and store blades in a protective container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Striking Needle

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Needle for Striking

What it is used for

To guide stitches or manipulate fabric during textile and handicraft work using a striking needle. To work carefully without damaging the material or injuring the hands.

Main parts

- Needle shaft
- Pointed tip
- Eye or thread opening, if fitted
- Handle or grip, if fitted

Before you start

1. Select a needle suitable for the material and task.
2. Check the needle for bends, cracks, or a damaged tip.
3. Secure the fabric or workpiece on a stable surface.
4. Keep fingers clear of the needle path.

How to use

1. Mark the intended stitch or insertion point.
2. Hold the needle securely by its handle or safe grip area.
3. Position the tip at the marked point.
4. Guide the needle through the material using controlled pressure.
5. Pull the needle through carefully without forcing it.
6. Check the result and repeat as needed.
7. Place the needle in a safe holder after use.

⚠ Safety

- Keep the sharp tip pointed away from the body.
- Do not use a bent or damaged needle.
- Avoid leaving needles loose in fabric or on work surfaces.
- Use a suitable needle holder when required.

Care & storage

- Wipe the needle clean after use.
- Keep it dry to prevent corrosion.

- Store in a labelled needle case or holder.
- Replace damaged needles promptly.

Common mistakes to avoid

- Using the wrong needle for the material.
- Forcing the needle through resistant material.
- Leaving the needle hidden in fabric.
- Storing the needle loose and exposed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tailor's Chalk

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 10 labs

Covers these lab items: Marking chalk; Marking Chalk (yellow/pink/green/blue coloured), Tailor's Chalk, Carbon paper (A4, yellow, red and white carbon); Tailor's Chalk

What it is used for

Tailor's chalk marks temporary cutting, stitching, and design guides on fabric.

How to handle

- Select a chalk colour visible against the fabric.
- Test the chalk on a fabric scrap to check removability.
- Mark lightly using suitable rulers or pattern templates.
- Avoid pressing too hard or damaging delicate fabric.
- Brush away markings or remove them according to fabric care instructions.

Storage

- Keep chalk dry in a labelled container.
- Store separately from fabrics to prevent unwanted marks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thimble

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Thimble

What it is used for

A thimble protects the fingertip while pushing a needle through fabric during hand sewing. Choose a size that fits securely without restricting finger movement.

How to use

1. Inspect the thimble for cracks, sharp edges, or deformation.
2. Choose a thimble that fits comfortably on the finger used to push the needle.
3. Place it securely on the fingertip before sewing.
4. Use the thimble to push the needle through fabric while guiding the needle with the other hand.
5. Remove the thimble after sewing and check it for damage.
6. Clean and store it in the sewing kit.

⚠ Safety

- Keep the needle pointed away from fingers and avoid forcing it through thick fabric.
- Do not use a cracked or poorly fitting thimble.
- Store needles separately and safely when not in use.

Care & storage

- Wipe the thimble clean and store it dry with sewing supplies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thread

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 13 labs

Covers these lab items: Embroidery thread; Embroidery Threads; Measuring thread; Sewing Threads; Thread; Zari; Zari thread

What it is used for

Threads are used for sewing, embroidery, decorative work, and textile assembly across different materials.

How to handle

- Select thread type, thickness, colour, and strength suitable for the fabric and task.
- Check thread for tangling, contamination, or damage before use.
- Thread needles or machines according to the equipment instructions and keep hands away from moving parts.
- Secure loose thread ends and avoid pulling thread forcefully.

Storage

- Store spools in labelled containers away from moisture, dust, and direct sunlight.
- Keep different colours and thread types organised to prevent mix-ups.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thread Cutter

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Thread Clipper; Thread Cutter

What it is used for

A thread cutter is used to trim loose threads and excess thread ends during sewing and textile work. Use it carefully to make clean cuts without damaging the fabric.

How to use

1. Inspect the cutter for clean blades, smooth movement, and damage.
2. Hold the fabric steady and identify the thread to be trimmed.
3. Position the blades close to the loose thread without touching the fabric.
4. Cut the thread with a controlled movement.
5. Check that the fabric and nearby stitches remain undamaged.
6. Clean the blades and store the cutter safely after use.

⚠ Safety

- Keep fingers away from the sharp cutting edges.
- Do not cut threads while pulling the fabric toward your hand.
- Keep the cutter closed or covered when not in use.

Care & storage

- Remove thread scraps and store the cutter in a dry, safe place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thread Stand

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Cone Thread Holder

What it is used for

A cone thread stand holds thread cones and guides thread smoothly to a sewing machine. Position the stand securely and ensure the thread path remains unobstructed during sewing.

How to use

1. Inspect the stand base, spindle, guide eyelets, and thread path for damage.
2. Place the stand on a stable surface behind or beside the sewing machine as designed.
3. Fit the thread cone securely onto the spindle.
4. Route the thread through the guide eyelets in the specified order.
5. Pull the thread gently to check that the cone rotates freely without snagging.
6. Thread the sewing machine according to its instructions and test on scrap fabric.
7. Remove loose thread and store the stand safely after use.

⚠ Safety

- Keep the stand clear of moving machine parts and walkways.
- Do not use a cracked base or bent spindle.
- Avoid pulling thread forcefully or allowing it to wrap around moving components.
- Switch off the sewing machine before correcting a jammed thread path.

Care & storage

- Remove lint and thread debris regularly and store the stand in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Thread Sucking Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Thread Sucking Machine

What it is used for

A thread sucking machine uses suction to collect loose thread ends, lint, and textile waste from garments or sewing work areas. Proper operation keeps the work area clean and helps improve finishing quality.

Main parts

- Machine body and suction motor
- Suction hose or nozzle
- Thread and lint collection container
- Air filter or dust filter
- Power switch and electrical cable
- Foot switch or suction control if fitted

Before you start

1. Read the operating instructions and confirm the machine is suitable for the intended textile work.
2. Inspect the power cable, plug, hose, nozzle, and collection container for damage.
3. Check that the filter is installed correctly and is not blocked.
4. Empty the collection container if it is near capacity.
5. Ensure the work area is dry and the machine is placed securely.
6. Connect the machine to the approved power supply and check that controls are accessible.

How to use

1. Switch on the machine using the designated power control.
2. Adjust the suction setting if the model provides adjustable airflow.
3. Position the nozzle near loose thread ends or lint without touching delicate fabric unnecessarily.
4. Move the nozzle slowly over the required area to collect loose threads and textile waste.
5. Keep fingers, fabric edges, and accessories away from the suction opening.
6. Check the workpiece regularly to confirm that loose threads have been removed without damaging seams.
7. Switch off the machine before emptying the collection container or checking a blockage.
8. Remove collected waste and dispose of it according to workshop procedures.
9. Clean the filter and nozzle as specified by the manufacturer.
10. Switch off, unplug, and store the machine safely after use.

⚠ Safety

- Keep hands, hair, jewellery, and loose clothing away from the suction opening.
- Do not suck liquids, hot material, or sharp objects unless the machine is specifically designed for them.
- Never operate the machine with a damaged cable, plug, or exposed electrical part.
- Disconnect the power supply before cleaning, clearing blockages, or servicing the machine.
- Do not remove filters or open the collection chamber while the machine is running.
- Keep the machine away from water and maintain a dry work area.

Care & storage

- Empty the collection container regularly to maintain suction performance.
- Clean the nozzle and filter according to the manufacturer's instructions.
- Inspect hoses, seals, switches, and cables for wear or damage.
- Store the machine in a clean, dry place with the hose protected from kinks.

Common mistakes to avoid

- Using the machine with a blocked filter reduces suction and can overheat the motor.
- Pressing the nozzle hard against fabric may damage delicate material or seams.
- Allowing the collection container to overfill can block airflow.
- Cleaning or clearing a blockage while connected to power can cause injury.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tracing Wheel

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Tracing Wheel

What it is used for

To transfer pattern lines or markings onto fabric or suitable pattern material using a tracing wheel and marking medium.

How to handle

- Inspect the wheel, teeth or edge, handle, and attachment for damage or looseness before use.
- Place the pattern and fabric on a suitable flat surface and secure them to prevent slipping.
- Use the appropriate tracing paper or marking medium for the fabric and follow the material's care instructions.
- Roll the wheel along the pattern lines with controlled pressure, keeping fingers clear of the wheel edge.
- Clean the wheel after use and test any marking medium on a small area first if staining or fabric damage is possible.

Storage

- Store clean and dry in a sewing kit or designated drawer.
- Protect the wheel edge from damage and keep marking materials sealed to prevent drying or contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Weaving Beater

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Picks and Beaters

What it is used for

To beat or compact inserted weft yarn evenly into the fabric during weaving.

How to handle

- Inspect the beater for cracks, loose parts, bent teeth, or sharp damage.
- Position the tool correctly for the loom or weaving frame being used.
- After inserting the weft yarn, move the beater evenly and gently to set the yarn in place.
- Keep fingers clear of moving loom parts and avoid excessive force that could damage yarn or fabric.
- Remove loose fibres and inspect the tool after use.

Storage

- Clean off fibres and dry the tool before storage.
- Store in a designated dry tool rack or container to prevent bending and damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Yarn

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Novelty Yarns; Yarn Spools (Set of 3)

What it is used for

Handle novelty yarns and yarn spools to preserve fibre quality and prevent tangling.

How to handle

- Check yarn type, colour, and thickness against the project requirement.
- Pull yarn smoothly from the spool and avoid tight knots or sudden tension.
- Keep yarn away from wet surfaces, dust, and sharp tools.
- Separate different yarn types and colours during work.

Storage

- Store spools in labelled bags, boxes, or racks.
- Keep dry and protected from sunlight, pests, and contamination.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Zipper

Category: **Sewing, Textile & Handicraft** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Zipper

What it is used for

Handle zippers carefully for sewing and garment assembly.

How to handle

- Check zipper length, teeth, slider, and material before use.
- Open and close the slider gently without forcing it.
- Keep the zipper aligned during stitching to prevent puckering.
- Use suitable tools to trim excess material without damaging the teeth.

Storage

- Store zippers in labelled packets sorted by size and type.
- Keep dry and away from dust, moisture, and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Zipper Guide

Category: **Sewing, Textile & Handicraft** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Zipper Guide

What it is used for

A zipper guide helps align fabric and zipper components during sewing or installation. Use the guide suited to the sewing method and zipper type.

How to use

1. Inspect the guide for cracks, burrs, or loose attachment points.
2. Select a guide compatible with the zipper and sewing machine or hand-sewing setup.
3. Position the zipper and fabric according to the guide instructions.
4. Secure the guide without obstructing the needle, presser foot, or zipper slider.
5. Test alignment slowly on scrap fabric before sewing the actual piece.
6. Remove the guide after sewing and check the zipper movement.

⚠ Safety

- Keep fingers away from the sewing needle and moving machine parts.
- Switch off the sewing machine before installing or adjusting the guide.
- Do not force fabric or zipper teeth through a misaligned guide.

Care & storage

- Remove lint and loose threads and store the guide in a dry container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Battery

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **High** 8 labs

Covers these lab items: Batteries for total station; Battery; Coin Cell; Spare Batteries and Bulbs

What it is used for

Batteries supply electrical energy to instruments, devices, and equipment. Identify the correct battery type, voltage, and polarity before installation or replacement.

How to use

1. Check the battery label, type, voltage, and required polarity for the device.
2. Inspect the battery for leakage, swelling, corrosion, or physical damage.
3. Switch off the device before replacing a removable battery.
4. Install the battery according to the positive and negative markings.
5. Use the approved charger only for rechargeable batteries.
6. Switch on the device and confirm normal operation or battery status.
7. Remove depleted batteries when appropriate and send them to an approved collection point.

⚠ Safety

- Never short-circuit battery terminals or connect them with metal objects.
- Do not recharge non-rechargeable batteries or use incompatible chargers.
- Do not open, crush, puncture, or burn batteries.
- Handle leaking or damaged batteries according to safety guidance and avoid skin contact.

Care & storage

- Store batteries in a cool, dry place away from metal objects and direct heat.
- Separate used batteries for approved recycling and replace them with the specified type.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Battery Charger

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **High** 4 labs

Covers these lab items: Battery Charger

What it is used for

A battery charger supplies controlled electrical energy to recharge compatible rechargeable batteries. Match the charger to the battery chemistry, voltage, and charging requirements before connecting it.

How to use

1. Inspect the charger, cable, plug, and connectors for damage.
2. Check the battery label and confirm that the charger is compatible.
3. Place the battery and charger in a dry, well-ventilated location.
4. Connect the battery with the correct polarity and follow the manufacturer's sequence.
5. Switch on the charger and check its status indicators.
6. Allow charging to finish according to the charger instructions and monitor for unusual heat or alarms.
7. Switch off the supply before disconnecting if required by the instructions.
8. Disconnect, inspect the battery, and store the charger safely after use.

⚠ Safety

- Never charge damaged, leaking, swollen, or incompatible batteries.
- Keep the charger away from water and flammable materials.
- Do not cover ventilation openings or leave charging equipment unattended where prohibited.
- Use only the specified power supply and approved charger.

Care & storage

- Keep the charger and connectors clean, dry, and free from dust.
- Inspect cables regularly and replace damaged parts before further use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Battery Storage Unit

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Battery Storage Unit (Lithium-ion)

What it is used for

A lithium-ion battery storage unit stores electrical energy and supplies power to connected equipment or loads through a battery management system and associated power electronics. Operate it within the manufacturer's voltage, temperature, current, and charging limits, and monitor alarms to reduce the risk of electric shock, fire, and thermal runaway.

Main parts

- Lithium-ion battery modules or cells
- Battery management system (BMS)
- Battery enclosure and protective covers
- Power conversion equipment or inverter, if fitted
- Electrical isolators, fuses, and circuit breakers
- Cooling or ventilation system and monitoring interface

Before you start

1. Confirm the unit's chemistry, rated voltage, capacity, operating limits, and approved application.
2. Check the enclosure, cables, terminals, connectors, isolators, and visible modules for damage, swelling, corrosion, or leakage.
3. Review the BMS display for state of charge, temperature, voltage, alarms, and fault history.
4. Verify that earthing, overcurrent protection, emergency isolation, and installation clearances meet the approved design.
5. Confirm that cooling, ventilation, fire detection, and other required protective systems are operational.
6. Ensure that only authorised personnel are present and that the emergency response procedure is available.

How to use

1. Inspect the installation and confirm that no active alarm or unresolved safety fault is present.
2. Verify that the connected load, inverter, and charging source are compatible with the battery unit.
3. Confirm that all covers are closed and protective interlocks are in their required operating state.
4. Energise the system using the manufacturer's prescribed startup sequence and approved isolator procedure.
5. Check the BMS and power conversion interface for normal voltage, current, temperature, and communication status.

6. Start charging or discharging only within the approved operating limits and site energy-management procedure.
7. Monitor cell or module temperatures, state of charge, current, and alarms during operation.
8. If a warning occurs, follow the approved response procedure and reduce or stop operation when required.
9. If smoke, unusual odour, hissing, swelling, rapid temperature rise, or fire is observed, keep clear, raise the alarm, and follow the emergency plan; do not open the enclosure.
10. For shutdown, stop the load or charging process and isolate the system using the approved sequence.
11. Allow the system to reach the required safe state before any inspection or maintenance; use the specified electrical isolation and verification procedure.
12. Record operating data, alarms, inspections, and maintenance actions in the site log.

⚠ Safety

- Only qualified and authorised personnel should install, commission, isolate, or service the battery unit.
- Battery terminals and internal circuits may remain electrically hazardous even when the inverter is off; follow the approved isolation procedure.
- Never open battery modules, bypass the BMS, short-circuit terminals, or use unapproved chargers.
- Do not operate a battery that is damaged, swollen, leaking, overheated, or showing a critical fault.
- Keep combustible materials clear and maintain required fire detection, ventilation, and emergency access.
- If thermal runaway or fire is suspected, evacuate the affected area and contact emergency responders; follow the site-specific battery fire plan.

Care & storage

- Inspect the enclosure, terminals, cables, cooling equipment, and protective devices according to the maintenance schedule.
- Review BMS alarms and trends in cell voltage, temperature, and capacity for abnormal changes.
- Keep air inlets and cooling paths clear; do not obstruct ventilation openings.
- Arrange qualified testing and servicing of protection devices, connections, and power electronics.
- Follow the manufacturer's requirements for storage state of charge, temperature, and long-term shutdown.

Common mistakes to avoid

- Operating outside the specified voltage, current, or temperature limits.
- Ignoring BMS alarms, unusual heating, swelling, or abnormal odours.
- Opening the enclosure or servicing electrical parts without proper isolation and authorisation.
- Blocking cooling paths or using incompatible charging equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrolyser

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Electrolyser

What it is used for

An electrolyser uses electrical energy to split water or another approved feedstock into chemical products, such as hydrogen and oxygen in water electrolysis. Correct fluid preparation, electrical control, gas handling, and ventilation help maintain process efficiency and reduce fire, explosion, chemical, and electrical hazards.

Main parts

- Electrolyser stack or cell assembly
- Power supply and current or voltage controls
- Feedwater or electrolyte reservoir and circulation system
- Gas outlets, separators, and collection lines
- Pressure, temperature, and flow sensors
- Ventilation, cooling system, protective enclosure, and emergency shutdown

Before you start

1. Operate the electrolyser only if trained and authorised for the specific technology and process.
2. Read the equipment manual, process diagram, safety data sheets, and site emergency procedures.
3. Confirm the feedwater or electrolyte meets the manufacturer's specified purity, concentration, and compatibility requirements.
4. Inspect the stack, hoses, fittings, valves, seals, electrical cables, sensors, and protective enclosure for leaks or damage.
5. Verify that ventilation, cooling, pressure relief, gas detection, and emergency shutdown systems are available and functional.
6. Check that hydrogen and oxygen outlets are correctly identified and routed to approved, separate handling systems as designed.
7. Ensure the work area is free of ignition sources and that required PPE and gas monitoring are available.

How to use

1. Confirm that the system is correctly assembled, all required guards are fitted, and access to emergency shutdown controls is clear.
2. Fill the feedwater or electrolyte reservoir using the approved material and specified procedure.
3. Open required valves and establish the prescribed circulation, cooling, and ventilation conditions before energising the stack.

4. Check that gas outlets, pressure relief paths, and collection or vent lines are open and correctly connected.
5. Switch on the control system and verify sensor readings, interlocks, alarms, and operating status.
6. Set the approved current, voltage, temperature, pressure, and flow limits for the equipment and process.
7. Energise the electrolyser using the manufacturer's startup sequence and increase load only as specified.
8. Monitor voltage, current, temperature, pressure, fluid level, flow, gas output, and alarms at the required intervals.
9. Check for leaks, unusual vibration, unexpected heating, abnormal gas flow, or signs of cross-contamination between gas streams.
10. If a leak, gas alarm, cooling failure, excessive pressure, or abnormal temperature occurs, initiate the approved shutdown and emergency response.
11. Do not tighten fittings or repair leaks while the system is energised or pressurised.
12. At the end of the run, reduce load and shut down according to the manufacturer's sequence; isolate electrical power when required.
13. Allow the system to reach its specified safe state, then isolate feed and gas lines according to the approved procedure.
14. Record operating parameters, alarms, maintenance observations, and any deviations from the approved process.

⚠ Safety

- Hydrogen is highly flammable and can form explosive mixtures with air; keep ignition sources away and maintain approved ventilation.
- Do not mix hydrogen and oxygen streams or modify gas piping without authorised engineering approval.
- Never block gas outlets or pressure-relief devices, and do not exceed rated pressure or temperature limits.
- Use only compatible materials and the specified electrolyte; corrosive electrolytes require appropriate chemical PPE and handling procedures.
- Treat the stack, terminals, and power connections as hazardous electrical equipment during operation and maintenance.
- Use approved leak-detection methods and follow site procedures for gas alarms, ventilation failure, or suspected leaks.
- Depressurise and isolate the system according to the manufacturer's procedure before maintenance; never open a pressurised cell or line.
- Follow applicable hazardous-area, gas-storage, fire-safety, and emergency-response requirements.

Care & storage

- Inspect stack seals, fittings, hoses, valves, and electrical connections at the prescribed intervals.

- Monitor feedwater or electrolyte quality and replace or replenish it according to the manufacturer's instructions.
- Check cooling, circulation, ventilation, pressure relief, and gas detection systems regularly.
- Keep the equipment exterior clean and dry without allowing cleaning fluids to enter electrical or gas components.
- Maintain operating logs and arrange scheduled servicing, leak checks, and sensor calibration.

Common mistakes to avoid

- Using feedwater or electrolyte outside the specified quality or concentration limits.
- Starting the stack before establishing circulation, cooling, ventilation, and safe gas routing.
- Ignoring pressure, temperature, gas alarms, or suspected leaks.
- Opening or servicing pressurised or energised equipment without approved isolation procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hydrogen Purification System

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Hydrogen purification systems with alert labels at important parts

What it is used for

To remove specified impurities from hydrogen gas using an approved hydrogen purification system. To maintain required gas quality while controlling pressure, monitoring alerts, and preventing leaks or fire hazards.

Main parts

- Hydrogen inlet and outlet connections
- Purification module or filter bed
- Pressure regulators and gauges
- Flow meter and control valves
- Gas-quality sensors and monitoring display
- Vent line, emergency shutoff, and alert labels

Before you start

1. Only trained and authorized personnel may operate the system.
2. Review the system manual, process flow diagram, safety data, and emergency procedures.
3. Verify that alert labels, flow-direction markings, and hazard signs are visible and legible.
4. Inspect tubing, fittings, seals, valves, and the enclosure for damage or signs of leakage.
5. Confirm that ventilation, gas detection, emergency shutoff, and required safety systems are functional.
6. Check that the inlet hydrogen supply and downstream equipment meet the system's pressure and purity requirements.
7. Verify that the vent outlet discharges to the approved safe location and is not blocked.

How to use

1. Confirm that the system is correctly installed and all required inspections are complete.
2. Keep ignition sources away and ensure the designated operating area is controlled.
3. Verify that downstream equipment is ready and the required outlet path is open.
4. Open the approved supply isolation valve according to the manufacturer's startup procedure.
5. Increase pressure and flow gradually within the specified operating limits.
6. Monitor inlet and outlet pressure, flow, temperature, and gas-quality readings as applicable.
7. Check system indicators and alarms throughout operation.
8. Confirm that purified hydrogen meets the specified quality criteria before sending it to the process.
9. Record operating readings, alarm events, and any required gas-quality test results.

10. If an alarm or abnormal reading occurs, follow the approved response procedure and isolate the system when safe.
11. At shutdown, stop the process and isolate the hydrogen supply in the sequence specified by the manufacturer.
12. Depressurize and purge only according to the approved procedure and designated vent arrangement.
13. Verify the system is in its required safe state before maintenance or opening any component.

⚠ Safety

- Hydrogen is highly flammable and can form explosive mixtures with air; eliminate ignition sources and follow the site's hydrogen safety rules.
- Never use flames to locate leaks; use approved hydrogen leak-detection methods.
- Do not bypass gas detectors, pressure relief devices, alarms, interlocks, or emergency shutoffs.
- Never exceed the rated pressure, temperature, flow, or gas-purity limits.
- Do not loosen fittings or open filters while the system is pressurized or contains hydrogen.
- Use only approved, compatible components and maintain suitable ventilation and hazardous-area controls.
- If a hydrogen leak is suspected, raise the alarm, avoid operating electrical switches in the affected area, evacuate as required, and follow the site emergency plan.

Care & storage

- Inspect labels, detectors, valves, fittings, and seals at the scheduled intervals.
- Replace purification media and filters according to the manufacturer's service instructions.
- Calibrate gas-quality sensors and pressure instruments as required.
- Keep the enclosure and ventilation paths clear of dust and obstructions.
- Document inspections, maintenance, alarms, and any replaced components.

Common mistakes to avoid

- Starting the system without checking ventilation, alarms, and leak-detection readiness.
- Opening valves too quickly and causing pressure surges.
- Ignoring gas-quality readings or repeated alarms.
- Using incompatible seals or unapproved replacement filters.
- Opening a pressurized component or attempting an improvised leak repair.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Load Bank

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Load (AC/DC)

What it is used for

To apply a controlled electrical load for testing generators, batteries, inverters, or other power systems.

How to handle

- Inspect the load bank, cables, terminals, cooling vents, switches, and protective devices for damage before use.
- Confirm the load bank's AC/DC rating, voltage, current, power capacity, and compatibility with the equipment under test.
- Connect the load bank according to the manufacturer's wiring diagram with the supply isolated and authorised personnel supervising the setup.
- Verify grounding, ventilation, protective settings, and cable ratings before energising; never touch exposed live terminals.
- Apply load in controlled steps, monitor voltage, current, temperature, and alarms, then reduce the load and isolate power before disconnecting.

Storage

- Store in a dry, secure, well-ventilated location according to the manufacturer's requirements.
- Keep cables coiled safely, terminals protected, and cooling openings free of dust and obstructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Portable Generator

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Portable Generator (Optional)

What it is used for

A portable generator produces electrical power from an engine-driven alternator for temporary use. Correct setup and operation help prevent electric shock, fire, carbon monoxide poisoning and equipment damage.

Main parts

- Identify the engine and alternator housing.
- Locate the fuel tank, fuel cap and fuel shut-off control.
- Identify the engine start and stop controls.
- Locate the AC outlets, circuit breakers and grounding terminal, where fitted.
- Identify the voltage and frequency indicators, if provided.
- Locate the engine oil filler, dipstick and air filter.
- Identify the exhaust outlet, frame and transport handles.

Before you start

1. Read the operating instructions and confirm the generator's rated output.
2. Operate the generator outdoors, far from doors, windows and air intakes.
3. Never run the generator inside homes, classrooms, garages or enclosed spaces.
4. Inspect fuel lines, electrical cables, outlets and housing for damage or leaks.
5. Check engine oil and fuel levels according to manufacturer instructions.
6. Place the generator on a firm, dry and level surface away from flammable materials.
7. Ensure connected loads do not exceed the generator's rated capacity.
8. Disconnect electrical loads before starting the generator.

How to use

1. Position the generator outdoors in a well-ventilated area with exhaust directed away from people.
2. Check that the generator is switched off and all connected loads are disconnected.
3. Refuel only when the engine is cool and switched off, following the instructions.
4. Open the fuel valve and set the choke or starting control as specified.
5. Start the engine using the designated starting method.
6. Allow the engine to stabilise and check the voltage or status indicators.
7. Connect suitable loads using correctly rated cables and protected outlets.
8. Switch on connected equipment in stages while monitoring the total load.
9. Watch for overheating, unusual noise, fuel leakage or unstable output.

10. Disconnect loads before switching off the generator and closing the fuel valve.

⚠ Safety

- Never operate a generator indoors or in partially enclosed spaces because carbon monoxide can be fatal.
- Keep the generator dry and never connect loads with wet hands.
- Never refuel a running or hot engine or smoke near fuel.
- Use correctly rated cables and suitable residual-current protection where required.
- Do not connect the generator directly to building wiring without an approved transfer arrangement installed by a qualified electrician.
- Stop the generator immediately if fuel leaks, sparks or dangerous electrical faults occur.

Care & storage

- Switch off the engine and allow it to cool before cleaning or servicing.
- Check engine oil, air filter and spark plug according to the maintenance schedule.
- Inspect outlets, cables, fuel lines and frame regularly for damage.
- Store the generator dry with fuel handled according to manufacturer guidance.

Common mistakes to avoid

- Running the generator indoors or near open windows and ventilation intakes.
- Overloading the generator with appliances beyond its rated output.
- Refuelling while the engine is hot or running.
- Connecting the generator to building wiring without approved isolation equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Pyranometer

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Pyranometer; Solar power meter (pyranometer)

What it is used for

To measure solar irradiance using a pyranometer. To obtain reliable readings by positioning the sensor correctly and following the instrument's measurement procedure.

Main parts

- Radiation sensor
- Protective dome or sensor surface
- Display or data logger
- Mounting base
- Signal cable or power connection

Before you start

1. Inspect the sensor surface, cable, and display for damage.
2. Check calibration status and operating instructions.
3. Place the sensor in an unobstructed location with the required sky view.
4. Ensure the sensor surface is clean and dry.
5. Confirm the correct units and data logging settings.

How to use

1. Mount the pyranometer level using its levelling feature where provided.
2. Connect the sensor to the approved display or data logger.
3. Switch on the system and allow the required stabilization time.
4. Check the sensor orientation and confirm that shadows do not fall on it.
5. Record irradiance readings under the required conditions.
6. Note the time, weather conditions, and measurement location.
7. Transfer or save the readings according to the procedure.
8. Clean the sensor surface and secure the instrument after use.

⚠ Safety

- Follow electrical safety instructions for powered data loggers.
- Use suitable precautions when working outdoors or at height.
- Do not touch hot or energized electrical connections.
- Avoid looking directly at the sun while positioning the instrument.

Care & storage

- Clean the sensor surface using the manufacturer's approved method.
- Inspect cables, seals, and mounting hardware regularly.
- Keep the instrument level and calibration current.
- Store it in a protective case when not installed.

Common mistakes to avoid

- Allowing shadows to fall on the sensor.
- Using an unlevel or dirty sensor.
- Ignoring calibration status or units.
- Recording readings without noting time and conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Charge Controller

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Charge controller

What it is used for

A solar charge controller regulates power from a photovoltaic array to charge a compatible battery safely and efficiently. Correct system matching, polarity checks, configuration, and monitoring help protect the battery and connected equipment from electrical faults and improper charging.

Main parts

- PV input terminals receive power from the solar panels.
- Battery terminals connect the controller to the battery bank.
- Load terminals supply controlled DC power where fitted.
- Charge regulation circuit controls battery charging.
- Display or indicator lights show charging status and faults.
- Voltage and current sensors monitor system conditions where fitted.
- Protection features help manage overvoltage, overcurrent, and reverse polarity where supported.
- Communication port connects compatible monitoring equipment where fitted.

Before you start

1. Only trained personnel should install or service the solar charge controller.
2. Read the controller manual and confirm compatibility with the PV array, battery chemistry, battery voltage, and system current.
3. Inspect the controller, terminals, wiring, connectors, and enclosure for damage or moisture.
4. Check the PV array open-circuit voltage and expected current against the controller's maximum ratings.
5. Identify the manufacturer's required connection and disconnection sequence before starting work.
6. Isolate PV input, battery, and loads according to the manual; remember that PV wiring can remain live in daylight.
7. Use suitable insulated tools and electrical PPE for the system voltage and work conditions.

How to use

1. Mount the controller in a dry, ventilated location with the clearances specified by the manufacturer.
2. Confirm that all power sources are isolated as required and verify terminal labels and polarity.
3. Connect the battery using the specified sequence, cable size, and protective devices.

4. Configure the battery type, charging voltage, charging current, and other settings according to battery manufacturer requirements.
5. Connect the PV array using the approved sequence and verify correct polarity before energising.
6. Connect DC loads to the designated load terminals only when supported and correctly rated.
7. Switch on the system according to the controller instructions and observe status indicators or display readings.
8. Check battery voltage, charging current, PV input, temperature, and any fault indicators.
9. Stop and isolate the system if abnormal heating, smoke, unexpected voltage, or a fault occurs.
10. After testing, secure terminal covers, route cables safely, and record settings and measured readings.

⚠ Safety

- Never exceed the controller's maximum PV voltage, input current, or battery current rating.
- Do not reverse battery or PV polarity; incorrect connections can damage equipment or create a fire hazard.
- PV modules can produce dangerous voltage in daylight; do not assume the array is de-energised.
- Never disconnect live PV connectors under load unless the equipment and procedure explicitly permit it.
- Use correctly rated fuses, circuit breakers, cables, and protective earth connections where required.
- Do not open or repair internal electronics unless qualified and authorised.

Care & storage

- Keep ventilation openings clear and remove dust from the enclosure using an approved method.
- Inspect terminals, cable insulation, protective devices, and mounting points regularly.
- Check fault logs and charging settings at the recommended service interval.
- Keep the controller dry and protect it from direct heat, moisture, and corrosive environments.
- Arrange qualified service if charging becomes irregular or a fault persists.

Common mistakes to avoid

- Connecting the PV array before the battery when the manual requires battery-first connection.
- Selecting the wrong battery chemistry or charging voltage.
- Ignoring PV voltage limits or polarity markings.
- Working on solar wiring in daylight without appropriate isolation and precautions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Charging Station

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Solar charging station

What it is used for

A solar charging station converts solar energy into electrical power for charging compatible batteries or devices. Operate it within its rated voltage and current limits and follow the installed system's electrical protection and charging instructions.

Main parts

- Solar PV panels and mounting structure
- Charge controller or power conversion unit
- Battery storage system, if fitted
- Charging outlets, connectors, or charging ports
- Protection devices, isolators, and circuit breakers
- Monitoring display and earthing system

Before you start

1. Confirm the station's output ratings and compatibility with the intended battery or device.
2. Inspect panels, mounting, cables, connectors, enclosures, and charging ports for damage.
3. Check that earthing, protective devices, and isolators are installed and accessible.
4. Ensure the area around the station is clear, dry, and adequately ventilated.
5. Check battery condition and follow the battery manufacturer's charging requirements.
6. Confirm that all connections and protection settings match the approved installation design.

How to use

1. Keep the charging output switched off while inspecting or connecting equipment.
2. Position the vehicle or device at the designated charging point.
3. Verify the battery or device voltage and charging requirements against the station rating.
4. Connect the approved charging cable securely to the correct port.
5. Switch on the required isolator or controller according to the installation procedure.
6. Start charging using the approved control or user interface.
7. Check the display for charging status, current, voltage, and warning indicators.
8. Monitor the system for unusual heat, smell, noise, or error messages.
9. Stop charging if a fault, damaged connector, or abnormal condition occurs.
10. Complete charging according to the battery or device instructions.
11. Stop the session using the approved control and disconnect the cable safely.
12. Switch off the charging output if required and return cables to their holders.
13. Record faults or abnormal readings and report them to the responsible technician.

⚠ Safety

- Only qualified personnel should install, repair, or service the electrical system.
- Solar panels may remain live in daylight; never touch exposed conductors or connectors.
- Do not charge damaged, swollen, leaking, or incompatible batteries.
- Keep connectors dry and do not plug or unplug damaged connectors under load.
- Never bypass fuses, circuit breakers, earthing, or other protective devices.
- Follow battery-specific fire procedures and keep the required emergency access clear.

Care & storage

- Clean panels using the approved method and safe isolation procedure.
- Inspect cables, connectors, mounting structures, and protective devices regularly.
- Keep charging ports clean, dry, and protected from impact.
- Check controller settings and battery condition according to the maintenance schedule.
- Arrange qualified service for faults, corrosion, overheating, or repeated charging errors.

Common mistakes to avoid

- Connecting a battery or device with incompatible charging requirements.
- Ignoring damaged cables, connectors, or warning indicators.
- Touching PV wiring without qualified electrical precautions.
- Bypassing protective devices or leaving faulty charging equipment in use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Connector

Category: **Solar, Power & Energy** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: MC4 connectors; Solar Cable Connectors

What it is used for

Solar connectors join compatible solar cables to create secure electrical connections in photovoltaic systems.

How to handle

- Confirm connector type, polarity, cable size, and manufacturer compatibility before use.
- Inspect housings, seals, and contacts for cracks, corrosion, or damage.
- Keep connectors clean and dry and avoid touching exposed conductive contacts.
- Use the specified crimping tools and follow the manufacturer's assembly instructions.
- Never disconnect connectors under load; isolate the system and follow electrical safety procedures before maintenance.

Storage

- Keep connectors in their original sealed packaging until needed.
- Store in a clean, dry area protected from dust, moisture, and physical damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Conversion Kit

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Solar conversion kits

What it is used for

A solar conversion kit provides components for converting sunlight into electrical power and connecting it to a compatible battery, load, or existing electrical system. Correct system design, component matching, wiring, and commissioning help achieve reliable operation while reducing shock, fire, and equipment damage risks.

Main parts

- Solar PV panel converts sunlight into DC electricity.
- Mounting structure secures the PV panel.
- Charge controller regulates battery charging where included.
- Battery stores electrical energy where included.
- Inverter converts DC power into AC power where included.
- Cables and connectors connect system components.
- Fuses, circuit breakers, and isolators provide protection and disconnection.
- Display or monitoring unit shows system status where supplied.

Before you start

1. Only trained personnel should install or commission the solar conversion kit.
2. Review the kit manual and confirm the intended system voltage, power rating, battery compatibility, and load requirements.
3. Check all components, cables, connectors, mounting parts, and protective devices for damage.
4. Confirm the PV panel ratings, controller limits, battery specifications, and inverter input requirements are compatible.
5. Choose a structurally sound mounting location with suitable sunlight exposure and safe cable routing.
6. Identify all required isolators, earthing arrangements, fuses, and circuit breakers before installation.
7. Remember that PV panels can produce live voltage in daylight; plan safe isolation and use suitable electrical PPE.

How to use

1. Lay out the kit components and identify each connection using the supplied wiring diagram.
2. Secure the PV panel and mounting structure according to the manufacturer's installation instructions.
3. Install the controller, battery, and inverter in suitable ventilated locations where included.

4. Keep all sources isolated and confirm cable sizes, connector types, and polarity before making connections.
5. Connect the battery and charge controller in the sequence specified by the kit manufacturer.
6. Connect the PV panel through the approved isolator or protection device and verify polarity.
7. Connect the inverter and loads only within their rated capacity and according to the wiring diagram.
8. Install and verify fuses, circuit breakers, earthing, and other required protective devices.
9. Energise the system in the specified sequence and observe indicators, voltage, charging current, and output readings.
10. Test the intended load under controlled conditions and check for abnormal heat, noise, smell, or fault messages.
11. Isolate the system if a fault occurs, correct the issue using an approved procedure, and retest before use.
12. Label circuits, secure cables, record commissioning readings, and explain basic operation to the responsible user.

⚠ Safety

- Never work on live wiring unless the task is specifically authorised and suitable controls are in place.
- Do not exceed the voltage, current, or power ratings of any kit component.
- Do not reverse polarity or connect components in a sequence contrary to the manufacturer instructions.
- Protect batteries from short circuits and use the specified fuses and cable sizes.
- Keep inverter and battery ventilation openings clear and keep equipment away from water unless rated for that environment.
- Use a qualified electrician for connection to building wiring or the utility grid where required.

Care & storage

- Inspect PV panels, mounting hardware, cables, connectors, and protective devices at the recommended interval.
- Keep panels clean using an approved method and follow safe access procedures.
- Check battery condition and controller or inverter fault indicators regularly.
- Keep cable routes secure and inspect for abrasion, corrosion, water entry, or loose connections.
- Arrange authorised maintenance if output drops unexpectedly or protection devices trip repeatedly.

Common mistakes to avoid

- Mixing components with incompatible voltage, current, or battery specifications.
- Connecting the PV array without checking polarity and isolation requirements.
- Using undersized cables or omitting required protective devices.
- Commissioning the system without recording readings or checking the intended load.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the

manufacturer's original manual / file.

Solar Cooker

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Solar cooker

What it is used for

A solar cooker uses sunlight to heat and cook food without conventional fuel. Position it correctly, use suitable cookware, and monitor food temperature and cooking progress.

How to use

1. Inspect the cooker, reflector, glazing, supports, and hinges for damage.
2. Place the cooker on stable ground in a location with unobstructed sunlight.
3. Adjust the reflector or orientation according to the manufacturer's instructions.
4. Prepare food in suitable cookware and place it securely inside the cooking area.
5. Use heat-resistant gloves when handling hot cookware or internal components.
6. Monitor cooking progress and check food is adequately cooked before serving.
7. Move or close the cooker safely when cooking is complete.
8. Allow components to cool before cleaning and storage.

⚠ Safety

- Never look directly at concentrated reflected sunlight.
- Use gloves and suitable utensils to prevent burns from hot cookware.
- Keep children away from hot surfaces and moving reflectors.
- Follow food hygiene practices and ensure food reaches a safe cooking temperature.

Care & storage

- Clean the cooking surface and reflector with suitable non-abrasive materials.
- Store the cooker dry and protect glazing and reflectors from impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Dryer

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Solar dryers

What it is used for

A solar dryer uses solar heat and airflow to reduce moisture in suitable agricultural or food products. Follow the approved drying conditions for the product to maintain quality and prevent contamination.

How to use

1. Inspect the dryer enclosure, transparent cover, trays, vents, and airflow components.
2. Place the dryer in a sunny, well-ventilated location according to its design.
3. Clean and dry all product-contact surfaces before loading.
4. Prepare the product uniformly and arrange it in a single suitable layer on the trays.
5. Adjust vents or airflow controls according to the manufacturer's instructions.
6. Monitor temperature, drying progress, and product condition at the required intervals.
7. Remove adequately dried product, protect it from recontamination, and clean the dryer.

⚠ Safety

- Do not exceed the recommended product temperature or drying conditions.
- Use heat-resistant gloves when handling hot trays or surfaces.
- Keep the dryer away from fire hazards and protect it from strong winds if required.
- Follow food hygiene requirements and do not use spoiled or contaminated raw material.

Care & storage

- Clean trays and vents after use and inspect the cover and airflow paths regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar E-Rickshaw

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Solar e-rickshaw

What it is used for

A solar e-rickshaw is an electric three-wheeler that may use solar panels to supplement battery charging or power auxiliary systems. Operate the vehicle according to its manufacturer instructions and treat the traction battery and electrical drive system with appropriate precautions.

Main parts

- Vehicle chassis, wheels, steering, and brakes
- Traction battery and battery management system
- Electric motor and motor controller
- Solar panels and solar charge controller, if fitted
- Charging port, cables, and electrical protection
- Dashboard, lights, and driver controls

Before you start

1. Inspect tyres, brakes, steering, lights, mirrors, and chassis for damage.
2. Check the traction battery status, dashboard warnings, and charging connections.
3. Inspect solar panels, mounting, cables, and controller for visible damage.
4. Confirm the battery charge level is adequate for the planned journey.
5. Check that the passenger and cargo load stays within the rated capacity.
6. Ensure the vehicle is in a safe condition and required documents and protective equipment are available.

How to use

1. Park the vehicle on level ground and apply the parking brake before checks.
2. Switch off the vehicle before inspecting electrical connectors or charging equipment.
3. Confirm that solar and mains charging connections, if fitted, are secure and undamaged.
4. Charge the battery only through the approved charging system and within its rated limits.
5. Check the dashboard for warnings before starting the vehicle.
6. Adjust mirrors and ensure passengers or cargo are secured correctly.
7. Release the parking brake and start moving gradually using the approved controls.
8. Drive at a safe speed appropriate to road conditions, load, and vehicle limits.
9. Monitor battery status, warning lights, steering, brakes, and unusual sounds during travel.
10. Stop safely if a critical warning, loss of power, or electrical fault occurs.
11. Park in a safe location, apply the parking brake, and switch off the vehicle.
12. Disconnect charging equipment only after the charging session has stopped.

3. Record defects and arrange qualified inspection before further use when safety is uncertain.

⚠ Safety

- Only trained and authorised drivers should operate the e-rickshaw.
- Follow local traffic rules, passenger limits, and the manufacturer's rated load.
- Do not drive with faulty brakes, steering, tyres, or critical warning indicators.
- Do not touch exposed high-voltage or traction-battery conductors; repairs require qualified personnel.
- Do not charge a damaged, swollen, or overheated battery.
- Keep the battery and charging area away from fire, water ingress, and unauthorised access.

Care & storage

- Inspect tyres, brakes, steering, lights, and suspension at recommended intervals.
- Keep solar panels clean using the approved method and safe procedure.
- Check battery condition, connectors, and charging equipment according to the maintenance schedule.
- Keep the vehicle dry where required and protect electrical connectors from corrosion.
- Arrange servicing for unusual battery heating, reduced range, or repeated electrical faults.

Common mistakes to avoid

- Exceeding the rated passenger or cargo capacity.
- Ignoring battery warnings or signs of electrical damage.
- Assuming solar panels replace the need for correct battery charging and maintenance.
- Attempting traction-battery repairs without appropriate qualifications.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Home Lighting System

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Home lighting system; Solar home lighting system

What it is used for

A solar home lighting system uses solar energy stored in a battery to power compatible lights and other approved loads. Operate the system within its rated capacity and follow the controller and battery instructions.

How to use

1. Inspect the solar panel, battery, controller, lights, cables, and connectors for damage.
2. Confirm that the system components and loads have compatible voltage ratings.
3. Ensure installation and wiring are performed by a qualified person using the approved design.
4. Check that the battery is connected with the correct polarity and controller sequence.
5. Position the solar panel securely where it receives adequate sunlight.
6. Switch on the system and verify that the lights and controller indicators operate normally.
7. Monitor battery status and avoid connecting loads beyond the system capacity.
8. Switch off the system safely before inspection or maintenance.

⚠ Safety

- Do not touch exposed conductors or damaged cables.
- Never short-circuit the battery or reverse its polarity.
- Use only compatible batteries and approved charging controllers.
- Keep the battery and electrical components dry and follow the required ventilation instructions.

Care & storage

- Clean the solar panel and check mounting, cables, and connectors regularly.
- Inspect the battery and controller according to the manufacturer's maintenance schedule.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Inverter

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Inverter (String / Hybrid); Solar photovoltaic inverter

What it is used for

A solar inverter converts DC electricity from solar panels or batteries into usable AC electricity for compatible loads or the grid. Correct installation checks, operating limits, and monitoring help maintain reliable performance and reduce electrical hazards.

Main parts

- DC input terminals receive power from the solar array or battery where supported.
- Power conversion circuit converts DC electricity into AC electricity.
- AC output terminals connect the inverter to approved loads or the grid.
- MPPT controller optimises solar panel operating conditions on equipped models.
- Display or monitoring interface shows operating values and fault codes.
- Cooling system removes heat from internal components.
- Protection circuits respond to electrical faults and abnormal conditions.
- Disconnect switches and protective devices isolate circuits where provided.

Before you start

1. Only qualified and authorised personnel should install or service the inverter.
2. Read the manual and confirm compatibility with the solar array, battery, and AC system.
3. Inspect the enclosure, cables, connectors, isolators, and visible wiring for damage.
4. Check that voltage, current, polarity, and power ratings are within the manufacturer's limits.
5. Confirm the mounting location is dry, ventilated, and suitable for the equipment rating.
6. Verify earthing, protective devices, and required isolation arrangements according to the approved installation design.

How to use

1. Keep the inverter isolated from all energy sources before making or changing connections.
2. Verify DC and AC cable identification, polarity, terminal ratings, and protective arrangements.
3. Connect the solar array, battery if supported, and AC circuits according to the approved wiring diagram.
4. Secure the terminals to the specified torque using the correct tools.
5. Check that covers, cable glands, earthing, and protective devices are correctly installed.
6. Restore power in the sequence specified by the manufacturer and approved installation procedure.
7. Check the display for normal startup, operating status, and any fault codes.
8. Confirm that measured voltage, current, and operating conditions are within the specified limits.

9. Monitor generation and alarms through the display or approved monitoring system.
10. If a fault occurs, follow the shutdown procedure and arrange qualified inspection before restarting.

⚠ Safety

- Solar DC conductors may remain live in daylight even when the inverter is switched off.
- Never open the enclosure or touch internal terminals while energised.
- Use appropriately rated PPE and test instruments for the electrical system.
- Never disconnect DC connectors under load unless the equipment procedure explicitly permits it.
- Do not bypass earthing, isolation, overcurrent protection, or other safety devices.
- Follow the manufacturer's shutdown sequence and wait the specified time before servicing.

Care & storage

- Keep ventilation openings clear and remove external dust using the approved method.
- Inspect visible cables, connectors, glands, and enclosure seals at recommended intervals.
- Review fault logs and generation data for abnormal changes.
- Arrange authorised servicing for repeated faults, overheating, unusual noise, or damaged components.

Common mistakes to avoid

- Connecting DC cables with incorrect polarity or unsuitable ratings.
- Ignoring earthing, isolation, or protective device requirements.
- Blocking ventilation openings and allowing excessive heat buildup.
- Opening the inverter before all energy sources are safely isolated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Lantern

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Lantern; Light and heat sources (solar lanterns, fire starters) for extended field visits; Solar lantern; Solar torch

What it is used for

A solar lantern uses stored solar energy to provide light without a direct mains connection. Charge and operate it according to the manufacturer's instructions.

How to use

1. Inspect the lantern, solar panel, charging port, battery compartment, and switch.
2. Place the solar panel in direct sunlight where it is not shaded.
3. Allow charging for the time recommended by the manufacturer.
4. Switch on the lantern and select the required brightness setting.
5. Check that the light works and the lantern remains stable during use.
6. Switch it off after use and store it in a dry, secure place.

⚠ Safety

- Do not use a lantern with a swollen battery, cracked casing, or damaged wiring.
- Keep the lantern and charging connections away from water unless rated for wet conditions.
- Do not open or puncture the battery compartment.

Care & storage

- Keep the solar panel clean and store the lantern away from excessive heat and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar PV System

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: A complete unit of Solar photovoltaic system; A complete unit of Solar photovoltaic system model of solar photovoltaic power plant; Solar power plant

What it is used for

A solar photovoltaic system converts sunlight into electrical energy and delivers it to approved loads, storage, or the grid through compatible equipment. Correct inspection, safe isolation, and regular performance monitoring help maintain system efficiency and reduce electrical risks.

Main parts

- Solar PV modules convert sunlight into DC electricity.
- Mounting structure supports the modules at the designed angle and position.
- DC cables and connectors carry electricity from the modules.
- Combiner box combines strings and may contain protective devices.
- Solar inverter converts DC electricity into AC electricity.
- AC distribution board routes power through suitable protective devices.
- Earthing and bonding components provide protective electrical paths.
- Monitoring equipment measures generation and system status; batteries may be included in hybrid systems.

Before you start

1. Only trained and authorised personnel should install, inspect, or service electrical parts of the system.
2. Read the system documentation and identify the approved layout and shutdown procedure.
3. Inspect modules, mounting structures, cables, connectors, and visible enclosures for damage.
4. Check that cables, protective devices, and inverter ratings match the approved design.
5. Confirm that the work area is safe, dry where required, and free from obstructions.
6. Identify all energy sources, including PV strings, batteries, and grid connections, before isolation.

How to use

1. Switch off and isolate the system according to the manufacturer's and site procedures before electrical work.
2. Check mounting hardware and module condition from a safe, authorised access position.
3. Inspect visible DC cables, connectors, combiner boxes, and AC distribution equipment for damage.
4. Verify that earthing, bonding, cable routing, and protective devices match the approved design.
5. Restore connections and covers only after the authorised inspection or maintenance is complete.
6. Restore power in the specified sequence and check the inverter and monitoring interface.
7. Confirm that the system starts normally and no unexpected fault indicators appear.

8. Record energy generation, system status, and relevant voltage or current readings where provided.
9. Compare performance with expected conditions and investigate unusual drops or repeated alarms.
10. Report defects and arrange qualified repair before operating equipment with unresolved safety faults.

⚠ Safety

- PV modules can generate DC voltage whenever illuminated; switching off the inverter does not make every conductor safe.
- Never disconnect PV connectors under load unless the equipment procedure explicitly permits it.
- Do not walk on modules or work on roofs without approved fall protection and access arrangements.
- Avoid electrical work in wet conditions unless the approved procedure specifically permits it.
- Use suitable PPE and rated test instruments; never bypass protective devices or earthing.
- Only qualified personnel should work on grid connections, battery circuits, or damaged high-voltage components.

Care & storage

- Keep modules free of excessive dust and shading using approved cleaning methods.
- Inspect mounting hardware, cable supports, connectors, and enclosures at scheduled intervals.
- Review inverter alarms and generation records for abnormal changes.
- Arrange authorised servicing after storms, physical damage, insulation faults, or repeated system trips.

Common mistakes to avoid

- Assuming the PV array is safe because the inverter has been switched off.
- Using incompatible connectors or mixing connector types without approval.
- Ignoring damaged cables, loose mounting hardware, or repeated inverter alarms.
- Cleaning modules or servicing electrical parts without the required isolation and safety precautions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Panel

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **High** 5 labs

Covers these lab items: Solar panels; Solar Panels for Demonstration

What it is used for

A solar panel converts sunlight into electrical energy for demonstrations or power applications. Handle the panel carefully and use suitable electrical equipment when checking its output.

How to use

1. Inspect the panel surface, frame, junction box, and cables for cracks or damage.
2. Place or mount the panel securely in a stable location facing the light source.
3. Keep the panel surface clean and remove shading from the active area.
4. Connect the panel to the approved demonstration load or measuring equipment using the correct polarity.
5. Check voltage and current with suitable rated instruments and the approved procedure.
6. Record the test conditions, including light conditions and measured output.
7. Disconnect the equipment safely and cover or store the panel when required.

⚠ Safety

- Do not touch exposed conductors or damaged cables.
- Avoid short-circuiting the panel output and follow the specified connection procedure.
- Secure panels against wind and accidental tipping.
- Use care when lifting panels and protect them from impact.

Care & storage

- Clean the panel with a soft cloth and suitable water-based cleaner when needed.
- Inspect cables, connectors, mounting points, and the panel surface regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Panel Mounting Structure

Category: **Solar, Power & Energy** Type: **Handling note** Priority: **Low** 7 labs

Covers these lab items: Solar Panel Clips; Solar Panel Mounting Brackets; Solar Panel Mounting Brackets & Rails; Solar Panel Mounting System; Solar Panel Support Blocks; Solar panels clips and fasterners (AC/DC)

What it is used for

Solar panel mounting structures secure photovoltaic modules and maintain the required orientation and support.

How to handle

- Verify structure material, dimensions, mounting compatibility, and design load requirements.
- Inspect rails, brackets, clips, fasteners, and support blocks for damage or corrosion.
- Follow approved installation drawings and manufacturer torque specifications.
- Use fall protection and appropriate PPE when working at height; isolate electrical hazards before nearby installation work.
- Do not climb on panels or use mounting parts outside their rated purpose.

Storage

- Store rails and brackets on stable supports and protect them from bending.
- Keep fasteners and clips labelled, dry, and separated by type.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Power Bank

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Solar power bank

What it is used for

A solar power bank stores electrical energy for charging compatible devices. Use suitable sunlight and compatible connections to prevent damage and overheating.

How to use

1. Inspect the power bank, casing, ports, cable, and battery for damage.
2. Read the device instructions and check its charging capacity and compatibility.
3. Place the solar panel in direct sunlight according to the manufacturer's instructions.
4. Connect a compatible device using the correct charging cable.
5. Check charging indicators and disconnect the device when charging is complete.
6. Store the power bank safely after use.

⚠ Safety

- Keep ports and connectors dry and protect the device from rain.
- Do not expose the power bank to excessive heat or open flames.
- Stop using it if the battery swells, leaks, or becomes unusually hot.

Care & storage

- Keep the solar panel clean and store the power bank in a cool, dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Street Light

Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Solar street light

What it is used for

A solar street light uses a solar panel and battery to power outdoor lighting, often through an automatic controller. Check the system components and operate it according to the installation and maintenance instructions.

How to use

1. Inspect the panel, light fixture, battery enclosure, controller, and cables for damage.
2. Confirm that the pole and mounting brackets are secure before inspection.
3. Check panel orientation and remove shading where safe and permitted.
4. Verify battery status and controller settings according to the manufacturer's instructions.
5. Test the light using the approved operating or test mode.
6. Confirm that automatic lighting functions respond correctly to the configured conditions.
7. Report faults and isolate the system before authorised maintenance.
8. Secure all covers and confirm the installation is stable after work.

⚠ Safety

- Do not climb poles or access elevated fixtures without approved training and fall protection.
- Keep away from exposed conductors and damaged electrical enclosures.
- Use qualified personnel for electrical work and battery replacement.
- Avoid working at height during strong winds, rain, or lightning.

Care & storage

- Clean the solar panel and inspect mounting hardware at scheduled intervals.
- Check battery condition, wiring, and lighting performance according to the maintenance plan.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Water Heater

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Solar water heater

What it is used for

A solar water heater uses solar collectors to heat water for domestic or other approved applications. Operate it within the system's pressure and temperature limits and follow safe procedures for hot water, plumbing, and electrical components.

Main parts

- Solar collector panels or evacuated tubes
- Insulated storage tank
- Water inlet, outlet, and circulation piping
- Temperature and pressure relief valve
- Mounting frame and insulation
- Controller, pump, sensors, and electrical connections, if fitted

Before you start

1. Identify the system type and read the manufacturer's operating instructions.
2. Inspect collectors, tank, piping, insulation, valves, and mounting structure for damage or leakage.
3. Check that the water supply and circulation path are available and unobstructed.
4. Confirm that pressure relief devices and isolation valves are correctly installed and accessible.
5. For pumped systems, inspect the pump, controller, sensors, and electrical connections.
6. Keep the service area accessible and use suitable PPE when inspecting hot components.

How to use

1. Ensure the system is correctly filled and vented according to its design before operation.
2. Check the water supply and open the designated valves in the specified sequence.
3. For active systems, switch on the controller and circulation pump as instructed.
4. Verify that temperature and pressure readings remain within the system's rated limits.
5. Inspect visible pipe joints and connections for leaks without touching hot surfaces.
6. Allow the system to heat water under normal operating conditions.
7. Monitor controller indications and check for abnormal noise, leakage, or pressure changes.
8. Use hot water cautiously and test its temperature before contact or use.
9. If a fault or overpressure warning occurs, follow the system's shutdown and service procedure.
10. Isolate the water or electrical supply only when safe and according to the instructions.
11. Allow components to cool before cleaning, inspection, or maintenance.
12. Record faults and arrange qualified service for leaks, relief-valve issues, or control failures.

⚠ Safety

- Installation and repair should be performed by qualified plumbing or electrical personnel as applicable.
- Water from the tank and outlet may be hot enough to cause scalds.
- Never block or disable the temperature and pressure relief valve.
- Do not open pressurised pipes or the tank while the system is hot.
- Use safe access equipment when working on roof-mounted collectors.
- Disconnect electrical power before servicing pumps or controllers and follow isolation procedures.

Care & storage

- Inspect collectors, tank insulation, pipes, supports, and valves at scheduled intervals.
- Clean collector surfaces using the manufacturer's approved method.
- Check for leaks, corrosion, scale buildup, and damaged insulation.
- Test or service relief valves according to the manufacturer's instructions and local requirements.
- Arrange qualified servicing for reduced heating performance, abnormal pressure, or recurring faults.

Common mistakes to avoid

- Operating the system without the required water fill or circulation.
- Ignoring leaks, corrosion, or damaged insulation.
- Blocking the pressure relief valve or exceeding rated limits.
- Touching hot pipes or working on roof collectors without safe access and protection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solar Water Pump

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Solar water pump

What it is used for

A solar water pump uses electricity generated by solar panels to move water for irrigation, livestock or other approved uses. Correct installation and operation help maintain water flow while protecting electrical components and the pump.

Main parts

- Identify the solar panels and their supporting mounting structure.
- Locate the pump unit installed in the water source or pipeline.
- Identify the controller or solar pump drive unit.
- Locate the DC or AC cables and electrical connectors.
- Identify the inlet strainer, suction pipe or water intake arrangement, as applicable.
- Locate the delivery pipe, outlet valve and water storage tank, if fitted.
- Identify the isolator, protective earthing and safety labels, where provided.

Before you start

1. Read the operating instructions and confirm the pump's intended application.
2. Inspect solar panels, mounting supports, cables and connectors for damage.
3. Ensure the pump and controller are suitable for the panel configuration.
4. Check water-source availability and confirm that the intake is clear.
5. Inspect pipes, valves and joints for leakage or blockage.
6. Keep electrical connections dry and confirm that required protective devices are installed.
7. Obtain qualified supervision before electrical wiring or controller adjustments.

How to use

1. Position the solar panels securely in the approved location with suitable sunlight exposure.
2. Check that the pump is correctly installed and the water intake is submerged when required.
3. Confirm that delivery pipes and valves are connected correctly.
4. Keep the pump and controller switched off while checking electrical connections.
5. Connect the solar panels, controller and pump according to the wiring diagram.
6. Open the required water valves and ensure the delivery path is clear.
7. Switch on the isolator and controller using the specified startup procedure.
8. Allow the pump to start under available sunlight and monitor water flow.
9. Check for unusual noise, leakage, overheating or controller warnings.
10. Stop the system before inspecting, adjusting or disconnecting components.
11. Switch off and isolate the system before maintenance or cleaning.

⚠ Safety

- Never touch exposed electrical conductors or wet electrical connections.
- Solar panels can generate electricity in daylight even when the pump is switched off.
- Never run a pump dry unless the manufacturer explicitly permits it.
- Keep hands away from rotating shafts and pump coupling components.
- Isolate all relevant power sources before electrical maintenance.
- Do not enter wells, tanks or confined water spaces without authorised procedures.

Care & storage

- Clean solar panels using the recommended method and avoid damaging their surfaces.
- Inspect cables, connectors, mounting structures and pipe joints regularly.
- Clean inlet strainers and check pump performance according to instructions.
- Protect the controller and electrical connections from water, dust and unauthorised access.

Common mistakes to avoid

- Running the pump without sufficient water at the intake.
- Connecting panels or controller terminals with incorrect polarity or configuration.
- Ignoring pipe leaks, blocked filters or controller warning indicators.
- Attempting electrical repairs without isolating power and qualified supervision.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

UPS

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 17 labs

Covers these lab items: Backup Power Supply (UPS); Online UPS 5 KVA; Power Protection; Uninterruptible Power Supply (UPS); UPS

What it is used for

A UPS provides temporary backup power and helps protect connected equipment during power interruptions. Correct installation, load management, and battery care support safe and reliable operation.

Main parts

- Input connection receives power from the mains supply.
- Output sockets or terminals supply backup power to connected equipment.
- Battery stores energy for use during power failure.
- Inverter converts battery DC power into usable AC power.
- Control panel displays operating status, alarms, and battery condition.
- Circuit breaker or protective device helps interrupt unsafe electrical conditions.
- Ventilation openings or cooling fans remove heat from internal components.

Before you start

1. Read the manufacturer's instructions and confirm the UPS rating suits the connected load.
2. Allow only trained and authorised personnel to install or service electrical connections.
3. Inspect the enclosure, power cord, plugs, terminals, and ventilation openings for damage.
4. Place the UPS on a stable, dry, well-ventilated surface away from heat sources.
5. Check that the battery is installed correctly and there are no warning alarms.
6. Confirm the connected equipment's total power demand stays within the UPS rating.

How to use

1. Switch off the UPS and connected equipment before making permitted external connections.
2. Connect the input supply to a properly rated and earthed outlet as instructed.
3. Connect compatible equipment to the designated UPS output sockets or terminals.
4. Switch on the mains supply and then start the UPS according to its instructions.
5. Check the display for normal operation, battery status, and active alarms.
6. Switch on connected equipment and verify that the UPS supports the load correctly.
7. During a power failure, save work and shut down equipment when backup time becomes limited.
8. Restore mains power and check that the UPS returns to normal charging operation.
9. Switch off and disconnect the UPS before cleaning its external surfaces.
10. Report unusual alarms, overheating, swelling batteries, or unexpected shutdowns to authorised personnel.

⚠ Safety

- Never open the UPS enclosure or touch internal terminals; dangerous voltage may remain present.
- Do not overload the UPS or connect equipment that exceeds its specified rating.
- Keep the UPS away from water, moisture, flammable materials, and blocked ventilation.
- Never short-circuit battery terminals or connect batteries with incorrect polarity.
- Do not remove or replace internal batteries unless trained and authorised to do so.
- Treat swollen, leaking, unusually hot, or damaged batteries as hazards and report them immediately.

Care & storage

- Keep ventilation openings clear and wipe the exterior with a dry cloth after disconnecting power.
- Inspect external cables and connections regularly for damage, looseness, or overheating.
- Follow manufacturer instructions for battery testing, replacement, and preventive maintenance.
- Arrange authorised servicing if backup time decreases significantly or alarms persist.

Common mistakes to avoid

- Connecting equipment whose total power demand exceeds the UPS rating.
- Blocking cooling vents or placing the UPS in a damp area.
- Opening the enclosure or handling internal batteries without authorisation.
- Ignoring warning alarms, unusual heat, or reduced backup time.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Agility Ladder

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Agility Ladder

What it is used for

To practise footwork, coordination, balance, and agility through structured movement drills.

How to handle

- Inspect the ladder, rungs, straps, and connectors for damage before use.
- Lay the ladder flat on a dry, level, non-slip surface and secure it to prevent movement.
- Complete a warm-up and demonstrate the drill before participants begin.
- Move through the ladder with controlled steps, keeping eyes forward and allowing enough space between participants.
- Stop if the ladder shifts or anyone feels pain or dizziness; collect the ladder after the session.

Storage

- Wipe off dirt and moisture before storage.
- Fold or roll the ladder as designed and store in a dry sports-equipment area away from sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Arrow

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Arrows

What it is used for

An arrow is a projectile designed for use with a compatible bow in supervised archery activities. Handle arrows carefully and follow the range rules at all times.

How to use

1. Inspect the shaft, point, fletching, and nock for cracks or damage.
2. Use only arrows compatible with the bow and approved for the activity.
3. Follow the instructor's commands and stand behind the designated shooting line.
4. Point the arrow only toward the approved target within the range.
5. Shoot only when the range is declared clear and the target area is safe.
6. Wait for the signal before retrieving arrows and check them again afterward.

⚠ Safety

- Never point an arrow at a person, even as a joke.
- Never shoot upward, toward an unknown area, or beyond the target zone.
- Do not use cracked, bent, or damaged arrows.
- Only retrieve arrows when the instructor confirms shooting has stopped.

Care & storage

- Wipe arrows clean, keep points protected, and store them securely in a suitable quiver or case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Balance Beam

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Balance Beam Set

What it is used for

A balance beam develops balance, coordination, and controlled movement during physical education. Use the beam at a suitable height with adequate supervision and a clear landing area.

How to use

1. Inspect the beam surface, supports, joints, and fasteners for damage.
2. Set up the beam on a level, non-slip surface with a clear surrounding area.
3. Check that all supports are locked and the beam does not wobble.
4. Warm up and wait for the instructor's instructions before starting.
5. Step onto the beam carefully and move slowly with eyes focused ahead.
6. Step down at the designated end and allow the next participant to begin.
7. Return portable components to storage after checking the equipment.

⚠ Safety

- Use mats and instructor supervision appropriate to the beam height and activity.
- Do not run, push others, or perform unapproved stunts on the beam.
- Stop using the beam if it slips, wobbles, or becomes damaged.

Care & storage

- Wipe the surface clean and store removable components in a dry, secure place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ball

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 8 labs

Covers these lab items: Basketball - Size 5; Handball; Plastic Balls; PU Hand Stitched Football - Size 4; Rugby Ball; Soft Ball (big); Soft Balls (small); Tennis Ball

What it is used for

Use a suitable ball for sports practice, drills, games, or physical education activities according to its design and size.

How to use

1. Choose the ball type and size suitable for the sport and players.
2. Inspect the ball for tears, punctures, loose seams, or surface damage.
3. Check inflation and pressure according to the manufacturer's guidance where applicable.
4. Clear the playing area of obstacles and confirm adequate space.
5. Explain the activity rules and demonstrate safe handling before starting.
6. Use the ball only for the intended sport and skill level.
7. Clean and store the ball after the activity.

⚠ Safety

- Use appropriate footwear and protective equipment for the activity.
- Do not use a damaged or excessively hard ball.
- Keep players clear of nearby traffic, glass, and other hazards.
- Follow age, skill, and activity-specific rules to reduce injury risk.

Care & storage

- Clean the ball with a suitable cloth and cleaner.
- Store it away from sharp objects, direct sunlight, and extreme temperatures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Ball Pump

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Exercise Ball Pump

What it is used for

A ball pump inflates compatible sports or exercise balls using a pump and suitable needle or nozzle. Use the correct attachment and follow the ball's recommended pressure.

How to use

1. Inspect the pump, hose, connector, and inflation needle or nozzle for damage.
2. Check the ball's valve type and recommended inflation pressure.
3. Attach the correct needle or nozzle securely to the pump.
4. Moisten the needle if the ball manufacturer recommends it and insert it gently.
5. Pump steadily while checking firmness or pressure with a suitable gauge.
6. Stop at the recommended pressure and withdraw the needle carefully.
7. Remove the attachment and store the pump after use.

⚠ Safety

- Do not force the needle into the valve or use a bent needle.
- Avoid overinflating the ball beyond its recommended pressure.
- Keep the needle pointed away from fingers and eyes.

Care & storage

- Clean the pump and attachment and store the needle in a protective holder.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Boxing Equipment

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Boxing Gloves, Punching Bag

What it is used for

Boxing gloves protect the hands during supervised boxing practice, while punching bags provide a target for training. Use correctly fitted equipment and follow the instructor's training plan.

How to use

1. Inspect gloves, wrist closures, bag straps, chains, and mounting points for damage.
2. Wear hand wraps if instructed and secure the gloves comfortably.
3. Check that the punching bag is stable and has enough surrounding clearance.
4. Warm up and practise the instructed stance and punching technique.
5. Strike the bag with controlled force while keeping wrists aligned.
6. Stop periodically to check equipment and your physical condition.
7. Remove gloves after training and clean equipment according to its instructions.

⚠ Safety

- Use boxing equipment only under suitable supervision and with proper technique.
- Never strike another person outside an approved, supervised activity.
- Stop if you experience pain, dizziness, or wrist discomfort.
- Keep bystanders clear of the swinging bag and training area.

Care & storage

- Air out gloves after use and inspect bag attachments regularly for wear.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Camping Equipment

Category: **Sports & Physical Education** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Eco-friendly ground covers

What it is used for

Handle eco-friendly ground covers safely for outdoor and camping activities.

How to handle

- Inspect the cover for tears, sharp debris, or damaged fasteners before use.
- Spread it on suitable ground away from open flames and hazardous areas.
- Secure edges to prevent wind from lifting the cover or creating trip hazards.
- Shake off soil and clean according to the material instructions after use.

Storage

- Dry completely before folding to prevent mould and odour.
- Store in a clean, dry bag away from pests and sharp equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Clip

Category: **Sports & Physical Education** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Joining Clips

What it is used for

Use joining clips safely to connect or secure materials during sports and practical activities.

How to handle

- Confirm that the clip type is suitable for the material and intended load.
- Inspect for cracks, sharp edges, or weak locking mechanisms before use.
- Keep fingers clear of pinch points when closing the clip.
- Do not use clips as safety restraints unless specifically rated for that purpose.

Storage

- Store by size and type in labelled containers.
- Keep dry and away from excessive heat or corrosive substances.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cricket Bat

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Foam Paddle Bat (Red, Green, Yellow); Plastic Cricket Bat; Wooden Cricket Bat

What it is used for

To practise batting, coordination, and basic cricket skills using a suitable wooden, plastic, or foam paddle bat. To use the bat in a controlled practice area and avoid collisions or injury.

Main parts

- Bat handle
- Bat blade or striking surface
- Grip
- Toe or lower edge
- Protective grip tape, if fitted

Before you start

1. Choose a bat suitable for the player's age, skill, and practice activity.
2. Inspect the handle, grip, blade, and edges for cracks or loose parts.
3. Use an appropriate ball for the bat material and training level.
4. Ensure the playing area is clear of people, obstacles, and breakable objects.
5. Wear suitable footwear and protective equipment for the activity.

How to use

1. Hold the handle with a secure, comfortable grip.
2. Adopt a balanced batting stance with feet stable.
3. Watch the ball and practise a controlled swing.
4. Keep the swing within the marked practice area.
5. Retrieve the ball only when the practice area is safe.
6. Check the bat for damage after practice.
7. Clean and store the bat in a dry, safe place.

⚠ Safety

- Never swing the bat near another person.
- Use foam or plastic bats with compatible balls as directed.
- Wear a helmet and other protective gear when required by the practice level.
- Do not use a cracked or splintered wooden bat.

Care & storage

- Wipe the bat clean after use.
- Keep wooden bats dry and away from excessive heat.
- Check the grip and handle regularly.
- Store the bat securely without placing heavy objects on it.

Common mistakes to avoid

- Using a bat that is too heavy or unsuitable.
- Swinging before checking that the area is clear.
- Using an incompatible ball with a foam or plastic bat.
- Ignoring cracks, splinters, or a loose handle.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cricket Stump Set

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Cricket Stump Set

What it is used for

A cricket stump set marks the wicket area for cricket practice and matches. Set the stumps securely at the required position and keep the playing area clear.

How to use

1. Inspect the stumps and base or ground spikes for cracks or damage.
2. Choose a level playing area away from hazards and obstructions.
3. Position the stumps according to the activity's rules or instructor's directions.
4. Insert or secure the stumps so they remain upright and stable.
5. Place the bails carefully on top if the set includes them.
6. Check the wicket stability before starting play.
7. Remove the stumps after use and store the set safely.

⚠ Safety

- Keep players clear while stumps are being installed or removed.
- Do not use splintered, cracked, or sharp-edged stumps.
- Ensure the wicket area has enough space for safe movement.

Care & storage

- Wipe off dirt, dry the set, and store it away from moisture and direct heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Exercise Ball

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Exercise/Swiss Balls

What it is used for

An exercise ball supports balance, flexibility, and core-strengthening exercises. Select the correct size, inflate it as specified, and use it on a clear, nonslip surface.

How to use

1. Inspect the ball for cuts, leaks, wear, or deformation.
2. Confirm that the ball size and inflation level suit the user and exercise.
3. Place the ball on a flat, nonslip surface away from sharp objects.
4. Sit or position yourself carefully and establish balance before exercising.
5. Perform only suitable exercises with controlled movements and proper posture.
6. Stop immediately if the ball slips, loses air, or causes discomfort.
7. Deflate or store the ball according to the manufacturer's instructions when needed.

⚠ Safety

- Keep the exercise area clear of furniture, sharp edges, and slippery objects.
- Do not exceed the ball's rated load or recommended inflation pressure.
- Use support or supervision for exercises requiring additional balance.
- Do not use a damaged or unstable ball.

Care & storage

- Clean with a compatible mild cleaner and store away from sharp objects and excessive heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Exercise Bench

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Gym Bench

What it is used for

An exercise bench supports strength training and other fitness exercises. Check its stability and adjust it correctly before placing weights or body weight on it.

How to use

1. Inspect the bench frame, padding, feet, adjustment pins, and fasteners for damage.
2. Place the bench on a flat, nonslip floor.
3. Adjust the backrest or seat to the required position and secure the locking pins.
4. Check that the bench remains stable before starting the exercise.
5. Use correct lifting technique and position your body securely on the bench.
6. Keep weights controlled and return them to their designated storage after use.
7. Wipe down the bench and report any loose or damaged parts.

⚠ Safety

- Do not exceed the bench's rated weight capacity.
- Never use a bench with loose fasteners, unstable feet, or damaged padding.
- Use a spotter when required for heavy lifting.
- Keep hands clear of adjustment hinges and locking mechanisms.

Care & storage

- Clean the padding and frame after use and inspect fasteners regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Foam Roller

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Foam Roller

What it is used for

A foam roller is used to apply controlled pressure to muscles for warm-up, mobility, and recovery exercises. Select a suitable roller and use slow, comfortable movements.

How to use

1. Inspect the roller for cracks, tears, deformation, or surface damage.
2. Choose a flat, nonslip exercise area with enough space to move.
3. Position the roller under the muscle group being exercised.
4. Support your body with your hands or feet and maintain controlled movement.
5. Roll slowly over the target area without applying excessive pressure to joints.
6. Stop if you experience sharp pain, numbness, or unusual discomfort.
7. Wipe the roller clean after use and return it to storage.

⚠ Safety

- Do not roll directly over joints, bones, or injured areas.
- Avoid excessive pressure and sudden movements.
- Use support if balance is difficult and stop if pain occurs.

Care & storage

- Clean with a compatible mild cleaner and store away from direct sunlight and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Frisbee

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Frisbee; Soft Flyer

What it is used for

To practise throwing, catching, coordination, and recreational team activities with a flying disc.

How to handle

- Inspect the disc for cracks, sharp edges, or deformation before play.
- Choose an open, level playing area away from roads, glass, power lines, and other hazards.
- Explain the throwing and catching rules and ensure participants maintain safe spacing.
- Throw at a controlled speed and avoid aiming at faces or people who are not ready to receive the disc.
- Stop play if the area becomes unsafe and collect the disc after the session.

Storage

- Wipe the disc clean and dry after play.
- Store flat in a sports-equipment bin away from direct heat and heavy loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Goal Post

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Pop up Goal post

What it is used for

A pop-up goal post provides a portable target for football and other suitable sports activities. Set it up securely and check stability before play.

How to use

1. Inspect the frame, joints, net, pegs, and carry bag for damage.
2. Choose a level playing area clear of hazards and obstructions.
3. Unfold the frame according to the manufacturer's setup instructions.
4. Attach the net and secure all designated clips or fasteners.
5. Anchor the goal using the supplied pegs or approved weights where appropriate.
6. Check that the goal is stable and the net is properly attached.
7. After play, remove anchors, fold the goal as instructed, and store it.

⚠ Safety

- Do not climb, hang, or swing from the goal frame.
- Secure the goal against wind or accidental movement.
- Keep players clear while unfolding or folding the frame.
- Do not use damaged joints, sharp parts, or unstable anchors.

Care & storage

- Remove dirt and moisture, allow the net to dry, and store the goal in its bag.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hula Hoop

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Low** 1 labs

Covers these lab items: Multi-coloured Hula Hoops

What it is used for

A hula hoop supports coordination, flexibility, balance, and movement activities in sports and fitness sessions. Select a suitable hoop size and provide enough clear space for safe movement.

How to use

1. Inspect the hoop for cracks, sharp edges, or loose connectors.
2. Choose a hoop size suitable for the user's age, ability, and activity.
3. Select a flat, clear area away from furniture and other participants.
4. Demonstrate the intended movement and explain safe spacing before starting.
5. Use controlled movements and maintain awareness of nearby people and objects.
6. Stop if the hoop becomes damaged or the activity area becomes unsafe.
7. Collect, clean, and store hoops after the session.

⚠ Safety

- Do not use cracked or sharp-edged hoops.
- Maintain enough space to avoid striking other people or objects.
- Avoid swinging hoops toward faces or using them in crowded areas.
- Stop activity if pain, dizziness, or loss of balance occurs.

Care & storage

- Wipe hoops clean and store them away from heat or heavy objects that may deform them.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Hurdle

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: 6 Step Training Hurdle; 9 Step Hurdle; Hurdles

What it is used for

To support agility, coordination, speed, and obstacle-training exercises in sports and fitness sessions.

How to handle

- Inspect each hurdle for cracks, sharp edges, loose joints, and stable footing before use.
- Set hurdles on a flat, clear training surface and space them according to the activity and participant ability.
- Explain the exercise, demonstrate safe technique, and ensure participants warm up before starting.
- Maintain sufficient clearance between participants and supervise jumping or stepping drills to reduce collision and trip risks.
- Stop the activity if a hurdle becomes damaged or unstable, and collect equipment after the session.

Storage

- Store clean and dry in a designated sports-equipment area.
- Stack or fold only as designed by the manufacturer and keep heavy loads off the hurdles.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Marker Cone

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Low** 3 labs

Covers these lab items: Cones; Marking Cones; Saucer Cone

What it is used for

Marker cones define boundaries, lanes, targets, and movement paths during sports and physical training. Arrange cones with adequate spacing and keep the training area clear of hazards.

How to use

1. Inspect each cone for cracks, sharp edges, or deformation.
2. Choose a flat, clear surface suitable for the planned activity.
3. Place cones according to the training layout and required spacing.
4. Explain the course or boundary markings to participants before starting.
5. Monitor cone positions during the activity and replace any displaced cones.
6. Collect cones after the session and check for damage.
7. Clean and store the cones in a designated location.

⚠ Safety

- Do not use broken cones with sharp edges.
- Keep cones away from vehicle routes and emergency exits unless specifically required by an approved setup.
- Maintain enough space between participants to prevent collisions.
- Remove wet or slippery hazards from the activity area.

Care & storage

- Wipe off dirt and moisture and store cones away from excessive heat or crushing loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Medicine Ball

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: 1 kg Medicine Ball; Medicine Balls

What it is used for

To perform strength, coordination, balance, and conditioning exercises using a weighted medicine ball.

How to handle

- Inspect the ball surface, seams, and grip for cracks, leaks, or other damage.
- Choose a ball weight appropriate for the participant's ability and the planned exercise.
- Use a clear, non-slip exercise area and demonstrate correct posture and movement before starting.
- Keep a secure grip, use controlled movements, and maintain adequate distance from others during throws or partner drills.
- Do not use a damaged ball; stop if pain, dizziness, or loss of control occurs.

Storage

- Wipe the ball clean and dry after use.
- Store on a stable sports-equipment rack or in a designated bin away from sharp objects and excessive heat.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Plyometric Box

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Plyometric Box

What it is used for

A plyometric box is used for supervised jump, step-up, and lower-body conditioning exercises. Choose a stable box height appropriate to the user's ability and maintain a clear landing area.

How to use

1. Inspect the box top, sides, base, and joints for damage or instability.
2. Place the box on a flat, nonslip surface with clear space around it.
3. Select a box height suitable for the user's fitness and skill level.
4. Warm up and demonstrate the correct landing and stepping technique.
5. Perform the selected exercise with controlled movement and stable foot placement.
6. Step down carefully when appropriate and allow sufficient recovery between repetitions.
7. Check the box and exercise area after use and return equipment to storage.

⚠ Safety

- Do not use a cracked, slippery, or unstable box.
- Keep the landing zone clear of people and obstacles.
- Avoid jumps beyond the user's ability and do not jump down unnecessarily from high boxes.
- Stop if pain, dizziness, or loss of balance occurs.

Care & storage

- Wipe the surface clean, check stability regularly, and store the box in a dry designated area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Punching Pad

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Punching Pads, Kick Shields, Focus Mitts

What it is used for

Punching pads, kick shields, and focus mitts provide targets for supervised striking and kicking practice. Select suitable protective equipment and ensure the holder and participant use controlled techniques.

How to use

1. Inspect the pad, shield, mitts, straps, stitching, and padding for damage.
2. Choose equipment appropriate for the activity, participant size, and training level.
3. Clear the practice area and ensure adequate space around both participants.
4. Fit hand pads or mitts securely and establish agreed target positions.
5. Begin with controlled strikes while the holder maintains a stable stance.
6. Increase intensity only when technique, coordination, and supervision are adequate.
7. Stop the session if equipment shifts, damage appears, or discomfort occurs.

⚠ Safety

- Use suitable hand wraps, gloves, and other required protective equipment.
- Never strike the holder's unprotected body or head.
- Maintain clear communication and avoid uncontrolled high-force strikes.
- Do not use torn, loose, or inadequately padded equipment.

Care & storage

- Wipe equipment after use, allow it to dry fully, and inspect straps and seams regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Relay Baton

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: Batons; Relay Batons

What it is used for

To practise and conduct relay races using a baton for safe, controlled handovers between runners.

How to handle

- Inspect the baton for cracks, sharp edges, loose parts, or slippery surfaces.
- Use a baton of suitable size and weight for the activity and ensure the running lane is clear.
- Hold the baton securely while running and follow the coach's instructions for passing and receiving it.
- Pass the baton within the designated exchange zone without throwing it at another runner.
- Retrieve dropped batons safely after checking that the running path is clear.

Storage

- Wipe the baton clean and dry after practice or competition.
- Store in a sports-equipment rack or bag away from heavy loads, heat, and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sit and Reach Box

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Sit and Reach Box

What it is used for

A sit and reach box measures flexibility by recording how far a person can reach forward while seated. Use the same standardized posture and instructions for each test to support consistent results.

How to use

1. Inspect the box, measuring scale, sliding indicator, and base for damage.
2. Place the box on a flat, nonslip surface and align it securely.
3. Explain the test and demonstrate the required seated position.
4. Ask the participant to remove shoes if required by the testing protocol and sit with legs extended.
5. Position the feet against the designated surface and keep knees straight without forcing them.
6. Have the participant reach forward smoothly and hold the furthest comfortable position briefly.
7. Record the reading, allow recovery, and clean the equipment after testing.

⚠ Safety

- Do not force a stretch or bounce toward the measurement point.
- Stop if pain, dizziness, or unusual discomfort occurs.
- Adapt or avoid the test when contraindications or mobility limitations require professional advice.
- Keep the test area clear to prevent slips or falls.

Care & storage

- Wipe contact surfaces after use and store the box in a dry, protected area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Skipping Rope

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Skipping Ropes

What it is used for

A skipping rope supports cardiovascular fitness, coordination, and footwork training. Select an appropriate rope length and exercise on a clear, nonslip surface with controlled movements.

How to use

1. Inspect the rope, handles, bearings, and adjustment mechanism for wear or damage.
2. Adjust the rope length according to the user's height and the manufacturer's guidance.
3. Choose a flat, nonslip surface with sufficient overhead and surrounding clearance.
4. Warm up and hold one handle securely in each hand.
5. Rotate the rope using controlled wrist movements and begin with small, comfortable jumps.
6. Maintain a steady rhythm and stop to rest when necessary.
7. After use, inspect the rope and store it without tight kinks.

⚠ Safety

- Keep people, pets, and objects outside the rope's rotation area.
- Wear suitable footwear and avoid wet or slippery surfaces.
- Begin at a manageable pace and stop if pain, dizziness, or unusual breathlessness occurs.
- Do not use a frayed or damaged rope.

Care & storage

- Wipe handles and rope after use and store them in a dry, untangled condition.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sports Bib

Category: **Sports & Physical Education** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Nylon Bibs

What it is used for

Handle nylon sports bibs hygienically for team identification during activities.

How to handle

- Check bibs for tears, loose seams, and damaged elastic before use.
- Assign clean bibs to participants and avoid sharing unwashed items.
- Choose sizes that fit comfortably without restricting movement or breathing.
- Wash after use according to the fabric care instructions and dry fully.

Storage

- Store clean, dry bibs in labelled groups by colour or size.
- Keep separate from used or damp sportswear.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sports Protective Gear

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Head Guards, Shin Guards, Mouth Guards

What it is used for

Sports protective gear such as head guards, shin guards, and mouth guards helps reduce injury risk during training and competition. Select properly fitting equipment appropriate for the sport and participant.

How to use

1. Inspect each item for cracks, tears, loose straps, or damaged padding.
2. Choose gear of the correct size and approved type for the sport.
3. Fit the head guard securely without obstructing vision or hearing unnecessarily.
4. Position shin guards to cover the required area and secure the straps.
5. Fit the mouth guard according to its instructions and check that breathing remains comfortable.
6. Check movement and comfort before starting the activity.
7. Clean, dry, and store the equipment after use.

⚠ Safety

- Do not use cracked helmets, damaged guards, or torn straps.
- Never share mouth guards unless properly cleaned and approved for reuse.
- Replace equipment after significant impact or when its protective integrity is uncertain.
- Follow sport-specific rules and coach or clinician guidance.

Care & storage

- Clean according to manufacturer instructions and inspect fit and condition before every session.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sports Ring

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Flat Ring

What it is used for

A flat sports ring marks a target, boundary, or movement position during physical education and training activities. Place rings securely and arrange them to suit the activity and available space.

How to use

1. Inspect the ring for cracks, sharp edges, deformation, or slippery surfaces.
2. Choose a flat, clear area suitable for the planned activity.
3. Place the ring securely on the ground at the required position.
4. Arrange additional rings with sufficient spacing to prevent collisions.
5. Explain the activity and demonstrate how participants should move around or through the ring.
6. Monitor the activity and reposition rings if they move or create hazards.
7. Collect, clean, and store the rings after use.

⚠ Safety

- Do not use broken or sharp-edged rings.
- Keep rings away from walkways and emergency exits unless they are part of a controlled activity.
- Prevent participants from jumping onto unstable or stacked rings.
- Maintain adequate spacing between participants.

Care & storage

- Wipe rings clean and store them flat in a dry, designated area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Stopwatch

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 7 labs

Covers these lab items: Stop Watch

What it is used for

Use a stopwatch to measure elapsed time during sports, physical education, training, or other timed activities.

How to use

1. Inspect the stopwatch buttons, display, and battery condition.
2. Reset the display to zero before starting a new measurement.
3. Position yourself where the activity can be observed clearly.
4. Press start at the agreed beginning of the activity.
5. Press stop when the required interval ends and read the time.
6. Record the result and reset the stopwatch for the next trial.
7. Switch off or store the stopwatch after use if applicable.

⚠ Safety

- Keep attention on the activity and avoid standing in the participants' path.
- Do not use a damaged stopwatch with loose battery parts.
- Keep the device away from water unless it is water-resistant.

Care & storage

- Wipe the stopwatch with a soft, dry cloth after use.
- Store it safely and replace the battery when needed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Swimming Ring

Category: **Sports & Physical Education** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Swimming Ring

What it is used for

Handle swimming rings safely as supervised water-play aids.

How to handle

- Inspect the ring for punctures, leaks, damaged seams, or faulty valves before use.
- Use only as a buoyancy aid under direct adult supervision; it is not a life-saving device.
- Choose a size appropriate for the user and keep away from sharp objects and excessive heat.
- Do not use a damaged ring or rely on it in deep water or strong currents.

Storage

- Rinse off pool water or salt and dry fully before storage.
- Deflate only as recommended and store away from sunlight, heat, and sharp objects.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tent

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Camping tents

What it is used for

A camping tent provides temporary shelter during outdoor activities. Select a suitable location, assemble the tent according to its instructions, and secure it against expected weather conditions.

How to use

1. Inspect the tent fabric, poles, pegs, guy lines, zippers, and seams for damage.
2. Choose level ground away from flood channels, falling branches, and other hazards.
3. Clear sharp debris from the tent footprint and spread the groundsheet if supplied.
4. Assemble poles and connect them to the tent using the designated fittings.
5. Secure the tent with pegs and guy lines appropriate for the ground conditions.
6. Check doors, ventilation, and weather protection before placing equipment inside.
7. After use, dry the tent, pack all components, and store them together.

⚠ Safety

- Avoid pitching beneath unstable trees or near waterways that may rise suddenly.
- Keep flames, stoves, and heaters away from tent fabric.
- Secure the tent against wind and evacuate if conditions become unsafe.
- Do not enter a flooded or structurally damaged tent.

Care & storage

- Dry the tent completely before storage and repair damaged fabric or poles promptly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Tessel

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Tessel

What it is used for

A tessel is used according to its specific design and intended sports or training application. Confirm the equipment identity and follow the manufacturer's instructions before use.

How to use

1. Identify the exact equipment type and confirm its intended function.
2. Inspect the item for damage, loose parts, sharp edges, or wear.
3. Check the instructions for correct setup, adjustment, and operating limits.
4. Prepare a clear work or activity area suitable for the equipment.
5. Demonstrate the correct handling method before allowing participants to use it.
6. Use the equipment only for its approved purpose and monitor for unusual movement or damage.
7. Clean, inspect, and store the item after use.

⚠ Safety

- Do not use equipment with uncertain identity or unclear operating instructions.
- Keep the activity area clear of obstacles and bystanders.
- Do not modify the equipment or exceed its rated limits.
- Stop use immediately if a fault or injury risk is identified.

Care & storage

- Follow the manufacturer's cleaning instructions and store the item in a dry, designated location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Training Arch

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Training Arch

What it is used for

A training arch provides a target or passage for agility, coordination, and movement exercises. Set it up securely in a clear area and adapt its position to the participants and activity.

How to use

1. Inspect the arch, base, joints, and fastening parts for cracks or deformation.
2. Choose a flat surface with sufficient clearance around the arch.
3. Assemble the arch according to the manufacturer's instructions.
4. Secure the base and check that the structure remains stable when lightly tested.
5. Explain the activity and demonstrate safe movement through or around the arch.
6. Monitor participants and reposition the arch if it shifts or creates a hazard.
7. Disassemble, clean, and store the arch after use.

⚠ Safety

- Do not use a cracked, unstable, or poorly assembled arch.
- Keep adequate spacing between participants to prevent collisions.
- Do not hang or climb on the arch unless specifically designed for that purpose.
- Keep the exercise area clear of obstacles and slippery materials.

Care & storage

- Clean the arch after use and store its components in a dry, protected location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Whistle

Category: **Sports & Physical Education** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Whistle

What it is used for

A whistle produces an audible signal for instructions, timing, or emergency communication during sports and physical activities. Use clear, controlled signals and follow the activity's established whistle codes.

How to use

1. Inspect the whistle for cracks, blockage, or damage to its attachment.
2. Clean the mouthpiece before use, especially when the whistle is shared.
3. Attach the whistle securely to a lanyard or designated holder.
4. Confirm the signal pattern with participants before the activity begins.
5. Blow short, clear signals appropriate to the instruction or emergency.
6. Check that participants can hear the signal and respond correctly.
7. Clean and store the whistle after the activity.

⚠ Safety

- Avoid prolonged or excessively loud blowing close to people's ears.
- Do not share mouth-contact equipment without appropriate cleaning.
- Keep the lanyard clear of machinery and other entanglement hazards.
- Use emergency signals consistently and according to the site's procedures.

Care & storage

- Clean the mouthpiece after use and store the whistle in a clean, dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

AR Learning Kit

Category: **Teaching Aids, Charts, Books & Software** Type: **Short SOP** Priority: **Medium** 2 labs

Covers these lab items: AR Cubes-based; AR Globe Kit

What it is used for

To support interactive learning using augmented-reality cubes, globe kits, and compatible digital content.

How to handle

- Check the kit components, printed markers, globe or cubes, and any supplied device for damage and completeness.
- Use the approved app or software and follow the setup instructions for the specific kit.
- Place the learning objects on a stable surface and scan the correct markers using the device camera.
- Guide learners to view the digital overlays and relate them to the physical learning model.
- Avoid rough handling, liquids, and unauthorized app downloads; sign out of shared accounts when required.

Storage

- Return cubes, globe components, markers, and accessories to their labelled compartments.
- Store devices and printed materials in a clean, dry, secure cabinet; charge electronics according to manufacturer guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Anatomical Model

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: 3D models of Human Body organ; Human Body Skeleton; Models of the human body or specific organs (optional); Wound Care Model Anatomical

What it is used for

Anatomical models demonstrate human body structures, organs, skeletons, and basic care procedures for education.

How to handle

- Identify the model and review any instructions before demonstration.
- Handle joints, removable organs, and delicate parts carefully to avoid breakage.
- Use models for educational purposes and do not treat them as substitutes for clinical equipment or medical advice.
- Clean reusable models with manufacturer-approved products between sessions.
- Report missing parts or damage and avoid using unstable models.

Storage

- Return detachable parts to their correct positions or labelled compartments.
- Store models in a clean, dry cabinet away from direct sunlight and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Audio Guide Tool

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Use of QR-based storytelling or audio-guided walk tools

What it is used for

Use QR-based storytelling and audio-guided walk tools effectively and safely.

How to handle

- Test QR codes, audio links, and device volume before the activity.
- Scan only approved QR codes and use trusted links.
- Keep devices secure and avoid looking at screens while walking.
- Respect privacy and obtain permission before recording or sharing audio.

Storage

- Store devices and accessories in a secure, labelled location.
- Keep charging equipment dry and follow device-specific storage instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Audio-Visual Content

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 25 labs

Covers these lab items: Audio/video on English, Hindi or local language course; Audio-visual examples of professional guided walks and narration techniques; Audio-visual recordings of professional travel consultations and virtual tour presentations; AV Aids; AV Aids for understanding Human body structure and function; Samp

What it is used for

Audio-visual content supports teaching, demonstrations, language learning, and professional skills training.

How to handle

- Verify the content, language, relevance, and suitability for the intended learners.
- Use authorised media files and approved playback equipment or software.
- Check sound, display, captions, and accessibility before the session begins.
- Respect copyright, privacy, and permissions when recording or sharing content.

Storage

- Store files in approved folders with clear filenames and appropriate access permissions.
- Keep backup copies and protect media from accidental deletion or damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Book / Manual

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 139 labs

Covers these lab items: Accessories & Trims file; Activity Book for Students - English (Class 10); Activity Book for Students - English (Class 11); Activity Book for Students - English (Class 6); Activity Book for Students - English (Class 7); Activity Book for Students - English (Class 8); Activity Book for Students - Eng

What it is used for

Books and manuals provide learning content, technical instructions, reference information, and activity guidance.

How to handle

- Check the title, edition, completeness, and relevance before issuing or using the material.
- Handle pages and bindings carefully and avoid writing in shared copies unless permitted.
- Use the correct manual for the equipment or procedure and follow current approved instructions.
- Return borrowed material and report missing, damaged, or outdated copies.

Storage

- Store upright or flat on clean, dry shelves according to size and binding.
- Keep away from moisture, pests, direct sunlight, and food or liquids.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Brochure

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 9 labs

Covers these lab items: Bancassurance Product Brochures; Product Catalogues or Brochures; Sample conservation pamphlets and eco-awareness materials for guest education; Sample maps, brochures, and local area guides for guests; Templates/brochures of companies; Tour Brochures of travel companies and tourist sites; Travel In

What it is used for

Brochures provide concise information about products, services, locations, conservation, and educational topics.

How to handle

- Check content accuracy, print quality, dates, and contact details before distribution.
- Select brochures appropriate for the intended audience and purpose.
- Keep brochures clean, dry, and neatly arranged for easy access.
- Distribute current approved versions and remove outdated or incorrect copies.

Storage

- Store brochures in labelled holders, folders, or dry cabinets.
- Protect printed materials from moisture, sunlight, and physical damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Case Study

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Case studies and checklists on data protection and confidentiality in travel operations; Case Study Materials

What it is used for

Case studies and checklists support learning, decision-making, and review of workplace procedures and confidentiality requirements.

How to handle

- Use approved, current materials relevant to the learning objective.
- Remove or anonymise personal, customer, and commercially sensitive information unless authorised for training use.
- Explain the scenario and expected task before distributing materials.
- Keep completed exercises secure and discuss sensitive examples respectfully.
- Return, archive, or dispose of materials according to organisational document-handling rules.

Storage

- Store printed and digital materials in designated locations with appropriate access controls.
- Protect confidential records and avoid leaving documents unattended in public areas.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Computer Based Training

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 16 labs

Covers these lab items: CBTs; CBTs on working on computer; CBTs on working on Computer system

What it is used for

Computer-based training provides interactive lessons, demonstrations, and assessments through approved digital learning systems.

How to handle

- Use authorised accounts and select the correct course or training module.
- Check the computer, audio, display, and network connection before beginning.
- Follow the lesson instructions and complete activities without sharing login credentials.
- Save progress where supported and report technical issues through the designated support channel.

Storage

- Store training files and records in approved locations with appropriate access permissions.
- Log out after use and protect learner data from unauthorised access.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Demonstration Model

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 12 labs

Covers these lab items: Dummy Truck; Housing models; Models for clean-up drives and eco-friendly guest practices; Models of different breeds; MPFI working model system; Working model of EV Vehicle; Working model system Electric vehicle

What it is used for

Demonstration models illustrate systems, equipment, structures, and processes for teaching and practical learning.

How to handle

- Identify the model's purpose and review any setup or operating instructions before use.
- Inspect the base, joints, moving parts, electrical connections, and labels for damage.
- Place the model on a stable surface and demonstrate it only within its intended limits.
- Keep fingers away from moving parts and disconnect power before adjustments when applicable.

Storage

- Return detachable parts and accessories to their designated compartments after use.
- Store models securely in a dry area and protect delicate components from impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Display Material

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 13 labs

Covers these lab items: Cultural artifacts and props to enrich storytelling practice; Danglers; Different kinds of luggage (sizes, materials etc); Different kinds of luggage (sizes, materials etc) (pictures if actual items not available); Display Material; Display Photographs

What it is used for

Display materials and props support demonstrations, storytelling, product presentation, and educational activities.

How to handle

- Check items for cleanliness, damage, stability, and suitability for the intended display.
- Arrange props securely without obstructing walkways, emergency exits, or safety equipment.
- Handle cultural, fragile, or borrowed items respectfully and follow any special instructions.
- Remove damaged or unstable items and return props after the activity.

Storage

- Group items by category in labelled boxes or display storage areas.
- Protect fragile materials from moisture, dust, direct sunlight, and impact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Document

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 54 labs

Covers these lab items: BOM & documents required in store; Buyer Requirement/ Size Chart; Buyer Requirement/comment Sheet; Physical Bill Copy; Picklist, BOM; PUCs; Reports and Dashboards; Reports and Dashboards (financial, performance, compliance reports); Sample billing statements, invoices, receipts, and vouchers; Sample

What it is used for

Documents support inventory control, purchasing, dispatch, compliance, reporting, and other administrative or operational activities.

How to handle

- Verify document identity, version, completeness, dates, and required approvals before processing.
- Handle original records carefully and maintain traceability for receipts, transfers, and changes.
- Protect personal, financial, commercial, and confidential information from unauthorized access.
- File, scan, distribute, or dispose of documents according to the approved record procedure.

Storage

- Store physical records in labelled, access-controlled locations protected from moisture and fire hazards.
- Keep digital copies in approved systems with appropriate permissions and backups.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Drawing / Plan

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 8 labs

Covers these lab items: Instrumentation diagrams; Layout plans; Location Plans; Loop diagrams

What it is used for

Drawings and plans communicate layouts, locations, instrumentation arrangements, and system connections.

How to handle

- Confirm the drawing title, revision number, scale, symbols, and approval status before use.
- Use the correct plan for the site, system, or training task.
- Protect drawings from spills, tearing, and unauthorised changes.
- Verify critical dimensions and connections against approved documentation before practical work.
- Report unclear, outdated, or conflicting information to the responsible supervisor.

Storage

- Store current drawings in labelled folders or approved digital repositories.
- Archive superseded revisions separately and retain document control information.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Flash Cards

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Alphabet Cards Set (A - Z); Flashcards for architecture symbolism, food trail highlights, and cultural festivals

What it is used for

Use flash cards as clean, organised teaching aids.

How to handle

- Sort cards by subject, alphabet, or activity before use.
- Show cards clearly and avoid bending or marking them unnecessarily.
- Use age-appropriate cards and explain symbols or cultural references accurately.
- Return all cards to the correct set after the activity.

Storage

- Store in labelled envelopes, boxes, or sets.
- Keep dry and away from direct sunlight and food spills.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Flip Chart

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 34 labs

Covers these lab items: Flip Chart; Flip Chart Holder; Flip Chart on human skeleton systems; Flip chart with markers; Flip Charts; Flip charts and marker sets for group presentations; Flipchart Stand

What it is used for

Flip charts and holders support presentations, brainstorming, group exercises, and instructional demonstrations.

How to handle

- Check the stand, fasteners, paper, and writing surface for damage before use.
- Set the holder on stable ground and secure the chart pages properly.
- Use compatible markers and keep writing within the intended page area.
- Turn pages carefully and keep the stand clear of walkways and emergency exits.

Storage

- Fold or disassemble the holder according to its design and store it securely.
- Keep paper and markers dry and protected from damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Form

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 39 labs

Covers these lab items: Errand Card; Export Documentation; Feedback collection materials like digital surveys or feedback cards; Feedback/Complaint Forms; Fitness Assessment Forms; Forms and Templates; Forms and Templates (loan, insurance, account opening forms); Guest activity logs and attendance records for group managem

What it is used for

Forms collect structured information for feedback, assessment, documentation, requests, and administrative workflows.

How to handle

- Use the correct current form and verify that all required fields and instructions are present.
- Provide clear, accurate information and obtain necessary signatures or approvals.
- Protect personal and confidential details and share completed forms only with authorised recipients.
- Review forms for completeness before submission and correct errors through the approved process.

Storage

- Store blank forms in labelled, accessible locations and protect them from moisture or damage.
- Keep completed forms in access-controlled physical or digital records according to retention rules.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Map

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Topographic maps of local trails or parks; Tourist maps; Tourist maps (Site/India/World)

What it is used for

Maps show geographical locations, routes, terrain, and attractions for navigation, teaching, and planning.

How to handle

- Confirm the map title, area covered, scale, legend, and publication date before use.
- Choose a map appropriate for the route, activity, and required level of detail.
- Handle printed maps with clean hands and protect them from moisture and tearing.
- Cross-check critical route or safety information with current authoritative sources.
- Do not rely on an outdated map alone for navigation in unfamiliar or hazardous areas.

Storage

- Fold maps along existing creases or store them flat in protective sleeves.
- Keep maps dry and store digital versions with clear filenames and dates.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Picture / Photograph

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Photographs of Stacked Notes; Picture (Apparel magazines); Pictures of destinations; Pictures of tourism attractions

What it is used for

Pictures and photographs support teaching, identification, visual reference, and presentation of places or subjects.

How to handle

- Check images for relevance, accuracy, clarity, and appropriate usage rights.
- Handle printed photographs with clean, dry hands and avoid marking or folding them.
- Use clear, correctly labelled images for educational or display purposes.
- Protect personal or sensitive images from unauthorised access or distribution.

Storage

- Store printed images in labelled albums, folders, or protective sleeves.
- Keep digital copies in organised folders with appropriate access controls.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Presentation Slides

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 12 labs

Covers these lab items: Power Point Presentation; Presentation Slides; Presentations

What it is used for

Presentation slides organise visual information for teaching, reporting, demonstrations, and professional communication.

How to handle

- Verify content accuracy, slide order, readability, and suitability for the audience.
- Use authorised images and media and respect copyright and privacy requirements.
- Test fonts, links, audio, video, and display compatibility before presenting.
- Keep a backup copy and avoid sharing confidential content with unauthorised audiences.

Storage

- Save slides in approved folders with clear filenames and version details.
- Maintain backups and protect files with appropriate access permissions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Reference Sample

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Embroidery Samples; Examples of branded stationery and marketing material used by travel agencies; Flights tickets; Garment Samples; Garments Made Ups and Home Furnishing Articles

What it is used for

Reference samples provide examples for comparison, training, evaluation, and quality checks.

How to handle

- Identify each sample and record its source, description, and relevant reference details.
- Use samples only for the intended demonstration or comparison task.
- Handle garments, documents, tickets, and other samples carefully to prevent damage or loss.
- Protect personal, confidential, or proprietary information shown on samples.
- Return samples to their designated place after use.

Storage

- Label and organise samples by type, topic, or reference number.
- Keep paper samples dry and textile samples clean; secure valuable or confidential items.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Role-Play Material

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 14 labs

Covers these lab items: Mock customer data templates for role-playing secure information handling; Mock guest kits including feedback cards, info leaflets, and safety instructions; Mock guest kits with items like binoculars, magnifying glasses, and small field guides; Mock room keycards, key holders, and room assignment sh

What it is used for

Role-play materials help learners practise communication, customer service, safety, and workplace scenarios in a controlled setting.

How to handle

- Review the scenario, learning objectives, and materials before the activity begins.
- Use mock or anonymised data instead of real personal or confidential information.
- Explain roles, boundaries, and safety expectations to participants before starting.
- Collect and reset props and forms after the exercise, and discuss feedback respectfully.

Storage

- Store scenario kits, cards, and props in labelled containers by activity.
- Keep sensitive training records access-controlled and replace damaged materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Sheet Metal Sample

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Different types of sheet metal brackets

What it is used for

Handle sheet metal bracket samples safely for teaching and demonstration.

How to handle

- Inspect each sample for sharp edges, burrs, rust, or deformation before use.
- Wear suitable cut-resistant gloves when moving samples and safety glasses during demonstrations.
- Hold samples by safe edges and avoid passing them with sharp corners facing others.
- Do not use damaged or unstable samples for demonstrations.

Storage

- Store samples in labelled trays or racks with sharp edges protected.
- Keep dry and organised by type, size, or material.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Simulated Environment

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 4 labs

Covers these lab items: Simulated Environments for Practice

What it is used for

Simulated environments provide controlled practice for learning procedures, customer interactions, or workplace scenarios.

How to handle

- Review the scenario objectives, operating instructions, and safety rules before starting.
- Check props, equipment, and digital resources for damage or incorrect setup.
- Use the simulation only for its intended training purpose and follow the instructor's directions.
- Do not perform unsafe real-world actions merely because the environment is simulated.
- Reset equipment and report faults or hazards after the session.

Storage

- Return props and equipment to their designated locations.
- Keep digital resources organised and protect any recorded personal information.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Software

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low**

152 labs

Covers these lab items: 3D modeling software; 3D Modelling, Sculpting and Painting Software; Access to sample GDS/CRS interfaces (simulated environment); Accounting Software; Adobe Photoshop; Annotation Software; Anti Virus; Anti Virus Latest version; Anti-virus for Computer Systems; Antivirus software; Arduino IDE & Free

What it is used for

Software supports digital learning, design, modelling, accounting, simulation, and other approved instructional or workplace tasks.

How to handle

- Use authorized installations or accounts and follow the applicable licence terms.
- Check system requirements and save work regularly to approved storage locations.
- Protect passwords, personal information, and confidential files; avoid untrusted downloads and links.
- Report errors, suspicious activity, or access problems through the designated support process.

Storage

- Keep installation media, licence records, and manuals in controlled digital or physical storage.
- Back up important work to approved locations and maintain required updates.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Teaching Aid

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Educational materials such as whiteboard, projectors etc.

What it is used for

Handle educational materials such as whiteboards, projectors, and related teaching aids safely.

How to handle

- Inspect equipment, cables, and accessories before use and report visible damage.
- Follow operating instructions and switch off equipment before connecting or disconnecting cables.
- Keep liquids away from electrical equipment and ensure ventilation openings remain clear.
- Position teaching aids securely so they do not obstruct walkways or emergency exits.

Storage

- Clean whiteboards and equipment using manufacturer-approved methods.
- Store accessories in labelled locations and secure projectors and electronic devices after use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Time Management Tool

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 1 labs

Covers these lab items: Time Management Tools

What it is used for

Use time management tools to plan tasks and monitor activity duration.

How to handle

- Select a suitable planner, timer, schedule, or tracking tool for the activity.
- Enter realistic time estimates and communicate schedules clearly to participants.
- Use timers responsibly and avoid creating unsafe pressure during practical tasks.
- Review progress and adjust plans when priorities or conditions change.

Storage

- Keep paper planners and tools organised in a designated location.
- Charge and store electronic timers or devices according to manufacturer instructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Training Material

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 3 labs

Covers these lab items: Drill on field triage casualty evacuation techniques; Emergency response guides and role-play materials for mock scenarios; First-aid drill, drill including CPR (Cardio pulmonary resuscitation) and Choking

What it is used for

Training materials support practice in field triage, casualty evacuation, emergency response, and first-aid drills.

How to handle

- Review learning objectives, scenario instructions, and the limits of the exercise before starting.
- Use current, approved first-aid and emergency-response guidance.
- Clearly distinguish simulated casualties and props from real emergencies.
- Conduct CPR and evacuation practice under qualified supervision using appropriate training manikins and equipment.
- Stop the drill if an actual emergency, injury, or unsafe condition occurs and report any damaged materials.

Storage

- Keep printed guides, checklists, and role-play materials organised and dry.
- Clean and inspect reusable training equipment after sessions and store it in its designated area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wall Chart

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 159 labs

Covers these lab items: Anatomy and physiology charts; Artificial insemination Chart; Basic 5s charts; Big Poster; Big Poster (at POS) for offer related advertisement; Block diagram/flow chart of hydrogen compression; Block diagrams; Breed charts Cow,Goat,Bull ock; Chart and Posters on body systems; Chart of bamboo process

What it is used for

Wall charts and posters display educational, instructional, promotional, or safety information for learners and staff.

How to handle

- Verify the content, accuracy, readability, and relevance before displaying the chart.
- Use suitable mounting materials and secure the chart without damaging walls or equipment.
- Position charts at an accessible viewing height and keep exits, controls, and safety signs unobstructed.
- Replace outdated, damaged, or misleading displays promptly.

Storage

- Store rolled or flat charts in a clean, dry location to prevent creasing and fading.
- Keep charts away from moisture, direct sunlight, and damaging adhesives.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Whiteboard

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low** 69 labs

Covers these lab items: Blackboard/Whiteboard; Black/White board; Notice Board / Whiteboard; White Board; White Board/Black Board; White/Black Boards; Whiteboard or blackboard with markers/chalk; Whiteboard or smartboard

What it is used for

A whiteboard or blackboard provides a reusable surface for teaching, planning, notices, and group communication.

How to handle

- Inspect the board, frame, mounting, and writing surface for damage or instability.
- Use compatible markers, chalk, and erasers designed for the board surface.
- Write clearly and erase using the recommended method to avoid surface damage.
- Keep the board and nearby floor free from dust, clutter, and obstructions.

Storage

- Store removable boards securely and keep accessories in a designated container.
- Protect the writing surface from moisture, scratches, and unsuitable cleaning chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Whiteboard Duster

Category: **Teaching Aids, Charts, Books & Software** Type: **Handling note** Priority: **Low**

13 labs

Covers these lab items: Duster; Eraser

What it is used for

A whiteboard duster removes dry-erase marker or chalk residue from compatible teaching boards.

How to handle

- Check the duster for loose parts, excessive dirt, or damage before use.
- Use the duster only on the board type for which it is intended.
- Wipe the board gently and avoid spreading heavy dust into the breathing zone.
- Clean or replace the duster when it no longer removes residue effectively.

Storage

- Store the duster in a designated holder or clean container.
- Keep it dry and separate from food, chemicals, and other cleaning tools.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Cable Tester

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Cable testers, multimeters, and continuity testers; Lan Tester

What it is used for

A cable tester checks whether cable connections and conductors are correctly wired and continuous. Use the tester type suitable for the cable, such as Ethernet or telephone cable.

How to use

1. Inspect the tester, leads, adapters, and battery condition.
2. Identify the cable type and choose compatible tester ports or adapters.
3. Disconnect the cable from powered equipment before testing unless the tester is specifically designed otherwise.
4. Connect both ends of the cable to the correct tester ports.
5. Run the test and read the wiring, continuity, or fault indication.
6. Disconnect the cable and tester, then record any faults found.

⚠ Safety

- Never connect a basic cable tester to live mains wiring.
- Do not test cables connected to powered equipment unless the tester permits it.
- Keep connectors and hands dry, and replace damaged test leads.

Care & storage

- Keep connectors clean, replace weak batteries, and store the tester in its protective case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

EPABX

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: EPABX

What it is used for

To operate an EPABX system for internal extensions and external telephone calls. To route calls correctly, transfer them to the intended extension, and maintain clear communication.

Main parts

- EPABX control unit
- Telephone handsets or extensions
- Extension ports and line connections
- Power supply
- Programming or operator console, if fitted

Before you start

1. Inspect the control unit, handsets, and cables for damage.
2. Confirm that the system has power and the telephone lines are connected.
3. Check the extension directory and call-handling instructions.
4. Ensure the operator has the required access permissions.

How to use

1. Lift the handset or use the approved call interface.
2. Dial the required internal extension or external access code.
3. Wait for the call to connect and communicate clearly.
4. To transfer a call, use the system's transfer function and dial the target extension.
5. Confirm the transfer according to the system procedure before disconnecting.
6. End the call using the handset or interface control.
7. Report repeated line faults or extension failures to the administrator.

⚠ Safety

- Do not open or repair the control unit while powered unless authorized and following the service procedure.
- Keep liquids away from the equipment.
- Do not disclose confidential caller information to unauthorized people.
- Follow the organization's emergency-call procedure.

Care & storage

- Keep handsets and the console clean.
- Avoid pulling or sharply bending telephone cables.
- Check extension labels and directory information regularly.
- Report faults and protect the system from power disturbances as instructed.

Common mistakes to avoid

- Dialling the wrong extension or access code.
- Transferring a call without confirming the destination.
- Disconnecting before the transfer is complete.
- Changing system settings without authorization.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

GPS Device

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Advanced GPS devices and trail mapping tools; Advanced GPS units; Basic GPS devices

What it is used for

To determine geographic position and support navigation, route recording, and trail mapping.

How to handle

- Inspect the device, screen, buttons, antenna, and battery before use.
- Charge the battery and enable location services or satellite reception as specified by the device manual.
- Move to an area with a clear view of the sky and allow the device to acquire a reliable satellite fix.
- Select the correct coordinate system, map, or tracking mode and record position or route data as needed.
- Handle the device carefully, save important records, and switch it off when no longer required.

Storage

- Store in a protective case in a dry location within the recommended temperature range.
- Keep maps, cables, and accessories organised and protect the device from impact and moisture.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

IP Phone

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Communication systems such as VoIP phones and email management tools

What it is used for

An IP phone uses an internet protocol network to make and receive voice calls and may support messaging or directory features. Connect it to the approved network and verify its account and audio settings.

How to use

1. Inspect the phone, handset, cable, power adapter, and network port for damage.
2. Connect the phone to the approved Ethernet port or supported wireless network.
3. Provide power through the approved adapter or compatible Power over Ethernet connection.
4. Follow the organization's setup instructions to configure the extension or service account.
5. Check network status and confirm that the phone has registered successfully.
6. Make a test call and verify microphone, speaker, and call quality.
7. Sign out or secure the phone when required and report connection problems.

⚠ Safety

- Use only authorized network credentials and approved device configurations.
- Do not share passwords or expose confidential call information.
- Keep liquids away from the phone and electrical connections.
- Follow organizational privacy and acceptable-use policies when using communication tools.

Care & storage

- Clean the handset and keypad with approved materials and keep cables and firmware maintained.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Intercom

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Intercom

What it is used for

An intercom enables voice communication between rooms, offices, or designated stations. Connect and test the system according to its configuration and operating instructions.

How to use

1. Inspect the intercom units, cables, power supply, and mounting points for damage.
2. Confirm that the stations are connected to the correct ports or network.
3. Switch on the system and check the power or connection indicators.
4. Select the required station or extension using the designated controls.
5. Place a test call and verify microphone, speaker, and audio clarity.
6. Check any supported call, door-release, or alert functions using approved procedures.
7. Finish the test and report faults or abnormal operation.

⚠ Safety

- Use only the specified power supply and compatible connections.
- Keep electrical units away from liquids unless rated for the environment.
- Do not open powered equipment or touch exposed wiring.
- Follow privacy and access-control rules when using intercom features.

Care & storage

- Clean the handset and controls with approved materials and inspect cables periodically.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Internet Connection

Category: **Telecom & Networking** Type: **Handling note** Priority: **Low** 15 labs

Covers these lab items: Internet access; Internet connection; Internet Connection Setup; Internet connectivity; Internet with Wi-Fi; Telephone line (For Internet)

What it is used for

An internet connection provides network access for communication, research, online learning, and approved digital services.

How to handle

- Use authorised network credentials and connect only to the approved network.
- Check router, cable, Wi-Fi, and device settings without opening electrical equipment.
- Protect accounts with strong passwords and avoid sharing confidential information over unsecured connections.
- Report outages, suspicious activity, or configuration changes to the designated IT support team.

Storage

- Keep network devices ventilated and protected from moisture, dust, and physical damage.
- Secure router credentials and document approved network settings in a controlled location.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Internet Dongle

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Digital Dongle

What it is used for

An internet dongle provides network connectivity to a compatible computer or device through a supported mobile or data connection. Configure it using the approved network settings and verify connectivity before use.

How to use

1. Inspect the dongle, connector, SIM slot, and casing for damage.
2. Insert an activated compatible SIM card if required by the device.
3. Connect the dongle to a supported USB port or power source.
4. Install or open the required software according to the manufacturer's instructions.
5. Enter the approved network settings or access credentials if prompted.
6. Wait for network registration and confirm the connection using a permitted test.
7. Disconnect safely before removing the dongle or SIM card.

⚠ Safety

- Use trusted software and authorized network credentials.
- Do not expose the dongle to moisture, excessive heat, or physical impact.
- Avoid unknown configuration files or suspicious connection prompts.
- Follow organizational policies when connecting to workplace networks.

Care & storage

- Keep the connector clean, protect the dongle during transport, and store it in a dry place.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Network Cable

Category: **Telecom & Networking** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Coaxial cables, CAT5/CAT6 cables, HDMI/VGA cables; Network cable; UTP Cat. 5 cables; UTP Cat. 5E/ 6 cables; UTP Cat. 6 cables

What it is used for

Network and signal cables connect compatible networking, video, and communication equipment.

How to handle

- Identify the cable type, category, connector, and required application before use.
- Inspect the jacket and connectors for cuts, bent pins, or damaged clips.
- Connect to the correct port without forcing the connector.
- Maintain the recommended bend radius and keep cables away from sharp edges and heat.
- Label both ends where needed and test the connection after installation.

Storage

- Coil cables loosely without tight bends or knots.
- Store labelled cables in a dry location away from moisture and crushing loads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Network Switch

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 11 labs

Covers these lab items: 16 Port Switches; Hubs; Hubs/switches; Local Area Network; Local Area Network (LAN); Network Switch

What it is used for

Use a network switch to connect compatible wired devices within a local area network and enable communication between them.

How to use

1. Check the switch ports, power adapter, and casing for visible damage.
2. Place the switch in a ventilated location with suitable power access.
3. Connect the power adapter using the specified voltage and polarity.
4. Connect Ethernet cables from computers, routers, or other network devices to the switch ports.
5. Observe link and activity indicators to confirm physical connections.
6. Configure managed-switch settings only when authorized and according to the network plan.
7. Test connectivity and label cables or ports where required.

⚠ Safety

- Use only the power adapter specified for the switch.
- Do not connect damaged cables or expose the device to moisture.
- Disconnect power before cleaning or handling internal parts.
- Follow network security rules and do not change settings without authorization.

Care & storage

- Keep ventilation openings clear and dust the casing with power disconnected.
- Label connections and store spare cables neatly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Networking Equipment

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Network devices; Networking equipment; Networking Equipment (Router/Switch)

What it is used for

To connect, distribute, and manage data communication between computers and other network devices.

How to handle

- Inspect routers, switches, access points, cables, connectors, and power supplies for damage.
- Verify the device model, port requirements, network plan, and approved configuration before connecting it.
- Connect Ethernet and power cables to the correct ports and ensure ventilation is not blocked.
- Configure network settings using authorised credentials and test connectivity, link status, and required services.
- Do not expose passwords or change production settings without approval; report faults to the network administrator.

Storage

- Keep equipment in a clean, dry, ventilated, and access-controlled location.
- Label cables and accessories and protect devices from dust, moisture, static discharge, and unauthorised access.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

OTDR

Category: **Telecom & Networking** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: OTDR

What it is used for

An optical time-domain reflectometer (OTDR) tests fibre-optic links by sending light pulses into a fibre and analysing the returned signal. It helps locate breaks, splices, connectors, and loss events along the fibre route.

Main parts

- OTDR main unit and display
- Optical output port and connector adapter
- Launch fibre and receive fibre when required
- Test leads and approved fibre connectors
- Battery or power adapter
- Cleaning tools and optical inspection accessories

Before you start

1. Read the OTDR manual and identify the fibre type, wavelength, and test limits.
2. Confirm that the fibre link is disconnected from live transmitters and is safe to test.
3. Inspect the OTDR port, test leads, launch fibre, and connectors for damage or contamination.
4. Clean and inspect fibre connectors using approved fibre-optic cleaning tools.
5. Select the correct test wavelength and refractive index for the fibre under test.
6. Set the range, pulse width, averaging time, and launch fibre arrangement appropriate to the link.

How to use

1. Connect the launch fibre to the OTDR and the fibre link according to the test plan.
2. Connect the receive fibre at the far end when required to evaluate the final connector or link end.
3. Verify that all connections are secure and that the test path is correctly identified.
4. Start the OTDR acquisition and allow the instrument to collect the required trace.
5. Review the trace for events such as connectors, splices, bends, breaks, and attenuation changes.
6. Check event distance, reflectance, and loss values against the acceptance criteria.
7. Repeat the test at the specified wavelength or direction when required by the test plan.
8. Save the trace and record fibre identification, wavelength, settings, and test results.
9. Disconnect the test leads carefully and replace protective caps on clean connectors.
10. Switch off the OTDR and store the unit, launch fibre, and accessories in their protective case.

⚠ Safety

- Never look directly into a fibre end or OTDR optical port during testing.
- Verify that the fibre is disconnected from active optical transmitters before connecting test equipment.
- Use only compatible optical connectors and never force a connector into the port.
- Keep fibre ends and glass fragments away from skin and eyes; collect scraps in a designated sharps container.
- Clean connectors with approved tools and avoid touching the polished end face.
- Follow site procedures for working near live telecom equipment and elevated or confined routes.

Care & storage

- Inspect and clean optical connectors before and after each test.
- Protect the OTDR port and fibre ends with clean dust caps when not in use.
- Inspect launch fibres, test leads, adapters, and cables for damage or tight bends.
- Keep software, calibration status, battery condition, and test records up to date.

Common mistakes to avoid

- Testing a live fibre can expose the user to invisible laser radiation and damage the instrument.
- Dirty connectors can create false loss events and unreliable readings.
- Using an incorrect refractive index or pulse width can produce inaccurate event distances or resolution.
- Skipping the launch fibre can prevent reliable measurement of the near-end connector.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

RJ45 Connector

Category: **Telecom & Networking** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: BNC/RJ45 connectors; RJ 45 modular plug; RJ45 connecters

What it is used for

RJ45 connectors terminate Ethernet cables for compatible wired network connections.

How to handle

- Confirm the connector type and category match the cable and networking equipment.
- Inspect contacts and locking clips for damage before installation.
- Use the correct crimping tool and wiring standard for the application.
- Avoid touching contacts unnecessarily and ensure the cable is properly seated before crimping.
- Test the finished cable with a suitable cable tester.

Storage

- Keep connectors in closed, labelled packaging.
- Store in a dry, clean place away from dust and crushing pressure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Satellite Phone

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Satellite phones for areas with limited cell service

What it is used for

A satellite phone provides voice or messaging communication in areas with limited cellular coverage. Check service availability, antenna positioning, and emergency communication procedures before use.

How to use

1. Inspect the phone, battery, charger, antenna, and accessories for damage.
2. Charge the battery and activate the service according to provider instructions.
3. Move to an open area with a clear view of the sky.
4. Extend or position the antenna as specified by the manufacturer.
5. Wait for satellite registration and confirm signal availability.
6. Make a test call or send a permitted test message.
7. End the connection, power off if required, and store the phone securely.

⚠ Safety

- Use only approved batteries, chargers, and accessories.
- Keep the antenna clear of obstructions during communication.
- Follow local laws and service restrictions for satellite phone use.
- Do not rely on the phone alone when an emergency backup plan is required.

Care & storage

- Keep the phone dry, charge the battery as recommended, and protect the antenna from damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Telephone

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 6 labs

Covers these lab items: Telephone; Telephone and intercom systems for guest communication; Telephone or Intercom System; Telephones, headsets, and mock radio/walkie-talkie systems for communication practice

What it is used for

A telephone or intercom system is used for voice communication between people, rooms, or departments. Use the correct handset, headset, or extension for the intended call.

How to use

1. Inspect the handset, headset, keypad, cable, and power connection if applicable.
2. Lift the handset or activate the headset according to the device instructions.
3. Dial the required number or extension and wait for the connection.
4. Speak clearly and listen without covering the microphone.
5. End the call using the correct button or by replacing the handset.
6. Return the handset or headset to its holder and report any faults.

⚠ Safety

- Keep devices and connectors dry and handle cables without pulling them sharply.
- Use a comfortable listening volume and avoid prolonged high-volume headset use.
- Do not share personal or confidential information with an unverified caller.

Care & storage

- Wipe shared handsets and headsets regularly and keep cables neatly arranged.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Two-Way Radio

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 5 labs

Covers these lab items: Two Way Radios and Chargers; Two-way radios and emergency alert systems; Two-way radios and emergency communication devices; Walkie talky; Walkie-Talkie / Intercom

What it is used for

A two-way radio allows short-range voice communication between team members without using a telephone network. Use the assigned channel and follow communication procedures.

How to use

1. Inspect the radio, antenna, battery, buttons, and microphone for damage.
2. Charge the battery and attach the antenna if it is removable.
3. Switch on the radio and select the assigned channel or group.
4. Press and hold the push-to-talk button before speaking clearly.
5. Release the button to listen and wait for the other person to respond.
6. Switch off the radio after use and place it on charge or in storage as required.

⚠ Safety

- Use only authorized frequencies and follow local radio regulations.
- Do not transmit near explosive atmospheres unless the radio is approved for that environment.
- Keep the antenna intact and do not hold it against your face during transmission.
- Keep communication concise during emergencies and follow the site emergency plan.

Care & storage

- Keep the radio clean and dry, charge batteries correctly, and store it securely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wi-Fi Router

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 9 labs

Covers these lab items: Internet wifi setup; Modem/ router; Wifi; Wi-Fi systems

What it is used for

Use a Wi-Fi router to connect compatible devices to a local wireless network and, when configured, to the internet.

How to use

1. Place the router in a central, ventilated location away from moisture and heat.
2. Check the power adapter and network cables for damage.
3. Connect the internet or modem cable to the router's correct port.
4. Power on the router and wait for the status indicators to stabilize.
5. Open the router setup page or approved app using the manufacturer's instructions.
6. Set a strong Wi-Fi password and secure administrator credentials.
7. Connect a device and test network access before finishing setup.

⚠ Safety

- Use only the specified power adapter and voltage.
- Change default administrator credentials and use current security settings.
- Do not share Wi-Fi passwords with unauthorized users.
- Keep the router dry and disconnect power before cleaning cables or connectors.

Care & storage

- Keep ventilation openings clear and wipe the casing with a dry cloth.
- Install firmware updates from the manufacturer when appropriate and store setup details securely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Wildlife Tracking Device

Category: **Telecom & Networking** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Tracking devices and tags for wildlife monitoring (if applicable)

What it is used for

Wildlife tracking devices and tags help monitor animal movement for research and conservation when authorized and appropriate. Use approved equipment and follow species-specific welfare, permitting, and data-handling requirements.

How to use

1. Confirm the research purpose, target species, permits, and approved tracking method.
2. Inspect the device, tag, battery, attachment system, and sensor functions.
3. Check that the device size and weight are suitable for the target animal.
4. Have trained and authorized personnel handle animals and attach tags using approved methods.
5. Activate the device and verify its identification, signal, and data transmission.
6. Record deployment details securely and monitor device performance and animal welfare.
7. Retrieve or replace the device only under the approved monitoring and animal-care plan.

⚠ Safety

- Do not capture, restrain, or tag wildlife without appropriate permits and trained personnel.
- Follow ethical review requirements and species-specific welfare guidance.
- Avoid devices or attachment methods that restrict movement or cause injury.
- Protect sensitive location data from unauthorized access or disclosure.

Care & storage

- Check battery life, attachment condition, and data quality according to the approved monitoring plan.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Arc Welding Machine

Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Arc Welding Machine (Transfor mer); Arc Welding Machine (Transformer) NABL test certified; Welding Machine

What it is used for

An arc welding machine supplies electrical current to create an arc that melts and joins compatible metal parts. Correct current settings, secure work connections, and suitable protective equipment help produce sound welds while reducing electric shock, fire, and eye injury risks.

Main parts

- Power source or transformer supplies welding current.
- Current control adjusts the welding output.
- Electrode holder grips the consumable electrode.
- Welding electrode carries current and provides filler material where applicable.
- Work clamp connects the return cable to the workpiece.
- Welding cables carry current between the machine and work area.
- Power switch controls the machine supply.
- Cooling vents and protective casing support safe operation.

Before you start

1. Only trained personnel should perform arc welding under an approved workshop procedure.
2. Read the machine manual and confirm the process, electrode type, and current range for the material.
3. Inspect the machine, electrode holder, work clamp, cables, plug, and insulation for damage.
4. Ensure the workpiece is clean, securely positioned, and connected to the return clamp.
5. Remove flammable materials and check the area for fire hazards before welding.
6. Wear a suitable welding helmet with the correct shade, welding gloves, protective clothing, and safety footwear.

How to use

1. Position the machine on a stable, dry surface with adequate ventilation.
2. Connect the power supply and welding cables according to the manual.
3. Attach the work clamp securely to a clean metal area on the workpiece.
4. Set the welding current and polarity according to the electrode and material requirements.
5. Fit the correct electrode into the holder without touching exposed conductive parts unnecessarily.
6. Place suitable welding screens to protect nearby people from arc radiation and sparks.
7. Strike the arc using the approved technique and maintain a controlled arc length.

8. Move the electrode steadily along the joint while monitoring weld appearance and machine behaviour.
9. Stop welding if cables overheat, the machine behaves abnormally, or a fire hazard appears.
10. Switch off the machine after welding, allow hot parts to cool, and inspect the weld safely.

⚠ Safety

- Never weld with wet gloves, wet clothing, or in standing water.
- Do not touch the electrode or exposed conductive parts while the circuit is energised.
- Use suitable eye, face, hand, and body protection against arc radiation, sparks, and hot metal.
- Provide adequate ventilation and use appropriate fume extraction for the materials being welded.
- Keep a suitable fire extinguisher available and monitor the area for smouldering material after welding.
- Do not weld sealed containers or unknown materials without an approved assessment and preparation procedure.

Care & storage

- Switch off and isolate power before cleaning or inspecting the machine.
- Remove dust from vents and the exterior using an approved method.
- Inspect cables, holders, clamps, connectors, and insulation regularly for wear or damage.
- Arrange qualified servicing for overheating, abnormal output, damaged insulation, or repeated tripping.

Common mistakes to avoid

- Using an incorrect current setting or unsuitable electrode.
- Welding with a loose work clamp or damaged cable insulation.
- Working without a proper welding helmet or nearby fire precautions.
- Ignoring fumes, overheating, or damaged electrical components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Blow Lamp

Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Blow Lamp

What it is used for

A blow lamp uses a controlled flame to heat materials for suitable workshop tasks such as soldering, brazing, or local heating, depending on the model and fuel. Use the specified fuel and ignition procedure, maintain ventilation, and control the flame carefully to prevent burns, fires, explosions, and harmful fumes.

Main parts

- Fuel tank or gas cylinder connection
- Burner nozzle and mixing tube
- Fuel valve or flow regulator
- Ignition system or approved lighting method
- Handle and heat-resistant grip
- Stand, flame adjustment control, and protective components

Before you start

1. Confirm the blow lamp type, approved fuel, and intended heating task.
2. Inspect the fuel container, hose, valve, regulator, seals, burner, and connections for damage or leakage.
3. Check that the nozzle is clean and the correct burner assembly is fitted.
4. Work in a well-ventilated, fire-safe area away from combustible materials and flammable vapours.
5. Secure the workpiece on a stable, heat-resistant surface and protect nearby components from heat.
6. Wear safety glasses or a face shield, heat-resistant gloves, suitable clothing, and other PPE required for the task.

How to use

1. Ensure the fuel valve is closed before connecting or inspecting the fuel supply.
2. Connect the specified fuel source according to the manufacturer's instructions and check for leaks using an approved method.
3. Position the blow lamp on its stable stand or hold it in the designed grip, with the nozzle pointed away from people.
4. Open the fuel valve gradually using the specified startup procedure.
5. Ignite the burner with the approved ignition device or method; never use an improvised ignition source.

6. Adjust the flame to the required stable setting and check for abnormal flame colour, noise, or instability.
7. Direct the flame only at the intended work area and move it steadily to distribute heat as required.
8. Keep the fuel container and hose away from the flame, hot workpiece, sparks, and sharp edges.
9. Monitor the workpiece temperature and surrounding area throughout heating.
10. If the flame goes out, close the fuel valve and follow the manufacturer's relighting procedure after allowing any accumulated gas to disperse safely.
11. When finished, close the fuel valve fully and allow the burner and workpiece to cool in a safe location.
12. Disconnect and store the fuel source only as directed, inspect for leaks, and keep the equipment away from heat and ignition sources.

⚠ Safety

- Only trained users should operate a blow lamp; follow the fuel supplier's and manufacturer's instructions.
- Never use a leaking, damaged, or incorrectly assembled blow lamp.
- Do not heat sealed containers, pressurised vessels, or unknown materials that may release toxic fumes.
- Keep a suitable fire extinguisher nearby and never leave a lit flame unattended.
- Do not use the lamp near flammable liquids, combustible dust, or oxygen-enriched environments.
- Never point the flame toward people, gas cylinders, fuel hoses, or electrical cables.

Care & storage

- Allow the equipment to cool completely before cleaning or storing it.
- Clean the burner nozzle using only the approved method and tools.
- Inspect valves, seals, hoses, and connectors regularly for wear or leakage.
- Store fuel cylinders or containers upright and according to the supplier's safety instructions.
- Remove damaged equipment from service and arrange qualified repair; do not modify the burner or fuel system.

Common mistakes to avoid

- Using the wrong fuel or ignoring a suspected leak.
- Lighting the burner with an unapproved method or opening the valve too quickly.
- Leaving the flame unattended or directing heat toward combustible materials.
- Attempting to relight immediately after flame failure without safely closing the fuel and following the restart procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Brazing Torch

Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Brazing tool; Brazing torch

What it is used for

A brazing torch heats compatible metal parts and filler material so that the filler flows into a joint without melting the base metals. Correct flame adjustment, joint preparation, and ventilation help create reliable joints while reducing fire and burn risks.

Main parts

- Torch body and handle provide controlled gripping and operation.
- Fuel gas connection supplies the selected fuel gas.
- Oxygen or air inlet supports combustion depending on the torch design.
- Hoses carry gases from compatible supply equipment.
- Regulators control gas pressure where fitted.
- Torch tip or nozzle shapes the flame.
- Control valves regulate gas flow to the torch.
- Ignition device or approved striker lights the torch.

Before you start

1. Only trained personnel should operate a brazing torch under the approved workshop procedure.
2. Read the torch and gas supply instructions and confirm fuel, tip, and regulator compatibility.
3. Inspect hoses, connections, regulators, valves, torch body, and nozzle for damage or leaks.
4. Secure gas cylinders upright and away from heat, sparks, and traffic.
5. Clean and fit the joint parts, and select compatible filler metal and flux where required.
6. Remove combustible materials, provide ventilation, and prepare suitable fire protection and PPE.

How to use

1. Check that gas valves are closed before connecting the torch to the approved supply.
2. Connect the correct hoses and regulators and secure all fittings as instructed.
3. Check connections for leaks using an approved leak-detection method before ignition.
4. Set the gas supply and regulator settings according to the torch instructions.
5. Ignite the torch using the approved ignition device and adjust the flame as specified.
6. Heat the joint area evenly and avoid directing the flame toward hoses or cylinders.
7. Apply compatible filler metal and flux as required, allowing the filler to flow into the joint.
8. Stop heating when the joint is filled and avoid overheating the base metal.
9. Close the torch valves and gas supply in the sequence specified by the manufacturer.
10. Allow the joint and torch to cool safely, then inspect the joint and store equipment properly.

⚠ Safety

- Wear suitable eye protection, heat-resistant gloves, protective clothing, and safety footwear.
- Never use a torch with damaged hoses, leaking connections, or faulty valves.
- Keep cylinders upright, secured, and away from hot work and ignition sources.
- Never use oil or grease on oxygen fittings and keep gas equipment clean.
- Provide adequate ventilation and avoid inhaling flux fumes or heated-metal fumes.
- Keep a suitable fire extinguisher available and check the work area for hot spots after brazing.

Care & storage

- Close gas supplies and allow the torch to cool before cleaning or storage.
- Inspect hoses, regulators, valves, and the nozzle regularly for wear or blockage.
- Clean the tip using the recommended tools and avoid enlarging the nozzle opening.
- Store the torch and accessories in a dry, secure area and arrange qualified servicing for faults.

Common mistakes to avoid

- Using incompatible fuel, oxygen, regulator, or torch tip combinations.
- Lighting the torch before checking connections for leaks.
- Overheating the base metal instead of allowing filler metal to flow properly.
- Leaving the gas supply open or the work area unchecked after brazing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Desoldering Pump

Category: **Welding & Soldering** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: De-soldering pump

What it is used for

A desoldering pump removes molten solder from a joint to help release electronic components. Use it with a suitable soldering iron and protect nearby components from heat.

How to use

1. Inspect the pump plunger, nozzle, seal, and release mechanism for damage.
2. Switch off and prepare the soldering iron according to its instructions.
3. Position the pump nozzle near the solder joint without touching nearby components.
4. Heat the joint until the solder melts, avoiding prolonged heating.
5. Trigger the pump to draw molten solder into its chamber.
6. Allow the joint and components to cool before checking or removing the part.
7. Empty and clean the pump according to its design after use.

⚠ Safety

- Wear eye protection and work in a well-ventilated area.
- Keep the hot soldering iron and molten solder away from skin and cables.
- Do not touch the nozzle immediately after use.
- Disconnect power before cleaning or servicing the soldering iron.

Care & storage

- Clean solder residue from the pump and inspect the seal and nozzle regularly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Electrode Holder

Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Welding / Electrode Holder

What it is used for

A welding electrode holder securely grips a coated welding electrode and conducts current from the welding cable to the electrode during manual arc welding. Correct holder selection, insulation checks, secure cable connection, and safe electrode handling help reduce electric shock, burns, and fire hazards.

Main parts

- Insulated handle provides a safe grip.
- Electrode jaws hold the welding electrode.
- Cable connection terminal attaches the welding lead.
- Spring or clamping mechanism maintains electrode contact.
- Insulating covers protect conductive parts where fitted.
- Electrode angle positions the rod for welding.
- Cable lug and fasteners secure the electrical connection.
- Current rating label identifies the holder's approved capacity where provided.

Before you start

1. Only trained welding personnel should use the electrode holder.
2. Inspect the holder, handle insulation, jaws, cable, connector, and cable lug for cracks, burns, loose parts, or exposed conductors.
3. Confirm that the holder's current rating and duty cycle are suitable for the welding machine and process.
4. Check that the welding cable is correctly sized and securely connected with approved fittings.
5. Ensure the work area is dry, ventilated, and clear of flammable materials.
6. Wear a welding helmet with the correct shade, welding gloves, flame-resistant clothing, and safety footwear.
7. Confirm that the welding machine, work return connection, protective earth arrangements, and required safety controls are correctly set up.

How to use

1. Switch off and isolate the welding power source before inspecting or connecting the holder.
2. Attach the welding cable to the holder terminal using the approved connection method and tighten the fasteners securely.
3. Check that no bare conductor is exposed beyond the intended connection and that insulation is intact.

4. Select a compatible electrode and inspect it for damage or moisture according to the welding procedure.
5. Insert the electrode into the jaws at the intended gripping position and ensure it is held firmly.
6. Connect the work return lead to the workpiece or approved work table as specified by the welding setup.
7. Set welding current and polarity according to the electrode and welding procedure.
8. Put on the required PPE and confirm that nearby personnel are protected from arc radiation and sparks.
9. Energise the welding machine and strike the arc using the approved welding technique.
10. Maintain a controlled electrode angle and distance while keeping the holder clear of grounded conductive objects.
11. When the electrode is consumed or needs replacement, place the holder in an insulated holder stand or other designated safe location.
12. Switch off and isolate the welding source before replacing a damaged holder or repairing cable connections.
13. After work, allow the holder to cool, inspect it for damage, and store it in a dry, protected place.

⚠ Safety

- Never use a holder with damaged insulation, exposed conductors, loose jaws, or overheated connections.
- Do not touch the electrode, holder jaws, or electrical connections with wet gloves or wet hands.
- Never place the live holder on a conductive or flammable surface.
- Keep the electrode holder away from water, fuel, gas cylinders, and combustible materials.
- Do not exceed the holder's rated current or duty cycle.
- Disconnect and isolate the welding power source before maintenance or cable repairs; follow the welding machine's safety instructions.

Care & storage

- Inspect the holder and welding lead before each shift or welding session.
- Remove spatter and dirt using an approved method without damaging the insulation.
- Check jaws, springs, terminal fasteners, and insulating covers for wear or loosening.
- Replace damaged insulating parts or the complete holder with an approved compatible unit.
- Store the holder dry and protect it from impact, heat, and corrosive substances.

Common mistakes to avoid

- Using a holder with damaged insulation or an unsuitable current rating.
- Connecting the welding cable with loose or exposed conductors.
- Leaving the live holder on the workpiece or another conductive surface.
- Changing electrodes or repairing connections without isolating the welding power source.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Solder Wire

Category: **Welding & Soldering** Type: **Handling note** Priority: **Low** 5 labs

Covers these lab items: Lead Solder

What it is used for

Solder wire joins compatible electronic or electrical components using controlled heating and an appropriate soldering process.

How to handle

- Check the solder alloy and confirm whether it contains lead before use.
- Use a temperature-controlled soldering iron and work in effective local ventilation.
- Wear suitable eye protection and avoid inhaling solder fumes or touching molten solder.
- Wash hands after handling solder, especially lead-containing material; never eat or drink at the workbench.
- Keep the hot iron in its stand and disconnect it after use.

Storage

- Store solder wire in a labelled, closed container.
- Keep lead-containing solder clearly identified and separate from food or general-use materials.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Soldering Flux

Category: **Welding & Soldering** Type: **Handling note** Priority: **Low** 10 labs

Covers these lab items: Flux; Soldering Flux

What it is used for

Soldering flux removes or reduces surface oxides to support solder wetting and reliable electrical or metal joints.

How to handle

- Identify the flux type and confirm compatibility with the solder, materials, and process.
- Read the label and Safety Data Sheet; wear suitable eye protection and gloves as required.
- Use adequate fume extraction and avoid inhaling vapours or contacting skin and eyes.
- Apply only the required amount and clean residues according to the flux manufacturer's instructions.

Storage

- Keep flux in its original labelled, tightly closed container.
- Store under specified temperature conditions away from heat and incompatible chemicals.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Soldering Iron

Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 21 labs

Covers these lab items: Electric Soldering Iron; Soldering equipment; Soldering Iron; Soldering Iron (khadya); Soldering irons; Soldering kit; Soldering Kit with Accessories; Soldering Kits – Basic and Battery powered

What it is used for

A soldering iron heats solder to join suitable electrical components and wires. Correct temperature control, ventilation, and safe handling help prevent burns and poor connections.

Main parts

- Heated metal tip transfers heat to the joint.
- Heating element produces heat inside the iron.
- Insulated handle helps the user hold the tool safely.
- Power cord or battery supplies energy, depending on the model.
- Temperature control, if fitted, adjusts the tip temperature.
- Stand holds the hot iron when it is not being used.
- Sponge or brass wool helps clean the tip.

Before you start

1. Read the equipment instructions and confirm the iron suits the soldering task.
2. Wear safety goggles and work in a well-ventilated area with suitable fume extraction.
3. Inspect the cord, plug, handle, tip, stand, and temperature control for damage.
4. Place the stand on a stable, heat-resistant surface away from paper and plastics.
5. Prepare suitable solder, compatible flux, and the correct tip for the joint.
6. Secure the circuit or wire and switch off its power before soldering.

How to use

1. Place the soldering iron securely in its stand before connecting power.
2. Set the recommended temperature for the solder, tip, and components being used.
3. Switch on the iron and wait for it to reach working temperature.
4. Clean the tip using the recommended sponge or brass wool.
5. Apply a small amount of suitable solder to tin the clean tip when needed.
6. Heat the component lead and pad or wire joint together.
7. Feed solder into the heated joint rather than directly onto the tip.
8. Use only enough solder to form a neat, wetted connection.
9. Remove solder first, then lift the iron away and allow the joint to cool.
10. Switch off, unplug the iron, and leave it in the stand until fully cool.

⚠ Safety

- Never touch the metal tip or freshly soldered joint; both can cause serious burns.
- Keep the hot iron in its stand and never leave it unattended while powered.
- Use effective ventilation or fume extraction and avoid breathing soldering fumes.
- Keep the tool away from water, flammable materials, and trailing cables.
- Do not solder an energised circuit or a battery connected to the circuit.
- Wash hands after handling solder and flux, especially before eating.

Care & storage

- Switch off and unplug the iron before cleaning or replacing its tip.
- Clean and tin the tip as recommended to reduce oxidation.
- Do not file or scrape plated tips unless the manufacturer specifically permits it.
- Store the fully cooled iron, solder, and accessories in a dry, secure place.

Common mistakes to avoid

- Touching the hot tip to check whether the iron is ready.
- Heating the solder instead of heating the joint surfaces properly.
- Using excessive solder or keeping the iron on components too long.
- Leaving the iron powered on or resting it outside its stand.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Welding Cable

Category: **Welding & Soldering** Type: **Handling note** Priority: **Low** 2 labs

Covers these lab items: Cable & Lugs (Welding cable)

What it is used for

Handle welding cables and lugs safely to maintain electrical continuity and prevent burns or shock.

How to handle

- Inspect cable insulation, connectors, and lugs for wear, looseness, or overheating before use.
- Use cables with the correct current rating and compatible welding equipment.
- Switch off the welding power source before fitting or removing connections.
- Keep cables away from hot metal, sharp edges, water, and vehicle or foot traffic.

Storage

- Coil cables loosely after they have cooled and are safe to handle.
- Store in a dry, ventilated place away from oils, chemicals, and sharp tools.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.